

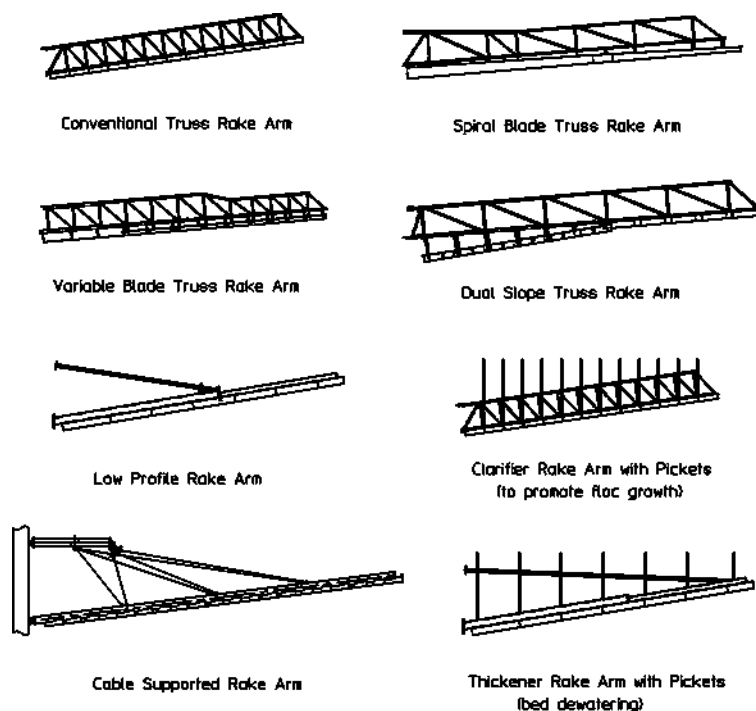


FIG. 18-104 Rake-mechanism designs for specific applications and duties. (Dorr-Oliver EIMCO.)

(25 to 40 ft/min); and for coarse solids or crystalline materials, 12 to 30 m/min (40 to 100 ft/min).

Feedwell The feedwell is designed to allow the feed to enter the thickener with minimum turbulence and uniform distribution while dissipating most of its kinetic energy. Feed slurry enters the feedwell, which is usually located in the center of the thickener, through a pipe or launder suspended from the bridge. To avoid excess velocity, an open launder normally has a slope no greater than 1 to 2 percent. Pulp should enter the launder at a velocity that prevents sanding at the inlet. With nonsanding pulps, the feed may also enter upward through the center column from a pipeline installed beneath the tank.

The standard feedwell for a thickener is designed for a maximum vertical outlet velocity of about 1.5 m/min (5 ft/min). High turbidity caused by short-circuiting the feed to the overflow can be reduced by increasing the depth of the feedwell. When overflow clarity is important or the solids specific gravity is close to the liquid specific gravity, deep feedwells of large diameter are used, and measures are taken to reduce the velocity of the entering feed slurry.

Shallow feedwells may be used when overflow clarity is not important, the overflow rate is low, and/or solids density is appreciably greater than that of water. Some special feedwell designs used to dissipate entrance velocity and create quiescent settling conditions split the feed stream and allow it to enter the feedwell tangentially on opposite sides. The two streams shear or collide with one another to dissipate kinetic energy.

When flocculants are used, often it will be found that the optimum solids concentration for flocculation is considerably less than the normal concentration, and significant savings in reagent cost will be made possible by dilution of the feed prior to flocculation. This can be achieved by recycling overflow or more efficiently by feedwell modifications, including openings in the feedwell rim. These will allow supernatant to enter the feedwell, and flocculant can be added at this point or injected below the surface of the pulp in the feedwell. Another effective means of achieving this dilution prior to flocculant

addition is illustrated in Fig. 18-105. This approach utilizes the energy available in the incoming feed stream to achieve the dilution by momentum transfer and requires no additional energy expenditure to dilute this slurry by as much as three to four times.

Overflow Arrangements Clarified effluent typically is removed in a peripheral launder located inside or outside the tank. The effluent enters the launder by overflowing a V-notch or level flat weir, or through submerged orifices in the bottom of the launder. Uneven overflow rates caused by wind blowing across the liquid surface in large thickeners can be better controlled when submerged orifices or V-notch weirs are used. Radial launders are used when uniform upward liquid flow is desired in order to improve clarifier detention efficiency. This arrangement provides an additional benefit in reducing the effect of wind, which can seriously impair clarity in applications that employ basins of large diameter.

The hydraulic capacity of a launder must be sufficient to prevent flooding, which can cause short-circuiting of the feed and deterioration of overflow clarity. Standards are occasionally imposed on weir overflow rates for clarifiers used in municipal applications; typical rates are 3.5 to 15 m³/(h·m) [7000 to 30,000 gal/(day·ft)], and they are highly dependent on clarifier side-water depth. Industrial clarifiers may have higher overflow rates, depending on the application and the desired overflow clarity. Launders can be arranged in a variety of configurations to achieve the desired overflow rate. Several alternatives to improve clarity include an annular launder inside the tank (the liquid overflows both sides), radial launders connected to the peripheral launder (providing the very long weir that may be needed when abnormally high overflow rates are encountered and overflow clarity is important), and Stamford baffles, which are located below the launder to direct flow currents back toward the center of the clarifier.

In many thickener applications, on the other hand, complete peripheral launders are not required, and no difference in either overflow clarity or underflow concentration will result through the use of launders extending over only a fraction (e.g., one-fifth) of the perimeter.



FIG. 18-105 E-Duc® Feed dilution system installed on a 122-m diameter thickener. (Dorr-Oliver EIMCO.)

For design purposes, a weir-loading rate in the range of 7.5 to 30.0 m³/(h·m) [10 to 40 gpm/ft] can be used, the higher values being employed with well-flocculated, rapidly settling slurries. The overflow launder required may occupy only a single section of the perimeter rather than consisting of multiple, shorter segments spaced uniformly around the tank.

Underflow Arrangements Concentrated solids are removed from the thickener by use of centrifugal or positive displacement pumps or, particularly with large-volume flows, by gravity discharge through a flow control valve or orifice suitable for slurry applications. Due to the risk to the thickener operation of a plugged underflow pipe, it is recommended that duplicate underflow pipes and pumps be installed in all thickening applications. Provision to recycle underflow slurry back to the feedwell is also useful, particularly if solids are to be stored in the thickener.

There are three basic underflow arrangements: (1) the underflow pump adjacent to the thickener sidewall with buried piping from the discharge cone, (2) the underflow pump under the thickeners or adjacent to the sidewall with the piping from the discharge cone in a tunnel, and (3) the underflow pump located in the center of the thickener on the bridge, or using piping up through the center column.

Pump Adjacent to Thickener with Buried Piping This arrangement of buried piping from the discharge cone is the least expensive system but the most susceptible to plugging. It is used only when the solids do not compact to an unpumpable slurry and can be easily backflushed if plugging occurs. Typically, two or more underflow pipes are installed from the discharge cone to the underflow pump so that solids removal can continue if one of the lines plugs. Valves should be installed to permit flushing with water and compressed air in both directions to remove blockages.

Tunnel A tunnel may be constructed under the thickener to provide access to the discharge cone when underflow slurries are difficult to pump and have characteristics that cause plugging. The underflow pump may be installed underneath the thickener or at the perimeter. Occasionally thickeners are installed on legs or piers, making tunneling for access to the center unnecessary. A tunnel or an elevated thickener is more expensive than the other underflow arrangements, but

there are certain operational and maintenance advantages. Of course, the hazards of working in a tunnel (flooding and interrupted ventilation, for example) and related safety regulations must be considered.

Center-Column Pumping This arrangement may be used instead of a tunnel. Several designs are available. One is a bridge-mounted pump with a suction line through a wet or dry center column. The pump selection may be limiting, requiring special attention to priming, net positive suction head, and the maximum density that the pump can handle. Another design has the underflow pump located in a room under the thickener mechanism and connected to openings in the column. Access is through the drive gear at the top of the column.

INSTRUMENTATION

Thickener Thickener control philosophies are usually based on the idea that the underflow density obtained is the most important performance criterion. The overflow clarity is also a consideration, but this is generally not as critical. Additional factors which must be considered are optimization of flocculant usage and protection of the raking mechanism.

Automated control schemes employ one or more sets of controls, which will fit into three categories: (1) control loops which are used to regulate the addition of flocculant, (2) control loops to regulate the withdrawal of underflow, and (3) rake drive controls. Frequently, the feed to a thickener is not controlled and most control systems have been designed with some flexibility to deal with changes in feed characteristics.

Flocculant addition rate can be regulated in proportion to the thickener volumetric feed rate or solids mass flow in a feed-forward mode, or in a feed-back mode on either rake torque, underflow density, settling solids (sludge) bed level, or solids settling rate.

Underflow is usually withdrawn continuously on the bases of bed level, rake torque, or underflow solids concentration in a feedback mode. Some installations incorporate two or more of these parameters in their underflow withdrawal control philosophy. For example, the

continuous withdrawal may be based on underflow solids density with an override to increase the withdrawal rate if either the rake torque or the bed level reaches a preset value. In some cases, underflow withdrawal has been regulated in a feed-forward mode on the basis of thickener feed solids mass flow rate. Any automated underflow pumping scheme should incorporate a lower limit on volumetric flow rate as a safeguard against line pluggage.

It is also important to consider the level of the sludge bed in the thickener. Although this can be allowed to increase or decrease within moderate limits, it must be controlled enough to prevent solids from overflowing the thickener or from falling so low that the underflow density becomes too dilute. The settling slurry within the sludge bed is normally free flowing and will disperse to a consistent level across the thickener diameter.

Rake drive controls protect the drive mechanism from damage and usually incorporate an alarm to indicate high torque with an interlock to shut down the drive at a higher torque level. They can have an automated rake raising and lowering feature with a device to indicate the elevation of the rakes.

A complete automated control scheme incorporates controls from each of the three categories. It is important to consider the interaction of the various controls, especially of the flocculant addition and underflow withdrawal control loops, when designing a system. The lag and dead times of any feedback loops as well as the actual response of the system to changes in manipulated variables must be considered. For example, in some applications it is possible that excessive flocculant addition may produce an increase in the rake torque (due to island formation or viscosity increase) without a corresponding increase in underflow density. Additionally, sludge bed level sensors generally require periodic cleaning to produce a reliable signal. In many cases, it has not been possible to effectively maintain the sludge bed level sensors, requiring a change in the thickener control logic after start-up. Some manufacturers offer complete thickener control packages.

Clarifiers Control philosophies for clarifiers are based on the idea that the overflow is the most important performance criterion. Underflow density or suspended solids content is a consideration, as is optimal use of flocculation and pH control reagents. Automated controls are of three basic types: (1) control loops that optimize coagulant, flocculant, and pH control reagent additions; (2) those that regulate underflow removal; and (3) rake drive controls. Equalization of the feed is provided in some installations, but the clarifier feed is usually not a controlled variable with respect to the clarifier operation.

Automated controls for flocculating reagents can use a feed-forward mode based on feed turbidity and feed volumetric rate, or a feed-back mode incorporating a streaming current detector on the flocculated feed. Attempts to control coagulant addition on the basis of overflow turbidity generally have been less successful. Control for pH has been accomplished by feed-forward modes on the feed pH and by feed-back modes on the basis of clarifier feedwell or external reaction tank pH. Control loops based on measurement of feedwell pH are useful for control in applications in which flocculated solids are internally recirculated within the clarifier feedwell.

Automated sludge withdrawal controls are usually based on the sludge bed level or pressure. These can operate in on-off or continuous modes and can use either single-point or continuous sludge level indication sensors. In many applications, automated control of underflow withdrawal does not provide an advantage, since so few settled solids are produced that it is only necessary to remove sludge for a short interval once a day, or even less frequently. In applications in which the underflow is recirculated internally within the feedwell, it is necessary to maintain sufficient sludge inventory for the recirculation turbine to pull from. This can be handled in an automated system with a single-point *low sludge* bed level sensor in conjunction with a low-level alarm or pump shutoff solenoid. Some applications require continuous external recirculation of the underflow direct to the feedwell or external reaction tanks, and an automated control loop can be used to maintain recirculation based on flow measurement, with a manually adjusted setpoint.

Control philosophies applied to continuous countercurrent decantation (CCD) thickeners are similar to those used for thickeners in

other applications, but have emphasis on maintaining the CCD circuit in balance. It is important to prevent any one of the thickeners from pumping out too fast, otherwise an upstream unit could be starved of wash liquor while at the same time too much underflow could be placed in a downstream unit, disrupting the operation of both units as well as reducing the circuit washing efficiency. Several control configurations have been attempted, and the more successful schemes have linked the solids mass flow rate of underflow pumping to that of the upstream unit or to the CCD circuit solids mass feed rate. Wide variations in the solids feed rate to a CCD circuit will require some means of dampening these fluctuations if design wash efficiency is to be maintained.

The following types of devices are commonly applied to measure the various operational parameters of thickeners and clarifiers. They have been used in conjunction with automatic valves and variable-speed pumps to achieve automatic operation as well as to simply provide local or remote indications.

INSTRUMENTATION AND CONTROLS

Torque Rake torque is an indication of the force necessary to rotate the rakes. Higher rake torque is an indication of higher underflow density or viscosity, deeper mud bed, higher fraction of coarse material, island formation, or heavy scale buildup on the rake arms.

Rake torque measurement is usually provided by the thickener manufacturer as part of the rake drive mechanism. Typical methods involve load cells, motor power measurement, hydraulic pressure, or mechanical displacement against a spring. Torque-measuring devices are designed to produce a signal that may be utilized for alarming or control.

Rake Height Rake lifting devices are used to minimize the torque on the arms by lifting them out of heavy bed solids and enable the rake to continue running during upset conditions. It is desirable to prevent the rake drives from running for extended periods at torques above 50 to 60 percent, to prevent accelerated wear on the drive. Lifting the rakes a small distance is usually effective in relieving the pressure on the rakes, thus reducing the torque. Because of this, in using "torque indication" in a control strategy one must also consider the rake height to effectively control the thickener. The two most common rake height indicators are the ultrasonic or potentiometer type with a reeling cable. Lifting of the rakes allows a short period of time to make corrections before one is forced to shut down the thickener.

Bed Level There are several general types of bed level detection instruments: ultrasonic, nuclear, float and rod, and reeling (with various sensors). Each has advantages and disadvantages, which are discussed below. There is not a standard bed level sensor that is recommended for all applications.

- **Ultrasonic** bed level sensors work by sending a pulse down from just under the surface, which bounces off the bed and back to the receiver. Elapsed time is used to calculate the distance. Advantages are noninterfering location, measurement over a large span, and relatively low cost. The downside is that they do not work on all applications. If the overflow is cloudy, it can interfere with the transmission or cause too much reflection to give a reliable signal. Sealing affects accuracy and can cause drifting or loss of signal. Using them on concentrate thickeners has proved to be particularly troublesome.
- **Nuclear** bed level sensors work by sensing either the background radiation level or attenuation between a source and detector, depending on whether the solids have a natural background radiation level. The sensor is comprised of a long rod that extends down into the bed with radiation detectors spaced along the length. If the ore changes from not having radiation to having it, there will be problems. The advantages are that it is relatively reliable when properly applied. The downside is that it measures over a limited range, may interfere with rakes (a hinged version that will swing out of the way when the rakes pass by is available), and is relatively expensive.
- **Float and rod** types work with a ball with a hollow sleeve that slides up and down on a rod that extends down into the bed. The ball weight can be adjusted to float on top of the bed of solids.

TABLE 18-7 Typical Thickener and Clarifier Design Criteria and Operating Conditions

	Percent solids		Unit area, m ² /(t/d)*†	Overflow rate, m ³ /(m ² ·h)*
	Feed	Underflow		
Alumina Bayer process				
Red mud, primary	3–4	10–25	2–5	
Red mud, washers	6–8	15–35	1–4	
Hydrate, fine or seed	1–10	20–50	1.2–3	0.07–0.12
Brine purification	0.2–2.0	8–15		0.5–1.2
Coal, refuse	0.5–6	20–40	0.5–1	0.7–1.7
Coal, heavy-media (magnetic)	20–30	60–70	0.05–0.1	
Cyanide, leached-ore	16–33	40–60	0.3–1.3	
Flue dust, blast-furnace	0.2–2.0	40–60		1.5–3.7
Flue dust, BOF	0.2–2.0	30–70		1–3.7
Flue-gas desulfurization sludge	3–12	20–45	0.3–3†	
High-density paste thickeners				
Red mud, washers†	3–4	45–50	0.05–0.08	
Coal, refuse†	6–8	50–54	0.08–0.1	
Cyanide, leached ore†	10–15	65–70	0.05–0.08	
Copper tailings†	10–20	50–75	0.07–0.15	
Tailings (magnetic)†	10–20	60–75	0.07–0.1	
Tailings (nonmagnetic)†	10–20	60–70	0.07–0.1	
Magnesium hydroxide from brine	8–10	25–40	5–10	
Magnesium hydroxide from seawater	1–4	15–20	3–10	0.5–0.8
Metallurgical				
Copper concentrates	14–50	40–75	0.2–2	
Copper tailings	10–30	45–65	0.4–1	
Iron ore				
Concentrate (magnetic)	20–35	50–70	0.01–0.08	
Concentrate (nonmagnetic), coarse: 40–65% –325	25–40	60–75	0.02–0.1	
Concentrate (nonmagnetic), fine: 65–100% –325	15–30	60–70	0.15–0.4	
Tailings (magnetic)	2–5	45–60	0.6–1.5	1.2–2.4
Tailings (nonmagnetic)	2–10	45–50	0.8–3	0.7–1.2
Lead concentrates	20–25	60–80	0.5–1	
Molybdenum concentrates	10–35	50–60	0.2–0.4	
Nickel, (NH ₄) ₂ CO ₃ leach residue	15–25	45–60	0.3–0.5	
Nickel, acid leach residue	20	60	0.8	
Zinc concentrates	10–20	50–60	0.3–0.7	
Zinc leach residue	5–10	25–40	0.8–1.5	
Municipal waste				
Primary clarifier	0.02–0.05	0.5–1.5		1–1.7
Thickening				
Primary sludge	1–3	5–10	8	
Waste-activated sludge	0.2–1.5	2–3	33	
Anaerobically digested sludge	4–8	6–12	10	
Phosphate slimes	1–3	5–15	1.2–18	
Pickle liquor and rinse water	1–8	9–18	3.5–5	
Plating waste	2–5	5–30		1.2
Potash slimes	1–5	6–25	4–12	
Potato-processing waste	0.3–0.5	5–6		1
Pulp and paper				
Green-liquor clarifier	0.2	5		0.8
White-liquor clarifier	8	35–45	0.8–1.6	
Kraft waste	0.01–0.05	2–5		0.8–1.2
Deinking waste	0.01–0.05	4–7		1–1.2
Paper-mill waste	0.01–0.05	2–8		1.2–2.2
Sugarcane defecation			0.5§	
Sugar-beet carbonation	2–5	15–20	0.03–0.07†	
Uranium				
Acid-leached ore	10–30	25–65	0.02–1	
Alkaline-leached ore	20	60	1	
Uranium precipitate	1–2	10–25	5–12.5	
Water treatment				
Clarification (after 30-min flocculation)				1–1.3
Softening lime-soda (high-rate, solids-contact clarifiers)				3.7
Softening lime-sludge	5–10	20–45	0.6–2.5	

*m²/(t/d) × 9.76 = ft²/(short ton/day); m³/(m²·h) × 0.41 = gal/(ft²·min); 1 t = 1 metric ton.

†High-rate thickeners using required flocculant dosages operate at 10 to 50 percent of these unit areas.

‡Typical design using high Density/Paste thickeners. Feed per cent solids are that diluted for flocculation.

§Basis: 1 t of cane or beets.

These are subject to fouling and sticking, and can be installed and measured only in the area above the rakes; however, they are relatively inexpensive.

- **Reeling devices** work by dropping a sensor down on a cable and sensing the bed level by optical or conductivity sensors. In theory they are nonfouling and get out of the way of the rakes, but in

practice, they frequently become entangled with the rakes. The price is mid-range. Freezing wind and cold temperatures can lead to icing problems.

- **Vibrating or tuning fork sensors** are designed to sense a difference in the vibrating frequency in different masses of solids. These are used in Europe and Africa.

- **Bubble tube or differential pressure** is an old, but tried and true, method of bed level detection. There may be some plugging or fouling of the tube over time.
- **External density** through sample ports. Slurry samples are taken from nozzles on the side of the tank and pass through a density meter to determine the presence of solids. This system can be set up with automated valves to measure several different sample points. This system requires external piping and disposal of the sample stream. Line pluggage is often a problem.

Bed Pressure Because thickeners maintain a constant liquid level, the pressure at the bottom of the thickener is an indication of the overall specific gravity in the tank. If the liquor specific gravity is constant, the overall specific gravity is an indication of the amount of solids in the tank and can be converted to a rough solids inventory. This can be a very effective tool for thickener control. Because of relative height-to-diameter ratios, it is considered less useful for large-diameter thickeners.

Differential pressure sensors are used to measure the bed pressure, leaving one leg open to the atmosphere to compensate for barometric pressure variations. Care must be taken in the installation to minimize the plugging with solids. This is normally done by tilting the tank nozzle on which the DP cell is mounted downward from the sensor, so the solids tend to settle away from the sensor surface. A shutoff valve and washer flush tap are recommended to allow easy maintenance.

Flow Rate Flow rates for feed and underflow lines are useful, particularly when combined with density measurements to generate solids mass flow rates. Since flocculant is usually dosed on a solids mass basis, knowing the mass flow rate is very useful for flocculant control, providing a fast response system. Flow rate measurement is an absolute necessity for the newer generation of ultrahigh-rate and ultrahigh-density thickeners. The streams being measured are usually slurries, and the flow rate is usually measured by either magnetic flowmeters or Doppler-type flowmeters. As long as these instruments are properly installed in suitable full straight pipe sections, avoiding air if possible, they are accurate and reliable. If the feed stream is in an open launder, flow measurement is more difficult but can be accomplished using ultrasonic devices.

Density Nuclear gauges are the norm for density measurement. Nuclear density instruments require nuclear handling permits in most countries. Note that there are now some types that use very low level sources that do not require nuclear licensing. Density gauges should be recalibrated regularly as they are subject to drifting. Small flow applications may be able to use a coriolis meter to measure both mass and percent solids with one instrument.

Settling Rate The settling rate in the feed well is a good indication of the degree of flocculation, and it can be used to maintain consistent flocculation over widely varying feed conditions. A settleometer is a device that automatically pulls a sample from the feed well and measures the settling rate. The flocculant can then be adjusted to maintain a constant settling rate.

Overflow Turbidity Overflow turbidity can be used to control flocculant or coagulant. There may be some significant lag time between the actual flocculation process and when the clarified liquor reaches the overflow discharge point where the sensor is typically positioned. These sensors and meters are generally used as alarms or for trim only.

Note: It is critical that all instruments be well maintained and serviced on a regular basis in order to get the best results.

CONTINUOUS COUNTERCURRENT DECANTATION

The system of separation of solid-phase material from an associated solution by repeated stages of dilution and gravity sedimentation is adapted for many industrial-processing applications through an operation known as continuous countercurrent decantation (CCD). The flow of solids proceeds in a direction countercurrent to the flow of solution diluent (water, usually), with each stage composed of a mixing step followed by settling of the solids from the suspension. The number of stages ranges from 2 to as many as 10, depending on the degree of separation required, the amount of wash fluid added

(which influences the final solute concentration in the first-stage overflow), and the underflow solids concentration attainable. Applications include processes in which the solution is the valuable component (as in alumina extraction), or in which purified solids are sought (magnesium hydroxide from seawater), or both (as frequently encountered in the chemical-processing industry and in base-metal hydrometallurgy).

The factors which may make CCD a preferred choice over other separation systems include the following: rapidly settling solids, assisted by flocculation; relatively high ratio of solids concentration between underflow and feed; moderately high wash ratios allowable (2 to 4 times the volume of liquor in the thickened underflows); large quantity of solids to be processed; and the presence of fine-size solids that are difficult to concentrate by other means. A technical feasibility and economic study is desirable in order to make the optimum choice.

Flow-Sheet Design Thickener-sizing tests, as described earlier, will determine unit areas, flocculant dosages, and underflow densities for the various stages. For most cases, unit areas will not vary significantly throughout the circuit; similarly, underflow concentrations should be relatively constant. In practice, the same unit area is generally used for all thickeners in the circuit to simplify construction. Serious consideration should be given to the design underflow density since operating at the higher, manageable densities will offer the benefits of improved wash efficiency. Many CCD installations, alumina in particular, have installed paste thickeners and reduced the number of stages or lowered the required volume of wash water.

Equipment The equipment selected for CCD circuits may consist of multiple-compartment washing-tray thickeners or a train of individual unit thickeners. The washing-tray thickener consists of a vertical array of coaxial trays connected in series, contained in a single tank. The advantages of this design are smaller floor-area requirements, less pumping equipment and piping, and reduced heat losses in circuits operating at elevated temperatures. However, operation is generally more difficult, and user preference has shifted toward ultrahigh-rate and ultrahigh-density thickeners.

Underflow Pumping Diaphragm pumps with open discharge are employed in some low-volume cases, primarily because underflow densities are readily controlled with these units. Disadvantages include the generally higher maintenance and initial costs than for other types and their inability to transfer the slurry any great distance. Large flows often are best handled with variable-speed, rubber-lined centrifugal pumps, utilizing automatic control to maintain the underflow rate and density.

Overflow Pumps These can be omitted if the thickeners are located at increasing elevations from first to last so that overflows are transferred by gravity or if the mixture of underflow and overflow is to be pumped. Overflow pumps are necessary, however, when maximum flexibility and control are sought.

Interstage Mixing Efficiencies Mixing or stage efficiencies rarely achieve the ideal 100 percent, in which solute concentrations in overflow and underflow liquor from each thickener are identical. Part of the deficiency is due to insufficient blending of the two streams, and attaining equilibrium will be hampered further by heavily flocculated solids. In systems in which flocculants are used, interstage efficiencies often will drop gradually from first to last thickener, and typical values will range from 98 percent to as low as 70 percent. In some cases, operators will add the flocculant to an overflow solution which is to be blended with the corresponding underflow. While this is very effective for good flocculation, it can result in reflocculation of the solids before the entrained liquor has had a chance to blend completely with the overflow liquor. The preferable procedure is to recycle a portion of the overflow back to the feed line of the same thickener, adding the reagent to this liquor.

The usual method of interstage mixing consists of a relatively simple arrangement in which the flows from preceding and succeeding stages are added to a feed box at the thickener periphery. A nominal detention time in this mixing tank of 30 to 60 s and sufficient energy input to avoid solids settling will ensure interstage efficiencies greater than 95 percent.

The performance of a CCD circuit can be estimated through use of the following equations, which assume 100 percent stage efficiency:

$$R = \frac{O/U[(O/U)^N - 1]}{[(O/U)^{N+1} - 1]} \quad (18-52)$$

$$R = 1 - \left(\frac{U}{O'}\right)^N \quad (18-53)$$

for O/U and $U/O' \neq 1$. R is the fraction of dissolved value in the feed which is recovered in the overflow liquor from the first thickener, O and U are the overflow and underflow liquor volumes per unit weight of underflow solids, and N is the number of stages. Equation (18-49) applies to a system in which the circuit receives dry solids with which the second-stage thickener overflow is mixed to extract the soluble component. In this instance, O' refers to the overflow volume from the thickeners following the first stage.

For more precise values, computer programs can be used to calculate soluble recovery as well as solution compositions for conditions that are typical of a CCD circuit, with varying underflow concentrations, stage efficiencies, and solution densities in each of the stages. The calculation sequence is easily performed by utilizing material-balance equations around each thickener.

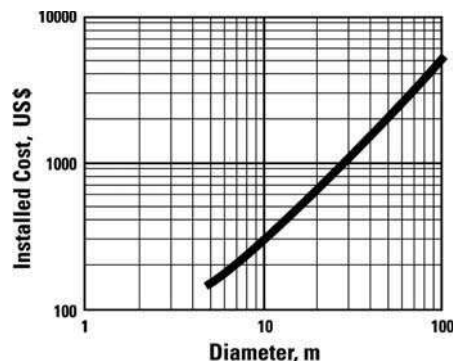


FIG. 18-106 Approximate installed cost of single-compartment thickeners (2005 US \$).

DESIGN SIZING CRITERIA

Table 18-7 has the typical design sizing criteria and operating conditions for a number of applications. It is presented for purposes of illustration or preliminary estimate. Actual thickening and classification performance is dependent on particle-size distribution, specific gravity, sludge bed compaction characteristics, and other factors. Final design should be based on bench scale tests.

THICKENER COSTS

Equipment Costs vary widely for a given diameter because of the many types of construction. As a general rule, the total installed cost will be about 3 to 4 times the cost of the raking mechanism (including drivehead and lift), plus walkways and bridge or centerpier cage, railings, and overflow launders. Figure 18-106 shows the approximate installed costs of thickeners up to 107 m (350 ft) in diameter. These costs are to be used only as a guide. They include the erection of mechanism and tank plus normal uncomplicated site preparation, excavation, reinforcing bar placement, backfill, and sur-

veying. The price does not include any electrical work, pumps, piping, instrumentation, walkways, or lifting mechanisms. Special design modifications, which are not in the price, could include elevated tanks (for underflow handling); special feedwell designs to control dilution, entrance velocity, and turbulence; electrical and drive enclosures required because of climatic conditions; and mechanism designs required because of scale buildup tendencies.

Operating Costs Power cost for a continuous thickener is an almost insignificant item. For example, a unit thickener 60 m (200 ft) in diameter with a torque rating of 1.0 MN·m (8.8 Mlbf·in) will normally require 12 kW (16 hp). The low power consumption is due to the very slow rotative speeds. Normally, a mechanism will be designed for a peripheral speed of about 9 m/min (0.5 ft/s), which corresponds to only 3 r/h for a 60-m (200-ft) unit. This low speed also means very low maintenance costs. Operating labor is low because little attention is normally required after initial operation has balanced the feed and underflow. If chemicals are required for flocculation, the chemical cost frequently dwarfs all other operating costs.

FILTRATION

GENERAL REFERENCES: Moir, *Chem. Eng.*, **89**(15), 46 (1982). Brown, *ibid.*, 58; also published as McGraw-Hill Repr. A078. Cheremisinoff and Azbel, *Liquid Filtration*, Ann Arbor Science, Woburn, Mass., 1983. Orr (ed.), *Filtration: Principles and Practice*, part I, Marcel Dekker, New York, 1977; part II, 1979. Purchas (ed.), *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977. Schweitzer (ed.), *Handbook of Separation Techniques for Chemical Engineers*, part 4, McGraw-Hill, New York, 1979. Shoemaker (ed.), "What the Filter Man Needs to Know about Filtration," *Am. Inst. Chem. Eng. Symp. Ser.*, **73**(171), (1977). Talcott et al., in *Kirk-Othmer Encyclopedia of Chemical Technology*, 3d ed., vol. 10, Wiley, New York, 1980, p. 284. Tiller et al., *Chem. Eng.*, **81**(9), 116-136 (1974); also published as McGraw-Hill Repr. R203.

DEFINITIONS AND CLASSIFICATION

Filtration is the separation of a fluid-solids mixture involving passage of most of the fluid through a porous barrier which retains most of the solid particulates contained in the mixture. This subsection deals only with the filtration of solids from liquids; gas filtration is treated in Sec. 17. **Filtration** is the term for the unit operation. A filter is a piece of unit-operations

equipment by which filtration is performed. The **filter medium** or **septum** is the barrier that lets the liquid pass while retaining most of the solids; it may be a screen, cloth, paper, or bed of solids. The liquid that passes through the filter medium is called the **filtrate**.

Filtration and filters can be classified several ways:

1. **By driving force.** The filtrate is induced to flow through the filter medium by hydrostatic head (gravity), pressure applied upstream of the filter medium, vacuum or reduced pressure applied downstream of the filter medium, or centrifugal force across the medium. Centrifugal filtration is closely related to centrifugal sedimentation, and both are discussed later under "Centrifuges."

2. **By filtration mechanism.** Although the mechanism for separation and accumulation of solids is not clearly understood, two models are generally considered and are the basis for the application of theory to the filtration process. When solids are stopped at the surface of a filter medium and pile upon one another to form a cake of increasing thickness, the separation is called **cake filtration**. When solids are trapped within the pores or body of the medium, it is termed **depth, filter-medium, or clarifying filtration**.

3. *By objective.* The process goal of filtration may be dry solids (the cake is the product of value), clarified liquid (the filtrate is the product of value), or both. Good solids recovery is best obtained by cake filtration, while clarification of the liquid is accomplished by either depth or cake filtration.

4. *By operating cycle.* Filtration may be intermittent (batch) or continuous. Batch filters may be operated with constant-pressure driving force, at constant rate, or in cycles that are variable with respect to both pressure and rate. Batch cycle can vary greatly, depending on filter area and solids loading.

5. *By nature of the solids.* Cake filtration may involve an accumulation of solids that is compressible or substantially incompressible, corresponding roughly in filter-medium filtration to particles that are deformable and to those that are rigid. The particle or particle-aggregate size may be of the same order of magnitude as the minimum pore size of most filter media (1 to 10 μm and greater), or may be smaller (1 μm down to the dimension of bacteria and even large molecules). Most filtrations involve solids of the former size range; those of the latter range can be filtered, if at all, only by filter-medium-type filtration or by ultrafiltration unless they are converted to the former range by aggregation prior to filtration.

These methods of classification are not mutually exclusive. Thus filters usually are divided first into the two groups of cake and clarifying equipment, then into groups of machines using the same kind of driving force, then further into batch and continuous classes. This is the scheme of classification underlying the discussion of filters of this subsection. Within it, the other aspects of operating cycle, the nature of the solids, and additional factors (e.g., types and classification of filter media) will be treated explicitly or implicitly.

FILTRATION THEORY

While research has developed a significant and detailed filtration theory, it is still so difficult to define a given liquid-solid system that it is both faster and more accurate to determine filter requirements by performing small-scale tests. Filtration theory does, however, show how the test data can best be correlated, and extrapolated when necessary, for use in scale-up calculations.

In cake or surface filtration, there are two primary areas of consideration: continuous filtration, in which the resistance of the filter cake (deposited process solids) is very large with respect to that of the filter media and filtrate drainage, and batch pressure filtration, in which the resistance of the filter cake is not very large with respect to that of the filter media and filtrate drainage. Batch pressure filters are generally fitted with heavy, tight filter cloths plus a layer of precoat and these represent a significant resistance that must be taken into account. Continuous filters, except for precoats, use relatively open cloths that offer little resistance compared to that of the filter cake.

Simplified theory for both batch and continuous filtration is based on the time-honored Hagen-Poiseuille equation:

$$\frac{1}{A} \frac{dV}{d\Theta} = \frac{P}{\mu(\alpha w V/A + r)} \quad (18-54)$$

where V is the volume of filtrate collected, Θ is the filtration time, A is the filter area, P is the total pressure across the system, w is the weight of cake solids/unit volume of filtrate, μ is the filtrate viscosity, α is the cake-specific resistance, and r is the resistance of the filter cloth plus the drainage system.

CONTINUOUS FILTRATION

Since testing and scale-up are different for batch and continuous filtration, discussion in this section will be limited to continuous filtration.

It is both convenient and reasonable in continuous filtration, except for precoat filters, to assume that the resistance of the filter cloth plus filtrate drainage is negligible compared to the resistance of the filter cake and to assume that both pressure drop and specific cake resistance remain constant throughout the filter cycle. Equation (18-54), integrated under these conditions, may then be manipulated to give the following relationships:

$$W = \sqrt{\frac{2wP\Theta_f}{\mu(\alpha w V/A + r)}} \quad (18-55)$$

$$V_f = \sqrt{\frac{2P\Theta_f}{\mu\alpha w}} \quad (18-56)$$

$$\Theta_w = \frac{WV_w\mu\alpha}{P_w} \quad (18-57)$$

$$\Theta_w \propto NW^2 \quad (18-58)$$

$$\frac{\Theta_w}{\Theta_f} = 2 \frac{V_w}{V_f} \quad (18-59)$$

where W is the weight of dry filter cake solids/unit area, V_f is the volume of cake formation filtrate/unit area, V_w is the volume of cake wash filtrate/unit area, Θ_f is the cake formation time, Θ_w is the cake wash time, and N is the wash ratio, the volume of cake wash/volume of liquid in the discharged cake.

As long as the suspended solids concentration in the feed remains constant, these equations lead to the following convenient correlations:

$$\log W \text{ vs. } \log \Theta_f \quad (18-60)$$

$$\log V_f \text{ vs. } \log \Theta_f \quad (18-61)$$

$$\Theta_w \text{ vs. } WV_w \quad (18-62)$$

$$\Theta_w \text{ vs. } NW^2 \quad (18-63)$$

$$\Theta_{w/f} \text{ vs. } V_w/V_f \quad (18-64)$$

There are two other useful empirical correlations as follows:

$$W \text{ vs. cake thickness} \quad (18-65)$$

$$\log R \text{ vs. } N \quad (18-66)$$

where R is percent remaining—the percent of solute in the unwashed cake that remains after washing.

FACTORS INFLUENCING SMALL-SCALE TESTING

[Purchas (ed.), *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977.]

Vacuum or Pressure The vast majority of all continuous filters use vacuum to provide the driving force for filtration. However, if the feed slurry contains a highly volatile liquid phase, or if it is hot, saturated, and/or near the atmospheric pressure boiling point, the use of pressure for the driving force may be required. Pressure filtration might also be used where the required cake moisture content is lower than that obtainable with vacuum.

The objective of most continuous filters is to produce a dry or handleable cake. Most vacuum filters easily discharge a “dry” consolidated cake as they are usually operated in an open or semiopen environment. However, whenever the filter must operate under pressure or within a vapor-tight enclosure, either because of the need for a greater driving force or because of the vapor pressure of the liquid phase, a dry cake discharge becomes difficult. The problem of removing a dry cake from a pressurized enclosure has precluded the use of continuous-pressure filters in many cases where this is a requirement. Applications which do discharge dry cake from a “sealed” enclosure are restricted to relatively dry, friable cakes that will “flow” through double valves which form a “vapor lock.”

Cake Discharge For any filter application to be practical, it must be possible to produce a cake thick enough to discharge. Table 18-8 tabulates the minimum acceptable cake thickness required for discharge for various types of filters and discharge mechanisms. The experimenter, when running small-scale tests, should decide early in the test program which type of discharge is applicable and then tailor the data collected to fit the physical requirements of that type of unit. Note, however, that the data correlations recommended later are sufficiently general in nature to apply to most equipment types.

TABLE 18-8 Minimum Cake Thickness for Discharge

Filter type	Minimum design thickness	
	mm	in
Drum		
Belt	3-5	$\frac{1}{8}$ - $\frac{3}{16}$
Roll discharge	1	$\frac{1}{32}$
Std. scraper	6	$\frac{1}{4}$
Coil	3-5	$\frac{1}{8}$ - $\frac{3}{16}$
String discharge	6	$\frac{1}{4}$
Precoat	0-3 max.	0- $\frac{1}{8}$ max.
Horizontal belt	3-5	$\frac{1}{8}$ - $\frac{3}{16}$
Horizontal table	20	$\frac{3}{4}$
Tilting pan	20-25	$\frac{3}{4}$ -1
Disc	10-13	$\frac{3}{8}$ - $\frac{1}{2}$

Feed Slurry Temperature Temperature can be both an aid and a limitation. As temperature of the feed slurry is increased, the viscosity of the liquid phase is decreased, causing an increase in filtration rate and a decrease in cake moisture content. The limit to the benefits of increased temperature occurs when the vapor pressure of the liquid phase starts to materially reduce the allowable vacuum. If the liquid phase is permitted to flash within the filter internals, various undesired results may ensue: disruption in cake formation adjacent to the medium, scale deposit on the filter internals, a sharp rise in pressure drop within the filter drainage passages due to increased vapor flow, or decreased vacuum pump capacity. In most cases, the vacuum system should be designed so that the liquid phase does not boil.

In some special cases, steam filtration can be used to gain the advantages of temperature without having to heat the feed slurry. Where applicable, dry steam is passed through the deliquored cake to raise the temperature of the residual moisture, reduce its viscosity, and lower its content. The final drying or cooling period which follows steam filtration uses the residual heat left in the cake to evaporate some additional moisture.

Cake Thickness Control Sometimes the rate of cake formation with bottom feed-type filters is rapid enough to create a cake too thick for subsequent operations. Cake thickness may be controlled by adjusting the bridge-blocks in the filter valve to decrease the effective submergence, by reducing the slurry level in the vat, and by reducing the vacuum level in the cake formation portion of the filter valve. If these measures are inadequate, it may be necessary to use a top-loading filter.

Cake thickness must frequently be restricted when cake washing is required or the final cake moisture content is critical. Where the time required for cake washing is the rate-controlling step in the filter cycle, maximum filtration rate will be obtained when using the minimum cake thickness that gives good cake discharge. Where minimum cake moisture content is the controlling factor, there is usually some leeway with respect to cake thickness, although the minimum required for cake discharge is controlling in some cases. Since a relatively constant quantity of moisture is transferred from the medium to the filter cake when the vacuum is released prior to cake discharge, very thin cakes will sometimes be wetter than thicker cakes.

The effect of an increase in cake thickness on the time required for washing is easy to see if one considers what happens when the cake thickness is doubled. Assume that two cakes have the same permeability and that the quantity of washing fluid to cake solids is to remain constant. Doubling of the cake thickness doubles the resistance to flow of the washing fluid through the cake. At the same time, the quantity of washing fluid per unit area is also doubled. Thus, the time required for the washing fluid to pass through the cake is increased by the square of the ratio of the cake thicknesses. In this particular example, the washing time would be increased by a factor of four, while cake production would only be doubled.

Filter Cycle Each filter cycle is composed of cake formation plus one or more of the following operations: deliquoring (dewatering or drying), washing, thermal drying, steam drying, and cake discharge. The number of these operations required by a given filtration operation depends upon the process flowsheet. It is neither possible nor necessary to consider all of these operations at once. The basic testing program is designed to look at each operation individually. The requirements for each of the steps are then fit into a single filter cycle.

All filters utilizing a rotary filter valve have their areas divided into a number of sections, sectors, or segments (see Fig. 18-134). When a drainage port passes from one portion of the filter valve to another, the change at the filter medium does not occur instantaneously nor does it occur at some precise location on the filter surface. The change is relatively gradual and occurs over an area, as the drainage port at the filter valve first closes by passing onto a stationary bridge-block and then opens as it passes off that bridge-block on the other side.

On a horizontal belt filter, the equivalent sections extend across the filter in narrow strips. Therefore, changes in vacuum do occur rapidly and may be considered as happening at a particular point along the length of the filter.

Representative Samples The results which are obtained in any bench-scale testing program can be only as good as the sample which is tested. It is absolutely essential that the sample used be representative of the slurry in the full-scale plant and that it be tested under the conditions that prevail in the process. If there is to be some significant time between taking or producing the sample and commencing the test program, due consideration must be given to what effect this time lapse may have on the characteristics of the slurry. If the slurry is at a temperature different from ambient, the subsequent heating and/or cooling could change the particle size distribution. Even sample age itself may exert a significant influence on particle size. If there is likely to be an effect, the bench-scale testing program should be carried out at the plant or laboratory site on fresh material.

Whenever a sample is to be held for some time or shipped to a distant laboratory for testing, some type of characterizing filtration test should be run on the fresh sample and then duplicated at the time of the test program. A comparison of the results of the two tests will indicate how much of a change there has been in the sample. If the change is too great, there would be no point in proceeding with the tests, and it would be necessary to make arrangements to work on a fresh sample. Any shipped sample, especially during the winter months, must be protected from freezing, as freezing can substantially change the filtration characteristics of a slurry, particularly hydrated materials.

The slurry should always be defined as completely as possible by noting suspended solids concentration, particle size distribution, viscosity, density of solids and liquid, temperature, chemical composition, and so on.

Feed Solids Concentration Feed slurries that are so dilute that they settle rapidly usually yield reduced solids filtration rates and produce stratified cakes with higher moisture contents than would normally be obtained with a homogeneous cake. It is well known that an increase in feed solids concentration is generally an effective means of increasing solids filtration rate, assisting in forming a homogeneous suspension and thereby minimizing cake moisture content, and so on. Equipment required to concentrate a slurry sample and the tests needed to predict how far a slurry will thicken are discussed elsewhere in this section.

Pretreatment Chemicals Even though the suspended solids concentration of the slurry to be tested may be correct, it is frequently necessary to modify the slurry in order to provide an acceptable filtration rate, washing rate, or final cake moisture content. The most common treatment, and one which may provide improvement in all three of these categories, is the addition of flocculating agents, either inorganic chemicals or natural or synthetic polymers. The main task at this point is to determine which is the most effective chemical and the quantity of chemical which should be used.

It is usually difficult to observe visually a change in floc structure in a concentrated slurry. The two best indications that an effective quantity of chemical has been added is a sudden thickening or increase in viscosity of the slurry and the formation of *riverlets* on the surface of a spatula when treated slurry is shaken from it. It is generally necessary to exceed a threshold quantity of chemical before there is a measurable improvement. The proper dosage becomes an economic balance between the cost of additional chemicals and the savings resulting from a reduction in filter area.

Screening tests are used to determine the best chemical and its approximate dosage. It is usually convenient to use small, graduated beakers and sample quantities in the range of 50 to 150 mL. The chemical may be added with a syringe or medicine dropper and a note made of the quantity used, together with the results. The experimenter should filter and wash the flocculated sample on any convenient, small,

top-loading filter with good filtrate drainage. The cake formation and wash times obtained from these micro tests are not intended to provide sizing data, but they do provide an excellent indication of the relative effectiveness of various chemicals and treatment levels.

With any chemical treatment system, the main task is one of getting the chemical thoroughly mixed with the solids without degrading the flocs which are formed. For those slurries that are relatively fluid, the chemical can frequently be added and mixed satisfactorily using a relatively wide spatula. However, for those thick, relatively viscous slurries, a power mixer will be required. In this case, the mixer should be stopped about one second after the last of the flocculant is added. Should this approach be required, it means that a suitably designed addition system must be supplied with the full-scale installation in order to do an effective job of flocculation.

While the volume of chemical used should be minimized, the experimenter must use good judgment based on the viscosities of both the slurry to be treated and the chemical used. If both are relatively viscous, use of a power mixer is indicated.

There are a number of commercially available surfactants that can be employed as an aid in filter cake moisture reduction. These reagents can be added to the filter feed slurry or to the filter cake wash water, if washing is used. Since these reagents have a dispersing effect, flocculation may be required subsequently. Typical moisture reductions of 2 to 4 percentage points are obtained at reagent dosages of 200 to 500 g/m² of solids.

Cloth Blinding Continuous filters, except for precoat, generally use some type of medium to effect the separation of the solid and filtrate phases. Since the medium is in contact with the process solids, there is always the danger, and almost invariably the actual occurrence, of medium blinding. The term **blinding** refers to blockage of the fabric itself, either by the wedging of process solids or by solids precipitated in and around the yarn.

The filter medium chosen should be as open as possible yet still able to maintain the required filtrate clarity. Those fabrics which will produce a clear filtrate and yet do not have rapid blinding tendencies are frequently light in weight (woven from thin filaments or yarn) and will not wear as long as some of the heavier, more open fabrics (woven from heavy filaments or yarn). Whenever the filter follows a gravity thickening or clarification step, it is advisable to return the filtrate to the thickener or clarifier so that the filtrate clarity requirements may be relaxed in favor of using a heavier, more open cloth with reduced blinding tendencies. Excessively dirty filtrates should be avoided as the solids may be abrasive and detrimental to the internals of the filter or perhaps may cut the fabric yarn.

It should be noted at this point that an **absolutely** clear filtrate can rarely be obtained on a cloth-covered continuous filter. The passages through the medium are invariably larger than some of the solids in the slurry, and there will be some amount of solids passing through the medium. Once the pores of the fabric have been bridged, the solids themselves form the septum for the remaining particles, and the filtrate becomes clear. It is this bridging action of the solids that permits the use of a relatively open filter medium, while at the same time maintaining a reasonably clear filtrate.

Filters with media in the form of an endless belt have greatly reduced the concern about blinding. Most synthetic fabrics can be successfully cleaned of process solids by washing the medium after cake discharge, and the rate of blinding due to chemical precipitation also can be drastically reduced. Current practice suggests that the belt-type filter with continuous-medium washing be the first choice unless experience has shown that medium blinding is not a factor or if the belt-type system cannot be successfully applied.

Sealing of the belt along the edges of the filter drum is never perfect, and some leakage should be expected. If good clarity is essential, it may be preferable to use a drum filter with the cloth caulked in place and sealing the system to contend with the effects of blinding.

The one exception to the points noted above is the continuous precoat filter. Here the purpose of the filter medium is to act as a support for the sacrificial bed of precoat material. Thus, the medium should be tight enough to retain the precoat solids and prevent bleeding of the precoat solids through the filter medium during operation, yet open enough to permit easy cleaning at the end of each cycle. Light-weight felt media work well in these respects.

Homogeneous Cake Accurate test results and optimum filter performance require the formation of a homogeneous cake and thus the maintenance of a similarly homogeneous suspension. Settling in the sample container during a bottom-feed test program can usually be detected by comparing the back-calculated feed solids concentration (based on filtrate, wet cake, and dry cake weights) with the slurry solids concentration as prepared. It is normal to find that the back-calculated concentration is slightly lower than the prepared concentration. This difference is normally within 2 percentage points and should never be greater than 5 percentage points. Since this difference does exist, it means that the slurry sample will concentrate to some extent as the tests continue. Adding fresh slurry to the sample container after each test can counteract this condition, as the system will reach an equilibrium similar to that found in a full-scale machine.

If a more positive check is required on the quality of the filter cake, particle size analyses may be run and compared with the sample as prepared.

In a top-feed filter test, the filter cake will contain all of the solids, provided they are all emptied from the sample container. The danger in this type of test is that the solids will stratify, particularly if the cake formation time is prolonged. Close examination of the filter cake will indicate whether or not this has happened. If there has been significant stratification, the feed slurry should be modified by thickening and/or flocculation in order to increase the dry solids filtration rate and permit formation of a homogeneous cake. Another possibility, but not necessarily the best, is to use a thinner, but still dischargeable, cake to avoid stratification.

Agitation of Sample All slurries used in bottom-feed tests must be agitated by hand (if slurry characteristics permit) to check whether or not the solids are settling out around the edges of the container and to determine the degree of agitation required to maintain the solids in suspension. Generally speaking, if the solids can be maintained in suspension by hand agitation, the slurry can be processed by a bottom-feed-type filter.

Agitation by a wide spatula may be substituted for hand agitation, but only after it has been determined by feel that the spatula will provide the needed agitation. If this cannot be done, then confirmation of proper agitation must be based on back-calculated feed solids concentrations and/or particle size analyses of the filter cakes.

If it is not possible to maintain a uniform suspension, the sample should be thickened and the flowsheet modified to provide the required thickening.

Mechanical agitation of a sample is very difficult to use effectively. Generally speaking, if enough room is left in the sample container for the leaf and the agitator, the agitation is not sufficient to prevent settling out in the corners of the container. If sufficient agitation is used to maintain suspension in all parts of the container, then it is highly probable that the velocity of the slurry across the face of the test leaf will wash the solids from the leaf and give very erroneous results. Furthermore, the tendency is to leave the agitator running continuously, or at least for so long a period of time that there is attrition of the solids and, therefore, inaccurate results. Mechanical agitation during testing can generally be justified only in a most unusual and exceptional circumstance.

Use of Steam or Hot Air It was indicated earlier that the cycle might include steam filtration or thermal drying using hot air. While effective use is made of both steam and hot air, the applications are rather limited, and testing procedures are difficult and specialized. As a general rule, steam application will reduce cake moisture 2 to 4 percentage points. Hot-air drying can produce a bone-dry cake, but generally it is practical only if the air rate is high, greater than about 1800 m³/m²-h (98 cfm/ft²). Both systems require a suitable hood which must contact the dam on the leaf during the drying cycle, allowing the steam or hot air to pass through the cake without dilution by cold air. The end of the operation can be determined by a noticeable increase in the temperature of the gas leaving the leaf.

SMALL-SCALE TEST PROCEDURES

[Purchas (ed.), *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977.]

Apparatus There are several variations of the bench-scale test leaf that may be used, but they all have features similar to the one discussed below.

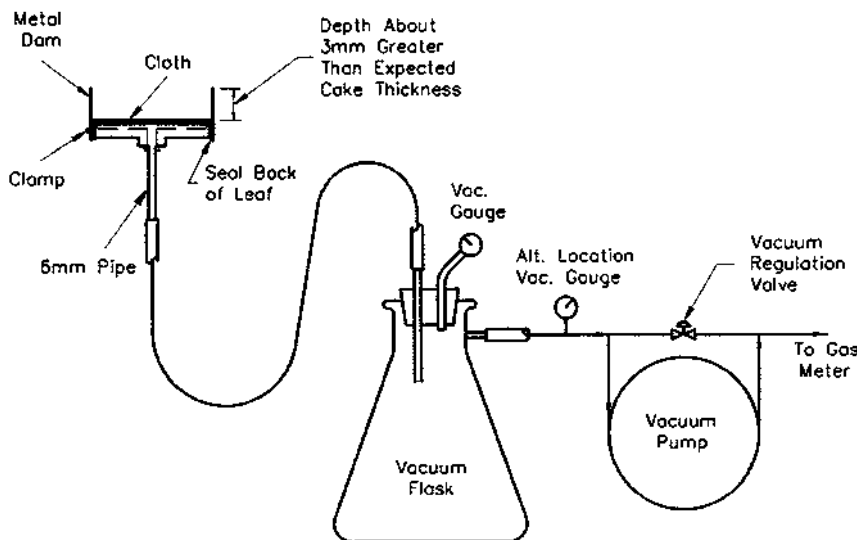


FIG. 18-107 Typical bottom-feed leaf test setup.

One typical test leaf is a circular disc with a plane area of 92.9 cm^2 (0.1 ft^2). One face of the leaf is grooved to provide large filtrate drainage passages and a support for the filter medium. A threaded drainage connection is provided on the center of the other face of the leaf. The test leaf is fitted with a filter medium and a dam, and the assembly clamped together as shown in Fig. 18-107. The depth of the dam for bottom-feed tests should be no greater than the depth of the maximum cake thickness, except where cake washing tests are to be performed. In this case, the dam depth should be about 3 mm ($\frac{1}{8} \text{ in}$) greater than the maximum expected cake thickness. Excessive dam depth will interfere with slurry agitation and can result in the formation of a nonhomogeneous cake.

It is absolutely necessary that a dam be used in all cases, except for roll discharge applications which do not involve cake washing or

where the maximum cake thickness is on the order of 2 mm or less. If a dam is not used, filter cake will form past the edge of the leaf in the general shape of a mushroom. When this happens, the total filter area is some unknown value, greater than the area of the leaf, that constantly increases with time during cake formation.

The back of the leaf assembly and the joint where the dam overlaps must be sealed with some suitable material so that the filtrate volume collected accurately represents the liquid associated with the deposited cake solids.

Figure 18-108 also contains a schematic layout of the equipment which is required for all bottom-feed leaf tests. Note that there are no valves in the drainage line between the test leaf and the filtrate receiver, nor between the filtrate receiver and the vacuum pump.

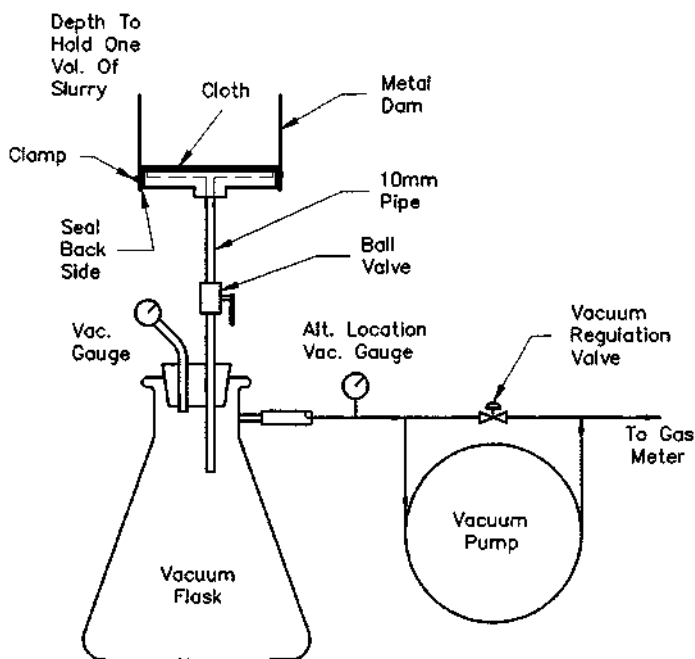


FIG. 18-108 Typical top-feed leaf test setup.

At the start of the leaf test run, the hose between the test leaf and filtrate receiver should be crimped by hand to bring the filtrate receiver to the operating vacuum level. The use of a valve at this point is not only less convenient but very frequently results in a hydraulic restriction. The net result, then, is a measurement of flow through the valve rather than the rate at which the filter cake is capable of forming. Hydraulic restriction is something which should always be kept in mind. If the filtrate runs at a high and full pipe flow rate into the filtrate receiver, it is quite likely that there is some degree of hydraulic restriction, and larger tubing and piping should be considered. When very high air flow rates are obtained, the experimenter must be satisfied that the rates being measured are limited by cake resistance and not by pressure drop through the equipment.

There will be many times when the quantity of sample is limited. While it is best to use the 92.9 cm² (0.1 ft²) area leaf in order to minimize edge effects and improve accuracy, when the sample volume is limited it is much better to have several data points with a smaller leaf than only one or two using the larger leaf. Data from leaves as small as 23.2 cm² (0.025 ft²) are reasonably accurate and can be used to scale up to commercially sized units. However, it is usually prudent to employ a more conservative scale-up factor.

For top-feed applications, the most convenient assembly is that shown in Fig. 18-108. The depth of the dam must, of course, be suffi-

cient to contain the total quantity of feed slurry required for the test. Since the test leaf is mounted on top of the vacuum receiver, it is necessary to provide a valve between the test leaf and the receiver so that the desired operating vacuum may be obtained in the receiver before the start of a test run. It is imperative, however, that there be no restriction in this valve. The preferred choice is a ball valve with the full bore of the drainage piping.

Test Program Figure 18-109 is a suggested data sheet which contains spaces for most of the information which should be taken during a leaf test program, together with space for certain calculated values. Additional data which may be required include variations in air flow rate through the cake during each dewatering period and chemical and physical data for those tests involving cake washing.

It is difficult to plan a filtration leaf test program until one test has been run. In the case of a bottom-feed test, the first run is normally started with the intention of using a 30-s cake formation time. However, if the filtrate rate is very high, it is usually wise to terminate the run at the end of 15 s. Should the filtrate rate be very low, the initial form period should be extended to at least 1 min. If cake washing is to be employed, it is useful to apply a quantity of wash water to measure its rate of passage through the cake. The results of this first run will give the experimenter an approximation of cake formation rate, cake washing rate, and the type of cake discharge that must be used. The

rest of the leaf test program can then be planned accordingly.

In any leaf test program there is always a question as to what vacuum level should be used. With very porous materials, a vacuum in the range of 0.1 to 0.3 bar (3 to 9 in Hg) should be used, and, except for thermal-drying applications using hot air, the vacuum level should be adjusted to give an air rate in the range of 450 to 900 m³/m²-h (30 to 40 cfm/ft²) measured at the vacuum.

For materials of moderate to low porosity, a good starting vacuum level is 0.6 to 0.7 bar (18 to 21 in Hg), as the capacity of most vacuum pumps starts to fall off rapidly at vacuum levels higher than 0.67 bar (20 in Hg). Unless there is a critical moisture content which requires the use of higher vacuums, or unless the deposited cake is so impervious that the air rate is extremely low, process economics will favor operation at vacuums below this level. When test work is carried out at an elevation above sea level different than that of the plant, the elevation at the plant should be taken into account when determining the vacuum system capacity for high vacuum levels (>0.5 bar).

Generalized correlations are available for each of the operations which make up the full filter cycle. This means that simulated operating conditions can be varied to obtain a maximum of information without requiring an excessive number of test runs. The minimum number of test runs required for a given feed will, of course, vary with the expertise of the experimenter and the number of operations performed during the filter cycle. If, for example, the operation involves only the dewatering of a slurry which forms a cake of relatively low to moderate porosity, frequently sufficient data can be obtained in as little as six runs. For more difficult tests, more runs are usually advisable, and the novice certainly should make a larger number of runs as there is likely to be more data scatter.

Bottom-Feed Test Procedure The procedure for collecting data using bottom-feed leaf test techniques is as follows:

1. Fit the test leaf with a filter cloth expected to give reasonable results and seal the back of the leaf and side of the dam with silicone or other suitable material.

2. Hand-crimp the hose in back of the test leaf, and then turn on the vacuum pump and regulate the bypass valve on the pump to give the desired vacuum level in the receiver.

3. Agitate the slurry by hand or with a wide spatula to maintain a homogeneous suspension. Immerse the test leaf face downward to approximately one-half the depth of the slurry.

4. Simultaneously start the timer and release the crimped hose to begin cake formation. Maintain agitation during cake formation and move the leaf as may be required to ensure that solids do not settle out in any part of the container. It is not necessary to try to simulate the velocity with which the full-scale unit's filtration surface passes through the slurry in the filter tank.

5. Remove the leaf from the slurry at the end of the cake-formation period and note the time. If the slurry is particularly thick and viscous, the leaf may be gently shaken to remove excess slurry and prevent the dam from scooping up extra material. Maintain the leaf in an upright position (cake surface on top) and elevated so that liquid within the drainage passages may pass to the receiver. Tilt and rotate the leaf to help the filtrate reach the drain outlet. Continue this dewatering period until:
 - a. the preselected time has elapsed, or
 - b. the cake cracks.

6. If the cake is to be washed, apply a measured quantity of wash fluid and note the time required for free fluid to disappear from the surface of the cake. Pour the wash fluid onto a deflecting baffle, such as a bent spatula, to prevent the cake from being gouged. Washing must begin before cake cracking occurs. In particular, observe that there is no crack along the edge between the cake and the dam.

7. Continue with the various operations in the predetermined sequence.

8. During each of the operations record all pertinent information such as vacuum level, temperature, time required for the cake to crack, filtrate foaming characteristics, air flow rate during the drying periods, etc.

9. At the end of the run, measure and record the filtrate volume (and weight, if appropriate), cake thickness, final cake temperature (if appropriate), wet cake weight, and note the cake discharge characteristics (roll, sticks to media, etc.).

10. For runs involving cake dewatering only, it is usually convenient to dry the total cake sample, if the associated solution contains little or no dissolved solids.

11. When cake washing is involved, it is usually convenient to weigh the wet cake and then repulp it in a known quantity of distilled water or in water at the same pH as the filtrate, if precipitation of solute could occur in distilled water. The resultant slurry is then filtered using a clean dry filter and flask and a sample of the clear liquid analyzed for the reference constituent.

Should the mother liquor contain a significant quantity of dissolved solids, the filter cake should be thoroughly washed (after the sample for analysis has been taken) so that the final dry weight of the cake will represent suspended solids only. The quantity of reference constituent in the final washed cake can be readily calculated from the wet and dry cake weights and the known amount of distilled water used for repulping.

In cake-washing tests, it is important that the feed slurry liquid be analyzed for total dissolved solids and density as well as the reference constituent.

Top-Feed Test Procedure The sequence of operations with a top-feed leaf test is the same as in a bottom-feed test, except that the leaf is not immersed in the slurry. The best method for transferring the slurry to the top-feed leaf is, of course, a function of the characteristics of the slurry. If the particles in the slurry do not settle rapidly, the feed can usually be transferred to the leaf from a beaker. If, however, the particles settle very rapidly, it is virtually impossible to pour the slurry out of a beaker satisfactorily. In this case, the best method is to make use of an Erlenmeyer flask, preferably one made of plastic. The slurry is swirled in the flask until it is completely suspended and then abruptly inverted over the leaf. This technique will ensure that all of the solids are transferred to the leaf.

When the solids involved are coarse and fast settling, the vacuum should be applied an instant after the slurry reaches the surface of the filter medium.

Precoat Test Procedure Precoat filtration tests are run in exactly the same manner as bottom-feed tests except that the leaf must first be precoated with a bed of diatomaceous earth, perlite, or other shaveable inert solids. Some trial and error is involved in selecting a grade of precoat material which will retain the filtered solids to be removed on the surface of the bed without any significant penetration. During this selection process, relatively thin precoat beds of 1 to 2 cm are satisfactory. After a grade has been selected, bench-scale tests should be run using precoat beds of the same thickness as expected on the full-scale unit.

Where the resistance of the precoat bed is significant in comparison to the resistance of the deposited solids, the thickness of the precoat bed effectively controls the filtration rate. In some instances, the resistance of the deposited solids is very large with respect to even a thick precoat bed. In this case, variations in thickness through the life of the precoat bed have relatively little effect on filtration rate. This type of information readily becomes apparent when the filtration rate data are correlated.

The depth of cut involved in precoat filtration is a very important economic factor. There is some disagreement as to the method required to accurately predict the minimum permissible depth of cut. Some investigators maintain that the depth of cut can be evaluated only in a qualitative manner during bench-scale tests by judging whether the process solids remain on the surface of the precoat bed. This being so, they indicate that it is necessary to run a continuous pilot-plant test to determine the minimum permissible depth of cut. The use of a continuous pilot-plant filter is a very desirable approach and will provide accurate information under a variety of operating conditions.

However, it is not always possible to run a pilot-plant test in order to determine the depth of cut. A well-accepted alternative approach makes use of the more sophisticated test leaf illustrated in Fig. 18-110. This test leaf is designed so that the cake and precoat are extruded axially out the open end of the leaf. The top of the retaining wall on this end of the leaf is a machined surface which serves as a support for a sharp discharge knife. This approach permits variable and known depths of cut to be made so that the minimum depth of cut may be determined. Test units are available from Betts Advanced Metal, hompoc, Calif., (805) 735-5130.

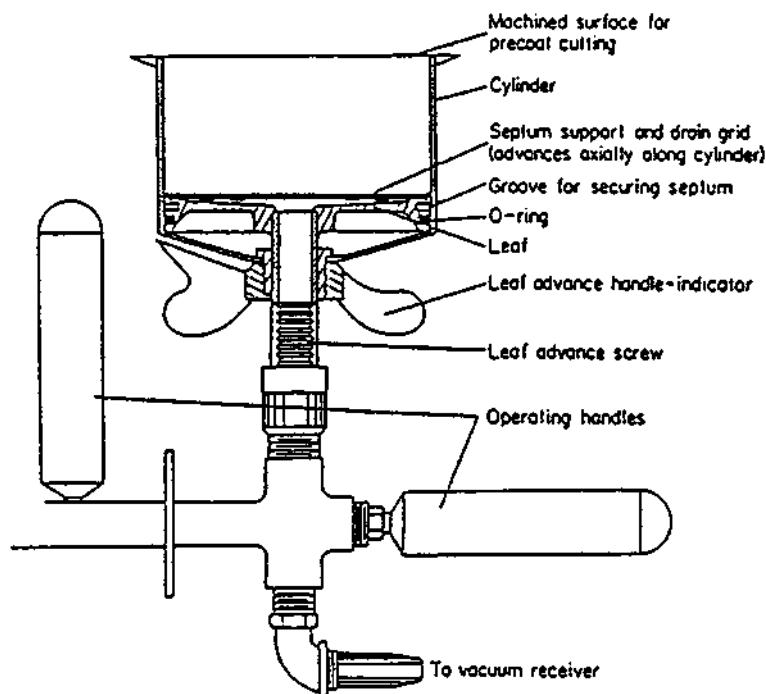


FIG. 18-110 Special test leaf for precoat filtration.

Lacking the above-described actual data, it is possible to estimate precoat consumption by using these values: nonpenetrating solids, 0.06-mm cut/drum revolution (0.0024 in); visible penetration, 0.15- to 0.20-mm cut/drum revolution (0.006 to 0.008 in); precoat bed density, 4.2 kg/m²-cm of bed depth (2.2 lb/ft²-in) for diatomaceous earth or 2.1 to 3.0 kg/m²-cm (1.1 to 1.6 lb/ft²-in) for perlite.

DATA CORRELATION

[Purchas (ed.), *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977.]

The correlations used are based partly on theoretical consideration and partly on empirical observations. The basic filtration data are correlated by application of the classic cake-filtration equation, aided by various simplifying assumptions which are sufficiently valid for many (but not all) situations. Washing and drying correlations are of a more empirical nature but with strong experimental justification. If steam or thermal drying is being examined, additional correlations are required beyond those summarized below; for such applications, it is advisable to consult an equipment manufacturer or refer to published technical papers for guidance.

Dry Cake Weight vs. Thickness It is convenient to convert the test dry cake weight to the weight of dry cake per unit area per cycle (W), and plot these values as a function of cake thickness (Fig. 18-111). Cake weight is measured quite accurately, while cake thickness measurements are subject to some variation. By plotting the data, variations in thickness measurements are averaged. The data usually give a straight line passing through the origin. However, with compressible material, sometimes a slightly curved line best represents the data, since thinner cakes are usually compressed more than thicker cakes.

Dry Solids or Filtrate Rate Filtration rate, expressed either in terms of dry solids or filtrate volume, may be plotted as a function of time on log-log paper. However, it is more convenient to delay the rate calculation until the complete cycle of operations has been defined.

It is most useful to plot either dry cake weight (weight of dry solids/unit area/cycle) or filtrate volume (volume/unit area/cycle) as a function of time on log-log paper. These data should give straight-line plots for constant operating conditions in accordance with Eqs. (18-55) and (18-56). The expected slope of the resultant rate/time plots is +0.50, as in Fig. 18-112. In practice, the vast majority of slopes range from +0.50 to +0.35. Slopes steeper than +0.5 indicate that there is some significant resistance other than that of the cake solids,

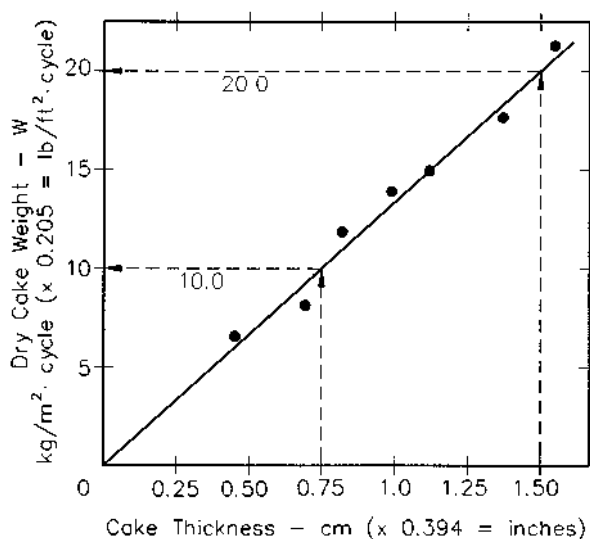


FIG. 18-111 Dry cake weight vs. cake thickness.

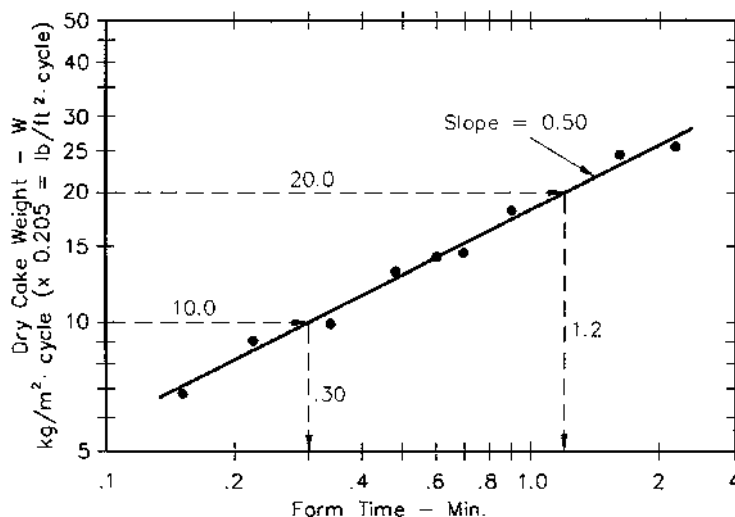


FIG. 18-112 Dry cake weight vs. form time.

such as a hydraulic restriction in the equipment or an exceptionally tight filter cloth.

Data from precoat tests, however, generally produce filtrate curves with much steeper slopes. The precoat bed has a greater resistance than most filter fabrics, and the particles which are separated on a continuous precoat usually form a cake which has a relatively low resistance when compared to that of the precoat bed. Once the thickness of the deposited solids becomes significant, their resistance increases. Thus, at very short form times, the slope of the filtrate curve may be close to 1.0, but as form time increases, the slope of the curve will decrease and will approach +0.5 (Fig. 18-113).

There are some solids, however, which form a less permeable cake, even in very thin layers. With these solids, the resistance of the

deposited cake will be very high when compared to that of the precoat bed, and the slope of the filtrate curve will be +0.5 for all values of form time.

Effect of Time on Flocculated Slurries Flocculated slurries usually show significant decreases in filterability with time (Fig. 18-114). The rate of degradation may be established by running a series of repetitive leaf tests at frequent intervals on a flocculated slurry, starting as soon as practical after the addition of the flocculant. If there is little change in the filtration rate, this factor need be given no more consideration. However, it is usually found that there is significant degradation.

When a flocculated feed is added to a filter tank, there is a definite time lag before this material reaches the surface of the filter medium.

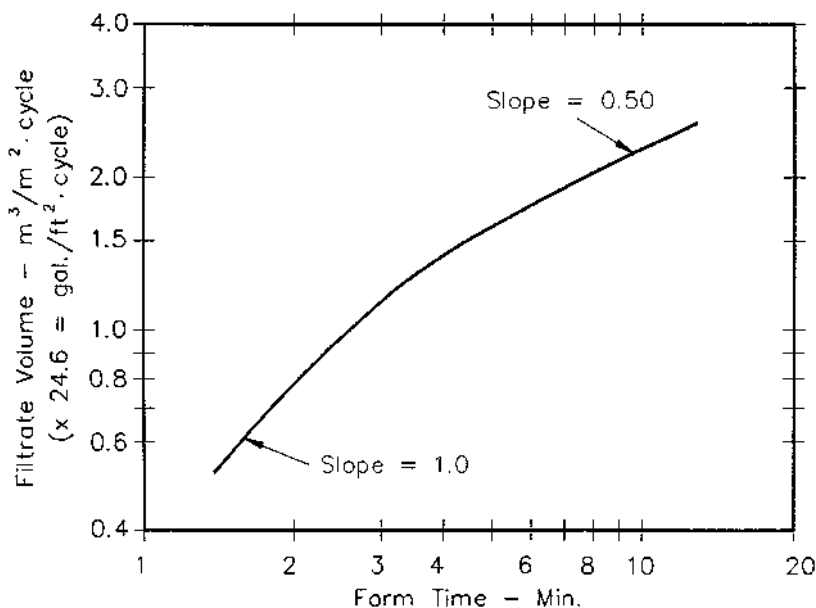


FIG. 18-113 Filtrate volume per cycle vs. form time.

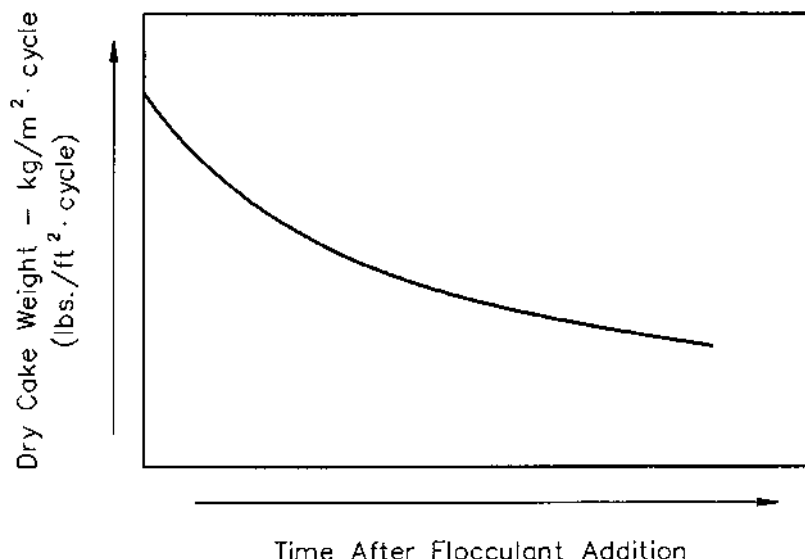


FIG. 18-114 Degradation of flocculation with time.

Since this lag time is not known at the time of testing, a lag time of 8 to 10 minutes should be allowed before starting the first leaf test on a flocculated slurry. Two, or perhaps three, tests can be run before the elapsed time exceeds the probable retention time in the full-scale filter tank. With knowledge of the elapsed time after flocculation and data relating to the rate of degradation, the rates obtained on the leaf test runs can be adjusted to some constant lag time consistent with the anticipated full-scale design.

Cake Moisture Results on a wide variety of materials have shown that the following factor is very useful for correlating cake moisture content data:

$$\text{Correlating factor} = (\text{m}^3/\text{m}^2\text{-h})(P_c/W)(\Theta_D/\mu), \quad (18-67)$$

where $\text{m}^3/\text{m}^2\text{-h}$ = air rate through filter cake measured at downstream pressure or vacuum

P_c = pressure drop across cake
 W = dry cake weight/unit area/cycle
 Θ_D = dry time per cycle
 μ = viscosity of liquid phase

For a more rigorous discussion of cake moisture correlation, the reader is referred to an earlier article by Nelson and Dahlstrom [*Chem. Eng. Progress*, **53**, 7, 1957]. Figure 18-115 shows the general shape of the curve obtained when using the cake moisture correlating factor. The value of the correlating factor chosen for design should be somewhere past the knee of the curve. Values at and to the left of the knee are in an unstable range where a small change in operating conditions can result in a relatively large change in cake moisture content.

It is not always necessary to use all of the terms in the correlating factor, and those conditions which are held constant throughout

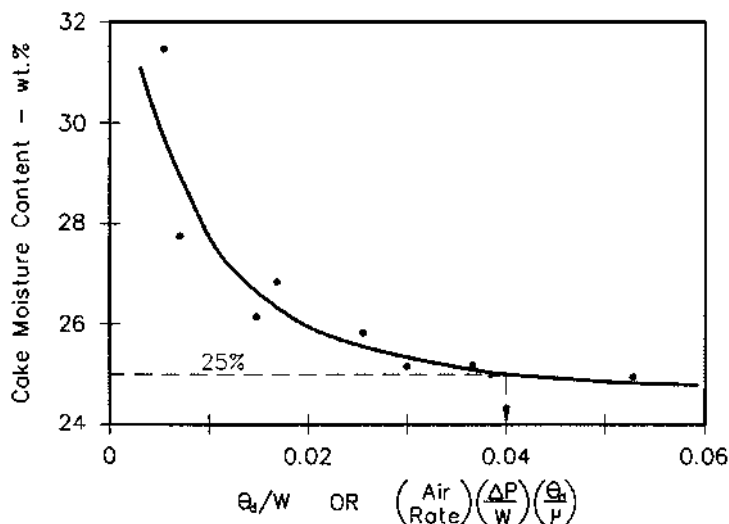


FIG. 18-115 Cake moisture correlation.

the testing may be dropped from the correlating factor. Many times, air rate data are not available and reasonable correlations can be obtained without this information, particularly if the cakes are relatively low in permeability. By dropping these terms, the correlating factor is reduced to the simplified version, Θ_d/W , involving only drying time and cake weight per unit area per revolution. While this is a very convenient factor to use, care must be taken in its application, as it is no longer a generalized factor, and there will be a tendency for the data to produce different curves for changes in operating conditions such as vacuum level and cake thickness.

Cake Washing Wash efficiency data are most conveniently represented by a semilog plot of percent remaining R as a function of wash ratio N as shown in Fig. 18-116. Percent remaining refers to that portion of the solute in the dewatered but unwashed cake which is left in the washed and dewatered cake. Since a cake-washing operation involves the displacement of one volume of liquid by another volume, the removal of solute is related to the ratio of the volume of washing fluid divided by the volume of liquid in the cake. This ratio is defined as wash ratio N .

Practical experience has shown that the most convenient and best means of expressing R is in terms of the solute concentrations in the washed cake liquid, the feed liquid (or unwashed cake liquid), and the cake wash liquid. Furthermore, the wash ratio N may also be expressed either as a volume or weight ratio.

Percent remaining is defined as follows:

$$\frac{R}{100} = \frac{C_2 - C_w}{C_1 - C_w} \quad (18-68)$$

where R = % solute remaining after washing
 C_2 = solute concentration in washed cake liquid
 C_1 = solute concentration in unwashed cake liquid
 C_w = solute concentration in wash liquid

If the cake is washed with solute-free liquid, percent remaining is readily calculated by dividing the solute concentration in the liquid remaining in the washed cake by the solute concentration in the liquid in the original feed.

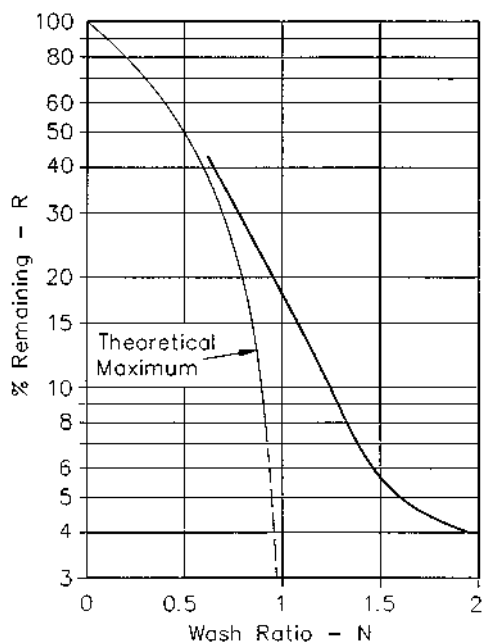


FIG. 18-116 Wash effectiveness.

The residence time of the cake-washing fluid within the cake is relatively short and is not normally considered useful for any kind of leaching operation. Therefore, it is assumed that all of the solute is in solution.

If it were possible to obtain a perfect slug displacement wash, the fraction remaining would be numerically equal to 1 minus the wash ratio. This ideal condition is represented by the maximum theoretical line as shown in Fig. 18-116. Since it represents the best that can be done, no data point should fall to the left of this curve. Most, but not all, cake-washing curves tend to fall along the heavy solid line shown. In the absence of actual data, one may estimate washing results by using this curve.

The quantity of wash water to be used in a given operation is dictated by flowsheet considerations and the required solute content of the washed cake. Generally speaking, the maximum wash water quantity should be equivalent to a wash ratio of 1.5 to 2.5. Where high solute removals are required, it is frequently necessary to use a two-stage filtration system with intermediate repulping. These two stages may involve countercurrent flow of wash water, or fresh wash water may be used on both filters and for the intermediate repulping step.

Wash Time Cake-washing time is the most difficult of the filtration variables to correlate. It is obviously desirable to use one which provides a single curve for all of the data. Filtration theory suggests three possible correlations [Eqs. (18-62) to (18-64)]. These are listed below, beginning with the easiest to use:

1. Wash time vs. WV_w
2. Wash time vs. NW^2
3. Wash time/form time vs. wash volume/form volume

where W = weight dry cake/unit area/cycle
 V_w = volume of cake wash/unit area/cycle
 N = wash ratio
 $=$ volume of wash/volume of liquid in discharged cake

Fortunately, the easiest correlation to use usually gives satisfactory results. This curve is usually a straight line passing through the origin, but frequently falls off as the volume of wash water increases (Fig. 18-117). If for some reason this correlation is not satisfactory, one of the other two should be tried.

Air Rate Air rate through the cake, and thus vacuum pump capacity, can be determined from measurements of the air flow for various lengths of dry time. Figure 18-118 represents instantaneous air rate data. The total volume of gas passing through the cake during a dry period is determined by integrating under the curve.

Note that the air rate at the beginning of each drying or dewatering period within a given cycle starts at zero and then increases with time.

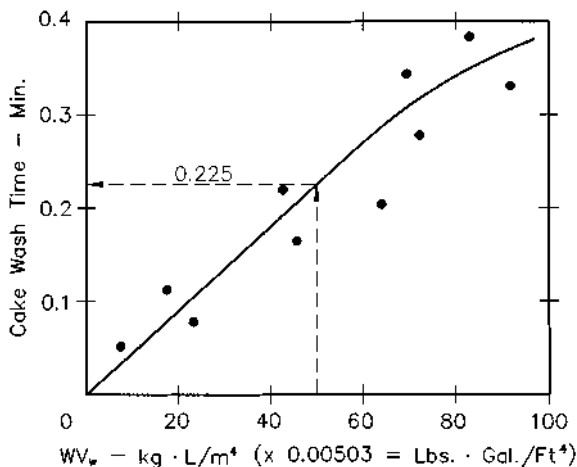


FIG. 18-117 Cake wash time correlation.

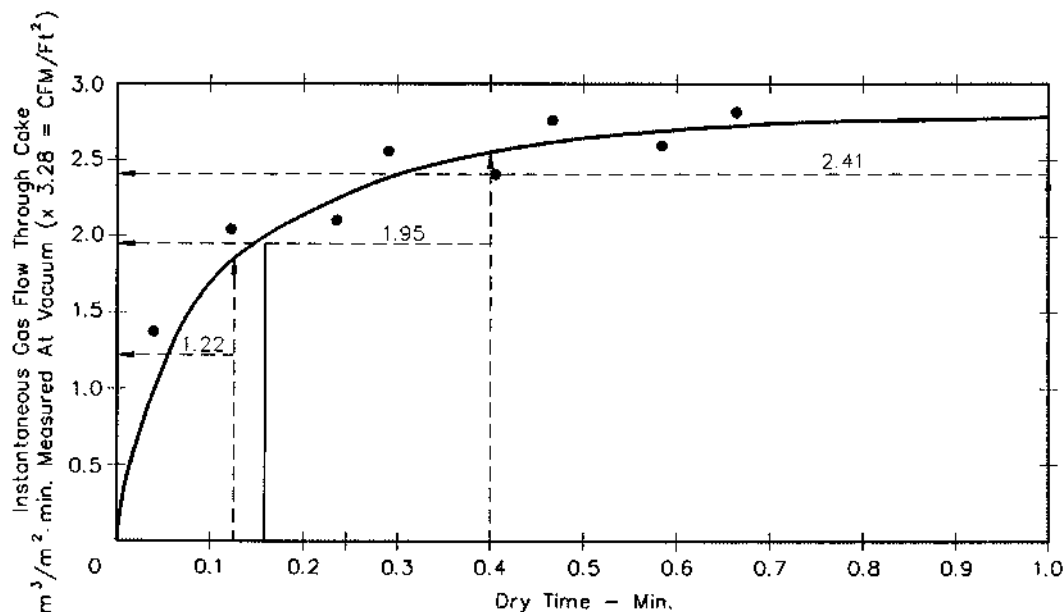


FIG. 18-118 Airflow through cake.

The shape of this curve will be a function of the permeability of the deposited cake.

Vacuum pump capacity is conventionally based on the total cycle and expressed as $\text{m}^3/\text{h}\cdot\text{m}^2$ (cfm/ft^2) of filter area measured at pump inlet conditions. Thus, the gas volumes per unit area passing during each dry period in the cycle are totaled and divided by the cycle time to arrive at the design air rate. Since air rate measurements in the test program are based on pressure drop across the cake and filter medium only, allowance must be made for additional expansion due to pressure drop within the filter and auxiliary piping system in arriving at vacuum pump inlet conditions.

Air rate measurements made during a leaf test program account only for gas flow through the cake. An operating filter has drainage passages that must also be evacuated. The extra air flow may be conveniently accounted for in most cases by multiplying the leaf test rate by 1.10. With horizontal belt filters, one must also add for the leakage that occurs along the sliding seal and, depending upon the type of filter cloth used, edge leakage due to lateral permeability of the cloth. Typically, the total leakage can be significant, amounting to about 35 to 50 $\text{m}^3/\text{h}\cdot\text{m}^2$ (2 to 3 cfm/ft^2).

Adjustment may also be required for differences in altitude between the test site and the commercial installation. In general terms, if the plant elevation is higher, the vacuum pump size must be increased, and conversely.

Darcy's law has been used to derive an expression which reflects not only the effect of a change in elevation, but also provides a means for estimating changes in air rate resulting from changes in vacuum level and cake thickness (or cake weight per unit area). In order for this relationship to hold for changes in vacuum and cake thickness, it must be assumed that both cakes have the same specific resistance.

The generalized equation is as follows:

$$(\text{Air rate})_{b2} = (\text{air rate})_{a2} \frac{(W_a)}{(W_b)} \frac{(P_{b1}^2 - P_{b2}^2)}{(P_{a1}^2 - P_{a2}^2)} \frac{(P_{a2})}{(P_{b2})} \quad (18-69)$$

where P_1 = inlet absolute pressure
 P_2 = outlet absolute pressure
 a = base condition
 b = revised condition
 W = weight of dry cake solids/unit area/cycle

SCALE-UP FACTORS

[Purchas (ed.), *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977.]

The overall scale-up factor used to convert a rate calculated from bench-scale data to a design rate for a commercial installation must incorporate separate factors for each of the following:

- Scale-up on rate
- Scale-up on cake discharge
- Scale-up on actual area

Special note should be made that these scale-up factors are *not* safety factors to allow for additional plant capacity at some future date. Their purpose is to account for differences in scale, including such things as minor deviations in the plant slurry from the sample tested, edge effects due to the size of the test equipment, close control of operating conditions during the leaf test program, and long-term medium blinding.

Scale-up on Rate Filtration rates calculated from bench-scale data should be multiplied by a factor of 0.8 for all types of commercial units which do not employ continuous washing of the filter medium and on which there is a possibility of filter-medium blinding. For those units which employ continuous filter-medium washing, belt-type drum and horizontal units, the scale-up factor may be increased to 0.9.

The use of this scale-up factor assumes the following:

- Complete cake discharge
- Nominal filter area approximately equal to actual area
- A representative sample
- Suitable choice of filter medium
- Operating conditions equal to those used in testing
- Normal cloth conditioning during testing and operation

The scale-up factor on rate specifically does not allow for:

- Changes in slurry filterability
- Changes in feed rate
- Changes in operating conditions
- Cloth blinding

Where there is any doubt about some of the conditions listed above (except for cake discharge and actual area which should be handled separately) a more conservative scale-up factor should be used.

Scale-up on Cake Discharge When a filter is selected for a particular application, it is intended that the unit be capable of discharging essentially 100 percent of the cake which is formed. There are,

however, many applications which are marginal, regardless of the type of discharge mechanism used. In these cases, the experimenter must judge the percent of cake discharge to be expected and factor the design rate accordingly.

Scale-up on Actual Area The nominal area of a filter as used by equipment manufacturers is based upon the overall dimensions of the filtering surface. The fraction of this total area that is active in filtration is a function of the filterability of the material being handled and any special treatment which the surface may receive.

The filtering surface is divided into a number of sections by division strips, radial rods, or some other impervious separator. Material which forms a thin, rather impervious cake will not form across the dividers, and thus the actual area is somewhat less than the nominal. Where relatively thick cakes of at least 1.5 cm are formed, the cake tends to form across the dividers due to cross-drainage in both the filter cake and the filter medium. In this case, the effective area is relatively close to the nominal area.

For most applications, the actual area of a drum filter will generally be no less than 94 to 97 percent of the nominal area, depending upon the size and number of sections. This variation is generally not accounted for separately and is assumed to be taken care of in the scale-up factor on filtration rate.

There are, however, certain special applications where the filter medium around the edge of the section may be deliberately blinded by painting in order to improve cake discharge. This technique is most frequently used on disc filters, with the result that the actual area may be only 75 to 85 percent of the nominal area. This is a significant deviation from the nominal area and must be considered separately.

Overall Scale-up Factor The final design filtration rate is determined by multiplying the bench-scale filtration rate by each of the scale-up factors discussed above. While this approach may seem to be ultraconservative, one must realize that the experimenter maintains careful control over the various steps during the filter cycle while running a bench-scale test, whereas a commercial filter operates with a minimum of attendance and at average conditions which are chosen to provide a satisfactory result in a production context.

FULL-SCALE FILTER PERFORMANCE EVALUATION

The correlations which have been presented for the evaluation of bench-scale data are the same correlations which should be used to evaluate the performance of a commercial installation.

A few random samples taken from a commercial installation most probably will not provide enough insight to determine that the filter is performing as expected. However, by making use of reasonable variations in the most important parameters, the desired correlations can be developed. Bench-scale tests should be run on representative feed samples taken at the same time test runs are made on the commercial unit. The bench-scale tests can be varied over a much wider range to provide a sound basis for both the location and shape of the appropriate correlation. A comparison of these results with the data taken from the commercial installation provides a good measure for efficiency of the commercial unit and a basis for identifying problem areas on the full-scale unit.

FILTER SIZING EXAMPLES

[Purchas (ed.), *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977.]

The examples which follow show how data from the correlations just presented and a knowledge of the physical characteristics of a particular filter are used to determine a filtration cycle and, subsequently, the size of the filter itself. The three examples which follow involve a disc, a drum belt, and a horizontal belt filter.

Example 5: Sizing a Disc Filter Equipment physical factors, selected from Table 18-9: Maximum effective submergence = 28%; maximum portion of filter cycle available for dewatering = 45%. (High submergence versions require trunion seals, and their use is limited to specific applications.)

Scale-up factors: On rate = 0.8; On area = 0.8; On discharge = 0.9. (Scale-up on discharge may be increased to 0.97 if based on previous experience or to 0.95 if the total filter area is based on the measured effective area of the disc.)

Objective: Determine the filter size and vacuum system capacity required to dewater 15 mtpd (metric tons per hour) of dry solids and produce a cake containing an average moisture content of 25 wt %.

Calculation procedure:

1. Choose cake thickness = 1.5 cm, slightly thicker than minimum value listed in Table 18-8.

2. From Fig. 18-111, $W = 20.0$ kg dry cake/(m² × rev.).

3. From Fig. 18-112, form time, $\tau = 1.20$ min.

4. Use simplified moisture content correlating factor in Fig. 18-115.

Choose $d/W = 0.04$ at avg. moisture content of 25 wt %.

Dry time = $d = 0.04 \times 20.0 = 0.80$ min.

5. Calculate cycle time CT both on the basis of form time and dry time to determine which is controlling:

$CT_{\text{form}} = 1.20/.28 = 4.29$ mpr (min/rev.).

$CT_{\text{dry}} = 0.80/.45 = 1.78$ mpr.

Therefore, cake formation rate is controlling and a cycle time of 4.29 mpr must be used.

6. Overall scale-up factor based on the factors presented previously = $0.8 \times 0.8 \times 0.9 = 0.58$

7. Design filtration rate = $(20.0/4.29)/(60 \times 0.58) = 162$ kg/h × m²

8. Area required to filter 15 metric tons of dry solids per hour = $15 \times 1000/162 = 92.6$ m². The practical choice would then be the nearest commercial size of filter corresponding to this calculated area.

9. Dry time = 45% of CT = $0.45 \times 4.29 = 1.93$ min. This is a much longer dry time than required. Therefore, reduce the dry time to 1.00 min by proper bridging in the filter valve.

10. From Fig. 18-118, the average gas flow rate during the 1.00 min drying period was found by graphical integration to be 2.41 m³/m² × min.

11. Total volume of air flowing during dry period = $2.41 \times 1.00 = 2.41$ m³ per m² per cycle. Add 10% to allow for evacuation of drainage passages. Total flow = 2.65 m³/m² × cycle.

12. Required vacuum pump capacity = $2.65/4.29 = 0.62$ m³/min × m² of total filter area. Allow for pressure drop within system when specifying the vacuum pump. See next example.

Example 6: Sizing a Drum Belt Filter with Washing Equipment physical factors, selected from Table 18-9: Maximum effective submergence = 30%; max. apparent subm. = 35%; max. arc for washing = 29%; portion of cycle under vacuum = 75%.

Scale-up factors: On rate = 0.9; On area = 1.0; On discharge = 1.0.

Process data: Sp. gr. of feed liquid = 1.0; TDS (total dissolved solids) in feed liquid = 4.0 wt %; fresh water used for washing; vacuum level = 18 in Hg; final cake liquid content = 25 wt %.

TABLE 18-9 Typical Equipment Factors for Cycle Design

Filter type	% of cycle				
	Submergence		Total under active vac. or pres.	Max. for washing	Max. for dewatering only
	Apparent	Max. effective			
Drum					
Standard scraper	35	30	80	29	50-60
Roll discharge	35	30	80	29	50-60
Belt	35	30	75	29	45-50
Coil or string	35	30	75	29	45-50
Precoat	35,55	35,55	93	30	60,40
Horizontal belt	As req'd.	As req'd.	Lengthen as req'd.	As req'd.	As req'd.
Horizontal table	As req'd.	As req'd.	80	As req'd.	As req'd.
Tilting pan	As req'd.	As req'd.	75	As req'd.	As req'd.
Disc	35	28	75	None	45

Objective: Determine the filter size and vacuum capacity required to dewater and wash 15 mtpd of dry solids while producing a final washed cake with a moisture content of 25 wt % and containing 0.10 wt % TDS based on dry cake solids.

Calculation procedure:

1. Choose cake thickness = 0.75 cm, slightly thicker than the minimum in Table 18-8.

2. From Fig. 18-111, $W = 10 \text{ kg/m}^2 \times \text{cycle}$.

3. From Fig. 18-112, form time = 0.30 min.

4. From Fig. 18-115, $d/W = 0.04$ for 25 wt % residual moisture.

5. Dry time $= d/W = 10.0 \times 0.04 = 0.40 \text{ min}$.

6. Determine required wash quantity:

Calculated TDS concentration in washed cake liquor:

Liquid in final cake = $10 \times 0.25/0.75 = 3.33 \text{ kg/m}^2 \times \text{cycle}$.

TDS in dry washed solids = $10 \times 0.001/0.999 = 0.010 \text{ kg/m}^2 \times \text{cycle}$.

TDS in final washed cake liquor = $(0.010/3.33)100 = 0.300 \text{ wt } \%$.

Percent remaining, $R = ((C_2 - C_w)/(C_1 - C_w))100$.

Since $C_w = 0$,

Required percent remaining, $R = (C_2/C_1)100 = (0.300/4.00)100 = 7.5\%$.

From Fig. 18-116, required wash ratio $N = 1.35$.

For design, add 10% $\rightarrow N = 1.35 \times 1.1 = 1.49$.

Wash vol. = $V_w = 1.49 \times 3.33/1.00 = 4.96 \text{ L/m}^2 \times \text{cycle}$.

7. Determine wash time:

$WV_w = 10.0 \times 4.96 = 49.6 \text{ kgL/m}^2$.

From Fig. 18-117, wash time $= w = 0.225 \text{ min}$.

8. Summary of minimum times for each operation:

Form (step 3) = 0.30 min.

Wash (step 6) = 0.225 min.

Final dry (step 5) = 0.40 min.

9. Maximum washing arc = horizontal centerline to 15° past top dead center, or 29% of total cycle. Minimum percent of cycle between end of form and earliest start of wash = area between horizontal centerline and maximum apparent submergence = $(50\% - 35\%)/2 = 7.5\%$.

10. Maximum percentage of cycle for wash + final dry = $75 - 30 - 7.5 = 37.5\%$.

11. Determine cycle time based on the rate-controlling operation:

a. $CT_{\text{form}} = 0.30/30 = 1.00 \text{ mpr}$.

b. $CT_{\text{wash}} = 0.225/29 = 0.77 \text{ mpr}$.

c. $CT_{\text{wash} + \text{dry}} = (0.225 + 0.40)/37.5 = 1.67 \text{ mpr}$.

Therefore, the cake wash + final dry rate is controlling and a cycle time of 1.67 mpr must be used.

12. Since (c) is larger than (a) in the previous step, too thick a cake will be formed and it will not wash or dry adequately unless the effective submergence is artificially restricted to yield the design cake thickness. This may be accomplished by proper bridge-block adjustment or by vacuum regulation within the form zone of the filter valve.

13. The required washing arc of $(0.225/1.67)360 = 48.5^\circ$ is assumed to start at the horizontal center line. Careful control of the wash sprays will be required to minimize runback into the slurry in the vat.

14. Overall scale-up factor = $0.9 \times 1.0 \times 1.0 = 0.9$.

15. Design filtration rate = $(10.0/1.67)/(60 \times 0.9)$.

$= 323.3 \text{ kg/h} \times \text{m}^2$.

16. Total filter area required = $15 \times 1000/323.3 = 46.4 \text{ m}^2$.

Nearest commercial size for a single unit could be a 10 ft dia. $\times$ 16 ft long with a total area of $502 \text{ ft}^2 = 46.7 \text{ m}^2$.

17. Determine required vacuum capacity:

Initial dry time = $1.67 \times 0.075 = 0.125 \text{ min}$.

Calculate gas vol. through cake using data from Fig. 18-118:

Initial dry = $0.125 \times 1.22 = 0.153 \text{ m}^3/\text{m}^2 \times \text{rev}$.

Final dry = $0.40 \times 1.95 = 0.780 \text{ m}^3/\text{m}^2 \times \text{rev}$.

Total, including 10% for evacuation of drainage passages = $0.933 \times 1.10 = 1.03 \text{ m}^3/\text{m}^2 \times \text{rev}$.

Air rate based on total cycle = $1.03/1.67 = 0.62 \text{ m}^3/\text{min} \times \text{m}^2$ measured at 18 in Hg vacuum.

If pressure drop through system = 1.0 in Hg and barometric pressure = 30 in Hg, design air rate = $0.62 \times 12/11 = 0.68 \text{ m}^3/\text{min} \times \text{m}^2$ measured at 19 in Hg vacuum.

Horizontal Belt Filter Since the total cycle of a horizontal belt filter occurs on a single, long horizontal surface, there is no restriction with respect to the relative portions of the cycle. Otherwise, scale-up procedures are similar.

BATCH FILTRATION

Since most batch-type filters operate under pressure rather than vacuum, the following discussion will apply primarily to pressure filtration and the various types of pressure filters.

To use Eq. (18-54) one must know the pattern of the filtration process, i.e., the variation of the flow rate and pressure with time. Generally the pumping mechanism determines the filtration flow

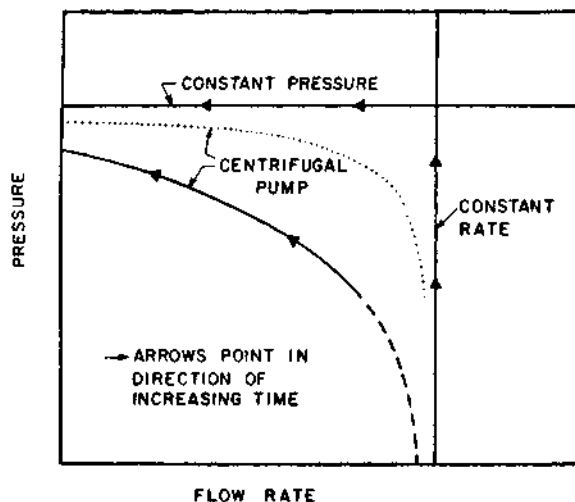


FIG. 18-119 Typical filtration cycles. [Tiller and Crump, *Chem. Eng. Prog.* 73(10), 72(1977), by permission.]

characteristics and serves as a basis for the following three categories* [Tiller and Crump, *Chem. Eng. Prog.*, 73(10), 65 (1977)]:

1. **Constant-pressure filtration.** The actuating mechanism is compressed gas maintained at a constant pressure.

2. **Constant-rate filtration.** Positive-displacement pumps of various types are employed.

3. **Variable-pressure, variable-rate filtration.** The use of a centrifugal pump results in this pattern: the discharge rate decreases with increasing back pressure.

Flow rate and pressure behavior for the three types of filtration are shown in Fig. 18-119. Depending on the characteristics of the centrifugal pump, widely differing curves may be encountered, as suggested by the figure.

Constant-Pressure Filtration For constant-pressure filtration Eq. (18-54) can be integrated to give the following relationships between total time and filtrate measurements:

$$\frac{\theta}{V/A} = \frac{\mu\alpha}{2P} \frac{W}{A} + \frac{\mu r}{P} \quad (18-70)$$

$$\frac{\theta}{V/A} = \frac{\mu\alpha w}{2P} \frac{V}{A} + \frac{\mu r}{P} \quad (18-71)$$

$$\frac{\theta}{V/A} = \frac{\mu\alpha pc}{2P(1-mc)} \frac{V}{A} + \frac{\mu r}{P} \quad (18-72)$$

For a given constant-pressure filtration, these may be simplified to

$$\frac{\theta}{V/A} = K_p \frac{W}{A} + C = K'_p \frac{V}{A} + C \quad (18-73)$$

where K_p , K'_p , and C are constants for the conditions employed. It should be noted that K_p , K'_p , and C depend on filtering pressure not only in the obvious explicit way but also in the implicit sense that α , m , and r are generally dependent on P .

Constant-Rate Filtration For substantially incompressible cakes, Eq. (18-54) may be integrated for a constant rate of slurry feed to the filter to give the following equations, in which filter-medium resistance is treated as the equivalent constant-pressure component to be deducted from the rising total pressure drop to give the variable pressure through the filter cake [Ruth, *Ind. Eng. Chem.*, 27, 717 (1935)]:

* A combination of category 2 followed by category 1 as parts of the same filtration cycle is considered by some as a fourth category. For a method of combining the constant-rate and constant-pressure equations for such a cycle, see Brown, loc. cit.

$$\frac{\theta}{V/A} = \frac{1}{\text{rate per unit area}} = \frac{\mu\alpha}{P - P_1} \frac{W}{A} \quad (18-74)$$

which may also be written

$$\frac{\theta}{V/A} = \frac{\mu\alpha w}{P - P_1} \frac{V}{A} = \frac{\mu\alpha\rho c}{(P - P_1)(1 - mc)} \frac{V}{A} \quad (18-75)$$

In these equations P_1 is the pressure drop through the filter medium.

$$P_1 = \mu r (V/A\theta)$$

For a given constant-rate run, the equations may be simplified to

$$V/A = P/K_r + C' \quad (18-76)$$

where K_r and C' are constants for the given conditions.

Variable-Pressure, Variable-Rate Filtration The pattern of this category complicates the use of the basic rate equation. The method of Tiller and Crump (loc. cit.) can be used to integrate the equation when the characteristic curve of the feed pump is available.

In the filtration of small amounts of fine particles from liquid by means of bulky filter media (such as absorbent cotton or felt) it has been found that the preceding equations based upon the resistance of a cake of solids do not hold, since no cake is formed. For these cases, in which filtration takes place on the surface or within the interstices of a medium, analogous equations have been developed [Hermans and Bredée, *J. Soc. Chem. Ind.*, **55T**, 1 (1936)]. These are usefully summarized, for both constant-pressure and constant-rate conditions, by Grace [*Am. Inst. Chem. Eng. J.*, **2**, 323 (1956)]. These equations often apply to the clarification of such materials as sugar solutions, viscose and other spinning solutions, and film-casting dopes.

If a **constant-pressure test** is run on a slurry, care being taken that not only the pressure but also the temperature and the solid content remain constant throughout the run and that time readings begin at the exact start of filtration, one can observe values of filtrate volume or weight and corresponding elapsed time. With the use of the known filtering area, values of $\theta/(V/A)$ can be calculated for various values of V/A which, when plotted with $\theta/(V/A)$ as the ordinate and V/A as the abscissa (Fig. 18-120a), result in a straight line having the slope $\mu\alpha w/2P$ and an intercept on the vertical axis of $\mu r/P$. Since μ , w , and P are known, α and r can be calculated from

$$\alpha = 2P/\mu w \times (\text{slope})$$

and

$$r = P/\mu \times (\text{vertical intercept})$$

The effect of the change of any variable not affecting α or r can now be estimated. It should be remembered that α and r usually depend on P and may be affected by w .

The symbol α represents the average specific cake resistance, which is a constant for the particular cake in its immediate condition. In the usual range of operating conditions it is related to the pressure by the expression

$$\alpha = \alpha' P^s \quad (18-77)$$

where α' is a constant determined largely by the size of the particles forming the cake; s is the cake compressibility, varying from 0 for rigid, incompressible cakes, such as fine sand and diatomite, to 1.0 for very highly compressible cakes. For most industrial slurries, s lies between 0.1 and 0.8. The symbol r represents the resistance of unit area of filter

medium but includes other losses (besides those across the cake and the medium) in the system across which P is the pressure drop.

It should be noted also that the intercept is difficult to determine accurately because of large potential experimental error in observing the time of the start of filtration and the time-volume correspondence during the first moments when the filtration rate is high. The value of r calculated from the intercept may vary appreciably from test to test, and will almost always be different from the value measured with clean medium in a permeability test.

To determine the effect of a change in pressure, it is necessary to run tests at three or more pressures, preferably spanning the range of interest. Plotting α or r against P on log-log paper (or log α or log r against the log P on cartesian coordinates) results in an approximate straight line (Fig. 18-120b) from which one may estimate values of α or r at interpolated or reasonably extrapolated magnitudes of P . The slope of the line is the index of a power relationship between α and P or r and P .

Not uncommonly r is found to be only slightly dependent on pressure. When this is true and especially when the filter-medium resistance is, as it should be, relatively small, an average value may be used for all pressures.

It is advisable to start a constant-pressure filtration test, like a comparable plant operation, at a low pressure, and smoothly increase the pressure to the desired operating level. In such cases, time and filtrate-quantity data should not be taken until the constant operating pressure is realized. The value of r calculated from the extrapolated intercept then reflects the resistance of both the filter medium and that part of the cake deposited during the pressure-buildup period. When only the total mass of dry cake is measured for the total cycle time, as is usually true in vacuum leaf tests, at least three runs of different lengths should be made to permit a reliable plot of θ/V against W . If rectification of the resulting three points is dubious, additional runs should be made.

Pressure Tests

Leaf Tests A bomb filter is used for small-scale leaf tests to simulate the performance of pressure-leaf (leaf-in-shell) filters. The equipment used is a small [50.8- by 50.8-mm (2- by 2-in.)] leaf, covered with appropriate filter medium, suspended in a cell large enough to contain sufficient slurry to form the desired cake (Fig. 18-121). The slurry may be agitated gently, for example, by an air sparger.

Although incremental time and filtrate volume may be taken during a cake-forming cycle at a selected pressure to permit a plot like Fig. 18-120a from a single run, it may be more satisfactory to make several successive quick runs at the same pressure but for different lengths of time, recording only the terminal values of filtrate volume, time, and cake mass. Operation of the commercial unit should be kept in mind when the test cycles are planned. Displacement washing and air blowing of the cake should be tried if appropriate. Wet discharge can be simulated by opening the cell and playing a jet of water on the cake; dry discharge, by applying a gentle air blast to the filtrate-discharge tube. Tests at several pressures must be conducted to determine the compressibility of the cake solids.

Plate-and-Frame Tests These tests should be conducted if the use of a filter press in the plant is anticipated; at least a few confirming tests are advisable after preliminary leaf tests, unless the slurry is very rapidly filtering. A laboratory-size filter press consisting of two plates and a single frame may be used. It will permit the observation of solids-settling, cake-packing, and washing behavior, which may be quite different for a frame than for a leaf.

Compression-Permeability Tests Instead of model leaf tests, compression-permeability experiments may be substituted with advantage for appreciably compressible solids. As in the case of constant-rate filtration, a single run provides data equivalent to those obtained from a series of constant-pressure runs, but it avoids the data-treatment complexity of constant-rate tests.

The equipment consists of a cylindrical cell with a permeable bottom and an open top, into which is fitted a close-clearance, hollow, cylindrical piston with a permeable bottom. Slurry is poured into the cell, and a cake is formed by applying gentle vacuum to the filtrate discharge line. The cell is then filled with filtrate, and the counter-weighted piston is allowed to descend to the cake level. Successive

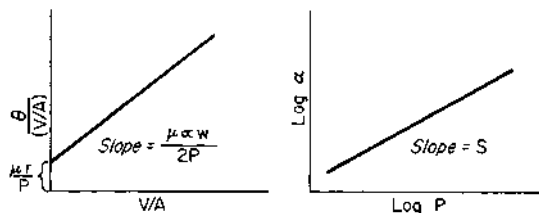


FIG. 18-120 Typical plots of filtration data.

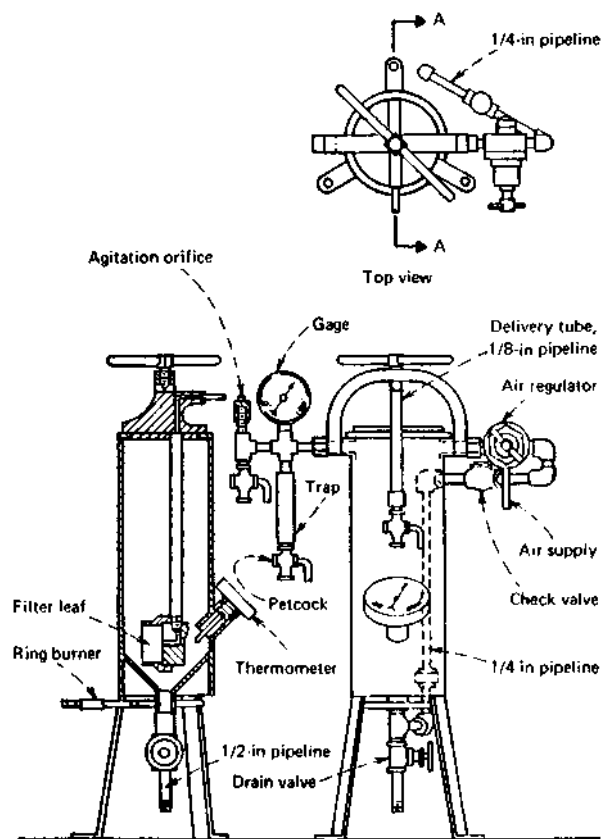


FIG. 18-121 Bomb filter for small-scale pressure filtration tests. [Silverblatt et al., *Chem. Eng.*, 81(9), 132 (1974), by permission.]

increments of mechanical stress are applied to the solids, at each of which the permeability of the cake is determined by passing filtrate through the piston under low head.

The experimental procedure and method of treatment of compression-permeability data have been explained by Grace [*Chem. Eng. Prog.*, 49, 303, 427 (1953)], who showed that the values of α measured in such a cell and in a pressure filter were the same, and by Tiller [*Filtr. Sep.*, 12, 386 (1975)].

Scaling Up Test Results The results of small-scale tests are determined as dry weight of solids or volume of filtrate per unit of area per cycle. This quantity multiplied by the number of cycles per day permits the calculation of either the filter area required for a stipulated daily capacity or the daily capacity of a specified plant filter. The scaled-up filtration area should be increased by 25 percent as a factor of uncertainty. In the calculation of cycle length, proper account must be made of the downtime of a batch filter.

FILTER MEDIA

All filters require a filter medium to retain solids, whether the filter is for cake filtration or for filter-medium or depth filtration. Specification of a medium is based on retention of some minimum particle size at good removal efficiency and on acceptable life of the medium in the environment of the filter. The selection of the type of filter medium is often the most important decision in success of the operation. For cake filtration, medium selection involves an optimization of the following factors:

1. Ability to bridge solids across its pores quickly after the feed is started (i.e., minimum propensity to bleed)

2. Low rate of entrapment of solids within its interstices (i.e., minimum propensity to blind)
3. Minimum resistance to filtrate flow (i.e., high production rate)
4. Resistance to chemical attack
5. Sufficient strength to support the filtering pressure
6. Acceptable resistance to mechanical wear
7. Ability to discharge cake easily and cleanly
8. Ability to conform mechanically to the kind of filter with which it will be used

9. Minimum cost

For filter-medium filtration, attributes 3, 4, 5, 8, and 9 of the preceding list apply and must have added to them (a) ability to retain the solids required, (b) freedom from discharge of lint or other adulterant into the filtrate, and (c) ability to plug slowly (i.e., long life).

Filter-medium selection embraces many types of construction: fabrics of woven fibers, felts, and nonwoven fibers, porous or sintered solids, polymer membranes, or particulate solids in the form of a permeable bed. Media of all types are available in a wide choice of materials.

Fabrics of Woven Fibers For cake filtration these fabrics are the most common type of medium. A wide variety of materials are available; some popular examples are listed in Table 18-10, with ratings for chemical and temperature resistance. In addition to the material of the fibers, a number of construction characteristics describe the filter cloth: (1) weave, (2) style number, (3) weight, (4) count, (5) ply, and (6) yarn number. Of the many types of weaves available, only four are extensively used as filter media: plain (square) weave, twill, chain weave, and satin.

All these weaves may be made from any textile fiber, natural or synthetic. They may be woven from spun staple yarns, multifilament continuous yarns, or monofilament yarns. The performance of the filter cloth depends on the weave and the type of yarn.

A recently developed medium known as a double weave incorporates different yarns in warp and fill in order to combine the specific advantages of each type. An example of this is Style 99FS, made by Madison Filtration, in which multifilament warp yarns provide good cake release properties and spun staple fill yarns contribute to greater retentivity.

Metal Fabrics or Screens These are available in several types of weave in nickel, copper, brass, bronze, aluminum, steel, stainless steel, Monel, and other alloys. In the plain weave, 400 mesh is the closest wire spacing available, thus limiting use to coarse crystalline slurries, pulps, and the like. The "Dutch weaves" employing relatively large, widely spaced, straight warp wires and relatively small crimped filling wires can be woven much more closely, providing a good medium for filtering fine crystals and pulps. This type of weave tends to plug readily when soft or amorphous particles are filtered and makes the use of filter aid desirable. Good corrosion and high temperature resistance of properly selected metals makes filtrations with metal media desirable for long-life applications. This is attractive for handling toxic materials in closed filters to which minimum exposure by maintenance personnel is desirable.

Pressed Felts and Cotton Batting These materials are used to filter gelatinous particles from paints, spinning solutions, and other viscous liquids. Filtration occurs by deposition of the particles in and on the fibers throughout the mat.

Nonwoven media consist of web or sheet structures which are composed primarily of fibers or filaments bonded together by thermal, chemical, or mechanical (such as needlepunching) means. Needled felts are the most commonly used nonwoven fabric for liquid filtration. Additional strength often is provided by including a scrim of woven fabric encapsulated within the nonwoven material. The surface of the medium can be calendered to improve particle retention and assist in filter cake release. Weights range from 270 to 2700 gm/m² (8 to 80 oz/yd²). Because of their good retentivity, high strength, moderate cost, and resistance to blinding, nonwoven media have found wide acceptance in filter press use, particularly in mineral concentrate filtration applications. They are used frequently on horizontal belt filters where their dimensional stability reduces or eliminates wrinkling and biasing problems often encountered with woven belts.

Filter Papers These papers come in a wide range of permeability, thickness, and strength. As a class of material, they have low strength, however, and require a perforated backup plate for support.

TABLE 18-10 Characteristics of Filter-Fabric Materials*

Generic name and description	Breaking tenacity, g/denier	Abrasion resistance	Resistance to acids	Resistance to alkalis	Resistance to oxidizing agents	Resistance to solvents	Specific gravity	Maximum operating temperature, °F†
Acetate—cellulose acetate. When not less than 92% of the hydroxyl groups are acetylated, "triacetate" may be used as a generic description.	1.2–1.5	G	F	P	G	G	1.33	210
Acrylic—any long-chain synthetic polymer composed of at least 85% by weight of acrylonitrile units.	2.0–4.8	G	G	F	G	E	1.18	300
Glass—fiber-forming substance is glass.	3.0–7.2	P	E	P	E	E	2.54	600
Metallic—composed of metal, metal-coated plastic, plastic-coated metal, or a core completely covered by metal.	—	G						
Modacrylic—fiber-forming substance is any long-chain synthetic polymer composed of less than 85% but at least 35% by weight of acrylonitrile units.	2.5–3.0	G	G	G	G	G	1.30	180
Nylon—any long-chain synthetic polyamide having recurring amide groups as an integral part of the polymer chain.	3.8–9.2	E	F–P	G	F–P	G	1.14	225
Polyester—any long-chain synthetic polymer composed of at least 85% by weight of an ester of a dihydric alcohol and terephthalic acid (p -HOOC—C ₆ H ₄ —COOH).	2.2–7.8	E–G	G	G–F	G	G	1.38	300
Polyethylene—long-chain synthetic polymer composed of at least 85% weight of ethylene.	1.0–7.0	G	G	G	F	G	0.92	165‡
Polypropylene—long-chain synthetic polymer composed of at least 85% by weight of propylene.	4.8–8.5	G	E	E	G	G	0.90	250§
Cotton—natural fibers.	3.3–6.4	G	P	F	G	E–G	1.55	210
Fluorocarbon—long-chain synthetic polymer composed of tetrafluoroethylene units.	1.0–2.0	F	E	E	E	G	2.30	550¶

*Adapted from Mais, *Chem. Eng.*, **78**(4), 51 (1971). Symbols have the following meaning: E = excellent, G = good, F = fair, P = poor.

†°C = (°F – 32)/1.8; K = (°F + 459.7)/1.8.

‡Low-density polymer. Up to 230°F for high-density.

§Heat-set fabric; otherwise lower.

¶Requires ventilation because of release of toxic gases above 400°F.

Rigid Porous Media These are available in sheets or plates and tubes. Materials used include sintered stainless steel and other metals, graphite, aluminum oxide, silica, porcelain, and some plastics—a gamut that allows a wide range of chemical and temperature resistance. Most applications are for clarification.

Polymer Membranes These are used in filtration applications for fine-particle separations such as microfiltration and ultrafiltration (clarification involving the removal of 1- μ m and smaller particles). The membranes are made from a variety of materials, the commonest being cellulose acetates and polyamides. Membrane filtration, discussed in Sec. 22, has been well covered by Porter (in Schweitzer, op. cit., sec. 2.1).

Media made from woven or nonwoven fabrics coated with a polymeric film, such as Primapor, and Primapor II made by Madison Filtration, Gore-Tex, made by W. L. Gore and Associates, and Tetratex, made by Donaldson Company, combine the high retentivity characteristics of a membrane with the strength and durability of a thick filter cloth. These media are used on both continuous and batch filters where excellent filtrate clarity is required.

Granular Beds of Particulate Solids Beds of solids like sand or coal are used as filter media to clarify water or chemical solutions containing small quantities of suspended particles. Filter-grade grains of desired particle size can be purchased. Frequently beds will be constructed of layers of different materials and different particle sizes.

Various types of filter media and the materials of which they are constructed are surveyed extensively by Purchas (*Industrial Filtration of Liquids*, CRC Press, Cleveland, 1967, chap. 3), and characterizing measurements (e.g., pore size, permeability) are reviewed in detail by Rushton and Griffiths (in Orr, op. cit., chap. 3). Briefer summaries of classification of media and of practical criteria for the selection of a filter medium are presented by Shoemaker (op. cit., p. 26) and Purchas [*Filtr. Sep.*, **17**, 253, 372 (1980)].

FILTER AIDS

Use of filter aids is a technique frequently applied for filtrations in which problems of slow filtration rate, rapid medium blinding, or unsatisfactory filtrate clarity arise. Filter aids are granular or fibrous solids capable of forming a highly permeable filter cake in which very fine solids or slimy, deformable flocs may be trapped. Application of filter aids may allow the use of a much more permeable filter medium than the clarification would require to produce filtrate of the same quality by depth filtration.

Filter aids should have low bulk density to minimize settling and aid good distribution on a filter-medium surface that may not be horizontal. They should also be porous and capable of forming a porous cake to minimize flow resistance, and they must be chemically inert to the filtrate. These characteristics are all found in the two most popular

commercial filter aids: diatomaceous earth (also called diatomite), which is an almost pure silica prepared from deposits of diatom skeletons; and expanded perlite, particles of "puffed" lava that are principally aluminum alkali silicate. Cellulosic fibers (ground wood pulp) are sometimes used when siliceous materials cannot be used but are much more compressible. The use of other less effective aids (e.g., carbon and gypsum) may be justified in special cases. Sometimes a combination of carbon and diatomaceous earth permits adsorption in addition to filter-aid performance. Various other materials, such as salt, fine sand, starch, and precipitated calcium carbonate, are employed in specific industries where they represent either waste material or inexpensive alternatives to conventional filter aids.

Diatomaceous Earth Filter aids of diatomaceous earth have a dry bulk density of 128 to 320 kg/m³ (8 to 20 lb/ft³), contain particles mostly smaller than 50 μ m, and produce a cake with porosity in the range of 0.9 (volume of voids/total filter-cake volume). The high porosity (compared with a porosity of 0.38 for randomly packed uniform spheres and 0.2 to 0.3 for a typical filter cake) is indicative of its filter-aid ability. Different methods of processing the crude diatomite result in a series of filter aids having a wide range of permeability.

Perlite Perlite filter aids are somewhat lower in bulk density (48 to 96 kg/m³, or 3 to 6 lb/ft³) than diatomaceous silica and contain a higher fraction of particles in the 50- to 150- μ m range. Perlite is also available in a number of grades of differing permeability and cost, the grades being roughly comparable to those of diatomaceous earth. Diatomaceous earth will withstand slightly more extreme pH levels than perlite, and it is said to be somewhat less compressible.

Filter aids are used in two ways: (1) as a precoat and (2) mixed with the slurry as a "body feed." Precoat filtration, employing a thin layer of about 0.5 to 1.0 kg/m² (0.1 to 0.2 lb/ft²) deposited on the filter medium prior to beginning feed to the filter, is in wide use to protect the filter medium from fouling by trapping solids before they reach the medium. It also provides a finer matrix to trap fine solids and assure filtrate clarity. Body-feed application is the continuous addition of filter aid to the filter feed to increase the porosity of the cake. The amount of addition must be determined by trial, but in general, the quantity added should at least equal the amount of solids to be removed. For solids loadings greater than 1000 ppm this may become a significant cost factor. An acceptable alternative might be to use a rotary vacuum precoat filter [Smith, *Chem. Eng.*, **83**(4), 84 (1976)]. Further details of filter-aid filtration are set forth by Cain (in Schweitzer, op. cit., sec. 4.2) and Hutto [*Am. Inst. Chem. Eng. Symp. Ser.*, **73**(171), 50 (1977)]. Figure 18-122 shows a flow sheet indicating arrangements for both precoat and body-feed applications. Most filter aid is used on a one-time basis, although some techniques have been demonstrated to reuse precoat filter aid on vertical-tube pressure filters.

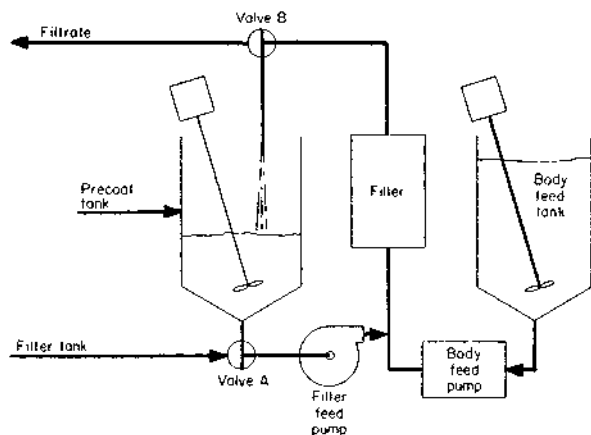


FIG. 18-122 Filter-aid filtration system for precoat or body feed. (Schweitzer, Handbook of Separation Techniques for Chemical Engineers, p. 4-12. Copyright 1979 by McGraw-Hill, Inc. and used with permission.)

FILTRATION EQUIPMENT

Cake Filters Filters that accumulate appreciable visible quantities of solids on the surface of a filter medium are called cake filters. The slurry feed may have a solids concentration from about 1 percent to greater than 40 percent. The filter medium on which the cake forms is relatively open to minimize flow resistance, since once the cake forms, it becomes the effective filter medium. The initial filtrate therefore may contain unacceptable solids concentration until the cake is formed. This situation may be made tolerable by recycling the filtrate until acceptable clarity is obtained or by using a downstream polishing filter (clarifying type).

Cake filters are used when the desired product of the operation is the solids, the filtrate, or both. When the filtrate is the product, the degree of removal from the cake by washing or blowing with air or gas becomes an economic optimization. When the cake is the desired product, the incentive is to obtain the desired degree of cake purity by washing, blowing, and sometimes mechanical expression of residual liquid.

Implicit in cake filtration is the removal and handling of solids, since the cake is usually relatively dry and compacted. Cakes can be sticky and difficult to handle; therefore, the ability of a filter to discharge the cake cleanly is an important equipment-selection criterion.

In the operational sense, some filters are batch devices, whereas others are continuous. This difference provides the principal basis for classifying cake filters in the discussion that follows. The driving force by which the filter functions—hydrostatic head ("gravity"), pressure imposed by a pump or a gas blanket, or atmospheric pressure ("vacuum")—will be used as a secondary criterion.

Batch Cake Filters

Nutsche Filters A nutsche is one of the simplest batch filters. It is a tank with a false bottom, perforated or porous, which may either support a filter medium or act as the filter medium. The slurry is fed into the filter vessel, and separation occurs by gravity flow, gas pressure, vacuum, or a combination of these forces. The term "nutsche" comes from the German term for sucking, and vacuum is the common operating mode.

The design of most nutsche filters is very simple, and they are often fabricated by the user at low cost. The filter is very frequently used in laboratory, pilot-plant, or small-plant operation. For large-scale processing, however, the excessive floor area encumbered per unit of filtration area and the awkwardness of cake removal are strong deterrents. For small-scale operations, cake is manually removed. For large-scale applications, cake may be further processed by reslurrying or redissolving; or it may be removed manually (by shovel) or by mechanical discharge arrangements such as a movable filter medium belt.

Thorough displacement washing is possible in a nutsche if the wash solvent is added before the cake begins to be exposed to air displacement of filtrate. If washing needs to be more effective, an agitator can be provided in the nutsche vessel to reslurry the cake to allow adequate diffusion of solute from the solids.

Horizontal Plate Filter The horizontal multiple-plate pressure filter consists of a number of horizontal circular drainage plates and guides placed in a stack in a cylindrical shell (Fig. 18-123). In normal practice the filtering pressure is limited to 345 kPa (50 psig), although special filters have been designed for shell pressures of 2.1 MPa (300 psig) or higher.

Filter Press The filter press, one of the most frequently used filters in the early years of the chemical industry, is still widely employed. Often referred to generically (in error) as the plate-and-frame filter, it has probably over 100 design variations. Two basic popular designs are the flush-plate, or plate-and-frame, design and the recessed-plate press. Both are available in a wide range of materials: metals, coated metals, plastics, or wood.

Plate-and-frame press. This press is an alternate assembly of plates covered on both sides with a filter medium, usually a cloth, and hollow frames that provide space for cake accumulation during filtration. The frames have feed and wash manifold ports, while the plates have filtrate drainage ports. The plates and frames usually are

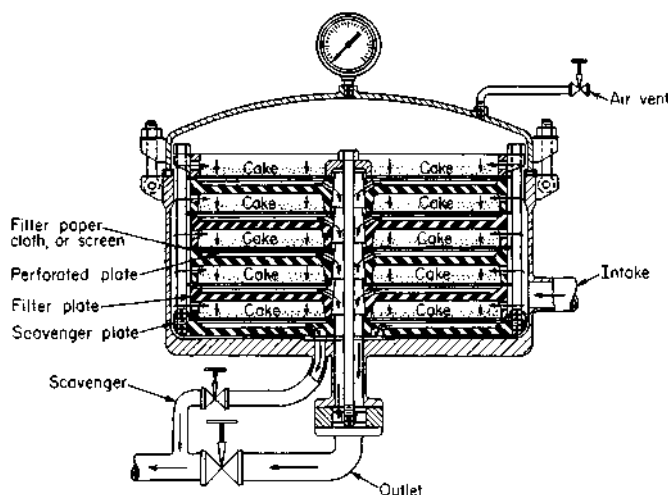


FIG. 18-123 Elevation section of a Sparkler horizontal plate filter. (Sparkler Filters, Inc.)

rectangular, although circles and other shapes also are used (Fig. 18-124). They are hung on a pair of horizontal support bars and pressed together during filtration to form a watertight closure between two end plates, one of which is stationary. The press may be closed manually, hydraulically, or by a motor drive. Several feed and filtrate discharge arrangements are possible. In the most popular, the feed and discharge of the several elements of the press are manifolded via some of the holes that are in the four corners of each plate and frame (and filter cloth) to form continuous longitudinal channels from the stationary end plate to the other end of the press. Alternatively, the filtrate may be drained from each plate by an individual valve and spigot (for open discharge) or tubing (for closed). Top feed to and bottom discharge from the chambers provide maximum recovery of filtrate and maximum mean cake dryness. This arrangement is especially suitable for heavy fast-settling solids. For most slurries, bottom feed and top filtrate discharge allow quick air displacement and produce a more uniform cake.

Two wash techniques are used in plate-and-frame filter presses, illustrated in Fig. 18-125. In simple washing, the wash liquor follows the same path as the filtrate. If the cake is not extremely uniform and highly permeable, this type of washing is ineffective in a well-filled

press. A better technique is thorough washing, in which the wash is introduced to the faces of alternate plates (with their discharge channels valved off). The wash passes through the entire cake and exits through the faces of the other plates. This improved technique requires a special design and the assembly of the plates in proper order. Thorough washing should be used only when the frames are well filled, since an incomplete fill of cake will allow cake collapse during the wash entry. The remainder of the wash flow will bypass through cracks or channels opened in the cake.

Filter presses are made in plate sizes from 10 by 10 cm (4 by 4 in) to 2.4 by 2.4 m (94 by 94 in). Frame thickness ranges from 0.3 to 20 cm (0.125 to 8 in). Operating pressures up to 689 kPa (100 psig) are common, with some presses designed for 6.9 MPa (1000 psig). Some metal units have cored plates for steam or refrigerant. Maximum pressure for wood or plastic frames is 410 to 480 kPa (60 to 70 psig).

The filter press has the advantage of simplicity, low capital cost, flexibility, and ability to operate at high pressure in either a cake-filter or a clarifying-filter application. Floor-space and headroom needs per unit of filter area are small, and capacity can be adjusted by adding or removing plates and frames. Filter presses are cleaned easily, and the filter medium is easily replaced. With proper operation a denser, drier cake compared with that of most other filters is obtained.

There are several serious disadvantages, including imperfect washing due to variable cake density, relatively short filter-cloth life due to the mechanical wear of emptying and cleaning the press (often involving scraping the cloth), and high labor requirements. Presses frequently drip or leak and thereby create housekeeping problems, but the biggest problem arises from the requirement to open the filter for cake discharge. The operator is thus exposed routinely to the contents of the filter, and this is becoming an increasingly severe disadvantage as more and more materials once believed safe are given restricted exposure limits.

Recessed-plate filter press. This press is similar to the plate-and-frame press in appearance but consists only of plates (Fig. 18-126). Both faces of each plate are hollowed to form a chamber for cake accumulation between adjacent plates. This design has the advantage of about half as many joints as a plate-and-frame press, making a tight closure more certain. Figure 18-127 shows some of the features of one type of recessed-plate filter which has a gasket to further minimize leaks. Air can be introduced behind the cloth on both sides of each plate to assist cake removal.

Some interesting variations of standard designs include the ability to roll the filter to change from a bottom to a top inlet or outlet and the

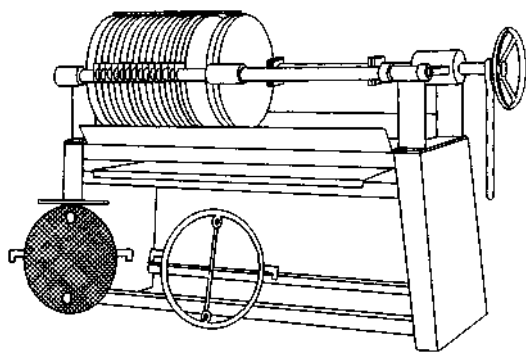


FIG. 18-124 Circular-plate fabricated-metal filter press. (Star Systems Filtration Division.)

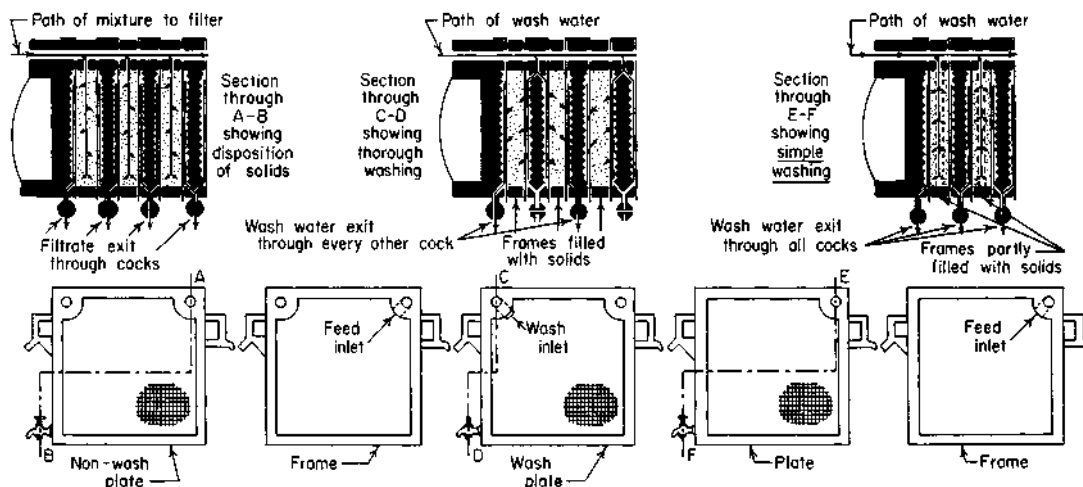


FIG. 18-125 Filling and washing flow patterns in a filter press. (D. R. Sperry & Co.)

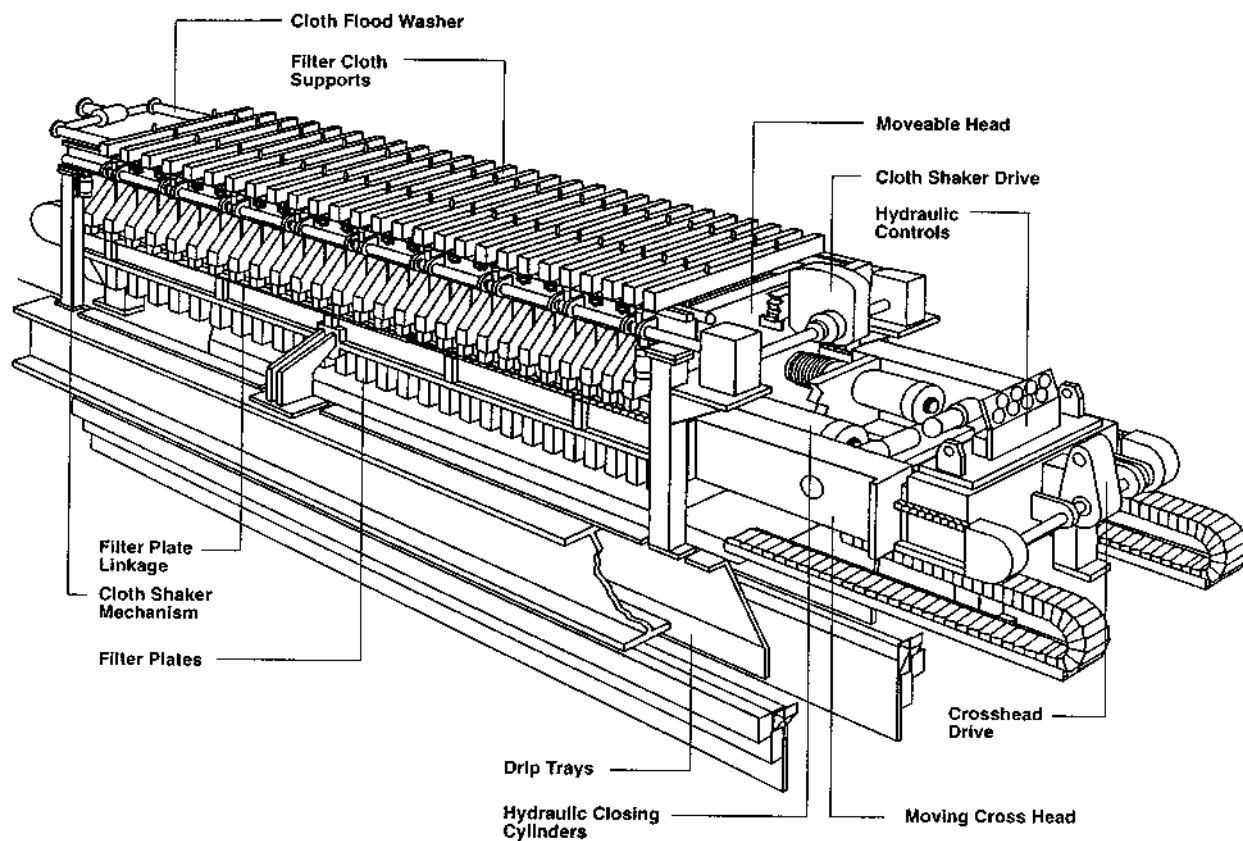


FIG. 18-126 Automated recessed-plate filter press used in mineral applications. (Dorr-Oliver EIMCO.)

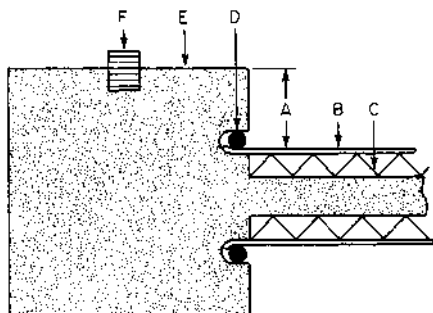


FIG. 18-127 Section detail of a caulked-gasketed-recessed filter plate: (a) cake recess; (b) filter cloth; (c) drainage surface of plate; (d) caulking strip; (e) plate joint; (f) sealing gasket. (Dorr-Oliver EIMCO)

ability to add blank dividers to convert a press to a multistage press for further clarification of the filtrate or to do two separate filtrations simultaneously in the same press. Some designs have rubber membranes between plates which can be expanded when filtration is finished to squeeze out additional moisture. Some designs feature automated opening and cake-discharge operations to reduce labor requirements. Examples of this type of pressure filter include Larox, Vertipress, and Oberlin.

Internal Cake Tube Filters or Liquid Bag Filters This type of filter, such as manufactured by Industrial Filter and Pump Mfg. Co., Rosedale, Illinois, and many others, utilizes one or more perforated tubes supported by a tube sheet or by the lip of the pressure vessel. A cylindrical filter bag sealed at one end is inserted into the perforated tube. The open end of the filter bag generally has a flange or special seal ring to prevent leakage.

Slurry under pressure is admitted to the chamber between the head of the shell and the tube sheet, whence it enters and fills the tubes. Filtration occurs as the filtrate passes radially outward through the filter medium and the wall of each tube into the shell and on out the filtrate discharge line, depositing cake on the medium. The filtration cycle is ended when the tubes have filled with cake or when the media have become plugged. The cake can be washed (if it has not been allowed to fill the tubes completely) and air-blown. The filter has a removable head to provide easy access to the tube sheet and mouth of the tubes; thus "sausages" of cake can be removed by taking out the filter bags or each tube and bag assembly together. The tubes themselves are easily removed for inspection and cleaning.

The advantages of the tubular filter are that it uses an easily replaced filter medium, its filtration cycle can be interrupted and the shell can be emptied of prefiltrate at any time without loss of the cake, the cake is readily recoverable in dry form, and the inside of the filter is conveniently accessible. There is also no unfiltered heel. Disadvantages are the necessity and attendant labor requirements of emptying by hand and replacing the filter media and the tendency for heavy solids to settle out in the header chamber. Applications are as a scavenger filter to remove fines not removed in a prior-filtration stage with a different kind of equipment, to handle the runoff from other filters, and in semiworks and small-plant operations in which the filter's size, versatility, and cleanliness recommend it.

External-Cake Tubular Filters Several filter designs are available with vertical tubes supported by a filtrate-chamber tube sheet in a vertical cylindrical vessel (Fig. 18-128). The tubes may be made of wire cloth; porous ceramic, carbon, plastic, or metal; or closely wound wire. The tubes may have a filter cloth on the outside. Frequently a filter-aid precoat will be applied to the tubes. The prefiltrate slurry is fed near the bottom of the vertical vessel. The filtrate passes from the outside to the inside of the tubes and into a filtrate chamber at the top or the bottom of the vessel. The solids form a cake on the outside of the tubes with the filter area actually increasing as the cake builds up, partially compensating for the increased flow resistance of the thicker cake. The filtration cycle continues until the differential pressure reaches a specified level, or until about 25 mm (1 in) of cake thickness is obtained.

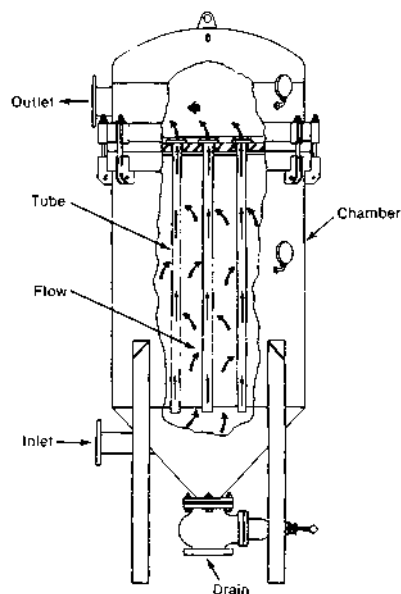


FIG. 18-128 Top-outlet tubular filter. (Industrial Filter & Pump Mfg. Co.)

Cake-discharge methods are the chief distinguishing feature among the various designs. That of the Industrial Filter & Pump Hydra-Shoc, for example, removes cake from the tubes by filtrate backflushing assisted by the "shocking" action of a compressed-gas pocket formed in the filtrate chamber at the top of the vertical vessel. Closing the filtrate outlet valve while continuing to feed the filter causes compression of the gas volume trapped in the dome of the vessel until, at the desired gas pressure, quick-acting valves stop the feed and open a bottom drain. The compressed gas rapidly expands, forcing a rush of filtrate back across the filter medium and dislodging the cake, which drains out the bottom with the flush liquid. Of course, this technique may be used only when wet-cake discharge is permitted.

Dry cake discharge can be achieved with a Fundabac candle-type filter manufactured by DrM, Dr. Müller, AG, of Switzerland. This filter uses a candle made up of six small-diameter tubes around a central filtrate delivery tube. This design allows the filter cloth to be flexed outward upon blowback, easily achieving an effective dry cake discharge (Fig. 18-129).

Pressure Leaf Filters Sometimes called tank filters, they consist of flat filtering elements (leaves) supported in a pressure shell. The leaves are circular, arc-sided, or rectangular, and they have filtering surfaces on both faces. The shell is a cylindrical or conical tank. Its axis may be horizontal or vertical, and the filter type is described by its shell axis orientation.

A filter leaf consists of a heavy screen or grooved plate over which a filter medium of woven fabric or fine wire cloth may be fitted. Textile fabrics are more commonly used for chemical service and are usually applied as bags that may be sewed, zippered, stapled, or snapped. Wire-screen cloth is frequently used for filter-aid filtrations, particularly if a precoat is applied. It may be attached by welding, riveting, bolting, or caulking or by the clamped engagement of two 180° bends in the wire cloth under tension, as in Multi Metal's Rim-Lok leaf. Leaves may also be of all-plastic construction. The filter medium, regardless of material, should be as taut as possible to minimize sagging when it is loaded with a cake; excessive sag can cause cake cracking or dropping. Leaves may be supported at top, bottom, or center and may discharge filtrate from any of these locations. Figure 18-130 shows the elevation section of a precoated bottom-support wire leaf.

Pressure leaf filters are operated batchwise. The shell is locked, and the prefiltrate slurry is admitted from a pressure source. The slurry enters in such a way as to minimize settling of the suspended solids. The shell

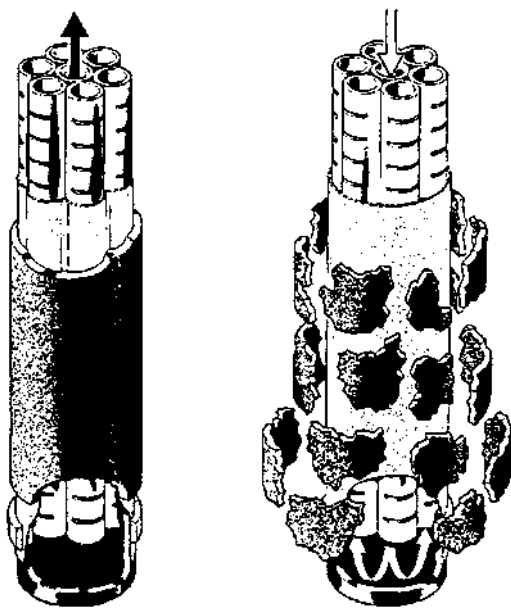


FIG. 18-129 Cake formation and discharge with the Fundabac filter element. (DrM, Dr. Müller AG, Switzerland.)

is filled, and filtration occurs on the leaf surfaces, the filtrate discharging through an individual delivery line or into an internal manifold, as the filter design dictates. Filtration is allowed to proceed only until a cake of the desired thickness has formed, since to overfill will cause cake consolidation with consequent difficulty in washing and discharge. The decision of when to end the filtering cycle is largely a matter of experience, guided roughly by the rate in a constant-pressure filter or pressure drop in a constant-rate filter. This judgment may be supplanted by the use of a detector which "feels" the thickness of cake on a representative leaf.

If the cake is to be washed, the slurry heel can be blown from the filter and wash liquor can be introduced to refill the shell. If the cake tends to crack during air blowing, it may be necessary to displace the slurry heel with wash gradually so as never to allow the cake to dry. Upon the completion of filtration and washing, the cake is discharged by one of several methods, depending on the shell and leaf configuration.

Horizontal pressure leaf filters. In these filters the leaves may be rectangular leaves which run parallel to the axis and are of varying sizes since they form chords of the shell; or they may be circular or square elements parallel to the head of the shell, and all of the same dimension. The leaves may be supported in the shell from an independent rack, individually from the shell, or from a filtrate manifold. Horizontal filters are particularly suited to dry-cake discharge.

Most of the currently available commercial horizontal pressure filters have leaves parallel to the shell head. Cake discharge may be wet or dry; it can be accomplished by sluicing with liquid sprays, vibration of the leaves, or leaf rotation against a knife, wire, or brush. If a wet-cake discharge is allowable, the filters will probably be sluiced with high-pressure liquid. If the filter has a top or bottom filtrate manifold, the leaves are usually in a fixed position, and the spray header is rotated to contact all filter surfaces. If the filtrate header is center-mounted, the leaves are generally rotated at about 3 r/min and the spray header is fixed. Some units may be wet-cake-discharged by mechanical vibration of the leaves with the filter filled with liquid. Dry-cake discharge normally will be accomplished by vibration if leaves are top- or bottom-manifolded and by rotation of the leaves against a cutting knife, wire, or brush if they are center-manifolded.

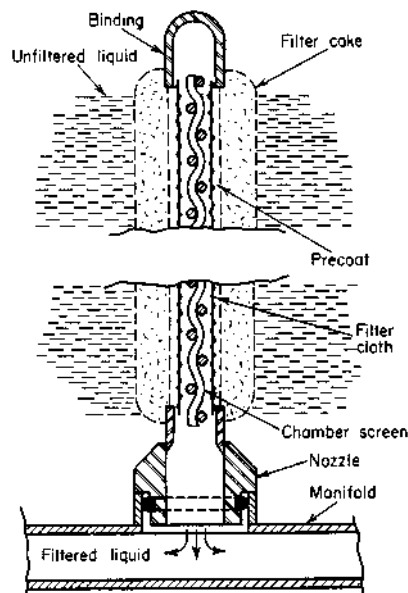


FIG. 18-130 Section of precoated wire filter leaf.

In many designs the filter is opened for cake discharge, and the leaf assembly is separated from the shell by moving one or the other on rails (Fig. 18-131). For processes involving toxic or flammable materials, a closed filter system can be maintained by sloping the bottom of the horizontal cylinder to the drain nozzle for wet discharge or by using a screw conveyor in the bottom of the shell for dry discharge.

Vertical pressure leaf filters. These filters have vertical, parallel, rectangular leaves mounted in an upright cylindrical pressure tank. The leaves usually are of such different widths as to allow them to conform to the curvature of the tank and to fill it without waste space. The leaves often rest on a filtrate manifold, the connection being sealed by an O ring, so that they can be lifted individually from the top of the filter for inspection and repair. A scavenger leaf frequently is installed in the bottom of the shell to allow virtually complete filtration of the slurry heel at the end of a cycle.

Vertical filters are not convenient for the removal of dry cake, although they can be used in this service if they have a bottom that can be retracted to permit the cake to fall into a bin or hopper below. They are adapted rather to wet-solids discharge, a process that may be assisted by leaf vibration, air or steam sparging of a filter full of water, sluicing from fixed, oscillating, or traveling nozzles, and blowback. They are made by many companies, and they enjoy their widest use for filter-aid precoat filtration.

Advantages and uses. The advantages of pressure leaf filters are their considerable flexibility (up to the permissible maximum, cakes of various thickness can be formed successfully), their low labor charges, particularly when the cake may be sluiced off or the dry cake discharged cleanly by blowback, the basic simplicity of many of the designs, and their adaptability to quite effective displacement washing. Their disadvantages are the requirement of exceptionally intelligent and watchful supervision to avoid cake consolidation or dropping, their inability to form as dry a cake as a filter press, their tendency to classify vertically during filtration and to form misshapen nonuniform cakes unless the leaves rotate, and the restriction of most models to 610 kPa (75 psig) or less.

Pressure leaf filters are used to separate much the same kinds of slurries as are filter presses and are used much more extensively than filter presses for filter-aid filtrations. They should be seriously considered whenever uniformity of production permits long-time operation

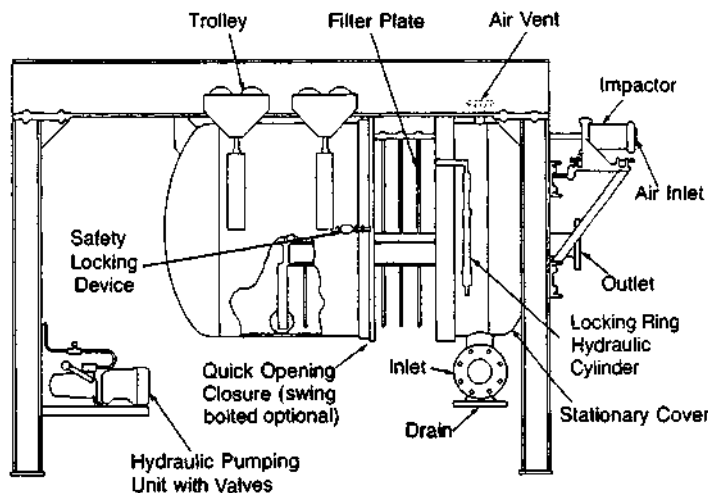


FIG. 18-131 Horizontal-tank pressure leaf filter designed for dry cake discharge. (Sparkler Filter, Inc.)

under essentially constant filtration conditions, when thorough washing with a minimum of liquor is desired, or when vapors or fumes make closed construction desirable. Under such conditions, if the filter medium does not require frequent changing, they may show a considerable advantage in cycle and labor economy over a filter press, which has a lower initial cost, and advantages of economy and flexibility over continuous vacuum filters, which have a higher first cost.

Pressure leaf filters are available with filtering areas of 930 cm^2 (1 ft^2) (laboratory size) up to about 440 m^2 (4734 ft^2) for vertical filters and 158 m^2 (1700 ft^2) for horizontal ones. Leaf spacings range from 5 to 15 cm (2 to 6 in) but are seldom less than 7.5 cm (3 in) since 1.3 to 2.5 cm (0.5 to 1 in) should be left open between surfaces.

Centrifugal-Discharge Filter Horizontal top-surface filter plates may be mounted on a hollow motor-connected shaft that serves both as a filtrate-discharge manifold and as a drive shaft to permit centrifugal removal of the cake. An example is the Funda filter (marketed in the United States by Steri Technologies), illustrated schematically in Fig. 18-132. The filtering surface may be a textile fabric or a wire screen, and the use of a precoat is optional. The Funda filter is driven from the top, leaving the bottom unobstructed for inlet and drainage lines; a somewhat similar machine that employs a bottom drive, pro-

viding a lower center of mass and ground-level access to the drive system, is the German-made **Schenk filter** (marketed in the United States by Pall Seitz Schenk Filtersystems).

During filtration, the vessel that coaxially contains the assembly of filter plates is filled with prefilter under pressure, the filtrate passes through the plates and out the hollow shaft, and cake is formed on the top surfaces of the plates. After filtration, the vessel is drained, or the heel may be filtered by recirculation through a cascade ring at the top of the filter. The cake may be washed—or it may be extracted, steamed, air-blown, or dried by hot gas. It is discharged, wet or dry, by rotation of the shaft at sufficiently high speed to sling away the solids. If flushing is permitted, the discharge is assisted by a backwash of appropriate liquid.

The operating advantages of the centrifugal-discharge filter are those of a horizontal-plate filter and, further, its ability to discharge cake without being opened. It is characterized by low labor demand, easy adaptability to automatic control, and amenability to the processing of hazardous, noxious, or sterile materials. Its disadvantages are its complexity and maintenance (stuffing boxes, high-speed drive) and its cost. The Funda filter is made in sizes that cover the filtering area range of 1 to 50 m^2 (11 to 537 ft^2). The largest Schenk filter provides 100 m^2 (1075 ft^2) of area.

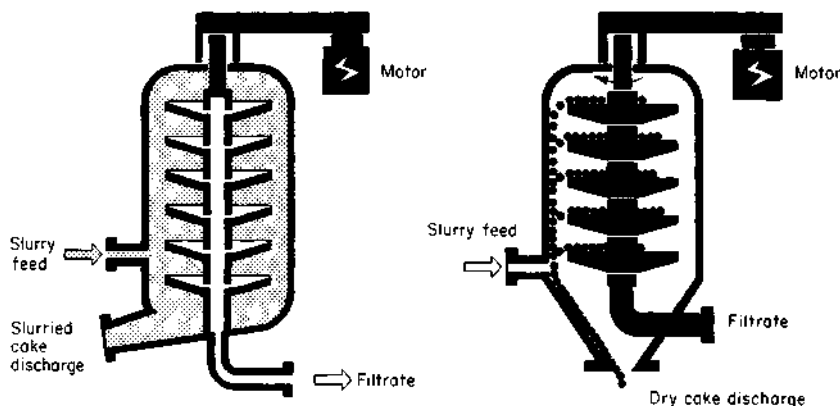


FIG. 18-132 Schematic of a centrifugal-discharge filter. (Steri Technologies.)

Continuous Cake Filters Continuous cake filters are applicable when cake formation is fairly rapid, as in situations in which slurry flow is greater than about 5 L/min (1 to 2 gal/min), slurry concentration is greater than 1 percent, and particles are greater than 0.5 μm in diameter. Liquid viscosity below 0.1 Pa·s (100 cP) is usually required for maintaining rapid liquid flow through the cake. Some designs of continuous filters can compromise some of these guidelines by sacrificial use of filter aid when the cake is not the desired product.

Rotary Drum Filters The rotary drum filter is the most widely used of the continuous filters. There are many design variations, including operation as either a pressure filter or a vacuum filter. The major difference between designs is in the technique for cake discharge, to be discussed later. All the alternatives are characterized by a horizontal-axis drum covered on the cylindrical portion by filter medium over a grid support structure to allow drainage to manifolds. Basic materials of construction may be metals or plastics. Sizes (in terms of filter areas) range from 0.37 to 186 m^2 (4 to 2000 ft^2).

All drum filters (except the single-compartment filter) utilize a rotary-valve arrangement in the drum-axis support trunnion to facilitate removal of filtrate and wash liquid and to allow introduction of air or gas for cake blowback if needed. The valve controls the relative duration of each cycle as well as providing "dead" portions of the cycle through the use of bridge blocks. A typical valve design is shown in Fig. 18-133. Internal piping manifolds connect the valve with various sections of the drum.

Most drum filters are fed by operating the drum with about 35 percent of its circumference submerged in a slurry trough, although submergence can be set for any desired amount between zero and almost total. Some units contain an oscillating rake agitator in the trough to aid solids suspension. Others use propellers, paddles, or no agitator.

Slurries of free-filtering solids that are difficult to suspend are sometimes filtered on a top-feed drum filter or filter-dryer. An example application is in the production of table salt. An alternative for slurries of extremely coarse, dense solids is the internal drum filter. In the chemical-process industry both top-feed and internal drums (which are described briefly by Emmett in Schweitzer, *op. cit.*, p. 4-41) have largely been displaced by the horizontal vacuum filter (q.v.).

Most drum filters operate at a rotation speed in the range of 0.1 to 10 r/min. Variable-speed drives are usually provided to allow adjustment for changing cake-formation and drainage rates.

Drum filters commonly are classified according to the feeding arrangement and the cake-discharge technique. They are so treated in this subsection. The characteristics of the slurry and the filter cake usually dictate the cake-discharge method.

Scraper-Discharge Filter The filter medium is usually caulked into grooves in the drum grid, with cake removal facilitated by a scraper

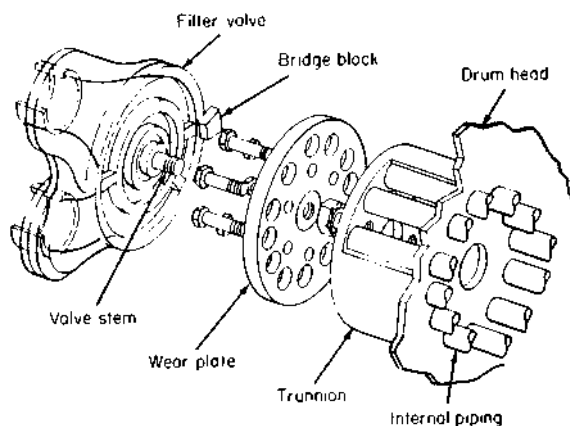


FIG. 18-133 Component arrangement of a continuous-filter valve. (Dorr-Oliver EIMCO.)

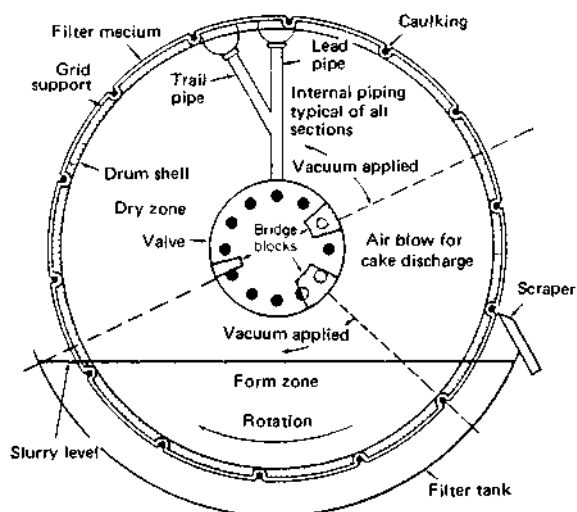


FIG. 18-134 Schematic of a rotary-drum vacuum filter with scraper discharge, showing operating zones. (Schweitzer, *Handbook of Separation Techniques for Chemical Engineers*, p. 4-38. Copyright 1979 by McGraw-Hill, Inc. and used with permission.)

blade just prior to the resubmergence of the drum (Fig. 18-134). The scraper serves mainly as a deflector to direct the cake, dislodged by an air blowback, into the discharge chute, since actual contact with the medium would cause rapid wear. In some cases the filter medium is held by circumferentially wound wires spaced 50 mm (2 in) apart, and a flexible scraper blade may rest lightly against the wire winding. A taut wire in place of the scraper blade may be used in some applications in which physical dislodging of sticky, cohesive cakes is needed.

For a given slurry, the maximum filtration rate is determined by the minimum cake thickness which can be removed—the thinner the cake, the less the flow resistance and the higher the rate. The minimum thickness is about 6 mm (0.25 in) for relatively rigid or cohesive cakes of materials such as mineral concentrates or coarse precipitates like gypsum or calcium citrate. Solids that form friable cakes composed of less cohesive materials such as salts or coal will usually require a cake thickness of 13 mm (0.5 in) or more. Filter cakes composed of fine precipitates such as pigments and magnesium hydroxide, which often produce cakes that crack or adhere to the medium, usually need a thickness of at least 10 mm (0.38 in).

String-Discharge Filter A system of endless strings or wires spaced about 13 mm (0.5 in) apart pass around the filter drum but are separated tangentially from the drum at the point of cake discharge, lifting the cake off as they leave contact with the drum. The strings return to the drum surface guided by two rollers, the cake separating from the strings as they pass over the rollers. If it has the required body, a thinner cake (5 mm or about $\frac{3}{16}$ in) than can be handled by drum filters is feasible, allowing more difficult materials to be filtered. This is done at the expense of greater dead area on the drum. Success depends on the ability of the cake to be removed with the strings and must be determined experimentally. Applications are mainly in the starch and pharmaceutical industries, with some in the metallurgical field.

Removable-Medium Filters Some drum filters provide for the filter medium to be removed and reapplied as the drum rotates. This feature permits the complete discharge of thin or sticky cake and provides the regenerative washing of the medium to reduce blinding. Higher filtration rates are possible because of the thinner cake and clean medium, but this is compromised by a less pure filtrate than normally produced by a nonremovable medium.

Belt-discharge filter. This is a drum filter carrying a fabric that is removed, passed over rollers, washed, and returned to the drum. Figure 18-135 shows the path of the medium while it is off the drum.

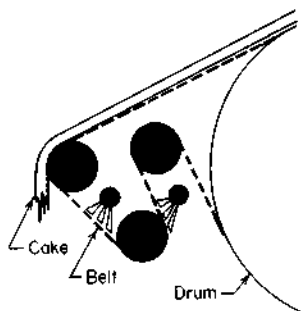


FIG. 18-135 Cake discharge and medium washing on an EIMCO belt filter. (Dorr-Oliver EIMCO.)

A special aligning device keeps the medium wrinkle-free and in proper line during its travel. Thin cakes of difficult solids which may be slightly soluble are good applications. When acceptable, a sluice discharge makes cakes as thin as 1.5 to 2 mm (about $\frac{1}{16}$ in) feasible. Several manufacturers offer belt-discharge filters.

Coilfilter. The **Coilfilter** (Komline-Sanderson Engineering Corp.) is a drum filter with a medium consisting of one or two layers of stainless-steel helically coiled springs, about 10 mm (0.4 in) in diameter, placed in a corduroy pattern around the drum. The springs follow the drum during filtration with cake forming the coils. They are separated from the drum to discharge the cake and undergo washing; if two layers are used, the coils of each layer are further separated from those of the other, passing over different sets of rolls. The use of stainless steel in spring form provides a relatively permanent medium that is readily cleaned by washing and flexing. Filtrate clarity is poorer than with most other media, and a relatively large vacuum pump is needed to handle greater air leakage than is characteristic of fabric media. Material forming a slimy, matlike cake (e.g., raw sewage) is the typical application.

Roll-Discharge Filters A roll in close proximity to the drum at the point of cake discharge rotates in the opposite direction at a peripheral speed equal to or slightly faster than that of the drum (Fig. 18-136). If the cake on the drum is adequately tacky and cohesive for this discharge technique, it adheres to cake on the smaller roll and separates from the drum. A blade or taut wire removes the material from the discharge roll. This design is especially good for thin, sticky cakes. If necessary, a slight air blow may be provided to help release the cake from the drum. Typical cake thickness is 1 to 10 mm (0.04 to 0.4 in).

Single-Compartment Drum Filter

Bird-Young filter. This filter (Bird Machine Co.) differs from most drum filters in that the drum is not compartmented and there is no internal piping or rotary valve. The entire inside of the drum is subjected to vacuum, with its surface perforated to pass the filtrate. Cake is discharged by an air blowback applied through a "shoe" that covers a narrow discharge zone on the inside surface of the drum to interrupt the vacuum, as illustrated in Fig. 18-137. The internal drum surface must be machined to provide close clearance of the shoe to avoid leakage. The filter is designed for high filtration rates with thin cakes. Rotation speeds to 40 r/min are possible with cakes typically 3 to 6 mm (0.12 to 0.24 in) thick. Filter sizes range from 930 cm² to 19 m² (1 to 207 ft²) with 93 percent of the area active. The slurry is fed into a conical feed tank designed to prevent solids from settling without the use of mechanical agitators. The proper liquid level is maintained by overflow, and submergence ranges from 5 to 70 percent of the drum circumference.

The perforated drum cylinder is divided into sections about 50 to 60 mm (2 to 2.5 in) wide. The filter medium is positioned into tubes between the sections and locked into place by round rods. No caulking, wires, or other fasteners are needed.

Wash sprays may be applied to the cake, with collection troughs or pans inserted inside the drum to keep the wash separate from the filtrate. Filtrate is removed from the lower section of the drum by a pipe passing through the trunnions.

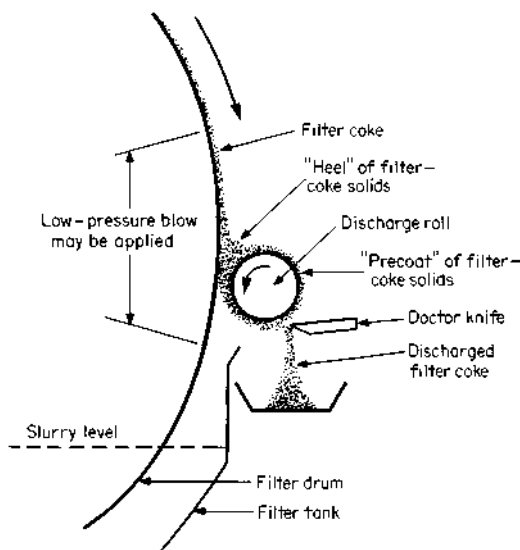


FIG. 18-136 Operating principles of a roll-discharge mechanism. (Schweitzer, Handbook of Separation Techniques for Chemical Engineers, p. 4-40. Copyright 1979 by McGraw-Hill, Inc. and used with permission.)

The major advantages of the Bird-Young filter are its ability to handle thin cakes and operate at high speeds, its washing effectiveness, and its low internal resistance to air and filtrate flow. An additional advantage is the possibility of construction as a pressure filter with up to 1.14 MPa (150-psig) operating pressure to handle volatile liquids. The chief disadvantages are its high cost and the limited flexibility imposed by not having an adjustable rotary valve. Best applications are on free-draining nonblinding materials such as paper pulp or crystallized salts.

Continuous Pressure Filters These filters consist of conventional drum or disc filters totally enclosed in pressure vessels. Filtration takes place with the vessel pressurized up to 6 bar and the filtrate discharging either at atmospheric pressure or into a receiver maintained at a suitable backpressure. Cake discharge is facilitated through a dual valve and lock-hopper arrangement in order to maintain vessel pressure. Alternatively, the discharged filter cake can be reslurried within the filter or in an adjoining pressure vessel and removed through a control valve.

One variation in design, the Ceramec, offered by Outokumpu Mintec, employs "gasless" ceramic media instead of traditional filter fabrics, relying partly on capillary action to achieve low moistures. This results in a significant drop in power consumption by greatly reducing the compressed air requirements.

Continuous Precoat Filters These filters may be operated as either pressure or vacuum filters, although vacuum operation is the prevailing one. The filters are really not continuous but have an extremely long batch cycle (1 to 10 days). Applications are for continuous clarification of liquids from slurries containing 50 to 5000 ppm of solids when only very thin unacceptable cakes would form on other filters and where "perfect" clarity is required.

Construction is similar to that of other drum filters, except that vacuum is applied to the entire rotation. Before feeding slurry a precoat layer of filter aid or other suitable solids, 75 to 125 mm (3 to 5 in) thick, is applied. The feed slurry is introduced and trapped in the outer surface of the precoat, where it is removed by a progressively advancing doctor knife which trims a thin layer of solids plus precoat (Fig. 18-138). The blade advances 0.05 to 0.2 mm (0.002 to 0.008 in) per revolution of the drum. When the precoat has been cut to a pre-defined minimum thickness, the filter is taken out of service, washed, and freshly precoat. This turnaround time may be 1 to 3 h.

Disc Filters A disc filter is a vacuum filter consisting of a number of vertical discs attached at intervals on a continuously rotating

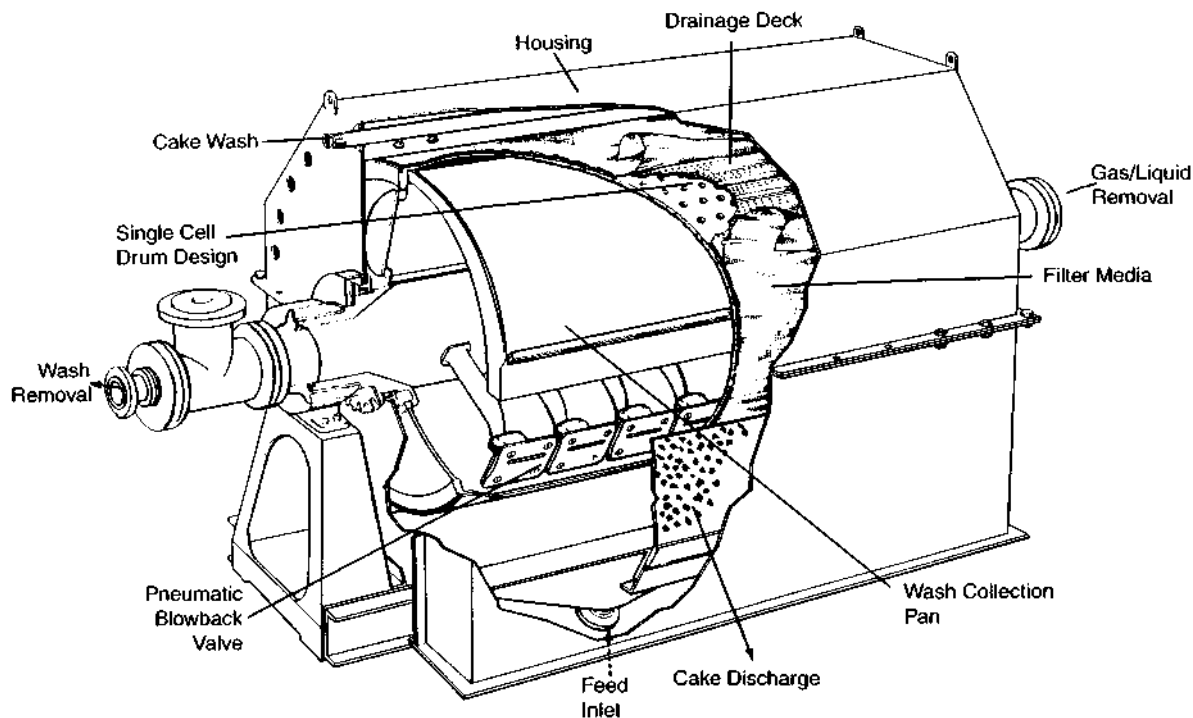


FIG. 18-137 Cutaway of the single-compartment drum filter. (Andritz Bird.)

horizontal hollow central shaft (Fig. 18-139). Rotation is by a gear drive. Each disc consists of 10 to 30 sectors of metal, plastic, or wood, ribbed on both sides to support a filter cloth and provide drainage via an outlet nipple into the central shaft. Each sector may be replaced individually. The filter medium is usually a cloth bag slipped over the sectors and sealed to the discharge nipple. For some heavy-duty applications on ores, stainless-steel screens may be used.

The discs are typically 30 to 50 percent submerged in a troughlike vessel containing the slurry. Another horizontal shaft running beneath the discs may contain agitator paddles to maintain suspension of the solids, as in the EIMCO **Agidisc filter**. In some designs, feed is distributed through nozzles below each disc. Vacuum is supplied to the sectors as they rotate into the liquid to allow cake formation. Vacuum is maintained as the sectors emerge from the liquid and are exposed to

air. Wash may be applied with sprays, but most applications are for dewatering only. As the sectors rotate to the discharge point, the vacuum is cut off, and a slight air blast is used to loosen the cake. This allows scraper blades to direct the cake into discharge chutes positioned between the discs. Vacuum and air blowback is controlled by an automatic valve as in rotary-drum filters.

Of all continuous filters, the vacuum disc is the lowest in cost per unit area of filter when mild steel, cast iron, or similar materials of construction may be used. It provides a large filtering area with minimum floor space, and it is used mostly in high-tonnage dewatering applications in sizes up to about 300 m² (3300 ft²) of filter area.

The main disadvantages are the inadaptability to have effective wash and the difficulty of totally enclosing the filter for hazardous-material operations.

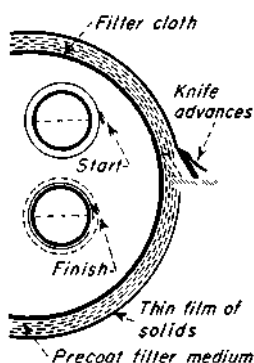


FIG. 18-138 Operating method of a vacuum precoat filter. (Dorr-Oliver EIMCO.)

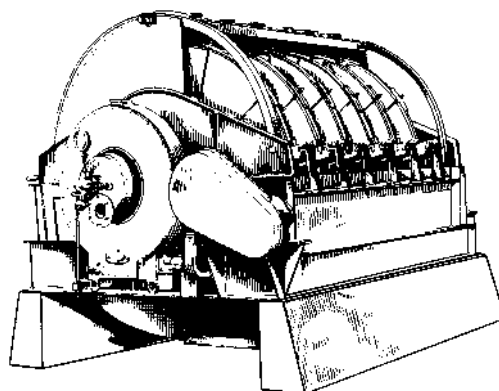


FIG. 18-139 Rotary disc filter. (Dorr-Oliver EIMCO.)

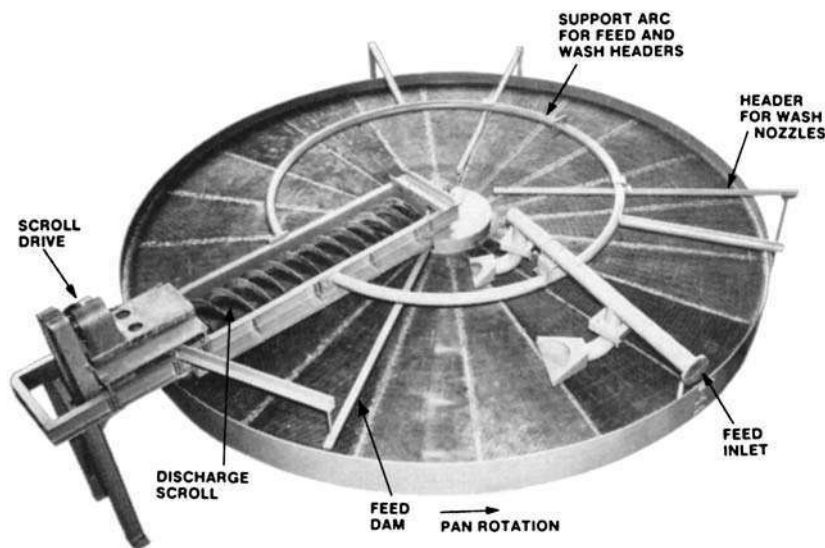


FIG. 18-140 Continuous horizontal vacuum table filter. (Dorr-Oliver EIMCO.)

Horizontal Vacuum Filters These filters are generally classified into two broad classes: rotary circular and belt-type units. Regardless of geometry, they have similar advantages and limitations. They provide flexibility of choice of cake thickness, washing time, and drying cycle. They effectively handle heavy, dense solids, allow flooding of the cake with wash liquor, and are easily designed for true counter-current leaching or washing. The disadvantages are they are more expensive to build than drum or disc filters, they use a large amount of floor space per filter area, and they are difficult to enclose for hazardous applications.

Horizontal-Table, Scroll-Discharge, and Pan Filters These are all basically revolving annular tables with the top surface a filter medium (Fig. 18-140). The table is divided into sectors, each of which is a separate compartment. Vacuum is applied through a drainage chamber beneath the table that leads to a large rotary valve. Slurry is fed at one point, and cake is removed after completing more than three-fourths of the circle, by a horizontal scroll conveyor which elevates the cake over the rim of the filter. A clearance of about 10 mm (0.4 in) is maintained between the scroll and the filter medium to prevent damage to the medium. Residual cake on the medium may be loosened by an air blow from below or with high-velocity liquid sprays from above. This residual cake is a disadvantage peculiar to this type of filter. With material that can cause blinding, frequent shutdowns for thorough cleaning may be needed. Unit sizes range from about 0.9 to 9 m (3 to 30 ft) in diameter, with about 80 percent of the surface available for filtration.

Tilting-Pan Filter This is a modification of the table or pan filter in which each of the sectors is an individual pan pivoted on a radial axis to allow its inversion for cake discharge, usually assisted by an air blast. Filter-cake thicknesses of 50 to 100 mm (2 to 4 in) are common. Most applications involve free-draining inorganic-salt dewatering. In addition to the advantages and disadvantages common to all horizontal continuous filters, tilting-pan filters have the relative advantages of complete wash containment per sector, good cake discharge, filter-medium washing, and feasibility of construction in very large sizes, up to about 25 m (80 ft) in diameter, with about 75 percent of the area usable. Relative disadvantages are high capital cost (especially in smaller sizes) and mechanical complexity leading to higher maintenance costs.

Horizontal-Belt Filter This filter consists of a slotted or perforated elastomer drainage belt driven as a conveyor belt carrying a filter fabric belt (Fig. 18-141). Both belts are supported by and pass across a lubricated support deck. A vacuum pan, aligned with the slots in the elastomer belt, forms a continuous vacuum surface which may include multiple zones for cake formation, washing and final dewater-

ing. Several manufacturers provide horizontal-belt filters, the major differences among which lie in the construction of the drainage belt, the method of retaining the slurry/cake on the belt, and the method of maintaining the alignment of the filter medium. The filters are rated according to the available active filtration area. *Indexing horizontal-belt filters* do away with the elastomer drainage belt of the original design in favor of large drainage pans directly beneath the filter medium. Either the pans or the filter medium is indexed to provide a pseudo-continuous filtration operation. The applied vacuum is cycled with the indexing operation to minimize wear to the sliding surfaces. As a result the indexing filter must be de-rated for the indexing cycle. The indexing horizontal-belt filter avoids the problem of process compatibility with the elastomer drainage belt. The major differences among the indexing machines of several manufacturers lie in the method of indexing and the method of cycling the applied vacuum.

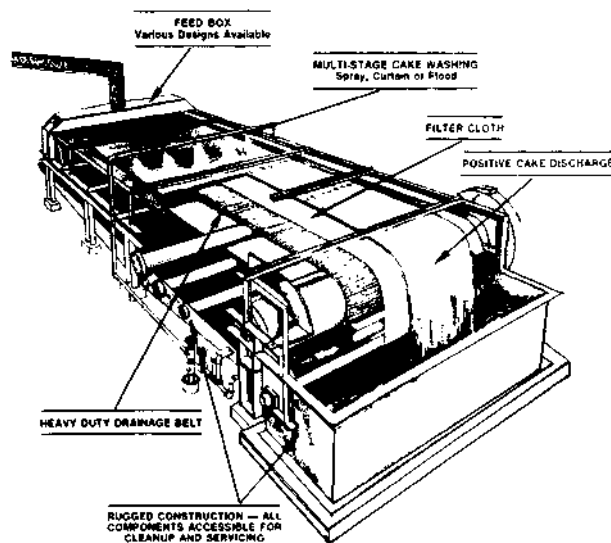


FIG. 18-141 Horizontal-belt filter. (Dorr-Oliver EIMCO.)

The method of feeding, washing, dewatering and discharging is essentially the same with all horizontal-belt filters. Slurry is fed at one end by overflow weirs or a fantail chute; wash liquor, if required, is applied by sprays or weirs at one or more locations as the formed cake moves along the filter. Wiping dams and separations in the drainage pan(s) provide controlled wash application. The cake is discharged as the filter-medium belt passes over the end pulley after separation from the drainage surface. Separating the filter-medium from the drainage surface allows thorough spray cleaning of the filter-medium belt. The duration of the filtration cycle is controlled by belt speed which may be as high as 1 m/s (3.3 ft/s) and is typically variable. The minimum possible cake thickness, at a given solids loading, which can be effectively discharged limits the belt speed from a process point of view. The maximum cake thickness is dependent on the method used to retain the slurry during cake formation and can be 100 to 150 mm (4 to 6 in) with fast draining materials.

Some of the advantages of horizontal-belt filters are the precise control of the filtration cycle including the capability for countercurrent washing of the cake, effective cake discharge and thorough cleaning of the filter medium belt. The horizontal-belt filter's primary disadvantage is that at least half of the filtration medium is always idle during the return loop. This contributes to a significantly higher capital cost which can be two to four times that of a drum or disc filter with equal area. Horizontal-belt filters with active filtration area ranging from 0.18 m² to 150 m² (2 ft² to 1600 ft²) on a single machine have been installed.

Additional equipment is sometimes integrated with horizontal-belt filters to further dewater the cake through expression. The addition of such equipment shouldn't be confused with expression equipment that utilizes filter medium belts. Belt type expression equipment is described later in the "Expression" subsection.

Filter Thickeners Thickeners are devices which remove a portion of the liquid from a slurry to increase the concentration of solids in suspension. Thickening is done to prepare a dilute slurry for more economical filtration or to change the consistency or concentration of the slurry for process reasons. The commonest method of thickening is by gravity sedimentation, discussed earlier in this section. Occasions may arise, however, in which a filter may be called upon for thickening service. Many of the filters previously discussed as cake filters can be operated as thickeners: the filter press with special plates containing flow channels that keep velocity high enough to prevent cake buildup, cycloned tube or candle filters with the cake discharge into the filter tank, and continuous leaf filters which use rotating elements adjacent to the filtering surfaces to limit filter cake buildup. Examples of these filters include the Shriver Thickener, the Industrial Hydra-Shoc Filter employing Back Pulse Technology, the DrM, Dr. Müller AG, Contibac Thickener, and the Ingersoll Rand Continuous Pressure Filter.

Clarifying Filters Clarifying filters are used to separate liquid mixtures which contain only very small quantities of solids. When the solids are finely divided enough to be observed only as a haze, the filter which removes them is sometimes called a polishing filter. The prefilter slurry generally contains no more than 0.10 percent solids, the size of which may vary widely (0.01 to 100 µm). The filter usually produces no visible cake, sometimes because the amount of solids removed is so small, sometimes because the particles are removed by being entrapped within rather than upon the filter medium. Compared with cake filters, clarifying filters are of minor importance to pure chemical-process work, their greatest use being in the fields of beverage and water polishing, pharmaceutical filtration, fuel- and lubricating-oil clarification, electroplating-solution conditioning, and dry-cleaning-solvent recovery. They are essential, however, to the processes of fiber spinning and film extrusion; the spinning solution or dope must be free of particles above a certain size to maintain product quality and to prevent the clogging of spinnerets.

Most cake filters can be so operated as to function as clarifiers, although not necessarily with efficiency. On the other hand, a number of clarifying filters which can be used for no purpose other than clarifying or straining have been developed. In general, clarifying filters are less expensive than cake filters. Clarifying filters may be classified as disc and plate presses, cartridge clarifiers, precoat pressure filters, deep-bed filters, and miscellaneous types. Membrane filters constitute a special class of plate presses and cartridge filters. Simple strainers sometimes are

used as clarifiers of liquids containing very large particles. Because they more closely resemble wet screens than filters and because they have little primary process application, they are not discussed here.

Disc Filters and Plate Presses Filters employing asbestos-pulp discs, cakes of cotton fibers (filtermasse), or sheets of paper or other media are used widely for the polishing of beverages, plating solutions, and other low-viscosity liquids containing small quantities of suspended matter. The term **disc filter** is applied to assemblies of pulp discs made of asbestos and cellulose fibers and sealed into a pressure case. The discs may be preassembled into a self-supporting unit (Fig. 18-142), or each disc may rest on an individual screen or plate against which it is sealed as the filter is closed (Fig. 18-143). The liquid flows through the discs, and into a central or peripheral discharge manifold. Flow rates are on the order of 122 L/(min·m²) [3 gal/(min·ft²)], and the operating pressure does not normally exceed 345 kPa (50 psig) (usually it is less). Disc filters are almost always operated as pressure filters. Individual units deliver up to 378 L/min (6000 gal/h) of low-viscosity liquid.

Disc-and-plate assemblies somewhat resemble horizontal-plate pressure filters, which, in fact, may be used for polishing. In one design (**Sparkler VR filter**) both sides of each plate are used as filtering surfaces, having paper or other media clamped against them.

Pulp filters. These filters employ one or more packs of filter-masse (cellulose fibers compressed to a compact cylinder) stacked into a pressure case. The packs are sometimes supported in individual trays which provide drainage channels and sometimes rest on one another with a loose spacer plate between each two packs and with a drainage screen buried in the center of each pack. The liquid being clarified flows under a pressure of 345 kPa (50 psig) or less through the pulp packs and into a drainage manifold. Flow rates are somewhat less than for disc filters, on the order of 20 L/(min·m²) [0.5 gal/(min·ft²)]. Pulp filters are used chiefly to polish beverages. The filtermasse may be washed in special washers and re-formed into new cakes.

Plate presses. Sometimes called sheet filters, these are assemblies of plates, sheets of filter media, and sometimes screens or frames. They are essentially modified filter presses with practically no cake-holding capacity. A press may consist of many plates or of a single filter sheet between two plates, the plates may be rectangular or circular, and the sheets may lie in a horizontal or vertical plane. The operation is similar to that of a filter press, and the flow rates are about the same as for disc filters. The operating pressure usually does not exceed 138 kPa (20 psig). The presses are used most frequently for low-viscosity liquids, but an ordinary filter press with thin frames is commonly used as a clarifier for 100-Pas (1000-P) rayon-spinning solution. Here the filtration pressure may be 6900 kPa (1000 psig).

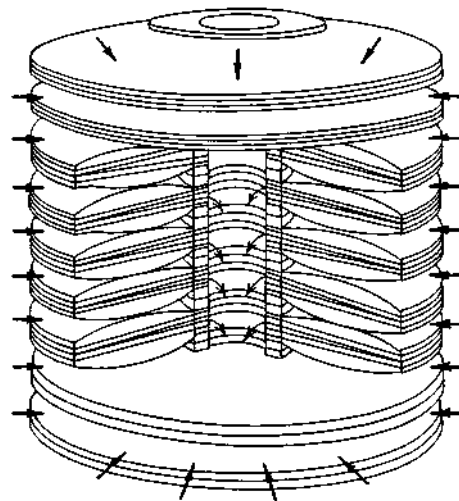


FIG. 18-142 Preassembled pack of clarifying-filter discs. (Ertel Alsop.)

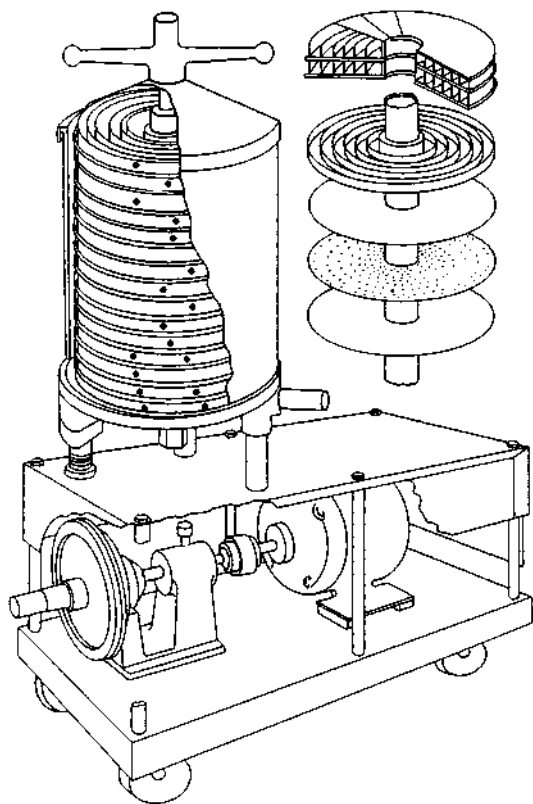


FIG. 18-143 Disc-and-plate clarifying-filter assembly. (Ertel Alsop.)

Disc, pulp, and sheet filters accomplish extreme clarification. Not infrequently their mission is complete removal of particles above a stipulated cut size, which may be much less than $1\ \mu\text{m}$. They operate over a particle-size range of four to five orders of magnitude, contrasting with two orders of magnitude for most other filters. It is not surprising, therefore, that they involve a variety of kinds and grades of filter media, often in successive stages. In addition to packs or discs of cellulosic, polymeric, or asbestos fiber, sheets of pulp, paper, asbestos, carded fiber, woven fabrics, and porous cellophane or polymer are employed. Sandwich-pack composites of several materials have been used for viscous-dope filtration.

The use of asbestos has been greatly diminished because of its identification with health hazards. There have been proposed replacement materials such as the **Zeta Plus** filter media from the AMF Cuno Division, consisting of a composite of cellulose and inorganic filter aids that have a positive charge and provide an electrokinetic attraction to hold colloids (usually negatively charged). These media therefore provide both mechanical straining and electrokinetic adsorption.

Cartridge Clarifiers Cartridge clarifiers are units which consist of or use one or more replaceable or renewable cartridges containing the active filter element. The unit usually is placed in a line carrying the liquid to be clarified; clarification thus occurs while the liquid is in transit.

Mechanical or edge filters. These consist of stacks of discs separated to precise intervals by spacer plates, or a wire wound on a cage in grooves of a precise pitch, or a combination of the two. The liquid to be filtered flows radially between the discs, wires, or layers of paper, and particles larger than the spacing are screened out. Edge filters can remove particles down to $0.001\ \text{in}$ ($25\ \mu\text{m}$) but more often have a minimum spacing of twice this value. They have small solids-retaining capacity and hence must be cleaned often to avoid plugging. Continuous cleaning is provided in some filters. For example, the Cuno **Auto-Klean**, a wire-wound unit, employs a slowly rotating scraper that fits

into the interdisc slots to comb away accumulated solids. In either case, the dislodged solids fall into a sump that may be drained at intervals.

Micronic clarifiers. The greatest number of cartridge clarifiers are of the micronic class, with elements of fiber, resin-impregnated filter paper, porous stone, or porous stainless steel of controlled porosity. Other rustless metals are also available. The elements may be chosen to remove particles larger than a fraction of a micrometer, although many are made to pass $10\text{-}\mu\text{m}$ solids and smaller. By proper choice of multiple-cylinder cartridges or multiple cartridges in parallel any desired flow rate can be obtained at a reasonable pressure drop, often less than $138\ \text{kPa}$ ($20\ \text{psig}$).

When the pressure rises to the permissible maximum, the cartridge must be opened and the element replaced. Micronic elements of the fiber type cannot be cleaned and are so priced that they can be discarded or the filter medium replaced economically. Stone elements usually must be cleaned, a process best accomplished by the manufacturer of the porous ceramic or in accordance with the manufacturer's directions. The user can clean stainless-steel elements by chemical treatment.

Flexibility. Cartridge filters are flexible: cartridges of different ratings and materials of construction can be interchanged, permitting ready accommodation to shifting conditions. They have the disadvantage of very limited solid-handling capability so that the maximum solid concentrations in the feed are limited to about 0.01 percent solids. The biggest limitation for modern process-plant operation is the need to open the filter to replace cartridges, which makes their use for the processing of hazardous materials undesirable. Some manufacturers—for example, the Hydraulic Research Division of Textron Inc. and the Fluid Dynamics Division of Brunswick Corp.—have designed cartridges of bonded metal fibers that can be back-flushed or chemically cleaned without opening the unit. These filters, which can operate at temperatures to 482°C (900°F) and at pressures of $33\ \text{MPa}$ ($325\ \text{atm}$) or greater, are particularly useful for filtering polymers.

Granular Media Filters Many types of granular media filters are used for clarification, operating either as gravity or pressure filters. Gravity filters rely on a difference in elevation between inlet and outlet to provide the driving force necessary to force the liquid through the granular media. Pressure filters employ enclosed vessels operating at relatively low pressure differentials, in the order of 50 to $70\ \text{kPa}$ (7 to $10\ \text{psig}$), which may function in either an upflow or a downflow mode.

The media may be a single material, such as sand, but more often will consist of two or even three layers of different materials, such as anthracite coal in the top layer and sand in the lower one. Solids are captured throughout the bed depth, rather than on the surface, and the gradient in void size provides substantially more solids-holding capacity. The anthracite layer, typically employing 1-mm grain size, serves as a roughing filter and also provides a flocculating action which helps the finer sand, $\sim 0.5\text{-mm}$ particle size, to serve as an effective polishing zone. Media depths vary, but 0.7 to $1.0\ \text{m}$ is typical of a dual media installation. Deeper beds of up to $2.5\ \text{m}$ ($8\ \text{ft}$) are employed in some cases involving special applications where greater solids-holding capacity is desired.

Filtrate is collected in the underdrain system, which may be as simple as a network of perforated pipes covered by graded gravel or a complex structure with slotted nozzles or conduits that will retain the finest sand media while maintaining high flow rates. This latter design allows the use of both air and liquid for the backwashing and cleaning operations.

Backwashing usually is carried out when a limiting pressure drop is reached and before the bed becomes nearly filled with solids, which would lead to a deterioration in filtrate clarity. Cleaning the media is greatly aided by the use of an air scour which helps break loose the trapped solids and provides efficient removal of this material in the subsequent backflushing step. The filtration action tends to agglomerate the filtered solids and, as a result, these generally will settle out readily from the backwash fluid. If the filter is handling a clarifier overflow, usually it is possible to discharge the backwash liquid into the clarifier without risk of these solids returning to the filter. Filter media consumption is low, with normal replacement usually being less than 5 percent per year.

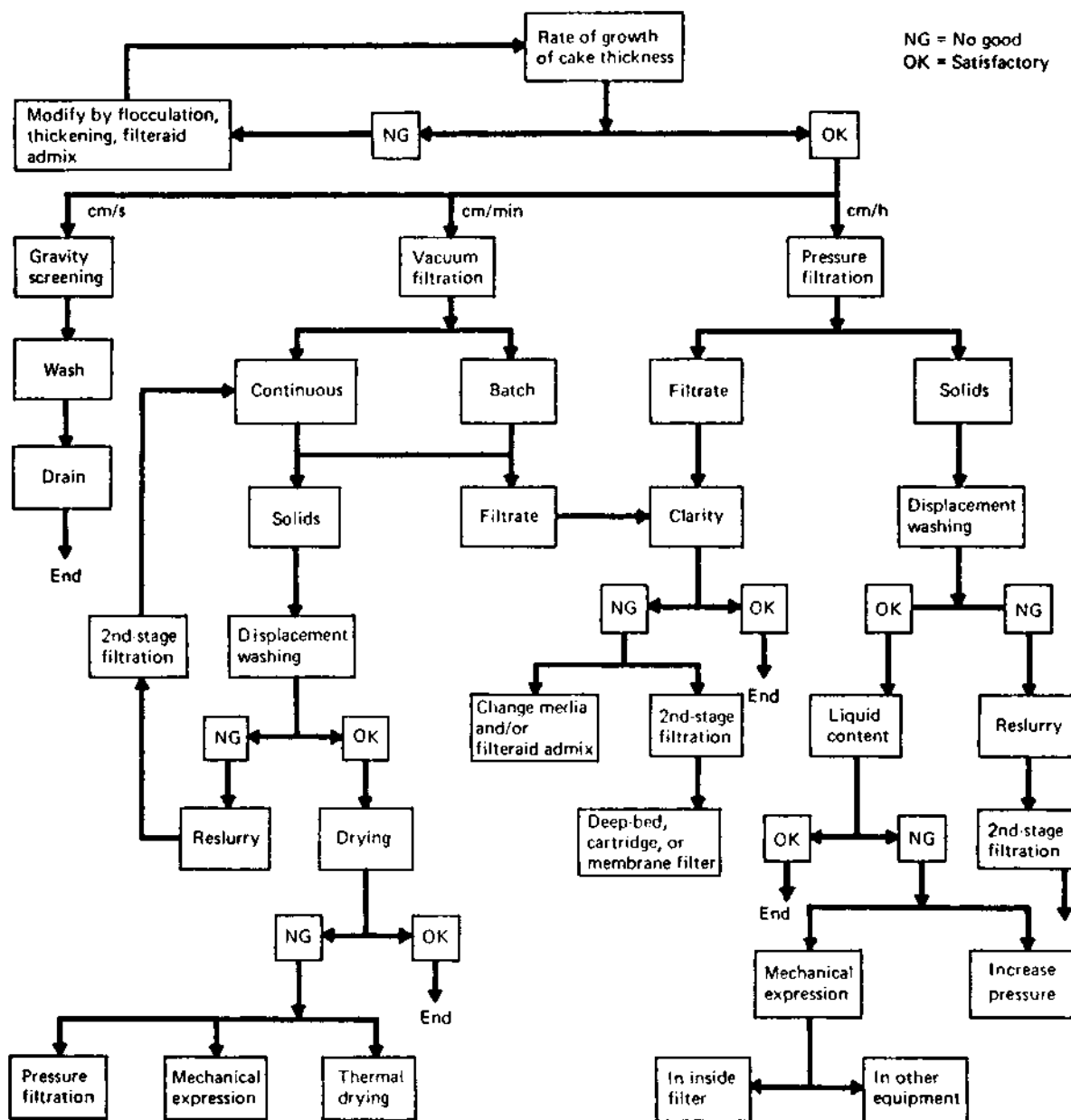


FIG. 18-144 Decision pattern for solving a filtration problem. [Tiller, Chem. Eng., 81(9), 118 (1974), by permission.]

These filters are best applied on relatively dilute suspensions, <150 mg/L suspended solids, allowing operation at relatively high rates, 7.5 to 15.0 m³/m²h (3 to 6 gpm/ft²). Solids capture will range from 90 to 98 percent in a well-designed system. Typical operating cycles range from 8 to 24 h of filtration (and up to 48 h in municipal water treatment), followed by a backwash interval of 15 to 30 min. Applications are principally in municipal and wastewater treatment, but granular media filters also have been employed in industrial uses such as pulp and paper plant inlet water treatment; removal of oil, grease, and scale from steelmaking process wastewater; and clarification of electrolyte in copper electro-winning operations.

United States Filter Corp. Maxi-Flo Filter. The **Maxi-Flo Filter** is an example of the upflow closed-vessel design. Filtration rates to 0.0081 m³/

(m²·s) [12 gal/(ft²·min)] and filter cross-section areas up to 10.5 m² (113 ft²) are possible. Deep-bed filtration has been reviewed by Tien and Payatakes [Am. Inst. Chem. Eng. J., 25, 737 (1970)] and by Oulman and Baumann [Am. Inst. Chem. Eng. Symp. Ser., 73(171), 76 (1977)].

Dyna Sand Filter. A filter that avoids batch backwashing for cleaning, the **Dyna Sand Filter** is available from Parkson Corporation. The bed is continuously cleaned and regenerated by recycling solids internally through an air-lift pipe and a sand washer. Thus a constant pressure drop is maintained across the bed, and the need for parallel filters to allow continued on-stream operation, as with conventional designs, is avoided.

Miscellaneous Clarifiers Various types of filters such as cartridge, magnetic, and bag filters are widely used in polishing operations,

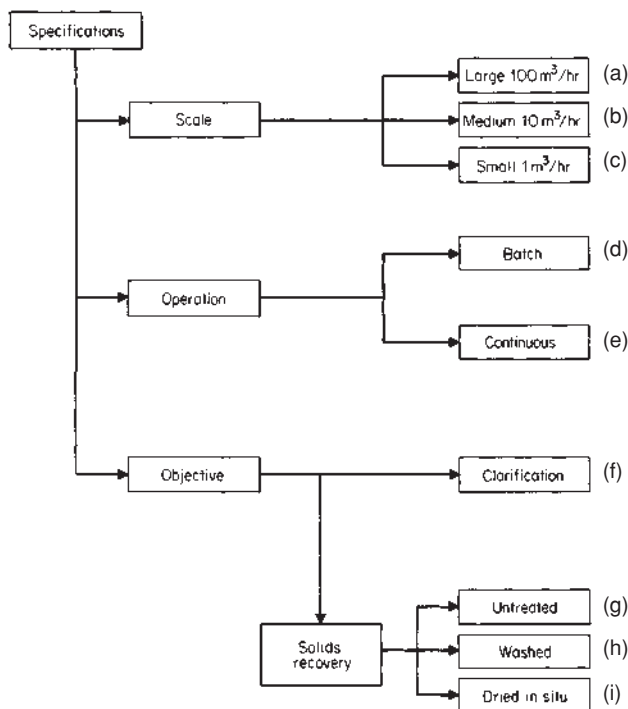


FIG. 18-145 Coding the problem specification. (Purchas, *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977, p. 10, by permission.)

generally to remove trace amounts of suspended solids remaining from prior unit operations. A thorough discussion of cartridge and felt strainer bag filters is available in Schweitzer, op. cit., Section 4.6 (Nickolaus) and Section 4.7 (Wrotnowski).

SELECTION OF FILTRATION EQUIPMENT

If a process developer who must provide the mechanical separation of solids from a liquid has cleared the first decision hurdle by determining that filtration is the way to get the job done (see the final subsection of Sec. 18, "Selection of a Solids-Liquid Separator")—or that it

must remain in the running until some of the details of equipment choice have been settled—choosing the right filter and right filtration conditions may still be difficult. Much as in the broader determination of which unit operation to employ, the selection of filtration equipment involves the balancing of process specifications and objectives against capabilities and characteristics of the various equipment choices (including filter media) available. The important process-related factors are slurry character, production throughput, process conditions, performance requirements, and permissible materials of construction. The important equipment-related factors are type of cycle (batch or continuous), driving force, production rates of the largest and smallest units, separation sharpness, washing capability, dependability, feasible materials of construction, and cost. The estimated cost must account for installed cost, equipment life, operating labor, maintenance, replacement filter media, and costs associated with product-yield loss (if any). In between the process and equipment factors are considerations of slurry preconditioning and use of filter aids.

Slurry characteristics determine whether a clarifying or a cake filter is appropriate; and if the latter, they determine the rate of formation and nature of the cake. They affect the choice of driving force and cycle as well as specific design of machine.

There are no absolute selection techniques available to come up with the "best" choice since there are so many factors involved, many of them difficult to make quantitative and, not uncommonly, some contradictory in their demands. However, there are some published general suggestions to guide the thinking of the engineer who faces the selection of filtration equipment. Figure 18-144 is a decision tree designed by Tiller [*Chem. Eng.*, **81**(9), 118 (1974)] to show the steps to be followed in solving a filtration problem. It is erected on the premise that rate of cake formation is the most important guide to equipment selection. A filter-selection process proposed by Purchas (op. cit., pp. 10–14) employs additional criteria and is based on a combination of process specifications and the results of simple tests. The application is coded by use of Figs. 18-145, 18-146, and 18-147, and the resulting codes are matched against Table 18-11 to identify possible filters. Information needed for Fig. 18-148 can be obtained by observing the settling of a slurry sample (Purchas suggests 1 L) in a graduated cylinder. Filter-cake-growth rate (Fig. 18-148) is determined by small-scale leaf or funnel tests as described earlier.

Almost all types of continuous filters can be adapted for cake washing. The effectiveness of washing is a function of the number of wash displacements applied, and this, in turn, is influenced by the ratio of wash time to cake-formation time. Countercurrent washing, particularly with three or more stages, is usually limited to horizontal filters, although a two-stage countercurrent wash sometimes can be applied on a drum filter handling freely filtering material, such as crystallized

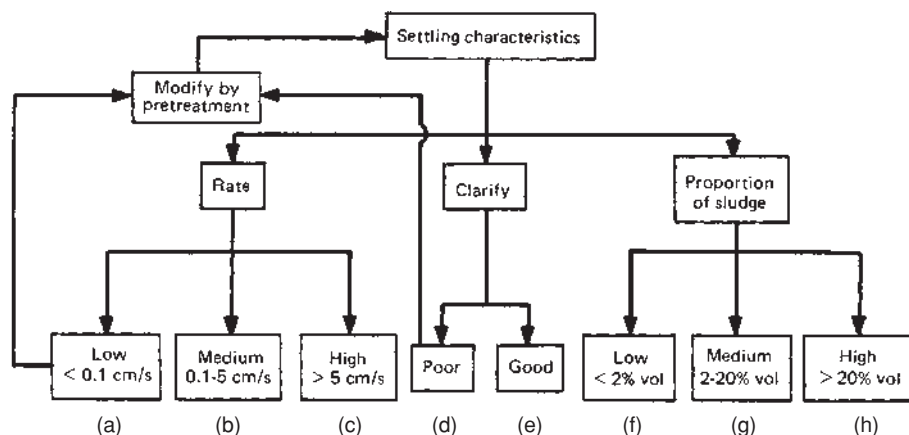


FIG. 18-146 Coding the settling characteristics of a slurry. (Purchas, *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977, p. 11, by permission.)

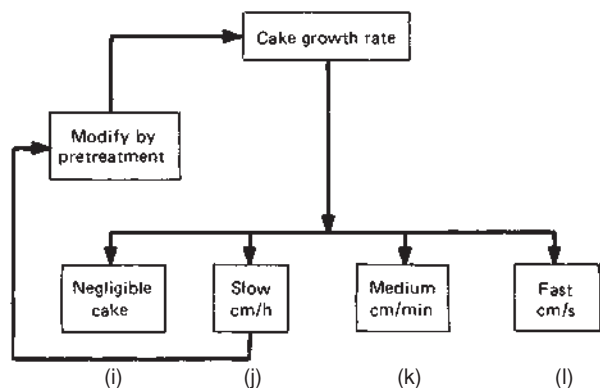


FIG. 18-147 Coding the filtration characteristics of a slurry. (Purchas, *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977, p. 12, by permission.)

salts. Cake washing on batch filters is commonly done, although, generally, a greater number of wash displacements may be required in order to achieve the same degree of washing obtainable on a continuous filter.

Continuous filters are most attractive when the process application is a steady-state continuous one, but the rate at which cake forms and the magnitude of production rate are sometimes overriding factors. A rotary vacuum filter, for example, is a dubious choice if a 3-mm (0.12-in) cake will not form under normal vacuum in less than 5 min and if less than 1.4 m³/h (50 ft³/h) of wet cake is produced. Upper production-rate limits to the practicality of batch units are harder to establish, but any operation above 5.7 m³/h (200 ft³/h) of wet cake should be considered for continuous filtration if it is at all feasible. Again, however, other factors such as the desire for flexibility or the need for high pressure may dictate batch equipment.

For estimating filtration rate (therefore, operating pressure and size of the filter), washing characteristics, and other important features,

TABLE 18-11 Classification of Filters according to Duty and Slurry-Separation Characteristics*

Type of equipment	Suitable for duty specification†	Required slurry-separation characteristics‡	
		Slurry-settling characteristics	Slurry-filtering characteristics
Deep-bed filters	a or b e f	A D F	T
Cartridges	b or c d f	A or B D or E F	
Batch filters			
Pressure vessel with vertical elements	a, b, or c d	A or B D or E	I or J
Pressure vessel with horizontal elements	f, g, h, or i b or c d	F or G A or B D or E	J or K
Filter presses	g or h a, b, or c d	F or G A (or B) D or E	I or J
Variable-volume filters	f, g, h, or i a, b, or c d or e g (or h)	F, G, or H A (or B) D or E G or H	J or K
Continuous filters			
Bottom-fed drum or belt drum	a, b, or c e	A or B D or E	I, J, K, or L
Top-fed drum	f, g, h, or i a, b, or c e	F, G, or H C E	L
Disc	g, i (or h) a, b, or c e	G or H A or B D or E	J or K
Horizontal belt, pan, or table	g a, b, or c d or e g or h	G or H A, B, or C D or E F, G, or H	J, K, or L

*Adapted from Purchas, *Solid/Liquid Separation Equipment Scale-Up*, Uplands Press, Croydon, England, 1977, p. 13, by permission.

†Symbols are identified in Fig. 18-135.

‡Symbols are identified in Figs. 18-136 and 18-137.

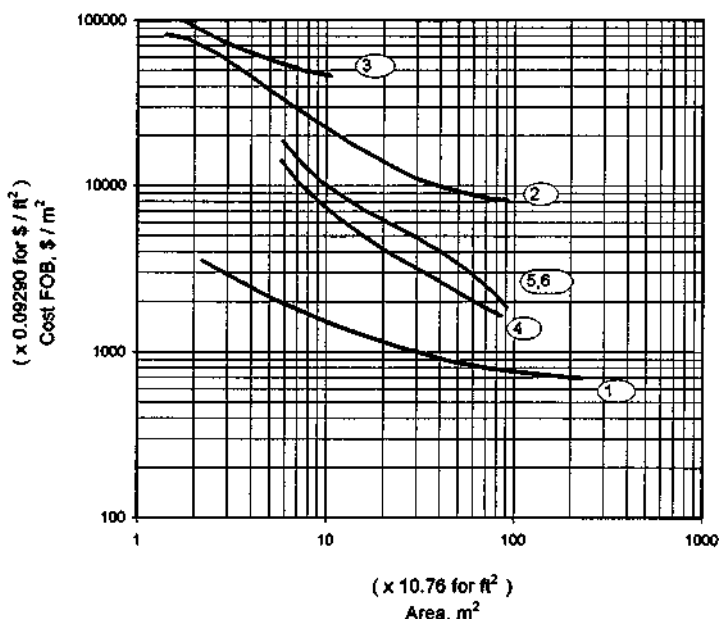


FIG. 18-148 Price of filters installed, FOB point of manufacture. (EIMCO Process Equipment Co.)

small-scale tests such as the leaf or pressure bomb tests described earlier are usually essential. In the conduct and interpretation of such tests, and for advice on labor requirements, maintenance schedule, and selection of accessory equipment the assistance of a dependable equipment vendor is advisable.

FILTER PRICES

As indicated, one of the factors affecting the selection of a filter is total cost of carrying out the separation with the selected machine. An important component of this cost item is the installed cost of the filter, which starts with the purchase price.

From a survey of early 1982, prices of a number of widely used types of process filter were collated by Hall and coworkers [*Chem. Eng.*, **89**(7), 80 (1982)]. These data are drawn together in Fig. 18-148, updated to 1995 prices. They have a claimed accuracy of ± 10 percent, but they should be used confidently only with study-level cost estimations (± 25 percent) at best. Cost of delivery to the plant can be approximated as 3 percent of the FOB price [Pikulik and Diaz, *Chem. Eng.*, **84**(21), 106 (1977)].

The cost of the filter station includes not only the installed cost of the filter itself but also that of all the accessories dedicated to the filtration operation. Examples are feed pumps and storage facilities, precoat tanks, vacuum systems (often a major cost factor for a vacuum filter station), and compressed-air systems. The delivered cost of the accessories plus the cost of installation of filter and accessories generally is of the same order of magnitude as the delivered filter cost and commonly is several times as large. Installation costs, of course, must be estimated with reference to local labor costs and site-specific considerations.

The relatively high prices of pulp and paper filters reflect the construction features that accommodate the very high hydraulic capacity that is required. The absence of data for some common types of filters, in particular the filter press, is explained by Hall as due to the complex variety of individual features and materials of construction. For information about missing filters and for firmer estimates for those types presented, vendors should be consulted. In all cases of serious interest, consultation should take place early in the evaluation procedure so that it can yield timely advice on testing, selection, and price.

CENTRIFUGES

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Nomenclature

Symbol	Definition	SI units	U.S. customary units
a	Acceleration	m/s ²	ft/s ²
B_s	Bond number	Dimensionless	Dimensionless
b	Basket axial length	m	ft
C_f	Frictional coefficient	Dimensionless	Dimensionless
D	Bowl/basket diameter	m	in
d	Particle diameter	m	ft
Ek	Ekman number	Dimensionless	Dimensionless
F	Cumulative fraction	Dimensionless	Dimensionless
G	Centrifugal gravity	m/s ²	ft/s ²
g	Earth gravity	m/s ²	ft/s ²
h	Cake height	m	ft
K	Cake permeability	m ²	ft ²
L	Length	m	ft
M	Mass	kg	lb
m	Bulk mass rate	kg/s	lb/h
N_c	Capillary number	Dimensionless	Dimensionless
P	Power	kw	hp
Q	Flow rate	L/s	gpm
R_m	Filter media resistance	m ⁻¹	ft ⁻¹
Ro	Rossby number	Dimensionless	Dimensionless
r	Radius	m	ft
Rec	Solids recovery	Dimensionless	Dimensionless
S	Liquid saturation in cake (=volume of liquid/volume cake void)	Dimensionless	Dimensionless

Nomenclature (Concluded)

Symbol	Definition	SI units	U.S. customary units
sg	Specific gravity	Dimensionless	Dimensionless
t	Time	s	s
t_d	Time	Dimensionless	Dimensionless
u	Velocity	m/s	ft/s
$V\theta$	Circumferential velocity	m/s	ft/s
V_c	Bulk cake volume	m ³	ft ³
V	Velocity	m/s	ft/s
W	Weight fraction of solids	Dimensionless	Dimensionless
γ	Yield	Dimensionless	Dimensionless
Z	Capture efficiency	Dimensionless	Dimensionless
Greek symbols			
ϵ	Cake void volume fraction	Dimensionless	Dimensionless
ϵ_s	Cake solids volume fraction	Dimensionless	Dimensionless
μ	Liquid viscosity	Pa·s	P
ρ_s	Solid density	kg/m ³	lb/ft ³
ρ_L	Liquid density	kg/m ³	lb/ft ³
σ	Surface tension	N/m	lbf/ft
σ_h	Hoop stress	Pa	psi
θ	Angle	Radian	degree
Σ	Scale-up factor (equivalent sedimentation area)	m ²	ft ²
ζ	Time	Dimensionless	Dimensionless
ϕ	Feed solids volume fraction	Dimensionless	Dimensionless
τ_y	Yield stress	Pa	psi
Δ	Differential speed	1/s	r/min
Ω	Angular speed	1/s	r/min
Subscripts			
b	Bowl or basket		
c	Cake		
e	Centrate		
f	Feed		
f	Filtrate		
acc	Acceleration		
p	Pool		
con	Conveyance		
t	Tangential		
L	Liquid		
s	Solid		

INTRODUCTION

Centrifuges for the separation of solids from liquids are of two general types: (1) sedimentation centrifuges, which require a difference in density between the two or three phases present (solid-liquid or liquid-liquid or liquid-liquid-solid or solid-liquid-solid) and (2) filtering centrifuges (for solid-liquid separation), in which the solid phase is retained by the filter medium through which the liquid phase is free to pass. The following discussion is focused on solid-liquid separation for both types of centrifuges; however, a dispersed liquid phase in another continuous liquid phase as used in sedimenting centrifuges exhibits similar behavior as solid in liquid, and therefore the results developed are generally applicable. The use of centrifuges covers a broad range of applications, from separation of fine calcium carbonate particles of less than 10 μm to coarse coal of 0.013 m ($\frac{1}{2}$ in).

GENERAL PRINCIPLES

Centripetal and Centrifugal Acceleration A centripetal body force is required to sustain a body of mass moving along a curve trajectory. The force acts perpendicular to the direction of motion and is directed radially inward. The centripetal acceleration, which follows the same direction as the force, is given by the kinematic relationship:

$$a = \frac{V_\theta^2}{r} \quad (18-78)$$

where V_θ is the tangential velocity at a given point on the trajectory and r is the radius of curvature at that point. This analysis holds for the motion of a body in an inertial reference frame, for example, a stationary laboratory. It is most desirable to consider the process in a centrifuge, and the dynamics associated with such, in a noninertial reference frame such as in a frame rotating at the same angular speed as the centrifuge. Here, additional forces and accelerations arise, some of which are absent in the inertial frame. Analogous to centripetal acceleration, an observer in the rotating frame experiences a centrifugal acceleration directed radially outward from the axis of rotation with magnitude:

$$a = \Omega^2 r \quad (18-79)$$

where Ω is the angular speed of the rotating frame and r is the radius from the axis of rotation.

Solid-Body Rotation When a body of fluid rotates in a solid-body mode, the tangential or circumferential velocity is linearly proportional to radius:

$$V_\theta = \Omega r \quad (18-80)$$

as with a system of particles in a rigid body. Under this condition, the magnitude of the centripetal acceleration, Eq. (18-78), equals that of the centrifugal acceleration, Eq. (18-79), despite the fact that these accelerations are considered in two different reference frames. Hereafter, the rotating frame attached to the centrifuge is adopted. Therefore, centrifugal acceleration is exclusively used.

G-Level Centrifugal acceleration G is measured in multiples of earth gravity g :

$$\frac{G}{g} = \frac{\Omega^2 r}{g} \quad (18-81)$$

With the speed of the centrifuge Ω in r/min and D the diameter of the bowl,

$$\frac{G}{g} = 0.000559 \Omega^2 D, D(m) \quad (18-82)$$

With D in inches, the constant in Eq. (18-82) is 0.0000142. G can be as low as 100g for slow-speed, large basket units to as much as 10,000g for high-speed, small decanter centrifuges and 15,000g for disc centrifuges. Because G is usually very much greater than g , the effect due to earth's gravity is negligible. In analytical ultracentrifuges used to

process small samples, G can be as much as 500,000g to effectively separate two phases with very small density difference.

Coriolis Acceleration The Coriolis acceleration arises in a rotating frame, which has no parallel in an inertial frame. When a body moves at a linear velocity u in a rotating frame with angular speed Ω , it experiences a Coriolis acceleration with magnitude:

$$a = 2\Omega u \quad (18-83)$$

The Coriolis vector lies in the same plane as the velocity vector and is perpendicular to the rotation vector. If the rotation of the reference frame is anticlockwise, then the Coriolis acceleration is directed 90° clockwise from the velocity vector, and vice versa when the frame rotates clockwise. The Coriolis acceleration distorts the trajectory of the body as it moves rectilinearly in the rotating frame.

Effect of Fluid Viscosity and Inertia The dynamic effect of viscosity on a rotating liquid slurry as found in a sedimenting centrifuge is confined in very thin fluid layers, known as **Ekman layers**. These layers are adjacent to rotating surfaces which are perpendicular to the axis of rotation, such as bowl heads, flanges, and conveyor blades, etc. The thickness of the Ekman layer δ is of the order

$$\delta = \sqrt{\frac{\mu/\rho}{\Omega}} \quad (18-84)$$

where μ/ρ is the kinematic viscosity of the liquid. For example with water at room temperature, μ/ρ is $1 \times 10^{-6} \text{ m}^2/\text{s}$ and for a surface rotating at $\Omega = 3000 \text{ r/min}$, δ is 0.05 mm! These layers are very thin; nevertheless, they are responsible for transfer of angular momentum between the rotating surfaces to the fluid during acceleration and deceleration. They worked together with the larger-scale inviscid bulk flow transferring momentum in a rather complicated way. This is demonstrated by the teacup example in which the content of the cup is brought to speed when it is stirred and it is brought to a halt after undergoing solid-body rotation. The viscous effect is characterized by the dimensionless Ekman number:

$$\text{Ek} = \frac{\mu/\rho}{\Omega L^2} \quad (18-85)$$

where L is a characteristics length. It measures the scale of the viscous effect to that of the bulk flow.

The effect of fluid inertia manifests during abrupt change in velocity of the fluid mass. It is quantified by the Rossby number:

$$\text{Ro} = \frac{u}{\Omega L} \quad (18-86)$$

Typically, Ro is small to the order of 1 with the high end of the range showing possible effect due to inertia, whereas the Ek number is usually very small, 10^{-6} or smaller. Therefore, the viscous effect is confined to thin boundary layers with thickness $\text{Ek}^{1/2} L$.

Sedimenting and Filtering Centrifuges Under centrifugal force, the solid phase assumed to be denser than the liquid phase settles out to the bowl wall—sedimentation. Concurrently, the lighter, more buoyant liquid phase is displaced toward the smaller diameter—flotation. This is illustrated in Fig. 18-149a. Some centrifuges run with an air core, i.e., with free surface, whereas others run with slurry filled to the center hub or even to the axis in which pressure can be sustained.

In a sedimenting centrifuge, the separation can be in the form of **clarification**, wherein solids are separated from the liquid phase in which clarity of the liquid phase is of prime concern. For biological sludge, polymers are used to agglomerate fine solids to facilitate clarification. Separation can also be in the form of **classification** and dewatering at which separation is effected by means of particle size and density. Typically, the finer solids (such as kaolin) of smaller size and/or density in the feed slurry are separated in the centrate stream as product (for example 90 percent of particles less than 1 μm , etc.), whereas the larger and/or denser solids are captured in cake as reject. Furthermore, separation can be in the form of **thickening**, where solids settle under centrifugal force to form a stream with concentrated solids. In

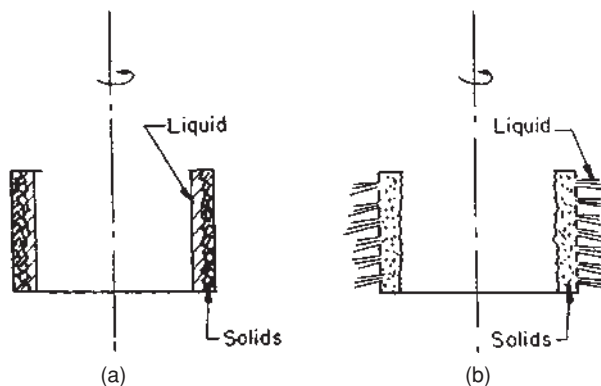


FIG. 18-149 Principles of centrifugal separation and filtration: (a) sedimentation in rotating impermeable bowl; (b) filtration in rotating perforate basket.

dewatering or deliquoring, the objective is to produce dry cake with high solids consistency by centrifugation.

In a filtering centrifuge, separating solids from liquid does not require a density difference between the two phases. Should a density difference exist between the two phases, sedimentation is usually at a much more rapid rate compared to filtration. In both cases, the solid and liquid phases move toward the bowl under centrifugal force. The solids are retained by the filter medium, while the liquid flows through the cake solids and the filter. This is illustrated in Fig. 18-149b.

Performance Criteria Separation of a given solid-liquid slurry is usually measured by the purity of the separated liquid phase in the centrate (or liquid effluent) in sedimenting mode or filtrate in filtering mode, and the separated solids in the cake. In addition, there are other important considerations. Generally, a selected subset of the following criteria are used, depending on the objectives of the process:

- Cake dryness or moisture content
- Total solids recovery
- Polymer dosage
- Size recovery and yield
- Volumetric and solids throughput
- Solid purity and wash ratio
- Power consumption

Cake Dryness In dewatering, usually the cake needs to be as dry as possible. Cake dryness is commonly measured by the solids fraction by weight W or by volume ϵ_s . The moisture content is measured by the complement of W or ϵ_s . The volume fraction of the pores and void in the wet cake is measured by the cake porosity $\epsilon (= 1 - \epsilon_s)$; whereas the volume fraction of the liquid in the pores of the cake is measured by the saturation S . For well-defined solids in the cake with solid density (bone dry) ρ_s and liquid density ρ_L , and given that the cake volume $\dot{V}_c$ and the mass of solids in the cake w_s are known, the cake porosity is determined by

$$\epsilon = 1 - \frac{w_s}{\rho_s \dot{V}_c} \quad (18-87)$$

For undersaturated cake with $S < 1$, saturation can be inferred from the weight fraction of solids and the porosity of the cake, together with the solid and liquid densities:

$$S = \left(\frac{1 - W}{W} \right) \left(\frac{1 - \epsilon}{\epsilon} \right) \frac{\rho_s}{\rho_L} \quad (18-88)$$

When the cake is saturated $S = 1$, the cake porosity can be determined from Eq. (18-89) as

$$\epsilon = \left(1 + \frac{\rho_L}{\rho_s} \frac{W}{1 - W} \right)^{-1} \quad (18-89)$$

Cake dewatering by compression and rearrangement of the solids in the cake matrix reduce ϵ , yet the cake is still saturated with $S = 1$. (Assuming cake solids are ideal spheres of uniform size, the maximum packing, in rhombohedral arrangement, is such that $\epsilon_s = 74$ percent or $\epsilon = 26$ percent.) Drainage of liquid within the cake by centrifugation further reduces S to be less than 1. There is a lower limit on S which is determined by the cake height, dewatering time, centrifugal force as compared to the capillary and surface forces, as well as the surface roughness and porosity of the particles.

Total Solids Recovery In clarification, the clarity of the effluent is measured indirectly by the total solids recovered in the cake as

$$\text{Rec} = \frac{m_c W_c}{m_f W_f} \quad (18-90)$$

where subscripts c and f denote, respectively, the cake and the feed. m is the bulk mass flow rate in kg/s (lb/h).

Under steady state, the mass balance on both solids and liquid yield, respectively:

$$m_f W_f = m_c W_c + m_e W_e \quad (18-91)$$

$$m_f = m_c + m_e \quad (18-92)$$

From the above, it follows that

$$\text{Rec} = \frac{1 - (W_e/W_f)}{1 - (W_e/W_c)} \quad (18-93)$$

where subscript e represents liquid centrate. Stringent requirements on centrate quality or capture of valuable solid product often require the recovery to exceed 90 percent and, in some cases, 99+ percent. In such cases, the centrate solids are typically measured in ppm.

Polymer Dosage Cationic and anionic polymers have been commonly used to coagulate and flocculate fine particles in the slurry. This is especially pertinent to biological materials such as are found in wastewater treatment. In the latter, cationic polymers are often used to neutralize the negative-charge ions left on the surface of the colloidal particles. Polymer dosage is measured by kg of dry polymer/1000 kg of dry solids cake (lbm of dry polymer/ton of dry solids cake). With liquid polymers, the equivalent (active) dry solid polymer is used to calculate the dosage. There is a minimum polymer dosage to agglomerate and capture the fines in the cake. Overdose can be undesirable to recovery and cake dryness. The range of optimal dosage is dictated by the type of solids in the slurry, slurry physical properties such as pH, ionic strength, etc., and the operating condition and characteristics of the centrifuge. It is known that flocculated particles or flocs obtained from certain polymers may be more sensitive to shear than others, especially during feed acceleration in the centrifuges. A more gentle feed accelerator is beneficial for this type of polymer. Also, polymers can be introduced to the feed at various locations either within or external to the centrifuge.

Size Recovery and Yield Centrifuges have been applied to classify polydispersed fine particles. The size distribution of the particles is quantified by the cumulative weight fraction F less than a given particle size d for both the feed and the centrate streams. It is measured by a particle size counter which operates based on principles such as sedimentation or optical scattering.

In kaolin classification, the product is typically measured with a certain percentage less than a given size (example 90 percent or 95 percent less than 1 or 2 μm). Each combination of percent and size cut represents a condition by which the centrifuge would have to tune to yield the product specification.

The yield Y is defined as the fraction of feed particles of a given size below which they report to the centrate product. Thus,

$$Y = \frac{m_c W_c F_c}{m_f W_f F_f} \quad (18-94)$$

From material balance, the particle size distribution of the feed and centrate, as well as the total solids recovery, determine the yield,

$$Y(d) = \frac{F_c(d)}{F_f(d)} (1 - \text{Rec}) \quad (18-95)$$

The complementary is the cumulative capture efficiency $Z = (1 - Y)$, which is defined as the feed particles of a given size and smaller which are captured in the cake, which in most dewatering applications is the product stream.

Volumetric and Solids Throughput The maximum volumetric and solids throughput to a centrifuge are dictated by one or several governing factors, the most common ones are the centrate solids, cake dryness, and capacity (torque and power) of drive/gear unit. The solids throughput is also governed by other factors such as solids conveyance and discharge mechanisms for continuous and batch centrifuges. The settling rate, as may be significantly reduced by increasing feed solids concentration, also becomes crucial to solids throughput, especially if it has to meet a certain specification on centrate quality.

Solid Purity and Wash Ratio Cake washing in a centrifuge is used to remove dissolved impurities on the solids particle surface. It is most effective in filtering centrifuges—typically with a wash ratio of 0.05 to 0.3 kg wash/kg solids in continuous centrifuges, although higher ratios can be achieved with derated capacities to provide sufficient residence time. Batch filtering centrifuges are unlimited in the wash quantity that can be applied. Solid soluble impurities generally cannot be washed out in situ due to insufficient contact time, in which case repulp washing may be more effective. Repulp is often utilized with sedimenting centrifuges where wash is required.

Power Consumption Power is consumed to overcome windage and bearing (and seal) friction, to accelerate feed stream from zero speed to full tangential speed at the pool so as to establish the required G -force for separation, and to convey and discharge cake. The power to overcome windage and bearing friction is usually established through tests for a given centrifuge geometry at different rotation speeds. It is proportional to the mass of the centrifuge, to the first power of the speed for the bearing friction, and to the second power of speed for windage. It is also related to the bearing diameter. The seal friction is usually small.

The horsepower for feed acceleration is given by

$$P_{acc} = 5.984(10^{-10})sgQ(\Omega r_p)^2 \quad (18-96)$$

where sg is the specific gravity of the feed slurry, Q the volumetric flow rate of feed in gpm(l/s), Ω the speed in r/min, r_p in meters corresponds to the radius of the pool surface for sedimenting centrifuge, or to the radius of the cake surface for filtering centrifuge. *Note:* To convert horsepower to kilowatts, multiply by 0.746.

The horsepower for cake conveyance for scroll centrifuge is

$$P_{con} = 1.587 (10^{-5}) T\Delta \quad (18-97)$$

where Δ is the differential speed in r/min (s^{-1}) between the scroll conveyor and the bowl, and T is the conveyance torque in in-lb_f (N-m). For centrifuge where cake is discharged differently, the conveyance power is simply

$$P_{con} = MGC_fV \quad (18-98)$$

where M is the mass of the cake, G the centrifugal acceleration, C_f is the coefficient of friction, and V is the cake velocity. Comparing Eqs. (18-97) and (18-98) the conveyance torque is inversely related to the differential speed and directly proportional to the G acceleration, cake velocity, and cake mass.

Stress in the Centrifuge Rotor The stress in the centrifuge rotor is quite complex. Analytical methods, such as the finite element method, are used to analyze the mechanical integrity of a given rotor design. Without getting into an involved analysis, some useful knowledge can be gained from a simple analysis of the hoop stress of a rotating bowl under load. At equilibrium, the tensile hoop stress σ_h of the cylindrical bowl wall with thickness t is balanced by the centrifugal body force due to the mass of the bowl wall with density ρ_m and its contents (cake or slurry or liquid) with equivalent density ρ_L . Consider a circular wall segment with radius r , unit subtended angle, and unit axial length. A force balance requires

$$\sigma_h = \rho_m V_t^2 \left[1 + 0.5 \frac{\rho_L}{\rho_m} \left(1 - \frac{r_s^2}{r_b^2} \right) \frac{r_b}{t} \right] \quad (18-99)$$

$V_t = \Omega r_b$ is the tip speed of the bowl. The term in the bracket is typically of order 1. If the maximum allowed σ_h is designed to be no more than 60 percent of the yield stress of the bowl material, which for steel is about $2.07(10^8) \text{ Nm}^{-2}(30,000 \text{ lb}_f/\text{in}^2)$. Given that the ρ_m of stainless steel is $7867 \text{ kg/m}^3 (0.284 \text{ lbm/in}^3)$, and there is no liquid load, then $(V_t)_{max} = \{\sigma_h/\rho_m\}^{1/2} = 126 \text{ m/s (412 ft/s)}$. With additional liquid load, $\rho_L = 1000 \text{ kg/m}^3 (0.0361 \text{ lbm/in}^3)$, $r_b/t = 10$, and further assuming the worst case with liquid filling to the axis, the term in the curly bracket is 1.636. Using Eq. (18-99), $(V_t)_{max} = \{\sigma_h/\rho_m/1.636\}^{1/2} = 98 \text{ m/s (322 ft/s)}$. Indeed, almost all centrifuges are designed with top rim speed about 91 m/s (300 ft/s). With special construction materials for the rotor, such as Duplex Ferritic/Austenitic stainless steel, with higher yield stress, the maximum rim speed under full load can be over 122 m/s (400 ft/s).

G-Force vs. Throughput The G -acceleration can be expressed as

$$G = \Omega^2 r_b = \frac{(V_t)_{max}^2}{r_b} \quad (18-100)$$

Figure 18-150 shows the range of diameter of commercial centrifuges and the range of maximum G developed in each type. It demonstrates an inverse relationship between G and r_b at $V_t = (V_t)_{max}$, which is constant for a given material. Figure 18-151 shows a log-log plot of G versus Ω for various bowl diameters, [Eq. (18-100)]. Also, the limiting conditions as delineated by $G = \Omega^2 R = \Omega(V_t)_{max}$ with various $(V_t)_{max}$ are superimposed on these curves. These two sets of curves dictate the operable speed and G for a given diameter and a given construction material for the bowl. The throughput capacity of a machine, depending on the process need, is roughly proportional to the n th power of the bowl radius,

$$Q = C_1 r_b^n \quad (18-101)$$

where n is normally between 2 to 3, depending on clarification, classification, thickening, or dewatering. Thus,

$$G = c_2 (V_t)_{max}^2 Q^{1/n} \quad (18-102)$$

where c_1 and c_2 are constants. It follows that large centrifuges can deliver high flow rate but separation is at lower G -force; vice versa, smaller centrifuges can deliver lower flow rate but separation is at higher G -force. Also, using higher-strength material for construction of the rotating assembly permits higher maximum tip speed, thus allowing higher G -force for separation at a given feed rate.

Centrifuge bowls are made of almost every machinable alloy of reasonably high strength. Preference is given to those alloys having 1 percent elongation to minimize the risk of cracking at stress-concentration points. Typically the list includes (in increasing cost) rubber-lined carbon steel, SS316L, SS317LMN, duplex SS (SAF2205), Alloy 904L, AL6 XN, Inconel, Hastelloy C22, Hastelloy B, nickel, and titanium Gr. 2. Most coatings (Halar, PTFE, etc.) cause problems with stressed components such as baskets, but can be utilized for static components such as housings. Vertical-basket centrifuges are frequently constructed of carbon steel or stainless steel coated with rubber, neoprene, Penton, or Kynar. Casings and feed, rinse, and discharge lines that are stationary and lightly stressed may be constructed of any suitable rigid corrosion-resistant material. Wear-resistant materials—tungsten and cermaic carbide, hard-facing, and others—are often used to protect the bare metal surfaces in high-wear areas such as the blade tips of the decanter centrifuge.

Critical Speeds In the design of any high-speed rotating machinery, attention must be paid to the phenomenon of critical speed. This is the speed at which the frequency of rotation matches the natural frequency of the rotating part. At this speed, any vibration induced by slight unbalance in the rotor is strongly reinforced, resulting in large deflections, high stresses, and even failure of the equipment. Speeds corresponding to harmonics of the natural frequency are also critical speeds but give relatively small deflections and are much less troublesome than the fundamental frequency. The critical speed of simple shapes may be calculated from the moment of inertia; with complex elements such as a loaded centrifuge bowl, it is best found by tests.

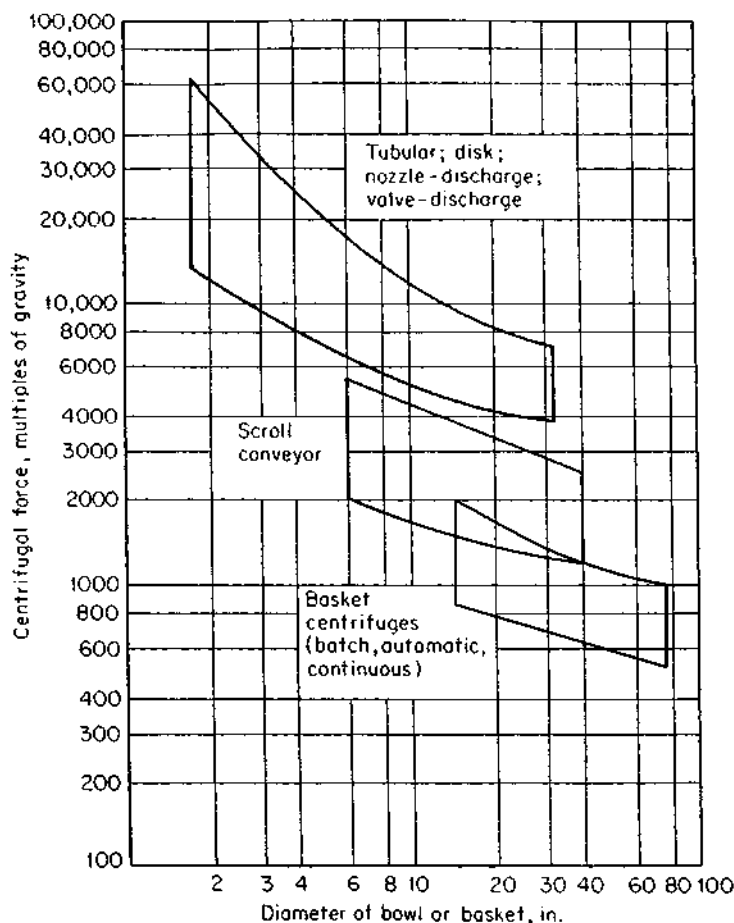


FIG. 18-150 Variation of centrifugal force with diameter in industrial centrifuges.

Nearly all centrifuges operate at speeds well above the primary critical speed and therefore must pass through this speed during acceleration and deceleration. To permit them to do so safely, some degree of damping in their mounting must be provided. This may result from the design of the spindle or driveshaft alone, spring-loading of the spindle bearing nearest the rotor, elastic loading of the suspension, or a combination of these. Smaller and medium-sized centrifuges of the cream-separator and bottle-centrifuge design are frequently mounted on elastic cushions. Horizontal decanter centrifuges are mounted on isolators with dampers to reduce vibration transmitted to the foundation.

SEDIMENTATION CENTRIFUGES

When a spherical particle of diameter d settles in a viscous liquid under earth gravity g , the terminal velocity V_s is determined by the weight of the particle-balancing buoyancy and the viscous drag on the particle in accordance to Stokes' law. In a rotating flow, Stokes' law is modified by the "centrifugal gravity" $G = \Omega^2 r$; thus

$$V_s = \frac{1}{18\mu} \Omega^2 r (\rho_s - \rho_L) d^2 \quad (18-103)$$

In order to have good separation or high settling velocity, a combination of the following conditions is generally sufficient:

1. High centrifuge speed
2. Large particle size

3. Large density difference between solid and liquid
4. Large separation radius
5. Low liquor viscosity

Among the five parameters, the settling velocity is very sensitive to change in speed and particle size. It varies as the square of both parameters. The maximum achievable rotational speed of a centrifuge is normally dictated by the stresses exerted by the processing medium on the bowl and the stresses of the bowl on periphery equipment, most notably the drive system, which consists of a gear unit or hydraulic pump. If the particles in the feed slurry are too small to be separated in the existing G -field, coagulation and flocculation by polymers are effective ways to create larger agglomerated particles for settling. Unlike separation under a constant gravitational field, the settling velocity under a centrifugal field increases linearly with the radius. The greater the radius at which the separation takes place in a given centrifuge at a given rotational speed, the better the separation. Sedimentation of particles is favorable in a less viscous liquid. Some processes are run under elevated temperature where liquid viscosity drops to a fraction of its original value at room temperature.

Laboratory Tests

Spin-Tube Tests The objective of the spin-tube test is to check the settleability of solids in a slurry under centrifugation. The clarity of the supernatant liquid and the solid concentration in the sediment can also be evaluated. A small and equal amount of feed is introduced into two diametrically opposite test tubes (typically plastic tubes in a stainless

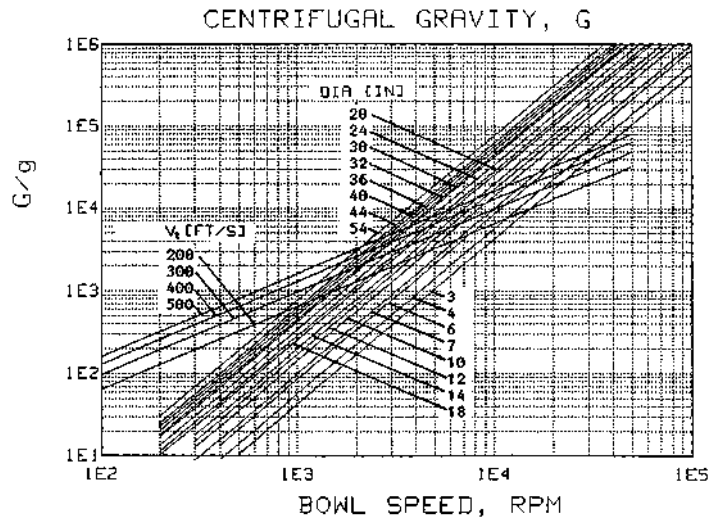


FIG. 18-151 Variation of centrifugal force with r/min.

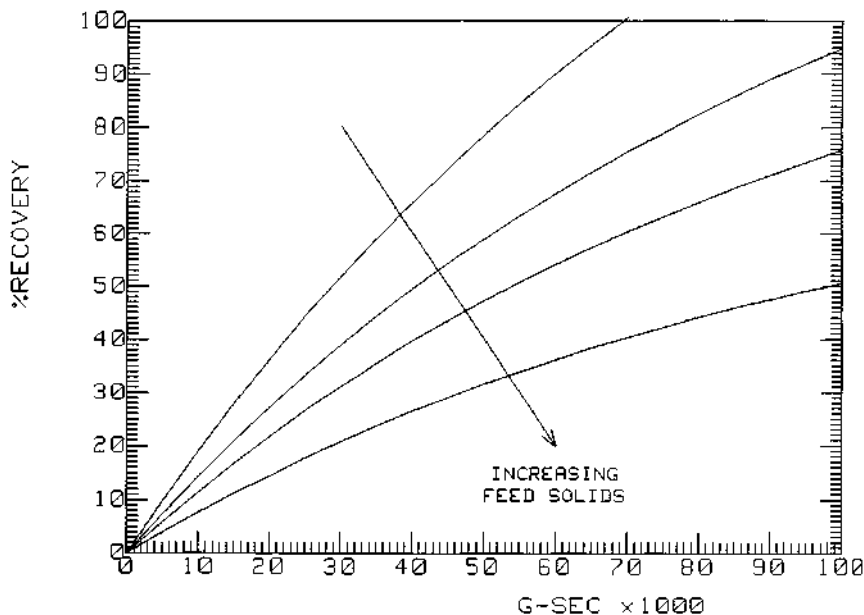
steel holder) with a volume of 15 to 50 mL. The samples are centrifuged at a given G and for a period of time t . The supernatant liquid is decanted off from the spin tubes from which the clarity (in the form of turbidity or any measurable solids, dissolved and suspended) is measured. The integrity—more precisely, the yield stress—of the cake can be determined approximately by the amount of penetration of a rod into the cake under its weight and accounting for the buoyancy effect due to the wet cake. It is further assumed that the rod does not lean on the sides of the tube. The yield stress τ_y of the centrifuged cake can be determined from:

$$\tau_y = \frac{1}{2} \left(\frac{\rho_r L}{h} - \rho_L \right) g r_d \quad (18-104)$$

where r_d and L are, respectively, the radius and length of the solid circular rod; ρ_r is the density of the rod; and h is the penetration of the rod into the cake. By using rods of various sizes and densities, yield stress which is indicative of cake handling and integrity can be measured for a wide range of conditions.

The solids recovery in the cake can be inferred from measurements using Eq. (18-93). It is shown as a function of G -seconds for different feed solids concentration in Fig. 18-152; see also theory discussed below.

Imperforate Bowl Tests The amount of supernatant liquid from spin tubes is usually too small to warrant accurate gravimetric analysis. A fixed amount of slurry is introduced at a controlled rate into a rotating imperforate bowl to simulate a continuous sedimentation centrifuge. The liquid is collected as it overflows the ring weir. The test is

FIG. 18-152 Recovery as a function of G -seconds for centrifugal sedimentation.

stopped when the solids in the bowl build up to a thickness which affects centrate quality. The solid concentration of the centrate is determined similarly to that of the spin tube.

Transient Centrifugation Theory As in gravitational sedimentation, there are three layers which exist during batch settling of a slurry in a centrifuge: a clarified liquid layer closest to the axis, a middle feed slurry layer with suspended solids, and a cake layer adjacent to the bowl wall with concentrated solids. Unlike with constant gravity g , the centrifugal gravity G increases linearly with radius. It is highest near the bowl and is zero at the axis of rotation. Also, the cylindrical surface area through which the particle has to settle increases linearly with radius. Both of these effects give rise to some rather unexpected results.

Consider the simple initial condition $t = 0$ where the solid concentration ϕ_{so} is constant across the entire slurry domain $r_L \leq r \leq r_b$ where r_L and r_b are, respectively, the radii of the slurry surface and the bowl. At a later time $t > 0$, three layers coexist: the top clarified layer, a middle slurry layer, and a bottom sediment layer. The air-liquid interface remains stationary at radius r_L , while the liquid-slurry interface with radius r_s expands radially outward, with t with r_s given by:

$$\frac{r_s}{r_L} = \sqrt{\frac{\phi_{so}}{\phi_s}} \quad (18-105)$$

Eq. (18-105) can be derived from conservation of angular momentum as applied to the liquid-slurry interface.

Interestingly, the solid concentration in the slurry layer ϕ_s does not remain constant with time as in gravitational sedimentation. Instead, ϕ_s decreases with time uniformly in the entire slurry layer in accordance to:

$$\frac{\phi_s}{\phi_{s\max}} = 1 - \left[1 - \left(\frac{\phi_{so}}{\phi_{s\max}} \right) \right] e^{2\xi} \quad (18-106)$$

where ξ is a dimensionless time variable:

$$\xi = \left(\frac{V_{go} t}{r_b} \right) \left(\frac{G}{g} \right) \quad (18-107)$$

In Eqs. (18-106) and (18-107), under hindered settling and 1 g , the solids flux $\phi_s V_s$ is assumed to be a linear function of ϕ_s , decreasing at a rate of V_{go} . Also, the solids flux is taken to be zero at the "maximum" solids concentration $\phi_{s\max}$. As $G/g \gg 1$, this solids flux behavior based on 1 g is assumed to be ratioed by G/g .

Concurrent with the liquid-slurry interface moving radially outward, the cake layer builds up with the cake-slurry interface moving radially inward, with radial position given by:

$$\frac{r_c}{r_b} = \sqrt{\frac{(\epsilon_s - \phi_{so})}{(\epsilon_s - \phi_s)}} \quad (18-108)$$

where ϵ_s is a constant cake solids concentration. Sedimentation stops when the growing cake-slurry interface meets the decreasing slurry-liquid interface with $r_c = r_s$. This point is reached at $\phi_s = \phi_s^*$ and $t = t^*$ when

$$\frac{1}{\phi_s^*} = \frac{1}{\epsilon_s} + \left(\frac{r_b}{r_L} \right)^2 \left(\frac{1}{\phi_{so}} - \frac{1}{\epsilon_s} \right) \quad (18-109)$$

$$t^* = \frac{1}{2} \left(\frac{g r_b}{G V_{go}} \right) \ln \left(\frac{\phi_{s\max} - \phi_s^*}{\phi_{s\max} - \phi_{so}} \right) \quad (18-110)$$

Example

Calcium carbonate–water slurry

$G/g = 2667$

$V_{go} = 1.31 \times 10^{-6}$ m/s (5.16×10^{-5} in/s)

$\phi_{s\max} = 0.26$ (with $\phi_s V_s = 0$)

$\phi_{so} = 0.13$

$r_L = 0.0508$ m (2 in)

$r_b = 0.1016$ m (4 in)

$\xi = 2667 (1.31 \times 10^{-6}) (1/0.1016) = 0.0344$

$\epsilon_s = 0.52$

t (s)	ξ	$\phi_s/\phi_{s\max}$	r_s (m)	r_c (m)
0.0	0	0.50	0.051	0.102
1.0	0.034	0.46	0.053	0.100
5.0	0.173	0.29	0.067	0.095
7.6	0.261	0.16	0.091	0.091

There are six types of industrial sedimenting centrifuges:

- Tubular-bowl centrifuges
- Multichamber centrifuges
- Skimmer pipe/knife-discharge centrifuges
- Disc centrifuges
- Decanter centrifuges
- Screenbowl centrifuges

The first three types, including the manual-discharge disc, are batch-feed centrifuges, whereas the latter three, including the intermittent and nozzle-discharge discs, are continuous centrifuges.

Tubular-Bowl Centrifuges The tubular-bowl centrifuge is widely employed for purifying used lubricating and other industrial oils and in the food, biochemical, and pharmaceutical industries. Industrial models have bowls 102 to 127 mm (4 to 5 in) in diameter and 762 mm (30 in) long (Table 18-12). It is capable of delivering up to 18,000g. The smallest size, 44 mm $\times$ 229 mm (1.75 in $\times$ 9 in bowl), is a laboratory model capable of developing up to 65,000g. It is also used for separating difficult-to-separate biological solids with very small density difference, such as cells and virus.

The bowl is suspended from an upper bearing and drive (electric or turbine motor) assembly through a flexible-drive spindle with a loose guide in a controlled damping assembly at the bottom. The unit finds its axis of rotation if it becomes slightly unbalanced due to process load.

The feed slurry is introduced into the lower portion of the bowl through a small orifice. Immediately downstream of the orifice is a distributor and a baffle assembly which distribute and accelerate the feed to circumferential speed. The centrate discharges from the top end of the bowl by overflowing a ring weir. Solids that have sedimented against the bowl wall are removed manually from the centrifuge when the buildup of solids inside the bowl is sufficient to affect the centrate clarity.

The liquid-handling capacity of the tubular-bowl centrifuge varies with use. The low end shown in Table 18-11 corresponds to stripping small bacteria from a culture medium. The high end corresponds to purifying transformer oil and restoring its dielectric value. The solids-handling capacity of this centrifuge is limited to 4.5 kg (10 lb) or less. Typically, the feed stream solids should be less than 1 percent in practice.

Multichamber Centrifuges While the tubular has a high aspect ratio (i.e., length-to-diameter ratio) of 5 to 7, the multichamber centrifuges have aspect ratios of 1 or less. The bowl driven from below consists of a series of short tubular sections of increasing diameter nested to form a continuous tubular passage of stepwise increasing diameter for the liquid flow. The feed is introduced at the center tube and gradually finds its way to tubes with larger diameters. The larger and denser particles settle out in the smaller-diameter tube, while the smaller and lighter particles settle out in the larger-diameter tubes. Classification of particles can be conveniently carried out. Clarification may be significantly improved by spacing especially the outer tubes more closely together to reduce the settling distance, a concept which is fully exploited by the disc-centrifuge design. This also serves to maintain a constant velocity of flow between adjacent tubes. As much as six chambers can be accommodated. The maximum solids-holding capacity is 0.064 m³ (17 gal). The most common use is for clarifying fruit juices, wort, and beer. For these services it is equipped with a centripetal pump at effluent discharge to minimize foaming and contact with air.

Knife-Discharge Centrifugal Clarifiers Knife-discharge centrifuges with solid instead of perforated bowls are used as sedimenting centrifuges. The liquid flow is usually continuous until the settled solids start to interfere with the effluent liquid. The feed enters the

TABLE 18-12 Specifications and Performance Characteristics of Typical Sedimenting Centrifuges

Type	Bowl diameter	Speed, r/min	Maximum centrifugal force $\times$ gravity	Throughput		Typical motor size, hp
				Liquid, gal/min	Solids, tons/h	
Tubular	1.75	50,000*	62,400	0.05–0.25		*
	4.125	15,000	13,200	0.1–10		2
	5	15,000	15,900	0.2–20		3
Disc	7	12,000	14,300	0.1–10		1/2
	13	7,500	10,400	5–50		6
	24	4,000	5,500	20–200		7 1/2
Nozzle discharge	10	10,000	14,200	10–40	0.1–1	20
	16	6,250	8,900	25–150	0.4–4	40
	27	4,200	6,750	40–400	1–11	125
	30	3,300	4,600	40–400	1–11	125
Scroll conveyor	6	8,000	5,500	To 20	0.03–0.25	5
	14	4,000	3,180	To 75	0.5–1.5	20
	18	3,500	3,130	To 100	1–3	50
	24	3,000	3,070	To 250	2.5–12	125
	30	2,700	3,105	To 350	3–15	200
	36	2,250	2,590	To 600	10–25	300
	44	1,600	1,600	To 700	10–25	400
	54	1,000	770	To 750	20–60	250
Knife discharge	20	1,800	920	†	1.0†	20
	36	1,200	740	†	4.1†	30
	68	900	780	†	20.5†	40

*Turbine drive, 100 lb/h (45 kg/h) of steam at 40 lbf/in² gauge (372 KPa) or equivalent compressed air.

†Widely variable.

‡Maximum volume of solids that the bowl can contain, ft³.

NOTE: To convert inches to millimeters, multiply by 25.4; to convert revolutions per minute to radians per second, multiply by 0.105; to convert gallons per minute to liters per second, multiply by 0.063; to convert tons per hour to kilograms per second, multiply by 0.253; and to convert horsepower to kilowatts, multiply by 0.746.

hub end and is accelerated to speed before introducing to the separation pool. The solids settle out to the bowl wall and the clarified liquid overflows the ring weir or discharges through a skimmer pipe. In some designs, internal baffles in the bowl are required to stop wave action primarily along the axial direction. When sufficient thick solid layer has built up inside the bowl, the supernatant liquid is skimmed off by moving the opening of the skimmer pipe radially inward. After the liquid is sucked out, the solids are knifed out as with centrifugal filters. However, unlike centrifugal filters, the cake is always fully saturated with liquid, $S = 100\%$. These centrifuges are used for coarse, fast-settled solids. When greater clarification effectiveness is required, the operation may be totally batchwise with prolonged spinning of each batch. If the solids content in the feed is low, several batches may be successively charged and the resulting supernatant liquor skimmed off before unloading of the accumulated solids.

Commercial centrifuges of this type have bowl diameters ranging from 0.3 to 2.4 m (12 to 96 in). The large sizes are used on heavy-duty applications such as coal dewatering and are limited by stress considerations to operate at 300g. The intermediate sizes for chemical process service develop up to 1000g (see Table 18-10).

Disc Stack Centrifuges One of the most common types of commercially utilized centrifuges is a vertically mounted disc machine, one type of which is shown in Fig. 18-153. Feed is introduced proximate to the axis of the bowl, accelerated to speed typically by a radial vane assembly, and flows through a stack of closely spaced conical discs in the form of truncated cones. Generally 50 to 150 discs are used. They are spaced 0.4 to 3 mm (0.015 to 0.125 in) apart to reduce the distance for solid/liquid separation. The angle made by conical discs with the horizontal is typically between 40 to 55° to facilitate solids conveyance. Under centrifugal force the solids settle against the underside of the disc surface and move down to the large end of the conical disc and subsequently to the bowl wall. Concurrently, the clarified liquid phase moves up the conical channel. Each disc carries several holes spaced uniformly around the circumference. When the disc stack is assembled, the holes provide a continuous upward passage for the lighter clarified liquid released from each conical channel. The liquid collects at the top of the disc stack and discharges through overflow ports. To recover the

kinetic energy and avoid foaming due to discharging of a high-velocity jet against a stationary casing, the rotating liquid is diverted to a stationary impeller from which the kinetic energy of the stream is converted to hydrostatic pressure. Unlike most centrifuges operating with a slurry pool in contact with a free surface, disc centrifuges with a rotary seal arrangement can operate under high pressure. The settled solids at the bowl wall are discharged in different forms, depending on the type of disc centrifuges.

Manual Discharge Disc Stack Centrifuges In the simplest design shown in Fig. 18-154a, the accumulated solids must be removed manually on a periodic basis, similar to that for the tubular-bowl centrifuge. This requires stopping and disassembling the bowl and removing the disc stack. Although the individual discs rarely require cleaning, manual removal of solids is economical only when the fraction of solids in the feed is very small.

Self-Cleaning Disc Centrifuges More commonly known as clarifiers (two-phase) and separators (three-phase), these centrifuges, which also contain a conical disc stack inside the bowl, automatically discharge accumulated solids on a timed cycle while the bowl is at full speed. Feed is introduced into the bowl via a nonrotating feed pipe and into a distributor which evenly distributes the slurry to the appropriate disc stack channels. Slurry is forced up through the disc stack where solids accumulate on the underside of the discs and slide down the discs, where they are forced to the sludge holding area just inside the maximum diameter of the double cone-shaped bowl, as shown in Fig. 18-153. When the solids chamber is full, the bottom of the bowl, which is held closed to the top portion hydraulically, drops by evacuating the hydraulic operating fluid. The solids are discharged at full speed in a very short time into an outer housing where they are diverted out of the machine. The liquid or liquids (in a three-phase separator) are normally discharged via stationary impellers under pressure. These types of centrifuges are commonly used in the clarification of beverages and the purification of mineral and edible oils.

Disc Nozzle Centrifuges In the nozzle-discharge disc centrifuge, solids are discharged continuously, along with a portion of the liquid phase, through nozzles spaced around the periphery of the bowl, which are tapered radially outward, providing a space for solids

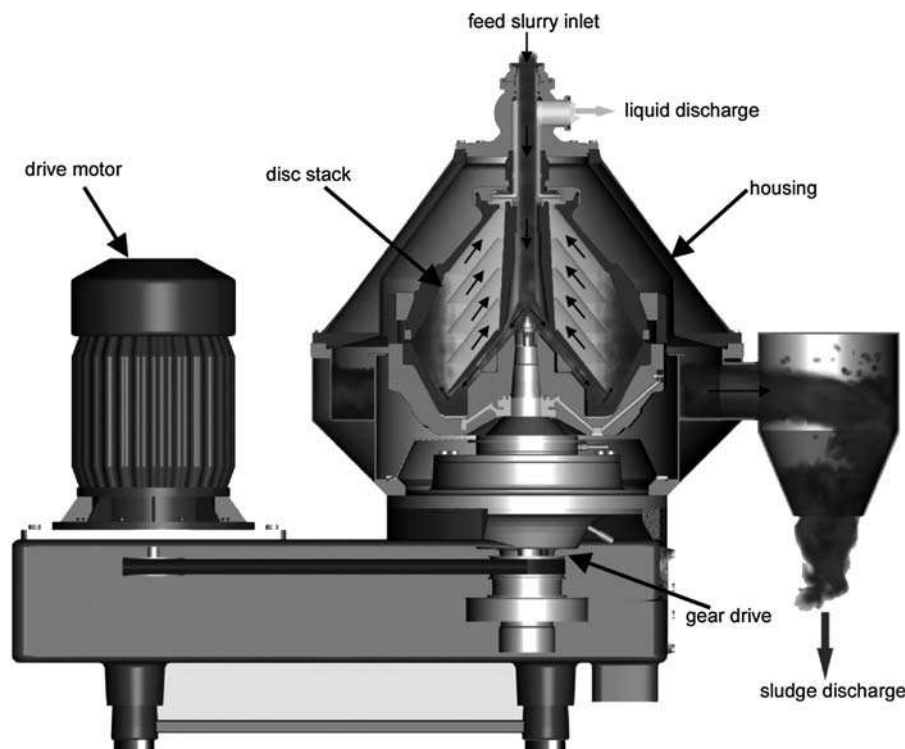


FIG. 18-153 Disc Stack Centrifuge. (Flottweg Separation Technology.)

storage (see Fig. 18-154*b*). The angle of repose of the sedimented solids determines the slope of the bowl walls for satisfactory operation. Clarification efficiency is seriously impaired if the buildup of solids between nozzles reaches into the disc stack. The nozzle diameter should be at least twice the diameter of the largest particle to be processed, and prescreening is recommended of extraneous solids. Typically, nozzle diameters range from 0.6 to 3 mm (0.25 to 0.125 in). Large disc centrifuges may have as much as 24 nozzles spaced out at the bowl.

For clarification of a single liquid phase with controlled concentration of the discharged slurry, a centrifuge which provides recirculation is used (see Fig. 18-154*b*). A fraction of the sludge discharged out of the machine is returned to the bowl to the area adjacent to the nozzles through lines external to the machine as well as built-in annular passages at the periphery of the bowl. This has the effect of preloading the nozzles with sludge which has already been separated and reduces the net flow of liquid with the newly sedimented solids from the feed. Increased concentration can be obtained alternatively by recycling a portion of the sludge to the feed, but this increases solids loading at the disc stack, with a corresponding sacrifice in the effluent clarity for a given feed rate.

With proper rotary seals, the pressure in the machine can be contained up to 1.1 MPa (150 psig) or higher. Also, operating temperature can be as high as 315°C (about 600°F). The rotating parts are made of stainless steel with the high-wear nozzles made of tungsten carbide. The bowls may be underdriven or suspended and range from several centimeters to over 1 m (3.3 ft) in outer diameter. The largest size capable of clarifying up to 1920 L/m (500 gpm) requires 112 kW (150 hp). These types of centrifuges are commonly used in applications including corn wet milling (starch separation, gluten thickening), the classification of kaolin clay particles, washing of terephthalic acid crystals, and dewaxing of lube oils.

Decanter Centrifuges The decanter centrifuge (also known as the solid-bowl or scroll centrifuge) consists of a solid exterior bowl with

an internal screw or scroll conveyor (see Fig. 18-155). Both the bowl and the conveyor rotate at a high speed, yet there is a difference in speed between the two, which is responsible for conveying the sediment along the machine from the cylinder to the conical discharge end. The rotating assembly is commonly mounted horizontally with bearings on each end. Some centrifuges are vertically mounted with the weight of the rotating assembly supported by a single bearing at the bottom or with the entire machine suspended from the top. With the former configuration, the weight of the rotating assembly provides a good sealing surface at the bearing for high-pressure applications. The bowl may be conical in shape or, in most instances, it has combined conical and cylindrical sections (see Fig. 18-155).

Slurry is fed through a stationary pipe into the feed zone located near the center of the scroll. The product is then accelerated circumferentially and passes through distribution ports into the bowl. The bowl has a cylindrical/conical shape and rotates at a preset speed optimal for the application. The slurry rotates with the bowl at the operating speed and forms a concentric layer at the bowl wall, as shown in Fig. 18-156*a*. In the separation pool or pond, under centrifugal gravity the solids which are heavier compared to the liquid settle toward the bowl wall, while the clarified liquid moves radially toward the pool surface. Subsequently, the liquid flows along the helical channel (or channels, if the screw conveyor has multiple leads) formed by adjacent blades of the conveyor to the liquid bowl head, from which it discharges over the weirs. The annular pool/pond height can be changed by adjusting the radial position of the weir openings, which take the form of circular holes or crescent-shaped slots or by adjusting a stationary impeller, which will discharge the liquid under pressure (see Fig. 18-156*b*).

The solids contained in the slurry are deposited against the bowl wall by centrifugal force. The length of the cylindrical bowl section and the cone angle are selected to meet the specific requirements of an application. The scroll conveyor rotates at a slightly different speed from the bowl, and conveys the deposited solids toward the conical

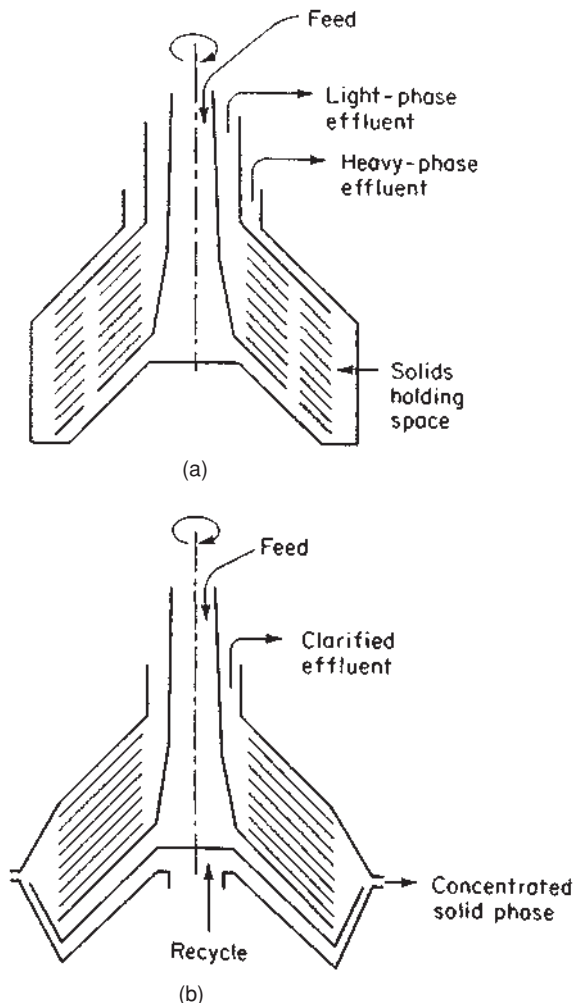


FIG. 18-154 Disc stack-centrifuge bowls: (a) separator, solid wall; (b) recycle clarifier, nozzle discharge.

end of the bowl, also known as the beach. The half cone angle ranges between 5° and 20° . The cake is submerged in the pool when it is in the cylinder and at the beginning of the beach. In this region, liquid buoyancy helps to reduce the effective weight of the cake under centrifugal gravity, resulting in lower conveyance torque. Farther up the

beach, the cake emerges above the pool and moves along the “dry beach,” where buoyancy force is absent, resulting in more difficult conveyance and higher torque. But it is also in this section that the cake is dewatered, with expressed liquid returned back to the pool. The centrifugal force helps to dewater, yet at the same time hinders the transport of the cake in the dry beach. Therefore, a balance in cake conveyance and cake dewatering is the key in setting the pool and the G-force for a given application. Also, clarification is important in dictating this decision.

The cylindrical section provides clarification under high centrifugal gravity. In some cases, the pool should be shallow to maximize the G-force for separation. In other cases, when the cake layer is too thick inside the cylinder, the settled solids—especially the finer particles at the cake surface—entrain into the fast-moving liquid stream above, which eventually ends up in the centrate. A slightly deeper pool becomes beneficial in these cases because there is a thicker buffer liquid layer to ensure settling of resuspended solids. This can be at the expense of cake dryness due to reduction of the dry beach. Consequently, there is again a compromise between centrate clarity and cake dryness. Another reason for the tradeoff of centrate clarity with cake dryness is that, in losing fine solids to the centrate (i.e., classification), the cake with larger particles, having less surface-to-volume ratio, can dewater more effectively, resulting in drier cake. It is best to determine the optimal pool for a given application through tests.

The speed with which the cake transports is controlled by the differential speed. High differential speed facilitates high solids throughput where the cake thickness is kept to a minimum so as not to impair centrate quality due to entrainment of fine solids. Also, cake dewatering is improved due to a reduction in the drainage path with smaller cake height; however, this is offset by the fact that higher differential speed also reduces cake residence time, especially in the dry beach. The opposite holds for low differential speed. Therefore, an optimal differential speed is required to balance centrate clarity and cake dryness. The desirable differential speed is usually maintained using a two-stage planetary gearbox, the housing of which rotates with the bowl speed, with a fixed first-stage pinion shaft. In some applications, the pinion is driven by an electrical backdrive (dc or ac), hydraulic backdrive, or braked by an eddy-current device at a fixed rotation speed. The differential speed is then the difference in speed between the bowl and the pinion divided by the gear ratio. This also applies to the case when the pinion arm is held stationary, in which the pinion speed is zero. The torque at the spline of the conveyor, conveyance torque, is equal to the product of the pinion torque and the gear ratio. Higher gear ratio gives lower differential speed, and vice versa; lower gear ratio gives higher differential for higher solids capacity. The torque at the pinion shaft has been used to control the feed rate or to signal an overload condition by shearing of a safety pin. Under this condition, both the bowl and the conveyor are bound to rotate at the same speed (zero differential) with no conveyance torque and no load at the pinion.

Soft solids, most of which are biological waste such as sewage, are difficult to convey up the beach. Annular baffles or dams have been commonly used to provide a pool-level difference wherein the pool is

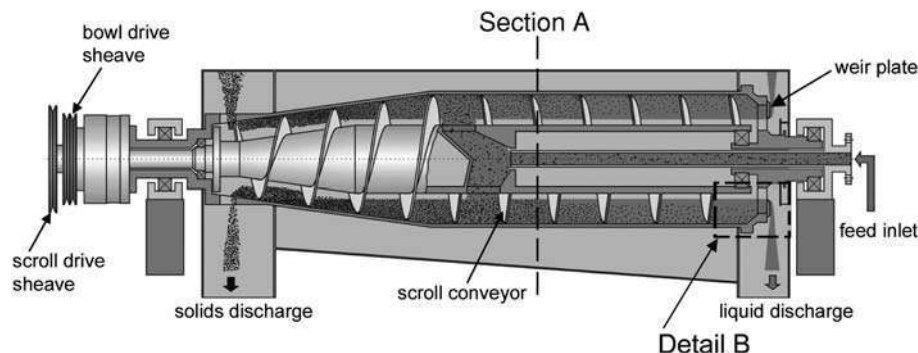


FIG. 18-155 Two-phase decanter centrifuge—gravity liquid discharge. (Flottweg Separation Technology.)

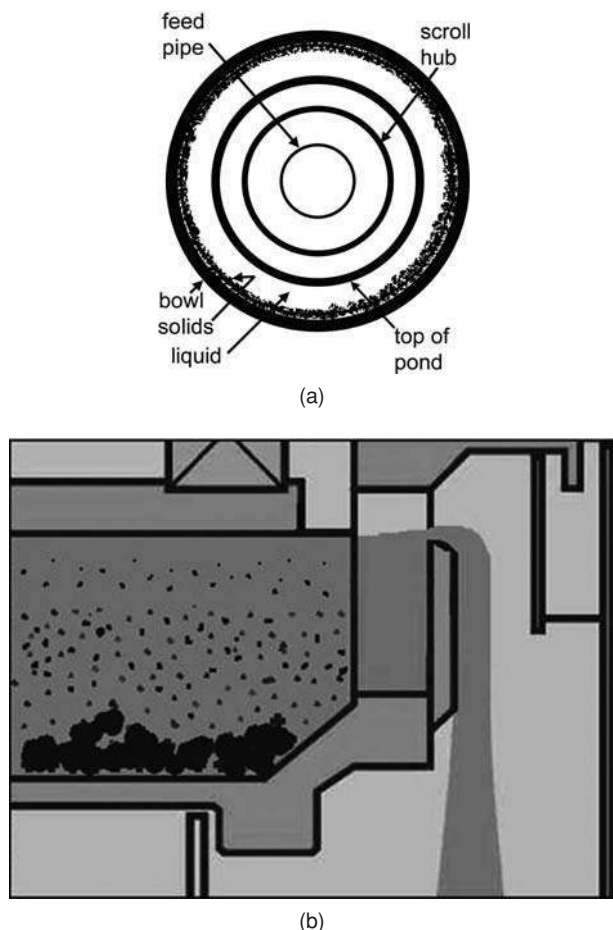


FIG. 18-156 (a) Section A from Fig. 18-155. (b) Detail B from Fig. 18-155. (Flottweg Separation Technology.)

deeper upstream of the baffle toward the clarifier and lower downstream of the baffle toward the beach. The pool-level difference across the baffle, together with the differential speed, provide the driving force to convey the compressible sludge up the beach. This has been used effectively in thickening of waste-activated sludge and in some cases of fine clay with dilatant characteristics.

High solids decanter centrifuges have been used to dewater mixed raw sewage sludge (with volume ratio of primary to waste-activated

sludge such as 50 percent to 50 percent or 40 percent to 60 percent, etc.), aerobically digested sludge, and anaerobically digested sludge. Cake solids as dry as 28 percent to 35 percent by weight are obtained for raw mixed sludge and 20 percent to 28 percent for the digested sludges, with the aerobic sludge at the lower end of the range. The typical characteristics of high-solids applications are: low differential speed (0.5 to 3 r/min), high conveyance torque, high polymer dosage (10 to 30 lb dry polymer/ton dry solids, depending on the feed sewage), and slightly lower volumetric throughput rate. An electrical (dc or ac with variable-frequency drive) or hydraulic backdrive on the conveyor with high torque capacity is essential to operate these conditions at steady state.

The horizontal decanter centrifuge is operated below its critical speed. The bowl is mounted between fixed bearings anchored to a rigid frame. The gearbox is cantilevered outboard of one of these bearings, and the feed pipe enters the rotating assembly through the other end. The frame is isolated from the support structure by spring-type or rubber vibration isolators. In the vertical configuration, the bowl and the gearbox are suspended from the drive head, which is connected to the frame and casing through vibration isolators. A clearance bushing at the bottom limits the excursion of the bowl during start-up and shutdown but does not provide the radial constraint of a bearing under normal operating conditions.

Decanter centrifuges with mechanical shaft-to-casing seals are available for pressure containment up to 1.1 MPa (150 psig), similar to the nozzle-disc centrifuge. They can be built to operate at temperatures from -87 to $+260^{\circ}\text{C}$ (-125 to $+500^{\circ}\text{F}$).

When abrasive solids are processed, the points of wear are protected with replaceable inserts/tiles made from silica carbide, tungsten carbide, ceramic, or other abrasive-resistant materials. These high-wear areas include the feed zone including feed ports; the conveyor blade tip, especially the pressure or pushing face; the conical beach; and the solids discharge ports. Transport of solids is encouraged in some applications by longitudinal strips or grooves at the inner diameter of the bowl, especially at the beach, to enhance the frictional characteristics between the sediment and the bowl surface, and by polished conveyor faces to reduce frictional drag. For fluidlike sediment cake, by using the strips in the beach, a much tighter gap between the conveyor blade tip and the bowl surface is possible with a cake heel layer trapped by the strips. This reduces leakage of the fluid sediment flowing through an otherwise larger gap opening to the pool. Gypsum coating on the bowl wall at the beach section has been used to achieve the same objective.

Various bowl configurations with a wide range of aspect ratios—i.e., length-to-diameter ratio—from less than 1 to 4 are available for specific applications, depending on whether the major objective is maximum clarification, classification, or solids dryness. Generally, the movement of liquid and solids is in countercurrent directions, but in the cocurrent design the movement of liquid is in the same direction as that of the solids. In this design, the feed is introduced at the large end of the machine and the centrate is taken by a skimmer at the beach-cylinder junction. The settled solids transverse the entire machine and discharge at the beach exit. Compound angle beaches are used in

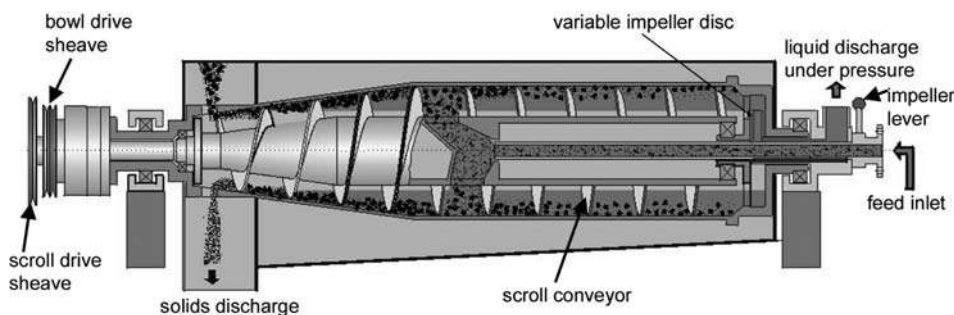


FIG. 18-157 Two-phase decanter centrifuge—pressurized liquid discharge. (Flottweg Separation Technology.)

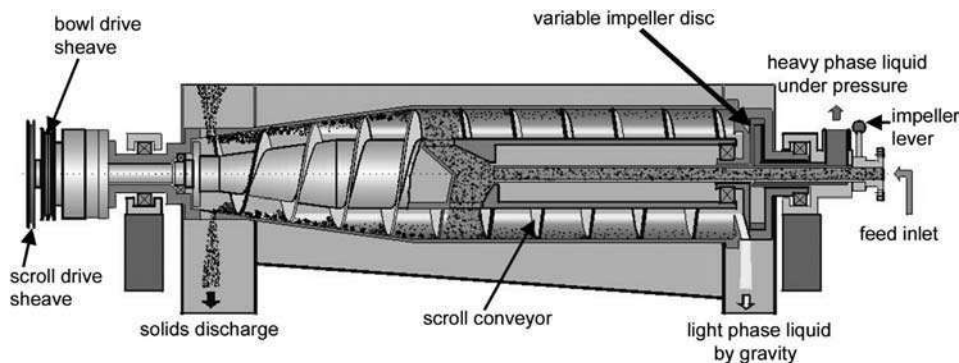


FIG. 18-158 Three-phase decanter centrifuge. (Flottweg Separation Technology.)

specific applications such as washing and drying of polystyrene beads. The pool level is located at the intersection of the two angles at the beach, the steeper angle being under the pool and the shallower angle above the pool (i.e., dry beach), allowing a longer dewatering time. The wash is applied at the pool side of the beach-angle intersection and functions as a continuously replenished annulus of wash liquid through which the solids are conveyed. The size of decanter centrifuge ranges from 6 in diameter to 54 in. The larger is the machine, the slower is the speed, and less is the G-force (Fig. 18-129). However, it provides a much higher throughput capacity, which cannot be accommodated with smaller machines (see Table 18-10). Decanter centrifuges are utilized in a broad range of industries and applications where a large amount of solids separation from liquids is required on a continuous basis. These industries include but are not limited to food, beverage including dairy, chemical, pharmaceutical, oil, edible oil, industrial, and municipal wastewater.

Three-Phase Decanter (Tricanter) Centrifuges Tricanter centrifuges are similar in principle to decanter centrifuges (see Fig. 18-158), but separate three phases that contain two immiscible liquids and one sedimenting/suspended solids phase. The sedimenting solids that collect on the bowl wall are conveyed out of the centrifuge and discharged similarly to a decanter centrifuge. The two liquid phases are discharged either via gravity over two sets of adjustable weir plates or rings or via a dual discharge system where the heavy liquid phase (typically water) is discharged via a stationary impeller under pressure and the light liquid phase (typically fat or oil) is discharged via gravity over a ring dam. The benefit of the dual discharge system is that the liquid interface zone (and ultimately the pool/pond height) is adjustable while the machine is operating at full speed. These types of centrifuges are commonly used in the fish, animal by-products, oil sludge, and edible oil (i.e., olive and palm) industries.

Specialty Decanter Centrifuges Decanter technology has evolved over the past 10 years to include machines suitable for separation applications normally not effective in standard decanter centrifuges. These specialty decanters/tricanters utilize the same basic premise of solids discharge via an internal scroll, but with specific machine geometries that allow for specialty separations. These specialty decanters include the Sedicenter (shown in Fig. 18-159) and Sorticanter (shown in Fig. 18-160). The Sedicenter, which has a double-cone cocurrent bowl design and specialized scroll geometry, is capable of achieving higher rotational speeds (up to 7750 rpm and 10,000G) and can, therefore, increase clarification efficiency and effectively discharge fine, pasty solids where a normal decanter is inefficient and ineffective. The Sedicenter is commonly used in certain biotechnology, vitamin, soy, and yeast separations.

The Sorticanter has a scroll with reversing pitch on one side, which skims a floating solids layer off the top of the carrier liquid, usually an aqueous brine of intermediate density. Sinking solids are scrolled out similarly to normal decanter centrifuges, and liquid is discharged under pressure via a stationary but variable impeller. The Sorticanter is utilized in the plastics recycling industry.

Screenbowl Centrifuges The screenbowl centrifuge consists of a solid-bowl decanter to which, at the smaller conical end, a cylindrical screen has been added (see Fig. 18-161). The scroll spans the entire bowl, conforming to the profile of the bowl. It combines a sedimenting centrifuge together with a filtering centrifuge. Therefore, the solids which are processed are typically larger than 23 to 44 μm .

As in a decanter, an accelerated feed is introduced to the separation pool. The denser solids settle toward the bowl wall and the effluent escapes through the ports at the large end of the machine. The sediment is scrolled toward the beach, typically with a steeper angle compared to the decanter centrifuge. As the solids are conveyed to the

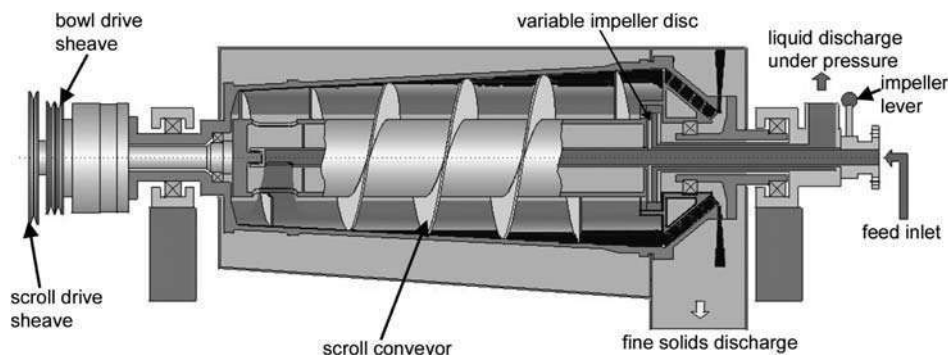


FIG. 18-159 Sedicenter centrifuge. (Flottweg Separation Technology.)

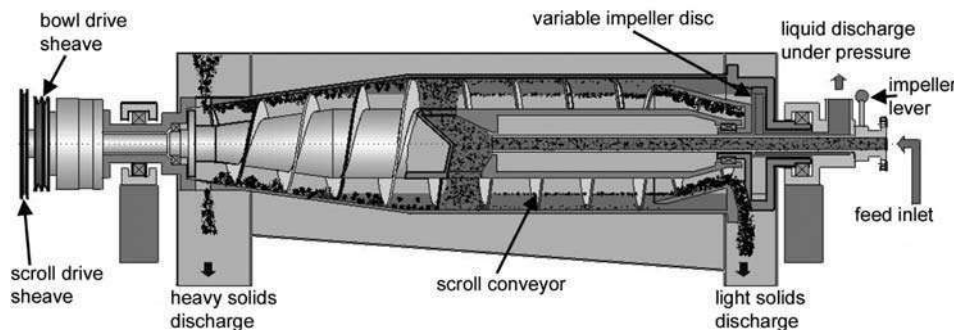


FIG. 18-160 Sorticanter centrifuge. (Flottweg Separation Technology.)

screen section, the liquid in the sediment cake further drains through the screen, resulting in drier cake. Washing of the sediment in the first half of the screen section is very effective in removing impurities, with the second half of the screen section reserved for dewatering of mother liquor and wash liquid.

The screen is typically constructed of a wedge-bar with an aperture between adjacent bars, which opens up to a larger radius. This prevents solids from blinding the screen as well as reduces conveyance torque. For abrasive materials such as coal, the screens are made of wear-resistant materials such as tungsten carbide.

Continuous Centrifugal Sedimentation Theory The Stokes settling velocity of a spherical particle under centrifugal field is given by Eq. (18-103). Useful relationships have been established on continuous sedimentation by studying the kinematics of settling of a spherical particle of diameter d in an annular rotating pool. Equating the time rate of change in a radial position to the settling velocity, and the rate of change in an axial position to bulk-flow velocity, thus gives

$$\frac{dr}{dt} = crd^2 \quad (18-111)$$

$$\frac{dx}{dt} = \frac{Q}{\pi(r_b^2 - r_p^2)} \quad (18-112)$$

where $c = (\rho_s - \rho_L) \Omega^2 / 18\mu$, x is distance along the axis of the bowl, Q is the volumetric feed rate, r_b and r_p are, respectively, the bowl and pool surface radii. For a particle located at one end of the bowl at radius r with $r_p < r < r_b$, after transverse the full bowl length, it set-

les out and is captured by the bowl wall. Solving the above equations with these boundary conditions, the limiting trajectory is:

$$\frac{r}{r_b} = \exp \left[\frac{-\pi c L (r_b^2 - r_p^2) d^2}{Q} \right] \quad (18-113)$$

If the same size particle d is located at an initial starting radius less than r given by Eq. (18-113) it is assumed to escape from being captured by the bowl, whereas it would have been captured if it had been at an initial radius greater than r . Assuming that the number of particles with size d is uniformly distributed across the annular pool, the recovery Rec_d (known also as **grade efficiency**) is the differential of the cumulative recovery $Z = 1 - Y$, with Y given in Eq. (18-95) for particles with size d , as the ratio of the two annular areas:

$$\text{Rec}_d = \frac{r_b^2 - r^2}{r_b^2 - r_p^2} \quad (18-114)$$

Combining Eqs. (18-113) and (18-114), the maximum Q to the centrifuge, so as to meet a given recovery Rec_d of particles with diameter d , is

$$\frac{Q_d}{2V_{gd}} = \left(\frac{\pi \Omega^2 L}{g} \right) \left(\frac{r_b^2 - r_p^2}{\ln [1 - \text{Rec}_d [1 - (r_p/r_b)^2]]} \right) = \Sigma_{\text{Rec}_d} \quad (18-115)$$

Note in Eq. (18-115) that V_{gd} is the settling rate under $1g$, and it is a function of the particle size and density and fluid properties. The ratio $Q_d/2V_{gd}$ is then related only to the operating speed and geometry of

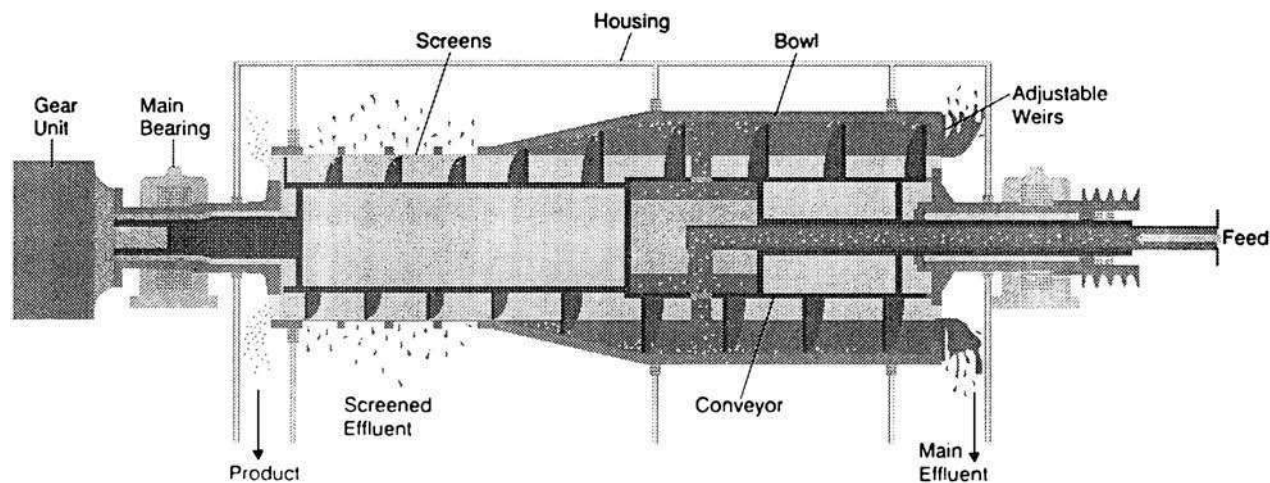


FIG. 18-161 Cylindrical-conical screen-bowl centrifuge. (Bird Machine Co.)

the centrifuge, as well as to the size recovery. It measures the *required* surface area for settling under centrifugal gravity to meet a specified Rec_d . When the size recovery Rec_d is set at 50 percent, the general result, Eq. (18-115), reduces to the special case, which is the well-known Ambler's Sigma factor, which for a straight rotating bowl (applicable to bottle centrifuge, decanter centrifuge, etc.) is:

$$\Sigma = \left[\frac{\pi \Omega^2 L}{g} \right] \left[\frac{r_b^2 - r_p^2}{\ln [2r_b^2 / (r_b^2 + r_p^2)]} \right] \quad (18-116)$$

It can be simplified to:

$$\Sigma = \pi \Omega^2 L \frac{(3r_b^2 + r_p^2)}{2g} \quad (18-117)$$

For a disc centrifuge, a similar derivation results in

$$\Sigma = \frac{2\pi \Omega^2 (N-1) (r_2^3 - r_1^3)}{3g \tan \theta} \quad (18-118)$$

where N is the number of discs in the stack, r_1 and r_2 are the outer and inner radii of the disc stack, and θ is the conical half-angle.

Typical Σ factors for the three types of sedimenting centrifuges are given in Table 18-13. In scale-up from laboratory tests, sedimentation performance should be the same if the value of $\dot{Q}/\Sigma$ is the same for the two machines. This is a widely used criterion for the comparison of centrifuges of similar geometry and liquid-flow patterns developing approximately the same G ; however, it should be used with caution when comparing centrifuges of different configurations. In general, the shortcomings of the theory are due to the oversimplified assumptions being made, such as (1) there is an idealized plug-flow pattern; (2) sedimentation abides by Stokes law as extended to many g 's; (3) feed solids are uniformly distributed across the surface of the bowl head at one end of the clarifier and capture implies that the particles' trajectory intersect the bowl wall; (4) the feed reaches full tangential speed as it is introduced to the pool; (5) the recovery of given-size particles is at 50 percent; (6) this does not account for possible entrainment of already settled particles in the liquid stream; (7) there is absence of entrance and exit effects.

Experience in using the Σ concept has demonstrated that the calculated Σ factor should be modified by an efficiency factor to account for

some of the aforementioned effects which are absent in the theory and, as such, this factor depends on the type of centrifuge. It is nearly 100 percent for simple spin-tube bottle centrifuge, 80 percent for tubular centrifuge, and less than 55 percent for disc centrifuges. The efficiency varies widely for decanter centrifuges, depending on cake conveyability and other factors.

FILTERING CENTRIFUGES

Filtering centrifuges are broadly categorized as continuous operating and batch operating. Both continuous and batch filtering centrifuges use some type of filtration media fitted against the basket (bowl) wall. As the solid-liquid mixture is introduced, the liquid filters through the solids, through the filter media, and typically through perforations in the basket shell (except with rotational siphon designs, discussed later). Filtering centrifuges are primarily chosen over sedimenting types where high cake purity through cake washing is a requirement, or where minimal residual cake moisture is desired. Typical solids retention times range from 5 to 45 s for continuous operating filtering centrifuges and 5 to 180 min for batch operating filtering centrifuges.

Usually, the solids phase is of a higher specific gravity than the liquid; but unlike with sedimenting centrifuges, this is not an absolute requirement. In the nontypical case where the opposite is true, then filtration, be it centrifugal, vacuum, or pressure, is the only option for solid-liquid separation. However, for batch filtering centrifuges with the solids phase lighter than the liquid, care must be taken that liquid is not allowed to build in the basket during the feed step, or else buoyancy forces may float the settled solids, resulting in uneven filtration and high vibration. This is not of concern for continuous filtering centrifuges since the filtration rate must inherently exceed the liquid feed rate for stable operation. Refer to Table 18-14 for typical operating ranges of filtering centrifuges.

BATCH FILTERING CENTRIFUGES

Although continuous centrifuges are often preferred for reasons of lowest capital cost, high unit capacity, and ease of integration into continuous upstream and downstream processes, batch filtering centrifuges with cyclic operation will always have a role in the CPI for reasons of highest possible product purity, lowest possible cake moisture, and highest product recovery. The slurry's physical properties such as particle size distribution or liquid viscosity may require long

TABLE 18-13 Scale-up Factors for Sedimenting Centrifuges

Type of centrifuge	Inside diameter, in	Disc diameter, in/number of discs	Speed, r/min	Σ value, units of 10^4 ft^2	Recommended scale-up factors*
Tubular	1.75	—	23,000	0.32	1†
Tubular	4.125	—	15,000	2.7	21
Tubular	4.90	—	15,000	4.2	33
Disc	—	4.1/33	10,000	1.1	1
Disc	—	9.5/107	6,500	21.5	15
Disc	—	12.4/98	6,250	42.5	30
Disc	—	13.7/132	4,650	39.3	25
Disc	—	19.5/144	4,240	105	73
Scroll conveyor	6	—	6,000	0.27	1
Scroll conveyor	14	—	4,000	1.34	5
Scroll conveyor	14‡	—	4,000	3.0	10
Scroll conveyor	18	—	3,450	3.7	12.0
Scroll conveyor	20	—	3,350	4.0	13.3
Scroll conveyor	25	—	3,000	6.1	22
Scroll conveyor	25	—	2,700	8.6	31

*These scale-up factors are relative capacities of centrifuges of the same type but different sizes when performing at the same level of separation achievement (e.g., same degree of clarification). These factors must not be used to compare the capacities of different types of centrifuges.

†Approaches 2.5 at rates below mL/min.

‡Long bowl configuration.

NOTE: To convert inches to millimeters, multiply by 25.4; to convert revolutions per minute to radians per second, multiply by 0.105; and to convert 10^4 square feet to square meters, multiply by 929.

TABLE 18-14 Operating Range of Filtering Centrifuges

Type of centrifuge	G/g	Minimum feed solids concentration by wt.	Minimum mean particle size, μm	Minimum V_{f0} , m/s
Vibratory	30–120	50	300	5×10^{-4}
Tumbler	50–300	50	200	2×10^{-4}
Screen scroll	500–2000	35	75	1×10^{-5}
Pusher	300–800	40	120	5×10^{-5}
Screen bowl	500–2000	20	45	2×10^{-6}
Peeler	300–2000	5	10	2×10^{-7}
Vertical	200–1000	5	5	1×10^{-7}
Inverting filter	300–1000	5	2	5×10^{-8}

retention time and drive the selection from continuous to batch. Generally, when the value of the product is high, batch operating centrifuges are often preferred.

A development over the last 10 to 20 years is the introduction of true cGMP designs suitable for high-purity fine chemicals and pharmaceutical applications. Requirements in this sector usually favor batch operation due to demands for batch identity; typical properties of the slurry dictate batch operation, high wash requirements, cleanability and inspectability, avoidance of cross-contamination for multipurpose applications, and elimination of operator exposure to the process.

Most modern batch filtering centrifuges are provided with ac variable-frequency drives (VFDs) for speed variation, often with power regenerative braking. Even in the case of peeler centrifuges which are capable of constant-speed operation throughout the cycle, in most cases they are still fitted with VFDs to meet starting requirements (accelerating a high inertial load), operating flexibility, and regenerative braking in hazardous areas. However, there are still units available that use either a two-speed motor (generally considered obsolete) or hydraulic drive.

The main subcategories of batch filtering centrifuges are vertical basket centrifuges (both top unloading and bottom unloading of various configurations), horizontal peeler centrifuges, and horizontal inverting filter centrifuges.

Vertical Basket Centrifuge—Operating Method and Mechanical Design The vertical basket centrifuge is equipped with a cylindrical basket rotating about the vertical axis. The basket shell is normally perforated and lined with filtration media consisting of filter cloth, and backing screens to provide liquid drainage paths to the basket holes. Securing the filter cloth may be by hook and loop attachment or with snap-in retainers.

Feed slurry is introduced into the basket through either single or multiple feed pipes, or by other means such as rotating feed cone to help distribute the solids on the basket wall. In most cases, feed slurry is introduced at an intermediate speed, although in some applications (FGD gypsum, for one), feeding is done at full speed. There are several methods available to control the feed and cake level such as mechanical, paddle-type feelers, capacitance probes, ultrasonic sensors, feed totalizer, or load cells.

The solids distribution profile may tend to be parabolic with thicker cakes near the bottom of the basket, tapering down toward the top, since the G-field is perpendicular to the force of gravity. This is especially true with fast-sedimenting solids that will settle toward the basket bottom before the slurry is fully accelerated by the basket. The coarser solids can settle toward the basket bottom, while the finer solids deposit preferentially toward the top. This can result in uneven filtration resistance in the cake, affecting the wash pattern and efficiency of the wash. In cases where this is a concern, a rotating feed cone may be better for even distribution, or a horizontal peeler centrifuge may be better suited to the application.

During and after feeding, filtrate passes through the cake, filter media, and out through the basket shell and is collected in a housing surrounding the basket and discharged through a tangential nozzle. The solids build up on the basket wall during the feed step until the desired loading is achieved.

After a spin time to filter the mother liquor through the solids, wash liquor is commonly applied in either a single step or various combinations, typically via a wash pipe with nozzles. The cake is spun for a

time at high speed, then the machine ramps down to discharge. Solids removal can be accomplished by one of several methods.

Top Unloading Vertical Basket Centrifuges This is one of the oldest types of centrifuges, dating back to about 1900 or even earlier. With this design, the perforated basket is fitted with either a filter cloth or a filter bag. The basket has a solid bottom. After the dry spin portion of the cycle is completed, the machine is stopped. Solids removal is either by manually digging out the cake or by removing (lifting) the filter bag from the top of the unit. Except for small pilot-scale units and some specialty applications, this design is no longer commonly marketed or desired due to labor intensiveness, incompatibility with solvent wet or toxic products (operator exposure), and overall inefficiency of operation. See Fig. 18-162.

One modern top unloading design is used for bulk pharmaceutical processing as part of a complete closed-loop dewatering and drying system including heaters, blowers, and sterile filters. This involves a two-motion discharge “mouth” to pneumatically remove thin layers of cake while blowing sterile, dehumidified hot nitrogen into the housing and into the discharge pipe. This method both pneumatically conveys and flash-dries the solids en route to the waiting conical mixer/dryer for subsequent vacuum drying.

Bottom Unloading Vertical Basket Centrifuges Most common for modern machines is the bottom discharge design, incorporating a swiveling scraper mechanism, typically cutting the cake in a single motion, or with a two-motion, oscillating scraper in the case of finer or stickier cakes or pharmaceuticals. In every case, solids discharge must be at low speed, which necessitates ramping the machine up and down every cycle. After discharge, a thin cake layer or heel remains on the filter cloth. See Fig. 18-163.

Heel removal can be automated by dissolving the heel, flushing the heel out the solids discharge chute with subsequent downstream diverting away from solids handling equipment, or pneumatically removing the heel with blowoff nozzles, discharging it out the solids discharge. Pneumatic heel removal can be accomplished either from within the basket (often incorporated with the knife) or from the outside of the basket.

There are different cover arrangements to access the interior, such as hinged or pivoting manway, half or full opening covers. Filter cloth maintenance or component adjustments usually require entering the unit in any case, except for small sizes. See Fig. 18-164.

Isolation of the load imbalances from the structure has historically been by link, or three-column suspension. This system is relatively inefficient and transmits substantial dynamic forces to the building foundation, limiting operating speeds and performance. In response to the common shortcomings of this design, some manufacturers redesigned the suspension/isolation system to a massive inertial base-plate and housing supported on tuned coil springs and dampers at each corner. This greatly improved the dynamic force attenuation and stability of the machine. This system has allowed designing very large machines (1600-mm-diameter by 1250-mm-high baskets) processing materials with high solids density such as wallboard-grade gypsum from FGD systems with a unit capacity in excess of 10 Mtons/h.

Top Suspended Vertical Centrifuges A special type of top suspended centrifuge is widely used in sugar processing and is shown in Fig. 18-165. Conventionally, the drive is suspended from a horizontal bar supported at both ends from two A-frames. The drive head, which is connected to the motor or a driven pulley through a flexible coupling, carries the thrust and radial bearings that support the basket,

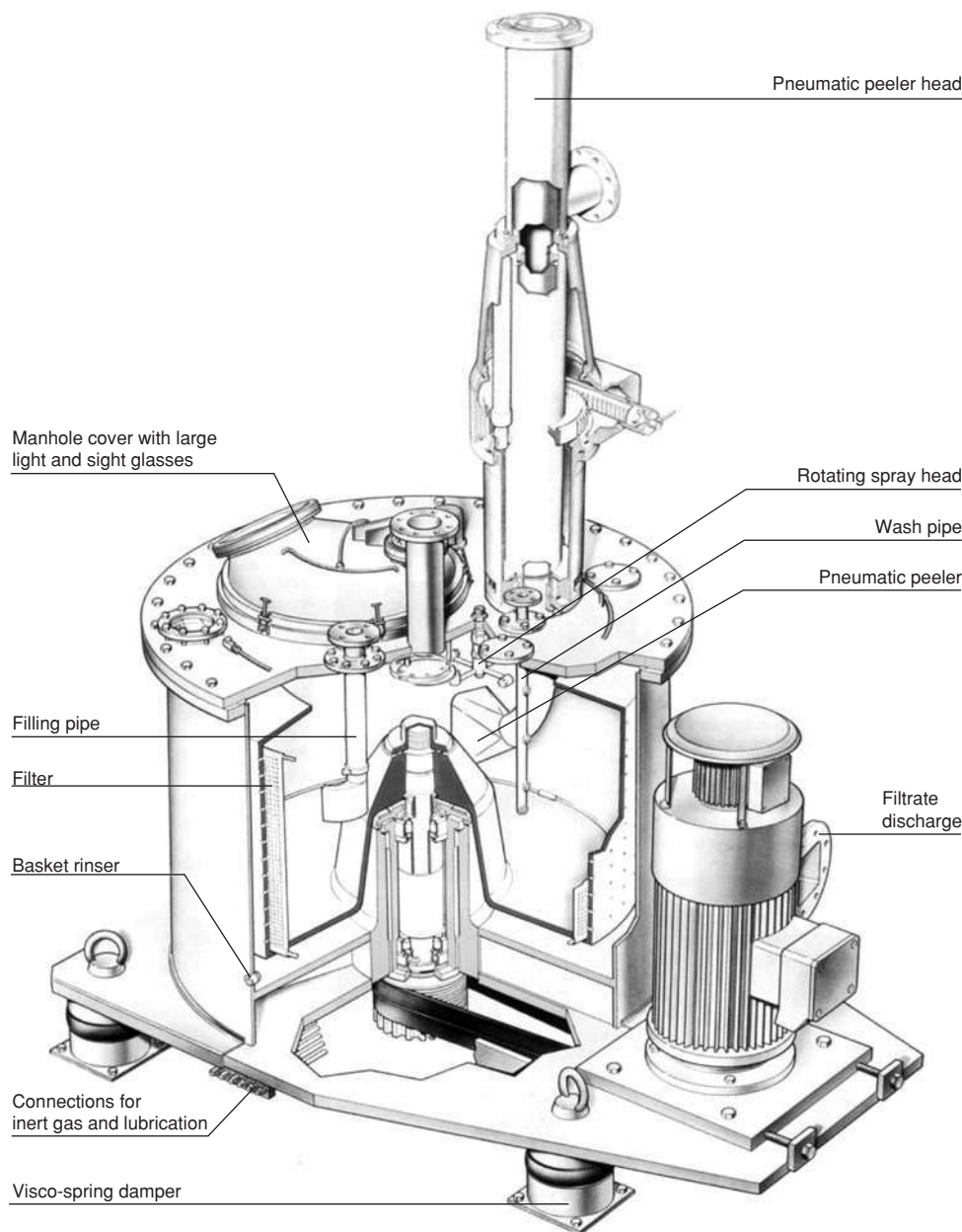


FIG. 18-162 Top unloading vertical basket centrifuge. (Krauss Maffei Process Technology.)

shaft, and load. These units can be equipped with large 112-kW (150-hp) drive motors, and on white sugar can process about 350 kg/cycle with 24 cycles/h.

Horizontal Peeler Centrifuge—Operating Method and Mechanical Design The chemical design peeler centrifuge was developed in the 1920s and has wide areas of application (Fig. 18-166). Like the vertical basket, it has a rotating filtration basket, except that it rotates about the horizontal axis. Early machines supported the basket from both ends, but virtually all modern machines are cantilever-supported for purposes of accessibility to the basket and internal components. Some machines are equipped with a full opening door which swings away with internal components, while other designs

incorporate a full opening housing that also provides access to the basket exterior. This unit has an extremely rugged construction compared to vertical baskets, required due to the full-speed feeding and discharge capability of the peeler centrifuge. It is often provided with high-power ac VFD drives for accelerating the feed slurry at full speed and for optimum operating flexibility. Casttight construction to 400-mm (16-in) water column is usually standard, and higher pressure ratings can easily be accomplished.

By reorienting the axis to horizontal, many advantages become possible such as superior wash capability with the more uniform solids distribution compared to the vertical basket, with the potential for uneven, parabolic cake profile resulting in uneven wash penetration.

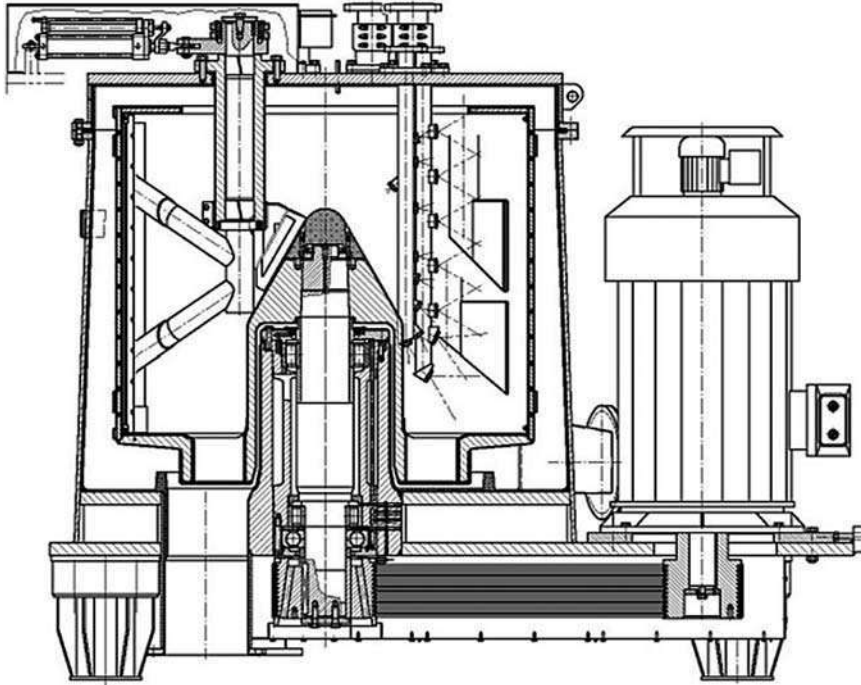


FIG. 18-163 Typical bottom unloading vertical basket centrifuge. (Krauss Maffei Process Technology.)

The peeler centrifuge costs more than a comparable basket size vertical machine, although often a smaller peeler centrifuge can outperform a larger vertical basket. In addition, the higher capital cost is offset by numerous process and mechanical advantages, such as these:

- Full opening door contains the feed, wash, feed control, and solids discharge components. Easily swung open, it then provides complete

access to the basket interior and all internal components mounted on the door. Filter cloth exchange does not require vessel entry.

- Isolation of dynamic forces is far superior with horizontal machines compared to vertical.
- The peeler centrifuge will distribute the solids more evenly since it is not feeding perpendicular to gravity as is the vertical basket. This

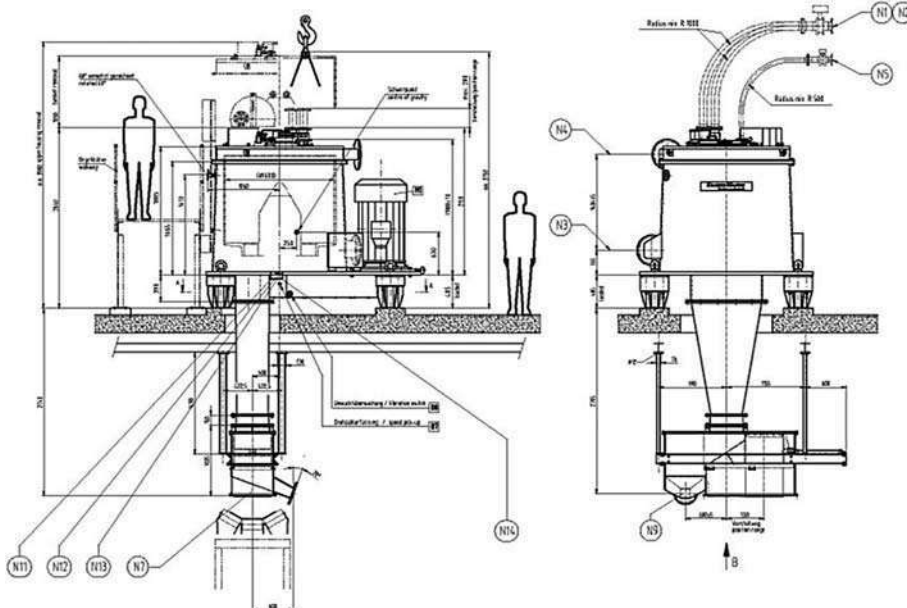


FIG. 18-164 Typical vertical basket centrifuge installation. (Krauss Maffei Process Technology.)

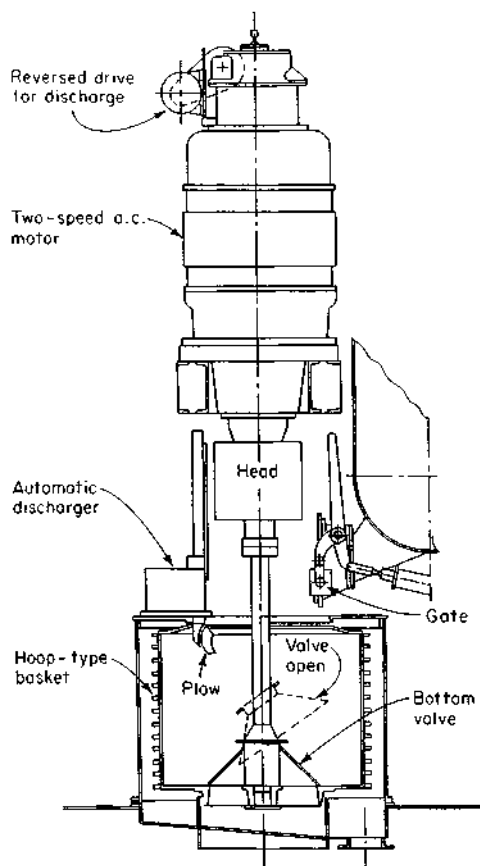


FIG. 18-165 Top-suspended vertical centrifuge. (Western States Machine Co.)

provides smoother operation, better wash effect, and an ability to handle faster-draining materials.

- Ability to discharge at high speed eliminates or minimizes dead cycle time required for acceleration and braking for higher capacity, lower power consumption, and lower wear and tear. The cycle time savings is particularly beneficial with short-cycle (fast-filtering) requirements.
- The peeler centrifuge can provide higher centrifugal forces than can vertical baskets, for increased performance and flexibility.
- Peeler centrifuges are available in larger sizes than vertical centrifuges—up to 2100-mm (83-in) diameter.
- The peeler centrifuge is also capable of automatic heel removal by several methods: dissolving the heel, reslurrying and discharging the heel wet and diverting downstream, or dry heel removal pneumatically. Pneumatic heel removal can be accomplished either from within the basket or from outside the basket.

Siphon Peeler Centrifuge The siphon peeler centrifuge (Fig. 18-167) was developed and patented by Krauss-Maffei in the 1970s. Instead of utilizing only centrifugal pressure as the driving force, as do all perforated units both vertical and horizontal, the rotational siphon centrifuge provides an increased pressure gradient by reducing the pressure behind the filter media and thereby increasing the driving force for filtration.

As in the perforated basket design, the liquid filters through the cake and filter media, but instead of discharging through perforations in the basket shell, the basket wall is solid and the liquid flows axially to the basket rear and into a separate chamber. At this point, the filtrate is skimmed out with a radially adjustable skimmer.

In perforated baskets, the driving force for filtration is approximately the hydrostatic pressure established by the liquid column. The driving force diminishes as the liquid column height decreases, often causing a wet layer near the base of the cake due to capillary pressure balancing the centrifugal pressure. In siphon baskets, in addition to the centrifugal pressure, by skimming at a radius greater than the filter cloth, a rotational siphon is established. Due to the gravitational field in which it is working, a height difference Δh of only 20 to 30 mm is sufficient to lower the pressure behind the cloth to the vapor pressure of the liquid. This additional vacuum remains in place until all the interstitial liquid is drawn through the cake, and will overcome the cake capillary pressure, thus preventing this wet layer. Once the supernatant and interstitial

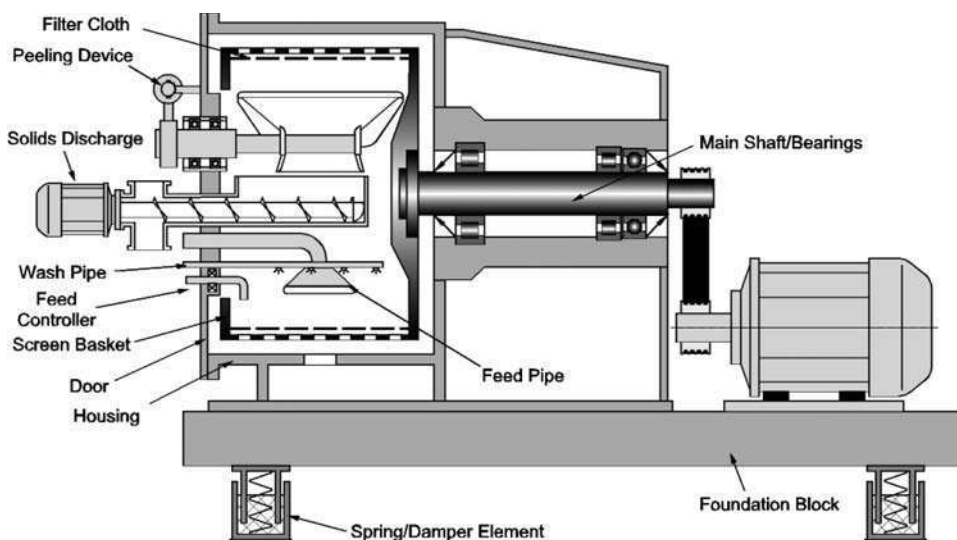


FIG. 18-166 Peeler centrifuge cross section. (Krauss Maffei Process Technology.)

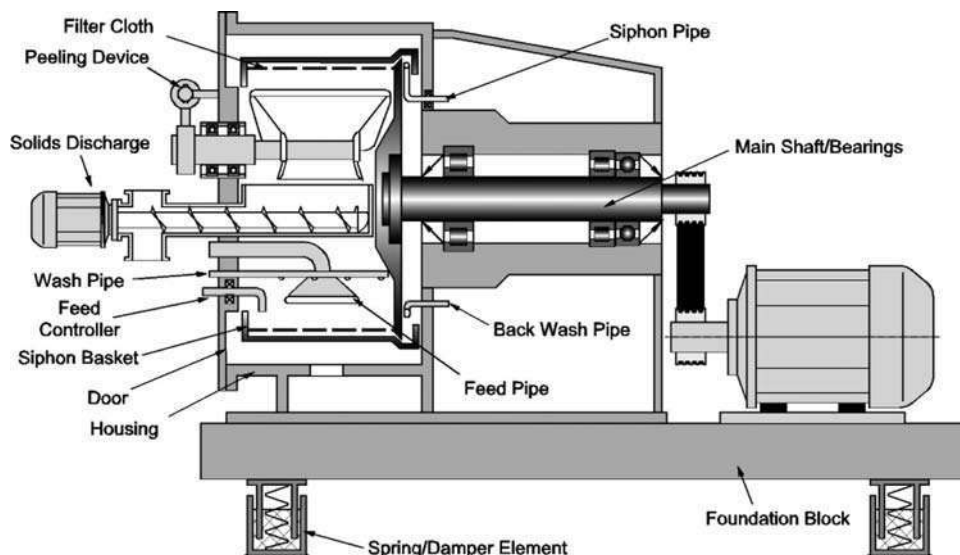


FIG. 18-167 Siphon peeler centrifuge cross section. (Krauss Maffei Process Technology.)

liquid drains from the cake, the siphon chamber behind the cloth drains and filtration characteristics are like those of a perforated basket. See Fig. 18-168.

To reestablish the siphon for the next cycle, a priming step precedes feeding where the siphon skimmer is pivoted inward near the rim of the siphon chamber, and liquid is introduced into the siphon chamber that backflows up through the heel cake, displacing gas from the chamber. Feeding then begins with the heel submerged. After a time delay, the siphon swivels downward to the working position (δh of +20 to 30 mm) where it remains for the remainder of the cycle. Besides increased driving force for filtration, other benefits of the rotational siphon include:

- Accurate control of the filtration rate is useful during feed and wash. For fast-filtering products, filtration rates can be throttled, ensuring even solids distribution.
- Backwashing the residual heel between each cycle rejuvenates the heel to maintain good permeability. Heel life is often extended.
- Feeding into a liquid bath helps lay down a more porous heel layer since the larger particles sediment faster than finer particles.

- Separate discharge of filtrate from splash/overflow provides better product yield.
- Deep siphon chambers with cake backwashing capability have been successfully utilized to completely submerge the cake and indefinitely increase wash contact time.

Pressurized Siphon Peeler Centrifuge Theoretically, the same principle can further increase driving force with overpressure in the process housing; for example, 3-bar overpressure would produce up to 4-bar pressure gradient across the cake. In practice, to date this has not been utilized due to the complexity of the installation.

Pharma Peeler Centrifuge For applications requiring hygienic operation, a special type of peeler centrifuge was developed in the 1990s (Fig. 18-169).

Primary applications for this type of machine are in fine chemicals and pharmaceuticals, often in multipurpose use where cross-contamination must be avoided. It provides for ease of cleanability and inspectability with automatic CIP/SIP, access to every wetted surface, pressure-tight construction suitable for steam sterilization, automatic heel removal, and separation of mechanical components from the

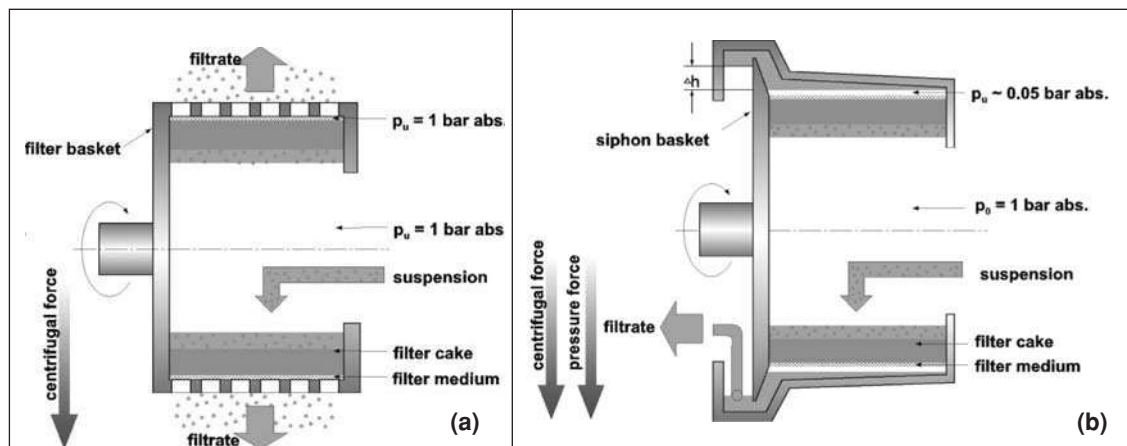


FIG. 18-168 Schematic representation of (a) perforate versus (b) siphon centrifuge. (Krauss Maffei Process Technology.)



FIG. 18-169 Pharma peeler centrifuge. (Krauss Maffei Process Technology.)

process end, making it suitable for through-the-wall clean-room installation. Operation is contained, thereby eliminating operator exposure.

Inverting Filter Centrifuge The inverting filter centrifuge was introduced in the late 1970s to provide a means of ensuring that all the filter cake is discharged from the filter medium. By turning the cloth inside out to achieve solids discharge, the problems of operator exposure and variable product quality associated with manual cake removal or residual-heel blinding were largely eliminated. By the late 1980s, this style of centrifuge had found widespread application in

pharmaceutical and agricultural chemical production. See Figs. 18-170 to 18-172.

The inverting filter design comprises a horizontal axis shaft with a two-part bowl attached. The perforated cylindrical bowl remains in a fixed axial position throughout the operation, while a bowl insert is able to move along the horizontal axis. The filter cloth is attached at one end to the axially fixed bowl and at the other to the axially movable bowl insert. Therefore, moving the bowl insert causes the filter cloth to turn inside out. In the filtering position, the bowl insert sits inside the bowl, with the filter cloth covering perforations. As with other filtering centrifuges, the cake builds up on the cloth during filling. It is washed using a true positive-displacement, plug-flow wash, and cake dewatering can be achieved simply by spinning (often at the maximum speed) for a time.

As this cake-discharging mechanism involves little or no risk of cloth blinding, inverting filter centrifuges usually operate at optimum conditions with relatively thin cakes and frequent discharges. (Cake thicknesses are typically 1 to 3 in, and cycle times are typically 8 to 14 min.) This style of operation is particularly effective with compressible materials where the filtration rate drops off dramatically with increasing cake thickness. By operating with thin cakes and short cycle times, the average filtration flux throughout the batch operation is maximized for these difficult applications.

Inverting filter centrifuges come in bowl diameters ranging from 300 to 1300 mm and achieve g -forces of 3000 – 900 $\times$ gravity.

CONTINUOUS FILTERING CENTRIFUGES

Where processing conditions and objectives allow, continuous filtering centrifuges offer the combination of high processing capacities and good wash capabilities. Inherently they are less flexible than batch filtering centrifuges, primarily constrained by much shorter retention time, and in some cases liquid handling capacity requires upstream

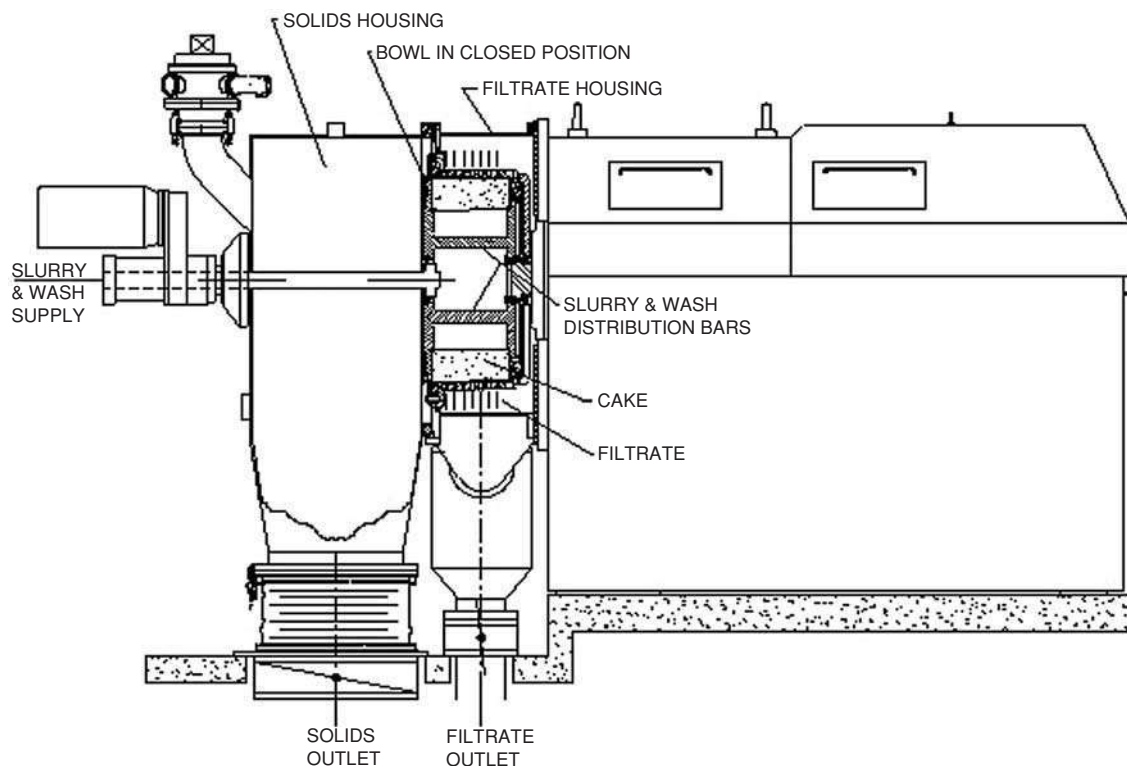


FIG. 18-170 Inverting filter centrifuge. (Heinkel USA.)

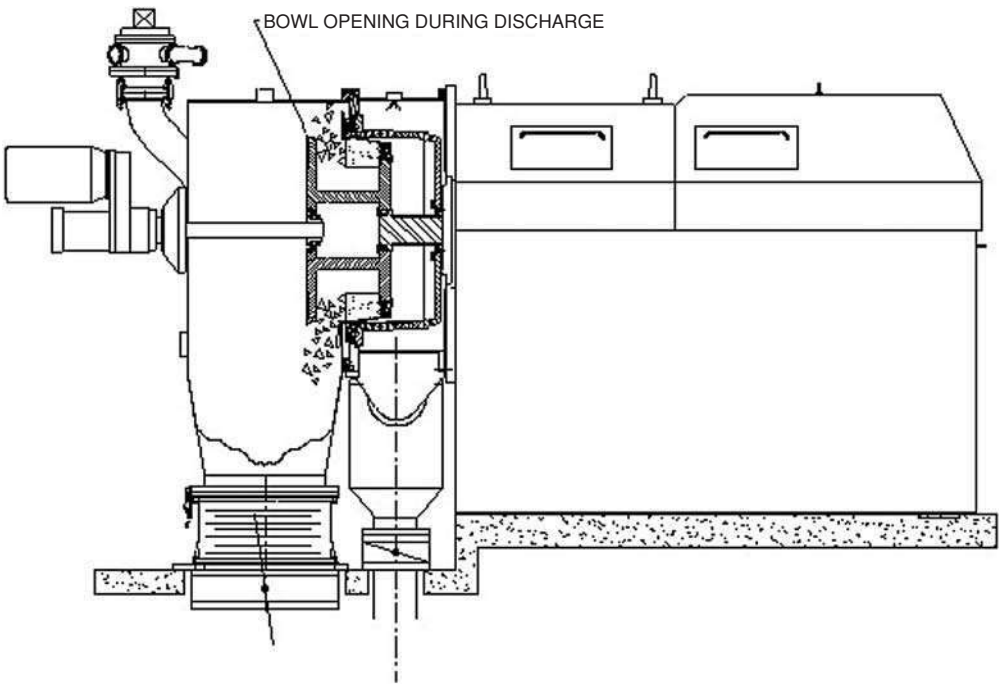


FIG. 18-171 Inverting filter centrifuge. (Heinkel USA.)

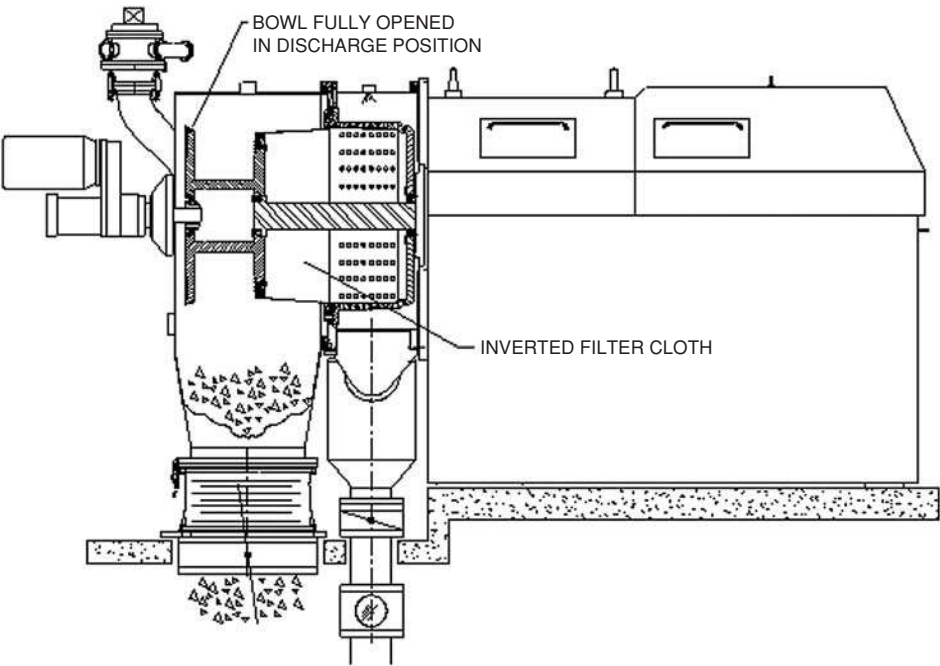


FIG. 18-172 Inverting filter centrifuge. (Heinkel USA.)

preconcentration of the slurry. Fines loss to the filtrate is also greater with continuous designs compared to batch.

Conical-Screen Centrifuges When a conical screen in the form of a frustum is rotated about its axis, the component of the centrifugal force normal to the screen surface impels the liquid to filter through the cake and the screen, whereas the component of the centrifugal force parallel to the screen in the longitudinal direction conveys the cake to the screen at a larger diameter. The sliding of the solids on the cone is favored by smooth perforated plates or wedge-wire sections with slots parallel to the axis of rotation, rather than woven wire mesh.

Wide-angle conical screen centrifuges. If the half-angle of the cone screen is greater than the angle of repose of the solids, the solids will slide across it with a velocity which depends on frictional properties of the cake but not on feed rate. The frictional property of the cake depends on the solid property, such as shape and size, as well as on moisture content. If the half-angle of the cone greatly exceeds the angle of repose, the cake slides across the screen at a high velocity, thereby reducing the retention time for dewatering. The angle selected is therefore highly critical with respect to performance on a specific application. Wide-angle and compound-angle centrifuges are used to dewater coarse coal and rubber crumb and to dewater and wash crude sugar and vegetable fibers such as from corn and potatoes.

Shallow-angle conical screen centrifuges. By selecting a half-angle for the conical screen that is less than the angle of repose of the cake and providing supplementary means for the controlled conveyance of the cake across the conical screen from the small to large diameter, longer retention time is available for cake dewatering. Three methods are in common use for cake conveyance:

1. **Vibrational conveyance.** This is referred as the vibratory centrifuge. A relatively high frequency force is superimposed on the rotating assembly. This can be either in-line with the axis of rotation or torsional, around the driveshaft. In either case, the cake under inertial force from the vibration is partly "fluidized" and propelled down the screen under a somewhat steady pace toward the large end, where it is discharged.

2. **Oscillating or "tumbling" conveyance.** This is commonly known as the tumbling centrifuge. The driveshaft is supported at its lower end on a pivot point. A supplementary power source causes the shaft and the rotating bracket it carries to gyrate about the pivot at a controlled amplitude and at a frequency lower than the rate of rotation of the basket. The inertia force generated also provides partial

fluidization of the bed of solids in the basket, causing the cake to convey toward the large end, as in the vibrational conveyance.

3. **Scroll conveyance.** Another type of continuous filtering centrifuge is the scroll screen centrifuge, as shown in Fig. 18-173. The scroll screen centrifuges are also sometimes called worm screen centrifuges. The design consists of a fixed-angle rotating basket and a concentric screw conveyor to control the transport and discharge of solids. Common applications include crystal, fiber, and mineral separations.

Scroll screen centrifuges are typically used for continuous feeds of slurries of at least 10 percent solids by volume, of materials with an average size of 100 μm or greater. This design offers some residency to process variation and typically removes the bulk of surface moisture.

The scroll and the screen are rotating in the same direction with a small differential speed of typically less than 100 rpm. The feed is deposited into the acceleration cone of the scroll, then passes through the feed openings of the scroll. The solids are retained on the screen; as the liquid migrates through, the cake passes the screen media and the basket. The discharge housing collects the liquid, and the solids are conveyed to the large diameter of the rotating basket and are continuously discharged.

An internal product wash is also available in the scroll screen centrifuges. Wash liquid is added in a chamber midway along the basket, and the wash liquid migrates through the cake prior to final drying and discharge.

The rotating basket is used to retain the screen media. Wedge-wire as well as sheet metal screens are available, but are typically limited to a minimum opening size of 70 μm or larger. Common basket designs include 10, 15 and 20 degrees.

The scroll acts as a screw conveyor and discharges the solids. The typical solids retention time in the centrifuge ranges from 0.5 to 6 s. A close tolerance, 0.3 to 1 mm, is common between the scroll and screen; therefore little material remains on the screen. This minimizes the potential for imbalances.

Pusher Centrifuges—Operating Method and Mechanical Design Pusher centrifuges (Fig. 18-174) are continuous filtering centrifuges used for dewatering and washing free-draining bulk crystalline, polymer, or fibrous materials. Where suited, they provide the best washing characteristics of any continuous centrifuge due to control of retention time, uniform cake bed, and essentially

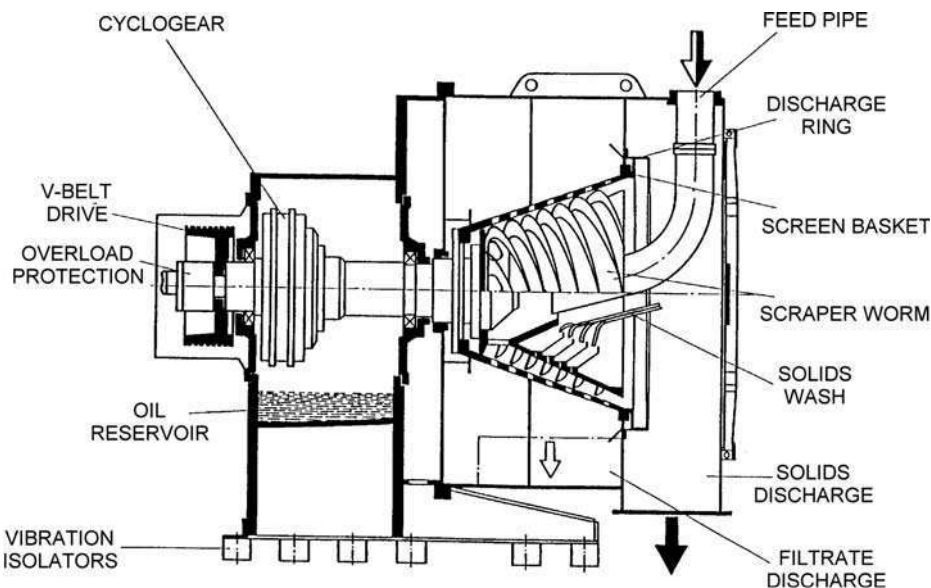


FIG. 18-173 Scroll screen centrifuge. (TEMA Systems, Inc.)

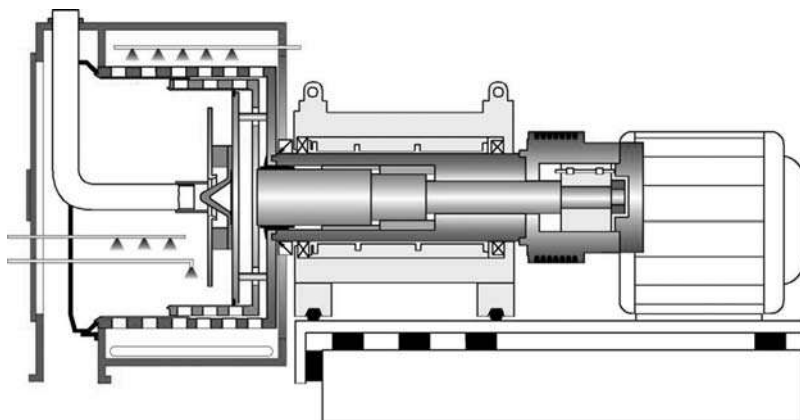


FIG. 18-174 Pusher centrifuge cross section. (Krauss Maffei Process Technology.)

plug flow of solids through the unit. For a typical application such as salt, they range in capacity from about 1 ton/h for small (250-mm-diameter) units up to about 120 tons/h or more in the largest units (1250-mm-diameter). They are generally applied where the mean particle size is at least 150 μm . Typical solids retention time is between 10 and 30 s. Normally the machine is fed by a feed pipe, but it can also be used as posttreatment of a prior dewatering step such as a vacuum filter. In this case it is fed by a feed screw. Due to the gentle handling of the product, pusher centrifuges are better suited for fragile crystals than are other types of continuous filtering centrifuges.

Generally there are three limitations to capacity in pusher centrifuges: (1) solids volumetric throughput, (2) liquid filtration capacity, and (3) retention time necessary to achieve desired objectives regarding cake purity and residual cake moisture. In most cases (2) dictates; therefore to optimize capacity and performance, preconcentrating the feed slurry as high as possible is desired. Some designs have a short conical section at the feed end for prethickening within the unit, but generally it is preferable to thicken ahead of the centrifuge with gravity settlers, hydrocyclones, or inclined screens.

As depicted schematically in Fig. 18-174, the rotating assembly consists of a belt-driven outer rotor that rotates at constant speed. The outer shaft (hollow shaft) is fixed to the main or outer basket. Within the hollow shaft is the pusher shaft which is keyed together with the hollow shaft but also oscillates. The reciprocal motion is provided by a mechanical gearbox for smaller units (400-mm-diameter and less) or hydraulically in larger units. The depicted schematic is of a two-stage design in which the pusher shaft is fixed to the inner basket and the pusher plate is attached to the outer basket by posts. The stroke length is between 30 and 85 mm depending on machine size, and stroke frequency is usually between 45 and 90 strokes per minute.

The feed slurry enters through a stationary central pipe into a feed accelerator/distributor, then is introduced onto the (in this case) oscillating inner basket just in front of the pusher plate. In the feed zone, most of the liquid is drained, forming a cake sufficiently stiff to transfer the push force through the bed of solids and transport the cake without shearing. This is why it requires fast-draining materials and is liquid-limited, since it must form a cake within the period of one stroke.

With designs that use a simple feed cone or plate for feed distribution, most of the slurry acceleration takes place on the screen surface, with lower effective slurry speed and driving force for filtration. More advanced designs utilize an impeller-type feed accelerator that largely preaccelerates the slurry prior to introduction on the screen for higher capacity and lower screen wear.

With each stroke of the pusher, the material in the feed zone is pushed up to a certain height primarily depending on the friction coefficient between the solids and the screen and the screen deck

length, and secondarily depending on G -force and loading. Once the cake in the feed zone is compressed and has formed a ring with this height, it transmits the push force to the stationary bed of cake in the basket which begins to move the cake bed forward until the forward end of the stroke. This cake height is often referred to as the natural cake height. The schematic in Fig. 18-175 shows what is taking place in the feed zone.

The distance the cake ring moves forward divided by the stroke length is defined as the *push efficiency*. The push efficiency varies with solids volumetric loading, resulting in a self-compensating control of varying rates. Depending on the cake properties, primarily compressibility, up to about 90 percent push efficiency is achievable. In some cases volumetric throughput can be further increased beyond the volumetric push capacity at the natural cake height, in which case the push efficiency remains almost constant and the cake height increases with increasing load, commonly referred to as the forced cake height. This realm of operation is usually only possible with multistage designs.

As the cake bed is transported through the basket, it passes through the various process steps shown in Fig. 18-176 with product moisture gradient as shown in Fig. 18-177.

Usually, cake wash ratios of about 0.1 to 0.3 kg wash/kg solids are possible within the normal residence time of the wash zone. This usually can displace at least 95 percent of the mother liquor and impurities. In some cases higher wash ratios or even multistage countercurrent washes are utilized, in which case sufficient residence time via throughput reduction must be considered.

Single-Stage versus Multistage Pusher centrifuges can be single-stage configuration with a single long basket and screen, two-stage (as shown schematically in Fig. 18-177), three-, or four-stage designs. Cake height and push force are primarily influenced by screen deck length and cake friction coefficient.

Single-Stage Where single-stage units are appropriate (ammonium sulfate is one example due to very large crystal size and good cake shear strength), the solids volumetric capacity can be maximized. However, because the push force requirement increases with screen length, cake shear or buckling can be the result with unstable operation. Because the average cake thickness in the feed zone is higher, filtration capacity may be slightly less than with multistage units. Fines losses can be slightly less with single-stage units since a smaller proportion of the cake bed is in contact with the slotted screen and there is no reorientation of crystals between stages. These units are often limited to low-speed operation for stability.

Two-Stage The majority of pusher centrifuges sold today are of this type. It provides greater flexibility compared to single stages in terms of greater filtration capacity, lower tendency for cake shear, and higher speed capability. When it is possible to operate with a forced cake, capacities can approach those of single-stage designs. Wash typically is applied on the latter portion of the first stage and through the

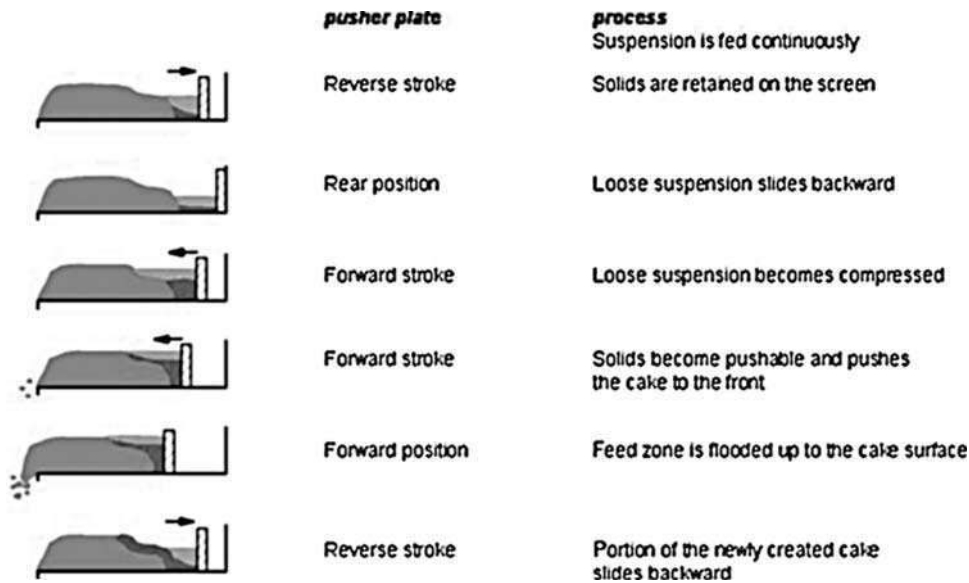


FIG. 18-175 Pusher centrifuge solids transport. (Krauss Maffei Process Technology.)

transition onto the second stage. During this transition the crystals are reoriented and the capillaries opened, which can enhance the wash effect. With even stage units, the feed acceleration system is not oscillating relative to the feed pipe. Some advanced designs of feed acceleration systems incorporating impellers benefit from this constant relationship.

Three- and Four-Stage These designs are generally reserved for the largest sizes that have long baskets that need to be subdivided

into reasonable-length stages as well as for very special applications with very high friction coefficients, low internal cake shear strength, or fairly high compressibility. For example, in processing high rubber ABS, four-stage units have been utilized; but the deck lengths are so short, with the corresponding thin cakes and short retention time, that capacity and performance are severely reduced. Other types of machine (such as peeler centrifuges or cylindrical conical pushers discussed below) can be better suited.

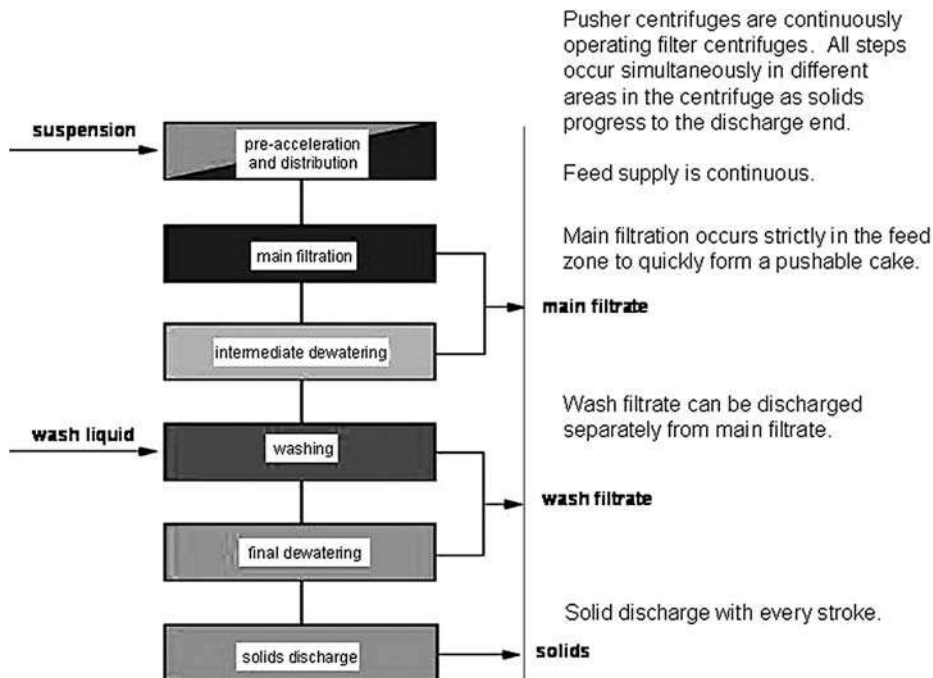


FIG. 18-176 Pusher centrifuge process steps. (Krauss Maffei Process Technology.)

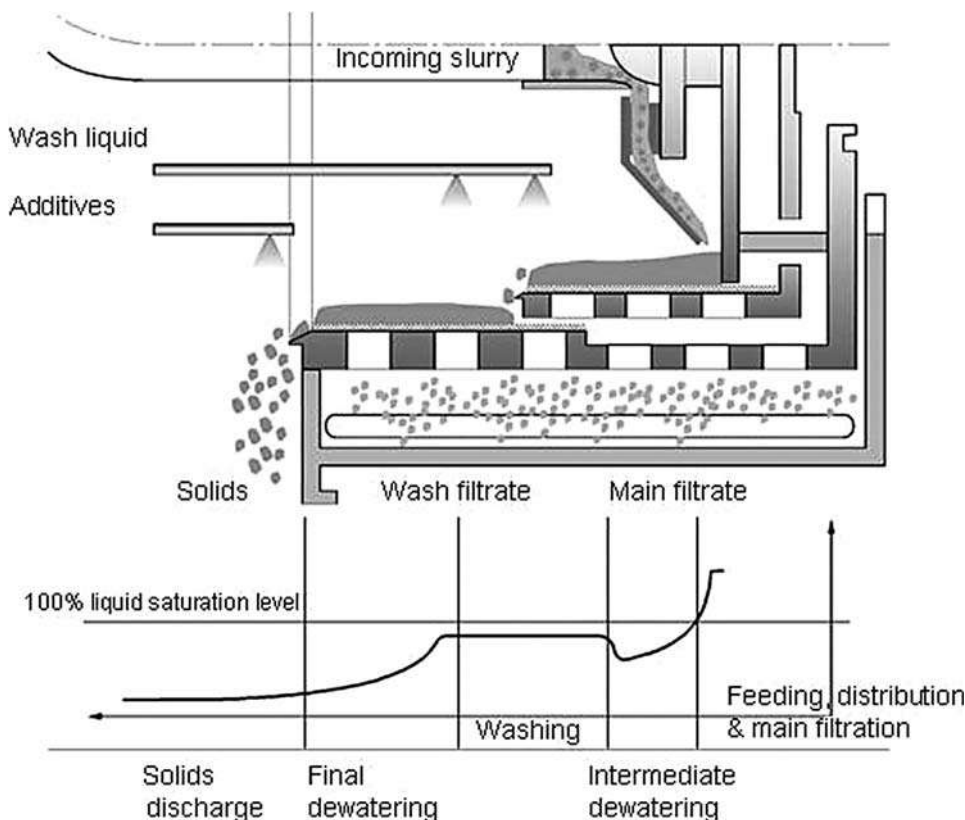


FIG. 18-177 Pusher centrifuge product moisture gradient. (Krauss Maffei Process Technology.)

Cylindrical/Conical A variation of single- and two-stage designs utilizes a cylindrical section or stage at the feed end followed by a conical section or stage sloping outward to the discharge end. The benefit of this design is that the axial component of force in the conical end assists with solids transport. Care must be taken that the cone angle not exceed the sliding friction angle of the cake, or else the cake will short-circuit the zone, resulting in poor performance and high vibration. Fabrication costs of the baskets are higher than those of cylindrical designs, and slotted screen construction is complicated with high replacement costs.

Theory of Centrifugal Filtration Theoretical predictions of the behavior of solid-liquid mixtures in a filtering centrifuge are more difficult compared to pressure and gravity filtration. The area of flow and driving force are both proportional to the radius, and the specific resistance and porosity may also change markedly within the cake. Filtering centrifuges are nearly always selected by scale-up from lab tests on materials to be processed, such as using bucket centrifuges where a wide range of test conditions (cake thickness, time, and G -force) can be controlled. Although tests with the bucket centrifuge provide some quantitative data to scale-up, the results include wall effect from buckets, which are not representative of actual cylindrical basket geometry, bucket centrifuges are not useful in quantifying filtration rates. A modified version of the buckets or even a cylindrical perforated basket can be used. In the latter, there is less control of cake depth and circumferential uniformity. The desired quantities to measure are filtration rate, washing rate, spinning time, and residual moisture. Also, with filtering centrifuges such as the screen-bowl centrifuge, screen-scroll centrifuge, and to some extent in multistage pushers, the cake is constantly disturbed by the scroll conveyor or conveyance mechanism; liquid saturation due to capillary rise as measured in bucket tests is absent. While bucket centrifuge tests are

very useful for first-look feasibility, it is always recommended to follow with pilot-scale testing of the actual equipment type being considered.

Filtration Rate When the centrifuge cake is submerged in a pool of liquid, as in the case of a fast-sedimenting, solids-forming cake almost instantly, and the rate of filtration becomes limiting, the bulk filtration rate $\bar{Q}$ for a basket with axial length b is:

$$\bar{Q} = \frac{\pi b \rho K \Omega^2 (r_b^2 - r_p^2)}{\mu \left(\ln \left[\frac{r_b}{r_c} \right] + \frac{K R_m}{r_b} \right)} \quad (18-119a)$$

where μ and ρ are, respectively, the viscosity and density of the liquid; Ω is the angular speed; K is the average permeability of the cake and is related to the specific resistance α by the relationship $\alpha K \rho = 1$, with ρ_s being the solids density; r_p , r_c , and r_b are, respectively, the radius of the liquid pool surface, the cake surface, and the filter medium adjacent to the perforated bowl. Here, the pressure drop across the filter medium, which also includes that from the cake heel, is $\Delta p_m = \mu R_m (\bar{Q}/A)$ with R_m being the combined resistance. The permeability K has a unit m^2 , α m/kg , and R_m m^{-1} . The driving force is due to the hydrostatic pressure difference across the bowl wall and the pool surface—i.e., the numerator of Eq. (18-119a), and the resistance is due to the cake layer and the filter medium—i.e., denominator of Eq. (18-119a). Fig. 18-178 shows the pressure distribution in the cake and the liquid layer above. The pressure (gauge) rises from zero to a maximum at the cake surface; thereafter, it drops monotonically within the cake in overcoming resistance to flow. There is a further pressure drop across the filter medium, the magnitude dependent on the combined resistance of the medium and the heel at a given flow rate. This

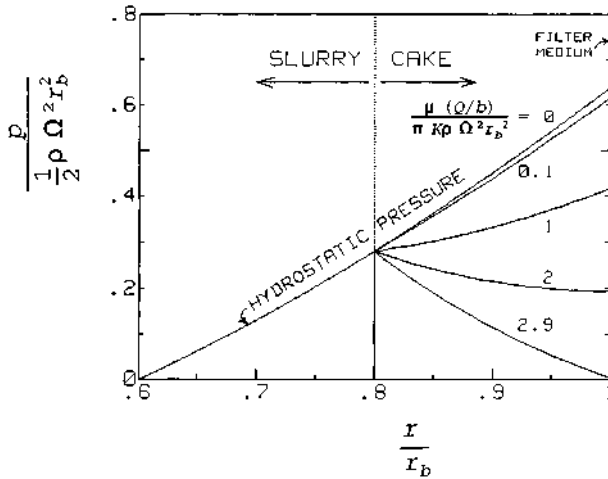


FIG. 18-178 Pressure distribution in a basket centrifuge under bulk filtration.

scenario holds, in general, for incompressible as well as for compressible cake. For the latter, the pressure distribution also depends on the compressibility of the cake.

For incompressible cake, the pressure distribution and the rate depend on the resistance of the filter medium and the permeability of the cake. Figure 18-178 shows several possible pressure profiles in the cake with increasing filtration rates through the cake. It is assumed that $r_c/r_b = 0.8$ and $r_p/r_b = 0.6$. The pressure at $r = r_b$ corresponds to pressure drop across the filter medium Δp_m with the ambient pressure taken to be zero. The filtration rate as well as the pressure distribution depend on the medium resistance and that of the cake. High medium resistance or blinding of the medium results in greater penalty on filtration rate.

In most filtering centrifuges, especially the continuous-feed ones, the liquid pool above the cake surface should be minimum to avoid liquid running over the cake. Setting $r_p = r_c$ in Eq. (18-119a), the dimensionless filtration flux is plotted in Fig. 18-179 against r_c/r_b for different ratios of filter-medium resistance to cake resistance, KR_m/r_b . For negligible medium resistance, the flux is a monotonic decreasing function with increasing cake thickness, i.e., smaller r_c . With finite medium resistance, the flux curve for a range of different cake thicknesses has a

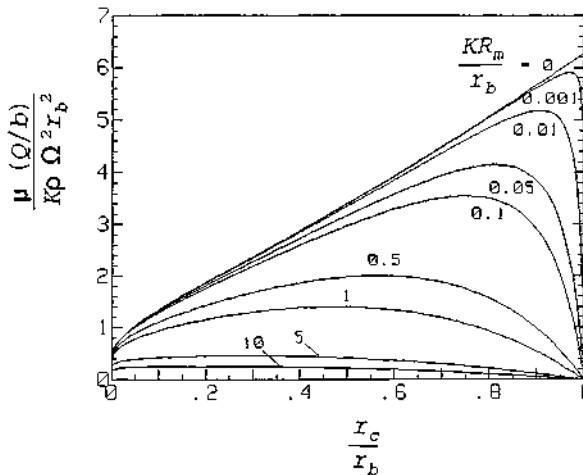


FIG. 18-179 Centrifugal filtration rate as a function of both cake and medium resistance.

maximum. This is because for thin cake the driving liquid head is small and the medium resistance plays a dominating role, resulting in lower flux. For very thick cake, despite the increased driving liquid head, the resistance of the cake becomes dominant; therefore, the flux decreases again. The medium resistance to cake resistance should be small, with $KR_m/r_b < 5$ percent. However, the cake thickness, which is directly proportional to the throughput, should not be too small, despite the fact that the machine may have to operate at somewhat less than the maximum flux condition.

It is known that the specific resistance for centrifuge cake, especially for compressible cake, is greater than that of the pressure or vacuum filter. Therefore, the specific resistance has to be measured from centrifuge tests for different cake thicknesses so as to scale up accurately for centrifuge performance. It cannot be extrapolated from pressure and vacuum filtration data. For cake thickness that is much smaller compared to the basket radius, Eq. (18-119a) can be approximated by

$$V_f = V_{f0} \left(\frac{h}{h_c} \right) \quad (18-119b)$$

where $h = r_b - r_p$ is the liquid depth, $h_c = r_b - r_c$ is the cake thickness, and $v_{f0} = (\rho G K / \mu)$ is a characteristic filtration velocity. Table 18-14 shows some common filtering centrifuges and the application with respect to the G -level, minimum feed-solids concentration, minimum mean particle size, and typical filtration velocity. The vibratory and tumbler centrifuges have the largest filtration rate of 5×10^{-4} m/s (0.02 in/s) for processing 200- μ m or larger particles, whereas the pendulum has the lowest filtration rate of 1×10^{-7} m/s (4×10^{-6} in/s) for processing 5- μ m particles with increased cycle time. The screen-sroll, pusher, screen-bowl, and peeler centrifuges are in between.

Film Drainage and Residual Moisture Content Desaturation of the liquid cake ($S < 1$) begins as the bulk filtration ends, at which point the liquid level starts to recede below the cake surface. Liquids are trapped in: (1) cake pores between particles that can be drained with time (free liquid), (2) particle contact points (pendular liquid), (3) fine pores forming continuous capillaries (capillary rise), (4) particle pores or they are bound by particles (bound liquid). Numbers (1) to (3) can be removed by centrifugation, and, as such, each of these components depends on G to a different extent. Only desaturation of the free liquid, and to a much lesser extent the liquid at contact points, is a function of time. The wet cake starts from a state of being fully saturated, $S = 1$, to a point where $S < 1$, depending on the dewatering time. At a very large amount of time, it approaches an equilibrium point S_∞ , which is a function of G , capillary force, and the amount of bound liquid trapped inside or externally attached to the particles.

The following equations, which have been tested in centrifugal dewatering of granular solids, prove useful:

Total saturation:

$$S_{\text{total}} = S_\infty + S_T(t) \quad (18-120)$$

Equilibrium component:

$$S_\infty = S_c + (1 - S_c) (S_p + S_z) \quad (18-121)$$

Transient component:

$$S_T(t) = (1 - S_c) (1 - S_p - S_z S_f(t)) \quad (18-122)$$

The details of the mathematical model of these four components are given below.

Drainage of free liquid in thin film:

$$S_f(t) = \left(\frac{4}{3} \right) \left(\frac{1}{t_d^n} \right), \quad t_d > 0 \quad (18-123)$$

where for smooth-surface particles, $n = 0.5$, and for particles with rough surfaces, n can be as low as 0.25.

Bound liquid saturation:

$$S_p = \text{function}(\text{particle characteristics}) \quad (18-124)$$

Pendular saturation:

$$S_z = 0.075, \quad N_c \leq 5 \quad (18-125a)$$

$$S_z = \frac{5}{(40 + 6N_c)}, \quad 5 \leq N_c \leq 10 \quad (18-125b)$$

$$S = \frac{0.5}{N_c}, \quad N_c \geq 10 \quad (18-125c)$$

Frequently when $N_c < 10$, S_p and S_z are combined for convenience, the sum of which is typically 0.075 for smooth particles and can be as high as 0.35 for rough-surface particles. This has to be determined from tests.

Saturation due to capillary rise:

$$S_c = \frac{4}{B_o} \quad (18-126)$$

where the dimensionless time t_d , capillary number N_c , and Bond number B_o are, respectively:

$$t_d = \frac{\rho G d_h^2 t}{\mu H} \quad (18-127)$$

$$N_c = \frac{\rho G d_h^2}{\sigma \cos \theta} \quad (18-128)$$

$$B_o = \frac{\rho G H d_h}{\sigma \cos \theta} \quad (18-129)$$

where ρ and μ are, respectively, the density and viscosity of the liquid; θ is the wetting angle of the liquid on the solid particles; σ is the interfacial tension; H is the cake height; d is the mean particle size; and t is the dewatering time. The hydraulic diameter of the particles can be approximated by either $d_h = 0.667 \epsilon d / (1 - \epsilon)$ or $d_h = 7.2(1 - \epsilon)K^{1/2}/\epsilon^{3/2}$, where ϵ is the cake porosity.

Example

Given: $\rho = 1000 \text{ kg/m}^3$, $\rho_s = 1200 \text{ kg/m}^3$, $\mu = 0.004 \text{ N-s/m}^2$, $\sigma \cos \theta = 0.068 \text{ N/m}$, $H = 0.0254 \text{ m}$, $d = 0.0001 \text{ m}$, $\epsilon = 0.4$, $G/g = 2000$, $t = 2 \text{ s}$, $S_p = 0.03$.

Calculate: $d_h = 4.4 \times 10^{-5} \text{ m}$, $t_d = 748$, $N_c = 0.56$, $B_o = 322$, $S_t = 0.048$, $S_z = 0.075$, $S_c = 0.012$, $S_{\infty} = 0.116$, $S_T = 0.043$, $S_{\text{total}} = 0.158$, $W = 0.919$.

Note that W is the solids fraction by weight and is determined indirectly from Eq. (18-89). The moisture weight fraction is 0.081.

The transient component depends not only on G , cake height, and cake properties, but also on dewatering time, which ties to solids throughput for a continuous centrifuge and cycle time for batch centrifuge. If the throughput is too high or the dewatering cycle is too short, the liquid saturation can be high and becomes limiting. Given that time is not the limiting factor, dewatering of the liquid lens at particle contact points requires a much higher G -force. The residual saturation depends on the G -force to the capillary force, as measured by N_c , the maximum of which is about 7.5%, which is quite significant. If the cake is not disturbed (scrolled and tumbled) during conveyance and dewatering, liquid can be further trapped in fine capillaries due to liquid rise, the amount of which is a function of B_o , which weighs the G -force to the capillary force. This amount of liquid saturation is usually smaller as compared to capillary force associated with liquid-lens (also known as pendular) saturation. Lastly, liquid can be trapped by chemical force at the particle surface or physical capillary or interfacial force in the pores within the particles. Because the required desaturating force is extremely high, this portion of moisture cannot be removed by mechanical centrifugation. Fortunately, for most applications it is a small percentage, if it exists.

SELECTION OF CENTRIFUGES

Table 18-15 summarizes the several types of commercial centrifuges, their manner of liquid and solids discharge, their unloading speed, and their relative volumetric capacity. When either the liquid or the solids discharge is not continuous, the operation is said to be cyclic. Cyclic or batch centrifuges are often used in continuous processes by providing appropriate upstream and downstream surge capacity.

Sedimentation Centrifuges These centrifuges frequently are selected on the basis of tests on tubular, disc, or helical-conveyor centrifuges of small size. The centrifuge should be of a configuration similar to that of the commercial centrifuge it is proposed to be used for. The results in terms of capacity for a given performance (effluent clarity and solids concentration) may be scaled up by using the sigma concept of Eqs. (18-117) to (18-119). Spin-tube tests may be used for information on systems containing well-dispersed solids. Such tests are totally unreliable on systems containing a dispersed phase that agglomerates or flocculates during the time of centrifugation.

Filtering Centrifuges These filters often can be selected on the basis of batch tests on a laboratory unit, preferably one at least 12 in (305 mm) in diameter. A bucket centrifuge test would be helpful to study the effect of G , cake height, and dewatering time, but not filtration rates. It is always recommended to follow bucket tests with pilot-scale testing of the actual equipment type being considered. Caution has to be taken in correcting for capillary saturation, which may be absent in large continuous centrifuges with scrolling conveyances.

Unless operating data on similar material are available from other sources, continuous centrifuges should be selected and sized only after tests on a centrifuge of identical configuration.

It seems needless to state but is frequently overlooked that test results are valid only to the extent that the slurry and the test conditions duplicate what will exist in the operating plant. This may involve testing on a small scale (or even on a large one) with a slipstream from an existing unit, but the dependability of the data is often worth the extra effort involved. Most centrifuge manufacturers provide testing services and demonstration facilities in their own plants and maintain a supply of equipment for field-testing in the customer's plant, such as with a pilot centrifuge module with associated peripheral equipment. Larger-scale pilot equipment provides better scale-up accuracy, e.g., in evaluating the effect of cake thickness in batch filtering centrifuges.

COSTS

Neither the investment cost nor the operating cost of a centrifuge can be directly correlated with any single characteristic of a given type of centrifuge. The costs depend on the features of the centrifuge tailored toward the physical and chemical nature of the materials being separated, the degree and difficulty of separation, the flexibility and capability of the centrifuge and its auxiliary equipment, the environment in which the centrifuge is located, and many other nontechnical factors, including market competition. The cost figures presented herewith represent centrifuges only for use in the process industries as of 2004. In any particular installations, the costs may be somewhat less or much greater than those presented here. The prices presented herewith are for rough guidance only. Substantial variations will be found due to volatility of currency exchange and material costs.

The useful parameter for value analysis is the installed cost of the number of centrifuges required to produce the demanded separative effect (end product) at the specified capacity of the plant. The possible benefits of adjustments in the upstream and downstream components of the plant and the process should be carefully examined in order to minimize the total overall plant costs; the systems approach should be used.

Purchase Price Typical purchase prices, including drive motors, of tubular and disc sedimenting centrifuges are given in Table 18-16. The price will vary upward with the use of more exotic materials of construction, the need for explosion-proof electrical gear, the type of enclosure required for vapor containment, and the degree of portability, and this holds for all types of centrifuges.

The average purchase prices of continuous-feed, solid-bowl centrifuges made, respectively, of 316 stainless steel and steel are shown in

TABLE 18-15 Characteristics of Commercial Centrifuges

Method of separation	Rotor type	Centrifuge type	Manner of liquid discharge	Manner of solids discharge or removal	Centrifuge speed for solids discharge	Capacity*
Sedimentation	Batch	Ultracentrifuge	Batch	Batch manual	Zero	1 mL
	Tubular	Laboratory, clinical	Continuous†	Batch manual	Zero	To 6 L
		Supercentrifuge	Continuous†	Batch manual	Zero	To 1,200 gal/h
	Disc	Multipass clarifier	Continuous†	Batch manual	Zero	To 3,000 gal/h
		Solid wall	Continuous†	Batch manual	Zero	To 30,000 gal/h
		Light-phase skimmer	Continuous	Continuous for light-phase solids	Full	To 1,200 gal/h
		Peripheral nozzles	Continuous	Continuous	Full	To 24,000 gal/h
		Peripheral valves	Continuous	Intermittent	Full	To 3,000 gal/h
		Peripheral annulus	Continuous	Intermittent	Full	To 12,000 gal/h
	Solid bowl (scroll conveyor)	Constant-speed horizontal	Continuous†	Cyclic	Full (usually)	To 60 ft ³
		Variable-speed vertical	Continuous†	Cyclic	Zero or reduced	To 16 ft ³
		Continuous decanter	Continuous	Continuous screw conveyor	Full	To 54,000 gal/h
						To 100 tons/h solids
Sedimentation and filtration		Screen-bowl decanter	Continuous	Continuous	Full	To 60,000 gal/h
Filtration	Conical screen	Wide-angle screen	Continuous	Continuous	Full	To 40 tons/h solids
		Differential conveyor	Continuous	Continuous	Full	To 80 tons/h solids
		Vibrating, oscillating, and tumbling screens	Continuous	Essentially continuous	Full	To 250 tons/h solids
	Cylindrical screen	Reciprocating pusher	Continuous	Essentially continuous	Full	Limited data
		Reciprocating pusher, single and multistage	Continuous	Essentially continuous	Full	To 100 tons/h solids
		Horizontal	Cyclic	Intermittent, automatic	Full (usually)	To 25 tons/h solids
		Vertical, underdriven	Cyclic	Intermittent, automatic, or manual	Zero or reduced	To 10 tons/h solids
		Vertical, suspended	Cyclic	Intermittent, automatic, or manual	Zero or reduced	To 10 tons/h solids

*To convert gallons per hour to liters per second, multiply by 0.00105; to convert tons per hour to kilograms per second, multiply by 0.253; and to convert cubic feet to cubic meters, multiply by 0.0283.

†Feed and liquid discharge interrupted while solids are unloaded.

Fig. 18-180. The average purchase prices of continuous-feed filtering centrifuges are shown in Fig. 18-181. This chart includes a comparison of prices on screen-bowl, pusher, screen-scroll, and oscillating conical baskets. On average, the screen bowl is approximately 10 percent higher in price than the solid bowl of the same diameter and length. This incremental cost results from the added complexity of the screen section, bowl configuration, and casing differences. Prices for both the solid-bowl and the filtering centrifuges do not include the drive motor, which typically adds another 5 to 25 percent to the cost. The higher end of this range represents a variable-speed-type drive. If a variable-speed backdrive is used instead of the gear unit, the incremental cost is about another 10 to 15 percent, depending on the capability.

The average prices of the batch centrifuge are shown in Fig. 18-182. All the models include the drive motor and control. In Fig. 18-182, the inverting filter, horizontal peeler, and the advanced vertical peeler are the premium baskets especially used for specialty chemicals and pharmaceuticals. Control versatility with the use of programmable

logic control (PLC), automation, and cake-heel removal are the key features which are responsible for the higher price. The underdriven, top-driven, and pendulum baskets are less expensive with fewer features.

Installation Costs Installation costs of centrifuges vary over an extremely wide range, depending on the type of centrifuge, on the area and kind of structure in which it is installed, and on the details of installation. Some centrifuges, such as portable tubular and disc oil purifiers, are shipped as package units and require no foundation and a minimum of connecting piping and electrical wiring. Others, such as large batch automatic and continuous scroll-type centrifuges, may require substantial foundations and even building reinforcement, extensive interconnecting piping with required flexibility, auxiliary feed and discharge tanks and pumps and other facilities, and elaborate electrical and process-control equipment. Minimum installation costs, covering a simple foundation and minimum piping and wiring, are about 5 to 10 percent of purchase price for tubular and disc centrifuges;

TABLE 18-16 Typical Purchase Prices, Including Drive Motors, of Tubular and Disc Sedimenting Centrifuges, 2004

Type	Bowl diameter, in (mm)	Approximate Σ value, units of 10 ⁴ ft ² (10 ³ m ²)	Designation	Purchase price, 2004 \$
Tubular	4 (102)	2.7 (2.5)	Oil purifier	60,000–80,000
	4 (102)	2.7 (2.5)	Chemical separation	60,000–80,000
	5 (127)	4.2 (3.9)	Blood fractionation	100,000–140,000
Manual discharge disc	13.5 (343)	21 (20)	Hermetic	100,000–130,000
	24 (610)	95 (88)	Centripetal pump	150,000–300,000
Continuous nozzle-discharge disc	12 (305)	12 (11)	Clarifier	100,000–130,000
	18 (457)	25 (23)	Separator	150,000–200,000
	30 (762)	100 (93)	Recycle clarifier	270,000–300,000
Self-cleaning disc	14 (356)	13 (12)	Centripetal pump	130,000–150,000
	18 (457)	22 (20)	Centripetal pump	170,000–200,000
	24 (610)	38 (35)	Centripetal pump	250,000–300,000

*NOTE: All prices quoted are for stainless steel construction with the exception of the oil purifier noted.

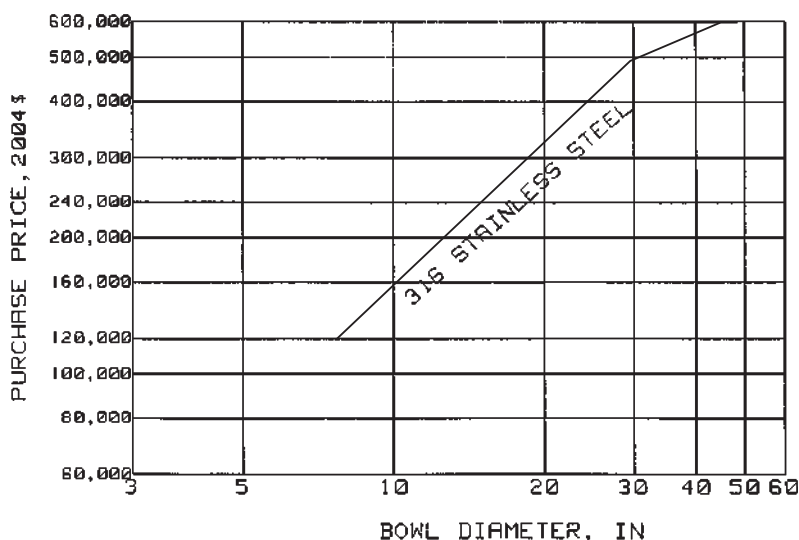


FIG. 18-180 Costs of continuous-feed solid-bowl.

10 to 25 percent for bottom drive, batch automatic, and continuous-scroll centrifuges; and up to 30 percent for top-suspended basket centrifuges. If the cost of all auxiliaries—special foundations, tanks, pumps, conveyors, electrical and control equipment, etc.—is included, the installation cost may well range from 1 to 2 times the purchase price of the centrifuge itself.

Maintenance Costs Because of the care with which centrifuges are designed and built, their maintenance costs are in line with those of other slower-speed separation equipment, averaging in the range of 1 to 4 percent for batch machines, 3 to 8 percent for pusher centrifuges, and 5 to 10 percent for decanters and disc centrifuges per year of the purchase price for centrifuges in light to moderate duty. For centrifuges in severe service and on highly corrosive fluids, the maintenance cost may be several times this value. Maintenance costs

are likely to vary from year to year, with lower costs for general maintenance and periodic large expenses for major overhaul. Centrifuges are subject to erosion from abrasive solids such as sand, minerals, and grits. When these solids are present in the feed, the centrifuge components are subject to wear. Feed and solids discharge ports, unloader knives, helical scroll blade tips, etc., should be protected with replaceable wear-resistant materials. Excessive out-of-balance forces strongly contribute to maintenance requirements and should be avoided.

Operating Labor Centrifuges run the gamut from completely manual control to fully automated operation. For the former, one operator can run several centrifuges, depending on their type and the application. Fully automatic centrifuges usually require little direct operation attention. In most production environments, PLC- or DCS-based automatic controls are the norm.

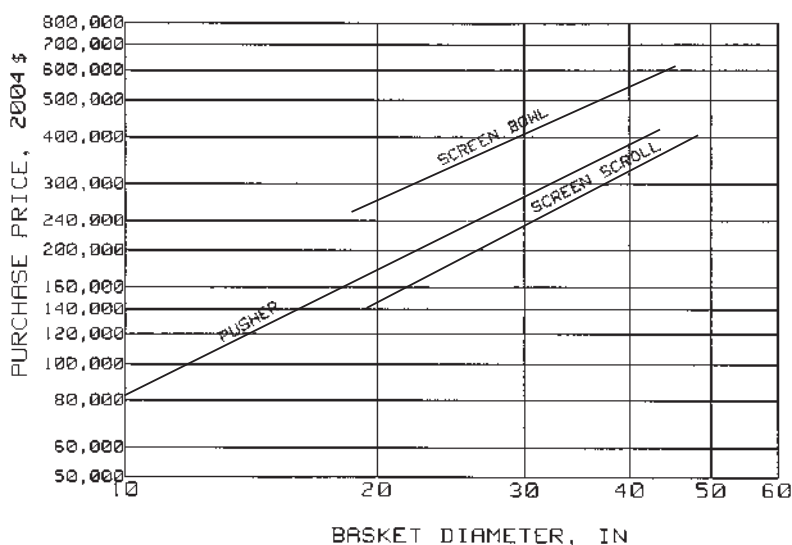


FIG. 18-181 Costs of continuous baskets (316 stainless steel).

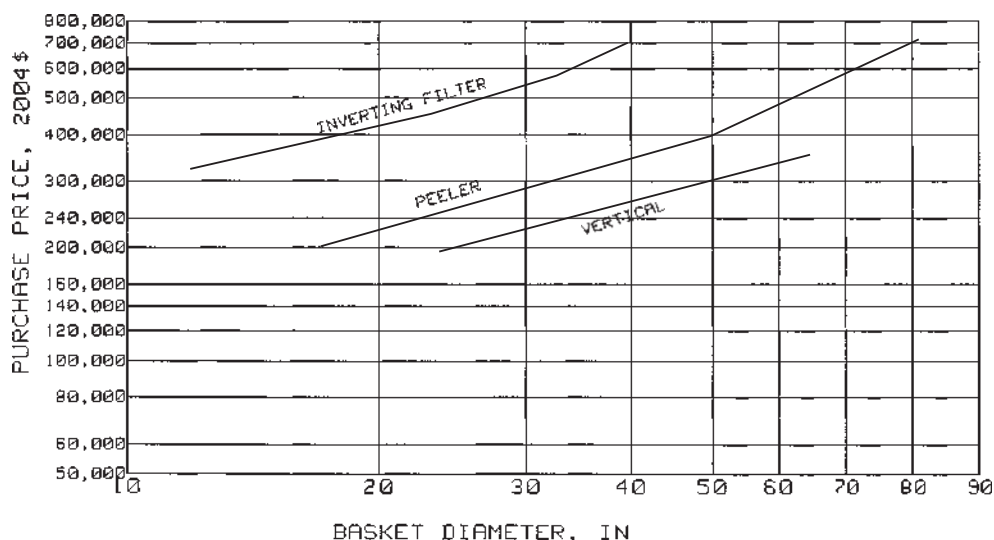


FIG. 18-182 Costs of batch baskets (316 stainless steel).

EXPRESSION

GENERAL REFERENCES: F. M. Tiller and L. L. Horng, "Hydraulic Deliquoring of Compressible Filter Cakes," *AIChE J.*, **29** (2) (1983). F. M. Tiller and C. S. Yeh, "The Role of Porosity in Filtration VI: Filtration Followed by Expression," *AIChE J.*, **33** (1987). F. M. Tiller and W. Li, "Dangers of Lab-Plant Scaleup for Solid/Liquid Separation Systems," *Chem. Eng. Commun.*, **190** (1) (2003). F. M. Tiller and T. C. Green, "Role of Porosity in Filtration IX: Skin Effect with Highly Compressible Materials," *AIChE J.*, **19** (1973). F. M. Tiller and W. Li, "Determination of the Critical Pressure Drop for Filtration of Supercompactible Cakes," *Water Sci. and Technol.*, **44** (10) (2001). M. Shirato, T. Murase, and T. Aragaki, "Slurry Deliquoring by Expression," *Progress in Filtration*, vol. 4, Elsevier, 1986. M. Shirato et al., "Internal Flow Mechanism in Filter Cakes," *AIChE J.*, **15** (1969). M. Shirato et al., "Analysis of Consolidation Process in Filter Cake Dewatering by Use of Difficult to Filter Slurries," *J. Chem. Eng. Japan*, **19** (6) (1986). F. M. Tiller and W. F. Leu, "Basic Data Fitting in Filtration," *J. Chinese Inst. Chem. Engr.*, **11** (1980). W. Chen, F. J. Parma and W. Schabel, "Testing Methods for Belt Press Biosludge Dewatering," *Filtration J.*, **5** (1) (2005).

FUNDAMENTALS OF EXPRESSION

Definition Deliquoring of filter cakes is one of the last stages of solid-liquid separations. It has been widely applied in a variety of fields, e.g., in food industries to increase product yield, in wastewater treatment plants to reduce transportation and disposal cost by decreasing sewage sludge moisture content, and in chemical processes to eliminate liquid content in the solid product prior to drying. The energy required to express liquid from solid-liquid mixtures is negligible compared to that of any thermal method. Deliquoring operations include hydraulic expression, mechanical expression, air or gas blowing, and gravity or centrifugal drainage. Hydraulic expression is provided by direct pump pressure or reversed or right-angled flow of liquid at the end of filtration (Tiller and Horng, 1983). The term *expression* used in this presentation refers to mechanical compression of a solid-liquid mixture by applying diaphragms, rolls, pistons, or screw presses on the surface of cakes.

Filtration and Expression of Compactible Filter Cakes

Filtration A filter cake can be incompactible, moderately compactible, highly compactible, or supercompactible (Tiller and Li, 2003). Tiller and Green (1973) showed that porosity or solidosity (volume fraction of cake solids ϵ_s , solidosity + porosity = 1) is not uniformly distributed in a compactible cake, and a skin cake of high resistance to

liquid flow is developed next to the filter medium, as shown in Fig. 18-183 (Tiller and Li, 2001). The skin deters frictional forces necessary to consolidate the cake and increase solidosity in a large portion of the cake. As a result, as illustrated by Fig. 18-184, increasing filtration pressure on highly compactible filter cakes cannot attain substantial deliquoring (floculated latex) while increasing filtration pressure does help to make a dryer cake on a less compactible material (Kaolin Flat D).

Expression Mechanical expression applies pressure directly on filter cakes rather than relying on flow frictions generated by hydraulic pressure drop to deliquor the cake. The effects of stress distribution in a compactible filter cake by these two different mechanisms are shown in Fig. 18-185. The stress distribution of an expression is more uniform than that of a pressure filtration, leading to a more uniform filter cake. Expression is therefore a better choice for deliquoring of compactible filter cakes.

Fundamental Theory A theoretical model was developed by Shirato (1969, 1986) based on Terzaghi's and Voigt's consolidation model in

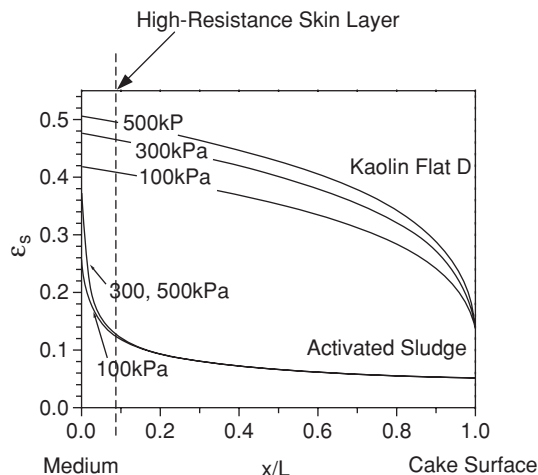


FIG. 18-183 Solidosity ϵ_s variations as a function of fractional distance throughout filter cake thicknesses.

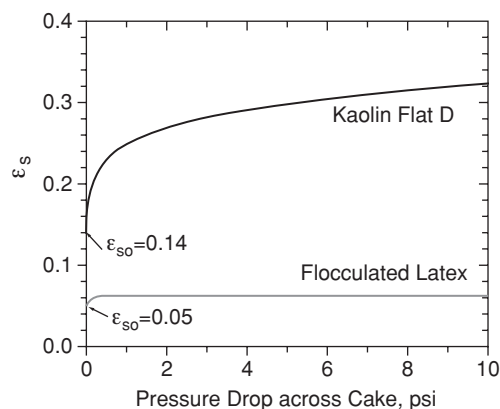


FIG. 18-184 Effect of filtration pressure on average solidosity ϵ_s .

soil mechanics. Shirato's expression theory includes a filtration stage followed by a consolidation. Average consolidation ratio U_c is given as a function of consolidation time θ_c and other characteristic parameters of an expression process including true solids density, liquid density, liquid viscosity, specific resistance (or permeability) versus pressure, porosity versus pressure, and frictional stress on solids throughout cake thickness versus applied pressure (Shirato et al., 1986). The relationships of specific resistance, and porosity versus pressure, and local frictional stress on solids throughout cake thickness during the primary consolidation stage are given by empirical constitutive equations (Tiller and Leu, 1980), and can be determined by a compression-permeability cell test (Tiller, 1977, 1980), as shown in Fig. 18-186.

Factors Affecting Expression Operations Based on fundamental theory, variables affecting expression include characteristics of suspending particles, properties of liquid, properties of filter cake, and expression operation conditions as summarized in Fig. 18-187. Expression efficiency is determined by the properties of the filter cake, which very much depend on characteristics of the suspending particle, properties of liquid, and operation conditions. Interrelationships of the above parameters are described by empirical equations covering restrictive ranges.

EXPRESSION EQUIPMENT

This type of equipment uses mechanical expression rather than pump pressure for cake compression. Dryer cakes and faster cycle rate can

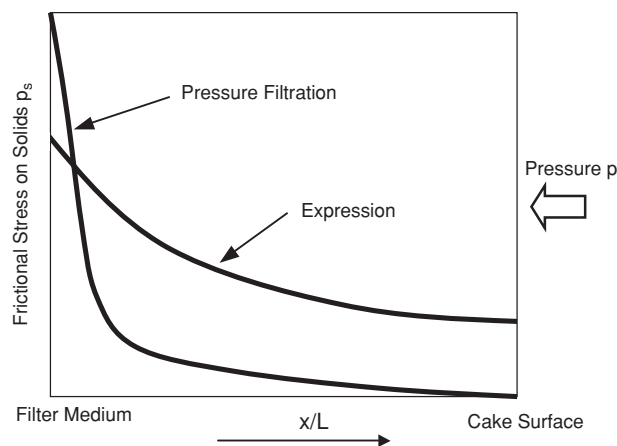


FIG. 18-185 Comparisons of frictional stress distributions in expression and pressure filtration.

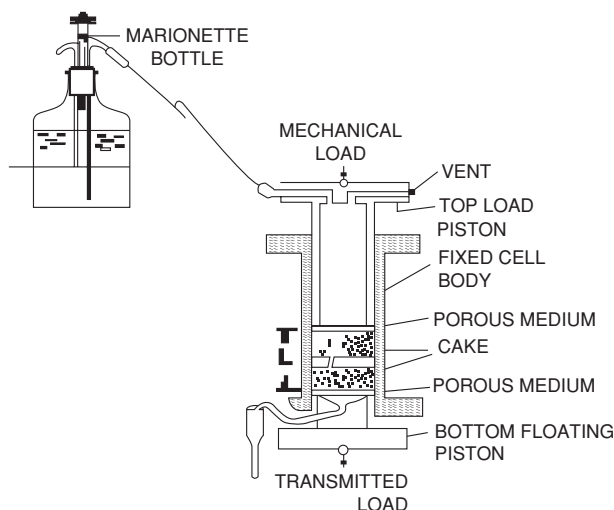


FIG. 18-186 Compression-permeability (C-P) cell.

be achieved compared to pressure filters. Low- to high-pressure (up to 2000 psi) units are available for expression equipment. They can be divided into two categories, batch expression equipment, which allows higher compression pressure and has lower slurry handling capacity, and continuous expression equipment, which uses lower compression pressure but offers higher slurry handling capacities.

Batch Expression Equipment In batch expression equipment, the cake is initially formed by pressure filtration just as in other pressure filters. After the filtration stage, a squeezing device such as a diaphragm is inflated with gas or liquid to compress the cake. Batch expression equipment allows longer compression time and higher compression pressure. The cake can be very dry.

Diaphragm Presses Diaphragm presses, also called membrane presses, are derived from filter presses, which were described in the pressure filtration section. In a diaphragm press, a diaphragm (Fig. 18-188a) is attached to the recessed chamber plate. The operation of a diaphragm press is the same as that of a chamber press during the filtration step. At the end of filtration, the diaphragm is inflated (Fig. 18-188b) to squeeze the filter cake to achieve the mechanical expression. After the squeezing, the diaphragm is deflated and the filter chamber opened to discharge the cake.

The diaphragm can be made of polypropylene or rubber, but polypropylene is most often used today. Both air and water can be used as the inflating medium for the diaphragm. As the inflating medium needs to be brought into the filter plates by hoses, a dangerous condition can exist if a hose is broken with air flowing in it. Therefore, hydraulic fluid (mostly water) is used to inflate the diaphragm to squeeze the cake. Air is only used occasionally in small pilot units.

As in filter presses, one disadvantage of the diaphragm press is the manual operation for filter cake discharge. With recent development, automatic cake discharge devices are available from most filter manufacturers. However, the reliability of an automatic cake discharge device needs to be verified by actual field operation. Normally, automatic cake discharge has a better chance of success in diaphragm presses than filter presses as the cakes are normally dryer in diaphragm presses.

The cake deliquoring is primarily done during the expression step so the cake formation period is normally carried out under low pressure and a high-pressure slurry pump is not necessary; it helps to reduce floc damage during pumping. The normal expression pressure used in a diaphragm press is 110 or 220 psi; in some designs pressure up to 800 psi can be used.

Diaphragm presses are superior to filter presses in deliquoring compactible cakes (such as biological sludge, pulps, or highly flocculated materials). As a diaphragm press is more expensive than a regular filter press, the use of a diaphragm press may not be advantageous

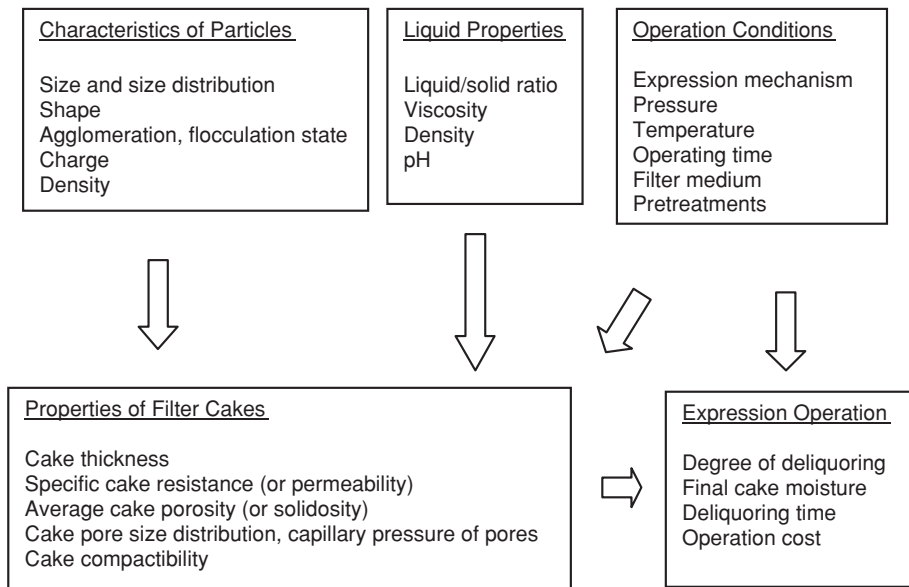


FIG. 18-187 Variables affecting expression.

if solids are not very compactible. There are laboratory and pilot tests available to determine the need for a diaphragm press.

The best way to evaluate diaphragm presses for an application is to run tests with a small pilot unit. Although smaller test units are available, pilot units with 1-ft² filter plate area are more common and are recommended.

A laboratory pressure filter (Fig. 18-189) equipped with a piston can provide a simple feasibility test. In this kind of device, the suspension is poured into the filter cylinder, and the first stage of the test is just like a pressure filtration test. After the filtration, compressed air or water is used to push the piston down to squeeze the filter cake. The filtration rate, final cake thickness and dryness are recorded for evaluation and comparison with the same test without the compression by the piston.

Horizontal Diaphragm Presses This is similar to the diaphragm press except the filter plates lay horizontally (while in diaphragm press, the filter plates are operated vertically). The press can be a single-chamber unit, or multiple chambers can be stacked to achieve greater filtration area.

In each filter plate, the filter medium is attached to a moving belt (Fig. 18-190). An elastomer seal is used at the edge of the filter chamber. The slurry is fed into the filter chamber, and the operation starts as a pressure filtration. After filtration, the diaphragm is inflated to squeeze on the

cake. At the end of expression, the filter chamber opens and the belt moves the cake out of the filter chamber for discharging. The filter chamber is then closed and ready for the next filtration cycle. Permanent filter belt or disposable medium can be used as filter media. The disposable media are especially useful when handling particles which have high tendency to foul the filter media. With the moving belt, the press operation is fully automatic and is another advantage of this equipment.

The testing for evaluating the horizontal diaphragm press is the same as that described above for the (vertical) diaphragm presses. To ensure automatic operation, the cake solids should not stick to the seal of the filter chamber and need to be carefully evaluated during testing.

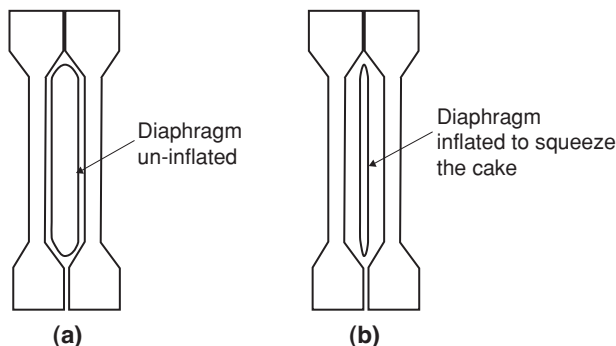


FIG. 18-188 Diaphragm press plate.

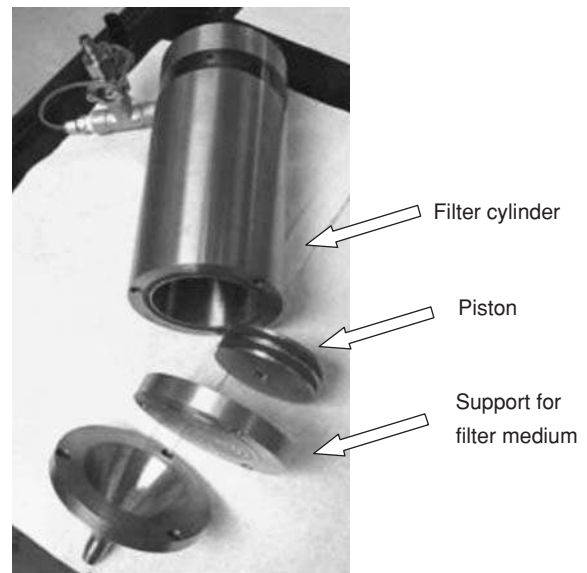


FIG. 18-189 Laboratory pressure filter with a piston to compress the cake.

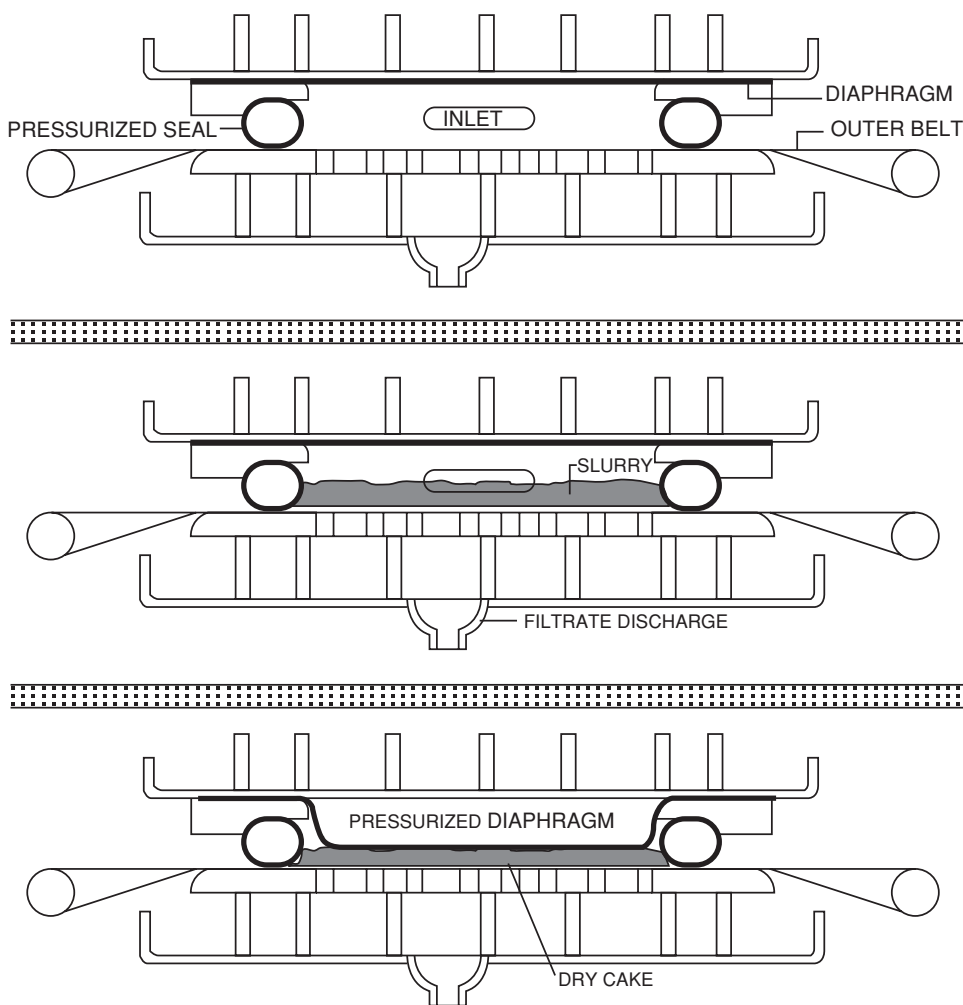


FIG. 18-190 A horizontal diaphragm press. (Courtesy of Filtra Systems.)

Tower Presses This press is similar to the stacked horizontal diaphragm presses, but only one filter belt is used (Fig. 18-191). The operation is also fully automatic. The primary applications are in chemical, mineral and pharmaceutical industries. The testing method is the same as the diaphragm presses. One important factor in designing a tower press is the solids need to be able to be cleared from the chamber seal, otherwise leakage will occur in the following filtration cycle.

Tubular Presses As the name implies, this press is composed of a candle filter inside a cylindrical hydraulic casing (Fig. 18-192). The filter cloth is wrapped around the filter candle, and a diaphragm is attached to the inner side of the outer casing. During the filtration step, the space in between two cylinders is filled with slurry, and pressure filtration is conducted. At the end of the filtration step, the diaphragm is inflated to squeeze the cake around the filter candle. At the end of expression, the bottom of the hydraulic casing tube is opened and the filter assembly is lowered. Air is then introduced to pulse the cake off the candle. After the cake is discharged, the inner filter candle moves back, and the bottom is closed for the next filtration cycle.

Tubular presses use the highest pressures among all expression equipment. The pressure can be as high as 1500 psi. With the high pressure, the cake can be very dry (> 95 percent dryness). This type of equipment normally has low capacity so multiple units are used.

Typical applications of tubular presses are for fine particle dewatering including minerals, talc, and CaCO_3 .

The same laboratory testing equipment as in the diaphragm press can be used but with a higher pressure. A commercially available piston press can also be used.

Continuous Expression Equipment Continuous expression equipment has the advantage of large capacity and automatic operation. Compared to batch expression equipment, lower pressure is used to squeeze the cake in the continuous expression equipment. As a result, the cakes are not as dry as those from the batch expression devices.

Belt Filter Presses Belt presses (Fig. 18-193) have two filter belts that move around rollers of different sizes to dewater the slurry. A typical belt press may have one or more of the following stages: a preconditioning zone, a gravity drainage zone, a linear compression zone (low pressure), and a roller compression zone (high pressure).

The conditioned slurry is fed into the belt press at the preconditioning zone (a tank or pipe), where coagulant and flocculant are added to condition the slurry. The slurry then goes to a horizontal section where the slurry is thickened by gravity drainage. At the end of the gravity drainage section, the thickened slurry (or dilute cake) drops into a wedge section where the wet cake starts to be squeezed by both belts under pressure. At the end of the wedge section, both

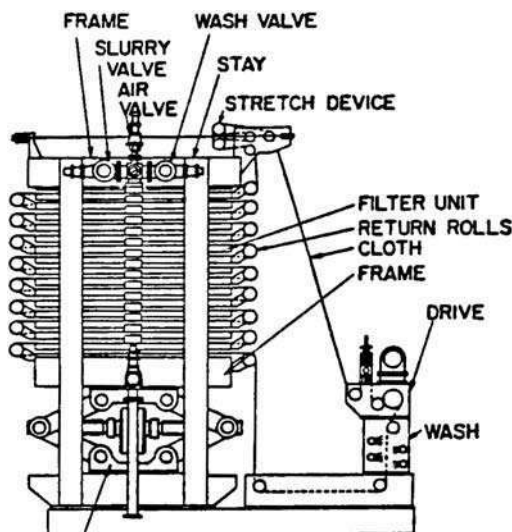


FIG. 18-191 Tower press. (Courtesy of Larox.)

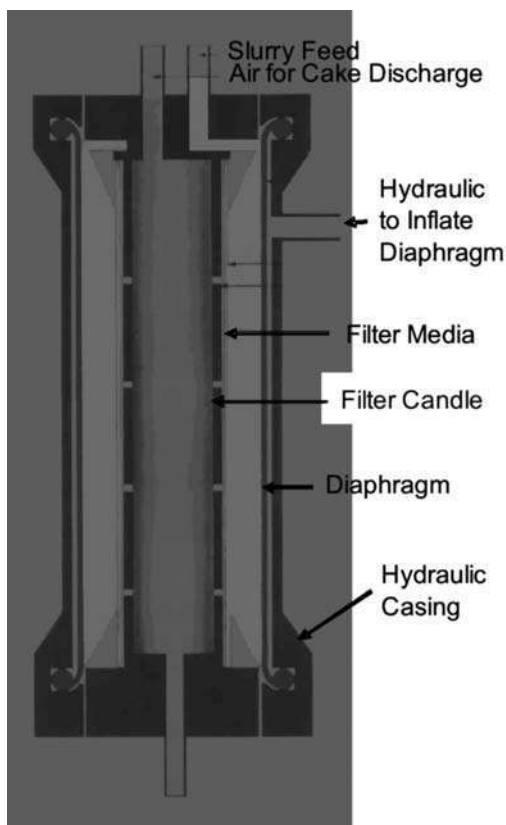


FIG. 18-192 Tubular press. (Courtesy of Metso Minerals.)

belts come together with the cake sandwiched in between and move through a series of rollers. The final dewatering is accomplished by moving the cake through these rollers in the order of decreasing roller diameters. While the roller diameter gets smaller, the pressure exerted on the cake gets higher. After the final roller, the two belts are separated to release the cake. Each belt goes through some washing nozzles to clean off any remaining solids on the belt.

It is important to condition the slurry by coagulation and/or flocculation before it is fed into the belt press. An insufficiently flocculated slurry will not dewater properly, and the cake might be squeezed out through the belts or from the side (both sides of a belt press are open). Good conditioned flocs look like cottage cheese, and it is a good field indication for troubleshooting. Most of the challenges in operating a belt press are in the slurry conditioning and the optimization of flocculant dosage. Flocculant consumption can contribute to a significant operation cost if proper control strategy is not used.

The pressure applied on the cake in a belt press operation is low compared to that in other compression filters. The applied pressures are commonly expressed in pli (pound per linear inch) which is not straightforward in translating to a commonly recognized pressure unit. As a rough comparison, the pressures used in belt presses are around 10 to 20 psi. This pressure can be controlled by the belt and roller tension but seldom is adjusted by operators in the field.

Belt presses have the advantage of large capacity and automatic operation. The initial capital cost is also low. They were originally developed in the pulp and paper industry. Any slurry with fibers will do well in a belt press, and high-fiber material can be added to the slurry as a filter aid for belt press operation. Today, in addition to pulp dewatering, the belt press is widely used in wastewater sludge dewatering.

Due to the relatively low pressure used, the final cakes are not very dry. The dryness of biological sludge cakes from a belt press ranges from 10 to 20 wt %. As fiber content goes up, the cake can be as dry as 40 wt % in dryness.

Testing for applications in belt presses is most commonly done by flocculation in beakers and visual observation of the size and strength of the formed flocs. The conditioned slurry can be poured into a filter for a gravity drainage test. These tests can be useful for an experienced person to evaluate if a slurry can be used in belt presses and to optimize an existing belt press. However, the simulation of the final cake dryness is not

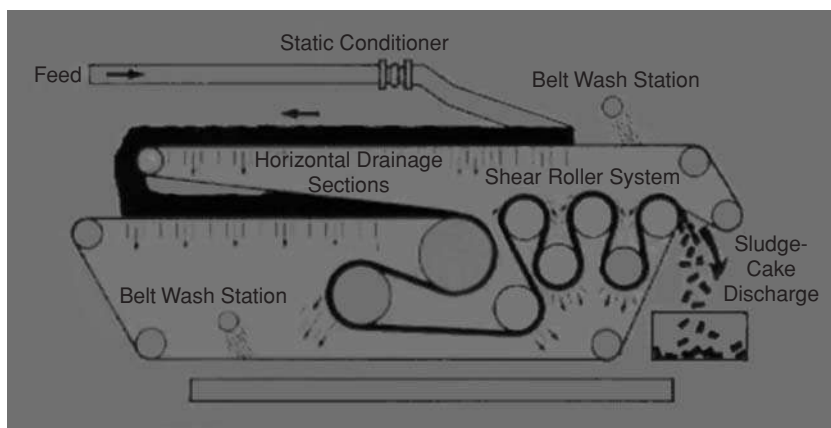


FIG. 18-193 A belt filter press. (Courtesy of Ashbrook.)

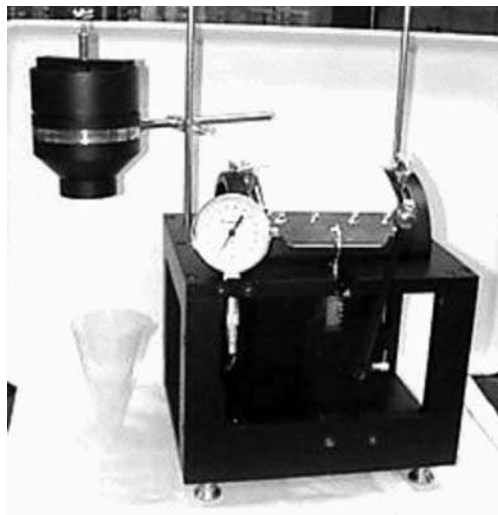


FIG. 18-194 The crown press.

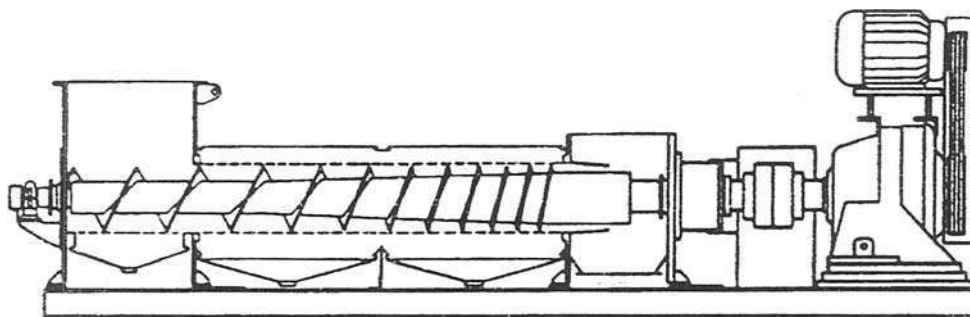


FIG. 18-195 A screw press.

possible with the above methods. The most effective testing is done with a commercially available apparatus called the crown press (Fig. 18-194). This device can simulate the roller actions on the actual belt press and can provide very accurate cake dryness predictions.

Screw Presses A typical screw press is shown in Fig. 18-195, where the slurry is fed into the feed tank at the left-hand side. The core of a screw press is a screw conveyor turning inside a perforated or slotted cylinder. The screw has a smaller diameter at the feed end, and the diameter gradually increases and the screw pitch is shortened toward the discharge end. This design allows gradually decreasing space for slurry/cake and also increasing squeezing pressure on the

cake. As the cake moves toward the outlet, the water is squeezed out through the perforated cylinder.

Screw presses also have the advantage of continuous and automatic operation. Screw presses are primarily used in the pulp and paper, citrus, and dairy industries. Applications also exist in many other industries such as dewatering of synthetic rubbers and wastewater sludge. Three pressure (high, medium, and low) ranges are used. High-pressure screw presses are used for vegetable and animal oil; the capacities are relatively smaller. Medium-pressure units are used to dewater deformable particles (such as plastic pellet and synthetic rubber) and paper pulp. Wastewater sludge applications normally use low-pressure options.

SELECTION OF A SOLIDS-LIQUID SEPARATOR

A good solids-liquid separator performs well in service, both initially and over time. It operates reliably day after day, with enough flexibility to accommodate to normal fluctuations in process conditions, and does not require frequent maintenance and repair. Selection of such a separator begins with a preliminary listing of a number of possible devices, which may solve the problem at hand, and usually ends with the purchase and installation of one or more commercially available machines of a specific type, size, and material of construction. Rarely is it worthwhile to develop a new kind of separator to fill a particular need.

In selecting a solids-liquid separator, it is important to keep in mind the capabilities and limitations of commercially available devices. Among the multiplicity of types on the market, many are designed for fairly specific applications, and unthinking attempts to apply them to other situations are likely to meet with failure. The danger is the more insidious because failure often is not of the clean no-go type; rather it is likely to be in the character of underproduction, subspecification product, or excessively costly operation—the kinds of limping failure that may be slowly detected and difficult to analyze for cause. In addition, it should be recognized that the performance of mechanical separators—more, perhaps, than most chemical-processing equipment—strongly depends on preceding steps in the process. A relatively minor upstream process change, one that might be inadvertent, can alter the optimal separator choice.

PRELIMINARY DEFINITION AND SELECTION

The steps in solving a solids-liquid separation problem, in general, are:

1. Define the overall problem, with expert assistance if necessary.
2. Establish process conditions.
3. Identify appropriate separator types; make preliminary selections.
4. Develop a test program.
5. Take representative samples.
6. Make simple tests.
7. Modify process conditions if necessary.
8. Consult equipment manufacturers.
9. Make final selection; obtain quotations.

Problem Definition Intelligent selection of a separator requires a careful and complete statement of the nature of the separation problem. Focusing narrowly on the specific problem, however, is not sufficient, especially if the separation is to be one of the steps in a new process. Instead, the problem must be defined as broadly as possible, beginning with the chemical reactor or other source of material to be separated and ending with the separated materials in their desired final form. In this way the influence of preceding and subsequent process steps on the separation step will be illuminated. Sometimes, of course, the new separator is proposed to replace an existing unit; the new separator must then fit into the current process and accept feed materials of more or less fixed characteristics. At other times the separator is only one item in a train of new equipment, all parts of which must work in harmony if the separator is to be effective.

Assistance in problem definition and in developing a test program should be sought from persons experienced in the field. If your organization has a consultant in separations of this kind, by all means make

use of the expertise available. If not, it may be wise to employ an outside consultant, whose special knowledge and guidance can save time, money, and headaches. It is important to do this early; after the separation equipment has been installed, there is little a consultant can do to remedy the sometimes disastrous effects of a poor selection. Often it is best to work with established equipment manufacturers throughout the selection process, unless the problem is unusually sensitive or confidential. Their experience with problems similar to yours may be most helpful and avoid many false starts.

Preliminary Selections Assembling background information permits tentative selection of promising equipment and rules out clearly unsuitable types. If the material to be processed is a slurry or pumpable suspension of solids in a liquid, several methods of mechanical separation may be suitable, and these are classified into settling and filtration methods as shown in Fig. 18-196. If the material is a wet solid, removal of liquid by various methods of expression should be considered.

Settling does not give a complete separation: one product is a concentrated suspension and the other is a liquid which may contain fine particles of suspended solids. However, settling is often the best way to process very large volumes of a dilute suspension and remove most of the liquid. The concentrated suspension can then be filtered with smaller equipment than would be needed to filter the original dilute suspension, and the cloudy liquid can be clarified if necessary. Settlers can also be used for classifying particles by size or density, which is usually not possible with filtration.

Screens may sometimes be used to separate suspensions of coarse particles, but are not widely applicable. For separating fine solids from liquids, cake filtration or the newer systems of crossflow filtration should be considered. Crossflow filtration includes ultrafiltration, where the solids are macromolecules or very fine solids ($D_p \leq 0.1 \mu\text{m}$), and microfiltration, where the particle size generally ranges from 0.1 to 5 μm . In microfiltration a suspension is passed at high velocity of 1 to 3 m/s (3 to 10 ft/s) and moderate pressure (10 to 30 lb/in² gauge) parallel to a semipermeable membrane in sheet or tubular form. Organic membranes are made of various polymers including cellulose acetate, polysulfone, and polyamide; and they are usually asymmetric, with a thin selective skin supported on a thicker layer that has larger pores. Inorganic membranes of sintered metal or porous alumina are also available in various shapes, with a range of average pore sizes and permeabilities. Most membranes have a wide distribution of pore sizes and do not give complete rejection unless the average pore size is much smaller than the average particle size in the suspension.

In microfiltration, particles too large to enter the pores of the membrane accumulate at the membrane surface as the liquid passes through. They form a layer of increasing thickness that may have appreciable hydraulic resistance and cause a gradual decrease in permeate flow. A decline in liquid flow may also result from small particles becoming embedded in the membrane or plugging some of the pore mouths. The particle layer may reach a steady-state thickness because of shear-induced migration of particles back into the mainstream, or the liquid flux may continue to decline, requiring frequent backwashing or other cleaning procedures. Because of the high velocities the change in solids concentration per pass is small, and the suspension is either recycled to the feed tank or sent through several

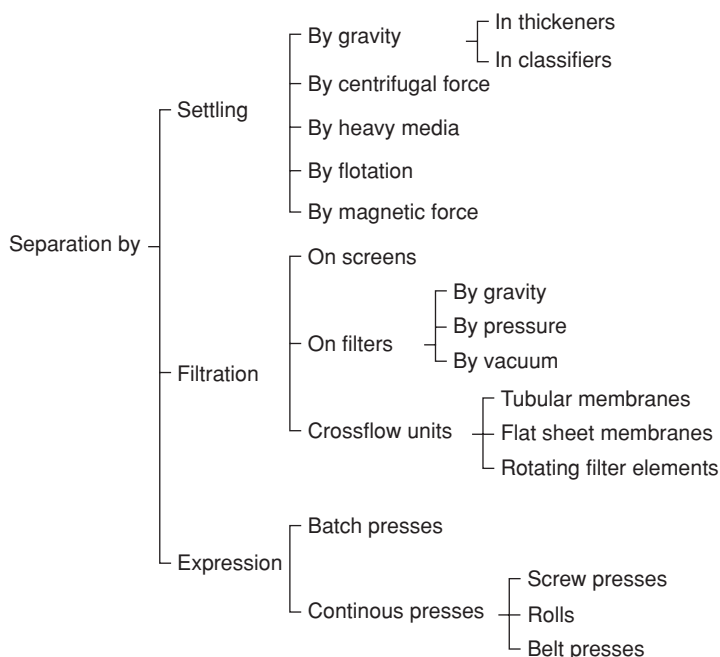


FIG. 18-196 Main paths to solids-liquid separation.

units in series to achieve the desired concentration. The products are a clear liquid and a concentrated suspension similar to those produced in a settling device, but the microfiltration equipment is much smaller for the same production rate.

SAMPLES AND TESTS

Once the initial choice of promising separator types is made, representative liquid-solid samples should be obtained for preliminary tests. At this point, a detailed test program should be developed, preferably with the advice of a specialist.

Establishing Process Conditions Step 2 is taken by defining the problem in detail. Properties of the materials to be separated, the quantities of feed and products required, the range of operating variables, and any restrictions on materials of construction must be accurately fixed, or reasonable assumptions must be made. Accurate data on the concentration of solids, the average particle size or size distribution, the solids and liquid densities, and the suspension viscosity should be obtained *before* selection is made, not after an installed separator fails to perform. The required quantity of the liquid and solid may also influence separator selection. If the solid is the valuable product and crystal size and appearance are important, separators that minimize particle breakage and permit nearly complete removal of fluid may be required. If the liquid is the more valuable product, can minor amounts of solid be tolerated, or must the liquid be sparkling clear? In some cases, partial or incomplete separation is acceptable and can be accomplished simply by settling or by crossflow filtration. Where clarity of the liquid is a key requirement, the liquid may have to be passed through a cartridge-type clarifying filter after most of the solid has been removed by the primary separator.

Table 18-17 lists the pertinent background information that should be assembled. It is typical of data requested by manufacturers when they are asked to recommend and quote on a solid-liquid separator. The more accurately and thoroughly these questions can be answered, the better the final choice is likely to be.

Representative Samples For meaningful results, tests must be run on representative samples. In liquid-solids systems good samples

are hard to get. Frequently a liquid-solid mixture from a chemical process varies significantly from hour to hour, from batch to batch, or from week to week. A well-thought-out sampling program over a prolonged period, with samples spaced randomly and sufficiently far apart, under the most widely varying process conditions possible, should be formulated. Samples should be taken from all shifts in a continuous process and from many successive batches in a batch process. The influence of variations in raw materials on the separating characteristics should be investigated, as should the effect of reactor or crystallizer temperature, intensity of agitation, or other process variables.

Once samples are taken, they must be preserved unchanged until tested. Unfortunately, cooling or heating the samples or the addition of preservatives may markedly change the ease with which solids may be separated from the liquid. Sometimes they make the separation easier, sometimes harder; in either case, tests made on deteriorated samples give a false picture of the capabilities of separation equipment. Even shipping of the samples can have a significant effect. Often it is so difficult to preserve liquid-solids samples without deterioration that accurate results can be obtained only by incorporating a test separation unit directly in the process stream.

Simple Tests It is usually profitable, however, to make simple preliminary tests, recognizing that the results may require confirmation through subsequent large-scale studies.

Preliminary gravity settling tests are made in a large graduated cylinder in which a well-stirred sample of slurry is allowed to settle, the height of the interface between clear supernatant liquid and concentrated slurry being recorded as a function of settling time. Centrifugal settling tests are normally made in a bottle centrifuge in which the slurry sample is spun at various speeds for various periods of time, and the volume and consistency of the settled solids are noted. In gravity settling tests in particular, it is important to evaluate the effects of flocculating agents on settling rates.

Preliminary filtration tests may be made with a Büchner funnel or a small filter leaf, covered with canvas or other appropriate medium and connected to a vacuum system. Usually the suspension is poured carefully into the vacuum-connected funnel, whereas the leaf is immersed

TABLE 18-17 Data for Selecting a Solids-Liquid Separator*

1. Process
 - a. Describe the process briefly. Make up a flowsheet showing places where liquid-solid separators are needed.
 - b. What are the objections to the present process?
 - c. Briefly, what results are expected of the separator?
 - d. Is the process batch or continuous?
 - e. Number the following objectives in order of importance in your problem: (a) separation of two different solids _____; (b) removal of solids to recover valuable liquor as overflow _____; (c) removal of solids to recover the solids as thickened underflow _____ or as "dry" cake _____; (d) washing of solids _____; (e) classification of solids _____; (f) clarification or "polishing" of liquid _____; (g) concentration of solids _____.
 - f. List the available power and current characteristics.
2. Feed
 - a. Quantity of feed:
Continuous process: _____ gal/min; _____ h/day; _____ lb/h of dry solids.
Batch process: volume of batch: _____; total batch cycle: _____ h.
 - b. Feed properties: temp. _____; pH _____; viscosity _____.
 - c. What maximum feed temperature is allowable?
 - d. Chemical analysis and specific gravity of carrying liquid.
 - e. Chemical analysis and specific gravity of solids.
 - f. Percentage of solids in feed slurry.
 - g. Screen analysis of solids: wet _____ dry _____.
 - h. Chemical analysis and concentration of solubles in feed.
 - i. Impurities: form and probable effect on separation.
 - j. Is there a volatile component in the feed? _____. Should the separator be vapor-tight? _____. Must it be under pressure? _____. If so, how much? _____.
3. Filtration and settling rates
 - a. Filtration rate on Büchner funnel: _____ gal/(min)(ft²) of filter area under a vacuum of _____ in Hg. Time required to form a cake _____ in thick: _____ s.
 - b. At what rate do the solids settle by gravity?
 - c. What percentage of the total feed volume do the settled solids occupy after settling is complete? After how long?
4. Feed preparation
 - a. If the feed tends to foam, can antifoaming agents be used? If so, what type?
 - b. Can flocculating agents be used? If so, what agents?
 - c. Can a filter aid be used?
 - d. What are the process steps immediately preceding the separation? Can they be modified to make the separation easier?
 - e. Could another carrying liquid be used?
5. Washing
 - a. Is washing necessary?
 - b. What are the chemical analysis and specific gravity of wash liquid?
 - c. Purpose of wash liquid: to displace residual mother liquor or to dissolve soluble material from the solids?
 - d. Temperature of wash liquid.
 - e. Quantity of wash allowable, in lb/lb of solids.
6. Separated solids
 - a. What percentage of solids is desired in the cake or thickened underflow?
 - b. Is particle breakage important?
 - c. Amount of residual solubles allowable in solids.
 - d. What further processing will have to be carried out on the solids?
7. Separated liquids
 - a. Clarity of liquor: what percentage of solids is permissible?
 - b. Must the filtrate and spent wash liquid be kept separate?
 - c. What further processing will be carried out on the filtrate and/or spent wash?
8. Materials of construction
 - a. What metals look most promising?
 - b. What metals must not be used?
 - c. What gasket and packing materials are suitable?

*U.S. customary engineering units have been retained in this data form. The following SI or modified-SI units might be used instead: centimeters = inches $\times$ 2.54; kilograms per kilogram = pounds per pound $\times$ 1.0; kilograms per hour = pounds per hour $\times$ 0.454; liters per minute = gallons per minute $\times$ 3.785; liters per second-square meter = gallons per minute-square foot $\times$ 0.679; and pascals = inches mercury $\times$ 3377.

in a sample of the slurry and vacuum is applied to pull filtrate into a collecting flask. The time required to form each of several cakes in the range of 3 to 25 mm ($\frac{1}{8}$ to 1 in) thick under a given vacuum is noted, as is the volume of the collected filtrate. Properly conducted tests with a Büchner or a vacuum leaf closely simulate the action of rotary vacuum filters of the top- and bottom-feed variety, respectively, and may give the experienced observer enough information for complete specification of a plant-size filter. Alternatively, they may point to pressure-filter tests or, indeed, to a search for an alternative to filtration. Centrifugal filter tests are made in a perforated basket centrifugal filter 254 or 305 mm (10 or 12 in) in diameter lined with a suitable filter medium. Slurry is poured into the rotating basket until an appropriately thick cake—say, 25 mm (1 in)—is formed. Filtrate is recycled to the basket at such a rate that a thin layer of liquid is just visible on the surface of the cake. The discharge rate of the liquor under these conditions is the draining rate. The test is repeated with cakes of other thicknesses to establish the productive capacity of the centrifugal filter.

Batch tests of microfiltration may be carried out in small pressurized cells with a porous membrane at the bottom and a magnetic stirrer to provide high shear at the membrane surface. These tests may quickly show what type of membrane, if any, gives satisfactory separation, but scaling up to large production units is difficult. Small modules with hollow-fiber, tubular, or spiral-wound membranes are available from equipment vendors, so that tests can be made with continuous flow at pressures and velocities likely to be used for large-scale operation. The permeate flux should be measured as a function of time for different slurry concentrations, pressure drops, and solution velocities or Reynolds numbers. Often a limiting flux will be reached as the pressure drop is increased, but operation at a lower pressure drop is often desirable since the flux decline may not be as great and the average permeation rate over a batch cycle may be greater.

More detailed descriptions of small-scale sedimentation and filtration tests are presented in other parts of this section. Interpretation of the results and their conversion into preliminary estimates of such quantities as thickener size, centrifuge capacity, filter area, sludge density, cake dryness, and wash requirements also are discussed. Both the tests and the data treatment must be in experienced hands if error is to be avoided.

Modification of Process Conditions Relatively small changes in process conditions often markedly affect the performance of specific solids-liquid separators, making possible their application when initial test results indicated otherwise or vice versa. Flocculating agents are an example; many gravity settling operations are economically feasible only when flocculants are added to the process stream. Changes in precipitation or crystallization steps may greatly enhance or diminish filtration rates and hence filter capacity. Changes in the temperature of the process stream, the solute content, or the chemical nature of the suspending liquid also influence solids-settling rates. Occasionally it is desirable to add a heavy, finely divided solid to form a pseudo-liquid suspending medium in which the particles of the desired solid will rise to the surface. Attachment of air bubbles to solid particles in a flotation cell, using a suitable flotation agent, is another way of changing the relative densities of liquid and solid.

Consulting the Manufacturer Early in the selection campaign—certainly no later than the time at which the preliminary tests are completed—manufacturers of the more promising separators should be asked for assistance. Additional tests may be made at a manufacturer's test center; again a major problem is to obtain and preserve representative samples. As much process information as tolerable should be shared with the manufacturers to make full use of their experience with their particular equipment. Full-scale plant tests, although expensive, may well be justified before final selection is made. Such tests demonstrate operation on truly representative feed, show up long-term operating problems, and give valuable operating experience.

In summary, separator selection calls for clear problem definition, in broad terms; thorough cataloging of process information; and preliminary and tentative equipment selection, followed by refinement of the initial selections through tests on an increasingly larger scale. Reliability, flexibility of operation, and ease of maintenance should be weighed heavily in the final economic evaluation; rarely is purchase price, by itself, a governing factor in determining the suitability of a liquid-solids separator.

SECTION 19

Reactors

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



CARMO J. PEREIRA
TIBERIU M. LEIB

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Reactors*

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*The contributions of Stanley M. Walas, Ph.D., Professor Emeritus, Department of Chemical and Petroleum Engineering, University of Kansas (Fellow, American Institute of Chemical Engineers), author of this section in the seventh edition, are acknowledged.

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SOME CASE STUDIES

Nomenclature and Units

In this section, the concentration is represented by C . Mass balance accounting in terms of the number of moles and the fractional conversion is discussed in Sec. 7 and can be very useful. The rate of reaction is r ; the flow rate in moles is N_a ; the volumetric flow rate is V' ; reactor volume is V_r . Several equations are presented without specification of units. Use of any consistent unit set is appropriate.

Following is a listing of typical nomenclature expressed in SI and U.S. Customary System units. Specific definitions and units are stated at the place of application in this section.

Symbol	Definition	SI units	U.S. Customary System Units	Symbol	Definition	SI units	U.S. Customary System Units
a	Surface area per volume	l/m	l/ft	V'	Volumetric flow rate	m ³ /s	ft ³ /s
A_k	Heat-transfer area	m ²	ft ²	V_r	Volume of reactor	m ³	ft ³
C	Concentration of substance	kg·mol/m ³	lb·mol/ft ³	w	Catalyst loading		
C_0	Initial mean concentration	kg·mol/m ³	lb·mol/ft ³	x	Axial position in a reactor, conversion		
c_p	Heat capacity at constant pressure	kJ/(kg·K)	Btu/(lbm·°F)	Greek letters			
CSTR	Ideal continuous stirred tank reactor			α	Fraction of feed that bypasses reactor		
d	Diameter	m	ft	β	Fraction of reactor volume that is stagnant, Prater number		
D	Diameter, diffusivity			$\delta(\tau)$	Unit impulse input, Dirac delta function		
D_{eff}	Effective diffusion coefficient	m ² /s	ft ² /s	δ_L	Distance or film thickness	m	ft
D_e	Effective dispersion coefficient	m ² /s	ft ² /s	ε	Void fraction in a packed bed, particle porosity		
E	Activation energy	kJ/(kg·mol)	Btu/(lb·mol)	η	Effectiveness factor of porous catalyst		
$E(t)$	Residence time distribution			λ	Thermal conductivity	kJ/(s·m·°C)	Btu/(h·ft·°F)
$E(t_r)$	Normalized residence time distribution			$\Lambda(t)$	Intensity function		
f_a	Fraction of A remaining unconverted, C_a/C_{a0} or $n_{d/a0}$			μ	Viscosity	Pa·s	lbm/(ft·s)
$F(t)$	Age function of tracer			ν	Kinematic viscosity, μ/ρ	m ² /s	ft ² /s
h	Heat-transfer coefficient	kJ/(s·m ² ·°C)	Btu/(h·ft ² ·°F)	ρ	Density	kg/m ³	lbm/ft ³
H	Height of tank	m	ft	$\sigma^2(\tau)$	Variance		
H_e	Henry constant	Pa·m ³ /(kg·mol)	atm·ft ³ /(lb·mol)	$\sigma^2(t_r)$	Normalized variance		
ΔH_r	Heat of reaction	kJ/(kg·mol)	Btu/(lb·mol)	ξ	Fractional conversion		
k	Specific rate constant for first-order reaction	l/s	l/s	τ	Tortuosity		
k_m	Mass-transfer coefficient	m/s	ft/s	ϕ	Thiele modulus		
L	Length of path in reactor	m	ft	ϕ_σ	Shape factor		
m	Magnitude of impulse	kg·mol	lb·mol	ϕ_E	Local rate of energy dissipation		
n	Number of stages in a CSTR battery or parameter of Erlang or gamma distribution			Subscripts			
Nu	Nusselt number			a	Agitator, axial, species A		
N	Speed of agitator	rpm	rpm	b	Bed, species B		
Ha	Hatta number			c	Critical value, catalyst, coolant, continuous phase		
p_a	Partial pressure of substance A	Pa	psi	cir	Circulation		
P	Total pressure	Pa	psi	d	Dispersed phase		
Pe	Peclet number for dispersion			f	Fluid, feed		
PFR	Plug flow reactor			G	Gas phase		
q	Heat flux, reaction order, or impeller-induced flow			i	Interface		
Q	Volumetric flow rate	m ³ /s	ft ³ /s	L	Liquid		
r	Rate of reaction per unit volume, radius			ma	Macro		
R	Radius	m	ft	me	Meso		
Re	Reynolds number			mi	Micro		
Sc	Schmidt number			p	Pellet		
Sh	Sherwood number			r	Reaction, reduced		
t	Time	s	s	R	Reactor		
$\bar{t}$	Mean residence time	s	s	s	Surface		
t_r	Reduced time, $t/\bar{t}$			t	Tank		
T	Temperature	°C	°F	u	Step function		
TFR	Tubular flow reactor			w	Wall		
u	Linear velocity	m/s	ft/s	0	Inlet		
$u(t)$	Unit step input			δ	Delta function		
U	Overall heat-transfer coefficient	kJ/(s·m ² ·°C)	Btu/(h·ft ² ·°F)	Superscripts			
v	Volumetric flow rate during semibatch operation	m ³ /s	ft ³ /s	0	Initial condition		
v_{ij}	Stoichiometric coefficients						

GENERAL REFERENCES: The General References listed in Sec. 7 are applicable for Sec. 19. References to specific topics are made throughout this section.

A chemical reactor is a controlled volume in which a chemical reaction can occur in a safe and controllable manner. A reactor typically is a piece of equipment; however, it can also be a product (such as a coating or a protective film). One or more reactants may react together at a desired set of operating conditions, such as temperature and pressure. There may be a need for appropriate mixing, control of flow distribution and residence time, contacting between the reactants (sometimes in the presence of a catalyst or biocatalyst), removal (or addition) of heat, and integration of the reactor with the rest of the downstream process. Depending on the nature of the rate-limiting step(s), a reactor may serve primarily as a holding tank, a heat exchanger, or a mass-transfer device. Chemical reactions generate desired products and also by-products that have to be separated and disposed. A successful commercial unit is an economic balance of all these factors. A variety of reactor types are used in the chemical, petrochemical, and pharmaceutical industries. Some of these reactors are listed in Table 19-1. They include gas, liquid, or multiphase batch reactors, stirred tank reactors, and tubular reactors.

There are a number of textbooks on chemical reaction engineering. Davis and Davis (*Fundamentals of Chemical Reaction Engineering*, McGraw-Hill, 2003) provide a lucid discussion of kinetics and principles. A more comprehensive treatment together with access to

CD-ROM and web resources is in the text by Fogler (*Elements of Chemical Reaction Engineering*, 3d ed., Prentice-Hall, 1999). A chemistry-oriented perspective is provided by Schmidt (*The Engineering of Chemical Reactions*, Oxford University Press, 1999). The book by Froment and Bischoff provides a thorough discussion of reactor analysis and design. A practical manual on reactor design and scale-up is by Harriott (*Chemical Reactor Design*, Marcel Dekker, 2003). Levenspiel (*Chemical Reaction Engineering*, 3d ed., Wiley, 1999) was among the first to present a phenomenological discussion of fundamentals. The mathematical underpinnings of reactor modeling are covered by Bird et al. (*Transport Phenomena*, 2d ed., Wiley, 2002). This section contains a number of illustrations and sketches from books by Walas (*Chemical Process Equipment Selection and Design*, Butterworths, 1990) and Ullmann [*Encyclopedia of Chemical Technology* (in German), vol. 3, Verlag Chemie, 1973, pp. 321–518].

Mathematical models may be used to design reactors and analyze their performance. Detailed models have mainly been developed for large-scale commercial processes. A number of software tools are now available. This chapter will discuss some of the reactors used commercially together with how mathematical models may be used. For additional details, a number of books on reactor analysis cited in this section are available. The discussion will indicate that logical choices aimed at maximizing reaction rate and selectivity for a given set of kinetics can lead to rational reactor selection. While there has been progress in recent years, reactor design and modeling are largely an art.

REACTOR CONCEPTS

Since a primary purpose of a reactor is to provide desirable conditions for reaction, the *reaction rate per unit volume* of reactor is important in analyzing or sizing a reactor. For a given production rate, it determines the reactor volume required to effect the desired transformation. The residence time in a reactor is inversely related to the term *space velocity* (defined as volumetric feed rate/reactor volume). The fraction of reactants converted to products and by-products is the *conversion*. The fraction of desired product in the material converted on a molar basis is referred to as *selectivity*. The product of conversion and the fractional selectivity provides a measure of the fraction of reactants converted to product, known as *yield*. The product yield provides a direct measure of the level of (atom) utilization of the raw materials and may be an important component of operating cost. A measure of reactor utilization called *space time yield (STY)* is the ratio of product generation rate to reactor volume. When a catalyst is used, the reactor has to make product without major process interruptions. The catalyst may be homogeneous or heterogeneous, and the latter can be a living biological cell. A key aspect of catalyst performance is the *durability* of the active site. Since a chemical or biochemical process has a number of unit operations around the reactor, it is often beneficial to minimize the variability of reactant and product flows. This typically means that the reactor is operated at a *steady state*. Interactions between kinetics, fluid flow, transport resistances, and heat effects sometimes result in *multiple steady states* and *transient (dynamic) behavior*. Reactor dynamics can also result in *runaway behavior*, where reactor temperature continues to increase until the reactants are depleted, or *wrong-way behavior*, where reducing inlet temperature (or reactant flow rate) can result in temperature increases farther downstream and a possible runaway. Since such behavior can result in large perturbations in the process and possibly safety issues, a reactor *control strategy* has to be implemented. The need to operate safely under all conditions calls for a thorough analysis to ensure that the reactor is *inherently safe* and that all possible unsafe outcomes have been considered and addressed. Since various solvents may be used in chemical processes and reactors generate both products and by-products, solvent and by-product emissions can cause *emission* and *environmental footprint* issues that must be considered.

Reactor design is often discussed in terms of independent and dependent variables. Independent variables are choices such as reactor type and internals, catalyst type, inlet temperature, pressure, and fresh feed composition. Dependent variables result from independent variable selection. They may be constrained or unconstrained. Con-

strained dependent variables often include pressure drop (limited due to compressor cost), feed composition (dictated by the composition of the recycle streams), temperature rise (or decline), and local and effluent composition. The reactor design problem is often aimed at optimizing independent variables (within constraints) to maximize an objective function (such as conversion and selectivity).

Since the reactor feed may contain inert species (e.g., nitrogen and solvents) and since there may be unconverted feed and by-products in the reactor effluent, a number of unit operations (distillation, filtration, etc.) may be required to produce the desired product(s). In practice, the flow of mass and energy through the process is captured by a process flow sheet. The flow sheet may require recycle (of unconverted feed, solvents, etc.) and purging that may affect reaction chemistry. Reactor design and operation influence the process and vice versa.

REACTOR TYPES

Reactors may be classified according to the mode of operation, the end-use application, the number of phases present, whether (or not) a catalyst is used, and whether some other function (e.g., heat transfer, separations, etc.) is conducted in addition to the reaction.

Classification by Mode of Operation

Batch Reactors A “batch” of reactants is introduced into the reactor operated at the desired conditions until the target conversion is reached. Batch reactors are typically tanks in which stirring of the reactants is achieved using internal impellers, gas bubbles, or a pump-around loop where a fraction of the reactants is removed and externally recirculated back to the reactor. Temperature is regulated via internal cooling surfaces (such as coils or tubes), jackets, reflux condensers, or pump-around loop that passes through an exchanger. Batch processes are suited to small production rates, to long reaction times, to achieve desired selectivity, and for flexibility in campaigning different products.

Continuous Reactors Reactants are added and products removed continuously at a constant mass flow rate. Large daily production rates are mostly conducted in continuous equipment.

A *continuous stirred tank reactor (CSTR)* is a vessel to which reactants are added and products removed while the contents within the vessel are vigorously stirred using internal agitation or by internally (or externally) recycling the contents. CSTRs may be employed in series or in parallel. An approach to employing CSTRs in series is to have a large

TABLE 19-1 Residence Times and/or Space Velocities in Industrial Chemical Reactors*

Product (raw materials)	Type	Reactor phase	Catalyst	T, °C	P, atm	Residence time or space velocity	Source and page†
Acetaldehyde (ethylene, air)	FB	L	Cu and Pd chlorides	50–100	8	6–40 min	[2] 1, [7] 3
Acetic anhydride (acetic acid)	TO	L	Triethylphosphate	700–800	0.3	0.25–5 s	[2]
Acetone (<i>i</i> -propanol)	MT	LG	Ni	300	1	2.5 h	[1] 1 314
Acrolein (formaldehyde, acetaldehyde)	FL	G	MnO ₂ , silica gel	280–320	1	0.6 s	[1] 1 384, [7] 33
Acrylonitrile (air, propylene, ammonia)	FL	G	Bi phosphomolybdate	400	1	4.3 s	[3] 684, [2] 47
Adipic acid (nitration of cyclohexanol)	TO	L	Co naphthenate	125–160	4–20	2 h	[2] 51, [7] 49
Adiponitrile (adipic acid)	FB	G	H ₃ BO ₃	370–410	1	3.5–5 s	[1] 2 152,
Alkylate (<i>i</i> -C ₄ , butenes)	CST	L	H ₃ PO ₄	5–10	2–3	350–500 GHSV	[7] 52
Alkylate (<i>i</i> -C ₄ , butenes)	CST	L	HF	25–38	8–11	5–40 min	[4] 223
Allyl chloride (propylene, Cl ₂)	TO	G	NA	500	3	5–25 min	[4] 223
Ammonia (H ₂ , N ₂)	FB	G	Fe	450	150	0.3–1.5 s	[1] 2 416, [7] 67
Ammonia (H ₂ , N ₂)	FB	G	Fe	450	225	28 s	[6] 61
Ammonia oxidation	Flame	G	Pt gauze	900	8	7,800 GHSV	[6] 115
Aniline (nitrobenzene, H ₂)	B	L	FeCl ₃ in H ₂ O	95–100	1	33 s	[1] 3 289
Aniline (nitrobenzene, H ₂)	FB	G	Cu on silica	250–300	1	0.0026 s	[7] 82
Aspirin (salicylic acid, acetic anhydride)	B	L	None	90	1	0.5–100 s	[7] 89
Benzene (toluene)	TU	G	None	740	38	>1 h	[6] 36,
Benzene (toluene)	TU	G	None	650	35	48 s	[9] 109
Benzoic acid (toluene, air)	SCST	LG	None	125–175	9–13	815 GHSV	[1] 4 183, [7] 98
Butadiene (butane)	FB	G	Cr ₂ O ₃ , Al ₂ O ₃	750	1	128 s	[7] 101
Butadiene (1-butene)	FB	G	None	600	0.25	0.2–2 h	[7] 118
Butadiene sulfone (butadiene, SO ₂)	CST	L	<i>t</i> -Butyl catechol	34	12	0.1–1 s	[3] 572
<i>i</i> -Butane (<i>n</i> -butane)	FB	L	AlCl ₃ on bauxite	40–120	18–36	34,000 GHSV	[1] 5 192
<i>i</i> -Butane (<i>n</i> -butane)	FB	L	Ni	370–500	20–50	0.2 LHSV	[4] 239, [7] 683
Butanols (propylene hydroformylation)	FB	L	PH ₃ -modified Co carbonyls	150–200	1,000	0.5–1 LHSV	[4] 239
Butanols (propylene hydroformylation)	FB	L	Fe pentacarbonyl	110	10	1–6 WHSV	[1] 5 373
Calcium stearate	B	L	None	180	5	100 g L-h	[7] 125
Caprolactam (cyclohexane oxime)	CST	L	Polyphosphoric acid	80–110	1	1 h	[7] 135
Carbon disulfide (methane, sulfur)	Furn.	G	None	500–700	1	1–2 h	[1] 6 73, [7] 139
Carbon monoxide oxidation (shift)	TU	G	Cu-Zn or Fe ₂ O ₃	390–220	26	0.25–2 h	[1] 6 322, [7] 144
Portland cement	Kiln	S		1,400–1,700	1	1.0 s	[6] 44
Chloral (Cl ₂ , acetaldehyde)	CST	LG	None	20–90	1	4.5 s	7,000 GHSV
Chlorobenzenes (benzene, Cl ₂)	SCST	LG	Fe	40	1	10 h	[11]
Coking, delayed (heater)	TU	LG	None	490–500	15–4	140 h	[7] 158
Coking, delayed (drum, 100 ft max height)	B	LG	None	500–440	4	24 h	[1] 8 122
Cracking, fluid catalytic	Riser	G	Zeolite	520–540	2–3	250 s	[1] 10 8
Cracking, hydro (gas oils)	FB	LG	Ni, SiO ₂ , Al ₂ O ₃	350–420	100–150	0.3–0.5 ft/s vapor	(14) 353
Cracking (visbreaking residual oils)	TU	LG	None	470–495	10–30	2–4 s	[11]
Cumene (benzene, propylene)	FB	G	H ₃ PO ₄	260	35	1–2 LHSV	[11]
Cumene hydroperoxide (cumene, air)	CST	L	Metal porphyrins	95–120	2–15	450 s, 8 LHSV	[11]
Cyclohexane (benzene, H ₂)	FB	G	Ni on Al ₂ O ₃	150–250	25–55	23 LHSV	[7] 191
Cyclohexanol (cyclohexane, air)	SCST	LG	None	185–200	48	1–3 h	[7] 201
Cyclohexanone (cyclohexanol)	CST	L	N.A.	107	1	0.75–2 LHSV	[7] 203
Cyclohexanone (cyclohexanol)	MT	G	Cu on pumice	250–350	1	2–10 min	[8] (1963)
Cyclopentadiene (dicyclopentadiene)	TJ	G	None	220–300	1–2	0.75 h	[8] (1963)
DDT (chloral, chlorobenzene)	B	L	Oleum	0–15	1	4–12 s	[7] 212
Dextrose (starch)	CST	L	H ₂ SO ₄	165	1	0.1–0.5 LHSV	[7] 213
Dextrose (starch)	CST	L	Enzyme	60	1	8 h	[7] 233
Dibutylphthalate (phthalic anhydride, butanol)	B	L	H ₂ SO ₄	150–200	1	20 min	[8] (1951)
Diethylketone (ethylene, CO)	TO	L	Co oleate	150–300	200–500	100 min	[7] 217
Dimethylsulfide (methanol, CS ₂)	FB	G	Al ₂ O ₃	375–535	5	1–3 h	[7] 227
Diphenyl (benzene)	MT	G	None	730	2	0.1–10 h	[7] 243
Dodecylbenzene (benzene, propylene tetramer)	CST	L	AlCl ₃	15–20	1	150 GHSV	[7] 266
Ethanol (ethylene, H ₂ O)	FB	G	H ₃ PO ₄	300	82	0.6 s	[7] 275,
Ethyl acetate (ethanol, acetic acid)	TU, CST	L	H ₂ SO ₄	100	1	3.3 LHSV	[8] (1938)
Ethyl chloride (ethylene, HCl)	TO	G	ZnCl ₂	150–250	6–20	1–30 min	[7] 283
Ethylene (ethane)	TU	G	None	860	2	1,800 GHSV	[2] 356, [7] 297
Ethylene (naphtha)	TU	G	None	550–750	2–7	0.5–0.8 LHSV	[10] 45, 52, 58
Ethylene, propylene chlorohydrins (Cl ₂ , H ₂ O)	CST	LG	None	30–40	3–10	2 s	[7] 305
Ethylene glycol (ethylene oxide, H ₂ O)	TO	LG	1% H ₂ SO ₄	50–70	1	1.03 s	[3] 411,
Ethylene glycol (ethylene oxide, H ₂ O)	TO	LG	None	195	13	1,880 GHSV	[6] 13
Ethylene oxide (ethylene, air)	FL	G	Ag	270–290	1	0.5–3 s	[7] 254
Ethyl ether (ethanol)	FB	G	WO ₃	120–375	2–100	0.5–5 min	[7] 310, 580
Fatty alcohols (coconut oil)	B	L	Na, solvent	142	1	30 min	[2] 398
Formaldehyde (methanol, air)	FB	G	Ag gauze	450–600	1	1 h	[2] 398
Glycerol (allyl alcohol, H ₂ O ₂)	CST	L	H ₂ WO ₄	40–60	1	1 s	[2] 409, [7] 322
						2 h	[7] 326
						2 h	[8] (1953)
						0.01 s	[2] 423
						3 h	[7] 347

TABLE 19-1 Residence Times and/or Space Velocities in Industrial Chemical Reactors (Concluded)

Product (raw materials)	Type	Reactor phase	Catalyst	T, °C	P, atm	Residence time or space velocity	Source and page†
Hydrogen (methane, steam)	MT	G	Ni	790	13	5.4 s	[6] 133
Hydrosulfurization of naphtha	TO	LG	Co-MO	315–500	20–70	3,000 GHSV 1.5–8 LHSV 125 WHSV	[4] 285, [6] 179, [9] 201
Hydrogenation of cottonseed oil	SCST	LG	Ni	130	5	6 h	[6] 161
Isoprene (<i>i</i> -butene, formaldehyde)	FB	G	HCl, silica gel	250–350	1	1 h	[7] 389
Maleic anhydride (butenes, air)	FL	G	V ₂ O ₅	300–450	2–10	0.1–5 s	[7] 406
Melamine (urea)	B	L	None	340–400	40–150	5–60 min	[7] 410
Methanol (CO, H ₂)	FB	G	ZnO, Cr ₂ O ₃	350–400	340	5,000 GHSV	[7] 421
Methanol (CO, H ₂)	FB	G	ZnO, Cr ₂ O ₃	350–400	254	28,000 GHSV	[3] 562
<i>o</i> -Methyl benzoic acid (xylene, air)	CST	L	None	160	14	0.32 h	[3] 732
Methyl chloride (methanol, Cl ₂)	FB	G	Al ₂ O ₃ gel	340–350	1	3.1 LHSV	[2] 533
Methyl ethyl ketone (2-butanol)	FB	G	ZnO	425–475	2–4	275 GHSV	[7] 437
Methyl ethyl ketone (2-butanol)	FB	G	Brass spheres	450	5	0.5–10 min 2.1 s 13 LHSV	[10] 284
Nitrobenzene (benzene, HNO ₃)	CST	L	H ₂ SO ₄	45–95	1	3–40 min	[7] 468
Nitromethane (methane, HNO ₃)	TO	G	None	450–700	5–40	0.07–0.35 s	[7] 474
Nylon-6 (caprolactam)	TU	L	Na	260	1	12 h	[7] 480
Phenol (cumene hydroperoxide)	CST	L	SO ₂	45–65	2–3	15 min	[7] 520
Phenol (chlorobenzene, steam)	FB	G	Cu, Ca phosphate	430–450	1–2	2 WHSV	[7] 522
Phosgene (CO, Cl ₂)	MT	G	Activated carbon	50	5–10	16 s	[11]
Phthalic anhydride (<i>o</i> -xylene, air)	MT	G	V ₂ O ₅	350	1	900 GHSV	[3] 482, 539, [7] 529
Phthalic anhydride (naphthalene, air)	FL	G	V ₂ O ₅	350	1	1.5 s	[9] 136, [10] 335
Polycarbonate resin (bisphenol-A, phosgene)	B	L	Benzyltriethylammonium chloride	30–40	1	5 s 0.25–4 h	[7] 452
Polyethylene	TU	L	Organic peroxides	180–200	1,000–1,700	0.5–50 min	[7] 547
Polyethylene	TU	L	Cr ₂ O ₃ , Al ₂ O ₃ , SiO ₂	70–200	20–50	0.1–1,000 s	[7] 549
Polypropylene	TO	L	R ₂ AlCl, TiCl ₄	15–65	10–20	15–100 min	[7] 559
Polyvinyl chloride	B	L	Organic peroxides	60	10	5.3–10 h	[6] 139
<i>i</i> -Propanol (propylene, H ₂ O)	TO	L	H ₂ SO ₄	70–110	2–14	0.5–4 h	[7] 393
Propionitrile (propylene, NH ₃)	TU	G	CoO	350–425	70–200	0.3–2 LHSV	[7] 578
Reforming of naphtha (H ₂ /hydrocarbon = 6)	FB	G	Pt	490	30–35	3 LHSV 8,000 GHSV	[6] 99
Starch (corn, H ₂ O)	B	L	SO ₂	25–60	1	18–72 h	[7] 607
Styrene (ethylbenzene)	MT	G	Metal oxides	600–650	1	0.2 s 7,500 GHSV	[5] 424
Sulfur dioxide oxidation	FB	G	V ₂ O ₅	475	1	2.4 s 700 GHSV	[6] 86
<i>t</i> -Butyl methacrylate (methacrylic acid, <i>i</i> -butene)	CST	L	H ₂ SO ₄	25	3	0.3 LHSV	[1] 5 328
Thiophene (butane, S)	TU	G	None	600–700	1	0.01–1 s	[7] 652
Toluene diisocyanate (toluene diamine, phosgene)	B	LG	None	200–210	1	7 h	[7] 657
Toluene diamine (dinitrotoluene, H ₂)	B	LG	Pd	80	6	10 h	[7] 656
Tricresyl phosphate (cresyl, POCl ₃)	TO	L	MgCl ₂	150–300	1	0.5–2.5 h	[2] 850, [7] 673
Vinyl chloride (ethylene, Cl ₂)	FL	G	None	450–550	2–10	0.5–5 s	[7] 699
Aldehydes (diisobutene, CO)	CST	LG	Co Carbonyl	150	200	1.7 h	[12] 173
Allyl alcohol (propylene oxide)	FB	G	Li phosphate	250	1	1.0 LHSV	[15] 23
Automobile exhaust	FB	G	Pt-Pd: 1–2 g/unit	400–600+	1		
Gasoline (methanol)	FB	G	Zeolite	400	20	2 WHSV	[13] 3 383
Hydrogen cyanide (NH ₃ , CH ₄)	FB	G	Pt-Rh	1150	1	0.005 s	[15] 211
Isoprene, polymer	B	L	Al(<i>i</i> -Bu) ₃ /TiCl ₄	20–50	1–5	1.5–4 h	[15] 82
NO _x pollutant (with NH ₃)	FB	G	V ₂ O ₅ -TiO ₂	300–400	1–10		[14] 332
Automobile emission control	M	G	Pt/Rh/Pd/Al ₂ O ₃	350–500	1	20,000 GHSV	[16] 69
Nitrogen oxide emission control	M	G	V ₂ O ₅ -WO ₃ /TiO ₂	300–400	1	4–10,000 GHSV	[16] 306
Carbon monoxide and hydrocarbon emission control	M	G	Pt-Pd/Al ₂ O ₃	500–600	1	80–120,000 GHSV	[16] 334
Ozone control from aircraft cabins	M	G	Pd/Al ₂ O ₃	130–170	1	~10 ⁶ GHSV	[16] 263
Vinyl acetate (ethylene + CO)	MT	LG	Cu-Pd	130	30	1 h L, 10 s G	[12] 140

* Abbreviations: reactors: batch (B), continuous stirred tank (CST), fixed bed of catalyst (FB), fluidized bed of catalyst (FL), furnace (Furn.), monolith (M), multi-tubular (MT), semicontinuous stirred tank (SCST), tower (TO), tubular (TU). Phases: liquid (L), gas (G), both (LG). Space velocities (hourly): gas (GHSV), liquid (LHSV), weight (WHSV). Not available, NA. To convert atm to kPa, multiply by 101.3.

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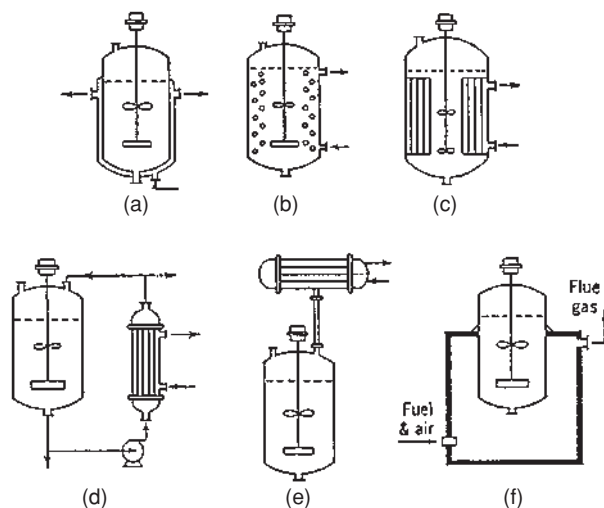


FIG. 19-1 Stirred tank reactors with heat transfer. (a) Jacket. (b) Internal coils. (c) Internal tubes. (d) External heat exchanger. (e) External reflux condenser. (f) Fired heater. (Walas, *Reaction Kinetics for Chemical Engineers*, McGraw-Hill, 1959.)

cylindrical tank with partitions: feed enters the first compartment and over (or under) flows to the next compartment, and so on. The composition is maintained as uniform as possible in each individual compartment; however, a stepped concentration gradient exists from one CSTR to the next. When the reactants have limited solubility (miscibility) and a density difference, the vertical staged reactor with counter-current operation may be used. Alternatively, each CSTR in a series or parallel configuration can be an independent vessel. Examples of stirred tank reactors with heat transfer are shown in Fig. 19-1.

A *tubular flow reactor* (TFR) is a tube (or pipe) through which reactants flow and are converted to product. The TFR may have a varying diameter along the flow path. In such a reactor, there is a continuous gradient (in contrast to the stepped gradient characteristic of a CSTR-in-series battery) of concentration in the direction of flow. Several tubular reactors in series or in parallel may also be used. Both horizontal and vertical orientations are common. When heat transfer is needed, individual tubes are jacketed or a shell-and-tube construction is used. The reaction side may be filled with solid catalyst or internals such as static mixers (to improve interphase contact in heterogeneous reactions or to improve heat transfer by turbulence). Tubes that have 3- to 4-in diameter and are several miles long may be used in polymerization service. Large-diameter vessels, with packing (or trays) used to regulate the residence time in the reactor, may also be used. Some of the configurations in use are axial flow, radial flow, multishell with built-in heat exchangers, and so on.

A reaction battery of CSTRs in series, although both mechanically and operationally more complex and expensive than a tubular reactor, provides flexibility. Relatively slow reactions are best conducted in a stirred tank reactor battery. A tubular reactor is used when heat transfer is needed, where high pressures and/or high (or low) temperatures occur, and when relatively short reaction times suffice.

Semibatch Reactors Some of the reactants are loaded into the reactor, and the rest of the reactants are fed gradually. Alternatively, one reactant is loaded into the reactor, and the other reactant is fed continuously. Once the reactor is full, it may be operated in a batch mode to complete the reaction. Semibatch reactors are especially favored when there are large heat effects and heat-transfer capability is limited. Exothermic reactions may be slowed down and endothermic reactions controlled by limiting reactant concentration. In bioreactors, the reactant concentration may be limited to minimize toxicity. Other situations that may call for semibatch reactors include control of undesirable by-products or when one of the reactants is a gas of limited solubility that is fed continuously at the dissolution rate.

Classification by End Use Chemical reactors are typically used for the synthesis of chemical intermediates for a variety of specialty (e.g., agricultural, pharmaceutical) or commodity (e.g., raw materials for polymers) applications. *Polymerization reactors* convert raw materials to polymers having a specific molecular weight and functionality. The difference between polymerization and chemical reactors is artificially based on the size of the molecule produced. *Bioreactors* utilize (often genetically manipulated) organisms to catalyze biotransformations either aerobically (in the presence of air) or anaerobically (without air present). *Electrochemical reactors* use electricity to drive desired reactions. Examples include synthesis of Na metal from NaCl and Al from bauxite ore. A variety of reactor types are employed for specialty materials synthesis applications (e.g., electronic, defense, and other).

Classification by Phase Despite the generic classification by operating mode, reactors are designed to accommodate the reactant phases and provide optimal conditions for reaction. Reactants may be fluid(s) or solid(s), and as such, several reactor types have been developed. *Single-phase* reactors are typically gas- (or plasma-) or liquid-phase reactors. *Two-phase* reactors may be gas-liquid, liquid-liquid, gas-solid, or liquid-solid reactors. *Multiphase* reactors typically have more than two phases present. The most common type of multiphase reactor is a gas-liquid-solid reactor; however, liquid-liquid-solid reactors are also used. The classification by phases will be used to develop the contents of this section.

In addition, a reactor may perform a function other than reaction alone. *Multifunctional* reactors may provide both reaction and mass transfer (e.g., reactive distillation, reactive crystallization, reactive membranes, etc.), or reaction and heat transfer. This coupling of functions within the reactor inevitably leads to additional operating constraints on one or the other function. Multifunctional reactors are often discussed in the context of process intensification. The primary driver for multifunctional reactors is functional synergy and equipment cost savings.

REACTOR MODELING

As discussed in Sec. 7, chemical kinetics may be mathematically described by rate equations. Reactor performance is also amenable to quantitative analysis. The quantitative analysis of reaction systems is dealt with in the field of chemical reaction engineering.

The level of mathematical detail that can be included in the analysis depends on the level of understanding of the physical and chemical processes that occur in a reactor. As a practical matter, engineering data needed to build a detailed model for some new chemistry typically are unavailable early in the design phase. Reactor designers may use similarity principles (e.g., dimensionless groups), rules of thumb, trend analysis, design of experiments (DOE), and principal-component analysis (PCA) to scale up laboratory reactors. For hazardous systems in which compositional measurements are difficult, surrogate indicators such as pressure or temperature may be used. As more knowledge becomes available, however, a greater level of detail may be included in a mathematical model. A detailed reactor model may contain information on vessel configuration, stoichiometric relationships, kinetic rate equations, correlations for thermodynamic and transport properties, contacting efficiency, residence time distribution, and so on.

Models may be used for analyzing data, estimating performance, reactor scale-up, simulating start-up and shutdown behavior, and control. The level of detail in a model depends on the need, and this is often a balance between value and cost. Very elaborate models are justifiable and have been developed for certain widely practiced and large-scale processes, or for processes where operating conditions are especially critical.

Modeling Considerations A useful reactor model allows the user to predict performance or to explore uncertainties not easily or cost-effectively investigated through experimentation. Uncertainties that may be explored through modeling may include scale-up options, explosion hazards, runaway reactions, environmental emissions, reactor internals design, and so on. As such, the model must contain an optimal level of detail (principle of *optimal sloppiness*) required to meet the desired objective(s). For example, if mixing is critical to performance, the model must include flow equations that reflect the role of mixing. If heat effects are small, an isothermal model may be used.

19-8 REACTORS

A key aspect of modeling is to derive the appropriate momentum, mass, or energy conservation equations for the reactor. These balances may be used in lumped systems or derived over a differential volume within the reactor and then integrated over the reactor volume. Mass conservation equations have the following general form:

$$\left[\begin{array}{c} \text{Amount of A} \\ \text{introduced} \\ \text{per unit time} \end{array} \right] - \left[\begin{array}{c} \text{Amount of A} \\ \text{leaving per unit} \\ \text{time} \end{array} \right] - \left[\begin{array}{c} \text{Amount of A} \\ \text{converted per} \\ \text{unit time} \end{array} \right] = \left[\begin{array}{c} \text{Amount of A} \\ \text{accumulated} \\ \text{per unit time} \end{array} \right] \quad (19-1)$$

The general form for the energy balance equation is

$$\left[\begin{array}{c} \text{Amount of} \\ \text{energy added} \\ \text{per unit time} \end{array} \right] - \left[\begin{array}{c} \text{Amount of} \\ \text{energy removed} \\ \text{per unit time} \end{array} \right] - \left[\begin{array}{c} \text{Energy} \\ \text{generated per} \\ \text{unit time} \end{array} \right] = \left[\begin{array}{c} \text{Accumulation} \\ \text{of energy} \\ \text{per unit time} \end{array} \right] \quad (19-2)$$

The model defines each of these terms. Solving the set of equations provides outputs that can be validated against experimental observations and then used for predictive purposes. Mathematical models for ideal reactors that are generally useful in estimating reactor performance will be presented. Additional information on these reactors is available also in Sec. 7.

Batch Reactor Since there is no addition or removal of reactants, the mass and energy conservation equations for a batch reactor with a constant reactor volume are

$$V_r r(C, T) + V_r \frac{dC}{dt} = 0 \quad (19-3)$$

$$-qA_k - V_r (-\Delta H_r) r(C, T) + V_r \rho c_p \frac{dT}{dt} = 0 \quad (19-4)$$

where qA_k is the addition (or removal) of heat from the reactor. Mean values of physical properties are used in Eqs. (19-3) and (19-4). For an isothermal first-order reaction $r(C, T) = kC$, the mass and energy equations can be combined and the solution is

$$C = C_0 e^{-kt} \quad (19-5)$$

Typically batch reactors may have complex kinetics, mixing, and heat-transfer issues. In such cases, detailed momentum, mass, and energy balance equations will be required.

Semibatch Reactor Feed is added for a fixed time, and the reaction proceeds as the feed is added. The reactor equations governing the feed addition portion of the process are

$$V_r r(C, T) + \frac{d(V_r C)}{dt} = V_r r(C, T) + C \frac{dV_r}{dt} + V_r \frac{dC}{dt} = 0 \quad (19-6)$$

$$-qA_k - V_r (-\Delta H_r) r(C, T) + \rho c_p \frac{d(V_r T)}{dt} = 0 \quad (19-7)$$

For a constant reactant flow rate v_0 ,

$$\frac{dV_r}{dt} = v_0 \quad (19-8)$$

Given an initial condition $V_r = V_r^0$ at $t = 0$,

$$V_r(t) = V_r^0 + v_0 t \quad (19-9)$$

For an isothermal first-order reaction, substitution of this relationship in Eq. (19-6) yields

$$C = \frac{C_0 v_0}{k(V_r^0 + v_0 t)} (1 - e^{-kt}) \quad (19-10)$$

After feed addition is completed, the reactor may be operated in a batch mode. In this case, Eqs. (19-3) and (19-4) may be used with the

concentration at the end of feed addition serving as the initial concentration for the batch reactor.

Ideal Continuous Stirred Tank Reactor In an ideal CSTR, reactants are fed into and removed from an ideally mixed tank. As a result, the concentration within the tank is uniform and identical to the concentration of the effluent. The mass and energy conservation equations for an ideal constant-volume or constant-density CSTR with constant volumetric feed rate V' may be written as

$$V' C_0 = V' C + V_r r(C, T) + V_r \frac{dC}{dt} \quad (19-11)$$

$$V' \rho c_p T_0 = -Q(T) + V' \rho c_p T - V_r (-\Delta H_r) r(C, T) + V_r \rho c_p \frac{dT}{dt} \quad (19-12)$$

where $Q(T)$ represents any addition or removal of heat from the reactor and mean values of physical properties are used. For example, if heat is transferred through the reactor wall, $Q(T) = A_k U (T_c - T)$, where A_k is the heat-transfer area, U is the overall heat-transfer coefficient, and T_c is the temperature of the heat-transfer fluid.

The above *ordinary differential equations* (ODEs), Eqs. (19-11) and (19-12), can be solved with an initial condition. For an isothermal first-order reaction and an initial condition, $C(0) = 0$, the linear ODE may be solved analytically. At steady state, the accumulation term is zero, and the solution for the effluent concentration becomes

$$\frac{C}{C_0} = \frac{1}{1 + kV_r/V'} = \frac{1}{1 + kt} \quad (19-13)$$

Since the contents of an ideal CSTR are perfectly mixed, the dispersion within the reactor is infinite. In practice, CSTRs may not be ideally mixed. In such cases, the reactor may be modeled as having a fraction of the feed α in bypass and a fraction β of the reactor volume stagnant. The material balance is

$$C = \alpha C_0 + (1 - \alpha) C_1 \quad (19-14)$$

$$(1 - \alpha) V' C_0 = (1 - \alpha) V' C_1 + (1 - \beta) k V_r C_1 \quad (19-15)$$

where C_1 is the concentration leaving the active zone of the tank. Elimination of C_1 will relate the input and overall output concentrations. For a first-order reaction,

$$\frac{C_0}{C} = 1 + \frac{kV_r(1 - \beta)}{V'(1 - \alpha)} \quad (19-16)$$

The two parameters α and β may be expected to depend on reactor internals and the amount of agitation.

Plug Flow Reactor A plug flow reactor (PFR) is an idealized tubular reactor in which each reactant molecule enters and travels through the reactor as a "plug," i.e., each molecule enters the reactor at the same velocity and has exactly the same residence time. As a result, the concentration of every molecule at a given distance downstream of the inlet is the same. The mass and energy balance for a differential volume between position V_r and $V_r + dV_r$ from the inlet may be written as *partial differential equations* (PDEs) for a constant-density system:

$$V' \frac{\partial C}{\partial V_r} + r(C, T) + \frac{\partial C}{\partial t} = 0 \quad (19-17)$$

$$\frac{-Q(T)}{V_r} + V' \rho c_p \frac{\partial T}{\partial V_r} - (-\Delta H_r) r(C, T) + c_p \rho \frac{\partial T}{\partial t} = 0 \quad (19-18)$$

where $Q(T)$ represents any addition of heat to (or removal from) the reactor wall and mean values of physical properties are used. The above PDEs can be solved with an initial condition, e.g., $C(x, 0) = C_{t=0}(x)$, and a boundary condition, e.g., $C(0, t) = C_0(t)$, which is the concentration at the inlet. At steady state, the accumulation term above is zero, and the solution for an isothermal first-order reaction is the same as that for a batch reactor, Eq. (19-5):

$$C = C_0 \exp\left(-k \frac{V_r}{V'}\right) = C_0 e^{-kt} \quad (19-19)$$

A tubular reactor will likely deviate from plug flow in most practical cases, e.g., due to backmixing in the direction of flow, reactor internals, etc. A way of simulating axial backmixing is to represent the reactor volume as a series of n stirred tanks in series. The steady-state solution for a single ideal CSTR may be extended to find the effluent concentration after two ideal CSTRs and then to n ideal stages as

$$\frac{C_n}{C_0} = \frac{1}{(1 + kV_r/V')^n} \quad (19-20)$$

In this case, V_r is the volume of each individual reactor in the battery. In modeling a reactor, n is empirically determined based on the extent of reactor backmixing obtained from tracer studies or other experimental data. In general, the number of stages n required to approach an ideal PFR depends on the rate of reaction (e.g., the magnitude of the specific rate constant k for the first-order reaction above). As a practical matter, the conversion for a series of stirred tanks approaches a PFR for $n > 6$.

An alternate way of generating backmixing is to recycle a fraction of the product from a PFR back to the inlet. This reactor, known as a recycle reactor, has been described in Sec. 7 of the Handbook. As the recycle ratio (i.e., recycle flow to product flow) is increased, the effective dispersion is increased and the recycle reactor approaches an ideal CSTR.

Tubular Reactor with Dispersion An alternative approach to describe deviation from ideal plug flow due to backmixing is to include a term that allows for axial dispersion D_e in the plug flow reactor equations. The reactor mass balance equation now becomes

$$V' \frac{\partial C}{\partial V_r} - D_e \frac{\partial^2 C}{\partial V_r^2} + r(C, T) + \frac{\partial C}{\partial t} = 0 \quad (19-21)$$

The model is referred to as a *dispersion model*, and the value of the dispersion coefficient D_e is determined empirically based on correlations or experimental data. In a case where Eq. (19-21) is converted to dimensionless variables, the coefficient of the second derivative is referred to as the Peclet number ($Pe = uL/D_e$), where L is the reactor length and u is the linear velocity. For plug flow, $D_e = 0$ ($Pe \Rightarrow \infty$) while for a CSTR, $D_e = \infty$ ($Pe = 0$). To solve Eq. (19-21), one initial condition and two boundary conditions are needed. The “closed-ends” boundary conditions are $uC_0 = (uC - D_e \partial C/\partial L)_{L=0}$ and $(\partial C/\partial L)_{L=L} = 0$ (e.g., see Wen and Fan, *Models for Flow Systems in Chemical Reactors*, Marcel Dekker, 1975). Figure 19-2 shows the performance of a tubular reactor with dispersion compared to that of a plug flow reactor.

Ideal chemical reactors typically may be modeled using a combination of ideal CSTR, PFR, and dispersion model equations. In the case of a single phase, the approach is relatively straightforward. In the case of two-phase flow, a bubble column (fluidized-bed) reactor may be modeled as containing an ideal CSTR liquid (emulsion) phase and a plug flow (with dispersion) gas phase containing bubbles. Given inlet gas conditions, the concentration in the liquid (emulsion) may be calculated using mass-transfer correlations from the bubbles to the liquid (emulsion) along with reaction in the liquid (emulsion) phase along the length of the reactor. In flooded gas-liquid reactors where the gas and liquid are countercurrent to each other, a plug flow (with dispersion) model may be used for both phases. The concentration of reactant in a phase at each end of the reactor is known. The concentration of the other phase is assumed at one end, and mass-transfer correlations and reaction kinetics are used together with a plug flow (with dispersion) model to get to the other exit. The iterative process continues until the concentrations at each end match the feed conditions.

Reactor Selection Ideal CSTR and PFR models are extreme cases of complete axial dispersion ($D_e = \infty$) and no axial dispersion ($D_e = 0$), respectively. As discussed earlier, staged ideal CSTRs may be used to represent intermediate axial dispersion. Alternatively, within the context of a PFR, the dispersion (or a PFR with recycle) model may be used to represent increased dispersion. Real reactors inevitably have a level of dispersion in between that for a PFR or an ideal CSTR. The level of dispersion may depend on fluid properties (e.g., is the fluid newtonian),

fluid flow (e.g., the level of mixing), transport properties (e.g., the diffusivity of reactants in the fluid), and reactor geometry. The effect of dispersion in a real reactor is discussed within the context of an ideal CSTR and PFR model in Fig. 19-2.

Figure 19-2a shows the effect of dispersion on the reactor volume required to achieve a certain exit concentration (or conversion). As Pe number increases (i.e., dispersion decreases), the reactor begins to approach plug flow and the reactor volume required to achieve a certain conversion approaches the volume for a PFR. At lower Pe numbers, reactor performance approaches that of an ideal CSTR and the reactor volume required to achieve a certain concentration is much higher than that of a PFR. This behavior can be observed in Fig. 19-2b that shows the effect of exit concentration on reaction rate. At a given rate, an ideal CSTR has the highest exit concentration (lowest conversion) and a PFR has the lowest exit concentration (highest conversion). As Fig. 19-2c shows, since the concentration in an ideal CSTR is the same as the exit concentration, there is a sharp drop in concentration from the inlet to the bulk concentration. In contrast, the concentration in the reactor drops continuously from the inlet to the outlet for a PFR. At intermediate values of Pe , the “closed-ends” boundary condition in the dispersion model causes a drop in concentration to levels lower than for an ideal CSTR.

As discussed in Fig. 19-2, for a given conversion, the reactor residence time (or reactor volume required) for a positive order reaction with dispersion will be greater than that of a PFR. This need for a longer residence time is illustrated for a first-order isothermal reaction in a PFR versus an ideal CSTR using Eqs. (19-13) and (19-19).

$$\frac{t_{\text{ideal CSTR}}}{t_{\text{PFR}}} = \frac{C/C_0 - 1}{(C/C_0) \ln(C/C_0)} \quad (19-22)$$

Equation (19-22) indicates that, for a nominal 90 percent conversion, an ideal CSTR will need nearly 4 times the residence time (or volume) of a PFR. This result is also worth bearing in mind when batch reactor experiments are converted to a battery of ideal CSTRs in series in the field. The performance of a completely mixed batch reactor and a steady-state PFR having the same residence time is the same [Eqs. (19-5) and (19-19)]. At a given residence time, if a batch reactor provides a nominal 90 percent conversion for a first-order reaction, a single ideal CSTR will only provide a conversion of 70 percent. The above discussion addresses conversion. Product selectivity in complex reaction networks may be profoundly affected by dispersion. This aspect has been addressed from the standpoint of parallel and consecutive reaction networks in Sec. 7.

Reactors may contain one or more fluid phases. The level of dispersion in each phase may be represented mathematically by using some of the above thinking.

In industrial practice, the laboratory equipment used in chemical synthesis can influence reaction selection. As issues relating to kinetics, mass transfer, heat transfer, and thermodynamics are addressed, reactor design evolves to commercially viable equipment. Often, more than one type of reactor may be suitable for a given reaction. For example, in the partial oxidation of butane to maleic anhydride over a vanadium pyrophosphate catalyst, heat-transfer considerations dictate reactor selection and choices may include fluidized beds or multi-tubular reactors. Both types of reactors have been commercialized. Often, experience with a particular type of reactor within the organization can play an important part in selection.

There are several books on reactor analysis and modeling including those by Froment and Bischoff (*Chemical Reactor Analysis and Design*, Wiley, 1990), Fogler (*Elements of Chemical Reaction Engineering*, Prentice-Hall International Series, 2005), Levenspiel (*Chemical Reaction Engineering*, Wiley, 1999), and Walas (*Modeling with Differential Equations in Chemical Engineering*, Butterworth-Heinemann, 1991).

Chemical Kinetics Reactor models include chemical kinetics in the mass and energy conservation equations. The two basic laws of kinetics are the *law of mass action* for the rate of a reaction and the *Arrhenius equation* for its dependence on temperature. Both of these strictly apply to elementary reactions. More often, laboratory data are

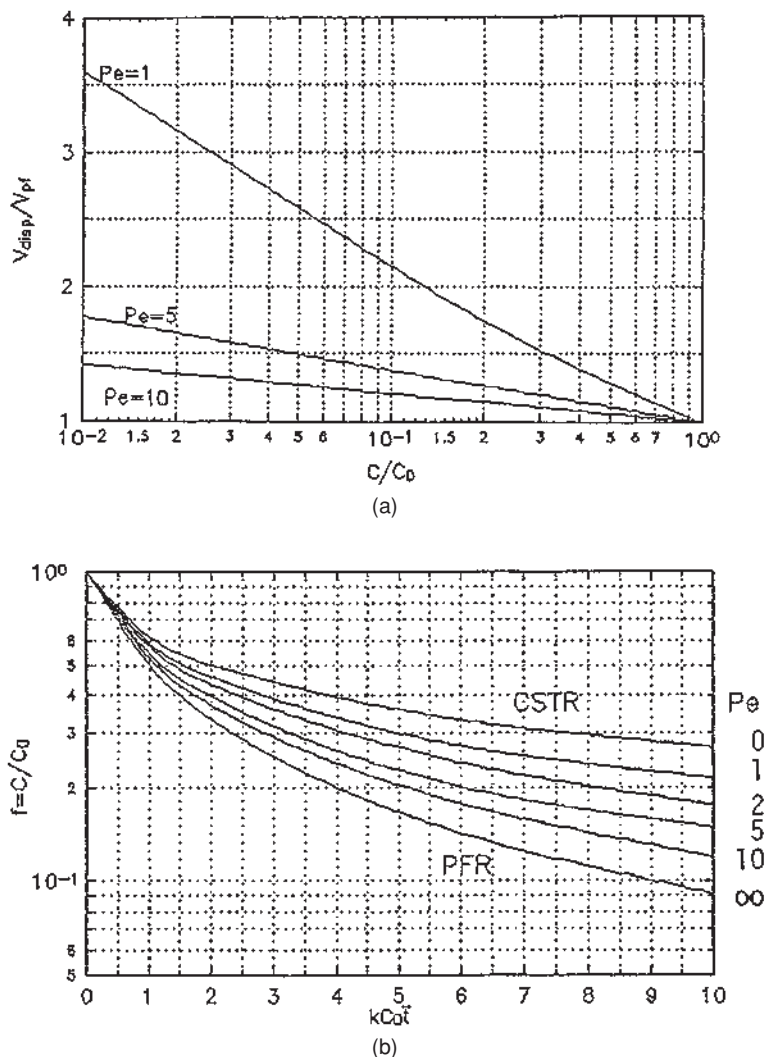


FIG. 19-2 Chemical conversion by the dispersion model. (a) Volume relative to plug flow against residual concentration ratio for a first-order reaction. (b) Residual concentration ratio against kC_0t for a second-order reaction. (c) Concentration profile at the inlet of a closed-end vessel with dispersion for a second-order reaction with $kC_0t = 5$.

used to develop mathematical relationships that describe reaction rates that are then used. These relationships require analysis of the laboratory reactor data, as discussed in Sec. 7. Reactor models will require that kinetic rate information be expressed on a unit reactor volume basis. Two-phase or multiphase reactors will require a level of detail (e.g., heat and mass transport between phases) to capture the relevant physical and chemical processes that affect rate.

Pressure Drop, Mass and Heat Transfer Pressure drop is more important in reactor design than in analysis or simulation. The size of the compressor is dictated by pressure drop across the reactor, especially in the case of gas recycle. Compressor costs can be significant and can influence the aspect ratio of a packed or trickle bed reactor. Pressure drop correlations often may depend on the geometry, the scale, and the fluids used in data generation. Prior to using literature correlations, it often is advisable to validate the correlation with measurements on a similar system at a relevant scale.

Depending on the type of reactor, appropriate mass-transfer correlations may have to be used to connect intrinsic chemical kinetics to

the reaction rate per unit reactor volume. A number of these correlations have already been discussed in Sec. 5 of the Handbook, "Heat and Mass Transfer." The determination of intrinsic kinetics has already been discussed in Sec. 7 of the Handbook. In the absence of a correlation validated for a specific use, the analogy between momentum, heat and mass transfer may often be invoked.

The local reactor temperature affects the rates of reaction, equilibrium conversion, and catalyst deactivation. As such, the local temperature has to be controlled to maximize reaction rate and to minimize deactivation. In the case of an exothermic (endothermic) reaction, higher (lower) local temperatures can cause suboptimal local concentrations. Heat will have to be removed (added) to maintain more uniform temperature conditions. The mode of heat removal (addition) will depend on the application and on the required heat-transfer rate.

Examples of stirred tank reactors with heat transfer are shown in Fig. 19-1. If the heat of reaction is not significant, an adiabatic reactor may be used. For modest heat addition (removal), a jacketed stirred tank is adequate (Fig. 19-1a). As the heat exchange requirements

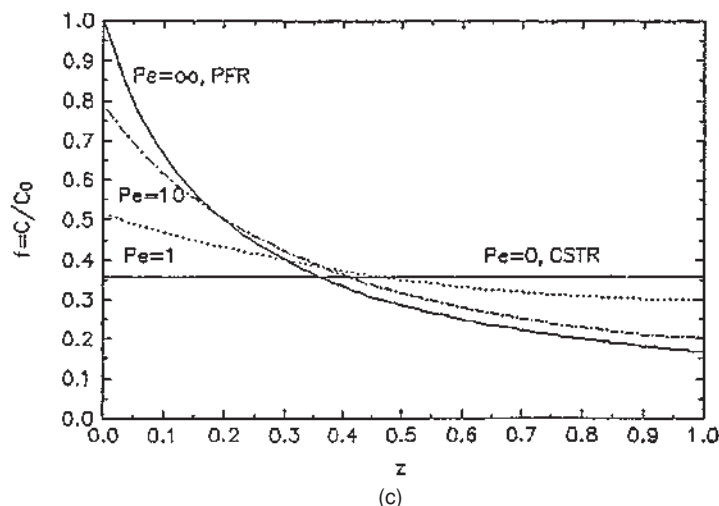


FIG. 19-2 (Continued)

increase, internal coils or internal tubes that contain a heat-transfer fluid may be required (Fig. 19-1b and c). In special cases, where the peak temperature has to be tightly controlled (e.g., in bioreactors) or where fouling may be an issue, the liquid may be withdrawn, circulated through an external heat exchanger, and returned to the reactor (Fig. 19-1d). In some cases, the vapor above the liquid may be passed through an external reflux condenser and returned to the reactor (Fig. 19-1e). In highly endothermic reactors, the entire reactor may be placed inside a fired heater (Fig. 19-1f), or the reactor shell may be heated to high temperatures by using induction heat.

Several of the heat-transfer options for packed beds are illustrated in Fig. 19-3. Again, if heat requirements are modest, an adiabatic reactor is adequate (Fig. 19-3a). If pressure drop through the reactor is an issue, a radial flow reactor may be used (Fig. 19-3b). There are few examples of radial flow reactors in industry. Potential problems include gas distribution in the case of catalyst attrition or settling. A common way of dealing with more exothermic (endothermic) reactions is to split the reactor into several beds and then provide interbed heat exchange (Fig. 19-3c). For highly exothermic (endothermic) reactors, a shell-and-tube multitubular reactor concept may be utilized (Fig. 19-3d). The reactor now begins to look more like a heat exchanger. If multiple beds are needed, rather than using interbed heat exchangers, cold feed may be injected (also called cold shot) in between beds (Fig. 19-3e). In some cases, the heat exchanger may be outside the reactor (Fig. 19-3f). The concept of a reactor as a heat exchanger may be extended to an autothermal multitubular reactor in which, for example, the reactants are preheated on the shell side with reaction occurring in the tubes (Fig. 19-3g). Such reactors can have control issues and are not widely used. A common approach is to have multiple adiabatic reactors with cooling in between reactors (Fig. 19-3h). If the reaction is endothermic, heat may be added by passing the effluents from each reactor through tubes placed inside a common process heater (as is the case for a petroleum reforming reactor shown in Fig. 19-3i). For highly endothermic reactions, a fuel-air mixture or raw combustion gases may be introduced into the reactor. In an extreme situation, the entire reactor may be housed within a furnace (as in the case of steam reforming for hydrogen synthesis or ethane cracking for ethylene production).

At times, the reaction may be exothermic with conversion being limited by thermodynamic equilibrium. In such cases, packed beds in series with interstage cooling may be used as well. The performance enhancement associated with this approach is shown for two cases in Table 19-2. Such units can take advantage of initial high rates at high temperatures and higher equilibrium conversions at lower tempera-

tures. For SO_2 oxidation, the conversion attained in the fourth bed is 97.5 percent, compared with an adiabatic single-bed value of 74.8 percent. With the three-bed ammonia reactor, final ammonia concentration is 18.0 percent, compared with the one-stage adiabatic value of 15.4 percent.

Since reactors come in a variety of configurations, use a variety of operating modes, and may handle mixed phases, design provisions for temperature control may draw on a large body of heat-transfer theory and data. These extensive topics are treated in other sections of this Handbook and in other references. Some of the high points pertinent to reactors are covered by Rase (*Chemical Reactor Design for Process Plants*, Wiley, 1977). Two encyclopedic references, *Heat Exchanger Design Handbook* (5 vols., Begell House, 1983–1998) and Chermisnoff (ed.) (*Handbook of Heat and Mass Transfer*, 4 vols., Gulf, 1986–1990), have several articles addressed specifically to reactors.

Reactor Dynamics Continuous reactors are designed to operate at or near a steady state by controlling the operating conditions. In addition, process control systems are designed to minimize fluctuations from the target conditions and for safety. Batch and semibatch reactors are designed to operate under predefined protocols based on the best understanding of the process. However, the potential for large and unexpected deviations from steady state as a result of process variable fluctuations is significant due to the complexity and nonlinearity of reaction kinetics and of the relevant mass- and heat-transfer processes. For a set of operating conditions (pressure, temperature, composition, and phases present), more than one steady state can exist. Which steady state is actually reached depends on the initial condition. Not all steady states are stable states, and only those that are stable can be reached without special control schemes. More complex behavior such as self-sustained oscillations and chaotic behavior has also been observed with reacting systems. Further, during start-up, shutdown, and abrupt changes in process conditions, the reactor dynamics may result in conditions that exceed reactor design limits (e.g., of temperature, pressure, materials of construction, etc.) and can result in a temperature runaway, reactor blowout, and even an explosion (or detonation). Parametric sensitivity deals with the analysis of reactor dynamics in response to abrupt changes.

Steady-State Multiplicity and Stability A simple example of steady-state multiplicity is due to the interaction between kinetics and heat transport in an adiabatic CSTR. For a first-order reaction at steady state, Eq. (19-13) gives

$$r(C, T) = kC = \frac{kC_f}{1 + k\bar{t}} = \frac{C_f \exp(a + b/T)}{1 + \bar{t} \exp(a + b/T)} \quad (19-23)$$

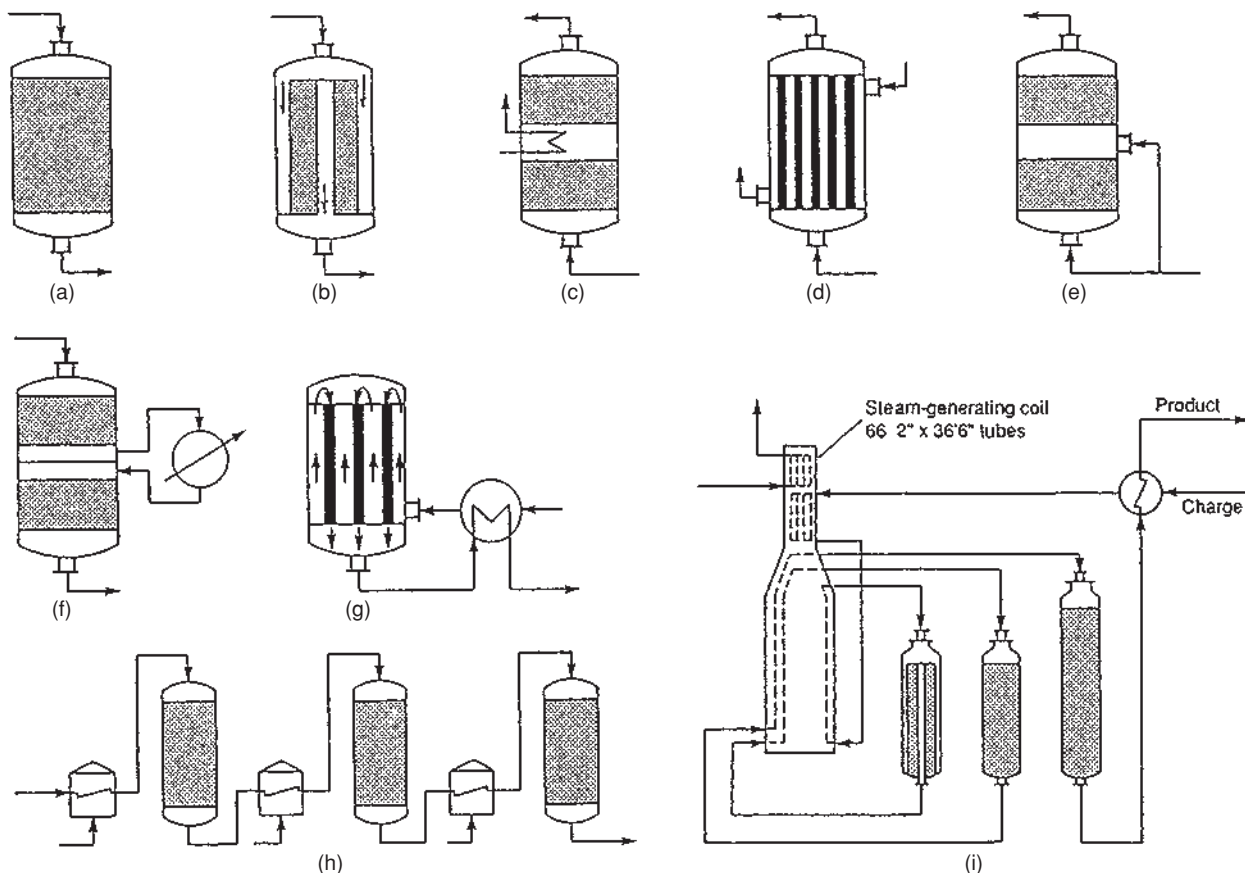


FIG. 19-3 Fixed-bed reactors with heat exchange. (a) Adiabatic downflow. (b) Adiabatic radial flow, low ΔP . (c) Built-in interbed exchanger. (d) Shell and tube. (e) Interbed cold-shot injection. (f) External interbed exchanger. (g) Autothermal shell, outside influent/effluent heat exchanger. (h) Multibed adiabatic reactors with interstage heaters. (i) Platinum catalyst, fixed-bed reformer for 5000 BPSD charge rates reactors 1 and 2 are 5.5 by 9.5 ft and reactor 3 is 6.5 by 12.0 ft; temperatures $502 \Rightarrow 433$, $502 \Rightarrow 471$, $502 \Rightarrow 496^\circ\text{C}$. To convert feet to meters, multiply by 0.3048; BPSD to m^3/h , multiply by 0.00662.

where C_f is the feed concentration and a and b are constants related to Arrhenius rate expression. The energy balance equation at steady state is given by

$$Q_C(T) = -\Delta H_r V_r r(C, T) = V_r \rho C_p (T - T_f) = Q_H(T) \quad (19-24)$$

where Q_C is the heat generation by reaction, Q_H is the heat removal by flow, T is the reactor temperature at steady state, and T_f is the feed temperature. Plotting the heat generation and heat removal terms versus temperature gives the result shown in Fig. 19-4. As shown, as many as three steady states are possible at the intersection of Q_C and Q_H .

Another example of multiplicity is shown in Fig. 19-15 for an adiabatic catalyst pellet, indicating that three effectiveness factor values can be obtained for a given Thiele modulus for a range of Prater numbers and Thiele modulus values, leading to three potential steady states. Multiple steady states can occur in different reactor types, including isothermal systems with complex nonlinear kinetics and systems with interphase transfer, the main requirement being the existence of a feedback mechanism—hence, a homogeneous PFR (without backmixing) will not exhibit multiplicity. Depending on the various physical and chemical interactions in a reactor, oscillatory and chaotic behavior can also occur.

There is a voluminous literature on steady-state multiplicity, oscillations (and chaos), and derivation of bifurcation points that define the conditions that lead to onset of these phenomena. For example, see Morbidelli et al. ["Reactor Steady-State Multiplicity and Stability," in *Chemical Reaction and Reactor Engineering*, Carberry and Varma (eds), Marcel Dekker, 1987], Luss ["Steady State Multiplicity and Uniqueness

Criteria for Chemically Reacting Systems," in *Dynamics and Modeling of Reactive Systems*, Stewart et al. (eds.), Academic Press, 1980], Schmitz [*Adv. Chem. Ser.*, **148**: 156, ACS (1975)], and Razon and Schmitz [*Chem. Eng. Sci.*, **42** (1987)]. However, many of these criteria for specific reaction and reactor systems have not been validated experimentally.

Linearized or asymptotic stability analysis examines the stability of a steady state to small perturbations from that state. For example, when heat generation is greater than heat removal (as at points A[−] and B⁺ in Fig. 19-4), the temperature will rise until the next stable steady-state temperature is reached (for A[−] it is A, for B⁺ it is C). In contrast, when heat generation is less than heat removal (as at points A⁺ and B[−] in Fig. 19-4), the temperature will fall to the next-lower stable steady-state temperature (for A⁺ and B[−] it is A). A similar analysis can be done around steady-state C, and the result indicates that A and C are stable steady states since small perturbations from the vicinity of these return the system to the corresponding stable points. Point B is an unstable steady state, since a small perturbation moves the system away to either A or C, depending on the direction of the perturbation. Similarly, at conditions where a unique steady state exists, this steady state is always stable for the adiabatic CSTR. Hence, for the adiabatic CSTR considered in Fig. 19-4, the slope condition $dQ_H/dT > dQ_C/dT$ is a necessary and sufficient condition for asymptotic stability of a steady state. In general (e.g., for an externally cooled CSTR), however, the slope condition is a necessary but not a sufficient condition for stability; i.e., violation of this condition leads to asymptotic instability, but its satisfaction does not ensure asymptotic stability. For example, in select reactor systems even

TABLE 19-2 Multibed Reactors, Adiabatic Temperature Rises and Approaches to Equilibrium*

 Oxidation of SO₂ at atmospheric pressure in a four-bed reactor. Feed 6.26% SO₂, 8.3% O₂, 5.74% CO₂, and 79.7% N₂.

°C		Conversion, %	
In	Out	Plant	Equilibrium
463.9	592.8	68.7	74.8
455.0	495.0	91.8	93.4
458.9	465.0	96.0	96.1
435.0	437.2	97.5	97.7

 Ammonia synthesis in a three bed reactor at 225 atm. Feed 22% N₂, 66% H₂, 12% inerts.

°C		Ammonia, %	
In	Out	Calculated	Equilibrium
399	518.9	13.0	15.4
427	488.9	16.0	19.0
427	470.0	18.0	21.7

*To convert atm to kPa multiply by 101.3.

 SOURCE: Plant data and calculated design values from Rase, *Chemical Reactor Design for Process Plants*, Wiley, 1977.

a unique steady state can become unstable, leading to oscillatory or chaotic behavior.

Local asymptotic stability criteria may be obtained by first solving the steady-state equations to obtain steady states and then linearizing the transient mass and energy balance equations in terms of deviations of variables around each steady state. The determinant (or slope) and trace conditions derived from the matrix A in the set of equations obtained are necessary and sufficient for asymptotic stability.

$$\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix} \quad x = C - C_{ss} \quad y = T - T_{ss}$$

$$\Delta = \det(A) > 0 \quad \sigma = \text{trace}(A) < 0 \quad (19-25)$$

where x and y are the deviation variables around the steady state (C_{ss} , T_{ss}). The approach may be extended to systems with multiple concentrations and complex nonlinear kinetics. For additional references on asymptotic stability analysis, see Denn (*Process Modeling*, Longman, 1986) and Morbidelli et al. ["Reactor Steady-State Multiplicity and Stability," in *Chemical Reaction and Reactor Engineering*, Carberry and Varma (eds.), Marcel Dekker, 1987].

Parametric Sensitivity and Dynamics The global stability and sensitivity to abrupt changes in parameters cannot be determined from an asymptotic analysis. For instance, for the simple CSTR, a key question is whether the temperature can run away from a lower stable

steady state to a higher one. The critical temperature difference ΔT_c is useful in designing for globally stable operation:

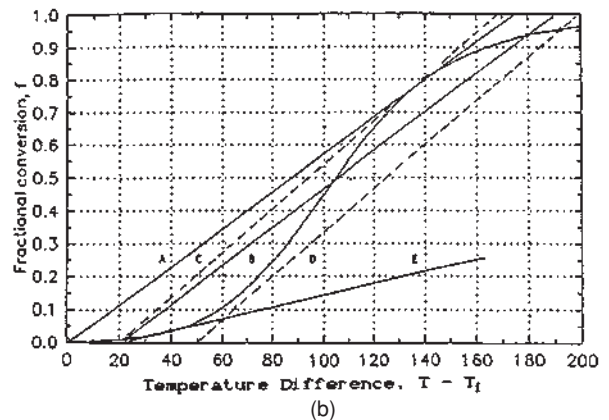
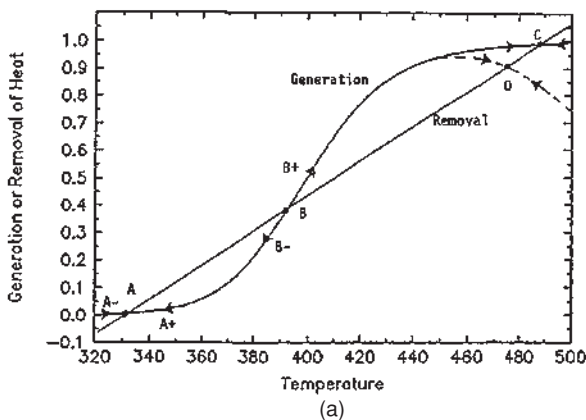
$$T - T_j < \Delta T_c = \left(\frac{RT^2}{E} \right) \quad (19-26)$$

where T is the reactor temperature, T_j is the cooling jacket temperature, E is the activation energy, and R is the universal gas constant. Similarly, for a jacketed PFR, a conservative criterion for stability is $T_{\max} - T_j < \Delta T_c$, where $T_{\max}$ is the temperature of the hot spot.

Another example of sensitivity to abrupt changes is the *wrong-way effect*, exhibited, for instance, in packed-bed reactors, where an abrupt reduction in feed rate or in feed temperature results in a dramatic increase in reactor peak temperature for exothermic reactions. Either the reactor may eventually return to the original steady state or, if a higher-temperature steady state exists, the reactor may establish a temperature profile corresponding to the new high steady state. Such a dynamic excursion can result in an increase of undesirable by-products concentration, catalyst deactivation, permanent reactor damage, and safety issues; e.g., see work by Luuss and coworkers ["Wrong-Way Behavior of Packed-Bed Reactors: I. The Pseudo-homogeneous Model," *AIChE J.* **27**: 234–246 (1981)]. For more complex systems, the transient model equations are solved numerically. A more detailed discussion of parametric sensitivity is provided by Varma et al. (*Parametric Sensitivity in Chemical Systems*, Cambridge University Press, 1999).

Reactor Models As discussed earlier, reactor models attempt to strike a balance between the level of detail included and the usefulness of the model. Too many details in the model may require a larger number of adjustable model parameters, increase computational requirements, and limit how widely the model may be used. Too few details, on the other hand, increase ease of implementation but may compromise the predictive or design capabilities of the model. Figure 19-5 is a schematic of the inherent tradeoff between ease of implementation and the insight that may be obtained from the model.

Increases in computational power are allowing a more cost-effective inclusion of a greater number of details. Computational fluid dynamics (CFD) models provide detailed flow information by solving the Navier-Stokes transport equations for mass, momentum, and heat balances. The user will, however, need to be familiar with the basic elements of the software and may need a license. A typical numerical solution of the governing transport equations is obtained within the eulerian framework, using a large number of computational cells (or finite volumes that represent reactor geometry). Current capabilities in commercial CFD software can be used to resolve the flow, concentration, and temperature patterns in a single phase with sufficient detail and reasonable accuracy for all length and time scales. The ability to visualize flow, concentration, and temperature inside a reactor is useful in understanding performance and in designing reactor internals.


FIG. 19-4 Multiple steady states of CSTRs, stable and unstable, adiabatic. (a) First-order reaction, A and C stable, B unstable, the dashed line is for a reversible reaction. (b) One, two, or three steady states depending on the combination (C_f , T_j).

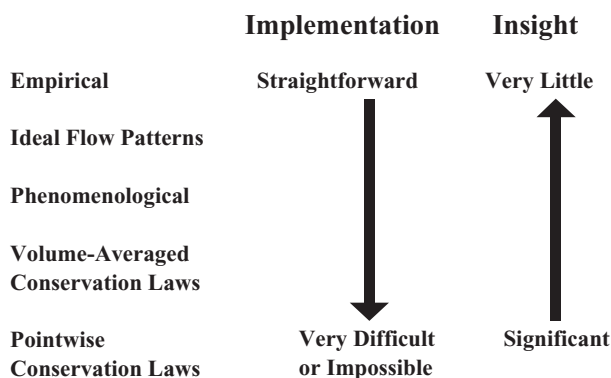


FIG. 19-5 Hierarchy of reactor models.

Addition of transport properties and more than one phase (as is the case with solid catalysts) within a CFD framework complicates the problem in that the other phase(s) also may have to be included in the calculations. This may require additional transport equations to address a range of complexities associated with the dynamics and physics of each phase, the interaction between and within phases, sub-grid-scale heterogeneities (such as size distributions within each phase), and coupling with kinetics at the molecular level. For example, one needs the bubble size distribution in a bubble column reactor to correctly model interfacial area and local mass-transfer coefficients, which can further affect the chemical kinetics. Although phenomenological models describing such physical effects have greatly improved over the years, this area still lacks reliable multiphase turbulence closures, or experimentally validated intraphase and interphase transport models. Mathematical modeling in industrial practice will continue to involve compromises between computational complexity, experimental data needs, ability to validate the model, cost, and the time frame in which the work may be useful to the organization.

RESIDENCE TIME DISTRIBUTION AND MIXING

The time spent by reactants and intermediates at reaction conditions determines conversion (and perhaps selectivity). It is therefore often important to understand the residence time distribution (RTD) of reaction species in the reactor. This RTD could be considerably different from what is expected. Reasons for the deviation could be channeling of fluid, recycling of fluid, or creation of stagnant regions in the reactor, as illustrated in Fig. 19-6.

This section introduces how tracers are used to establish the RTD in a reactor and to contrast against RTDs of ideal reactors. The section

ends with a discussion of how reactor performance may be connected to RTD information.

TRACERS

Tracers are typically nonreactive substances used in small concentration that can be easily detected. The tracer is injected at the inlet of the reactor along with the feed or by using a carrier fluid, according to some definite time sequence. The inlet and outlet concentrations of the tracer

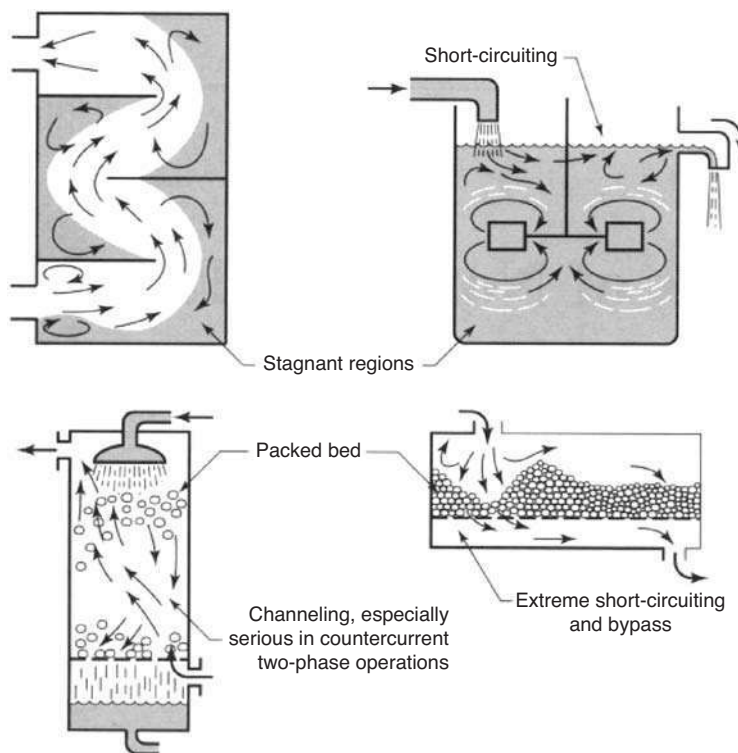


FIG. 19-6 Some examples of nonideal flow in reactors. (Fig. 11.1 in *Levenspiel, Chemical Reaction Engineering*, John Wiley & Sons, 1999.)

are recorded as a function of time. These data are converted to a residence time distribution of feed in the reactor vessel. Tracer studies may be used to detect and define regions of nonideal behavior, develop phenomenological zone models, calculate reactor performance (conversion, selectivity), and synthesize optimal reactor configurations for a given process. The RTD does not represent the mixing behavior in a vessel uniquely. Several arrangements of reactors or internals within a vessel may provide the same tracer response. For example, any series arrangement of the same number of CSTR and plug flow reactor elements will provide the same RTD. This lack of uniqueness may limit direct application of tracer studies to first-order reactions with constant specific rates. For other reactions, the tracer curve may determine the upper and lower limits of reactor performance. When this range is not too broad, or when the purpose of the tracer test is to diagnose maldistribution or bypassing in the reactor, the result can be useful. Tracer data also may be taken at several representative positions in the vessel in order to develop a better understanding for the flow behavior.

Inputs Although some arbitrary variation of input concentration with time may be employed, five mathematically simple tracer input signals meet most needs. These are *impulse*, *step*, *square pulse* (started at time a , kept constant for an interval, then reduced to the original value), *ramp* (increased at a constant rate for a period of interest), and *sinusoidal*. Sinusoidal inputs are difficult to generate experimentally.

Types of Responses The key relationships associated with tracers are provided in Table 19-3. Effluent concentrations resulting from impulse and step inputs are designated C_δ and C_u , respectively. The mean concentration resulting from an impulse of magnitude m into a vessel of volume V_r is $C^0 = m/V_r$. The mean residence time is the ratio of the vessel volume to the volumetric flow rate:

$$\bar{t} = \frac{V_r}{V'} \quad \text{or} \quad \bar{t} = \frac{\int_0^\infty t C_\delta dt}{\int_0^\infty C_\delta dt} \quad (19-27)$$

The reduced time is $t_r = t/\bar{t}$. Residence time distributions are used in two forms: normalized, $E(t_r) = C_\delta/C^0$; or plain, $E(t) = C_\delta/\int_0^\infty C_\delta dt$. The area under either RTD is unity: $\int_0^\infty E(t_r) dt_r = \int_0^\infty E(t) dt = 1$, and the relation between them is $E(t_r) = \bar{t}E(t)$. The area between the ordinates at t_1 and t_2 is the fraction of the total effluent that has spent the period between those times in the vessel. The age function is defined in terms of the step input as

$$F(t) = \frac{C_u}{C_f} = \int_0^t E(t) dt \quad (19-28)$$

Reactor Tracer Responses

Continuous Stirred Tank Reactor (CSTR) With a step input of magnitude C_f , the unsteady material balance of tracer

$$V_r \frac{dC}{dt} + V'C = V'C_f \quad (19-29)$$

can be integrated to yield

$$\frac{C}{C_f} = F(t_r) = 1 - \exp(-t_r) \quad (19-30)$$

With an impulse input of magnitude m or an initial mean concentration $C^0 = m/V_r$, the material balance is

$$\frac{dC}{dt_r} + C = 0 \quad \text{with} \quad C = C^0, t = 0 \quad (19-31)$$

And integration gives

$$\frac{C}{C^0} = E(t_r) = \exp(-t_r) \quad (19-32)$$

These results show that

$$E(t_r) = \frac{dF(t_r)}{dt_r} \quad (19-33)$$

TABLE 19-3 Tracer Response Functions

Mean residence time:

$$\bar{t} = \frac{\int_0^\infty t C_\delta dt}{\int_0^\infty C_\delta dt} = \frac{\int_0^\infty t C_u dt}{C_u}$$

Initial mean concentration with impulse input,

$$C^0 = \frac{m}{V_r} = \left(\frac{V'}{V_r}\right) \int_0^\infty C_\delta dt = \frac{\int_0^\infty C_\delta dt}{\bar{t}}$$

Reduced time:

$$t_r = \frac{t}{\bar{t}}$$

Residence time distribution:

$$E(t) = \frac{C_\delta}{\int_0^\infty C_\delta dt} = \frac{E(t_r)}{\bar{t}} = \frac{dF(t)}{dt}$$

Residence time distribution, normalized,

$$E(t_r) = \frac{\text{impulse output}}{\text{initial mean concentration}} = \frac{C_\delta}{C^0} = \frac{\bar{t} C_\delta}{\int_0^\infty C_\delta dt} = \bar{t} E(t) = \frac{dF(t)}{dt}$$

Age:

$$F(t) = \frac{\text{step output}}{\text{step input}} = \frac{C_u}{C_f} = \frac{\int_0^t C_\delta dt}{\int_0^\infty C_\delta dt} = F(t_r)$$

Internal age:

$$I(t) = 1 - F(t)$$

Intensity:

$$\Lambda(t) = \frac{E(t)}{1 - F(t)} = \frac{E(t)}{I(t)}$$

Variance:

$$\sigma^2(t) = \int_0^\infty (t - \bar{t})^2 E(t) dt = -\bar{t}^2 + \frac{\int_0^\infty t^2 C_\delta dt}{\int_0^\infty C_\delta dt}$$

Variance, normalized:

$$\sigma^2(t_r) = \frac{\sigma^2(t)}{\bar{t}^2} = -1 + \frac{\int_0^\infty t^2 C_\delta dt}{\int_0^\infty C_\delta dt} = \int_0^1 (t_r - 1)^2 dF(t_r)$$

Skewness, third moment:

$$\gamma^3(t_r) = \int_0^\infty (t_r - 1)^3 E(t_r) dt_r$$

Multistage CSTR Since tubular reactor performance can be simulated by a series of CSTRs, multistage CSTR tracer models are useful in analyzing data from empty tubular and packed-bed reactors. The solution for a tracer through n CSTRs in series is found by induction from the solution of one stage, two stages, and so on.

$$E(t_r) = \frac{C_n}{C^0} = \frac{n^n}{(n-1)!} t_r^{n-1} \exp(-nt_r) \quad (19-34)$$

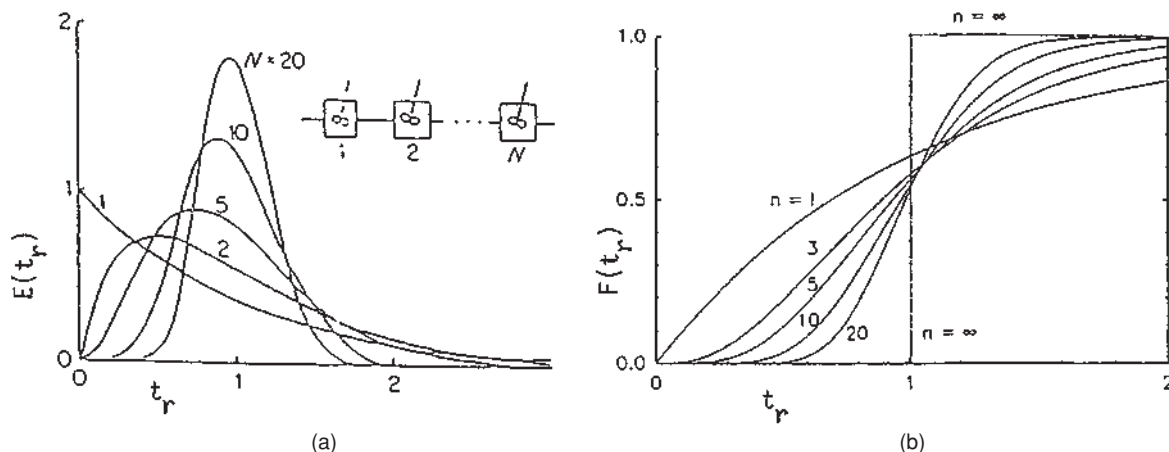


FIG. 19-7 Tracer responses to n -stage continuous stirred tanks in series: (a) Impulse inputs. (b) Step input.

The solution for a step response can be obtained by integration

$$F(t_r) = \int_0^{t_r} E(t_r) dt = 1 - \exp(-nt_r) \sum_{j=0}^{n-1} \frac{(nt_r)^j}{j!} \quad (19-35)$$

where $E(t_r)$ and $F(t_r)$ for various values of n are shown in Fig. 19-7.

The theoretical RTD responses in Fig. 19-7a are similar in shape to the experimental responses from pilot and commercial reactors shown in Fig. 19-8. The value of n in Fig. 19-8 represents the number of CSTRs in series that provide a similar RTD to that observed commercially. Although not shown in the figure, a commercial reactor having a similar space velocity as a pilot reactor and a longer length typically has a higher n value than a pilot reactor due to greater linear velocity.

The variance of the RTD of a series of CSTRs, σ^2 , is the inverse of n .

$$\sigma^2 = \int_0^{\infty} (t_r - 1)^2 E(t_r) dt_r = \frac{1}{n} \quad (19-36)$$

Plug Flow Reactor The tracer material balance over a differential reactor volume dV_r is

$$\frac{\partial C}{\partial t} + V_r \frac{\partial C}{\partial V_r} = 0 \quad (19-37)$$

With step input $u(t)$, the initial and boundary conditions are

$$C(0, t) = C_f u(t) \quad \text{and} \quad C(V_r, 0) = 0 \quad (19-38)$$

The solution is

$$\frac{C}{C_f} = F(t) = u(t - \bar{t}) = \begin{cases} 0 & \text{when } t \leq \bar{t} \\ 1 & \text{when } t > \bar{t} \end{cases} \quad (19-39)$$

As discussed earlier, the response to an impulse input is the derivative of $F(t)$.

$$\frac{C}{C_\delta} = E(t) = \delta(t_r - 1) \quad (19-40)$$

The effluent RTD is an impulse that is delayed from the input impulse by $t_r = 1$, or $t = \bar{t}$.

Tubular Reactor with Dispersion As discussed earlier, a multi-stage CSTR model can be used to simulate the RTD in pilot and commercial reactors. The *dispersion model*, similar to Fick's molecular diffusion law with an empirical dispersion coefficient D_e replacing the diffusion coefficient, may also be used.

$$\frac{\partial C}{\partial t} + V_r \frac{\partial C}{\partial V_r} - D_e \frac{\partial^2 C}{\partial V_r^2} = 0 \quad (19-41)$$

The above equation is often converted to dimensionless variables and solved. The solution of this partial differential equation is recorded in the literature [Otake and Kunigata, *Kagaku Kogaku*, **22**: 144 (1958)]. The plots of $E(t_r)$ versus t_r are bell-shaped, similar to the response for a series of n CSTRs model (Fig. 19-7). A relation between $\sigma^2(t_r)$, n , and Pe (for the closed-ends condition) is

$$\sigma^2(t_r) = \frac{2[Pe - 1 + \exp(-Pe)]}{Pe^2} = \frac{1}{n} \quad (19-42)$$

Examples of values of Pe are provided in Fig. 19-8. When Pe is large, $n \Rightarrow Pe/2$ and the dispersion model reduces to the PFR model. For small values of Pe , the above equation breaks down since the lower limit on n is $n = 1$ for a single CSTR. To better represent dispersion behavior, a series of CSTRs with backmixing may be used; e.g., see Froment and Bischoff (*Chemical Reactor Analysis and Design*, Wiley, 1990). A model analogous to the dispersion model may be used when there are velocity profiles across the reactor cross-section (e.g., for laminar flow). In this case, the equation above will contain terms associated with the radial position in the reactor.

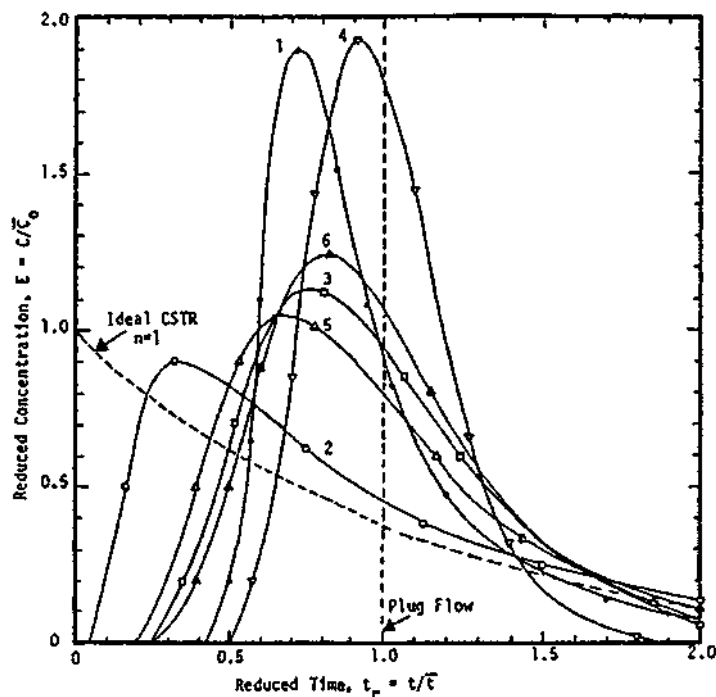
Understanding Reactor Flow Patterns As discussed above, a RTD obtained using a nonreactive tracer may not uniquely represent the flow behavior within a reactor. For diagnostic and simulation purposes, however, tracer results may be explained by combining the expected tracer responses of ideal reactors combined in series, in parallel, or both, to provide an RTD that matches the observed reactor response. The most commonly used ideal models for matching an actual RTD are PRF and CSTR models. Figure 19-9 illustrates the responses of CSTRs and PFRs to impulse or step inputs of tracers.

Since the tracer equations are linear differential equations, a Laplace transform $L\{f(t)\} = \int_0^\infty f(t)e^{-st} dt$ may be used to relate tracer inputs to responses. The concept of a transfer function facilitates the combination of linear elements.

$$\bar{C}_{\text{output}}(s) = (\text{transfer function}) \bar{C}_{\text{input}}(s) = G(s) \bar{C}_{\text{input}}(s) \quad (19-43)$$

Some common Laplace transfer functions are listed in Table 19-4.

The Laplace transform may be inverted to provide a tracer response in the time domain. In many cases, the overall transfer function cannot be analytically inverted. Even in this case, moments of the RTD may be derived from the overall transfer function. For instance, if G'_0 and G''_0 are the limits of the first and



No.	Code	Process	σ^2	n	Pe
1	○	Aldolization of butyraldehyde	0.050	20.0	39.0
2	●	Olefin oxonation pilot plant	0.663	1.5	1.4
3	□	Hydrodesulfurization pilot plant	0.181	5.5	9.9
4	▽	Low-temp hydroisomerization pilot	0.046	21.6	42.2
5	△	Commercial hydrofiner	0.251	4.0	6.8
6	▲	Pilot plant hydrofiner	0.140	7.2	13.2

FIG. 19-8 Residence time distributions of pilot and commercial reactors. σ^2 = variance of the residence time distribution, n = number of stirred tanks with the same variance, Pe = Peclet number. (Walas, Chemical Process Equipment, Butterworths, 1990.)

second derivatives of the transfer function $G(s)$ as $s \Rightarrow 0$, the mean residence time and variance are

$$\bar{t} = G'_0 \quad \text{and} \quad \sigma^2(t) = G''_0 - (G'_0)^2 \quad (19-44)$$

In addition to understanding the flow distribution, tracer experiments may be conducted to predict or explain reactor performance based on a particular RTD. To do this, a mathematical expression for the RTD is needed. A PFR, or a dispersion model with a small value of the dispersion coefficient, may be used to simulate an empty tubular reactor. Stirred tank performance often is nearly completely mixed (CSTR). In some cases, to fit the measured RTD, the model may have to be modified by taking account of bypass zones, stagnant zones, or other parameters associated with the geometry and operation of the reactor. Sometimes the vessel can be visualized as a zone of complete mixing in the vicinity of impellers followed by plug flow zones elsewhere, e.g., CSTRs followed by PFRs. Packed beds usually deviate substantially from plug flow. The dispersion model and some combination of PFRs and CSTRs or multiple CSTRs in series may approximate their behavior. Fluidized beds in small sizes approximate CSTR behavior, but large ones exhibit bypassing, stagnancy, nonhomogeneous regions, and several varieties of contact between particles and fluid. The additional parameters required to simulate such mixing behavior can increase the mathematical complexity of the model.

The characteristic bell shape of many RTDs can be fit to well-known statistical distributions. Hahn and Shapiro (*Statistical Models in Engineering*, Wiley, 1967) discuss many of the standard distributions and conditions for their use. The most useful distributions are the gamma (or Erlang) and the gaussian together with its Gram-Charlier extension. These distributions are represented by only a few parameters that can be used to determine, for instance, the mean and the variance.

Qualitative inspection of the tracer response can go a long way toward identifying flow distribution problems. Additional references on tracers are Wen and Fan (*Models for Flow Systems in Chemical Reactors*, Marcel Dekker, 1975) and Levenspiel (*Chemical Reaction Engineering*, 3d ed., Wiley, 1999).

CONNECTING RTD TO CONVERSION

When the flow pattern is known, the conversion for a given reaction mechanism may be evaluated from the appropriate material and energy balances. When only the RTD is known (or can be calculated from tracer response data), however, different networks of reactor elements can match the observed RTD. In reality, reactor performance for a given reactor network will be unique. The conversion obtained by matching the RTD is, however, unique only for linear kinetics. For nonlinear kinetics, two additional factors have to be

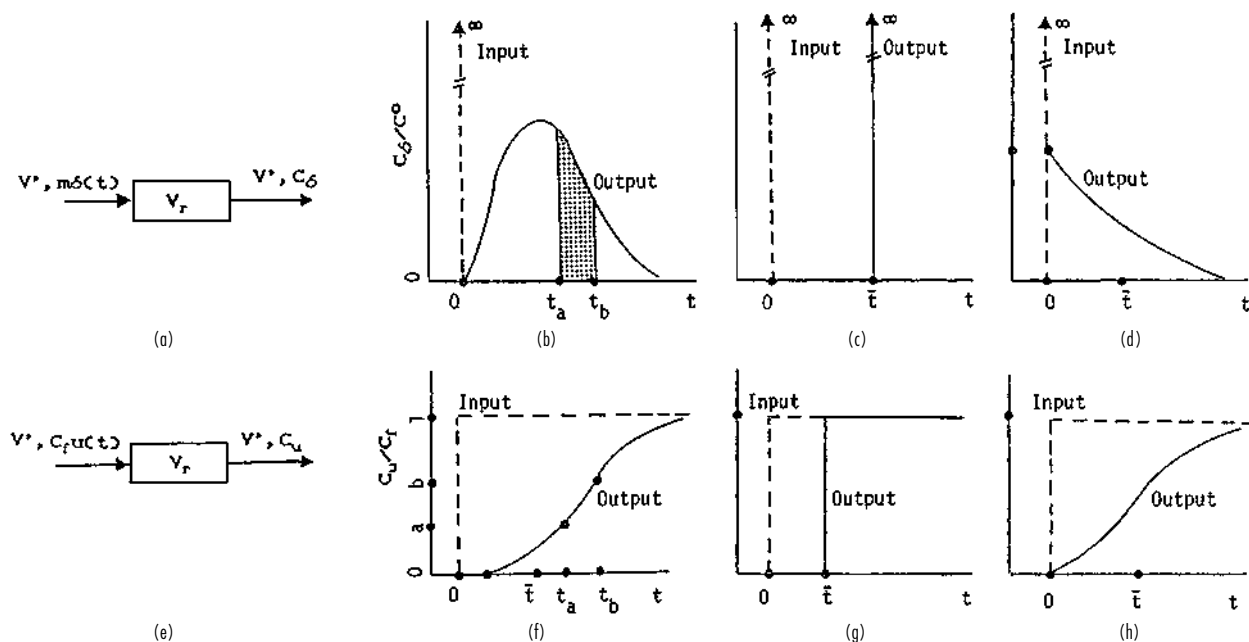


FIG. 19-9 Tracer inputs and responses for PFR and CSTR. (a) Experiment with impulse input of tracer. (b) Generic behavior; area between ordinates at t_a and t_b equals the fraction of the tracer with residence time in that range. (c) Plug flow behavior. (d) Completely mixed vessel. (e) Experiment with step input of tracer. (f) Generic behavior; fraction with ages between t_a and t_b equals the difference between the ordinates, $b - a$. (g) Plug flow behavior. (h) Completely mixed behavior.

accounted for to fully describe the contacting or flow pattern: the degree of segregation of the fluid and the earliness of mixing of the reactants.

Segregated Flow The degree of segregation relates to the tendency of fluid particles to move together as aggregates or clumps (e.g., bubbles in gas-liquid reactors, particle clumps in fluidized beds, polymer striations in high-viscosity polymerization reactors) rather than each molecule behaving independently (e.g., homogeneous gas, low-viscosity liquid). A system with no aggregates may be called a microfluid, and the system with aggregates a macrofluid (e.g., see Levenspiel, *Chemical Reaction Engineering*, 3d ed., Wiley, 1999). In an ideal plug flow or in an ideal batch reactor, the segregated particles in each clump spend an equal time in the reactor and therefore the behavior is no different from that of a microfluid that has individual molecules acting independently. The reactor performance is therefore unaffected by the degree of segregation, and the PFR or ideal batch model equations may be used to estimate performance. As shown below, however, this is not the case for a CSTR where the performance equation for a microfluid is the same as that of an ideal CSTR, while that of a CSTR with segregated flow is not.

In segregated flow the molecules travel as distinct groups. All molecules that enter the vessel together leave together. The groups are small enough that the RTD of the whole system is represented by a

smooth curve. Each group of molecules reacts independently of any other group, that is, as a batch reactor. For a batch reactor with a power law kinetics,

$$\left(\frac{C}{C_0}\right)_{\text{batch}} = \begin{cases} \exp(-kt) = \exp(-k\bar{t}t_r) & \text{first order} \\ \left[\frac{1}{1 + (q-1)kC^{q-1}\bar{t}t_r}\right]^{1/(q-1)} & \text{order } q \end{cases} \quad (19-45)$$

For other rate equations a numerical solution may be needed. The mean conversion of all the groups is the sum of the products of the individual conversions and their volume fractions of the total flow. Since the groups are small, the sum may be replaced by an integral. Thus,

$$\left(\frac{C}{C_0}\right)_{\text{segregated}} = \int_0^\infty \left(\frac{C}{C_0}\right)_{\text{batch}} E(t) dt = \int_0^\infty \left(\frac{C}{C_0}\right)_{\text{batch}} E(t_r) dt_r \quad (19-46)$$

When a conversion and an RTD are known, a value of k may be estimated by trial and error so the segregated integral is equal to the known value. If a series of conversions are known at several residence times, the order of the reaction that matches the data may be estimated by trial and error. One has to realize, however, that the RTD may change with residence time. Alternatively, for known intrinsic kinetics, a combination of ideal reactors that reasonably match both RTD and performance may be considered.

Early versus Late Mixing—Maximum Mixedness The concept of early versus late mixing may be illustrated using a plug flow reactor and an ideal CSTR in series. In one case, the ideal CSTR precedes the plug flow reactor, a case of early mixing. In the other case, the plug flow reactor precedes the CSTR, and this is a case of late mixing. Each of the two arrangements has the same RTD.

In maximum mixedness (or earliest possible mixing), the feed is intimately mixed with elements of fluid of different ages, for instance, using multiple side *inlets* at various points along a plug flow reactor. The

TABLE 19-4 Some Common Laplace Transform Functions

Element	Transfer function $G(s)$
Ideal CSTR	$\frac{1}{1 + ts}$
PFR	$\exp(-\bar{t}s)$
n -stage CSTR (Erlang)	$\frac{1}{(1 + ts)^n}$
Erlang with time delay	$\frac{\exp(-\bar{t}s)}{(1 + ts)^n}$

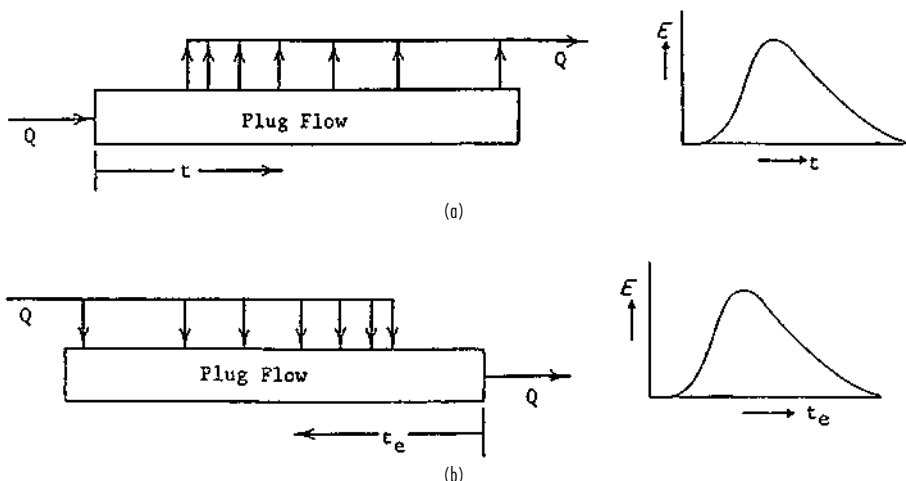


FIG. 19-10 Two limiting flow patterns with the same RTD. (a) Segregated flow. (b) Maximum mixedness flow.

amount and location of the inlet flows match the RTD. This means that each portion of fresh material is mixed with all the material that has the same life expectation, regardless of the actual residence time in the vessel up to the time of mixing. The life expectation under plug flow conditions is related to the distance remaining to be traveled before leaving the vessel. The concept of maximum mixedness and completely segregated flow is illustrated in Fig. 19-10. Segregated flow is represented as a plug flow reactor with multiple side outlets and has the same RTD.

In contrast to segregated flow, in which the mixing occurs only after each side stream leaves the vessel, under maximum mixedness flow, mixing of all molecules having a certain life expectancy occurs at the time of introduction of fresh material. These two mixing extremes—as late as possible and as soon as possible, both having the same RTD—correspond to extremes of reactor performance.

The mathematical model for maximum mixedness has been provided by Zwietering [*Chem. Eng. Sci.* **11**: 1 (1959)].

$$\frac{dC}{dt} = r_c - \frac{E(t)}{1 - F(t)} (C_0 - C) \quad (19-47)$$

where r_c is the chemical reaction rate; e.g., for an order q , $r_c = kC^q$. The above differential equation in dimensionless variable form (where $f = C/C_0$ and $t_r = t/t$) becomes

$$\frac{df}{dt_r} = k \bar{C}_0^{q-1} f^q - \frac{E(t_r)}{1 - F(t_r)} (1 - f) \quad (19-48)$$

with boundary condition

$$\frac{df}{dt_r} = 0 \quad \text{for } t_r \rightarrow \infty \quad (19-49)$$

which makes

$$k \bar{C}_0^{q-1} f_\infty^q - \frac{E(\infty)}{1 - F(\infty)} (1 - f_\infty) = 0 \quad (19-50)$$

The conversion achieved in the vessel is obtained by the solution of the differential equation at the exit of the vessel where the life expectation is $t = 0$. The starting point for the integration is (f_∞, t_∞) . When integrating numerically, however, the RTD becomes essentially 0 by the time t_r approaches 3 or 4. Accordingly, the integration interval is from $(f_\infty, t_r \leq 3$ or 4) to $(f_{\text{effluent}}, t_r = 0)$ with f_∞ obtained from Eq. (19-50).

The conversion is a maximum in *segregated flow* and a minimum under *maximum mixedness* conditions, for a given RTD and reaction orders >1 . A few comparisons are made in Fig. 19-11. In some ranges of the parameters n or r_c , the differences in reactor volume for a given conversion, when segregated or maximum mixedness flow is assumed,

are substantial. If only the RTD is known, these two extremes bracket reactor performance. As a general trend, for reaction orders >1 , conversion increases as maximum mixedness $<$ late mixing of microfluids $<$ segregated flow (and the opposite is the case for orders <1). Increased deviation from ideal plug flow increases the effect of segregation on conversion. At low conversion, the conversion is insensitive to the RTD and to the extent of segregation.

Novosad and Thyn [*Coll. Czech. Chem. Comm.* **31**: 3,710–3,720 (1966)] solved the maximum mixedness and segregated flow equations (fit with the Erlang model) numerically. There are few experimental confirmations of these mixing extremes. One study with a

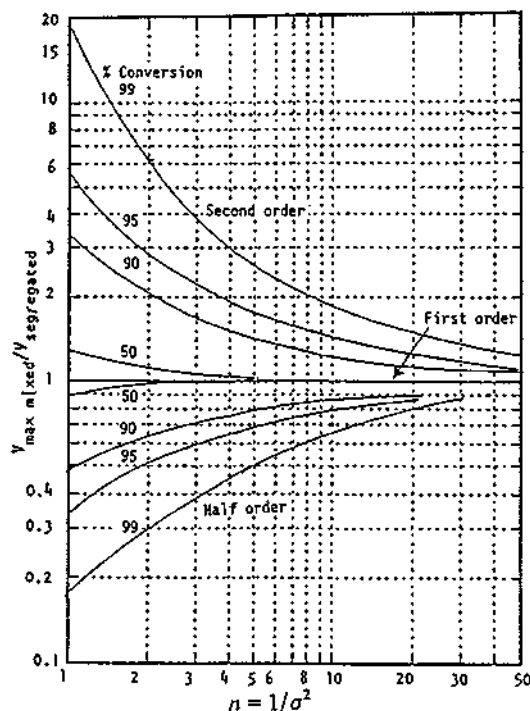


FIG. 19-11 Ratio of reactor volume for maximum mixedness and segregated flow models as a function of the variance (or n), for several reaction orders.

50-gal stirred tank reactor found segregation at low agitation and was able to correlate complete mixing and maximum mixedness in terms of the power input and recirculation within the vessel [Worrell and Eagleton, *Can. J. Chem. Eng.* pp. 254–258 (Dec. 1964)].

REACTION AND MIXING TIMES

Reactants may be premixed or fed directly into the reactor. To the extent that the kinetics are limiting (i.e., reaction rate is slow), the rate of mixing plays a minor role in determining conversion or selectivity. If the time to mix reactants is comparable to the reaction rate, however, mixing can have a significant impact.

The characteristic chemical reaction time t_r or characteristic time scale of the chemistry may be calculated from the reaction rate expression. For a single reaction,

$$t_r = \frac{C_0}{r(C_0, T_0)} \quad (19-51)$$

where C_0 is a reference concentration of the limiting reactant and T_0 is a reference temperature. For a first-order reaction, $t_r = 1/k$, where k (s^{-1}) is the rate constant.

Mixing may occur on several scales: on the reactor scale (macro), on the scale of dispersion from a feed nozzle or pipe (meso), and on a molecular level (micro). Examples of reactions where mixing is important include fast consecutive-parallel reactions where reactant concentrations at the boundaries between zones rich in one or the other reactant being mixed can determine selectivity.

Much of the literature around mixing times has been developed around the mixing of two liquids in agitated stirred tanks. The macromixing time t_{ma} can be defined as the time for the concentration to settle within, say, ± 2 percent of its final value (98 percent homogeneity). With a standard turbine in a baffled tank and $Re (= nD_a^2 \rho/\mu) > 5000$,

$$t_{ma} \cong \frac{4}{n} \left(\frac{D_t}{D_a} \right)^2 \left(\frac{H}{D_t} \right) \quad (19-52)$$

where n is the stirrer speed, D_t is the tank diameter, D_a is the agitator diameter, and H is the height of the tank; t_{ma} varies inversely with the stirrer speed. In a case of a tank with an aspect ratio of unity and $D_a/D_t = \frac{1}{3}$, $nt_{ma} \cong 36$. For a stirrer speed of 120 rpm, the macromixing time is 18 s.

The circulation time t_{cir} is the time to circulate the reactor contents once:

$$t_{cir} = \frac{V_r}{q} \quad (19-53)$$

where q is the flow induced by the impeller. The induced flow is about 2 times the direct discharge from the turbine, creating uncertainty in estimating q ; t_{cir} is roughly one-fourth of the macromixing time.

The micromixing time t_{mi} is the time required for equilibration of the smallest eddies by molecular diffusion, engulfment, and stretching. For liquid-liquid mixing, stretching and engulfment are limiting factors and t_{mi} depends on the kinematic viscosity (μ/ρ) and the local rate of energy dissipation $\phi\bar{\epsilon}$:

$$t_{mi} = 17 \left(\frac{\mu/\rho}{\phi\bar{\epsilon}} \right)^{1/2} \quad (19-54)$$

For a kinematic viscosity of $10^{-6} \text{ m}^2/\text{s}$ and an energy dissipation of 1.0 W/kg , $t_{mi} = 0.017 \text{ s}$. The local energy dissipation will vary greatly with position in the tank with its greatest value near the tip of the impeller. Injection of reactant at the point of greatest turbulence minimizes t_{mi} .

The mesomixing time t_{me} is the time for “significant mixing” of an incoming jet of feed liquid with the surrounding fluid. A formula for estimating t_{me} is the time for turbulent diffusion to transport liquid over a distance equal to the feed pipe diameter d_0 .

$$t_{me} \cong \frac{5.3d_0^2}{(\phi\bar{\epsilon})^{1/3}D_a^{4/3}} \quad (19-55)$$

If the diameter of the pipe is proportional to the agitator diameter, t_{me} increases as $d_0^{2/3}$. Since t_{me} depends on the local energy dissipation, it is sensitive to location. Typically, $t_{me} (> t_{mi})$ is a fraction of a second or so.

A parameter used to diagnose mixing issues for reactive systems is the *Damköhler number* Da which is the ratio of the mixing time to the reaction time, $Da = t_{mixing}/t_r$. Small Da numbers ($Da \ll 1$) indicate relatively rapid mixing compared to the reaction, so mixing is less important. In contrast, large Da numbers ($Da \gg 1$) indicate a need to consider mixing issues. A more complete discussion of the topic is provided in the appropriate section of the Handbook, in Baldyga and Bourne (*Turbulent Mixing and Chemical Reactions*, Wiley, 1998), and in Harriott (*Chemical Reactor Design*, Marcel Dekker, 2003).

SINGLE-PHASE REACTORS

Section 7 of this Handbook presents the theory of reaction kinetics that deals with homogeneous reactions in batch and continuous equipment. Single-phase reactors typically contain a liquid or a gas with (or without) a homogeneous catalyst that is processed in a reactor at conditions required to complete the desired chemical transformation.

LIQUID PHASE

Batch reactions of single or miscible liquids are often done in stirred or pump-around tanks. The agitation is needed to mix multiple feeds and to enhance heat exchange with cooling (or heating media) during the process. Topics that acquire special importance on an industrial scale are the quality of mixing in tanks and the residence time distribution in vessels where plug flow may be the goal. A special case is that of laminar and related flow distributions characteristic of nonnewtonian fluids, which often occurs in polymerization reactors. The information about agitation and heat transfer in tanks is described in the relevant Handbook section.

Homogeneous Catalysis A catalyst is a substance, usually used in small amounts relative to the reactants, that increases the rate of a reaction without being consumed in the process. Liquid-phase reactions are often conducted in the presence of homogeneous catalysts. Typically, homogeneous catalysts are ions or metal coordination

complexes or enzymes in aqueous solution. The specific action of a particular metal complex can be altered by varying the ligands (or coordination number) of the complex or the oxidation state of the central metal atom. Some examples of homogeneous catalysts in industrial practice include hydrolysis of esters by hydronium (H_3O^+) or hydroxyl (OH^-) ions, hydroformylation of olefins using Rh or Co carbonyls, decomposition of hydrogen peroxide by ferrous ions, decomposition of nitramides catalyzed by acetate ion, inversion of sucrose by HCl, halogenation of acetone by H^+ and OH^- , and hydration of isobutene by acids. A characteristic of homogeneous catalysis is that, compared to solid catalysis, the reaction(s) proceeds under relatively mild conditions. A key issue associated with homogeneous catalysis is the difficulty of separating product and catalyst.

In stirred tanks, the power input to agitate the tank will depend on the physical properties of the liquid. In tubular reactors, the axial dispersion in empty tubes may be estimated [e.g., Wen in Petho and Noble (eds.), *Residence Time Distribution Theory in Chemical Engineering*, Verlag Chemie, 1982] as

$$\frac{1}{Pe} = \frac{1}{(Re)(Sc)} + \frac{(Re)(Sc)}{192} \quad 1 \leq Re \leq 2000 \quad \text{and} \quad 0.2 \leq Sc \leq 1000 \quad (19-56)$$

$$\frac{1}{Pe} = \frac{3 \times 10^7}{(Re)^{2.1}} + \frac{1.35}{(Re)^{0.125}} \quad Re \geq 2000$$

In a general case, the velocity may also be a function of radius. One such case is that of laminar flow which is characterized by a parabolic velocity profile. The velocity at the wall is zero while that at the centerline is twice the average velocity. In such cases, a momentum balance equation is solved along with the equations for heat and mass transfer, and each equation contains terms for the radial contribution. Laminar flow can be avoided by mixing over the cross-section. For this purpose, *in-line* static mixers can be provided. For very viscous materials and pastes, screws of the type used for pumping and extrusion are used as reactors. When the temperature of the reactants changes during the course of the reaction (due to either the heat of reaction or the work required to keep the contents well mixed), material and energy balance equations have to be solved simultaneously.

Examples

- Crude oil is heated to temperatures at which it thermally cracks into gasoline and distillate products and lower-molecular-weight gases. This liquid cracking process is referred to as visbreaking. A schematic of the process and the effect of operating variables on performance is shown in Fig. 19-12.
- The Wacker process for the oxidation of ethylene to acetaldehyde with $PdCl_2/CuCl_2$ at $100^\circ C$ ($212^\circ F$) with 95 percent yield and 95 to 99 percent conversion per pass.
- The OXO process for higher alcohols: $CO + H_2 + C_3H_6 \Rightarrow n\text{-butanal} \Rightarrow$ further processing. The catalyst is a rhodium triphenylphosphine coordination compound at $100^\circ C$ ($212^\circ F$), 30 atm (441 psi).
- Acetic acid from methanol by the Monsanto process, $CH_3OH + CO \Rightarrow CH_3COOH$, rhodium iodide catalyst, 3 atm (44 psi), $150^\circ C$ ($302^\circ F$), 99 percent selectivity.

See a review of industrial processes that employ homogeneous catalysts by Jennings (ed.), *Selected Developments in Catalysis*, Blackwell Scientific, 1985.

GAS PHASE

There are few examples of industrial processes with pure gas-phase reactions. The most common and oldest example is combustion. Although termed *homogeneous*, most gas-phase reactions take place in contact with solids, either the vessel wall or particles as heat carriers. With inert solids, the only complication is with heat transfer. Several of these reactions are listed in Table 19-1. Whenever possible, liquefaction of gas-phase systems is considered to take advantage of the higher rates of liquid reactions, to utilize liquid homogeneous catalysts, or to keep equipment size down.

The specific type of equipment used for gas-phase reactions depends on the conditions required for undertaking the reaction. Examples of noncatalytic gas-phase reactions are shown in Fig. 19-13. In general, mixing of feed gases and temperature control are major process requirements. Gases are usually mixed by injecting one of the streams into the rest of the gases using a high-speed nozzle, as in the flame reactor (Fig. 19-13d).

Examples

- In the cracking of light hydrocarbons and naphtha to olefins, heat is supplied from combustion gases through tubes in fired heaters at $800^\circ C$ ($1472^\circ F$) and sufficiently above atmospheric pressure to overcome pressure drop. Superheated steam is injected to bring the temperature up quickly and retard coke deposition. The reaction time is 0.5 to 3.0 s, followed by rapid quenching. The total tube length of an industrial furnace may be more than 1000 m. Some other important gas-phase cracking processes include conversion of toluene to benzene, diphenyl to benzene, dicyclopentadiene to cyclopentadiene, and 1-butene to butadiene. Figure 19-13a shows a cracking furnace.
- The Wulf process for acetylene by pyrolysis of natural gas utilizes a heated brick checkerwork on a 4-min cycle of heating and reacting. Heat is transferred by direct contact with solids that have been pre-

heated by combustion gases. The process is a cycle of alternate heating and reacting periods. The temperature play is $15^\circ C$ ($27^\circ F$), peak temperature is $1200^\circ C$ ($2192^\circ F$), residence time is 0.1 s of which 0.03 s is near the peak (Faith, Keyes, and Clark, *Industrial Chemicals*, vol. 27, Wiley, 1975).

- The Wisconsin process for the fixation of nitrogen from air operates at $2200^\circ C$ ($3992^\circ F$), followed by extremely rapid quenching to freeze the small equilibrium content of nitrogen oxide that is made [Ermenc, *Chem. Eng. Prog.* **52**: 149 (1956)]. A pebble heater recirculates refractory pebbles continuously through heating and reaction zones. Such moving-bed units have been proposed for cracking to olefins but have been obsolesced like most moving-bed reactors.
- Acetylene may be produced from light hydrocarbons and naphtha by injecting inert combustion gases directly into the reacting stream in a flame reactor. Figure 19-13a and d shows two such devices; Fig. 19-13e shows a temperature profile (with reaction times in milliseconds).
- Oxidative pyrolysis of light hydrocarbons to acetylene is conducted in a special burner, at 0.001- to 0.01-s reaction time, peak at $1400^\circ C$ ($2552^\circ F$), followed by rapid quenching with oil or water. A portion of a combustible reactant is burned by adding a small amount of air or oxygen to generate the reaction temperatures needed.
- Chlorination reactions of methane and other hydrocarbons typically result in a mixture of products whose relative amounts can be controlled by varying the Cl/hydrocarbon ratio and recycling unwanted derivatives. For example, one can recycle the mono and di derivatives when only the tri and tetra derivatives are of value or keep the chlorine ratio low when emphasizing the lower derivatives. Temperatures are normally kept in the range of 230 to $400^\circ C$ (446 to $752^\circ F$) to limit carbon formation but may be raised to $500^\circ C$ ($932^\circ F$) when favoring CCl_4 . Exothermic processes utilize cooling through heat-transfer surfaces or cold shots. Shell-and-tube reactors with small-diameter tubes, towers with internal recirculation of gases, or multiple stages with intercooling may be used for these reactions.

SUPERCRITICAL CONDITIONS

At near-critical or supercritical conditions, a heterogeneous reaction mixture (e.g., of water, organic compounds, and oxygen) becomes homogeneous and has some liquid and gaseous properties. The rate of reaction may be considerably accelerated because of (1) the higher gas-phase diffusivity, (2) increase of concentration due to liquidlike density, (3) enhanced solubility, and (4) increase of the specific rate of reaction by pressure. The mole fraction solubility of naphthalene in ethylene at $35^\circ C$ ($95^\circ F$) goes from 0.004 at 20 atm (294 psi) to 0.02 at 100 atm (1470 psi) and 0.05 at 300 atm (4410 psi). High destructive efficiencies (above 99.99 percent) of complex organic pollutant compounds in water can be achieved with residence times of under 5 min at near-critical conditions. The critical properties of water are $374^\circ C$ ($705^\circ F$) and 218 atm (3205 psi).

We are not aware of any industrial implementation of supercritical conditions in reactors. Two areas of potential interest are wastewater treatment (for instance, removal of phenol or organic compounds) and reduction of coke on refining catalysts by keeping heavy oil decomposition products in solution. A pertinent reference is by Kohnstam ("The Kinetic Effects of Pressure," in *Progress in Reaction Kinetics*, Pergamon, 1970). More recent reviews of research progress are by Bruno and Ely (eds.), *Supercritical Fluid Technology*, CRC Press, 1991; Kiran and Brennecke (eds.), *Supercritical Engineering Science*, ACS, 1992.

POLYMERIZATION REACTORS

Polymerization reactors contain one or more phases. There are examples using solvents in which the reactants and products are in the liquid phase, the reactants are fed as a liquid (gas) but the products are solid, or the reactants are a slurry and the products are soluble. Phase transformations can occur, and polymers that form from the liquid phase may remain dissolved in the remaining monomer or solvent, or they may precipitate. Sometimes beads are

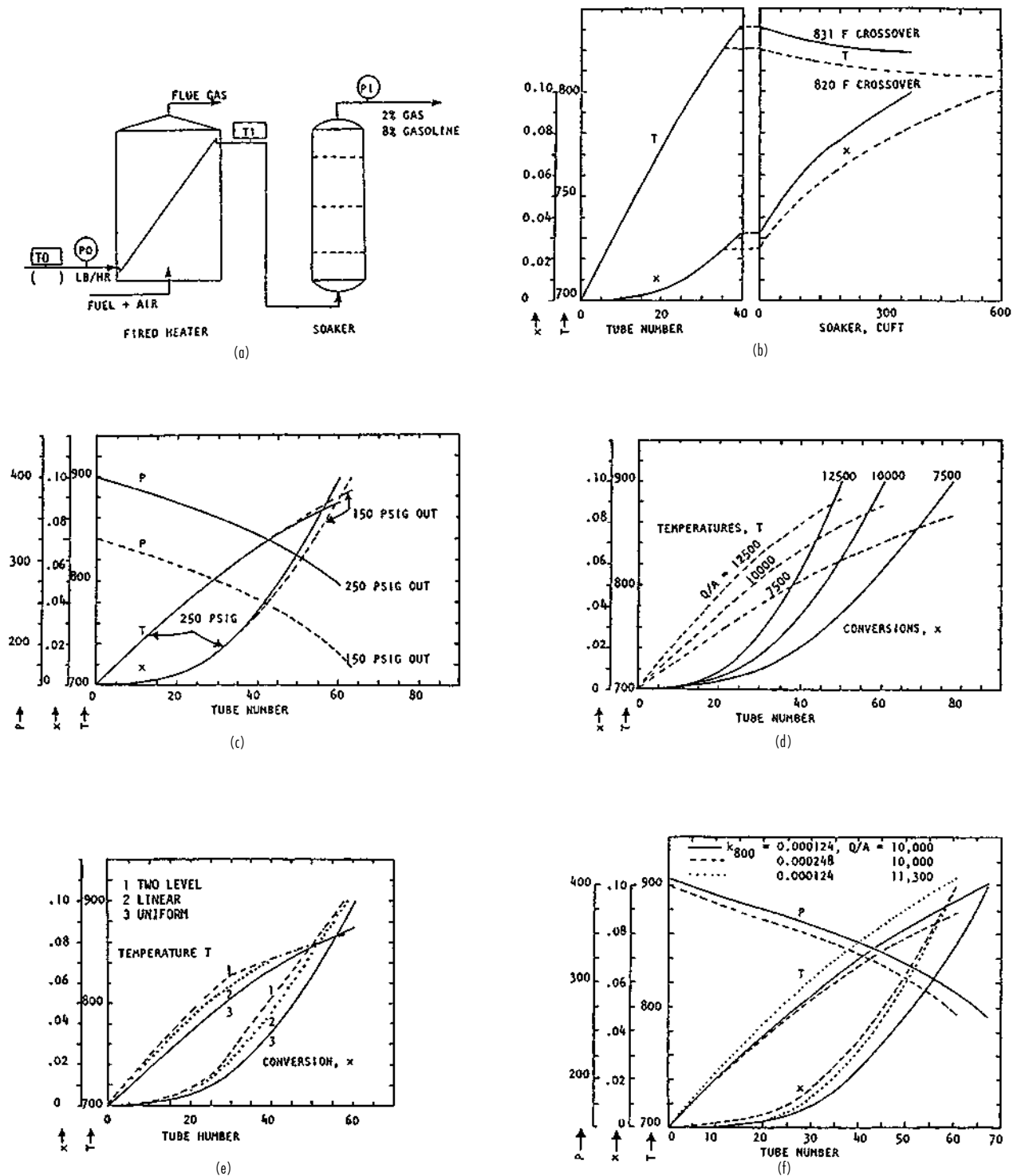


FIG. 19-12 (a) Visbreaking flow sketch, feed 160,000 lbm/h, $k_{800} = 0.000248/s$, tubes 5.05-in ID by 40 ft. (b) $Q/A = 10,000$ Btu/(ft²·h), $P_{out} = 250$ psig. (c) $Q/A = 10,000$ Btu/(ft²·h), $P_{out} = 150$ or 250 psig. (d) Three different heat fluxes, $P_{out} = 250$ psig. (e) Variation of heat flux, average 10,000 Btu/(ft²·h), $P_{out} = 250$ psig. (f) Halving the specific rate. T in °F. To convert psi to kPa, multiply by 6.895; ft to m, multiply by 0.3048; in to cm, multiply by 2.54.

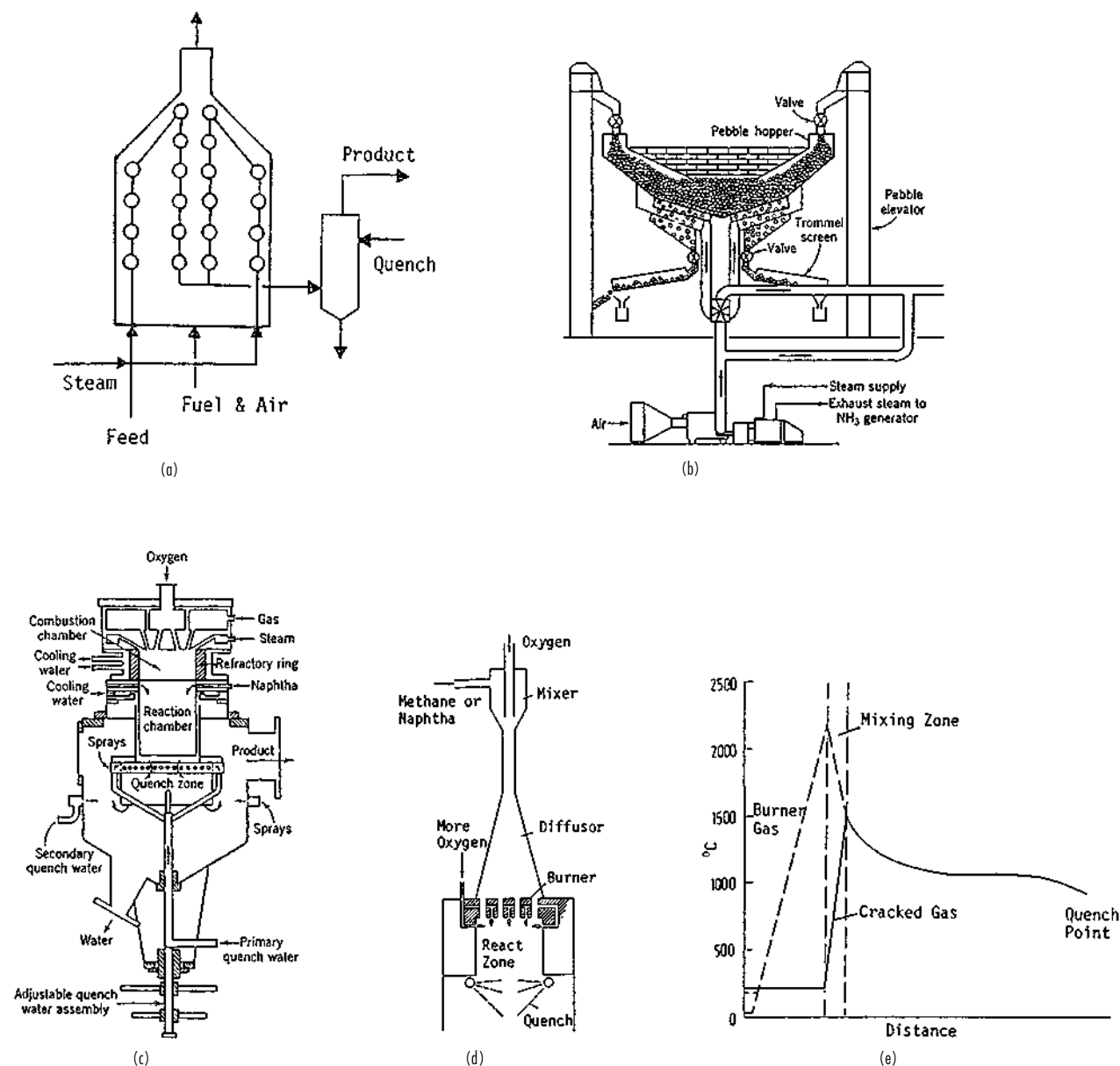


FIG. 19-13 Noncatalytic gas-phase reactions. (a) Steam cracking of light hydrocarbons in a tubular fired heater. (b) Pebble heater for the fixation of nitrogen from air. (c) Flame reactor for the production of acetylene from hydrocarbon gases or naphthas. [Patton, Grubb, and Stephenson, *Pet. Ref.* **37**(11): 180 (1958).] (d) Flame reactor for acetylene from light hydrocarbons (BASF). (e) Temperature profiles in a flame reactor for acetylene (Ullmann *Encyclopadie der Technischen Chemie*, vol. 3, Verlag Chemie, 1973, p. 335).

formed and remain in suspension; sometimes emulsions form. In some processes, solid polymers precipitate from a gas phase into a fluidized bed containing product solids. Polymers are thought of as organic materials; however, inorganic polymers may be also synthesized (e.g., using crystallization and precipitation). Examples of inorganic polymers are zeolites.

The structure of the polymer determines its physical properties, e.g., crystallinity, refractive index, tensile strength, glass transition temperature (at which the specific volume changes slope), and processability. The average molecular weight can cover a wide range between 10^4 to 10^7 . Given the change in molecular weight, the viscosity can change dramatically as conversion increases. For example,

in styrene polymerization, the viscosity increases by a factor of 10^6 as conversion increases from 0 to 60 percent. Initiators of chain polymerization reactions have concentration as low as 10^{-8} g-mol/L so they are highly sensitive to small concentrations of poisons and impurities. The reaction time can also vary. Reaction times for butadiene-styrene rubbers are 8 to 12 h; polyethylene molecules continue to grow for 30 min, whereas ethyl acrylate in 20 percent emulsion reacts in less than 1 min, so monomer must be added gradually to keep the temperature within limits. In some cases, the adiabatic temperature rise may be very high. For example, in polymerization of ethylene, a high adiabatic temperature rise may lead to reactor safety issues by initiating runaway ethylene decomposition reactions. The reactor

19-24 REACTORS

operating conditions have to be controlled such that the possibility of *ethylene decomposition* is eliminated.

Since it is impractical to fractionate the products and reformulate them into desirable ranges of molecular weights, immediate attainment of desired properties must be achieved through the correct choice of reactor type and operating conditions, notably of distributions of residence time and temperature. Reactor selection may be made on rational grounds, for historical reasons, or to obtain a proprietary position.

Each reactor is designed based on the need for mass transfer, heat transfer, and reaction. Stirred batch (autoclave) and continuous tubular reactors are widely used because of their flexibility. In stirred tanks, ideal mixing is typically not achieved, wide variations in temperatures may result, and stagnant zones and bypassing may exist. Devices that counteract these unfavorable characteristics include inserts that cause radial mixing, scraping impellers, screw feeders, hollow-shaft impellers (with coolant flow through them), recirculation using internal and external draft tubes, and so on. The high viscosity of bulk and melt polymerization

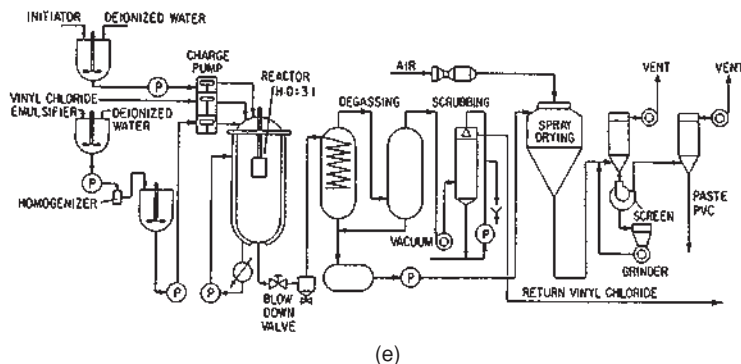
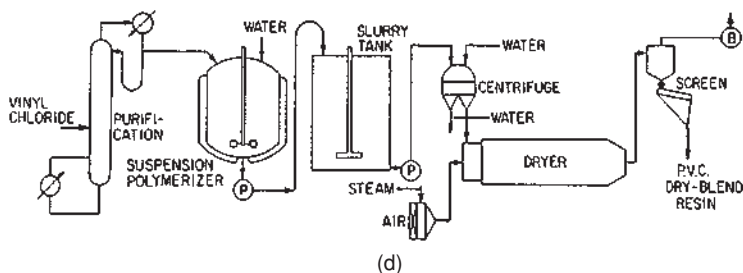
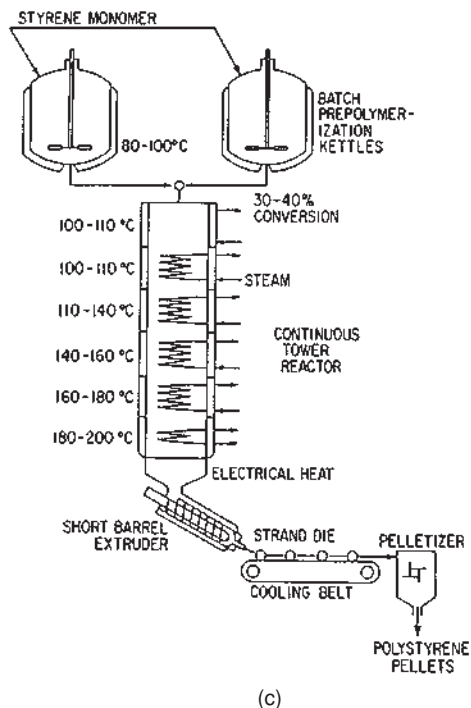
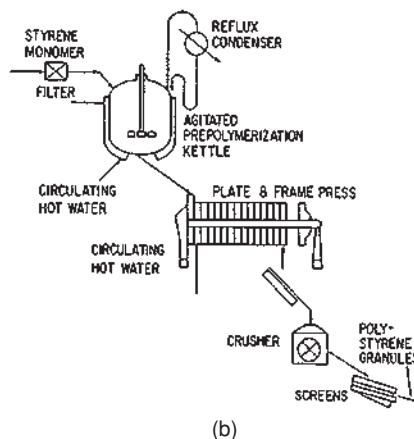
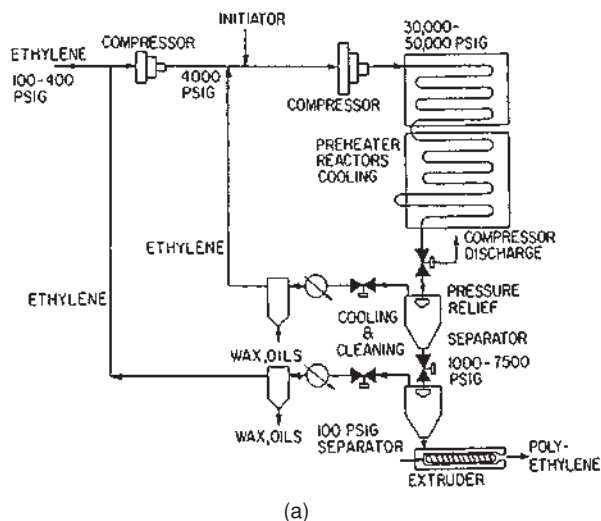


FIG. 19-14 Batch and continuous polymerizations. (a) Polyethylene in a tubular flow reactor, up to 2 km long by 6.4-cm ID. (b) Batch process for polystyrene. (c) Batch-continuous process for polystyrene. (d) Suspension (bead) process for polyvinylchloride. (e) Emulsion process for polyvinylchloride. (Ray and Laurence, in Lapidus and Amundson (eds.), *Chemical Reactor Theory Review*, Prentice-Hall, 1977.)

reactions is avoided with solution, bead, or emulsion polymerization, and more favorable RTDs are obtained. In tubular reactors, such as for low-density polyethylene production, there are strong temperature gradients in the radial direction and cooling may become an issue. These reactors are operated in a single phase, often with multiple catalyst injection points, and the reactor can be several miles in length. Examples of polymerization reactors are illustrated in Fig. 19-14.

FLUID-SOLID REACTORS

A number of industrial reactors involve contact between a fluid (either a gas or a liquid) and solids. In these reactors, the fluid phase contacts the solid catalyst which may be either stationary (in a fixed bed) or in motion (particles in a fluidized bed, moving bed, or a slurry). The solids may be a catalyst or a reactant (product). Catalyst and reactor selection and design largely depend upon issues related to heat transfer, pressure drop and contacting of the phases. In many cases, continuous regeneration or periodic replacement of deteriorated or deactivated catalyst may be needed.

HETEROGENEOUS CATALYSTS

Solid catalysts may have a homogeneous catalyst (or enzyme) or catalytic ingredients dispersed on a support. The support may be organic or inorganic in nature. For example, a catalyst metal atom may be anchored to the polymer (e.g., polystyrene) through a group that is chemically bound to the polymer with a coordinating site such as $-P(C_6H_5)_2$ or $-C_5H_4$ (cyclopentadienyl). Immobilized catalysts have applications in hydrogenation, hydroformylation, and polymerization reactions [Lieto and Gates, *ChemTech*, pp. 46–53 (Jan. 1983)]. Metal or mixed metal oxides may be dispersed on amorphous materials (such as carbon, silica, or alumina) or exchanged into the cages of a zeolite. Expensive catalytic metal ingredients, such as Pt or Pd, may be <1 percent of catalyst weight. Catalysts may be shaped as monoliths, shaped pellets, spheres, or powders. Some exceptions are bulk catalysts such as platinum gauzes for the oxidation of ammonia and synthesis of hydrogen cyanide, which are in the form of several layers of fine-mesh catalyst gauze.

The catalyst support may either be inert or play a role in catalysis. Supports typically have a high internal surface area. Special shapes (e.g., trilobed particles) are often used to maximize the geometric surface area of the catalyst per reactor volume (and thereby increase the reaction rate per unit volume for diffusion-limited reactions) or to minimize pressure drop. Smaller particles may be used instead of shaped catalysts; however, the pressure drop increases and compressor costs become an issue. For fixed beds, the catalyst size range is 1 to 5 mm (0.04 to 0.197 in). In reactors where pressure drop is not an issue, such as fluidized and transport reactors, particle diameters can average less than 0.1 mm (0.0039 in). Smaller particles improve fluidization; however, they are entrained and have to be recovered. In slurry beds the diameters can be from about 1.0 mm (0.039 in) down to 10 μ m or less.

The support has an internal pore structure (i.e., pore volume and pore size distribution) that facilitates transport of reactants (products) into (out of) the particle. Low pore volume and small pores limit the accessibility of the internal surface because of increased diffusion resistance. Diffusion of products outward also is decreased, and this may cause product degradation or catalyst fouling within the catalyst particle. As discussed in Sec. 7, the effectiveness factor η is the ratio of the actual reaction rate to the rate in the absence of any diffusion limitations. When the rate of reaction greatly exceeds the rate of diffusion, the effectiveness factor is low and the internal volume of the catalyst pellet is not utilized for catalysis. In such cases, expensive catalytic metals are best placed as a shell around the pellet. The rate of diffusion may be increased by optimizing the pore structure to provide larger pores (or macropores) that transport the reactants (products) into (out of) the pellet and smaller pores (micropores) that provide the internal surface area needed for effective catalyst dispersion. Micropores typically have volume-averaged diameters of 50 to

200 Å with macropore diameters of 1000 to 5000 Å. The pore volume and the pore size distribution within a porous support determine its surface area. The surface area of supports can range from 0.06 m²/mL (18,300 ft²/ft³) to 600 m²/mL (1.83×10^8 ft²/ft³) and above. Higher pore volume catalysts have higher diffusion rate at the expense of reduced crush strength and increased particle attrition.

The effective diffusion coefficient D_{eff} determines the rate of diffusion and therefore the volume of the catalyst utilized. The coefficient is determined by the nature of the diffusing species and the pore structure of the catalyst. It has been found to be directly proportional to the product of diffusivity and porosity ϵ and inversely proportional to the tortuosity τ (that is empirically determined). In large pores of >1000 Å, where molecules collide with one another and the interaction with the pore walls is minimal, molecular (or bulk) diffusion is important. For pore diameters in the range of 50 to 200 Å, collision with the pore walls becomes more important, and this regime is called the Knudsen diffusion regime. In an extreme case where the size of the molecule is comparable to the size of the pore, the size and configuration of the pores themselves affect diffusivity. This happens when the diffusing molecule is very large (as in transporting large organometallic molecules through catalyst pores in heavy oil hydrotreating) or the pore is very small (as in diffusion in zeolites), or both (e.g., see Sec. 7 for diffusion regimes). ϵ ranges from 0.1 to 0.5 and τ ranges from 1 to 7. In the absence of other information, a τ value of 3 to 4 may be used; however, it is best measured for the catalyst of interest. Expressions for estimating the effective diffusion coefficient are available in textbooks such as Satterfield (*Heterogeneous Catalysis in Practice*, McGraw-Hill, 1991).

The effectiveness factor η is the ratio of the rate of reaction in a porous catalyst to the rate in the absence of diffusion (i.e., under bulk conditions). The theoretical basis for η in a porous catalyst has been discussed in Sec. 7. For example, for an isothermal first-order reaction

$$r_c = k\eta C_i \quad (19-57)$$

where C_i is the bulk concentration of the reactant. As discussed previously, η is a function of the ratio of the rate of reaction to diffusion, also called the Thiele modulus ϕ . As the rate constant increases, η decreases and eventually reaches an asymptotic value (that depends on ϕ). Under these conditions, $k\eta$ increases as $k^{1/2}$. The role of diffusion and reaction in porous catalysts, however, is more complicated in a case where heat effects are present. In addition to the mass conservation equation around the pellet, an energy balance equation is required. Two additional dimensionless parameters are needed for estimating an effectiveness factor:

$$\beta = -\frac{\Delta H_r D_{eff} C_0}{\lambda T_s} \quad \text{and} \quad \gamma = \frac{E}{RT_s} \quad (19-58)$$

where ΔH_r is the heat of reaction, λ is the thermal conductivity of the catalyst, E is the activation energy, and R is the universal gas constant. The dimensionless parameter β , known as the Prater number, is the ratio of the heat generation to heat conduction within the pellet and is a measure of the intra-particle temperature increase; γ is the dimensionless activation energy for the reaction. For an exothermic reaction, the temperature inside the catalyst pellet is greater than or equal to the surface temperature. The maximum steady-state temperature inside the pellet is $T_s(1 + \beta)$. Figure 19-15 is one of several cases examined by Weisz and Hicks for a first-order reaction in an adiabatic

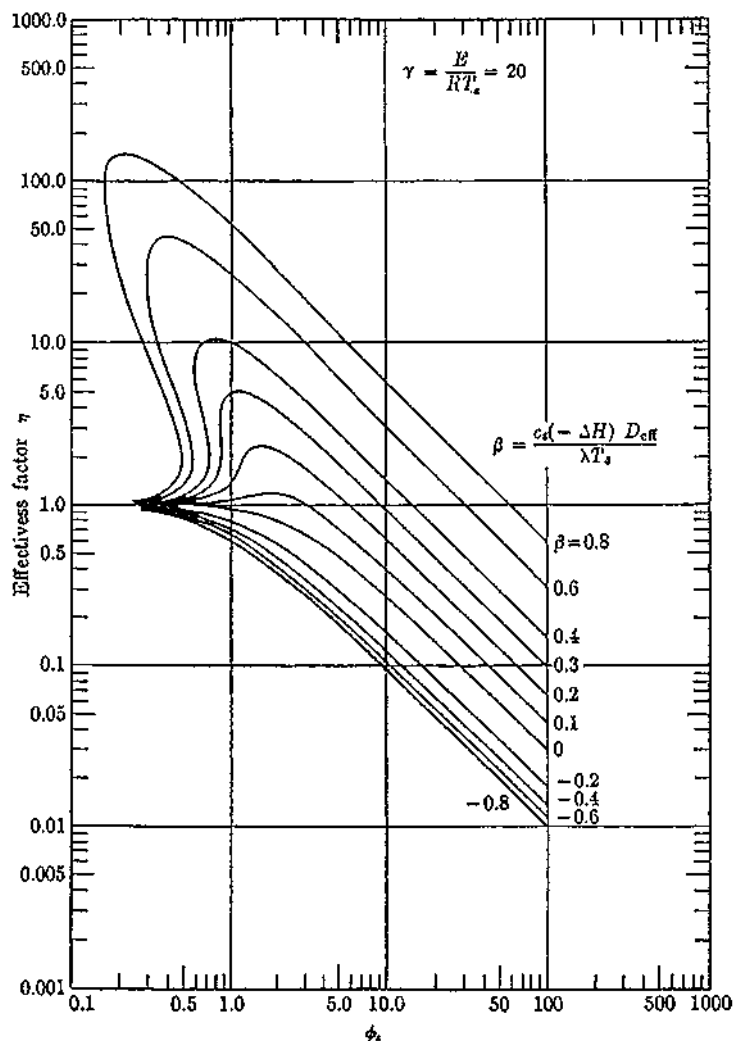


FIG. 19-15 Effectiveness factors versus Thiele modulus for a first-order reaction in spheres under adiabatic conditions. [Weisz and Hicks, *Chem. Eng. Sci.*, **17**: 265 (1962).]

catalyst pellet [*Chem. Eng. Sci.* **17**: 263 (1962)]. Although this predicts some very large values of η in some ranges of the parameters, these values are often not realized in commercial reactors (see Table 19-5). The modified Lewis number defined as $Lw' = \lambda_s / \rho_s C_{ps} D_{\text{eff}}$ can determine the transient temperature inside the pellet, which can be much larger than the steady-state temperature.

The concept of an effectiveness factor is useful in estimating the reaction rate per catalyst pellet (volume or mass). It is, however, mainly useful for simple reactions and simple kinetics. When there are complex reaction pathways, the concept of effectiveness factor is no longer easily applicable, and species and energy balance equations inside the particle may have to be solved to obtain the reaction rates per unit volume of

TABLE 19-5 Parameters of Some Exothermic Catalytic Reactions

Reaction	β	γ	$\gamma\beta$	Lw'	ϕ
NH ₃ synthesis	0.000061	29.4	0.0018	0.00026	1.2
Synthesis of higher alcohols from CO and H ₂	0.00085	28.4	0.024	0.00020	—
Oxidation of CH ₃ OH to CH ₂ O	0.0109	16.0	0.175	0.0015	1.1
Synthesis of vinyl chloride from acetylene and HCl	0.25	6.5	1.65	0.1	0.27
Hydrogenation of ethylene	0.066	23–27	2.7–1	0.11	0.2–2.8
Oxidation of H ₂	0.10	6.75–7.52	0.21–2.3	0.036	0.8–2.0
Oxidation of ethylene to ethylenoxide	0.13	13.4	1.76	0.065	0.08
Dissociation of N ₂ O	0.64	22.0	1.0–2.0	—	1–5
Hydrogenation of benzene	0.12	14–16	1.7–2.0	0.006	0.05–1.9
Oxidation of SO ₂	0.012	14.8	0.175	0.0415	0.9

SOURCE: After Hlavacek, Kubicek, and Marek, *J. Catal.*, **15**, 17, 31 (1969).

catalyst. Dumesic et al. (*The Microkinetics of Heterogeneous Catalysis*, American Chemical Society, 1993) use microkinetic analysis to elucidate reaction pathways of several commercial catalysts.

Another complication is the fact that Fig. 19-15 was developed for the constant-concentration boundary condition, $C|_{r=R} = C_0$. In a more general case, external mass-transfer limitations will need to be included.

$$k_m a (C_0 - C_i) = r_c(C_i) = \eta k C_i \quad (19-59)$$

where k_m is the external mass-transfer coefficient obtained from literature correlations and a is the external surface area per unit pellet volume. The above equation will have to be solved for C_i , the concentration of the reactant on the external surface of the catalyst, so that the rate per pellet can be obtained. The reaction rate per unit reactor volume then becomes $r_c(1 - \epsilon_b)$, where ϵ_b is the bed void fraction.

A further complication is that catalyst activity declines with time. Catalysts may deactivate chemically (via poisons and masking agents), thermally (via support sintering), or mechanically (through attrition). Commercial catalyst life can range from a second to several years. For example, in refinery fluid catalytic cracking, the catalyst may lose most of its activity in less than 10 s, and a transport bed reactor coupled with a fluidized-bed regenerator is used to circulate catalyst. In contrast, a refinery hydroprocessing catalyst deactivates very slowly and a fixed-bed reactor may be used without catalyst replacement for one or more years. The deactivation rate expression may often be inferred from aging experiments undertaken under pilot-plant conditions of constant temperature or conversion. Since accelerated-aging experiments are often difficult (especially when the concentration of reactant or products affects the deactivation rate), reactor designs where the catalyst charge provides the required performance between regeneration cycles is typically based on good basic data and experience. The literature describes approaches aimed at managing deactivation. In the case of platinum reforming with fixed beds, a large recycle of hydrogen prevents coke deposition while a high temperature compensates for the retarding effect of hydrogen on this essentially dehydrogenating process. Fluidized beds are largely isothermal and can be designed for continuous regeneration; however, they are more difficult to operate, require provisions for dust recovery, suffer from backmixing, and are more expensive. Catalyst deactivation mechanisms and kinetics are discussed in detail in Sec. 7 of the Handbook.

A catalyst for a particular chemical transformation is selected using knowledge of similar chemistry and some level on empirical experimentation. Solid catalysts are widely used due to lower cost and ease of separation from the reaction medium. Their drawbacks include a possible lack of specificity and deactivation that can require reactor shutdown for catalyst regeneration or replacement.

There are number of useful books on catalysis. Information on catalysts and processes is presented by Thomas (*Catalytic Processes and Proven Catalysts*, Academic Press, 1970), Pines (*Chemistry of Catalytic Conversions of Hydrocarbons*, Academic Press, 1981), Gates et al. (*Chemistry of Catalytic Processes*, McGraw-Hill, 1979), Matar et al. (*Catalysis in Petrochemical Processes*, Kluwer Academic Publishers, 1989), and Satterfield (*Heterogeneous Catalysis in Industrial Practice*, McGraw-Hill, 1991). The books by Thomas (*Catalytic Processes and Proven Catalysts*, Academic Press, 1970), Butt and Petersen (*Activation, Deactivation and Poisoning of Catalyst*, Academic Press, 1988), and Delmon and Froment (*Catalyst Deactivation*, Elsevier, 1980) provide several examples of catalyst deactivation. Catalyst design is discussed by Trimm (*Design of Industrial Catalysts*, Elsevier, 1980), Hegedus et al. (*Catalyst Design Progress and Perspectives*, Wiley, 1987), and Becker and Pereira (*Catalyst Design*, Marcel Dekker, 1993). A thorough review of catalytic reactions and catalysts arranged according to the periodic table is in a series by Roiter (ed.) (*Handbook of Catalytic Properties of Substances*, in Russian, 1968). Stiles (*Catalyst Manufacture*, Dekker, 1983) discusses catalyst manufacture.

CATALYTIC REACTORS

Due to the considerations noted above, reactor selection will depend on the type of catalyst chosen and its activity, selectivity, and deactivation

behavior. Some reactors with solid catalysts are represented in Fig. 19-16.

Wire Gauzes Wire screens are used for very fast catalytic reactions or reactions that require a bulk noble metal surface for reaction and must be quenched rapidly. The nature and morphology of the gauze or the finely divided catalyst are important in reactor design. Reaction temperatures are typically high, and the residence times are on the order of milliseconds.

Since noble metals are expensive, the catalyst cost is typically high. The physical properties of the gauze pack are important to determine performance, selectivity, and catalyst replacement strategy. The gauze is typically mounted over the top of a heat exchanger tube sheet or over porous ceramic bricks that are laid over the tube sheet. The gauze pack may be covered with a ceramic blanket to minimize radiation losses. From a modeling standpoint, the external surface area per gauze volume and the external mass-transfer coefficient for each component are important parameters, and the reaction rate per unit volume of catalyst may be limited by the rate of external mass transfer. The reaction rate can then be included into a corresponding PFR or dispersion model to obtain estimates of conversion and selectivity.

Examples

- In ammonia oxidation, a 10 percent NH_3 concentration in air is oxidized by flow through a fine-gauze catalyst made of 2 to 10 percent Rh in Pt, 10 to 30 layers, 0.075-mm-diameter (0.0030-in) wire. Contact time is 0.0003 s at 750°C (1382°F) and 7 atm (103 psi) followed by rapid quenching.
- In hydrogen cyanide synthesis using the Andrussov process, air, methane, and ammonia are fed over 15 to 50 layers of noble metal gauze at 1050 to 1150°C at near atmospheric pressure.

Monolith Catalysts For fast reactions that may require a slightly higher residence time than gauzes or that do not benefit from the bulk noble metal gauze structure, monoliths may be used. Most often, the monolith catalyst is an extruded ceramic honeycomb structure that has discrete channels that traverse its length. The catalytic ingredients may be dispersed on a high surface area support and coated on an inert honeycomb. In some cases, the catalyst paste itself may be extruded into a monolith catalyst. Monoliths may also be made of metallic supports. Stainless steel plates (or wire mesh) with ridges may be coated with catalysts and stacked one against the other in a reactor. Corrugated stainless steel layers may alternate in between flat sheets to form the structure. A variant is a stainless steel sheet that is corrugated in a herringbone pattern, coated with catalyst and then rolled (or folded back and forth onto itself) into a reactor module. Examples of cross-sections of the types of monoliths used in industry are shown in Fig. 19-17.

The thickness of monolith walls is adjusted according to the materials of construction (ceramic honeycombs have thicker walls to provide mechanical strength). The size of the channels is selected according to the application. For example, for particulate-laden gases, a larger channel size ceramic monolith and a higher linear velocity allow the particles to pass through the catalyst without plugging the channel. In contrast, for feed that does not contain particles, smaller channel monoliths may be used. The cell density of the monolith may vary between 9 and 600 cells per square inch.

A monolith catalyst has a much higher void fraction (between 65 and 91 percent) than does a packed bed (which is between 36 and 45 percent). In the case of small channels, monoliths have a high geometric surface area per unit volume and may be preferred for mass-transfer-limited reactions. The higher void fraction provides the monolith catalyst with a pressure drop advantage compared to fixed beds.

A schematic of a monolith catalyst is shown in Fig. 19-18a. In cases where pressure drop is limiting, such as for CO oxidation in cogeneration power plant exhausts, monolith catalyst panels may be stacked to form a thin (3- to 4-in-thick) wall. The other dimensions of the wall can be on the order of 35 × 40 ft. CO conversion is over 90 percent with a pressure drop across the catalyst of 1.5 in of water. Alternatively, the monolith may be used as a catalyst and filter, as is the case for a diesel particulate filter. In this case, monolith channels are blocked and the exhaust gases from a diesel truck are forced through the walls (Fig. 19-18b). The filter is a critical component in a continuous regenerable trap. NO in the exhaust

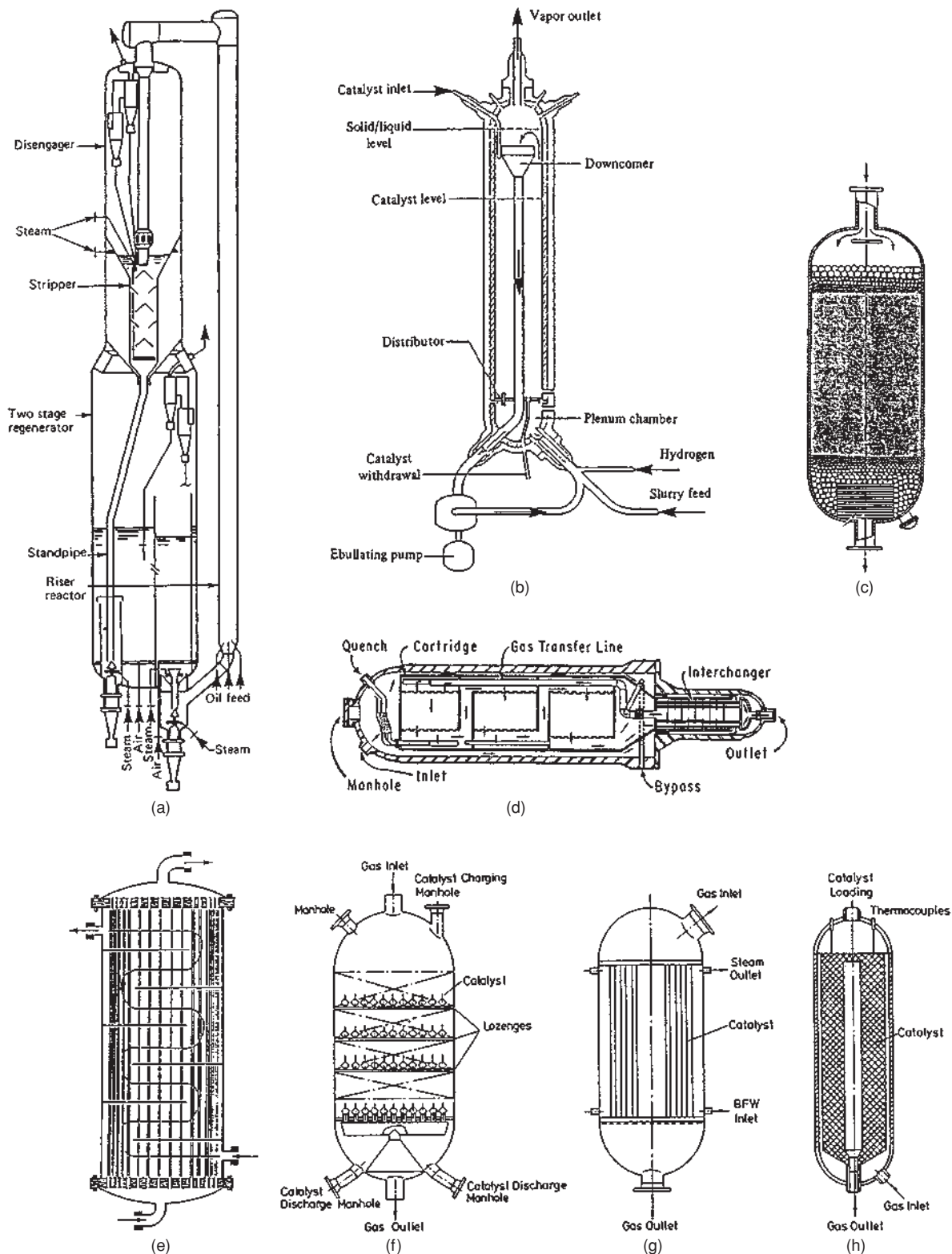


FIG. 19-16 Reactors with solid catalysts. (a) Fluid catalytic cracking riser-regenerator with fluidized zeolite catalyst, 540°C. (b) Ebullating fluidized bed for conversion of heavy stocks to gas and light oils. (c) Fixed-bed unit with support and hold-down zones of larger spheres. (d) Horizontal ammonia synthesizer, 26 m long without the exchanger (*M W Kellogg Co.*). (e) Shell-and-tube vessel for hydrogenation of crotonaldehyde has 4000 packed tubes, 30-mm ID, 10.7 m long [after Berty, in *Leach (ed.)*, *Applied Industrial Catalysis*, vol. 1, Academic Press, 1983, p. 51]. (f), (g), (h) Methanol synthesizers, 50 to 100 atm, 230 to 300°C, Cu catalyst; ICI quench type, Lurgi tubular, Haldor Topsoe radial flow (*Marschner and Moeller*, in *Leach, loc. cit.*). To convert atm to kPa, multiply by 101.3.

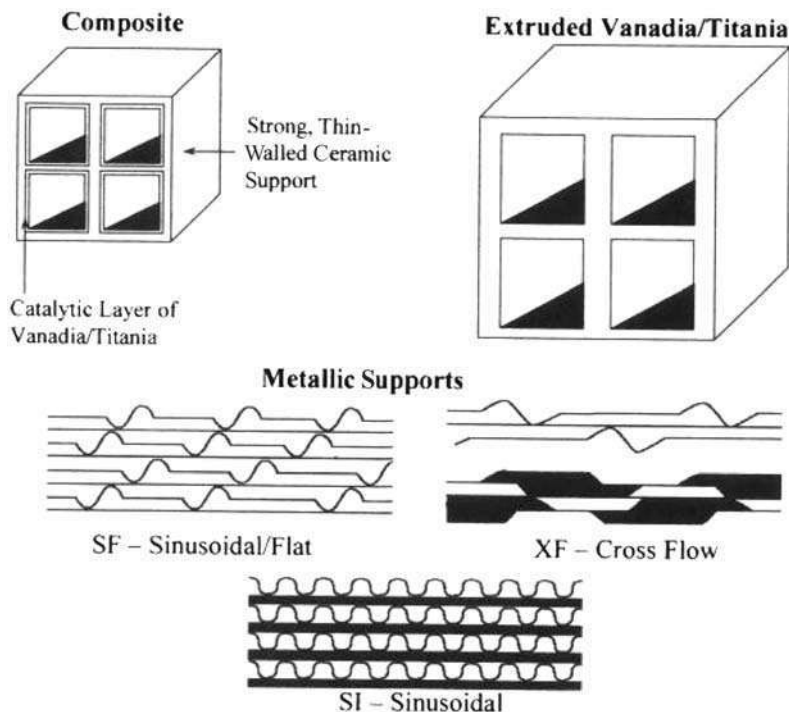


FIG. 19-17 Types of monolith catalysts. (Fig. 12.9 in Heck, Farrauto, and Gulati, *Catalytic Air Pollution Control: Commercial Technology*, Wiley-Interscience, 2002.)

gases is oxidized into NO_2 that reacts with the soot trapped in the walls of the filter to regenerate it in situ.

Modeling considerations for monoliths are similar to those of gauze catalysts; however, since the flow and temperature in each channel may be assumed to be identical to those in the next channel, the solution for a single channel may reflect the performance of the reactor. For an application in which the reaction rate is mass-transfer-limited, the reactant concentration at the wall of the catalyst is much lower than in the bulk and may be neglected. In such a case, the fractional conversion ξ is

$$\xi = 1 - e^{-k_a a t} = 1 - \exp\left(-\frac{\text{Sh } aL}{\text{Sc Re}}\right) \quad (19-60)$$

where $\text{Sh} (= k_m d_{ch}/D)$ is the Sherwood number, $\text{Sc} (= \mu/\rho D)$ is the Schmidt number, and $\text{Re} (= u d_{ch} \rho/\mu)$ is the channel Reynolds number; a is the geometric surface area per unit volume of monolith. A number of correlations for Sh are available for various types of monoliths. For example, in the case of extruded ceramic monoliths, a correlation for estimating the external mass-transfer coefficient is provided by Uberoi and Pereira (*Ind. Eng. Chem. Res.* **35**: 113–116 (1996)):

$$\text{Sh} = 2.696 \left(1 + 0.139 \text{ Sc Re} \frac{d}{L}\right)^{0.81} \quad (19-61)$$

Since typical monolith catalysts have a thin coating of catalytic ingredients on the channel walls, they can be susceptible to poisoning.

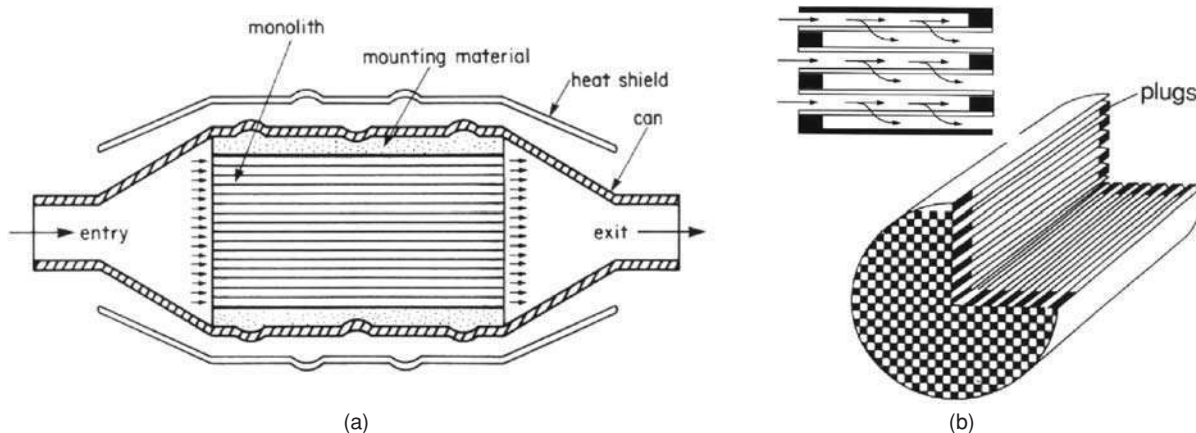


FIG. 19-18 Monolith catalysts. (a) Schematic of an automobile catalytic converter for the three-way removal of CO , hydrocarbons, and NO_x . (b) Schematic of a diesel trap. (Figs. 7.10 and 9.6 in Heck, Farrauto, and Gulati, *Catalytic Air Pollution Control: Commercial Technology*, Wiley-Interscience, 2002.)

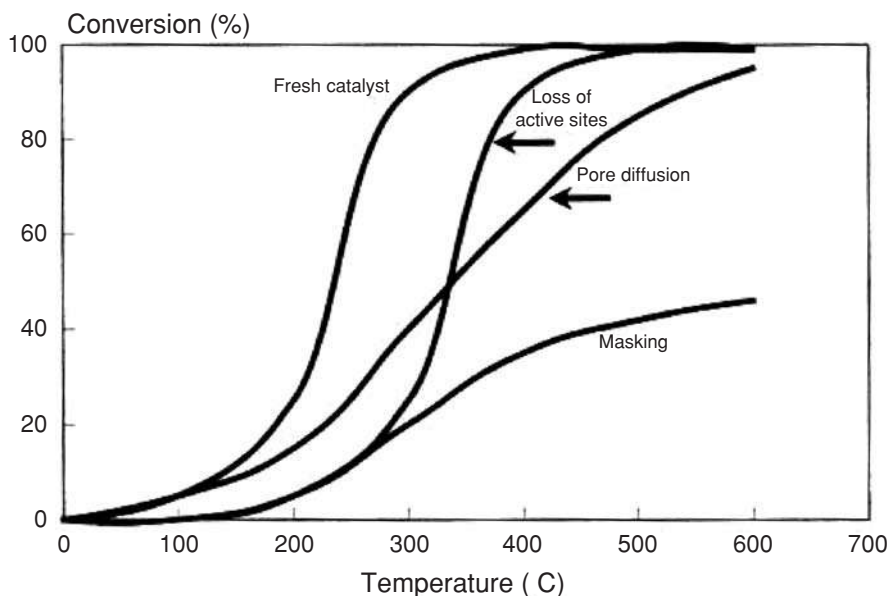


FIG. 19-19 Relative changes in conversion versus temperature behavior for various deactivation models. (Fig. 5.4 in Heck, Farrauto, and Gulati, *Catalytic Air Pollution Control: Commercial Technology*, Wiley-Interscience, 2002.)

The various mechanisms for catalyst poisoning have been discussed in Sec. 7 of the Handbook. The nature and shape of a monolith light-off curve for a facile hydrocarbon oxidation often indicate the poisoning mechanism, as shown in Fig. 19-19. The figure shows the light-off curve for a fresh catalyst. A reduction in the number of active sites (due to either poisoning or sintering of the catalytic metal) results in movement of the curve to the right. In contrast, when the pores within the catalyst become plugged with reactants or products (such as coke), the light-off curve shifts to the right and downward. In the case of deactivation due to masking, the active sites are covered with masking agents that may also plug the pores (such as in the case of silica deposition), resulting in more severe deactivation. Understanding the root cause of deactivation may allow for the design of improved catalysts, contaminant guard beds, catalyst regeneration procedures, and catalyst replacement protocols.

A good reference on monolith applications is by Heck, Farrauto, and Gulati (*Catalytic Air Pollution Control: Commercial Technology*, Wiley-Interscience, 2002).

Examples

- For the control of carbon monoxide, hydrocarbon, and nitrogen oxide emissions from automobiles, oval-shaped extruded cordierite or metal monolith catalysts are wrapped in ceramic wool and placed inside a stainless steel casing (Fig. 19-18a). The catalytic metals are Pt-Rh or Pd-Rh, or combinations. Cell sizes typically range between 400 and 600 cells per square inch. The catalysts achieve over 90 percent reduction in all three pollutants.
- Monolith catalysts are used for the control of carbon monoxide and hydrocarbon (known as volatile organic compounds or VOCs) emissions from chemical plants and cogeneration facilities. In this case, square bricks are stacked on top of one another in a wall perpendicular to the flow of exhaust gases at the appropriate temperature location within the heat recovery boiler. The size of the brick can vary from 6 in (ceramic) to 21 ft (metal). Pt and Pd catalysts are used at operating temperatures between 600 and 1200°F. Cell sizes typically range between 100 and 400 cells per square inch. Typical pressure drop requirements for monoliths are less than 2 in of water.
- *Selective catalytic reduction* (SCR) catalysts are used for controlling nitrogen oxide emissions from power plants. The reducing agent is

ammonia, and the active ingredients are $V_2O_5/WO_3/TiO_2$. Operating temperatures are 300 to 450°C. Cell sizes vary between 9 and 50 cells per square inch. The paper by Beeckman and Hegedus [*Ind. Chem. Eng. Res.* **30**: 969 (1991)] is a good reaction engineering reference on SCR catalysts.

Fixed Beds A fixed-bed reactor typically is a cylindrical vessel that is uniformly packed with catalyst pellets. Nonuniform packing of catalyst may cause channeling that could lead to poor heat transfer, poor conversion, and catalyst deactivation due to hot spots. The bed is loaded by pouring and manually packing the catalyst or by sock loading. As discussed earlier, catalysts may be regular or shaped porous supports, uniformly impregnated with the catalytic ingredient or containing a thin external shell of catalyst. Catalyst pellet sizes usually are in the range of 0.1 to 1.0 cm (0.039 to 0.39 in).

Packed-bed reactors are easy to design and operate. The reactor typically contains a manhole for vessel entry and openings at the top and bottom for loading and unloading catalyst, respectively. A metal support grid is placed near the bottom, and screens are placed over the grid to support the catalyst and prevent the particles from passing through. In some cases, inert ceramic balls are placed above and below the catalyst bed to distribute the feed uniformly and to prevent the catalyst from passing through, respectively. One has to guard the bed from sudden pressure surges as they can disturb the packing and cause maldistribution and bypassing of feed.

As discussed earlier, heat management is an important issue in the design of fixed-bed reactors. A series of adiabatic fixed beds with interbed cooling (heating) may be used. For very highly exothermic (endothermic) reactions, a multitubular reactor with catalyst packed inside the tubes and cooling (heating) fluids on the shell side may be used. The tube diameter is typically greater than 8 times the diameter of the pellets (to minimize flow channeling), and the length is limited by allowable pressure drop. The heat transfer required per volume of catalyst may impose an upper limit on diameter as well. Multitubular reactors require special procedures for catalyst loading that charge the same amount of catalyst to each tube at a definite rate to ensure uniform loading, which in turn ensures uniform flow distribution from the common header. After filling, each tube is checked for pressure

drop. In addition to the high surface area for heat transfer/volume, the advantage of a multitubular fixed-bed reactor is its easy scalability. A bench-scale unit can be a full-size single tube, a pilot plant can be several dozen tubes, and a large-scale commercial reactor can have thousands of tubes. Disadvantages include high cost and a limit on maximum size (tube length and diameter, and number of tubes).

As discussed in Sec. 7, the intrinsic reaction rate and the reaction rate per unit volume of reactor are obtained based on laboratory experiments. The kinetics are incorporated into the corresponding reactor model to estimate the required volume to achieve the desired conversion for the required throughput. The acceptable pressure drop across the reactor often can determine the reactor aspect ratio. The pressure drop may be estimated by using the Ergun equation

$$\frac{\Delta P}{L} = \frac{150u_0\mu}{(\phi_p d_p)^2} \frac{(1-\epsilon_b)^2}{\epsilon_b^3} + \frac{1.75\rho u_0^2}{\phi_p d_p} \left(\frac{1-\epsilon_b}{\epsilon_b^3} \right) \quad (19-62)$$

where u_0 is the superficial velocity, ϵ_b is the bed porosity, ϕ_p is the shape factor, and d_p is the particle diameter. Correlations that provide estimates for the heat-transfer and mass transport properties are available in the literature. For example, if the dispersion model is used to simulate concentration and temperature profiles along the reactor, the axial dispersion coefficient may be estimated from Wen [in Petho and Noble (eds.), *Residence Time Distribution Theory in Chemical Engineering*, Verlag Chemie, 1982].

$$\frac{1}{Pe} = \frac{0.3}{(Re)(Sc)} + \frac{0.5}{3.8} \quad 0.008 \leq Re \leq 400 \quad \text{and} \quad 0.28 \leq Sc \leq 2.2 \quad (19-63)$$

where $Pe = d_p u_0 / (\epsilon_b D_e)$, $Re = d_p \rho u_0 / \mu$, u_0 is the superficial velocity, and d_p is the particle diameter.

Mathematical Models Catalytic packed-bed reactors are used for exothermic (e.g., hydrogenations, Fischer-Tropsch synthesis, oxidations) and endothermic (e.g., steam reforming and ammonia synthesis) reactions. The two primary modes of heat management are (1) adiabatic operation usually in a single or series of packed zones, the latter with interstage cooling or heating and (2) multitubular reactors with cooling (e.g., shell-and-tube heat exchange with a coolant) or heating (e.g., locating the tubes in a furnace with heat supplied by combustion of a fuel). Other more complex schemes can include heat exchange between the feed and the effluent, reverse flow operation, etc., and are discussed in the multifunctional reactors section.

The mechanism for heat transfer includes the following steps: (1) conduction in the catalyst particle; (2) convection from the particle to the gas phase; (3) conduction at contact points between particles; (4) convection between the gas and vessel wall; (5) radiation heat transfer between the particles, the gas, and the vessel wall; (6) conduction in the wall; and (7) convection to the coolant. There are a number of ways, through reactor models, that these steps are correlated to provide design and analysis estimates and criteria for preventing runaway in exothermic reactors.

The temperature profile depends on the relative rates of heat generation by reaction and heat transfer. The temperature rise or drop in a reactor affects catalyst life, product selectivity, and equilibrium conversion, and excessive heat release can lead to reaction runaway. Hence, reactor design and analysis requires good understanding of the coupling of reaction and heat transfer. Mathematical models for fixed-bed reactors can vary in the level of detail depending on the end use. For more details see Froment and Bischoff (*Chemical Reactor Analysis and Design*, Wiley, 1990) and Harriott (*Chemical Reactor Design*, Marcel Dekker, 2003).

Homogeneous one-dimensional model This is the simplest description of a packed bed, with an overall heat-transfer coefficient U . The particle and gas temperatures are identical, and only axial variation in temperature is considered, giving the following mass and energy balance equations for any species C_i :

$$\frac{d(uC_i)}{dz} = \sum_j v_{ij} r_j \quad C_i = C_{i0} \quad \text{at } z = 0$$

$$u\rho c_p \frac{dT}{dz} = \sum_j v_{ij} (-\Delta H)_{Tj} - \frac{4U}{d_R} (T - T_c) \quad (19-64)$$

$$T = T_0 \quad \text{at } z = 0$$

Equation (19-64) is similar to generic PFR Eqs. (19-17) and (19-18). The overall heat-transfer coefficient U is based on the bed side heat-transfer area A_R and includes three terms: heat transfer on the bed side b , thermal conduction in the vessel wall w , and heat transfer on the coolant side c :

$$\frac{1}{U} = \frac{1}{h_b} + \frac{d_R}{k_w} \frac{A_R}{A_m} + \frac{1}{h_c} \frac{A_R}{A_c} \quad (19-65)$$

$$A_m = \log \text{mean}(A_R, A_c) = \frac{A_c - A_R}{\ln(A_c/A_R)}$$

Here, h_i are the heat-transfer coefficients in the bed side and the coolant side, k_w is the wall thermal conductivity, and A_i are the heat-transfer areas. The coolant side heat-transfer coefficient can be obtained from general heat-transfer correlations in tubes (see any heat-transfer text and the relevant sections in this Handbook). For the process-side heat-transfer coefficient, there is a large body of literature with a variety of correlations. There is no clear advantage of one correlation over another, as these depend on the particle and fluid properties, temperature range, etc.; e.g., see the correlation of Leva, *Chem. Eng.* **56**: 115 (1949):

$$Nu_R = \begin{cases} 0.813 Re_p^{0.9} e^{6d_p/d_s} & \text{for heating} \\ 3.5 Re_p^{0.7} e^{4.6d_p/d_s} & \text{for cooling} \end{cases} \quad (19-66)$$

$$Re_p = \frac{\rho u d_p}{\mu} \quad Nu_R = \frac{h_i d_R}{\lambda_f}$$

The one-dimensional homogeneous model is useful for first-order estimates and when lab (or pilot-plant) data for the same diameter tube are available. This simple model does not provide information on the effect of the tube diameter on the effective radial temperature gradients.

Fixed-bed reactors may exhibit axial dispersion. If axial dispersion is important for reactor simulation, analysis, or design, a variant of the one-dimensional homogeneous model that contains an axial dispersion term may be used. Approximate criteria to determine if mass and heat axial dispersion have to be considered are available (see, e.g., Froment and Bischoff, *Chemical Reactor Analysis and Design*, Wiley, 1990).

Homogeneous two-dimensional model This model accounts for radial variation of composition and temperature in the bed that may be present for large heats of reaction. Corresponding material and energy balances are:

$$D_{er} \left(\frac{\partial^2 C_i}{\partial r^2} + \frac{1}{r} \frac{\partial C_i}{\partial r} \right) - \frac{\partial(uC_i)}{\partial z} + \sum_j v_{ij} r_j = 0$$

$$k_{er} \left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} \right) - u\rho c_p \frac{dT}{dz} - \sum_j v_{ij} (-\Delta H)_{Tj} = 0 \quad (19-67)$$

$$C_i = C_{i0} \quad T = T_0 \quad \text{at } z = 0$$

$$\frac{\partial C_i}{\partial r} = 0 \quad \frac{\partial T}{\partial r} = 0 \quad \text{at } r = 0$$

$$\frac{\partial C_i}{\partial r} = 0 \quad \frac{\partial T}{\partial r} = -\frac{h_w}{k_{er}} (T_R - T_w) \quad \text{at } r = R$$

The effective radial diffusivity D_{er} is normally different from the axial diffusivity. It is often safe to neglect the radial variation of species concentration due to the relatively fast radial mixing. The effective conductivity k_{er} has to be determined from heat-transfer experiments preferably with the actual bed and fluids. This coefficient can be

either a constant (averaged radially) or a function of the radial position with a higher value for the core and a lower value near the wall due to different velocity and void fraction near the wall. There are a number of correlations for k_{er} and h_w , and there is significant variability in their utility; e.g., see the correlation of De Wash and Froment, *Chem. Eng. Sci.* **27**: 567 (1972):

$$k_{er} = k_{er0} + \frac{0.0105 \text{ Re}}{1 + 46 (dp/d_R)^2} \quad (19-68)$$

$$h_w = h_{w0} + 0.0481 \text{ Re} \frac{d_R}{d_p}$$

The static contribution k_{er0} incorporates heat transfer by conduction and radiation in the fluid present in the pores, conduction through particles, at the particle contact points and through stagnant fluid zones in the particles, and radiation from particle to particle. Figure 19-20 compares various literature correlations for the effective thermal conductivity and wall heat-transfer coefficient in fixed beds [Yagi and Kunii, *AIChE J.* **3**: 373(1957)].

The two-dimensional model can be used to develop an equivalent one-dimensional model with a bed-side heat-transfer coefficient defined as [see, e.g., Froment, *Chem. Eng. Sci.* **7**: 29 (1962)]

$$\frac{1}{h_b} = \frac{1}{h_w} + \frac{R}{4k_{er}} \quad (19-69)$$

The objective is to have the radially averaged temperature profile of the 2D model match the temperature profile of the 1D model.

Heterogeneous one-dimensional model The heterogeneous model allows resolution of composition and temperature differences between the catalyst particle and the fluid.

$$\frac{d(uC_i)}{dz} = k_G a (C_i - C_{is})$$

$$k_G a (C_i - C_{is}) = \sum_j v_{ij} r_j$$

$$u\rho c_p \frac{dT}{dz} = ha(T_s - T) - \frac{4U}{d_R} (T - T_c) \quad (19-70)$$

$$ha(T_s - T) = \sum_j v_{ij}(-\Delta H)_j r_j$$

$$C_i = C_{i0} \quad T = T_0 \quad \text{at } z = 0$$

Mears developed a criterion that provides conditions for limiting interphase temperature gradients [Mears, *J. Catal.* **20**: 127 (1971) and *I&EC Proc. Des. Dev.* **10**: 541 (1971)]

$$\frac{d_p \sum_j v_{ij}(-\Delta H)_j r_j}{2hT} < 0.15 \frac{RT}{E} \quad (19-71)$$

An extension of this one-dimensional heterogeneous model is to consider intraparticle diffusion and temperature gradients, for which the lumped equations for the solid are replaced by second-order diffusion/conduction differential equations. Effectiveness factors can be used as applicable and discussed in previous parts of this section and in Sec. 7 of this Handbook (see also Froment and Bischoff, *Chemical Reactor Analysis and Design*, Wiley, 1990).

Typically the interphase temperature gradients are substantially smaller than the radial and axial temperature gradients, being on the order of 1 to 3°C, and can often be neglected.

Heterogeneous two-dimensional model Two-dimensional heterogeneous models have been developed, e.g., De Wash and Froment, *Chem. Eng. Sci.* **27**:567 (1972). Figure 19-21 compares the various models. The results indicate that the homogeneous and heterogeneous models predict similar temperature profiles; however, the heterogeneous model contains additional information on interparticle concentration and temperature gradients that may be useful in catalyst or reactor design. The 2D models predict substantially higher peak temperatures than the corresponding 1D models. The pseudohomogeneous 2D model may contain valuable information on radial temperature profiles, especially in the case of exothermic reactions. The heterogeneous 2D model also contains additional radial interparticle mass and heat-transfer information. The heterogeneous 2D model with no heat transfer through the solid shows

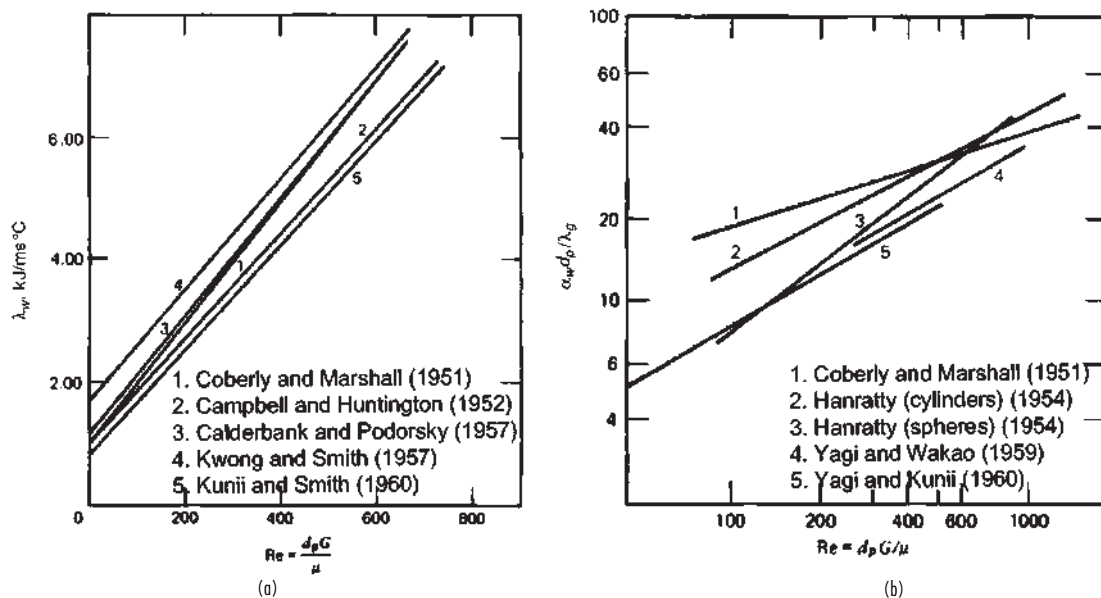


FIG. 19-20 Thermal conductivity and wall heat transfer in fixed beds. (a) Effective thermal conductivity. (b) Nusselt number for wall heat transfer. (Figs. 11.7.1-2 and 11.7.1-3 in Froment and Bischoff, *Chemical Reactor Analysis and Design*, Wiley, 1990.)

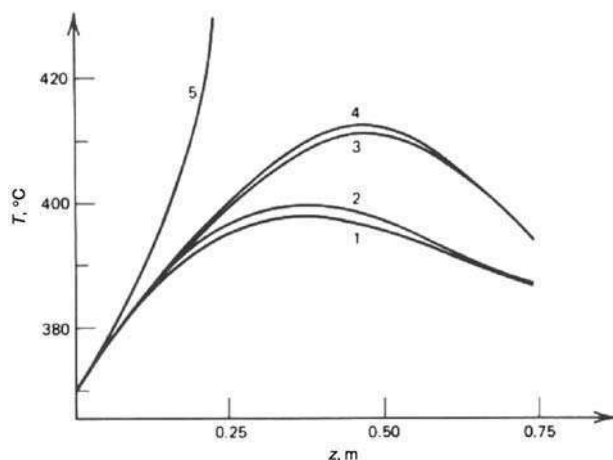


FIG. 19-21 Comparison of model predictions for radial mean temperature as a function of bed length. (1) Basic pseudohomogeneous one-dimensional model. (2) Heterogeneous model with interfacial gradients. (3) Pseudohomogeneous two-dimensional model. (4) Two-dimensional heterogeneous model with appropriate boundary conditions. (5) Two-dimensional heterogeneous model with no heat transfer through solid. (Fig. 11.10-1 in Froment and Bischoff, *Chemical Reactor Analysis and Design*, Wiley, 1990.)

a very steep temperature rise. This case illustrates the notion that a reasonably complicated model may indeed provide unrealistic results if inappropriate assumptions are made.

Examples

- Oxidation of SO_2 in large adiabatic packed-bed reactors (Fig. 19-22a). The catalyst is nominally $\frac{1}{4}$ -in Cs-promoted $\text{V}_2\text{O}_5/\text{SiO}_2$ pellets. The inlet temperature is between 350 and 450°C. The temperature increase across the bed is 50 to 100°C. The oxidation is thermodynamically limited, with lower temperatures favoring SO_3 . After each bed, the exhaust is cooled via heat exchange or injection of a cold shot. The advantage of interstage cooling is shown in Table 19-2.
- Phosgene synthesis from CO and Cl_2 in a multitubular reactor (Fig. 19-22b). The activated carbon catalyst is packed inside the tubes with water on the shell side. Reaction by-products include CCl_4 . The temperature profile in a tube (shown in the figure) is characterized by a hot spot. The position of the hot spot moves toward the exit of the reactor as the catalyst deactivates.
- Production of cumene from benzene and propylene using a phosphoric acid on quartz catalyst (Fig. 19-22c). There are four reactor beds with interbed cooling with cold feed. The reactor operates at 260°C.
- Vertical ammonia synthesizer at 300 atm with five cold shots and an internal exchanger (Fig. 19-22d). The nitrogen and hydrogen feeds are reacted over an Al_2O_3 -promoted spongy iron catalyst. The concentration of ammonia is also shown in the figure.
- Vertical methanol synthesizer at 300 atm (Fig. 19-22e). A Cr_2O_3 -ZnO catalyst is used with six cold shots totaling 10 to 20 percent of the fresh feed.
- Methanol is oxidized to formaldehyde in a thin layer of finely divided silver or a multilayer screen, with a contact time of 0.01 s at 450 to 600°C (842 to 1112°F). A shallow bed of silver catalyst is also used in the DuPont process for in situ production of methyl isocyanate by reacting monomethylformamide and oxygen at 500°C.
- The Sohio process for vapor-phase oxidation of propylene to acrylic acid uses two beds of bismuth molybdate at 20 to 30 atm (294 to 441 psi) and 290 to 400°C (554 to 752°F).
- Oxidation of ethylene to ethylene oxide also is done in two stages with supported silver catalyst, the first stage to 30 percent conversion, the second to 76 percent, with a total of 1.0-s contact time.
- Steam reforming reactors have the supported nickel catalyst packed in tubes and the endothermic heat of reaction supplied from a

furnace on the shell side. The feed is natural gas (or naphtha) and water vapor heated to over 800°C (1056°F).

- Vinyl acetate reactors that have a supported Pd/Au catalyst packed in ~25-mm (0.082-ft) ID tubes and exothermic heat of reaction removed by generating steam on the shell side. The feed contains ethylene, oxygen, and acetic acid in the vapor phase at 150 to 175°C (302 to 347°F).
- Maleic anhydride is made by oxidation of benzene with air above 350°C (662°F) with V-Mo catalyst in a multitubular reactor with 2-cm tubes. The heat-transfer medium is a molten salt eutectic mixture at 375°C (707°F). Even with small tubes, the heat transfer is so limited that a peak temperature 100°C (212°F) above the shell side is developed and moves along the tubes.
- Butanol is made by the hydrogenation of crotonaldehyde in a reactor with 4000 tubes, 28-mm (0.029-ft) ID by 10.7 m (35.1 ft) long [Berty, in Leach (ed.), *Applied Industrial Catalysis*, vol. 1, Academic Press, 1983, p. 51].
- Vinyl chloride is made from ethylene and chlorine with Cu and K chlorides. The Stauffer process employs three multitubular reactors in series with 25-mm (0.082-ft) ID tubes [Naworski and Velez, in Leach (ed.), *Applied Industrial Catalysis*, vol. 1, Academic Press, 1983, p. 251].

Moving Beds In a moving-bed reactor, the catalyst, in the form of large granules, circulates by gravity and gas lift between reaction and regeneration zones (Fig. 19-23). The first successful operation was the Houdry cracker that replaced a plant with fixed beds that operated on a 10-min cycle between reaction and regeneration. Handling of large (hot) solids is difficult. The Houdry process was soon made obsolete by FCC units. The only currently publicized moving-bed process is a UOP platinum reformer (Fig. 19-23c) that regenerates a controlled quantity of catalyst on a continuous basis.

Fluidized Beds Fluidized beds are reactors in which small particles (with average size below 0.1 mm) are fluidized by the reactant gases or liquids. When the linear velocity is above the minimum required for fluidization, a dense fluidized bed is obtained. As the superficial velocity increases, the bed expands and becomes increasingly dilute. At a high enough linear velocity, the smallest particles entrain from the bed and have to be separated from the exhaust gases and recycled.

Advantages of fluidized beds are temperature uniformity, good heat transfer, and the ability to continuously remove catalyst for regeneration. Disadvantages are solids backmixing, catalyst attrition, and recovery of fines. Baffles have been used often to reduce backmixing.

Fluidized beds contain a bottom support plate over which the solids reside. The reactant gases typically are fed through a sparging system placed very near the bottom of the plate. These reactors employ a wide range of particle sizes and densities. Geldart (*Gas Fluidization Technology*, Wiley, 1986) developed a widely accepted particle classification system based on fluidization characteristics. Type A powders are employed in many refinery and chemical processes, such as catalytic cracking, acrylonitrile synthesis, and maleic anhydride synthesis. Type B powders are also utilized, e.g., in fluidized-bed combustion. The properties of different powders are summarized in the fluidized bed subsection of Sec. 17. Good distributor design and the presence of a substantial fraction of fines (mainly for processes employing group A powders) are essential for good fluidization, to eliminate maldistribution, and for good performance. Internals for heat transfer (e.g., cooling tubes) and other baffling for improved performance provide design challenges as their effect is not yet well understood (in spite of the voluminous literature).

The *particle size distribution* and the linear velocity are important in reactor design. The *minimum fluidization velocity* is the velocity at the onset of fluidization while the *terminal velocity* is the velocity above which a particle can become entrained from the bed. The nature of the particles and the linear velocity determine bed properties such as *gas holdup*, *equilibrium bubble size* (for bubbling systems), *entrainment rate* of particles from the bed, and the flow regime *transition velocities*. The height beyond which the concentration of entrained particles does not vary significantly is called the *transport disengagement height*. Knowledge of this height is required for the design and location of *cyclones* for solids containment. In addition to the velocity and the nature of the particles, the layout of the equipment can determine the *particle attrition rate*.

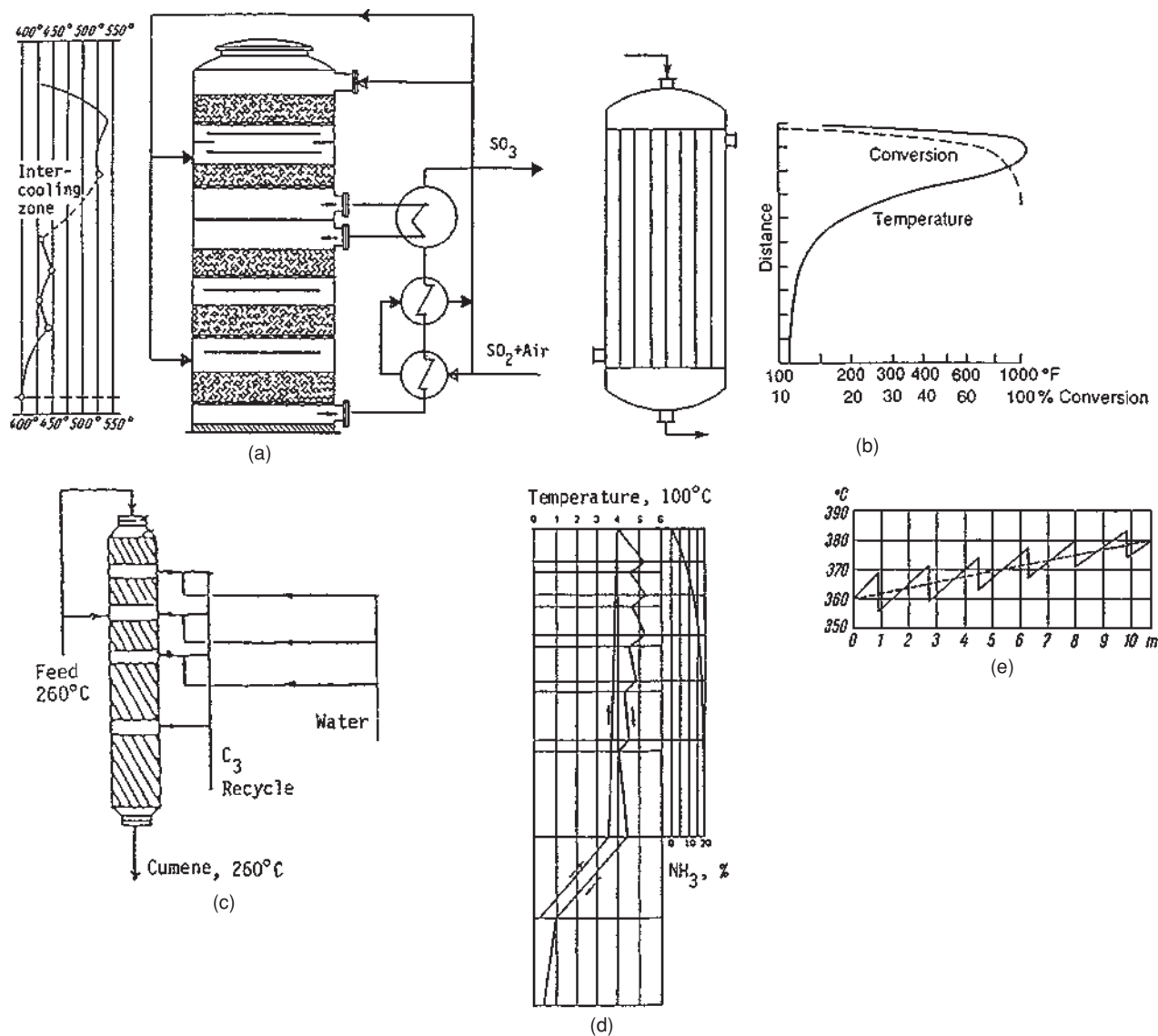


FIG. 19-22 Temperature and composition profiles. (a) Oxidation of SO_2 with intercooling and two cold shots. (b) Phosgene from CO and Cl_2 , activated carbon in 2-in tubes, water-cooled. (c) Cumene from benzene and propylene, phosphoric acid on quartz with four quench zones, 260°C . (d) Vertical ammonia synthesizer at 300 atm, with five cold shots and an internal exchanger. (e) Vertical methanol synthesizer at 300 atm, $\text{Cr}_2\text{O}_3\text{-ZnO}$ catalyst, with six cold shots totaling 10 to 20 percent of the fresh feed. To convert psi to kPa, multiply by 6.895; atm to kPa, multiply by 101.3.

The two-phase theory of fluidization has been extensively used to describe fluidization (e.g., see Kunii and Levenspiel, *Fluidization Engineering*, 2d ed., Wiley, 1990). The fluidized bed is assumed to contain a bubble and an emulsion phase. The bubble phase may be modeled by a plug flow (or dispersion) model, and the emulsion phase is assumed to be well mixed and may be modeled as a CSTR. Correlations for the size of the bubbles and the heat and mass transport from the bubbles to the emulsion phase are available in Sec. 17 of this Handbook and in textbooks on the subject. Davidson and Harrison (*Fluidization*, 2d ed., Academic Press, 1985), Geldart (*Gas Fluidization Technology*, Wiley, 1986), Kunii and Levenspiel (*Fluidization Engineering*, Wiley, 1969), and Zenz (*Fluidization and Fluid-Particle Systems*, Penn-Corp Publications, 1989) are good reference books.

Examples

- The original fluidized-bed reactor was the Winkler coal gasifier (patented 1922), followed in 1940 by the Esso cracker that has now been replaced by riser reactors with zeolite catalysts.
- Transport fluidized bed reactor is used for the Sasol Fischer-Tropsch process (Fig. 19-23a).
- Esso-type stable fixed fluidized-bed reactor/regenerator is used for cracking petroleum oils (Fig. 19-23b).
- UOP uses a reformer with moving bed of platinum catalyst and continuous regeneration of a controlled quantity of catalyst (Fig. 19-23c).
- Acrylonitrile is made in a fixed fluidized bed by reacting propylene, ammonia, and oxygen at 400 to 510°C (752 to 950°F) over a Bi-Mo oxide catalyst. The good temperature control with embedded heat exchangers permits catalyst life of several years.

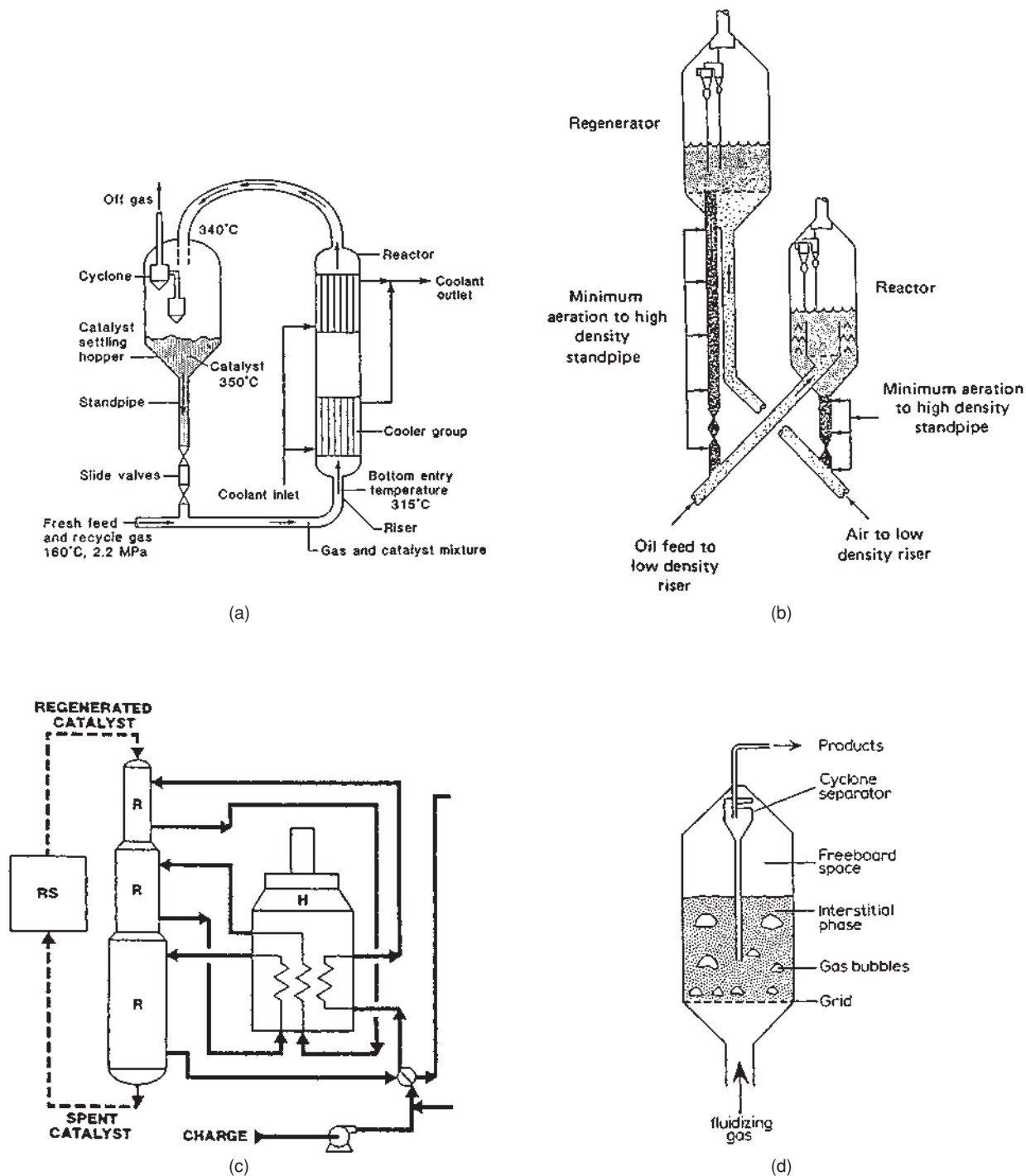


FIG. 19-23 Reactors with moving catalysts. (a) Transport fluidized type for the Sasol Fischer-Tropsch process, nonregenerating. (b) Esso type of stable fluidized-bed reactor/regenerator for cracking petroleum oils. (c) UOP reformer with moving bed of platinum catalyst and continuous regeneration of a controlled quantity of catalyst. (d) Flow distribution in a fluidized bed; the catalyst rains through the bubbles.

- Vinyl chloride is produced by chlorination of ethylene at 200 to 300°C (392 to 572°F), 2 to 10 atm (29.4 to 147 psi), with a supported cupric chloride catalyst in a fluidized bed.

Slurry Reactors Slurry reactors are akin to fluidized beds except the fluidizing medium is a liquid. In some cases (e.g., for hydrogenation), a limited amount of hydrogen may be dissolved in the liquid feed. The solid material is maintained in a fluidized state by agitation, internal or external recycle of the liquid using pipe spargers or distributor plates with perforated holes at the bottom of the reactor. Most industrial processes with slurry reactors also use a gas in reactions such as chlorination, hydrogenation, and oxidation, so the discussion will be deferred to the multiphase reactor section of slurry reactors.

Transport Reactors The superficial velocity of the gas exceeds the terminal velocity of the solid particles, and the particles are transported along with the gas. Usually, there is some "slip" between the gas and the solids—the solid velocity is slightly lower than the gas velocity. Transport reactors are typically used when the required residence time is small and the fluid reactant (or the solid reactant) can be substantially converted (consumed). They may also be used when the catalyst is substantially deactivated during its time in the reactor and has to be regenerated.

Advantages of transport reactors include low gas and solid backmixing (compared to fluidized beds) and the ability to continuously remove deactivated catalyst (and add fresh catalyst), thereby maintaining catalyst activity. The fluid and catalyst are separated downstream by using settlers, cyclones, or filters.

Transport reactors are typically cylindrical pipes. The reactants may be injected at a tee or by using injection pipes at the bottom of the reactor. The size of the pipe may be increased along the reaction path to accommodate volumetric changes that may occur during reaction. Both solid and gas phases may be modeled using a PFR model with exchange between the gas and solid phases. A core-annular concept is often used to describe transport or riser reactors, with most of the particles rising at the center and some flowing back down along the walls.

Examples

- A transport reactor is also used in the Sasol Fischer-Tropsch process. The catalyst is promoted iron. It circulates through the 1.0-m (3.28-ft) ID riser at 72,600 kg/h (160,000 lbm/h) at 340°C (644°F) and 23 atm (338 psi) and has a life of about 50 days. Figure 19-23a shows an in-line heat exchanger in the Sasol unit.
- The fluid catalytic cracking unit (FCCU) riser cracks crude oil into gasoline and distillate range products in a transport bed reactor using a zeolite-Y catalyst. The riser residence time is 4 to 10 s. The riser top temperature is between 950 and 1050°F. The ratio of catalyst to crude oil is between 4 and 8 on a weight basis. During its stay in the riser, the catalyst is deactivated by coke which is burned in the regenerator. The heat generated by burning the coke heats the catalyst and is used to vaporize the crude oil feed. A schematic of the FCCU is shown in Fig. 19-23b.

Multifunctional Reactors Reaction may be coupled with other unit operations to reduce capital and/or operating costs, increase selectivity, and improve safety. Examples are reaction and distillation and reaction with heat transfer. Concepts that combine reaction with membrane separation, extraction, and crystallization are also being explored. In each case, while possibly reducing cost, the need to accommodate both reaction and the additional operation constrains process flexibility by reducing the operating envelope.

Examples

- The Eastman process for reacting methanol with acetic acid to produce methyl acetate and water in one column. Product separation (instead of increased feed concentration) is used to drive the equilibrium to the right.
- Methyl *tert*-butyl ether (MTBE) has been produced by reactive distillation of isobutylene and methanol. The reaction is conducted in a distillation column loaded with socks containing a solid acid catalyst.
- VOC emissions from printing and chemical plants are oxidized in reverse flow reactors that couple reaction with regenerative heat transfer. The concept here is to maintain a catalyst zone in the center of a packed bed with inert heat-transfer packing on either side.

Feed is heated to the desired temperature as it travels through the hot inert bed to the catalyst zone. After the catalyst, the outlet gases lose heat to the cooler packing downstream as they leave the reactor. When the exit temperature of the gases exceeds a certain threshold temperature, the flow is reversed.

NONCATALYTIC REACTORS

These reactors may be similar to the gas-solid catalytic reactors, except for the fact that there is no catalyst involved. The gas and/or the solid may be reactants and/or products. Section 7 of this Handbook provides greater discussion on reaction types and corresponding kinetics for a range of gas-solid reactions. The oldest examples of gas-solid noncatalytic reactors are kilns. A solid is heated with hot combustion gases (that may contain a reactant) to form a desired product. Some of the equipment in use is represented in Fig. 19-24. Temperatures are usually high so the equipment is refractory-lined. The solid is in granular form, at most a few millimeters or centimeters in diameter. Historically, much of the equipment was developed for the treatment of ores and the recovery of metals. In recent years, gas-solid reactions are practiced in the electronics industry. In chemical vapor deposition (CVD), gases react to form solid films in microelectronic chips and wear protective coatings.

Rotary Kilns A rotary kiln is a long, narrow cylinder inclined 2 to 5° to the horizontal and rotated at 0.25 to 5 rpm. It is used for the decomposition of individual solids, for reactions between finely divided solids, and for reactions of solids with gases or even with liquids. The length/diameter ratio ranges from 10 to 35, depending on the reaction time needed. The solid is in granular form and may have solid fuel mixed in. The granules are scooped into the vapor space and are heated as they cascade downward. Holdup of solids is 8 to 15 percent of the cross-section. For most free-falling materials, the solids pattern approaches plug flow axially and complete mixing laterally. Rotary kilns can tolerate some softening and partial fusion of the solid. For example, CaF_2 with SO_3 is reacted in a rotary kiln to make hydrofluoric acid. The morphology of the CaF_2 solids can change considerably as they travel downward through the kiln. Approximate ranges of space velocities in rotary kilns are shown in Table 19-6.

Vertical Kilns Vertical kilns are used primarily where no fusion or softening occurs, as in the burning of limestone or dolomite, although rotary kilns may also be used for these operations. A cross-section of a continuous 50,000-kg/d (110,000-lbm/d) lime kiln is shown in Fig. 19-24c. The diameter range of these kilns is 2.4 to 4.5 m (7.9 to 14.8 ft), and height is 15 to 24 m (49 to 79 ft). Peak temperatures in lime calcination are 1200°C (2192°F), although decomposition proceeds freely at 1000°C (1832°F). Fuel supply may be coke mixed and fed with the limestone or other fuel. Space velocity of the kiln is 14 to 25 kg $\text{CaO}/(\text{m}^3\cdot\text{h})$ [0.87 to 1.56 lbm/(ft³·h)] or 215 to 485 kg $\text{CaO}/(\text{m}^3\cdot\text{h})$ [44 to 99 lbm/(ft³·h)]. Factors that influence kiln size include its vintage, the method of firing, and the lump size, which is in the range of 10 to 25 cm (3.9 to 9.8 in). A five-stage fluidized-bed calciner is sketched in Fig. 19-24d. Such a unit 4 m (13 ft) in diameter and 14 m (46 ft) high has a production of 91,000 kg CaO/d (200,000 lbm/d).

The blast furnace (Fig. 19-24f) is a vertical kiln in which fusion takes place in the lower section. This is a vertical moving-bed device; iron oxides and coal are charged at the top and flow countercurrently to combustion and reducing gases. Units of 1080 to 4500 m³ (38,000 to 159,000 ft³) may produce up to 9×10^6 kg (20×10^6 lbm) of molten iron per day. Figure 19-24f identifies the temperature and composition profiles. Reduction is with CO and H₂ that are made from coal, air, and water within the reactor.

In addition to rotary and vertical kilns, hearth furnaces or fluidized-bed reactors may be used. These high-temperature reactors convert minerals for easier separation from gangue or for easier recovery of metal. Fluidized beds are used for the combustion of solid fuels, and some 30 installations are listed in *Encyclopedia of Chemical Technology* (vol. 10, Wiley, 1980, p. 550). The roasting of iron sulfide in fluidized beds at 650 to 1100°C (1202 to 2012°F) is analogous. The pellets have 10-mm (0.39-in) diameter. There are numerous plants, but they are threatened with obsolescence because cheaper sources of sulfur are available for making sulfuric acid.

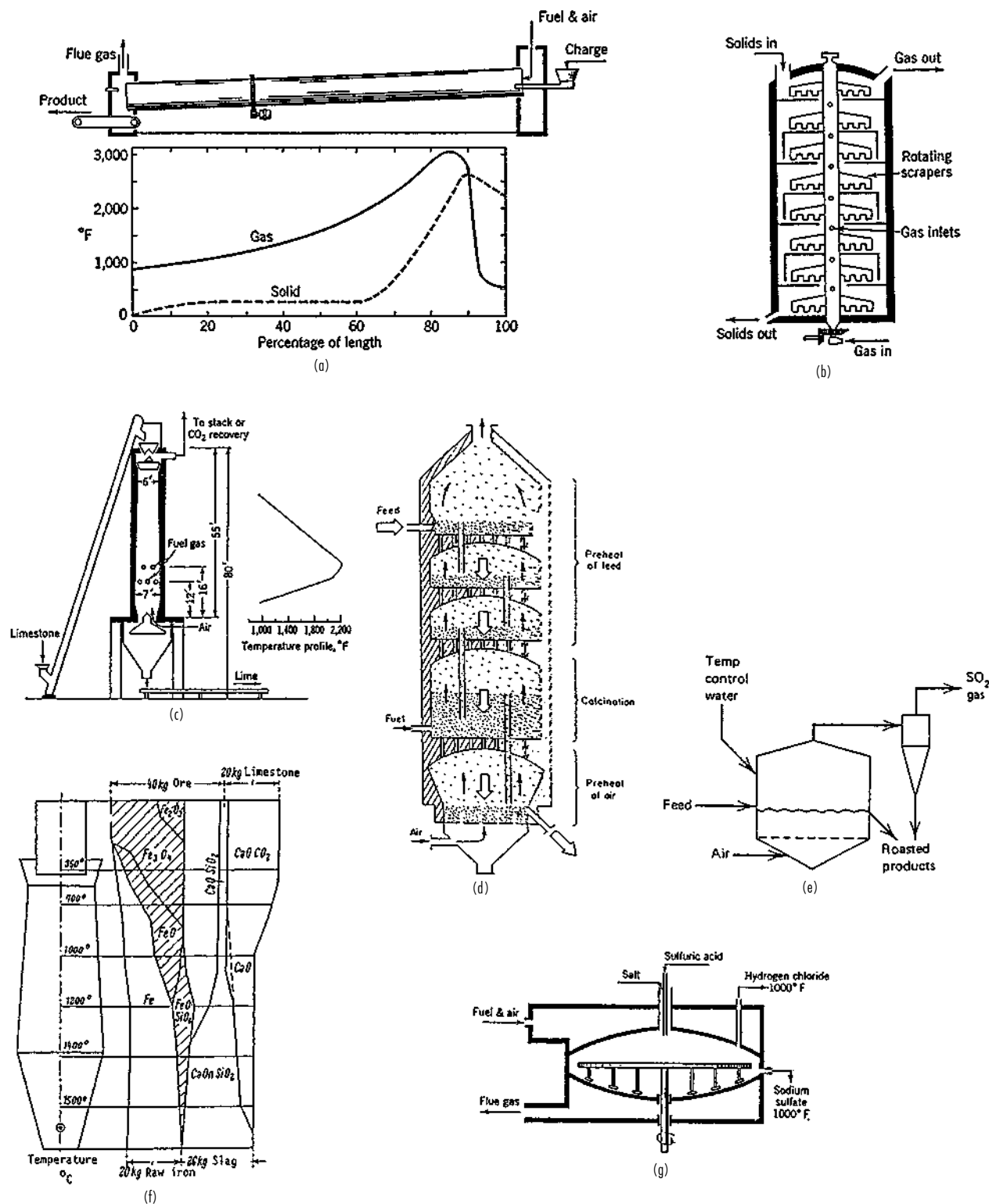


FIG. 19-24 Reactors for solids. (a) Temperature profiles in a rotary cement kiln. (b) A multiple-hearth reactor. (c) Vertical kiln for lime burning, 55 ton/d. (d) Five-stage fluidized-bed lime burner, 4 by 14 m, 100 ton/d. (e) A fluidized bed for roasting iron sulfides. (f) Conditions in a vertical moving bed (blast furnace) for reduction of iron oxides. (g) A mechanical salt cake furnace. To convert ton/d to kg/h, multiply by 907.

TABLE 19-6 Approximate Ranges of Space Velocities in Rotary Kilns

Process	Space velocity, m tons/(m ³ d)
Cement, dry process	0.4–1.1
Cement, wet process	0.4–0.8
Cement, with heat exchange	0.6–1.9
Lime burning	0.5–0.9
Dolomite burning	0.4–0.6
Pyrite roasting	0.2–0.35
Clay calcination	0.5–0.8
Magnetic roasting	1.5–2.0
Ignition of inorganic pigments	0.15–2.0
Barium sulfide preparation	0.35–0.8

There are a number of references on gas-solid noncatalytic reactions, e.g., Brown, Dollimore, and Galwey [*Reactions in the Solid State*, in Bamford and Tipper (eds.), *Comprehensive Chemical Kinetics*, vol. 22, Elsevier, 1980], Galwey (*Chemistry of Solids*, Chapman and Hall, 1967), Sohn and Wadsworth (eds.) (*Rate Processes of Extractive Metallurgy*, Plenum Press, 1979), Szekely, Evans, and Sohn (*Gas-Solid Reactions*, Academic Press, 1976), and Ullmann (*Enzyklopaedie der technischen Chemie*, "Uncatalyzed Reactions with Solids," vol. 3, 4th ed., Verlag Chemie, 1973, pp. 395–464).

Examples

- Cement kilns are up to 6 m (17 ft) in diameter and 200 m (656 ft) long. Inclination is 3 to 4° and rotation is 1.2 to 2.0 rpm. Typical temperature profiles are shown in Fig. 19-24a. Near the flame the temperature is 1800 to 2000°C (3272 to 3632°F). The temperature of the solid reaches 1350 to 1500°C (2462 to 2732°F) which is necessary for clinker formation. In one smaller kiln, a length of 23 m (75 ft) was allowed for drying, 34 m (112 ft) for preheating, 19 m (62 ft) for calcining, and 15 m (49 ft) for clinkering. Total residence time is 40 min to 5 h, depending on the type of kiln. The time near the clinkering temperature of 1500°C (2732°F) is 10 to 20 min. Subsequent cooling is as rapid as possible. A kiln 6 m (20 ft) in diameter by 200 m (656 ft) can produce 2.7×10^6 kg/d (6×10^6 lbm/d) of cement. For production rates less than 270,000 kg/d (600,000 lbm/d), shaft kilns are used. These are vertical cylinders 2 to 3 m (6.5 to 10 ft) by 8 to 10 m (26 to 33 ft) high, fed with pellets and finely ground coal.
- Chlorination of ores ($\text{MeO} + \text{Cl}_2 + \text{C} \Rightarrow \text{MeCl}_2 + \text{CO}$, where Me is Ti, Mg, Be, U, and Zr, whose chlorides are water-soluble). For titanium, carbon is roasted with ore and chlorine is sparged through the bed ($\text{TiO}_2 + \text{C} + 2\text{Cl}_2 \Rightarrow \text{TiCl}_4 + \text{CO}_2$). The chlorine can be supplied indirectly, as in $\text{Cu}_2\text{S} + 2\text{NaCl} + \text{O}_2 \Rightarrow 2\text{CuCl} + \text{Na}_2\text{SO}_4$.
- Oxidation of sulfide ores ($\text{MeS} + 1.5\text{O}_2 \Rightarrow \text{MeO} + \text{SO}_2$, where Me is Fe, Mo, Pb, Cu, or Ni). Iron sulfide (pyrite) is burned with air for recovery of sulfur and to make the iron oxide from which the metal is more easily recovered. Sulfides of other metals also are roasted. A multiple-hearth furnace, as shown in Fig. 19-24b, is used. In some designs, the plates rotate; in others, the scraper arms rotate or oscillate and discharge the material to lower plates. Material charged at the top drops to successively lower plates while reactant and combustion

gases flow upward. A reactor with 9 trays 5 m (16 ft) in diameter and 12 m (39 ft) high can roast about 600 kg/h (1300 lbm/h) of pyrite. A major portion of the reaction is found to occur in the vapor space between trays. A unit in which most of the trays are replaced by empty space is called a *flash roaster*; its mode of operation is like that of a spray dryer. Molybdenum sulfide is roasted at the rate of 5500 kg/d (12,000 lbm/d) in a unit with 9 stages, 5-m (16-ft) diameter, at $630 \pm 15^\circ\text{C}$ ($1166 \pm 27^\circ\text{F}$) and the sulfur is reduced from 35.7 percent to 0.006 percent. A Dorr-Oliver fluidized-bed roaster is 5.5 m (18 ft) in diameter, 7.6 m (25 ft) high, with a bed height of 1.2 to 1.5 m (3.9 to 4.9 ft). It operates at 650 to 700°C (1200 to 1300°F) and has a capacity of 154,000 to 200,000 kg/d (340,000 to 440,000 lbm/d) (Kunii and Levenspiel, *Fluidization Engineering*, Butterworth, 1991). Two modes of operation can be used for a fluidized-bed unit like that shown in Fig. 19-24e. In one mode, a stable fluidized-bed level is maintained. The superficial gas velocity of 0.48 m/s (1.6 ft/s) is low. A reactor is 4.8 m (16 ft) in diameter, 1.5 m (4.9 ft) bed depth, 3 m (9.8 ft) freeboard. The capacity is 82,000 kg/d (180,000 lbm/d) pyrrhotite of 200 mesh. It operates at 875°C (1600°F) and 53 percent of the solids are entrained. In the other mode, the superficial gas velocity of 1.1 m/s (3.6 ft/s) is higher and results in 100 percent entrainment. This reactor is known as a *transfer line* or *pneumatic transport reactor*; a unit 6.6 m (22 ft) in diameter by 1.8 m (5.9 ft) can process 545,000 kg/d (1.2×10^6 lbm/d) of 200 mesh material at 780°C (1436°F).

- Sodium sulfate. A single-hearth furnace like that shown in Fig. 19-24g is used. Sodium chloride and sulfuric acid are charged continuously to the center of the pan, and the rotating scrapers gradually work the reacting mass to the periphery, where the sodium sulfate is discharged at 540°C (1000°). Pans are 3.3 to 5.5 m (11 to 18 ft) in diameter and can handle 5500 to 9000 kg/d (12,000 to 20,000 lbm/d) of salt. Rotary kilns also are used for this purpose. Such a unit 1.5 m (4.9 ft) in diameter by 6.7 m (22 ft) has a capacity of 22,000 kg/d (48,000 lbm/d) of salt cake. A pan furnace also is used, for instance, in the Leblanc soda ash process and for making sodium sulfide from sodium sulfate and coal.
- Magnetic roasting. In this process ores containing Fe_2O_3 are reduced with CO to Fe_3O_4 , which is magnetically separable from gangue. Rotary kilns are used, with temperatures of 700 to 800°C (1292 to 1472°F). Higher temperatures form FeO. The CO may be produced by incomplete combustion of a fuel. A unit for 2.3×10^6 kg/d (5×10^6 lbm/d) has a power consumption of 0.0033 to 0.0044 kWh/kg (3 to 4 kWh/ton) and a heat requirement of 180,000 to 250,000 kcal/ton (714,000 to 991,000 Btu/ton). The magnetic concentrate can be agglomerated for further treatment by pelletizing or sintering.
- Other examples include *calcination reactions* ($\text{MeCO}_3 \Rightarrow \text{MeO} + \text{CO}_2$, where Me is Ca, Mg, and Ba), *sulfating reactions* ($\text{CuS} + 2\text{O}_2 \Rightarrow \text{CuSO}_4$, of which the sulfate is water-soluble), and *reduction reactions* ($\text{MeO} + \text{H}_2 \Rightarrow \text{Me} + \text{H}_2\text{O}$, $\text{MeO} + \text{CO} \Rightarrow \text{Me} + \text{CO}_2$, where Me is Fe, W, Mo, Ge, and Zn).
- The deposition of polycrystalline silicon in microelectronic circuit fabrication ($\text{SiH}_4 \Rightarrow \text{Si} + 2\text{H}_2$) or the deposition of hard TiC films on machine tool surfaces ($\text{TiCl}_4 + \text{CH}_4 \Rightarrow \text{TiC} + 4\text{HCl}$).
- In reactive etching, a patterned film is selectively etched by reacting it with a gas such as chlorine ($\text{Si} + 2\text{Cl}_2 \Rightarrow \text{SiCl}_4$).

FLUID-FLUID REACTORS

Industrial fluid-fluid reactors may broadly be divided into gas-liquid and liquid-liquid reactors. Gas-liquid reactors typically may be used for the manufacture of pure products (such as sulfuric acid, nitric acid, nitrates, phosphates, adipic acid, and other chemicals) where all the gas and liquid react. They are also used in processes where gas-phase reactants are sparged into the reactor and the reaction takes place in the liquid phase (such as hydrogenation, halogenation, oxidation, nitration, alkylation, fermentation, oxidation of sludges, production of proteins, biochemical oxidations, and so on). Gas purification (in which relatively small amounts of impurities such as CO_2 , CO, COS, SO_2 , H_2S , NO, and

others are removed from reactants) is also an important class of gas-liquid reactions. Liquid-liquid reactors are used for synthesis of chemicals (or fuels). One of the liquids may serve as the catalyst, or the liquids may react with one another across the interface. In the latter case, the product may be soluble in one of the liquids or precipitate out as a solid.

GAS-LIQUID REACTORS

Since the reaction rate per unit reactor volume depends on the transfer of molecules from the gas to the liquid, the mass-transfer coefficient is

important. As discussed in Sec. 7, the mass-transfer coefficient in a nonreacting system depends on the physical properties of the gas and liquid and the prevailing hydrodynamics. Here D_G and D_L are diffusivities of the absorbing species in the gas and liquid phases, respectively; $p_i = f(C_i)$ or $p_i = \text{He}C_{Li}$ are the equilibrium relation at the gas-liquid interface; a = interfacial area/unit volume; and δ_G , δ_L are film thicknesses on the gas and liquid sides, respectively. The steady rates of solute transfer are

$$r = k_G a (p_G - p_i) = k_{La} (C_{Li} - C_L) \quad (19-72)$$

where $k_G = D_G/\delta_G$ and $k_L = D_L/\delta_L$ are the mass-transfer coefficients of the individual films. Overall coefficients are defined by

$$r = K_G a (p_G - p_L) = K_{La} (C_G - C_L) \quad (19-73)$$

Upon introducing the equilibrium relation at the interface, the relation between the various mass-transfer coefficients is

$$\frac{1}{K_G a} = \frac{\text{He}}{K_{La} a} = \frac{1}{k_G a} + \frac{\text{He}}{k_{La} a} \quad (19-74)$$

When the solubility is low, the Henry constant He is large and $k_L \Rightarrow K_L$; when the solubility is high, He is small and $k_G \Rightarrow K_G$. The reaction rate in the liquid phase determines the relative importance of the mass-transfer coefficient. For slow reactions, reaction rate in the liquid phase determines overall rate. In contrast, for fast reactions, transport of reactant from the gas to the liquid across the gas-liquid interface is rate-determining. If the reaction is fast, reaction also occurs in the film along with diffusion, thus enhancing the mass transfer. The relative role of mass transfer (across the gas-liquid interface) versus kinetics is important in gas-liquid reactor selection and design. Three modes of contacting gas with liquid are possible: (1) The gas is dispersed as bubbles in the liquid; (2) the liquid is dispersed as droplets in the gas; and (3) the liquid and gas are brought together as thin films over a packing or wall. Considerations that influence reactor selection include the magnitude and distribution of the residence times of the phases, the power requirements, the scale of the operation, the opportunity for heat transfer, and so on.

As indicated above, for purely physical absorption, the mass-transfer coefficients depend on the hydrodynamics and the physical properties of the phases. The literature contains measured values of mass-transfer coefficients and correlations (see discussion on agitated tanks and bubble columns below). Tables 19-7 and 19-8 present experimental information on apparent mass-transfer coefficients for absorption of select gases. On this basis, a tower for absorption of SO_2 with NaOH is smaller than that with pure water by a factor of roughly $0.317/7.0 = 0.045$. Table 19-9 lists the main factors that are needed for

TABLE 19-7 Typical Values of $K_G a$ for Absorption in Towers Packed with 1.5-in Intalox Saddles at 25% Completion of Reaction*

Absorbed gas	Absorbent	$K_G a$, lb mol/(h-ft ³ ·atm)
Cl_2	$\text{H}_2\text{O-NaOH}$	20.0
HCl	H_2O	16.0
NH_3	H_2O	13.0
H_2S	$\text{H}_2\text{O-MEA}$	8.0
SO_2	$\text{H}_2\text{O-NaOH}$	7.0
H_2S	$\text{H}_2\text{O-DEA}$	5.0
CO_2	$\text{H}_2\text{O-KOH}$	3.10
CO_2	$\text{H}_2\text{O-MEA}$	2.50
CO_2	$\text{H}_2\text{O-NaOH}$	2.25
H_2S	H_2O	0.400
SO_2	H_2O	0.317
Cl_2	H_2O	0.138
CO_2	H_2O	0.072
O_2	H_2O	0.0072

*To convert in to cm, multiply by 2.54; lb mol/(h-ft³·atm) to kg mol/(h-m³·kPa), multiply by 0.1581.

SOURCE: From Eckert et al., *Ind. Eng. Chem.*, **59**, 41 (1967).

TABLE 19-8 Selected Absorption Coefficients for CO_2 in Various Solvents in Towers Packed with Raschig Rings*

Solvent	$K_G a$, lb mol/(h-ft ³ ·atm)
Water	0.05
1-N sodium carbonate, 20% Na as bicarbonate	0.03
3-N diethanolamine, 50% converted to carbonate	0.4
2-N sodium hydroxide, 15% Na as carbonate	2.3
2-N potassium hydroxide, 15% K as carbonate	3.8
Hypothetical perfect solvent having no liquid-phase resistance and having infinite chemical reactivity	24.0

*Basis: $L = 2,500$ lb/(h-ft²); $G = 300$ lb/(h-ft²); $T = 77^\circ\text{F}$; pressure, 1.0 atm. To convert lb mol/(h-ft³·atm) to kg mol/(h-m³·kPa) multiply by 0.1581.

SOURCE: From Sherwood, Pigford, and Wilke, *Mass Transfer*, McGraw-Hill, 1975, p. 305.

mathematical representation of $K_G a$ in a typical case of the absorption of CO_2 by aqueous monoethanolamine. Other than Henry's law, $p = \text{He}C$, which holds for some fairly dilute solutions, there is no general simple form of equilibrium relation. A typically complex equation is that for CO_2 in contact with sodium carbonate solutions [Harte, Baker, and Purcell, *Ind. Eng. Chem.* **25**: 528 (1933)], which is

$$p_{\text{CO}_2} = \frac{137f^2N^{1.29}}{S(1-f)(365-T)^2} \quad (19-75)$$

where f = fraction of total base present as bicarbonate

N = normality, 0.5 to 2.0

S = solubility of CO_2 in water at 1 atm, g-mol/L

T = temperature, 65 to 150°F

The mass-transfer coefficient with a reactive solvent can be represented by multiplying the purely physical mass-transfer coefficient by an *enhancement factor* E that depends on a parameter called the *Hatta number* (analogous to the *Thiele* modulus in porous catalyst particles).

$$\text{Ha}^2 = \frac{\text{maximum possible reaction in film}}{\text{maximum diffusional transport through film}} \quad (19-76)$$

TABLE 19-9 Correlation of $K_G a$ for Absorption of CO_2 by Aqueous Solutions of Monoethanolamine in Packed Towers*

$K_G a = F \left(\frac{L}{\mu} \right)^{2/3} [1 + 5.7(C_e - C)M e^{0.0067T - 3.4p}]$		
where $K_G a$ = overall gas-film coefficient, lb mol/(h-ft ³ ·atm)		
μ = viscosity, centipoises		
C = concentration of CO_2 in the solution, mol/mol monoethanolamine		
M = amine concentration of solution (molarity, g mol/L)		
T = temperature, °F		
p = partial pressure, atm		
L = liquid-flow rate, lb/(h-ft ²)		
C_e = equilibrium concentration of CO_2 in solution, mol/mol monoethanolamine		
F = factor to correct for size and type of packing		
Packing	F	Basis for calculation of F
5- to 6-mm glass rings	7.1×10^{-3}	Shneerson and Leibush data, 1-in column, atmospheric pressure
3/4-in ceramic rings	3.0×10^{-3}	Unpublished data for 4-in column, atmospheric pressure
3/4- by 2-in polyethylene Tellerettes	3.0×10^{-3}	Teller and Ford data, 8-in column, atmospheric pressure
1-in steel rings	2.1×10^{-3}	
1-in ceramic saddles	$0.4-0.6 \times 10^{-3}$	Gregory and Scharmann and unpublished data for two commercial plants, pressures 30 to 300 psig

*To convert in to cm multiply by 2.54.

SOURCE: From Kohl and Riesenfeld, *Gas Purification*, Gulf, 1985.

For example, for the reaction $A(g) + bB(l) \rightarrow P$ with liquid reactant B in excess,

$$Ha^2 = \frac{kC_{aLi}C_{bLi}\delta_L}{D_{aL}(C_{aLi} - 0)/\delta_L} = \frac{kC_{bLi}\delta_L^2}{D_{aL}} = \frac{kC_{bLi}D_{aL}}{k_L^2} \quad (19-77)$$

When $Ha \gg 1$, all the reaction occurs in the film and the amount of interfacial area is controlling, necessitating equipment that generates a large interfacial area. When $Ha \ll 1$, no reaction occurs in the film and the bulk volume is controlling. As guidance the following criteria may be used:

$Ha < 0.3$	Reaction needs large bulk liquid volume.
$0.3 < Ha < 3.0$	Reaction needs large interfacial area and large bulk liquid volume.
$Ha > 3.0$	Reaction needs large interfacial area.

Of the parameters making up the Hatta number, liquid diffusivity and mass-transfer coefficient data and measurement methods are well reviewed in the literature.

As discussed in Sec. 7, the factor E represents an enhancement of the rate of transfer of A caused by the reaction compared with physical absorption, i.e., K_G is replaced by EK_G . The theoretical variation of E with Hatta number for a first- and second-order reaction in a liquid film is shown in Fig. 19-25. The uppermost line on the upper right represents the pseudo first-order reaction, for which $E = Ha \coth(Ha)$. Three regions are identified with different requirements of liquid holdup ϵ and interfacial area a , and for which particular kinds of contacting equipment may be best:

Region I, $Ha > 2$. Reaction is fast and occurs mainly in the liquid film so $C_{aL} \approx 0$. The rate of reaction $r_a = k_L a E C_{aLi}$ will be large when a is large, but liquid holdup is not important. Packed towers or stirred tanks will be suitable.

Region II, $0.02 < Ha < 2$. Most of the reaction occurs in the bulk of the liquid. Both interfacial area and holdup of liquid should be high. Stirred tanks or bubble columns will be suitable.

Region III, $Ha < 0.02$. Reaction is slow and occurs in the bulk liquid. Interfacial area and liquid holdup should be high, especially the latter. Bubble columns will be suitable.

The above analysis and Fig. 19-25 provide a theoretical foundation similar to the Thiele-modulus effectiveness factor relationship for fluid-solid systems. However, there are no generalized closed-form expressions of E for the more general case of a complex reaction network, and its value has to be determined by solving the complete diffusion-reaction equations for known intrinsic mechanism and kinetics, or alternatively estimated experimentally.

Some of this theoretical thinking may be utilized in reactor analysis and design. Illustrations of gas-liquid reactors are shown in Fig. 19-26. Unfortunately, some of the parameter values required to undertake a rigorous analysis often are not available. As discussed in Sec. 7, the intrinsic rate constant k_c for a liquid-phase reaction without the complications of diffusional resistances may be estimated from properly designed laboratory experiments. Gas- and liquid-phase holdups may be estimated from correlations or measured. The interfacial area per unit reactor volume a may be estimated from correlations or measurements that utilize techniques of transmission or reflection of light, though these are limited to small diameters. The combined volumetric mass-transfer coefficient $k_L a$, can be also directly measured in reactive or nonreactive systems (see, e.g., Charpentier, *Advances in Chemical Engineering*, vol. 11, Academic Press, 1981, pp. 2-135). Mass-transfer coefficients, interfacial areas, and liquid holdup typical for various gas-liquid reactors are provided in Tables 19-10 and 19-11.

There are numerous examples of commercial gas-liquid reactions in the literature. These include common operations such as absorption of ammonia to make fertilizers and of carbon dioxide to make soda ash. Other examples are recovery of phosphine from off-gases of phosphorous plants; recovery of HF; oxidation, halogenation, and hydrogenation of various organics; hydration of olefins to alcohols; oxo reaction for higher aldehydes and alcohols; ozonolysis of oleic acid; absorption of carbon monoxide to make sodium formate; alkylation of acetic acid with isobutylene to make *tert*-butyl acetate, absorption of olefins to make various products; HCl and HBr plus higher alcohols to make alkyl halides; and so on. By far the greatest number of applications is for the removal or recovery of mostly small concentrations of acidic and other components from air, hydrocarbons, and hydrogen. Two lists of gas-liquid reactions of industrial importance have been compiled. The literature survey by Danckwerts (*Gas-Liquid*

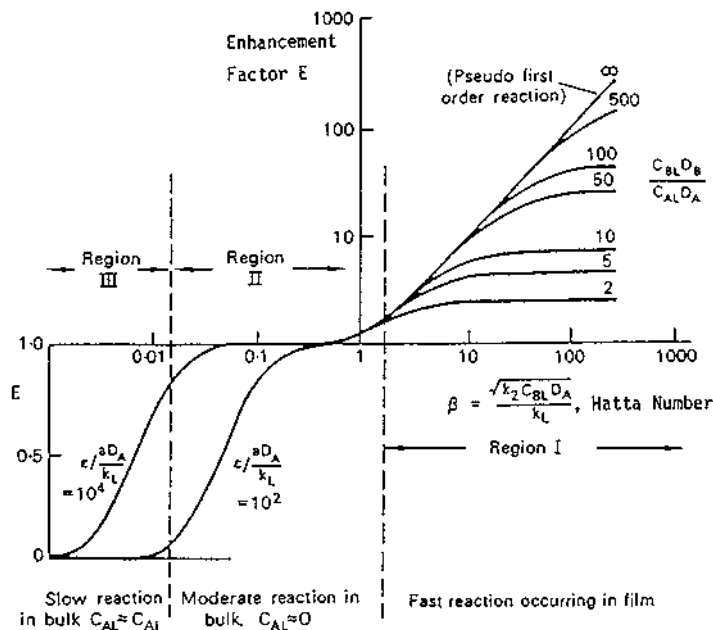


FIG. 19-25 Enhancement factor E and Hatta number of first- and second-order gas-liquid reactions. (Coulson and Richardson, *Chemical Engineering*, vol. 3, Pergamon, 1971, p. 80.)

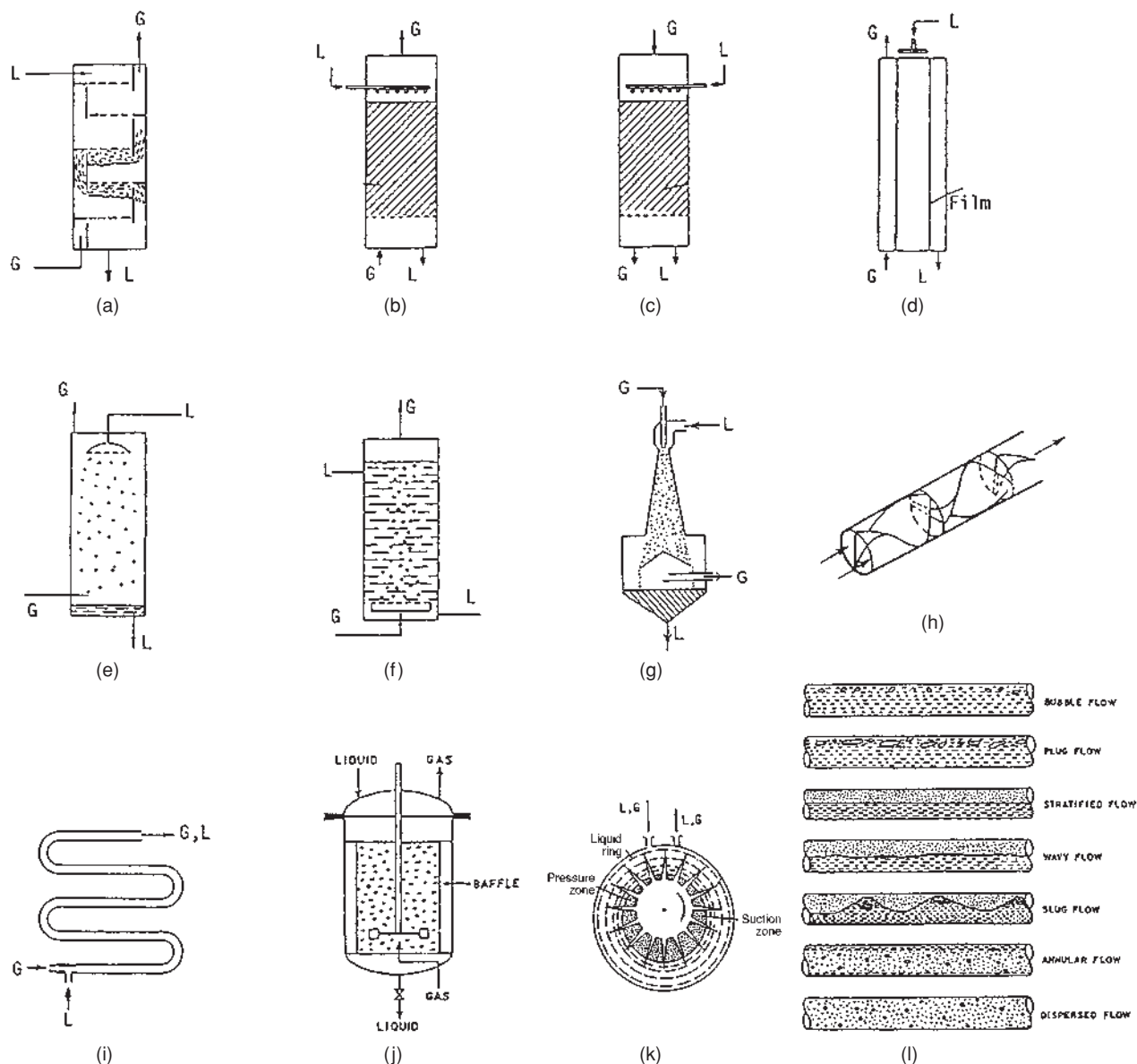


FIG. 19-26 Types of industrial gas-liquid reactors. (a) Tray tower. (b) Packed, countercurrent. (c) Packed, co-current. (d) Falling liquid film. (e) Spray tower. (f) Bubble tower. (g) Venturi mixer. (h) Static in-line mixer. (i) Tubular flow. (j) Stirred tank. (k) Centrifugal pump. (l) Two-phase flow in horizontal tubes.

Reactions, McGraw-Hill, 1970) cites 40 different systems. A supplementary list by Doraiswamy and Sharma (*Heterogeneous Reactions: Fluid-Fluid-Solid Reactions*, Wiley, 1984) cites another 50 cases and indicates the most suitable kind of reactor to be used for each. A number of devices have been in use for estimating mass-transfer coefficients, and correlations are available. This topic is reviewed in books, for example, by Danckwerts (*Gas-Liquid Reactions*, McGraw-Hill, 1970) and Charpentier [in Ginetto and Silveston (eds.), *Multiphase Chemical Reactor Theory, Design, Scaleup*, Hemisphere, 1986]. One of the issues associated with designing commercial reactors is to properly understand whether data obtained on the laboratory scale are applicable or whether larger scale data are needed to reduce the scale-up risk.

LIQUID-LIQUID REACTORS

Much of the thinking on gas-liquid reactors is also applicable to liquid-liquid reactors. The liquids are usually not miscible, and the transport of reactants can determine the specific reaction rate. Liquid-liquid reactors require dispersion of one of the liquid phases to provide sufficient interfacial area for mass transfer. This can be achieved by the use of static mixers, jets, or mechanical means such as in a CSTR.

In a stirred tank, either liquid can be made continuous by charging that liquid first, starting the agitator, and introducing the liquid to be dispersed. For other reactor types, the choice of which phase is continuous and which is dispersed will depend on the physicochemical properties of the phases and operating conditions (such as temperature,

TABLE 19-10 Mass-Transfer Coefficients, Interfacial Areas, and Liquid Holdup in Gas-Liquid Reactions

Type of reactor	ϵ_L , %	k_G , gm mol/(cm ² ·s·atm) × 10 ⁴	k_L , cm/s × 10 ²	a , cm ² /cm ³ reactor	$k_L a$, s ⁻¹ × 10 ²
Packed columns					
Countercurrent	2–25	0.03–2	0.4–2	0.1–3.5	0.04–7
Cocurrent	2–95	0.1–3	0.4–6	0.1–17	0.04–102
Plate columns					
Bubble cap	10–95	0.5–2	1–5	1–4	1–20
Sieve plates	10–95	0.5–6	1–20	1–2	1–40
Bubble columns	60–98	0.5–2	1–4	0.5–6	0.5–24
Packed bubble columns	60–98	0.5–2	1–4	0.5–3	0.5–12
Tube reactors					
Horizontal and coiled	5–95	0.5–4	1–10	0.5–7	0.5–70
Vertical	5–95	0.5–8	2–5	1–20	2–100
Spray columns	2–20	0.5–2	0.7–1.5	0.1–1	0.07–1.5
Mechanically agitated bubble reactors	20–95	—	0.3–4	1–20	0.3–80
Submerged and plunging jet	94–99	—	0.15–0.5	0.2–1.2	0.03–0.6
Hydrocyclone	70–93	—	10–30	0.2–0.5	2–15
Ejector reactor	—	—	—	1–20	—
Venturi	5–30	2–10	5–10	1.6–25	8–25

SOURCE: From Charpentier, *Advances in Chemical Engineering*, vol. 11, Academic Press, 1981, pp. 2–135.

pressure, and flow rates). Equipment suitable for reactions between liquids is represented in Fig. 19-27. Almost invariably, one of the phases is aqueous and the other organic, with reactants distributed between phases. Such reactions can be carried out in any kind of equipment that is suitable for physical extraction, including mixer-settlers and towers of various kinds: empty or packed, still or agitated, either phase dispersed, provided that adequate heat transfer can be incorporated. Mechanically agitated tanks are favored because the interfacial area can be made large, as much as 100 times that of spray towers, for instance. Power requirements for liquid-liquid mixing are about 5 hp/1000 gal. Agitator tip speed of turbine-type impellers is 4.6 to 6.1 m/s (15 to 20 ft/s). Table 19-12 provides data for common types of liquid-liquid contactors. As shown, the given range of $k_L a$ is more than 100/1 even for the same equipment. It is provided merely for guidance, and correlations need to be validated with data at some reasonable scale.

Efficiencies of several kinds of small-scale extractors are shown in Fig. 19-28. Larger-diameter equipment may have less than one-half these efficiencies. Spray columns are inefficient and are used only when other kinds of equipment may become clogged. Packed columns as liquid-liquid reactors are operated at 20 percent of flooding. Their *height equivalent to theoretical stage* (HETS) range is from 0.6 to 1.2 m (1.99 to 3.94 ft). Sieve trays minimize backmixing and provide repeated coalescence and redispersion. Mixer-settlers provide approximately one theoretical stage, but several stages can be incorporated in a single shell, although with some loss of operating flexibility. The HETS of rotating disk contactor (RDC) is 1 to 2 m (3.2 to 6.4 ft). More elaborate staged extractors bring this down to 0.35 to 1.0 m (1.1 to 3.3 ft).

When liquid-liquid contactors are used as reactors, values of their mass-transfer coefficients may be enhanced by reaction, analogously to those of gas-liquid processes. Reactions can occur in either or both phases or near the interface. Nitration of aromatics with HNO₃-H₂SO₄ occurs in the aqueous phase [Albright and Hanson (eds.), *Industrial and Laboratory Nitrations*, ACS Symposium Series 22

(1975)]. An industrial example of reaction in both phases is the oximation of cyclohexanone, a step in the manufacture of caprolactam for nylon (Rod, *Proc. 4th Int./6th European Symp. Chemical Reactions*, Heidelberg, Pergamon, 1976, p. 275). The formation of dioxane from isobutene in a hydrocarbon phase and aqueous formaldehyde occurs preponderantly in the aqueous phase where the rate equation is first-order in formaldehyde, although the specific rate is also proportional to the concentration of isobutene in the organic phase [Hellin et al., *Genie. Chim.* 91: 101 (1964)]. Doraiswamy and Sharma (*Heterogeneous Reactions*, Wiley, 1984) have compiled a list of 26 classes of reactions. The reactions include examples such as making soap with alkali, nitration of aromatics to make explosives, and alkylation of C₄s with sulfuric acid to make gasoline alkylate.

REACTOR TYPES

The discussion is centered around gas-liquid reactors. If the dissolved gas content exceeds the amount needed for the reaction, the liquid may be first saturated with gas and then sent through a stirred tank or tubular reactor as a single phase. If the residence times for the liquid and gas are comparable, both gas and liquid may be pumped in and out of the reactor together. If the gas has limited solubility, it is bubbled through the reactor and the residence time for gas is much smaller. Figure 19-29 provides examples of gas-liquid reactors for specific processes.

Agitated Stirred Tanks Stirred tanks are common gas-liquid reactors. Reaction requirements dictate whether the gas and liquid are in a batch or continuous mode. For a liquid-phase reaction with a long time constant, a batch mode may be used. The reactor is filled with liquid, and gas is continuously fed into the reactor to maintain pressure. If by-product gases form, these gases may need to be purged continuously. If gas solubility is limiting, a higher-purity gas may be continuously fed (and, if required, recycled). As the liquid residence time decreases, product may be continuously removed as well. A

TABLE 19-11 Order-of-Magnitude Data of Equipment for Contacting Gases and Liquids

Device	$k_L a$, s ⁻¹	V , m ³	$k_L a V$, m ³ /s (duty)	a , m ⁻¹	ϵ_L	Liquid mixing	Gas mixing	Power per unit volume, kW/m ³
Baffled agitated tank	0.02–0.2	0.002–100	10 ⁻⁴ –20	~200	0.9	~Backmixed	Intermediate	0.5–10
Bubble column	0.05–0.01	0.002–300	10 ⁻⁵ –3	~20	0.95	~Plug	Plug	0.01–1
Packed tower	0.005–0.02	0.005–300	10 ⁻⁵ –6	~200	0.05	Plug	~Plug	0.01–0.2
Plate tower	0.01–0.05	0.005–300	10 ⁻⁵ –15	~150	0.15	Intermediate	~Plug	0.01–0.2
Static mixer (bubble flow)	0.1–2	Up to 10	1–20	~1000	0.5	~Plug	Plug	10–500

SOURCE: From J. C. Middleton, in Hamby, Edwards, and Nienow, *Mixing in the Process Industries*, Butterworth, 1985.

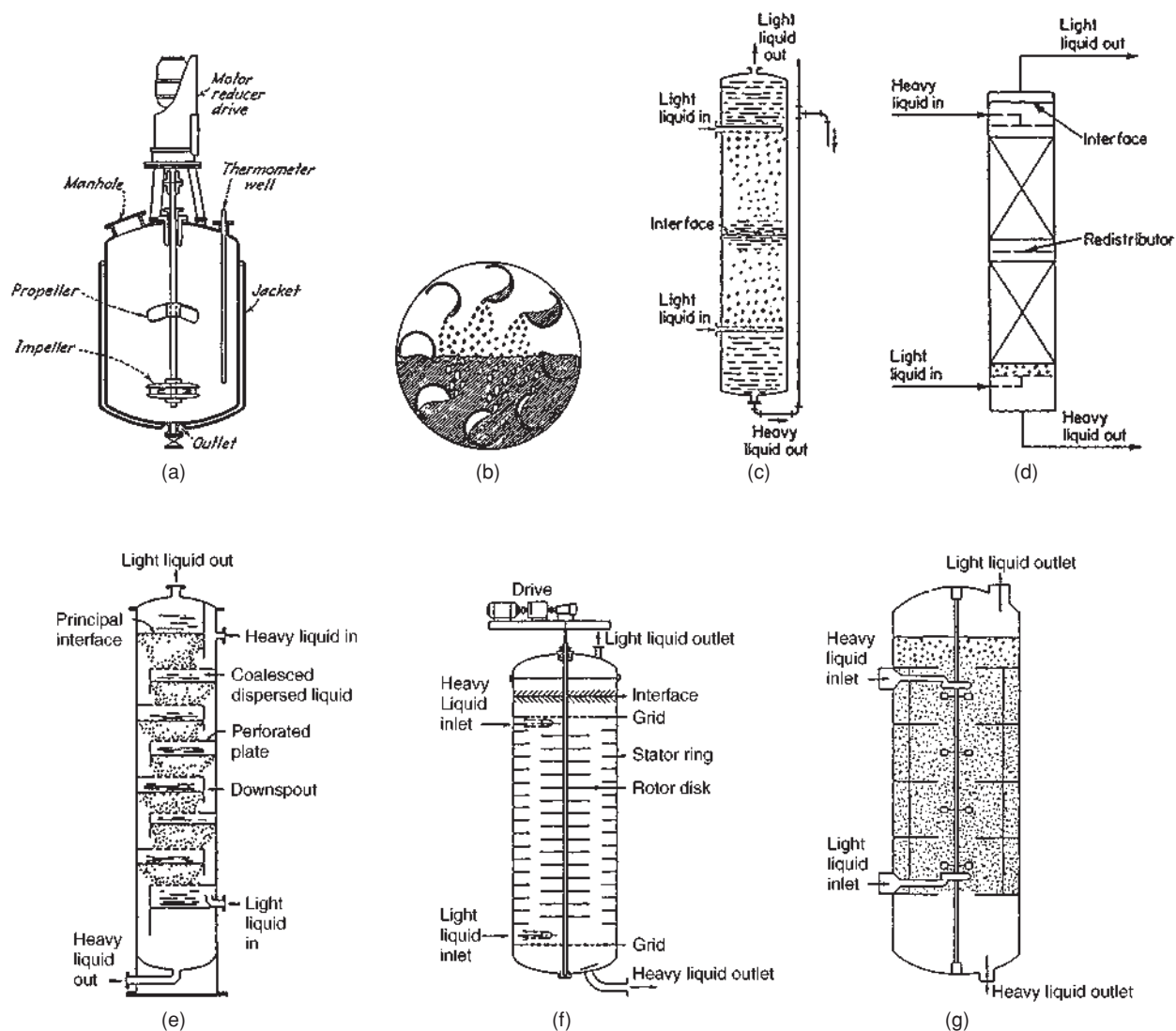


FIG. 19-27 Equipment for liquid-liquid reactions. (a) Batch stirred sulfonator. (b) Raining bucket (RTL S A, London). (c) Spray tower with both phases dispersed. (d) Two-section packed tower with light phase dispersed. (e) Sieve tray tower with light phase dispersed. (f) Rotating disk contactor (RDC) (Escher B V, Holland). (g) Oldshue-Rushton extractor (Mixing Equipment Co.).

TABLE 19-12 Continuous-Phase Mass-Transfer Coefficients and Interfacial Areas in Liquid-Liquid Contactors*

Type of equipment	Dispersed phase	Continuous phase	ϵ_D	τ_D	$k_L \times 10^3$, cm/s	a , cm ² /cm ³	$k_{La} \times 10^2$, s ⁻¹
Spray columns	P	M	0.05–0.1	Limited	0.1–1	1–10	0.1–10
Packed columns	P	P	0.05–0.1	Limited	0.3–1	1–10	0.3–10
Mechanically agitated contactors	PM	M	0.05–0.4	Can be varied over a wide range	0.3–1	1–800	0.3–800
Air-agitated liquid/liquid contactors	PM	M	0.05–0.3	Can be varied over a wide range	0.1–0.3	10–100	1.0–30
Two-phase cocurrent (horizontal) contactors	P	P	0.05–0.2	Limited	0.1–1.0	1–25	0.1–25

*P = plug flow, M = mixed flow, ϵ_D = fractional dispersed phase holdup, τ_D = residence time of the dispersed phase.

SOURCE: From Doraiswamy and Sharma, *Heterogeneous Reactions*, Wiley, 1984.

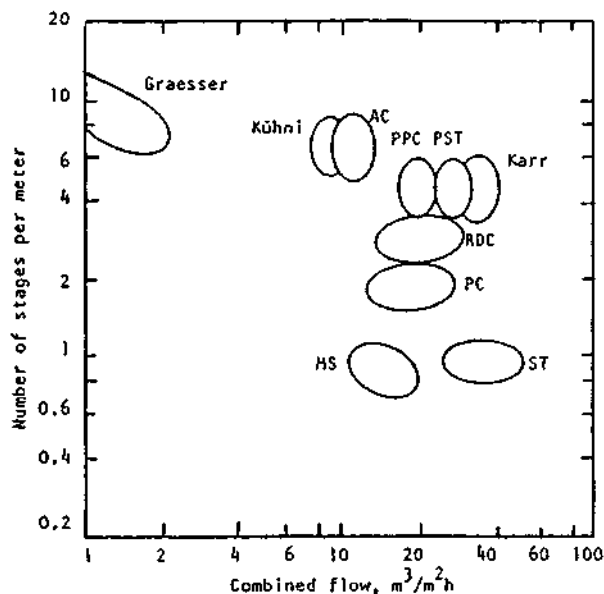


FIG. 19-28 Efficiency and capacity range of small-diameter extractors, 50- to 150-mm diameter. Acetone extracted from water with toluene as the disperse phase, $V_d/V_c = 1.5$. Code: AC = agitated cell; PPC = pulsed packed column; PST = pulsed sieve tray; RDC = rotating disk contactor; PC = packed column; MS = mixer-settler; ST = sieve tray. [Stichlmair, Chem. Ing. Tech. 52(3): 253-255 (1980).]

hybrid reactor type is the semibatch reactor. Gas and liquid are continuously fed to the reactor until the reactor is full of liquid. The reactor then operates as a batch reactor.

Agitated stirred tanks are preferred when high gas-liquid interfacial area is needed. Disadvantages include maintenance of the motor and seals, potential for contamination in biological and food applications, and higher cost.

A basic stirred tank design is shown in Fig. 19-30. Height/diameter ratio is $H/D = 1$ to 3. Heat transfer may be provided through a jacket or internal coils. Baffles prevent movement of the mass as a whole. A draft tube can enhance vertical circulation. The vapor space is about 20 percent of the total volume. A hollow shaft and impeller increase gas circulation by entraining the gas from the vapor space into the liquid. A splasher can be attached to the shaft at the liquid surface to improve entrainment of gas. A variety of impellers is in use. The pitched propeller moves the liquid axially, the flat blade moves it radially, and inclined blades move it both axially and radially. The anchor and some other designs are suited to viscous liquids. For gas dispersion, the six-bladed turbine is preferred. When the ratio of liquid height to diameter is $H/D \leq 1$, a single impeller suffices, and in the range $1 \leq H/D \leq 1.8$ two are needed.

Gases may be dispersed in liquids by spargers or nozzles. However, more intensive dispersion and redispersion are obtained by mechanical agitation. The gas is typically injected at the point of greatest turbulence near the injector tip. Agitation also provides the heat transfer and, if needed, keeps catalyst particles (in a three-phase or slurry reactor) in suspension. Power inputs of 0.6 to 2.0 kW/m³ (3.05 to 10.15 hp/1000 gal) are suitable. Bubble sizes depend on agitation as well as on the physical properties of the liquid. They tend to be greater than a minimum size regardless of power input due to coalescence. Pure liquids are of a coalescing type; solutions with electrolytes are noncoalescing. Agitated bubble size in air/water is about 0.5 mm (0.020 in), holdup fractions are about 0.10 for coalescing and 0.25 for noncoalescing liquids; however, more elaborate correlations are available and required for reactor sizing. The reactor may be modeled as two ideal reactors, one for each phase, with mass transfer between the phases. More elaborate models that utilize CFD have

also been used. For example, if the gas has limited solubility and is sparged through a liquid, the gas may be modeled as a PFR and the liquid as a CSTR. Mass-transfer coefficients vary, e.g., as the 0.7 exponent on the power input per unit volume (with the dimensions of the vessel and impeller and the superficial gas velocity as additional factors). A survey of such correlations is made by van't Riet [Ind. Eng. Chem. Proc. Des. Dev. 18: 357 (1979)]. Also, Charpentier [in Gianetto and Silveston (eds.), *Multiphase Chemical Reactors*, Hemisphere, 1986, pp. 104-151] discusses hydrodynamic parameters for stirred tank (and other) reactors, and typical values are shown in Tables 19-10 and 19-11.

Examples

- **Production of penicillin.** An agitated stirred tank is used for the large-scale aerobic fermentation of penicillin by the growth of a specific mold. Commercial vessel sizes are 40,000 to 200,000 L (1400 to 7000 ft³). The operation is semibatch in that the lactose or glucose nutrient and air are charged at controlled rates to a precharged batch of liquid nutrients and cell mass. Reaction time is 5 to 6 d. The broth is limited to 7 to 8 percent sugars, which is all the mold will tolerate. Solubility of oxygen is limited, and air must be supplied over a long period as it is used up. The air is essential to the growth. Dissolved oxygen must be kept at a high level for the organism to survive. Air also serves to agitate the mixture and to sweep out the CO₂ and any noxious by-products that are formed. Air supply is in the range of 0.5 to 1.5 volumes/(volume of liquid)(min). For organisms grown on glucose, the oxygen requirement is 0.4 g/g dry weight; on methanol it is 1.2 g/g. The pH is controlled at about 6.5 and the temperature at 24°C (75°F). The heat of reaction requires cooling water at the rate of 10 to 40 L/(1000 L holdup)(h). Vessels under about 500 L (17.6 ft³) are provided with jackets, larger ones with coils. For a 55,000-L vessel, 50 to 70 m² may be taken as average. Mechanical agitation is needed to break up the gas bubbles but must avoid rupturing the cells. The disk turbine with radial action is most suitable. It can tolerate a superficial gas velocity up to 120 m/h (394 ft/h) without flooding [whereas the propeller is limited to about 20 m/h (66 ft/h)]. When flooding occurs, the impeller is working in a gas phase and cannot assist the transfer of gas to the liquid phase. Power input by agitation and air sparger is 1 to 4 W/L [97 to 387 Btu/(ft³·h)] of liquid.

- **Refinery alkylation.** C3-C4 olefins are reacted with isobutane in the presence of concentrated acid to form higher-molecular-weight hydrocarbons that may be blended into the gasoline pool. Commercial alkylation processes are catalyzed by either sulfuric or hydrofluoric acid. For both processes, alkylate product quality and acid consumption are impacted by temperature, isobutene/olefin ratio, space velocity, and acid concentration. DuPont Stratco's contactor reactor is a horizontal pressure vessel containing an inner circulation tube, a tube bundle to remove the heat of reaction, and a mixing impeller. The hydrocarbon feed and sulfuric acid enter on the suction side of the impeller inside the circulation tube, producing an emulsion. The reaction emulsion is partially separated in a settler, and the acid emulsion is recycled to the contactor's shell side. The hydrocarbon effluent is directed to the contactor's tube bundle where flash vaporization removes the heat of reaction. Contactor arrangements are also utilized when the alkylation reaction is conducted using hydrofluoric acid.

Bubble Columns Nozzles or spargers disperse the gas. The mixing is due to rising bubbles, not mechanical agitation. Bubble action provides agitation about equivalent to that of mechanical stirrers (and similar mass- and heat-transfer coefficients) at the same power input per volume. The reaction medium may be a liquid (or slurry containing a heterogeneous catalyst). To improve the operation, redispersion of gas in liquid or an approach to plug flow may be achieved by using static mixers (such as perforated plates) at regular intervals. Because of their large volume fraction of liquid, bubble column reactors are suited to slow reactions where the rate of reaction is limiting. Major advantages are an absence of moving parts, the ability to handle solid particles without erosion or plugging, good heat transfer at the wall or coils, high interfacial area, and high mass-transfer coefficients. A disadvantage is backmixing in the liquid phase and some backmixing in the gas phase. The static head of the liquid will increase gas pressure drop, and this may be undesirable.

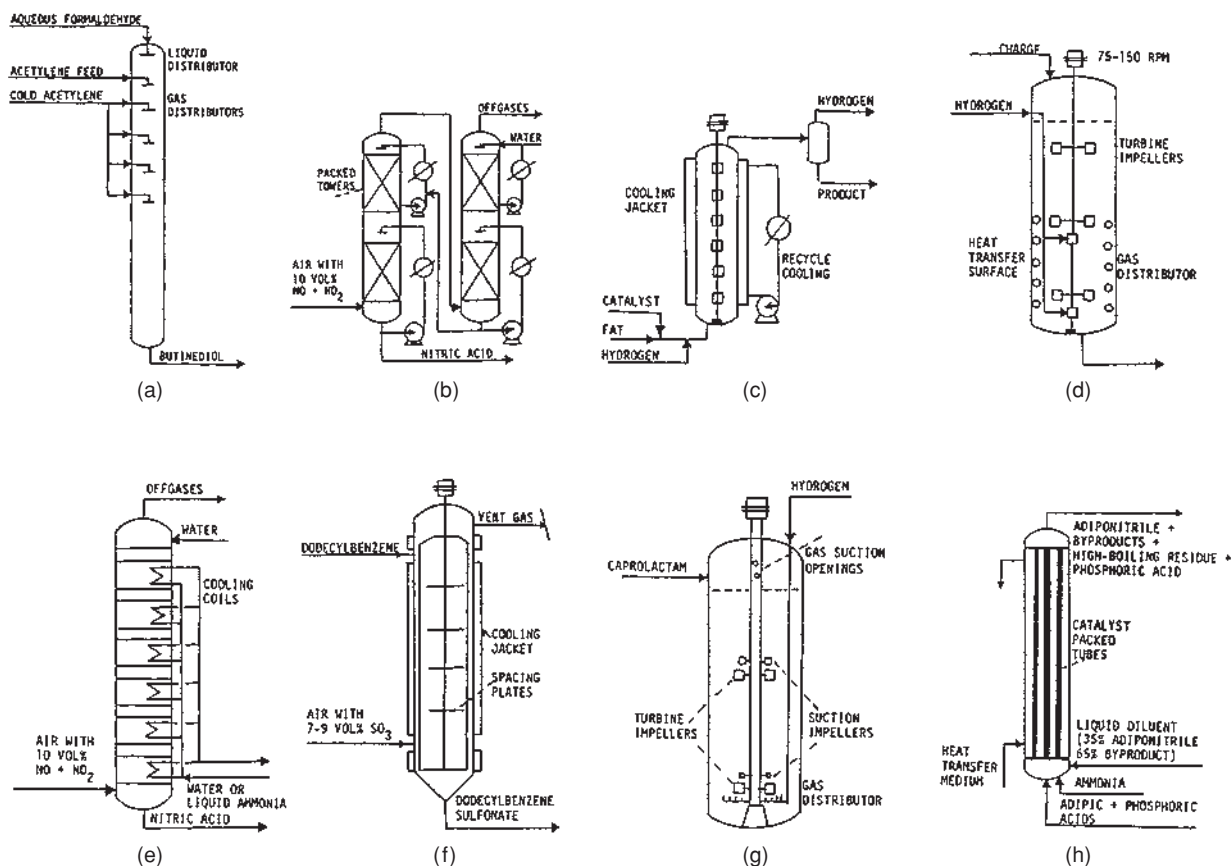


FIG. 19-29 Examples of reactors for specific gas-liquid processes. (a) Trickle reactor for synthesis of butanediol, 1.5-m diameter by 18 m high. (b) Nitrogen oxide absorption in packed columns. (c) Continuous hydrogenation of fats. (d) Stirred tank reactor for batch hydrogenation of fats. (e) Nitrogen oxide absorption in a plate column. (f) A thin-film reactor for making dodecylbenzene sulfonate with SO₃. (g) Stirred tank reactor for the hydrogenation of caprolactam. (h) Tubular reactor for making adiponitrile from adipic acid in the presence of phosphoric acid.

Generally, the bubble column height can be greater than for tray or packed towers.

From a mechanical standpoint, a bubble column reactor is a vertical cylindrical vessel with nozzles or a sparger grid at the bottom. The sparger grid is an array of parallel pipes connected to a manifold or several radial arms in a spider pattern or concentric circles, all with downward-facing holes every few inches or so. The holes are sized to give exit velocities of 100 to 300 ft/s, and the gas enters the liquid as jets that break up into bubbles after a short distance. The height/diameter ratio of the vessel is at least 1.5 and may be as large as 20. Depending on the heat-transfer requirements, coils or a jacket may be needed.

The liquid may be in batch mode or enter from the top or bottom. The simplest mathematical model may assume that the liquid is well mixed and the gas is in plug flow. Liquid backmixing may have a detrimental effect on selectivity. In the oxidation of liquid *n*-butane, for instance, the ratio of methyl ethyl ketone to acetic acid is much higher in plug flow than in backmixed reactors. Similarly, in the air oxidation of isobutane to *tert*-butyl hydroperoxide, where *tert*-butanol also is obtained, plug flow is more desirable. Backmixing in the liquid may be reduced with packing or perforated plates. Packed bubble columns operate with flooded packing, in contrast with normal packed columns that usually operate below 70 percent of the flooding point. With packing, liquid backmixing is reduced and interfacial area is increased 15 to 80 percent, but the true mass-transfer coefficient remains the same. At relatively high superficial

gas velocities [10 to 15 cm/s (0.33 to 0.49 ft/s)] and for taller columns, backmixing is reduced so the vessel performs as a CSTR battery. Radial baffles (also called disk-and-doughnut baffles) are also helpful. A rule of thumb is that the hole should be about 0.7 times the vessel diameter, and the spacing should be 0.8 times the diameter.

The literature may provide guidance on several parameters: bubble diameter and bubble rise velocity, gas holdup, interfacial area, mass-transfer coefficient k_L , axial liquid-phase and gas-phase dispersion coefficients, and heat-transfer coefficient to the wall. The key design variable is the superficial velocity of the gas that affects the gas holdup, the interfacial area, and the mass-transfer coefficient. Each of these has been described in some detail by Deckwer (*Bubble Column Reactors*, Wiley, 1992). The effect of vessel diameter on these parameters is not well understood beyond $D \geq 0.15$ to 0.3 m (0.49 to 1 ft), the range for most of the existing literature correlations. From a qualitative standpoint, increasing the superficial gas velocity increases the holdup of gas, the interfacial area, and the overall mass-transfer coefficient. The ratio of height to diameter is not very important in the range of 4 to 10. Decreasing viscosity and decreasing surface tension increase the interfacial area. Electrolyte solutions have smaller bubbles, higher gas holdup, and higher interfacial area. Sparger design is unimportant for superficial gas velocities > 5 to 10 cm/s (0.16 to 0.32 ft/s) and tall columns. Liquid entrainment considerations (discussed in the appropriate section of the Handbook) provide an upper bound on gas superficial velocity; however, gas conversion falls off at higher

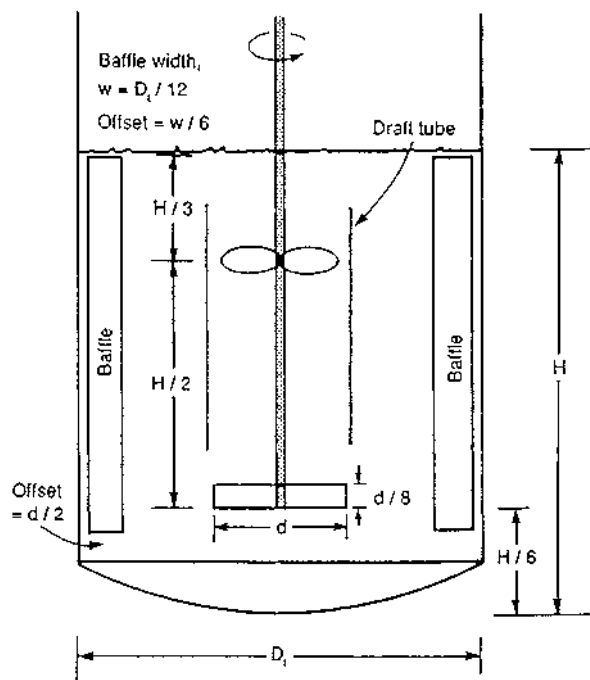


FIG. 19-30 A basic stirred tank design, not to scale, showing a lower radial impeller and an upper axial impeller housed in a draft tube. Four equally spaced baffles are standard. H = height of liquid level, D_i = tank diameter, d = impeller diameter. For radial impellers, $0.3 \leq d/D_i \leq 0.6$.

superficial velocities, so values under 10 cm/s (0.32 ft/s) are often desirable. Some examples of bubble column reactor types are illustrated in Fig. 19-31. Figure 19-31a is a conventional bubble column with no internals. Figure 19-31b is a tray bubble column. The trays are used to redistribute the gas into the liquid and to induce staging to approximate plug flow. Figure 19-31c is a packed bubble column with the packing being either an inert or a catalyst. Bubble columns are further discussed in the multiphase reactor section.

An excellent reference is Deckwer (*Bubble Column Reactors*, Wiley, 1992). Two complementary reviews of this subject are by Shah et al. [AIChE J. **28**: 353–379 (1982)] and Deckwer [in de Lasa (ed.), *Chemical Reactor Design and Technology*, Martinus Nijhoff, 1985, pp. 411–461]. Useful comments are made by Doraiswamy and Sharma (*Heterogeneous Reactions*, Wiley, 1984).

Examples

- A number of reactions in the production of pharmaceuticals or crop protection chemicals are conducted in bubble columns. Oxygen, chlorine, etc., may be the reactant gas.
- Hydrogenation reactions may be carried out in bubble column reactors. Often a slurry catalyst may be used which makes it a multiphase reactor.
- Aerobic fermentations are carried out in bubble columns when scale advantage is required, and the cells can be considered a third phase, making these multiphase reactors.

Tubular Reactors In a tubular or pipeline reactor, gas and liquid flow concurrently. A variety of flow patterns, ranging from a small quantity of bubbles in the liquid to small quantities of droplets in the gas, are possible, depending on the flow rate of the two streams. Figure 19-26f shows the patterns in horizontal flow; those in vertical flow are a little different.

Two-phase tubular reactors offer opportunities for temperature control, accommodate wide ranges of T and P , and approach plug flow, and the high velocities prevent settling of slurries or accumulations on the walls. Mixing of the phases may be improved by helical

in-line static mixing inserts. Idealized models use a PFR for both gas and liquid phases.

Depending on the gas and liquid residence times required, the reactor could be operated horizontally or vertically with either downflow or upflow. Weikard (in Ullmann, *Enzyklopaedie*, 4th ed., vol. 3, Verlag Chemie, 1973, p. 381) discusses possible reasons for operating an upflow concurrent flow tubular reactor for the production of adipic acid nitrile (from adipic acid and ammonia). The reactor has a liquid holdup of 20 to 30 percent and a residence time of 1.0 s for gas and 3 to 5 min for liquid.

1. The process has a large Hatta number; that is, the rate of reaction is much greater than the rate of diffusion, so a large interfacial area is desirable for carrying out the reaction.

2. With normal excess ammonia the gas/liquid ratio is about 3500 m³/m³. At this high ratio there is danger of fouling the surface with tarry reaction products. The ratio is brought down to a more satisfactory value of 1000 to 1500 by recycle of some of the effluent.

3. High selectivity of the nitrile is favored by short contact time.

4. The reaction is highly endothermic so heat input must be at a high rate.

Points 2 and 4 are the main ones governing the choice of reactor type. The high gas/liquid ratio restricts the choice to types d , e , i , and k in Fig. 19-26. Due to the high rate of heat transfer needed, the choice is a falling film or tubular reactor.

A loop reactor is used for the bioconversion of methane to produce biomass used, e.g., as fish meal. This is a large-diameter pipe operated at high liquid circulation velocity with the O₂/CH₄ feed injected at several locations along the reactor. Cooling of the exothermic aerobic fermentation is accomplished by external heat exchangers. Static mixers are used to maintain gas dispersion in the liquid.

Packed, Tray, and Spray Towers Packed and tray towers have been discussed in the subsection "Mass Transfer" in Sec. 5. Typically, the gas and liquid are countercurrent to each other, with the liquid flowing downward. Each phase may be modeled using a PFR or dispersion (series of stirred tanks) model. The model is solved numerically.

Spray columns are used with slurries or when the reaction product is a solid. The coefficient k_L in spray columns is about the same as in packed columns, but the spray interfacial area is much lower. Considerable backmixing of the gas also takes place, which makes the spray volumetrically inefficient. An entrainment control device (e.g., mist eliminator) usually is needed at the outlet. In the treatment of phosphate rock with sulfuric acid, off-gases contain HF and SiF₄. In a spray column with water, solid particles of fluorosilic acid are formed but do not harm the spray operation.

In venturi scrubbers the gas is the motive fluid. This equipment is of simple design and is able to handle slurries and large volumes of gas, but the gas pressure drop may be high. When the reaction is slow, further holdup in a spray chamber is necessary.

In liquid ejectors or aspirators, the liquid is the motive fluid, so the gas pressure drop is low. Flow of slurries in the nozzle may be erosive. Otherwise, the design is as simple as that of the venturi. Kohl and Riesenfeld (*Gas Purification*, Gulf, 1985, pp. 268–288) describe the application of liquid dispersion reactors to the absorption of fluorine gases.

Examples

- Process effluent gas emissions of CO₂ and H₂S are controlled in packed or tray towers. Aqueous solutions of monethanolamine (MEA), diethanolamine (DEA), and K₂CO₃ are the principal reactive solvents for the removal of acidic constituents from gas streams (Danckwerts and Sharma, *The Chemical Engineer*, 202, 1966, CE244). These solvents are all regenerable. Absorption proceeds at a lower temperature or higher pressure and regeneration is done in a subsequent vessel at higher temperature or lower pressure, usually with some assistance from stripping steam. The CO₂ can be recovered to make dry ice. H₂S is treated for recovery of the sulfur. Vessel diameters and allowable gas and liquid flow rates are established by the same correlations as for physical absorption. The calculation of tower heights utilizes vapor-liquid equilibrium data and enhanced mass-transfer coefficients for the

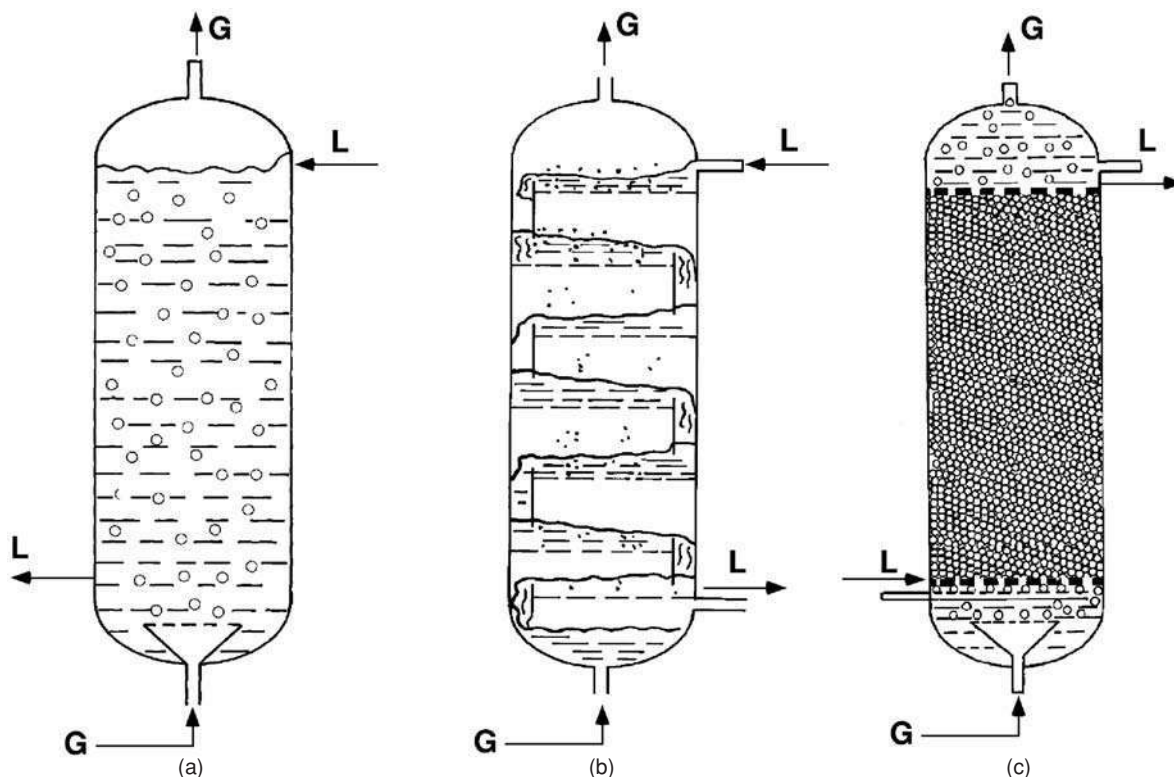


FIG. 19-31 Some examples of bubble column reactor types. (a) Conventional bubble column with no internals. (b) Tray bubble column. (c) Packed bubble column with the packing being either an inert or a catalyst. [From Mills, Ramachandran, and Chaudhari, "Multiphase Reaction Engineering for Fine Chemicals and Pharmaceuticals," *Reviews in Chemical Engineering*, 8(1-2), 1992, Figs. 2, 3, and 4.]

particular system. Such calculations are complex enough to warrant the use of the professional methods of tower design that are available from a number of service companies. Partly because of their low cost, aqueous solutions of sodium or potassium carbonate also are used for CO_2 and H_2S removal. Potassium bicarbonate has the higher solubility so the potassium salt is preferred. In view of the many competitive amine and carbonate plants that are in operation, the economics of alternative options have to be reviewed rather carefully. Additives are often used to affect equilibria and enhance absorption coefficients. Sodium arsenite is the major additive in use; however, sodium hypochlorite and small amounts of amines also are effective. *Sterically hindered amines* as promoters are claimed by Say et al. [*Chem. Eng. Prog.* 80(10): 72-77 (1984)] to result in 50 percent more capacity than ordinary amine promoters of carbonate solutions. Kohl and Riesenfeld (*Gas Purification*, Gulf, 1985) cite operating data for carbonate plants. Pilot-plant tests are reported on 0.10- and 0.15-m (4- and 6-in) columns packed to depths of 9.14 m (30 ft) of Raschig rings by Benson et al. [*Chem. Eng. Prog.* 50: 356 (1954)].

- SO_2 emissions from sulfuric acid plants are controlled in spray towers. Effluent gases contain less than 0.5 percent SO_2 . The SO_2 emissions have to be controlled (or recovered as elemental sulfur by, for example, the Claus process). An approach is to absorb the SO_2 in a lime (or limestone) slurry (promoted by small amounts of carboxylic acids, such as adipic acid). Flow is in parallel downward. The product calcium salt is sent to a landfill or sold as a by-product. Limestone is pulverized to 80 to 90 percent through 200 mesh. Slurry concentrations of 5 to 40 percent have been used in pilot plants.

Rotary wheel atomizers require 0.8 to 1.0 kWh/1000 L. The lateral throw of a spray wheel requires a large diameter to prevent accumulation on the wall; the ratio of length to diameter of 0.5 to 1.0 is in use in such cases. The downward throw of spray nozzles permits smaller diameters but greater depths; L/D ratios of 4 to 5 or more are used. Spray vessel diameters of 15 m (50 ft) or more are known. Liquid/gas ratios are 0.2 to 0.3 gal/MSCF. Flue gas enters at 149°C (300°F) at a velocity of 2.44 m/s (8 ft/s). Utilization of 80 percent of the solid reagent may be approached. Residence times are 10 to 12 s. At the outlet the particles are made just dry enough to keep from sticking to the wall, and the gas is within 11 to 28°C (20 to 50°F) of saturation. The fine powder is recovered with fabric filters. In one test facility, a gas with 4000 ppm SO_2 had 95 percent removal with lime and 75 percent removal with limestone.

- A study on the hydrolysis of fats with water was conducted at 230 to 260°C (446 to 500°F) and 41 to 48 atm (600 to 705 psi) in a continuous commercial spray tower. A small amount of water dissolved in the fat and reacted to form an acid and glycerine. Most of the glycerine migrated to the water phase. The tower was operated at about 18 percent of flooding, at which condition the HETS was found to be about 9 m (30 ft) compared with an expected 6 m (20 ft) for purely physical extraction [Jeffreys, Jensen, and Miles, *Trans. Inst. Chem. Eng.* 39: 389-396 (1961)].
- There are instances where an extractive solvent is employed to force completion of a reversible homogeneous reaction by removing the reaction product. In the production of KNO_3 from KCl and HNO_3 , for instance, the HCl can be removed continuously from the aqueous phase by contact with amyl alcohol, thus forcing completion [Baniel and Blumberg, *Chim. Ind.* 4: 27 (1957)].

SOLIDS REACTORS

Reactions of solids are typically feasible only at elevated temperatures. High temperatures are achieved by direct contact with combustion gases. Often, the product of reaction is a gas. The gas has to diffuse away from the reactant, sometimes through a solid product. Thermal and mass-transfer resistances are major factors in the performance of solids reactors. There are a number of commercial processes that utilize solid reactors. Reactor analysis and design appear to rely on empirical models that are used to fit the kinetics of solids decomposition. Most of the information on commercial reactors is proprietary.

General references on solids reactions include Brown, Dollimore, and Galwey ["Reactions in the Solid State," in Bamford and Tipper (eds.), *Comprehensive Chemical Kinetics*, vol. 22, Elsevier, 1980], Galwey (*Chemistry of Solids*, Chapman and Hall, 1967), Sohn and Wadsworth (eds.) (*Rate Processes of Extractive Metallurgy*, Plenum Press, 1979), Szekeley, Evans, and Sohn (*Gas-Solid Reactions*, Academic Press, 1976), and Ullmann (ed.) (*Enzyklopaedie der technischen Chemie*, "Uncatalyzed Reactions with Solids," vol. 3, 4th ed., Verlag Chemie, 1973, pp. 395-464).

THERMAL DECOMPOSITION

Thermal decompositions may be exothermic or endothermic. Solids that decompose on heating without melting often form gaseous products. When the product is a gas, the reaction rate can be affected by diffusion so particle size can be important. Aging of solids can result in crystallization of the surface. Annealing reduces strains and slows the decomposition rate. The decomposition of some fine powders follows a first-order rate law. Otherwise, empirical rate equations are available (e.g., in Galwey, *Chemistry of Solids*, Chapman and Hall, 1967).

A few organic compounds decompose before melting. These decomposition processes are highly exothermic and may cause explosions. Decomposition kinetics may follow an autocatalytic law. The temperature range for decomposition is 100 to 200°C (212 to 392°F). The decomposition of oxalic acid (m.p. 189°C) obeyed a zero-order law at 130 to 170°C (266 to 338°F). The decomposition of malonic acid has been measured for both the solid and the supercooled liquid.

Exothermic decompositions are nearly always irreversible. When several gaseous products are formed, the reverse reaction would require that these products all combine together, which is unlikely. Commercial interest in such materials lies more in their energy storage properties than as a source of desirable products. These are often nitrogen-containing compounds such as azides, diazo compounds, and nitramines. Ammonium nitrate, an important explosive, decomposes into nitrous oxide and water. In the solid phase, decomposition begins at about 150°C (302°F) but becomes extensive only above its melting point (170°C) (338°F). The reaction is first-order, with activation energy of about 40 kcal/(g-mol) [72,000 Btu/(lb-mol)]. Traces of moisture and Cl⁻ lower the decomposition temperature. Many investigations have reported on the decomposition of azides of barium, calcium, strontium, lead, copper, and silver in the range of 100 to 200°C (212 to 392°F). Activation energies were found to be 30 to 50 kcal/(g-mol) [54,000 to 90,000 Btu/(lb-mol)] or so. Some difficulties with data reproducibility were encountered with these hazardous materials. Lead styphnate (styphnic acid contains nitrogen) monohydrate was found to detonate at 229°C (444°F). The course of decomposition could be followed at 228°C and below. Sodium azide is a propellant in most motor vehicle SRS systems (airbags). Silver oxalate decomposes smoothly and completely in the range of 100 to 160°C (212 to 320°F). Ammonium chromates and some other solids exhibit aging effects. Material that has been stored for months or years follows a different decomposition rate than a fresh material. Examples of such materials are available in the review by Brown et al. ("Reactions in the Solid State," in Bamford and Tipper, *Comprehensive Chemical Kinetics*, vol. 22, Elsevier, 1980).

Endothermic decompositions are generally reversible. Hydroxides (which give off water) and carbonates (which give off CO₂) have been the most investigated compounds. Activation energies are nearly the same as reaction enthalpies. As the reaction proceeds, the rate of reaction may be limited by diffusion of the water through the product layer.

Since a particular compound may have several hydrates, the level of dehydration will depend on the partial pressure of water vapor in the gas. For example, FeCl₂ combines with 4, 5, 7, or 12 molecules of water with melting points ranging from about 75 to 40°C (167 to 104°F). The dehydration of CuSO₄ pentahydrate at 53 to 63°C (127 to 145°F) and of the trihydrate at 70 to 86°C (158 to 187°F) obeys the Avrami-Erofev equation $[-\ln(1-x) = kt^n]$, $n = 3.5, 4$. The rate of water loss from Mg(OH)₂ at lower temperatures is sensitive to the partial pressure of water. Its decomposition above 297°C (567°F) yields appreciable amounts of hydrogen and is not reversible. Carbonates decompose at relatively high temperatures, e.g., 660 to 740°C (1220 to 1364°F) for CaCO₃. When deep beds are used, the rate of heat transfer or the rate of CO₂ removal controls the decomposition rate. Some ammonium salts decompose reversibly and release ammonia, e.g., $(\text{NH}_4)_2\text{SO}_4 \rightleftharpoons \text{NH}_4\text{HSO}_4 + \text{NH}_3$ at 250°C (482°F). Further heating can release SO₃ irreversibly. The decomposition of silver oxide was one of the earliest solid reactions studied. It is smoothly reversible below 200°C (392°F). The reaction is sensitive to the presence of metallic silver at the start (indicating autocatalysis) and to the presence of silver carbonate, which was accidentally present in some investigations.

SOLID-SOLID REACTIONS

In solid-solid reactions, ions or molecules in solids diffuse to the interface prior to reaction. This diffusion takes place through the normal crystal lattices of reactants and products as well as in channels and fissures of imperfect crystals. Solid diffusion is slow compared to liquids even at the elevated temperatures at which these reactions have to be conducted. Solid-solid reactions are conducted in powder metallurgy. Typical particle sizes are 0.1 to 1000 μm and pressures are 138 to 827 MPa (20,000 to 60,000 psi). Reactions of solids occur in ceramic, metallurgical, and other industries. Even though cement manufacture has been discussed in the gas-solid reactor section, solid-solid reactions take place as well. Large contact areas between solid phases are essential. These may be obtained by forming and mixing fine powders and compressing them. Reaction times are 2 to 3 h at 1200 to 1500°C (2192 to 2732°F) even with 200-mesh particles.

The literature reports several examples of laboratory solid-solid reactions. The mechanism of zinc ferrite formation ($\text{ZnO} + \text{Fe}_2\text{O}_3 \Rightarrow \text{ZnFe}_2\text{O}_4$) has been studied up to temperatures of 1200°C (2192°F). At lower temperatures, ZnO is the mobile phase that migrates and coats the Fe₂O₃ particles. Similarly, MgO is the mobile phase in the $\text{MgO} + \text{Fe}_2\text{O}_3 \Rightarrow \text{MgFe}_2\text{O}_4$ reaction. Smaller particles (< 1 μm) obey the power law $x = k \ln t$, but larger ones have a more complex behavior. In the reaction $2\text{AgI} + \text{HgI}_2 \Rightarrow \text{Ag}_2\text{HgI}_4$, nearly equivalent amounts of the ions Ag⁺ and Hg₂²⁺ were found to migrate in opposite directions and arrive at their respective interfaces after 66 days at 65°C (149°F).

Several reactions that yield gaseous products have attracted attention because their progress is easily followed. Examples include $\text{MnO}_3 + 2\text{MoO}_3 \Rightarrow 2\text{MnMoO}_4 + 0.5\text{O}_2$ (where MoO₃ was identified as the mobile phase) and $\text{Ca}_3(\text{PO}_4)_2 + 5\text{C} \Rightarrow 3\text{CaO} + \text{P}_2 + 5\text{CO}$. For the reaction $\text{KClO}_4 + 2\text{C} \Rightarrow \text{KCl} + \text{CO}_2$, fine powders were compressed to 69 MPa (10,000 psi) and reacted at 350°C (662°F), well below the 500°C (932°F) melting point. The reaction $\text{CuCr}_2\text{O}_4 + \text{CuO} \Rightarrow \text{Cu}_3\text{Cr}_2\text{O}_4 + 0.5\text{O}_2$ eventually becomes diffusion-controlled and is described by the relationship $[1 - (1-x)^{1/3}]^2 = k \ln t$. In the reaction, $\text{CsCl} + \text{NaI} \Rightarrow \text{CsI} + \text{NaCl}$, two solid products are formed. The rate-controlling step is the diffusion of iodide ion in CsCl.

Carbothermic reactions are solid-solid reactions with carbon that apparently take place through intermediate CO and CO₂. The reduction of iron oxides has the mechanism $\text{Fe}_x\text{O}_y + y\text{CO} \Rightarrow x\text{Fe} + y\text{CO}_2$, $\text{CO}_2 + \text{C} \Rightarrow 2\text{CO}$. The reduction of hematite by graphite at 907 to 1007°C in the presence of lithium oxide catalyst was correlated by the equation $1 - (1-x)^{1/3} = kt$. The reaction of solid ilmenite ore and carbon has the mechanism $\text{FeTiO}_3 + \text{CO} \Rightarrow \text{Fe} + \text{TiO}_2 + \text{CO}_2$, $\text{CO}_2 + \text{C} \Rightarrow 2\text{CO}$. A similar case is the preparation of metal carbides from metal and carbon, $\text{C} + 2\text{H}_2 \Rightarrow \text{CH}_4$, $\text{Me} + \text{CH}_4 \Rightarrow \text{MeC} + 2\text{H}_2$.

Self-Propagating High-Temperature Synthesis (SHS) Conventional methods of synthesizing materials via solid reactions involve

multiple grinding, heating, and cooling of suitable precursor compounds. Reactions need extended time periods mainly because interdiffusion in solids is slow, even at high temperatures. By contrast, in SHS, highly reactive metal particles ignite in contact with boron, carbon, nitrogen, and silica to form boride, carbide, nitride, and silicide ceramics. Since the reactions are extremely exothermic, the reaction fronts propagate rapidly through the precursor powders. Usually, the

ultimate particle size can be controlled by the particle size of the precursors. In recent years, several commercial and semicommercial facilities have been built (in Russia, the United States, Spain, and Japan) to synthesize TiC powders, nitrided ferroalloys, silicon nitride (β -phase) and titanium hydride powders, high-temperature insulators, lithium niobate, boron nitride, etc. (e.g., Weimer, *Carbide, Nitride and Boride Materials Synthesis and Processing*, Chapman & Hall, 1997).

MULTIPHASE REACTORS

Multiphase reactors include, for instance, gas-liquid-solid and gas-liquid-liquid reactions. In many important cases, reactions between gases and liquids occur in the presence of a porous solid catalyst. The reaction typically occurs at a catalytic site on the solid surface. The kinetics and transport steps include dissolution of gas into the liquid, transport of dissolved gas to the catalyst particle surface, and diffusion and reaction in the catalyst particle. Say the concentration of dissolved gas A in equilibrium with the gas-phase concentration of A is C_{aL} . Neglecting the gas-phase resistance, the series of rates involved are from the liquid side of the gas-liquid interface to the bulk liquid where the concentration is C_{aL} , and from the bulk liquid to the surface of catalyst where the concentration is C_{as} and where the reaction rate is $\eta w k C_{as}^m$. At steady state,

$$r_a = k_L a (C_{aL} - C_{aL}) = k_s a_s (C_{aL} - C_{as}) = \eta w k C_{as}^m \quad (19-78)$$

where w is the catalyst loading (mass of catalyst per slurry volume). For a first-order reaction, $m = 1$, the catalyst effectiveness η is independent of C_{as} , so that after elimination of C_{aL} and C_{as} the explicit solution for the observed specific rate is

$$r_{a,\text{observed}} = C_{aL} \left(\frac{1}{k_L a} + \frac{1}{k_s a_s} + \frac{1}{\eta w k} \right)^{-1} \quad (19-79)$$

More complex chemical rate equations will require numerical solution. Ramachandran and Chaudhari (*Three-Phase Chemical Reactors*, Gordon and Breach, 1983) apply such rate equations to the sizing of plug flow, CSTR, and dispersion reactors. They list 75 reactions and identify reactor types, catalysts, temperature, and pressure for processes such as hydrogenation of fatty oils, hydrodesulfurization, Fischer-Tropsch synthesis, and miscellaneous hydrogenations and oxidations. A list of 74 gas-liquid-solid reactions with literature references has been compiled by Shah (*Gas-Liquid-Solid Reactions*, McGraw-Hill, 1979), classified into groups where the solid is a reactant, a catalyst, or an inert. Other references include de Lasa (*Chemical Reactor Design and Technology*, Martinus Nijhoff, 1986), Gianetto and Silveston (eds.) (*Multiphase Chemical Reactors*, Hemisphere, 1986), Ramachandran et al. (eds.) (*Multiphase Chemical Reactors*, vol. 2, Sijthoff & Noordhoff, 1981) and Satterfield [*Trickle Bed Reactors*, *AIChE J.* **21**: 209–228 (1975)]. Some contrasting charac-

teristics of the main kinds of three-phase reactors are summarized in Table 19-13.

BIOREACTORS

Bioreactors use live cells or enzymes to perform biochemical transformations of feedstocks to desired products. Bioreactor operation is restricted to conditions at which these biological systems can function. Most plant and animal cells live at moderate temperatures and do not tolerate extremes of pH. The vast majority of microorganisms also prefer mild conditions, but some thrive at temperatures above the boiling point of water or at pH values far from neutral. Some can endure concentrations of chemicals that most other cells find highly toxic. Commercial operations depend on having the correct organisms or enzymes and preventing death (or deactivation) or the entry of foreign organisms that could harm the process.

The pH, temperature, redox potential, and nutrient medium may favor certain organisms and discourage the growth of others. In mixed culture systems, especially those for biological waste treatment, there is an ever-shifting interplay between microbial populations and their environments that influences performance and control. Although open systems may be suitable for hardy organisms or for processes in which the conditions select the appropriate culture, many bioprocesses are closed and elaborate precautions including sterilization and cleaning are taken to prevent contamination. The optimization of the complicated biochemical activities of isolated strains, of aggregated cells, of mixed populations, and of cell-free enzymes or components presents engineering challenges. Performance of a bioprocess can suffer from changes in any of the many biochemical steps functioning in concert, and genetic controls are subject to mutation. Offspring of specialized mutants, especially bioengineered ones that yield high concentrations of product, tend to revert during propagation to less productive strains—a phenomenon called *rundown*.

Developments such as immobilized enzymes and cells have been exploited partially, and genetic manipulations through recombinant DNA techniques are leading to practical processes for molecules that could previously be found only in trace quantities in plants or animals.

Bioreactors may have either two phases (liquid-solid, e.g., in anaerobic processes) or three phases (gas-liquid-solid, e.g., aerobic processes). The solid phase typically contains cells that serve as the biocatalyst. The

TABLE 19-13 Characteristics of Gas-Liquid-Solid Reactors

Property	Trickle bed	Flooded	Stirred tank	Entrained solids	Fluidized bed
Gas holdup	0.25–0.45	Small	0.2–0.3		
Liquid holdup	0.05–0.25	High	0.7–0.8		
Solid holdup	0.5–0.7		0.01–0.10		0.5–0.7
Liquid distribution	Good only at high liquid rate		Good	Good	Good
RTD, liquid phase	Narrow	Narrower than for entrained solids reactor	Wide	Wide	Narrow
RTD, gas phase	Nearly plug flow		Backmixed	Backmixed	Narrow
Interfacial area	20–50% of geometrical	Like trickle bed reactor	100–1500 m ² /m ³	100–400 m ² /m ³	Less than for entrained solids reactor
MTC, gas/liquid	High		Intermediate		
MTC, liquid/solid	High		High		
Radial heat transfer	Slow		Fast	Fast	Fast
Pressure drop	High with small d_p		Hydrostatic head		

RTD = residence time distribution; MTC = mass-transfer coefficient.

solid can be either the free biocatalyst (bacteria, fungi, algae, etc.), also called the biotic phase (with density close to water), or an immobilized version, in which case the cells are immobilized on a solid structure (e.g., porous particles). The liquid is primarily water with dissolved feed (usually a sugar together with mineral salts and trace elements) and products (referred to as metabolites). In aerobic bioreactors, the gas phase is primarily air with the product gas containing product CO_2 produced by the organism and evaporated water. Bioreactors are mainly mechanically agitated tanks, bubble columns and air lift reactors. For low biomass concentrations (e.g., less than 60 g/L) bioreactor design is similar to that of a gas-liquid reactor. For some specialized applications, such as in some wastewater treatment processes, packed beds or slurry reactors with immobilized biocatalyst are used. Figure 19-32 shows some typical bioreactors.

While bioreactors do not differ fundamentally from other two- and three-phase reactors, as indicated above, there are more stringent requirements regarding control of temperature, pH, contamination (presence and growth of other microorganisms or phage), and toxicity (that may result from high feed and product concentrations). In aerobic processes, since O_2 is required for respiration, it must be properly distributed and managed. Whereas bacteria and yeast cells are very robust, cultivations of filamentous fungi and especially animal cell cultures and plant cell cultures are quite shear-sensitive. To maintain a robust culture of animal and plant cells, very gentle stirring either by a mechanical stirrer or by gas sparging is usually necessary. Unlike chemical catalysis, one of the (main) bioreaction products is biomass (new cells), leading to autocatalytic behavior; i.e., the rate of production of new cells per liquid volume is proportional to the cell concentration. Section 7 of this Handbook presents more details on the kinetics of bioreactions.

Bioreactors mainly operate in batch or semibatch mode, which allows better control of the key variables. However, an increasing number of bioprocesses are operated in continuous mode, typically processes for treating wastewater, but also large-scale processes such as lactic acid production, conversion of natural gas to biomass (single-cell protein production), and production of human insulin using genetically engineered yeast. Continuous operation requires good process control, especially of the sterility of the feed, but also that the biocatalyst be robust and its traits (especially for bioengineered strains) persist over many generations.

Several special terms are used to describe traditional reaction engineering concepts. Examples include *yield coefficients* for the generally fermentation environment-dependent stoichiometric coefficients, *metabolic network* for reaction network, *substrate* for feed, *metabolite* for secreted bioreaction products, *biomass* for cells, *broth* for the fermenter medium, *aeration rate* for the rate of air addition, *vvm* for volumetric airflow rate per broth volume, *OUR* for O_2 uptake rate per broth volume, and *CER* for CO_2 evolution rate per broth volume. For continuous fermentation, *dilution rate* stands for feed or effluent rate (equal at steady state), *washout* for a condition where the feed rate exceeds the cell growth rate, resulting in washout of cells from the reactor. Section 7 discusses a simple model of a CSTR reactor (called a chemostat) using empirical kinetics.

The mass conservation equations for a batch reactor are as follows:

$$\text{Cells: } V_r \frac{dC_x}{dt} = (r_g - r_d)V_r \quad (19-80)$$

$$\text{Substrate: } V_r \frac{dC_s}{dt} = Y_{xs}(-r_g)V_r - r_{sm}V_r \quad (19-81)$$

$$\text{Product: } V_r \frac{dC_p}{dt} = Y_{sp}(r_g)V_r \quad (19-82)$$

Several of the terms above have been discussed in Sec. 7: r_g and r_d are the specific rates (per broth volume) for cell growth and death, respectively; r_{sm} is the specific rate of substrate consumed for cell maintenance, and Y_{xi} are the stoichiometric yield coefficient of species i relative to biomass x . The maintenance term in Eq. (19-81) can result

also in an increased production of product p [additional term required in Eq. (19-82)] for metabolites such as lactic acid, but not for protein production. In many cases, a semibatch reactor is used, where the reactants are added with an initial cells and sugar concentration, and a certain feed profile or recipe is used—this is also called *fed batch* operation mode.

Further modeling details are available in the books by Nielsen, Villadsen, and Liden (*Bioreaction Engineering Principles*, Kluwer Academic/Plenum Press, 2003) and Fogler (*Elements of Chemical Reaction Engineering*, 3d ed., Prentice-Hall, 1999). Bioreactors and bioreaction engineering are discussed in detail by Bailey and Ollis (*Biochemical Engineering Fundamentals*, 2d ed., McGraw-Hill, 1986), Clark (*Biochemical Engineering*, Marcel Dekker, 1997), and Schugerl and Bellgardt (*Bioreaction Engineering, Modeling and Control*, Springer, 2000).

ELECTROCHEMICAL REACTORS

Electrochemical reactors are used for electrolysis (conversion of electric energy to chemicals, e.g., chlor-alkali), power generation (conversion of chemicals to electric energy, e.g., batteries or fuel cells), or for chemical separations (electrodialysis). An electrochemical cell contains at least two electronically conducting electrode phases and one ionic conducting electrolyte phase. The electrolyte phase separates the two electrode phases. The electrode phases are also connected to each other through an electronically conducting pathway, typically external of the electrochemical cell; but in the case of corrosion, the electrode phases may be localized regions on the same piece of metal, the bulk metal allowing electron flow between the regions. Thus a series electric circuit is completed beginning at one electrode through the electrolyte to the second electrode and then out of the reactor through the external circuit back into the starting electrode.

An electrochemical cell reaction involves the transfer of electrons across an electrode/electrolyte interface. There are two types of electrochemical cell reactions. In one reaction the electron transfer is from an electrode to a chemical species within the electrolyte, resulting in a reduction process, and in this case the electrode is defined as the cathode. The second electrochemical reaction involves the electron transfer from a chemical species within the electrolyte to an electrode, resulting in an oxidation process; in this case the electrode is defined as the anode. Each of these cathode (reduction) or anode (oxidation) electrochemical reactions is considered a half-cell reaction. Since an electrochemical cell requires a complete series electric circuit, the overall electrochemical cell reaction is the stoichiometric sum of the electrochemical half-cell reactions, and all electrochemical cell reactions are close-coupled to maintain the conservation of electric charge. Electrochemical cell reactions are considered heterogeneous reactions since they occur at the interface of the electrode surface and electrolyte. Sometimes the electrochemical product species is employed, in turn, as a reducing or oxidizing species, either in the bulk electrolyte or in a separate external process vessel. Subsequently, the spent reducing or oxidizing species is regenerated within the electrochemical reactor. This augmentation is known as a mediated (or indirect) electrochemical process. More details on the mechanism and kinetics of electrochemical reactions are given in Sec. 7.

An electrochemical reactor is a controlled volume containing the electrolyte and two electrodes. The electrode phases may be a solid, e.g., carbon or metal, or a liquid, e.g., mercury. The geometry of the electrodes is optimized to maximize energy efficiency and/or cell life and usually consists of parallel plates or concentric cylinders. The electrolyte may be a liquid (such as concentrated brine in the production of caustic or a molten salt in the production of aluminum) or a solid (such as a proton-conducting Nafion® membrane in fuel cells). As the electric current passes through the electrolyte, a voltage drop occurs that represents an energy loss; therefore, the gap or spacing between the electrodes is usually minimized. The electrodes may also be separated by a membrane, a diaphragm, or a separator so as to prevent the unwanted mixing of chemical species, ensure process safety, and maintain product purity and yield. One or both of the electrodes may evolve a gas (e.g., chlorine); or alternatively, one or both of the electrodes may be fed with a gas (e.g., hydrogen or oxygen) to reduce cell voltage or utilize gaseous fuels. Examples of electrochemical reactors are shown in Fig. 19-33.

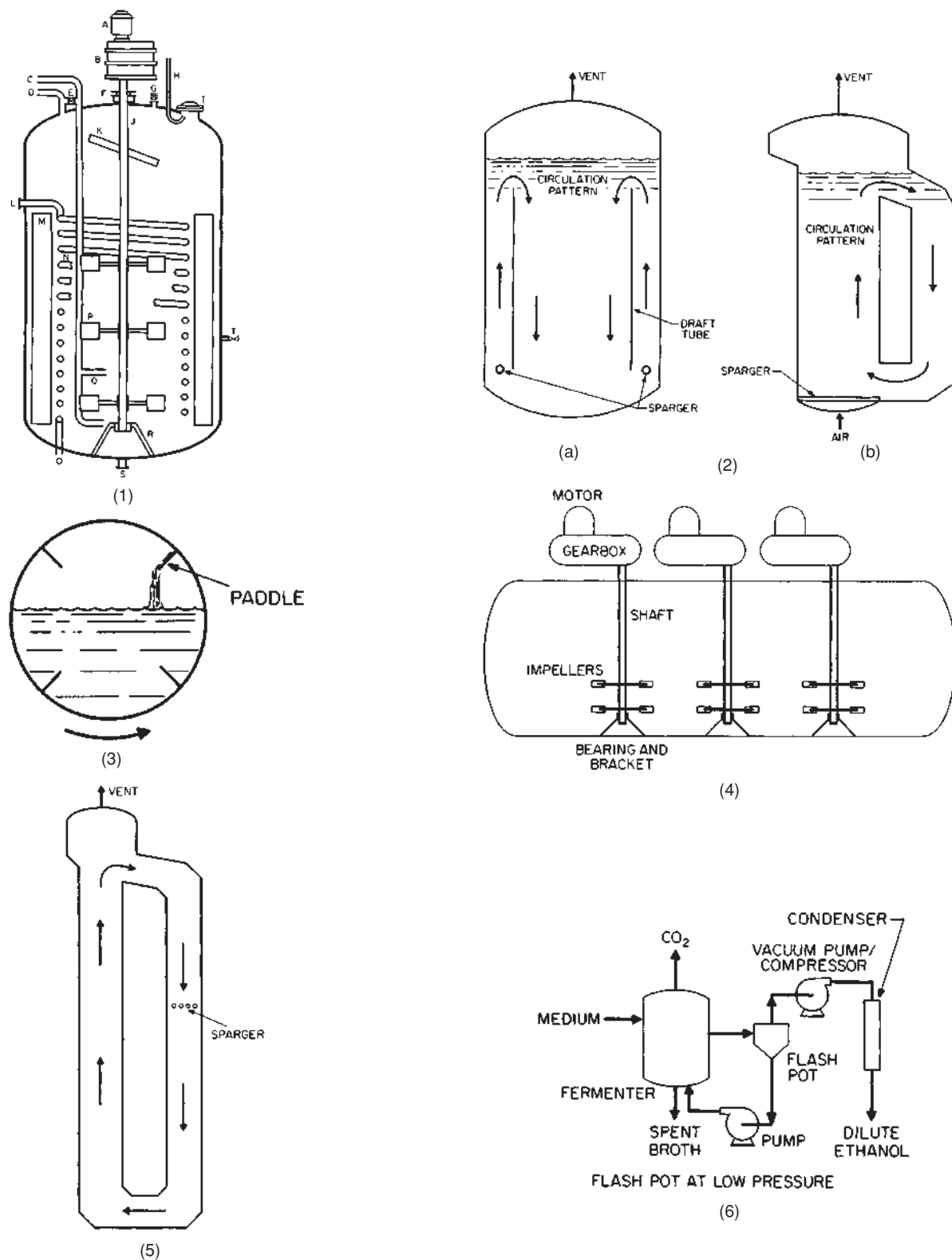


FIG. 19-32 Some examples of fermenters. (1) Conventional batch fermenter. (2) Air lift fermenters: (a) Concentric cylinder or bubble column with draft tube; (b) external recycle. (3) Rotating fermenter. (4) Horizontal fermenter. (5) Deep-shaft fermenter. (6) Flash-pot fermenter.

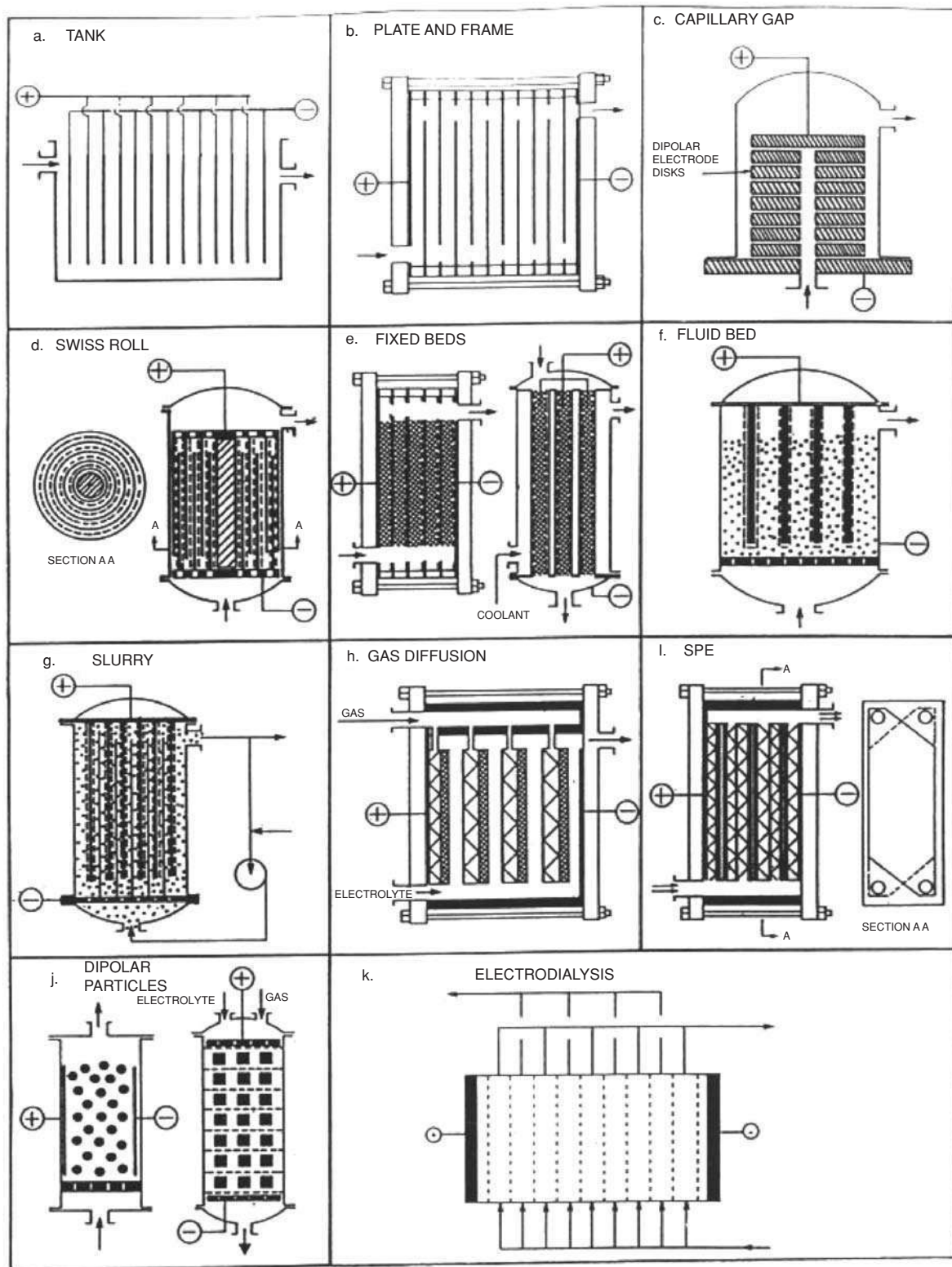


FIG. 19-33 Electrochemical reactor configurations. [From Oloman, *Electrochemical Processing for the Pulp and Paper Industry*, The Electrochemical Consultancy, 1999, p. 79, Fig. 2.10; printed in Great Britain by Alresford Press Ltd. Referring to "Tutorial Lectures in Electrochemical Engineering and Technology" (D. Chin and R. Alkire, eds.), AIChE Symposium Series 229, vol. 79, 1983; reproduced with permission.]

The size of an electrochemical reactor may be determined by evaluating the capital costs and the operating costs (on a dollar per unit mass basis) as a function of the operating current density (production rate per unit electrode area basis). Typically, the capital costs decrease with increasing current density, and the operating cost increase with current density, thus, a minimum in the total costs may be observed and serve as a basis for the sizing of the electrochemical reactor. Given an optimal current density, the electrochemical reactor design is refined to minimize voltage losses and maximize current efficiency. This is done by taking into consideration the component availability (e.g., membrane widths), the management of the excess heat removal, the minimization of pressure drops (due to liquid and gas traffic within the electrochemical reactor), and the maintenance costs (associated with reactor rebuilding). The largest, cost-effective reactor size is then replicated to meet production capacity needs. An electrochemical reactor usually has shorter operating life than the rest of the plant facility, requiring the periodic rebuilding of the reactors.

In electrochemical engineering, several terms share similar definitions to traditional reaction engineering. These include fractional conversion, yield, selectivity, space velocity, and space time yield. Several terms are unique to electrochemical reaction engineering such as *cell voltage* (the electric potential difference between the two electrodes within the electrochemical cell) and *cell overpotentials* (voltage losses within the electrochemical cell). Voltage losses include (1) *ohmic overpotential* (associated with passage of electric current in the bulk of the electrolyte phase and the bulk electrode phases and the electrical conductors between the electrochemical cell and the power supply or electrical load); (2) *activation overpotential* (associated with the limiting rates at which some steps in the electrode reactions can proceed); and (3) *concentration overpotential* (generated from the local depletion of reactants and accumulation of products at the electrode/electrolyte interface relative to the bulk electrolyte phase due to mass transport limitations). The *current density* is the current per unit surface area of the electrode. Typically, the geometric or projected area is utilized since the true electrode area is usually difficult to estimate due to surface roughness and/or porosity. It is related to the production rate of the electrolytic cell through the Faraday constant. The *current efficiency* is the ratio of the theoretical electric charge (coulombs) required for the amount of product obtained to the total amount of electric charge passed through the electrochemical cell. Many of these and other terms are discussed in Sec. 7, in Pletcher and Walsh (*Industrial Electrochemistry*, 2d ed., Chapman and Hall, 1984) and in Gritzner and Kreysa ["Nomenclature, Symbols and Definitions in Electrochemical Engineering," *Pure & Appl. Chem.* **65**: 5, 1009–1020 (1993)].

A discussion of electrochemical reactors is available in books by Prentice (*Electrochemical Engineering Principles*, Prentice-Hall, 1991), Hine (*Electrode Processes and Electrochemical Engineering*, Plenum Press, 1985), Oloman (*Electrochemical Processing for the Pulp and Paper Industry*, The Electrochemical Consultancy, 1996), and Goodridge and Scott (*Electrochemical Process Engineering: A Guide to the Design of Electrolytic Plant*, Plenum, 1995).

REACTOR TYPES

Multiphase reactors are typically mechanically agitated vessels, bubble columns, trickle bed, flooded fixed beds, gas-liquid-solid fluidized beds, and entrained solids reactors. *Agitated reactors* keep solid catalysts in suspension mechanically; the overflow may be a clear liquid or slurry, and the gas disengages from the vessel. *Bubble column reactors* keep the solids in suspension as a result of agitation caused by the sparging gas. In *trickle bed reactors* both gas and liquid phases flow down through a packed bed of catalyst. The reactor is gas continuous with liquid "trickling" as a film over the solid catalyst. In *flooded reactors*, the gas and liquid flow upward through a fixed bed. The reactor is liquid continuous. As the superficial velocity is increased, the solids first become suspended (as a dense fluidized bed) and may eventually be entrained and the effluent separated into its phases in downstream equipment. When the average residence time of solids approaches that of the liquid, the reactor becomes an *entrained solids reactor*.

Agitated Slurry Reactors The gas reactant and solid catalyst are dispersed in a continuous liquid phase by mechanical agitation using stirrers. Most issues associated with gas-liquid-solid stirred tanks are analogous to the gas-liquid systems. In addition to providing good

gas-liquid contacting, the agitation has to be sufficient to maintain the solid phase suspended. Catalytic reactions in stirred gas-liquid-solid reactors are used in a large number of applications including hydrogenations, oxidations, halogenations, and fermentations.

The benefits of using a mechanically agitated tank include nearly isothermal operation, excellent heat transfer, good mass transfer, and use of high-activity powder catalyst with minimal intraparticle diffusion limitations. The reactors may be operated in a batch, semibatch, or continuous mode; and catalyst deactivation may be managed by on-line catalyst makeup. Scale-up is relatively straightforward through geometric similarity and by providing the agitator power/volume required to produce the same volumetric mass-transfer coefficient at different scales. The hydrodynamics are decoupled from the gas flow rate. Some downsides of stirred gas-liquid-solid reactors include difficulty with catalyst/liquid product separation and lower volumetric productivity than fixed beds (due to lower catalyst loading per reactor volume). In addition, the reactor size may be limited due to high power consumption (due to horsepower limitations on agitator motor)—typically the limit is at around 50 m³. Sealing of the agitator system can also be challenging for large reactors (magnetic coupling is used for small to midrange units). These result in increased capital and operating costs.

Solid particles are in the range of 0.01 to 1.0 mm (0.0020 to 0.039 in), the minimum size limited by filterability. Small diameters are used to provide as large an interface as possible to minimize the liquid-solid mass-transfer resistance and intraparticle diffusion limitations. Solids concentrations up to 30 percent by volume may be handled; however, lower concentrations may be used as well. For example, in hydrogenation of oils with Ni catalyst, the solids content is about 0.5 percent. In the manufacture of hydroxylamine phosphate with Pd-C, the solids content is 0.05 percent.

The hydrodynamic parameters that are required for stirred tank design and analysis include phase holdups (gas, liquid, and solid); volumetric gas-liquid mass-transfer coefficient; liquid-solid mass-transfer coefficient; liquid, gas, and solid mixing; and heat-transfer coefficients. The hydrodynamics are driven primarily by the stirrer power input and the stirrer geometry/type, and not by the gas flow. Hence, additional parameters include the power input of the stirrer and the pumping flow rate of the stirrer.

The reactant gas either is sparged below the stirrer or is induced from the vapor space by a *gas-inducing* agitator which has a hollow shaft with suction orifices on the shaft and discharge orifices on the impeller. Impellers vary with applications. For low-viscosity applications, flat-bladed Rushton turbines are widely used and provide radial mixing and gas dispersion. Pitched-blade turbines may also be used to induce axial flow. Often multiple impellers are provided on one shaft, sometimes with a mix of flat blade and pitched-blade type agitators. Additional information may be obtained from the corresponding section in this Handbook and from Baldyga and Bourne (*Turbulent Mixing and Chemical Reactions*, Wiley, 1998).

As the stirrer speed is increased, different flow regimes are observed depending on the stirrer type/geometry and the nature of the gas-liquid system considered. For example, for a Rushton turbine with a low-viscosity liquid, three primary flow regimes are observed (Fig. 19-34). Regime I (Fig. 19-34a) has single bubbles that rise, and the gas is not dispersed uniformly. Regime II (Fig. 19-34b) has the gas dispersed radially as the bubbles ascend. Regime III (Fig. 19-34c) has the gas recirculated to the stirrer in an increasingly complex pattern [see, e.g., Baldi, *Hydrodynamics and Gas-Liquid Mass Transfer in Stirred Slurry Reactors*, in Gianetto and Silveston (eds.), *Multiphase Chemical Reactors*, Hemisphere, 1986].

For gas-liquid systems, the power dissipated by the stirrer at the same stirrer speed N is lower than the corresponding power input for liquid systems due to reduced drag on the impeller. The power of the gassed system P_G is related to that of the ungassed system P_0 by using the power number N_p correlation with the aeration number N_a :

$$N_p = \frac{P_G}{P_0} \quad (19-83)$$

$$N_a = \frac{Q_G}{ND_i^3} \quad (19-84)$$

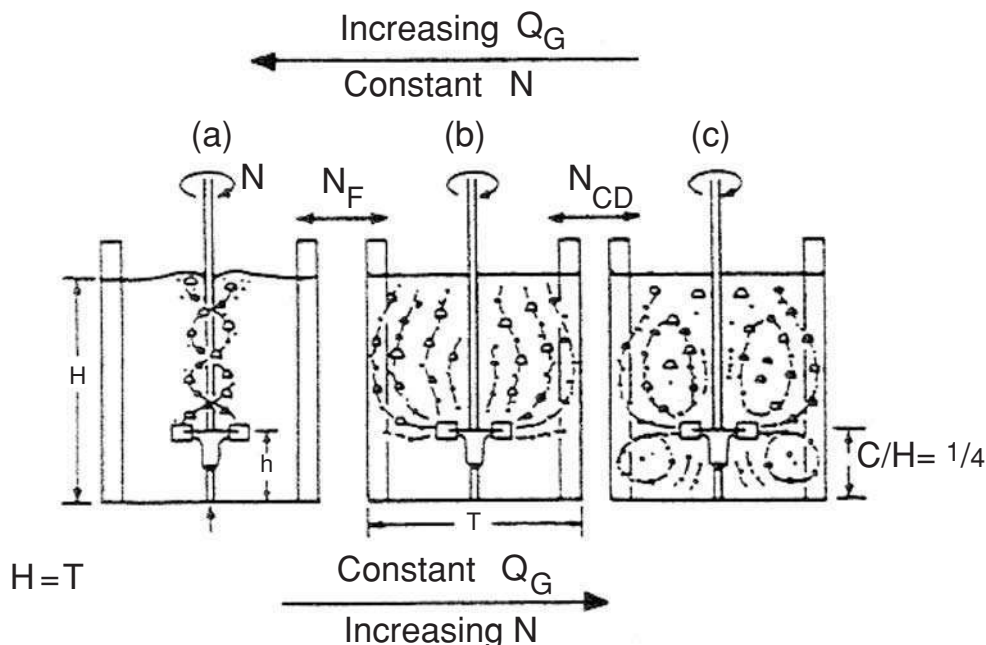


FIG. 19-34 Gas circulation as a function of stirrer speed. (From Nienow *et al.*, 5th European Conference on Mixing, Wurzburg, 1985; published by BHRA, The Fluid Engineering Centre, Cranfield, England; Fig. 1.)

The power number is a decreasing function of the aeration rate, as shown in Fig. 19-35.

For instance, Hughmark [*Ind. Eng. Chem. Proc. Des. Dev.* **19**: 638 (1980)] developed a correlation for the power number of Rushton turbines that correlates a large database:

$$N_p = 0.1 N_a^{-0.25} \left(\frac{D_f^3}{V_L} \right)^{-0.25} \left(\frac{N^2 D^4}{g H V_L^{2/3}} \right)^{-0.2} \quad (19-85)$$

Increasing the solids content increases the power number, as indicated, e.g., by Wiedman *et al.* [*Chem. Eng. Comm.* **6**: 245 (1980)].

With solids present, a minimum agitator speed is required to suspend all the solids, e.g., the correlation of Baldi *et al.* [*Chem. Eng. Sci.* **33**: 21 (1978)]:

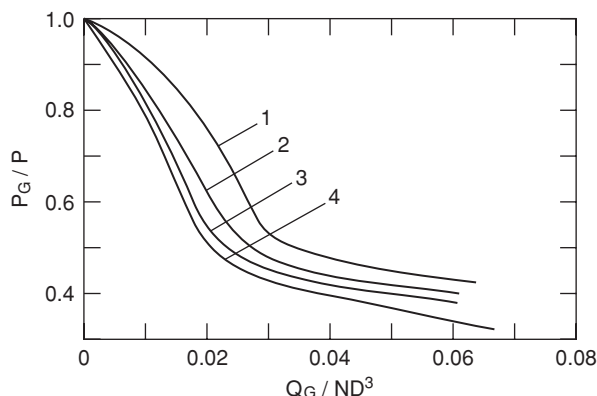


FIG. 19-35 Effect of aeration number and stirrer speed on the power number— N increases in order of $N_1 < N_2 < N_3 < N_4$. [Adapted from Baldi, "Hydrodynamics and Mass Transfer in Stirred-Slurry Reactors," in Gianetto and Silveston (eds.), *Multiphase Chemical Reactors*, Hemisphere Publishing Corp., 1986, Fig. 14.8.]

$$N_m = \frac{\beta_2 \mu_L^{0.17} [g(\rho_p - \rho_L)]^{0.42} d_p^{0.14} w^{0.125}}{\rho_L^{0.58} D_i^{0.89}} \quad (19-86)$$

where w is the catalyst loading in weight percent and parameter β_2 depends on reactor/impeller ratio, e.g., from Nienow [*Chem. Eng. J.* **9**: 153 (1975)], $\beta_2 = 2(d_R/D_i)^{1.33}$.

Gas holdup and volumetric gas-liquid mass-transfer coefficients are correlated with the gassed power input/volume and with the aeration rate (actual gas superficial velocity), e.g., the correlation of van't Riet [*Ind. Eng. Chem. Proc. Des. Dev.* **18**: 357 (1979)] for the volumetric mass-transfer coefficient of coalescing and noncoalescing systems:

$$k_L a = \begin{cases} 2.6 \times 10^{-2} \left(\frac{P_G}{V_L} \right)^{0.4} u_G^{0.5} & \text{for coalescing nonviscous liquids} \\ 2.0 \times 10^{-3} \left(\frac{P_G}{V_L} \right)^{0.7} u_G^{0.2} & \text{for noncoalescing nonviscous liquids} \end{cases} \quad (19-87)$$

For the gas holdup a similar correlation was developed by Loiseau *et al.* [*AIChE J.* **23**: 931 (1977)]:

$$\epsilon_G = \begin{cases} 0.011 \sigma^{-0.36} \mu_L^{-0.056} \left(\frac{P'_G}{V_L} \right)^{0.27} u_G^{0.36} & \text{for nonfoaming systems} \\ 0.0051 \left(\frac{P'_G}{V_L} \right)^{0.57} u_G^{0.24} & \text{for nonfoaming system} \end{cases} \quad (19-88)$$

$$\frac{P'_G}{V_L} = \frac{P_G}{V_L} + 0.03 \frac{Q_G \rho_G u_0^2}{V_L} + \frac{Q_G \rho_G R T}{V_L M W_G} \ln \frac{P_1}{P_2}$$

The last two terms of the power/volume equation include the power/volume from the isothermal expansion of the gas through the gas distributor holes having a velocity u_0 and the power/volume to transfer the gas across the hydrostatic liquid head.

Increasing the solids loading leads to a decrease in gas holdup and gas-liquid volumetric mass-transfer coefficient at the same power/volume [e.g., Inga and Morsi, *Can. J. Chem. Eng.* **75**: 872 (1997)].

Liquid-solid mass transfer is typically not limiting due to the small particle size resulting in large particle surface area/volume of reactor, unless the concentration of the particles is very low, and/or larger particles are used. In the latter case, intraparticle mass-transfer limitations would also occur. Ramachandran and Chaudhari (*Three-Phase Catalytic Reactors*, Gordon and Breach, 1983) present several correlations for liquid-solid mass transfer, typically as a Sherwood number versus particle Reynolds and Schmidt numbers, e.g., the correlation of Levins and Glastonbury [*Trans. Inst. Chem. Engrs.* **50**: 132 (1972)]:

$$\text{Sh} = \begin{cases} 2 + 0.44 \text{Re}_p^{0.5} \text{Sc}^{0.38} \\ \frac{k_d d_p}{D} \end{cases} \quad \text{Re}_p = \frac{\rho_L u_c d_p}{\mu_L} \quad \text{Sc} = \frac{\nu_L}{D} \quad (19-89)$$

Here u_c is a characteristic velocity, and the velocity terms composing it are estimated from additional correlations.

There is good heat transfer in agitated gas-liquid-solid slurry reactors; see, e.g., van't Riet and Tramper for correlations (*Basic Bioreactor Design*, Marcel Dekker, 1991).

Additional information on mechanically agitated gas-liquid-solid reactors can be obtained in van't Riet and Tramper (*Basic Bioreactor Design*, Marcel Dekker, 1991), Ramachandran and Chaudhari (*Three-Phase Catalytic Reactors*, Gordon and Breach, 1983), and Gianetto and Silveston (*Multiphase Chemical Reactors*, Hemisphere, 1986).

Examples

- Liquid benzene is chlorinated in the presence of metallic iron turnings or Raschig rings at 40 to 60°C (104 to 140°F).
- Carbon tetrachloride is made from CS₂ by bubbling chlorine into it in the presence of iron powder at 30°C (86°F).
- Substances that have been hydrogenated in slurry reactors include nitrobenzene with Pd-C, butynediol with Pd-CaCO₃, chlorobenzene with Pt-C, toluene with Raney® Ni, and acetone with Raney® Ni.
- Some oxidations in slurry reactors include cumene with metal oxides, cyclohexene with metal oxides, phenol with CuO, and *n*-propanol with Pt.
- Aerobic fermentations.

For many hydrogenations, semibatch operations often are preferred to continuous ones because of the variety of feedstocks or product specifications, or long reaction times, or small production rates. A sketch of a batch hydrogenator is shown in Fig. 19-36.

The vegetable oil hydrogenator, which is to scale, uses three impellers. The best position for inlet of gas is at a point of maximum turbulence near the impeller, or at the bottom of the draft tube. A sparger is desirable; however, an open pipe is often used. A two-speed motor is desirable to prevent overloading. Since the gassed power requirement is significantly less than ungassed, the lower speed is used when the gas supply is cut off but agitation is to continue. In tanks of 5.7 to 18.9 m³ (1500 to 5000 gal), rotation speeds are from 50 to 200 rpm and power requirements are 2 to 75 hp; both depend on superficial velocities of gas and liquid [Hicks and Gates, *Chem. Eng.*, pp. 141–148 (July 1976)]. As a rough guide, power requirements and impeller tip speeds are shown in Table 19-14.

Edible oils are mixtures of unsaturated compounds with molecular weights in the vicinity of 300. The progress of the hydrogenation reaction is expressed in terms of iodine value (IV), which is a measure of unsaturation. The IV is obtained by a standardized procedure in which the iodine adds to the unsaturated double bond in the oil. IV is the ratio of the amount of iodine absorbed per 100 g of oil.

To start a hydrogenation process, the oil and catalyst are charged, then the vessel is evacuated for safety and hydrogen is continuously added and maintained at some fixed pressure, usually in the range of 1 to 10 atm (14.7 to 147 psi). Internal circulation of hydrogen is provided by axial and radial impellers or with a hollow impeller that throws the gas out centrifugally and sucks gas in from the vapor space through the hollow shaft. Some plants have external gas circulators. Reaction times are 1 to 4 h. For edible oils, the temperature is kept at about 180°C (356°F). Since the reaction is exothermic and because space for heat-transfer coils in the vessel is limited, the process is organized to give a maximum IV drop of about 2.0/min. The rate of

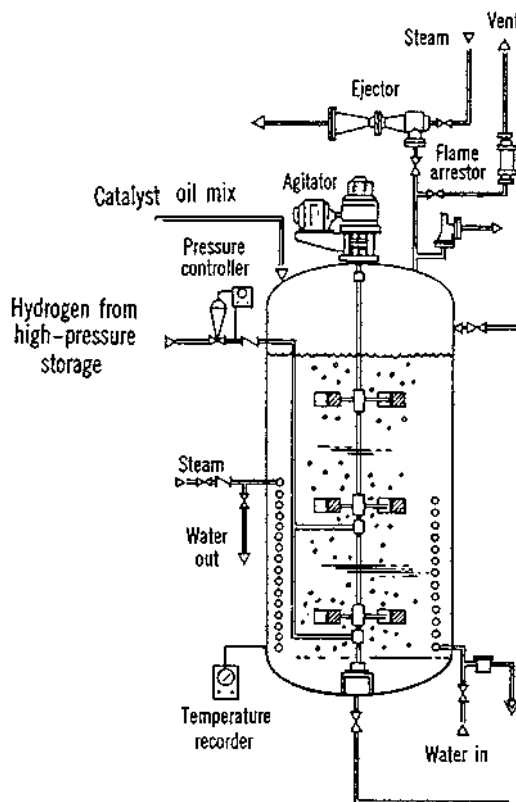


FIG. 19-36 Stirred tank hydrogenator for edible oils. (Votator Division, Chemetron Corporation.)

reaction drops off rapidly as the reaction proceeds, so a process may take several hours. The endpoint of a hydrogenation is a specified IV of the product. Hardness or refractive index also can be measured to follow reaction progress.

Saturation of the oil with hydrogen is maintained by agitation. The rate of reaction depends on agitation and catalyst concentration. Beyond a certain agitation rate, resistance to mass transfer is eliminated, and the rate becomes independent of pressure. The effect of catalyst concentration also reaches limiting values. The effects of pressure and temperature on the rate are indicated by Fig. 19-37.

A supported nickel catalyst (containing 20 to 25 weight percent Ni on a porous silica particle) is typically used. The pores allow access of the reactants to the extended pore surface, which is in the range of 200 to 600 m²/g (977 × 10³ to 2931 × 10³ ft²/lbm) of which 20 to 30 percent is catalytically active. The concentration of catalyst in the slurry can vary over a wide range but is usually under 0.1% Ni. After the reaction is complete, the catalyst can be easily separated from the product. Catalysts are subject to degradation and poisoning, particularly by sulfur compounds. Accordingly, 10 to 20 percent of the recovered catalyst is replaced by fresh catalyst before reuse. Other catalysts are applied in

TABLE 19-14 Power Requirements and Impeller Tip Speed Guidelines

Operation	hp/1000 gal ^a	Tip speed, ft/s
Homogeneous reaction	0.5–1.5	7.5–10
With heat transfer	1.5–5	10–15
Liquid-liquid mixing	5	15–20
Gas-liquid mixing	5–10	15–20

^a 1 hp/1000 gal = 0.197 kW/m³.

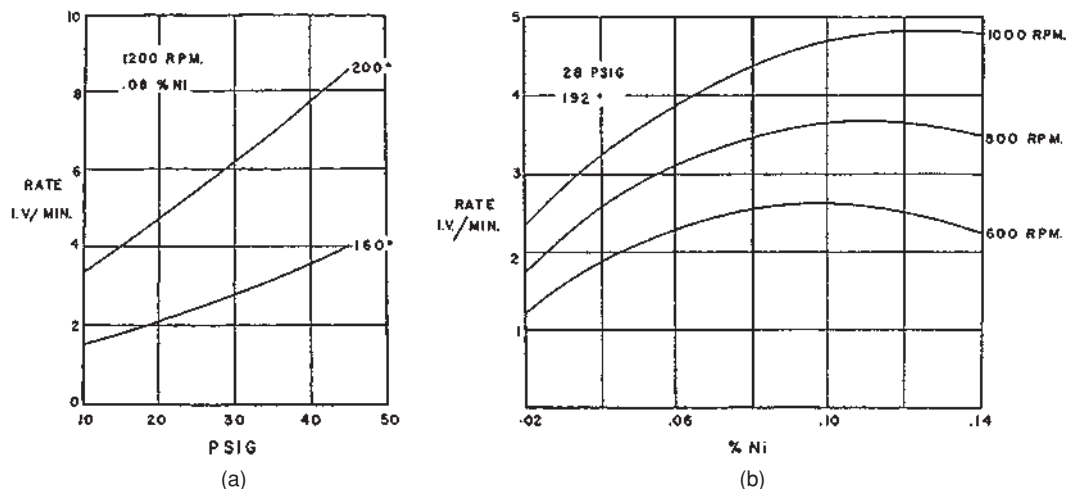


FIG. 19-37 Hydrogenation of soybean oil. (a) Effect of reaction pressure and temperature on rate. (b) Effect of catalyst concentration and stirring rate on hydrogenation. [Sivern (ed.), Bailey's Industrial Oil and Fat Products, vol. 2, Wiley, 1979.]

special cases. Expensive palladium has about 100 times the activity of nickel and is effective at lower temperatures. A case study of the hydrogenation of cottonseed oil was made by Rase (*Chemical Reactor Design for Process Plants*, vol. 2, Wiley, 1977, pp. 161–178).

Slurry Bubble Column Reactors As in the case of gas-liquid slurry agitated reactors, bubble column reactors may also be used when solids are present. Most issues associated with multiphase bubble columns are analogous to the gas-liquid bubble columns. In addition, the gas flow and/or the liquid flow have to be sufficient to maintain the solid phase suspended. In the case of a bubble column fermenter, the sparged oxygen is partly used to grow biomass that serves as the catalyst in the system. Many bubble columns operate in semibatch mode with gas sparged continuously and liquid and catalyst in batch mode.

The benefits of using slurry bubble columns include nearly isothermal operation, excellent heat transfer, good mass transfer, and use of high-activity powder catalyst with minimal intraparticle diffusion limitations. The reactors may be operated in a batch, semibatch, or continuous mode and require less power input than mechanically agitated reactors. Catalyst deactivation may be managed by on-line catalyst makeup. The reactor (essentially an empty shell with a sparger grid at the bottom) is easy to design, and the capital investment can be low. Some downsides of slurry bubble column reactors include catalyst/liquid product separation difficulty and lower volumetric productivity than fixed beds (due to lower catalyst loading per reactor volume), and catalyst distribution can be skewed with higher concentration at the bottom than at the top of the reactor. Also, accounting for the effect of internals (e.g., heat exchange tubes) and of increased diameter on the hydrodynamics is not well understood. Hence gradual scale-up is often required over multiple intermediate scales before commercialization. Cold flow models can also be useful in determining hydrodynamics in the absence of reaction.

As is the case for reactors with two or more mobile phases, a variety of flow regimes exist depending primarily on the gas superficial velocity (the driver for bubble column hydrodynamics) and column diameter. A qualitative flow regime map is shown in Fig. 19-38.

In the homogeneous flow regime at low gas superficial velocity, bubbles are relatively small and rise at constant rate (about 20 to 25 cm/s). As the flow rate is increased, bubbles become larger and irregular in shape, they frequently coalesce and break up, and the transition to churn turbulent regime is obtained. In small-diameter columns, the larger bubbles may bridge the column, creating slugs—hence the slug flow regime. The large transition zones in Fig. 19-38 are indicative of the lack of accurate knowledge and of the dependence of the transition region on conditions (temperature, pressure) and physical properties of the gas and liquid.

Hydrodynamic parameters that are required for bubble column design and analysis include phase holdups (gas, liquid, and solid for

slurry bubble columns); volumetric gas-liquid mass-transfer coefficient; liquid-solid mass-transfer coefficient; liquid, gas, and solid axial and radial mixing; and heat-transfer coefficients. These parameters depend strongly on the prevailing flow regime.

Correlations for gas holdup and the volumetric gas-liquid mass-transfer coefficient can have the general form

$$\epsilon_G = \alpha u_G^\beta \quad k_{LG} = \gamma u_G^\delta \quad (19-90)$$

where u_G is the superficial gas velocity, ϵ_G is the gas holdup (fraction of gas volume), k_L is the liquid-side gas-liquid mass-transfer coefficient, and a is the interfacial area per volume of either the liquid or the expanded liquid (liquid + gas). The exponents are $\beta, \delta \sim 1$ for the homogeneous bubbly flow regime and $\beta, \delta < 1$ for heterogeneous turbulent flow regime. The correlations depend on the gas-liquid-solid system properties. Gas-liquid systems can be classified as coalescing leading to increased bubble size, and noncoalescing, leading to larger gas holdup and volumetric mass-transfer coefficients for the latter. There is a voluminous literature for these parameters, and there is substantial variability in estimated values—one should be careful to validate the parameters with data applicable to the real system considered. For instance, for gas holdup see the correlation of Yoshida

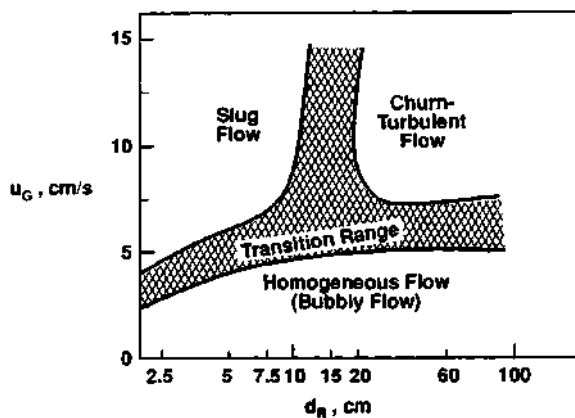


FIG. 19-38 Flow regime map for gas-liquid bubble columns. [Fig. 16 in Deckwer et al., Ind. Eng. Chem. Process Des. Dev. 19:699–708 (1980).]

and Akita [*AIChE J.* **11**: 9 (1965)]

$$\frac{\epsilon_G}{(1 - \epsilon_G)^4} = \alpha \left(\frac{\rho_L g d_R^2}{\sigma} \right)^{1/8} \left(\frac{\rho_L^2 g d_R^3}{\mu_L^2} \right)^{1/12} \frac{u_G}{\sqrt{g d_R}}$$

$$\alpha = \begin{cases} 0.2 & \text{for pure liquids and nonelectrolytes} \\ 0.25 & \text{for salt solutions} \end{cases} \quad (19-91)$$

and for volumetric gas-liquid mass-transfer coefficient, see the correlation of Akita and Yoshida [*IEC Proc. Des. Dev.*, **12**: 76 (1973)]:

$$k_{La} = 0.6 \frac{D}{d_R^2} \left(\frac{\mu_L}{\rho_L D} \right)^{0.5} \left(\frac{\rho_L g d_R^2}{\sigma} \right)^{0.62} \left(\frac{\rho_L^2 g d_R^3}{\mu_L^2} \right)^{0.31} \epsilon_G^{1.1} \quad (19-92)$$

More recent correlations for gas holdup and mass transfer include the effect of pressure and bimodal bubble size distribution (small and large bubbles), in a manner analogous to the treatment of dilute and dense phases in fluidized beds [see, e.g., Letzel et al., *Chem. Eng. Sci.*, **54**: (13): 2237 (1999)].

Increasing the catalyst loading decreases the gas holdup and the volumetric gas-liquid mass transfer coefficient [see, e.g., Maretti and Krishna, *Catalysis Today*, **52**: 279 (1999)].

Axial mixing in the liquid, induced by the upflow of the gas bubbles, can be substantial in commercial-scale bubble columns, especially in the churn turbulent regime. Due to typically small particle size, the axial dispersion of the solid catalyst in slurry bubble columns is expected to follow closely that of the liquid; exceptions are high-density particles. The liquid axial mixing can be represented by an *axial dispersion coefficient*, which typically has the form

$$D_{aL} = \alpha u_G^\beta d_R^\gamma \quad (19-93)$$

Based on theoretical considerations (Kolmogoroff's theory of isotropic turbulence), $\beta = 1/3$ and $\gamma = 4/3$. For example, Deckwer et al. [*Chem. Eng. Sci.* **29**: 2177 (1973)] developed the following correlation:

$$D_{aL} = 2.7 u_G^{0.3} d_R^{1.4} \quad (19-94)$$

It is expected that the strong dependence on reactor diameter only extends up to a maximum diameter beyond which there is no effect of diameter; however, there is disagreement among experts as to what that maximum diameter may be. There are a large number of correlations for liquid axial dispersion with widely different predictions, and care must be exerted to validate the predictions with data at some significant scale, even if only in a cold flow mockup.

The gas axial mixing is due to the bubble size distribution resulting in a distribution of bubble rise velocities, which varies along the column due to bubble breakup and coalescence. There are a variety of correlations in the literature, with varying results and reliability, for instance, the correlation of Mangartz and Pilhofer [*Verfahrenstechn.*, **14**: 40 (1980)].

$$D_{aG} = 5 \times 10^{-4} \left(\frac{u_G}{\epsilon_G} \right)^3 d_R^{1.5} \quad (19-95)$$

This equation is dimensional, and cm/s for u_G , cm for d_R , and cm²/s for D_{aG} should be used. The radial mixing can be represented by radial dispersion coefficients for the gas and the liquid. For instance, the liquid radial dispersion coefficient is estimated at less than one-tenth of the axial one.

Correlations for the heat-transfer coefficient have the general form

$$St = f(Re \, Fr \, Pr^2)$$

$$Re \, Fr = \frac{u_G^3 \rho_L}{\mu_L g} \quad Pr = \frac{c_{pL} \mu_L}{\lambda_L} \quad St = \frac{h_w}{u_G \rho_L c_{pL}} \quad (19-96)$$

For instance, see the correlation of Deckwer et al. [*Chem. Eng. Sci.* **35**(6): 1341–1346 (1980)].

$$St = 0.1 (Re \, Fr \, Pr^2)^{-1/4} \quad (19-97)$$

Additional information on hydrodynamics of bubble columns and slurry bubble columns can be obtained from Deckwer (*Bubble Column Reactors*, Wiley, 1992), Nigam and Schumpe (*Three-Phase Sparged Reactors*, Gordon and Breach, 1996), Ramachandran and Chaudhari (*Three-Phase Catalytic Reactors*, Gordon and Breach, 1983), and Gianetto and Silvestro (*Multiphase Chemical Reactors*, Hemisphere, 1986). Computational fluid mechanics approaches have also been recently used to estimate mixing and mass-transfer parameters [e.g., see Gupta et al., *Chem. Eng. Sci.* **56**(3): 1117–1125 (2001)].

Examples There are a number of examples including Fischer-Tropsch synthesis in the presence of Fe or Co catalysts, methanol synthesis in the presence of Cu/Zn solid catalyst, and hydrocracking in the presence of zeolite catalyst. Fermentation reactions are conducted in bubble column reactors when there is a benefit for increased scale and for reduced cost. The oxygen is sparged from the bottom, and the liquid reactants are added in a semibatch mode. The absence of reactor internals is an advantage as it prevents contamination. Heat transfer has to be managed through a cooling jacket. If heat removal is an issue, cooling coils may be installed.

Fluidized Gas-Liquid-Solid Reactors In a gas-liquid-solid fluidized bed reactor, only the fluid mixture leaves the vessel. Gas and liquid enter at the bottom. Liquid is continuous, gas is dispersed. Particles are larger than in bubble columns, 0.2 to 1.0 mm (0.008 to 0.04 in). Bed expansion can be small. Bed temperatures are uniform within 2°C (3.6°F) in medium-size beds, and heat transfer to embedded surfaces is excellent. Catalyst may be bled off and replenished continuously, or reactivated continuously.

Figure 19-39 shows examples of gas-liquid-solid fluidized-bed reactors. Figure 19-39a illustrates a conventional gas-liquid-solid fluidized bed reactor. Figure 19-39b shows an *ebullating bed* reactor for the hydroprocessing of heavy crude oil. A stable fluidized bed is maintained by recirculation of the mixed fluid through the bed and a draft tube. Reactor temperatures may range from 350 to 600°C (662 to 1112°F) and 200 atm (2940 psi). An external pump sometimes is used instead of the built-in impeller shown. Such units were developed for the liquefaction of coal.

A biological treatment process (Dorr-Oliver Hy-Flo) employs a vertical column filled with sand on which bacterial growth takes place while waste liquid and air are charged. A large interfacial area for reaction is provided, about 33 cm²/cm³ (84 in²/in³). BOD removal of 85 to 90 percent is claimed in 15 min compared with 6 to 8 h in conventional units.

In entrained beds, the three-phase mixture flows through the vessel and is separated downstream. These reactors are used in preference to fluidized beds when catalyst particles are very fine or subject to disintegration or if the catalyst deactivates rapidly in the process.

Trickle Bed Reactors Reactant gas and liquid flow cocurrently downward through a packed bed of solid catalyst particles. The most common use of trickle bed reactors is for hydrogenation reactions. The solubility of feed hydrogen in the liquid even at the higher pressure is insufficient to provide the stoichiometric needs of the reaction, and a gas flow exceeding the need is fed into the reactor. High hydrogen partial pressures can prevent catalyst deactivation due to undesirable reactions, such as coking. Cooling (or heating) is typically done between stages either with heat transfer to a coolant outside the reactor or through direct cooling with a cold reactant gas or liquid.

Advantages of a trickle bed are ease of installation, low liquid holdup (and therefore less undesirable homogeneous reactions), minimal catalyst handling issues, low catalyst attrition, and catalyst life of 1 to 4 years. The liquid and gas flow in trickle beds approaches plug flow (leading to higher conversion than slurry reactors for the same reactor volume). Downsides of trickle beds include flow maldistribution (bypassing), sensitivity to packing uniformity and prewetting (leading to hot spots), incomplete contacting/wetting, intraparticle diffusion resistance, potential for fouling and bed plugging due to particulate matter in the feed, and high pressure drop. A significant fraction of the flow is gas that has to be compressed and recycled (i.e., increased compressor costs).

A schematic of a trickle bed reactor is shown in Fig. 19-40. The reactor is a high-pressure vessel equipped with a drain and a manhole for vessel entry. Typical vessel diameters may range from 3 to 30 ft with height from 6 to 100 ft. The liquid enters the reactor and is

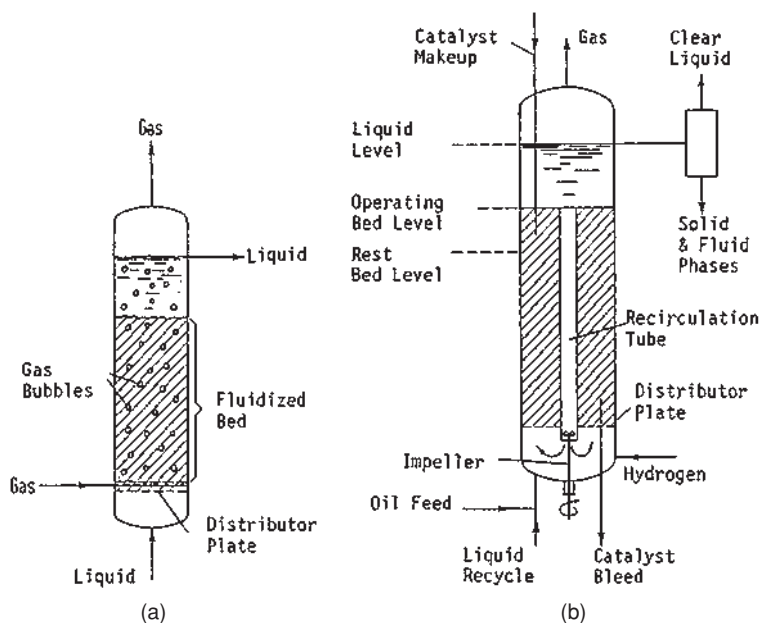


FIG. 19-39 Gas-liquid-solid reactors. (a) Three-phase fluidized-bed reactor. (b) Ebullating bed reactor for hydroliquefaction of coal. (Kampiner, in Winnacker-Keuchler, *Chemische Technologie*, vol. 3, Hanser, 1972, p. 252.)

distributed across the cross-section by a distributor plate. The liquid feed flows downward due to gravity helped along by the drag of the gas at such a low rate that it is distributed over the catalyst as a thin film. The gas enters at the top and is distributed along with the liquid. In the simplest arrangement, the liquid distributor is a perforated plate with about 10 openings/dm² (10 openings/15.5 in²), and the gas enters

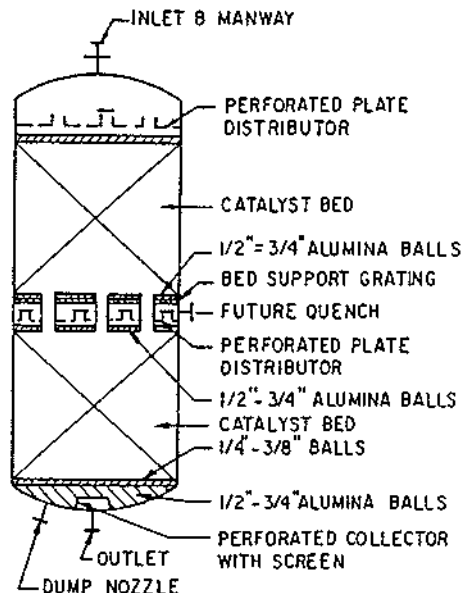


FIG. 19-40 Trickle bed reactor for hydrotreating 20,000 bbl/d of light catalytic cracker oil at 370°C and 27 atm. To convert atm to kPa, multiply by 101.3. (Baldi in Gianetto and Silveston, *Multiphase Chemical Reactors*, Hemisphere, 1986, pp. 533-563.)

through several risers about 15 cm (5.9 in) high. More elaborate distributor caps also are used. Uniform distribution of liquid across the reactor is critical to reactor performance. The aspect ratio of the reactor can vary between 1 and 10 depending on the pressure drop that can be accommodated by the compressor. It is not uncommon to redistribute the liquid using a redistribution grid every 8 to 15 ft.

The catalyst is often loaded on screens supported by a stainless steel grid near the bottom of the reactor. Often, large inert ceramic balls are loaded at the very bottom, with slightly smaller ceramic balls above the first layer, and then the catalyst. Smaller inert ceramic balls can also be loaded above the catalyst bed and topped off with the larger balls. The layer of inert balls can be 6 in to 2 ft in depth. The balls restrict the movement of the bed and distribute the liquid across the catalyst.

As is the case when two or more mobile phases are present, cocurrent gas-liquid downflow through packed beds produces a variety of flow regimes depending on the gas and liquid flow rates and the physical properties of the gas and the liquid. In Fig. 19-41, a flow regime map for trickle beds of foaming and nonfoaming systems is presented. Here L and G are the liquid and gas fluxes (mass flow rate per total flow cross-sectional area). In the low interaction or trickle flow regime, gas is the continuous phase and the liquid is flowing as rivulets. Increasing the liquid and gas flow results in high interaction or pulse flow, with the liquid and gas alternatively bridging the bed voids. At high liquid flow and low gas flow, the liquid becomes the continuous phase and the gas is the dispersed phase, called dispersed bubble flow. Finally at high gas flow and low liquid flow, the spray flow regime exists with liquid being the dispersed phase.

The literature contains a number of references to other flow regime maps; however, there is no clear advantage of using one map versus another. Wall effects can also have a major effect on the hydrodynamics of trickle bed reactors. Most of the data reported in the literature are for small laboratory units of 2-in diameter and under.

Hydrodynamic parameters that are required for trickle bed design and analysis include *bed void fraction*, *phase holdups* (gas, liquid, and solid), *wetting efficiency* (fraction of catalyst wetted by liquid), *volumetric gas-liquid mass-transfer coefficient*, *liquid-solid mass-transfer coefficient* (for the wetted part of the catalyst particle surface), *gas-solid*

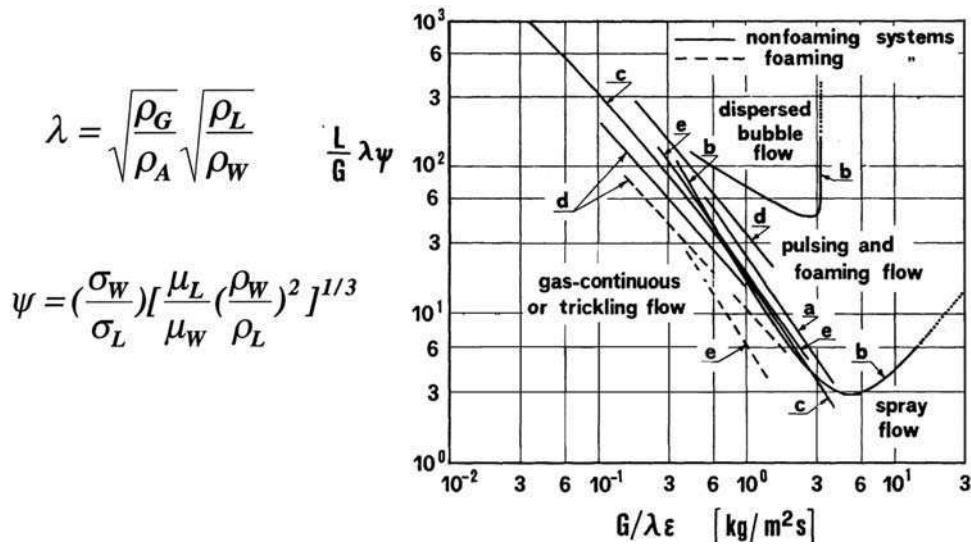


FIG. 19-41 Trickle bed flow regime map. [From Gianetto *et al.*, *AIChE J.* **24**(6):1087-1104 (1978); reproduced with permission.]

mass-transfer coefficient (for the unwetted part of the catalyst particle surface), liquid and gas axial mixing, pressure drop, and heat-transfer coefficients. These parameters vary with the flow regime (i.e., for the low and high interaction regimes).

There are a number of pressure drop correlations for two-phase flow in packed beds originating from the Lockhart-Martinelli correlation for two-phase flow in pipes. These correlate the two-phase pressure drop to the single-phase pressure drops of the gas and the liquid obtained from the Ergun equation. See, for instance, the Larkins correlation [Larkins, White, and Jeffrey, *AIChE J.* **7**: 231 (1967)]

$$\ln \frac{\Delta P_{GL}}{\Delta P_L + \Delta P_G} = \frac{5.0784}{3.531 + (\ln X)^2}$$

$$\text{where } X = \sqrt{\frac{\Delta P_L}{\Delta P_G}} \quad 0.05 \leq X \leq 30 \quad (19-98)$$

Since some of the published pressure drop correlations can differ by an order of magnitude, it is best to verify the relationship with actual data before designing a reactor. Other approaches to two-phase pressure drop include the relative permeability method of Saez and Carbonell [*AIChE J.* **31**(1): 52-62 (1985)].

The bed void volume available for flow and for gas and liquid holdup is determined by the particle size distribution and shape, the particle porosity, and the packing effectiveness. The total voidage and the total liquid holdup can be divided into external and internal terms corresponding to interparticle (bed) and intraparticle (porosity) voidage. The external liquid holdup is further subdivided into *static holdup* ϵ_{Ls} (holdup remaining after bed draining due to surface tension forces) and *dynamic holdup* ϵ_{Ld} . Additional expressions for the liquid holdup are the pore fillup F_i and the liquid saturation S_L :

$$\begin{aligned} \epsilon_t &= \epsilon_B + \epsilon_p(1 - \epsilon_B) && \text{total voidage} \\ \epsilon_L &= \epsilon_{Le} + \epsilon_{Li} && \text{total liquid holdup} \\ \epsilon_{Le} &= \epsilon_{Ld} + \epsilon_{Ls} && \text{external liquid holdup} \\ \epsilon_{Li} &= F_i \epsilon_p(1 - \epsilon_B) && \text{internal liquid holdup} \end{aligned} \quad (19-99)$$

$$S_L = \frac{\epsilon_L}{\epsilon_B} \quad \text{liquid saturation}$$

The static holdup can be correlated with the Eotvos number N_{Eo} as it results from a balance of surface tension and gravity forces on the liquid held up in the pores in absence of flow:

$$N_{Eo} = \frac{\text{gravity force}}{\text{surface tension force}} = \frac{\rho_L g d_p^2}{\sigma_L} \quad (19-100)$$

For instance Fig. 19-42 illustrates the dependence of the static holdup on the Eotvos number for porous and nonporous packings.

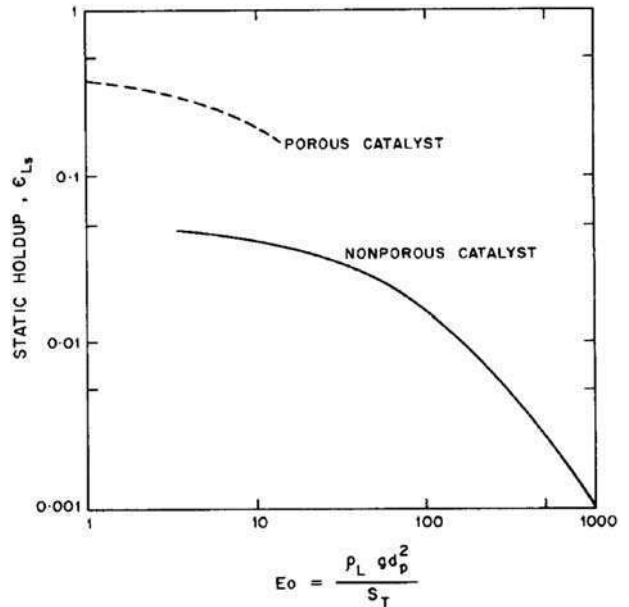


FIG. 19-42 The static liquid holdup for porous and nonporous solids. (Fig. 7.7 in Ramachandran and Chaudhari, *Three-Phase Catalytic Reactors*, Gordon and Breach, 1983.)

A variety of correlations have been developed for the total and the dynamic liquid holdup. For instance, the total liquid holdup has been correlated with the Lockhardt-Martinelli parameter X for spherical and cylindrical particles [Midou, Favier, and Charpentier, *J. Chem. Eng. Japan*, **9**: 350 (1976)]

$$\frac{\epsilon_L}{\epsilon_b} = \frac{0.66X^{0.81}}{1 + 0.66X^{0.81}} \quad (19-101)$$

Correlations for the dynamic liquid holdup have also been developed as function of various dimensionless numbers including the liquid and gas Reynolds number, and the two-phase pressure drop [see, e.g., Ramachandran and Chaudhari, *Three-Phase Catalytic Reactors*, Gordon and Breach, 1983; and Hofmann, *Hydrodynamics and Hydrodynamic Models of Fixed Bed Reactors*, in Gianetto and Silveston (eds.), *Multiphase Chemical Reactors*, Hemisphere 1986].

The various volumetric mass-transfer coefficients are defined in a manner similar to that discussed for gas-liquid and fluid-solid mass transfer in previous sections. There are a large number of correlations obtained from different gas-liquid-solid systems. For more details see Shah (*Gas-Liquid-Solid Reactor Design*, McGraw-Hill, 1979), Ramachandran and Chaudhari (*Three-Phase Catalytic Reactors*, Gordon and Breach, 1983), and Shah and Sharma [*Gas-Liquid-Solid Reactors* in Carberry and Varma (eds.), *Chemical Reaction and Reactor Engineering*, Marcel Dekker, 1987].

Axial mixing of the liquid is an important factor in the design of trickle bed reactors, and criteria were proposed to establish conditions that limit axial mixing. Mears [*Chem. Eng. Sci.* **26**: 1361 (1971)] developed a criterion that when satisfied, ensures that the conversion will be within 5 percent of that predicted by plug flow:

$$Pe = \frac{u_L L}{D} > 20n \ln \frac{1}{1-x} \quad (19-102)$$

where n is the order of the reaction with respect to the limiting reactant and x is the fractional conversion of that reactant. Correlations for axial dispersion can be found in Ramachandran and Chaudhari, *Three-Phase Catalytic Reactors*, Gordon and Breach, 1983.

Incomplete wetting can be also a critical factor in reactor design and analysis, leading usually to lower performance due to incomplete utilization of the catalyst bed. In a few select cases, the opposite may be the case, e.g., when a volatile reactant reacts faster than its liquid phase because it is not limited by the gas-liquid mass-transfer resistance and higher gas diffusivity. Correlations for the fraction of catalyst surface wetted are available, although not very reliable and strongly system-dependent (e.g., Shah, *Gas-Liquid-Solid Reactor Design*, McGraw-Hill, 1979).

Due to the complex hydrodynamics and the dependence of the hydrodynamic parameters on the flow regime, trickle beds are notoriously difficult to scale up. Laboratory units (used for kinetics and process development) and commercial units typically are operated at the same *liquid hourly space velocity* (LHSV). Since the LHSV represents the ratio of the superficial liquid velocity to the reactor length, the superficial velocity in a laboratory reactor will be lower than in a commercial reactor by the ratio of reactor lengths, which is often well over an order of magnitude. This means that heat and mass transport parameters may be considerably different in laboratory reactors operated at the target LHSV. This also shifts the flow regime from trickle flow (low interaction) in the lab and small pilot plants to the high-interaction regime in large-scale commercial reactors.

Wall effects in lab units of 50-mm (1.97-in) diameter can be important while these are negligible for commercial reactors of 1 m or more diameter. Wall effects in the lab can be reduced by using reactor/particle diameter ratios greater than 8. If that is not possible, inert fines are added in the lab to reduce wall effects. Also, in large-diameter beds, uniform liquid distribution is difficult, even with a large number of distributor nozzles, and unless the flow is redistributed, the nonuniformity can persist along the bed, leading to potential hot spots that can cause by-products and fast catalyst deactivation. In trickle beds that are not prewetted, a *hysteresis* phenomenon related to wetting

occurs, where the behavior with increasing flow of the liquid phase is not retraced with decreasing liquid flow. This can often be avoided by prewetting the reactor before start-up.

In practice, the thickness of liquid films in trickle beds has been estimated to vary between 0.01 and 0.2 mm (0.004 and 0.008 in). The dynamic liquid holdup fraction is 0.03 to 0.25, and the static fraction is 0.01 to 0.05. The high end of the static fraction includes the liquid that partially fills the pores of the catalyst. The effective gas-liquid interface is 20 to 50 percent of the geometric surface of the particles, but it can approach 100 percent at high liquid loading. This results in an increase of reaction rate as the amount of wetted surface increases (i.e., when the gas-solid reaction rate is negligible).

Examples Hydrodesulfurization of petroleum oils was the first large-scale application of trickle bed reactors commercialized in 1955. In this application, organosulfur species contained in refinery feeds are removed in the presence of hydrogen and a catalyst and released as hydrogen sulfide. Conditions depend on the quality and boiling range of the oil. The reactor pressure is optimized to increase the solubility of the hydrogen and minimize catalyst deactivation due to coking. Over the life of the catalyst, the temperature is increased to maintain a constant conversion. Temperatures are in the range of 345 to 425°C (653 to 797°F) with pressures of 34 to 102 atm (500 to 1500 psi). A large commercial reactor may have 20 to 25 m (66 to 82 ft) of total depth of catalyst, and may be up to 3-m (9.8-ft) diameter or above in several beds of 3- to 6-m (9.8- to 19.7-ft) depth. Bed depth is often limited by pressure drop, the catalyst crush strength, and the maximum adiabatic temperature increase for stable operation. The need to limit pressure drop is driven by the capital and operating costs associated with the hydrogen recycle compressor. Catalyst granules are 1.5 to 3.0 mm (0.06 to 0.12 in), sometimes a little more. Catalysts are 10 to 20 percent Co and Mo (or Ni and W) on alumina. The adiabatic temperature rise in each bed usually is limited to 30°C (86°F) by injection of cold hydrogen between beds. Since the liquid trickles over the catalyst, the wetting efficiency of the catalyst is important in determining the volumetric reaction rate. As expected, wetting efficiency increases with increasing liquid rate. Catalyst effectiveness of particles 3 to 5 mm (0.12 to 0.20 in) in diameter has been found to be about 40 to 60 percent.

Packed Bubble Columns (Cocurrent Upflow) These reactors are also called flooded-bed reactors. In contrast to trickle beds, both gas and liquid flow up cocurrently. A screen is needed at the top to retain the catalyst particles. Such a unit has been used for the hydrogenation of nitro and double-bond compounds and nitriles [Ovcinnikov et al., *Brit. Chem. Eng.* **13**: 1367 (1968)]. High gas rates can cause movement and attrition of the particles. Accordingly, such equipment is restricted to low gas flow rates, for instance, where a hydrogen atmosphere is necessary but the consumption of hydrogen is slight. The liquid is the continuous phase, and the gas, the dispersed phase. Benefits of cocurrent upflow versus trickle (cocurrent downflow) include high wetting efficiency (resulting in good liquid-solid contacting), good liquid distribution, and better heat and mass transfer. Disadvantages include higher pressure drop and liquid backmixing, the latter resulting in increased extent of undesirable homogeneous reactions.

A number of flow regime maps are available for packed bubble columns [see, e.g., Fukushima and Kusaka, *J. Chem. Eng. Japan*, **12**: 296 (1979)]. Correlations for the various hydrodynamic parameters can be found in Shah (*Gas-Liquid-Solid Reactor Design*, McGraw-Hill, 1979), Ramachandran and Chaudhari (*Three-Phase Catalytic Reactors*, Gordon and Breach, 1983), and Shah and Sharma [*Gas-Liquid-Solid Reactors* in Carberry and Varma (eds.), *Chemical Reaction and Reactor Engineering*, Marcel Dekker, 1987].

Countercurrent Flow The gas flows up countercurrent with the downflow liquid. This mode of operation is not as widely used for catalytic reactions since operation is limited by flooding at high gas velocity: at flooding conditions increasing the liquid flow does not result in increase of the liquid holdup.

For more details see Shah (*Gas-Liquid-Solid Reactor Design*, McGraw-Hill, 1979) and Hofmann [*Hydrodynamics and Hydrodynamic Models of Fixed Bed Reactors*, in Gianetto and Silveston (eds.), *Multiphase Chemical Reactors*, Hemisphere 1986].

SOME CASE STUDIES

The literature contains case studies that may be useful for analysis or design of new reactors. Several of these are listed for reference.

Rase (*Case Studies and Design Data*, vol. 2 of *Chemical Reactor Design for Process Plants*, Wiley, 1977):

- Styrene polymerization
- Cracking of ethane to ethylene
- Quench cooling in the ethylene process
- Toluene dealkylation
- Shift conversion
- Ammonia synthesis
- Sulfur dioxide oxidation
- Catalytic reforming
- Ammonia oxidation
- Phthalic anhydride production
- Steam reforming
- Vinyl chloride polymerization
- Batch hydrogenation of cottonseed oil
- Hydrodesulfurization

Rase (*Fixed Bed Reactor Design and Diagnostics*, Butterworths, 1990) has several case studies and a general computer program for reactor design:

- Methane-steam reaction
- Hydrogenation of benzene to cyclohexane
- Dehydrogenation of ethylbenzene to styrene

Tarhan (*Catalytic Reactor Design*, McGraw-Hill, 1983) has computer programs and results for these cases:

- Toluene hydrodealkylation to benzene and methane
- Phthalic anhydride by air oxidation of naphthalene
- Trickle bed reactor for hydrodesulfurization

Ramage et al. (*Advances in Chemical Engineering*, vol. 13, Academic Press, 1987, pp. 193–266):

- Mobil's kinetic reforming model Dente and Ranzi [in Albright et al. (eds.), *Pyrolysis Theory and Industrial Practice*, Academic Press, 1983, pp. 133–175]:
- Mathematical modeling of hydrocarbon pyrolysis reactions Shah and Sharma [in Carberry and Varma (eds.), *Chemical Reaction and Reaction Engineering Handbook*, Marcel Dekker, 1987, pp. 713–721]:
- Hydroxylamine phosphate manufacture in a slurry reactor

Exploration for an acceptable or optimum design for a new reactor may require consideration of several feed and product specifications, reactor types, catalysts, operating conditions, and economic evaluations. Modifications to an existing process likewise may need to consider many cases. Commercial software may be used to facilitate examination of options. A typical package can handle a number of reactions in various ideal reactors under isothermal, adiabatic, or heat-transfer conditions in one or two phases. Outputs can provide profiles of composition, pressure, and temperature as well as vessel size.

Thermodynamic software packages may be used to find equilibrium compositions at prescribed temperatures and pressures. Such calculations require knowledge of feed components and products and their thermodynamic properties and are based on Gibbs free energy minimization techniques. Examples of thermodynamic packages may be found in Smith and Missen (*Chemical Reaction Equilibrium Analysis Theory and Algorithms*, Wiley, 1982) and in Walas (*Phase Equilibria in Chemical Engineering*, Butterworths, 1985).

For some widely practiced processes, especially in the petroleum industry, computer models are available from a number of vendors or, by license, from proprietary sources. Such processes include fluid catalytic cracking, hydrotreating, hydrocracking, alkylation with HF or H₂SO₄, reforming with Pt or Pt-Re catalysts, tubular steam cracking of hydrocarbon fractions, noncatalytic pyrolysis to ethylene, and ammonia synthesis. Catalyst vendors may sometimes also provide simple process models. The reader is advised to peruse some of the process simulation packages listed for sale in the *CEP Software Directory* (e.g., AIChE, 1994) that gets periodically updated with new offerings.

SECTION 20

Alternative
Separation
Processes

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



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Alternative Separation Processes*

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CRYSTALLIZATION FROM THE MELT

GENERAL REFERENCES: Van't Land, *Industrial Crystallization of Melts*, Taylor & Francis, New York, 2004. Mullin, *Crystallization*, 4th ed., Butterworth-Heinemann, 2001. Myerson, *Handbook of Industrial Crystallization*, 2d ed., Butterworth-Heinemann, 2001. Plann, *Zone Melting*, 2d ed., Wiley, New York, 1966. U.S. Patents 3,621,664 and 3,796,060. Zief and Wilcox, *Fractional Solidification*, Marcel Dekker, New York, 1967.

INTRODUCTION

Purification of a chemical species by solidification from a liquid mixture can be termed either **solution** crystallization or crystallization from the **melt**. The distinction between these two operations is somewhat subtle. The term **melt crystallization** has been defined as the separation of components of a binary mixture without addition of solvent, but this definition is somewhat restrictive. In **solution crystallization** a diluent solvent is added to the mixture; the solution is then directly or indirectly cooled, and/or solvent is evaporated to effect crystallization. The solid phase is formed and maintained somewhat below its pure-component freezing-point temperature. In melt crystallization no diluent solvent is added to the reaction mixture, and the solid phase is formed by cooling of the melt. Product is frequently maintained near or above its pure-component freezing point in the refining section of the apparatus.

A large number of techniques are available for carrying out crystallization from the melt. An abbreviated list includes partial freezing and solids recovery in cooling crystallizer-centrifuge systems, partial melting (e.g., sweating), staircase freezing, normal freezing, zone melting, and column crystallization. A description of all these methods is not within the scope of this discussion. Zief and Wilcox (op. cit.) and Myerson (op. cit.) describe many of these processes. Three of the more common methods—progressive freezing from a falling film, zone melting, and melt crystallization from the bulk—are discussed here to illustrate the techniques used for practicing crystallization from the melt.

High or ultrahigh product purity is obtained with many of the melt-purification processes. Table 20-1 compares the product quality and product form that are produced from several of these operations. Zone refining can produce very pure material when operated in a batch mode; however, other melt crystallization techniques also provide high purity and become attractive if continuous high-capacity processing is desired. Comparison of the features of melt crystallization and distillation are shown on Table 20-2.

A brief discussion of solid-liquid phase equilibrium is presented prior to discussing specific crystallization methods. Figures 20-1 and 20-2 illustrate the phase diagrams for binary solid-solution and eutectic systems, respectively. In the case of binary solid-solution systems, illustrated in Fig. 20-1, the liquid and solid phases contain equilibrium quantities of both components in a manner similar to vapor-liquid phase behavior. This type of behavior causes separation difficulties since multiple stages are required. In principle, however, high purity

TABLE 20-2 Comparison of Features of Melt Crystallization and Distillation

Distillation	Melt crystallization
Phase equilibria	
Both liquid and vapor phases are totally miscible.	Liquid phases are totally miscible; solid phases are not.
Conventional vapor/liquid equilibrium.	Eutectic system.
Neither phase is pure.	Solid phase is pure, except at eutectic point.
Separation factors are moderate and decrease as purity increases.	Partition coefficients are very high (theoretically, they can be infinite).
Ultrahigh purity is difficult to achieve.	Ultrahigh purity is easy to achieve.
No theoretical limit on recovery.	Recovery is limited by eutectic composition.
Mass-transfer kinetics	
High mass-transfer rates in both vapor and liquid phases.	Only moderate mass-transfer rate in liquid phase, zero in solid.
Close approach to equilibrium.	Slow approach to equilibrium; achieved in brief contact time. Included impurities cannot diffuse out of solid.
Adiabatic contact assures phase equilibrium.	Solid phase must be remelted and refrozen to allow phase equilibrium.
Phase separability	
Phase densities differ by a factor of 100–10,000:1.	Phase densities differ by only about 10%.
Viscosity in both phases is low.	Liquid phase viscosity moderate, solid phase rigid.
Phase separation is rapid and complete.	Phase separation is slow; surface-tension effects prevent completion.
Countercurrent contacting is quick and efficient.	Countercurrent contacting is slow and imperfect.

Wynn, *Chem. Eng. Prog.*, **88**, 55 (1992). Reprinted with permission of the American Institute of Chemical Engineers. Copyright © 1992 AIChE. All rights reserved

and yields of both components can be achieved since no eutectic is present.

If the impurity or minor component is completely or partially soluble in the solid phase of the component being purified, it is convenient to define a distribution coefficient k , defined by Eq. (20-1):

$$k = C_s/C_\ell \quad (20-1)$$

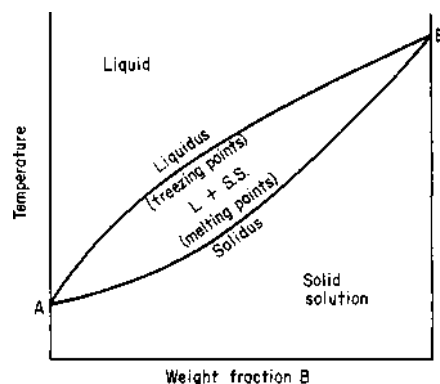


FIG. 20-1 Phase diagram for components exhibiting complete solid solution. (Zief and Wilcox, *Fractional Solidification*, vol. 1, Marcel Dekker, New York, 1967, p. 31.)

TABLE 20-1 Comparison of Processes Involving Crystallization from the Melt

Processes	Approximate upper melting point, °C	Materials tested	Minimum purity level obtained, ppm, weight	Product form
Progressive freezing	1500	All types	1	Ingot
Zone melting				
Batch	3500	All types	0.01	Ingot
Continuous	500	SiI ₄	100	Melt
Melt crystallization				
Continuous	300	Organic	10	Melt
Cyclic	300	Organic	10	Melt

Abbreviated from Zief and Wilcox, *Fractional Solidification*, Marcel Dekker, New York, 1967, p. 7.

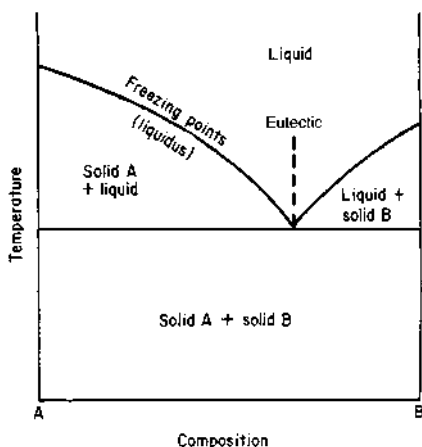


FIG. 20-2 Simple eutectic-phase diagram at constant pressure. (Zief and Wilcox, *Fractional Solidification*, vol. 1, Marcel Dekker, New York, 1967, p. 24.)

C_s is the concentration of impurity or minor component in the solid phase, and C_l is the impurity concentration in the liquid phase. The distribution coefficient generally varies with composition. The value of k is greater than 1 when the solute raises the melting point and less than 1 when the melting point is depressed. In the regions near pure A or B the liquidus and solidus lines become linear; i.e., the distribution coefficient becomes constant. This is the basis for the common assumption of constant k in many mathematical treatments of fractional solidification in which ultrapure materials are obtained.

In the case of a simple eutectic system shown in Fig. 20-2, a pure solid phase is obtained by cooling if the composition of the feed mixture is not at the eutectic composition. If liquid composition is eutectic, then separate crystals of both species will form. In practice it is difficult to attain perfect separation of one component by crystallization of a eutectic mixture. The solid phase will always contain trace amounts of impurity because of incomplete solid-liquid separation, slight solubility of the impurity in the solid phase, or volumetric inclusions. It is difficult to generalize on which of these mechanisms is the major cause of contamination because of analytical difficulties in the ultrahigh-purity range.

The distribution-coefficient concept is commonly applied to fractional solidification of eutectic systems in the ultrapure portion of the phase diagram. If the quantity of impurity entrapped in the solid phase for whatever reason is proportional to that contained in the melt, then assumption of a constant k is valid. It should be noted that the theoretical yield of a component exhibiting binary eutectic behavior is fixed by the feed composition and position of the eutectic. Also, in contrast to the case of a solid solution, only one component can be obtained in a pure form.

There are many types of phase diagrams in addition to the two cases presented here; these are summarized in detail by Zief and Wilcox (op. cit., p. 21). Solid-liquid phase equilibria must be determined experimentally for most binary and multicomponent systems. Predictive methods are based mostly on ideal phase behavior and have limited accuracy near eutectics. A predictive technique based on extracting liquid-phase activity coefficients from *vapor-liquid* equilibria that is useful for estimating nonideal binary or multicomponent *solid-liquid* phase behavior has been reported by Muir (Pap. 71f, 73d ann. meet., AIChE, Chicago, 1980).

PROGRESSIVE FREEZING

Progressive freezing, sometimes called *normal* freezing, is the slow, directional solidification of a melt. Basically, this involves slow solidification at the bottom or sides of a vessel or tube by indirect cooling. The impurity is rejected into the liquid phase by the advancing solid

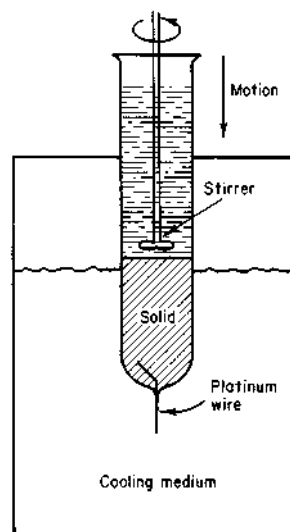


FIG. 20-3 Progressive freezing apparatus.

interface. This technique can be employed to concentrate an impurity or, by repeated solidifications and liquid rejections, to produce a very pure ingot. Figure 20-3 illustrates a progressive freezing apparatus. The solidification rate and interface position are controlled by the rate of movement of the tube and the temperature of the cooling medium. There are many variations of the apparatus; e.g., the residual-liquid portion can be agitated and the directional freezing can be carried out vertically as shown in Fig. 20-3 or horizontally (see Richman et al., in Zief and Wilcox, op. cit., p. 259). In general, there is a solute redistribution when a mixture of two or more components is directionally frozen.

Component Separation by Progressive Freezing When the distribution coefficient is less than 1, the first solid which crystallizes contains less solute than the liquid from which it was formed. As the fraction which is frozen increases, the concentration of the impurity in the remaining liquid is increased and hence the concentration of impurity in the solid phase increases (for $k < 1$). The concentration gradient is reversed for $k > 1$. Consequently, in the absence of diffusion in the solid phase a concentration gradient is established in the frozen ingot.

One extreme of progressive freezing is equilibrium freezing. In this case the freezing rate must be slow enough to permit diffusion in the solid phase to eliminate the concentration gradient. When this occurs, there is no separation if the entire tube is solidified. Separation can be achieved, however, by terminating the freezing before all the liquid has been solidified. Equilibrium freezing is rarely achieved in practice because the diffusion rates in the solid phase are usually negligible (Pfann, op. cit., p. 10).

If the bulk-liquid phase is well mixed and no diffusion occurs in the solid phase, a simple expression relating the solid-phase composition to the fraction frozen can be obtained for the case in which the distribution coefficient is independent of composition and fraction frozen [Pfann, *Trans. Am. Inst. Mech. Eng.*, **194**, 747 (1952)].

$$C_s = kC_0(1 - X)^{k-1} \quad (20-2)$$

C_0 is the solution concentration of the initial charge, and X is the fraction frozen. Figure 20-4 illustrates the solute redistribution predicted by Eq. (20-2) for various values of the distribution coefficient.

There have been many modifications of this idealized model to account for variables such as the freezing rate and the degree of mixing in the liquid phase. For example, Burton et al. [*J. Chem. Phys.*, **21**, 1987 (1953)] reasoned that the solid rejects solute faster than it can diffuse into the bulk liquid. They proposed that the effect of the

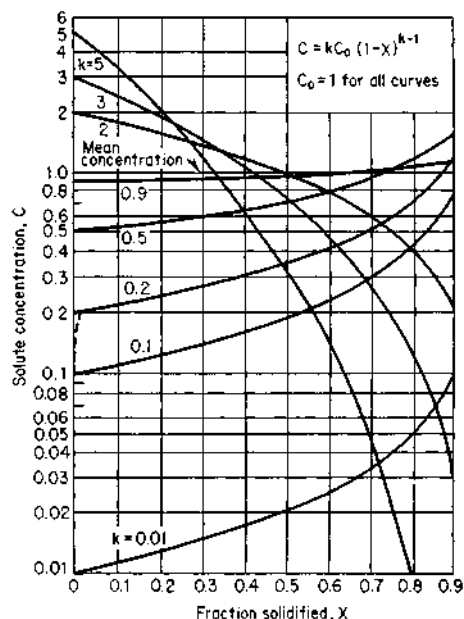


FIG. 20-4 Curves for progressive freezing, showing solute concentration C in the solid versus fraction-solidified X . (Pfann, *Zone Melting*, 2d ed., Wiley, New York, 1966, p. 12.)

freezing rate and stirring could be explained by the diffusion of solute through a stagnant film next to the solid interface. Their theory resulted in an expression for an effective distribution coefficient k_{eff} which could be used in Eq. (20-2) instead of k .

$$k_{\text{eff}} = \frac{1}{1 + (1/k - 1)e^{-f\delta/D}} \quad (20-3)$$

where f = crystal growth rate, cm/s; δ = stagnant film thickness, cm; and D = diffusivity, cm²/s. No further attempt is made here to summarize the various refinements of Eq. (20-2). Zief and Wilcox (op. cit., p. 69) have summarized several of these models.

Pertinent Variables in Progressive Freezing The dominant variables which affect solute redistribution are the degree of mixing in the liquid phase and the rate of solidification. It is important to attain sufficient mixing to facilitate diffusion of the solute away from the solid-liquid interface to the bulk liquid. The film thickness δ decreases as the level of agitation increases. Cases have been reported in which essentially no separation occurred when the liquid was not stirred. The freezing rate which is controlled largely by the lowering rate of the tube (see Fig. 20-3) has a pronounced effect on the separation achieved. The separation is diminished as the freezing rate is increased. Also fluctuations in the freezing rate caused by mechanical vibrations and variations in the temperature of the cooling medium can decrease the separation.

Applications Progressive freezing has been applied to both solid solution and eutectic systems. As Fig. 20-4 illustrates, large separation factors can be attained when the distribution coefficient is favorable. Relatively pure materials can be obtained by removing the desired portion of the ingot. Also in some cases progressive freezing provides a convenient method of concentrating the impurities; e.g., in the case of $k < 1$ the last portion of the liquid that is frozen is enriched in the distributing solute.

Progressive freezing has been applied on the commercial scale. For example, aluminum has been purified by continuous progressive freezing [Dewey, *J. Metals*, **17**, 940 (1965)]. The Proabd refiner described by Molinari (Zief and Wilcox, op. cit., p. 393) is also a commercial example of progressive freezing. In this apparatus the mixture is directionally solidified on cooling tubes. Purification is achieved

because the impure fraction melts first; this process is called sweating. This technique has been applied to the purification of naphthalene and *p*-dichlorobenzene and commercial equipment is available from BEFS PROKEM, Houston, Tx.

ZONE MELTING

Zone melting also relies on the distribution of solute between the liquid and solid phases to effect a separation. In this case, however, one or more liquid zones are passed through the ingot. This extremely versatile technique, which was invented by W. G. Pfann, has been used to purify hundreds of materials. Zone melting in its simplest form is illustrated in Fig. 20-5. A molten zone can be passed through an ingot from one end to the other by either a moving heater or by slowly drawing the material to be purified through a stationary heating zone.

Progressive freezing can be viewed as a special case of zone melting. If the zone length were equal to the ingot length and if only one pass were used, the operation would become progressive freezing. In general, however, when the zone length is only a fraction of the ingot length, zone melting possesses the advantage that a portion of the ingot does not have to be discarded after each solidification. The last portion of the ingot which is frozen in progressive freezing must be discarded before a second freezing.

Component Separation by Zone Melting The degree of solute redistribution achieved by zone melting is determined by the zone length l , ingot length L , number of passes n , the degree of mixing in the liquid zone, and the distribution coefficient of the materials being purified. The distribution of solute after one pass can be obtained by material-balance considerations. This is a two-domain problem; i.e., in the major portion of the ingot of length $L - l$ zone melting occurs in the conventional sense. The trailing end of the ingot of length l undergoes progressive freezing. For the case of constant-distribution coefficient, perfect mixing in the liquid phase, and negligible diffusion in the solid phase, the solute distribution for a single pass is given by Eq. (20-4) [Pfann, *Trans. Am. Inst. Mech. Eng.*, **194**, 747 (1952)].

$$C_s = C_0[1 - (1 - k)e^{-kx/l}] \quad (20-4)$$

The position of the zone x is measured from the leading edge of the ingot. The distribution for multiple passes can also be calculated from a material balance, but in this case the leading edge of the zone encounters solid corresponding to the composition at the point in question for the previous pass. The multiple-pass distribution has been numerically calculated (Pfann, *Zone Melting*, 2d ed., Wiley, New York, 1966, p. 285) for many combinations of k , L/l , and n . Typical solute-composition profiles are shown in Fig. 20-6 for various numbers of passes.

The ultimate distribution after an infinite number of passes is also shown in Fig. 20-6 and can be calculated for $x < (L - l)$ from the following equation (Pfann, op. cit., p. 42):

$$C_s = Ae^{Bx} \quad (20-5)$$

where A and B can be determined from the following relations:

$$k = B\ell/(e^{B\ell} - 1) \quad (20-6)$$

$$A = C_0BL/(e^{BL} - 1) \quad (20-7)$$

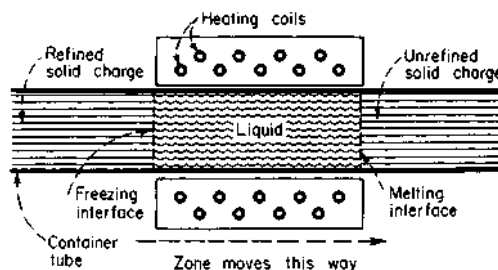


FIG. 20-5 Diagram of zone refining.

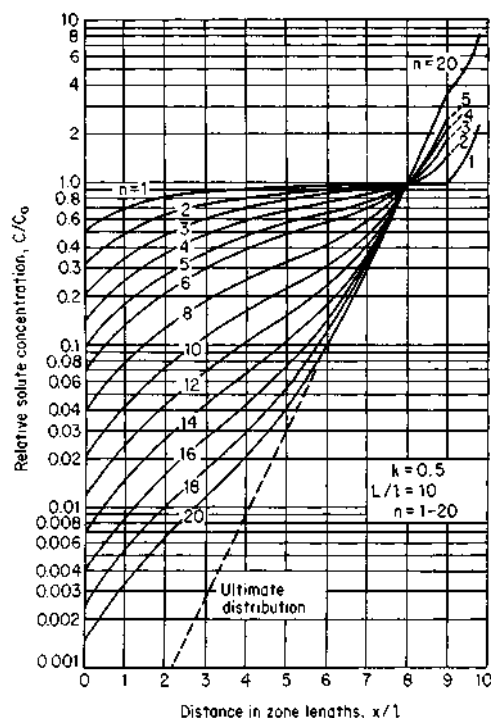


FIG. 20-6 Relative solute concentration C/C_0 (logarithmic scale) versus distance in zone lengths x/l from beginning of charge, for various numbers of passes n . L denotes charge length. (Pfann, *Zone Melting*, 2d ed., Wiley, New York, 1966, p. 290.)

The ultimate distribution represents the maximum separation that can be attained without cropping the ingot. Equation (20-5) is approximate because it does not include the effect of progressive freezing in the last zone length.

As in progressive freezing, many refinements of these models have been developed. Corrections for partial liquid mixing and a variable distribution coefficient have been summarized in detail (Zief and Wilcox, op. cit., p. 47).

Pertinent Variables in Zone Melting The dominant variables in zone melting are the number of passes, ingot-length-zone-length ratio, freezing rate, and degree of mixing in the liquid phase. Figure 20-6 illustrates the increased solute redistribution that occurs as the number of passes increases. Ingot-length-zone-length ratios of 4 to 10 are commonly used (Zief and Wilcox, op. cit., p. 624). An exception is encountered when one pass is used. In this case the zone length should be equal to the ingot length; i.e., progressive freezing provides the maximum separation when only one pass is used.

The freezing rate and degree of mixing have effects in solute redistribution similar to those discussed for progressive freezing. Zone travel rates of 1 cm/h for organic systems, 2.5 cm/h for metals, and 20 cm/h for semiconductors are common. In addition to the zone-travel rate the heating conditions affect the freezing rate. A detailed summary of heating and cooling methods for zone melting has been outlined by Zief and Wilcox (op. cit., p. 192). Direct mixing of the liquid region is more difficult for zone melting than progressive freezing. Mechanical stirring complicates the apparatus and increases the probability of contamination from an outside source. Some mixing occurs because of natural convection. Methods have been developed to stir the zone magnetically by utilizing the interaction of a current and magnetic field (Pfann, op. cit., p. 104) for cases in which the charge material is a reasonably good conductor.

Applications Zone melting has been used to purify hundreds of inorganic and organic materials. Many classes of inorganic compounds including semiconductors, intermetallic compounds, ionic salts, and

oxides have been purified by zone melting. Organic materials of many types have been zone-melted. Zief and Wilcox (op. cit., p. 624) have compiled tables which give operating conditions and references for both inorganic and organic materials with melting points ranging from -115°C to over 3000°C .

Some materials are so reactive that they cannot be zone-melted to a high degree of purity in a container. Floating-zone techniques in which the molten zone is held in place by its own surface tension have been developed by Keck et al. [*Phys. Rev.*, **89**, 1297 (1953)].

Continuous-zone-melting apparatus has been described by Pfann (op. cit., p. 171). This technique offers the advantage of a close approach to the ultimate distribution, which is usually impractical for batch operation.

Performance data have been reported by Kennedy et al. (*The Purification of Inorganic and Organic Materials*, Marcel Dekker, New York, 1969, p. 261) for continuous-zone refining of benzoic acid.

MELT CRYSTALLIZATION FROM THE BULK

Conducting crystallization inside a vertical or horizontal column with a countercurrent flow of crystals and liquid can produce a higher product purity than conventional crystallization or distillation. The working concept is to form a crystal phase from the bulk liquid, either internally or externally, and then transport the solids through a countercurrent stream of enriched reflux liquid obtained from melted product. The problem in practicing this technology is the difficulty of controlling solid-phase movement. Unlike distillation, which exploits the specific-gravity differences between liquid and vapor phases, melt crystallization involves the contacting of liquid and solid phases that have nearly identical physical properties. Phase densities are frequently very close, and gravitational settling of the solid phase may be slow and ineffective. The challenge of designing equipment to accomplish crystallization in a column has resulted in a myriad of configurations to achieve reliable solid-phase movement, high product yield and purity, and efficient heat addition and removal.

Investigations Crystallization conducted inside a column is categorized as either **end-fed** or **center-fed** depending on whether the feed location is upstream or downstream of the crystal forming section. Figure 20-7 depicts the features of an end-fed commercial column described by McKay et al. [*Chem. Eng. Prog. Symp. Ser.*, no. 25, 55, 163 (1969)] for the separation of xylenes. Crystals of *p*-xylene are formed by indirect cooling of the melt in scraped-surface heat exchangers, and the resultant slurry is introduced into the column at the top. This type of column has no mechanical internals to transport solids and instead relies upon an imposed hydraulic gradient to force the solids through the column into the melting zone. Residue liquid is removed through a filter directly above the melter. A pulse piston in the product discharge improves washing efficiency and column reliability.

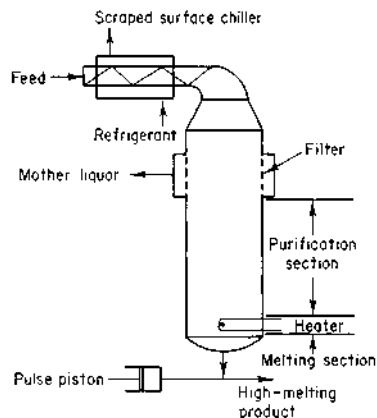


FIG. 20-7 End-fed column crystallizer. (Phillips Petroleum Co.)

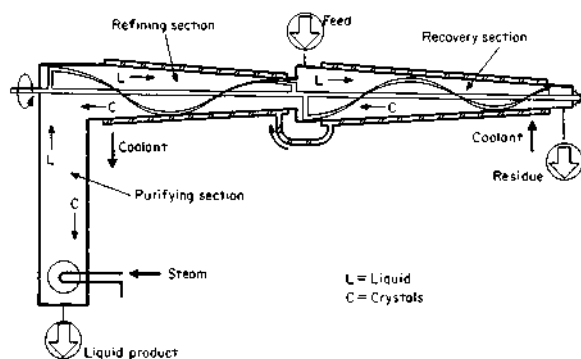


FIG. 20-8 Horizontal center-fed column crystallizer. (The C. W. Nofsinger Co.)

Figure 20-8 shows the features of a horizontal center-fed column [Brodie, *Aust. Mech. Chem. Eng. Trans.*, **37** (May 1979)] which has been commercialized for continuous purification of naphthalene and *p*-dichlorobenzene. Liquid feed enters the column between the hot purifying section and the cold freezing or recovery zone. Crystals are formed internally by indirect cooling of the melt through the walls of the refining and recovery zones. Residue liquid that has been depleted of product exits from the coldest section of the column. A spiral conveyor controls the transport of solids through the unit.

Another center-fed design that has only been used on a preparative scale is the vertical spiral conveyor column reported by Schildknecht [*Angew. Chem.*, **73**, 612 (1961)]. In this device, a version of which is shown on Fig. 20-9, the dispersed-crystal phase is formed in the freezing section and conveyed downward in a controlled manner by a rotating spiral with or without a vertical oscillation.

Differences have been observed in the performance of end- and center-fed column configurations. Consequently, discussions of center- and end-fed column crystallizers are presented separately. The design and operation of both columns are reviewed by Powers (Zief and Wilcox, op. cit., p. 343). A comparison of these devices is shown on Table 20-3.

Center-Fed Column Crystallizers Two types of center-fed column crystallizers are illustrated on Figs. 20-8 and 20-9. As in a simple distillation column, these devices are composed of three distinct sections: a freezing or recovery section, where solute is frozen from the

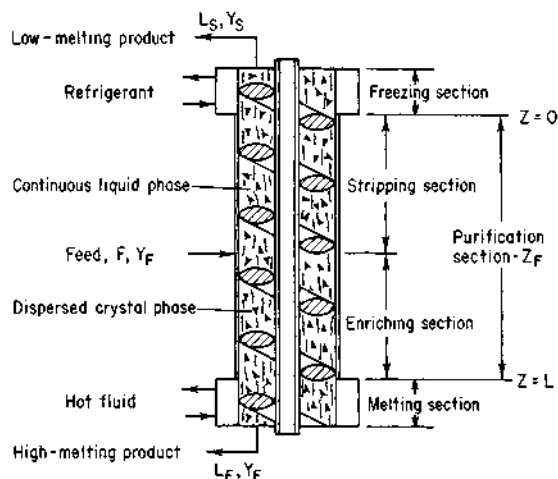


FIG. 20-9 Center-fed column crystallizer with a spiral-type conveyor.

TABLE 20-3 Comparison of Melt-Crystallizer Performance

Center-fed column	End-fed column
Solid phase is formed internally; thus, only liquid streams enter and exit the column.	Solid phase is formed in external equipment and fed as slurry into the purifier.
Internal reflux can be controlled without affecting product yield.	The maximum internal liquid reflux is fixed by the thermodynamic state of the feed relative to the product stream. Excessive reflux will diminish product yield.
Operation can be continuous or batchwise at total reflux.	Total reflux operation is not feasible.
Center-fed columns can be adapted for both eutectic and solid-solution systems.	End-fed columns are inefficient for separation of solid-solution systems.
Either low- or high-porosity-solids-phase concentrations can be formed in the purification and melting zones.	End-fed units are characterized by low-porosity-solids packing in the purification and melting zones.
Scale-up depends on the mechanical complexity of the crystal-transport system and techniques for removing heat. Vertical oscillating spiral columns are likely limited to about 0.2 m in diameter, whereas horizontal columns of several meters are possible.	Scale-up is limited by design of melter and/or crystal-washing section. Vertical or horizontal columns of several meters in diameter are possible.

impure liquor; the purification zone, where countercurrent contacting of solids and liquid occurs; and the crystal-melting and -refluxing section. Feed position separates the refining and recovery portions of the purification zone. The section between feed location and melter is referred to as the refining or enrichment section, whereas the section between feed addition and freezing is called the recovery section. The refining section may have provisions for sidewall cooling. The published literature on column crystallizers connotes stripping and refining in a reverse sense to distillation terminology, since refined product from a melt crystallizer exits at the hot section of the column rather than at the cold end as in a distillation column.

Rate processes that describe the purification mechanisms in a column crystallizer are highly complex since phase transition and heat- and mass-transfer processes occur simultaneously. Nucleation and growth of a crystalline solid phase along with crystal washing and crystal melting are occurring in various zones of the apparatus. Column hydrodynamics are also difficult to describe. Liquid- and solid-phase mixing patterns are influenced by factors such as solids-transport mechanism, column orientation, and, particularly for dilute slurries, the settling characteristics of the solids.

Most investigators have focused their attention on a differential segment of the zone between the feed injection and the crystal melter. Analysis of crystal formation and growth in the recovery section has received scant attention. Table 20-4 summarizes the scope of the literature treatment for center-fed columns for both solid-solution and eutectic forming systems.

The dominant mechanism of purification for column crystallization of solid-solution systems is recrystallization. The rate of mass transfer resulting from recrystallization is related to the concentrations of the solid phase and free liquid which are in intimate contact. A model based on height-of-transfer-unit (HTU) concepts representing the composition profile in the purification section for the high-melting component of a binary solid-solution system has been reported by Powers et al. (in Zief and Wilcox, op. cit., p. 363) for total-reflux operation. Typical data for the purification of a solid-solution system, azobenzene-stilbene, are shown in Fig. 20-10. The column crystallizer was operated at total reflux. The solid line through the data was computed by Powers et al. (op. cit., p. 364) by using an experimental HTU value of 3.3 cm.

TABLE 20-4 Column-Crystallizer Investigations

	Treatments	
	Theoretical	Experimental
Solid solutions		
Total reflux—steady state	1, 2, 4, 6	1, 4, 6
Total reflux—dynamic	2	
Continuous—steady state	1, 4	4, 8, 9
Continuous—dynamic		
Eutectic systems		
Total reflux—steady state	1, 3, 4, 7	1, 3, 6
Total reflux—dynamic		
Continuous—steady state	1, 5, 10, 11, 12	5, 8, 9, 10, 11, 13
Continuous—dynamic		

1. Powers, *Symposium on Zone Melting and Column Crystallization*, Karlsruhe, 1963.
2. Anikin, *Dokl. Akad. Nauk SSSR*, **151**, 1139 (1969).
3. Albertins et al., *Am. Inst. Chem. Eng. J.*, **15**, 554 (1969).
4. Gates et al., *Am. Inst. Chem. Eng. J.*, **16**, 648 (1970).
5. Henry et al., *Am. Inst. Chem. Eng. J.*, **16**, 1055 (1970).
6. Schildknecht et al., *Angew. Chem.*, **73**, 612 (1961).
7. Arkenbout et al., *Sep. Sci.*, **3**, 501 (1968).
8. Betts et al., *Appl. Chem.*, **17**, 180 (1968).
9. McKay et al., *Chem. Eng. Prog. Symp. Ser.*, no. 25, 55, 163 (1959).
10. Bolsaitis, *Chem. Eng. Sci.*, **24**, 1813 (1969).
11. Moyers et al., *Am. Inst. Chem. Eng. J.*, **20**, 1119 (1974).
12. Griffin, M.S. thesis in chemical engineering, University of Delaware, 1975.
13. Brodie, *Aust. Mech. Chem. Eng. Trans.*, **37** (1971).

Most of the analytical treatments of center-fed columns describe the purification mechanism in an adiabatic oscillating spiral column (Fig. 20-9). However, the analyses by Moyers (op. cit.) and Griffin (op. cit.) are for a nonadiabatic dense-bed column. Differential treatment of the horizontal-purifier (Fig. 20-8) performance has not been reported; however, overall material and enthalpy balances have been described by Brodie (op. cit.) and apply equally well to other designs.

A dense-bed center-fed column (Fig. 20-11) having provision for internal crystal formation and variable reflux was tested by Moyers et al. (op. cit.). In the theoretical development (ibid.) a nonadiabatic, plug-flow axial-dispersion model was employed to describe the performance of the entire column. Terms describing interphase transport of impurity between adhering and free liquid are not considered.

A comparison of the axial-dispersion coefficients obtained in oscillating-spiral and dense-bed crystallizers is given in Table 20-5. The dense-bed column approaches axial-dispersion coefficients similar to those of densely packed ice-washing columns.

The concept of minimum reflux as related to column-crystallizer operation is presented by Brodie (op. cit.) and is applicable to all types

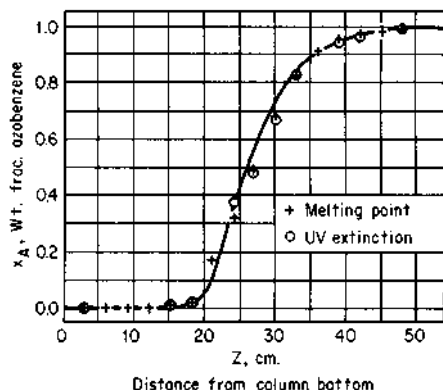


FIG. 20-10 Steady-state separation of azobenzene and stilbene in a center-fed column crystallizer with total-reflux operation. To convert centimeters to inches, multiply by 0.3937. (Zief and Wilcox, *Fractional Solidification*, vol. 1, Marcel Dekker, New York, 1967, p. 356.)

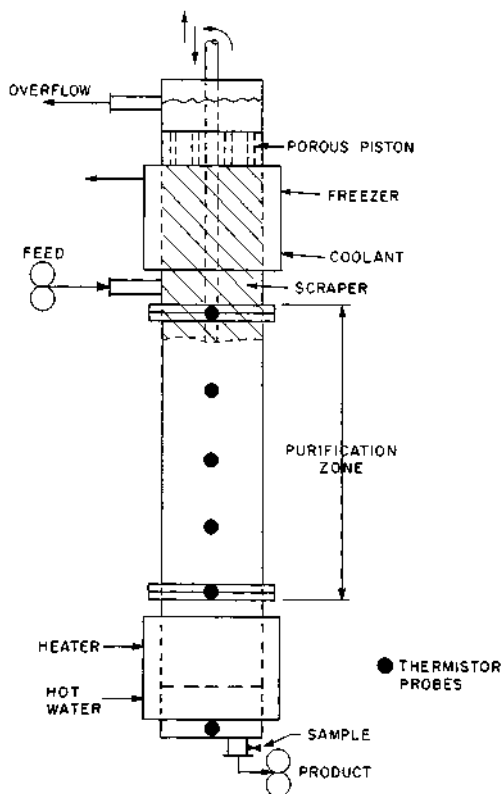


FIG. 20-11 Dense-bed center-fed column crystallizer. [Moyers et al., *Am. Inst. Chem. Eng. J.*, **20**, 1121 (1974).]

of column crystallizers, including end-fed units. In order to stabilize column operation the sensible heat of subcooled solids entering the melting zone should be balanced or exceeded by the heat of fusion of the refluxed melt. The relationship in Eq. (20-8) describes the minimum reflux requirement for proper column operation.

$$R = (T_p - T_F) C_p / \lambda \quad (20-8)$$

R = reflux ratio, g reflux/g product; T_p = product temperature, °C; T_F = saturated-feed temperature, °C; C_p = specific heat of solid crystals, cal/(g·°C); and λ = heat of fusion, cal/g.

All refluxed melt will refreeze if reflux supplied equals that computed by Eq. (20-8). When reflux supplied is greater than the minimum, jacket cooling in the refining zone or additional cooling in the

TABLE 20-5 Comparison of Axial-Dispersion Coefficients for Several Liquid-Solid Contactors

Column type	Dispersion coefficient, cm ² /s	Reference
Center-fed crystallizer (oscillating spiral)	1.6–3.5	1
Center-fed crystallizer (oscillating spiral)	1.3–1.7	2
Counter-current ice-washing column	0.025–0.17	3
Center-fed crystallizer	0.12–0.30	4

References:

1. Albertins et al., *Am. Inst. Chem. Eng. J.*, **15**, 554 (1969).
2. Gates et al., *Am. Inst. Chem. Eng. J.*, **16**, 648 (1970).
3. Ritter, Ph.D. thesis, Massachusetts Institute of Technology, 1969.
4. Moyers et al., *Am. Inst. Chem. Eng. J.*, **20**, 1119 (1974).

recovery zone is required to maintain product recovery. Since high-purity melts are fed near their pure-component freezing temperatures, little refreezing takes place unless jacket cooling is added.

To utilize a column-crystallizer design or rating model, a large number of parameters must be identified. Many of these are empirical in nature and must be determined experimentally in equipment identical to the specific device being evaluated. Hence macroscopic evaluation of systems by large-scale piloting is the rule rather than the exception. Included in this rather long list of critical parameters are factors such as impurity level trapped in the solid phase, product quality as a function of reflux ratio, degree of liquid and solids axial mixing in the equipment as a function of solids-conveyor design, size and shape of crystals produced, and ease of solids handling in the column. Heat is normally removed through metal surfaces; thus, the stability of the solution to subcooling can also be a major factor in design.

End-Fed Column Crystallizer End-fed columns were developed and successfully commercialized by the Phillips Petroleum Company in the 1950s. The sections of a typical end-fed column, often referred to as a Phillips column, are shown on Fig. 20-7. Impure liquor is removed through filters located between the product-freezing zone and the melter rather than at the end of the freezing zone, as occurs in center-fed units. The purification mechanism for end-fed units is basically the same as for center-fed devices. However, there are reflux restrictions in an end-fed column, and a high degree of solids compaction exists near the melter of an end-fed device. It has been observed that the free-liquid composition and the fraction of solids are relatively constant throughout most of the purification section but exhibit a sharp discontinuity near the melting section [McKay et al., *Ind. Eng. Chem.*, **52**, 197 (1969)]. Investigators of end-fed column behavior are listed in Table 20-6. Note that end-fed columns are adaptable only for eutectic-system purification and cannot be operated at total reflux.

Performance information for the purification of *p*-xylene indicates that nearly 100 percent of the crystals in the feed stream are removed as product. This suggests that the liquid which is refluxed from the melting section is effectively refrozen by the countercurrent stream of sub-cooled crystals. A high-melting product of 99.0 to 99.8 weight percent *p*-xylene has been obtained from a 65 weight percent *p*-xylene feed. The major impurity was *m*-xylene. Figure 20-12 illustrates the column-cross-section-area-capacity relationship for various product purities.

Column crystallizers of the end-fed type can be used for purification of many eutectic-type systems and for aqueous as well as organic systems (McKay, loc. cit.). Column crystallizers have been used for xylene isomer separation, but recently other separation technologies including more efficient melt crystallization equipment have tended to supplant the Phillips style crystallizer.

Commercial Equipment and Applications In the last two decades the practice of melt crystallization techniques for purification of certain organic materials has made significant commercial progress. The concept of refining certain products by countercurrent staging of crystallization in a column has completed the transition from laboratory and pilot equipment to large-scale industrial configurations. Chemicals which have been purified by suspension crystallization-purifier column techniques are listed on Table 20-7. The practice of crystal formation and growth from the bulk liquid (as is practiced in suspension crystallization techniques described in Sec. 18 of this handbook) and subsequent crystal melting and refluxing in a purifier

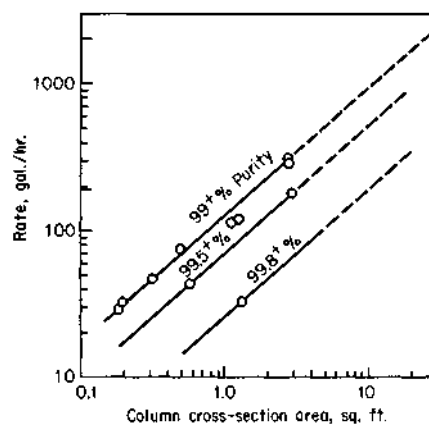


FIG. 20-12 Pulsed-column capacity versus column size for 65 percent *p*-xylene feed. To convert gallons per hour to cubic meters per hour, multiply by 0.9396; to convert square feet to square meters, multiply by 0.0929. (McKay et al., prepr., 59th nat. meet. AIChE, East Columbus, Ohio.)

column has evolved into two slightly different concepts: (1) the horizontal continuous crystallization technique with vertical purifier invented by Brodie (op. cit.) and (2) the continuous multistage or stepwise system with vertical purifier developed by Tsukishima Kikai Co., Ltd. (TSK). A recent description of these processes has been published by Meyer [*Chem. Proc.*, **53**, 50 (1990)].

The horizontal continuous Brodie melt crystallizer is basically an indirectly cooled crystallizer with an internal ribbon conveyor to transport crystals countercurrent to the liquid and a vertical purifier for final refining. Figure 20-8 describes the operation of a single tube unit and Fig. 20-13 depicts a multitube unit. The multitube design has been successfully commercialized for a number of organic chemicals. The Brodie purifier configuration requires careful control of process and equipment temperature differences to eliminate internal encrustations and is limited by the inherent equipment geometry to capacities of less than 15,000 tons per year per module.

In the multistage process described on Fig. 20-14 feed enters one of several crystallizers installed in series. Crystals formed in each crystallizer are transferred to a hotter stage and the liquid collected in the clarified zone of the crystallizer is transferred to a colder stage and eventually discharged as residue. At the hot end, crystals are transferred to a vertical purifier where countercurrent washing is performed by pure, hot-product reflux. TSK refers to this multistage process as the countercurrent cooling crystallization (CCCC) process. In

TABLE 20-6 End-Fed-Crystallizer Investigations

Eutectic systems	Treatments	
	Theoretical	Experimental
Continuous—steady state	1, 2, 4	1, 4
Batch	3	3

1. McKay et al., *Ind. Eng. Chem. Process Des. Dev.*, **6**, 16 (1967).
2. Player, *Ind. Eng. Chem. Process Des. Dev.*, **8**, 210 (1969).
3. Yagi et al., *Kagaku Kogaku*, **72**, 415 (1963).
4. Shen and Meyer, Prepr. 19F, AIChE Symp., Chicago, 1970.

TABLE 20-7 Chemicals Purified by TSK CCCC Process (The C. W. Nofsinger Co.)

Acetic acid	Isophthaloyl chloride
Acrylic acid	Isopregol
Adipic acid	Lutidine
Benzene	Maleic anhydride
Biphenyl	Naphthalene
Bisphenol-A	<i>p</i> -Nitrochloro benzene
Caprolactam	<i>p</i> -Nitrotoluene
Chloroacetic acid	Phenol
<i>p</i> -Chloro toluene	<i>b</i> -Picoline
<i>p</i> -Cresol	<i>g</i> -Picoline
Combat (proprietary)	Pyridine
Dibutyl hydroxy toluene (BHT)	Stilbene
<i>p</i> -Dichloro benzene	Terephthaloyl chloride
2,5 Dichlorophenol	Tertiary butyl phenol
Dicumyl peroxide	Toluene diisocyanate
Diene	Trioxane
Heliotropin	<i>p</i> -Xylene
Hexachloro cyclo butene	3,4 Xylidine
Hexamethylene diamine	

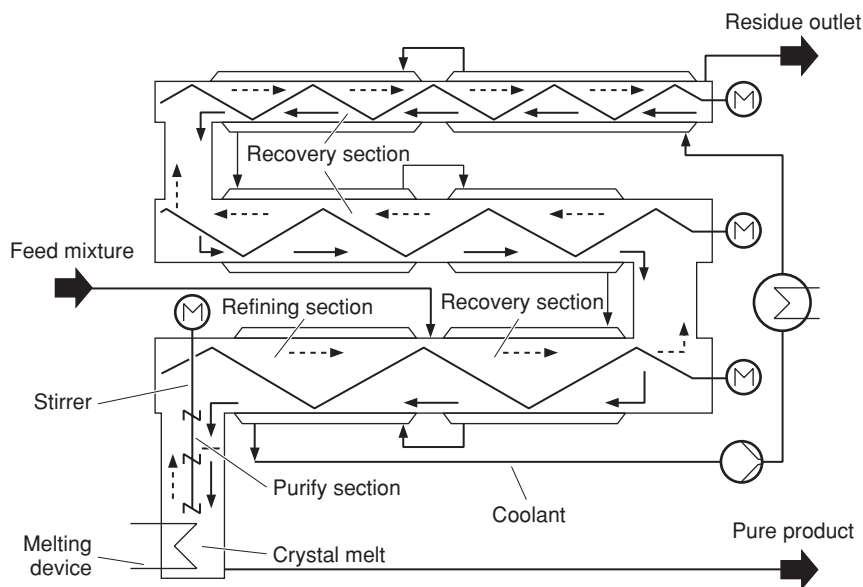


FIG. 20-13 Horizontal continuous Brodie melt crystallizer—multitube unit. (The C. W. Nofsinger Co.)

principle any suitable type of crystallizer can be used in the stages as long as the crystals formed can be separated from the crystallizer liquid and settled and melted in the purifier.

Commercial applications for both the Brodie and CCCC process are indicated on Table 20-8. Both the Brodie Purifier and the CCCC processes are available from The C. W. Nofsinger Company, PO Box 419173, Kansas City, MO 64141-0173.

FALLING-FILM CRYSTALLIZATION

Falling-film crystallization utilizes progressive freezing principles to purify melts and solutions. The technique established to practice the process is inherently cyclic. Figure 20-15 depicts the basic working concept. First a crystalline layer is formed by subcooling a liquid film on a vertical surface inside a tube. This coating is then

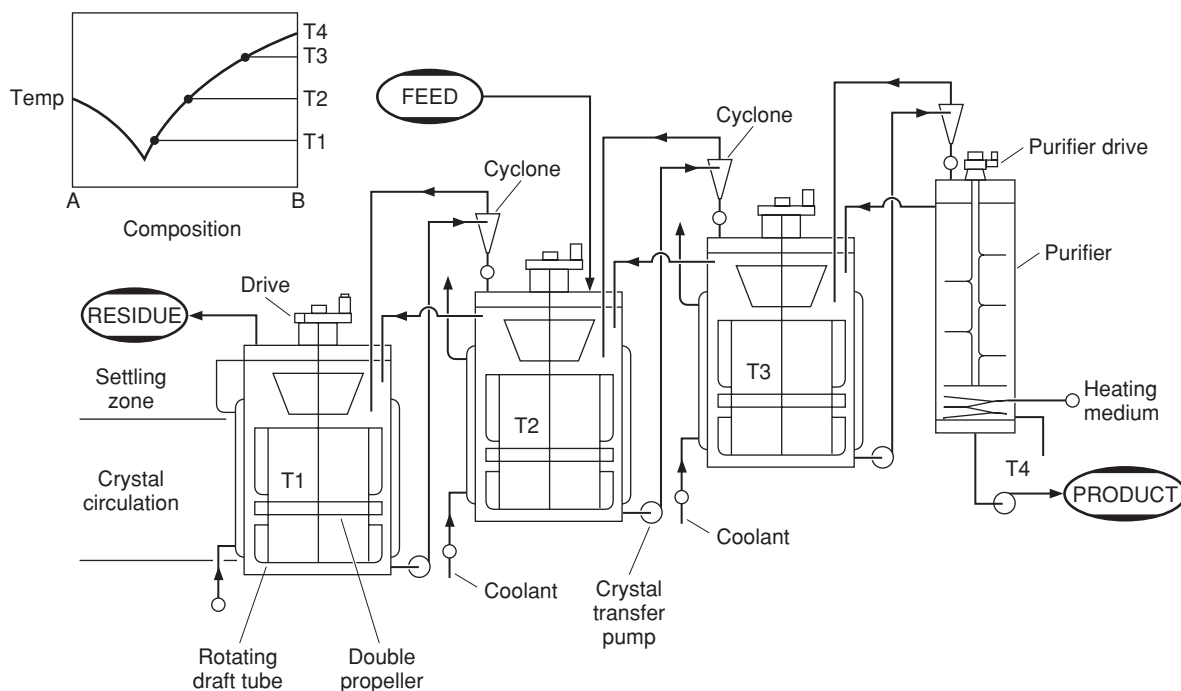


FIG. 20-14 Crystallizer—multistage process. (The C. W. Nofsinger Co.)

TABLE 20-8 Commercial TSK Crystallization Operating Plants

	Capacity MM lb/yr	Company	Date & location
Countercurrent Cooling Crystallization (CCCC) Process			
Nofsinger license			
Nofsinger design & construct			
<i>p</i> -Dichlorobenzene	Confidential	Monsanto Co.	1989—Sauget, IL
TSK license			
TSK design & construct			
Confidential ¹	Confidential	Confidential	1988—Japan
<i>p</i> -Xylene	137	MGC	1986—Mizushima, Japan
Confidential ²	Confidential	Confidential	1985—Japan
<i>p</i> -Xylene	132	MGC	1983—Mizushima, Japan
	expanded to 160		1984—Japan
<i>p</i> -Xylene	26.5	MGC	1981—Mizushima, Japan
Brodie			
TSK license			
TSK design & construct			
Naphthalene	8	SHSM	1985—China
<i>p</i> -DCB	13	Hodogaya	1981—Japan
<i>p</i> -DCB	5.5	Sumitomo	1978—Japan
UCAL license & design			
TSK hardware			
Naphthalene	16	Nippon Steel	1974—Japan
<i>p</i> -DCB	13	Hodogaya	1974—Japan
UCAL license & design			
Naphthalene	10	British Tar	1972—U.K.
<i>p</i> -DCB	3	UCAL	1969—Australia

ABBREVIATIONS:

Nofsinger	The C. W. Nofsinger Company
TSK	Tsukishima Kikai Co., Ltd.
SHSM	Shanghai Hozan Steel Mill
UCAL	Union Carbide Australia, Ltd.
MGC	Mitsubishi Gas Chemical Co., co-developer with TSK of the application for <i>p</i> -Xylene

1. Commercial scale plant started up in the spring of 1988 purifying a bulk chemical. This is the first application of the CCCC process on this bulk chemical.

2. This small unit is operating in Japan with an 800-mm crystallizer and 300-mm purifier. Because of confidentiality, we cannot disclose the company, capacity, or product.

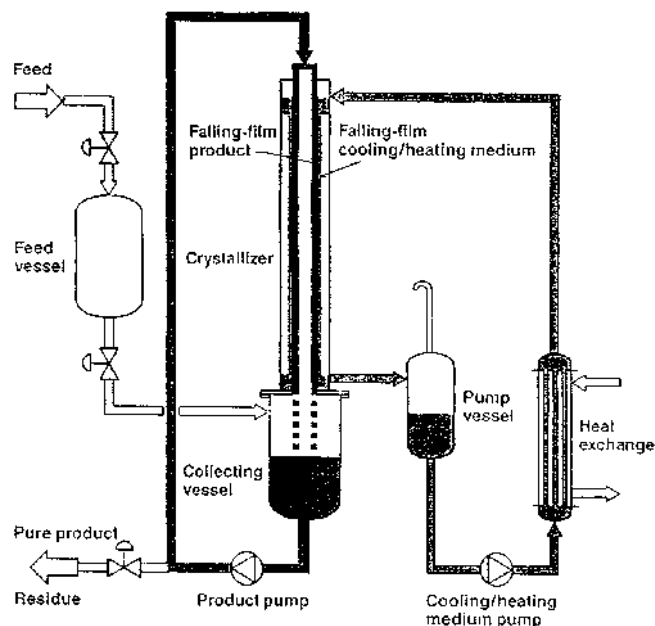


FIG. 20-15 Dynamic crystallization system. (Sulzer Chemtech.)

20-12 ALTERNATIVE SEPARATION PROCESSES

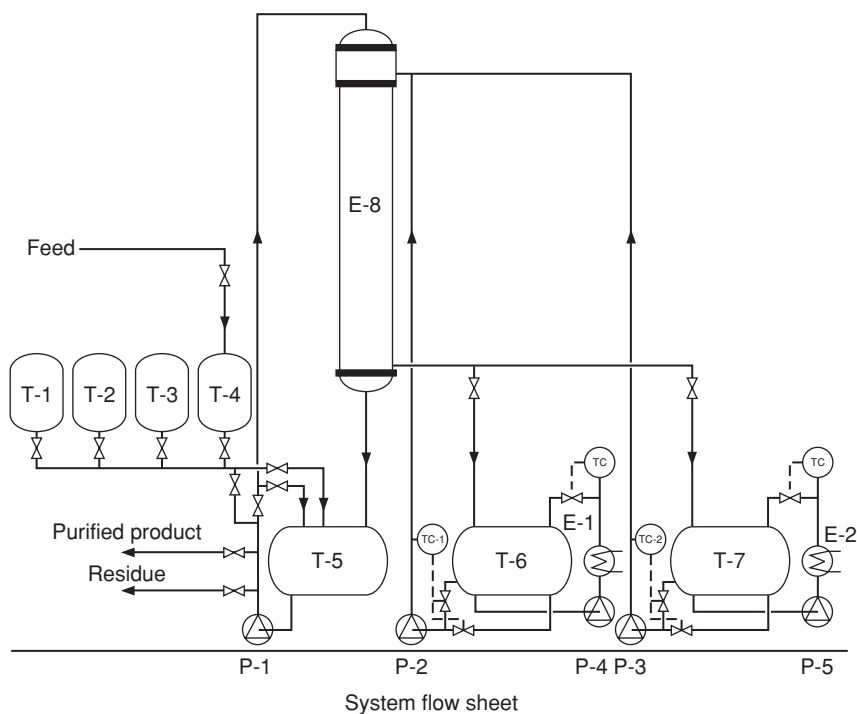
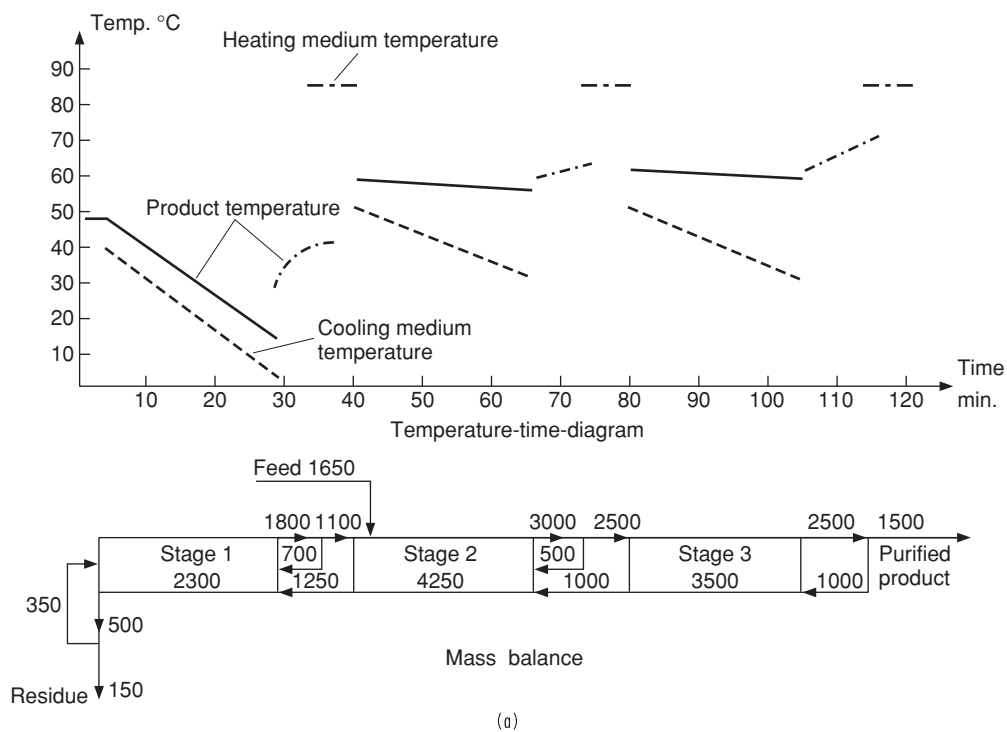


FIG. 20-16 Sulzer MWB-crystallization process. (a) Stepwise operation of the process. (b) System flow sheet. (Sulzer Chemtech.)

grown by extracting heat from a falling film of melt (or solution) through a heat transfer surface. Impure liquid is then drained from the crystal layer and the product is reclaimed by melting. Variants of this technique have been perfected and are used commercially for many types of organic materials. Both static and falling-film techniques have been described by Wynn [*Chem. Eng. Progr.*, (1992)]. Mathematical models for both static and dynamic operations have been presented by Gilbert [*AIChE J.*, **37**, 1205 (1991)].

Principles of Operation Figure 20-16 describes a typical three-stage falling-film crystallization process for purification of MCA (monochloro acetic acid). Crystallizer E-8 consists of a number of vertical tubular elements working in parallel enclosed in a shell. Normal tube length is 12 meters with a 50- to 75-millimeter tube inside diameter. Feed enters stage two of the sequential operation, is added to the kettle (T-5), and is then circulated to the top of the crystallizer and distributed as a falling film inside the tubes. Nucleation is induced at the inside walls and a crystal layer starts to grow. Temperature of the coolant is progressively lowered to compensate for reduced heat transfer and lower melt freezing point

until the thickness inside the tube is between 5 and 20 millimeters depending on the product. Kettle liquid is evacuated to the first-stage holding tank (T-3) for eventual recrystallization at a lower temperature to maximize product yield and to strip product from the final liquid residue. Semirefined product frozen to the inside of the tube during operation of stage two is first heated above its melting point and slightly melted (sweated). This semipurified melted material (sweat) is removed from the crystallizer kettle, stored in a stage tank (T-4), and then added to the next batch of fresh feed. The remaining material inside the crystallizer is then melted, mixed with product sweat from stage three, recrystallized, and sweated to upgrade the purity even further (stage 3).

Commercial Equipment and Applications The falling-film crystallization process was invented by the MWB company in Switzerland. The process is now marketed by Sulzer Chemtech. Products successfully processed in the falling-film crystallizer are listed on Table 20-9. The falling-film crystallization process is available from the Chemtech Div. of Sulzer Canada Inc., 60 Worcester Rd., Rexdale, Ontario N9W 5X2 Canada.

TABLE 20-9 Fractional Crystallization Reference List

Product	Main characteristics	Capacity, tons/year	Purity	Type of plant	Country	Client
Acrylic acid	Very low aldehyde content, no undesired polymerization in the plant	Undisclosed	99.95% 99.9%	Falling film Falling film	Undisclosed Undisclosed	Undisclosed Undisclosed
Benzoic acid	Pharmaceutical grade, odor- and color free	4,500	99.97%	Falling film	Italy	Chimica del Friuli
Bisphenol A	Polycarbonate grade, no solvent required	150,000	Undisclosed	Falling film	USA	General Electric
Carbonic acid		1,200	Undisclosed	Falling film	Germany	Undisclosed
Fatty acid	Separation of tallow fatty acid into saturated and unsaturated fractions	20,000	Stearic acid: Iodine no. 2 Oleic acid: Cloud pt 5°C	Falling film	Japan	Undisclosed
Fine chemicals		<1,000 <1,000 <1,000 <1,000 <1,000 <1,000 <1,000	Undisclosed Undisclosed Undisclosed Undisclosed Undisclosed Undisclosed Undisclosed	Falling film Static Falling film Falling film Falling film Falling film Falling film	GUS Switzerland Switzerland Switzerland USA Germany Japan	Undisclosed Undisclosed Undisclosed Undisclosed Undisclosed Undisclosed Undisclosed
Hydrazine	Satellite grade	3	>99.9%	Falling film	Germany	ESA
Monochloro acetic acid (MCA)	Low DCA content	6,000	>99.2%	Falling film	USA	Undisclosed
Multipurpose	Separation or purification of two or more chemicals, alternatively	1,000 1,000	Various grades Undisclosed	Falling film Falling film	Belgium Belgium	UCB Reibelco
Naphthalene	Color free and color stable with low thionaphthene content	60,000 20,000 10,000 12,000	99.5% 99.5% 99.8% Various grades	Falling film Falling film/static Falling film/static Falling film	Germany P.R. China P.R. China The Netherlands	Rütgers-Werke Anshan Jining Cindu Chemicals
<i>p</i> -Dichlorobenzene	No solvent washing required	40,000 5,000 4,000 3,000 3,000	99.95% 99.98% 99.8% 99.95% >97%	Falling film Falling film Falling film/distillation Falling film Static	USA Japan Brazil P.R. China P.R. China	Standard Chlorine Toa Gosei Nitroclor Fuyang Shandong
<i>p</i> -Nitrochlorobenzene		18,000 10,000	99.3% 99.5%	Falling film/distillation Static	P.R. China India	Jilin Chemical Mardia
Toluene diisocyanate (TDI)	Separation of TDI 80 into TDI 100 & TDI 65	20,000	Undisclosed	Falling film	Undisclosed	Undisclosed
Trioxan		<1,000	99.97%	Falling film	Undisclosed	Undisclosed

SUPERCRITICAL FLUID SEPARATION PROCESSES

GENERAL REFERENCES: Yeo and Kiran, *J. Supercritical Fluids*, **34**, 287–308 (2005). York, Kompella, and Shekunov, *Supercritical Fluid Technology for Drug Product Development*, Marcel Dekker, New York, 2004. Shah, Hanraath, Johnston, and Korgel, *J. Physical Chemistry B*, **108**, 9574–9587 (2004). Eckert, Liotta, Bush, Brown, and Hallett, *J. Physical Chemistry B*, **108**, 18108–18118 (2004). DeSimone, *Science*, **297**, 799–803 (2002). Arai, Sako, and Takebayashi, *Supercritical Fluids: Molecular Interactions, Physical Properties, and New Applications*, Springer, New York, 2002. Kiran, Debenedetti, and Peters, *Supercritical Fluids: Fundamentals and Applications*, Kluwer Academic, Dordrecht, 2000. McHugh and Krukonis, *Supercritical Fluid Extraction Principles and Practice*, 2d ed., Butterworth, Stoneham, Mass., 1994. Brunner, *Gas Extraction: An Introduction to Fundamentals of Supercritical Fluids and the Application to Separation Processes*, Springer, New York, 1994. Gupta and Shim, *Solubility in Supercritical Carbon Dioxide*, CRC Press, Boca Raton, Fla., 2007. Gupta and Kompella, *Nanoparticle Technology for Drug Delivery*, Taylor & Francis, New York, 2006.

INTRODUCTION

Fluids above their critical temperatures and pressures, called supercritical fluids (SCFs), exhibit properties intermediate between those of gases and liquids. Consequently, each of these two boundary conditions sheds insight into the nature of these fluids. Unlike gases, SCFs possess a considerable solvent strength, and transport properties are more favorable. In regions where a SCF is highly compressible, its density and hence its solvent strength may be adjusted over a wide range with modest variations in temperature and pressure. This tunability may be used to control phase behavior, separation processes (e.g., SCF extraction), rates and selectivities of chemical reactions, and morphologies in materials processing. For SCF separation processes to be feasible, the advantages (Table 20-10) must compensate for the costs of high pressure; examples of commercial applications are listed in Table 20-11.

The two SCFs most often studied—CO₂ and water—are the two least expensive of all solvents. CO₂ is nontoxic and nonflammable and has a near-ambient critical temperature of 31.1°C. CO₂ is an environmentally friendly substitute for organic solvents including chlorocarbons and chlorofluorocarbons. Supercritical water ($T_c = 374^\circ\text{C}$) is of interest as a substitute for organic solvents to minimize waste in extraction and reaction processes. Additionally, it is used for hydrothermal oxidation of hazardous organic wastes (also called supercritical water oxidation) and hydrothermal synthesis. (See also Sec. 15 for additional discussion of supercritical fluid separation processes.)

PHYSICAL PROPERTIES OF PURE SUPERCRITICAL FLUIDS

Thermodynamic Properties The variation in solvent strength of a SCF from gaslike to liquidlike values (see Table 20-12) may be described qualitatively in terms of the density ρ , as shown in Fig. 20-17, or the solubility parameter.

Similar characteristics are observed for other density-dependent variables including enthalpy, entropy, viscosity, and diffusion coefficient. Above the critical temperature, it is possible to tune the solvent

TABLE 20-10 Advantages of Supercritical Fluid Separations

Solvent strength is adjustable to tailor selectivities and yields.
Diffusion coefficients are higher and viscosities lower, compared with liquids.
Low surface tension favors wetting and penetration of small pores.
There is rapid diffusion of CO ₂ through condensed phases, e.g., polymers and ionic liquids.
Solvent recovery is fast and complete, with minimal residue in product.
Collapse of structure due to capillary forces is prevented during solvent removal.
Properties of CO ₂ as a solvent:
Environmentally acceptable solvent for waste minimization, nontoxic, nonflammable, inexpensive, usable at mild temperatures.
Properties of water as a solvent:
Nontoxic, nonflammable, inexpensive substitute for organic solvents.
Extremely wide variation in solvent strength with temperature and pressure.

TABLE 20-11 Commercial Applications of Supercritical Fluid Separations Technology

Extraction of foods and pharmaceuticals
Caffeine from coffee and tea
Flavors, cholesterol, and fat from foods
Nicotine from tobacco
Solvents from pharmaceutical compounds and drugs from natural sources
Extraction of volatile substances from substrates
Drying and aerogel formation
Cleaning fabrics, quartz rods for light guide fibers, residues in microelectronics
Removal of monomers, oligomers, and solvent from polymers
Fractionation
Residuum oil supercritical extraction (ROSE) (petroleum deasphalting)
Polymer and edible oils fractionation
CO ₂ enhanced oil recovery
Analytical SCF extraction and chromatography
Infusion of materials into polymers (dyes, pharmaceuticals)
Reactive separations
Extraction of <i>sec</i> -butanol from isobutene
Polymerization to form Teflon
Depolymerization, e.g., polyethylene terephthalate and cellulose hydrolysis
Hydrothermal oxidation of organic wastes in water
Crystallization, particle formation, and coatings
Antisolvent crystallization, rapid expansion from supercritical fluid solution (RESS)
Particles from gas saturated solutions
Crystallization by reaction to form metals, semiconductors (e.g., Si), and metal oxides including nanocrystals
Supercritical fluid deposition

TABLE 20-12 Physical Properties of a Supercritical Fluid Fall between Those of a Typical Gas and Liquid

	Liquid	Supercritical fluid	Gas
Density, g/mL	1	0.05 – 1	10 ⁻³
Viscosity, Pa·s	10 ⁻³	10 ⁻⁴ – 10 ⁻⁵	10 ⁻⁵
Diffusion coefficients, cm ² /s	10 ⁻⁵	10 ⁻³	10 ⁻¹
Surface tension, mN/m	20–50	0	0

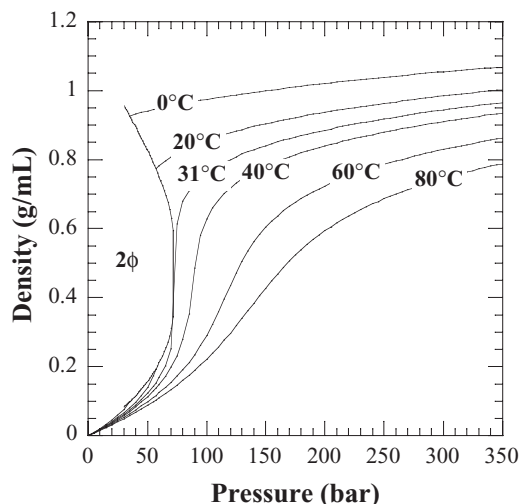


FIG. 20-17 Density versus pressure and temperature for CO₂. ($T_c = 31.1^\circ\text{C}$, $P_c = 73.8$ bar.)

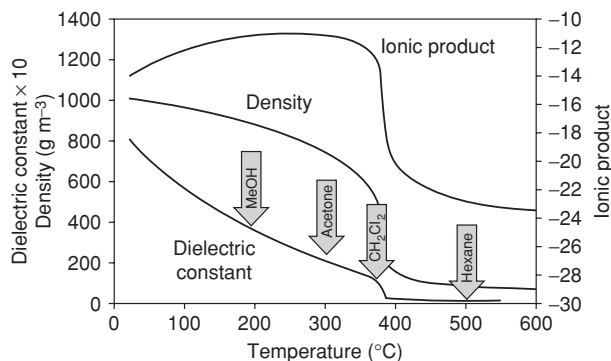


FIG. 20-18 Physical properties of water versus temperature at 240 bar. [Reprinted from Kritzer and Dinjus, "An Assessment of Supercritical Water Oxidation (SCWO): Existing Problems, Possible Solutions and New Reactor Concepts," Chem. Eng. J., vol. 83(3), pp. 207–214, copyright 2001, with permission from Elsevier.]

strength continuously over a wide range with either a small isothermal pressure change or a small isobaric temperature change. This unique ability to tune the solvent strength of a SCF may be used to extract and then recover selected products. A good indicator of the van der Waals forces contributed by a SCF is obtained by multiplying ρ by the molecular polarizability α , which is a constant for a given molecule. CO_2 's small $\alpha\rho$ and low solvent strength are more like those of a fluorocarbon than those of a hydrocarbon.

Water, a key SCF, undergoes profound changes upon heating to the critical point. It expands by a factor of 3, losing about two-thirds of the hydrogen bonds, the dielectric constant drops from 80 to 5 (Shaw et al., op. cit.), and the ionic product falls several orders of magnitude (see Fig. 20-18). At lower densities, supercritical water (SCW) behaves as a "nonaqueous" solvent, and it dissolves many organics and even gases such as O_2 . Here it does not solvate ions significantly.

Transport Properties Although the densities of SCFs can approach those of conventional liquids, transport properties are more favorable because viscosities remain lower and diffusion coefficients remain higher. Furthermore, CO_2 diffuses through condensed-liquid phases (e.g., adsorbents and polymers) faster than do typical solvents which have larger molecular sizes. For example, at 35°C the estimated pyrene diffusion coefficient in polymethylmethacrylate increases by 4 orders of magnitude when the CO_2 content is increased from 8 to 17 wt % with pressure [Cao, Johnston, and Webber, *Macromolecules*, 38(4), 1335–1340 (2005)].

PHASE EQUILIBRIA

Liquid-Fluid Equilibria Nearly all binary liquid-liquid phase diagrams can be conveniently placed in one of six classes (Prausnitz, Lichtenthaler, and de Azevedo, *Molecular Thermodynamics of Fluid Phase Equilibria*, 3d ed., Prentice-Hall, Upper Saddle River, N.J., 1998). Two-phase regions are represented by an area and three-phase regions by a line. In class I, the two components are completely miscible, and a single critical mixture curve connects their critical points. Other classes may include intersections between three phase lines and critical curves. For a ternary system, the slopes of the tie lines (distribution coefficients) and the size of the two-phase region can vary significantly with pressure as well as temperature due to the compressibility of the solvent.

Solid-Fluid Equilibria The solubility of the solid is very sensitive to pressure and temperature in compressible regions, where the solvent's density and solubility parameter are highly variable. In contrast, plots of the log of the solubility versus density at constant temperature often exhibit fairly simple linear behavior (Fig. 20-19). To understand the role of solute-solvent interactions on solubilities and selectivities, it is instructive to define an enhancement factor E as the actual solubility y_2 divided by the solubility in an ideal gas, so that $E = y_2 P / P_2^{\text{sat}}$, where P_2^{sat} is the vapor pressure. The solubilities in CO_2 are governed primarily by vapor pressures, a property of the solid

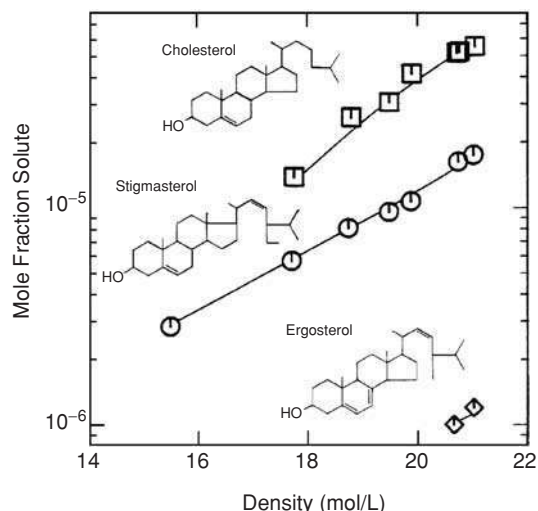


FIG. 20-19 Solubility of sterols in pure CO_2 at 35°C [Wong and Johnston, *Biotech. Prog.*, 2, 29 (1986)].

crystals, and only secondarily by solute-solvent interactions in the SCF phase. For example, for a given fluid at a particular T and P , the E 's are similar for the three sterols, each containing one hydroxyl group, even though the actual solubilities vary by many orders of magnitude (Fig. 20-19).

Polymer-Fluid Equilibria and the Glass Transition Most polymers are insoluble in CO_2 , yet CO_2 can be quite soluble in the polymer-rich phase. The solubility in CO_2 may be increased by a combination of lowering the cohesive energy density (which is proportional to the surface tension of the polymer [O'Neill et al., *Ind. Eng. Chem. Res.*, 37, 3067–79 (1998)]), branching, and the incorporation of either acetate groups in the side chain or carbonate groups in the backbone of the polymer [Sarbu, Styrane, and Beckman, *Nature*, 405, 165–168 (2000)]. Polyfluoromethacrylates are extremely soluble, and functionalized polyethers and copolymers of cyclic ethers and CO_2 have been shown to be more soluble than most other nonfluorinated polymers. The sorption of CO_2 into silicone rubber is highly dependent upon temperature and pressure, since these properties have a large effect on the density and activity of CO_2 . For glassy polymers, sorption isotherms are more complex, and hysteresis between the pressurization and depressurization steps may appear. Furthermore, CO_2 can act as a plasticizer and depress the glass transition temperature by as much as 100°C or even more, producing large changes in mechanical properties and diffusion coefficients. This phenomenon is of interest in conditioning membranes for separations and in commercial foaming of polymers to reduce VOC emissions.

Cosolvents and Complexing Agents Many nonvolatile polar substances cannot be dissolved at moderate temperatures in nonpolar fluids such as CO_2 . Cosolvents (also called entrainers) such as alcohols and acetone have been added to fluids to raise the solvent strength for organic solutes and even metals. The addition of only 2 mol % of the complexing agent tri-*n*-butyl phosphate (TBP) to CO_2 increases the solubility of hydroquinone by a factor of 250 due to Lewis acid-base interactions.

Surfactants and Colloids in Supercritical Fluids Because very few nonvolatile molecules are soluble in CO_2 , many types of hydrophilic or lipophilic species may be dispersed in the form of polymer latexes (e.g., polystyrene), microemulsions, macroemulsions, and inorganic suspensions of metals and metal oxides (Shah et al., op. cit.). The environmentally benign, nontoxic, and nonflammable fluids water and CO_2 are the two most abundant and inexpensive solvents on earth. Fluorocarbon and hydrocarbon-based surfactants have been used to form reverse micelles, water-in- CO_2

microemulsions (2- to 10-nm droplets) and macroemulsions (50-nm to 5- μm droplets) in SCFs including CO_2 . These organized molecular assemblies extend SCF technology to include nonvolatile hydrophilic solutes and ionic species such as amino acids and even proteins. Surfactant micelles or microemulsions are used commercially in dry cleaning and have been proposed for applications including polymerization, formation of inorganic and pharmaceutical particles, and removal of etch/ash residues from low- k dielectrics used in microelectronics. CO_2 -in-water macroemulsions, stabilized by surfactants with the proper hydrophilic- CO_2 -philic balance, are used in enhanced oil recovery to raise the viscosity of the flowing CO_2 phase for mobility control. Alkane ligands with various head groups have been used to stabilize inorganic nanocrystals in SCW and to stabilize Si and Ge nanocrystals in SCF hydrocarbons and CO_2 at temperatures from 350 to 500°C. Furthermore, colloids may be stabilized by electrostatic stabilization in CO_2 [Smith, Ryoo, and Johnston, *J. Phys. Chem. B*, **109**(43), 20155 (2005)].

Phase Equilibria Models Two approaches are available for modeling the fugacity of a solute in a SCF solution. The compressed gas approach includes a fugacity coefficient which goes to unity for an ideal gas. The expanded liquid approach is given as

$$f_i^L = x_i \gamma_i(P^\circ, x_i) f_i^{\text{OL}}(P^\circ) \exp\left(\int_{P^\circ}^P \frac{\bar{v}_i}{RT} dP\right) \quad (20-9)$$

where x_i is the mole fraction, γ_i is the activity coefficient, P° and f_i° are the reference pressure and fugacity, respectively, and $\bar{v}_i$ is the partial molar volume of component i . A variety of equations of state have been applied in each approach, ranging from simple cubic equations such as the Peng-Robinson equation of state to the more complex statistical associating fluid theory (SAFT) (Prausnitz et al., op. cit.). SAFT is successful in describing how changes in H bonding of SCF water influence thermodynamic and spectroscopic properties.

MASS TRANSFER

Experimental gas-solid mass-transfer data have been obtained for naphthalene in CO_2 to develop correlations for mass-transfer coefficients [Lim, Holder, and Shah, *Am. Chem. Soc. Symp. Ser.*, **406**, 379 (1989)]. The mass-transfer coefficient increases dramatically near the critical point, goes through a maximum, and then decreases gradually. The strong natural convection at SCF conditions leads to higher mass-transfer rates than in liquid solvents. A comprehensive mass-transfer model has been developed for SCF extraction from an aqueous phase to CO_2 in countercurrent columns [Seibert and Moosberg, *Sep. Sci. Technol.*, **23**, 2049 (1988); Brunner, op. cit.].

PROCESS CONCEPTS IN SUPERCRITICAL FLUID EXTRACTION

Figure 20-20 shows a one-stage extraction process that utilizes the adjustability of the solvent strength with pressure or temperature. The solvent flows through the extraction chamber at a relatively high pressure to extract the components of interest from the feed. The products are then recovered in the separator by depressurization, and the solvent is recompressed and recycled. The products can also be precipitated from the extract phase by raising the temperature after the extraction to lower the solvent density. Multiple extractions or multiple stages may be used with various profiles, e.g., successive increases in pressure or decreases in pressure. Solids may be processed continuously or semicontinuously by pumping slurries or by using lock hoppers. For liquid feeds, multi-stage separation may be achieved by continuous countercurrent extraction, much as in conventional liquid-liquid extraction. In SCF chromatography, selectivity may be tuned with pressure and temperature programming, with greater numbers of theoretical stages than in liquid chromatography and lower temperatures than in gas chromatography.

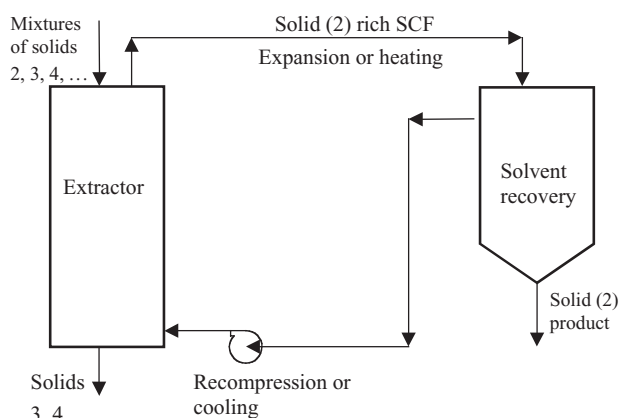


FIG. 20-20 Idealized diagram of a supercritical fluid extraction process for solids.

APPLICATIONS

Decaffeination of Coffee and Tea This application is driven by the environmental acceptability and nontoxicity of CO_2 as well as by the ability to tailor the extraction with the adjustable solvent strength. It has been practiced industrially for more than two decades. Caffeine may be extracted from green coffee beans, and the aroma is developed later by roasting. Various methods have been proposed for recovery of the caffeine, including washing with water and adsorption.

Extraction of Flavors, Fragrances, Nutraceuticals, and Pharmaceuticals Flavors and fragrances extracted by using supercritical CO_2 are significantly different from those extracted by using steam distillation or solvent extraction. The SCF extract can almost be viewed as a new product due to changes in composition associated with the greater amounts of extraction, as shown in Table 20-13. In many instances the flavor or fragrance of the extract obtained with CO_2 is closer to the natural one relative to steam distillation.

Temperature-Controlled Residue Oil Supercritical Extraction (ROSE) The Kerr-McGee ROSE process has been used worldwide for over two decades to remove asphaltenes from oil. The extraction step uses a liquid solvent that is recovered at supercritical conditions to save energy, as shown in Fig. 20-21. The residuum is contacted with butane or pentane to precipitate the heavy asphaltene fraction. The extract is then passed through a series of heaters, where it goes from the liquid state to a lower-density SCF state. Because the entire process is carried out at conditions near the critical point, a relatively small temperature change is required to produce a fairly large density change. After the light oils have been removed, the solvent is cooled back to the liquid state and recycled.

Polymer Devolatilization, Fractionation, and Plasticization Supercritical fluids may be used to extract solvent, monomers, and oligomers from polymers, including biomaterials. After extraction the pressure is reduced to atmospheric, leaving little residue in the

TABLE 20-13 Comparison of Percent Yields of Flavors and Fragrances from Various Natural Products*

Natural substance	Steam distillation (% yield)	Supercritical CO_2 (% yield)
Ginger	1.5–3.0	4.6
Garlic	0.06–0.4	4.6
Pepper	1.0–2.6	8–18
Rosemary	0.5–1.1	7.5

*Mukhopadhyay, *Natural Extracts Using Supercritical Carbon Dioxide*, CRC Press, Boca Raton, Fla., 2000; Moyler, Extraction of flavours and fragrances with compressed CO_2 , in *Extraction of Natural Products Using Near-Critical Solvents*, King and Bott (eds.), Blackie Academic & Professional, London, 1993.

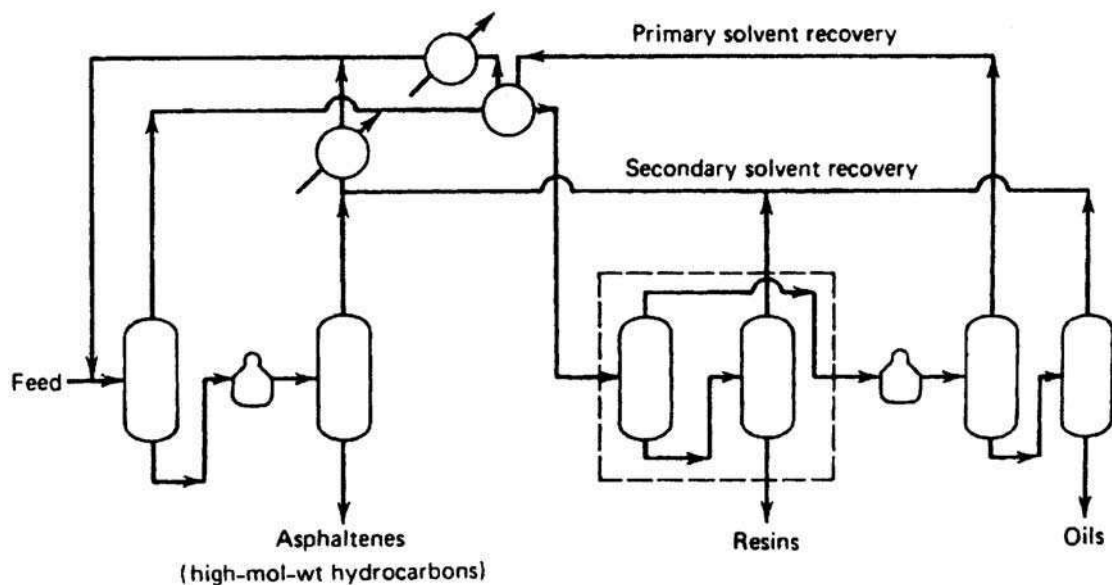


FIG. 20-21 Schematic diagram of the Kerr-McGee ROSE process.

substrate; furthermore, the extracted impurities are easily recovered from the SCF. The swelling and lowering of the glass transition temperature of the polymer by the SCF can increase mass-transfer rates markedly. This approach was used to plasticize block copolymer templates for the infusion of reaction precursors in the synthesis of porous low- k dielectrics. For homopolymers, plasticization may be used to infuse dyes, pharmaceuticals, etc., and then the SCF may be removed to trap the solute in the polymer matrix. SCFs may be used to fractionate polymers on the basis of molecular weight and/or composition with various methods for programming pressure and/or temperature (McHugh, op. cit.).

Drying and Aerogel Formation One of the oldest applications of SCF technology, developed in 1932, is SCF drying. The solvent is extracted from a porous solid with a SCF; then the fluid is depressurized. Because the fluid expands from the solid without crossing a liquid-vapor phase transition, capillary forces that would collapse the structure are not present. Using SCF drying, aerogels have been prepared with densities so low that they essentially float in air and look like a cloud of smoke. Also, the process is used in a commercial instrument to dry samples for electron microscopy without perturbing the structure.

Cleaning SCFs such as CO_2 can be used to clean and degrease quartz rods utilized to produce optical fibers, products employed in the fabrication of printed-circuit boards, oily chips from machining operations, precision bearings in military applications, and so on. Research is in progress for removing residues in etch/ash processes in microelectronics.

Microelectronics Processing SCF CO_2 is proposed as a "dry," environmentally benign processing fluid enabling replacement of aqueous and organic solvents in microelectronics fabrication (Desimone, op. cit.). Proposed applications include drying, lithography, solvent spin coating, stripping, cleaning, metal deposition, and chemical mechanical planarization. Due to its low surface tension, tunable solvent strength, and excellent mass-transfer properties, CO_2 offers advantages in wetting of surfaces and small pores, and in removal of contaminants at moderate temperatures.

Precipitation/Crystallization to Produce Nano- and Microparticles Because fluids such as CO_2 are weak solvents for many solutes, they are often effective antisolvents in fractionation and precipitation. In general, a fluid antisolvent may be a compressed gas, a gas-expanded liquid, or a SCF. Typically a liquid solution is sprayed through a nozzle into CO_2 to precipitate a solute. As CO_2 mixes with the liquid phase, it

decreases the cohesive energy density (solvent strength) substantially, leading to precipitation of dissolved solutes (e.g., crystals of progesterone). The high diffusion rates of the organic solvent into CO_2 and vice versa can lead to rapid phase separation, and the supersaturation curve may be manipulated to vary the crystalline morphology (Yeo and Kiran, op. cit.).

Nanoparticles of controllable size can be obtained in the supercritical antisolvent-enhanced mass-transfer (SAS-EM) process, which can produce commercial quantities of pharmaceuticals (see Fig. 20-22) [Chattopadhyay and Gupta, *Ind. Engr. Chem. Res.*, **40**, 3530–3539 (2001)]. Here, the solution jet is injected onto an ultrasonic vibrating surface H inside the antisolvent chamber to aid droplet atomization. The particle size is controlled by varying the vibration intensity. For most pharmaceuticals, organic compounds, proteins, and polymers, average particle diameters range from 100 to 1000 nm; even smaller particles may be obtained for certain inorganic compounds.

Rapid Expansion from Supercritical Solution and Particles from Gas Saturated Solutions Rapid expansion from supercritical solution (RESS) of soluble materials may be used to form microparticles or microfibers. A variety of inorganic crystals have been formed naturally and synthetically in SCF water, and organic crystals have been formed in SCF CO_2 . Recently, the addition of a solid cosolvent (e.g., menthol, which can be removed later by sublimation) has overcome key limitations by greatly enhancing solubilities in CO_2 and producing smaller nanoparticles by reducing particle-particle coagulation [Thakur and Gupta, *J. Sup. Fluids*, **37**, 307–315 (2006)]. Another approach is to expand the solutions into aqueous solutions containing soluble surfactants to arrest growth due to particle collisions. RESS typically uses dilute solutions. For concentrated solutions, the process is typically referred to as particle formation from gas saturated solutions (PGSS). Here CO_2 lowers the viscosity of the melt to facilitate flow. Union Carbide developed the commercial UNICARB process to replace organic solvents with CO_2 as a diluent in coating applications to reduce volatile organic carbon emissions and form superior coatings. For aqueous solutions, the expansion of CO_2 facilitates atomization, and the resulting cooling may be used to control the freezing of the solute.

Reactive Separations Reactions may be integrated with SCF separation processes to achieve a large degree of control for producing a highly purified product. Reaction products may be recovered by

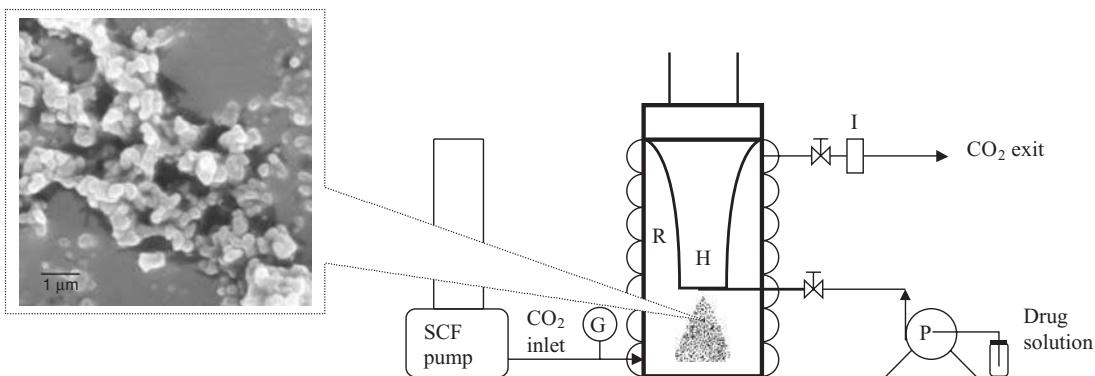


FIG. 20-22 Schematic of supercritical antisolvent with enhanced mass-transfer process to produce nanoparticles of controllable size. R, precipitation chamber; SCF pump, supply of supercritical CO₂; I, inline filter; H, ultrasonic horn; P, pump for drug solution; G, pressure gauge.

volatilization into, or precipitation from, a SCF phase. A classic example is the high-pressure production of polyethylene in SCF ethylene. The molecular weight distribution may be controlled by choosing the temperature and pressure for precipitating the polymer from the SCF phase. Over a decade ago, Idemitsu commercialized a 5000 metric ton per/year (t/yr) integrated reaction and separation process in SCF isobutene. The reaction of isobutene and water produces *sec*-butanol, which is extracted from water by the SCF solvent. SCF solvents have been tested for reactive extractions of liquid and gaseous fuels from heavy oils, coal, oil shale, and biomass. In some cases the solvent participates in the reaction, as in the hydrolysis of coal and heavy oils with SCW. Related applications include conversion of cellulose to glucose in water, delignification of wood with ammonia, and liquefaction of lignin in water.

Gas-expanded liquids (GXLs) are emerging solvents for environmentally benign reactive separation (Eckert et al., op. cit.). GXLs, obtained by mixing supercritical CO₂ with normal liquids, show intermediate properties between normal liquids and SCFs both in solvation power and in transport properties; and these properties are highly tunable by simple pressure variations. Applications include chemical reactions with improved transport, catalyst recycling, and product separation.

Hydrothermal oxidation (HO) [also called supercritical water oxidation (SCWO)] is a reactive process to convert aqueous wastes to water, CO₂, O₂, nitrogen, salts, and other by-products. It is an enclosed and complete water treatment process, making it more desirable to the public than incineration. Oxidation is rapid and efficient in this one-phase solution, so that wastewater containing 1 to 20 wt % organics may be oxidized rapidly in SCW with the potential for higher energy efficiency and less air pollution than in conventional incineration. Temperatures range from about 375 to 650°C and pressures from 3000 to about 5000 psia.

Crystallization by Chemical Reaction

Supercritical Fluid Deposition (SFD) Metal films may be grown from precursors that are soluble in CO₂. The SFD process yields copper films with fewer defects than those possible by using chemical vapor deposition, because increased precursor solubility removes mass-transfer limitations and low surface tension favors penetration of high-aspect-ratio features [Blackburn et al., *Science*, **294**, 141–145 (2001)].

High-Temperature Crystallization The size-tunable optical and electronic properties of semiconductor nanocrystals are attractive for a variety of optoelectronic applications. In solution-phase crystallization, precursors undergo chemical reaction to form nuclei, and particle growth is arrested with capping ligands that

passivate the surface. However, temperatures above 350°C are typically needed to crystallize the group IV elements silicon and germanium, due to the covalent network structure. Whereas liquid solvents boil away at these elevated temperatures, SCFs under pressure are capable of solvating the capping ligands to stabilize the nanocrystals (Shah et al., op. cit.). Crystalline Si and Ge nanocrystals, with an average size of 2 to 70 nm, may be synthesized in supercritical CO₂, hexane, or octanol at 400 to 550°C and 20 MPa in a simple continuous flow reactor. UV-visible absorbance and photoluminescence (PL) spectra of Ge nanocrystals of 3- to 4-nm diameter exhibit optical absorbance and PL spectra blue-shifted by approximately 1.7 eV relative to the band gap of bulk Ge, as shown in Fig. 20-23. One-dimensional silicon nanowires may be grown from relatively monodisperse gold nanocrystals stabilized with dodecanethiol ligands, as shown in Fig. 20-24. The first crystalline silicon nanowires with diameters smaller than 5 nm and lengths greater than 1 μm made by any technique were produced in SCF hexane. Hydrothermal crystallization has also been used to produce metal oxide nano- and microcrystals by rapid generation of supersaturation during hydrolysis of precursors, such as metal nitrates, during rapid heating of aqueous solutions.

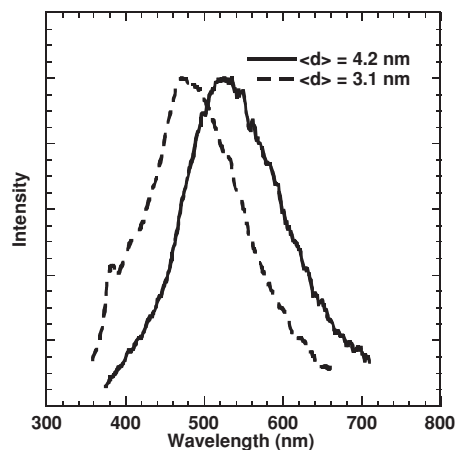


FIG. 20-23 Normalized photoluminescence spectra of 3.1-nm ($\lambda_{\text{excitation}} = 320 \text{ nm}$) and 4.2-nm ($\lambda_{\text{excitation}} = 340 \text{ nm}$) Ge nanoparticles dispersed in chloroform at 25°C with quantum yields of 6.6 and 4.6 percent, respectively. [Reprinted with permission from Lu et al., *Nano Lett.*, **4**(5), 969–974 (2004). Copyright 2004 American Chemical Society.]

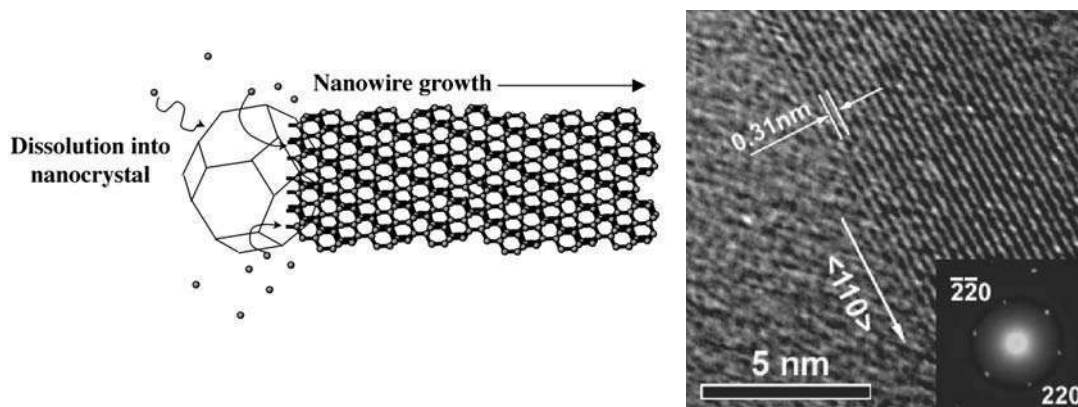


FIG. 20-24 High-resolution TEM image of Si nanowires produced at 500°C and 24.1 MPa in supercritical hexane from gold seed crystals. Inset: Electron diffraction pattern indexed for the <111> zone axis of Si indicates <110> growth direction. [Reprinted with permission from Lu *et al.*, Nano Lett., **3**(1), 93–99 (2003). Copyright 2003 American Chemical Society.]

ALTERNATIVE SOLID/LIQUID SEPARATIONS

SEPARATION PROCESSES BASED PRIMARILY ON ACTION IN AN ELECTRIC FIELD

Differences in mobilities of ions, molecules, or particles in an electric field can be exploited to perform useful separations. Primary emphasis is placed on electrophoresis and dielectrophoresis. Analogous separation processes involving magnetic and centrifugal force fields are widely applied in the process industry (see Secs. 18 and 19).

Theory of Electrical Separations

GENERAL REFERENCES: Newman, *Adv. Electrochem. Electrochem. Eng.*, **5**, 87 (1967); *Ind. Eng. Chem.*, **60**(4), 12 (1968). Ptasiński and Kerkhof, *Sep. Sci. Technol.*, **27**, 995 (1992).

For electrolytic solutions, migration of charged species in an electric field constitutes an additional mechanism of mass transfer. Thus the flux of an ionic species N_i in (g-mol)/(cm²·s) in dilute solutions can be expressed as

$$N_i = -z_i u_i \mathcal{F} c_i \nabla E - D_i \nabla c_i + c_i v \quad (20-10)$$

The ionic mobility u_i is the average velocity imparted to the species under the action of a unit force (per mole). v is the stream velocity, cm/s. In the present case, the electrical force is given by the product of the electric field ∇E in V/cm and the charge $z_i \mathcal{F}$ per mole, where $\mathcal{F}$ is the Faraday constant in C/g equivalent and z_i is the valence of the i th species. Multiplication of this force by the mobility and the concentration c_i [(g-mol)/cm³] yields the contribution of migration to the flux of the i th species.

The diffusive and convective terms in Eq. (20-10) are the same as in nonelectrolytic mass transfer. The ionic mobility u_i , (g-mol·cm²)/(J·s), can be related to the ionic-diffusion coefficient D_i , cm²/s, and the ionic conductance of the i th species λ_i , cm²/(Ω·g equivalent):

$$u_i = D_i / RT = \lambda_i / |z_i| \mathcal{F}^2 \quad (20-11)$$

where T is the absolute temperature, K; and R is the gas constant, 8.3143 J/(K·mol). Ionic conductances are tabulated in the literature (Robinson and Stokes, *Electrolyte Solutions*, Academic, New York, 1959). For practical purposes, a bulk electrolytic solution is electrically neutral.

$$\sum_i z_i c_i = 0 \quad (20-12)$$

since the forces required to effect an appreciable separation of charge are prohibitively large.

The **current density** (A/cm²) produced by movement of charged species is described by summing the terms in Eq. (20-13) for all species:

$$i = \mathcal{F} \sum_i z_i N_i = -\kappa \nabla E - \mathcal{F} \sum_i z_i D_i \nabla c_i \quad (20-13)$$

where the electrical conductivity κ in S/cm is given by

$$\kappa = \mathcal{F}^2 \sum_i z_i^2 u_i c_i \quad (20-14)$$

In solutions of uniform composition, the diffusional terms vanish and Eq. (20-13) reduces to Ohm's law.

Conservation of each species is expressed by the relation

$$\partial c_i / \partial t = -\nabla \cdot N_i \quad (20-15)$$

provided that the species is not produced or consumed in homogeneous chemical reactions. In two important cases, this conservation law reduces to the equation of convective diffusion:

$$(\partial c_i / \partial t) + \mathbf{v} \cdot \nabla c_i = D \nabla^2 c_i \quad (20-16)$$

First, when a large excess of inert electrolyte is present, the electric field will be small and migration can be neglected for minor ionic components; Eq. (20-16) then applies to these minor components, where D is the ionic-diffusion coefficient. Second, Eq. (20-16) applies when the solution contains only one cationic and one anionic species. The electric field can be eliminated by means of the electroneutrality relation.

In the latter case the diffusion coefficient D of the electrolyte is given by

$$D = (z_+ u_+ D_- - z_- u_- D_+) / (z_+ u_+ - z_- u_-) \quad (20-17)$$

which represents a compromise between the diffusion coefficients of the two ions. When Eq. (20-16) applies, many solutions can be obtained by analogy with heat transfer and nonelectrolytic mass transfer.

Because the solution is electrically neutral, conservation of charge is expressed by differentiating Eq. (20-13):

$$\nabla \cdot \mathbf{i} = 0 = -\kappa \nabla^2 E - \mathcal{F} \sum_i z_i D_i \nabla^2 c_i \quad (20-18)$$

For solutions of uniform composition, Eq. (20-18) reduces to Laplace's equation for the potential:

$$\nabla^2 E = 0 \quad (20-19)$$

This equation is the starting point for determination of the current-density distributions in many electrochemical cells.

Near an interface or at solution junctions, the solution departs from electroneutrality. Charges of one sign may be preferentially adsorbed at the interface, or the interface may be charged. In either case, the charge at the interface is counterbalanced by an equal and opposite charge composed of ions in the solution. Thermal motion prevents this countercharge from lying immediately adjacent to the interface, and the result is a "diffuse-charge layer" whose thickness is on the order of 10 to 100 Å.

A tangential electric field ∇E_t acting on these charges produces a relative motion between the interface and the solution just outside the diffuse layer. In view of the thinness of the diffuse layer, a balance of the tangential viscous and electrical forces can be written

$$\mu(\partial^2 v_t / \partial y^2) + \rho_e \nabla E_t = 0 \quad (20-20)$$

where μ is the viscosity and ρ_e is the electric-charge density, C/cm³. Furthermore, the variation of potential with the normal distance satisfies Poisson's equation:

$$\partial^2 E / \partial y^2 = -(\rho_e / \epsilon) \quad (20-21)$$

with ϵ defined as the **permittivity** of the solution. [The relative dielectric constant is ϵ/ϵ_0 , where ϵ_0 is the permittivity of free space; $\epsilon_0 = 8.8542 \times 10^{-14}$ C/(V·cm).] Elimination of the electric-charge density between Eqs. (20-20) and (20-21) with two integrations, gives a relation between ∇E_t and the velocity v_0 of the bulk solution relative to the interface.

$$\mu[v_t(\infty) - v_t(0)] = \epsilon \nabla E_t [E(\infty) - E(0)] \quad (20-22)$$

$$\text{or} \quad v_0 = -(\epsilon \nabla E_t \zeta / \mu) \quad (20-23)$$

The potential difference across the mobile part of the diffuse-charge layer is frequently called the **zeta potential**, $\zeta = E(0) - E(\infty)$. Its value depends on the composition of the electrolytic solution as well as on the nature of the particle-liquid interface.

There are four related electrokinetic phenomena which are generally defined as follows: *electrophoresis*—the movement of a charged surface (i.e., suspended particle) relative to a stationary liquid induced by an applied electrical field, *sedimentation potential*—the electric field which is created when charged particles move relative to a stationary liquid, *electroosmosis*—the movement of a liquid relative to a stationary charged surface (i.e., capillary wall), and *streaming potential*—the electric field which is created when liquid is made to flow relative to a stationary charged surface. The effects summarized by Eq. (20-23) form the basis of these electrokinetic phenomena.

For many particles, the diffuse-charge layer can be characterized adequately by the value of the zeta potential. For a spherical particle of radius r_0 which is large compared with the thickness of the diffuse-charge layer, an electric field uniform at a distance from the particle will produce a tangential electric field which varies with position on the particle. Laplace's equation [Eq. (20-19)] governs the distribution of potential outside the diffuse-charge layer; also, the Navier-Stokes equation for a creeping-flow regime can be applied to the velocity distribution. On account of the thinness of the diffuse-charge layer, Eq. (20-23) can be used as a local boundary condition, accounting for the effect of this charge in leading to movement of the particle relative to the solution. The result of this computation gives the velocity of the particle as

$$v = \epsilon \zeta \nabla E / \mu \quad (20-24)$$

and it may be convenient to tabulate the mobility of the particle

$$U = v / \nabla E = \epsilon \zeta / \mu \quad (20-25)$$

rather than its zeta potential. Note that this mobility gives the velocity of the particle for unit electric field rather than for unit force on the particle. Related equations can be developed for the velocity of electroosmotic flow. The subsections presented below ("Electrophoresis," "Electrofiltration," and "Cross-Flow-Electrofiltration") represent both established and emerging commercial applications of electrokinetic phenomena.

Electrophoresis

GENERAL REFERENCE: Wankat, *Rate-Controlled Separations*, Elsevier, London, 1990.

Electrophoretic Mobility Macromolecules move at speeds measured in tenths of micrometers per second in a field (gradient) of 1 V/cm. Larger particles such as bubbles or bacteria move up to 10 times as fast because U is usually higher. To achieve useful separations, therefore, voltage gradients of 10 to 100 V/cm are required. High voltage gradients are achieved only at the expense of power dissipation within the fluid, and the resulting heat tends to cause undesirable convection currents.

Mobility is affected by the dielectric constant and viscosity of the suspending fluid, as indicated in Eq. (20-25). The ionic strength of the fluid has a strong effect on the thickness of the double layer and hence on ζ . As a rule, mobility varies inversely as the square root of ionic strength [Overbeek, *Adv. Colloid Sci.*, **3**, 97 (1950)].

Modes of Operation There is a close analogy between sedimentation of particles or macromolecules in a gravitational field and their electrophoretic movement in an electric field. Both types of separation have proved valuable not only for analysis of colloids but also for preparative work, at least in the laboratory. Electrophoresis is applicable also for separating mixtures of simple cations or anions in certain cases in which other separating methods are ineffectual.

Electrodecentration or electroconvection is one of several operations in which one mobile component (or several) is to be separated out from less mobile or immobile ones. The mixture is introduced between two vertical semipermeable membranes; for separating cations, anion membranes are used, and vice versa. When an electric field is applied, the charged component migrates to one or another of the membranes; but since it cannot penetrate the membrane, it accumulates at the surface to form a dense concentrated layer of particles which will sink toward the bottom of the apparatus. Near the top of the apparatus immobile components will be relatively pure. Murphy [*J. Electrochem. Soc.*, **97**(11), 405 (1950)] has used silver-silver chloride electrodes in place of membranes. Frilette [*J. Phys. Chem.*, **61**, 168 (1957)], using anion membranes, partially separated H⁺ and Na⁺, K⁺ and Li⁺, and K⁺ and Na⁺.

Countercurrent electrophoresis can be used to split a mixture of mobile species into two fractions by the electrical analog of elutriation. In such countercurrent electrophoresis, sometimes termed an ion still, a flow of the suspending fluid is maintained parallel to the direction of the voltage gradient. Species which do not migrate fast enough in the applied electric field will be physically swept out of the apparatus. An apparatus based mainly on this principle but using also natural convection currents has been developed (Bier, *Electrophoresis*, vol. II, Academic, New York, 1967).

Membrane electrophoresis which is based upon differences in ion mobility, has been studied by Glueckauf and Kitt [*J. Appl. Chem.*, **6**, 511 (1956)]. Partial exclusion of coions by membranes results in large differences in coion mobilities. Superposing a cation and an anion membrane gives high transference numbers (about 0.5) for both cations and anions while retaining the selectivity of mobilities. Large voltages are required, and flow rates are low.

In continuous-flow zone electrophoresis the "solute" mixture to be separated is injected continuously as a narrow source within a body of carrier fluid flowing between two electrodes. As the "solute" mixture passes through the transverse field, individual components migrate sideways to produce zones which can then be taken off separately downstream as purified fractions.

Resolution depends upon differences in mobilities of the species. Background electrolyte of low ionic strength is advantageous, not only to increase electrophoretic (solute) mobilities, but also to achieve low electrical conductivity and thereby to reduce the thermal-convection current for any given field [Finn, in Schoen (ed.), *New Chemical Engineering Separation Techniques*, Interscience, New York, 1962].

The need to limit the maximum temperature rise has resulted in two main types of apparatus, illustrated in Fig. 20-25. The first consists of multicomponent ribbon separation units—apparatus capable of separating small quantities of mixtures which may contain few or many species. In general, such units operate with high voltages, low

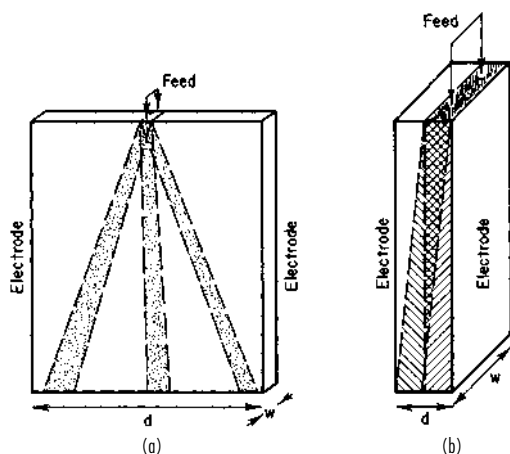


FIG. 20-25 Types of arrangement for zone electrophoresis or electrochromatography. (a) Ribbon unit, with $d > w$; cooling at side faces. (b) Block unit, with $w > d$; cooling at electrodes.

currents, a large transverse dimension, and a narrow thickness between cooling faces. Numerous units developed for analytic chemistry, generally with filter-paper curtains but sometimes with granular "anticonvectant" packing, are of this type. The second type consists of block separation units—apparatus designed to separate larger quantities of a mixture into two (or at most three) species or fractions. Such units generally use low to moderate voltages and high currents, with cooling by circulation of cold electrolyte through the electrode compartments. Scale-up can readily be accomplished by extending the thickness dimension w .

Both types of units have generally been operated in trace mode; that is, "background" or "elutant" electrolyte is fed to the unit along with the mixture to be separated. A desirable and possible means of operation for preparative applications is in bulk mode, in which one separated component follows the other without background electrolyte being present, except that other ions may be required to bracket the separated zones. Overlap regions between components should be recycled, and pure components collected as products.

For block units, the need to stabilize flow has given rise to a number of distinct techniques.

Free flow. Dobry and Finn [*Chem. Eng. Prog.*, **54**, 59 (1958)] used upward flow, stabilized by adding methyl cellulose, polyvinyl alcohol, or dextran to the background solution. Upward flow was also used in the electrode compartments, with cooling efficiency sufficient to keep the main solution within 1°C of entering temperature.

Density gradients to stabilize flow have been employed by Philpot [*Trans. Faraday Soc.*, **36**, 38 (1940)] and Mel [*J. Phys. Chem.*, **31**, 559 (1959)]. Mel's Staflo apparatus [*J. Phys. Chem.*, **31**, 559 (1959)] has liquid flow in the horizontal direction, with layers of increasing density downward produced by sucrose concentrations increasing to 7.5 percent. The solute mixture to be separated is introduced in one such layer. Operation at low electrolyte concentrations, low voltage gradients, and low flow rates presents no cooling problem.

Packed beds. A packed cylindrical electrochromatograph 9 in (23 cm) in diameter and 48 in (1.2 m) high, with operating voltages in the 25- to 100-V range, has been developed by Hybarger, Vermeulen, and coworkers [*Ind. Eng. Chem. Process Des. Dev.*, **10**, 91 (1971)]. The annular bed is separated from inner and outer electrodes by porous ceramic diaphragms. The unit is cooled by rapid circulation of cooled electrolyte between the diaphragms and the electrodes.

An interesting modification of zone electrophoresis resolves mixtures of ampholytes on the basis of **differing isoelectric points** rather than differing mobilities. Such **isoelectric spectra** develop when a pH gradient is established parallel to the electric field. Each species then migrates until it arrives at the region of pH where it possesses no net surface charge. A strong focusing effect is thereby

achieved [Kolin, in Glick (ed.), *Methods of Biochemical Analysis*, vol. VI, Interscience, New York, 1958].

Electrofiltration

GENERAL REFERENCE: P. Krishnaswamy and P. Klinkowski, "Electrokinetics and Electrofiltration," in *Advances in Solid-Liquid Separation*, H. S. Muradhi (ed.), Battelle Press, Columbus, OH, 1986.

Process Concept The application of a direct electric field of appropriate polarity when filtering should cause a net charged-particle migration relative to the filter medium (electrophoresis). The same direct electric field can also be used to cause a net fluid flow relative to the pores in a fixed filter cake or filter medium (electroosmosis). The exploitation of one or both of these phenomena form the basis of conventional electrofiltration.

In conventional filtration, often the object is to form a high-solids-content filter cake. At a single-filter surface, a uniform electric field can be exploited in one of two ways. The first method of exploitation occurs when the electric field is of a polarity such that the charged-particle migration occurs toward the filter medium. In this case, the application of the electric field increases the velocity of the solid particles toward the filter surface (electrosedimentation), thereby hastening the clarification of the feed suspension and, at the same time, increasing the compaction of the filter cake collected on the filter surface. In this first case, electroosmotic flow occurs in a direction away from the filter media. The magnitude of the pressure-driven fluid flow toward the filter surface far exceeds the magnitude of the electroosmotic flow away from the surface so that the electroosmotic flow results in only a minor reduction of the rate of production of filtrate. The primary benefits of the applied electric field in this case are increased compaction, and hence increased dewatering, of the filter cake and an increased rate of sedimentation or movement of the particles in bulk suspension toward the filter surface.

The second method of exploitation occurs when the electric field is of a polarity such that the charged-particle migration occurs away from the filter medium. The contribution to the net-particle velocity of the electrophoretically induced flow away from the filter medium is generally orders of magnitude less than the contribution to the net-particle velocity of the flow induced by drag due to the pressure-induced flow of the bulk liquid toward the filter media. (In conventional or cake filtration, the velocity of liquid in dead-end flow toward the filter is almost always sufficient to overcome any electrophoretic migration of particles away from the filter media so that the prevention of the formation of filter cake is not an option. This will not necessarily be the case for cross-flow electrofiltration.) The primary enhancement to filtration caused by the application of an electrical field in this manner is the increase in the filtrate flux due to electroosmotic flow through the filter cake. This electroosmotic flow is especially beneficial during the latter stages of filtration when the final filter-cake thickness has been achieved. At this stage, electroosmosis can be exploited to draw filtrate out from the pore structure of the filter cake. This type of drying of the filter cake is sometimes called **electroosmotic dewatering**.

Commercial Applications Krishnaswamy and Klinkowski, op. cit., describe the Dorr-Oliver EAVF[®]. The EAVF[®] combines vacuum filtration with electrophoresis and electroosmosis and has been described as a series of parallel platelike electrode assemblies suspended in a tank containing the slurry to be separated. When using the EAVF[®], solids are collected at both electrodes, one collecting a compacted cake simply by electrophoretic attraction and the second collecting a compacted cake though vacuum filtration coupled with electroosmotic dewatering. Upon the completion of a collection cycle, the entire electrode assembly is withdrawn from the slurry bath and the cake is removed. The EAVF[®] is quoted as being best suited for the dewatering of ultrafine slurries (particle sizes typically less than $10\ \mu\text{m}$).

Cross-Flow-Electrofiltration

GENERAL REFERENCES: Henry, Lawler, and Kuo, *Am. Inst. Chem. Eng. J.*, **23**(6), 851 (1977). Kuo, Ph.D. dissertation, West Virginia University, 1978.

Process Concept The application of a direct electric field of appropriate polarity when filtering should cause a net charged-particle

migration away from the filter medium. This electrophoretic migration will prevent filter-cake formation and the subsequent reduction of filter performance. An additional benefit derived from the imposed electric field is an electroosmotic flux. The presence of this flux in the membrane and in any particulate accumulation may further enhance the filtration rate.

Cross-flow–electrofiltration (CF-EF) is the multifunctional separation process which combines the electrophoretic migration present in electrofiltration with the particle diffusion and radial-migration forces present in cross-flow filtration (CFF) (microfiltration includes cross-flow filtration as one mode of operation in “Membrane Separation Processes” which appears later in this section) in order to reduce further the formation of filter cake. Cross-flow–electrofiltration can even eliminate the formation of filter cake entirely. This process should find application in the filtration of suspensions when there are charged particles as well as a relatively low conductivity in the continuous phase. Low conductivity in the continuous phase is necessary in order to minimize the amount of electrical power necessary to sustain the electric field. Low-ionic-strength aqueous media and nonaqueous suspending media fulfill this requirement.

Cross-flow–electrofiltration has been investigated for both aqueous and nonaqueous suspending media by using both rectangular- and tubular-channel processing configurations (Fig. 20-26). Henry, Lawler, and Kuo (op. cit.), using a rectangular-channel system with a 0.6- μ -pore-size polycarbonate Nuclepore filtration membrane, investigated CF-EF for 2.5- μ m kaolin-water and 0.5- to 2- μ m oil-in-water emulsion systems. Kuo (op. cit.), using similar equipment, studied 5- μ m kaolin-water, ~100- μ m Cr_2O_3 -water, and ~6- μ m Al_2O_3 -methanol and/or -butanol systems. For both studies electrical fields of 0 to 60 V/cm were used for aqueous systems, and to 5000 V/cm were used for nonaqueous systems. The studies covered a wide range of processing variables in order to gain a better understanding of CF-EF fundamentals. Lee, Gidaspow, and Wasan [Ind. Eng. Chem. Fundam., 19(2), 166 (1980)] studied CF-EF by using a porous stainless-steel tube (pore size = 5 μ m) as the filtration medium. A platinum wire running down the center of the tube acted as one electrode, while the porous steel tube itself acted as the other electrode. Nonaqueous suspensions of 0.3- to 2- μ m Al_2O_3 -tetralin and a coal-derived liquid diluted with xylene and tetralin were studied. By operating with applied electric fields (1000 to 10,000 V/cm) above the critical voltage, clear particle-free filtrates were produced. It should be noted that the

pore size of the stainless-steel filter medium (5 μ m) was greater than the particle size of the suspended Al_2O_3 solids (0.3 to 2 μ m). Cross-flow–electrofiltration has also been applied to biological systems. Brors, Kroner, and Deckwer [ECB6: Proc. 6th Eur. Cong. Biotech., 511 (1994)] separated malate dehydrogenase from the cellular debris of Baker's yeast using CF-EF. A two- to fivefold increase in the specific enzyme transport rate was reported when electric field strengths of 20 to 40 V/cm were used.

Theory Cross-flow–electrofiltration can theoretically be treated as if it were cross-flow filtration with superimposed electrical effects. These electrical effects include electroosmosis in the filter medium and cake and electrophoresis of the particles in the slurry. The addition of the applied electric field can, however, result in some qualitative differences in permeate-flux-parameter dependences.

The membrane resistance for CF-EF can be defined by specifying two permeate fluxes as

$$J_{om} = \Delta P / R_{om} \quad (20-26)$$

$$J_m = \Delta P / R_m \quad (20-27)$$

where J_{om} is the flux through the membrane in the absence of an electric field and any other resistance, m/s; J_m is the same flux in the presence of an electric field; and R_{om} is the membrane resistance in the absence of an electric field, (N·s)/m². When electroosmotic effects do occur,

$$J_m = J_{om} + K_m E \quad (20-28)$$

where K_m is the electroosmotic coefficient of the membrane, m²/(V·s); and E is the applied-electric-field strength, V/m. Equations (20-26), (20-27), and (20-28) can be combined and rearranged to give Eq. (20-29), the membrane resistance in the presence of an electric field.

$$R_m = \frac{R_{om}}{1 + \left(\frac{K_m E}{J_{om}} \right)} \quad (20-29)$$

Similarly, cake resistance can be represented as

$$R_c = \frac{R_{oc}}{1 + \left(\frac{K_c E}{J_{oc}} \right)} \quad (20-30)$$

where J_{oc} is the flux through the cake in the absence of an electric field or any other resistance, R_{oc} is the cake resistance in the absence of an electric field, and K_c is the electroosmotic coefficient of the cake. The cake resistance is not a constant but is dependent upon the cake thickness, which is in turn a function of the transmembrane pressure drop and electrical-field strength.

Particulate systems require the addition of the term $\mu_e E$ in order to account for the electrophoretic migration of the particle. The constant μ_e is the electrophoretic mobility of the particle, m²/(V·s). For the case of the CF-EF, the film resistance R_f can be represented as

$$R_f = \frac{\Delta P}{k \ln \left(\frac{C_s}{C_b} \right) + U_r + \mu_e E} \quad (20-31)$$

The resistances, when incorporated into equations descriptive of cross-flow filtration, yield the general expression for the permeate flux for particulate suspensions in cross-flow–electrofiltration systems.

There are three distinct regimes of operation in CF-EF. These regimes (Fig. 20-27) are defined by the magnitude of the applied electric field with respect to the critical voltage E_c . The critical voltage is defined as the voltage at which the net particle migration velocity toward the filtration medium is zero. At the critical voltage, there is a balance between the electrical-migration and radial-migration velocities away from the filter and the velocity at which the particles are swept toward the filter by bulk flow. There is no diffusive transport at $E = E_c$ (Fig. 20-27b) because there is no gradient in the particle concentration normal to the filter surface. At field strengths below the critical voltage (Fig. 20-27a), all migration velocities occur in the same direction as in the cross-flow-filtration systems discussed earlier. At values of applied voltage above the critical voltage (Fig. 20-27c) qualitative differences

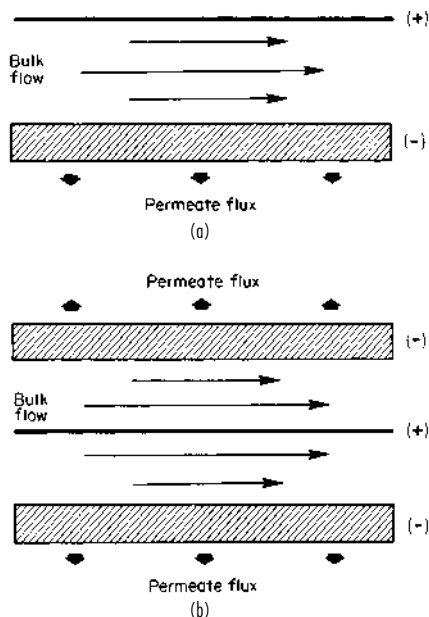


FIG. 20-26 Alternative electrode configurations for cross-flow–electrofiltration.

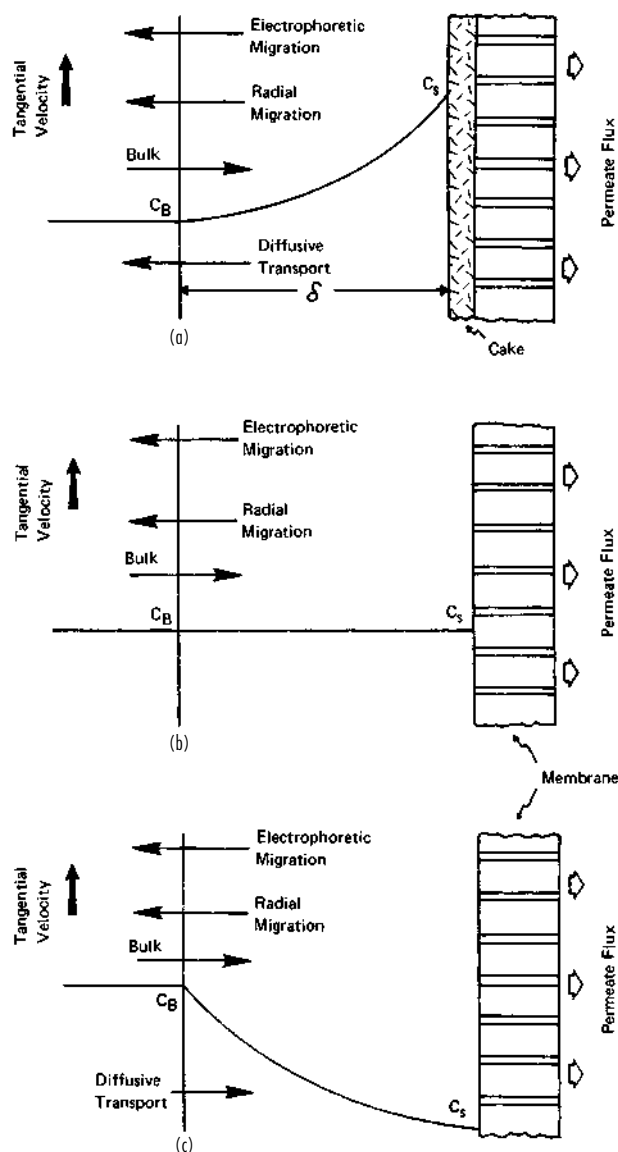


FIG. 20-27 Regimes of operation of cross-flow-electrofiltration: (a) voltage less than critical, (b) voltage equal to the critical voltage, (c) voltage greater than critical.

are observed. In this case, the electrophoretic-migration velocity away from the filter medium is greater than the velocity caused by bulk flow toward the filtration medium. Particles concentrate away from the filter medium. This implies that particle concentration is lowest next to the filter medium (in actuality, a clear boundary layer has been observed). The influence of fluid shear still improves the transfer of particles down the concentration gradient, but in this case it is toward the filtration medium. When the particles are small and diffusive transport dominates radial migration, increasing the circulation velocity will decrease the permeate flux rate in this regime. When the particles are large and radial migration dominates, the increase in circulation velocity will still improve the filtration rate. These effects are illustrated qualitatively in Fig. 20-28a. The solid lines represent systems in which the particle diffusive effect dominates the radial-migration effect, while the dashed lines represent the inverse. Figure 20-28b illustrates the increase in filtration rate with increasing electric field strength. For field strengths

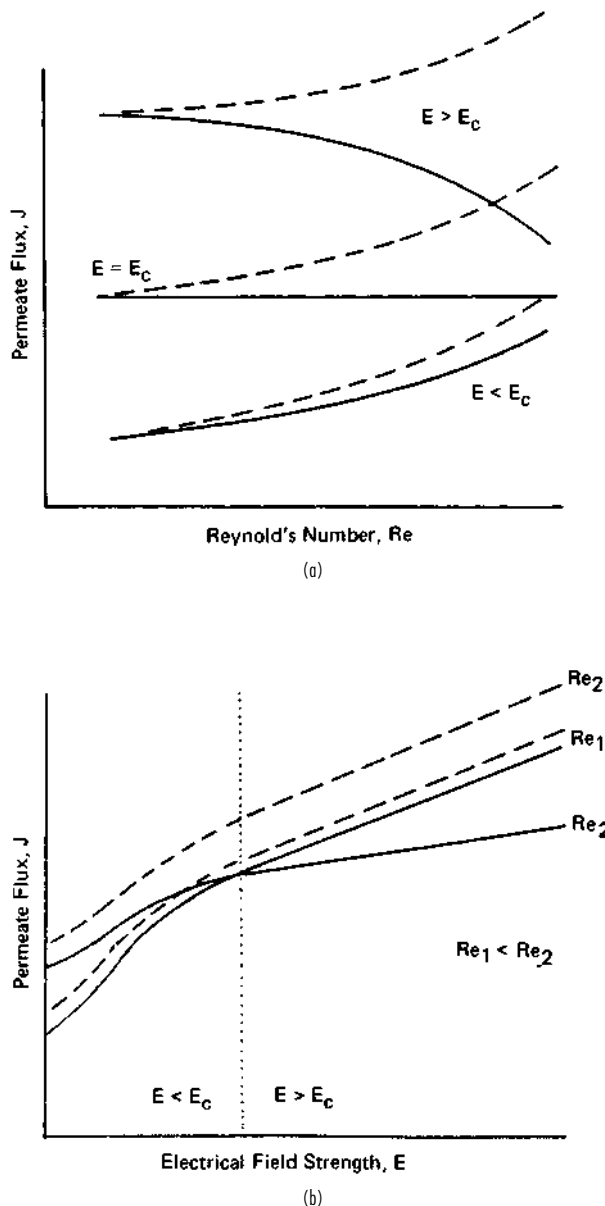


FIG. 20-28 Qualitative effects of Reynolds number and applied-electric-field strength on the filtration permeate flux J . Dashed lines indicate large particles (radial migration dominates); solid lines, small particles (particle diffusion dominates).

$E > E_c$, increases in permeate flux rate are due only to electroosmosis in the filtration medium.

One potential difficulty with CF-EF is the electrodeposition of the particles at the electrode away from the filtration medium. This phenomenon, if allowed to persist, will result in performance decay of CF-EF with respect to maintenance of the electric field. Several approaches such as momentary reverses in polarity, protection of the electrode with a porous membrane or filter medium, and/or utilization of a high fluid shear rate can minimize electrodeposition.

Dielectrophoresis

GENERAL REFERENCES: Gascoyne and Vykoukal, *Electrophoresis*, **23**, 1973–1983 (2002). Pohl, in Moore (ed.), *Electrostatics and Its Applications*,

Wiley, New York, 1973, chap. 14 and chap. 15 (with Crane). Pohl, in Catsim-poolas (ed.), *Methods of Cell Separation*, vol. I, Plenum Press, New York, 1977, chap. 3. Pohl, *Dielectrophoresis: The Behavior of Matter in Nonuniform Electric Fields*, Cambridge, New York, 1978.

Introduction Dielectrophoresis (DEP) is defined as the motion of neutral, polarizable matter produced by a nonuniform electric (ac or dc) field. DEP should be distinguished from electrophoresis, which is the motion of charged particles in a uniform electric field (Fig. 20-29).

The DEP of numerous particle types has been studied, and many applications have been developed. Particles studied have included aerosols, glass, minerals, polymer molecules, living cells, and cell organelles. Applications developed include filtration, orientation, sorting or separation, characterization, and levitation and materials handling. Effects of DEP are easily exhibited, especially by large particles, and can be applied in many useful and desirable ways. DEP effects can, however, be observed on particles ranging in size even down to the molecular level in special cases. Since thermal effects tend to disrupt DEP with molecular-sized particles, they can be controlled only under special conditions such as in molecular beams.

Principle The principle of particle and cell separation, control, or characterization by the action of DEP lies in the fact that a net force can arise upon even neutral particles situated in a nonuniform electric field. The force can be thought of as rising from the imaginary two-step process of (1) induction or alignment of an electric dipole in a particle placed in an electric field followed by (2) unequal forces on the ends of that dipole. This arises from the fact that the force of an electric field upon a charge is equal to the amount of the charge and to the local field strength at that charge. Since the two (equal) charges of the (induced or oriented) dipole of the particle lie in unequal field strengths of the diverging field, a net force arises. If the particle is suspended in a fluid, then the polarizability of that medium enters, too.

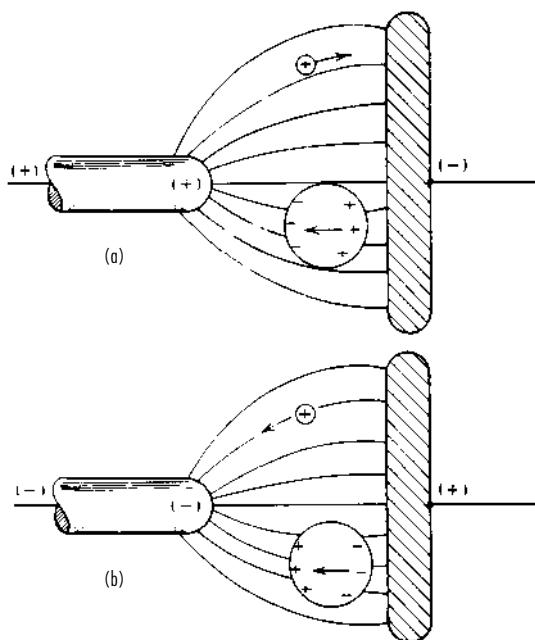


FIG. 20-29 Comparison of behaviors of neutral-charged bodies in an alternating nonuniform electric field. (a) Positively charged body moves toward negative electrode. Neutral body is polarized, then is attracted toward point where field is strongest. Since the two charge regions on the neutral body are equal in amount of charge but the force is proportional to the local field, a net force toward the region of more intense field results. (b) Positively charged body moves toward the negative electrode. Again, the neutral body is polarized, but it does not reverse direction although the field is reversed. It still moves toward the region of highest field intensity.

If, for example, the particle is more polarizable than the fluid, then the net force is such as to impel the particle to regions of greater field strength. Note that this statement implies that the effect is independent of the absolute sign of the field direction. This is found to be the case. Even rapidly alternating (ac) fields can be used to provide *unidirectional* motion of the suspended particles.

Formal Theory A small neutral particle at equilibrium in a static electric field experiences a net force due to DEP that can be written as $\mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E}$, where $\mathbf{p}$ is the dipole moment vector and $\mathbf{E}$ is the external electric field. If the particle is a simple dielectric and is isotropically, linearly, and homogeneously polarizable, then the dipole moment can be written as $\mathbf{p} = \alpha\mathbf{E}$, where α is the (scalar) polarizability, v is the volume of the particle, and $\mathbf{E}$ is the external field. The force can then be written as:

$$\mathbf{F} = \alpha v (\mathbf{E} \cdot \nabla) \mathbf{E} = \frac{1}{2} \alpha v \nabla |\mathbf{E}|^2 \quad (20-32)$$

This force equation can now be used to find the force in model systems such as that of an ideal dielectric sphere (relative dielectric constant K_2) in an ideal perfectly insulating dielectric fluid (relative dielectric constant K_1). The force can now be written as

$$\mathbf{F} = 2\pi a^3 \epsilon_0 K_1 \left(\frac{K_2 - K_1}{K_2 + 2K_1} \right) \nabla |\mathbf{E}|^2 \quad (20-33)$$

(ideal dielectric sphere in ideal fluid).

Heuristic Explanation As we can see from Fig. 20-30, the DEP response of *real* (as opposed to perfect insulator) particles with frequency can be rather complicated. We use a simple illustration to account for such a response. The force is proportional to the difference between the dielectric permittivities of the particle and the surrounding medium. Since a part of the polarization in real systems is thermally activated, there is a delayed response which shows as a phase lag between $\mathbf{D}$, the dielectric displacement, and $\mathbf{E}$, the electric-field intensity. To take this into account we may replace the simple (absolute) dielectric constant ϵ by the complex (absolute) dielectric constant $\hat{\epsilon} = \epsilon' - i\epsilon'' = \epsilon' - i\sigma/\omega$, where ω is the angular frequency of the applied field. For treating spherical objects, for example, the replacement

$$F \propto \frac{\epsilon_1(\epsilon_2 - \epsilon_1)}{\epsilon_2 + 2\epsilon_1} \rightarrow \text{Re} \left\{ \frac{\hat{\epsilon}_1^* (\hat{\epsilon}_2 - \hat{\epsilon}_1)}{\hat{\epsilon}_2 + 2\hat{\epsilon}_1} \right\} \quad (20-34)$$

can be made, where $\hat{\epsilon}^*$ is the complex conjugate of $\hat{\epsilon}$.

With this force expression for real dielectrics, we can now explain the complicated DEP response with the help of Fig. 20-30.

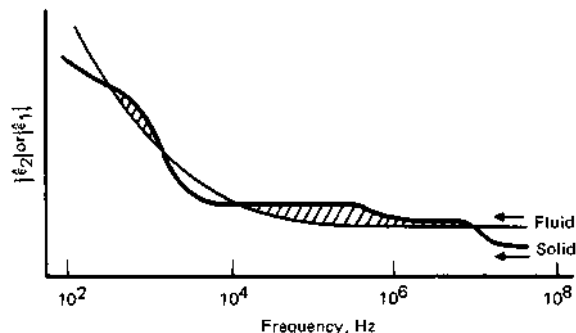


FIG. 20-30 A heuristic explanation of the dielectrophoretic-collection-rate (DCR)-frequency spectrum. The curves for the absolute values of the complex permittivities of the fluid medium and of the suspended particles are shown lying nearly, but not entirely, coincident over the frequency range of the applied electric field. When the permittivity (dielectric constant) of the particles exceeds that of the suspending medium, the collection, or "positive dielectrophoresis," occurs. In the frequency ranges in which the permittivity of the particles is less than that of the suspending medium no collection at the regions of higher field intensity occurs. Instead there is "negative dielectrophoresis," i.e., movement of the particles into regions of lower field intensity.

A particle, such as a living cell, can be imagined as having a number of different frequency-dependent polarization mechanisms contributing to the total effective polarization of the particle $|\epsilon_2|$. The heavy curve in Fig. 20-30 shows that the various mechanisms in the particle drop out stepwise as the frequency increases. The light curve in Fig. 20-30 shows the polarization for a simple homogeneous liquid that forms the surrounding medium. This curve is a smooth function which becomes constant at high frequency. As the curves cross each other (and hence $|\epsilon_2| = |\epsilon_1|$), various responses occur. The particle can thus be attracted to the strongest field region, be repelled from that region, or experience no force depending on the frequency.

Limitations It is desirable to have an estimate for the smallest particle size that can be effectively influenced by DEP. To do this, we consider the force on a particle due to DEP and also due to the osmotic pressure. This latter diffusional force will randomize the particles and tend to destroy the control by DEP. Figure 20-31 shows a plot of these two forces, calculated for practical and representative conditions, as a function of particle radius. As we can see, the smallest particles that can be effectively handled by DEP appear to be in range of 0.01 to 0.1 μm (100 to 1000 Å).

Another limitation to be considered is the volume that the DEP force can affect. This factor can be controlled by the design of electrodes. As an example, consider electrodes of cylindrical geometry. A practical example of this would be a cylinder with a wire running down the middle to provide the two electrodes. The field in such a system is proportional to $1/r$. The DEP force is then $F_{\text{DEP}} \propto \nabla |E|^2 \propto 1/r^3$, so that any differences in particle polarization might well be masked merely by positional differences in the force. At the outer cylinder the DEP force may even be too small to affect the particles appreciably. The

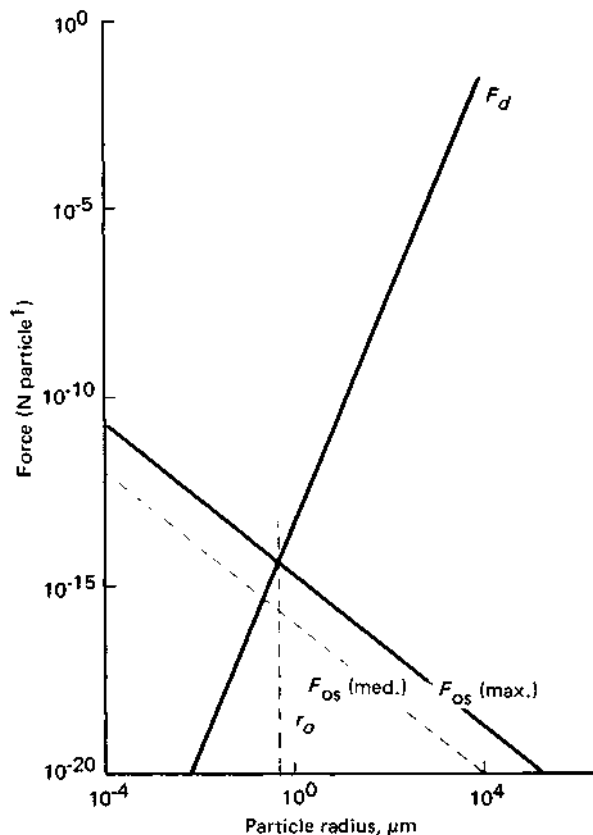


FIG. 20-31 Comparison of the dielectrophoretic (F_d) and osmotic (F_{os}) forces as functions of the particle size.

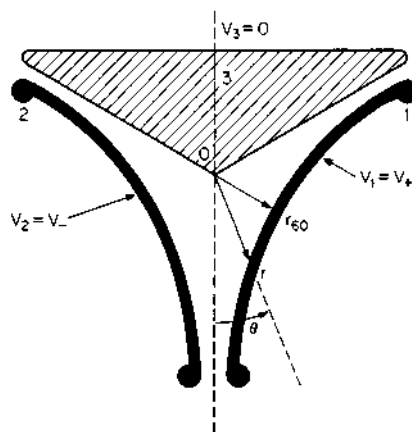


FIG. 20-32 A practical isomotive field geometry, showing r_{00} , the critical radius characterizing the isomotive electrodes. Electrode 3 is at ground potential, while electrodes 1 and 2 are at $V_1 = V_+$ and $V_2 = V_- = -V_+$ respectively. The inner faces of electrodes 1 and 2 follow $r = r_0 [\sin (3\theta/2)]^{-2/3}$, while electrode 3 forms an angle of 120° about the midline.

most desirable electrode shape is one in which the force is independent of position within the nonuniform field. This "isomotive" electrode system is shown in Fig. 20-32.

Applications of Dielectrophoresis Over the past 20 years the use of DEP has grown rapidly to a point at which it is in use for biological [Hughes, *Electrophoresis*, **23**, 2569–2582 (2002)], colloidal, and mineral materials studies and handling. The effects of nonuniform electric fields are used for handling particulate matter far more often than is usually recognized. This includes the removal of particulate matter by "electrofiltration," the sorting of mixtures, or its converse, the act of mixing, as well as the coalescence of suspensions. In addition to these effects involving the translational motions of particles, some systems apply the orientational or torsional forces available in nonuniform fields. One well-known example of the latter is the placing of "tip-up" grit on emery papers commercially. Xerography and many other imaging processes are examples of multibillion-dollar industries which depend upon DEP for their success.

A clear distinction between **electrophoresis** (field action on an object carrying excess free charges) and **dielectrophoresis** (field gradient action on neutral objects) must be borne in mind at all times.

A dielectrofilter [Lin and Benguigui, *Sep. Purif. Methods*, **10**(1), 53 (1981); Sisson et al., *Sep. Sci. Technol.*, **30**(7–9), 1421 (1995)] is a device which uses the action of an electric field to aid the filtration and removal of particulates from fluid media. A dielectrofilter can have a very obvious advantage over a mechanical filter in that it can remove particles which are much smaller than the flow channels in the filter. In contrast, the ideal mechanical filter must have all its passages smaller than the particles to be removed. The resultant flow resistance can be use-restrictive and energy-consuming unless a phenomenon such as dielectrofiltration is used.

Dielectrofiltration can (and often does) employ both electrophoresis and dielectrophoresis in its application. The precise physical process which dominates depends on a number of physical parameters of the system. Factors such as field intensity and frequency and the electrical conductivity and dielectric constants of the materials present determine this. Although these factors need constant attention for optimum operation of the dielectrofilter, this additional complication is often more than compensated for by the advantages of dielectrofiltration such as greater throughput and lesser sensitivity to viscosity problems, etc. To operate the dielectrofilter in the dominantly electrophoretic mode requires that excess free charges of one sign or the other reside on the particulate matter. The necessary charges can be those naturally present, as upon a charged sol; or they may need to be artificially implanted such as by passing the particles through a corona discharge.

Dielectrofiltration by the corona-charging, electrophoresis-dominated Cottrell technique is now widely used.

To operate the dielectrofilter (dominantly dielectrophoretic mode), on the other hand, one must avoid the presence of free charge on the particles. If the particles can become charged during the operation, a cycle of alternate charging and discharging in which the particles dash to and from the electrodes can occur. This is most likely to occur if static or very low frequency fields are used. For this reason, corona and like effects may be troublesome and need often to be minimized. To be sure, the DEP force is proportional to the field applied [actually to $\nabla(E)^2$], but fields which are too intense can produce such troublesome charge injection. A compromise for optimal operation is necessary between having $\nabla(E)^2$ so low that DEP forces are insufficient for dependable operation, on the one hand, and having E so high that troublesome discharges (e.g., coronalike) interfere with dependable operation of the dielectrofilter. In insulative media such as air or hydrocarbon liquids, for example, one might prefer to operate with fields in the range of, say, 10 to 10,000 V/cm. In more conductive media such as water, acetone, or alcohol, for example, one would usually prefer rather lower fields in the range of 0.01 to 100 V/cm. The higher field ranges cited might become unsuitable if conductive sharp asperities are present.

Another factor of importance in dielectrofiltration is the need to have the DEP effect firmly operative upon *all* portions of the fluid passing through. Oversight of this factor is a most common cause of incomplete dielectrofiltration. Good dielectrofilter design will emphasize this crucial point. To put this numerically, let us consider the essential field factor for DEP force, namely $\nabla(E_0)^2$. Near sharp points, e.g., E , the electric field varies with the radial distance r as $E \propto r^{-2}$, hence our DEP force factor will vary as $\nabla(E)^2 \propto r^{-5}$. In the neighborhood of sharp "line" sources such as at the edge of electrode plates, $E \propto r^{-1}$, hence, $\nabla(E)^2 \propto r^{-3}$. If, for instance, the distance is varied by a factor of 4 from the effective field source in these cases, the DEP force can be expected to weaken by a factor of 1024 or 64 respectively for the point source and the line source. The matter is even more keenly at issue when field-warping dielectrics (defined later) are used to effect maximal filtration. In this case the field-warping material is made to produce dipole fields as induced by the applied electric field. If we ask how the crucial factor, $\nabla(E)^2$, varies with distance away from such a dipole, we find that since the field E_d about a dipole varies approximately as r^{-3} , then $\nabla(E)^2$ can be expected to vary as r^{-7} . It then becomes critically important that the particles to be removed from the passing fluid do, indeed, pass very close to the surface of the field-warping material, or it will not be effectively handled. Clearly, it would be difficult to maintain successfully uniform dielectrofiltration treatment of fluid passing through such wildly variant regions. The problems can be minimized by ensuring that all the elements of the passing fluid go closely by such field sources in the dielectrofilter. In practice this is done by constructing the dielectrofilter from an assembly of highly comminuted electrodes or else by a set of relatively simple and widely spaced metallic electrodes between which is set an assembly of more or less finely divided solid dielectric material having a complex permittivity different from that of the fluid to be treated. The solid dielectric (fibers, spheres, chunks) serves to produce field nonuniformities or **field warpings** to which the particles to be filtered are to be attracted. In treating fluids of low dielectric constant such as air or hydrocarbon fluids, one sees field-warping materials such as sintered ceramic balls, glass-wool matting, open-mesh polyurethane foam, alumina, chunks, or BaTiO_3 particles.

An example of a practical dielectrofilter which uses both of the features described, namely, sharp electrodes *and* dielectric field-warping filler materials, is that described in Fig. 20-33 [H. J. Hall and R. F. Brown, *Lubric. Eng.*, **22**, 488 (1966)]. It is intended for use with hydraulic fluids, fuel oils, lubricating oils, transformer oils, lubricants, and various refinery streams. Performance data are cited in Fig. 20-34. It must be remarked that in the opinion of Hall and Brown the action of the dielectrofilter was "electrostatic" and due to free charge on the particles dispersed in the liquids. It is the present authors' opinion, however, that both electrophoresis and dielectrophoresis are operative here but that the dominant mechanism is that of DEP, in which neutral particles are polarized and attracted to the regions of highest field intensity.

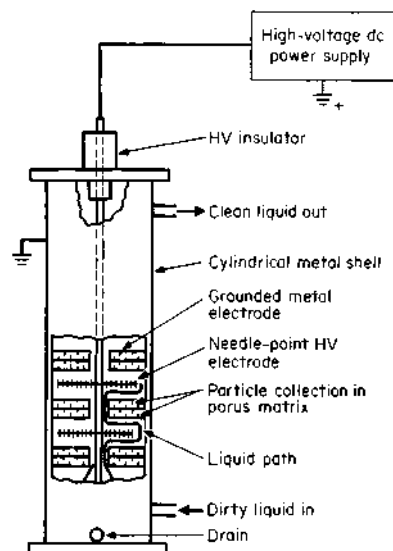


FIG. 20-33 Diagram illustrating the function of an electrostatic liquid cleaner.

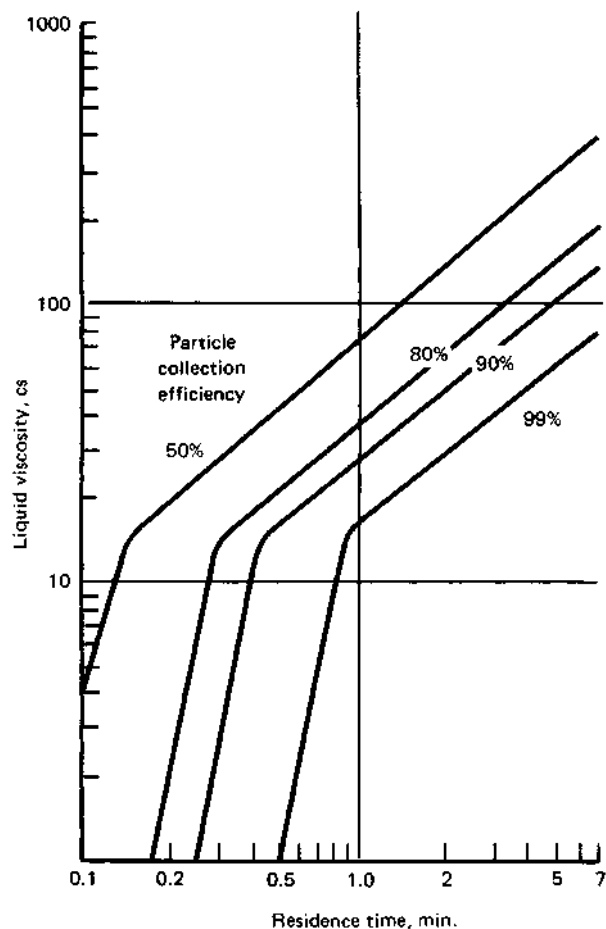


FIG. 20-34 Performance data for a typical high-efficiency electrostatic liquid cleaner.

A second commercial example of dielectric filtration is the Gulftronic® separator [G. R. Fritzsche, *Oil & Gas J.*, **75**, 73 (1977)] which was commercialized in the late 1970s by Gulf Science and Technology Company. Instead of using needle-point electrodes as shown in Fig. 20-33, the Gulftronic® separator relied on the use of a bed of glass beads to produce the field nonuniformities required for dielectric filtration. Either ac or dc electric fields could be used in this separator. The Gulftronic® separator has been used primarily to remove catalyst fines from FCC decant oils and has been reported to exhibit removal efficiencies in excess of 80 percent for this fine-particle separation problem.

Another example of the commercial use of DEP is in polymer clarification [A. N. Wennerberg, U.S. Patent 2,914,453, 1959; assignor to Standard Oil Co. (Indiana)]. Here, either ac or dc potentials were used while passing suspensions to be clarified through regions with an area-to-electrode-area ratio of 10:1 or 100:1 and with fields in the order of 10 kV/cm. Field warping by the presence of various solid dielectrics was observed to enhance filtration considerably, as expected for DEP. The filtration of molten or dissolved polymers to free them of objectionable quantities of catalyst residues, for example, was more effective if a solid dielectric material such as Attapulgus clay, silica gel, fuller's earth, alumina, or bauxite was present in the region between the electrodes. The effectiveness of percolation through such absorptive solids for removing color bodies is remarkably enhanced by the presence of an applied field. A given amount of clay is reported to remove from 4 to 10 times as much color as would be removed in the absence of DEP. Similar results are reported by Lin et al. [Lin, Yaniv, and Zimmels, *Proc. XIIIth Int. Miner. Process. Congr.*, Wrocław, Poland, 83-105 (1979)].

The instances cited were examples of the use of DEP to filter liquids. We now turn to the use of DEP to aid in dielectrofiltration of gases. Fielding et al. observe that the effectiveness of high-quality fiberglass air filters is dramatically improved by a factor of 10 or more by incorporating DEP in the operation. Extremely little current or power is required, and no detectable amounts of ozone or corona need result. The DEP force, once it has gathered the particles, continues to

act on the particles already sitting on the filter medium, thereby improving adhesion and minimizing blowoff.

The degree by which the DEP increases the effectiveness of gas filtration, or the dielectrophoretic augmentation factor (DAF), is definable. It is the ratio of the volumes of aerosol-laden gas which can be cleaned effectively by the filter with and without the voltage applied. For example, the application of 11 kV/cm gave a DAF of 30 for 1.0- μ m-diameter dioctyl phthalate particles in air, implying that the penetration of the glass filter is reduced thirtyfold by the application of a field of 1100 kV/m. Similar results were obtained by using "standard" fly ash supplied by the Air Pollution Control Office of the U.S. Environmental Protection Agency. The data obtained for several aerosols tested are shown in Table 20-14 and in Fig. 20-35. The relation $DAF = kV^2/v$ is observed to hold approximately for each aerosol. Here, the DEP augmentation factor DAF is observed to depend upon a constant K , a characteristic of the material, upon the square of the applied voltage, and upon the inverse of the volume flow rate v through the filter.

It is worth noting that in the case of the air filter described DEP serves as an augmenting rather than as an exclusive mechanism for the removal of particulate material. It is a unique feature of the dielectrophoretic gas filter that the DEP force is maximal when the particulates are at or on the fiber surface. This causes the deposits to be strongly retained by this particular filtration mechanism. It thus contrasts importantly with other types of gas filter in which the filtration mechanism no longer acts after the capture of the particle. In particular, in the case of the older electrostatic mechanisms involving only coulombic attraction, a simple charge alternation on the particle, such as caused by normal conduction, often evokes disruption of the filter operation because of particle repulsion from the contacting electrode. On the other hand, ordinary mechanical filtration depends upon the action of adventitious particle trapping or upon van der Waals forces, etc., to hold the particles. The high efficiency possible with electrofilters suggests their wider use.

TABLE 20-14 Dielectrophoretic Augmentation of Filtration of a Liquid Aerosol*

Air speed, cm/s	DAF at			
	2 kV	3.5 kV	5 kV	7 kV
0.3- μ m-diameter dioctyl phthalate aerosol				
3	8	19	95	330
6	3	13	39	120
9	3	11	28	100
15	2	6	13	42
20	2	5	9	27
28	2	4	6	14
39	2	3	4	9
50	1	2	3	6
1.0- μ m-diameter dioctyl phthalate aerosol				
3	30	110	300	1100
6	6	3	95	360
9	4	18	50	170
15	3	10	20	50
20	2	6	13	35
28	2	4	8	18
39	2	3	5	11
50	1	2	3	7
Fly-ash aerosol				
6	10	30	80	
10	8	30	80	
14	5	20	40	
20	4	10	30	70
35	3	7	10	20
45	1	2	6	
53	1	2	7	10

*Experimentally measured dielectrophoretic augmentation factor DAF as a function of air speed and applied voltage for a glass-fiber filter (HP-100, Farr Co.). Cf. Fielding, Thompson, Bogardus, and Clark, *Dielectrophoretic Filtration of Solid and Liquid Aerosol Particulates*, Prepr. 75-32.2, 68th ann. meet., Air Pollut. Control Assoc., Boston, June 1975.

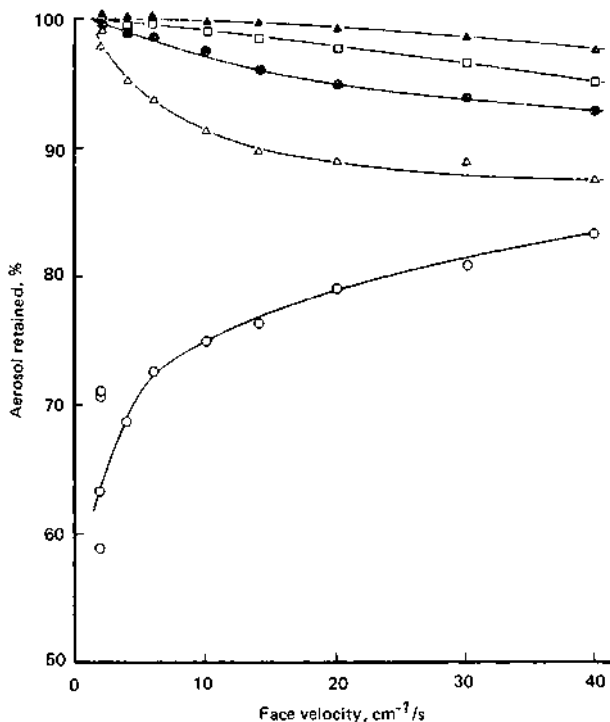


FIG. 20-35 Efficiency of an electrofilter as a function of gas flow rate at 5 different voltages. Experimental materials: 1- μ m aerosol of dioctyl phthalate; glass-fiber filter. Symbols: $\circ$, no voltage applied; Δ , 2 kV; $\bullet$, 3.5 kV; $\square$, 5 kV; $\blacktriangle$, 7 kV. (After Fielting et al., *Dielectrophoretic Filtration of Solid and Liquid Aerosol Particulates*, Prepr. 75-32.2, 68th ann. meet., Air Pollut. Control Assoc., Boston, June 1975.)

SURFACE-BASED SOLID-LIQUID SEPARATIONS INVOLVING A SECOND LIQUID PHASE

GENERAL REFERENCES: Fuerstenau, "Fine Particle Flotation," in Somasundaran (ed.), *Fine Particles Processing*, vol. 1, American Institute of Mining, Metallurgical, and Petroleum Engineers, New York, 1980. Henry, Prudich, and Lau, *Colloids Surf.*, **1**, 335 (1980). Henry, Prudich, and Vaidyanathan, *Sep. Purif. Methods*, **8**(2), 31 (1979). Jacques, Hovavongkura, and Henry, *Am. Inst. Chem. Eng. J.*, **25**(1), 160 (1979). Stratton-Crawley, "Oil Flotation: Two Liquid Flotation Techniques," in Somasundaran and Arbiter (eds.), *Beneficiation of Mineral Fines*, American Institute of Mining, Metallurgical, and Petroleum Engineers, New York, 1979.

Process Concept Three potential surface-based regimes of separation exist when a second, immiscible liquid phase is added to another, solids-containing liquid in order to effect the removal of solids. These regimes (Fig. 20-36) are:

1. Distribution of the solids into the bulk second liquid phase
2. Collection of the solids at the liquid-liquid interface
3. Bridging or clumping of the solids by the added fluid in order to form an agglomerate followed by settling or filtration

These separation techniques should find particular application in systems containing fine particles. The surface chemical differences involved among these separation regimes are only a matter of degree; i.e., all three regimes require the wetting of the solid by the second liquid phase. The addition of a surface-active agent is sometimes needed in order to achieve the required solids wettability. In spite of this similarity, applied processing (equipment configuration, operating conditions, etc.) can vary widely. Collection at the interface would normally be treated as a flotation process (see also "Adsorptive-Bubble Separation Methods" in Sec. 20), distribution to the bulk liquid as a liquid-liquid extraction analog, and particle bridging as a settling (sedimentation) or filtration process.

Even though surface-property-based liquid-solid-liquid separation techniques have yet to be widely used in significant industrial applications, several studies which demonstrate their effectiveness have appeared in literature.

Albertsson (*Partition of Cell Particles and Macromolecules*, 3d ed., Wiley, New York, 1986) has extensively used particle distribution to

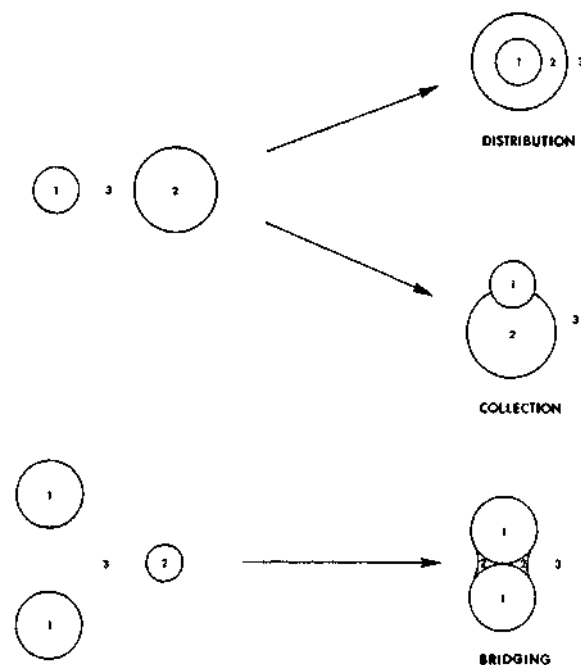


FIG. 20-36 Regimes of separation in a liquid-solid-liquid system. Phase 1 = particle; phase 2 = liquid (dispersed); phase 3 = liquid (continuous).

TABLE 20-15 Separations of Particles between Two Phases

System	Top phase	Bottom phase
Polyethylene glycol salt	Polystyrene	All others
PEG	Algae	All others
Dextran; 20,000 MW	Red cells	All others
PEG	Red cells	All others
Dextran; 200,000 MW	Cellulose particles	Starch
Methyl cellulose	Cellulose particles	Starch
Dextran	Cellulose particles	Starch

fractionate mixtures of biological products. (See also Sec. 15.) In order to demonstrate the versatility of particle distribution, he has cited the example shown in Table 20-15. The feed mixture consisted of polystyrene particles, red blood cells, starch, and cellulose. Liquid-liquid particle distribution has also been studied by using mineral-matter particles (average diameter = 5.5 μm) extracted from a coal liquid as the solid in a xylene-water system [Prudich and Henry, *Am. Inst. Chem. Eng. J.*, **24**(5), 788 (1978)]. By using surface-active agents in order to enhance the water wettability of the solid particles, recoveries of better than 95 percent of the particles to the water phase were observed. All particles remained in the xylene when no surfactant was added.

Particle collection at a liquid-liquid interface is a particularly favorable separation process when applied to fine-particle systems. Advantages of this type of processing include:

- Decreased liquid-liquid interfacial tension (when compared with a gas-liquid system) results in higher liquid-liquid interfacial areas, which favor solid-particle droplet collisions.
- Liquid-solid interactions due to long-range intermolecular forces are much larger than are gas-solid interactions. This means that it is easier to collect fine particles at a liquid-liquid interface than at a gas-liquid interface.
- The increased momentum of liquid droplets (when compared with gas) should favor solid-particle collection.

Fuerstenau [Lai and Fuerstenau, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **241**, 549 (1968); Raghavan and Fuerstenau, *Am. Inst. Chem. Eng. Symp. Ser.*, **71**(150), 59 (1975)] has studied this process with respect to the removal of alumina particles (0.1 μm) and hematite particles (0.2 μm) from an aqueous solution by using isooctane. The use of isooctane as the collecting phase for the hematite particles resulted in an increase in particle recovery of about 50 percent over that measured when air was used as the collecting phase under the same conditions. The effect of the wettability of the solid particles (as measured by the three-phase contact angle) on the recovery of hematite in the water-isooctane system is shown in Fig. 20-37. This behavior is typical of particle collection. Particle collection at an oil-water interface has also been studied with respect to particle removal from a coal liquid. Particle removals averaging about 80 percent have been observed when water is used as the collecting phase (Lau, master's thesis, West Virginia University, 1979). Surfactant addition was necessary in order to control the wettability of the solids.

Particle bridging has been chiefly investigated with respect to spherical agglomeration. Spherical agglomeration involves the collecting or transferring of the fine particles from suspension in a liquid phase into spherical aggregates held together by a second liquid phase. The aggregates are then removed from the slurry by filtration or settling. Like the other liquid-solid-liquid separation techniques, the solid must be wet by the second liquid phase. The spherical agglomeration process has resulted in the development of a pilot unit called the Shell Pelletizing Separator [Zuiderweg and Van Lookeren Campagne, *Chem. Eng. (London)*, **220**, CE223 (1968)].

The ability to determine in advance which of the separation regimes is most advantageous for a given liquid-solid-liquid system would be desirable. No set of criteria with which to make this determination presently exists. Work has been done with respect to the

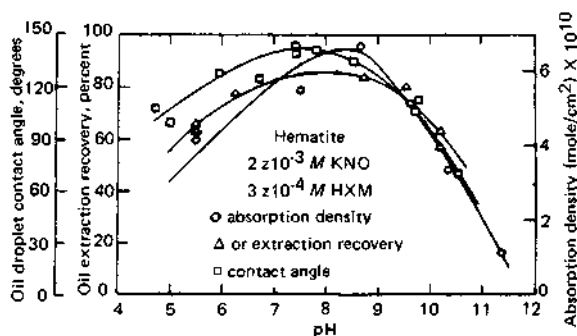


FIG. 20-37 The variation of adsorption density, oil-droplet contact angle, and oil-extraction recovery of hematite as a function of pH. To convert gram-moles per square centimeter to pound-moles per square foot, multiply by 2.048. [From Raghavan and Fuerstenau, *Am. Inst. Chem. Eng. Symp. Ser.*, **71**(150), 59 (1975).]

identification of system parameters which make these processes technically feasible. The results of these studies can be used to guide the selection of the second liquid phase as well as to suggest approximate operating conditions (dispersed-liquid droplet size, degree and type of mixing, surface-active-chemical addition, etc.).

Theory Theoretical analyses of spherical particles suspended in a planar liquid-liquid interface have appeared in literature for some time, the most commonly presented forms being those of a free energy and/or force balance made in the absence of all external body forces. These analyses are generally used to define the boundary criteria for the shift between the collection and distribution regimes, the bridging regime not being considered. This type of analysis shows that for a spherical particle possessing a three-phase contact angle between 0 and 180°, as measured through the receiving or collecting phase, collection at the interface is favored over residence in either bulk phase. These equations are summarized, using a derivation of Young's equation, as

$$\frac{\gamma_{s2} - \gamma_{s1}}{\gamma_{12}} > 1 \text{ particles wet to phase 1} \quad (20-35)$$

$$\frac{\gamma_{s2} - \gamma_{s1}}{\gamma_{12}} < -1 \text{ particles wet to phase 2} \quad (20-36)$$

$$\left| \frac{\gamma_{s2} - \gamma_{s1}}{\gamma_{12}} \right| \leq 1 \text{ particle at interface} \quad (20-37)$$

where γ_{ij} is the surface tension between phases i and j , N/m (dyn/cm); s indicates the solid phase; and subscripts 1 and 2 indicate the two liquid phases.

Several additional studies [Winitzer, *Sep. Sci.*, **8**(1), 45 (1973); *ibid.*, **8**(6), 647 (1973); Maru, Wasan, and Kintner, *Chem. Eng. Sci.*, **26**, 1615 (1971); and Rapacchietta and Neumann, *J. Colloid Interface Sci.*, **59**(3), 555 (1977)] which include body forces such as gravitational acceleration and buoyancy have been made. A typical example of a force balance describing such a system (Fig. 20-38) is summarized in Eq. (20-38):

$$[(\gamma_{s1} - \gamma_{s2}) \cos \delta + \gamma_{12} \cos B]L = g[V_{\text{total}}\rho_s - V_1\rho_1 - V_2\rho_2] \quad (20-38)$$

where V_i is the volume of the particle in fluid phase 1, V_2 is the volume in fluid phase 2, L is the particle circumference at the interface between the two liquid phases, ρ_i is the density of phase i , and g is the gravitational constant. The left-hand side of the equation represents the surface forces acting on the solid particle, while the right-hand side includes the gravitational and buoyancy forces. This example illustrates the fact that body forces can have a significant effect on system behavior. The solid-particle size as well as the densities of the solid and both liquid phases are introduced as important system parameters.

A study has also been performed for particle distribution for cases in which the radii of curvature of the solid and the liquid-liquid interface

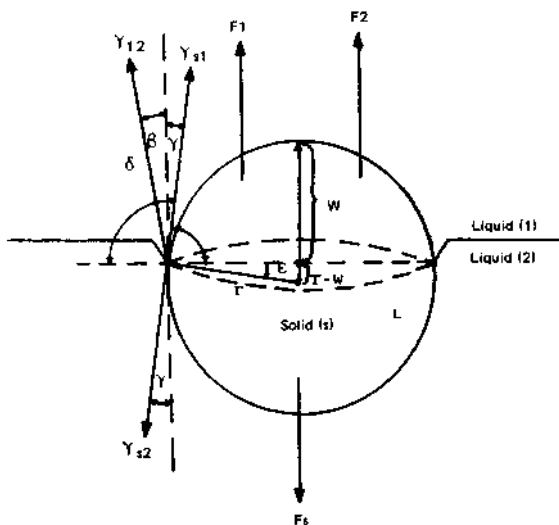


FIG. 20-38 Solid sphere suspended at the liquid-liquid interface. F_1 and F_2 are buoyancy forces; F_g is gravity. [From Winitzer, *Sep. Sci.*, **8**(1), 45 (1973).]

are of the same order of magnitude [Jacques, Hovarrongkura, and Henry, *Am. Inst. Chem. Eng. J.*, **25**(1), 160 (1979)]. Differences between the final and initial surface free energies are used to analyze this system. Body forces are neglected. Results (Fig. 20-39) demonstrate that n , the ratio of the particle radius to the liquid-liquid-interface radius, is an important system parameter. Distribution of the particle from one phase to the other is favored over continued residence in the original phase when the free-energy difference is negative. For a solid particle of a given size, these results show that as the second-phase droplet size decreases, the contact angle required in order to effect distribution decreases (the required wettability of the solid by the second phase increases). The case of particle collection at a curved liquid-liquid interface has also been studied in a similar manner [Smith and Van de Ven, *Colloids Surf.*, **2**, 387 (1981)]. This study shows that collection is preferred over distribution for any n in systems without external body forces when the contact angle lies between 0 and 180°.

While thermodynamic-stability studies can be valuable in evaluating the technical feasibility of a process, they are presently inadequate in determining which separation regime will dominate a particular liquid-solid-liquid system. These analyses ignore important processing phenomena such as the mechanism of encounter of the dispersed-phase liquid with the solid particles, the strength of particle attachment, and the mixing-energy input necessary to effect the separation. No models of good predictive value which take all these variables into account have yet been offered. Until the effects of these and other system variables can be adequately understood, quantified, and combined into such a predictive model, no a priori method of performance prediction will be possible.

ADSORPTIVE-BUBBLE SEPARATION METHODS

GENERAL REFERENCES: Lemlich (ed.), *Adsorptive Bubble Separation Techniques*, Academic, New York, 1972. Carleson, "Adsorptive Bubble Separation Processes" in Scaemhorn and Harwell (eds.), *Surfactant-Based Separation Processes*, Marcel Dekker, New York, 1989.

Principle The adsorptive-bubble separation methods, or adsorbable methods for short [Lemlich, *Chem. Eng.*, **73**(21), 7 (1966)], are based on the selective adsorption or attachment of material on the surfaces of gas bubbles passing through a solution or suspension. In most of the methods, the bubbles rise to form a foam or froth which carries the material off overhead. Thus the material (desirable or undesirable) is removed from the liquid, and not vice versa as in, say,

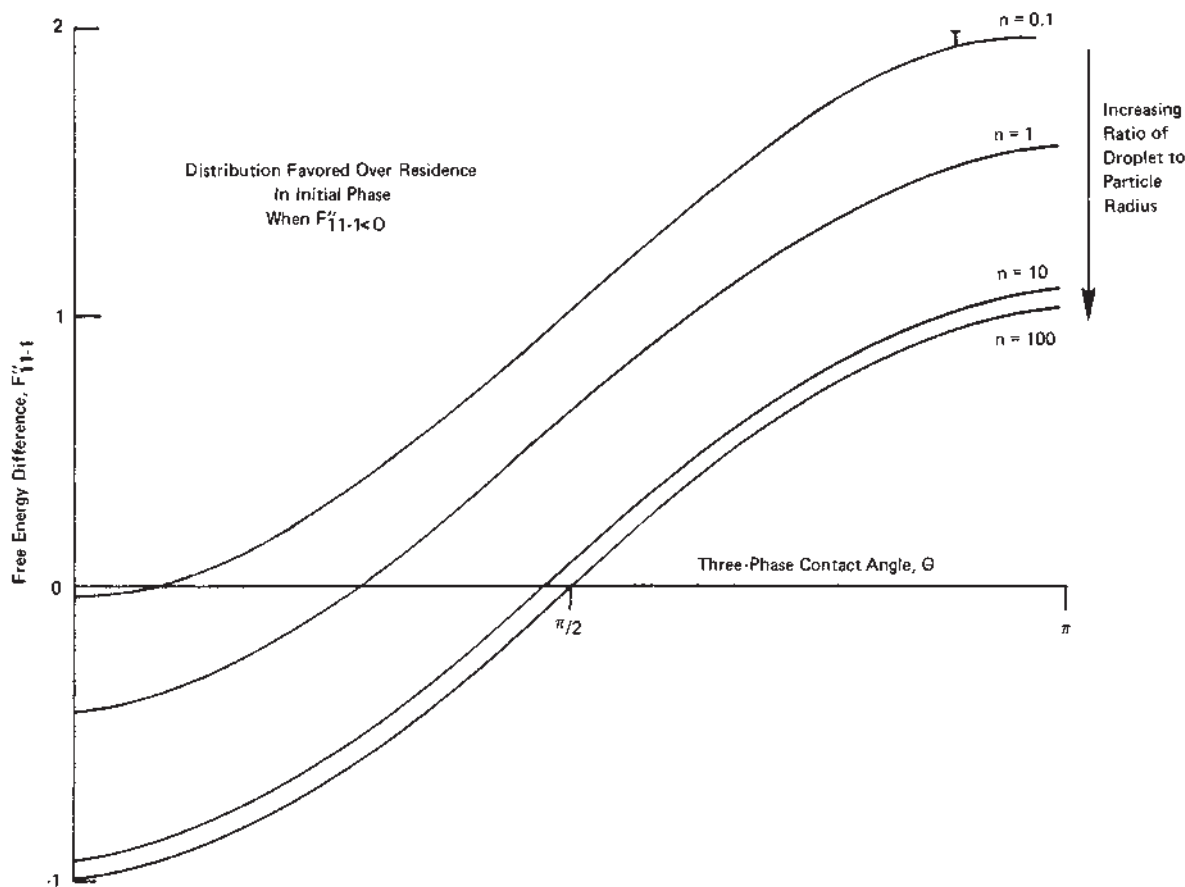


FIG. 20-39 Normalized free-energy difference between distributed (II) and nondistributed (I) states of the solid particles versus three-phase contact angle (collection at the interface is not considered). A negative free-energy difference implies that the distributed state is preferred over the nondistributed state. Note especially the significant effect of n , the ratio of the liquid droplet to solid-particle radius. [From Jacques, Hovarongkura, and Henry, *Am. Inst. Chem. Eng. J.*, **25**(1), 160 (1979).]

filtration. Accordingly, the foaming methods appear to be particularly (although not exclusively) suited to the removal of small amounts of material from large volumes of liquid.

For any adsorbable method, if the material to be removed (termed the **colligend**) is not itself surface-active, a suitable surfactant (termed the **collector**) may be added to unite with it and attach or adsorb it to the bubble surface so that it may be removed (Sebba, *Ion Flotation*, Elsevier, New York, 1962). The union between colligend and collector may be by chelation or other complex formation. Alternatively, a charged colligend may be removed through its attraction toward a collector of opposite charge.

Definitions and Classification Figure 20-40 outlines the most widely accepted classification of the various adsorbable methods [Karger, Grieves, Lemlich, Rubin, and Sebba, *Sep. Sci.*, **2**, 401 (1967)]. It is based largely on actual usage of the terms by various workers, and so the definitions include some unavoidable inconsistencies and overlap.

Among the methods of foam separation, **foam fractionation** usually implies the removal of dissolved (or sometimes colloidal) material. The overflowing foam, after collapse, is called the *foamate*. The solid lines of Fig. 20-41 illustrate simple continuous foam fractionation. (Batch operation would be represented by omitting the feed and bottoms streams.)

On the other hand, **flotation** usually implies the removal of solid particulate material. Most important under the latter category is **ore flotation**.

Also under the category of flotation are to be found **macroflotation**, which is the removal of macroscopic particles; **microflotation** (also called **colloid flotation**), which is the removal of microscopic particles, particularly colloids or microorganisms [Dognon and Dumontet, *Comptes Rendus*, **135**, 884 (1941)]; **molecular flotation**, which is the removal of surface-inactive molecules through the use of a collector (surfactant) which yields an insoluble product; **ion flotation**, which is the removal of surface-inactive ions via a collector which yields an insoluble product, especially a removable scum [Sebba, *Nature*, **184**, 1062 (1959)]; **adsorbing colloid flotation**, which is the removal of dissolved material in piggyback fashion by adsorption on colloidal particles; and **precipitate flotation**, in which a precipitate is removed by a collector which is not the precipitating agent [Baarson and Ray, "Precipitate Flotation," in Wadsworth and Davis (eds.), *Unit Processes in Hydrometallurgy*, Gordon and Breach, New York, 1964, p. 656]. The last definition has been narrowed to precipitate flotation of the first kind, the second kind requiring no separate collector at all [Mahne and Pinfold, *J. Appl. Chem.*, **18**, 52 (1968)].

A separation can sometimes be obtained even in the absence of any foam (or any floated floc or other surrogate). In **bubble fractionation** this is achieved simply by lengthening the bubbled pool to form a vertical column [Dorman and Lemlich, *Nature*, **207**, 145 (1965)]. The ascending bubbles then deposit their adsorbed or attached material at the top of the pool as they exit. This results in a concentration gradient which can serve as a basis for separation. Bubble fractionation can operate either alone or as a booster section below a foam

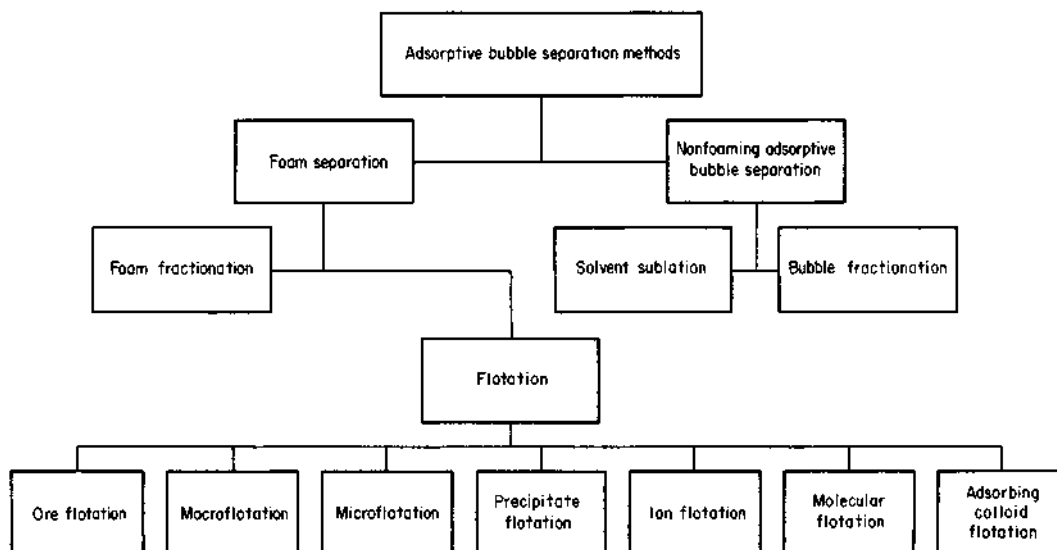


FIG. 20-40 Classification for the adsorptive-bubble separation methods.

fractionator, perhaps to raise the concentration up to the foaming threshold.

In **solvent sublation** an immiscible liquid is placed atop the main liquid to trap the material deposited by the bubbles as they exit (Sebba, *Ion Flotation*, Elsevier, New York, 1962). The upper liquid should dissolve or at least wet the material. With appropriate selectivity, the separation so achieved can sometimes be much greater than that with bubble fractionation alone.

The droplet analogs to the adsorbable methods have been termed the **adsoplet methods** (from adsorptive droplet separation methods) [Lemlich, "Adsorptive Bubble Separation Methods," *Ind. Eng. Chem.*, **60**(10), 16 (1968)]. They are omitted from Fig. 20-40, since they involve adsorption or attachment at liquid-liquid interfaces. Among them are **emulsion fractionation** [Eldib, "Foam and Emulsion Fractionation," in Kobe and McKetta (eds.), *Advances in Petroleum Chemistry and Refining*, vol. 7, Interscience, New York, 1963, p. 66], which is the analog of foam fractionation; and **droplet fractionation** [Lemlich, loc.

cit.; and Strain, *J. Phys. Chem.*, **57**, 638 (1953)], which is the analog of bubble fractionation. Similarly, the old beneficiation operation called bulk oil flotation (Gaudin, *Flotation*, 2d ed., McGraw-Hill, New York, 1957) is the analog of modern ore flotation. By and large, the adsoplet methods have not attracted the attention accorded to the adsorbable methods.

Of all the adsorbable methods, foam fractionation is the one for which chemical engineering theory is the most advanced. Fortunately, some of this theory also applies to other adsorbable methods.

Adsorption The separation achieved depends in part on the selectivity of adsorption at the bubble surface. At equilibrium, the adsorption of dissolved material follows the Gibbs equation (Gibbs, *Collected Works*, Longmans Green, New York, 1928).

$$d\gamma = -RT\Sigma\Gamma_i d \ln a_i \quad (20-39)$$

Γ_i is the surface excess (Davies and Rideal, *Interfacial Phenomena*, 2d ed., Academic, New York, 1963). For most purposes, it is sufficient to view Γ_i as the concentration of adsorbed component i at the surface in units of, say (g-mol)/cm². R is the gas constant, T is the absolute temperature, γ is the surface tension, and a_i is the activity of component i . The minus sign shows that material which concentrates at the surface generally lowers the surface tension, and vice versa. This can sometimes be a guide in determining preliminarily what materials can be separated.

When applied to a nonionic surfactant in pure water at concentrations below the critical micelle concentration, Eq. (20-39) simplifies into Eq. (20-40)

$$\Gamma_s = -\frac{1}{RT} \frac{d\gamma}{d \ln C_s} \quad (20-40)$$

C is the concentration in the bulk, and subscript s refers to the surfactant. Under some conditions, Eq. (20-40) may apply to an ionic surfactant as well (Lemlich, loc. cit.).

The major surfactant in the foam may usually be considered to be present at the bubble surfaces in the form of an adsorbed monolayer with a substantially constant Γ_s , often of the order of 3×10^{-10} (g-mol)/cm², for a molecular weight of several hundred. On the other hand, trace materials follow the linear-adsorption isotherm $\Gamma_i = K_i C_i$ if their concentration is low enough. For a wider range of concentration a Langmuir or other type of isotherm may be applicable (Davies and Rideal, loc. cit.).

Factors Affecting Adsorption K_i for a colligend can be adversely affected (reduced) through an insufficiency of collector. It can also be reduced through an excess of collector, which competes

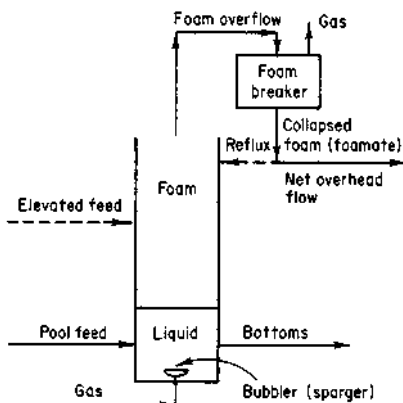


FIG. 20-41 Four alternative modes of continuous-flow operation with a foam-fractionation column: (1) The simple mode is illustrated by the solid lines. (2) Enriching operation employs the dashed reflux line. (3) In stripping operation, the elevated dashed feed line to the foam replaces the solid feed line to the pool. (4) For combined operation, reflux and elevated feed to the foam are both employed.

for the available surface against the collector-colligend complex [Schnepf, Gaden, Mirocznik, and Schonfeld, *Chem. Eng. Prog.*, **55**(5), 42 (1959)].

Excess collector can also reduce the separation by forming micelles in the bulk which adsorb some of the colligend, thus keeping it from the surface. This effect of the micelles on K_i for the colligend is given theoretically [Lemlich, "Principles of Foam Fractionation," in Perry (ed.), *Progress in Separation and Purification*, vol. 1, Interscience, New York, 1968, chap. 1] by Eq. (20-41) [Lemlich (ed.), *Adsorptive Bubble Separation Techniques*, Academic, New York, 1972] if Γ_s is constant when $C_s > C_{sc}$:

$$\frac{1}{K_2} = \frac{1}{K_1} + \frac{C_s - C_{sc}}{\Gamma_s E} \quad (20-41)$$

K_1 is K_i just below the collector's critical micelle concentration, C_{sc} . K_2 is K_i at some higher collector concentration, C_s . E is the relative effectiveness, in adsorbing colligend, of surface collector versus micellar collector. Generally, $E > 1$. Γ_s is the surface excess of collector. More about each K is available [Lemlich, "Adsorbable Methods," in Li (ed.), *Recent Developments in Separation Science*, vol. 1, CRC Press, Cleveland, 1972, pp. 113-127; Jashnani and Lemlich, *Ind. Eng. Chem. Process Des. Dev.*, **12**, 312 (1973)].

The controlling effect of various ions can be expressed in terms of thermodynamic equilibria [Karger and DeVivo, *Sep. Sci.*, **3**, 393 (1968)]. Similarities with ion exchange have been noted. The selectivity of counterion adsorption increases with ionic charge and decreases with hydration number [Jorne and Rubin, *Sep. Sci.*, **4**, 313 (1969); and Kato and Nakamori, *J. Chem. Eng. Japan*, **9**, 378 (1976)]. By analogy with other separation processes, the relative distribution in multicomponent systems can be analyzed in terms of a selectivity coefficient $\alpha_{mn} = \Gamma_m C_n / \Gamma_n C_m$ [Rubin and Jorne, *Ind. Eng. Chem. Fundam.*, **8**, 474 (1969); *J. Colloid Interface Sci.*, **33**, 208 (1970)].

Operation in the Simple Mode If there is no concentration gradient within the liquid pool and if there is no coalescence within the rising foam, then the operation shown by the solid lines of Fig. 20-41 is truly in the simple mode, i.e., a single theoretical stage of separation. Equations (20-42) and (20-43) will then apply to the steady-flow operation.

$$C_Q = C_W + (GS\Gamma_w/Q) \quad (20-42)$$

$$C_W = C_F - (GS\Gamma_w/F) \quad (20-43)$$

C_F , C_W , and C_Q are the concentrations of the substance in question (which may be a colligend or a surfactant) in the feed stream, bottoms stream, and foamate (collapsed foam), respectively. G , F , and Q are the volumetric flow rates of gas, feed, and foamate, respectively. Γ_w is the surface excess in equilibrium with C_W . S is the surface-to-volume ratio for a bubble. For a spherical bubble, $S = 6/d$, where d is the bubble diameter. For variation in bubble sizes, d should be taken as $\Sigma n_i d_i^3 / \Sigma n_i d_i^2$, where n_i is the number of bubbles with diameter d_i in a representative region of foam.

Finding Γ Either Eq. (20-42) or Eq. (20-43) can be used to find the surface excess indirectly from experimental measurements. To assure a close approach to operation as a single theoretical stage, coalescence in the rising foam should be minimized by maintaining a proper gas rate and a low foam height [Brunner and Lemlich, *Ind. Eng. Chem. Fundam.*, **2**, 297 (1963)]. These precautions apply particularly with Eq. (20-42).

For laboratory purposes it is sometimes convenient to recycle the foamate directly to the pool in a manner analogous to an equilibrium still. This eliminates the feed and bottoms streams and makes for a more reliable approach to steady-state operation. However, this recycling may not be advisable for colligend measurements in the presence of slowly dissociating collector micelles.

To avoid spurious effects in the laboratory, it is advisable to employ a prehumidified chemically inert gas.

Bubble Sizes Subject to certain errors (de Vries, *Foam Stability*, Rubber-Stichting, Delft, 1957), foam bubble diameters can be measured photographically. Some of these errors can be minimized by taking pains to generate bubbles of fairly uniform size, say, by using a bubbler with identical orifices or by just using a bubbler with a single orifice (gas rate permitting). Otherwise, a correction for planar statistical sampling bias in

the foam should be incorporated with actual diameters [de Vries, op. cit.] or truncated diameters [Lemlich, *Chem. Eng. Commun.*, **16**, 153 (1982)]. Also, size segregation can reduce mean mural bubble diameter by roughly half the standard deviation [Cheng and Lemlich, *Ind. Eng. Chem. Fundam.*, **22**, 105 (1983)]. Bubble diameters can also be measured in the liquid pool, either photographically or indirectly via measurement of the gas flow rate and stroboscopic determination of bubble frequency [Leonard and Lemlich, *Am. Inst. Chem. Eng. J.*, **11**, 25 (1965)].

Bubble sizes at formation generally increase with surface tension and orifice diameter. Prediction of sizes in swarms from multiple orifices is difficult. In aqueous solutions of low surface tension, bubble diameters of the order of 1 mm are common. Bubbles produced by the more complicated techniques of pressure flotation or vacuum flotation are usually smaller, with diameters of the order of 0.1 mm or less.

Enriching and Stripping Unlike truly simple foam fractionation without significant changes in bubble diameter, coalescence in a foam column destroys some bubble surface and so releases adsorbed material to trickle down through the rising foam. This downflow constitutes internal reflux, which enriches the rising foam by countercurrent action. The result is a richer foamate, i.e., higher C_Q than that obtainable from the single theoretical stage of the corresponding simple mode. Significant coalescence is often present in rising foam, but the effect on bubble diameter and enrichment is frequently overlooked.

External reflux can be furnished by returning some of the externally broken foam to the top of the column. The concentrating effect of reflux, even for a substance which saturates the surface, has been verified [Lemlich and Lavi, *Science*, **134**, 191 (1961)].

Introducing the feed into the foam some distance above the pool makes for stripping operation. The resulting countercurrent flow in the foam further purifies the bottoms, i.e., lowers C_W .

Enriching, stripping, and combined operations are shown in Fig. 20-41.

Foam-Column Theory The counterflowing streams within the foam are viewed as consisting *effectively* of a descending stream of interstitial liquid (equal to zero for the simple mode) and an ascending stream of interstitial liquid plus bubble surface. (By considering this ascending surface as analogous to a vapor, the overall operation becomes analogous in a way to distillation *with entrainment*.)

An effective concentration $\bar{C}$ in the ascending stream at any level in the column is defined by Eq. (20-44):

$$\bar{C} = C + (GS\Gamma/U) \quad (20-44)$$

where U is the volumetric rate of interstitial liquid upflow, C is the concentration in this ascending liquid at that level, and Γ is the surface excess in equilibrium with C . Any effect of micelles should be included.

For simplicity, U can usually be equated to Q . An effective equilibrium curve can now be plotted from Eq. (20-44) in terms of $\bar{C}$ (or rather C°) versus C .

Operating lines can be found in the usual way from material balances. The slope of each such line is $\Delta C / \Delta C = L/U$, where L is the downflow rate in the particular column section and C is now the concentration in the descending stream.

The number of theoretical stages can then be found in one of the usual ways. Figure 20-42 illustrates a graphical calculation for a stripper.

Alternatively, the number of transfer units (NTU) in the foam based on, say, the ascending stream can be found from Eq. (20-45):

$$NTU = \int_{C_w}^{C_Q} \frac{d\bar{C}}{C^\circ - C} \quad (20-45)$$

C° is related to C by the effective equilibrium curve, and C_w° is similarly related to C_w . $\bar{C}$ is related to C by the operating line.

To illustrate this integration analytically, Eq. (20-45) becomes Eq. (20-46) for the case of a stripping column removing a colligend which is subject to the linear-equilibrium isotherm $\Gamma = KC$.

$$NTU = \frac{F}{GSK - W} \ln \frac{FW + F(GSK - W)C_F/C_W}{GSK(GSK + F - W)} \quad (20-46)$$

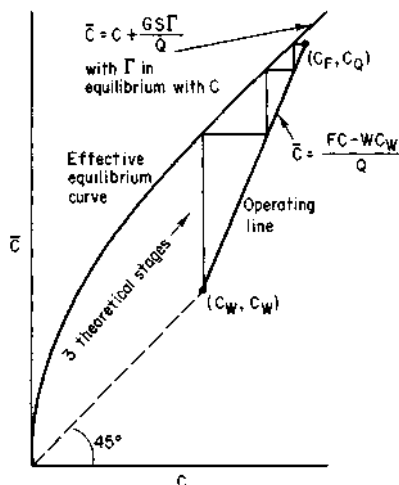


FIG. 20-42 Graphical determination of theoretical stages for a foam-fractionation stripping column.

As another illustration, Eq. (20-45) becomes Eq. (20-47) for an enriching column which is concentrating a surfactant with a constant Γ :

$$NTU = R \ln \frac{RG\Gamma(F - D)}{(R + 1)G\Gamma(F - D) - (R + 1)FD(C_D - C_F)} \quad (20-47)$$

Unless the liquid pool is purposely lengthened vertically in order to give additional separation via bubble fractionation, it is usually taken to represent one theoretical stage. A bubbler submergence of 30 cm or so is usually ample for a solute with a molecular weight that does not exceed several hundred.

In a colligend stripper, it may be necessary to add some collector to the pool as well as the feed because the collector is also stripped off.

Limiting Equations If the height of a foam-fractionation column is increased sufficiently, a concentration pinch will develop between the counterflowing interstitial streams (Brunner and Lemlich, loc. cit.). For an enricher, the separation attained will then approach the predictions of Eq. (20-48) and, interestingly enough, Eq. (20-43).

$$C_D = C_W + (GS\Gamma_W/D) \quad (20-48)$$

D is the volumetric rate at which net foamate (net overhead liquid product) is withdrawn. $D = Q/(R + 1)$. The concentration in the net foamate is C_D . In the usual case of total foam breakage (no dephlegmation), $C_D = C_Q$.

If the tall column is a stripper, the separation will approach that of Eqs. (20-49) and (20-50):

$$C_Q = C_F + (GS\Gamma_F/Q) \quad (20-49)$$

$$C_W = C_F - (GS\Gamma_F/W) \quad (20-50)$$

For a sufficiently tall combined column, the separation will approach that of Eqs. (20-51) and (20-50):

$$C_D = C_F + (GS\Gamma_F/D) \quad (20-51)$$

The formation of micelles in the foam breaker does not affect the limiting equations because of the theoretically unlimited opportunity in a sufficiently tall column for their transfer from the reflux to the ascending stream [Lemlich, "Principles of Foam Fractionation," in Perry (ed.), *Progress in Separation and Purification*, vol. 1, Interscience, New York, 1968, chap. 1].

In practice, the performance of a well-operated foam column several feet tall may actually approximate the limiting equations, provided there is little channeling in the foam and provided that reflux is either absent or is present at a low ratio.

Column Operation To assure intimate contact between the counterflowing interstitial streams, the volume fraction of liquid in the foam should be kept below about 10 percent—and the lower the better. Also, rather uniform bubble sizes are desirable. The foam bubbles will thus pack together as blunted polyhedra rather than as spheres, and the suction in the capillaries (Plateau borders) so formed will promote good liquid distribution and contact. To allow for this desirable deviation from sphericity, $S = 6.3/d$ in the equations for enriching, stripping, and combined column operation [Lemlich, *Chem. Eng.*, **75**(27), 95 (1968); **76**(6), 5 (1969)]. Diameter d still refers to the sphere.

Visible channeling or significant deviations from plug flow of the foam should be avoided, if necessary by widening the column or lowering the gas and/or liquid rates. The superficial gas velocity should probably not exceed 1 or 2 cm/s. Under proper conditions, HTU values of several cm have been reported [Hastings, Ph.D. dissertation, Michigan State University, East Lansing, 1967; and Jashnani and Lemlich, *Ind. Eng. Chem. Process Des. Dev.*, **12**, 312 (1973)]. The foam column height equals $NTU \times HTU$.

For columns that are wider than several centimeters, reflux and feed distributors should be used, particularly for wet foam [Haas and Johnson, *Am. Inst. Chem. Eng. J.*, **11**, 319 (1965)]. Liquid content within the foam can be monitored conductometrically [Chang and Lemlich, *J. Colloid Interface Sci.*, **73**, 224 (1980)]. See Fig. 20-43. Theoretically, as the limit $\mathcal{D} = K = 0$ is very closely approached, $\mathcal{D} = 3K$ [Lemlich, *J. Colloid Interface Sci.*, **64**, 107 (1978)].

Wet foam can be handled in a bubble-cap column (Wace and Banfield, *Chem. Process Eng.*, **47**(10), 70 (1966)] or in a sieve plate column [Aguayo and Lemlich, *Ind. Eng. Chem. Process Des. Dev.*, **13**, 153 (1974)]. Alternatively, individual short columns can be connected in countercurrent array [Banfield, Newson, and Alder, *Am. Inst. Chem. Eng. Symp. Ser.*, **1**, 3 (1965); Leonard and Blacyki, *Ind. Eng. Chem. Process Des. Dev.*, **17**, 358 (1978)].

A high gas rate can be used to achieve maximum throughput in the simple mode (Wace, Alder, and Banfield, AERE-R5920, U.K. Atomic Energy Authority, 1968) because channeling is not a factor in that mode. A horizontal drainage section can be used overhead [Haas and Johnson, *Ind. Eng. Chem. Fundam.*, **6**, 225 (1967)]. The highly mobile dispersion produced by a very high gas rate is not a true foam but is

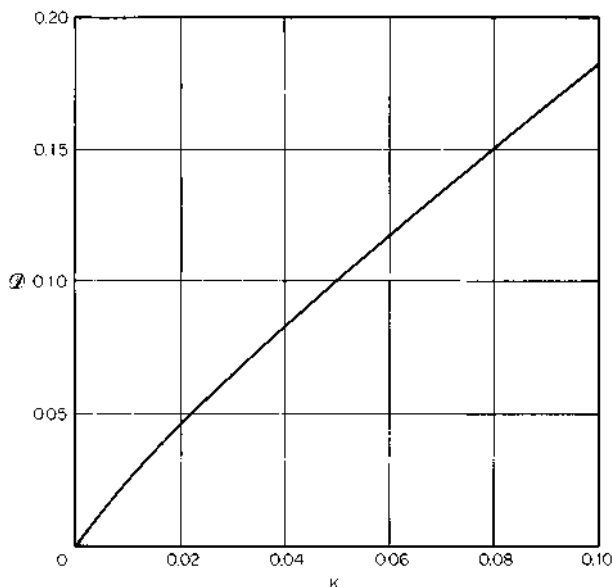


FIG. 20-43 Empirical relationship between $\mathcal{D}$, the volumetric fraction of liquid in common polydisperse foam, and K , the electrical conductivity of the foam divided by the electrical conductivity of the liquid. [Chang and Lemlich, *J. Colloid Interface Sci.*, **73**, 224 (1980).]

rather a so-called **gas emulsion** [Bikerman, *Ind. Eng. Chem.*, **57**(1), 56 (1965)].

A very low gas rate in a column several feet tall with internal reflux can sometimes be used to effect difficult multicomponent separations in batch operation [Lemlich, "Principles of Foam Fractionation," in Perry (ed.), *Progress in Separation and Purification*, vol. 1, Interscience, New York, 1968, chap. 1].

The same end may be achieved by continuous operation at total external reflux with a small U bend in the reflux line for foamate holdup [Rubin and Melech, *Can. J. Chem. Eng.*, **50**, 748 (1972)].

The slowly rising foam in a tall column can be employed as the sorbent for continuous chromatographic separations [Talman and Rubin, *Sep. Sci.*, **11**, 509 (1976)]. Low gas rates are also employed in short columns to produce the scumlike froth of batch-operated ion flotation, microflotation, and precipitate flotation.

Foam Drainage and Overflow The rate of foam overflow on a gas-free basis (i.e., the total volumetric foamate rate Q) can be estimated from a detailed theory for foam drainage [Leonard and Lemlich, *Am. Inst. Chem. Eng. J.*, **11**, 18 (1965)]. From the resulting relationship for overflow [Fanlo and Lemlich, *Am. Inst. Chem. Eng. Symp. Ser.*, **9**, 75, 85 (1965)], Eq. (20-52) can be employed as a convenient approximation to the theory so as to avoid trial and error over the usual range of interest for foam of low liquid content ascending in plug flow:

$$\frac{Q}{G} = 22 \left(\frac{v_g^3 \mu \mu_s^2}{g^3 \rho^3 d^8} \right)^{1/4} \quad (20-52)$$

The superficial gas velocity v_g is G/A , where A is the horizontal cross-sectional area of the empty vertical foam column. Also, g is the acceleration of gravity, ρ is the liquid density, μ is the ordinary liquid viscosity, and μ_s is the effective surface viscosity.

To account for inhomogeneity in bubble sizes, d in Eq. (20-52) should be taken as $\sqrt{\Sigma n d_i^3 / \Sigma n d_i}$ and evaluated at the top of the vertical column if coalescence is significant in the rising foam. Note that this average d for overflow differs from that employed earlier for S . Also, see "Bubble Sizes" regarding the correction for planar statistical sampling bias and the presence of size segregation at a wall.

For theoretical reasons, Q determined from Eq. (20-52) should be multiplied by the factor $(1 + 3Q/G)$ to give a final Q . However, for foam of sufficiently low liquid content this multiplication can be omitted with little error.

The effective surface viscosity is best found by experiment with the system in question, followed by back calculation through Eq. (20-52). From the precursors to Eq. (20-52), such experiments have yielded values of μ_s on the order of 10^{-4} (dyn·s)/cm for common surfactants in water at room temperature, which agrees with independent measurements [Lemlich, *Chem. Eng. Sci.*, **23**, 932 (1968); and Shih and Lemlich, *Am. Inst. Chem. Eng. J.*, **13**, 751 (1967)]. However, the expected high μ_s for aqueous solutions of such skin-forming substances as saponin and albumin was not attained, perhaps because of their non-newtonian surface behavior [Shih and Lemlich, *Ind. Eng. Chem. Fundam.*, **10**, 254 (1971); and Jashnani and Lemlich, *J. Colloid Interface Sci.*, **46**, 13 (1974)].

The drainage theory breaks down for columns with tortuous cross section, large slugs of gas, or heavy coalescence in the rising foam.

Foam Coalescence Coalescence is of two types. The first is the growth of the larger foam bubbles at the expense of the smaller bubbles due to interbubble gas diffusion, which results from the smaller bubbles having somewhat higher internal pressures (Adamson and Gast, *Physical Chemistry of Surfaces*, 6th ed., Wiley, New York, 1997). Small bubbles can even disappear entirely. In principle, the rate at which this type of coalescence proceeds can be estimated [Ranadive and Lemlich, *J. Colloid Interface Sci.*, **70**, 392 (1979)].

The second type of coalescence arises from the rupture of films between adjacent bubbles [Vrij and Overbeek, *J. Am. Chem. Soc.*, **90**, 3074 (1968)]. Its rate appears to follow first-order reaction kinetics with respect to the number of bubbles [New, *Proc. 4th Int. Congr. Surf. Active Substances*, Brussels, 1964, **2**, 1167 (1967)] and to decrease with film thickness [Steiner, Hunkeler, and Hartland, *Trans.*

Inst. Chem. Eng., **55**, 153 (1977)]. Many factors are involved [Bikerman, *Foams*, Springer-Verlag, New York, 1973; and Akers (ed.), *Foams*, Academic, New York, 1976].

Both types of coalescence can be important in the foam separations characterized by low gas flow rate, such as batchwise ion flotation producing a scum-bearing froth of comparatively long residence time. On the other hand, with the relatively higher gas flow rate of foam fractionation, the residence time may be too short for the first type to be important, and if the foam is sufficiently stable, even the second type of coalescence may be unimportant.

Unlike the case for Eq. (20-52), when coalescence is significant, it is better to find S from d evaluated at the feed level for Eqs. (20-49) to (20-51) and at the pool surface for Eqs. (20-43) and (20-48).

Foam Breaking It is usually desirable to collapse the overflowing foam. This can be accomplished by chemical means (Bikerman, op. cit.) if external reflux is not employed or by thermal means [Kishimoto, *Kolloid Z.*, **192**, 66 (1963)] if degradation of the overhead product is not a factor.

Foam can also be broken with a rotating perforated basket [Lemlich, "Principles of Foam Fractionation," in Perry (ed.), *Progress in Separation and Purification*, vol. 1, Interscience, New York, 1968, chap. 1]. If the foamate is aqueous (as it usually is), the operation can be improved by discharging onto Teflon instead of glass [Haas and Johnson, *Am. Inst. Chem. Eng. J.*, **11**, 319 (1965)]. A turbine can be used to break foam [Ng, Mueller, and Walden, *Can. J. Chem. Eng.*, **55**, 439 (1977)]. Foam which is not overly stable has been broken by running foamate onto it [Brunner and Stephan, *Ind. Eng. Chem.*, **57**(5), 40 (1965)]. Foam can also be broken by sound or ultrasound, a rotating disk, and other means [Ohkawa, Sakagawa, Sakai, Futai, and Takahara, *J. Ferment. Technol.*, **56**, 428, 532 (1978)].

If desired, dephlegmation (partial collapse of the foam to give reflux) can be accomplished by simply widening the top of the column, provided the foam is not too stable. Otherwise, one of the more positive methods of foam breaking can be employed to achieve dephlegmation.

Bubble Fractionation Figure 20-44 shows continuous bubble fractionation. This operation can be analyzed in a simplified way in terms of the adsorbed carry-up, which furthers the concentration gradient, and the dispersion in the liquid, which reduces the gradient [Lemlich, *Am. Inst. Chem. Eng. J.*, **12**, 802 (1966); **13**, 1017 (1967)].

To illustrate, consider the limiting case in which the feed stream and the two liquid takeoff streams of Fig. 20-44 are each zero, thus resulting in batch operation. At steady state the rate of adsorbed carry-up will equal the rate of downward dispersion, or $af\Gamma = \bar{D}AdC/dh$. Here a is the surface area of a bubble, f is the frequency of bubble formation, $\bar{D}$ is the dispersion (effective diffusion) coefficient based on the column cross-sectional area A , and C is the concentration at height h within the column.

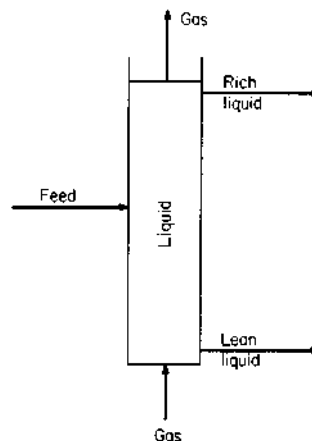


FIG. 20-44 Continuous bubble fractionation.

There are several possible alternative relationships for Γ (Lemlich, op. cit.). For simplicity, consider $\Gamma = K'C$, where K' is not necessarily the same as the equilibrium constant K . Substituting and integrating from the boundary condition of $C = C_B$ at $h = 0$ yield

$$C/C_B = \exp(Jh) \quad (20-53)$$

C_B is the concentration at the bottom of the column, and parameter $J = K'af/DA$. Combining Eq. (20-53) with a material balance against the solute in the initial charge of liquid gives

$$\frac{C}{C_i} = \frac{JH \exp(Jh)}{\exp(JH) - 1} \quad (20-54)$$

C_i is the concentration in the initial charge, and H is the total height of the column.

The foregoing approach has been extended to steady continuous flow as illustrated in Fig. 20-44 [Cannon and Lemlich, *Chem. Eng. Prog. Symp. Ser.*, **68**(124), 180 (1972); Bruin, Hudson, and Morgan, *Ind. Eng. Chem. Fundam.*, **11**, 175 (1972); and Wang, Granstrom, and Kown, *Environ. Lett.*, **3**, 251 (1972), **4**, 233 (1973), **5**, 71 (1973)]. The extension includes a rough method for estimating the optimum feed location as well as a very detailed analysis of column performance which takes into account the various local phenomena around each rising bubble (Cannon and Lemlich, op. cit.).

Uraizee and Narsimhan [*Sep. Sci. Technol.*, **30**(6), 847 (1995)] have provided a model for the continuous separation of proteins from dilute solutions. Although their work is focused on protein separation, the model should find general application to other separations.

In agreement with experiment [Shah and Lemlich, *Ind. Eng. Chem. Fundam.*, **9**, 350 (1970); and Garmendia, Perez, and Katz, *J. Chem. Educ.*, **50**, 864 (1973)], theory shows that the degree of separation that is obtained increases as the liquid column is made taller. But unfortunately it decreases as the column is made wider. In simple terms, the latter effect can be attributed to the increase in the dispersion coefficient as the column is widened.

In this last connection it is important that the column be aligned precisely vertically (Valdes-Krieg, King, and Sephton, *Am. Inst. Chem. Eng. J.*, **21**, 400 (1975)). Otherwise, the bubbles with their dragged liquid will tend to rise up one side of the column, thus causing liquid to flow down the other side, and in this way largely destroy the concentration gradient. A vertical foam-fractionation column should also be carefully aligned to be plumb.

The escaping bubbles from the top of a bubble-fractionation column can carry off an appreciable quantity of adsorbed material in an aerosol of very fine film drops [various papers, *J. Geophys. Res., Oceans Atmos.*, **77**(27), (1972)]. If the residual solute is thus appreciably depleted, C_i in Eq. (20-54) should be replaced with the average residual concentration.

This carry-off of film drops, which may also occur with breaking foam, in certain cases can partially convert water pollution into air pollution. If such is the case, it may be desirable to recirculate the gas. Such recirculation is also indicated if hydrocarbon vapors or other volatiles are incorporated in the gas stream to improve adsorptive selectivity [Maas, *Sep. Sci.*, **4**, 457 (1969)].

A small amount of collector (surfactant) or other appropriate additive in the liquid may greatly increase adsorption (Shah and Lemlich, op. cit.). Column performance can also be improved by skimming the surface of the liquid pool or, when possible, by removing adsorbed solute in even a tenuous foam overflow. Alternatively, an immiscible liquid can be floated on top. Then the concentration gradient in the tall pool of main liquid, plus the trapping action of the immiscible layer above it, will yield a combination of bubble fractionation and solvent sublation.

Systems Separated Some of the various separations reported in the literature are listed in Rubin and Gaden, "Foam Separation," in Schoen (ed.), *New Chemical Engineering Separation Techniques*, Interscience, New York, 1962, chap. 5; Lemlich, *Ind. Eng. Chem.*, **60**(10), 16 (1968); Pushkarev, Egorov, and Khristalev, *Clarification and Deactivation of Waste Waters by Frothing Flotation*, in Russian, Atomizdat, Moscow, 1969; Kuskin and Golman, *Flotation of Ions and Molecules*, in Russian, Nedra, Moscow, 1971; Lemlich (ed.), *Adsorptive Bubble*

Separation Techniques, Academic, New York, 1972; Lemlich, "Adsorbable Methods," in Li (ed.), *Recent Developments in Separation Science*, vol. 1, CRC Press, Cleveland, 1972, chap. 5; Grieves, *Chem. Eng. J.*, **9**, 93 (1975); Valdes-Krieg King, and Sephton, *Sep. Purif. Methods*, **6**, 221 (1977); Clarke and Wilson, *Foam Flotation*, Marcel Dekker, New York, 1983; and Wilson and Clarke, "Bubble and Foam Separations in Waste Treatment," in Rousseau (ed.), *Handbook of Separation Processes*, Wiley, New York, 1987.

Of the numerous separations reported, only a few can be listed here. Except for minerals beneficiation [ore flotation], the most important industrial applications are usually in the area of pollution control.

A pilot-sized foaming unit reduced the alkyl benzene sulfonate concentration of 500,000 gal of sewage per day to nearly 1 mg/L, using a G/F of 5 and producing a Q/F of no more than 0.03 [Brunner and Stephan, *Ind. Eng. Chem.*, **57**(5), 40 (1965); and Stephan, *Civ. Eng.*, **35**(9), 46 (1965)]. A full-scale unit handling over 45,420 m³/day (12 million gal/day) performed nearly as well. The foam also carried off some other pollutants. However, with the widespread advent of biodegradable detergents, large-scale foam fractionation of municipal sewage has been discontinued.

Other plant-scale applications to pollution control include the flotation of suspended sewage particles by depressurizing so as to release dissolved air [Jenkins, Scherfig, and Eckhoff, "Applications of Adsorptive Bubble Separation Techniques to Wastewater Treatment," in Lemlich (ed.), *Adsorptive Bubble Separation Techniques*, Academic, New York, 1972, chap. 14; and Richter, *Internat. Chem. Eng.*, **16**, 614 (1976)]. Dissolved-air flotation is also employed in treating wastewater from pulp and paper mills [Coertze, *Prog. Water Technol.*, **10**, 449 (1978); and Severeid, *TAPPI* **62**(2), 61, 1979]. In addition, there is the flotation, with electrolytically released bubbles [Chambers and Cottrell, *Chem. Eng.*, **83**(16), 95 (1976)], of oily iron dust [Ellwood, *Chem. Eng.*, **75**(16), 82 (1968)] and of a variety of wastes from surface-treatment processes at the maintenance and overhaul base of an airline [Roth and Ferguson, *Desalination*, **23**, 49 (1977)].

Fats and, through the use of lignosulfonic acid, proteins can be floated from the wastewaters of slaughterhouses and other food-processing installations [Hopwood, *Inst. Chem. Eng. Symp. Ser.*, **41**, M1 (1975)]. After further treatment, the floated sludge has been fed to swine.

A report of the recovery of protein from potato-juice wastewater by foaming [Weijenberg, Mulder, Drinkenberg, and Stemerding, *Ind. Eng. Chem. Process Des. Dev.*, **17**, 209 (1978)] is reminiscent of the classical recovery of protein from potato and sugar-beet juices [Ostwald and Siehr, *Kolloid Z.*, **79**, 11 (1937)]. The isoelectric pH is often a good choice for the foam fractionation of protein (Rubin and Gaden, loc. cit.). Adding a salt to lower solubility may also help. Additional applications of foam fractionation to the separation of protein have been reported by Uraizee and Narsimhan [*Enzyme Microb. Technol.*, **12**, 232 (1990)].

With the addition of appropriate additives as needed, the flotation of refinery wastewaters reduced their oil content to less than 10 mg/L in pilot-plant operation [Steiner, Bennett, Mohler, and Clere, *Chem. Eng. Prog.*, **74**(12), 39 (1978)] and full-scale operation [Simonsen, *Hydrocarb. Process. Pet. Refiner*, **41**(5), 145, 1962]. Experiments with a cationic collector to remove oils reportedly confirmed theory [Angelidon, Keskavarz, Richardson, and Jameson, *Ind. Eng. Chem. Process Des. Dev.*, **16**, 436 (1977)].

Pilot-plant [Hyde, Miller, Packham, and Richards, *J. Am. Water Works Assoc.*, **69**, 369 (1977)] and full-scale [Ward, *Water Serv.*, **81**, 499 (1977)] flotation in the preparation of potable water is described.

Overflow at the rate of 2700 m³ (713,000 gal) per day from a zinc-concentrate thickener is treated by ion flotation, precipitate flotation, and ultrafine-particle flotation [Nagahama, *Can. Min. Metall. Bull.*, **67**, 79 (1974)]. In precipitate flotation only the surface of the particles need be coated with collector. Therefore, in principle less collector is required than for the equivalent removal of ions by foam fractionation or ion flotation.

By using an anionic collector and external reflux in a combined (enriching and stripping) column of 3.8-cm (1.5-in) diameter with a feed rate of 1.63 m/h [40 gal/(h-ft²)] based on column cross section,

D/F was reduced to 0.00027 with C_w/C_F for Sr^{2+} below 0.001 [Shonfeld and Kibbey, *Nucl. Appl.*, **3**, 353 (1967)]. Reports of the adsorbable separation of 29 heavy metals, radioactive and otherwise, have been tabulated [Lemlich, "The Adsorptive Bubble Separation Techniques," in Sabadell (ed.), *Proc. Conf. Traces Heavy Met. Water*, 211–223, Princeton University, 1973, EPA 902/9-74-001, U.S. EPA, Reg. II, 1974). Some separation of ^{15}N from ^{14}N by foam fractionation has been reported [Hitchcock, Ph.D. dissertation, University of Missouri, Rolla, 1982].

The numerous separations reported in the literature include surfactants, inorganic ions, enzymes, other proteins, other organics, biological cells, and various other particles and substances. The scale of the systems ranges from the simple Crits test for the presence of surfactants in water, which has been shown to operate by virtue of transient foam fractionation [Lemlich, *J. Colloid Interface Sci.*, **37**, 497 (1971)], to the natural adsorbable processes that occur on a grand scale in the ocean [Wallace and Duce, *Deep Sea Res.*, **25**, 827 (1978)]. For further information see the reviews cited earlier.

MEMBRANE SEPARATION PROCESSES

GENERAL REFERENCES: Cheryan, *Ultrafiltration and Microfiltration Handbook*, Technomic Publishing Company, Pa., 1988. Ho and Sirkar (eds.), *Membrane Handbook*, Van Nostrand Reinhold, New York, 1992. Baker, *Membrane Technology and Applications*, 2d ed., Wiley, 2000. Eykamp, Sec. 22: Membrane Separation Processes, in *Perry's Chemical Engineers' Handbook*, 7th ed., Perry and Green (eds.), McGraw-Hill, New York, 1997. Millipore Corporation, *Protein Concentration and Diafiltration by Tangential Flow Filtration*, Lit. No. TB032 Rev. B, 1999. Zeman and Zydney, *Microfiltration and Ultrafiltration: Principles and Applications*, Marcel Dekker, New York, 1996. A review of membrane retention mechanisms can be found in Deen, *AIChE J.*, **33**, 1409–1425 (1987).

New developments in membranes are found in journals and trade magazines (e.g., *J. Membrane Sci.*, *BioPharm International*), vendor communications (e.g., websites), patent filings, and conference presentations (e.g., annual ACS or NAMS meetings). Areas of active research include new membrane polymers and surface modification with accompanying diagnostic methods (to reduce fouling, increase flux and retention, improve consistency), new module designs (to improve flux, cleanability, ease of use, scalability, reliability), new processing skids (better components, recovery, less holdup, better mixing, disposability, software for automated processing and archiving), new processing methods (diafiltration strategies, turbulence enhancements), and new applications (e.g., protein-protein separations, plasmids).

Topics Omitted from This Section In order to concentrate on the membrane processes of widest industrial interest, several are left out.

Dialysis and Hemodialysis Historically, dialysis has found some industrial use. Today, much of that is supplanted by ultrafiltration. Donan dialysis is treated briefly under electrodialysis. Hemodialysis is a huge application for membranes, and it dominates the membrane field in area produced and in monetary value. This medical application is omitted here.

An excellent description of the engineering side of both topics is provided by Kessler and Klein [in Ho and Sirkar (eds.), op. cit., pp. 163–216]. A comprehensive treatment of diffusion appears in: Von Halle and Shachter, "Diffusional Separation Methods," in *Encyclopedia of Chemical Technology*, pp. 149–203, Wiley, 1993.

Facilitated Transport Transport by a reactive phase through a membrane is promising but problematic. Way and Noble [in Ho and Sirkar (eds.), op. cit., pp. 833–866] have a description and a complete bibliography.

Liquid Membranes Several types of liquid membranes exist: molten salt, emulsion, immobilized/supported, and hollow-fiber-contained liquid membranes. Araki and Tsukube (*Liquid Membranes: Chemical Applications*, CRC Press, 1990) and Sec. IX and Chap. 42 in Ho and Sirkar (eds.) (op. cit., pp. 724, 764–808) contain detailed information and extensive bibliographies.

Catalytic Membranes Falconer, Noble, and Sperry (Chap. 14—"Catalytic Membrane Reactors" in Noble and Stern, op. cit., p. 669–712) give a detailed review and an extensive bibliography. Additional information can be found in a work by Tsotsis et al. ["Catalytic Membrane Reactors," pp. 471–551, in Becker and Pereira (eds.), *Computer-Aided Design of Catalysts*, Dekker, 1993].

GENERAL BACKGROUND AND DEFINITIONS

Applications Membranes create a boundary between different bulk gas or liquid mixtures. Different solutes and solvents flow through membranes at different rates. This enables the use of membranes in separation processes. Membrane processes can be operated at moderate temperatures for sensitive components (e.g., food, pharmaceuticals). Membrane processes also tend to have low relative capital and energy costs. Their modular format permits reliable scale-up and operation. This unit operation has seen widespread commercial adoption since the 1960s for component enrichment, depletion, or equilibration. Estimates of annual membrane module sales in 2005 are shown in Table 20-16. Applications of membranes for diagnostic and bench-scale use are not included. Natural biological systems widely employ membranes to isolate cells, organs, and nuclei.

Common Definitions Membrane processes have been evolving since the 1960s with each application tending to generate its own terminology. Recommended nomenclature is provided along with alternatives in current use.

Fluid Stream Designations For the generalized membrane module shown in Fig. 20-45, a feed stream enters a membrane module while both a permeate and a retentate stream exit the module. The permeate (or filtrate) stream flows through the membrane and has been depleted of retained components. The term *filtrate* is commonly used for NFF operation while permeate is used for TFF operation. The retentate (or concentrate) stream flows through the module, not the membrane, and has been enriched in retained components.

TABLE 20-16 Membrane Market in 2005

Segment	\$M/yr Size	Applications	Characteristics
Dialysis	~2,000	Medical	Mature growing 5%
Reverse osmosis	~500	Water treatment	Growing 10%
Microfiltration	~500	Water, food, pharm.	
Ultrafiltration	~400	Water, food, pharm.	Growing 10%
Gas separation	~500	Nitrogen	
Electrodialysis	~100	Water	
Pervaporation	~5	Solvent/water	Nascent
Facilitated transport	0	None	In development

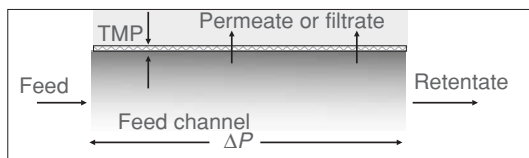


FIG. 20-45 Fluid stream schematic.

Flow Types: Normal Flow and Tangential Flow Filtration If the retentate flow in Fig. 20-45 is zero and all the feed stream flows as a velocity vector normal to the membrane surface, this type of filtration is referred to as *normal flow filtration (NFF)*, also called dead-end flow. If there is a stream that flows as a velocity vector tangent to the membrane surface, creating a velocity gradient at the membrane surface, and that exits the module as a retentate stream, this is referred to as *tangential flow filtration (TFF)*, also called crossflow filtration. A retentate stream that flows through a module without creating a surface velocity gradient is merely a bypass and not TFF.

TFF mitigates the accumulation of retained components on the membrane surface, reducing the plugging of the membrane and permitting a more steady-state operation. TFF is used in processing feeds with a high concentration of retained components.

Flow: Flux, Permeability, Conversion The productivity of a membrane module is described by its flux J = volumetric permeate flow rate/membrane area with units of volume per area per time. Relatively high flux rates imply that relatively small membrane areas are required. The permeate volume is usually greater than the feed volume for a given process. Flux is also the magnitude of the normal flow velocity with units of distance per time.

The sensitivity of productivity or flux to *transmembrane pressure (TMP)* is referred to as the permeability L = flux/transmembrane pressure. TMP refers to a module average. Pure-component permeability (e.g., water permeability) refers to membrane properties while the more industrially relevant process permeability includes fouling and polarization effects.

The efficiency of a membrane module is characterized by the recovery or conversion ratio CR = permeate flow rate/feed flow rate. Low conversion means that fluid has to be repeatedly cycled past the TFF module to generate permeate. High-efficiency NFF has $CR = 1$.

Flux Decline: Plugging, Fouling, Polarization Membranes operated in NFF mode tend to show a steady flux decline while those operated in TFF mode tend to show a more stable flux after a short initial decline. Irreversible flux decline can occur by membrane compression or retentate channel spacers blinding off the membrane. Flux decline by fouling mechanisms (molecular adsorption, precipitation on the membrane surface, entrapment within the membrane structure) are amenable to chemical cleaning between batches. Flux decline amenable to mechanical disturbance (such as TFF operation) includes the formation of a secondary structure on the membrane surface such as a static cake or a fluid region of high component concentration called a polarization layer.

Understanding polarization and controlling its effects are key to implementing a good TFF process. Solutes entrained by the permeate flow are retained by the membrane. They accumulate on the membrane surface and form a region of high concentration called the polarization boundary layer. A steady state is reached between back transport away from the membrane surface, tangential convective transport along the membrane surface, and normal convective flow toward the membrane. The back transport leading to steady-state operation gives TFF a high capacity. Plugging is commonly used to describe flux decline through fouling and caking mechanisms.

Single-Component Separation: Passage, Retention, LRV The passage of a component through a membrane (also called the sieving coefficient or transmission) is calculated as the ratio $S = c_P/c_F$, where c_P and c_F refer to the permeate concentration, and the feed concentration, respectively. These concentrations may change with position within a module. In industrial practice, it is common to measure these concentrations in the permeate stream exiting a module and the feed stream entering a module to give an observed passage or S_{obs} .

Membrane vendors and researchers may use the feed concentration at the membrane surface or c_w to calculate an intrinsic passage $S_{int} = c_P/c_w$. The intrinsic passage characterizes the membrane while the observed passage characterizes the module.

Complementary to the passage, one can also consider the *retention* of a component as $R = 1 - S$ (also called rejection). Retention can also be either an observed or an intrinsic measurement. Retention is useful in considering retained products during concentration mode operation. Other component separation characterizations include the *log reduction value* $LRV = -\log S$ which is used to characterize high-efficiency separations with permeate products (sterilization). The beta ratio $\beta = 1/S$ is sometimes used in NFF for clarification applications.

Multiple-Component Separation: Separation Factor Consistent with the characterization of different separation methods, one can define a separation factor α_{ij} (also called selectivity) for components i and j that compares their relative concentrations in the permeate stream to those in the feed stream:

$$\alpha_{ij} = \frac{c_{iP}/c_{jP}}{c_{iF}/c_{jF}} = \frac{S_i}{S_j} \quad (20-55)$$

The convention is to designate the component with higher passage as component i so that $\alpha_{ij} > 1$.

Membrane Types Key membrane properties include their size rating, selectivity, permeability, mechanical robustness (to allow module fabrication and withstand operating conditions), chemical robustness (to fabrication materials, process fluids, cleaners, and sanitizers), low extractibles, low fouling characteristics, high capacity, low cost, and consistency.

Size Ratings The relative sizes of common components and the associated membrane classes capable of retaining them are shown in Fig. 20-46.

Vendors characterize their filters with ratings indicating the approximate size (or corresponding molecular weight) of components retained by the membrane. This rating should be used as a rough guide only and followed up with retention testing. Among the factors affecting retention are the application-specific retention requirements, variable component size and shapes depending on solution environment, membrane fouling and compaction, degree of module polarization, and interaction between feed components.

Composition and Structure Commercial membranes consist primarily of polymers and some ceramics. Other membrane types include sintered metal, glass, and liquid film. Polymeric membranes are formed by precipitating a 5 to 25 wt % casting solution (lacquer) into a film by solvent evaporation (air casting), liquid extraction (immersion casting), or cooling (melt casting or thermally induced phase separation). Membranes can be cast as flat sheets on a variety of supports or as fibers through a die. Ceramic membranes are formed by depositing successive layers of smaller and smaller inorganic particles on a monolith substrate, to create smaller and smaller interstices or "pores" between particles. These layers are then sintered below the melting point to create a rigid structure.

Membranes may be surface-modified to reduce fouling or improve chemical resistance. This can involve adding surface-modifying agents directly to the lacquer or modifying the cast membrane through chemical or physical treatment. Membranes can also be formed by selective etching or track-etching (radiation treatment followed by etching). Stretching is used to change pore morphology.

Liquid film membranes consist of immiscible solutions held in membrane supports by capillary forces. The chemical composition of these solutions is designed to enhance transport rates of selected components through them by solubility or coupled chemical reaction.

Membrane Morphology—Pores, Symmetric, Composite Only nucleopore and anodized membranes have relatively uniform pores. Reverse osmosis, gas permeation, and pervaporation membranes have nonuniform angstrom-sized pores corresponding to spaces in between the rigid or dynamic membrane molecules. Solute-membrane molecular interactions are very high. Ultrafiltration membranes have nonuniform nanometer sized pores with some solute-membrane interactions. For other microfiltration membranes with nonuniform pores on the submicrometer to micrometer range, solute-membrane interactions are small.

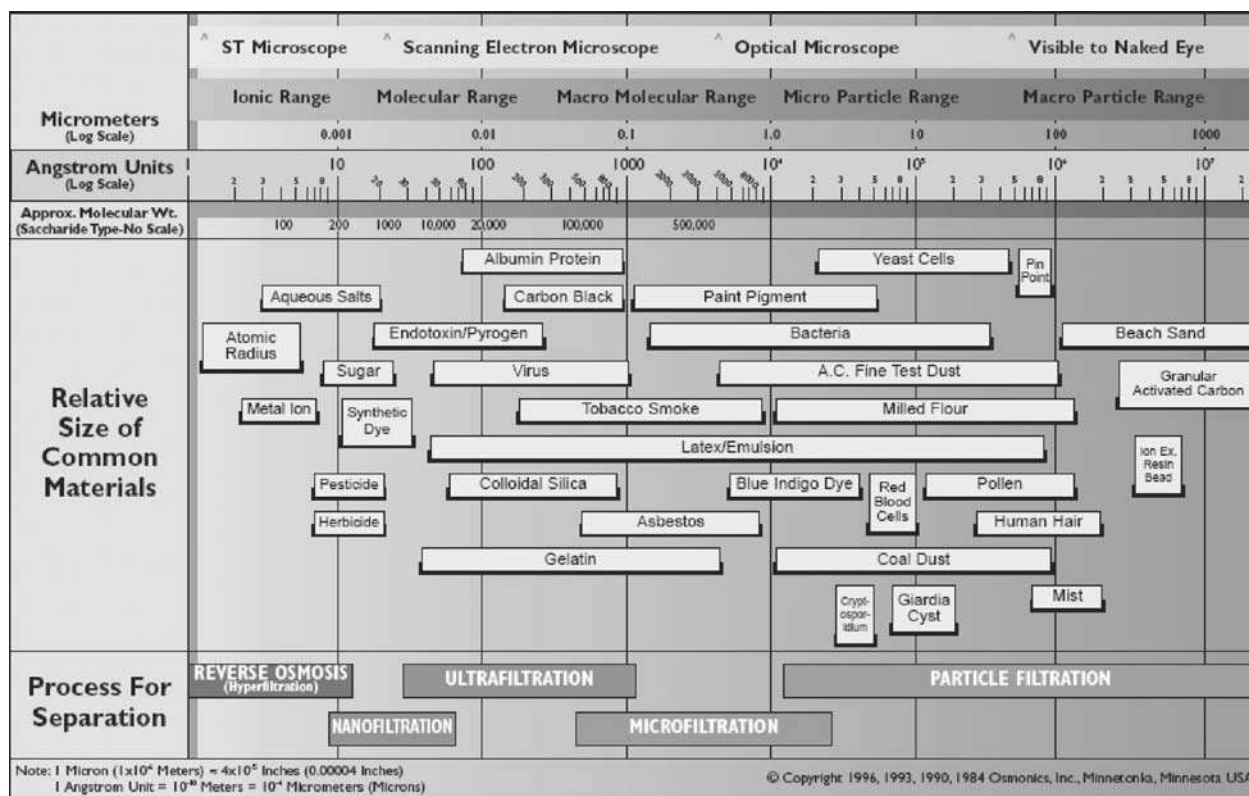


FIG. 20-46 Size spectrum. (Copyright © 1996. Reprinted by permission of Osmonics, Minnetonka, Minn.)

Membranes with a relatively uniform pore size distribution throughout the thickness of the membrane are referred to as symmetric or homogeneous membranes. Others may be formed with tight skin layers on the top or on both the top and bottom of the membrane surfaces. These are referred to as asymmetric or nonhomogeneous membranes. In addition, membranes can be cast on top of each other to form a composite membrane.

Asymmetric membranes have a "tight," low-permeability, retentive zone that performs the desired separation and a more open, high-permeability zone that provides mechanical strength to the overall membrane. This structure is particularly critical to the economic viability of reverse-osmosis membranes. Asymmetric membranes operated in TFF mode must have the tight side facing the feed channel so that particles are retained on its surface and can be acted upon by the tangential flow. Asymmetric membranes operated in NFF mode can

be operated with the open side facing the feed flow so that retained particles can penetrate the membrane and be dispersed without blinding off the membrane surface and plugging it quickly.

Component Transport Transport through membranes can be considered as mass transfer in series: (1) transport through a polarization layer above the membrane that may include static or dynamic cake layers, (2) partitioning between the upstream polarization layer and membrane phases at the membrane surface, (3) transport through the membrane, and (4) partitioning between the membrane and downstream fluid.

Surface Polarization in TFF The simplified model of polarization shown in Fig. 20-47 is used as a basis for analyzing more complex systems. Consider a single component with no reaction in a thin, two-dimensional boundary layer near the membrane surface. Axial diffusion is negligible along the membrane surface compared to convection.

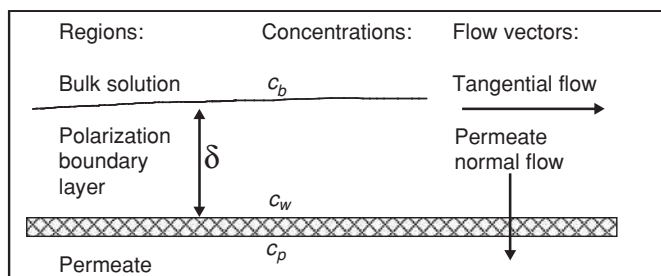


FIG. 20-47 Polarization in tangential flow filtration.

Fluid density and component brownian diffusivity D are also assumed constant. A steady-state component mass balance can be written for component concentration c :

$$u \frac{\partial c}{\partial x} + v \frac{\partial c}{\partial y} = D \frac{\partial^2 c}{\partial y^2} \quad u \text{ and } v \text{ are normal and tangential velocities} \quad (20-56)$$

Further neglecting the first term allows integration from $y = 0$ at the wall (membrane surface) into the boundary layer. At the wall, the net flux is represented by convection into the permeate

$$vc - D \frac{dc}{dy} = vc_p \quad (20-57)$$

This can be further integrated from the wall to the boundary layer thickness $y = \delta$, where the component is at the bulk concentration c_b . Substituting $J = -v$ and $k = D/\delta$, the mass-transfer coefficient yields the stagnant film model [Brian, *Desalination by Reverse Osmosis*, Merten (ed.), M.I.T. Press, Cambridge, Mass., 1966, pp. 161–292]:

$$\frac{v\delta}{D} = \ln\left(\frac{c_b - c_p}{c_w - c_p}\right) = -\frac{J}{k} \quad (20-58)$$

Although allowing for axial variations in v with a constant wall concentration c_w yields Eq. (20-59), as a more rigorous expression applicable to higher concentrations [Trettin et al., *Chem. Eng. Comm.*, **4**, 507 (1980)] the form of Eq. (20-58) is convenient in analyzing a variety of complex behavior.

$$J = 2.046k \left(\frac{c_w - c_p}{c_b - c_p} \right)^{1/3} \quad (20-59)$$

Substituting the intrinsic passage into Eq. (20-58) and rearranging yields an expression for the polarization modulus c_w/c_b :

$$\frac{c_w}{c_b} = \frac{e^{J/k}}{1 + S_{\text{int}}(e^{J/k} - 1)} \quad (20-60)$$

Figure 20-48 shows Wijmans's plot [Wijmans et al., *J. Membr. Sci.*, **109**, 135 (1996)] along with regions where different membrane processes operate (Baker, *Membrane Technology and Applications*, 2d ed., Wiley, 2004, p. 177). For RO and UF applications, $S_{\text{int}} < 1$, and $c_w > c_b$. This may cause precipitation, fouling, or product denaturation. For gas separation and pervaporation, $S_{\text{int}} > 1$ and $c_w < c_b$. MF is not shown since other transport mechanisms besides Brownian diffusion are at work.

Substituting the observed passage into Eq. (20-60) and rearranging yields Eq. (20-61). A plot of the LHS versus J data yields the mass-transfer coefficient from the slope, similar to the Wilson plot for heat transfer-coefficient determination:

$$\ln\left(1 - \frac{1}{S_{\text{obs}}}\right) = \ln\left(1 - \frac{1}{S_{\text{int}}}\right) - \frac{J}{k} \quad (20-61)$$

A generalized model of transport allowing for component interactions is provided by nonequilibrium thermodynamics where the flux of component i through the membrane J_i [gmol/(cm²·s)] is written as a first-order perturbation of the chemical potential $d\mu_i/dx$ [cal/(gmol·cm)]:

$$J_i = \sum_i L_i \frac{d\mu_i}{dx} \quad (20-62)$$

and L_i [1/(cal·cm·s)] is a component permeability. The chemical potential gradient provides the driving force for flow and may include contributions from pressure, concentration, electrical field, and temperature. A complementary description of transport is obtained by modeling each transport mechanism in some detail. See each application section for appropriate models.

Application of Eq. (20-62) to the polarization for two components. A and B requires solution of the component balances (Saksena, Ph.D. dissertation, University of Delaware, 1995):

$$u \frac{\partial c_A}{\partial x} + v \frac{\partial c_A}{\partial y} = \frac{\partial}{\partial y} \left(D_A \frac{\partial c_A}{\partial y} + D_{AB} \frac{\partial c_B}{\partial y} \right) \quad (20-63a)$$

$$u \frac{\partial c_B}{\partial x} + v \frac{\partial c_B}{\partial y} = \frac{\partial}{\partial y} \left(D_B \frac{\partial c_B}{\partial y} + D_{AB} \frac{\partial c_A}{\partial y} \right) \quad (20-63b)$$

Larger components with smaller brownian diffusivity polarize more readily. They can exclude smaller components, reducing their concentration at the membrane surface and increasing their retention by the membrane.

Solvent Transport through the Membrane The fluid flux J through a membrane is commonly modeled as an assembly of n_p parallel tubes per unit area with a uniform radius r and length l displaying Hagen-Poiseuille flow:

$$J = \frac{n_p \pi r^4}{8\eta l} \Delta P \quad (20-64)$$

where η is the fluid viscosity and ΔP the pressure drop. The dependence of flow on r^4 means that larger pores take most of the flow.

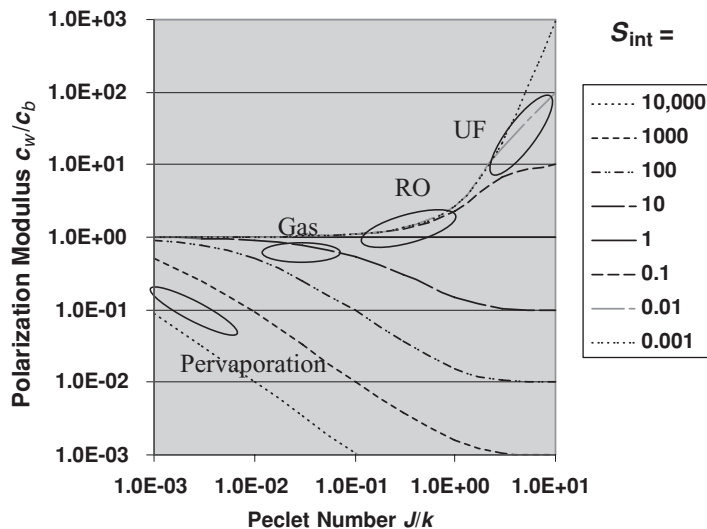


FIG. 20-48 TFF polarization.

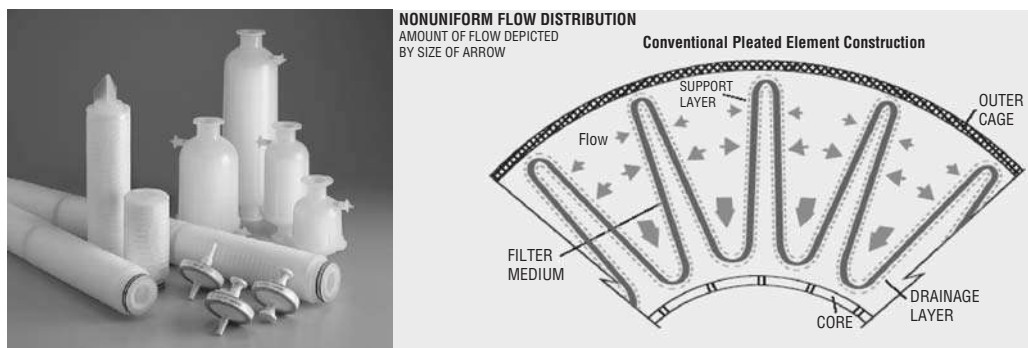


FIG. 20-49 NFF pleated cartridges and capsules. (Courtesy Millipore Corporation, Pall Corporation.)

MODULES AND MEMBRANE SYSTEMS

Membranes are packaged into modules for convenient operation. Key module properties include mechanical robustness (providing good sealing between fluid streams and membrane support), good flow distribution (low pressure drops, high TFF membrane shear, low dead volumes, and low sensitivity to feed channel plugging), cleanability, chemical robustness, low extractibles, low cost, ease of assembly, scalability, high product recovery, and high consistency. Cleanability considerations evolved from the dairy industry (3A Sanitary Standards for Crossflow Membrane Modules, USPHS No. 45-00, 1990). Modules may be integrity-tested for the absence of punctures, improper sealing, or other defects.

NFF modules (Figs. 20-49 and 20-50) include cartridges or capsules, stacked disks, and flat sheet. Pleated cartridges, the most common format, are available in a variety of sizes and are installed into multi-round stainless-steel housings. As shown in Fig. 20-49, multiple layers of materials such as support layers or membranes with different pore ratings may be pleated together. Capsules are made by sealing individual cartridges within their own plastic housing. Pleated cartridge manufacturing is shown in Fig. 20-51.

TFF module types include plate-and-frame (or cassettes), hollow fibers, tubes, monoliths, spirals, and vortex flow. Figures 20-52 and 20-53 show several common module types and the flow paths within each. Hollow fiber or tubular modules are made by potting the cast membrane fibers or tubes into end caps and enclosing the assembly in a shell. Similar to fibers or tubes, monoliths have their retentive layer coated on the inside of tubular flow channels or lumens with a high-permeability porous structure on the shell side.

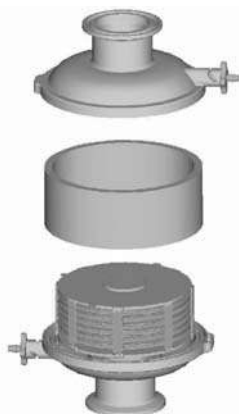


FIG. 20-50 NFF stacked disk modules. (Courtesy Millipore Corporation.)

Spiral-wound modules are made by assembling a membrane packet or leaf consisting of a permeate channel spacer sheet with two flat sheets of membranes on either side (facing outward from the permeate sheet). The sides of this packet are glued together, and the top is glued to a central core collection tube containing small holes that allow passage of permeate from the spacer channel into the collection tube. A feed-side spacer is placed on the packet, and the packet is wound around the core to create the module. The wound packet assembly is sometimes inserted into a shell. Spiral modules frequently use a multileaf construction in which several packets are glued to the central core before winding. This design reduces the pressure drop associated with a long permeate path length.

Cassettes use feed and permeate spacers with one set of precut holes for the feed/retentate and another, smaller set of holes for the permeate. The feed spacer has a raised-edge seal around the permeate hole to prevent flow, and the permeate spacer has a similar seal around the feed/retentate hole. Similar to spiral assembly, membrane packets are glued together around their edges and assembled into a stack, and the stack is glued together around the edges.

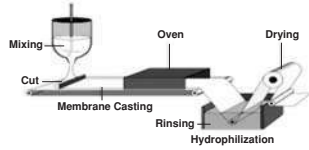
For fibers or tubular modules, the feed is generally introduced into the inside of the tubes, or lumen, while permeate is withdrawn from the shell side. This flow orientation enhances the shear at the membrane surface for TFF operation. These modules may also be run at high conversion or NFF mode with the feed introduced on the outside of the tubes or shell side. In this case, the shell side offers greater surface area.

Table 20-17 summarizes TFF module characteristics. The preferred TFF module type depends on the characteristics of each application.

In designing TFF modules, the polarization equation [Eq. (20-58)] indicates that for given fixed concentrations, increases in the mass-transfer coefficient k will increase the flux. This can be accomplished by (1) increasing the shear rate at the wall through higher tangential flow velocities and smaller channel heights, or mechanically moving the membrane by spinning or vibration; (2) altering the geometry of the feed channel to increase turbulent mixing in the normal direction by using feed channel screens or curving channels to introduce Dean or Taylor vortices; (3) introducing pulsatile flow in the feed channel or periodic bursts of gas bubbles; and (4) introducing body forces (centrifugal, gravitational, or electromagnetic) to augment transport away from the membrane and introducing large particles in the feed (e.g., $>0.5 \mu\text{m}$ PVC latex beads) that disrupt boundary layers and cause shear-induced diffusion.

Another device that finds frequent use is the stirred cell shown in Fig. 20-54. This device uses a membrane coupon at the bottom of the reservoir with a magnetic stir bar. Stirred cells use low fluid volumes and can be used in screening and R&D studies to evaluate membrane types and membrane properties. The velocity profiles have been well defined (Schlichting, *Boundary Layer Theory*, 6th ed., McGraw-Hill, New York, 1968, pp. 93–99).

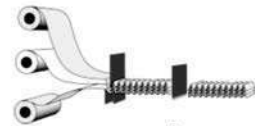
Membrane Manufacturing



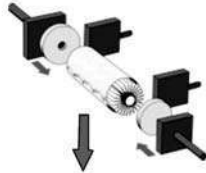
Lamination



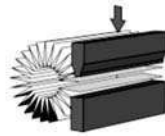
Pleating



End Cap Welding



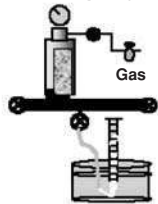
Ultrasonic Seaming



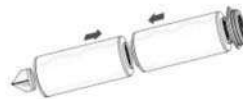
Serial No.



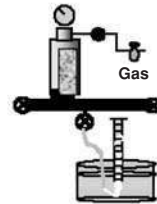
100% Integrity Testing



Stacking



100% Physical Testing



Audit Testing

- Toxicity
- Gravimetric Extractables
- Multiple Steaming

Packaging

Lot Release Testing



- Water Flow
- Bubble Point
- Diffusional Flow
- *B. diminuta* Spike
- USP Oxidizables
- USP Endotoxin
- Mechanical Stress
- Thermal Stress

FIG. 20-51 Pleated cartridge manufacturing.

Mass-transfer coefficients for a single newtonian component in various module types and flow regimes can be correlated by Eq. (20-65) with values for the constants in Table 20-18:

$$Sh = aRe^b Sc^c (d_h/L)^e \quad (20-65)$$

where $Sh = kD/d_h$

$$Re = \frac{\rho u d_h}{\mu} \quad (\text{tube, slit or spacer with hydraulic diameter } d_h)$$

$$Re = \frac{\rho \omega d^2}{\mu} \quad (\text{stirred cell diameter } d)$$

$$Re = \frac{2\rho \omega R d}{\mu} \quad (\text{rotating-cylinder inner radius } R \text{ and gap diameter } d)$$

$$Sc = \frac{\mu}{\rho D}$$

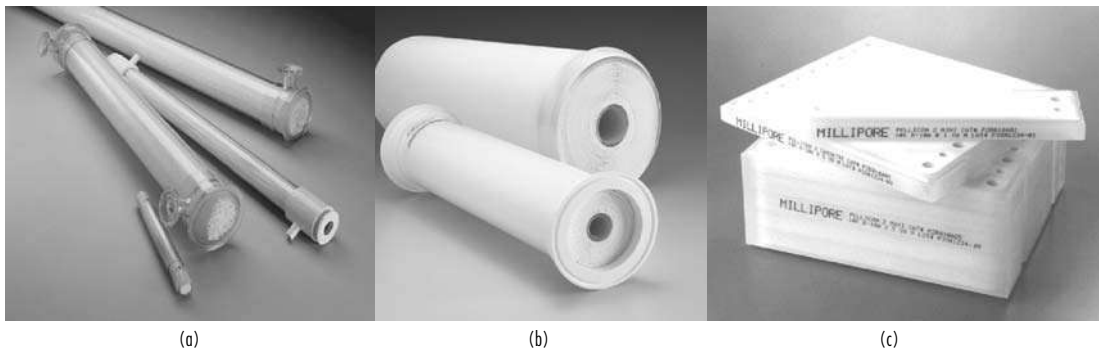


FIG. 20-52 Commercial TFF modules: (a) hollow fibers; (b) spirals; (c) cassettes. (Courtesy Millipore Corporation.)

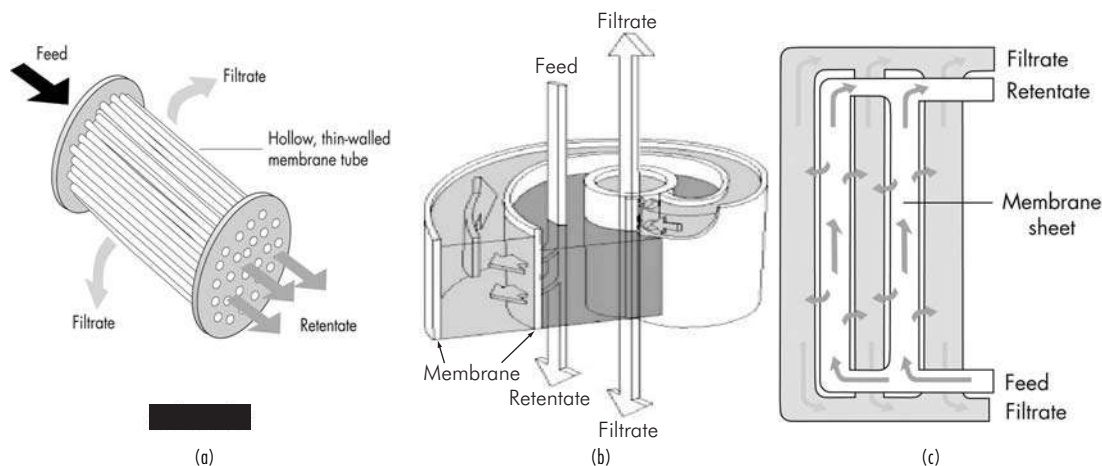


FIG. 20-53 TFF module flow paths: (a) hollow fibers; (b) spirals; (c) cassettes. (Courtesy Millipore Corporation.)

Large-Area Configurations NFF membrane systems consist primarily of large, multi-round housings that contain pleated cartridges inserted into a common base plate. These cartridges share a common feed and collect a common filtrate.

TFF membrane systems generally use a common feed distributed among parallel modules with a collection of common retentate and common permeate streams. In some applications, it is also useful to plumb TFF modules with the retentate in series where the retentate flow from one module provides the feed flow to the next module. This type of configuration is equivalent to increasing the length of the retentate channel. Permeate flows may or may not be plumbed together.

Process Configurations Basic membrane process configurations shown in Fig. 20-55 include single-pass, batch, fed-batch, and continuous.

Single-Pass Operation A single-pass configuration is used for NFF system operation (no retentate). Component concentrations change along the length of the retentate channel. For dilute streams such as water, assemblies of TFF modules that run in NFF mode (retentate closed) are also used. For nondilute streams, a single pass through a module may be insufficient to generate either the desired permeate flow or a concentrated retentate.

Steady-state component and solvent mass balances can be written for single-pass operation by considering an incremental area element dA in the axial or flow direction for a feed channel, module, or membrane

assembly of constant width. For volumetric crossflow Q in the feed channel and observed solute passage S_i ,

$$\frac{d(c_i Q)}{dA} = c_i \frac{dQ}{dA} + Q \frac{dc_i}{dA} = -J c_i S_i \quad (20-66)$$

$$\frac{dQ}{dA} = -J \quad (20-67)$$

Combining these equations and integrating yield $c_i = c_{i0} X^{1-S_i}$ for a volume reduction factor $X = Q/Q_0$ and the observed component passage S_i . This allows one to determine either final concentrations from crossflow rates or the reverse. For a fully retained product ($S_i = 0$), a 10-fold volume reduction ($X = 10$) produces a 10-fold more concentrated product. However, if the product is only partially retained, the volume reduction does not proportionately increase the final concentration due to losses through the membrane.

The area required for processing $A = (Q_0 - Q)/\bar{J}$, where $Q_0 - Q$ is the permeate volumetric flow, can be estimated by using the approximation $\bar{J} \approx 0.33J_{\text{initial}} + 0.67J_{\text{final}}$ (Cheryan, *Ultrafiltration Handbook*, Technomic, Lancaster, Pa., 1986) and a suitable flux model. An appropriate model relating flux to crossflow, concentration, and pressure is then applied. Pressure profiles along the retentate channel are empirically correlated with flow for spacer-filled channels to obtain $\Delta P = \Delta P_0 (Q/A)^n$.

TABLE 20-17 Commercial TFF Modules

	Spiral	Fiber	Cassette
Screens/spacers	Yes	No	Yes/no
Typical no. in series	1–2	1–2	1–2
Packing density, m ² /m ³	800	1000–6000	500
Feed flow, L/m ² /h (LMH)	700–5000	500–18,000	400
Feed pressure drop, psi/module	5–15	1–5	10–50
Channel height, mm	0.3–1	0.2–3	0.3–1
Plugging sensitivity	High	Moderate	High
Working volume, L/m ²	1	0.5	0.4
Holdup volume, L/m ²	0.03	0.03	0.02
Module cost, \$/m ²	40–200	200–900	500–1000
Ruggedness	Moderate	Low-moderate	High
Module areas, m ²	0.1–35	0.001–5	0.05–2.5
Membrane types	RO-UF	RO-UF-MF	UF-MF
Relative mass-transfer efficiency*	6	4	10
Ease of use	Moderate	High	Moderate
Scalability†	Fair	Moderate	Good

* Qualitative, based on relative fluxes.

† Cassettes keep retentate path length constant and require lower feed flow rates.



FIG. 20-54 Stirred cell TFF module. (Courtesy Millipore Corporation.)

Batch and Fed-Batch Operation Batch operation involves recycling the retentate to the feed tank to create a multipass flow of the feed through the module. The compositions in batch systems change over time. Recycling is necessary for nondilute streams to generate either the desired permeate flow or a concentrated retentate. Multipass operation increases residence times and pump passes that may degrade retentate components. For systems requiring high pressures to generate permeate flow, it is useful to run the entire system at high pressure to save energy.

An unsteady-state component mass balance, Eq. (20-68), can be written for batch operation by assuming a uniform average retentate concentration c_i within the system. Assuming a constant solvent concentration and a 100 percent passage, the solvent balance becomes Eq. (20-69).

$$\frac{d(c_i V)}{dt} = -J A c_i S_i \quad (20-68)$$

$$\frac{dV}{dt} = -J A \quad (20-69)$$

Dividing Eq. (20-68) by Eq. (20-69) and integrating yield the relationship in Table 20-19 between retentate concentration, the volume reduction factor X = (initial retentate volume)/(final retentate volume), and the observed component passage S_i . For a fully retained product ($S_i = 0$), a 10-fold volume reduction ($X = 10$) produces a 10-fold more concentrated product. However, if the product is only partially retained, the volume reduction does not proportionately increase the

final concentration due to losses through the membrane. The component mass in the retentate is obtained as the product of the retentate concentration and retentate volume, as shown in Table 20-19.

TFF systems have a maximum volume reduction capability, typically about 40X. This arises because tanks must be large enough to hold the batch volume while allowing operation at a minimum working volume. The minimum working volume may be limited by air entrainment into the feed pump, mixing in the feed tank, or level measurement capabilities. Fed-batch operation extends the volume reduction capability to 100X and provides some flexibility in processing a variety of batch volumes in a single skid. For a fed-batch operation, the retentate is returned to a smaller tank, not the large feed tank. Feed is added to the small retentate tank as permeate is withdrawn, so the system volume remains constant. The smaller retentate tank can allow a smaller working volume without entrainment. While one could make the retentate tank a section of pipe that returns the retentate directly into the pump feed (bypass line), such a configuration can be problematic to flush, vent, and drain, leading to cleaning and product recovery issues.

Equation (20-70) is the unsteady-state component mass balance for fed-batch concentration at constant retentate volume. Integration yields the equations for concentration and yield in Table 20-19.

$$V \frac{dc_i}{dt} = J A c_F - J A c_i S_i \quad (20-70)$$

Retentate concentration over the course of the process is shown in Fig. 20-56 for a fully retentive product ($S_i = 0$). The tank ratio = V_0 /retentate tank volume, C_0 is the feed concentration, and V_0 is the feed volume. The benefits sought from higher concentrations can lead to other problems such as reduced fluxes, larger area and pumps, possible denaturation, and extra lines that may have issues with cleaning and product recovery. This not only has caused significant commissioning and validation delays but also has led to the scrapping of a process skid as unworkable. The number of pump passes will also be higher, leading to greater potential protein degradation. Fed batch and bypass should be used only when necessary.

Diafiltration After concentration has removed permeable components from the system, a new buffer can be added to dilute the retained product back to its original volume. Repeated application of this procedure, known as batch diafiltration, is used to exchange one buffer for another and is conveniently implemented at lab scale.

An operating mode known as constant volume diafiltration involves adding a new buffer to the system while withdrawing permeate at the same rate. Figure 20-57 shows the relationship between retentate concentration and diavolumes N = (buffer volumes added)/(fixed retentate volume) for batch and constant volume diafiltration with $S = 1$ (Cheryan, 1998). For a fully passing solute ($S = 1$) such as a buffer, the retentate concentration decays 10-fold with each 2.3 diavolumes. Partially retained solutes do not decay as quickly and require more diavolumes to reach a final concentration target. A fully retained solute ($S = 0$) maintains its retentate concentration constant at the initial value. For fully passing solutes, > 4.6 diavolumes are needed to achieve a specification of < 1 percent of the original buffer

TABLE 20-18 Mass-Transfer Correlations

Geometry	Flow	a	b	c	e	Comments
Tube	Laminar	1.62	0.33	0.33	0.33	Theoretical ¹
	Laminar	1.86	0.33	0.33	0.33	Theoretical ¹
	Turbulent	0.023	0.80	0.33	—	Theoretical ²
	Turbulent	0.023	0.875	0.25	—	Theoretical ³
Spacer	—	0.664	0.50	0.33	0.50	Experimental ⁴
Stirred cell	Laminar	0.23	0.567	0.33	—	Experimental ⁵
Rotating	Laminar	0.75	0.50	0.33	0.42	Experimental ⁶

¹Leveque, *Ann. Mines*, **13**, 201 (1928).

²Gekas et al., *J. Membr. Sci.*, **30**, 153 (1987).

³Deissler in *Advances in Heat and Mass Transfer*, Harnett (ed.), McGraw-Hill, New York, 1961.

⁴DaCosta et al., *J. Membr. Sci.*, **87**, 79 (1994) with L = spacer mesh length/2, $d_h = 4e/[2/h + (1 - e)s]$ for spacer length channel height h , porosity e , and specific area s . See Costa for additional correction k_{dc} .

⁵Smith et al., *Chem. Eng. Prog. Symp. Ser.*, **64**, 45 (1968).

⁶Holeschovsky et al., *AIChE J.*, **37**, 1219 (1991).

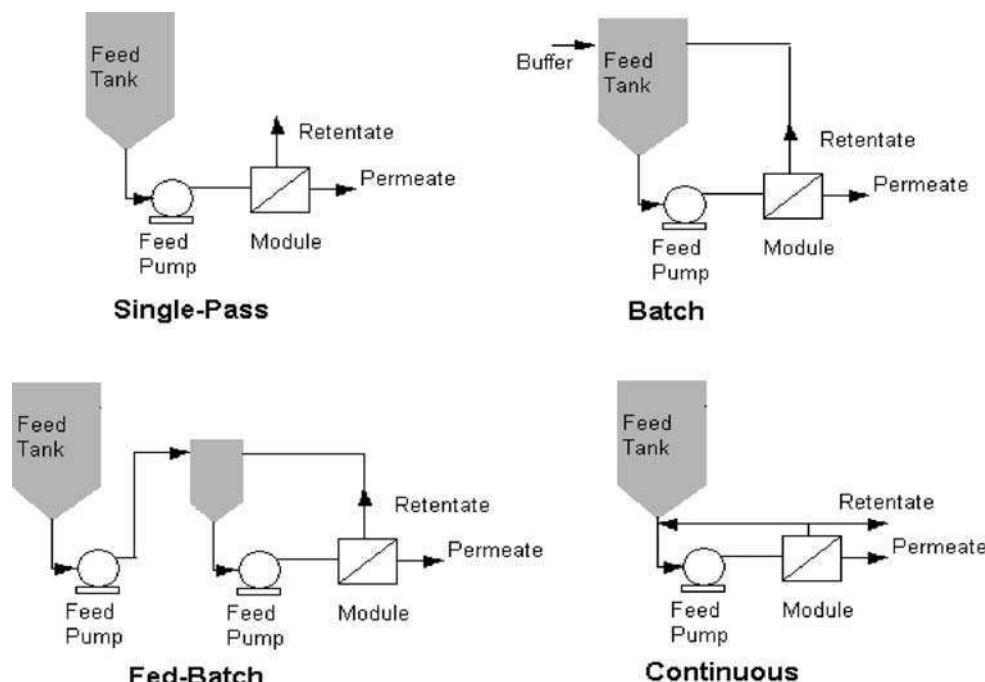


FIG. 20-55 Process configurations.

components. It is common to add an extra 1 to 2 diavolumes as a safety factor to ensure complete buffer exchange. Note that incomplete mixing (due to dead legs and liquid droplets on tank walls) becomes significant at high diavolumes ($N > 10$) and causes the curves in Fig. 20-57 to flatten out.

System sizing involves integration of Eq. (20-69) using a flux model to give Eq. (20-71), where V_p is the permeate volume and J is the average flux. Note the direct tradeoff between area and process time. Table 20-19 shows the concentration and diafiltration steps separated and processing time.

$$A = \frac{V}{Jt} \quad \text{where} \quad \frac{V_p}{Jt} = \int_0^{V_p} \frac{dV}{J} \quad (20-71)$$

The formulas for concentration and diafiltration can be combined for the entire process to derive an expression for the loss of product in the permeate. This loss is shown in Fig. 20-58 and Table 20-19 for

different levels of processing and membrane retention characteristics ($R = 1 - S$). Note that a membrane with 1 percent passage (99 percent retention) can have yield losses much higher than 1 percent because the protein is repeatedly cycled past the membrane during the entire process with losses at every pass. A high-yielding process (< 1 percent loss) requires membrane passage of < 0.1 percent (retention of > 99.9 percent).

The purification of a product p from an impurity i by an ultrafiltration process is shown in Fig. 20-59 and Eq. (20-72), where C_{i0} and C_{p0} are the initial concentrations (g/L) of the two components, Y_p is the yield of product in the retentate, and $\Psi = \text{selectivity} = S_i/S_p$, the ratio of passages. High yields are obtained in purifying out small solutes (high selectivity) but are compromised in removing larger impurities with similar passages to the product.

$$\text{Purification factor PF} = \frac{c_i/c_{i0}}{c_p/c_{p0}} = \left(\frac{\text{ppm in product}}{\text{ppm in feed}} \right) = Y_p^{\Psi-1} \quad (20-72)$$

TABLE 20-19 Batch and Fed-Batch Performance Equations

	Batch concentration and diafiltration	Fed-batch concentration
Retentate concentration	$c_i = c_{i0} X^{1-S_i} e^{-S_i N}$	$c_i = \frac{c_{i0}}{S_i} [1 - (1 - S_i) e^{-S_i(X-1)}]$
Retentate mass	$M_i = M_{i0} e^{-S_i(N+\ln X)}$	$M_i = \frac{M_{i0}}{S_i} [1 - (1 - S_i) e^{-S_i(X-1)}]$
Sizing	$A = \frac{V_0}{t} \left(\frac{1 - 1/X}{J_{\text{conc}}} + \frac{N/X}{J_{\text{diaf}}} \right)$	$A = \frac{V_0}{t} \left(\frac{1 - 1/X}{J_{\text{conc}}} \right)$
Comments	- Simplest system and control scheme - Smaller area or process time - Limited to $X < 40$	- Required for $40 < X < 100$ - Minimize fed-batch ratio (= total V_0 / recycle tank V) to minimize area - Can require larger area and time - Integrate for J or use Cheryan approximation - Test process performance using proposed fed-batch ratio

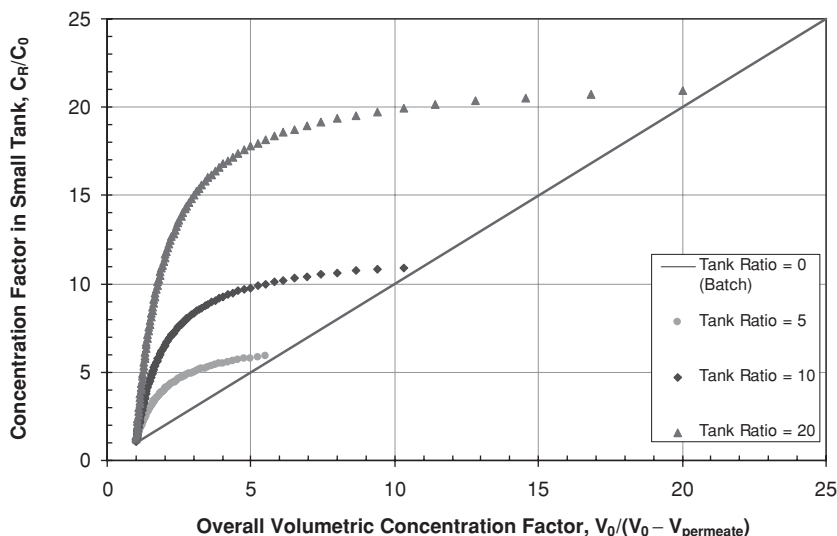


FIG. 20-56 Batch and fed-batch concentration.

Continuous Operation Continuous operation (also called feed-and-bleed) involves the partial recycle of the retentate. The residence time and number of pump passes are in between those of single-pass and batch operation, depending on the fraction of retentate recycled. Several continuous units can be plumbed with the retentate flow feeding the next consecutive system. This configuration is commonly used for large-scale membrane systems.

System Operation NFF system requirements depend on each application but should generally allow for installation, flushing, integrity testing, cleaning or sanitization, processing, recovery, and change-out.

Continued decline in performance indicates a membrane cleaning or compatibility issue. The adequacy of the cleaning step is determined by the recovery of at least 80 percent of the initial normalized water flux. Although some variability in water flux is typical, any consistent decline reflects an inadequate cleaning procedure.

Process Development/Scale-up Feasibility is evaluated by using small single-module systems. This involves selection of membrane type, operating processing conditions, processing strategies (concentration, diafiltration, etc.), cleaning procedures, performance evaluation, and sensitivity to feedstock variations. Pilot operation employs a larger skid, representative of planned commercial operation, running representative feedstock over extended periods to assess longer-term repeatable performance (product quality and retention, costs, yields, process flux, and membrane integrity).

Module change-out is based on preset number of uses, time, or when performance (retention, flux, integrity) drops below preset specifications.

Fluxes scale directly and volumes scale proportionately with the feed volume. A safety factor will be built into the scale-up design so that process times will not be the same.

REVERSE OSMOSIS (RO) AND NANOFILTRATION (NF)

RO and NF processes employ pressure driving forces of 0.3 to 10.5 MPa to drive liquid solvents (primarily water) through membranes while retaining small solutes. Reverse-osmosis (RO) membranes are commonly rated by a > 90 percent NaCl retention and retention of > 50 Da neutral organics while passing smaller liquid solvents. Nanofiltration membranes (sometimes called low-pressure RO, loose RO, or hyperfiltration) have 20 to 80 percent NaCl retention and retain > 200 to 1000 Da neutral organics. Dissolved gases have low retention. Neutral or undissociated solutes have lower retention than charged or dissociated

solutes. Although virus-retaining membranes are sometimes referred to as nanofilters, in keeping with membrane scientist classification they are described under ultrafiltration. RO membranes will remove bacteria and viruses. Lab-scale sample preparation in dilute solutions can be run as NFF in centrifuge tubes. TFF is used in large commercial processes and small home and shipboard water purification systems.

Commercial interest in RO began with the first high-flux, high-NaCl-retention Loeb-Sourirajan anisotropic cellulose acetate membrane. Practical application began with the *thin film composite (TFC)* membrane and implementation for seawater desalination at Jeddah, Saudi Arabia [Muhurji et al., *Desalination*, **76**, 75 (1975)].

Applications RO is primarily used for water purification: seawater desalination (35,000 to 50,000 mg/L salt, 5.6 to 10.5 MPa operation), brackish water treatment (5000 to 10,000 mg/L, 1.4 to 4.2 MPa operation), and *low-pressure RO (LPRO)* (500 mg/L, 0.3 to 1.4 MPa operation). A list of U.S. plants can be found at www2.hawaii.edu, and a 26 Ggal/yr desalination plant is under construction in Ashkelon, Israel. Purified water product is recovered as permeate while the concentrated retentate is discarded as waste. Drinking water specifications of total dissolved solids (TDS) ≤ 500 mg/L are published by the U.S. EPA and of ≤ 1500 mg/L by the WHO [Williams et al., chap. 24 in *Membrane Handbook*, Ho and Sirkar (eds.), Van Nostrand, New York, 1992]. Application of RO to drinking water is summarized in Eisenberg and Middlebrooks (*Reverse Osmosis Treatment of Drinking Water*, Butterworth, Boston, 1986).

RO competes well with multistage flash evaporation in desalination applications and is the current benchmark due to advantages in both capital and operating costs. It may be too costly to compete against freshwater sourced from wells and pipelines. LPRO and electrodialysis compete for brackish water treatment applications. Compact home kitchen RO runs on 0.4-MPa line pressure for < 2000-mg/L TDS feed solutions to reduce salts, odor, and trace metals such as As. These units range from \$200 to \$500 (U.S. 2005) to produce water at \$0.06 to \$0.25 per gallon. RO processes were estimated to represent 23 percent of the 3 Ggal/day of worldwide potable water production in 1986 [Wangnick, *IDA Magazine*, **1**, 53 (1986)].

Useful conversions are as follows:

Volume: $1 \text{ m}^3 = 1000 \text{ L} = 264.2 \text{ U.S. gal} = 8.11 \times 10^{-4} \text{ acre-ft} = 35.32 \text{ ft}^3$
 Pressure: 1 bar = 14.5 psi = 0.1 MPa
 Flow: 1 U.S. gal/min = 1440 gal/day = 227 L/h = 5.47 m^3/day
 Concentration: 1 mg/L = 1 ppm = 17.1 grains/gal

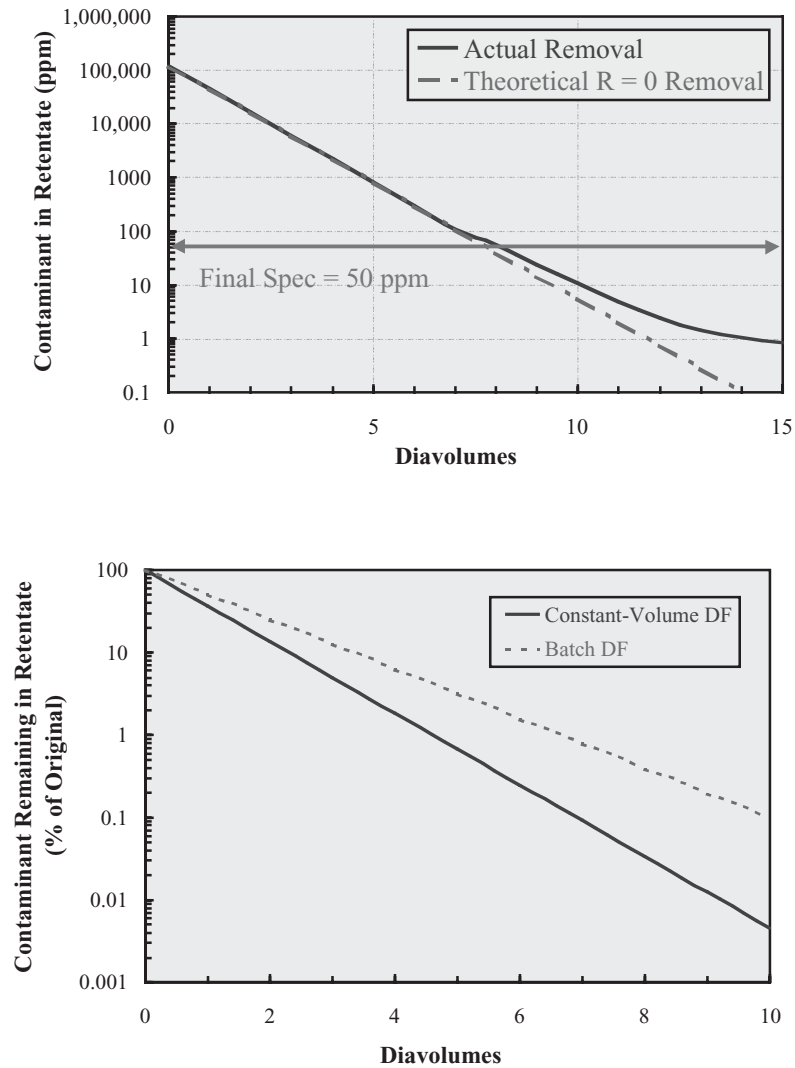


FIG. 20-57 Diafiltration performance (Cheryan, 1998).

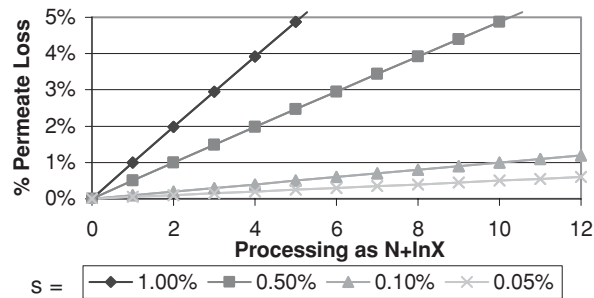


FIG. 20-58 Permeate losses.

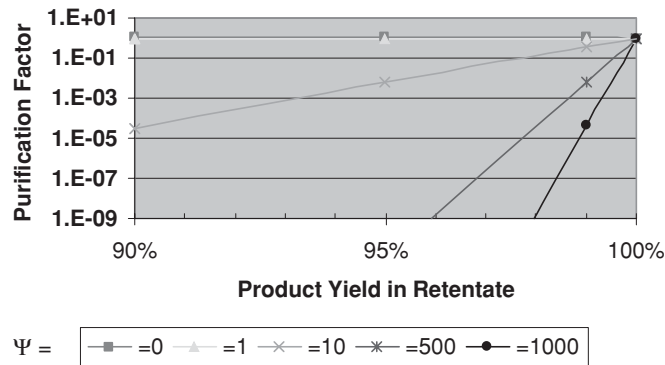


FIG. 20-59 Purification versus yield.

Process water applications include boiler water feed pretreatment before ion exchange or electrodialysis. RO is also used for ultrahigh-purity water production for use in laboratories, medical devices (kidney dialysis), microelectronic manufacturing (rinse fluids per ASTM D-19 D5127-90, 1990), and pharmaceutical manufacturing (purified water or water for injection as specified by USP).

RO applications recovering valuable product in the retentate include tomato, citrus, and apple juice dewatering, sometimes in combination with thermal evaporators at \$0.15 to \$0.18 per gallon [Short, chap. 9 in *Bioseparation Processes in Foods*, Singh and Rezi (eds.), Marcel Dekker, New York, 1995]. Dealcologization of wine or beer involves diafiltering alcohol through RO membranes while retaining flavor compounds in the retentate (Meier, *Wine East*, Nov./Dec., 10, 1991). LPRO is used to desalt soy sauce and dairy whey. Small-molecule drugs (e.g., penicillin) can be recovered by RO membranes.

RO can recover metals, antifreeze, paint, dyes, and oils in the retentate while generating cleaned up wastewater permeate for disposal. RO is also used to reduce the volume of waste liquids (e.g., spent sulfite liquor in paper manufacturing). Wastewater treatment application removals of 95 percent TOC, > 90 percent COD, > 98 percent PAH compounds, and pesticides > 99 percent have been seen [Williams et al., chap. 24 in *Membrane Handbook*, Sirkar and Ho (eds.), Van Nostrand, 1992].

Membranes, Modules, and Systems

Membranes RO membranes are designed for high salt retention, high permeability, mechanical robustness (to allow module fabrication and withstand operating conditions), chemical robustness (to fabrication materials, process fluids, cleaners, and sanitizers), low extractables, low fouling characteristics, high capacity, low cost, and consistency.

The predominant RO membranes used in water applications include cellulose polymers, *thin film composites (TFCs)* consisting of aromatic polyamides, and crosslinked polyetherurea. Cellulosic membranes are formed by immersion casting of 30 to 40 percent polymer lacquers on a web immersed in water. These lacquers include cellulose acetate, triacetate, and acetate-butyrate. TFCs are formed by interfacial polymerization that involves coating a microporous membrane substrate with an aqueous prepolymer solution and immersing in a water-immiscible solvent containing a reactant [Petersen, *J. Membr. Sci.*, **83**, 81 (1993)]. The Dow FilmTec FT-30 membrane developed by Cadotte uses 1-3 diaminobenzene prepolymer crosslinked with 1-3 and 1-4 benzenedicarboxylic acid chlorides. These membranes have NaCl retention and water permeability claims.

Cellulose membranes can generally tolerate a pH of 3 to 6 and 0.3 to 1.0 ppm of chlorine while TFC membranes can generally tolerate a pH of 3 to 11 and < 0.05 ppm of chlorine. Membrane tolerance is also described as the permitted cumulative ppm-hours of membrane exposure to chlorine. TFC membranes can range from 1000 to 12,000 ppm-h. Always check specific filter specifications. RO membranes

approved for food use by the FDA include aromatic polyamides, polypiperazinamide, and aryl-alkyl polyetherurea (Code of Federal Regulations, Title 21, Part 177, U.S. GPO, Washington, D.C., annually).

Nanofiltration membranes are negatively charged and reject multivalent anions at a much higher level than monovalent anions, an effect described as Donnan exclusion. Nanofiltration membranes have MgSO_4 retention and water permeability claims.

Modules RO modules are available in spiral, hollow fiber, tubular, and plate and frame formats. A comparison of characteristics is shown in Table 20-20. Spirals are used the most since they are compact, easy-to-use modules; have low feed-side pressure drops; are less prone to clogging; and are easily cleaned, mechanically robust, and economical. The standard spiral module has an 8-in diameter, is 40 in long with about 30-m^2 area. An anti-telescoping end cap is used to prevent the leaves from being pushed out of alignment by axial retentate flow. Hollow fibers are run with the feed on the shell side, flowing radially inward and out along the lumen. This places less mechanical stress on the fibers. Tubular modules are used with high-suspended-solids and high-viscosity fluids.

System Configurations Figure 20-60 shows common RO system configurations for permeate product recovery. For single-pass operation, the feed passes once through a membrane to generate acceptable permeate product. Double-pass operation uses permeate staging (the permeate from the first module feeds the second module) with allowance for blending to generate acceptable product. This configuration may be needed for high-concentration, hard-to-remove impurities, high product purities, and lower-retention modules. Double-pass operation has been recommended for desalination of > 38,000 mg/L [Klinko et al., *Desalination*, **54**, 3 (1985)] or modules with salt retention < 99.2 percent [Kaschemekat et al., *Desalination*, **46**, 151 (1983)]. It is also common in ultrahigh-purity water systems.

Spirals are normally packed at 5 to 7 modules per cylindrical housing and operate with staged retentates where the retentate from the first module feeds the second module. Retentate staging increases the length of the feed channel. Retentate staging of housings is used to increase the conversion ratio from around 25 to 50 percent in one housing stage to 50 to 75 percent for two housing stages and 75 to 90 percent in three housing stages. Since the feed flow is reduced by the

TABLE 20-20 RO Module Performance

	Spiral	Hollow fiber	Tubular	Plate and frame
Packing density, m^2/m^3	800	6000	70	500
Feed flow, $\text{m}^3/(\text{m}^2\cdot\text{s})$	0.3–0.5	0.005	1–5	0.3–0.5
Feed ΔP , psi	40–90	1.5–4.5	30–45	45–90
Plugging sensitivity	High	High	Low	Moderate
Cleanability	Fair	Poor	Excellent	Good
Relative cost	Low	Low	High	High
Chemistries available	C and TFC	C and TFC	C only	C and TFC

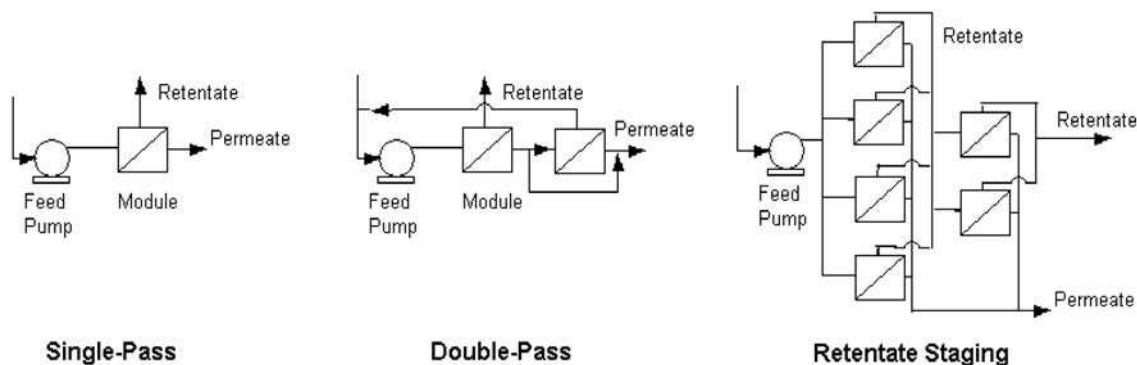


FIG. 20-60 RO system configurations.

conversion ratio after passing through the first module stage, maintaining a minimum crossflow rate requires that the second module stage have less membrane area in what has been called a “Christmas tree” design. Interstage pumping between housings may be required to maintain pressures and flows.

Retentate product recovery employs the batch, fed-batch, or continuous processing systems, and concentration or diafiltration modes of operation (see general membrane section). Beer and wine dealcoholization uses batch diafiltration.

Component Transport in Membranes RO is analyzed as a rate, not equilibrium, process. Polarization of retained salts and other low-MW components is described in the general membrane section. Solvent flux J_w and solute flux J_i through RO membranes can be described by using Eq. (20-73), the solution diffusion imperfection model [Sherwood et al., *Ind. Eng. Chem. Fund.*, **6**, 2 (1967)]. This model considers transport through the membrane in parallel with transport through a small number of nonretentive defects. Transport of solvent or solute through the membrane occurs by partitioning between the upstream polarization layer and membrane phases at the membrane surface, diffusion through the membrane, and partitioning between the membrane and downstream fluid. This process is reflected in permeabilities P_w and P_2 . Dissolution in the membrane and diffusion through it can be modeled as an activated process with an associated inverse exponential temperature dependence of the permeabilities. Transport of solvent or solute through defects occurs by convective flow. Water viscosity increases roughly 2 percent/°C near 25°C.

$$J_w = \frac{P_w}{l} (\Delta P - \Delta \pi) + \frac{P_3}{l} \Delta P \quad J_i = \frac{P_2}{l} (c_w - c_p) + \frac{P_3}{l} \Delta P c_w \quad (20-73)$$

In accordance with observed data, this model shows that water flux J_w increases linearly with applied pressure ΔP , decreases with higher salt concentration through its impact on osmotic pressure π , increases with a smaller membrane thickness l , and increases with temperature through the temperature dependence of the water permeability P_w . The model also demonstrates that the solute or salt flux J_i increases linearly with applied pressure ΔP , increases with higher salt concentration c_w , increases with a smaller membrane thickness l , and increases with temperature through the temperature dependence of the solute permeability P_2 . Polarization, as described early in this section, causes the wall concentration c_w to exceed the bulk concentration c_b .

Osmotic Pressure The osmotic pressure π of salt solutions is calculated from

$$\pi = -\frac{RT}{V_w} \ln a_w = \frac{RTM_w}{1000V_w} \sum_i m_i \phi_i \quad (20-74)$$

for water activity a_w , water partial molal volume $\bar{V}_w$, gas constant R , absolute temperature T , water molecular weight M_w , ion molality m_i , and osmotic coefficient ϕ_i for each ionic species i . Note that both Na^+

and Cl^- are counted as separate species for NaCl. For seawater at 35,000 to 50,000 mg/L (ppm) and 25°C, the osmotic pressure is in the range of 2.7 to 3.8 MPa (~11 kPa/1000 ppm).

Solute Retention Retention is determined by the relative rates of solute and solvent transport through the membrane. The impact of operation on solute retention $R_i = 1 - c_p/c_w$ can be evaluated from Eq. (20-73) by using the mass balance $J_i = c_p J_w$. Assuming high retention, $c_p \ll c_w$, and the solvent flux through defects is negligible, so the retention is calculated as

$$R_i = 1 - \frac{J_i}{J_w c_w} = 1 - \frac{P_2 + P_3 \Delta P}{P_w (\Delta P - \Delta \pi)} \quad (20-75)$$

At low pressures where $\Delta P \sim \Delta \pi$, solvent flux is low but solute flux is high (it is insensitive to pressure and occurs under a concentration gradient). This results in low solute retention. Pressures of 1 MPa over the feed osmotic pressure are needed to maintain high retention. Increasing the applied pressure ΔP increases the solvent flux, diluting the solute flux with solvent and resulting in a higher retention. Increasing the feed solute concentration lowers the solvent flux through its impact on osmotic pressure π , and decreases retention. Increasing temperature increases the solvent permeability P_w , the solute permeability P_2 , and the osmotic pressure. Solute retention tends to drop slightly at higher temperature.

Ionic solutions require charge neutrality to prevent a higher flux of species of one charge causing an associated electric current and voltage. This may limit the net salt flux through the membrane by the ion with the lowest permeability so that the sodium ion will show different retentions depending on the anion it is paired with. Charge neutrality can also enrich the permeate concentration of a salt relative to the feed to create a negative retention. For example, a feed solution containing Na^+ , Li^+ , Cl^- , and SO_4^{2-} ions would show a permeate with enriched levels of Cl^- to balance permeated Na^+ and Li^+ ions.

Nanofilters incorporate negative membrane charge for higher anion rejection. High feed salinities can passivate these charges and reduce anion retention.

Pretreatment and Cleaning Pretreatment is commonly used to extend membrane life and increase recovery. The representative pretreatment train for water purification applications in Fig. 20-61 controls feed channel clogging, mineral scaling, fouling by organic films and microorganisms, and oxidants that can degrade the membranes.

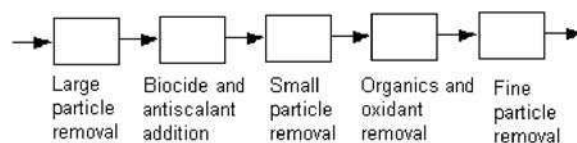


FIG. 20-61 RO pretreatment schematic.

TABLE 20-21 SDI Interpretation

SDI value	Operability
<1	Can run several years without colloidal fouling
1-3	Can run several months between cleaning
3-5	Frequent cleaning required
>5	Pretreatment required to reduce colloids

Particle Removal Solid particles can plug feed channel spacer channels and deposit as a cake layer on the membrane surface, causing increased flow resistance and lowering fluxes. Table 20-19 indicates the sensitivity of different module types to feed channel plugging. Particles may consist of organic colloids, rust flakes, precipitated iron hydroxide, algae, clay, colloidal silica, or silt. Pretreatment includes using a combination of coarse screening, hydrocyclones, flocculent addition (e.g., alum, ferric sulfate, polyelectrolytes), a sand or multimedia filtration bed with periodic backflushing, and depth filter cartridges or ultrafiltration systems.

The ability of a feed solution to plug a membrane is determined by the *silt density index* (SDI) employing a 0.45- μ m PVDF microfilter at 30-psi differential pressure (ASTM Standard D-4189-82, 1987).

$$SDI = \frac{100(1 - T_f/T_i)}{15} \quad (20-76)$$

where T_i is the time it takes to filter the first 0.5 L and T_f is the time it takes to filter 0.5 L starting when 15 min of test time has elapsed. Table 20-21 is an interpretation of SDI values.

Antiscalants Precipitation of salts can form a scale on the membrane surface when solubility limits (Table 20-22) are exceeded in the retentate. This decreases flux.

Metal hydroxides (e.g., Fe, Mn, Al) can also be a problem (Rautenbach and Albrecht, *Membrane Processes*, Wiley, New York, 1989). A chemical analysis of the feed solution composition along with consideration of solubility products allows one to determine the significance of precipitation. Solubility products can be affected by temperature, pH, and ionic strength. Seasonal temperature variations must be considered. Concentrations of silica need to be < 120 mg/L in the feed.

Water hardness in milligrams per liter of total Ca^{2+} and Mg^{2+} is considered soft at 0 to 17 mg/L, moderately hard at 60 to 120 mg/L, and very hard at > 180 mg/L. The Langelier saturation index, a measure of $CaCO_3$ solubility, should be < 0 in the retentate to prevent precipitation (*Pure Water Handbook*, Osmonics, 1997).

Scale prevention methods include operating at low conversion and chemical pretreatment. Acid injection to convert CO_3^{2-} to CO_2 is commonly used, but cellulosic membranes require operation at pH 4 to 7 to prevent hydrolysis. Sulfuric acid is commonly used at a dosing of 0.24 mg/L while hydrochloric acid is to be avoided to minimize corrosion. Acid addition will precipitate aluminum hydroxide. Water softening upstream of the RO by using lime and sodium zeolites will precipitate calcium and magnesium hydroxides and entrap some silica. Antiscalant compounds such as sodium hexametaphosphate, EDTA, and polymers are also commonly added to encapsulate potential precipitants. Oxidant addition precipitates metal oxides for particle removal (converting soluble ferrous Fe^{2+} ions to insoluble ferric Fe^{3+} ions).

Oxidant Removal The presence of oxidizers such as chlorine or ozone can degrade polyamide RO membranes, causing a drop in salt retention. Cellulosic membranes are less sensitive to attack. Addition of 1.5 to 6 mg sodium bisulfite/ppm chlorine or contacting with activated carbon will remove oxidizers. Vacuum degassing with a hydrophobic filter module is also used.

Biocides Microorganisms can form biofilms on membrane surfaces, causing lower fluxes [Flemming in *Reverse Osmosis*, Z. Amjad (ed.), Van Nostrand, New York, 1993, p. 163]. Cellulosic membranes

can be consumed by bacteria, causing lower retention. Control involves pretreatment with chemicals or UV light and membrane chemical sanitization on a regular or periodic basis. Chemical sanitants include chlorine, bromine, ozone and peroxide/peracetic acid oxidants (subject to membrane limits), formaldehyde, sodium bisulfite, copper sulfate for algae and plankton control, or chloramines. The use of oxidizers will also remove dissolved hydrogen sulfide. Cellulosic membranes are commonly sanitized with 0.2 mg/L of free chlorine.

Organic Fouls Organic components can range from 1 mg/L in the open ocean to 80 to 100 mg/L in coastal waters. Components include humic and fulvic acids (800 to 50,000 kDa) from decayed vegetation, or contaminating petroleum that can adsorb to the membranes and reduce their flux. Carbon beds used for dechlorination can also remove these organics by adsorption. For wastewater purification to remove organics, high pH and ozonation improve retention and reduce flux decline (Bhattacharyya et al., *Separation of Hazardous Organics by LPRO—Phase II*, U.S. EPA, 1990).

Cleaning/Sanitization Membrane cleaning restores initial membrane flux without compromising retention to extend membrane life. Appropriate treatment depends on the nature of the foulants, membrane compatibility, and compatibility of the chemicals with the application. Cleaning generally uses acid flushing (for scalants) but may include base or enzyme flushing (for organics) and 80°C hot water flushing (for sanitization/microorganism control).

Design Considerations and Economics

Osmotic Pinch As the feed travels along the feed channels, friction losses lower the feed pressure. Solvent flux along the feed channel dewater the feed and increases solute concentration and osmotic pressure. For a recovery or conversion ratio CR, a module solute retention R , and an inlet solute concentration of c_o , the bulk concentration c_b is

$$c_b = c_o(1 - CR)^{-R} \quad (20-77)$$

Both the feed pressure drop and the osmotic pressure increase lower the driving force for solvent flow $\Delta P - \Delta \pi$ along the feed channel, and the point at the end of the feed channel, having the lowest driving force, is the *osmotic pinch point*. This point has both the lowest solvent flux and the lowest solute retention. If the driving force falls below 0 along the feed channel, the solvent flux will reverse (from permeate into the feed) and subsequent membrane area will perform "negative work." Economic justification of membrane at the pinch point requires that it generate adequate flux to pay for itself. This, in turn, requires that the driving force be about 1 MPa.

Process Economics and Optimization Table 20-23 shows some representative process economics for several large 1 Mgal/day system designs [adapted from Ray, chap. 25 in *Membrane Handbook*, Ho and Sirkar (eds.), Van Nostrand, New York, 1992]. These costs are a useful guide, within ± 25 percent, but specific quotes are needed to account for site-specific differences between projects. No energy recovery technologies are included. For capacities above 50 to 100 gal/day, system costs scale almost linearly, as (gal/day capacity)^{0.85-0.95}, and can be conveniently expressed as dollars per gallon per day. Total costs are split evenly between capital and operating costs.

Energy use (pumping high flows to high pressures) represents the biggest operating cost while membrane replacement operating costs are only 10 to 15 percent. These operating costs can be represented as

$$\begin{aligned} \$E (\$/1000 \text{ gal}) &= (0.724 \text{ kWh/psig} \cdot 1000 \text{ gal}) \\ &\times (\Delta P \text{ psid}) (\$/\text{kWh}) / CR \cdot Ef \end{aligned} \quad (20-78)$$

where CR is the fractional conversion ratio (recovery), Ef is the pump efficiency, and

$$\$A (\$/1000 \text{ gal}) = (0.432) (\$/\text{m}^2) / CR \cdot J \cdot T \quad (20-79)$$

where J is the average system flux, L/(m²·h), and T is the years of membrane life.

By incorporating a flux model [Eq. (20-73)], solute concentration [Eq. (20-77)], and polarization model [Eq. (20-60)] into the area cost

TABLE 20-22 Solubility Limits at 30°C

CaCO ₃	MgSO ₄	CaSO ₄	BaSO ₄	SiO ₂	SrSO ₄
65 mg/L	9 mg/L	2090 mg/L	2 mg/L	120 mg/L	114 mg/L

TABLE 20-23 Representative RO Process Costs

Costs	Seawater	Brackish water	Softening	Notes
Operating conditions				
Inlet pressure	6.9 MPa	3.1 MPa	1.4 MPa	
Flux	25 LMH*	42 LMH	42 LMH	
Conversion	40%	60%	75%	
Total cost, \$/1000 gal	4.7	2.3	1.3	15-yr life
Capital cost	2.1	1.1	0.6	12% interest
Operating cost	2.6	1.2	0.7	15% downtime
Total capital cost, \$/(gal/day)	4.5	2.3	1.3	
Direct costs	3.7	1.9	1.1	
Equipment	3.3	1.6	0.8	
Indirect costs	0.8	0.4	0.2	
Total operating cost, \$/1000 gal	2.6	1.2	0.7	
Energy	1.6	0.4	0.2	\$0.075/kWh
Membrane replacement	0.4	0.2	0.2	3-yr life
Chemicals	0.2	0.2	0.1	2 per shift @ \$14/h
Labor	0.3	0.3	0.2	
Other	0.1	0.1	0.0	

*LMH = liters per square meter per hour.

[Eq. (20-79)], the operating cost term ($\$A + \E) varies with ΔP and CR. The osmotic pinch constraint requires a minimum ΔP to overcome the average osmotic pressure and requires a maximum CR where the concentrated retentate osmotic pressure exceeds the feed pressure. Assuming that solute retention is sufficient to maintain permeate quality, there exist optimum conversion ratios and ΔP that depend on the feed (osmotic pressure), module (mass transfer, permeability, life, cost), and pump (energy cost, efficiency). Conversion ratios per module stage range from 25 to 50 percent, and typical inlet pressures are shown in Table 20-23. No provision has been made here for costs of disposing of the retentate.

One can also rearrange Eq. (20-79) to solve for ΔP in terms of area and substitute in Eq. (20-78) to express the operating costs in terms of membrane area. The energy operating cost is inverse in area, while the membrane replacement cost is linear, reflecting the existence of an optimum area. Addition of capital costs leads to total costs that scale as $b(\text{area}^{0.95}) + c(\text{area}^{-0.6})$. Module vendors provide good general recommendations and support to design and economic evaluation.

Recognizing that the high-pressure retentate contains considerable kinetic energy before it is passed through a retentate valve provides an opportunity for energy recycling. A variety of energy recovery technologies have been proposed including Pelton wheels, reverse-running pumps, hydraulic turbochargers, and pressure exchangers with pumping energy reductions of 22 to 60 percent [Darwish et al., *Desalination*, **75**, 143 (1989)]. Energy recovery technologies are most useful for high-pressure desalination but may not be worth the investment for brackish water and LPRO applications [Glueckstein et al., chap. 12 in *Synthetic Membrane Processes*, Academic Press, New York, 1984].

The capital cost of a 15 kgal/day RO skid, uninstalled with instrumentation, is about \$35,000 (U.S., 2005) and scales with gallons-per-day capacity to the 0.52 power [Ray, chap. 25 in *Membrane Handbook*, Sirkar and Ho (eds.), Van Nostrand, New York, 1992]. Approximately 25 to 30 percent of this cost is associated with membranes and housings. Operating costs are roughly \$0.30/1000 gal for energy and membrane replacement, and \$0.09/1000 gal for pretreatment, maintenance, and cleaning. High-purity water systems employing double-pass staging operate at conversions of 70 to 80 percent in the first pass and 70 to 75 percent in the second pass. Home units using line pressure run at 5 to 15 percent conversion.

System Operation For corrosive pressurized seawater, 316L stainless steel is commonly used for piping and housings although specialized alloys may be needed [Oldfield et al., *Desalination*, **55**, 261 (1985)]. Sensors are used to measure inlet pH and temperature (to control scaling), pump pressure (to prevent surges and ensure sufficient driving force), pump flow (to ensure crossflow to prevent excessive polarization), permeate flow (to monitor performance), and permeate conductivity (to ensure module integrity and product

quality). Control systems are used to monitor and display sensor values, track and optimize operation over time, trigger alarms, and store the data. Provisions for module replacement and start-up are needed.

For batch processing, systems comparable to ultrafiltration are used.

Posttreatment of the permeate for potable water use can include dissolved CO₂ removal to prevent corrosion (by aeration, lime treatment), chlorination for microbial control, and oxygenation to improve taste.

ULTRAFILTRATION

Ultrafiltration processes (commonly UF or UF/DF) employ pressure driving forces of 0.2 to 1.0 MPa to drive liquid solvents (primarily water) and small solutes through membranes while retaining solutes of 10 to 1000 Å diameter (roughly 300 to 1000 kDa). Commercial operation is almost exclusively run as TFF with water treatment applications run as NFF. Virus-retaining filters are on the most open end of UF and can be run as NFF or TFF. Small-scale sample preparation in dilute solutions can be run as NFF in centrifuge tubes.

The first large commercial application of UF was paint recycling, followed by dairy whey recovery in the mid-1970s. UF applications are enabled by low-temperature and low-cost operation.

Applications Electrodeposition of cationic paint resin on automobiles (connected to the cathode) provides a uniform, defect-free coating with high corrosion resistance, but carries with it about 50 percent excess paint that must be washed off. UF is used to maintain the paint concentration in the paint bath while generating a permeate that is used for washing. The spent wash is fed back into the paint path [Zeman et al., *Microfiltration and Ultrafiltration*, Marcel Dekker, New York, 1996].

During cheese making, the coagulated milk or curd is used to make cheese while the supernatant whey is a waste product rich in salts, proteins, and lactose. Whey concentration and desalting by UF produce a retentate product that can be used as an animal feed supplement or food additive. The MMV process (Mauvois et al., French Patent 2,052,121) involves concentrating the milk by UF after centrifugation to remove the cream and before coagulation to improve yields and reduce disposal costs.

UF is used in biopharmaceutical purification (proteins, viral and bacterial vaccines, nucleic acids) for initial concentration of clarified fluids or lysates (to reduce subsequent column sizes and increasing binding), to change buffer conditions (for loading on to columns), and to concentrate and change buffers (for final formulation into a storage buffer) [Lutz et al., in *Process Scale Bioseparations for the Biopharmaceutical Industry*, Shukla, et al. (eds.), CRC Press, New York, 2006]. For these applications, the drug product is recovered in the retentate. For drug products with potential viral contaminants

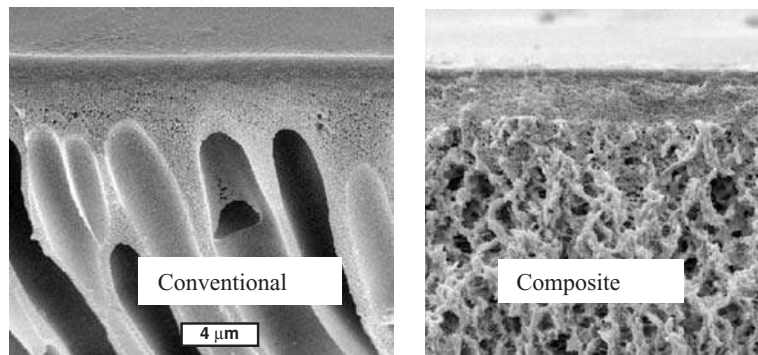


FIG. 20-62 Ultrafiltration membranes. (Courtesy Millipore Corporation.)

(sourced from mammalian cells or human and animal blood plasma), virus removal filters employed in either a TFF or NFF mode retain the viruses while the drug product is recovered in the permeate.

UF is used to clarify various fruit juices (apple, grape, pear, pineapple, cranberry, orange, lemon) which are recovered as the permeate [Blanck et al., *AIChE Symp. Ser.* **82**, 59 (1986)]. UF has also been used to remove pigments and reducing browning in wine production [Kosikowski in *Membrane Separations in Biotechnology*, McGregor (ed.), Marcel Dekker, New York, 1986].

Wastewater treatment and water purification applications employ UF in a TFF or NFF mode to produce permeate product with reduced colloids, pyrogens, and viruses. Oil droplets in wastewater are retained by UF for recycle or disposal at a significantly reduced volume.

Membranes, Modules, and Systems

Membranes UF membranes consist primarily of polymeric structures (polyethersulfone, regenerated cellulose, polysulfone, polyamide, polyacrylonitrile, or various fluoropolymers) formed by immersion casting on a web or as a composite on a MF membrane. Hydrophobic polymers are surface-modified to render them hydrophilic and thereby reduce fouling, reduce product losses, and increase flux [Cabasso in *Ultrafiltration Membranes and Applications*, Cooper (ed.), Plenum Press, New York, 1980]. Some inorganic UF membranes (alumina, glass, zirconia) are available but only find use in corrosive applications due to their high cost.

Early ultrafiltration membranes had thin surface retentive layers with an open structure underneath, as shown in Fig. 20-62. These membranes were prone to defects and showed poor retention and consistency. In part, retention by these membranes would rely on large retained components in the feed that polarize or form a cake layer that plugs defects. Composite membranes have a thin retentive layer cast on top of a microfiltration membrane in one piece. These composites demonstrate consistently high retention and can be integrity-tested by using air diffusion in water.

Table 20-24 compares properties of commonly used polyethersulfone (PES) or regenerated cellulose membranes. Membrane selection is based on experience with vendors, molecular weight rating for high

yields, chemical and mechanical robustness during product processing and clean-in-place (extractables, adsorption, swelling, shedding, class VI), process flux for sizing and costing, and the quality/consistency (ISO, cGMP) of the vendor and the membrane. Membranes are described by their normal water permeability (NWP), the ratio of their flux at 25°C to the average transmembrane pressure (TMP). Measurements taken at different temperatures can be adjusted to 25°C using a viscosity ratio correction. Process fluxes will be somewhat lower due to fouling. Membranes containing a hydrophilic surface generally have lower fouling and produce higher process fluxes.

UF membranes are assigned a nominal molecular weight limit (NMWL) corresponding roughly to their ability to retain a solute of a particular molecular weight. While tight membranes (i.e., low-molecular-weight cutoff ratings) provide high retention, they have corresponding low flow rates requiring greater membrane area to do the job. This leads to larger pumps with large holdup volumes and potential negative impact on product quality. A rule of thumb for selecting membrane NMWL is to take 0.2 to 0.3 of the MW of the solute that is to be retained by the membrane. However, the NMWL values should only be used as a very rough guide or screening method to select membranes for testing. Retention is based on hydrodynamic size, not molecular weight, so linear chain dextrans show a higher passage than globular proteins of the same molecular weight. In addition, there is no standardization of rating methods so specific retention properties vary considerably among membranes and vendors depending on the marker solute selected (e.g., protein, dextran in a particular buffer) and the level of retention selected for the marker solute (90, 95, 99 percent). Retention is affected by fouling due to adsorbed components and polarized solutes on the membrane surface. See retention below for additional discussion.

Modules and Systems UF modules include cassettes, spirals, hollow fibers, tubes, flat sheets, and inorganic monoliths. These are primarily run in TFF operation to increase flux by reducing plugging. Exceptions run in NFF operation include virus removal and water treatment applications where both cartridges and hollow fibers are used. Continuous operation provides lower-cost operation, but the biologicals process requires batch processing to meet regulatory requirements. Applications with high solids require multiple-pass operation to obtain a significant conversion ratio, but water treatment

TABLE 20-24 Ultrafiltration Membrane Properties

Material	Advantages	Disadvantages
Polyethersulfone (PES)	Resistance to temperature, Cl ₂ , pH, easy fabrication	Hydrophobic
Regenerated cellulose	Hydrophilic, low-fouling	Sensitive to temperature, pH, Cl ₂ , microbial attack, mechanical creep
Polyamide		Sensitive to Cl ₂ , microbial attack
Inorganic	Resistance to temperature, Cl ₂ , pH, high pressure, long life	Cost, brittleness, high crossflow rates

TABLE 20-25 UF Modules and Systems

Application	Paint recovery	Whey processing	Biologicals	Water treatment	Juice processing
Membrane	PES	PES	Cellulose, PES, PVDF	PES	PES, PVDF
Module	Fibers	Spirals	Cassette	Fibers	Tubes
System	Continuous	Continuous	Batch, NFF	Continuous, single-pass	Batch
Characteristics	Cost, plugging	Cost	Recovery	Cost	Plugging, cost

can achieve high conversion in one pass. The properties of these modules and system operation are described in the general section. Table 20-25 indicates where these modules are used.

Component Transport in Membranes UF is analyzed as a rate, not equilibrium, process. Pure solvent transport in UF is characterized by $J = \text{TMP}/R_{\text{membrane}}$, where TMP (transmembrane pressure) is the pressure difference across the membrane and R is the membrane resistance. Resistance R can be related to membrane properties by using Eq. (20-64) with pore size distributions following a lognormal distribution [Zydney et al., *J. Membr. Sci.*, **91**, 293 (1994)]. At ionic strengths $< 0.1 M$, charged membranes generate a streaming potential across their thickness which can lead to a 15 percent or more diminished solvent flux by counterelectroosmosis (Newman, *Electrochemical Systems*, Prentice-Hall, Englewood Cliffs, N.J., 1973). Most UF process fluids are aqueous with higher ionic strengths, but permeability measurements using low-conductivity water can encounter this effect.

Figure 20-63 shows that in the presence of retained solutes, the solvent flux flattens out at high pressures. UF has a high level of polarization where the membrane surface concentration can range up to 100 times the bulk concentration. Retained antibodies at a wall concentration of 191 g/L have an osmotic pressure Π of 30 psi (Mitra, 1978). That is, an elevated pressure of 30 psig must be applied to the protein-rich retentate side of a water-permeable membrane containing 191 g/L of antibody in order to prevent water backflow from the permeate side of the membrane containing water at 0 psig. This diminishes the driving force for flow and leads to the mechanistic-based osmotic flux model (Vilker, 1984):

$$\text{Polarization flux model } J = \frac{\text{TMP} - R \cdot \Delta\Pi(c_w)}{\mu \cdot (R_{\text{membrane}} + R_{\text{fouling}})} \quad (20-80)$$

where R is the intrinsic membrane retention and $\Delta\Pi(c_w)$ is the osmotic pressure at the wall concentration. An empirically based flux model can also be defined by omitting the osmotic term and adding a compressible polarization resistance term in the denominator (Cheryan, 1998).

Equation (20-80) requires a mass transfer coefficient k to calculate c_w and a relation between protein concentration and osmotic pressure. Pure water flux obtained from a plot of flux versus pressure is used to calculate membrane resistance (typically small). The LMH/psi slope is referred to as the NWP (normal water permeability). The membrane plus fouling resistances are determined after removing the reversible polarization layer through a buffer flush. To illustrate the components of the osmotic flux model, Fig. 20-63 shows flux versus TMP curves corresponding to just the membrane in buffer ($R_{\text{fouling}} = 0$, $c_w = 0$), fouled membrane in buffer ($c_w = 0$), and fouled membrane with osmotic pressure.

The region at low flux/low TMP is called the linear region and is dependent on TMP but independent of crossflow and bulk concentration. The region at high flux/high TMP is called the polarized region and is independent of TMP but dependent on crossflow and bulk concentration. This extremely counterintuitive result is the consequence of polarization. In between these two regions lies what is termed the "knee" of the flux curve. Increasing pressure beyond the knee gives diminishing returns in improving flux.

In the polarized region, the empirical gel model of Eq. (20-81) can provide a good description of the flux behavior. The gel model is based on polarization [Eq. (20-58)] with a constant wall concentration C_g and complete retention. The mass-transfer coefficient k is dependent on crossflow velocity, as shown in Table 20-18 for different geometries and flow regimes. Concentration-dependent transport (viscosity and diffusivity) will alter the velocity exponents. Operating at the knee of the flux curve, either by design or due to fouling that causes a shift in the flux curve to the right, will decrease the velocity dependence. Note that the empirical gel point concentration does not correspond to an actual gel on the membrane surface. As the feed concentration rises to within 5 to 10 percent of the gel point, the flux tends to drop off less rapidly than Eq. (20-81) would predict. The gel point is insensitive to crossflow but can increase with temperature. Cheryan summarizes gel point and velocity exponent values for a variety of solutions.

$$J = k \ln c_g/c \quad (20-81)$$

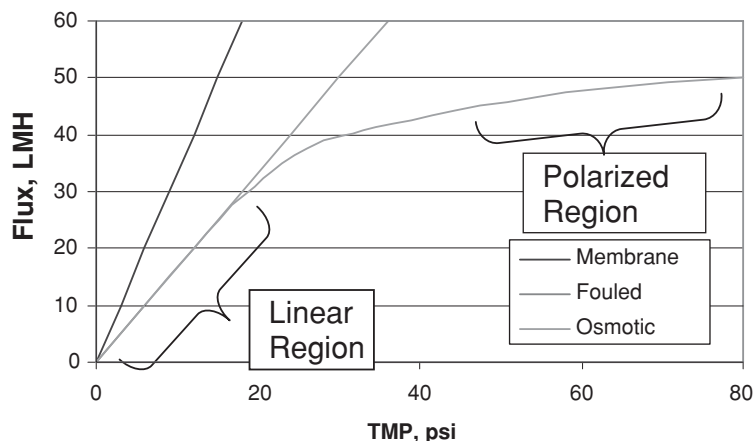


FIG. 20-63 Ultrafiltration flux behavior.

Solute Flux Solute partitioning between the upstream polarization layer and the solvent-filled membrane pores can be modeled by considering a spherical solute and a cylindrical pore. The equilibrium partition coefficient ϕ (pore/bulk concentration ratio) for steric exclusion (no long-range ionic or other interactions) can be written as

$$\text{Solute partitioning} \quad \phi = (1 - \lambda)^2 \quad (20-82)$$

where λ = solute radius/pore radius.

Solute flux within a pore can be modeled as the sum of hindered convection and hindered diffusion [Deen, *AIChE J.*, **33**, 1409 (1987)]. Diffusive transport is seen in dialysis and system start-up but is negligible for commercially practical operation. The steady-state solute convective flux in the pore is $J_s = K_c J_c = \phi K_c J_{c_w}$, where c is the radially averaged solute concentration and K_c is the convective hindrance factor (from pore wall drag, solute lag, and long-range solute pore wall and solute-solute potential interactions). For uncharged cylindrical pores $\phi K_c = \phi(2 - \phi) \exp(-0.7146\lambda^2)$ to within 2 percent [Zeman et al., in *Synthetic Membranes*, vol. 2, Turbak (ed.), *ACS Symp. Ser.*, **54**, ACS, Washington, D.C. (1981), p. 412].

A charged solute in a pore has a higher potential energy due to the distortion of its electrical field when it penetrates the pore wall into the polymer. This effect increases solute retention. The apparent size of solutes can be magnified by more than an order of magnitude by this effect. See Deen for other pore geometries and the impacts of long-range solute pore wall and solute-solute interactions. The use of electrostatic interactions to separate solutes is discussed by van Reis [*Biotechn. and Bioeng.*, **56**(1), 71 (1997)].

Solute Passage/Retention The intrinsic passage (c_p/c_w) is determined by the ratio of the solute flux to the solvent flux as $s_i = \phi K_c$. The intrinsic passage is inherent to the membrane and solute. As shown in Fig. 20-64 for a single pore, solute passage (or retention) does not follow a step change. Figure 20-65 shows passage for a membrane with a pore size distribution [Tkacik et al., *Biotechnology*, **9**, 941 (1991)].

The observed passage (c_p/c_b) varies with both the intrinsic passage and the extent of polarization, as shown in Eq. (20-83):

$$\text{Observed passage} \quad s_o = \frac{1}{1 + \left(\frac{1}{s_i} - 1 \right) \cdot \exp(-J/k)} \quad (20-83)$$

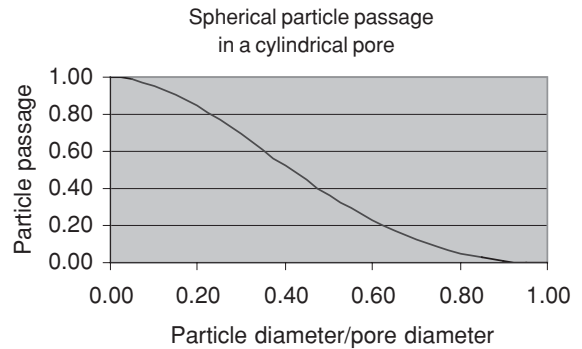


FIG. 20-64 Solute passage.

At low flux these passages are equivalent. At high fluxes the wall concentration is high due to polarization and the observed passage increases, approaching 100 percent, regardless of the intrinsic passage.

In multicomponent systems, large solutes with lower diffusivity, polarize more and exclude smaller solutes from the membrane surface, decreasing their passage. Operation at the “knee” of the flex curve reduces this effect.

Design Considerations and Economics The selection among batch, fed-batch, single-pass, or continuous operation has been described earlier in this section. Important performance considerations for UF applications involve product purity and/or concentration, product yield, and cost/sizing for a given production rate.

Batch or Fed-Batch Operation This mode of operation is typical of biologicals and juice processing where high solids and low fluxes require multiple passes, and batch operation is characteristic of the manufacturing process. Formulas in Table 20-19 can be used to calculate the required volume reduction factor X , diafiltration volumes

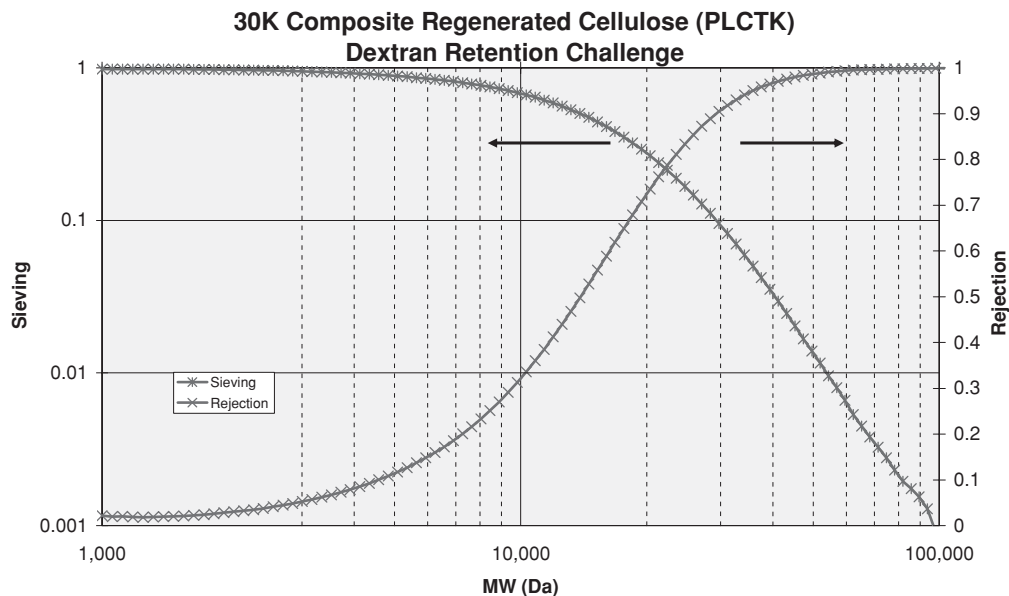


FIG. 20-65 Intrinsic membrane retention.

N , and solute passage S_i needed to produce desired retentate product with impurity concentrations c_i and retentate product yield M_i/M_{i0} . Permeate product characteristics for batch operation can be determined by mass balances using a permeate volume of $V_p = V_0(1 - 1/X + N/X)$, a mass of solute i in the permeate as $M_{i, \text{permeate}} = M_{i0}(1 - e^{-S_i(N+1/X)})$ and the permeate concentration as the ratio of the mass to volume. Fed-batch equations are obtained by using the same approach.

The system area A needed for batch or fed-batch operation can be calculated by using the formulas in Table 20-19 for production rates V_0/t based on feed volume V_0 and average fluxes during process steps. The final retentate batch volume $V_R = V_0/X$ or permeate batch volume $V_p = V_0(1 - 1/X + N/X)$ can be used to restate the production rate on other bases. Although experience can be used to estimate solute passage and process fluxes, they should be determined experimentally for each application.

Single-Pass TFF Single-pass TFF or NFF operation is typical of water treatment. Low solids allow high fluxes with high permeate product recovery and low retentate disposal costs. NFF operation is also used for virus retention in small-batch biologicals processing where low NFF fluxes (high areas) are compensated by ease of use, low residence time, and low capital costs.

Equations (20-66) and (20-67) present single-pass formulas relating retentate solute concentration, retentate crossflow, permeate flow, and membrane area. For relevant low-feed-concentration applications, polarization is minimal and the flux is mainly a function of pressure. Spiral or hollow fiber modules with low feed channel and permeate pressure drops are preferred.

Continuous Operation This mode of operation is typical of paint recovery, whey processing, and wastewater processing where high solids and low fluxes require multiple passes, and continuous operation is allowed by the manufacturing process. For a feed concentration c_F , a volumetric concentration factor X (= retentate flow/feed flow), and retention R , the outlet retentate concentration is

$$c = c_F \left[\frac{X}{X - R(X - 1)} \right] \quad (20-84)$$

In comparison to batch processing, this process operates at the final concentration with associated low fluxes. The use of multiple stages where the retentate outlet from one stage feeds the next allows successive systems to operate at different fluxes and will improve membrane productivities.

MICROFILTRATION

Process Description Microfiltration (MF) separates particles from true solutions, be they liquid or gas phase. Alone among the membrane processes, microfiltration may be accomplished without the use of a membrane. The usual materials retained by a microfiltration membrane range in size from several μm down to $0.2 \mu\text{m}$. At the low end of this spectrum, very large soluble macromolecules are retained by a microfilter. Bacteria and other microorganisms are a particularly important class of particles retained by MF membranes. Among membrane processes, dead-end filtration is uniquely common to MF, but cross-flow configurations are often used.

Brief Examples Microfiltration is the oldest and largest membrane field. It was important economically when other disciplines were struggling for acceptance, yet because of its incredible diversity and lack of large applications, it is the most difficult to categorize. Nonetheless, it has had greater membrane sales than all other membrane applications combined throughout most of its history. The early success of microfiltration was linked to an ability to separate microorganisms from water, both as a way to detect their presence, and as a means to remove them. Both of these applications remain important.

Laboratory Microfiltration membranes have countless laboratory uses, such as recovering biomass, measuring particulates in water, clarifying and sterilizing protein solutions, and so on. There are countless examples for both general chemistry and biology, especially for analytical procedures. Most of these applications are run in dead-end flow, with the membrane replacing a more conventional medium such as filter paper.

Medical MF membranes provide a convenient, reliable means to sterilize fluids without heat. Membranes are used to filter injectable fluids during manufacture. Sometimes they are inserted into the tube leading to a patient's vein.

Process Membrane microfiltration competes with conventional filtration, particularly with diatomaceous earth filtration in general-process applications. A significant advantage for membrane MF is the absence of a diatomaceous earth residue for disposal. Membranes have captured most of the final filtration of wine (displacing asbestos), are gaining market share in the filtration of gelatin and corn syrup (displacing diatomaceous earth), are employed for some of the cold pasteurization of beer, and have begun to be used in the pasteurization of milk. Wine and beer filtration operate dead-end; gelatin, corn syrup, and milk are cross-flow operations. MF is used to filter all fluid reactants in the manufacture of microcircuits to ensure the absence of particulates, with point-of-use filters particularly common.

Gas Phase Microfiltration plays an important and unique role in filtering gases and vapors. One important example is maintaining sterility in tank vents, where incoming air passes through a microfilter tight enough to retain any microorganisms, spores, or viruses. A related application is the containment of biological activity in purge gases from fermentation. An unrelated application is the filtration of gases, even highly reactive ones, in microelectronics fabrication to prevent particulates from contaminating a chip.

Downstream Processing Microfiltration plays a significant role in downstream processing of fermentation products in the pharmaceutical and bioprocessing industry. Examples are clarification of fermentation broths, sterile filtration, cell recycle in continuous fermentation, harvesting mammalian cells, cell washing, mycelia recovery, lysate recovery, enzyme purification, vaccines, and so forth.

MF Membranes Microfiltration is a mature field that has proliferated and subdivided. The scope and variety of MF membranes far exceeds that in any other field. A good overview is given by Strathmann [in Porter (ed.), op. cit., pp. 1-78]. MF membranes may be classified into those with tortuous pores or those with capillary pores. Tortuous-pore membranes are far more common, and are sponge-like structures. The pore openings in MF are much larger than those in any other membrane. Surface pores may be observed by electron microscope, but tortuous pores are much more difficult to observe directly. Membranes may be tested by bubble-point techniques. Many materials not yet useful for tighter membranes are made into excellent MF membranes. Retention is the primary attribute of an MF membrane, but important as well are permeability, chemical and temperature resistance, dirt capacity (for dead-end filters), FDA-USP approval, inherent strength, adsorption properties, wetting behavior, and service life.

Membrane-production techniques listed below are applicable primarily or only to MF membranes. In addition, the Loeb-Sourirajan process, used extensively for reverse osmosis and ultrafiltration membranes, is used for some MF membranes.

Membranes from Solids Membranes may be made from micro-particles by sintering or agglomeration. The pores are formed from the interstices between the solid particles. The simplest of this class of membrane is formed by sintering metal, metal oxide, graphite, ceramic, or polymer. Silver, tungsten, stainless steel, glass, several ceramics, and other materials are made into commercial membranes. Sintered metal may be coated by TiO_2 or zirconium oxide to produce MF and UF membranes. Membranes may be made by the careful winding of microfibers or wires.

Ceramic Ceramic membranes are made generally by the sol-gel process, the successive deposition of ever smaller ceramic precursor spheres, followed by firing to form multitube monoliths. The diameter of the individual channels is commonly about 2 to 6 mm. Monoliths come in a variety of shapes and sizes. A 19-channel design is common. One manufacturer makes large monoliths with square channels.

Track-Etched Track-etched membranes (Fig. 20-66) are now made by exposing a thin polymer film to a collimated beam of radiation strong enough to break chemical bonds in the polymer chains. The film is then etched in a bath which selectively attacks the damaged polymer. The technique produces a film with photogenic pores,

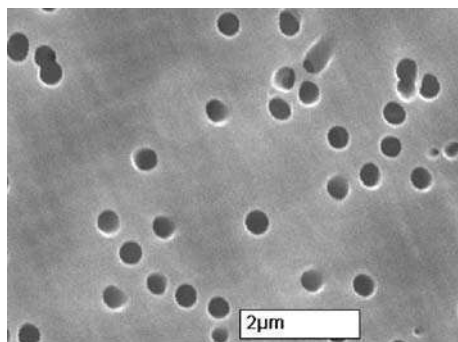


FIG. 20-66 Track-etched 0.4- μm polycarbonate membrane. (Courtesy Millipore Corporation.)

whose diameter may be varied by the intensity of the etching step. Commercially available membranes have a narrow pore size distribution and are reportedly resistant to plugging. The membranes have low flux, because it is impossible to achieve high pore density without sacrificing uniformity of diameter.

Chemical Phase Inversion Symmetrical phase-inversion membranes (Fig. 20-67) remain the most important commercial MF membranes produced. The process produces tortuous-flow membranes. It involves preparing a concentrated solution of a polymer in a solvent. The solution is spread into a thin film, then precipitated through the slow addition of a nonsolvent, usually water, sometimes from the vapor phase. The technique is impressively versatile, capable of producing fairly uniform membranes whose pore size may be varied within broad limits.

Thermal Phase Inversion Thermal phase inversion is a technique which may be used to produce large quantities of MF membrane economically. A solution of polymer in poor solvent is prepared at an elevated temperature. After being formed into its final shape, a sudden drop in solution temperature causes the polymer to precipitate. The solvent is then washed out. Membranes may be spun at high rates using this technique.

Stretched Polymers MF membranes may be made by stretching (Fig. 20-68). Semicrystalline polymers, if stretched perpendicular to the axis of crystallite orientation, may fracture in such a way as to make reproducible microchannels. Best known are Goretex[®] produced from Teflon[®], and Celgard[®] produced from polyolefin. Stretched polymers have unusually large fractions of open space, giving them very high fluxes in the microfiltration of gases, for example. Most such materials are very hydrophobic.

Membrane Characterization MF membranes are rated by flux and pore size. Microfiltration membranes are uniquely testable by

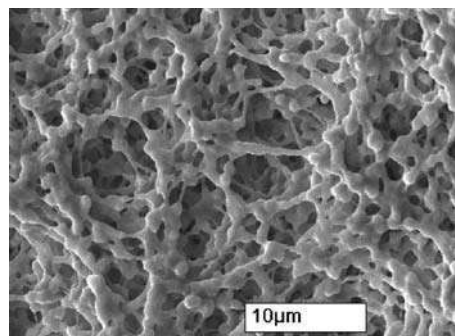


FIG. 20-67 Chemical phase inversion 0.45- μm polyvinylidene fluoride membrane. (Courtesy Millipore Corporation.)

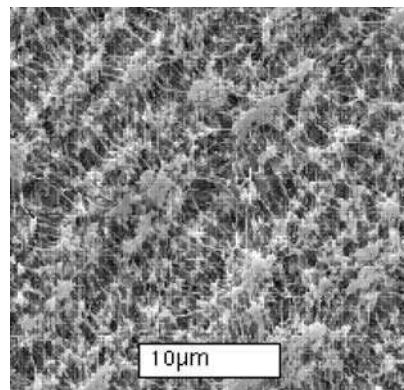


FIG. 20-68 Stretched polytetrafluoroethylene membrane. (Courtesy Millipore Corporation.)

direct examination, but since the number of pores that may be observed directly by microscope is so small, microscopic pore size determination is mainly useful for membrane research and verification of other pore-size-determining methods. Furthermore, the most critical dimension may not be observable from the surface. Few MF membranes have neat, cylindrical pores. Indirect means of measurement are generally superior. Accurate characterization of MF membranes is a continuing research topic for which interested parties should consult the current literature.

Bubble Point Large areas of microfiltration membrane can be tested and verified by a bubble test. Pores of the membrane are filled with liquid, then a gas is forced against the face of the membrane. The Young-Laplace equation, $\Delta P = (4\gamma \cos \Theta)/d$, relates the pressure required to force a bubble through a pore to its radius, and the interfacial surface tension between the penetrating gas and the liquid in the membrane pore. γ is the surface tension (N/m), d is the pore diameter (m), and P is transmembrane pressure (Pa). Θ is the liquid-solid contact angle. For a fluid wetting the membrane perfectly, $\cos \Theta = 1$.

By raising the gas pressure on a wet membrane until the first bubble appears, the largest pore may be identified, and its size computed. This is a good test to run on a membrane apparatus used to sterilize a fluid, since bacteria larger than the identified largest pore (or leak) cannot readily penetrate the assembly. Pore-size distribution may also be run by bubble point. Bubble-point testing is particularly useful in assembled microfilters, since the membrane and all seals may be verified. Periodic testing ensures that the assembly retains its integrity. Diffusional flow of gas is a complication in large MF assemblies. It results from gas dissolving in pore liquid at the high-pressure side, and desorbing at the low-pressure side. If the number of pores and the average pore length are known, the effect can be computed. Special protocols are used when this method is used for critical applications. Detail is provided in ASTM F316-86, "Standard test method for pore size characteristics of membrane filters by bubble point and mean flow pore test." The bubble-point test may also be run using two liquids. Because interfacial surface tensions of liquids can be quite low, this technique permits measurements on pores as small as 10 nm.

Charged Membranes The use of tortuous-flow membranes containing a positive electrical charge may reduce the quantity of negatively charged particles passing even when the pore size is much larger than the particle. The technique is useful for making prefilters or layered membranes that withstand much higher solids loadings before becoming plugged.

Bacteria Challenge Membranes are further tested by challenge with microorganisms of known size: their ability to retain all of the organisms is taken as proof that all pores are smaller than the organism. The best-known microorganism for pore-size determination is *Pseudomonas diminuta*, an asporogenous gram-negative rod with a mean diameter of 0.3 μm . Membranes with pore size smaller than that are used to ensure sterility in many applications. Leahy and

Sullivan [*Pharmaceutical Technology*, 2(11), 65 (1978)] provide details of this validation procedure.

Membrane thickness is a factor in microbial retention. Tortuous-pore membranes rated at 0.22 μm typically have surface openings as large as 1 μm (Fig. 20-67). Narrower restrictions are found beneath the surface. In challenge tests, *P. diminuta* organisms are found well beneath the surface of a 0.2- μm membrane, but not in the permeate.

Latex Latex particles of known size are available as standards. They are useful to challenge MF membranes.

Process Configuration As befits a field with a vast number of important applications and a history of innovation, there are countless variations on how an MF process is run.

Dead-end versus Cross-flow Conventional filtration is usually run dead-end, and is facilitated by amendments that capture the particulates being removed. Membranes have very low dirt capacity, so only applications with very low solids to be removed are run in conventional dead-end flow. A rough upper limit to solids content is about 0.5 percent; streams containing <0.1 percent are almost always processed by dead-end devices. Since dead-end membrane equipment is much less expensive than cross-flow, great ingenuity is applied to protecting the critical membrane pores by structured prefilters to remove larger particles and debris. The feed may also be pretreated. It is common practice to dispose of the spent membrane rather than clean it. The membrane may be run inverted. A review of dead-end membrane filtration is given by Davis and Grant [in Ho and Sirkar (eds.), op. cit., pp. 461–479].

Cross-flow is the usual case where cake compressibility is a problem. Cross-flow microfiltration is much the same as cross-flow ultrafiltration in principle. In practice, the devices are often different. As with UF, spiral-wound membranes provide the most economical configuration for many large-scale installations. However, capillary devices and cassettes are widely employed, especially at smaller scale. A detailed description of cross-flow microfiltration had been given by Murkes and Carlsson [*Crossflow Filtration*, Wiley, New York (1988)].

Membrane Inverted Most membranes have larger openings on one face than on the other. Common practice is to run the tightest face against the feed in order to avoid plugging of the backing by particles. The rationale is that anything that makes it past the “skin” will have relatively unimpeded passage into the backing and out with the permeate. For very low solids this convention is reversed, the rationale being that the porous backing provides a trap for particulates, rather like filter aid.

If the complete passage of soluble macromolecules is required, a highly polarized membrane is an advantage. The upside-down membrane hinders the back diffusion of macrosolutes. Countering the tendency of the retained particulates to “autofilter” soluble macrosolutes, the inhibition of back diffusion raises the polarization and thus the passage of macrosolutes such as proteins. The particulates are physically retained by the membrane. Blinding and plugging can be controlled if the membrane is backwashed frequently. This technique has been demonstrated at high solids loadings in an application where high passage of soluble material is critical, the microfiltration of beer [Wenten, Rasmussen, and Jonsson, *North Am. Membrane Soc. Sixth Annual Meeting*, Breckenridge, CO (1994)].

Liquid Backpulse Solid membranes are backwashed by forcing permeate backward through the membrane. Frequent pulsing seems to be the key.

Air Backflush A configuration unique to microfiltration feeds the process stream on the shell side of a capillary module with the permeate exiting the tube side. The device is run as an intermittent dead-end filter. Every few minutes, the permeate side is pressurized with air. First displacing the liquid permeate, a blast of air pushed backward through the membrane pushes off the layer of accumulated solids. The membrane skin contacts the process stream, and while being backwashed, the air simultaneously expands the capillary and membrane pores slightly. This momentary expansion facilitates the removal of imbedded particles.

Process Limitations The same sorts of process limitations affecting UF apply to MF. The following section will concentrate on the differences.

Concentration Polarization The equations governing cross-flow mass transfer are developed in the section describing ultrafiltration.

The velocity, viscosity, density, and channel-height values are all similar to UF, but the diffusivity of large particles (MF) is orders-of-magnitude lower than the diffusivity of macromolecules (UF). It is thus quite surprising to find the fluxes of cross-flow MF processes to be similar to, and often higher than, UF fluxes. Two primary theories for the enhanced diffusion of particles in a shear field, the inertial-lift theory and the shear-induced theory, are explained by Davis [in Ho and Sirkar (eds.), op. cit., pp. 480–505], and Belfort, Davis, and Zydney [*J. Membrane Sci.*, 96, 1–58 (1994)]. While not clear-cut, shear-induced diffusion is quite large compared to Brownian diffusion except for those cases with very small particles or very low cross-flow velocity. The enhancement of mass transfer in turbulent-flow microfiltration, a major effect, remains completely empirical.

Fouling Fouling affects MF as it affects all membrane processes. One difference is that the fouling effect caused by deposition of a foulant in the pores or on the surface of the membrane can be confounded by a rearrangement or compression of the solids cake which may form on the membrane surface. Also, the high, open space found in tortuous-pore membranes makes them slower to foul and harder to clean.

Equipment Configuration Since the early days when membrane was available only in flat-sheet form, the variety of offerings of various geometry and fabricated filter component types has grown geometrically. An entire catalog is devoted just to list the devices incorporating membranes whose area ranges from less than 1 cm^2 up to 3 m^2 . Microfiltration has grown to maturity selling these relatively small devices. Replacement rather than reuse has long been the custom in MF, and only with later growth of very large applications, such as water, sewage, and corn sweeteners, has long membrane life become an economic necessity on a large scale.

Conventional Designs Designs familiar from other unit operations are also used in microfiltration. Cartridge-filter housings may be fitted with pleated MF membrane making a high-area dead-end membrane filter. Plate-and-frame type devices are furnished with MF sheet stock, and are common in some applications. Capillary bundles with tube-side feed are used for cross-flow applications, and are occasionally used in dead-end flow. A few tubular membranes are used. Spiral-wound modules are becoming increasingly important for process applications where economics are paramount. Belt filters have been made using MF membrane.

Ceramics Ceramic microfilters for commercial applications are almost always employed as tube-side feed multitube monoliths. They are also available as flat sheet, single tubes, discs, and other forms primarily suited to lab use. They are used for a few high-temperature applications, in contact with solvents, and particularly at very high pH.

Cassettes *Cassette* is a term used to describe two different cross-flow membrane devices. The less-common design is a usually large stack of membrane separated by a spacer, with flow moving in parallel across the membrane sheets. This variant is sometimes referred to as a **flat spiral**, since there is some similarity in the way feed and permeate are handled. The more common cassette has long been popular in the pharmaceutical and biotechnical field. It too is a stack of flat-sheet membranes, but the membrane is usually connected so that the feed flows across the membrane elements in series to achieve higher conversion per pass. Their popularity stems from easy direct scale-up from laboratory to plant-scale equipment. Their limitation is that fluid management is inherently very limited and inefficient. Both types of cassette are very compact and capable of automated manufacture.

Representative Process Applications

Pharmaceutical Removal of suspended matter is a frequent application for MF. Processes may be either clarification, in which the main product is a clarified liquid, or solids recovery. Separating cells or their fragments from broth is the most common application. Clarification of the broth in preparation for product recovery is the usual objective, but the primary goal may be recovery of cells. Cross-flow microfiltration competes well with centrifugation, conventional filtration by rotary vacuum filter or filter press and decantation. MF delivers a cleaner permeate, an uncontaminated, concentrated cell product

which may be washed in the process, and generally gives high yields. There is no filter-aid disposal problem. Microfiltration has higher capital costs than the other processes, although total cost may be lower. The recovery of penicillin is an example of a process for which cross-flow microfiltration is generally accepted.

Water and Wastewater Microfiltration is beginning to be applied to large-scale potable-water treatment. Its major advantage is positive removal of cryptosporidium and giardia cysts, and its major disadvantage is cost. MF is used in a few large sewage-treatment facilities, where its primary advantage is that it permits a major reduction in the physical size of the facility.

Chemical MF is used in several applications to recover caustic values from cleaning or processing streams. An example is the caustic solution used to clean dairy evaporators, which may be cleaned for reuse by passing it through a microfilter. Significant savings in caustic purchase and disposal costs provide the incentive. Acids are also recovered and reused. Ceramic microfilters are most commonly used in these applications.

Food and Dairy Microfiltration has many applications in the food and dairy industries. An innovative dairy application uses MF membranes to remove bacteria as a nonthermal means of disinfection for milk. A special flow apparatus maintains a carefully controlled transmembrane pressure as the milk flows across the membrane. The concentrate contains the bacteria and spores, as well as any fat. The concentrate may be heat sterilized and recombined with the sterile permeate. In another milk application, some success is reported in separating fat from milk or other dairy streams by cross-flow microfiltration instead of centrifugation. Transmembrane pressure must be kept very low to prevent fat penetration into the membrane. In the food industry, MF membranes are replacing diatomaceous earth filtration in the processing of gelatin. The gelatin is passed with the permeate, but the haze producing components are retained. UF may be used downstream to concentrate the gelatin.

Flow Schemes The outline of batch, semibatch, and stages-in-series is given in the section describing ultrafiltration. Diafiltration is also described there. All these techniques are common in MF, except for stages-in-series, used rarely. MF features uses of special techniques to control transmembrane pressure in some applications. An example is one vendor's device for the microfiltration of milk. In most devices the permeate simply leaves by the nearest exit, but for this application the permeate is pumped through the device in such a way as to duplicate the pressure drop in the concentrate side, thus maintaining a constant transmembrane-pressure driving force. In spite of the low-pressure driving force, the flux is extremely high.

Limitations Some applications which seem ideal for MF, for example the clarification of apple juice, are done with UF instead. The reason is the presence of deformable solids which easily plug and blind an MF membrane. The pores of an ultrafiltration membrane are so small that this plugging does not occur, and high fluxes are maintained. UF can be used because there is no soluble macromolecule in the juice that is desired in the filtrate. There are a few other significant applications where MF seems obvious, but is not used because of deformable particle plugging.

Economics Microfiltration may be the triumph of the Lilliputians; nonetheless, there are a few large-industrial applications. Dextrose plants are very large, and as membrane filtration displaces the precoat filters now standard in the industry, very large membrane microfiltration equipment will be built.

Site Size Most MF processes require a smaller footprint than competing processes. Reduction in total-area requirements are sometimes a decisive economic advantage for MF. It may be apparent that the floor-space costs in a pharmaceutical facility are high, but municipal facilities for water and sewage treatment are often located on expensive real estate, giving MF an opportunity despite its higher costs otherwise.

Large Plants The economics of microfiltration units costing about \$10⁶ is treated under ultrafiltration. When ceramic membranes are used, the cost optimum may shift energy consumption upward to as much as 10 kWh/m³.

Disposables For smaller MF applications, short membrane life is a traditional characteristic. In these applications, costs are dominated by the disposables, and an important characteristic of equipment design is the ease, economy, and safety of membrane replacement.

Hygiene and Regulation Almost unique to MF is the influence of regulatory concerns in selection and implementation of a suitable microfilter. Since MF is heavily involved with industries regulated by the Food and Drug Administration, concerns about process stability, consistency of manufacture, virus reduction, pathogen control, and material safety loom far larger than is usually found in other membrane separations.

GAS-SEPARATION MEMBRANES

Process Description Gas-separation membranes separate gases from other gases. Some gas filters, which remove liquids or solids from gases, are microfiltration membranes. Gas membranes generally work because individual gases differ in their solubility and diffusivity through nonporous polymers. A few membranes operate by sieving, Knudsen flow, or chemical complexation.

Selective gas permeation has been known for generations, and the early use of palladium silver-alloy membranes achieved sporadic industrial use. Gas separation on a massive scale was used to separate U²³⁵ from U²³⁸ using porous (Knudsen flow) membranes. An upgrade of the membranes at Oak Ridge cost \$1.5 billion. Polymeric membranes became economically viable about 1980, introducing the modern era of gas-separation membranes. H₂ recovery was the first major application, followed quickly by acid gas separation (CO₂/CH₄) and the production of N₂ from air.

The more permeable component is called the *fast gas*, so it is the one enriched in the permeate stream. Permeability through polymers is the product of solubility and diffusivity. The diffusivity of a gas in a membrane is inversely proportional to its kinetic diameter, a value determined from zeolite cage exclusion data (see Table 20-26 after Breck, *Zeolite Molecular Sieves*, Wiley, New York, 1974, p. 636). Tables 20-27, 20-28, and 20-29 provide units conversion factors useful for calculations related to gas-separation membrane systems.

Leading Examples These applications are commercial, some on a very large scale. They illustrate the range of application for gas-separation membranes. Unless otherwise specified, all use polymeric membranes.

Hydrogen Hydrogen recovery was the first large commercial membrane gas separation. Polysulfone fiber membranes became available in 1980 at a time when H₂ needs were rising, and these novel membranes quickly came to dominate the market. Applications include recovery of H₂ from ammonia purge gas, and extraction of H₂ from petroleum cracking streams. Hydrogen once diverted to low-quality fuel use is now recovered to become ammonia, or is used to desulfurize fuel, etc. H₂ is the fast gas.

Carbon Dioxide-Methane Much of the natural gas produced in the world is coproduced with an acid gas, most commonly CO₂ and/or H₂S. While there are many successful processes for separating the gases, membrane separation is a commercially successful competitor, especially for small installations. The economics work best for feeds with very high or very low CH₄ content. Methane is a slow gas; CO₂, H₂S, and H₂O are fast gases.

Oxygen-Nitrogen Because of higher solubility, in many polymers, O₂ is faster than N₂ by a factor of 5. Water is much faster still. Since simple industrial single-stage air compressors provide sufficient pressure to drive an air-separating membrane, moderate purity N₂ (95–99.5%) may be produced in low to moderate quantities quite

TABLE 20-26 Kinetic Diameters for Important Gases

Penetrant	He	H ₂	NO	CO ₂	Ar	O ₂	N ₂	CO	CH ₄	C ₂ H ₄	Xe	C ₃ H ₈
Kinetic dia, nm	0.26	0.289	0.317	0.33	0.34	0.346	0.364	0.376	0.38	0.39	0.396	0.43

TABLE 20-27 Gas-Permeation Units

Quantity	Engineering units	Literature units	SI units
Permeation rate	Standard cubic feet/minute		kmol/s
Permeation flux	ft ³ /ft ² ·day	cm/sec (STP)	kmol/m ² ·s
Permeability	ft ³ ·ft/ft ² ·day·psi	Barrers	kmol/m·s·Pa
Permeance	ft ³ /ft ² ·day·psi	Barrers/cm	kmol/m ² ·s·Pa

economically by membrane separation. (Argon is counted as part of the nitrogen.) An O₂-enriched stream is a coproduct, but it is rarely of economic value. The membrane process to produce O₂ as a primary product has a limited market.

Helium Helium is a very fast gas, and may be recovered from natural gas through the use of membranes. More commonly, membranes are used to recover He after it has been used and become diluted.

Gas Dehydration Water is extremely permeable in polymer membranes. Dehydration of air and other gases is a growing membrane application.

Vapor Recovery Organic vapors are recovered from gas streams using highly permeable rubbery polymer membranes which are generally unsuitable for permanent gas separations because of poor selectivity. The high sorption of vapors in these materials makes them ideal for stripping and recovering vapors from gases.

Competing Processes Membranes are not the only way to make these separations, neither are they generally the dominant way. In many applications, membranes compete with cryogenic distillation and with pressure-swing adsorption; in others, physical absorption is the dominant method. The growth rate for membrane capacity is higher than that for any competitor.

Basic Principles of Operation Gas-separation literature often uses nomenclature derived from distillation, a practice that will generally be followed here. L is the molar feed rate, V is the molar permeate rate, R is the molar residue ($L - V$). Mole fractions of components i, j , in the feed-residue phase will be $x_i, x_j, \dots$ and in the permeate phase $y_i, y_j, \dots$. Stage cut, Θ , is permeate volume/feed volume, or V/L .

Basic Equations In "Background and Definitions," the basic equation for gas permeation was derived with the major assumptions noted.

$$J_i \sim N_i = (p_i/z)(p_{i,\text{feed}} - p_{i,\text{permeate}}) \quad (20-85)$$

where p_i is the permeability of component i through the membrane, J_i is the flux of component i through the membrane for the *partial pressure difference* (Δp) of component i . z is the effective thickness of the membrane. By choosing units appropriately, $J = p\Delta p$.

A similar equation may be written for a second component, j , and any additional number of components, employing partial pressures:

$$J_j = (p_j/z)(p_{j,\text{feed}} - p_{j,\text{permeate}}) \quad (20-86)$$

The total pressure is the sum of the partial pressures:

$$P_{\text{feed}} = \sum (p_{\text{feed}})_{i,j,\dots} \quad (20-87)$$

$$P_{\text{permeate}} = \sum (p_{\text{permeate}})_{i,j,\dots} \quad (20-88)$$

For simplicity, consider a two-component system. The volume fraction of a component is

$$x_i = \frac{p_{\text{feed}}}{P_{\text{feed}}} \quad (20-89)$$

$$y_i = \frac{p_{\text{permeate}}}{P_{\text{permeate}}} \quad (20-90)$$

TABLE 20-28 Barrer Conversion Factors

Quantity	Multiply	By	To get
Permeability	Barrers	3.348×10^{-19}	kmol/m·s·Pa
Permeability	Barrers	4.810×10^{-8}	ft ³ (STP)/ft·psi·day
Permeance	Barrers/cm	3.348×10^{-17}	kmol/m ² ·s·Pa
Permeance	Barrers/cm	1.466×10^{-6}	ft ³ (STP)/ft ² ·psi·day

When two species are permeating through a membrane, the ratio of their fluxes can be written following Eqs. (20-85) and (20-86) as:

$$\frac{J_i}{J_j} = \frac{\frac{p_i}{z}(p_{i,\text{feed}} - p_{i,\text{permeate}})}{\frac{p_j}{z}(p_{j,\text{feed}} - p_{j,\text{permeate}})} \quad (20-91)$$

Recalling Eq. (20-55), and restating it in the nomenclature for gas membranes:

$$\alpha = \frac{y_i/y_j}{x_i/x_j} \quad (20-92)$$

Defining the pressure ratio $\Phi = P_{\text{feed}}/P_{\text{permeate}}$ and applying Eqs. (20-87) through (20-90) gives:

$$\frac{J_i}{J_j} = \alpha \left[\frac{x_i - (y_i/\Phi)}{x_j - (y_j/\Phi)} \right] \quad (20-93)$$

Combining these equations and rearranging, the permeate composition may be solved explicitly:

$$y_i = \left(\frac{\Phi}{2} \right) \left[x_i + \frac{1}{\Phi} + \frac{1}{\alpha - 1} - \sqrt{\left(x_i + \frac{1}{\Phi} + \frac{1}{\alpha - 1} \right)^2 - \frac{4\alpha x_i}{(\alpha - 1)\Phi}} \right] \quad (20-94)$$

Equation (20-94) gives a permeate concentration as a function of the feed concentration at a stage cut, $\Theta = 0$. To calculate permeate composition as a function of Θ , the equation may be used iteratively if the permeate is unmixed, such as would apply in a test cell. The calculation for real devices must take into account the fact that the driving force is variable due to changes on both sides of the membrane, as partial pressure is a point function, nowhere constant. Using the same caveat, permeation rates may be calculated component by component using Eq. (20-86) and permeance values. For any real device, both concentration and permeation require iterative calculations dependent on module geometry.

Driving Force Gas moves across a membrane in response to a difference in chemical potential. Partial pressure is sufficiently proportional to be used as the variable for design calculations for most gases of interest, but fugacity must be used for CO₂ and usually for H₂ at high pressure. Gas composition changes as a gas passes along a membrane. As the fast gas passes through the membrane, x_i drops. Total pressure on the upstream side of the membrane drops because of frictional losses in the device. Frictional losses on the permeate side will affect the permeate pressure. The partial pressure of a component in the permeate may thus rise rapidly. Permeation rate is a point function, dependent on the difference in partial pressures at a point on the membrane. Many variables affect point partial pressures, among them are membrane structure, module design, and permeate gas-sweep rates. Juxtaposition of feed and permeate is a function of permeator design, and a rapid decline in driving force may result when it is not expected (see "Membrane System Design Features").

TABLE 20-29 Industry-Specific Gas Measures

Industry-unit	How measured	Cubic feet per pound mole	kmol per mscf
STP, Mscf	1000 ft ³ at 32°F	359.3	1.262
Gas industry, Mscf	1000 ft ³ at 60°F	379.8	1.194
Air industry, Mnsf	1000 ft ³ at 70°F	387.1	1.172

An additional complication may arise in a few cases from the Joule-Thompson effect during expansion of a gas through a membrane changing the temperature. High-pressure CO₂ is an example.

Plasticization Gas solubility in the membrane is one of the factors governing its permeation, but the other factor, diffusivity, is not always independent of solubility. If the solubility of a gas in a polymer is too high, plasticization and swelling result, and the critical structure that controls diffusion selectivity is disrupted. These effects are particularly troublesome with condensable gases, and are most often noticed when the partial pressure of CO₂ or H₂S is high. H₂ and He do not show this effect. This problem is well known, but its manifestation is not always immediate.

Limiting Cases Equation (20-94) has two limiting cases for a binary system. First, when $\alpha \gg \phi$. In this case, selectivity is no longer very important.

$$y_i \cong x_i \frac{P_{\text{feed}}}{P_{\text{permeate}}} = x_i \Phi \quad (20-95)$$

Module design is very important for this case, as the high α may result in high permeate partial pressure. An example is the separation of H₂O from air.

Conversely, when $\Phi \gg \alpha$, pressure ratio loses its importance, and permeate composition is:

$$y_i \cong \frac{x_i \alpha}{[1 + x_i(\alpha - 1)]} \quad (20-96)$$

Module design for this case is of lesser importance.

Selectivity and Permeability

State of the Art A desirable gas membrane has high separating power (α) and high permeability to the fast gas, in addition to critical requirements discussed below. The search for an ideal membrane produced copious data on many polymers, neatly summarized by Robeson [*J. Membrane Sci.*, **62**, 165 (1991)]. Plotting log permeability versus log selectivity (α), an "upper bound" is found (see Fig. 20-69) which all the many hundreds of data points fit. The data were taken between 20 and 50°C, generally at 25 or 35°C.

The lower line in Fig. 20-69 shows the upper bound in 1980. Although no breakthrough polymers have been reported in the past

TABLE 20-30 High-Performance Polymers for O₂/N₂*

Polymer	α (O ₂ /N ₂)	ρ (O ₂)—Barrers
Poly (trimethylsilylpropyne)	2.0	4000
Tetrabromo Bis A polycarbonate	7.47	1.36
Poly (tert-butyl acetylene)	3.0	300
Vectra polyester	15.3	0.00046
Poly (triazole)	9.0	1.2
Polypyrrolone	6.5	7.9

*Polymers that are near the upper bound and their characteristics.

few years, it would be surprising if these lines remain the state of the art forever. Performance parameters for polymers used for O₂/N₂ separations are given in Table 20-30.

The upper-bound line connects discontinuous points, but polymers exist near the bound for separations of interest. Whether these will be available as membranes is a different matter. A useful membrane requires a polymer which can be fabricated into a device having an active layer around 50 nm thick. At this thickness, membrane properties may vary significantly from bulk properties, although not by a factor of 2.

The data reported are *permeabilities*, not *fluxes*. Flux is proportional to permeability/thickness. The separations designer must deal with real membranes, for which thickness is determined by factors outside the designer's control. It is vital that flux data are used in design.

Glassy polymers are significantly overrepresented in the high-selectivity region near the upper bound, and rubbery polymers are overrepresented at the high-permeability end, although the highest-permeability polymer discovered, poly(trimethylsilylpropyne) is glassy. Most of the polymers with interesting properties are noncrystalline. Current membrane-materials research is strongly focused on glassy materials and on attempts to improve diffusivity, as it seems more promising than attempts to increase solubility [Koros, *North Am. Membrane Soc. Sixth Annual Meeting*, Breckenridge, CO (1994)].

Robeson [*J. Membrane Sci.*, **62**, 165 (1991); *Polymer*, **35**, 4970 (1994)] has determined upper-bound lines for many permeant pairs in hundreds of polymers. These lines may be drawn from Eq. (20-97) and the data included in Table 20-31. These values will give ρ_i in

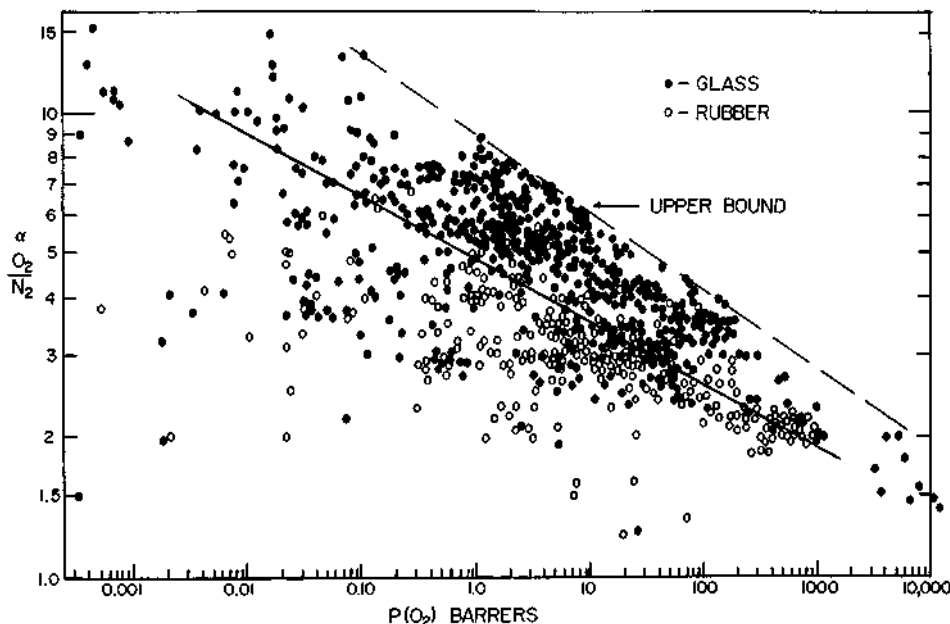


FIG. 20-69 Plot of separation factor versus permeability for many polymers, O₂/N₂. Abscissa—"Fast Gas Permeability, ρ (O₂) Barrers." Ordinate—"Selectivity, α (O₂/N₂)."

TABLE 20-31 Upper-Bound Coordinates for Gas Pairs

Gas pair	$\log k$	m
He/N ₂	4.0969	1.0242
H ₂ /N ₂	4.7236	1.5275
He/CH ₄	3.6991	0.7857
H ₂ /CH ₄	4.2672	1.2112
O ₂ /N ₂	5.5902	5.800
He/O ₂	3.6628	1.295
H ₂ /O ₂	4.5534	2.277
CO ₂ /CH ₄	6.0309	2.6264
He/H ₂	2.9823	4.9535
He/CO ₂	2.8482	1.220
H ₂ /CO ₂	3.0792	1.9363

Barrers; α is dimensionless. Robeson [op.cit., (1991); op. cit., (1994)] lists high-performance polymers for most of these gas pairs, like Table 20-31.

$$\log p_i = \log k - m \log \alpha_{ij} \quad (20-97)$$

Temperature Effects A temperature increase in a polymer membrane permits larger segmental motions in the polymer, producing a dramatic increase in diffusivity. Countering this is a decrease in solubility. It increases the size of the gaps in the polymer matrix, decreasing diffusivity selectivity. The net result is that for a glassy polymer, permeability rises while selectivity declines. For organic permeants in rubbery polymers, this trend is often reversed.

Plasticization and Other Time Effects Most data from the literature, including those presented above are taken from experiments where one gas at a time is tested, with α calculated as a ratio of the two permeabilities. If either gas permeates because of a high-sorption coefficient rather than a high diffusivity, there may be an increase in the permeability of all gases in contact with the membrane. Thus, the α actually found in a real separation may be much lower than that calculated by the simple ratio of permeabilities. The data in the literature do not reliably include the plasticization effect. If present, it results in the sometimes slow relaxation of polymer structure giving a rise in permeability and a dramatic decline in selectivity.

Other Caveats Transport in glassy polymers is different from transport in rubbery ones. In glassy polymers, there are two sites in which sorption may occur, and the literature dealing with dual-mode sorption is voluminous. The simplest case describes behavior when the downstream pressure is zero. It is of great help in understanding the theory but of limited value in practice. There are concerns about permeation of mixtures in glassy polymers with reports of crowding out and competitive sorption. Practical devices are built and operated for many streams, and the complications are often minor. But taking data independently determined for two pure gases and dividing them to obtain α in the absence of other facts is risky.

Gas-Separation Membranes

Organic Organic polymer membranes are the basis for almost all commercial gas-separation activity. Early membranes were cellulose esters and polysulfone. These membranes have a large installed base. New installations are dominated by specialty polymers designed for the purpose, including some polyimides and halogenated polycarbonates. In addition to skinned membranes, composites are made from "designer" polymers, requiring as little as 2 g/1000 m². The rapid rise of N₂/O₂ membranes in particular is the result of stunning improvements in product uniformity and quality. A few broken fibers in a 100 m² module results in the module's being scrapped.

Caulked Membrane manufacturing defects are unavoidable, and pinholes are particularly deleterious in gas-separation membranes. A very effective remedy is to **caulk** the membrane by applying a highly permeable, very thin topcoat over the finished membrane. While the coating will have poor selectivity, it will plug up the gross leaks while impairing the fast gas permeance only slightly. Unless the as-cast membrane is almost perfect, caulking dramatically improves membrane performance.

Metallic Palladium films pass H₂ readily, especially above 300°C. α for this separation is extremely high, and H₂ produced by purification through certain Pd alloy membranes is uniquely pure. Pd alloys are used to overcome the crystalline instability of pure Pd during heating-cooling cycles. Economics limit this membrane to high-purity applications.

Advanced Materials Experimental membranes have shown remarkable separations between gas pairs such as O₂/N₂ whose kinetic diameters (see Table 20-26) are quite close. Most prominent is the carbon molecular sieve membrane, which operates by ultramicro-porous molecular sieving. Preparation of large-scale permeators based on ultramicro-porous membranes has proven to be a major challenge.

Catalytic A catalytic-membrane reactor is a combination heterogeneous catalyst and permselective membrane that promotes a reaction, allowing one component to permeate. Many of the reactions studied involve H₂. Membranes are metal (Pd, Ag), nonporous metal oxides, and porous structures of ceramic and glass. Falconer, Noble, and Sperry [in Noble and Stern (eds.), op. cit., pp. 669–709] review status and potential developments.

Membrane System Design Features For the rate process of permeation to occur, there must be a driving force. For gas separations, that force is partial pressure (or fugacity). Since the ratio of the component fluxes determines the separation, the partial pressure of each component at each point is important. There are three ways of driving the process: Either high partial pressure on the feed side (achieved by high total pressure), or low partial pressure on the permeate side, which may be achieved either by vacuum or by introduction of a sweep gas. Both of the permeate options have negative economic implications, and they are less commonly used.

Figure 20-70 shows three of the principal operating modes for gas membranes. A critical issue is the actual partial pressure of permeant at a point on the membrane. Flow arrangements for the permeate are very important in determining the efficiency of the separation, in rough analogy to the importance of arrangements in heat exchangers.

Spiral membranes are the usual way to form flat sheet into modules. They have the characteristic that the feed and the permeate move at

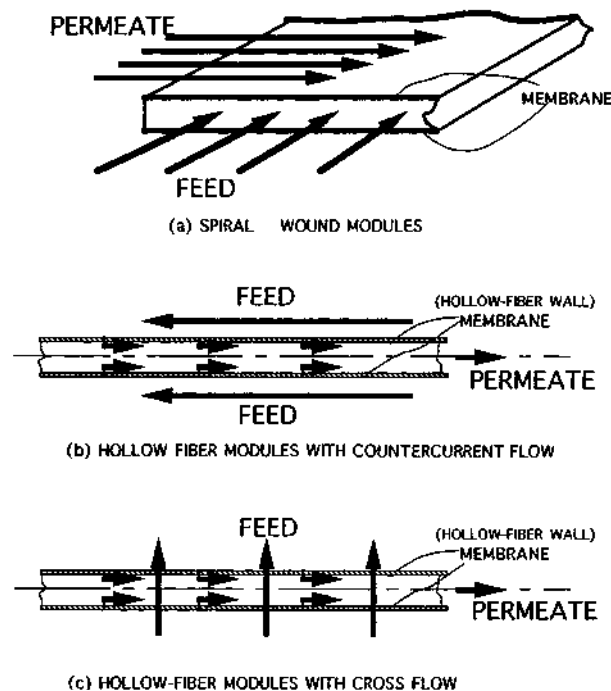


FIG. 20-70 Flow paths in gas permeators. (Courtesy Elsevier.)

right angles. Since the membrane is always cast on a porous support, point-permeate values are influenced by the substrate.

Hollow-fiber membranes may be run with shell-side or tube-side feed, cocurrent, countercurrent or in the case of shell-side feed and two end permeate collection, co- and countercurrent. Not shown is the scheme for feed inside the fiber, common practice in lower-pressure separations such as air.

The design of the membrane device will influence whether the membrane is operating near its theoretical limit. Sengupta and Sirkar [in Noble and Stern (eds.), op. cit., pp. 499–552] treat module design thoroughly (including numerical examples for most module configurations) and provide an extensive bibliography.

For a hollow-fiber device running with shell-side feed with the membrane on the outside, Giglia et al. [*Ind. Eng. Chem. Research*, **29**, 1239–1248 (1991)] analyzed the effect of membrane-backing porosity on separation efficiency. The application is production of N_2 from air, the desired result being the lowest possible O_2 content in the retentate at a given stage cut. The modules used were operated cocurrent and countercurrent. If the porous-membrane backing prevented the permeate from mixing with the gas adjacent to the membrane, a result approximating cross-flow is expected. For the particular membrane structure used, the experimental result for cocurrent flow was quite close to the calculated cocurrent value, while for countercurrent flow, the experimental data were between the values calculated for the crosscurrent model and the countercurrent model. For a membrane to be commercially useful in this application, the mass transfer on the permeate side must exceed the crosscurrent model.

Air is commonly run with tube-side feed. The permeate is run countercurrent with the separating skin in contact with the permeate. (The feed gas is in contact with the macroporous back side of the membrane.) This configuration has proven to be superior, since the permeate-side mass-transfer problem is reduced to a minimum, and the feed-side mass-transfer problem is not limiting.

Partial Pressure Pinch An example of the limitations of the partial pressure pinch is the dehumidification of air by membrane. While O_2 is the fast gas in air separation, in this application H_2O is faster still. Special dehydration membranes exhibit $\alpha = 20,000$. As gas passes down the membrane, the partial pressure of H_2O drops rapidly in the feed. Since the H_2O in the permeate is diluted only by the O_2 and N_2 permeating simultaneously, p_{H_2O} rises rapidly in the permeate. Soon there is no driving force. The commercial solution is to take some of the dry air product and introduce it into the permeate side as a countercurrent sweep gas, to dilute the permeate and lower the H_2O partial pressure. It is in effect the introduction of a leak into the membrane, but it is a controlled leak and it is introduced at the optimum position.

Fouling Industrial streams may contain condensable or reactive components which may coat, solvate, fill the free volume, or react with the membrane. Gases compressed by an oil-lubricated compressor may contain oil, or may be at the water dew point. Materials that will coat or harm the membrane must be removed before the gas is treated. Most membranes require removal of compressor oil. The extremely permeable poly(trimethylsilylpropyne) may not become a practical membrane because it loses its permeability rapidly. Part of the problem is pore collapse, but it seems extremely sensitive to contamination even by diffusion pump oil and gaskets [Robeson, op. cit., (1994)].

A foulinglike problem may occur when condensable vapors are left in the residue. Condensation may result which in the best case results in blinding of the membrane, and in the usual case, destruction of the membrane module. Dew-point considerations must be part of any gas-membrane design exercise.

Modules and Housings Modern gas membranes are packaged either as hollow-fiber bundles or as spiral-wound modules. The former uses extruded hollow fibers. Tube-side feed is preferable, but it is limited to about 1.5 MPa. Higher-pressure applications are usually fed on the shell side. A large industrial permeator contains fibers 400 μm by 200 μm i.d. in a 6-inch shell 10 feet long. Flat-sheet membrane is wound into spirals, with an 8- by 36-inch permeator containing 25 m^2 of membrane. Both types of module are similar to those illustrated in "Background and Definitions." Spiral modules are useful when feed

flows are very high and especially in vapor-permeation applications. Otherwise, fiber modules have a large and growing share of the market.

Energy Requirements The thermodynamic minimum energy requirement to separate a metric ton of N_2 from air and compress it to atmospheric pressure at 25°C is independent of separation method, 20.8 MJ or 5.8 kWh. In practice, a cryogenic distillation plant requires twice this energy, and it produces a very pure product as a matter of course. The membrane process requires somewhat more energy than distillation at low purity and much more energy at high purity. Membranes for O_2 -reduced air are economical and are a rapidly growing application, but not because of energy efficiency. Figure 20-71 may be used to estimate membrane energy requirements. From the required purity, locate the stage cut on either of the countercurrent flow curves. The compressor work required is calculated using the pressure and the term (product flow rate)/(1 - Θ). The N_2 -rich product is produced at pressure, and the O_2 -enriched permeate is vented. For example, if gas is required containing 97 percent inerts, assume the product composition would be 96 percent N_2 , 2 percent O_2 , 1 percent Ar, and 3 percent O_2 , giving a calculated molar mass of 28.2. One tonne would thus contain 35.4 kmol. For 98 percent inerts, Fig. 20-71 shows a stage cut of 67 percent when operating at 690 kPa (abs) and 25°C. Therefore, $35.5/(1 - 0.67) = 108$ kmol of air are required as feed. The adiabatic compression of 108 kmol of air from atmospheric pressure and 300 K (it would subsequently be cooled to $273 + 25 = 298$ K) requires 192 kWh. Assuming 75 percent overall compressor + driver efficiency, 256 kWh are required. For comparison, a very efficient, large N_2 distillation plant would produce 99.99 percent N_2 at 690 kPa for 113 kWh/metric ton. The thermodynamic minimum to separate ($N_2 + Ar$) and deliver it at the given P and T is 72 kWh.

Economics It is ironic that a great virtue of membranes, their versatility, makes economic optimization of a membrane process very difficult. Designs can be tailored to very specific applications, but each design requires a sophisticated computer program to optimize its costs. Spillman [in Noble and Stern (eds.), op. cit., pp. 589–667] provides an overall review and numerous specific examples including circa 1989 economics.

Rules of Thumb With a few notable exceptions such as H_2 through Pd membranes, membrane separations are not favored when a component is required at high purity. Often, membranes serve these needs by providing a moderate purity product which may be inexpensively upgraded by a subsequent process. Increasing the purity of N_2 by the introduction of H_2 or CH_4 to react with unwanted O_2 is a good example. Unless permeates are recycled, high product purity is accompanied by lower product recovery.

Pressure ratio (Φ) is quite important, but transmembrane ΔP matters as well. Consider the case of a vacuum permeate ($\Phi = \infty$): The membrane area will be an inverse function of P . Φ influences separation and area, transmembrane ΔP influences area.

In a binary separation, the highest purity of integrated permeate occurs at $\Theta = 0$. Purity decreases monotonically until it reaches the feed purity at $\Theta = 1$. In a ternary system, the residue concentration of the gas with the intermediate permeability will reach a maximum at some intermediate stage cut.

Concentration polarization is a significant problem only in vapor separation. There, because the partial pressure of the penetrant is normally low and its solubility in the membrane is high, there can be depletion in the gas phase at the membrane. In other applications it is usually safe to assume bulk gas concentration right up to the membrane.

Another factor to remember is that for $\alpha = 1$, or for $\Phi = 1$, or for $(1/\Theta) = 1$ there is no separation at all. Increasing any of the quantities as defined make for a better separation, but the improvement is diminishing in all cases as the value moves higher. An example of the economic tradeoff between permeability and α is illustrated in Fig. 20-72 where the economics are clearly improved by sacrificing selectivity for flux.

Compression If compression of either feed or permeate is required, it is highly likely that compression capital and operating costs will dominate the economics of the gas-separation process. In some applications, pressure is essentially "free," such as when removing small quantities of CO_2 from natural gas. The gas is often

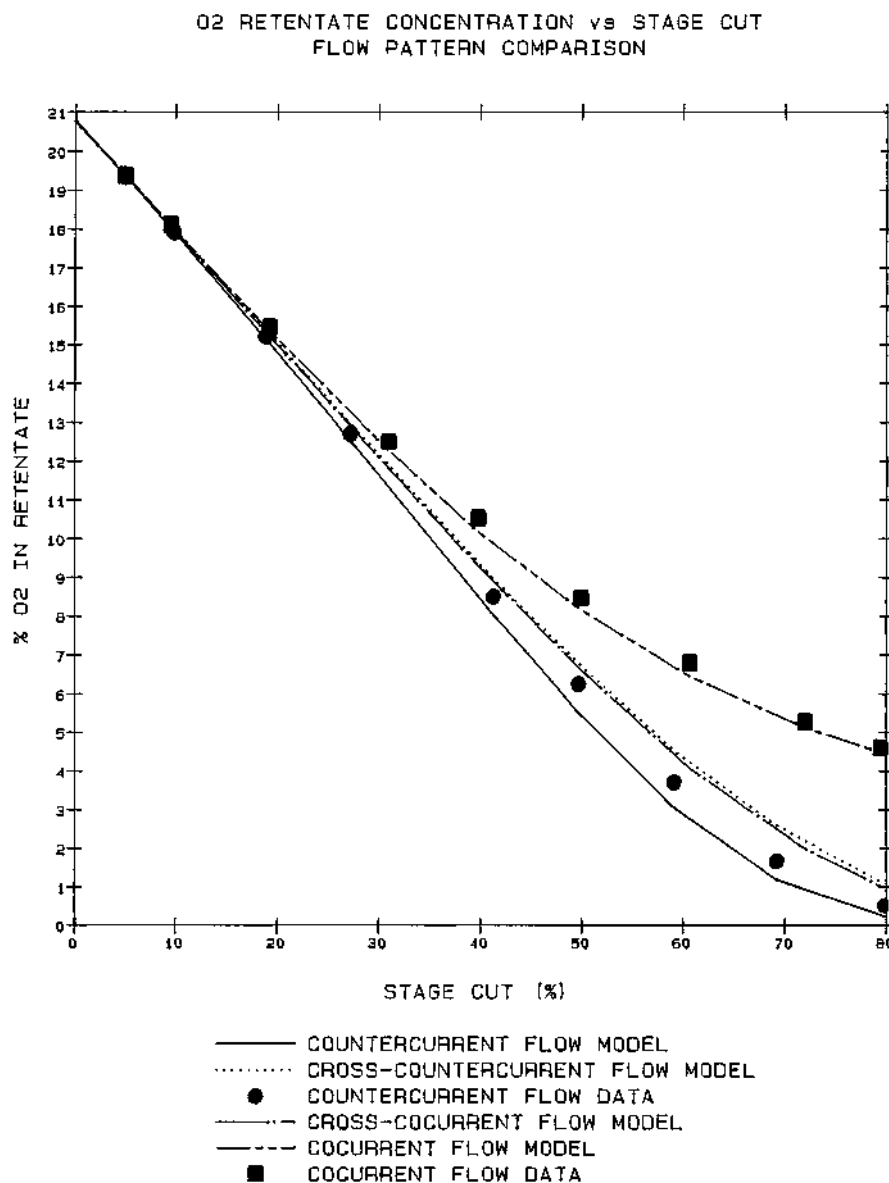


FIG. 20-71 Air fractionation by membrane. O₂ in retentate as a function of feed fraction passed through the membrane (stage cut) showing the different result with changing process paths. Process has shell-side feed at 690 kPa (abs) and 298 K. Module comprised of hollow fibers, diameter 370 μm od $\times$ 145 μm id $\times$ 1500 mm long. Membrane properties $\alpha = 5.7$ (O₂/N₂), permeance for O₂ = 3.75×10^{-6} Barrer/cm. (Courtesy Innovative Membrane Systems/Praxair.)

produced at pressure, but is compressed for transmission anyway, and since the residue constitutes the product, it continues downstream at pressure. H₂ frequently enters the separation process at pressure, an advantage for membranes. Unlike CH₄, the H₂ is in the permeate, and recompression may be a significant cost. A relative area cure for CO₂/CH₄ is shown in Fig. 20-73. When the permeate is the product (H₂, CO₂) the increasing membrane area shown in Fig. 20-74 is largely the effect of more gas to pass through the membrane, since the curve is based on a constant volume of feed gas, not a constant output of H₂. The facts of life in compressor economics are in painful opposition to the desires of the membrane designer. Pressure ratios higher than six become expensive; vacuum is very expensive, and scale is important.

Because of compressor economics, staging membranes with recompression is unusual. Designers can assume that a flow sheet that mixes unlike streams or reduces pressure through a throttling valve will increase cost in most cases.

Product Losses Account must be taken of the product loss, the slow gas in the permeate (such as CH₄), or the fast gas in the residue (such as H₂). Figure 20-75 illustrates the issue for a membrane used to purify natural gas from 93 percent to pipeline quality, 98 percent. In the upper figure, the gas is run through a permeator bank operating as a single stage. For the membrane and module chosen, the permeate contains 63 percent CH₄. By dividing the same membrane area into two stages, two permeates (or more) may be produced, one of which

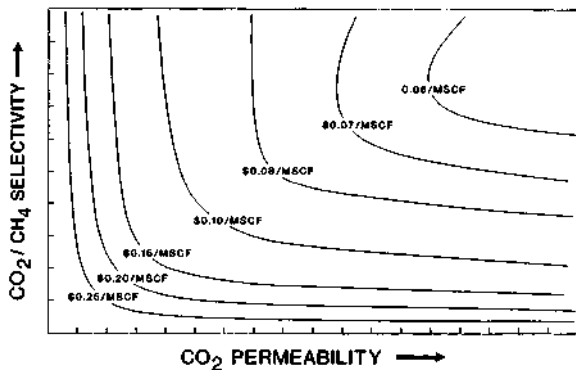


FIG. 20-72 Constant-cost lines as a function of permeability and selectivity for CO_2/CH_4 . Cellulose-acetate membrane "mscf" is one thousand standard cubic feet. (Courtesy W. R. Grace.)

may have significantly higher economic value than the single mixed permeate. In fact, where CH_4 is involved, another design parameter is the local economic value of various waste streams as fuel.

Membrane Replacement Membrane replacement is a significant cost factor, but membrane life and reliability are now reasonable. Membranes are more susceptible to operating upsets than more traditional equipment, but their field-reliability record in properly engineered, properly maintained installations is good to excellent. In N_2 separations, membrane life is very long.

Competing Technologies The determination of which separation technique is best for a specific application is a dynamic function of advances in membranes and several other technologies. At this writing, very small quantities of pure components are best obtained by purchase of gas in cylinders. For N_2 , membranes become competitive at moderate flow rates where purity required is <99.5 percent. At higher flow rates and higher purity, *pressure swing adsorption* (PSA) is better (see Sec. 16: "Adsorption and Ion Exchange"). At still higher volumes, delivered liquid N_2 , pipeline N_2 , or on-site distillation will be superior. For O_2 , membranes have little economic importance. The problem is the economic cost of using vacuum on the permeate side, or the equally unattractive prospect of compressing the feed and operating at low stage cut. For H_2 , membranes are dominant when the feed is at high pressure, except for high purity (excepting Pd noted above) and very large volume. Higher purity or lower feed pressure favor PSA. Very high volume favors cryogenic separation. For CH_4 , membranes compete at low purity and near pipeline (98 percent)

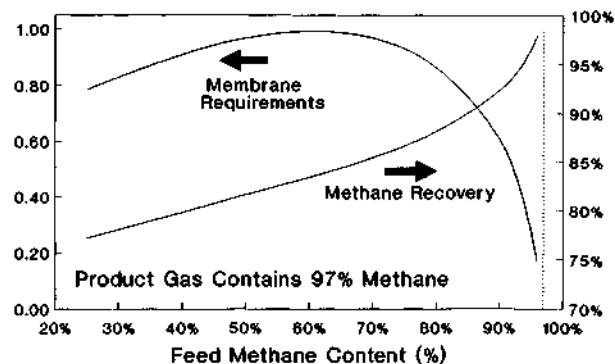


FIG. 20-73 Influence of feed purity on total membrane area when the residue gas at fixed purity is the product. Feed-gas volume is constant. CO_2/CH_4 cellulose-acetate membrane, $\alpha = 21$. (Courtesy W. R. Grace.)

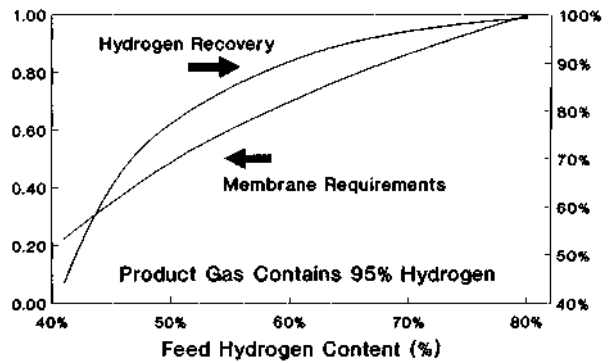


FIG. 20-74 Influence of feed purity on total membrane area when the permeate gas at fixed purity is the product. Feed-gas volume is constant. H_2/CH_4 cellulose-acetate membrane, $\alpha = 45$. (Courtesy W. R. Grace.)

purity, but distillation and absorption are very competitive at large scale and for intermediate CO_2 contamination levels.

Membranes are found as adjuncts to most conventional processes, since their use can improve overall economics in cases where membrane strength coincides with conventional process weakness.

PERVAPORATION

Process Description Pervaporation is a separation process in which a liquid mixture contacts a nonporous permselective membrane. One component is transported through the membrane preferentially. It evaporates on the downstream side of the membrane leaving as a vapor. The name is a contraction of permeation and evaporation. Permeation is induced by lowering partial pressure of the permeating component, usually by vacuum or occasionally with a sweep gas. The permeate is then condensed or recovered. Thus, three steps are necessary: Sorption of the permeating components into the membrane, diffusive transport across the nonporous membrane, then desorption into the permeate space, with a heat effect. Pervaporation membranes are chosen for high selectivity, and the permeate is often highly purified.

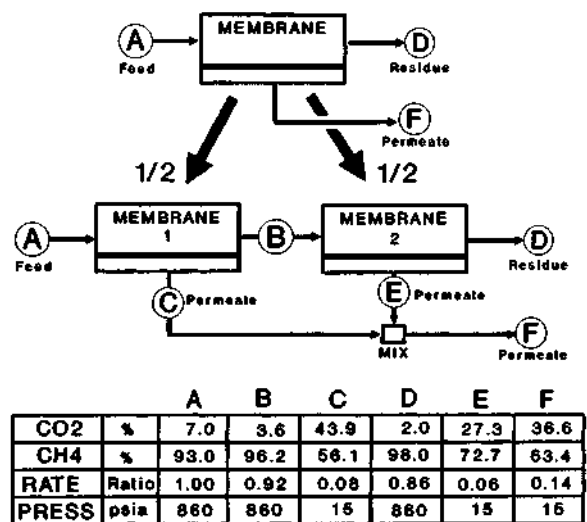


FIG. 20-75 Effect on permeate of dividing a one-stage separation into two equal stages having the same total membrane area. Compositions of A, D, and F are equal for both cases. (Courtesy W. R. Grace.)

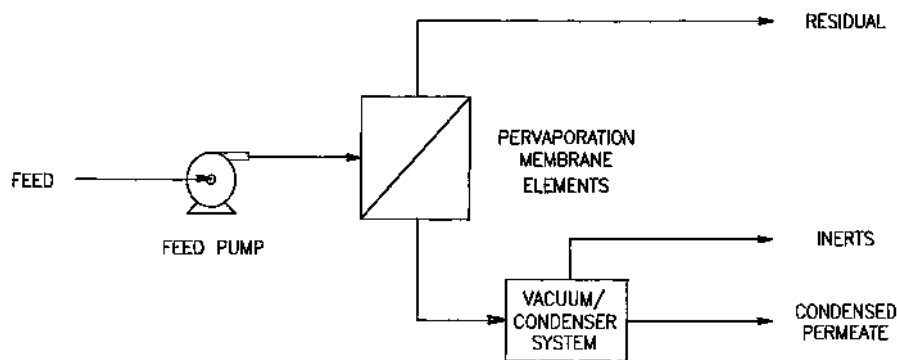


FIG. 20-76 Simplified flow schematic for a pervaporation system. Heated feed enters from left through a feed pump. Heaters in a recirculating feed loop may be required (not shown). Stripped liquid exits at the top of the pervaporation membrane. Vapor exits at the bottom to a condenser. Liquid and noncondensibles are removed under vacuum. (Courtesy Hoechst Celanese.)

In the flow schematic (Fig. 20-76), the condenser controls the vapor pressure of the permeating component. The vacuum pump, as shown, pumps both liquid and vapor phases from the condenser. Its major duty is the removal of noncondensibles. Early work in pervaporation focused on organic-organic separations. Many have been demonstrated; few if any have been commercialized. Still, there are prospects for some difficult organic separations.

An important characteristic of pervaporation that distinguishes it from distillation is that it is a rate process, not an equilibrium process. The more permeable component may be the less-volatile component. Pervaporation has its greatest utility in the resolution of azeotropes, as an adjunct to distillation. Selecting a membrane permeable to the minor component is important, since the membrane area required is roughly proportional to the mass of permeate. Thus pervaporation devices for the purification of the ethanol-water azeotrope (95 percent ethanol) are always based on a hydrophilic membrane.

Pervaporation membranes are of two general types. **Hydrophilic membranes** are used to remove water from organic solutions, often from azeotropes. **Hydrophobic membranes** are used to remove organic compounds from water. The important operating characteristics of hydrophobic and hydrophilic membranes differ. Hydrophobic membranes are usually used where the solvent concentration is about 6 percent or less—above this value, other separation methods are usually cheaper unless the flow rate is small. At low-solvent levels the usual membrane (silicone rubber) is not swollen appreciably, and movement of solvent into the membrane makes depletion of solvent in the boundary layer (concentration polarization) an important design problem. Hydrophilic pervaporation membranes operate such that the upstream portion is usually swollen with water, while the downstream face is low in water concentration because it is being depleted by vaporization. Fluxes are low enough ($<5 \text{ kg/m}^2\text{-hr}$) that boundary layer depletion (liquid side) is not limiting.

The simplifying assumptions that make Fick's law useful for other processes are not valid for pervaporation. The activity gradient across the membrane is far more important than the pressure gradient. Equation (20-98) is generally used to describe the pervaporation process:

$$J_i = -D_i C \frac{d \ln a_i}{dz} \quad (20-98)$$

where a_i is the activity coefficient of component i .

This equation is not particularly useful in practice, since it is difficult to quantify the relationship between concentration and activity. The Flory-Huggins theory does not work well with the crosslinked semicrystalline polymers that comprise an important class of pervaporation membranes. Neel (in Noble and Stern, op. cit., pp. 169–176) reviews modifications of the Stefan-Maxwell approach and other equations of state appropriate for the process.

A typical permeant-concentration profile in a pervaporation membrane is shown in Fig. 20-77. The concentration gradient at the permeate (vapor) side of the hydrophilic membrane is usually rate controlling. Therefore the downstream pressure (usually controlled by condenser temperature) is very important for flux and selectivity. Since selectivity is the ratio of fluxes of the components in the feed, as downstream pressure increases, membrane swelling at the permeate interface increases, and the concentration gradient at the permeate interface decreases. The permeate flux drops, and the more swollen membrane is less selective. A rise in permeate pressure may result in a drastic drop in membrane selectivity. This effect is diminished at low water concentrations, where membrane swelling is no longer dominant. In fact, when the water concentration drops far enough, permeate backpressure loses its significance. (See Fig. 20-78.)

For rubbery membranes (hydrophobic), the degree of swelling has less effect on selectivity. Thus the permeate pressure is less critical to the separation, but it is critical to the driving force, thus flux, since the vapor pressure of the organic will be high compared to that of water.

Definitions Following the practice presented under "Gas-Separation Membranes," distillation notation is used. Literature articles often use mass fraction instead of mole fraction, but the substitution of one to the other is easily made.

$$\beta = \frac{y_i}{x_i} \quad (20-99)$$

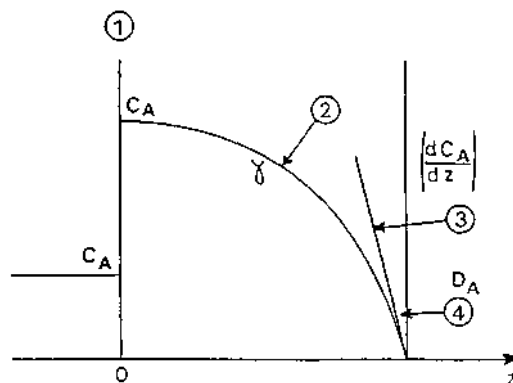


FIG. 20-77 Permeant-concentration profile in a pervaporation membrane. 1—Upstream side (swollen). 2—Convex curvature due to concentration-dependent permeant diffusivity. 3—Downstream concentration gradient. 4—Exit surface of membrane, depleted of permeant, thus unswollen. (Courtesy Elsevier.)

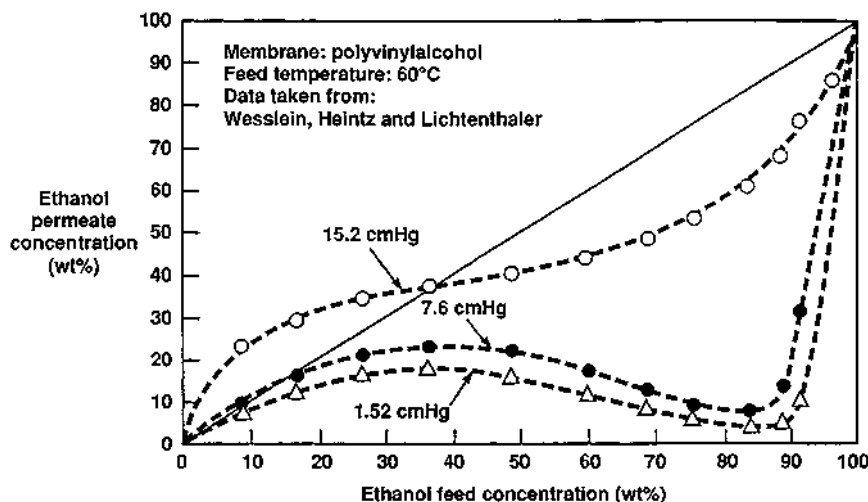


FIG. 20-78 Pervaporation schematic for ethanol-water. The illustration shows the complex behavior for a simple system at three pressures. Only the region above 90 percent is of commercial interest. (Courtesy Elsevier.)

where β is the enrichment factor. β is related to α [Eq. (20-92)], the separation factor, by

$$\beta = \frac{\alpha}{1 + (\alpha - 1)x_i} \quad \alpha = \frac{\beta(1 - x_i)}{1 - x_i\beta} \quad (20-100)$$

α is larger than β , and conveys more meaning when the membrane approaches the ideal. β is preferred in pervaporation because it is easier to use in formulations for cost, yield, and capacity. In fact, note that neither factor is constant, and that both generally change with x_i and temperature.

Operational Factors In industrial use, pervaporation is a continuous-flow single-stage process. Multistage cascade devices are unusual. Pervaporation is usually an adjunct separation, occasionally a principal one. It is used either to break an azeotrope or to concentrate a minor component. Large stand-alone uses may develop in areas where distillation is at a disadvantage, such as with closely boiling components, but these developments, have not yet been made. Notable exceptions occur when a stream is already fairly pure, such as a contaminated water source, or isopropanol from microelectronics fabrication washing containing perhaps 15 percent water.

In continuous flow, the feed will begin with concentration ($x_{i,feed}$) and be stripped until it reaches some desired ($x_{i,residue}$). If significant mass is removed from the feed, it will cool, since the general rule for liquids is that the latent heat exceeds the specific heat per °C by two orders of magnitude (more for water). So the membrane has a concentration gradient and a temperature gradient both across it and along it. Pervaporation is a complex multigradient process.

The term β is not constant for some important separations. Even worse, it can exhibit maxima that make analytic treatment difficult. Operating diagrams are often used for preliminary design rather than equations. Because of the very complex behavior in the membrane as concentrations change, all design begins with an experiment. For water removal applications, design equations often mispredict the rate constants as the water content of the feed approaches zero. The estimates tend to be lower than the experimental values, which would lead to oversize. Therefore it is necessary to obtain experimental data over the entire range of water concentrations encountered in the separation. Once the kinetic data are available, heat transfer and heat capacity are the problem. It is general practice to pilot the separation on a prototype module to measure the changes due to thermal effects. This is particularly true for water as the permeant, given its high latent heat.

For removal of an organic component from water, swelling of the organophilic membrane would result in a higher flux and lower α . At organic levels below about 10 percent, that has not been a major

problem. In most applications, boundary-layer mass-transfer limitations become limiting. Pilot data must therefore be taken with hydrodynamic similarity to the module that will be used and the actual organic permeation rate may become limited by the boundary layer. In organic-organic pervaporation, membrane swelling is a major concern.

Vapor Feed A variant on pervaporation is to use vapor, rather than liquid, as a feed. While the resulting process could be classified along with gas-separation membranes, it is customarily regarded as pervaporation.

The residue at the top of a distillation column is a vapor, so there is logic in using it as the feed to a membrane separator. Weighed against the obvious advantage are disadvantages of vapor handling (compressors cost more than pumps) and equipment size to handle the larger vapor volume. When the more volatile component permeates, heat must be added to maintain superheat. The vapor feed technique has been used in a few large installations where the advantages outweigh the disadvantages.

Leading Examples

Dehydration The growing use of isopropanol as a clean-rinse fluid in microelectronics produces significant quantities of an 85–90 percent isopropanol waste. Removing the water and trace contaminants is required before the alcohol can be reused. Pervaporation produces a 99.99 percent alcohol product in one step. It is subsequently polished to remove metals and organics. In Europe, dehydration of ethanol is the largest pervaporation application. For the very large ethanol plants typical of the United States, pervaporation is not competitive with thermally integrated distillation.

Organic from Water An area where pervaporation may become important is in flavors, fragrances, and essential oils. Here, high-value materials with unique properties are recovered from aqueous or alcohol solutions.

Pollution Control Pervaporation is used to reduce the organic loading of a waste stream, thus effecting product recovery and a reduction in waste-treatment costs. An illustration is a waste stream containing 11 percent (wt) *n*-propanol. The residue is stripped to 0.5 percent and 96 percent of the alcohol is recovered in the permeate as a 45 percent solution. This application uses a hydrophobic, rubbery membrane. The residue is sent to a conventional waste-treatment plant.

Pervaporation Membranes Pervaporation has a long history, and many materials have found use in pervaporation experiments. Cellulosic-based materials have given way to polyvinyl alcohol and blends of polyvinyl alcohol and acrylics in commercial water-removing membranes. These membranes are typically solution cast (from

water) on ultrafiltration membrane substrates. It is important to have enough crosslinking in the final polymer to avoid dissolution of the membrane in use. A very thin membrane with little penetration into the UF substrate is required. The substrate can be a problem, as it provides significant pressure drop to the vapor passing through it which, as mentioned above, has serious ramifications for flux and separation efficiency. Many new membranes are under development. Ion-exchange membranes, and polymers deposited by or polymerized by plasma, are frequently mentioned in the literature.

Modules Every module design used in other membrane operations has been tried in pervaporation. One unique requirement is for low hydraulic resistance on the permeate side, since permeate pressure is very low (0.1–1 Pa). The rule for near-vacuum operation is the bigger the channel, the better the transport. Another unique need is for heat input. The heat of evaporation comes from the liquid, and intermediate heating is usually necessary. Of course economy is always a factor. Plate-and-frame construction was the first to be used in large installations, and it continues to be quite important. Some smaller plants use spiral-wound modules, and some membranes can be made as capillary bundles. The capillary device with the feed on the inside of the tube has many advantages in principle, such as good vapor-side mass transfer and economical construction, but it is still limited by the availability of membrane in capillary form.

ELECTRODIALYSIS

GENERAL REFERENCES: This section is based on three publications by Heiner Strathmann, to which the interested reader is referred for greater detail [chap. 6, pp. 213–281, in Noble and Stern (eds.), op. cit.; sec. V, pp. 217–262, in Ho and Sirkar, op. cit.; chap. 8, pp. 8-1–8-53, Baker et al., op. cit.].

Process Description *Electrodialysis* (ED) is a membrane separation process in which ionic species are separated from water, macrosolutes, and all uncharged solutes. Ions are induced to move by an electrical potential, and separation is facilitated by ion-exchange membranes. Membranes are highly selective, passing either anions or cations and very little else. The principle of ED is shown in Fig. 20-79.

The feed solution containing ions enters a compartment whose walls are a cation-exchange and an anion-exchange membrane. If the

anion-exchange membrane is in the direction of the anode, as shown for the middle feed compartment, anions may pass through that membrane in response to an electrical potential. The cations can likewise move toward the cathode. When the ions arrive in the adjacent compartments, however, their further progress toward the electrodes is prevented by a membrane having the same electrical charge as the ion. The two feed compartments to the left and right of the central compartment are concentrate compartments. Ions entering these two compartments, either in the feed or by passing through a membrane, are retained, either by a same-charged membrane or by the EMF driving the operation. The figure shows two cells (four membranes) between anode and cathode. In an industrial application, a membrane stack can be composed of hundreds of cells, where mobile ions are alternately being depleted and concentrated.

Many related processes use charged membranes and/or EMF. Electrodialytic water dissociation (water splitting), diffusion dialysis, Donnan dialysis, and electrolysis are related processes. **Electrolysis** (chlorine-caustic) is a process of enormous importance much of which is processed through very special membranes.

Leading Examples Electrodialysis has its greatest use in removing salts from brackish water, where feed salinity is around 0.05–0.5 percent. For producing high-purity water, ED can economically reduce solute levels to extremely low levels as a hybrid process in combination with an ion-exchange bed. ED is not economical for the production of potable water from seawater. Paradoxically, it is also used for the concentration of seawater from 3.5 to 20 percent salt. The concentration of monovalent ions and selective removal of divalent ions from seawater uses special membranes. This process is unique to Japan, where by law it is used to produce essentially all of its domestic table salt. ED is very widely used for deashing whey, where the desalted product is a useful food additive, especially for baby food.

Many ED-related processes are practiced on a small scale, or in unique applications. Electrodialysis may be said to do these things well: separate electrolytes from nonelectrolytes and concentrate electrolytes to high levels. It can do this even when the pH is very low. ED does not do well at: removing the last traces of salt (although the hybrid process, electrodeionization, is an exception), running at high pH, tolerating surfactants, or running under conditions where solubility

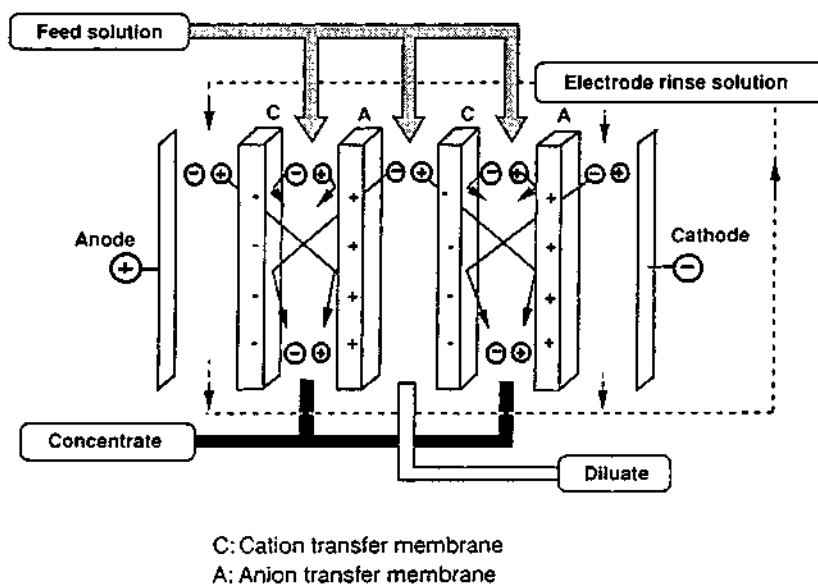


FIG. 20-79 Schematic diagram of electrodialysis. Solution containing electrolyte is alternately depleted or concentrated in response to the electrical field. Feed rates to the concentrate and diluate cells need not be equal. In practice, there would be many cells between electrodes.

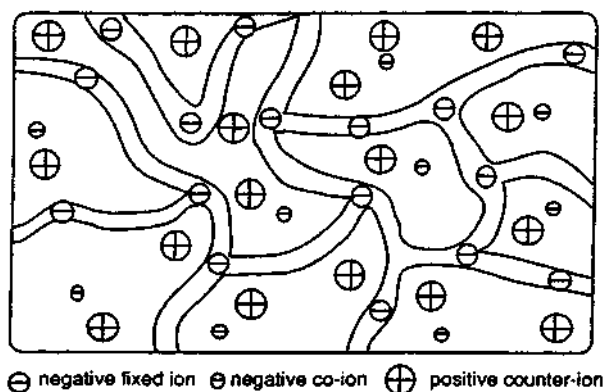


FIG. 20-80 Schematic diagram of a cation-exchange membrane showing the polymer matrix with fixed negative charges and mobile positive counterions. The density of fixed negative charges is sufficient to prevent the passage (exchange) of anions.

limits may be exceeded. Hydroxyl ion and especially hydrogen ion easily permeate both types of ED membrane. Thus, processes that generate a pH gradient across a membrane are limited in their scope.

Water splitting, a closely related process, is useful for reconstituting an acid and a base out of a salt. It is used to reclaim salts produced during neutralization.

Membranes Ion-exchange membranes are highly swollen gels containing polymers with a fixed ionic charge. In the interstices of the polymer are mobile counterions. A schematic diagram of a cation-exchange membrane is depicted in Fig. 20-80.

Figure 20-80 is a schematic diagram of a cation-exchange membrane. The parallel, curved lines represent the polymer matrix composed of an ionic, crosslinked polymer. Shown in the polymer matrix are the fixed negative charges on the polymer, usually from sulfonate groups. The spaces between the polymer matrix are the water-swollen interstices. Positive ions are mobile in this phase, but negative ions are repelled by the negative charge from the fixed charges on the polymer.

In addition to high permselectivity, the membrane must have low-electrical resistance. That means it is conductive to counterions and does not unduly restrict their passage. Physical and chemical stability are also required. Membranes must be mechanically strong and robust, they must not swell or shrink appreciably as ionic strength changes, and they must not wrinkle or deform under thermal stress. In the course of normal use, membranes may be expected to encounter the gamut of pH, so they should be stable from $0 < \text{pH} < 14$ and in the presence of oxidants.

Optimization of an ion-exchange membrane involves major trade-offs. Mechanical properties improve with crosslink density, but so does high electrical resistance. High concentration of fixed charges favors low electrical resistance and high selectivity, but it leads to high swelling, thus poor mechanical stability. Membrane developers try to combine stable polymeric backbones with stable ionic functional groups. The polymers are usually hydrophobic and insoluble. Polystyrene is the major polymer used, with polyethylene and polysulfone finding limited application.

Most commercial ion-exchange membranes are homogeneous, produced either by polymerization of functional monomers and crosslinking agents, or by chemical modification of polymers. Many heterogeneous membranes have been prepared both by melting and pressing a mixture of ion-exchange resin and nonfunctional polymer, or by dissolving or dispersing both functional and support resins in a solvent and casting into a membrane. Microheterogeneous membranes have been made by block and graft polymerization.

No membrane and no set of membrane properties has universal applicability. Manufacturers who service multiple applications have a variety of commercial membranes. One firm lists 20 different membranes having a broad spectrum of properties.

Cation-Exchange Membranes Polystyrene copolymerized with divinylbenzene, then sulfonated, is the major building block for cation-exchange membranes. These membranes have reasonable stability and versatility and are highly ionized over most of the pH range. Other chemistries mentioned in the literature include carboxylic acid membranes based on acrylic acid, PO_3^{2-} , HPO_2^- , AsO_3^{2-} , and SeO_3^- . Many specialty membranes have been produced for electrodialysis applications. A notable example is a membrane selective to monovalent cations made by placing a thin coating of positive charge on the cation-exchange membrane. Charge repulsion for polyvalent ions is much higher than that for monovalent ions, but the resistance of the membrane is also higher.

Anion-Exchange Membranes Quaternary amines are the major charged groups in anion-exchange membranes. Polystyrene-divinylbenzene polymers are common carriers for the quaternary amines. The literature mentions other positive groups based on N, P, and S. Anion-exchange membranes are problematic, for the best cations are less robust chemically than their cation exchange counterparts. Since most natural foulants are colloidal polyanions, they adhere preferentially to the anion-exchange membrane, and since the anion-exchange membrane is exposed to higher local pH there is a greater likelihood that precipitates will form there.

Membrane Efficiency The permselectivity of an ion-exchange membrane is the ratio of the transport of electric charge through the membrane by specific ions to the total transport of electrons. Membranes are not strictly semipermeable, for coions are not completely excluded, particularly at higher feed concentrations. For example, the Donnan equilibrium for a univalent salt in dilute solution is:

$$C_{\text{Co}}^M = \left(\frac{(C_{\text{Co}}^F)^2}{C_R^M} \right) \left(\frac{\gamma_{\pm}^F}{\gamma_{\pm}^M} \right)^2 \quad (20-101)$$

where C_{Co}^M is the concentration of coions (the ions having the same electrical charge as the fixed charges on the membrane); C_{Co}^F is the concentration of such coions in the ambient solution; C_R^M is the concentration of fixed charges in the gel water of the membrane; and $\gamma_{\pm}^F$ and $\gamma_{\pm}^M$ are, respectively, the geometric mean of the activity coefficients of the salt ions in the ambient solution and in the membrane. Equation (20-101) is applicable only when $C_{\text{Co}}^F \ll C_R^M$. Since the membrane properties are constant, coion transport rises roughly with the square of concentration.

Process Description Figure 20-81 gives a schematic view of an ED cell pair, showing the salt-concentration profile. In the solution compartment on the left, labeled "Diluate," anions are being attracted to the right by the anode. The high density of fixed cations in membrane "A" is balanced by a high mobile anion concentration within it. Cations move toward the cathode at the left, and there is a similar high mobile cation concentration in the fixed anion membrane "C" separating the depleting compartment from the concentrating compartment further left (not shown). For the anion permeable membrane "A," two boundary layers are shown. They represent depletion in the boundary layer on the left, and an excess in the boundary layer on the right, both due to the concentration polarization effects common to all membrane processes—ions must diffuse through a boundary layer whether they are entering or leaving a membrane. That step must proceed down a concentration gradient.

With every change in ion concentration, there is an electrical effect generated by an electrochemical cell. The anion membrane shown in the middle has three cells associated with it, two caused by the concentration differences in the boundary layers, and one resulting from the concentration difference across the membrane. In addition, there are ohmic resistances for each step, resulting from the E/I resistance through the solution, boundary layers, and the membrane. In solution, current is carried by ions, and their movement produces a friction effect manifested as a resistance. In practical applications, I^2R losses are more important than the power required to move ions to a compartment with a higher concentration.

Transfer of Ions Mass transfer of ions in ED is described by many electrochemical equations. The equations used in practice are empirical. If temperature, the flux of individual components, electroosmotic effects, streaming potential and other indirect effects are

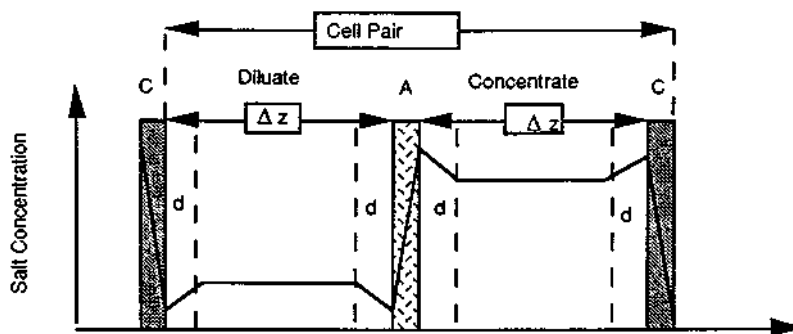


FIG. 20-81 Concentration profile of electrolyte across an operating ED cell. Ion passage through the membrane is much faster than in solution, so ions are enriched or depleted at the cell-solution interface. "d" is the concentration boundary layer. The cell gap Δz should be small. The ion concentration in the membrane proper will be much higher than shown. (Courtesy Elsevier.)

minor, an equation good to a reasonable level of approximation is:

$$J_n = C_{mn} U_{mn} \Delta\phi / \Delta x \quad (20-102)$$

where J is the component flux through the membrane $\text{kmol}/\text{m}^2 \cdot \text{s}$, C_{mn} is the concentration in phase m of component n , kmol/m^3 , U_{mn} is the ion mobility of n in m , $\text{m}^2/\text{v} \cdot \text{s}$, ϕ is the electrical potential, volts, and x is distance, m .

Equation (20-102) assumes that all mass transport is caused by an electrical potential difference acting only on cations and anions. Assuming the transfer of electrical charges is due to the transfer of ions.

$$i = F \sum_n |z_n J_n| \quad (20-103)$$

where i is the current density, amperes/ m^2 , F is the Faraday constant and z is the valence. The transport number, T_n , is the ratio of the current carried by an ion to the current carried by all ions.

$$T_n = \frac{J_n z_n}{\sum_n J_n z_n} \quad (20-104)$$

The transport number is a measure of the permselectivity of a membrane. If, for example, a membrane is devoid of coions, then all current through the membrane is carried by the counterion, and the transport number = 1. The transport numbers for the membrane and the solution are different in practical ED applications.

Concentration Polarization As is shown in the flow schematic, ions are depleted on one side of the membrane and enriched on the other. The ions leaving a membrane diffuse through a boundary layer into the concentrate, so the concentration of ions will be higher at the membrane surface, while the ions entering a membrane diffuse through a boundary layer from the diluate, so the bulk concentration in the diluate must be higher than it is at the membrane. These effects occur because the transport number of counterions in the membrane is always very much higher than their transport number in the solution. Were the transport numbers the same, the boundary layer effects would vanish. This concentration polarization is similar to that experienced in reverse osmosis, except that it has both depletion and enrichment components. The equations governing concentration polarization and depolarization of a membrane are given in the section describing ultrafiltration. The depolarizing strategies used for ED are similar to those employed in other membrane processes, as they involve induced flow past the membrane.

Two basic flow schemes are used: tortuous path flow and sheet flow (Fig. 20-82). Tortuous path spacers are cut to provide a long path between inlet and outlet, providing a relatively long residence time and high velocity past the membrane. The flow channel is open. Sheet flow units have a net spacer separating the membranes. Mass transfer is enhanced either by the spacer or by higher velocity.

Enhanced depolarization requires capital equipment and energy, but it achieves savings in overall capital costs (permits the use of a smaller stack) and energy (permits lower voltage.) The designer's task is to achieve an optimum balance in these requirements. Sheet-flow units have lower capital and operating costs in general, yet both sheet-flow and tortuous-flow units remain competitive. The fact that most ED reversal units (see below) are tortuous flow, and that ED reversal is the dominant technology for water and many waste treatment applications may explain the paradox.

Limiting Current Density As the concentration in the diluate becomes ever smaller, or as the current driving the process is increased, eventually a situation arises in which the concentration of ions at the membranes surrounding the diluate compartment approaches zero. When that occurs, there are insufficient ions to carry additional current, and the cell has reached the limiting current. Forcing the voltage higher results in the dissociation of water at the membrane, giving rise to a dramatic increase in pH due to OH^- ions emerging

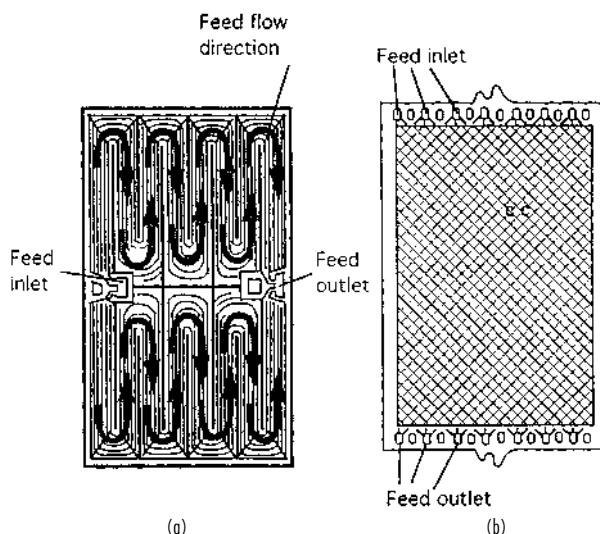


FIG. 20-82 Schematic of two ways to pass solution across an ED membrane. Tortuous flow (left) uses a special spacer to force the solution through a narrow, winding path, raising its velocity, mass transfer, and pressure drop. Sheet feed (right) passes the solution across the plate uniformly, with lower pressure drop and mass transfer. (Courtesy Elsevier.)

from the anion-exchange membrane. The high pH promotes precipitation of metal hydroxides and CaCO_3 on the membrane surface leading to flow restrictions, poor mass transfer, and subsequent membrane damage. Once a precipitate forms, its presence initiates a vicious cycle fostering the formation of more precipitates. The general expression for the limiting current density is $i_{\text{lim}} \sim (C_{\text{bulk diluate}})(\text{diluate velocity})^m$, where m is an experimental constant (often 0.6). Thus, the concentration at the membrane is limiting, and at constant current that is proportional to the bulk concentration and the mass-transfer coefficient. Flow in the compartment is laminar but mass transfer is enhanced by the spacer. A normal operating practice is to operate a stack at around 75 percent of the limiting current.

Process Configuration Figure 20-79 shows a basic cell pair. A stack is an assembly of many cell pairs, electrodes, gaskets, and manifolds needed to supply them. An exploded schematic of a portion of a sheet-flow stack is shown in Fig. 20-83.

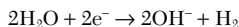
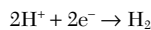
Gaskets are very important, since they not only keep the streams separated and prevent leaks from the cell, they have the manifolds to conduct feeds, both concentrate and diluate, built into them. No other practical means of feeding the stack is used in the very cramped space required by the need to keep cells thin because the diluate has very low conductivity. The manifolds are formed by aligning holes in membrane and gasket.

The membranes are supported and kept apart by feed spacers. A typical cell gap is 0.5–2 mm. The spacer also helps control solution distribution and enhances mass transfer to the membrane. Given that an industrial stack may have up to 500 cell pairs, assuring uniform flow distribution is a major design requirement.

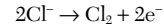
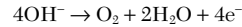
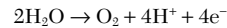
Since electrodialysis membranes are subject to fouling, it is sometimes necessary to disassemble a stack for cleaning. Ease of reassembly is a feature of ED.

Process Flow The schematic in Fig. 20-79 may imply that the feed rates to the concentrate and diluate compartments are equal. If they are, and the diluate is essentially desalted, the concentrate would leave the process with twice the salt concentration of the feed. A higher ratio is usually desired, so the flow rates of feed for concentrate and feed for diluate can be independently controlled. Since sharply differing flow rates lead to pressure imbalances within the stack, the usual procedure is to recirculate the brine stream using a feed-and-bleed technique. This is usually true for ED reversal plants. Some nonreversal plants use slow flow on the brine side avoiding the recirculating pumps. Diluate production rates are often $10\times$ brine-production rates.

Electrodes No matter how many cells are put in series, there will be electrodes. The more cells, the less the relative importance of the electrodes. The cathode reactions are relatively mild, and depending on the pH:



Anode reactions can be problematic. The anode may dissolve or be oxidized. Or, depending on the pH and the chloride concentration:



Dissolution of metal is avoided by selecting a resistant material such as Pt, Pt coated on Ti, or Pt on Nb. Base metals are sometimes used, as are graphite electrodes.

Electrode isolation is practiced to minimize chlorine production and to reduce fouling. A flush solution free of chlorides or with reduced pH is used to bathe the electrodes in some plants. Further information on electrodes may be found in a work by David ["Electrodialysis," pp. 496–499, in Porter (ed.), op. cit.].

Peripheral Components In addition to the stack, a power supply, pumps for diluate and concentrate, instrumentation, tanks for cleaning, and other peripherals are required. Safety devices are mandatory given the dangers posed by electricity, hydrogen, and chlorine.

Pretreatment Feed water is pretreated to remove gross objects that could plug the stack. Additives that inhibit the formation of scale, frequently acid, may be introduced into the feed.

Electrodialysis Reversal Two basic operating modes for ED are used in large-scale installations. **Unidirectional operation** is the mode described above in the general explanation of the process. The electrodes maintain their polarity and the ions always move in a constant direction. **ED reversal** is an intermittent process in which the polarity in the stack is reversed periodically. The interval may be from several minutes to several hours. When the polarity is reversed, the identity of compartments is also reversed, and diluate compartments become concentrate compartments and vice versa. The scheme requires instruments and valves to redirect flows appropriately after a reversal. The advantages that often justify the cost are a major reduction in membrane scaling and fouling, a reduction in feed additives required to prevent scaling, and less frequent stack-cleaning requirements.

Water Splitting A modified electrodialysis arrangement is used as a means of regenerating an acid and a base from a corresponding salt. For instance, NaCl may be used to produce NaOH and HCl. Water splitting is a viable alternative to disposal where a salt is produced by neutralization of an acid or base. Other potential applications include the recovery of organic acids from their salts and the treating of effluents from stack gas scrubbers. The new component required is a bipolar membrane, a membrane that splits water into H^+ and OH^- . At its simplest, a bipolar membrane may be prepared by laminating a cation and an anion membrane. In the absence of mobile ions, water sorbed in the membrane splits into its components when a sufficient electrical gradient is applied. The intimate contact of the two membranes minimizes the problem of the low ionic conductivity of ion-depleted water. As the water is split, replacement water readily

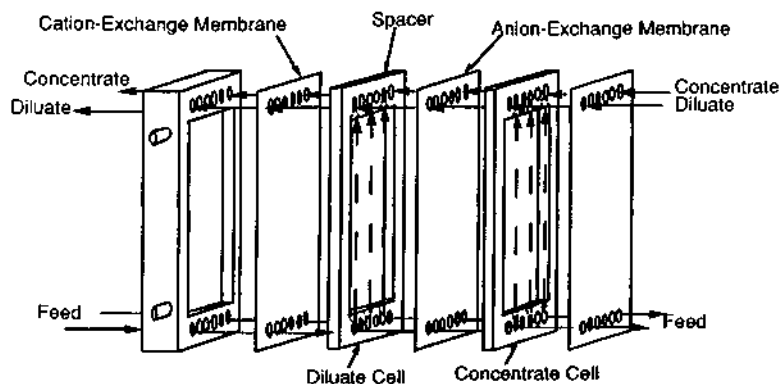


FIG. 20-83 Exploded view of a sheet-feed ED stack. Manifolds are built into the membranes and spacers as the practical way to maintain a narrow cell gap. (Courtesy Elsevier.)

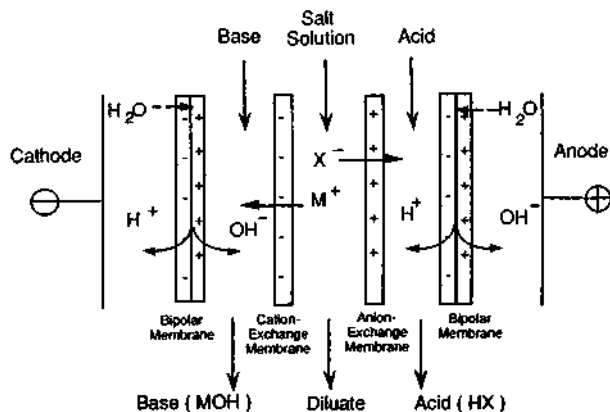


FIG. 20-84 Electrodialysis water dissociation (water splitting) membrane inserted into an ED stack. Starting with a salt, the device generates the corresponding acid and base by supplying H^+ and OH^- from the dissociation of water in a bipolar membrane. (Courtesy Elsevier.)

diffuses from the surrounding solution. Properly configured, the process is energy efficient.

A schematic of the production of acid and base by electrodialytic water dissociation is shown in Fig. 20-84. The bipolar membrane is inserted in the ED stack as shown. Salt is fed into the center compartment, and base and acid are produced in the adjacent compartments. The bipolar membrane is placed so that the cations are paired with OH^- ions and the anions are paired with H^+ . Neither salt ion penetrates the bipolar membrane. As is true with conventional electrodialysis, many cells may be stacked between the anode and the cathode.

If recovery of both acid and base is unnecessary, one membrane is left out. For example, in the recovery of a weak acid from its salt, the anion-exchange membrane may be omitted. The process limitations relate to the efficiency of the membranes, and to the propensity for H^+ and OH^- to migrate through membranes of like fixed charge, limiting the attainable concentrations of acid and base to 3–5 N. The problem is at its worst for HCl and least troublesome for organic acids. Ion leakage limits the quality of the products, and the regenerated acids and bases are not of high enough quality to use in regenerating a mixed-bed ion-exchange resin.

Diffusion Dialysis The propensity of H^+ and OH^- to penetrate membranes is useful in diffusion dialysis. An anion-exchange membrane will block the passage of metal cations while passing hydrogen ions. This process uses special ion-exchange membranes, but does not employ an applied electric current.

As an example, in the aircraft industry heavy-aluminum sections are shaped as airfoils, then masked. The areas where the metal is not required to be strong are then unmasked and exposed to NaOH to etch away unneeded metal for weight reduction. Sodium aluminate is generated, a potential waste problem. Cation-exchange membranes leak OH^- by a poorly understood mechanism that is not simply the transport of OH^- with its waters of hydration. The aluminate anion is retained in the feed stream while the caustic values pass through. NaOH recovery is high, because all the Na^+ participates in the driving force. There is considerable passage of water due to the osmotic pressure difference as well. This scheme operates efficiently only because aluminum hydroxide forms highly supersaturated solutions. Hydroxide precipitation within the apparatus is reported to be a minor problem. $Al(OH)_3$ is precipitated in a downstream crystallizer, and is reported to be of high quality.

Donnan Dialysis Another nonelectrical process using ED membranes is used to exchange ions between two solutions. The common application is to use H^+ to drive a cation from a dilute compartment to a concentrated one. A schematic is shown in Fig. 20-85. In the right compartment, the pH is 0, thus the H^+ concentration is 10^7 higher

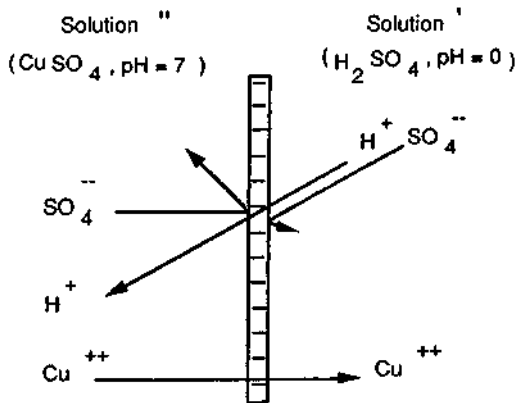


FIG. 20-85 Schematic of Donnan dialysis using a cation-exchange membrane. Cu^{2+} is "pumped" from the lower concentration on the left to a higher concentration on the right maintaining electrical neutrality accompanying the diffusion of H^+ from a low pH on the right to a higher pH on the left. The membrane's fixed negative charges prevent mobile anions from participating in the process. (Courtesy Elsevier.)

than in the pH 7 compartment on the left. H^+ diffuses leftward, creating an electrical imbalance that can only be satisfied by a cation diffusing rightward through the cation-selective membrane. By this scheme, Cu^{2+} can be "pumped" from left to right against a significant concentration difference.

Electrodialysis-Moderated Ion Exchange The production of ultrapure water is facilitated by incorporating a mixed-bed ion-exchange resin between the membranes of an ion-exchange stack. Already pure water is passed through the bed, while an electric current is passed through the stack. Provided the ion-exchange beads are in contact with each other and with the membranes, an electrical current can pass through the bed even though the conductivity of the very pure water is quite low. In passing, the current conducts any ions present into adjacent compartments, simultaneously and continuously regenerating the resin in situ.

Energy Requirements The thermodynamic limit on energy is the ideal energy needed to move water from a saline solution to a pure phase. The theoretical minimum energy is given by:

$$\Delta G = RT \ln (a/a_s) \quad (20-105)$$

where ΔG is the Gibbs free energy required to move one mole of water from a solution, a is the activity of pure water (≈ 1), and a_s is the activity of water in the salt solution. In a solution, the activity of water is approximately equal to the molar fraction of water in the solution. So that approximate activity is

$$a_s = \frac{n_s}{n_s + v \cdot s_s} \quad \frac{1}{a_s} = 1 + \frac{v \cdot s_s}{n_s} \quad (20-106)$$

where n_s is the number of moles of water in the salt solution, v is the number of atoms in the salt molecule (2 for NaCl, 3 for $CaCl_2$) and s_s is the number of moles of salt in the salt solution. The ratio of moles of salt in the salt solution to the number of moles of water in the salt solution is a very small number for a dilute solution. This permits using the approximation $\ln(1+x) = x$, when x is of the magnitude 0.01, making this an applicable approximation for saline water. That permits rewriting Eq. (20-105) as

$$\Delta G = vRT(s_s/n_s) \quad (20-107)$$

where ΔG is still the free energy required to move one mole of water from the saline solution to the pure water compartment.

The conditions utilized in the above development of minimum energy are not sufficient to describe electrodialysis. In addition to the desalination of water, salt is moved from a saline feed to a more concentrated compartment. That free-energy change must be added to the free energy given in Eq. (20-107), which describes the movement

of water from salt solution, the reverse of the actions in the diluate compartment (but having equal free energy). Schaffer & Mintz develop that change, and after solving the appropriate material balances, they arrive at a practical simplified equation for a monovalent ion salt, where activities may be approximated by concentrations:

$$\Delta G = RT(C_f - C_d) \left(\frac{\ln C_{fc}}{C_{fc} - 1} - \frac{\ln C_{fd}}{C_{fd} - 1} \right); C_{fc} = \frac{C_f}{C_c}; C_{fd} = \frac{C_d}{C_d} \quad (20-108)$$

where C_f is the concentration of ions in the feed, C_d is the concentration in the diluate, and C_c is the concentration in the concentrate, all in kmol/m^3 . When $C_{fc} \rightarrow 1$ and $C_{fd} \rightarrow \infty$, the operation is one approximating the movement of salt from an initial concentration into an unlimited reservoir of concentrate, while the diluate becomes pure. This implies that the concentrate remains at a constant salt concentration. In that case, Eq. (20-108) reduces to $RT(C_f - C_d)$. As a numerical example of Eq. (20-108) consider the desalting of a feed with initial concentration 0.05 M to 0.005 M, roughly approximating the production of drinking water from a saline feed. If 10 ℓ of product are produced for every 3 ℓ of concentrate, the concentrate leaves the process at 0.2 M. The energy calculated from Eq. (20-108) is 0.067 kWh/m^3 at 25°C. If the concentrate flow is infinite, $C_c = 0.05$ M, and the energy decreases to 0.031 kWh/m^3 .

This minimum energy is that required to move ions only, and that energy will be proportional to the ionic concentration in the feed. It assumes that all resistances are zero, and that there is no polarization. In a real stack, there are several other important energy dissipaters. One is overcoming the electrical resistances in the many components. Another is the energy needed to pump solution through the stack to reduce polarization and to remove products. Either pumping or desalting energy may be dominant in a working stack.

Energy Not Transporting Ions Not all current flowing in an electrodialysis stack is the result of the transport of the intended ions. Current paths that may be insignificant, minor, or significant include electrical leakage through the brine manifolds and gaskets, and transport of co-ions through a membrane. A related indirect loss of current is water transport through a membrane either by osmosis or with solvated ions, representing a loss of product, thus requiring increased current.

Pump Energy Requirements If there is no forced convection within the cells, the polarization limits the current density to a very uneconomic level. Conversely, if the circulation rate is too high, the

energy inputs to the pumps will dominate the energy consumption of the process. Furthermore, supplying mechanical energy to the cells raises the pressure in the cells, and raises the pressure imbalance between portions of the stack, thus the requirements of the confining gear and the gaskets. Also, cell plumbing is a design problem made more difficult by high circulation rates.

A rule of thumb for a modern ED stack is that the pumping energy is roughly 0.5 kWh/m^3 , about the same as is required to remove 1700 mg/l dissolved salts.

Equipment and Economics A very large electrodialysis plant would produce 500 ℓ/s of desalted water. A rather typical plant was built in 1993 to process 4700 m^3/day (54.4 ℓ/s). Capital costs for this plant, running on low-salinity brackish feed were \$1,210,000 for all the process equipment, including pumps, membranes, instrumentation, and so on. Building and site preparation cost an additional \$600,000. The building footprint is 300 m^2 . For plants above a threshold level of about 40 m^3/day , process-equipment costs usually scale at around the 0.7 power, not too different from other process equipment. On this basis, process equipment (excluding the building) for a 2000 m^3/day plant would have a 1993 predicted cost of \$665,000.

The greatest operating-cost component, and the most highly variable, is the charge to amortize the capital. Many industrial firms use capital charges in excess of 30 percent. Some municipalities assign long amortization periods and low-interest rates, reflecting their cost of capital. Including buildings and site preparation, the range of capital charges assignable to 1000 m^3 of product is \$90 to \$350.

On the basis of 1000 m^3 of product water, the operating cost elements (as shown in Table 20-32) are anticipated to be:

TABLE 20-32 Electrodialysis Operating Costs

\$ 66	Membrane-replacement cost (assuming seven-year life)
32	Plant power
16	Filters and pretreatment chemicals
11	Labor
8	Maintenance
\$133	Total

These items are highly site specific. Power cost is low because the salinity removed by the selected plant is low. The quality of the feed water, its salinity, turbidity, and concentration of problematic ionic and fouling solutes, is a major variable in pretreatment and in conversion.

SELECTION OF BIOCHEMICAL SEPARATION PROCESSES

GENERAL REFERENCES: Ahuja (ed.), *Handbook of Bioseparations*, Academic Press, London, 2000. Albertsson, *Partition of Cell Particles and Macromolecules*, 3d ed., Wiley, New York, 1986. Belter, Cussler, and Hu, *Bioseparations*, Wiley Interscience, New York, 1988. Cooney and Humphrey (eds.), *Comprehensive Biotechnology*, vol. 2, Pergamon, Oxford, 1985. Flickinger and Drew (eds.), *The Encyclopedia of Bioprocess Technology: Fermentation, Biocatalysis, and Bioseparations*, Wiley, New York, 1999. Harrison, Todd, Scott, and Petrides, *Bioseparations Science and Engineering*, Oxford University Press, Oxford, 2003. Hatti-Kaul and Mattiasson (eds.), *Isolation and Purification of Proteins*, Marcel Dekker, New York, 2003. Janson and Ryden (eds.), *Protein Purification*, VCH, New York, 1989. Ladisch, *Bioseparations Engineering: Principles, Practice, and Economics*, Wiley, New York, 2001. Scopes, *Protein Purification*, 2d ed., Springer-Verlag, New York, 1987. Stephanopoulos (ed.), *Biotechnology*, 2d ed., vol. 3, VCH, Weinheim, 1993. Subramanian (ed.), *Bioseparation and Bioprocess—A Handbook*, Wiley, New York, 1998. Zaslavsky, *Aqueous Two-Phase Partitioning—Physical Chemistry and Bianalytical Applications*, Marcel Dekker, New York, 1995.

GENERAL BACKGROUND

The biochemical industry derives its products from two primary sources. Natural products are yielded by plants, animal tissue, and fluids, and they are obtained via fermentation from bacteria, molds, and fungi and from mammalian cells. Products can also be obtained by

recombinant methods through the insertion of foreign DNA directly into the hosting microorganism to allow overproduction of the product in this unnatural environment. The range of bioproducts is enormous, and the media in which they are produced are generally complex and ill-defined, containing many unwanted materials in addition to the desired product. The product is almost invariably at low concentration to start with. The goals of downstream processing operations include removal of these unwanted impurities, bulk-volume reduction with concomitant concentration of the desired product, and, for protein products, transfer of the protein to an environment where it will be stable and active, ready for its intended application. This always requires a multistage process consisting of multiple-unit operations. A general strategy for downstream processing of biological materials and the types of operations that may be used in the different steps is shown in Fig. 20-86 [see also Ho, in M. R. Ladisch et al. (eds.), *Protein Purification from Molecular Mechanisms to Large-Scale Processes*, ACS Symp. Ser., 427, ACS, Washington, D.C. (1990), pp. 14-34]. Low-molecular-weight products, generally secondary metabolites such as alcohols, carboxylic and amino acids, antibiotics, and vitamins, can be recovered by using many of the standard operations such as liquid-liquid partitioning, adsorption, and ion exchange, described elsewhere in this handbook. Biofuel molecules also belong

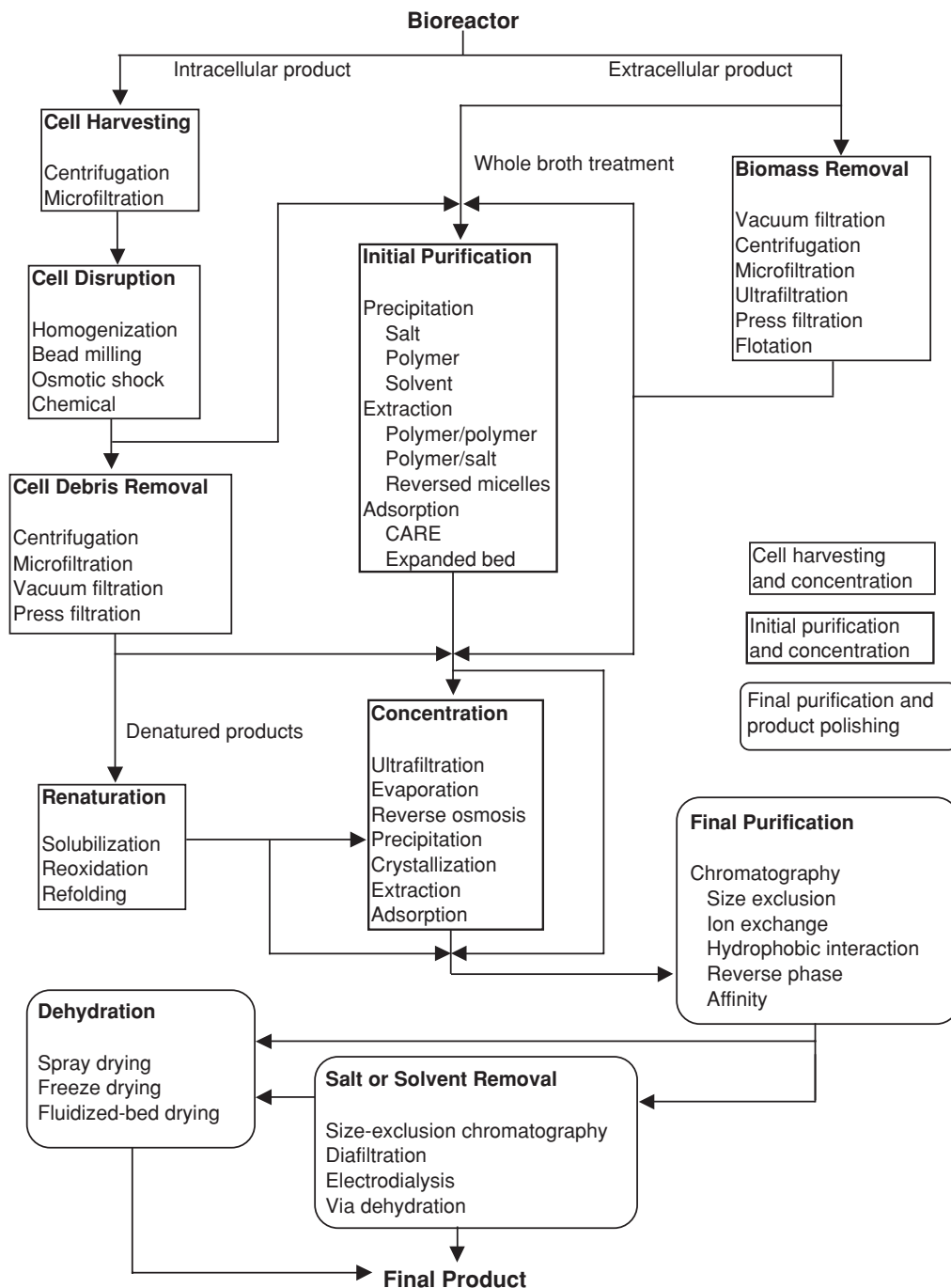


FIG. 20-86 General stages in downstream processing for protein production indicating representative types of unit operations used at each stage.

to this category. They are becoming more and more important. Less expensive and high-throughput unit operations are needed to make a biorefinery process economically feasible. Proteins require special attention, however, as they are sufficiently more complex, their function depending on the integrity of a delicate three-dimensional tertiary structure that can be disrupted if the protein is not handled correctly. For this reason, this section focuses primarily on protein separations.

Techniques used in bioseparations depend on the nature of the product (i.e., the unique properties and characteristics which provide a "handle" for the separation) and on its state (i.e., whether soluble or insoluble, intra- or extracellular, etc.). All early isolation and recovery steps remove whole cells, cellular debris, suspended solids, and colloidal particles; concentrate the product; and in many cases, achieve some degree of purification, all the while maintaining high yield. For intracellular compounds, the initial harvesting of the cells is important for their concentration prior to release of the product. Following this phase, a range of purification steps are employed to remove the remaining impurities and enhance the product purity; this purification phase, in turn, is followed by polishing steps to remove the last traces of contaminating components and process-related additions (e.g., buffer salts, detergents) and to prepare the product for storage and/or distribution. The prevention and/or avoidance of contamination is another important goal of downstream processing. Therapeutic proteins require very high purities that cannot be measured by weight percentages alone. To avoid potential harmful effects in humans, the levels of pyrogens and microbial and viral contaminants in final products must meet stringent safety and regulatory requirements.

Even for good yields of 80 to 95 percent per step, the overall yield can be poor for any process that requires a large number of steps. Thus, careful consideration must be given to optimization of the process in terms of both the unit operations themselves and their sequencing. A low-yield step should be replaced if improvement is unattainable to eliminate its impact on the overall yield. It is usually desirable to reduce the process volume early in the downstream processing, and to remove any components that can be removed fairly easily (particulates, small solutes, large aggregates, nucleic acids, etc.), so as not to overly burden the more refined separation processes downstream. Possible shear and temperature damage, and deactivation by endogenous proteases, must be considered in the selection of separation processes. Protein stability in downstream processing was discussed in depth by Hejnaes et al. [in Subramanian (ed.), vol. 2, op. cit., pp. 31–66].

INITIAL PRODUCT HARVEST AND CONCENTRATION

The initial processing steps are determined to a large extent by the location of the product species, and they generally consist of cell/broth separation and/or cell debris removal. For products retained within the biomass during production, it is first necessary to concentrate the cell suspension before homogenization or chemical treatment to release the product. Clarification to remove the suspended solids is the process goal at this stage.

Regardless of the location of the protein and its state, cell separation needs to be inexpensive, simple, and reliable, as large amounts of fermentation-broth dilute in the desired product may be handled. The objectives are to obtain a well-clarified supernatant and solids of maximum dryness, avoiding contamination by using a contained operation. Centrifugation or crossflow filtration is typically used for cell separation, and both unit operations can be run in a continuous-flow mode [Datar and Rosen, in Stephanopoulos (ed.), op. cit., pp. 369–503]. In recent years, expanded-bed adsorption has become an alternative. It combines broth clarification and adsorption separation in a single step.

Intracellular products can be present either as folded, soluble proteins or as dense masses of unfolded protein (inclusion bodies). For these products, it is first necessary to concentrate the cell suspension before effecting release of the product. Filtration can result in a suspension of cells that can be of any desired concentration up to 15 to 17 percent and that can be diafiltered into the desired buffer system. In contrast, the cell slurry that results from centrifugation will be that of

either a dry mass (requiring resuspension but substantially free of residual broth, i.e., from a tubular bowl centrifuge) or a wet slurry (containing measurable residual broth and requiring additional resuspension). During the separation, conditions that result in cell lysis (such as extremes in temperature) must be avoided. In addition, while soluble protein is generally protected from shear and external proteolysis, these proteins are still subject to thermal denaturation.

Cell Disruption Intracellular protein products are present as either soluble, folded proteins or inclusion bodies. Intracellular protein products are very common because *Escherichia coli* is a main workhorse for recombinant proteins. *E. coli* is a gram-negative bacterium that precipitates recombinant proteins in the form of inclusion bodies. Release of folded proteins must be carefully considered. Active proteins are subject to deactivation and denaturation and thus require the use of "gentle" conditions. In addition, due consideration must be given to the suspending medium; lysis buffers are often optimized to promote protein stability and protect the protein from proteolysis and deactivation. Inclusion bodies, in contrast, are protected by virtue of the protein agglomeration. More stressful conditions are typically employed for their release, which includes going to higher temperatures if necessary. For "native" proteins, gentler methods and temperature control are required.

The release of intracellular protein product is achieved through rupture of the cell walls, and release of the protein product to the surrounding medium, through either mechanical or nonmechanical means, or through chemical, physical, or enzymatic lysis [Engler, in Cooney and Humphrey (eds.), op. cit., pp. 305–324; Schutte and Kula, in Stephanopoulos (ed.), op. cit., pp. 505–526]. Mechanical methods use pressure, as in the Manton/APV-Gaulin/French Press, or the Microfluidizer, or mechanical grinding, as in ball mills, the latter being used typically for flocs and usually only for natural products. Nonmechanical means include use of desiccants or solvents, while cell lysis can also be achieved through physical means (osmotic shock, freeze/thaw cycles), chemical (detergents, chaotropes), or enzymatic (lysozyme, phages).

In a high-pressure homogenizer, a pressurized cell suspension is forced through a valve and undergoes a rapid pressure change from up to 50 MPa to the atmospheric pressure. This results in the instant rupture of cells. Product release, which generally follows first-order kinetics, occurs through impingement of the high-velocity cell suspension jet on the stationary surfaces, and possibly also by the high-shear forces generated during the acceleration of the liquid through the gap. While sufficiently high pressures can be attained using commercially available equipment to ensure good release in a single pass, the associated adiabatic temperature increases ($\sim 1.8^\circ\text{C}/1000$ psig) may cause unacceptable activity losses for heat-labile proteins. Further denaturation can occur on exposure to the lysis medium. Thus, multiple passes may be preferred, with rapid chilling of the processed cell suspension between passes. The number of passes and the heat removal ability should be carefully optimized. The efficiency of the process depends on the homogenizing pressure and the choice of the valve unit, for which there are many designs available. Materials of construction are important to minimize erosion of the valve, to provide surface resistance to aggressive cleaning agents and disinfectants, and to permit steam cleaning and sanitization.

The release of inclusion bodies, in contrast, may follow a different strategy. Since inclusion bodies are typically recovered by centrifugation, it is often advantageous to send the lysate through the homogenizer with multiple passes to decrease the particle size of the cell debris. Since the inclusion bodies are much denser than the cell debris, the debris, now much reduced in size, can be easily separated from the inclusion bodies by centrifugation at low speeds. The inclusion bodies may be resuspended and centrifuged multiple times (often in the presence of low concentrations of denaturants) to clean up these aggregates. Since the inclusion bodies are already denatured, temperature control is not as important as in the case of native proteins.

Another popular method for cell disruption is to use a bead mill. In a bead mill, a cell suspension is mixed with glass or metal beads and agitated by using a rotating agitator at high speed. Bead mills have a controllable residence time compared with high-pressure homogenization.

However, they are susceptible to channeling and also fracturing of the beads. Tough cells require multiple passes to achieve a desired yield.

Chemical lysis, or solubilization of the cell wall, is typically carried out by using detergents such as Triton X-100, or the chaotropes urea, and guanidine hydrochloride. This approach does have the disadvantage that it can lead to some denaturation or degradation of the product. While favored for laboratory cell disruption, these methods are not typically used at the larger scales. Enzymatic destruction of the cell walls is also possible, and as more economical routes to the development of appropriate enzymes are developed, this approach could find industrial application. Again, the removal of these additives is an issue.

Physical methods such as osmotic shock, in which the cells are exposed to high salt concentrations to generate an osmotic pressure difference across the membrane, can lead to cell wall disruption. Similar disruption can be obtained by subjecting the cells to freeze/thaw cycles, or by pressurizing the cells with an inert gas (e.g., nitrogen) followed by a rapid depressurization. These methods are not typically used for large-scale operations.

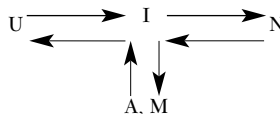
On homogenization, the lysate may drastically increase in viscosity due to DNA release. This can be ameliorated to some extent by using multiple passes to reduce the viscosity. Alternatively, precipitants or nucleic acid digesting enzymes can be used to remove these viscosity-enhancing contaminants.

For postlysis processing, careful optimization must be carried out with respect to pH and ionic strength. Often it is necessary to do a buffer exchange. Cell debris can act as an ion exchanger and bind proteins ionically, thus not allowing them to pass through a filtration device or causing them to be spun out in a centrifuge. Once optimal conditions are found, these conditions can be incorporated in the lysis buffer by either direct addition (if starting from cell paste) or diafiltration (if starting from a cell concentrate).

Protein Refolding Although protein refolding is not a bioseparation operation, it is an integral part of a downstream process for the production of an inactive (typically intracellular) protein. So far, it still remains a challenging art casting an uncertainty on the success of a process. Early in the product development stage, animal cell culture may be required to produce many bioactive candidates for initial screening of efficacy. To commercialize the product protein, it may be reexpressed in a far more economic and productive host such as *E. coli*. The commercial products of recombinant DNA technology are frequently not produced in their native, biologically active form, because the foreign hosts such as *E. coli* in which they are produced lack the appropriate apparatus for the folding of the proteins. Thus, the overproduced proteins are generally recovered as refractile or inclusion bodies, or aggregates, typically 1 to 3 μm in size, and all cysteine residues are fully reduced. It is necessary at some stage in the processing to dissolve the aggregates and then refold them to obtain the desired biologically active product [Cleland and Wang, in Stephanopoulos (ed.), op. cit., pp. 527–555].

Advantages of inclusion bodies in the production stage are their ease of separation by centrifugation following cell disruption, because of their size and density, and their provision of excellent initial-purification possibilities, as long as impurities are not copurified to any significant extent with the inclusion bodies. They also provide a high expression level and prevent endogenous proteolysis. There can be, however, significant product loss during protein refolding to the

active form. Figure 20-87 shows a typical process for refolding. The inclusion body which is released from the host cell by cell disruption is washed and then solubilized by using a denaturant such as guanidine hydrochloride (4 to 9 M), urea (7 to 8 M), sodium thiocyanate (4 to 9 M), or detergents such as Triton X-100 or sodium dodecyl sulfate. This step disrupts the hydrogen and ionic bonds to obtain fully denatured and stretched peptide chains. For proteins with disulfide bonds, addition of appropriate reducing agents (e.g., beta mercaptoethanol) is required to break all incorrectly formed intramolecular disulfide bonds. To permit proper refolding of the protein, it is necessary to remove the denaturant or detergent molecules from the surroundings of the stretched and solubilized peptides. This will initiate self-refolding of the protein molecules. For proteins with disulfide bonds, an oxidative reaction with oxygen or other oxidants is required to join two free SH groups to form an S—S covalent bond. This in vitro refolding operation is traditionally achieved by dilution with a refolding buffer. However, misfolding or aggregation is usually found in the refolding process. Analysis of in vitro refolding kinetics shows that there is at least an intermediate (I) between the unfolded protein (U) and the fully active refolded native protein (N), as illustrated below [Kuwajima and Arai, in R. H. Pain (ed.), *Mechanism of Protein Folding*, 2d ed., Oxford University Press, Oxford, 2000, pp. 138–171; Tsumoto et al., *Protein Expr. Purif.*, **28**, 1–8 (2003)].



The process to convert U to I may be fast. The process for I to N, however, may be slow and highly reversible. Some intermediate molecules may form aggregates (A) or misfolded proteins (M). Figure 20-87 is a simplified refolding pathway. In reality, the situation can be more complicated where several intermediates (I_1 , I_2 , I_3 , etc.) are present with numerous possibilities of aggregation and misfolding. For most proteins, refolding is a self-assembly process that follows a first-order kinetics. Aggregation, on the other hand, involves interactions between two or more molecules and follows the second- or higher-order kinetics. Therefore, in vitro refolding at higher protein concentrations would lead to the formation of more aggregates. Many observations have shown that a low final protein concentration, usually 10 to 100 $\mu\text{g/mL}$, is required for dilution refolding [Schlegl et al., *Chem. Engng. Sci.*, **60**, 5770–5780 (2005)]. In an industrial process, this strategy generally features large volumes of buffers, exerting an extra burden for subsequent purification steps because the concentration is low. Optimization of dilution strategy, such as the way of dilution, the speed of dilution, and the solution composition of a refolding buffer, is beneficial for an increased refolding yield.

A number of studies [Kuwajima and Arai, loc. cit.; Sadana, *Biotech. Bioeng.*, **48**, 481–489 (1995)] demonstrated that the presence of some molecules in the refolding buffer may suppress misfolding or aggregation. Molecular chaperones such as GroES and GroEL can promote correct refolding, but they are very expensive. Other molecules have been tried as artificial chaperones. The most commonly used molecules are polyethylene glycol (PEG) of various molecular weights and

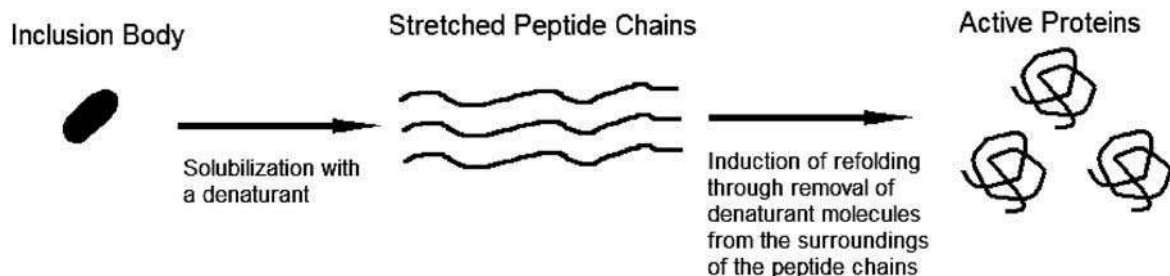


FIG. 20-87 Illustration of a refolding process for a protein from inclusion body.

concentrations, L-arginine (0.4 to 1 M), low concentrations of denaturants such as urea (1 to 2 M) and guanidine hydrochloride (0.5 to 1.5 M), and some detergents such as SDS, CTAB, and Triton X-100. The effects of these additives on protein refolding are still under investigation. A new direction of protein refolding involves the use of chromatographic techniques [Li et al., *Protein Expr. Purif.*, **33**, 1–10 (2004)]. Size exclusion, ion exchange, hydrophobic interaction, and metal chelate affinity chromatographic techniques have been studied with successful results. Chromatographic refolding explores the interaction between the protein molecule to be refolded and the packing medium in the column. It may reduce the interaction between protein molecules and increase the chance of self-assembly with the aid of the functional groups and pores in the matrix of the packing medium. An advantage for chromatographic refolding is the availability of gradient elution that creates a gradual change of the solution environment, leading to a gentle removal of the denaturant and a gradual change of favorable conditions such as pH and the artificial chaperone concentration. Simultaneous refolding and partial purification are possible with this new technique.

For extracellular products, which are invariably water-soluble, the first step is the removal of cells and cell debris using a clarification method and, in the case of typical protein products, the removal of dissolved low-molecular-weight compounds. This must be done under relatively gentle conditions to avoid undesired denaturation of the product. Again, either filtration or centrifugation can be applied. Filtration results in a cell-free supernatant with dilution associated with the diafiltration of the final cell slurry. Centrifugation, regardless of the mode, will result in a small amount of cells in the centrate, but there is no dilution of the supernatant. During the process development careful studies should be conducted to examine the effects of pH and ionic strength on the yield, as cells and cell debris may retain the product through charge interactions. If the broth or cell morphology does not allow for filtration or if dry cell mass is required, tubular-bowl centrifugation is typically utilized. Note that plant and animal cells cannot sustain the same degree of applied shear as can microbial cells, and thus crossflow filtration or classical centrifugation may not be applicable. Alternatives using low-shear equipment under gentle conditions are often employed in these situations.

For whole broths the range of densities and viscosities encountered affects the concentration factor that can be attained in the process, and can also render crossflow filtration uneconomical because of the high pumping costs, and so on. Often, the separation characteristics of the broth can be improved by broth conditioning using physicochemical or biological techniques, usually of a proprietary nature. The important characteristics of the broth are rheology and conditioning.

Clarification Using Centrifugation Centrifugation relies on the enhanced sedimentation of particles of density different from that of the surrounding medium when subjected to a centrifugal force field [Axelsson, in Cooney and Humphrey (eds.), op. cit., pp. 325–346] (see also Sec. 18, “Liquid-Solid Operations and Equipment”). Advantages of centrifugal separations are that they can be carried out continuously and have short retention times, from a fraction of a second to seconds, which limit the exposure time of sensitive biologicals to shear stresses. Yields are high, provided that temperature and other process conditions are adequately controlled. They have small space requirements, and an adjustable separation efficiency makes them a versatile unit operation. They can be completely closed to avoid contamination, and, in contrast to filtration, no chemical external aids are required that can contaminate the final product. The ability now to contain the aerosols typically generated by centrifuges adds to their operability and safety.

Sedimentation rates must be sufficiently high to permit separation, and they can be enhanced by modifying solution conditions to promote the aggregation of proteins or impurities. An increase in precipitation of the contaminating species can often be accomplished by a reduction in pH or an elevation in temperature. Flocculating agents, which include polyelectrolytes, polyvalent cations, and inorganic salts, can cause a 2000-fold increase in sedimentation rates. Some examples are polyethylene imine, EDTA, and calcium salts. Cationic bioprocessing aids (cellulosic or polymeric) reduce pyrogen, nucleic acid, and acidic protein loads which can foul chromatography columns. The removal of these additives during both centrifugation and subsequent processing must be clearly demonstrated.

There are many different types of centrifuges, classified according to the way in which the transport of the sediment is handled [Medronho, in Hatti-Kaul and Mattiasson (eds.), op. cit., pp. 131–190]. The selection of a particular centrifuge type is determined by its capacity for handling sludge; the advantages and disadvantages of various separator types are discussed by Axelsson [in Cooney and Humphrey (eds.), op. cit., pp. 325–346]. Solids-retaining centrifuges are operated in a semibatch mode, as they must be shut down periodically to remove the accumulated solids; they are primarily used when solids concentrations are low, and they have found application during the clarification and simultaneous separation of two liquids. In solids-ejecting centrifuges, the solids are removed intermittently either through radial slots or axially while the machine is running at full speed. These versatile machines can be used to handle a variety of feeds, including yeast, bacteria, mycelia, antibiotics, enzymes, and so on. Solids-discharging nozzle centrifuges have a large capacity and can accommodate up to 30 percent solids loading. Decanter centrifuges consist of a drum, partly cylindrical and partly conical, and an internal screw conveyor for transport of the solids, which are discharged at the conical end; liquids are discharged at the cylindrical end. Levels within the drum are set by means of external nozzles.

Continuous-flow units, the scroll decanter and disk-stack centrifuges, are easiest to use from an operational perspective; shutdown of the centrifuge during the processing of a batch is not expected. While the disk-stack centrifuge enjoys popularity as a process instrument within the pharmaceutical and biotechnology industries, the precise timing of solids ejection and the continuous high-speed nature of the device make for complex equipment and frequent maintenance. It is often used to harvest cells, since the solids generated are substantially wet and could lead to measurable yield losses in extracellular product systems. For intracellular product processing, the wet cell sludge is easily resuspended for use in subsequent processing.

The tubular bowl, in contrast, is a semibatch processing unit owing to the limited solids capacity of the bowl. The use of this unit requires shutdown of the centrifuge during the processing of the batch. The semibatch nature of these centrifuges can thus greatly increase processing cycle times. The introduction of disposable sheets to act as bowl liners has significantly impacted turnaround times during processing. The dry nature of the solids generated makes the tubular-bowl centrifuge well suited for extracellular protein processing, since losses to the cell sludge are minimal. In contrast, the dry, compact nature of the sludge can make the cells difficult to resuspend. This can be problematic for intracellular protein processing where cells are homogenized in easily clogged, mechanical disrupters.

Clarification Using Microfiltration Crossflow filtration (microfiltration includes crossflow filtration as one mode of operation in “Membrane Separation Processes,” which appears earlier in this section) relies on the retention of particles by a membrane. The driving force for separation is pressure across a semipermeable membrane, while a tangential flow of the feed stream parallel to the membrane surface inhibits solids settling on and within the membrane matrix [Datar and Rosen, loc. cit.].

Microfiltration is used for the removal of suspended particles, recovery of cells from fermentation broth, and clarification of homogenates containing cell debris. Particles removed by microfiltration typically average greater than 500,000 nominal molecular weight [Tutunjian, in Cooney and Humphrey (eds.), op. cit., pp. 367–381; Gobler, in Cooney and Humphrey (eds.), op. cit., pp. 351–366]. Ultrafiltration focuses on the removal of low-molecular-weight solutes and proteins of various sizes, and it operates in the less than 100,000 nominal-molecular-weight cutoff (NMWCO) range [Le and Howell, in Cooney and Humphrey (eds.), op. cit., pp. 383–409]. Both operations consist of a concentration segment (of the larger particles) followed by diafiltration of the retentate [Tutunjian, in Cooney and Humphrey (eds.), op. cit., pp. 411–437].

Generally, the effectiveness of the separation is determined not by the membrane itself, but rather by the formation of a secondary or dynamic membrane caused by interactions of the solutes and particles with the membrane. The buildup of a gel layer on the surface of an ultrafiltration membrane owing to rejection of macromolecules can provide the primary separation characteristics of the membrane. Similarly, with colloidal suspensions, pore blocking and bridging of

pore entries can modify the membrane performance, while molecules of size similar to the membrane pores can adsorb on the pore walls, thereby restricting passage of water and smaller solutes. Media containing poorly defined ingredients may contain suspended solids, colloidal particles, and gel-like materials that prevent effective microfiltration. In contrast to centrifugation, specific interactions can play a significant role in membrane separation processes.

The factors to consider in the selection of crossflow filtration include the flow configuration, tangential linear velocity, transmembrane pressure drop (driving force), separation characteristics of the membrane (permeability and pore size), size of particulates relative to the membrane pore dimensions, low protein-binding ability, and hydrodynamic conditions within the flow module. Again, since particle-particle and particle-membrane interactions are key, broth conditioning (ionic strength, pH, etc.) may be necessary to optimize performance.

Selection of Cell-Separation Unit Operation The unit operation selected for cell separations can depend on the subsequent separation steps in the train. In particular, when the operation following cell separation requires cell-free feed (e.g., chromatography), filtration is used, since centrifugation is not absolute in terms of cell separation. In addition, if cells are to be stored (i.e., they contain the desired product) because later processing is more convenient (e.g., only two-shift operation, facility competes for equipment with other products, batch is too big for single pass in equipment), it is generally better to store the cells as a frozen concentrate than a paste, since the concentrate thaws more completely, avoiding small granules of unfrozen cell solids that can foul homogenizers, columns, and filters. Here the retentate from filtration is desired, although the wet cell mass from a disc stack-type centrifuge may be used.

Centrifugation is generally necessary for complex media used to make natural products, for while the media components may be sifted prior to use, they can still contain small solids that can easily foul filters. The medium to be used should be tested on a filter first to determine the fouling potential. Some types of organisms, such as filamentous organisms, may sediment too slowly owing to their larger cross sections, and they are better treated by filtration (mycelia have the potential to easily foul tangential-flow units; vacuum-drum filtration using a filter aid, e.g., diatomaceous earth, should also be considered). Often the separation characteristics of the broth can be improved by broth conditioning using physicochemical or biological techniques, usually of a proprietary nature.

Regardless of the machine device, centrifuges are typically maintenance-intensive. Filters can be cheaper in terms of capital and maintenance costs and should be considered first unless centrifugal equipment already exists. Small facilities (< 1000 L) use filtration, since centrifugation scale-down is constrained by equipment availability. Comparative economics of the two classes of operations are discussed by Datar and Rosen (loc. cit.).

INITIAL PURIFICATION

Initial purification is the rough purification (considered by many people as isolation) to prepare a feed for subsequent high-resolution steps. In initial purification steps the goal is to obtain concentration with partial purification of the product, which is recovered as a precipitate (precipitation), a solution in a second phase (liquid-liquid partitioning), or adsorbed to solids (adsorption, chromatography).

Precipitation Precipitation of products, impurities, or contaminants can be induced by the addition of solvents, salts, or polymers to the solution; by increasing temperature; or by adjusting the solution pH (Scopes, op. cit., pp. 41–71; Ersson et al., in Janson and Ryden, op. cit., pp. 3–32). This operation is used most often in the early stages of the separation sequence, particularly following centrifugation, filtration, and/or homogenization steps. Precipitation is often carried out in two stages, the first to remove bulk impurities and the second to precipitate and concentrate the target protein. Generally, amorphous precipitates are formed, owing to occlusion of salts or solvents, or to the presence of impurities.

Salts can be used to precipitate proteins by “salting out” effects. The effectiveness of various salts is determined by the Hofmeister series, with anions being effective in the order citrate > PO₄²⁻ > SO₄²⁻

> CH₃COO⁻ > Cl⁻ > NO₃⁻, and cations according to NH₄⁺ > K⁺ > Na⁺ (Ersson et al., op. cit., p. 10; Belter et al., op. cit., pp. 221–236).

Salts should be inexpensive owing to the large quantities used in precipitation operations. Ammonium sulfate appears to be the most popular precipitant because it has an effective cation and an effective anion, high solubility, easy disposal, and low cost. Drawbacks to this approach include low selectivity, high sensitivity to operating conditions, and downstream complications associated with salt removal and disposal of the high-nitrogen-content stream. Generally, aggregates formed on precipitation with ammonium sulfate are fragile, and are easily disrupted by shear. Thus, these precipitation operations are, following addition of salt, often aged without stirring before being fed to a centrifuge by gravity feed or using low-shear pumps (e.g., diaphragm pumps).

The organic solvents most commonly used for protein precipitation are acetone and ethanol (Ersson et al., op. cit.). These solvents can cause some denaturation of the protein product. Temperatures below 0°C can be used, since the organic solvents depress the freezing point of the water. The precipitate formed is often an extremely fine powder that is difficult to centrifuge and handle. With organic solvents, in-line mixers are preferred, as they minimize solvent-concentration gradients and regions of high-solvent concentrations, which can lead to significant denaturation and local precipitation of undesired components typically left in the mother liquors. In general, precipitation with organic solvents at lower temperature increases yield and reduces denaturation. It is best carried out at ionic strengths of 0.05 to 0.2 M.

Water-soluble polymers and polyelectrolytes (e.g., polyethylene glycol, polyethylene imine polyacrylic acid) have been used successfully in protein precipitations, and there has been some success in affinity precipitations wherein appropriate ligands attached to polymers can couple with the target proteins to enhance their aggregation. Protein precipitation can also be achieved by using pH adjustment, since proteins generally exhibit their lowest solubility at their isoelectric point. Temperature variations at constant salt concentration allow for fractional precipitation of proteins.

Precipitation is typically carried out in standard cylindrical tanks with low-shear impellers. If in-line mixing of the precipitating agent is to be used, this mixing is employed just prior to the material's entering the aging tank. Owing to their typically poor filterability, precipitates are normally collected by using a centrifugal device.

Liquid-Liquid Partitioning Liquid-liquid partitioning (see also Sec. 15 on liquid-liquid extraction) involving an organic solvent is commonly known as solvent extraction or extraction. Solvent extraction is routinely used to separate small biomolecules such as antibiotics and amino acids. However, it is typically not suitable for protein fractionation with only a few exceptions because organic solvents may cause protein denaturation or degradation. A recent review of solvent extraction for bioseparations including a discussion on various parameters that can be controlled for solvent extraction was given by Gu [in Ahuja (ed.), op. cit., pp. 365–378]. As a replacement of solvent extraction, aqueous two-phase partitioning is typically used for protein purification. It uses two water-soluble polymers (and sometimes with some salts when polyelectrolytes are involved) to form two aqueous phases [Albertsson, op. cit.; Kula, in Cooney and Humphrey (eds.), op. cit., pp. 451–471]. Both phases contain water but differ in polymer (and salt) concentration(s). The high water content, typically greater than 75 percent, results in a biocompatible environment not attainable with traditional solvent extraction systems. Biomolecules such as proteins have different solubilities in the two phases, and this provides a basis for separation. Zaslavsky (op. cit., pp. 503–667) listed 163 aqueous two-phase systems including PEG-dextran-water, PEG-polyvinylmethylether-water, PEG-salt-water, polyvinylpyrrolidone-dextran-water, polyvinylalcohol-dextran-water, and Ficoll-dextran-water systems. Partitioning between the two aqueous phases is controlled by the polymer molecular weight and concentration, protein net charge and size, and hydrophobic and electrostatic interactions. Aqueous two-phase polymer systems are suitable for unclarified broths since particles tend to collect at the interface between the two phases, making their removal very efficient. They can also be used early on in the processing train for initial bulk-volume reduction and partial purification. One of the drawbacks of these systems is the subsequent need for the removal of phase-forming reagents.

Affinity partitioning is carried out by adding affinity ligands to an aqueous two-phase partitioning system. The biospecific binding of a biomolecule with the ligand moves the biomolecule to a preferred phase that enhances the partitioning of the biomolecule [Johansson and Tjerneld, in Street (ed.), *Highly Selective Separations in Biotechnology*, Blackie Academic & Professional, London, 1994, pp. 55–85]. Diamond and Hsu [in Fiechter (ed.), *Advances in Biochemical Engineering/Biotechnology*, vol. 47, Springer-Verlag, Berlin-New York, 2002, pp. 89–135] listed several dozens of biomolecules, many of which are proteins that have been separated by using affinity partitioning. Fatty acids and triazine are the two common types of affinity ligand while metallated iminodiacetic acid (IDA) derivatives of PEG such as Cu(II)IDA-PEG can be used for binding with proteins rich in surface histidines. The drawbacks of affinity partitioning include the costs of ligands and the need to couple the ligands to the polymers used in the aqueous two-phase partitioning.

Product recovery from these systems can be accomplished by changes in either temperature or system composition. Composition changes can be affected by dilution, backextraction, and micro- and ultrafiltration. As the value of the product decreases, recovery of the polymer may take on added significance. A flow diagram showing one possible configuration for the extraction and product and polymer recovery operations is shown in Fig. 20-88 [Greve and Kula, *J. Chem. Tech. Biotechnol.*, **50**, 27–42 (1991)]. The phase-forming polymer and salt are added directly to the fermentation broth. The cells or cell debris and contaminating proteins report to the salt-rich phase and are discarded. Following pH adjustment of the polymer-rich phase, more salt is added to induce formation of a new two-phase system in which the product is recovered in the salt phase, and the polymer can be recycled. In this example, disk-stack centrifuges are used to enhance the phase separation rates. Other polymer recycling options include extraction with a solvent or supercritical fluid, precipitation, or diafiltration. Electrodialysis can be used for salt recovery and recycling.

Reversed micellar solutions can also be used for the selective extraction of proteins [Kelley and Hatton, in Stephanopoulos (ed.), op. cit., pp. 593–616]. In these systems, detergents soluble in an oil phase aggregate to stabilize small water droplets having dimensions similar to those of the proteins to be separated. These droplets can host hydrophilic species such as proteins in an otherwise inhospitable organic solvent, thus enabling these organic phases to be used as protein extractants. Factors affecting the solubilization effectiveness of the solvents include charge effects, such as the net charge determined

by the pH relative to the protein isoelectric point; charge distribution and asymmetry on the protein surface; and the type (anionic or cationic) of the surfactant used in the reversed micellar phase. Ionic strength and salt type affect the electrostatic interactions between the proteins and the surfactants, and affect the sizes of the reversed micelles. Attachment of affinity ligands to the surfactants has been demonstrated to lead to enhancements in extraction efficiency and selectivity [Kelley et al., *Biotech. Bioeng.*, **42**, 1199–1208 (1993)].

Product recovery from reversed micellar solutions can often be attained by simple backextraction, by contacting with an aqueous solution having salt concentration and pH that disfavors protein solubilization, but this is not always a reliable method. Addition of cosolvents such as ethyl acetate or alcohols can lead to a disruption of the micelles and expulsion of the protein species, but this may also lead to protein denaturation. These additives must be removed by distillation, e.g., to enable reconstitution of the micellar phase. Temperature increases can similarly lead to product release as a concentrated aqueous solution. Removal of the water from the reversed micelles by molecular sieves or silica gel has also been found to cause a precipitation of the protein from the organic phase.

Extraction using liquid emulsion membranes involves the use of a surface-active agent such as a surfactant to form dispersed droplets that encapsulate biomolecules. Its economic viability for large-scale applications is still weak [Patnaik, in Subramanian (ed.), op. cit., vol. 1, pp. 267–303]. Another less-known method involving the use of a surfactant is the foam fractionation method that has seen limited applications. Ionic fluids have found commercial applications in chemical reactions by replacing volatile solvents. They are emerging as an environmentally friendly solvent replacement in liquid-liquid phase partitioning. Room-temperature ionic liquids are low-melting-point salts that stay as liquids at room temperature. Partition behavior in a system involving a room-temperature ionic fluid and an aqueous phase is influenced by the type of ionic liquid used as well as pH change [Visser et al., *Green Chem.*, Feb. 1–4, 2000]. An ionic fluid was also reportedly used in the mobile phase for liquid chromatography [He et al., *J. Chromatogr. A*, **1007**, 39–45 (2003)]. More research needs to be done in this area to develop this new green technology [Freemantle, *C&EN*, **83**, 33–38 (2005)].

Aqueous-detergent solutions of appropriate concentration and temperature can phase-separate to form two phases, one rich in detergents, possibly in the form of micelles, and the other depleted of the detergent [Pryde and Phillips, *Biochem. J.*, **233**, 525–533 (1986)].

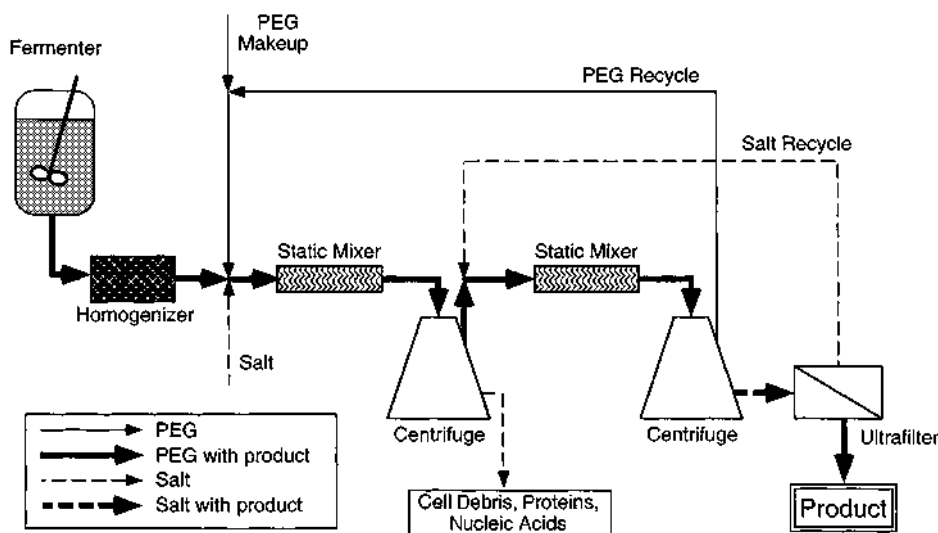


FIG. 20-88 Process scheme for protein extraction in aqueous two-phase systems for the downstream processing of intracellular proteins, incorporating PEG and salt recycling. [Reprinted from Kelly and Hatton in Stephanopoulos (ed.), op. cit.; adapted from Greve and Kula, op. cit.]

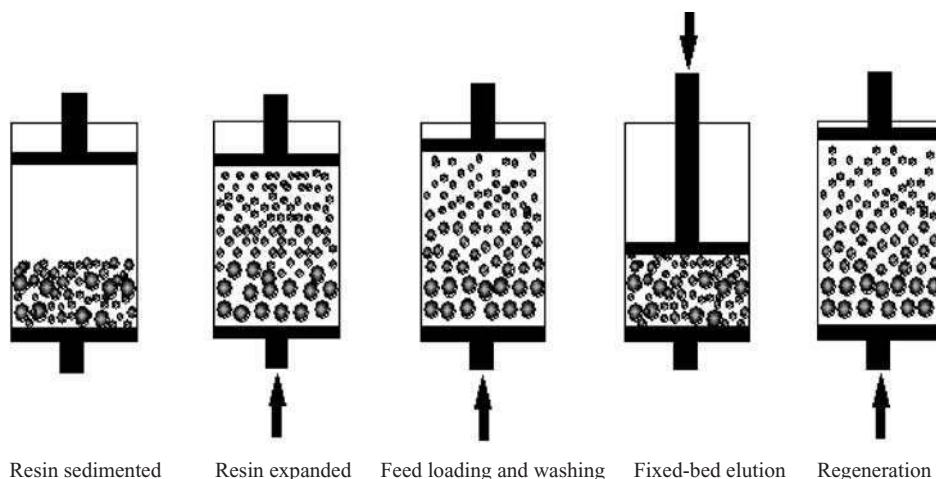


FIG. 20-89 An operation cycle in expanded-bed adsorption.

Proteins distribute between the two phases, hydrophobic (e.g., membrane) proteins reporting to the detergent-rich phase and hydrophilic proteins to the detergent-free phase. Indications are that the size-exclusion properties of these systems can also be exploited for viral separations. These systems would be handled in the same way as the aqueous two-phase systems.

On occasion, for extracellular products, cell separation can be combined with an initial volume reduction and purification step by using liquid-liquid extraction. This is particularly true for low-molecular-weight products, and it has been used effectively for antibiotic and vitamin recovery. Often scroll decanters can be used for this separation. The solids are generally kept in suspension (which requires that the solids be denser than heavy phase), while the organic phase, which must be lighter than water (cells typically sink in water), is removed. Experience shows that scrolls are good for handling the variability seen in fermentation feedstock. Podbielniak rotating-drum extraction units have been used often, but only when solids are not sticky, gummy, or flocculated, as they can get stuck in perforations of the concentric drums, but will actually give stages to the extraction in short-residence time (temperature-sensitive product). The Karr reciprocating-plate column can handle large volumes of whole-broth materials efficiently, and it is amenable to ready scale-up from small laboratory-scale systems to large plant-scale equipment.

Adsorption Adsorption (see also Sec. 16, "Adsorption and Ion Exchange") can be used for the removal of pigments and nucleic acids, e.g., or can be used for direct adsorption of the desired proteins. Stirred-batch or expanded-bed operations allow for presence of particulate matter, but fixed beds are not recommended for unclarified broths owing to fouling problems. These separations can be effected through charge, hydrophobic, or affinity interactions between the species and the adsorbent particles, as in the chromatographic steps outlined below. The adsorption processes described here are different from those traditionally ascribed to chromatography in that they do not rely on packed-bed operations.

In continuous affinity recycle extraction (CARE) operations, the adsorbent beads are added directly to the cell homogenate, and the mixture is fed to a microfiltration unit. The beads loaded with the desired solute are retained by the membrane, and the product is recovered in a second stage by changing the buffer conditions to disfavor binding.

Expanded-bed adsorption (EBA) has gained popularity in bioprocessing since its commercial introduction in the 1990s because of its ability to handle a crude feedstock that contains cells or other particulates. EBA eliminates the need for a dedicated clarification step by combining solid-liquid separation and adsorption into a single-unit operation [Hjorth et al., in Subramanian (ed.), op. cit., vol. 1, pp. 199–226; Mattiasson et al., in Ahuja (ed.), op. cit., pp. 417–451]. A

typical EBA operation cycle is illustrated in Fig. 20-89. A bed packed with an adsorption medium (or a resin), usually spherical particles of different sizes, is expanded by an upward-flow liquid stream from the bottom. An unclarified feed is introduced after a stable expansion of the bed is achieved. Particulates pass through the void spaces between the resin particles, while the soluble product molecules are adsorbed by the resin and retained in the column. After a washing step, the resin particles are left to settle in the column to form a fixed bed. The product molecules are then eluted out with a mobile phase entering from the top of the column in a way similar to that in conventional fixed-bed chromatography, to achieve a high-resolution separation. The elution can also be performed in expanded mode if needed. The regeneration step in the expanded mode flushes away residual particulates and refreshes the media for the next cycle. The difference between EBA and conventional fluidized-bed adsorption lies in the adsorption resin. In conventional fluidized-bed adsorption, the resin particles are randomly distributed in the column. In EBA, however, the resin particles are distributed vertically with large ones near the column bottom and the small ones near the top. There is no backmixing along the axial direction of the vertically standing column, thus achieving adsorption similar to that in a fixed-bed column. The resin particles have to be prepared to possess a suitable size distribution, or alternatively a distribution based on density differences if the particles sizes are uniform. An EBA column should have a length typically 3 to 4 times of the settled bed height to allow for bed expansion. An adjustable adapter at the top is needed to push the resin downward for elution in the fixed-bed mode. A proper design of the bottom frit is critical. Its holes must be smaller enough to retain the smallest resin particles but large enough to allow the particulates to enter and exit the column freely. In practical applications, plugging of the frit and nonuniform upward flow tend to be problematic, especially in columns with large diameters. To mitigate this problem, some EBA columns use an optimized inlet design and a mechanical stirrer at the bottom.

The advantages of EBA are its ability to adsorb the soluble product molecules directly from a cell suspension, a cell homogenate, or a crude biological fluid containing various particulates, thus making the "whole-broth processing" concept a reality. EBA eliminates the solid-liquid separation step (such as microfiltration and centrifugation) and enables a fast, more compact process requiring fewer steps and less time. It performs solid removal, concentration, and purification, all in a single-unit operation. By doing so, it can also minimize the risk of proteolytic breakdown of the product.

Membrane Ultrafiltration Membrane ultrafiltration is often one of the favored unit operations used for the isolation and concentration of biomolecules because they can be easily scaled up to process large feed volumes at low costs. Toward the end of an ultrafiltration operation, additional water or buffer is added to facilitate the passage of

smaller molecules. This is known as diafiltration. Diafiltration is especially helpful in the removal of small contaminating species such as unspent nutrients including salts and metabolites. Salt removal is usually necessary if the next step is ion-exchange or reverse-phase chromatography. For a protein that is not very large, two ultrafiltration steps can be used in sequence. In the first one, the protein ends up in the permeate, allowing the removal of large contaminating molecules including pyrogens and also viral particles. In the second ultrafiltration, the protein stays in the retentate. This removes small contaminating molecules and concentrates the feed up to the protein's solubility limit. Membrane materials, configurations, and design considerations were discussed earlier in this section. Proper membrane materials must be selected to avoid undesirable binding with proteins. External fouling, pore blockage, and internal fouling were discussed by Ghosh (*Protein Bioseparation Using Ultrafiltration*, Imperial College Press, London, 2003).

FINAL PURIFICATION

The final purification steps are responsible for the removal of the last traces of impurities. The volume reduction in the earlier stages of the separation train is necessary to ensure that these high-resolution operations are not overloaded. Generally, chromatography is used in these final stages. Electrophoresis can also be used, but since it is rarely found in process-scale operations, it is not addressed here. The final product preparation may require removal of solvent and drying, or lyophilization of the product.

Chromatography Liquid chromatography steps are ubiquitous in the downstream processing. It is the most widely used downstream processing operation because of its versatility, high selectivity, and efficiency, in addition to its adequate scale-up potential based on wide experience in the biochemical processing industries. As familiarity is gained with other techniques such as liquid-liquid extraction, they will begin to find greater favor in the early stages of the separation train, but are unlikely to replace chromatography in the final stages, where high purities are needed.

Chromatography is typically a fixed-bed adsorption operation, in which a column filled with chromatographic packing materials is fed with the mixture of components to be separated. Apart from size-exclusion (also known as gel-permeation or gel-filtration) chromatography, in the most commonly practiced industrial processes the solutes are adsorbed strongly to the packing materials until the bed capacity has been reached. The column may then be washed to remove impurities in the interstitial regions of the bed prior to elution of the solutes. This latter step is accomplished by using buffers or solvents which weaken the binding interaction of the proteins with the packings, permitting their recovery in the mobile phase. To minimize product loss of a high-value product, a small load far below the saturation capacity is applied to the column. A complete baseline separation can then be achieved after elution. Gradient elution uses varied modifier strength in the mobile phase to achieve better separations of more chemical species.

Types of Chromatography Practiced Separation of proteins by using chromatography can exploit a range of different physical and chemical properties of the proteins and the chromatography adsorption media [Janson and Ryden, op. cit.; Scopes, op. cit.; Egerer, in Finn and Prave (eds.), *Biotechnology Focus 1*, Hanser Publishers, Munich, 1988, pp. 95–151]. Parameters that must be considered in the selection of a chromatographic method include composition of the feed, the chemical structure and stability of the components, the electric charge at a defined pH value and the isoelectric point of the proteins, the hydrophilicity and hydrophobicity of the components, and molecular size. The different types of interactions are illustrated schematically in Fig. 20-90.

Ion-exchange chromatography relies on the coulombic attraction between the ionized functional groups of proteins and oppositely charged functional groups on the chromatographic support. It is used to separate the product from contaminating species having different charge characteristics under well-defined eluting conditions, and for concentration of the product, owing to the high-adsorptive capacity of most ion-exchange resins and the resolution attainable. Elution is carried out by using a mobile phase with competing ions or varied pH. Ion-exchange chromatography is used effectively at the front

end of a downstream processing train for early volume reduction and purification.

The differences in sizes and locations of hydrophobic pockets or patches on proteins can be exploited in *hydrophobic interaction chromatography (HIC)* and *reverse-phase chromatography (RPC)*; discrimination is based on interactions between the exposed hydrophobic residues and hydrophobic ligands which are distributed evenly throughout a hydrophilic porous matrix. As such, the binding characteristics complement those of other chromatographic methods, such as ion-exchange chromatography. In HIC, the hydrophobic interactions are relatively weak, often driven by salts in high concentration, and depend primarily on the exposed residues on or near the protein surface; preservation of the native, biologically active state of the protein is a desirable feature of HIC. HIC's popularity is on the rise in recent years because of this feature. Elution can be achieved differentially by decreasing salt concentration or increasing the concentration of polarity perturbants (e.g., ethylene glycol) in the eluent.

Reverse-phase chromatography relies on significantly stronger hydrophobic interactions than in HIC, which can result in unfolding and exposure of the interior hydrophobic residues, i.e., leads to protein denaturation and irreversible inactivation; as such, RPC depends on total hydrophobic residue content. Elution is effected by organic solvents applied under gradient conditions. RPC is the most commonly used analytical chromatographic method due to its ability to separate a vast array of chemicals with high resolutions. Denaturation of proteins does not influence the analytical outcome unless protein precipitation in the mobile phase becomes a problem.

HIC typically uses polymer-based resins with phenyl, butyl, or octyl ligands while RPC uses silica beads with straight-chain alkanes with 4, 8, or 18 carbons. Larger ligands provide stronger interactions. Polymeric beads are used in RPC when basic pH is involved because silica beads are unstable at such pH. In HIC, the mobile phase remains an aqueous salt solution, while RPC uses solvent in its mobile phase to regulate binding. Raising the temperature increases the hydrophobic interactions at the temperatures commonly encountered in biological processing.

HIC is most effective during the early stages of a purification strategy and has the advantage that sample pretreatment such as dialysis or desalting after salt precipitation is not usually required. It is also finding increased use as the last high-resolution step to replace gel filtration. It is a group separation method, and generally 50 percent or more of extraneous impurities are removed. This method is characterized by high adsorption capacity, good selectivity, and satisfactory yield of active material.

Despite the intrinsically nonspecific nature of ion-exchange and reversed-phase/hydrophobic interactions, it is often found that chromatographic techniques based on these interactions can exhibit remarkable resolution. This is attributed to the dynamics of multisite interactions being different for proteins having differing surface distributions of hydrophobic and/or ionizable groups.

Size-exclusion chromatography's (SEC's) separation mechanism is based on the sizes and shapes of proteins and impurities. The effective size of a protein is determined by its steric geometry and solvation characteristics. Smaller proteins are able to penetrate the small pores in the beads while large proteins are excluded, making the latter elute out of column more quickly. To suppress the ion-exchange side effects, a salt is typically added to the mobile phase. Ammonium carbonate or bicarbonate is used if the salt is to be removed by sublimation alone during lyophilization. In rare cases, a solvent at a low concentration is added to the mobile phase to suppress hydrophobic interactions between the protein molecules and the stationary phase. In industrial production processes, SEC columns are used to separate small molecules from proteins. It is also a choice for desalting and buffer exchange in the product polishing stage. SEC columns are typically very large because the feed loads to SEC columns are limited to 3 to 5 percent of the bed volumes. This loading capacity is far less than those with packings that have binding interactions.

Protein affinity chromatography can be used for the separation of an individual compound, or a group of structurally similar compounds from crude-reaction mixtures, fermentation broths, or cell lysates by exploiting very specific and well-defined molecular interactions

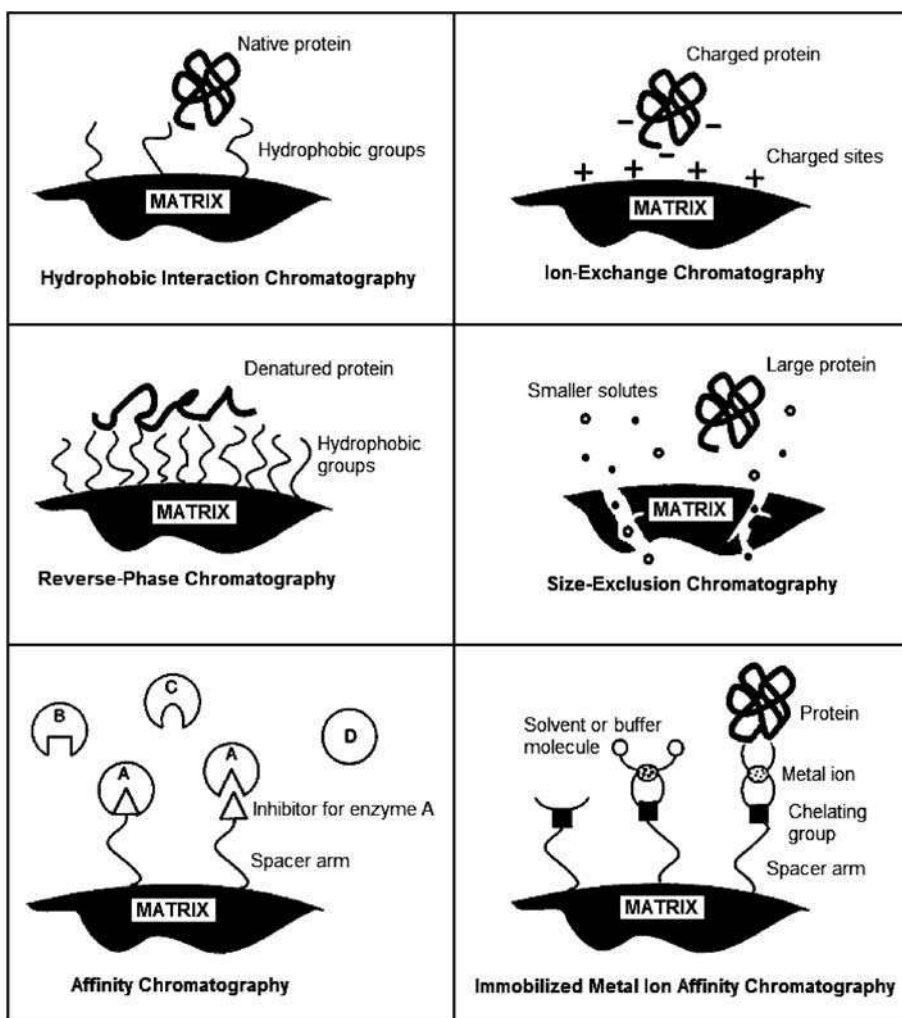


FIG. 20-90 Schematic illustration of the chromatographic methods most commonly used in downstream processing of protein products.

between the protein and affinity groups immobilized on the packing-support material. Examples of affinity interactions include antibody-antigen, hormone-receptor, enzyme-substrate/analog/inhibitor, metal ion-ligand, and dye-ligand pairs. Monoclonal antibodies are particularly effective as biospecific ligands for the purification of pharmaceutical proteins. Affinity chromatography may be used for the isolation of a pure product directly from crude fermentation mixtures in a single chromatographic step. Immunosorbents should not be subjected to crude extracts, however, as they are particularly susceptible to fouling and inactivation. Despite its high resolution and the ability to treat a very dilute feed, affinity chromatography is still costly on the process scale if protein ligands such as protein A or protein G is to be used, or a custom affinity matrix is required. Considerable research efforts are devoted to its development in part due to the increased number of protein pharmaceuticals produced at low concentrations.

After more than two decades of development, membrane chromatography has emerged as an attractive alternative to packed column chromatography. Using a porous membrane as the stationary phase in liquid chromatography has several potential advantages that include a very high flow rate through a very short and wide bed with only a modest transmembrane pressure drop. Elimination or minimization of diffusional mass-transfer resistance shifts the rate-controlling step to faster-binding kinetics, resulting in adsorptive separation of proteins

in a fraction of the time required by conventional packed columns. To achieve sufficient adsorptive separation, it is necessary to use a medium that binds strongly with target molecules when a very short flow path is involved. Thus, membrane chromatography typically uses an affinity membrane, and the combination of membrane chromatography with affinity interaction provides high selectivity and fast processing for the purification of proteins from dilute feeds. To a much less extent, ion-exchange, hydrophobic interaction, and reverse-phase membrane chromatography have also been reported [Charcosset, *J. Chem. Technol. Biotechnol.* **71**, 95–110 (1998)].

Figure 20-91 shows the interaction between proteins in the mobile phase and the affinity membrane matrix. The mechanical strength, hydrophobicity, and ligand density of the membranes can be engineered through chemical modifications to make them suitable for affinity membrane chromatography. A preferred membrane medium to be prepared for affinity membrane chromatography should provide (1) desirable physical characteristics such as pore structure and mechanical strength, allowing fast liquid flow at a small pressure loss; (2) reactive groups (such as $-\text{OH}$, $-\text{NH}_2$, $-\text{SH}$, $-\text{COOH}$) for coupling ligands or spacer arms; (3) physical and chemical stabilities to endure heat or chemical sterilization; and (4) a nondenaturing matrix to retain protein bioactivity [Zou et al., *J. Biochem. Biophys. Methods*, **49**, 199–240 (2001)]. Base membranes

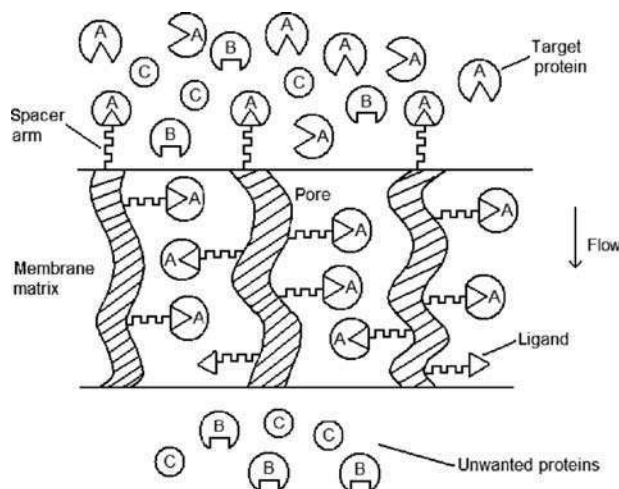


FIG. 20-91 Illustration of the interactions between proteins and membrane matrix in affinity membrane chromatography.

can be chosen from commercial membrane materials including organic, inorganic, polymeric, and composite materials. The selected membrane is first activated to create functional groups for chemical attachment of affinity ligands. If there is steric hindrance to binding between the immobilized ligand and the target molecule, a suitable spacer arm is used to bridge the activated membrane surface and the ligand. The ligand should retain its reversible binding capacity after immobilization onto the support membrane. Affinity ligands typically fall into two categories: (1) those derived from enzyme/substrate, antibody/antigen pairs that are capable of very strong and highly biospecific binding and (2) protein A and protein G, coenzyme, lectin, dyes, and metal chelates, etc., each capable of binding with a whole class of molecules. A highly specific ligand provides an unsurpassed resolution and an ability to handle a large volume of a dilute feed. However, they are typically fragile and expensive and may be unavailable off the shelf due to their narrow applications involving just one or a few molecules that can bind. Elution can also prove to be a difficult task because some biospecific bindings can be extremely tenacious. In contrast, a somewhat less specific ligand has a much wider market and thus is considerably less expensive. Various membrane cartridges have been used for affinity membrane chromatography including those with multiple layers of flat sheet membranes, hollow fibers, and spiral-wound and Chromarod membranes (Zou et al., loc. cit.).

Immobilized metal-ion affinity chromatography (IMAC) relies on the interaction of certain amino acid residues, particularly histidine, cysteine, and tryptophan, on the surface of the protein with metal ions fixed to the support by chelation with appropriate chelating compounds, invariably derivatives of iminodiacetic acid. Commonly used metal ions are Cu^{2+} , Zn^{2+} , Ni^{2+} , and Co^{2+} . Despite its relative complexity in terms of the number of factors that influence the process, IMAC is beginning to find industrial applications. The choice of chelating group, metal ion, pH, and buffer constituents will determine the adsorption and desorption characteristics. Elution can be effected by several methods, including pH gradient, competitive ligands, organic solvents, and chelating agents.

Following removal of unbound materials in the column by washing, the bound substances are recovered by changing conditions to favor desorption. A gradient or stepwise reduction in pH is often suitable. Otherwise, one can use competitive elution with a gradient of increasing concentration. IMAC eluting agents include ammonium chloride, glycine, histamine, histidine, or imidazol. Inclusion of a chelating agent such as EDTA in the eluent will allow all proteins to be eluted indiscriminately along with the metal ion.

Chromatographic Development The basic concepts of chromatographic separations are described elsewhere in this handbook. Proteins differ from small solutes in that the large number of charged and/or hydrophobic residues on the protein surface provide multiple

binding sites, which ensure stronger binding of the proteins to the adsorbents, as well as some discrimination based on the surface distribution of amino acid residues. The proteins are recovered by elution with a buffer that reduces the strength of this binding and permits the proteins to be swept out of the column with the buffer solution. In isocratic elution, the buffer concentration is kept constant during the elution period. Since the different proteins may have significantly different adsorption isotherms, the recovery may not be complete, or it may take excessive processing time and cause excessive band spreading to recover all proteins from the column. In gradient elution operations, the composition of the mobile phase is changed during the process to decrease the binding strength of the proteins successively, the more loosely bound proteins being removed first before the eluent is strengthened to enable recovery of the more strongly adsorbed species. The change in eluent composition can be gradual and continuous, or it can be stepwise. Industrially, in large-scale columns it is difficult to maintain a continuous gradient owing to difficulties in fluid distribution, and thus stepwise changes are still used. In some adsorption modes, the protein can be recovered by the successive addition of competing compounds to displace the adsorbed proteins. In all cases, the product is eluted as a chromatographic peak, with some possible overlap between adjacent product peaks.

Displacement chromatography relies on a different mode of elution. Here a displacer that is more strongly adsorbed than any of the proteins is introduced with the mobile phase. As the displacer concentration front develops, it pushes the proteins ahead of itself. The more strongly adsorbed proteins then act as displacers for the less strongly bound proteins, and so on. This leads to the development of a displacer train in which the different molecules are eluted from the column in abutting roughly rectangular peaks in the reverse order of their binding strength with the column's stationary phase. Many displacers have been developed in the past two decades including both high- and low-molecular-weight molecules. Displacement chromatography may be practical as an earlier chromatography step in a downstream process when baseline separations are not necessary. It has been considered for some industrial processes. However, displacer reuse and possible contamination of the protein product in addition to its inherent inability to achieve a clear-cut baseline separation make the use of displacement chromatography still a challenge. So far, no FDA-approved process exists [Shukla and Cramer, in Ahuja (ed.), op. cit., pp. 379-415].

For efficient adsorption it is advisable to equilibrate both the column and the sample with the optimum buffer for binding. Prior to this, the column must be cleaned to remove tightly bound impurities by increasing the salt concentration beyond that used in the product elution stages. At the finish of cleaning operation, the column should be washed with several column volumes of the starting buffer to remove remaining adsorbed material. In desorption, it is necessary to drive the favored binding equilibrium for the adsorbed substance from the stationary to the mobile phase. Ligand-protein interactions are generally a combination of electrostatic, hydrophobic, and hydrogen bonds, and the relative importance of each of these and the degree of stability of the bound protein must be considered in selecting appropriate elution conditions; frequently compromises must be made. Gradient elution often gives good results.

Changes in pH or ionic strength are generally nonspecific in elution performance; ionic strength increases are effective when the protein binding is predominantly electrostatic, as in IEC. Polarity changes are effective when hydrophobic interactions play the primary role in protein binding. By reducing the polarity of the eluting mobile phase, this phase becomes a more thermodynamically favorable environment for the protein than adsorption to the packing support. A chaotropic salt (KSCN , KCNO , KI in range of 1 to 3 M) or denaturing agent (urea, guanidine HCl; 3 to 4 M) in the buffer can also lead to enhanced desorption. For the most hydrophobic proteins (e.g., membrane proteins) one can use detergents just below their critical micelle concentrations to solubilize the proteins and strip them from the packing surface. Specific elution requires more selective eluents. Proteins can be desorbed from ligands by competitive binding of the eluting agent (low concentration of 5 to 100 mM) either to the ligand or to the protein.

Specific eluents are most frequently used with group-specific adsorbents since selectivity is greatly increased in the elution step.

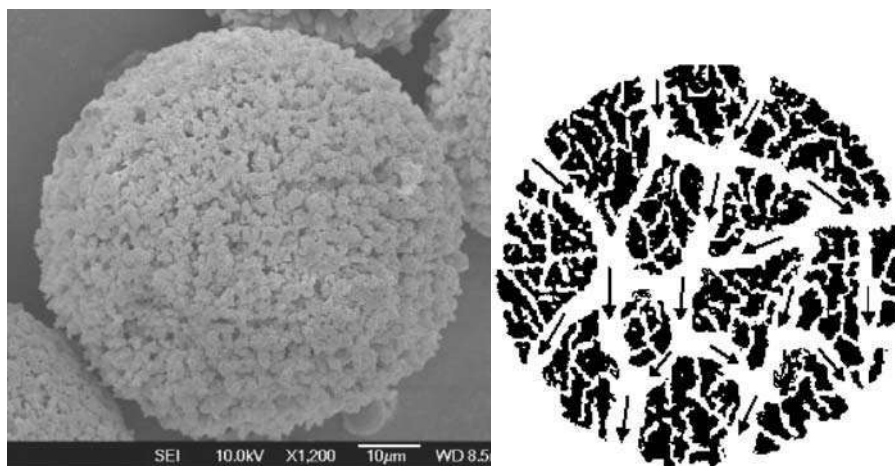


FIG. 20-92 SEM image of a poly(styrene-co-divinylbenzene) gigaporous particle synthesized from suspension polymerization and schematic of a gigaporous particle showing through-pores and diffusion pores [Gu et al., *China Particology*, 3, 349 (2005)].

The effectiveness of the elution step can be tailored by using a single eluent, pulses of different eluents, or eluent gradients. These systems are generally characterized by mild desorption conditions. If the eluting agent is bound to the protein, it can be dissociated by desalting on a gel filtration column or by diafiltration.

Column Packings The quality of the separation obtained in chromatographic separations will depend on the capacity, selectivity, and hydraulic properties of the stationary phase, which usually consists of porous beads of hydrophilic polymers filled with the solvent. The xerogels (e.g., crosslinked dextran) shrink and swell depending on solvent conditions, while aerogels have sizes independent of solution conditions. A range of materials are used for the manufacture of gel beads, classified according to whether they are inorganic, synthetic, or polysaccharides. The most widely used materials are based on neutral polysaccharides and polyacrylamide. Cellulose gels, such as crosslinked dextran, are generally used as gel filtration media, but can also be used as a matrix for ion exchangers. The primary use of these gels is for desalting and buffer exchange of protein solutions, as nowadays fractionation by gel filtration is performed largely with composite gel matrices. Agarose, a low-charge fraction of the seaweed polysaccharide agar, is a widely used packing material.

Microporous gels made by point crosslinking dextran or polyacrylamides are used for molecular-sieve separations such as size-exclusion chromatography and gel filtration, but are generally too soft at the porosities required for efficient protein chromatography. Macroporous gels are most often obtained from aggregated and physically crosslinked polymers. Examples include agarose, macroreticular polyacrylamide, silica, and synthetic polymers. These gels are good for ion-exchange and affinity chromatography as well as for other adsorption chromatography techniques. Composite gels, in which the microporous gel is introduced into the pores of macroreticular gels, combine the advantages of both types.

High matrix rigidity is offered by porous inorganic silica, which can be derivatized to enhance its compatibility with proteins, but it is unstable at alkaline pH. Hydroxyapatite particles have high selectivity for a wide range of proteins and nucleic acids.

Traditional porous media have pore sizes typically in the range of 100 to 300 Å. To reduce intraparticle diffusion for fast chromatography, column packing materials can be made either nonporous or extremely porous. Gustavsson and Larsson [in Hatti-Kaul and Mattiasson (eds), op. cit., pp. 423–454] discussed various chromatography media for fast chromatography of proteins. Nonporous particles such as the popular 5-µm modified silica beads for reverse-phase chromatography are excellent for analytical applications. However, due to their limited binding sites that exist only on the outer surface of the particles, nonporous particles are not suitable for preparative- or large-scale applications. Gigaporous media are gaining momentum in

recent years. They are particles with pore sizes above 1000 Å. Some have large enough interconnecting pores in the range of 4000 to 8000 Å that allow even convective flow inside the particles. POROS® perfusion chromatography media are the first commercial products in this category introduced in the 1980s. A few more products are being commercialized. POROS® media are synthesized in two steps. Nano-size subparticles are first synthesized and then polymerized to form 10- to 50-µm particles in a second step. In recent years, advances in suspension polymerization produced a new type of integral gigaporous media with improved physical strength. The spherical particles are formed in a single polymerization step. An oil phase (dispersed phase) consisting of a monomer (such as styrene), a crosslinking agent (such as divinylbenzene), an initiator, a diluent, and a special porogen is used. By dispersing the oil phase in a water phase containing a stabilizer, a suspension can be obtained. The suspension polymerization is carried out at an elevated temperature above the decomposition temperature of the initiator to obtain the polymer particles. The diluent and porogen in the particles form smaller diffusion pores and much larger through-pores, respectively. Figure 20-92 shows such a particle with both conventional diffusion pores and gigaporous through-pores. The through-pores allow convective flow to improve mass transfer, and the diffusion pores provide an overall large surface area required for a large binding capacity.

In the scale-up of a chromatographic column with conventional packing media, the column length scale-up is limited because of the subsequently increased column pressure. To increase the feed load, the column is typically scaled up by enlarging the column diameter after the column length reaches a certain limit. The resultant pancake-shaped column leads to a deteriorating resolution due to poor flow distribution in the column cross-section. With rigid gigaporous particles, axial direction scale-up becomes possible because the flow rate can be up to 40 times higher than that for the conventional media. This kind of scale-up is far superior because of the enhanced resolution together with increased feed capacity when the column axial length is increased. With the advances made in rigid gigaporous media, chromatography columns can potentially process large volumes of very dilute feeds when combined with strong binding kinetics such as those involved in affinity chromatography. This may have a significant impact on the overall downstream process design.

Alternative Chromatographic Columns Commercial radial-flow chromatography (RFC) columns first appeared in the mid-1980s. RFC is an alternative to the conventional axial-flow chromatography for preparative- and large-scale applications. In a RFC column, the mobile phase flows in the radial direction rather than the axial direction. Computer simulation proves that RFC is somewhat equivalent to a pancakelike axial-flow column [Gu, in Flickinger and Drew (eds),

op. cit., pp. 627–639]. Both configurations offer a short flow path and thus a low pressure drop.

Continuous chromatography can be achieved by using a simulated moving-bed (SMB) process shown in Sec. 16, in which several columns are linked with a switching device to simulate a continuous countercurrent flow process. This design maximizes productivity while minimizing eluent consumption [Nicoud, in Ahuja (ed.), op. cit., pp. 475–509]. It is suitable for a simple feed that results in only a limited fraction [Imamoglu, in Freitag (ed.), op. cit., pp. 212–231]. By switching the feed, eluant input positions and raffinate, extract output positions periodically in a series of columns to simulated countercurrent movement of the liquid phase and the solid (resin) phase, continuous chromatography is obtained through the use of multiple columns in series. A four-zone SMB consisting of four columns is capable of producing two fractions as a pair of raffinate and extract. To have two pairs (with a total of four fractions) of raffinate and extract outputs an eight-zone SMB system is needed. Another alternative design described in Sec. 16 is the so-called annual-flow column that rotates continuously in the angular direction. Despite its obvious advantage of being straightforwardly continuous, it suffers from angular dispersion and reduced bed volume. This design so far has seen very limited application since its commercial introduction in 1999 [Wolfgang and Prior, in Freitag (ed.), op. cit., pp. 233–255].

In the past decade, monolithic columns have gained popularity for analytical applications. Instead of using discrete packing particles, a whole polymer block is used as a column. Their continuous homogeneous structure provides fast mass-transfer rates and very high flow rates inside the column [Strancar et al., in Freitag (ed.), op. cit., pp. 50–85]. Thin monolithic disks with affinity binding are capable of fast chromatographic separations. They act much as affinity membrane chromatography cartridges do. To utilize long monolithic columns for process-scale separations, a breakthrough in column fabrication is needed to produce large columns suitable for commercial applications.

Sequencing of Chromatography Steps The sequence of chromatographic steps used in a protein purification train should be designed such that the more robust techniques are used first, to obtain some volume reduction (concentration effect) and to remove major impurities that might foul subsequent units; these robust units should have high chemical and physical resistance to enable efficient regeneration and cleaning, and they should be of low material cost. These steps should be followed by the more sensitive and selective operations, sequenced such that buffer changes and concentration steps between applications to chromatographic columns are avoided. Frequently, ion-exchange chromatography is used as the first step. The elution peaks from such columns can be applied directly to hydrophobic interaction chromatographic columns or to a gel filtration unit, without the need for desalting of the solution between applications. These columns can also be used as desalting operations, and the buffers used to elute the columns can be selected to permit direct application of the eluted peaks to the next chromatographic step.

Factors to be considered in making the selection of chromatography processing steps are cost, sample volume, protein concentration and sample viscosity, degree of purity of protein product, presence of nucleic acids, pyrogens, and proteolytic enzymes. Ease with which different types of adsorbents can be washed free from adsorbed contaminants and denatured proteins must also be considered.

Scale-up of Liquid Chromatography The chromatography columns in downstream processing typically are operated in the non-linear region due to a concentrated or overloaded feed. Their scale-up remains a challenging task. There are two general approaches: (1) the rule-based method using equations for column resizing [Ladisch, op. cit., pp. 299–448] and (2) the computer simulation method using rate models [Gu, *Mathematical Modeling and Scale-Up of Liquid Chromatography*, Springer-Verlag, Berlin-New York, 1995] with simulation software such as Chromulator® to predict column performance for a particular column setting and operating conditions.

PRODUCT POLISHING AND FORMULATION

The product from the final purification unit operation is typically in a liquid fraction containing water, a solvent, or a buffer. Based on the

requirement for the final product, they may need to be removed. A solid protein is usually far more stable with a much longer shelf life. Product formulation may also require an excipient to be added. Thus, additional unit operations are needed after the final purification step.

Lyophilization and Drying After the last high-performance purification steps it is usually necessary to prepare the finished product for special applications. For instance, final enzyme products are often required in the form of a dry powder to provide for stability and ease of handling, while pharmaceutical preparations also require high purity, stability during formulation, absence of microbial load, and extended shelf life. This product formulation step may involve drying of the final products by freeze drying, spray drying, fluidized-bed drying, or crystallization [Golker, in Stephanopoulos, op. cit., pp. 695–714]. Crystallization can also serve as an economical purification step [Lee and Kim, in Hatti-Kaul and Mattiasson (ed.), op. cit., pp. 277–320] in addition to its role as a unit operation for product polishing.

Freeze drying, or lyophilization, is normally reserved for temperature-sensitive materials such as vaccines, enzymes, microorganisms, and therapeutic proteins, as it can account for a significant portion of total production cost. This process is characterized by three distinct steps, beginning with freezing of the product solution, followed by water removal by sublimation in a primary drying step, and ending with secondary drying by heating to remove residual moisture.

Freezing is carried out on cooled plates in trays or with the product distributed as small particles on a drum cooler; by dropping the product solution in liquid nitrogen or some other cooling liquid; by cospraying with liquid CO₂ or liquid nitrogen; or by freezing with circulating cold air. The properties of the freeze-dried product, such as texture and ease of rehydration, depend on the size and shape of the ice crystals formed, which in turn depend on the degree of undercooling. It is customary to cool below the lowest equilibrium eutectic temperature of the mixture, although many multicomponent mixtures do not exhibit eutectic points. Freezing should be rapid to avoid effects from local concentration gradients. Removal of water from solution by the formation of ice crystals leads to changes in salt concentration and pH, as well as enhanced concentration of the product, in the remaining solution; this in turn can enhance reaction rates, and even reaction order can change, resulting in cold denaturation of the product. If the feed contains a solvent and an acid, the solvent tends to sublimate faster than the acid, causing acidic damage to the protein. With a high initial protein concentration the freeze concentration factor and the amount of ice formed will be reduced, resulting in greater product stability. For aseptic processing, direct freezing in the freeze-drying plant ensures easier loading of the solution after filtration than if it is transferred separately from remote freezers.

In the primary drying step, heat of sublimation is supplied by contact, conduction, or radiation to the sublimation front. It is important to avoid partial melting of the ice layer. Many pharmaceutical preparations dried in ampoules are placed on heated shelves. The drying time depends on the quality of ice crystals, indicating the importance of controlling the freezing process; smaller crystals offer higher interfacial areas for heat and mass transfer, but larger crystals provide pores for diffusion of vapor away from the sublimation front.

A high percentage of water remains after the sublimation process, present as adsorbed water, water of hydration, or dissolved in the dry amorphous solid; this is difficult to remove. Usually, shelf temperature is increased to 25 to 40°C and chamber pressure is lowered as far as possible. This still does not result in complete drying, however, which can be achieved only by using even higher temperatures, at which point thermally induced product degradation can occur.

Excipients can be used to improve stability and prevent deterioration and inactivation of biomolecules through structural changes such as dissociation from multimeric states into subunits, decrease in α -helical content accompanied by an increase in β -sheet structure, or complete unfolding of helical structure. These are added prior to the freeze-drying process. Examples of these protective agents include sugars, sugar derivatives, and various amino acids, as well as polymers such as dextran, polyvinyl pyrrolidone, hydroxyethyl starch, and polyethylene glycol. Some excipients, the lyoprotectants, provide protection during freezing, drying, and storage, while others, the cryoprotectants, offer protection only during the freezing process. Spray drying can use up to 50 percent less energy than freeze-drying operations and finds applica-

tion in the production of enzymes used as industrial catalysts, as additives for washing detergents, and as the last step in the production of single-cell protein. The product is usually fed to the dryer as a solution, a suspension, or a free-flowing wet substance.

Spray drying is an adiabatic process, the energy being provided by hot gas (usually hot air) at temperatures between 120 and 400°C. Product stability is ensured by a very short drying time in the spray-drying equipment, typically in the subsecond to second range, which limits exposure to the elevated temperatures in the dryer. Protection can be offered by addition of additives (e.g., galactomannan, polyvinyl pyrrolidone, methyl cellulose, cellulose). The spray-drying process requires dispersion of the feed as small droplets to provide a large heat- and mass-transfer area. The dispersion of liquid is attained by using rotating disks, different types of nozzles, or ultrasound, and is affected by interfacial tension, density, and dynamic viscosity of the feed solution, as well as the temperature and relative velocities of the liquid and air in the mixing zone. Rotating-disk atomizers operate at 4000 to 50,000 rpm to generate the centrifugal forces needed for dispersion of the liquid phase; typical droplet sizes of 25 to 950 μm are obtained. These atomizers are especially suitable for dispersing suspensions that would tend to clog nozzles. For processing under aseptic conditions, the spray drier must be connected to a filling line that allows aseptic handling of the product.

Thickeners and binders such as acacia, agar, starch, sodium alginate, gelatin, methyl cellulose, bentonite, and silica are used to improve product stability and enhance the convenience of the administration of a liquid formulation. Surface-active agents, colors, flavors; and preservatives may also be used in the final formulation (Garcia et al., *Bioseparation Process Science*, Blackwell Science, Malden, Mass., 1999, p. 374).

INTEGRATION OF UNIT OPERATIONS IN DOWNSTREAM PROCESSING

Generally speaking, a typical downstream process consisting of four stages: removal of insolubles, isolation, purification, and polishing (Belter et al., op. cit., p. 5). Cell disruption is required for intracellular products. One or two, and sometimes more chromatography steps will serve as the center-stage unit operations. The steps before them serve the purposes of feed volume reduction and removal of the majority of impurities. The steps after them are polishing and formulation operations. Based on these general outlines, a few rules of thumb may be used (Harrison et al., op. cit., p. 322; Garcia et al., op. cit., p. 358): (1) Reduce the feed volume early in the process, (2) remove the most abundant impurities or the easy-to-remove impurities first, (3) reduce the amount of impurities as much as possible before the delicate high-resolution chromatography steps, and (4) sequence the unit operations that exploit different separation mechanisms.

The purification of proteins to be used for therapeutic purposes presents more than just the technical problems associated with the separation process. Owing to the complex nature and intricate three-dimensional structure, the routine determination of protein structure as a quality control tool, particularly in its final medium for use, is not well established. In addition, the complex nature of the human immune system allows for even minor quantities of impurities and contaminants to be biologically active. Thus, regulation of biologics production has resulted in the concept of the process defining the product since even small and inadvertent changes in the process may affect the safety and efficacy of the product. Indeed, it is generally acknowledged that even trace amounts of contaminants introduced from other processes, or contaminants resulting from improper equipment cleaning, can compromise the product. From a regulatory perspective, then, operations should be chosen for more than just efficiency. The consistency of the unit operation, particularly in the face of potentially variable feed from the culture/fermentation process, is the cornerstone of the process definition. Operations that lack robustness or are subject to significant variation should not be considered. Another aspect of process definition is the ability to quantify the operation's performance. Finally, the ease with which the equipment can

be cleaned in a verifiable manner should play a role in unit operation selection. Obviously, certain unit operations are favored over others because they are easier to validate. Process validation was covered in a book edited by Subramanian (op. cit., vol. 2, pp. 379–460).

Keep in mind that some unit operations are not as scalable as others. The evolution of a bench-scale process to production scale will see changes in the types and number of unit operations selected. To configure an effective and efficient bioseparations process, a thorough understanding of the various unit operations in downstream processing described above is a prerequisite. Different process-scale and purity requirements can necessitate changes and may result in quite different configurations for the same product. Existing process examples and past experiences help greatly. Due to regulatory restrictions, once a process is approved for a biopharmaceutical, any change in unit operations requires time-consuming and costly new regulatory approval. This means that an optimized process design is much desired before seeking approval. The involvement of biochemical engineers early in the design process is highly recommended.

INTEGRATION OF UPSTREAM AND DOWNSTREAM OPERATIONS

Upstream fermentation (or cell culture) has a direct impact on the design and optimization of its downstream process. Different media or different operating conditions in fermentation result in a different feed for the downstream process. In the development of new products, optimization of the fermentation medium for titer only often ignores the consequences of the medium properties on subsequent downstream processing steps such as filtration and chromatography. It is imperative, therefore, that there be effective communication and understanding between workers on the upstream and downstream phases of the product development if rational tradeoffs are to be made to ensure overall optimality of the process. One example is to make the conscious decision, in collaboration with those responsible for the downstream operations, of whether to produce a protein in an unfolded form or in its native folded form. The purification of the aggregated unfolded proteins is simpler than that of the native protein, but the refolding process itself to obtain the product in its final form may lack scalability or certainty in its method development.

In some instances, careful consideration of the conditions used in the fermentation process, or manipulation of the genetic makeup of the host, can simplify and even eliminate some unit operations in the downstream processing sequence [Kelley and Hatton, *Bioseparation*, 1, 303–349 (1991)]. Some of the advances made in this area are the engineering of strains of *E. coli* to allow the inducible expression of lytic enzymes capable of disrupting the wall from within for the release of intracellular protein products, the use of secretion vectors for the expression of proteins in bacterial production systems. Fusion proteins can be genetically engineered to attach an extra peptide or protein that can bind with an affinity chromatography medium [Whitmarsh and Hornby, in Street (ed.), op. cit., pp. 163–177]. This can enhance the purification of an otherwise difficult to purify protein greatly. The cell culture medium can be selected to avoid components that can hinder subsequent purification procedures. Integration of the fermentation and initial separation/purification steps in a single operation can also lead to enhanced productivity, particularly when the product can be removed as it is formed to prevent its proteolytic destruction by the proteases which are frequently the by-product of fermentation processes. The introduction of a solvent directly to the fermentation medium (e.g., phase-forming polymers), the continuous removal of products by using ultrafiltration membranes, and the use of continuous fluidized-bed operations are examples of this integration.

Process economics for biological products was discussed by Harrison et al. (op. cit., pp. 334–369) and Datar and Rosen [in Asenjo (ed.), *Separation Processes in Biotechnology*, Dekker, New York, 1990, pp. 741–793] at length, and also by Ladisch (op. cit., pp. 401–430). They provided some illustrative examples with cost analyses. Bioprocess design software can also prove helpful in the overall design process (Harrison et al., op. cit., pp. 343–369).

SECTION 21

Solid-Solid
Operations and
Processing

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



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Solid-Solid Operations and Processing

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Nomenclature and Units for Particle-Size Analysis

Symbol	Definition	SI units	U.S. customary units	Symbol	Definition	SI units	U.S. customary units
A	Empirically determined constant	—	—	$q_p^*(z)$	Logarithmic normal distribution	—	—
a	Distance from the scatterer to the detector	m	ft	q_r^*	Logarithmic density distribution of dimension r	—	—
a_s	Specific surface per mass unit	m ² /g	ft ² /s	$q_0(x)$	Number density distribution	1/m	1/in
B	Empirically determined constant	—	—	$q_l(x)$	Length density distribution	1/m	1/in
C	Empirically determined constant	—	—	$q_z(x)$	Area density distribution	1/m	1/in
C	BET number	—	—	$q_z(x)$	Volume or mass density distribution	1/m	1/in
C_{PF}	Area concentration	1/cm ²	1/in ²	$q_{3,i}$	Volume density distribution of class i	1/m	1/in
D	Translational diffusion coefficient	m ² /s	ft ² /s	$\bar{q}_{3,i}^*$	Logarithmic volume density distribution of class i	1/m	1/in
D_m	Concentration undersize	—	—	r, r_i	Measurement radius	m	in
e	Elementary charge	C	C	s	Dimensionless standard deviation	—	—
f_i	Frequency i	Hz	Hz	s, s_i	Surface radius of a centrifuge	m	in
g	Acceleration due to gravity	m/s ²	ft/s ²	S_v	Volume specific surface	m ² /m ³	—
I_0	Illuminating intensity	W/m ²	fc	$S_1(\theta)$	Dimensionless, complex functions describing the change and amplitude in the perpendicular and the parallel polarized light	—	—
i	Index of size class	—	—	$S_2(\theta)$	—	—	—
I	Measured sound intensity	W/m ²	W/ft ²	T	Absolute temperature	K	K
I	Measured sound intensity	W/m ²	W/ft ²	t	Time	s	s
I_0	Illuminating intensity	W/m ²	fc	u	Settling velocity of particles	m/s	ft/s
I_0	Primary sound intensity	W/m ²	W/ft ²	v_1, v_2	Particle velocities	m/s	ft/s
$I(\theta)$	Total scattered intensity	W/m ²	W/ft ²	W	Weight undersize	g	lb
K	Related extinction cross section	—	—	x_{EQPC}	Particle size of the equivalent projection area of a circle	m	in
Kn	Knudsen number	—	—	$\bar{x}_F$	Average Feret diameter	m	in
$k \rightarrow k_B$	Wave number	—	—	$x_{F, \max}$	Maximum Feret diameter	m	in
k_1, k_2	Boltzmann constant	J/K	J/K	$x_{F, \max 90}$	Feret diameter measured 90° to the maximum Feret diameter	m	in
L	Incident illumination vectors	1/m	1/ft	$x_{F, \min}$	Minimum Feret diameter	m	in
L	Loschmidt number	1/mol	1/mol	x_i	Size of class i	m	in
l	Mean path of gas molecules	m	ft	$\bar{x}_{k,0}$	Arithmetic average particle size for a number distribution	m	in ²
$M_{k,r}$	k th moment of dimension r	—	—	$\bar{x}_{k,r}$	Average particle size	m	in
m	Refractive index	—	—	$x_{\min}$	Minimum particle size	m	in
n	Real part of the refractive index	—	—	x_d	Stokes diameter	m	in
n	Number of classes	—	—	$\bar{x}_{1,r}$	Weighted average particle size	m	in
n_a	Amount of absorbed gas	mol/g	mol/lb	$\bar{x}_{1,2}$	Sauter diameter	m	in
n_m	Monolayer capacity	mol/g	mol/lb	$x_{50,r}$	Mean size of dimension r	m	in
P	Settled weight	g	lb	z	Integration variable	m	in
p	Number of elementary charges	—	—	$Z(x)$	Electrical mobility of particle size x	—	—
p	Pressure	Pa	psi			C	C
p_o	Starting pressure	Pa	psi			Pa·s·m	Pa·s·m
$Q_0(x)$	Cumulative number distribution	—	—				
$Q_l(x)$	Cumulative length distribution	—	—				
$Q_z(x)$	Cumulative area distribution	—	—				
$Q_z(x)$	Cumulative volume or mass distribution	—	—				
$Q_{3,i}$	Cumulative volume distribution till class i	—	—				
q	Modulus of the scattering vector	1/m	1/ft				
$\bar{q}$	Scattering vector	1/m	1/ft				

Greek Symbols

Δl	Thickness of the suspension layer	m	in	ρ_l	Density of the liquid	g/cm ³	lb/in ³
$\Delta Q_{3,i}$	Normalized volume fraction in size class i	—	—	ρ_s	Density of the particle	g/cm ³	lb/in ³
Δx_i	Width of size class i	m	in	θ	Scattering angle	rad	deg
ε	Extension of a particle ensemble in the direction of a camera	m	in	σ	Dimensionless wave number	—	—
Γ	Decay rate	1/s	1/s	ω	Radial velocity of an agglomerate	rad/s	rad/s
η	Hydrodynamic viscosity of the dispersing liquid	Pa s	Poise	ω	Radial velocity of a centrifuge	rad/s	rad/s
κ	Imaginary part of refractive index	—	—	ψ_s	Sphericity	—	—
				ψ_A	Aspect ratio	—	—
				ψ_C	Convexity	—	—

Nomenclature and Units for Solids Mixing

Symbol	Definition	Units	Symbol	Definition	Units
d	Mixer diameter	m	t_f, t_m, t_e, t_i	Filling, mixing, discharging, and idle time	s
D	Axial coefficient of dispersion	m^2/s	$\bar{t}^*$	Mixing time	—
E_{Mix}	Mixing energy	W	T_p	Feed fluctuation period	s
g	Gravitational acceleration	m^2/s	v	Axial velocity	m/s
H	Height of fluidized bed	m	VRR	Variance reduction ratio	—
L	Mixer length	m	$W\{ \}$	Probability	—
m_p, m_q	Average particle weight of two components p and q in mixture	kg	x	Concentration of tracer component	—
M	Coefficient of mixing	m^2/s	x_i	Concentration in i sample	—
M	Mass of a sample	kg	Greek Symbols		
M	Mass of a batch	kg	μ	Mean concentration	—
n	Random sampling scope	—	ρ	Density of solids	kg/m^3
n	Rotational frequency	Hz	ρ_{bulk}	Bulk density	kg/m^3
Ne	Newton number	—	ρ_s	Density of solids	kg/m^3
N_g	Number of samples in basic whole	—	σ_p, σ_q	Standard deviation of particle weight for two ingredients in mix	kg
p	Tracer component concentration in basic whole	—	σ^2	Variance	—
p_g	Proportional mass volume of coarse ingredient	—	σ_z^2	Variance of a random mix	—
P	Power	W	$\Phi(\chi^2)$	Cumulative function of	—
q	$1-p$	—	χ^2	Chi square distribution	—
r	Mixer radius	m	χ^2_l, χ^2_u	Chi square distribution variables. In a two-sided confidence interval, l stands for lower and u for upper limit.	—
RSD	Relative standard deviation	—	ω	Angular velocity	1/s
S	Empirical standard deviation	—			
S^2	Random sample variance	—			
t, t'	Time	s			
t_e	Mean residence times	s			

21-6 SOLID-SOLID OPERATIONS AND PROCESSING

Nomenclature and Units for Size Enlargement and Practice

Symbol	Definition	SI units	U.S. customary units	Symbol	Definition	SI units	U.S. customary units
A	Parameter in Eq. (21-108)			k	Coalescence rate constant	1/s	1/s
A	Apparent area of indenter contact	cm ²	in ²	K	Agglomerate deformability		
A	Attrition rate	cm ³ /s	in ³ /s	K_c	Fracture toughness	MPa·m ^{1/2}	MPa·m ^{1/2}
A_i	Spouted-bed inlet orifice area	cm ²	in ²	l	Wear displacement of indenter	cm	in
B	Nucleation rate	cm ³ /s	in ³ /s	L	Roll loading	dyn	lbf
B_f	Fragmentation rate	g/s	lb/s	$(\Delta L/L)_c$	Critical agglomerate deformation strain		
B_f	Wear rate	g/s	lb/s	N_t	Granules per unit volume	1/cm ³	1/ft ³
c	Crack length	cm	in	n	Feed droplet size	cm	in
δ_c	Effective increase in crack length due to process zone	cm	in	$n(v,t)$	Number frequency size distribution by size volume	1/cm ⁶	1/ft ⁶
c	Unloaded shear strength of powder	kg/cm ²	psf	N_c	Critical drum or disc speed	rev/s	rev/s
d	Harmonic average granule diameter	cm	in	P	Applied load	dyn	lbf
d	Primary particle diameter	cm	in	P	Pressure in powder	kg/cm ²	psf
d	Impeller diameter	cm	in	Q	Maximum compressive force	kg/cm ²	psf
d	Roll press pocket depth	cm	in	$\dot{Q}$	Granulator flow rate	cm ³ /s	ft ³ /s
d_i	Indenter diameter	cm	in	r_p	Process zone radius	cm	in
d_p	Average feed particle size	cm	in	R	Capillary radius	cm	in
D	Die diameter	cm	in	S	Volumetric spray rate	cm ³ /s	ft ³ /s
D	Disc or drum diameter	cm	in	St	Stokes number, Eq. (21-48)		
D	Roll diameter	cm	in	St°	Critical Stokes number representing energy required for rebound		
D_c	Critical limit of granule size	cm	in	St_0	Stokes number based on initial nuclei diameter		
e_r	Coefficient of restitution			t	Time	s	s
E	Strain energy stored in particle	J	J	u, v	Granule volumes	cm ³	in ³
E^*	Reduced elastic modulus	kg/cm ²	psf	u_0	Relative granule collisional velocity	cm/s	in/s
f_c	Unconfined yield stress of powder	kg/cm ²	psf	U	Fluidization gas velocity	cm/s	ft/s
g	Acceleration due to gravity	cm/s ²	ft/s ²	U_{mf}	Minimum fluidization gas velocity	cm/s	ft/s
G_c	Critical strain energy release rate	J/m ²	lbf/ft ²	U_i	Spouted-bed inlet gas velocity	cm/s	ft/s
F	Indentation force	dyn	lbf	V	Volumetric wear rate	cm ³ /s	in ³ /s
F	Roll separating force	dyn	lbf	V_R	Mixer swept volume ratio of impeller	cm ³ /s	ft ³ /s
G	Layering rate	cm ³ /s	in ³ /s	V	Volume of granulator	cm ³	ft ³
h	Height of liquid capillary rise	cm	in	w	Weight fraction liquid		
h	Roll press gap distance	cm	in	w	Granule volume	cm ³	in ³
h	Binder liquid layer thickness	cm	in	w°	Critical average granule volume	cm ³	in ³
h_b	Fluid-bed height	cm	in	W	Roll width	cm	in
h_a	Height of surface asperities	cm	in	x	Granule or particle size	cm	in
h_c	Maximum height of liquid capillary rise	cm	in	y	Liquid loading		
H	Individual bond strength	dyn	lbf	Y	Calibration factor		
H	Hardness of agglomerate or compact	kg/cm ²	psf				

Greek Symbols

$\beta(u, v)$	Coalescence rate constant for collisions between granules of volumes u and v	1/s	1/s	$\Delta\rho$	Relative fluid density with respect to displaced gas or liquid	g/cm ³	
ε	Porosity of packed powder			ρ	Apparent agglomerate or granule density	g/cm ³	lb/ft ³
ε_b	Interagglomerate bed voidage			ρ_a	Apparent agglomerate or granule density	g/cm ³	lb/ft ³
ε_g	Intraagglomerate granule porosity			ρ_b	Bulk density	g/cm ³	lb/ft ³
κ	Compressibility of powder			ρ_g	Apparent agglomerate or granule density	g/cm ³	lb/ft ³
ϕ	Disc angle to horizontal	deg	deg	ρ_l	Liquid density	g/cm ³	lb/ft ³
ϕ	Internal angle of friction	deg	deg	ρ_s	True skeletal solids density	g/cm ³	lb/ft ³
ϕ_c	Effective angle of friction	deg	deg	σ_0	Applied axial stress	kg/cm ²	psf
ϕ_w	Wall angle of friction	deg	deg	σ_c	Resulting axial stress in powder	kg/cm ²	psf
ϕ_w	Roll friction angle	deg	deg	σ	Powder normal stress during shear	kg/cm ²	psf
$\phi(\eta)$	Relative size distribution			σ_c	Powder compaction normal stress	kg/cm ²	psf
γ^{lc}	Liquid-vapor interfacial energy	dyn/cm	dyn/cm	σ_f	Fracture stress under three-point bend loading	kg/cm ²	psf
γ^{sl}	Solid-liquid interfacial energy	dyn/cm	dyn/cm	σ_T	Granule tensile strength	kg/cm ²	psf
γ^{sv}	Solid-vapor interfacial energy	dyn/cm	dyn/cm	σ_y	Granule yield strength	kg/cm ²	psf
μ	Binder or fluid viscosity	poise		τ	Powder shear stress	kg/cm ²	psf
μ	Coefficient of internal friction			θ	Contact angle	deg	deg
ω	Impeller rotational speed	rad/s	rad/s	ζ	Parameter in Eq. (21-108)		
				η	Parameter in Eq. (21-108)		

Nomenclature and Units for Size Reduction and Size Enlargement

Symbol	Definition	SI units	U.S. customary units	Symbol	Definition	SI units	U.S. customary units
A	Coefficient in double Schumann equation			q_f	Fine-fraction mass flow rate	g/s	lb/s
a	Constant			q_o	Feed mass flow rate	g/s	lb/s
$a_{k,k}$	Coefficient in mill equations			q_p	Mass flow rate of classifier product	g/s	lb/s
$a_{k,n}$	Coefficient in mill equations			q_R	Mass flow rate of classifier tailings	g/s	lb/s
$\mathbf{B}$	Matrix of breakage function			q_R	Recycle mass flow rate to a mill	g/s	lb/s
$\Delta B_{k,n}$	Breakage function			R	Recycle		
b	Constant			R	Reid solution		
C	Constant			r	Dimensionless parameter in size-distribution equations		
C_s	Impact-crushing resistance	kWh/cm	ft-lb/in	S	Rate function	S^{-1}	S^{-1}
D	Diffusivity	m ² /s	ft ² /s	$\bar{S}$	Corrected rate function	S^{-1}	S^{-1}
D	Mill diameter	m	ft	$\mathbf{S'}$	Matrix of rate function	Mg/kWh	ton/(hp-h)
D_b	Ball or rod diameter	cm	in	$S_G(X)$	Grindability function	S^{-1}	S^{-1}
D_{mill}	Diameter of mill	m	ft	S_u	Grinding-rate function		
d	Differential			s	Parameter in size-distribution equations		
d	Distance between rolls of crusher	cm	in	s	Peripheral speed of rolls	cm/min	in/min
E	Work done in size reduction	kWh	hp-h	t	Time	s	s
E	Energy input to mill	kW	hp	u	Settling velocity of particles	cm/s	ft/s
E_i	Bond work index	kWh/Mg	hp-h/ton	$\mathbf{W}$	Vector of differential size distribution of a stream		
E_i	Work index of mill feed			w_k	Weight fraction retained on each screen		
E_2	Net power input to laboratory mill	kW	hp	w_u	Weight fraction of upper-size particles		
erf	Normal probability function			w_u	Material holdup in mill	g	lb
F	As subscript, referring to feed stream			X	Particle size or sieve size	cm	in
F	Bonding force	kg/kg	lb/lb	X'	Parameter in size-distribution equations	cm	in
g	Acceleration due to gravity	cm/s ²	ft/s ²	ΔX_i	Particle-size interval	cm	in
$\mathbf{I}$	Unit matrix in mill equations			X_i	Midpoint of particle-size interval ΔX_i	cm	in
i	Tensile strength of agglomerates	kg/cm ²	lb/in ²	X_0	Constant, for classifier design		
K	Constant			X_f	Feed-particle size	cm	in
k	Parameter in size-distribution equations	cm	in	X_m	Mean size of increment in size-distribution equations	cm	in
k	As subscript, referring to size of particles in mill and classifier parameters			X_p	Product-particle size	cm	in
L	As subscript, referring to discharge from a mill or classifier			X_p	Size of coarser feed to mill	cm	in
L	Length of rolls	cm	in	X_{25}	Particle size corresponding to 25 percent classifier-selectivity value	cm	in
L	Inside length of tumbling mill	m	ft	X_{50}	Particle size corresponding to 50 percent classifier-selectivity value	cm	in
$\mathbf{M}$	Mill matrix in mill equations			X_{75}	Particle size corresponding to 75 percent classifier-selectivity value	cm	in
m	Dimensionless parameter in size-distribution equations			ΔX_k	Difference between opening of successive screens	cm	in
N	Mean-coordination number			x	Weight fraction of liquid		
N_c	Critical speed of mill	r/min	r/min	Y	Cumulative fraction by weight undersize in size-distribution equations		
ΔN	Incremental number of particles in size-distribution equation			Y	Cumulative fraction by weight undersize or oversize in classifier equations		
n	Dimensionless parameter in size-distribution equations			ΔY	Fraction of particles between two sieve sizes		
n	Constant, general			ΔY	Incremental weight of particles in size-distribution equations	g	lb
n_c	Percent critical speed of mill			ΔY_{ci}	Cumulative size-distribution intervals of coarse fractions	cm	in
O	As subscript, referring to inlet stream			ΔY_{fi}	Cumulative size-distribution intervals of fine fractions	cm	in
P	As subscript, referring to product stream			$\mathbf{Z}$	Matrix of exponentials		
P_k	Fraction of particles coarser than a given sieve opening						
p	Number of short-time intervals in mill equations						
Q	Capacity of roll crusher	cm ³ /min	ft ³ /min				
q	Total mass throughput of a mill	g/s	lb/s				
q_c	Coarse-fraction mass flow rate	g/s	lb/s				
q_F	Mass flow rate of fresh material to mill	g/s	lb/s				

Greek Symbols

β	Sharpness index of a classifier			ρ_ℓ	Density of liquid	g/cm ³	lb/in ³
δ	Angle of contact	rad	0	ρ_s	Density of solid	g/cm ³	lb/in ³
ε	Volume fraction of void space			Σ	Summation		
Z	Residence time in the mill	s	s	σ	Standard deviation		
η_c	Size-selectivity parameter			σ	Surface tension	N/cm	dyn/cm
μ	Viscosity of fluid	N-S/m ²	P	υ	Volumetric abundance ratio of gangue to mineral		
ρ_f	Density of fluid	g/cm ³	lb/in ³				

PARTICLE-SIZE ANALYSIS

GENERAL REFERENCES: Allen, *Particle Size Measurement*, 4th ed., Chapman and Hall, 1990. Bart and Sun, Particle Size Analysis Review, *Anal. Chem.*, **57**, 151R (1985). Miller and Lines, *Critical Reviews in Analytical Chemistry*, **20**(2), 75–116 (1988). Herdan, *Small Particles Statistics*, Butterworths, London. Orr and DalleValle, *Fine Particle Measurement*, 2d ed., Macmillan, New York, 1960. Kaye, *Direct Characterization of Fine Particles*, Wiley, New York, 1981. Van de Hulst, *Light Scattering by Small Particles*, Wiley, New York, 1957. K. Leschonski, Representation and Evaluation of Particle Size Analysis Data, *Part. Part. Syst. Charact.*, **1**, 89–95 (1984). Terence Allen, *Particle Size Measurement*, 5th ed., Vol. 1, Springer, 1996. Karl Sommer, *Sampling of Powders and Bulk Materials*, Springer, 1986. M. Alderliesten, Mean Particle Diameters, Part I: Evaluation of Definition Systems, *Part. Part. Syst. Charact.*, **7**, 233–241 (1990); Part II: Standardization of Nomenclature, *Part. Part. Syst. Charact.*, **8**, 237–241 (1991); Part III: An Empirical Evaluation of Integration and Summation Methods for Estimating Mean Particle Diameters from Histogram Data, *Part. Part. Syst. Charact.*, **19**, 373–386 (2002); Part IV: Empirical Selection of the Proper Type of Mean Particle Diameter Describing a Product or Material Property, *Part. Part. Syst. Charact.*, **21**, 179–196 (2004); Part V: Theoretical Derivation of the Proper Type of Mean Particle Diameter Describing a Product or Process Property, *Part. Part. Syst. Charact.*, **22**, 233–245 (2005). ISO 9276, Representation of Results of Particle Size Analysis. H. C. van de Hulst, *Light Scattering by Small Particles*, Structure of Matter Series, Dover, 1981. Craig F. Bohren and Donald R. Huffman, *Absorption and Scattering of Light by Small Particles*, Wiley-Interscience, new edition. Bruce J. Berne and Robert Pecora, *Dynamic Light Scattering: With Applications to Chemistry, Biology, and Physics*, unabridged edition, Dover, 2000. J. R. Allegra and S. A. Hawley, Attenuation of Sound in Suspensions and Emulsions: Theory and Experiment, *J. Acoust. Soc. America* **51**, 1545–1564 (1972).

PARTICLE SIZE

Specification for Particulates The behavior of dispersed matter is generally described by a large number of parameters, e.g., the powder's bulk density, flowability, and degree of aggregation or agglomeration. Each parameter might be important for a specific application. In solids processes such as comminution, classification, agglomeration, mixing, crystallization, or polymerization, or in related material handling steps, particle size plays an important role. Often it is the dominant quality factor for the suitability of a specific product in the desired application.

Particle Size As particles are extended three-dimensional objects, only a perfect spherical particle allows for a simple definition of the particle size x , as the diameter of the sphere. In practice, spherical particles are very rare. So usually **equivalent diameters** are used, representing the diameter of a sphere that behaves as the real (nonspherical) particle in a specific sizing experiment. Unfortunately, the measured size now depends on the method used for sizing. So one can only expect identical results for the particle size if either the particles are spherical or similar sizing methods are employed that measure the same equivalent diameter.

In most applications more than one particle is observed. As each individual may have its own particle size, methods for data reduction have been introduced. These include the particle-size distribution, a variety of model distributions, and moments (or averages) of the distribution. One should also note that these methods can be extended to other particle attributes. Examples include pore size, porosity, surface area, color, and electrostatic charge distributions, to name but a few.

Particle-Size Distribution A particle-size distribution (PSD) can be displayed as a table or a diagram. In the simplest case, one can divide the range of measured particle sizes into size intervals and sort the particles into the corresponding size class, as displayed in Table 21-1 (shown for the case of volume fractions).

Typically the fractions $\Delta Q_{r,i}$ in the different size classes i are summed and normalized to 100 percent, resulting in the **cumulative distribution** $Q(x)$, also known as the percentage undersize. For a

TABLE 21-1 Tabular Presentation of Particle-Size Data

1	2	3	4	5	6	7
i	x_i , μm	$\Delta Q_{3,i}$	Δx_i , μm	$\bar{q}_{3,i} = \Delta Q_{3,i} / \Delta x_i$ $1/\mu\text{m}$	$Q_{3,i}$	$\bar{q}_{3,i}^*$
0	0.063				0.0000	
1	0.090	0.0010	0.027	0.0370	0.0010	0.0028
2	0.125	0.0009	0.035	0.0257	0.0019	0.0027
3	0.180	0.0016	0.055	0.0291	0.0035	0.0044
4	0.250	0.0025	0.070	0.0357	0.0060	0.0076
5	0.355	0.0050	0.105	0.0476	0.0110	0.0143
6	0.500	0.0110	0.145	0.0759	0.0220	0.0321
7	0.710	0.0180	0.210	0.0857	0.0400	0.0513
8	1.000	0.0370	0.290	0.1276	0.0770	0.1080
9	1.400	0.0610	0.400	0.1525	0.1380	0.1813
10	2.000	0.1020	0.600	0.1700	0.2400	0.2860
11	2.800	0.1600	0.800	0.2000	0.4000	0.4755
12	4.000	0.2100	1.200	0.1750	0.6100	0.5888
13	5.600	0.2400	1.600	0.1500	0.8500	0.7133
14	8.000	0.1250	2.400	0.0521	0.9750	0.3505
15	11.20	0.0240	3.200	0.0075	0.9990	0.0713
16	16.000	0.0010	4.800	0.0002	1.0000	0.0028

given particle size x , the Q value represents the percentage of the particles finer than x .

If the quantity measure is “number,” $Q_n(x)$ is called a **cumulative number distribution**. If it is length, area, volume, or mass, then the corresponding length $[Q_l(x)]$, area $[Q_a(x)]$, volume, or mass distributions are formed $[Q_v(x)]$; mass and volume are related by the specific density ρ . The index r in this notation represents the quantity measure (ISO 9276, *Representation of Results—Part 1 Graphical Representation*). The choice of the quantity measured is of decisive importance for the appearance of the PSD, which changes significantly when the dimension r is changed. As, e.g., one 100- μm particle has the same volume as 1000 10- μm particles or 10^6 1- μm particles, a number distribution is always dominated by and biased to the fine fractions of the sample while a volume distribution is dominated by and biased to the coarse.

The normalization of the fraction $\Delta Q_{r,i}$ to the size of the corresponding interval leads to the **distribution density** $\bar{q}_{r,i}$, or

$$\bar{q}_{r,i} = \frac{\Delta Q_{r,i}}{\Delta x_i} \quad \text{and} \quad \sum_{i=1}^n \Delta Q_{r,i} = \sum_{i=1}^n \bar{q}_{r,i} \Delta x_i = 1 = 100\% \quad (21-1)$$

If $Q_r(x)$ is differentiable, the distribution density function $q_r(x)$ can be calculated as the first derivative of $Q_r(x)$, or

$$q_r(x) = \frac{dQ_r(x)}{dx} \quad \text{or} \quad Q_r(x_i) = \int_{x_{\min}}^{x_i} q_r(x) dx \quad (21-2)$$

It is helpful in the graphical representation to identify the distribution type, as shown for the cumulative volume distribution $Q_3(x)$ and volume distribution density $q_3(x)$ in Fig. 21-1. If $q_r(x)$ displays one maximum only, the distribution is called a **monomodal size distribution**. If the sample is composed of two or more different-size regimes, $q_r(x)$ shows two or more maxima and is called a **bimodal** or **multimodal size distribution**.

PSDs are often plotted on a logarithmic abscissa (Fig. 21-2). While the $Q_r(x)$ values remain the same, care has to be taken for the transformation of the distribution density $q_r(x)$, as the corresponding areas under the distribution density curve must remain constant (in particular

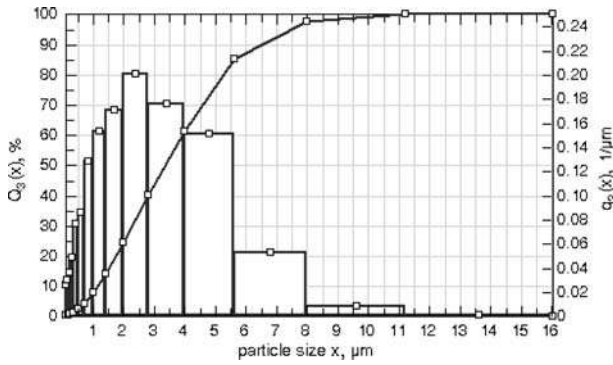


FIG. 21-1 Histogram $\bar{q}_3(x)$ and $Q_3(x)$ plotted with linear abscissa.

the total area remains 1, or 100 percent) independent of the transformation of the abscissa. So the transformation has to be performed by

$$\bar{q}_r^*(\ln x_{i-1}, \ln x_i) = \frac{\Delta Q_{r,i}}{\ln(x_i/x_{i-1})} \quad (21-3)$$

This equation also holds if the natural logarithm is replaced by the logarithm to base 10.

Example 1: From Table 21-1 one can calculate, e.g.,

$$\begin{aligned} \bar{q}_{3,11} &= \frac{\Delta Q_{3,11}}{\Delta x_{11}} = \frac{0.16}{0.8 \mu\text{m}} = 0.2 \mu\text{m}^{-1} \\ \bar{q}_{3,11}^* &= \bar{q}_3^*(\ln x_{10}, \ln x_{11}) = \frac{\Delta Q_{3,11}}{\ln(x_{11}/x_{10})} = \frac{0.16}{\ln(2.8 \mu\text{m}/2.0 \mu\text{m})} \\ &= \frac{0.16}{\ln 1.4} = 0.4755 \end{aligned}$$

Model Distribution While a PSD with n intervals is represented by $2n + 1$ numbers, further data reduction can be performed by fitting the size distribution to a specific mathematical model. The logarithmic normal distribution or the logarithmic normal probability function is one common model distribution used for the distribution density, and it is given by

$$q_r^*(z) = \frac{1}{\sqrt{2\pi}} e^{-0.5z^2} \quad \text{with} \quad z = \frac{1}{s} \ln \left[\frac{x}{x_{50,r}} \right] \quad (21-4)$$

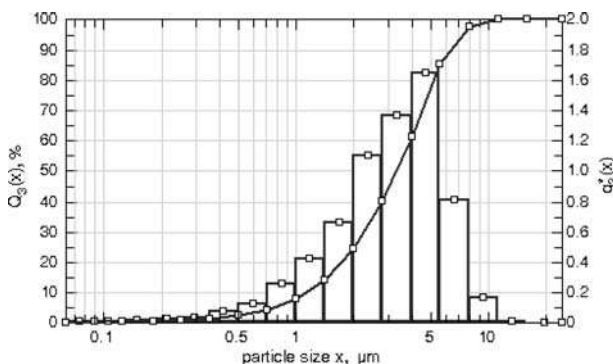


FIG. 21-2 Histogram $\bar{q}_3(x)$ and $Q_3(x)$ plotted with a logarithmic abscissa.

The PSD can then be expressed by two parameters, namely, the mean size $x_{50,r}$ and, e.g., by the dimensionless standard deviation s (ISO 9276, Part 5: *Methods of Calculations Relating to Particle Size Analysis Using Logarithmic Normal Probability Distribution*). The data reduction can be performed by plotting $Q_r(x)$ on logarithmic probability graph paper or using the fitting methods described in upcoming ISO 9276-3, *Adjustment of an Experimental Curve to a Reference Model*. This method is mainly used for the analysis of powders obtained by grinding and crushing and has the advantage that the transformation between PSDs of different dimensions is simple. The transformation is also log-normal with the same slope s .

Other model distributions used are the **normal distribution** (Laplace-Gauss), for powders obtained by precipitation, condensation, or natural products (e.g., pollens); the **Gates-Gaudin-Schuhmann distribution** (bilogarithmic), for analysis of the extreme values of fine particle distributions (Schuhmann, *Am. Inst. Min. Metall. Pet. Eng., Tech. Paper 1189 Min. Tech.*, 1940); or the **Rosin-Rammler-Sperling-Bennet distribution** for the analysis of the extreme values of coarse particle distributions, e.g., in monitoring grinding operations [Rosin and Rammler, *J. Inst. Fuel*, **7**, 29–36 (1933); Bennett, *ibid.*, **10**, 22–29 (1936)].

Moments Moments represent a PSD by a single value. With the help of moments, the average particle sizes, volume specific surfaces, and other mean values of the PSD can be calculated. The general definition of a moment is given by (ISO 9276, Part 2: *Calculation of Average Particle Sizes/Diameters and Moments from Particle Size Distributions*)

$$M_{k,r} = \int_{x_{\min}}^{x_{\max}} x^k q_r(x) dx \quad (21-5)$$

where $M_{k,r}$ is the k th moment of a $q_r(x)$ distribution density and k is the power of x .

Average Particle Sizes A PSD has many average particle sizes. The general equation is given by

$$\bar{x}_{k,r} = \sqrt[k]{M_{k,r}} \quad (21-6)$$

Two typically employed average particle sizes are the **arithmetic average particle size** $\bar{x}_{k=0} = M_{k=0}$ [e.g., for a number distribution ($r=0$) obtained by counting methods], and the **weighted average particle size** $\bar{x}_{1,r} = \bar{M}_{1,r}$ [e.g., for a volume distribution ($r=3$) obtained by sieve analysis], where $\bar{x}_{1,r}$ represents the center of gravity on the abscissa of the $q_r(x)$ distribution.

Specific Surface The **specific surface area** can be calculated from size distribution data. For spherical particles this can simply be calculated by using moments. The volume specific surface is given by

$$S_V = \frac{6}{\bar{x}_{1,2}} \quad \text{or} \quad S_V = \frac{6}{M_{1,2}} = \frac{M_{2,0}}{M_{3,0}} = 6 \cdot M_{-1,3} \quad (21-7)$$

where $\bar{x}_{1,2}$ is the weighted average diameter of the area distribution, also known as **Sauter mean diameter**. It represents a particle having the same ratio of surface area to volume as the distribution, and it is also referred to as a **surface-volume average diameter**. The Sauter mean is an important average diameter used in solids handling and other processing applications where aspects of two-phase flow become important, as it appropriately weights the contributions of the fine fractions to surface area. For nonspherical particles, a shape factor has to be considered.

Example 2: The Sauter mean diameter and the volume weighted particle size and distribution given in Table 21-1 can be calculated by using FDIS-ISO 9276-2, *Representation of Results of Particle Size Analysis—Part 2: Calculation of Average Particle Sizes/Diameters and Moments from Particle Size Distributions* via Table 21-2.

The Sauter mean diameter is

$$\bar{x}_{1,2} = M_{1,2} = \frac{M_{3,0}}{M_{2,0}} = \frac{1}{M_{-1,3}} \quad \text{with} \quad M_{-1,3} = \sum_{i=1}^n \Delta Q_{3,i} \frac{\ln(x_i/x_{i-1})}{x_i - x_{i-1}}$$

TABLE 21-2 Table for Calculation of Sauter Mean Diameter and Volume Weighted Particle Size

I	$x_i, \mu\text{m}$	$\Delta Q_{3,i}$	$\ln(x_i/x_{i-1})$	$\ln(x_i/x_{i-1}) / (x_i - x_{i-1})$	$\Delta Q_{3,i}^s / \ln(x_i/x_{i-1})$	$\Delta Q_{3,i}^s (x_i + x_{i-1}), \mu\text{m}$
0	0.0630					
1	0.0900	0.0010	0.3567	13.2102	0.013210	0.000153
2	0.1250	0.0009	0.3285	9.3858	0.008447	0.000194
3	0.1800	0.0016	0.3646	6.6299	0.010608	0.000488
4	0.2500	0.0025	0.3285	4.6929	0.011732	0.001075
5	0.3550	0.0050	0.3507	3.3396	0.016698	0.003025
6	0.5000	0.0110	0.3425	2.3620	0.025982	0.009405
7	0.7100	0.0180	0.3507	1.6698	0.030056	0.021780
8	1.0000	0.0370	0.3425	1.1810	0.043697	0.063270
9	1.4000	0.0610	0.3365	0.8412	0.051312	0.146400
10	2.0000	0.1020	0.3567	0.5945	0.060635	0.346800
11	2.8000	0.1600	0.3365	0.4206	0.067294	0.768000
12	4.0000	0.2100	0.3567	0.2972	0.062418	1.428000
13	5.6000	0.2400	0.3365	0.2103	0.050471	2.304000
14	8.0000	0.1250	0.3567	0.1486	0.018577	1.700000
15	11.2000	0.0240	0.3365	0.1051	0.002524	0.460800
16	16.0000	0.0010	0.3567	0.0743	0.000074	0.027200
$\Sigma 0.473736$						7.280590

which yields

$$\bar{x}_{1,2} = \frac{1}{0.473736} = 2.110882$$

The volume weighted average particle size is

$$\bar{x}_{1,3} = M_{1,3} = \frac{1}{2} \sum_{i=1}^n \Delta Q_{3,i} (x_i + x_{i-1})$$

which yields

$$\bar{x}_{1,3} = \frac{1}{2} \cdot 7.280590 = 3.640295$$

PARTICLE SHAPE

For many applications not only the particle size but also the shape are of importance; e.g., toner powders should be spherical while polishing powders should have sharp edges. Traditionally in microscopic methods of size analysis, direct measurements are made on enlarged images of the particles by using a calibrated scale. While such measurements are always encouraged to gather a direct sense of the particle shape and size, care should be taken in terms of drawing general conclusions from limited particle images. Furthermore, with the strong progress in computing power, instruments have become available that acquire the projected area of many particles in short times, with a significant reduction in data manipulation times. Although a standardization of shape parameters is still in preparation (upcoming ISO 9276, *Part 6: Descriptive and Qualitative Representation of Particle Shape and Morphology*), there is wide agreement on the following parameters.

Equivalent Projection Area of a Circle Equivalent projection area of a circle (Fig. 21-3) is widely used for the evaluation of particle sizes from the projection area A of a nonspherical particle.

$$x_{EQPC} = 2\sqrt{A/\pi} \quad (21-8)$$

Feret's Diameter Feret's diameter is determined from the projected area of the particles by using a slide gauge. In general it is defined as the distance between two parallel tangents of the particle at an arbitrary angle. In practice, the **minimum** $x_{F,min}$ and **maximum** **Feret** diameters $x_{F,max}$, the **mean Feret diameter** $\bar{x}_F$, and the Feret diameters obtained at 90° to the direction of the minimum and maximum Feret diameters $x_{F,max90}$ are used. The minimum Feret diameter is often used as the diameter equivalent to a sieve analysis.

Other diameters used in the literature include **Martin's diameter** or the edges of an **enclosing rectangle**. Martin's diameter is a line, parallel to a fixed direction, which divides the particle profile into two equal areas.

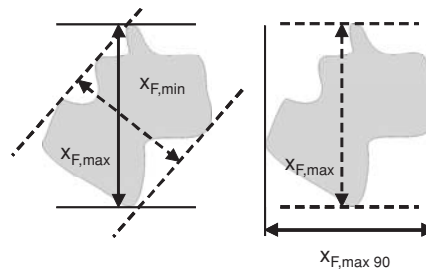


FIG. 21-3 Definition of Feret diameters.

These diameters offer an extension over volume equivalent diameters to account for shape deviations from spherical. As with any other quality measure of size, many particles must be measured to determine distributions of these particle-size diameters.

Sphericity, Aspect Ratio, and Convexity Parameters describing the shape of the particles include the following:

The **sphericity** ψ_s ($0 < \psi_s \leq 1$) is defined by the ratio of the perimeter of a circle with diameter x_{EQPC} to the perimeter of the corresponding projection area A . And $\psi_s = 1$ represents a sphere.

The **aspect ratio** ψ_A ($0 < \psi_A \leq 1$) is defined by the ratio of the minimum to the maximum Feret diameter $\psi_A = x_{Feret,min}/x_{Feret,max}$. It gives an indication of the **elongation** of the particle. Some literature also used $1/\psi_A$ as the definition of sphericity.

The **convexity** ψ_C ($0 < \psi_C \leq 1$) is defined by the ratio of the projection area A to the convex hull area $A + B$ of the particle, as displayed in Fig. 21-4.

In **Fourier techniques** the shape characteristic is transformed to a signature waveform. Beddow and coworkers (Beddow, *Particulate Science and Technology*, Chemical Publishing, New York, 1980) take the particle centroid as a reference point. A vector is then rotated about this centroid with the tip of the vector touching the periphery. A plot of the magnitude of the vector versus its angular position is a wave-type function. This waveform is then subjected to Fourier analysis. The lower-frequency harmonics constituting the complex wave correspond to the gross external morphology, whereas the higher frequencies correspond to the texture of the fine particle.

Fractal Logic This was introduced into fine particles science by Kaye and coworkers (Kaye, op. cit., 1981), who show that the noneuclidean logic of Mandelbrot can be applied to describe the ruggedness of a particle profile. A combination of fractal dimension and geometric shape factors such as the aspect ratio can be used to describe a population of fine particles of various shapes, and these can be related to the functional properties of the particle.

SAMPLING AND SAMPLE SPLITTING

As most of the sizing methods are limited to small sample sizes, an important prerequisite to accurate particle-size analysis is proper powder sampling and sample splitting (upcoming ISO 14488, *Particulate Materials—Sampling and Sample Splitting for the Determination of Particulate Properties*).

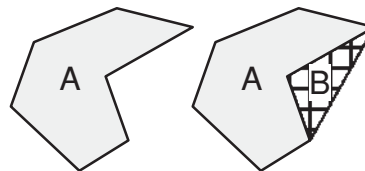


FIG. 21-4 Definition of the convex hull area $A + B$ for the projection area A of a particle.

When determining particle size (or any other particle attribute such as chemical composition or surface area), it is important to recognize that the error associated in making such a measurement can be described by its variance, or

$$\sigma_{\text{observed}}^2 = \sigma_{\text{actual}}^2 + \sigma_{\text{measurement}}^2 \quad (21-9)$$

$$\sigma_{\text{measurement}}^2 = \sigma_{\text{sampling}}^2 + \sigma_{\text{analysis}}^2 \quad (21-10)$$

That is, the *observed* variance in the particle-size measurement is due to both the actual physical variance in size as well as the variance in the measurement. More importantly, the variance in measurement has two contributing factors: variance due to sampling, which would include systematic errors in the taking, splitting, and preparation of the sample; and variance due to the actual sample analysis, which would include not only the physical measurement at hand, but also how the sample is *presented* to the measuring zone, which can be greatly affected by instrument design and sample dispersion (discussed below). Successful characterization of the sample (in this discussion, taken to be measurement of particle size) requires that the errors in measurement be much less than actual physical variations in the sample itself, especially if knowledge of sample deviations is important. In this regard, great negligence is unfortunately often exhibited in sampling efforts. Furthermore, measured deviations in particle size or other properties are often incorrectly attributed to and reflect upon the measuring device, where in fact they are caused by inattention to proper sampling and sample splitting. Worse still, such deviations caused by poor sampling may be taken as true sample deviations, causing undue and frequent process corrections.

Powders may be classified as nonsegregating (cohesive) or segregating (free-flowing). Representative samples can be more easily taken from **cohesive powders**, provided that they have been properly mixed. For wet samples a sticky paste should be created and mixed from which the partial sample is taken.

In the case of **free-flowing powders**, four key rules should be followed, although some apply or can be equally employed for cohesive materials as well. These rules are especially important for in-line and on-line sampling, discussed below. As extended from Allen, *Powder Sampling and Particle Size Determination* (Elsevier, 2003):

1. The particles should be sampled while in motion. Transfer points are often convenient and relevant for this. Sampling a stagnant bed of segregating material by, e.g., thieves disrupts the state of the mixture and may be biased to coarse or fines.

2. The whole stream of powder should be taken in many short time intervals in preference to part of the stream being taken over the whole time, i.e., a complete slice of the particle stream. Furthermore, any mechanical collection point should not be allowed to overfill, since this will make the sample bias toward fines, and coarse material rolls off formed heaps.

3. The entire sample should be analyzed, splitting down to a smaller sample if necessary. In many cases, segregation of the sample will not affect the measurement, provided the entire sample is analyzed. There are, however, exceptions in that certain techniques may only analyze one surface of the final sample. In the case of chemical analysis, an example would be near infrared spectroscopy operated in reflectance mode as opposed to transmission. Such a technique may still be prone to segregation during the final analysis. (See the subsection "Material Handling: Impact of Segregation on Measurements.")

4. A minimum sample size exists for a given size distribution, generally determined by the sample containing a minimum number of coarse particles representative of the customer application. While many applications involving fine pharmaceuticals may only require milligrams to establish a representative sample, other cases such as detergents and coffee might require kilograms. Details are given in the upcoming standard ISO/DIS14488, *Particulate Materials—Sampling and Sample Splitting for the Determination of Particulate Properties*.

In this regard, one should keep in mind that the sample size may also reflect variation in the degree of mixing in the bed, as opposed to true size differences. (See also the subsection "Solids Mixing: Measuring the Degree of Mixing.") In fact, larger samples in this case help minimize the impact of segregation on measurements.

TABLE 21-3 Reliability of Selected Sampling Method

Method	Estimated maximum sampling error, %
Cone and quartering	22.7
Scoop sampling	17.1
Table sampling	7.0
Chute splitting	3.4
Spinning riffling	0.42

The estimated maximum sampling error on a 60:40 blend of free-flowing sand using different sampling techniques is given in Table 21-3.

The **spinning riffler** (Fig. 21-5) generates the most representative samples. In this device a ring of containers rotates under the powder feed. If the powder flows a long time with respect to the period of rotation, each container will be made up of many small fractions from all parts of the bulk. Many different configurations are commercially available. Devices with small numbers of containers (say, 8) can be cascaded n times to get higher splitting ratios $1:8^n$. This usually creates smaller sampling errors than does using splitters with more containers. A splitter simply divides the sample into two halves, generally pouring the sample into a set of intermeshed chutes. Figure 21-6 illustrates commercial rifflers and splitters.

For reference materials sampling errors of less than 0.1 percent are achievable (S. Röthele and W. Witt, Standards in Laser Diffraction, PARTEC, 5th European Symposium Particle Characterization, Nürnberg, 1992, pp. 625–642).

DISPERSION

Many sizing methods are sensitive to the agglomeration state of the sample. In some cases, this includes primary particles, possibly with some percentage of such particles held together as weak agglomerates by interparticle cohesive forces. In other cases, strong aggregates of

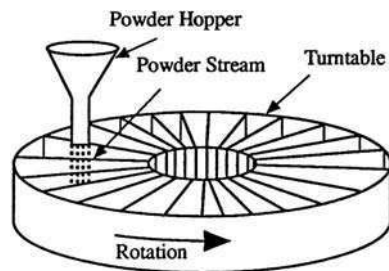


FIG. 21-5 Spinning riffler sampling device.



FIG. 21-6 Examples of commercial splitting devices. Spinning riffler and standard splitters. (Courtesy of Retsch Corporation.)

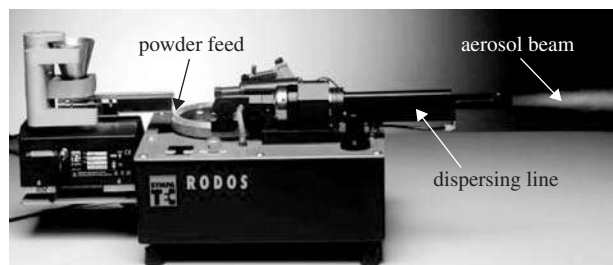


FIG. 21-7 Dry disperser RODOS with vibratory feeder VIBRI creating a fully dispersed aerosol beam from dry powder. (Courtesy of Sympatec GmbH.)

the primary particles may also exist. Generally, the size of either the primary particles or the aggregates is the matter of greatest interest. In some cases, however, it may also be desirable to determine the level of agglomerates in a sample, requiring that the intensity of dispersion be controlled and variable. Often the agglomerates have to be dispersed smoothly without comminution of aggregates or primary particles. This can be done either in gas (dry) or in liquid (wet) by using a suitable dispersion device which is stand-alone or integrated in the particle-sizing instrument. If possible, dry particles should be measured in gas and wet particles in suspension.

Wet Dispersion Wet dispersion separates agglomerates down to the primary particles by a suitable liquid. Dispersing agents and optional cavitation forces induced by ultrasound are often used. Care must be taken that the particles not be soluble in the liquid, or that they not flocculate. Microscopy and zeta potential measurements may be of utility in specifying the proper dispersing agents and conditions for dispersion.

Dry Dispersion Dry dispersion uses mechanical forces for the dispersion. While a simple fall-shaft with impact plates may be sufficient for the dispersion of coarse particles, say, $>300\ \mu\text{m}$, much higher forces have to be applied to fine particles.

In Fig. 21-7 the agglomerates are sucked in by the vacuum generated through expansion of compressed gas applied at an injector. They arrive at low speed in the dispersing line, where they are strongly accelerated. This creates three effects for the dispersion, as displayed in Fig. 21-8.

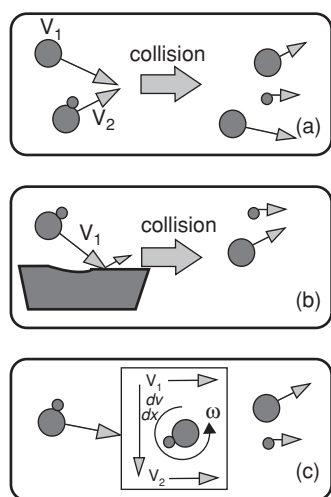


FIG. 21-8 Interactions combined for dry dispersion of agglomerates. (a) Particle-to-particle collisions. (b) Particle-to-wall collisions. (c) Centrifugal forces due to strong velocity gradients.

With suitable parameter settings agglomerates can be smoothly dispersed down to $0.1\ \mu\text{m}$ [K. Leschonski, S. R  thele, and U. Menzel, Entwicklung und Einsatz einer trockenen Dosier-Dispergiereinheit zur Messung von Partikelgr  ssenverteilungen in Gas-Feststoff-Freistrahl aus Laser-Beugungsspektren; *Part. Charact.*, **1**, 161–166 (1984)] without comminution of the primary particles.

PARTICLE-SIZE MEASUREMENT

There are many techniques available to measure the particle-size distribution of powders or droplets. The wide size range, from nanometers to millimeters, of particulate products, however, cannot be analyzed by using only a single measurement principle.

Added to this are the usual constraints of capital costs versus running costs, speed of operation, degree of skill required, and, most important, the end-use requirement.

If the particle-size distribution of a powder composed of hard, smooth spheres is measured by any of the techniques, the measured values should be identical. However, many different size distributions can be defined for any powder made up of nonspherical particles. For example, if a rod-shaped particle is placed on a sieve, then its diameter, not its length, determines the size of aperture through which it will pass. If, however, the particle is allowed to settle in a viscous fluid, then the calculated diameter of a sphere of the same substance that would have the same falling speed in the same fluid (i.e., the Stokes diameter) is taken as the appropriate size parameter of the particle. Since the Stokes diameter for the rod-shaped particle will obviously differ from the rod diameter, this difference represents added information concerning particle shape. The ratio of the diameters measured by two different techniques is called the **shape factor**.

While historically mainly methods using mechanical, aerodynamic, or hydrodynamic properties for discrimination and particle sizing have been used, today methods based on the interaction of the particles with electromagnetic waves (mainly light), ultrasound, or electric fields dominate.

Laser Diffraction Methods Over the past 30 years **laser diffraction** has developed into a leading principle for particle-size analysis of all kinds of aerosols, suspensions, emulsions, and sprays in laboratory and process environments.

The scattering of unpolarized laser light by a single spherical particle can be mathematically described by

$$I(\theta) = \frac{I_0}{2k^2a^2} \{ [S_1(\theta)]^2 + [S_2(\theta)]^2 \} \quad (21-11)$$

where $I(\theta)$ is the total scattered intensity as function of angle θ with respect to the forward direction; I_0 is the illuminating intensity; k is the wave number $2\pi/\lambda$; a is the distance from the scatterer to the detector; and $S_1(\theta)$ and $S_2(\theta)$ are dimensionless, complex functions describing the change and amplitude in the perpendicular and parallel polarized light. Different algorithms have been developed to calculate $I(\theta)$. The **Lorenz-Mie theory** is based on the assumption of spherical, isotropic, and homogenous particles and that all particles can be described by a common complex refractive index $m = n - i\kappa$. Index m has to be precisely known for the evaluation, which is difficult in practice, especially for the imaginary part κ , and inapplicable for mixtures with components having different refractive indices.

The **Fraunhofer theory** considers only scattering at the contour of the particle and the near forward direction. No preknowledge of the refractive index is required, and $I(\theta)$ simplifies to

$$I(\theta) = \frac{I_0}{2k^2a^2} \alpha^4 \left[\frac{J_1(\alpha \sin \theta)}{\alpha \sin \theta} \right]^2 \quad (21-12)$$

with J_1 as the Bessel function of first kind and the dimensionless size parameter $\alpha = \pi r/\lambda$. This theory does not predict polarization or account for light transmission through the particle.

For a single spherical particle, the diffraction pattern shows a typical ring structure. The distance r_0 of the first minimum to the center depends on the particle size, as shown in Fig. 21-9a. In the

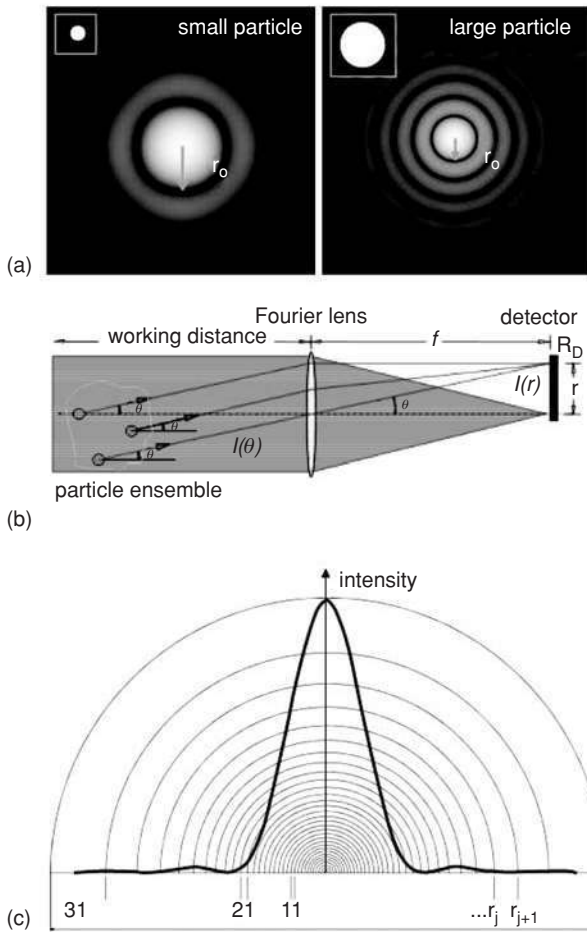


FIG. 21-9 (a) Diffraction patterns of laser light in forward direction for two different particle sizes. (b) The angular distribution $I(\theta)$ is converted by a Fourier lens to a spatial distribution $I(r)$ at the location of the photodetector. (c) Intensity distribution of a small particle detected by a semicircular photodetector.

particle-sizing instrument, the acquisition of the intensity distribution of the diffracted light is usually performed with the help of a multielement photodetector.

Diffraction patterns of static nonspherical particles are displayed in Fig. 21-10. As all diffraction patterns are symmetric to 180° , semicircular detector elements integrate over 180° and make the detected intensity independent of the orientation of the particle.

Simultaneous diffraction on more than one particle results in a superposition of the diffraction patterns of the individual particles, provided that particles are moving and diffraction between the particles is averaged out. This simplifies the evaluation, providing a parameter-free and model-independent mathematical algorithm for the inversion process (M. Heuer and K. Leschonski, Results Obtained with a New Instrument for the Measurement of Particle Size Distributions from Diffraction Patterns, *Part. Part. Syst. Charact.* **2**, 7–13, 1985).

Today the method is standardized (ISO 13320-1, 1999, *Particle Size Analysis—Laser Diffraction Methods—Part 1: General Principles*), and many companies offer instruments, usually with the choice of Fraunhofer and/or Mie theory for the evaluation of the PSD. The size ranges of the instruments have been expanded by combining low-angle laser light scattering with 90° or back scattering, the use of different wavelengths, polarization ratio, and white light scattering, etc.

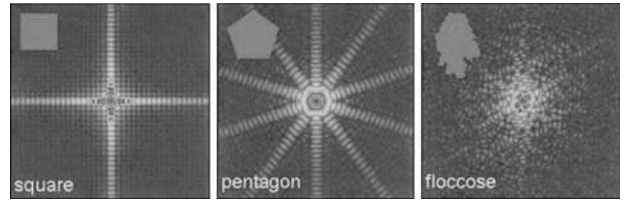


FIG. 21-10 Calculated diffraction patterns of laser light in forward direction for nonspherical particles: square, pentagon, and floccose. All diffraction patterns show a symmetry to 180° .

It is now ranging from below $0.1 \mu\text{m}$ to about 1 cm . Laser diffraction is currently the fastest method for particle sizing at highest reproducibility. In combination with dry dispersion it can handle large amounts of sample, which makes this method well suited for process applications.

Instruments of this type are available, e.g., from Malvern Ltd. (Mastersizer), Sympatec GmbH (HELOS, MYTOS), Horiba (LA, LS series), Beckmann Coulter (LS 13320), or Micromeritics (Saturn).

Image Analysis Methods The extreme progress in image capturing and exceptional increase of the computational power within the last few years have revolutionized microscopic methods and made image analysis methods very popular for the characterization of particles, especially since, in addition to size, relevant shape information becomes available by the method. Currently, mainly instruments creating a 2D image of the 3D particles are used. Two methods have to be distinguished.

Static image analysis is characterized by nonmoving particles, e.g., on a microscope slide (Fig. 21-11). The depth of sharpness is well defined, resulting in a high resolution for small particles. The method is well established and standardized (ISO 13322-1:2004, *Particle Size Analysis—Image Analysis Methods, Part 1: Static Image Analysis Methods*), but can handle only small amounts of data. The particles are oriented by the base; overlapping particles have to be separated by time-consuming software algorithms, and the tiny sample size creates a massive sampling problem, resulting in very low statistical relevance of the data. Commercial systems reduce these effects by using large or even stepping microscopic slides and the deposition of the particles via a dispersing chamber. As all microscopic techniques can be used, the size range is only defined by the microscope used.

Dynamic image analysis images a flow of moving particles. This allows for a larger sample size. The particles show arbitrary orientation, and the number of overlapping particles is reduced. Several companies offer systems which operate in either reflection or transmission, with wet dispersion or free fall, with matrix or line-scan cameras. The free-fall systems are limited to well flowing bulk materials. Systems with wet dispersion only allow for smallest samples sizes and slow particles. As visible light is used for imaging, the size range is

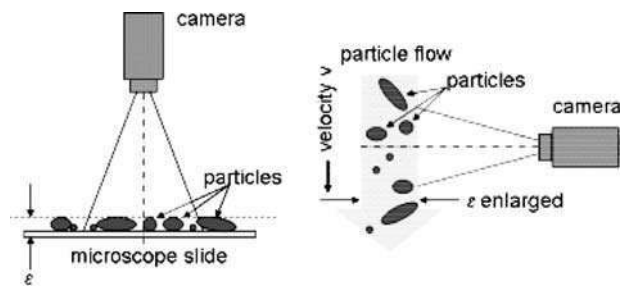


FIG. 21-11 Setup of static (left) and dynamic (right) image analysis for particle characterization.

limited to about 1 μm at the fine end. This type of instruments has been standardized (ISO/FDIS 13322-2:2006, *Particle Size Analysis—Image Analysis Methods, Part 2: Dynamic Methods*).

Common to all available instruments are small particle numbers, which result in poor statistics. Thus recent developments have yielded a combination of powerful dry and wet dispersion with high-speed image capturing. Particle numbers up to 10^7 can now be acquired in a few minutes. Size and shape analysis is available at low statistical errors [W. Witt, U. Köhler, and J. List, Direct Imaging of Very Fast Particles Opens the Application of the Powerful (Dry) Dispersion for Size and Shape Characterization, PARTEC 2004, Nürnberg].

Dynamic Light Scattering Methods Dynamic light scattering (DLS) is now used on a routine basis for the analysis of particle sizes in the submicrometer range. It provides an estimation of the average size and its distribution within a measuring time of a few minutes.

Submicrometer particles suspended in a liquid are in constant **brownian motion** as a result of the impacts from the molecules of the suspending fluid, as suggested by W. Ramsay in 1876 and confirmed by A. Einstein and M. Smoluchowski in 1905/06.

In the Stokes-Einstein theory of brownian motion, the particle motion at very low concentrations depends on the viscosity of the suspending liquid, the temperature, and the size of the particle. If viscosity and temperature are known, the particle size can be evaluated from a measurement of the particle motion. At low concentrations, this is the **hydrodynamic diameter**.

DLS probes this motion optically. The particles are illuminated by a coherent light source, typically a laser, creating a diffraction pattern, showing in Fig. 21-12 as a fine structure from the diffraction between the particles, i.e., its near-order. As the particles are moving from impacts of the thermal movement of the molecules of the medium, the particle positions change with the time, t .

The change of the position of the particles affects the phases and thus the fine structure of the diffraction pattern. So the intensity in a certain point of the diffraction pattern fluctuates with time. The fluctuations can be analyzed in the time domain by a correlation function analysis or in the frequency domain by frequency analysis. Both methods are linked by Fourier transformation.

The measured decay rates Γ are related to the translational diffusion coefficients D of spherical particles by

$$\Gamma = Dq^2 \quad \text{with} \quad q = \frac{4\pi}{\lambda_0} \sin \frac{\theta}{2} \quad \text{and} \quad D = \frac{k_B T}{2\pi\eta x} \quad (21-13)$$

where q is the modulus of the scattering vector, k_B is the Boltzmann constant, T is the absolute temperature, and η is the hydrodynamic viscosity of the dispersing liquid. The particle size x is then calculated by the Stokes-Einstein equation from D at fixed temperature T and η known.

DLS covers a broad range of diluted and concentrated suspension. As the theory is only valid for light being scattered once, any contribution of multiple scattered light leads to erroneous PCS results and misinterpretations. So different measures have been taken to minimize the influence of multiple scattering.

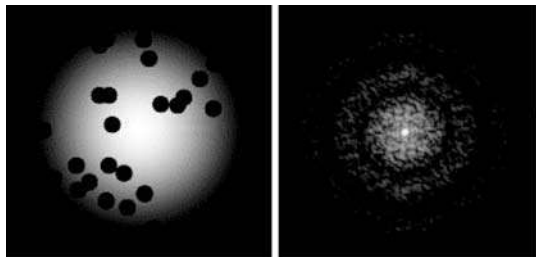


FIG. 21-12 Particles illuminated by a gaussian-shaped laser beam and its corresponding diffraction pattern show a fine structure.

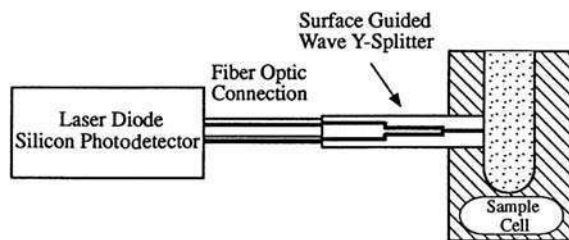


FIG. 21-13 Diagram of Leeds and Northrup Ultrafine Particle Size Analyzer (UPA), using fiber optics in a backscatter setup.

The well-established **photon correlation spectroscopy** (PCS) uses highly diluted suspensions to avoid multiple scattering. The low concentration of particles makes this method sensitive to impurities in the liquid. So usually very pure liquids and a clean-room environment have to be used for the preparation and operation (ISO 13321:1996, *Particle Size Analysis—Photon Correlation Spectroscopy*).

Another technique (Fig. 21-13) utilizes an optical system which minimizes the optical path into and out of the sample, including the use of backscatter optics, a moving cell assembly, or setups with the maximum incident beam intensity located at the interface of the suspension to the optical window (Trainer, Freud, and Weiss, Pittsburgh Conference, Analytical and Applied Spectroscopy, *Symp. Particle Size Analysis*, March 1990; upcoming ISO 22412, *Particle Size Analysis—Dynamic Light Scattering*).

Photon cross-correlation spectroscopy (PCCS) uses a novel three-dimensional cross-correlation technique which completely suppresses the multiple scattered fractions in a special scattering geometry. In this setup two lasers A and B are focused to the same sample volume, creating two sets of scattering patterns, as shown in Fig. 21-14. Two intensities are measured at different positions but with identical scattering vectors.

$$\vec{q} = \vec{k}_A - \vec{k}_1 = \vec{k}_B - \vec{k}_2 \quad (21-14)$$

Subsequent cross-correlation of these two signals eliminates any contribution of multiple scattering. So highly concentrated, opaque suspensions can be measured as long as scattered light is observed. High count rates result in short measuring times. High particle concentrations reduce the sensitivity of this method to impurities, so standard liquids and laboratory environments can be used, which simplifies the application [W. Witt, L. Aberle, and H. Geers, Measurement of Particle Size and Stability of Nanoparticles in Opaque Suspensions and Emulsions with Photon Cross Correlation Spectroscopy, *Particulate Systems Analysis*, Harrogate (UK), 2003].

Acoustic Methods **Ultrasonic attenuation spectroscopy** is a method well suited to measuring the PSD of colloids, dispersions, slurries, and emulsions (Fig. 21-15). The basic concept is to measure the frequency-dependent attenuation or velocity of the ultrasound as it passes through the sample. The attenuation includes

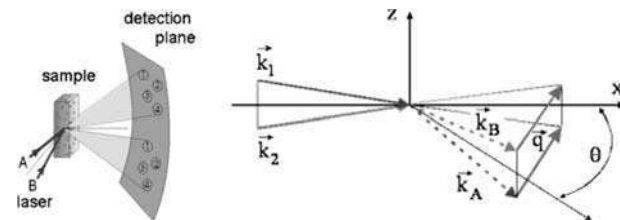


FIG. 21-14 Scattering geometry of a PCCS setup. The sample volume is illuminated by two incident beams. Identical scattering vectors $\vec{q}$ and the scattering volumes are used in combination with cross-correlation to eliminate multiple scattering.

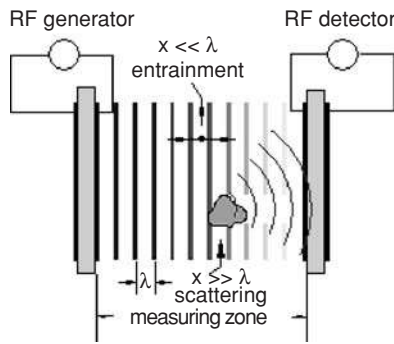


FIG. 21-15 Setup of an ultrasonic attenuation system for particle-size analysis.

contributions from the scattering or absorption of the particles in the measuring zone and depends on the size distribution and the concentration of the dispersed material. (ISO 20998:2006, *Particle Characterization by Acoustic Methods, Part 1: Ultrasonic Attenuation Spectroscopy*).

In a typical setup (see Fig. 21-15) an electric high-frequency generator is connected to a piezoelectric ultrasonic transducer. The generated ultrasonic waves are coupled into the suspension and interact with the suspended particles. After passing the measuring zone, the ultrasonic plane waves are received by an ultrasonic detector and converted to an electric signal, which is amplified and measured. The attenuation of the ultrasonic waves is calculated from the ratio of the signal amplitudes on the generator and detector sides.

PSD and concentration can be calculated from the attenuation spectrum by using either complicated theoretical calculations requiring a large number of parameters or an empirical approach employing a reference method for calibration. Following U. Riebel (Die Grundlagen der Partikelgrößenanalyse mittels Ultraschallspektrometrie, PhD-Thesis, University of Karlsruhe), the ultrasonic extinction of a suspension of monodisperse particles with diameter x can be described by Lambert-Beer's law. The extinction $-\ln(I/I_0)$ at a given frequency f is linearly dependent on the thickness of the suspension layer Δl , the projection area concentration C_{PF} , and the related extinction cross section K . In a polydisperse system the extinctions of single particles overlay:

$$-\ln\left(\frac{I}{I_0}\right)_{f_i} \cong \Delta l \cdot C_{PF} \cdot \sum_j K(f_i, x_j) \cdot q_2(x_j) \Delta x \quad (21-15)$$

When the extinction is measured at different frequencies f_i , this equation becomes a linear equation system, which can be solved for C_{PF} and $q_2(x)$. The key for the calculation of the particle-size distribution is the knowledge of the related extinction cross section K as a function of the dimensionless size parameter $\sigma = 2\pi x/\lambda$. For spherical particles K can be evaluated directly from the acoustic scattering theory. A more general approach is an empirical method using measurements on reference instruments as input.

This disadvantage is compensated by the ability to measure a wide size range from below 10 μm to above 3 mm and the fact that PSDs can be measured at very high concentrations (0.5 to >50 percent of volume) without dilution. This eliminates the risk of affecting the dispersion state and makes this method ideal for in-line monitoring of, e.g., crystallizers (A. Pankewitz and H. Geers, LABO, "In-line Crystal Size Distribution Analysis in Industrial Crystallization Processes by Ultrasonic Extinction," May 2000).

Current instruments use different techniques for the attenuation measurement: with *static* or *variable* width of the measuring zone, measurement in *transmission* or *reflection*, with *continuous* or *swept* frequency generation, with *frequency burst* or *single-pulse* excitation.

For process environment, probes are commercially available with a frequency range of 100 kHz to 200 MHz and a dynamic range of

>150 dB, covering 1 to 70 percent of volume concentration, 0 to 120°C, 0 to 40 bar, pH 1 to 14, and hazardous areas as an option.

Vendors of this technology include Sympatec GmbH (OPUS), Malvern Instruments Ltd. (Ultrazizer), Dispersion Technology Inc. (DT series), and Colloidal Dynamics Pty Ltd. (AcoustoSizer).

Single-Particle Light Interaction Methods Individual particles have been measured with light for many years. The measurement of the particle size is established by (1) the determination of the *scattered light* of the particle, (2) the measurement of the amount of *light extinction* caused by the particle presence, (3) the measurement of the *residence time* during motion through a defined distance, or (4) *particle velocity*.

Many commercial instruments are available, which vary in optical design, light source type, and means, and how the particles are presented to the light.

Instruments using **light scattering** cover a size range of particles of 50 nm to about 10 μm (liquid-borne) or 20 μm (gas-borne), while instruments using **light extinction** mainly address liquid-borne particles from 1 μm to the millimeter size range. The size range capability of any single instrument is typically 50 : 1. International standards are currently under development (ISO 13323-1:2000, *Determination of Particle Size Distribution—Single-Particle Light Interaction Methods, Part 1: Light Interaction Considerations*; ISO/DIS 21501-2, *Determination of Particle Size Distribution—Single Particle Light-Interaction Methods, Part 2: Light-Scattering Liquid-Borne Particle Counter*; ISO/DIS 21501-3, *Part 3: Light-Extinction Liquid-Borne Particle Counter*; ISO/DIS 21501-4, *Part 4: Light-Scattering Airborne Particle Counter for Clean Spaces*).

Instruments using the **residence time**, such as the aerodynamic particle sizers, or the **particle velocity**, as used by the phase Doppler particle analyzers, measure the particle size primarily based on the aerodynamic diameter.

Small-Angle X-Ray Scattering Method Small angle X-ray scattering can be used in a size range of about 1 to 300 nm. Its advantage is that the scattering mainly results from the differences in the electron density between the particles and their surrounding. As internal crystallites of external agglomerates are not visible, the measured size always represents the size of the primary particles and the requirement for dispersion is strongly reduced [Z. Jinyuan, L. Chulan, and C. Yan, *Stability of the Dividing Distribution Function Method for Particle Size Distribution Analysis in Small Angle X-Ray Scattering*, *J. Iron & Steel Res. Inst.*, 3(1), (1996); ISO/TS 13762:2001, *Particle Size Analysis—Small Angle X-Ray Scattering Method*].

Focused-Beam Techniques These techniques are based on a focused light beam, typically a laser, with the focal point spinning on a circle parallel to the surface of a glass window. When the focal point passes a particle, the reflected and/or scattered light of the particle is detected. The focal point moves along the particle on circular segments, as displayed in Fig. 21-16. Sophisticated threshold algorithms are used to determine the start point and endpoint of the chord, i.e., the edges of the particle. The chord length is calculated from the time interval and the track speed of the focal point. Focused-beam techniques measure a chord length distribution, which corresponds to the size and shape information of the particles typically in a complicated way (J. Worlsche, T. Hocker, and M. Mazzoti, *Restoration of PSD from Chord Length Distribution Data Using the Method of Projections onto Convex Sets, Part. Part. Syst. Char.*, 22, 81 ff.). So often the chord length distribution is directly used as the fingerprint information of the size, shape, and population status.

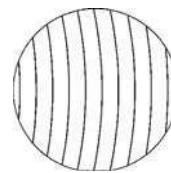


FIG. 21-16 Different chords measured on a constantly moving single spherical particle by focused-beam techniques.



FIG. 21-17 Multisizer™ 3 COULTER COUNTER® from Beckman Coulter, Inc., uses the electrical sensing zone method.

Instruments of this type are commercially available as robust finger probes with small probe diameters. They are used in on-line and preferably in in-line applications, monitoring the chord length distribution of suspensions and emulsions. Special flow conditions are used to reduce the sampling errors. Versions with fixed focal distance [Focused Beam Reflectance Measurement (FBRM®)] and variable focal distance (3D ORM technology) are available. The latter improves this technique for high concentrations and widens the dynamic range, as the focal point moves horizontally and vertically with respect to the surface of the window. For instruments refer, e.g., to Mettler-Toledo International Inc. (Lasentec FBRM probes) and Messtechnik Schwartz GmbH (PAT).

Electrical Sensing Zone Methods In the electric sensing zone method (Fig. 21-17), a well-diluted and -dispersed suspension in an electrolyte is caused to flow through a small aperture [Kubitschek, *Research*, **13**, 129 (1960)]. The changes in the resistivity between two electrodes on either side of the aperture, as the particles pass through, are related to the volumes of the particles. The pulses are fed to a pulse-height analyzer where they are counted and scaled. The method is limited by the resolution of the pulse-height analyzer of about 16,000:1 (corresponding to a volume diameter range of about 25:1) and the need to suspend the particles in an electrolyte (ISO 13319:2000, *Determination of Particle Size Distributions—Electrical Sensing Zone Method*).

Gravitational Sedimentation Methods In gravitational sedimentation methods, the particle size is determined from the settling velocity and the undersize fraction by changes of concentration in a settling suspension. The equation relating particle size to settling velocity is known as **Stokes' law** (ISO 13317, *Part 1: General Principles and Guidelines*):

$$x_{st} = \sqrt{\frac{18\eta u}{(\rho_s - \rho_l)g}} \quad (21-16)$$

where x_{st} is the Stokes diameter, η is viscosity, u is the particle settling velocity under gravity, ρ_s is the particle density, ρ_l is the liquid density, and g is the gravitational acceleration.

The **Stokes diameter** is defined as the diameter of a sphere having the same density and the same velocity as the particle settling in a liquid of the same density and viscosity under laminar flow conditions. Corrections for the deviation from Stokes' law may be necessary at the coarse end of the size range. Sedimentation methods are limited to sizes above 1 μm due to the onset of thermal diffusion (brownian motion) at smaller sizes.

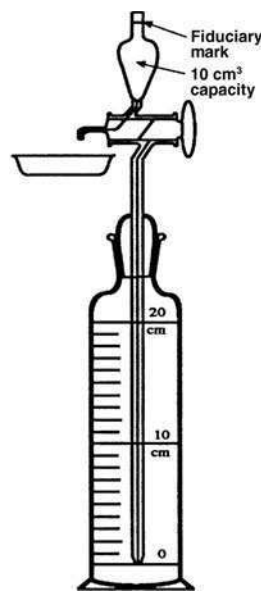


FIG. 21-18 Equipment used in the pipette method of size analysis.

An experimental problem is to obtain adequate dispersion of the particles prior to a sedimentation analysis. For powders that are difficult to disperse, the addition of dispersing agents is necessary, along with ultrasonic probing. It is essential to examine a sample of the dispersion under a microscope to ensure that the sample is fully dispersed. (See "Wet Dispersion.")

Equations to calculate size distributions from sedimentation data are based on the assumption that the particles sink freely in the suspension. To ensure that particle-particle interaction can be neglected, a volume concentration below 0.2 percent is recommended.

There are various procedures available to determine the changing solid concentration of a sedimenting suspension:

In the **pipette method**, concentration changes are monitored by extracting samples from a sedimenting suspension at known depths and predetermined times. The method is best known as Andreasen modification [Andreasen, *Kolloid-Z.*, **39**, 253 (1929)], shown in Fig. 21-18. Two 10-mL samples are withdrawn from a fully dispersed, agitated suspension at zero time to corroborate the 100 percent concentration given by the known weight of powder and volume of liquid making up the suspension. The suspension is then allowed to settle in a temperature-controlled environment, and 10-mL samples are taken at time intervals in geometric 2:1 time progression starting at 1 min (that is, 1, 2, 4, 8, 16, 32, 64 min). The amount of powder in the extracted samples is determined by drying, cooling in a desiccator, and weighing. Stokes diameters are determined from the predetermined times and the depth, with corrections for the changes in depth due to the extractions. The cumulative mass undersize distribution comprises a plot of the normalized concentration versus the Stokes diameter. A reproducibility of ± 2 percent is possible by using this apparatus. The technique is versatile in that it is possible to analyze most powders dispersible in liquids; its disadvantages are that it is a labor-intensive procedure, and a high level of skill is needed (ISO 13317, *Part 2: Fixed Pipette Method*).

The **hydrometer method** is simpler in that the density of the suspension, which is related to the concentration, is read directly from the stem of the hydrometer while the depth is determined by the distance of the hydrometer bulb from the surface (ASTM Spec. Pub. 234, 1959). The method has a low resolution but is widely used in soil science studies.

In **gravitational photo sedimentation methods**, the change of the concentration with time and depth of sedimentation is monitored



FIG. 21-19 The Sedigraph III 5120 Particle Size Analysis System determines particle size from velocity measurements by applying Stokes' law under the known conditions of liquid density and viscosity and particle density. Settling velocity is determined at each relative mass measurement from knowledge of the distance the X-ray beam is from the top of the sample cell and the time at which the mass measurement was taken. It uses a narrow, horizontally collimated beam of X-rays to measure directly the relative mass concentration of particles in the liquid medium.

by using a light point or line beam. These methods give a continuous record of changing optical density with time and depth and have the added advantage that the beam can be scanned to the surface to reduce the measurement time. A correction needs to be applied to compensate for a deviation from the laws of geometric optics (due to diffraction effects the particles cut off more light than geometric optics predicts). The normalized measurement is a $Q_3(x)$ distribution (coming ISO 1337, *Part 4: Photo Gravitational Method*).

In **gravitational X-ray sedimentation methods**, the change of the concentration with time and depth of sedimentation is monitored by using an X-ray beam. These methods give a continuous record of changing X-ray density with time and depth and have the added advantage that the beam can be scanned to the surface to reduce the measurement time. The methods are limited to materials having a high atomic mass (i.e., X-ray-opaque material) and give a $Q_3(x)$ distribution directly (ISO 13317, *Part 3: X-ray Gravitational Technique*). See Fig. 21-19.

Sedimentation Balance Methods In sedimentation balances the weight of sediment is measured as it accumulates on a balance pan suspended in an initial homogeneous suspension. The technique is slow due to the time required for the smallest particle to settle out over a given height. The relationship between settled weight P , weight undersize W , and time t is given by

$$P = W - \frac{dP}{dt} \quad (21-17)$$

Centrifugal Sedimentation Methods These methods extend sedimentation methods well into the submicrometer size range. Alterations of the particle concentration may be determined space- and time-resolved during centrifugation (T. Detloff and D. Lerche, "Determination of Particle Size Distributions Based on Space and Time Resolved Extinction Profiles in Centrifugal Field," Proceedings of Fifth World Congress on Particle Technology, Session Particle

Measurement, Orlando, Fla., April 23–27, 2006). Sizes are calculated from a modified version of the Stokes equation:

$$x_{st} = \sqrt{\frac{18\eta u}{(\rho_s - \rho_f)\omega^2}} \quad (21-18)$$

where ω is the radial velocity of the centrifuge. The concentration calculations are complicated due to radial dilution effects (i.e., particles do not travel in parallel paths as in gravitational sedimentation but move away from each other as they settle radially outward). Particle velocities are given by

$$u = \frac{\ln(r/s)}{t} \quad (21-19)$$

where both the measurement radius r and the surface radius s can be varying. The former varies if the system is a scanning system, and the latter if the surface falls due to the extraction of samples.

Concentration undersize D_m is determined by

$$D_m = \int_{x_{min}}^x \exp(-2ktz^2) q_3(x) dz \quad (21-20)$$

with
$$k = \frac{\rho_s - \rho_f}{18\eta} \omega^2 \quad (21-21)$$

where $q_3(x) = dQ_3(x)/dx$ is the volume or mass density distribution and z is the integration variable.

The solution of the integral for measuring the concentration at constant position over time is only approximately possible. A common way uses Kamack's equation [Kamack, *Br. J. Appl. Phys.*, **5**, 1962–1968 (1972)] as recommend by ISO 13318 (*Part 1: Determination of Particle Size by Centrifugal Liquid Sedimentation Methods*).

An analytical solution is provided by measuring the concentration to at least one time at different sedimentation heights:

$$Q_3(x) = \int_1^{D_m} \left(\frac{r_i}{s} \right)^2 dD_m \quad (21-22)$$

where r_i is the measurement position and s the surface radius; $Q_3(x)$ is the cumulative mass or volume concentration, and $(r_i/s)^2$ is the radial dilution correction factor.

The **disc centrifuge**, developed by Slater and Cohen and modified by Allen and Svarovsky [Allen and Svarovsky, *Dechema Monograph*, Nuremberg, Nos. 1589–1625, pp. 279–292 (1975)], is essentially a centrifugal pipette device. Size distributions are measured from the solids concentration of a series of samples withdrawn through a central drainage pillar at various time intervals.

In the **centrifugal disc photodensitometer**, concentration changes are monitored by a light point or line beam. In one high-resolution mode of operation, the suspension under test is injected into clear liquid in the spinning disc through an entry port, and a layer of suspension is formed over the free surface of liquid (the line start technique). The analysis can be carried out using a homogeneous suspension. Very low concentrations are used, but the light-scattering properties of small particles make it difficult to interpret the measured data.

Several **centrifugal cuvette photocentrifuges** are commercially available. These instruments use the same theory as the photocentrifuges but are limited in operation to the homogeneous mode of operation (ISO 13318:2001, *Determination of Particle Size Distribution by Centrifugal Liquid Sedimentation Methods—Part 1: General Principles and Guidelines; Part 2: Photocentrifuge Method*).

The **X-ray disc centrifuge** is a centrifuge version of the gravitational instrument and extends the measuring technique well into the submicrometer size range (ISO 13318-3:2004, *Part 3: Centrifugal X-ray Method*).

Sieving Methods Sieving is probably the most frequently used and abused method of analysis because the equipment, analytical procedure, and basic concepts are deceptively simple. In sieving, the particles are presented to equal-size apertures that constitute a series of go-no go gauges. Sieve analysis implies three major difficulties: (1) with woven-wire sieves, the weaving process produces three-dimensional apertures with considerable tolerances, particularly for fine-woven mesh; (2) the mesh is easily damaged in use; (3) the particles must be efficiently presented to the sieve apertures to prevent blinding.

Sieves are often referred to their mesh size, which is a number of wires per linear unit. Electroformed sieves with square or round apertures and tolerances of $\pm 2 \mu\text{m}$ are also available (ISO 3310, *Test Sieves—Technical Requirements and Testing*, 2000/2004: Part 1: *Test Sieves of Metal Wire Cloth*; 1999; Part 2: *Test Sieves of Perforated Metal Plate*; 1990; Part 3: *Test Sieves of Electroformed Sheets*).

For coarse separation, dry sieving is used, but other procedures are necessary for finer and more cohesive powders. The most aggressive agitation is performed with Pascal Inclyno and Tyler Ro-tap sieves, which combine gyratory and jolting movement, although a simple vibratory agitation may be suitable in many cases. With Air-Jet sieves, a rotating jet below the sieving surface cleans the apertures and helps the passage of fines through the apertures. The sonic sifter combines two actions, a vertical oscillating column of air and a repetitive mechanical pulse. Wet sieving is frequently used with cohesive powders.

Elutriation Methods and Classification In gravity elutriation the particles are classified in a column by a rising fluid flow. In centrifugal elutriation the fluid moves inward against the centrifugal force. A cyclone is a centrifugal elutriator, although it is not usually so regarded. The cyclosizer is a series of inverted cyclones with added apex chambers through which water flows. Suspension is fed into the largest cyclone, and particles are separated into different size ranges.

Differential Electrical Mobility Analysis (DMA) Differential electrical mobility analysis uses an electric field for the classification and analysis of charged aerosol particles ranging from about 1 nm to about 1 μm in a gas phase. It mainly consists of four parts: (1) A *pre-separator* limits the upper size to a known cutoff size. (2) A *particle charge conditioner* charges the aerosol particles to a known electric charge (a function of particle size). A bipolar diffusion particle charger is commonly used. The gas is ionized either by radiation from a radioactive source (e.g., ^{85}Kr), or by ions emitted from a corona electrode. Gas ions of either polarity diffuse to the aerosol particles until charge equilibrium is reached. (3) A *differential electrical mobility spectrometer* (DEMS) discriminates particles with different electrical mobility by particle migration perpendicular to a laminar sheath flow. The voltage between the inner cylinder and the outer cylinder (GND) is varied to adjust the discrimination level. (4) An *aerosol particle detector* uses, e.g., a continuous-flow condensation particle counter (CPC) or an aerosol electrometer (AE).

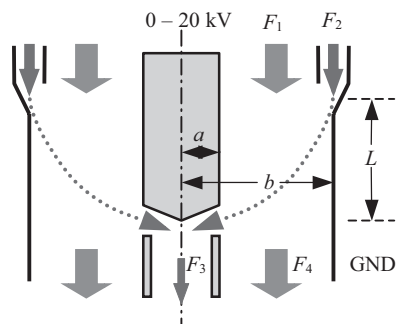


FIG. 21-20 Schematic of a differential electrical mobility analyzer.

A typical setup of the DEMS is shown in Fig. 21-20. It shows the flow rates of the sheath flow F_1 , the polydisperse aerosol sample F_2 , the monodisperse (classified) aerosol exiting the DEMS F_3 , and the excess air F_4 .

The electrical mobility Z depends on the particle size x and the number of elementary charges e :

$$Z(x) = \frac{p \cdot e}{3\pi\eta x} [1 + \text{Kn}(A + B e^{C/\text{Kn}})] \quad (21-23)$$

with the number of elementary charges p , the Knudsen number Kn of $2l/x$, the mean path l of the gas molecule, η the dynamic fluid viscosity, and numeric constants A , B , C determined empirically.

Commercial instruments are available for a variety of applications in aerosol instrumentation, production of materials from aerosols, contamination control, etc. (ISO/CD 15900 2006, *Determination of Particles Size Distribution—Differential Electrical Mobility Analysis for Aerosol Particles*).

Surface Area Determination The surface-to-volume ratio is an important powder property since it governs the rate at which a powder interacts with its surroundings, e.g., in chemical reactions. The surface area may be determined from size-distribution data or measured directly by flow through a powder bed or the adsorption of gas molecules on the powder surface. Other methods such as gas diffusion, dye adsorption from solution, and heats of adsorption have also been used. The most commonly used methods are as follows:

In **mercury porosimetry**, the pores are filled with mercury under pressure (ISO 15901-1:2005, *Pore Size Distribution and Porosity of Solid Materials—Evaluation by Mercury Porosimetry and Gas Adsorption—Part 1: Mercury Porosimetry*). This method is suitable for many materials with pores in the diameter range of about 3 nm to 400 μm (especially within 0.1 to 100 μm).

In **gas adsorption** for micro-, meso- and macropores, the pores are characterized by adsorbing gas, such as nitrogen at liquid-nitrogen temperature. This method is used for pores in the ranges of approximately <2 nm (*micropores*), 2 to 50 nm (*mesopores*), and >50 nm (*macropores*) (ISO/FDIS 15901-2, *Pore Size Distribution and Porosity of Solid Materials—Evaluation by Mercury Porosimetry and Gas Adsorption, Part 2: Analysis of Meso-pores and Macro-pores by Gas Adsorption*; ISO/FDIS 15901-3, *Part 3: Analysis of Micro-pores by Gas Adsorption*). An isotherm is generated of the amount of gas adsorbed versus gas pressure, and the amount of gas required to form a monolayer is determined.

Many theories of gas adsorption have been advanced. For mesopores the measurements are usually interpreted by using the BET theory [Brunauer, Emmet, and Teller, *J. Am. Chem. Soc.*, **60**, 309 (1938)]. Here the amount of absorbed n_a is plotted against the relative pressure p/p_0 . The monolayer capacity n_m is calculated by the BET equation:

$$\frac{p/p_0}{n_a(1 - p/p_0)} = \frac{1}{n_m C} + \frac{C - 1}{n_m C} \cdot \frac{p}{p_0} \quad (21-24)$$

The specific surface per unit mass of the sample is then calculated by assessing a value a_m for the average area occupied by each molecule in the complete monolayer (say, $a_m = 0.162 \text{ nm}^2$ for N_2 at 77 K) and the Loschmidt number L :

$$a_s = n_m \cdot a_m \cdot L \quad (21-25)$$

PARTICLE-SIZE ANALYSIS IN THE PROCESS ENVIRONMENT

The growing trend toward automation in industry has resulted in the development of particle-sizing equipment suitable for continuous

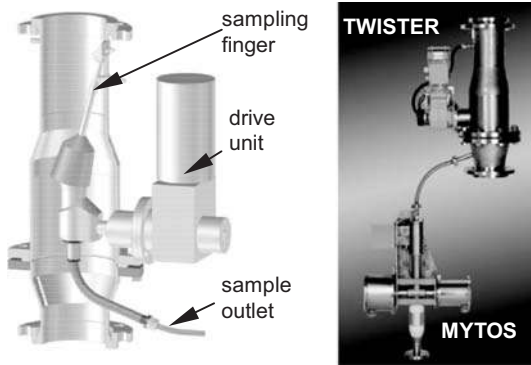


FIG. 21-21 A typical on-line application with a representative sampler (TWISTER) in a pipe of 150-mm, which scans the cross section on a spiral line, and dry disperser with particle-sizing instrument (MYTOS) based on laser diffraction. (Courtesy of Sympatec GmbH.)

work under process conditions—even in hazardous areas (Fig. 21-21). The acquisition of particle-size information in real time is a prerequisite for feedback control of the process.

Today the field of particle sizing in process environment is subdivided into three branches of applications.

At-line At-line is the fully automated analysis in a laboratory. The sample is still taken manually or by stand-alone devices. The sample is transported to the laboratory, e.g., by pneumatic delivery. Several hundred samples can be measured per day, allowing for precise quality control of slow processes. At-line laser diffraction is widely used for quality control in the cement industry. See Fig. 21-22.

On-line On-line places the measuring device in the process environment close to, but not in, the production line. The fully automated system includes the sampling, but the sample is transported to the measuring device. Mainly laser diffraction, ultrasonic extinction, and dynamic light scattering are used. See Fig. 21-23.

In-line In-line implements sampling, sample preparation, and measurement directly in the process, keeping the sample inside the production line. This is the preferred domain of laser diffraction

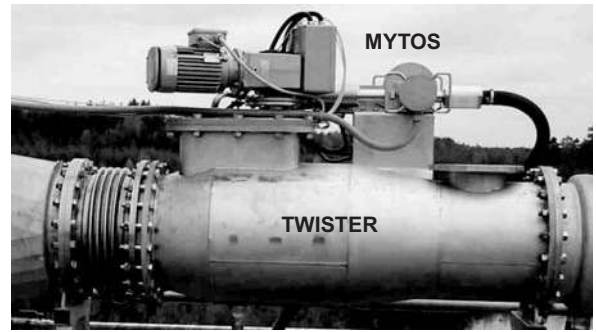


FIG. 21-23 Typical on-line outdoor application with a representative sampler TWISTER 440, which scans the cross section on a spiral line in a pipe of 440-mm, and a hookup dry disperser with laser diffraction particle sizer MYTOS. (Courtesy of Sympatec GmbH.)

(mainly dry), image analysis, focused-beam techniques, and ultrasonic extinction devices (wet). See Fig. 21-24.

VERIFICATION

The use of **reference materials** is recommended to verify the correct function of the particle-sizing equipment. A simple electrical, mechanical, or optical test is generally not sufficient, as all functions of the measuring process, such as dosing, transportation, and dispersion, are only tested with sample material applied to the instrument.

Reference Materials Many vendors supply certified standard reference materials which address either a single instrument or a group of instruments. As these materials are expensive, it is often advisable to perform only the primary tests with these materials and perform secondary tests with a stable and well-split material supplied by the user. For best relevance, the size range and distribution type of this material should be similar to those of the desired application. It is essential that the total operational procedure be adequately described in full detail (S. Röthele and W. Witt, Standards in Laser Diffraction, 5th European Symposium Particle Char., Nuremberg, March 24–26, 1992).

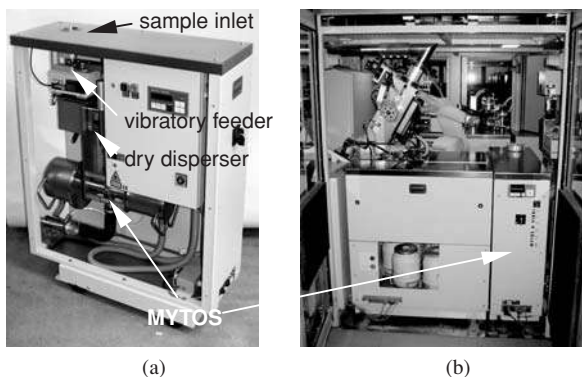


FIG. 21-22 (a) At-line particle sizing MYTOS module (courtesy of Sympatec GmbH) based on laser diffraction, with integrated dosing and dry dispersion stage. (b) Module integrated into a Polysius Polab® AMT for lab automation in the cement industry.

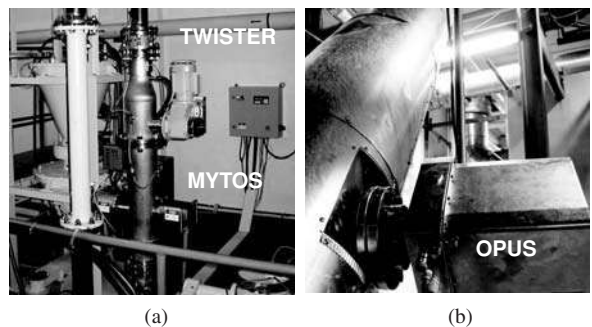


FIG. 21-24 (a) Typical in-line laser diffraction system with a representative sampler (TWISTER and MYTOS), all integrated in a pipe of 100-mm. (b) In-line application of an ultrasonic extinction (OPUS) probe monitoring a crystallization process in a large vessel. (Both by courtesy of Sympatec GmbH.)

SOLIDS HANDLING: BULK SOLIDS FLOW CHARACTERIZATION

GENERAL REFERENCES: Nedderman, *Statics and Kinematics of Granular Materials*, Cambridge University Press, 1992. Wood, *Soil Behavior and Critical State Soil Mechanics*, Cambridge University Press, 1990. J. F. Carr and D. M. Walker, *Powder Technology*, **1**, 369 (1967). Thompson, Storage of Particulate Solids, in *Handbook of Powder Science and Technology*, Fayed and Otten (eds.), Van Nostrand Reinhold, 1984. Brown and Richards, *Principles of Powder Mechanics*, Pergamon Press, 1970. Schofield and Wroth, *Critical State Soil Mechanics*, McGraw-Hill, 1968. M. J. Hvorslev, On the Physical Properties of Disturbed Cohesive Soils, *Ingeniørvidenskabelige Skrifter A*, no. 45, 1937. Janssen, *Zeits. D. Vereins Deutsch Ing.*, **39**(35), 1045 (1895). Jenike, Storage and Flow of Bulk Solids, *Bull.* 123, Utah Eng. Expt. Stn., 1964. O. Reynolds, On the Dilatancy of Media Composed of Rigid Particles in Contact: With Experimental Illustrations, *Phil. Mag.*, Series 5, **20**, 269 (1885). K. H., Roscoe, A. N. Schofield, and C. P. Wroth, On the Yielding of Soils, *Geotechnique*, **8**, 22 (1958). Dhodapkar et al., Fluid-Solid Transport in Ducts, in *Multiphase Flow Handbook*, Crowe (ed.), Taylor and Francis, 2006. Sanchez et al., *Powder Technology*, **138**, 93 (2003). Geldart, *Powder Technology*, **7**, 285 (1973). Kaye, *Powder Technology*, **1**, 11 (1967).

AN INTRODUCTION TO BULK POWDER BEHAVIOR

Bulk solids flow affects nearly all solids processing operations through material handling problems and mechanical behavior. Measurements of powder flow properties date back to Reynolds (loc. cit. 1885), Gibbs, Prandtl, Coulomb, and Mohr. However, the term **flowability** is rarely defined in an engineering sense. This often leads to a number of misleading analogies being made with fluid behavior. Unique features with regard to powder behavior are as follows:

1. Powders can withstand stress without flowing, in contrast to most liquids. The strength or **yield stress** of this powder is a function of previous compaction, and is not unique, but depends on stress application. Powders fail only under applied shear stress, and not isotropic load, although they do compress. For a given applied horizontal load, failure can occur by either raising or lowering the normal stress, and two possible values of failure shear stress are obtained (**active** versus **passive failure**).

2. When failure does occur, the flow is **frictional** in nature and often is a weak function of strain rate, depending instead on **shear strain**. Prior to failure, the powder behaves as an elastic solid. In this sense, bulk powders do *not* have a viscosity in the bulk state.

3. Powders do not readily transmit stress. In the case of columns, normal stress or weight of the bulk solid is held by wall friction. In addition, normal stress is not isotropic, with radial stress being only a fraction of normal stress. In fact, the end result is that stress in silos scales with diameter rather than bed height, a most obvious manifestation of this being the narrow aspect ratio of a corn silo.

4. A powder will not necessarily maintain a shear stress-imposed strain rate gradient in the fluid sense. Due to force instabilities, it will search for a characteristic slip plane, with one mass of powder flowing against the next, an example being rat-hole discharge from a silo.

5. Bulk solids are also capable of two-phase flow, with large gas interactions in silo mass discharge, fluidization, pneumatic conveying, and rapid compression and mixing. Under fluidized conditions, the bulk solid may now obtain traditional fluid behavior, e.g., pressure scaling with bed height. But there are other cases where fluidlike rheology is misinterpreted, and is actually due to time-dependent compression of interstitial fluid. After characteristic time scales related to permeability, stresses are transmitted to the solid skeleton. It may not be of utility to combine the rheology of the solid and interstitial fluids, but rather to treat them as separate, as is often done in soil mechanics.

PERMEABILITY AND AERATION PROPERTIES

Permeability and Deaeration Various states of fluidization and pneumatic conveying exist for bulk solid. Fluidization and aeration behavior may be characterized by a fluidization test rig, as illustrated in Fig. 21-25. A loosely poured powder is supported by a porous or perforated distributor plate. The quality and uniformity of this plate are critical to the design. Various methods of filling have been explored to include vibration and vacuum filling of related **permeameters**

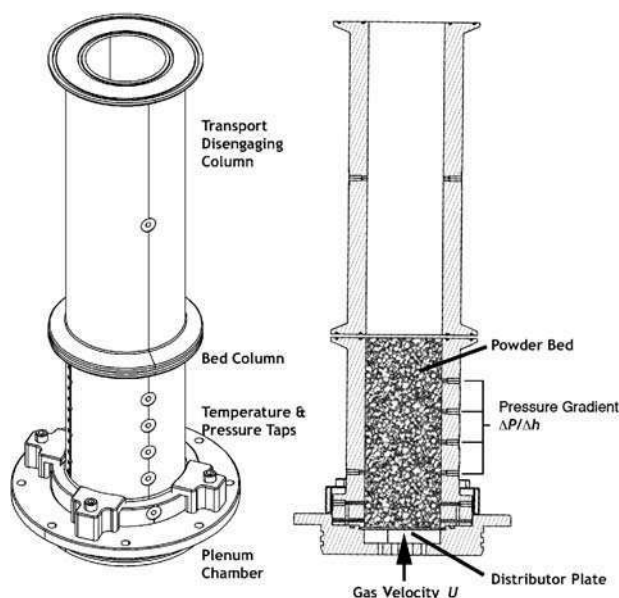


FIG. 21-25 iFluid™ fluidization permeameter, illustrating powder bed supported by distributor plate fluidized at a gas velocity U , with associated pressure taps for multiple pressure gradient measurements dP/dh . (Courtesy iPowder Systems, E&G Associates, Inc.)

[Kaye, *Powder Technology*, **1**, 11 (1967); Juhasz, *Powder Technology*, **42**, 123 (1985)]. Two key types of measurements may be performed. In the first, air or gas is introduced through the distributor, and the pressure drop across the bed is measured as a function of flow rate or superficial gas velocity (Fig. 21-26). In the second, the gas flow is stopped to an aerated bed, and the pressure drop or bed height is measured as a function of time, as the bed collapses and deaerates (Fig. 21-27).

For the first **fluidization measurement**, pressure drop will increase with gas velocity while powder remains in a **fixed-bed** state until it reaches a maximum plateau, after which the pressure drop equals the weight of the bed, provided the bed becomes uniformly fluidized. Bed expansion will also occur. The point of transition is referred to as the **minimum fluidization velocity** U_{mf} . Various states of a **fluidized bed** occur. For fine materials of limited cohesion, the bed will initially undergo **homogeneous fluidization** (also referred to as particulate fluidization), where bed expansion occurs without the formation of bubbles, and with further increases in gas velocity, it will transition to a bubbling bed, or **heterogeneous fluidization** (also referred to as bubbling or aggregative fluidization). Coarse materials do not expect the initial state of homogeneous fluidization, and $U_{mb} = U_{mf}$. The point at which bubbles form in the bed is referred to as the **minimum bubbling velocity** U_{mb} . The various stages of fluidization are described in detail in Sec. 17. In addition, for fine, cohesive powders, channeling may occur instead of uniform fluidization, resulting in lower, more erratic pressure drops. Various states of fluidization are indicated in Fig. 21-26. Lastly, mixing, bed expansion, heat and mass transport, and forces acting in fluidized beds scale with **excess gas velocity**, or $U - U_{mf}$.

Prior to reaching minimum bubbling, a homogeneous fluidized powder will undergo a peak in pressure prior to settling down to its plateau. This peak represents a measure of **aerated cohesion**, and it ranges from 10 percent for fine, low-cohesion powders capable of homogeneous fluidization, to 50 percent for fine, extremely cohesive material, which generally undergoes channeling when fluidized.

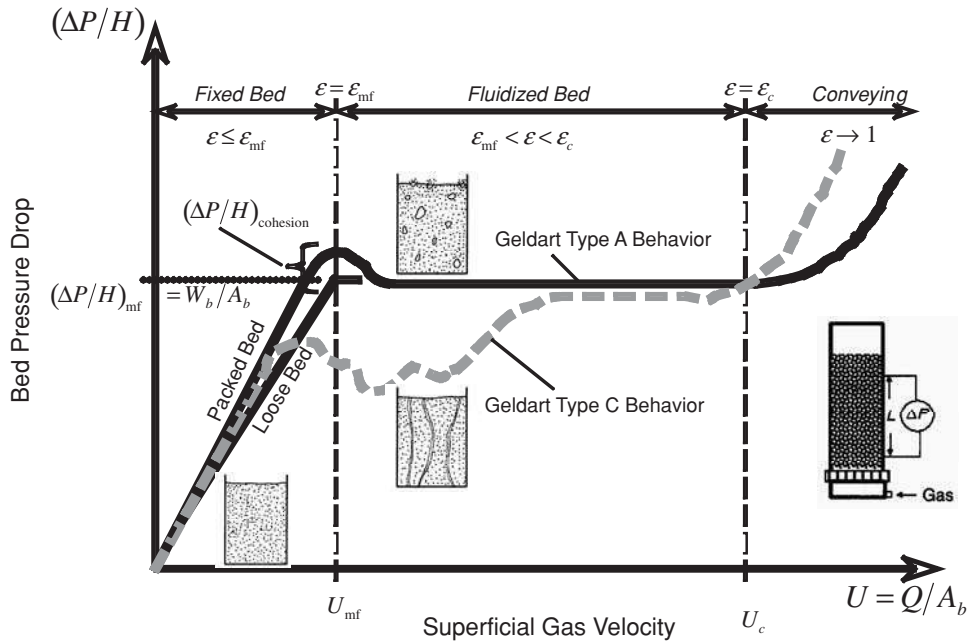


FIG. 21-26 Fluidization measurement of permeability and fluidization behavior. Bed pressure drop $\Delta P/H$ for fixed and fluidized beds as a function of gas velocity U . (After Rumpf, Particle Technology, Kluwer Academic, 1990.)

The pressure drop across the initial fixed bed (or final previously aerated bed) is a measure of **permeability** k_p as defined by Darcy (1856), given by

$$U = \frac{Q}{A_b} = k_p \frac{\Delta P/H_b}{\mu_g} \quad (21-26)$$

otherwise known as **Darcy's law**, which is strictly only valid for low Reynolds number. Comparing to the **Kozeny-Carman relation** [Kozeny (1927); Carman (1937)], permeability may be predicted from particle size (surface-volume average) and packing voidage:

$$k_{p0} = \frac{d_p^2 \epsilon^3}{C_{p1}(1 - \epsilon)^2} \quad (21-27)$$

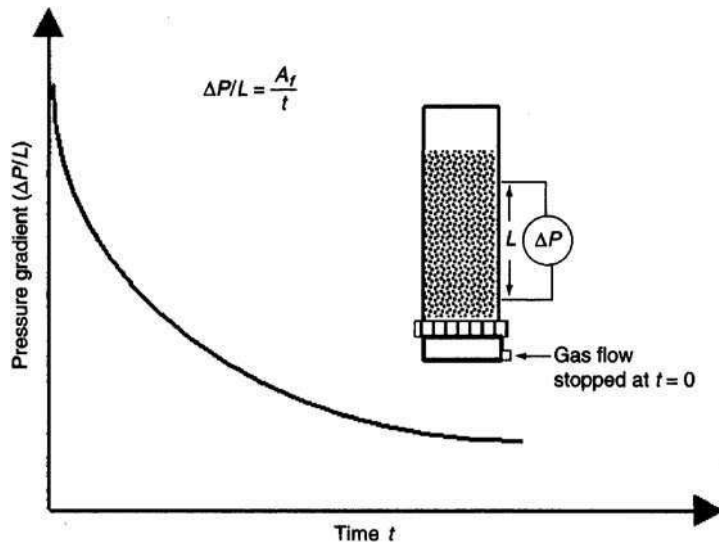


FIG. 21-27 Deaeration measurement of deaeration time and constant. Bed pressure drop ($\Delta P/H$) decay following fluidization as a function of time. [Dhodapkar et al., Fluid-Solid Transport in Ducts, in Multiphase Flow Handbook, Crowe (ed.), Taylor & Francis, 2006, with permission.]

which is valid for low Reynolds number and loose packing. C_{p1} equals 180 from the Kozeny-Carman relation and 150 from the Ergun relation. For a wider range of gas velocities, **Ergun's relation** should be utilized instead, where the pressure drop is given by

$$\frac{dP}{dh} = \frac{\mu_g U}{k_p} (1 + 1.75 \text{Re}_p) \quad \text{where} \quad \text{Re}_p = \frac{\rho_g U d_p}{\mu_g (1 - \epsilon)} \quad (21-28)$$

which can be rewritten to give

$$\frac{1}{p_f} = \frac{\mu_g}{k_p} = \frac{dP/dh}{U} = E_1 + E_2 U \quad (21-29)$$

where E_1 and E_2 may be determined from plotting the slope in the fixed-bed region divided by velocity [or $(dP/dh)/U$] versus gas velocity. Theoretically, these constants are given by

$$E_1 = \frac{\mu_g}{k_{p0}} = \frac{C_{p1} \mu_g (1 - \epsilon)^2}{d_p^3 \epsilon^3} \quad \text{and} \quad E_2 = \frac{C_{p2} \rho_g (1 - \epsilon)}{d_p \epsilon^3} \quad (21-30)$$

where $C_{p1} = 150$ and $C_{p2} = 1.75$ based on Ergun's relation. The standard value of permeability is then related to the intercept E_1 , but a velocity dependence can be determined as well for high velocity related to conveying. And p_f is another common definition of permeability, or **permeability factor**, which incorporates gas viscosity.

As with bulk density, permeability is a function of packing voidage and its uniformity, and in practice, it is best measured. It can vary substantially with previous compaction of the sample. An example is the change in bulk density—and therefore interstitial voidage—that occurs with a material as it moves through a hopper. By applying a load to the upper surface of the bed, permeability may be also determined as a function of solids consolidation pressure (see "Bulk Flow Properties"). Permeability is a decreasing function of applied solids pressure, and bulk density is often written in log form, or

$$k_p = k_{p0} \left(\frac{\rho_{bo}}{\rho_b} \right)^m \quad (21-31)$$

From the second **deaeration measurement**, pressure drop is measured as a decaying function of time, given by one of the forms (Fig. 21-27)

$$\frac{\Delta P}{dh} = ae^{-t/t_d} \quad \text{or} \quad \frac{\Delta P}{dh} = \frac{A_d}{t} \quad (21-32)$$

where t_d and A_d are a characteristic **deaeration time** and **deaeration factor**, respectively. Large deaeration time or factor implies that the powder retains air for long times. Also an additional deaeration factor has been defined to account for particle density, or

$$X_d = \frac{A_d / \rho_s}{(\Delta P/H)_{mf}} \quad (21-33)$$

Permeability and deaeration control both fluidization and pneumatic conveying. In addition, they impact the gas volume and pressure requirement for **air-augmented flow** in hoppers and feeders. Materials of low permeability have lower mass discharge rates from hopper openings (see "Mass Discharge Rates") and limit the rate of production in roll pressing, extrusion, and tableting, requiring vacuum to speed deaeration ("Compaction Processes"). Lastly permeability impacts wetting phenomena and the rate of drop uptake in granulation ("Wetting and Nucleation").

Classifications of Fluidization Behavior Geldart [Powder Technology, 7, 285 (1973)] and later Dixon [Pneumatic Conveying, Plastics Conveying and Bulk Storage, Butters (ed.), Applied Science Publishers, 1981] developed a classification of fluidization/aeration behavior from studies of fluidized beds and slugging in vertical tubes,

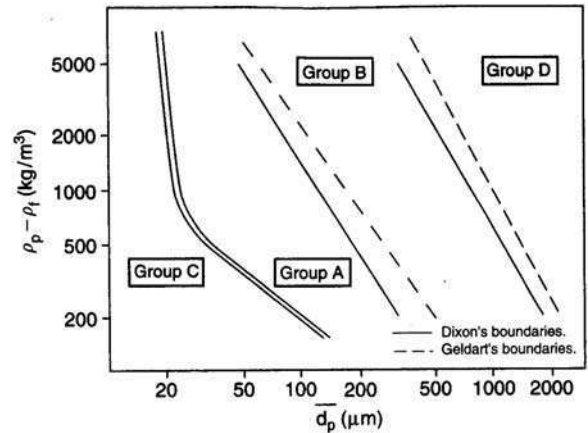


FIG. 21-28 Geldart's classification of aeration behavior with Dixon and Geldart boundaries. (From Mason, Ph.D. thesis, Thames Polytechnic, London, 1991, with permission.)

respectively (Fig. 21-28). The classification is based on particle size (surface-volume average for wide size distributions) and relative particle density. Particle size controls interparticle cohesive forces, whereas density controls the driving force to be overcome by drag. A summary of aeration behavior is provided in Table 21-4, where from **Geldart's classification** powders are broken down into group A (aeratable) for fine materials of low cohesion, which can exhibit homogeneous fluidization; group B (bubbling) for coarser material, which immediately bubbles upon fluidization; group C (cohesive), which typically channels and retains air for long periods; and group D (spoutable), which is coarse material of high permeability with no air retention capability.

Classifications of Conveying Behavior Aeration behavior also impacts mode and ease of pneumatic conveying [Dhodapkar et al., Fluid-Solid Transport in Ducts, in Multiphase Flow Handbook, Crowe (ed.), Taylor & Francis, 2006]. Figure 21-29 illustrates the impact of decreasing conveying velocity on flow pattern. At high gas flow, ideal dilute, homogeneous solids flow may occur (1). As gas velocity is reduced past some characteristic velocity, the solids can no longer be uniformly suspended and increasing amounts of solid will form on the bottom of the pipe, forming a moving stand of solids (2,3). With further decrease of gas velocity and deposited solids, moving dunes (4,5) and later slugs (6,7,8) will form which completely fill the pipe. Finally, ripple flow (9) and pipe pluggage (10) will occur. **Dilute-phase conveying** encompassed patterns 1 to 3, where **dense-phase conveying** includes the remainder of 4 to 10. Dhodapkar et al. (loc. cit.) further classified conveying patterns according to particle size. Fine materials (plastic powder, fly ash, cement, fine coal, carbon fines) may be transported in all patterns, with a smooth, predictable transition between regimes. At intermediate gas velocities, two-phase strand flow (2,3) is observed followed by dune flow at lower velocities (4–8), where the solid flow can appear as turbulent or fast-moving bed, wave, or fluidized-bed modes. Conveying might also be achieved in patterns 9 and 10 for materials that readily aerate and retain air, in which case they are conveyed as a fluidized plug. Coarse materials (pellets, grains, beans, large granules), however, form slugs when conveyed at low velocities, which form on a regular, periodic basis. The transition from dilute- to dense-phase conveying for coarse material is unstable and occurs under dune flow. Some coarse materials with substantial fines exhibit fine conveying modes.

Figure 21-30 provides classifications of conveying ability, where permeability and deaeration factor are plotted against pressure drop at minimum fluidization for a variety of materials [Mainwaring and Reed, Bulk Solids Handling, 7, 415 (1987)]. Lines of constant minimum fluidization $U_{mf} = 0.05$ m/s and deaeration factor $X_d = 0.001$ m³ · s/kg are shown. From Fig. 21-30a, materials which lie above the line of high permeability can be conveyed in plug or slug form as they do

TABLE 21-4 Characteristics of Geldart (1973) or Dixon (1981) Classification

Properties	Group A	Group B	Group C	Group D
Material	Fine/medium powder Fly ash, pulverized coal, plastic powders, alumina, granular sugar, pharma excipients	Course powder Sand, salt, granules, mineral powders, glass beads	Cohesive fine powder Cement, corn starch, titanium dioxide, carbon-black powder, many pharma actives	Granular Plastic pellets, wheat, large glass beads, tablets, coarse sand, seeds
Fluidization characteristics	Good air retention, small bubble size, considerable bed expansion	Poor air retention, low bed expansion, large bubble, asymmetric slugging at higher velocity or small beds	Cohesive and difficult to fluidize, tends to channel, retains gas for extended period once aerated, adhesion to walls and surfaces	Highly permeable, negligible expansion and no air retention, large bubbles, spouts, or axisymmetric slugs can form
Conveying characteristics	Can be conveyed in fluidized- or moving-bed mode, easy to convey, does not form slugs naturally	Unlikely to convey in conventional dense phase, unsteady and unpredictable plug formation, large pipe vibrations	Difficult but possible to convey in dense phase, forms impermeable plugs that break up, requires special conveying	Natural slugging ability and high permeability aid in slug or plug flow conveying; operationally easiest to dense phase convey
Pressure drop at U_{mf} : $(\Delta P/H)_{mf}$ [mbar/m]	<50	>80	50–130	5–150
Permeability factor $(k_p/\mu_g) = [m^2/(bar \cdot s)]$	0.1	0.01–0.1 to 1	0.1 to 1	>1
Deaeration	Collapses slowly, air retention	Collapses rapidly	Collapses slowly, long air retention	Collapses very rapidly

Adapted from Dhodapkar et al., Fluid-Solid Transport in Ducts, in *Multiphase Flow Handbook*, Crowe (ed.), Taylor & Francis, 2006; and Sanchez et al., *Powder Technology*, **138**, 93 (2003).

not readily retain air, whereas those below the line of low permeability can be conveyed by moving-bed flow, as they more readily retain air, or by dilute-phase flow. Similarly from Fig. 21-30b, materials which lie *below* the line with small deaeration constant (or time) can be conveyed in plug or slug form, whereas those above the line with large deaeration constant (or time) can be conveyed by moving-bed flow or dilute-phase flow, as they retain air. Jones and Miller [*Powder Handling and Processing*, **2**, 117 (1990)] combined deaeration behavior and permeability in a single classification, as shown in Fig. 21-31. Group 1 includes Geldart type A powders of low permeability and

large deaeration time which conveyed as moving-bed flow, whereas at the other extreme, group 3 includes Geldart type D materials with high permeability and short deaeration time conveyed as plug-type flow. Dilute- and dense-phase conveying is possible for group 2 or typically type B powder with (1) intermediate permeability and deaeration time, (2) small deaeration time and permeability, or (3) large deaeration time and permeability. Type C material exhibits all three forms of conveying. Sanchez et al. [*Powder Technology*, **138**, 93 (2003)] and Dhodapkar et al. (loc. cit.) provide current summaries of these classifications.

BULK FLOW PROPERTIES

Shear Cell Measurements The yield or flow behavior of bulk solids may be measured by **shear cells**. Figure 21-32 illustrates these principles for the case of a direct rotary split cell. For **flow measurements**, powder is contained within two sets of rings. **Normal stress** is applied to the powder bed through a horizontal roughened or patterned lid. The upper ring containing approximately one-half of the powder is sheared with respect to the lower ring, forming a **shear plane** or lens between the two halves of powder. This is accomplished by rotating the lower half of the powder mounted to a motorized base, which in turn attempts to rotate the upper half of powder through rotational shear stress transmitted through the shear plane. The upper half of powder is instead held fixed by the upper lid, which transmits this shear stress through an air bearing to a force transducer. Through this geometry, the **shear stress** between the two halves of powder, measured as a torque by the force transducer, is measured versus time or displacement as a function of applied normal stress. In addition, any corresponding changes in powder density are measured by changes in vertical displacement for a linear voltage displacement transducer.

For **wall friction measurements**, a wall coupon is inserted between the rings, and powder in the upper ring alone is sheared against a coupon of interest. Wall friction and adhesion, both static and dynamic, may be assessed against different materials of construction or surface finish.

Shear cell testing of powders has its basis in the more comprehensive field of **soil mechanics** (Schofield and Wroth, *Critical State Soil Mechanics*, McGraw-Hill, 1968), which may be further considered a subset of **solid mechanics** (Nadia, *Theory of Flow and Fracture of Solids*, vols. 1 and 2, McGraw-Hill, 1950). The most comprehensive testing of the shear and flow properties of soils is accomplished in

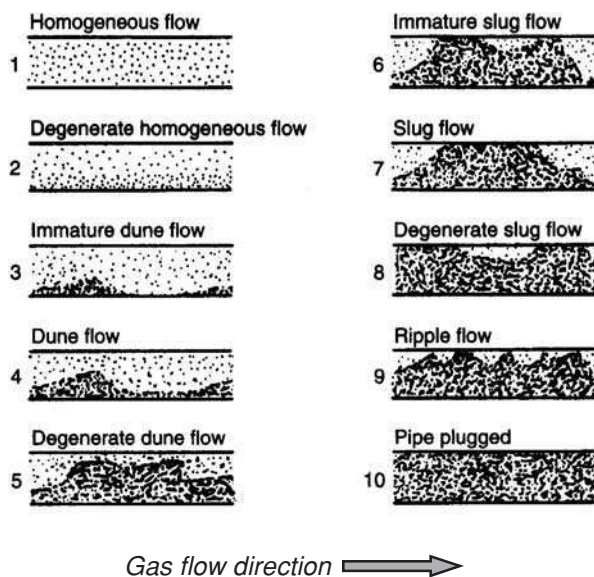


FIG. 21-29 Pattern of solids flow in pneumatic conveying. [From Wen, U.S. Dept. of Interior, Bureau of Mines, PA, IC 8314 (1959) with permission.]

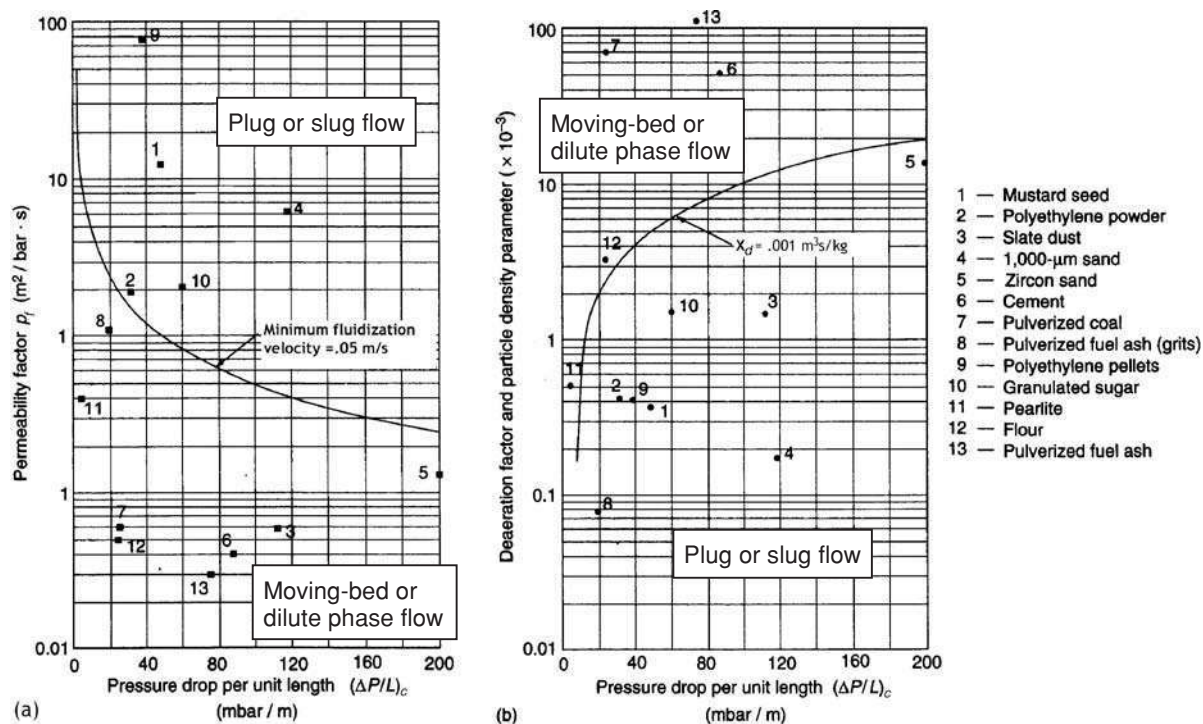


FIG. 21-30 Classification of pneumatic conveying based on (a) permeability factor and (b) deaeration factor. [From Mainwaring and Reed, Bulk Solids Handling, 7, 415 (1987) with permission.]

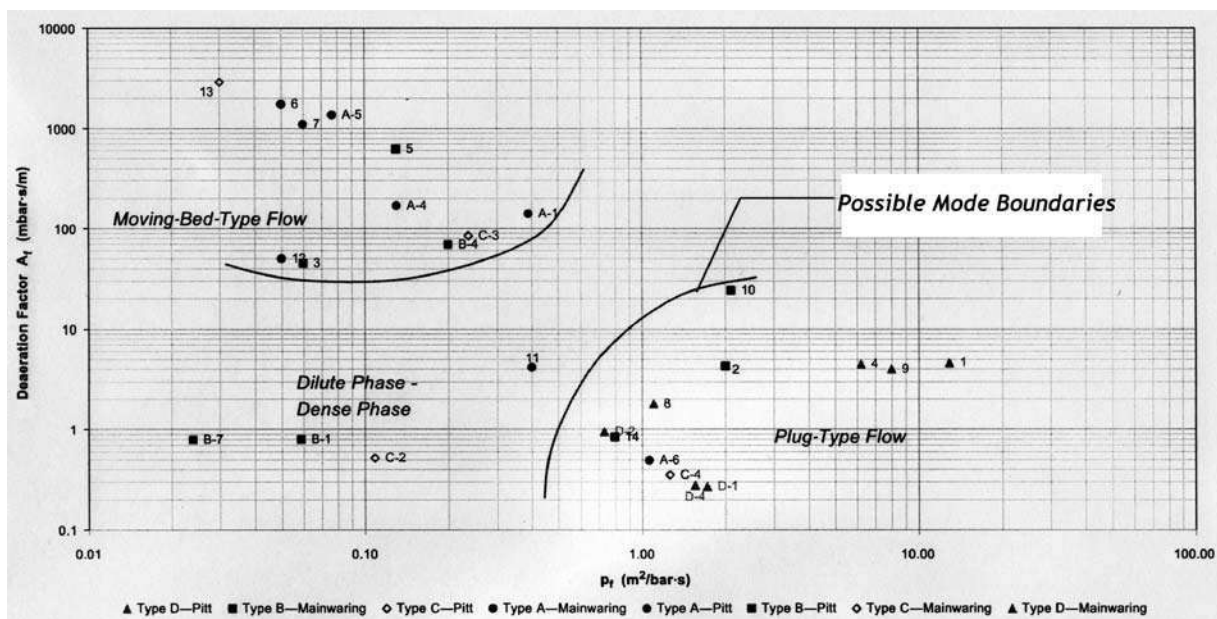


FIG. 21-31 Classification of pneumatic conveying based on combined permeability and deaeration factors, based on Jones and Miller. [Sanchez *et al.*, Powder Technology, 138, 93 (2003), with permission.]

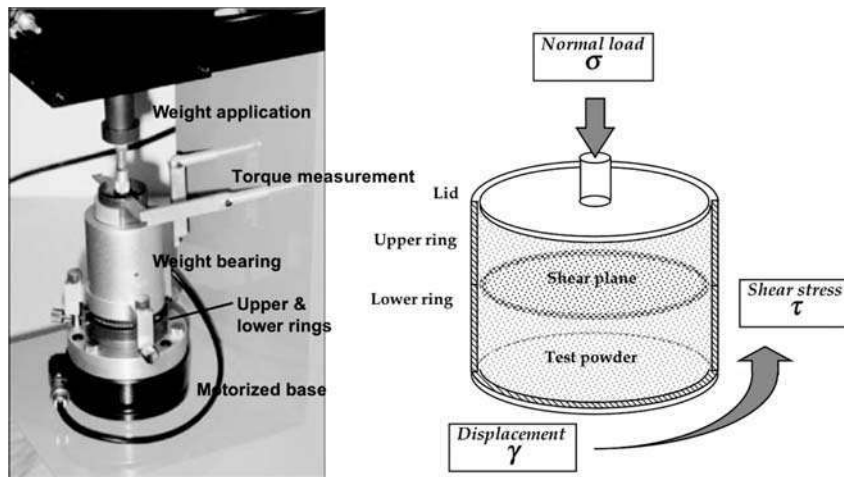


FIG. 21-32 iShear™ rotary, full annulus split cell, illustrating normal load weight application, rotational base, and shear stress/torque measurement. Vertical displacement of lid is monitored by displacement transducer. (Courtesy E&G Associates, Inc.)

triaxial shear cells (Fig. 21-33). There are two such types of triaxial shear cells. In the traditional cylindrical triaxial cell, the axial and radial pressures acting on the sample contained within a rubber membrane are directly controlled through applied axial force and radial hydraulic oil pressure. Deviatoric stress, i.e., shear stress due to difference in axial and radial pressure, is applied to the sample until failure. In a true triaxial cell, all three principal stresses may be varied; whereas only the major and minor principal stresses are controlled in the traditional cylindrical triaxial cell. Lastly, shear displacements are measured through a variety of strain gauges, and both the drained and undrained tests are possible. Such tests refer to simultaneous measurement of pressure of any interstitial fluid or gas. Interstitial fluid can have pronounced effects on mitigating powder friction and changing flow properties. While triaxial cells are not typically employed for powder characterization in industrial processing, they do provide the most comprehensive information as well as a knowledge base of application in such results for bulk solids flow, including detailed simulations of multiphase flow of such systems. Their disadvantage is their difficulty of use and time required to perform measurements. Future advances in employing these designs are likely.

Direct shear cells were introduced due to drastically reduced testing times, although the exact nature of stresses in the failure zone is not as precisely defined as with triaxial cells. Direct cells have undergone substantial automation in the last two decades. All have as a common feature that only the applied axial force or axial stress is controlled (Fig. 21-33). The shear stress required to accomplish failure is measured as a function of the applied axial stress, where translational or rotational motion is employed. Both **cup** and **split cell** designs are available. Rotational cells include both full **annulus** and **ring cells**. For a properly designed direct shear cell, failure occurs within a specific region, in which both the plane of failure and the acting stresses may be clearly defined. In addition, direct shear cells may be validated against an independent vendor standard, or the BCR116 limestone powder (see "Shear Cell Standards and Validation"). Rotary split cell designs have two possible advantages: (1) Unlimited displacement of the sample is possible, allowing ease of sample conditioning and repeated sample shear on a single sample. (2) The shear plane is induced in a defined region between the two cell halves, allowing unconfined expansion in the shear plane (Fig. 21-32).

Yield Behavior of Powders The yield behavior of a powder depends on the existing **state of consolidation** within the powder bed when it is caused to flow or yield under a given state of stress, defined by the acting normal and shear stresses. The consolidation state controls the current bed voidage or **porosity**. Figure 21-34 illustrates a times series of shears occurring for the BCR116 limestone standard for a rotary shear cell. For each shear step, torque is applied

to the sample by cell rotation until sample failure; the cell is then reversed until the shear force acting on the sample is removed. Two stages of a typical experiment may be noted. The first is a **consolidation stage** wherein repeated shears take place on the sample until the shear stress τ reaches a steady state, defined by either the maximum value or the steady value occurring after an initial peak. This occurs with a constant normal consolidation stress $\sigma = \sigma_c$ acting on the sample. During this step, the sample reaches a characteristic or **critical density** or **critical porosity** ϵ_c related to the consolidation normal stress. A set of shear steps is then performed during a **shear stage** with progressively smaller normal loads. In all cases, each shear step is preceded by a shear at the original consolidation normal stress.

Three characteristic displacement profiles may be observed during shear for shear stress and density (Fig. 21-35), which are unique to the state of consolidation:

1. **Critically consolidated.** If a powder is sheared sufficiently, it will obtain a constant density or critical porosity ϵ_c for this consolidation normal stress σ_c . This is defined as the critical state of the powder, discussed below. If a powder in such a state is sheared, initially the material will deform elastically, with shear forces increasing linearly with displacement or strain. Beyond a certain shear stress, the material will fail or flow, after which the shear stress will remain approximately constant as the bulk powder deforms plastically. Depending on the type of material, a small peak may be displayed originating from differences between static and dynamic density. Little change in density is observed during shear, as the powder has already reached the desired density for the given applied normal consolidation stress σ_c .

2. **Overconsolidated.** If the same sample is sheared, but at a lower normal stress of $\sigma < \sigma_c$, the shear stress will increase elastically to a peak and then fail, with this peak being less than that observed for the critically consolidated state, as the applied normal stress is lower. After the failure peak, the shear stress will decrease as the powder expands due to **dilation** and density decreases, eventually leveling off to a lower shear stress and lower density. Overconsolidated shears are observed during the shear stage of a shear cell experiment.

3. **Underconsolidated.** If the same sample is sheared, but at a higher normal stress of $\sigma > \sigma_c$, the shear stress will progressively increase to some value, while the material simultaneously densifies. Such underconsolidated responses are observed in the consolidation stage of an experiment.

In practice, following the filling of a cell, the powder is in an underconsolidated state. A set of shear steps is performed under a chosen consolidation stress in the consolidation stage to increase its density and bring it into a critical state. A set of shears is then performed at small normal stresses in the shear stage to determine the strength of

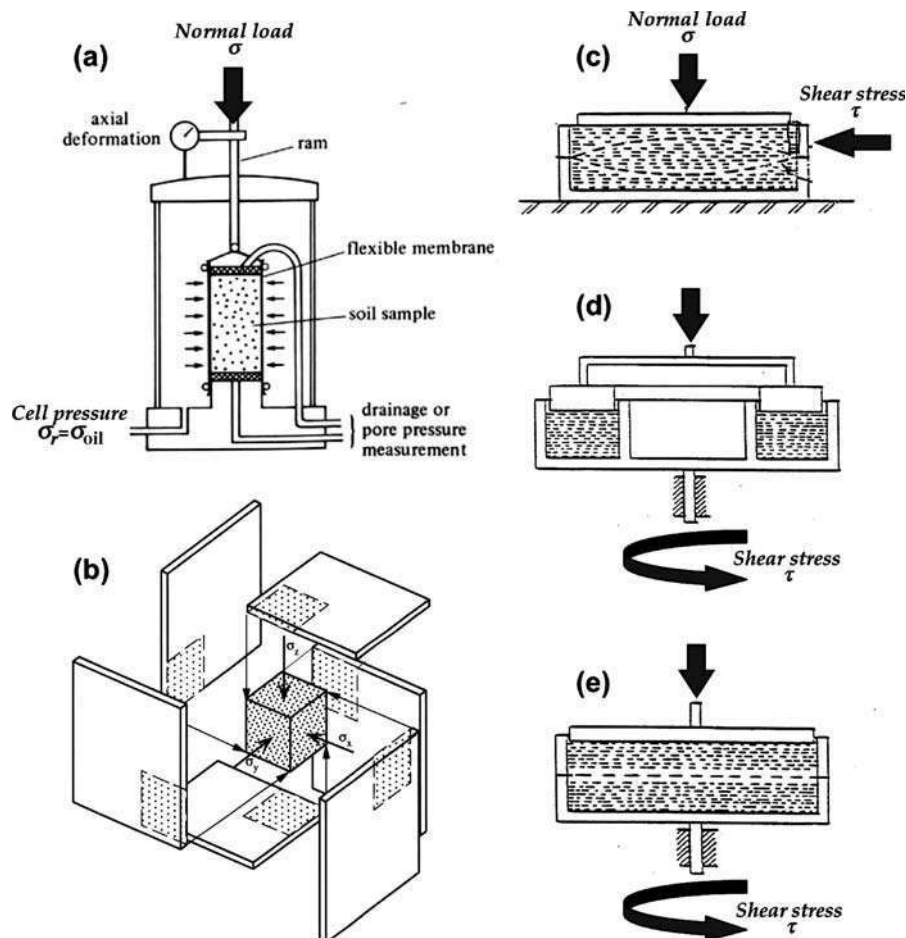


FIG. 21-33 Examples of powder shear cells. **Triaxial cells:** (a) Traditional triaxial cell; (b) true triaxial. **Direct shear cells:** (c) Translational split, Collin (1846), Jenike™ (1964); (d) rotational annulus, Carr and Walker (1967), Schulze™ (2000); (e) rotational split, Peschl and Colijn (1976), iShear™ (2003). (From *Measuring Powder Flowability and Its Applications*, E&G Associates, 2006, with permission.)

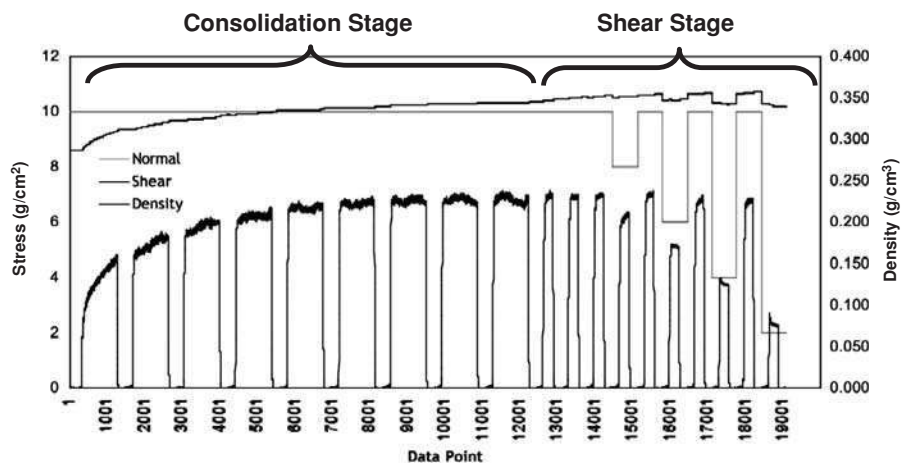


FIG. 21-34 Time-series shearing profile for the BCR116 limestone validation powder, in an iShear rotary split cell. (From *Measuring Powder Flowability and Its Applications*, E&G Associates, 2006, with permission.)

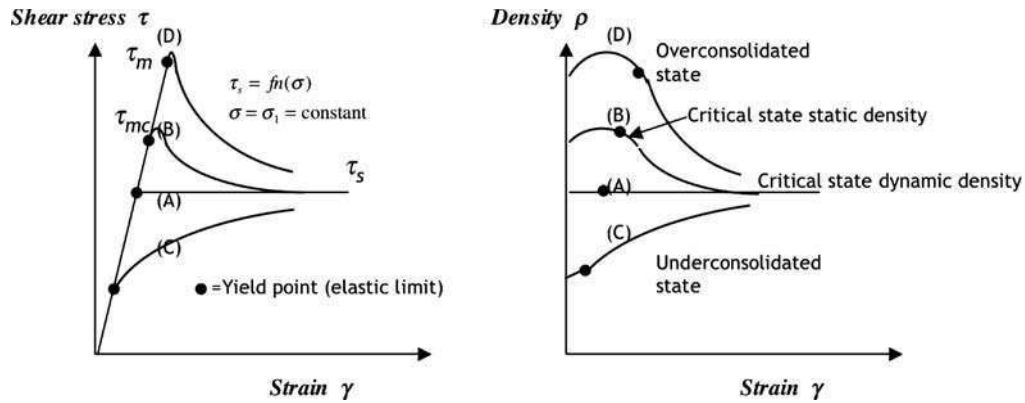


FIG. 21-35 Examples of yield behavior. (From Measuring Powder Flowability and Its Applications, E&G Associates, 2006, with permission.)

the powder as a function of normal load, in the overconsolidated or overcompacted state, each time reconsolidating the powder before performing the next shear step.

Powder Yield Loci For a given shear step, as the applied shear stress is increased, the powder will reach a maximum sustainable shear stress τ , at which point it yields or flows. The functional relationship between this limit of shear stress τ and applied normal load σ is referred to as a **yield locus**, i.e., a locus of yield stresses that may result in powder failure beyond its elastic limit. This functional relationship can be quite complex for powders, as illustrated in both principal stress space and shear versus normal stress in Fig. 21-36. See Nadia (loc. cit.), Stanley-Wood (loc. cit.), and Nedderman (loc. cit.) for details. Only the most basic features for **isotropic hardening** of the yield surface are mentioned here.

1. There exists a **critical state line**, also referred to as the **effective yield locus**. The effective yield locus represents the relationship between shear stress and applied normal stress for powders always in a critically consolidated state. That is, the powder is not over- or undercompacted but rather has obtained a steady-state density. This density increases along the line with increases in normal stress, and bed porosity decreases.

2. A given yield locus generally has an envelope shape; the initial density for all points forming this locus *prior* to shear is constant. That is, the locus represents a set of points all beginning at the porosity; this critical state porosity is determined by the intersection with the effective yield locus.

3. Points to the left of the effective yield locus are in a state of overconsolidation, and they dilate upon shear. If sheared long enough, the density and shear stress will continue to drop until reaching the effective yield locus. Points to the right are underconsolidated and compact with shear.

4. For negative normal stresses, a state of tension exists in the sample along the yield locus. This area is generally not measured by direct shear cells, but can be measured by triaxial shear and tensile split cells.

5. Multiple yield loci exist. As a powder is progressively compacted along the effective yield locus, it gains strength as density rises, reaching progressively higher yield loci. Yield loci of progressively larger envelope size have higher critical density and lower critical voidage, as shown in Fig. 21-36. Therefore, the shear strength of a powder τ is a function of the current normal stress σ , as well as its consolidation history or stress σ_c , which determined the starting density prior to shear.

Currently in industrial practice, we are most concerned with the overcompacted state of the powder, and applications of the undercompacted and tensile data are less common, although they are finding applications in compaction processes of size enlargement (see "Powder Compaction").

Although the yield locus in the overcompacted state may possess significant curvature, especially for fine materials, a common **Mohr-Coulomb** linear approximation to the yield locus as shown in Fig. 21-37 is given by

$$\tau = c + \mu\sigma = c + \sigma \tan \phi \quad (21-34)$$

Here, μ is the **coefficient of internal friction**, ϕ is the **internal angle of friction**, and c is the shear strength of the powder in the absence of any applied normal load. Overcompacted powders dilate when sheared, and the ability of powders to change volume with shear results in the powder's shear strength τ being a strong function of previous compaction. There are therefore a series of yield loci (YL), as illustrated in Fig. 21-37, for increasing previous consolidation stress. The individual yield loci terminate at a critical state line or effective yield locus (EYL) defined early, which typically passes through the stress-strain origin, or

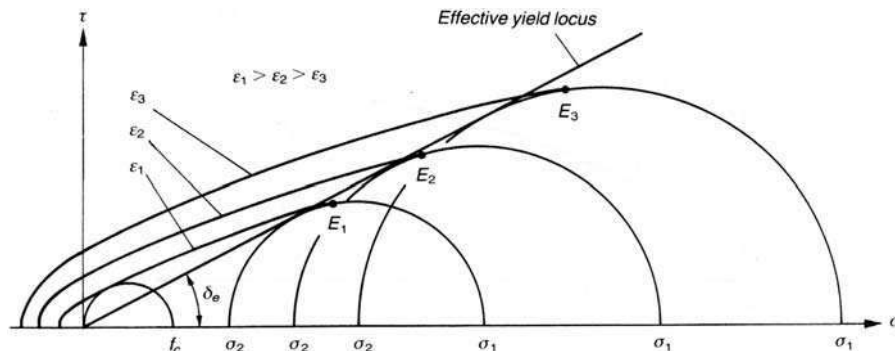


FIG. 21-36 Family of yield loci for a typical powder. (Rumpf, loc. cit., with permission.)

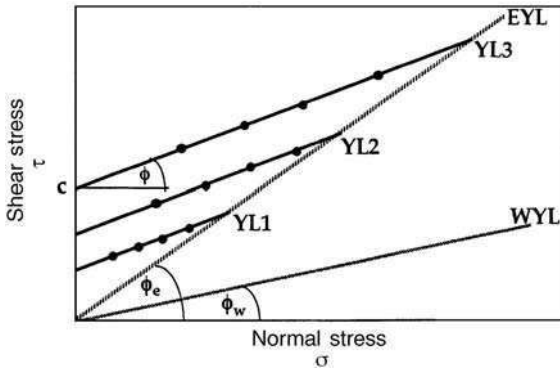


FIG. 21-37 The yield loci of a powder, reflecting the increased shear stress required for flow as a function of applied normal load. YL1 through YL3 represent yield loci for increasing previous compaction stress. EYL and WYL are the effective and wall yield loci, respectively.

$$\tau = \mu_e \sigma = \sigma \tan \phi_e \quad (21-35)$$

where μ_e is the **effective coefficient of powder friction** and ϕ_e is the **effective angle of powder friction** of the powder. In practice, there may be a small cohesion offset in the effective yield locus, in which case the effective angle is determined from a line intercepting an origin and touching the effective yield locus. In this case, the effective angle of friction is an asymptotic function of normal stress.

When sheared powders also experience friction along a wall, this relationship is described by the **wall yield locus**, or

$$\tau = \mu_w \sigma = \sigma \tan \phi_w \quad (21-36)$$

where μ_w is the **effective coefficient of wall friction** and ϕ_w is the **effective angle of wall friction**, respectively. In practice, there is a small **wall adhesion** offset, making the effective angle of wall friction an asymptotic function of normal stress, as with effective powder friction.

Lastly, both static (incipient powder failure) and dynamic (continued-flow) yield loci may be measured, giving both static and dynamic values of wall and powder friction angles as well as wall adhesion.

Flow Functions and Flowability Indices Consider a powder compacted in a mold at a **compaction pressure** σ_1 . When it is removed from the mold, we may measure the powder's strength, or **unconfined uniaxial compressive yield stress** f_c (Fig. 21-38). The unconfined yield and compaction stresses are determined directly from Mohr circle constructions to yield loci measurements (Fig. 21-36). This strength increases with increasing previous compaction, with this relationship referred to as the powder's **flow function** FF.

The flow function is the paramount characterization of powder strength and **powder flowability**. Common examples are illustrated in Fig. 21-38. Typically the flow function curves toward the normal stress axis with increasing load (A). An upward shift in the flow function indicates an overall gain of strength (B). If one were comparing the flowability of two lots of material, this would indicate a decrease in flowability. In other words, greater stresses would be required in processing for lot B than for lot A (e.g., hoppers, feeders, mixers) to overcome the strength of the powder and to induce flow of the mass. An upward shift also occurs with **time consolidation**, where a specified time of consolidation is allowed prior to each shear step of the yield locus. The resulting flow function is a **time flow function**, and it indicates the effect of prolonged storage on flow. Flow functions often cross (C vs. A), indicating lot C is more flowable at low pressure than lot A, but less flowable at high pressure. An upward curvature of the flow function is indicative of powder or granule degradation (C), with large gains of strength as breakdown of the material occurs, raising powder density and interparticle contacts.

Under the linear Mohr-Coloumb approximation, if parallel yield loci are assumed with constant angle of internal friction, and with zero intercept of the effective yield locus, the flow function is a straight line through the origin D, given by

$$f_c = f_{co} + A\sigma_1 = \left[\frac{2}{\cos \phi} \left(\frac{1 + \sin \phi}{1 + \sin \phi_e} \right) (\sin \phi_e - \sin \phi) \right] \sigma_1 \quad \text{where} \quad f_{co} = 0 \quad (21-37)$$

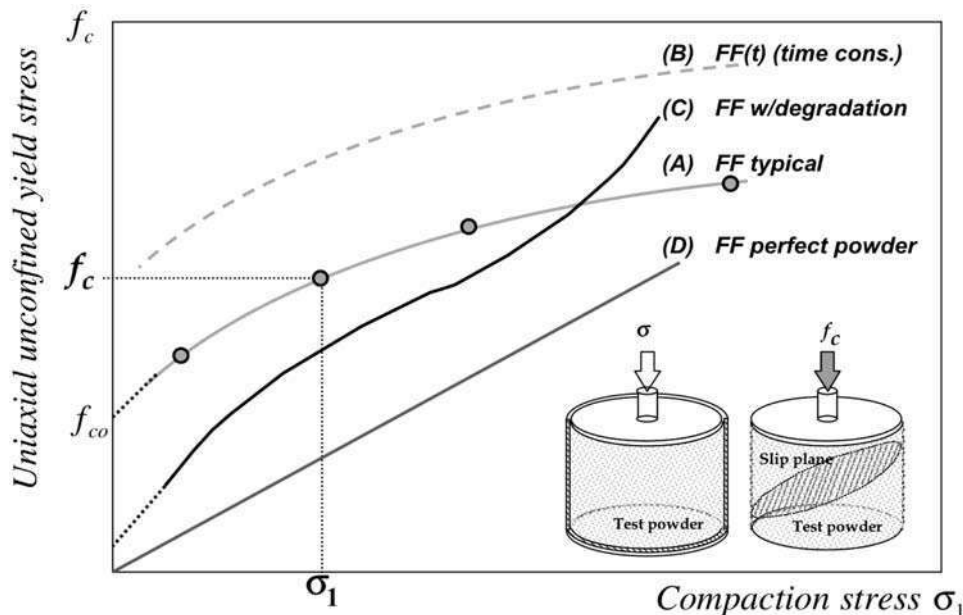


FIG. 21-38 Common flow functions of powder. (From Measuring Powder Flowability and Its Applications, E&G Associates, 2006, with permission.)

Other workers assume a linear form with a nonzero intercept f_c . This implies a minimum powder strength in the absence of gravity or any other applied consolidation stresses. As described above, the flow function is often curved, likely due to the angles of friction being a function of applied stress, and various fitting relations are extrapolated to zero to determine f_c . While this is a typical practice, it has questionable basis as the flow function may have pronounced curvature at low stress.

The flow function and powder strength have a large impact on minimum discharge opening sizes of hoppers to prevent arching and rat holing, mass discharge rates, mixing and segregation, and compact strength.

One may compare the flowability of powders at similar pressures by comparing their unconfined yield stress f_c at a single normal stress, or one point off a flow function. In this case one should clearly state the pressure of comparison. **Flow indices** have been defined to aid such one-point comparisons, given by the ratio of normal stress to strength, or

$$\text{Rel}_p = \frac{\sigma_1 - \sigma_3}{f_c} \quad \text{or} \quad \text{Rel}_f = \frac{\sigma_1}{f_c} \quad (21-38)$$

The first is due to Peschl (Peschl and Colijn, New Rotational Shear Testing Technique, Bulk Solids Handling and Processing Conference, Chicago, May 1976). For powders in the absence of caking it has a minimum value of 1 for a perfectly plastic, cohesive powder. The second definition is due to Jenike (Jenike, Storage and Flow of Bulk Solids, *Bull. 123*, Utah Eng Expt. Stn., 1964). The reciprocal of these **relative flow indices** represents a normalized yield strength of the powder, normalized by maximum consolidation shear in the case of Peschl and consolidation stress in the case of Jenike. Flowability increases with decreasing powder strength, or increasing flow index. Table 21-5 provides typical ranges of behavior for varying flow index. For powders of varying bulk density, **absolute flow indices** should be used, or

$$\text{Abs}_{\text{PorJ}} = \text{Rel}_{\text{PorJ}} \times (\rho_b/\rho_{\text{H}_2\text{O}}) \quad (21-39)$$

Therefore, for powders of *equal* powder strength, flowability increases with increasing bulk density for gravity-driven flow.

Shear Cell Standards and Validation While shear cells vary in design, and may in some cases provide differing values of powder strength, the testing does have an engineering basis in geotechnical engineering, and engineering properties are measured, i.e., yield stresses of a powder versus consolidation. As opposed to other phenomenological, or *instrument-specific*, characterizations of powder flowability, shear cells generally provide a common reliable ranking of flowability, and such data are directly used in design, as discussed below. (See also "Solids Handling: Storage, Feeding, and Weighing.") Rotary split cells (ASTM D6682-01), translation Jenike cells (ASTM D6128-97), and rotary annular ring cells (ASTM D6682-01) all have ASTM test methods. In addition, units may be validated against an independent, international powder standard, namely, the BCR-116 limestone validation powder for shear cell testing (Commission of the European Communities: Community Bureau of Reference). Table 21-6 provides an excerpt of shear values expected for the standard, and Fig. 21-39 provides a yield loci comparison between differing cell designs and a comparison to the standard values.

Stresses in Cylinders Bulk solids do not uniformly transmit stress. Consider the forces acting on a differential slice of material in,

TABLE 21-5 Typical Ranges of Flowability for Varying Flow Index, Modified after Peschl

Flow index	Level of cohesion	Example
$\text{Rel}_p < 1$	Bonding, solid	Caked material, time consolidated
$= 1$	Plastic material	Wet mass
1–2	Extremely cohesive	Magstearate, starch (nongravity)
2–4	Very cohesive	Coarse organics
4–10	Cohesive	Granules inorganics
10–15	Slightly cohesive	Hard silica, sand
15–25	Cohesionless	If fine, floodable

From Measuring Powder Flowability and Its Applications, E&G Associates, 2006, with permission.

TABLE 21-6 BCR-116 Limestone Validation Powder for Shear Cell Testing

COMMISSION OF THE EUROPEAN COMMUNITIES CERTIFIED REFERENCE MATERIAL CERTIFICATE OF MEASUREMENT CRM 116 LIMESTONE POWDER FOR JENIKER SHEAR TESTING			
CONSOLIDATION NORMAL STRESS kPa	SHEAR NORMAL STRESS kPa	MEAN SHEAR STRESS kPa	UNCERTAINTY kPa
3.0	3.0	2.14	± 0.31
3.0	2.0	1.75	± 0.19
3.0	1.75	1.64	± 0.17
3.0	1.5	1.54	± 0.14
3.0	1.25	1.41	± 0.13
3.0	1.0	1.27	± 0.10

say, a cylindrical bin (Fig. 21-40). Prior to failure or within the elastic limit, the axial stresses σ_z and radial stresses σ_r , under the assumption they are principal stresses, are related by

$$\sigma_r = \frac{\nu}{1 - \nu} \sigma_z \quad (21-40)$$

where ν is the Poisson ratio. Under active incipient failure, the axial and radial stresses are related by a lateral stress coefficient K_a given by

$$K_a = \frac{\sigma_r}{\sigma_z} = \frac{1 - \sin \phi_c}{1 + \sin \phi_c} \quad (\text{active}) \quad (21-41)$$

In the case of wall friction, the axial and radial stresses differ somewhat from the true principal stresses, and the stress coefficient becomes

$$K_a = \frac{\sigma_r}{\sigma_z} = \frac{1 - \sin \phi_c \cos(\omega - \phi_w)}{1 + \sin \phi_c \cos(\omega - \phi_w)} \quad \text{where} \quad \sin \omega = \frac{\sin \phi_w}{\sin \phi_c} \quad (21-42)$$

This may be contrasted to, e.g., the isotropic pressure developed in a fluid under pressure, with only nonnewtonian fluids able to develop and sustain a nonisotropic distribution of normal stress. In addition, the radial normal stress acting at the wall develops a wall shear stress that opposes gravity and helps support the weight of the powder. As originally developed by Janssen [*Zeits. D. Vereins Deutsch Ing.*, **39**(35), 1045 (1895)], from a balance of forces on a differential slice, the axial stress σ_z as a function of depth z is given by

$$\sigma_z = \frac{\rho_b g D}{4\mu_w K_a} (1 - e^{-(4\mu_w K_a/D)z}) \quad (21-43)$$

where D is the diameter of the column. Several comments may be made of industrial practicality:

1. Pressure initially scales with height as one would expect for a fluid, which may be verified by expanding Eq. (21-35) for small z . Or $\sigma_z \approx \rho_b g z$.

2. For sufficient depth (at least one diameter), the pressure reaches a maximum value given by $\sigma_z = \rho_b g D / (4\mu_w K_a)$. Note that this pressure scales with cylinder *diameter*, and *not* height. This is a critical property to keep in mind in processing; that diameter often controls pressure in a powder rather than depth. A commonplace example would be comparing the tall aspect ratio of a corn silo to that of a liquid storage vessel. The maximum pressure in the base of such a silo is controlled by diameter, which is kept small.

3. The exact transition to constant pressure occurs at roughly $2z_c$, where $z_c = D / (4\mu_w K_a)$.

Stress transmission in powders controls flow out of hoppers, feeders, filling of tubes, and compaction problems such as tableting and roll pressing. (See "Powder Compaction.")

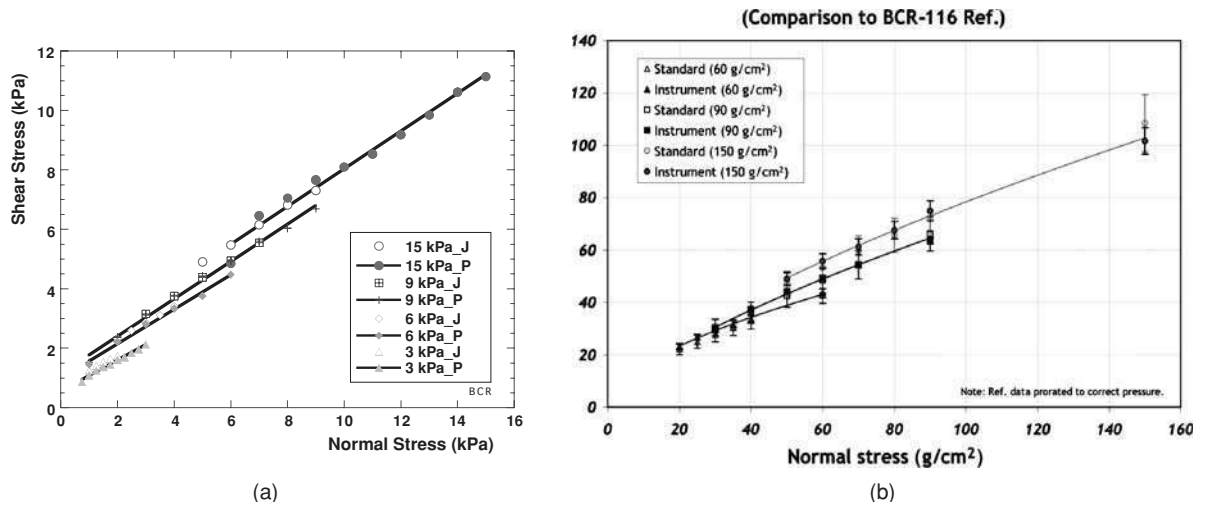


FIG. 21-39 Shear cell BCR-116 limestone validation yield loci. (a) Comparison of Jenike translational to Peschl rotary shear cell data (DuPont, 1994, used with permission). (b) Typical validation set performed on an iShearTM rotary shear cell as compared to BCR standard (2005). (Courtesy E&G Associates, Inc.)

Mass Discharge Rates for Coarse Solids The mass discharge rate from a flat-bottom bin with a circular opening of diameter B has been shown experimentally to be independent of bin diameter D and bed fill height H , for $H > 2B$. Dimensional analysis then indicates that the mass discharge rate W must be of the form $W = C\rho\sqrt{g}B^{3/2}$, where C is a constant function of powder friction. Such a form was verified by Beverloo [Beverloo et al., *Chem. Eng. Sci.*, **15**, 260 (1961)] and Hagen (1856), leading to the **Beverloo equation** of mass discharge, or

$$W_o = C\rho_b\sqrt{g(B - kd_p)^{2.5}} \approx 0.52\rho_b A\sqrt{2gB} \quad \text{for } B \gg d_p \quad (21-44)$$

Here, ρ_b is loose poured bulk density, $C \sim 0.58$ and is nearly independent of friction, $k = 1.5$ for spherical particles and is somewhat larger for angular powders, d_p is particle size, and A is the area of the opening. The correction term of particle size represents an excluded annulus effective lowering the opening diameter. See Nedderman (*Statics and Kinematics of Granular Materials*, Cambridge University Press, 1992) and Brown and Richards (*Principles of Powder Mechanics*, Pergamon Press, 1970) for reviews.

The Beverloo relation for solids discharge may be contrasted with the mass flow rate of an inviscid fluid from an opening of area A , or

$$W = 0.64\rho_b A\sqrt{2gH} \quad (21-45)$$

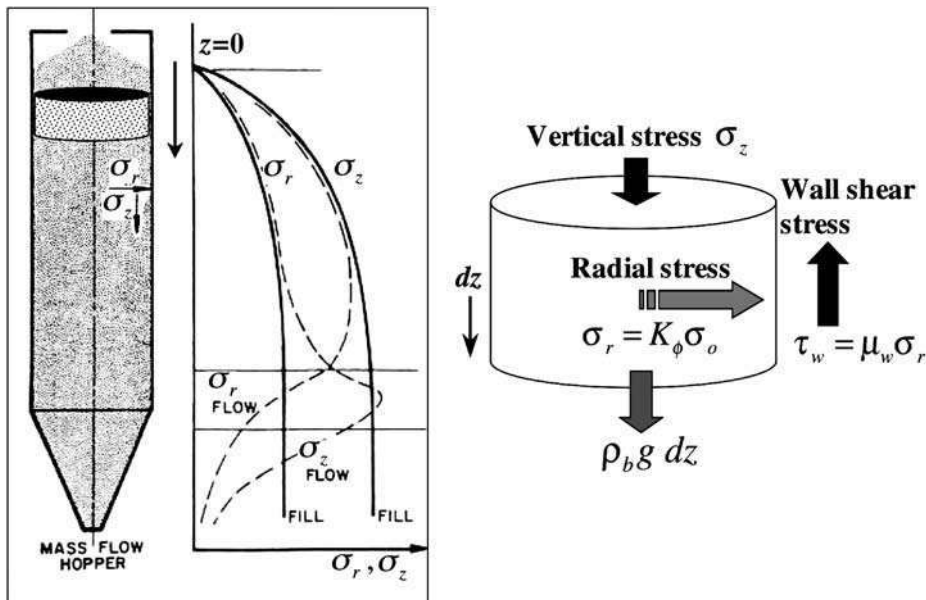


FIG. 21-40 Stresses in a vertical cylinder. [From *Measuring Powder Flowability and Its Applications*, E&G Associates, 2006, with permission.]

where ρ_p is fluid density. Note that mass flow rate scales with height, which controls fluid pressure, compared to mass discharge rates, which scale with orifice diameter.

For coarse materials of typical friction, discharge rates predicted by the Beverloo relation are within 5 percent for experimental values for discharge from flat-bottom bins or from hoppers emptying by **funnel flow**, and are most reliable for material of low powder cohesion, in the range of $400 \mu\text{m} < d_p < B/6$. However, for fine materials less than $100 \mu\text{m}$ or materials large enough to give mechanical interlocking, the Beverloo relation can substantially overpredict discharge.

Equation (21-45) may be generalized for noncircular openings by replacing diameter by hydraulic diameter, given by 4 times the opening area divided by the perimeter. The excluded annulus effect can be incorporated by subtracting kd_p from all dimensions. For slot opening of length $L \gg B$ with B as slot width, discharge rates have been predicted to within 1 percent for coarse materials (Myers and Sellers, Final Year Project, Department of Chemical Engineering, University of Cambridge, UK, 1977) by

$$W_o = \frac{4\sqrt{2}C}{\pi} \rho_b \sqrt{g(L - kd_p)(B - kd_p)^{2.5}} \quad (21-46)$$

Through solutions of radial stress fields acting at the opening, the discharge rate for smooth, wedge-shaped hoppers emptying by **mass flow** is given by the **hourglass theory** of discharge [Savage, *Br. J. Mech. Sci.*, **16**, 1885 (1967); Sullivan, Ph.D. thesis, California Institute of Technology, 1972; Davidson and Nedderman, *Trans. Inst. Chem. Eng.*, **51**, 29 (1973)]:

$$W = \frac{W_o}{\sin^{1/2} \alpha} \quad \text{where} \quad C(K_p) = \frac{\pi}{4} \sqrt{\frac{1 + K_p}{2(2K_p - 3)}} \quad (21-47)$$

where α is the vertical hopper half-angle, $C = f_n(K_p)$, and K_p is the passive Rankine stress coefficient given by

$$K_p = \frac{1 + \sin \phi_e}{1 - \sin \phi_e} \quad (21-48)$$

Here C is a decreasing function of powder friction, ranging from 0.64 to 0.47 for values of ϕ_e ranging from 30° to 50° . Equation (21-46) generally overpredicts wedge hopper rates by a factor of 2, primarily due to neglect of wall friction. The impact of wall friction may be incorporated through the work of Kaza and Jackson [*Powder Technology*, **39**, 915 (1984)] by replacing K_p with a modified coefficient κ given by

$$\kappa = K_p + \frac{(\omega + \phi_w) \sin \phi_e}{\alpha(1 - \sin \phi_e)} \quad (21-49)$$

From Eqs. (21-46) to (21-48), the mass flow discharge rate from wedge hopper increases with increasing orifice diameter $B^{2.5}$, increasing bulk density, decreasing powder friction and wall friction, and decreasing vertical hopper half-angle, and is independent of bed height.

Extensions to Mass Discharge Relations Johanson (*Trans. Soc. Min. Eng.*, March 1965) extended the Beverloo relations to include the effect of powder cohesion, with mass discharge rate given by

$$W_{sc} = 1.354 W_o \sqrt{\frac{1}{2m \tan \alpha} \left(1 - \frac{f_c}{\sigma_{1a}}\right)} \quad (21-50)$$

Here W_{sc} is the steady-state discharge rate for a cohesive powder for unconfined uniaxial compressive strength f_c , and $m = 1$ or 2 for a slot hopper or a conical hopper, respectively. σ_{1a} is the major consolidation stress acting at the hopper opening. Note that the discharge rate increases with increasing stress at the opening and decreasing powder strength, and that the major stress σ_{1a} must exceed the powder's strength f_c for flow to occur. In addition, Johanson determined an initial dynamic mass discharge rate given by

$$W_{dc} = W_{sc} \left(1 - 1.39 \frac{T}{t}\right) \quad \text{where} \quad T = \frac{W_{sc}}{2\rho_p g A} \left(\frac{1}{1 - f_c/\sigma_{1a}}\right) \quad (21-51)$$

where T is the period required to achieve steady-state state flow, which increases with the increases in the required steady discharge rate and increasing powder cohesion f_c .

It is also especially critical to note that an **applied surface pressure** to the top of the powder bed will *not* increase the flow rate. In fact, it is more likely to decrease the flow rate by increasing powder cohesive strength f_c . Similarly, **vibration** will increase flow rate only if the powder is in motion, primarily by lowering wall friction. If discharge is halted, vibration can lower or stop the discharge rate by compacting and raising powder strength.

Stresses in powders are an increasing function of diameter [cf. Eq. (21-43)]. Therefore, as a powder moves toward the opening, the stress acting upon it decreases and the powder undergoes a decrease in bulk density. The displaced solids volume due to the corresponding increase in powder voidage must be matched by an inflow of gas. For coarse solids governed by the Beverloo relation, this inflow of gas occurs with little air pressure change with negligible effect on mass discharge. However, for fine powders of low permeability defined above, large gas pressure gradients will be created at the opening, which opposes solids discharge. There is therefore a decrease in mass discharge with decreasing powder **permeability**, or decreasing particle size of the bulk solid. Verghese (Ph.D. thesis, University of Cambridge, UK, 1991) proposed an initial relation of the form

$$W = W_o \left(1 - \frac{\lambda}{\rho_b g d_p^2}\right)^{1/2} \frac{\lambda}{\rho_b g} \approx 1.48 \times 10^{-8} \text{ m}^3 \quad (21-52)$$

The decrease in mass discharge rate from the Beverloo relation for decreasing particle size is illustrated in Fig. 21-41. For fine enough materials, bubbling and fluidization actually halt flow from the orifice, after which a gain in bulk density will again initialize flow. This may be witnessed with fine sands discharging from hourglasses. A similar relation based on venting required predicted from the Carman-Kozeny equation gives a fine powder mass discharge rate of

$$W \approx \frac{2\pi(B/\sin \alpha)^3 \rho_b^2 d_p^2 g (1 - \cos \alpha)^3}{180\mu_g (1 - \epsilon)^3} \quad (21-53)$$

where μ_g is gas viscosity and ϵ is the bed voidage.

Gas venting may be used to increase discharge rate, either through venting in the hopper wall or through imposed pressure gradients. The involved pressure drops or required air volumes may be calculated from standard pressure drop correlations, based on, e.g., Darcy's law or the Ergun equation. For air-augmented flow, discharge rates are given by

$$W = W_o \left[1 + \left(\frac{2\kappa - 3}{2\kappa - 1}\right) \frac{\Delta P}{\rho g r_o}\right]^{1/2} \quad \text{for } Re_o < 10 \quad (21-54)$$

$$W = W_o \left[1 + \left(\frac{2\kappa - 3}{2\kappa - 1}\right) \frac{3 \Delta P}{\rho g r_o}\right]^{1/2} \quad \text{for } Re_o \text{ large} \quad (21-55)$$

$$W = W_o \left[1 + \left(\frac{150 + 5.25 Re_o}{150 + 1.75 Re_o}\right) \left(\frac{2\kappa - 3}{2\kappa - 1}\right) \left(\frac{\Delta P}{\rho g r_o}\right)\right]^{1/2} \quad \text{for intermediate } Re_o \quad (21-56)$$

where r_o is the radial distance from the hopper apex, $\Delta P/r_o$ is the pressure drop imposed across the orifice, and Re_o is the gas Reynolds number acting at the orifice (see Nedderman, *Statics and Kinematics of Granular Materials*, Cambridge University Press, 1992).

Other Methods of Flow Characterization A variety of other test methods to characterize flowability of powders have been proposed, which include density ratios, flow from funnels and orifices, angles of repose and sliding, simplified indicizer flow testing, and

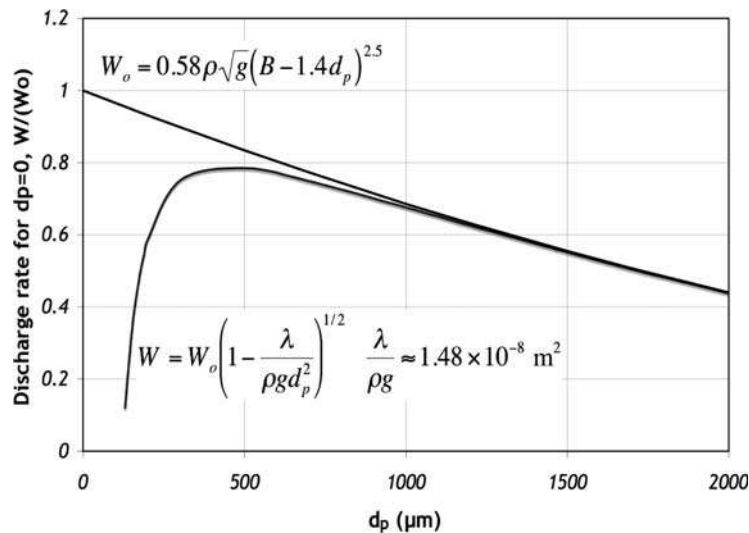


FIG. 21-41 The impact of decreasing particle size and bulk permeability on mass discharge rate.

tumbling avalanche methods. These methods should be used with caution, as (1) they are often a strong function of the test method and instrument itself, (2) engineering properties useful for either scale-up or a priori design are not measured, (3) they are only a crude characterization of flowability, and often suffer from lack of reproducibility, (4) they lack a fundamental basis of use, and (5) they suffer from the absence of validation powders and methods. The first two points are particularly crucial, the end result of which is that the ranking of powders determined by the apparatus cannot be truly linked to process performance, as the states of stress in the process may differ from the apparatus, and further, the ranking of powders may very well change with scale-up. In contrast, shear cells and permeability properties may be used directly for design, with no need for arbitrary scales of behavior, and the effect of changing stress state with scale-up can be predicted. Having said this, many of these methods have found favor due to the misleading ease of use. In some defined cases they may be useful for quality control, but should not be viewed as a replacement for more rigorous flow testing offered by shear cell and permeability testing.

Various **angles of repose** may be measured, referring to the horizontal angle formed along a powder surface. These include the angle of a heap, the angle of drain for material remaining in a flat-bottom bin, the angle of sliding occurring when a dish of powder is inclined, rolling angles in cylinders, and dynamic and static discharge angles onto vibrating feed chutes (Thompson, *Storage of Particulate Solids*, in *Handbook of Powder Science and Technology*, Fayed and Otten (eds.), Van Nostrand Reinhold Co., 1984). From Eq. (21-37) describing the impact of the angles of friction—as measured by shear cell—on cohesive strength, the angle of repose may be demonstrated to lack a true connection to flowability. For cohesive powders, there will be large differences between the internal and effective angles of friction, and the unconfined strength increases with an increase in the difference in sine of the angles. When one is measuring the angle of repose in this case, wide variations in the angle of the heap will be observed, and it likely varies between the angles of friction, making the measurement of little utility in a practical, measurement sense. However, when the difference in the angles of friction approaches zero, the angle of repose will be equal to both the internal and effective angle of friction. But at that point, the cohesive strength of the powder is zero [Eq. (21-37)], regardless of the angle of repose.

It is likely the above has formed the basis for the use of rotating **avalanche testers**, where the size and frequency of avalanches formed on the sliding, rotating bed are analyzed as a deviation of the time between avalanches, as well as strange attractor diagrams. This

approach is more consistent with the *variation* in the angle of repose being related to powder strength [Eq. (21-37)].

The typical density ratios are the **Carr** and **Hausner ratios**, given by

$$FI_{\text{Carr}}[\%] = 100 \frac{\rho_{b(\text{tapped})} - \rho_{b(\text{loose})}}{\rho_{b(\text{tapped})}} \quad \text{and} \quad FI_H[-] = \frac{\rho_{b(\text{tapped})}}{\rho_{b(\text{loose})}} \quad (21-57)$$

where $\rho_{b(\text{tapped})}$ is the equilibrium packed bulk density achieved under tapping. It could equally be replaced with a bulk density achieved under a given pressure. The Carr index is a measure of **compressibility**, or the gain in bulk density under stress, and is directly related to gain in powder strength. Large gains in density are connected to differences in the state of packing in the over- and critically consolidated state defined above (see “Yield Behavior of Powders”), which in turn results in differences between the internal and effective angles of friction, leading to a gain in unconfined yield strength [Eq. (21-37)]. However, the results are a function of the method and may not be discriminating for free-flowing materials. Lastly, changes in density are only one of many contributions to unconfined yield stress and powder flowability. Hence, Carr and Hausner indices may incorrectly rank flowability across ranges of material class that vary widely in particle mechanical and surface properties.

Two methods of **hopper flow characterization** are used. The first is the Flowdex™ tester, which consists of a cup with interchangeable bottoms of varying orifice size. The cup is filled from a funnel, and the covering lid then drops from the opening. The minimum orifice in millimeters required for flow to occur is determined as a ranking of flowability. This minimum orifice is analogous to the minimum orifice diameter determined from shear cell data for hopper design. Alternatively, the mass discharge rate out of the cup or from a funnel may be determined. Various methods of vibration both before and after initiation of flow may be utilized. Mass discharge rates, as expected, rank with the correlations described above. The disadvantage of this characterization method is that it is a direct function of hopper/cup geometry and wall friction, and has a low state of stress that may differ from the actual process. If a process hopper differs in vertical half-angle, wall friction, opening size, solids pressure, filling method, or a range of other process parameters, the ranking of powder behavior in practice may differ from the lab characterization, since scalable engineering properties are not measured.

The last set of tests includes **solid indicizers** pioneered by Johanson. These include the Flow Rate and Hang-Up Indicizers™ [cf. Bell et al., Practical Evaluation of the Johanson Hang-Up Indicizer, *Bulks Solids Handling*, 14(1), 117 (Jan. 1994)]. They represent simplified versions of permeability and shear cell tests. Assumptions are made with regard to typical pressures and wall frictions, and based on these, a

flow ranking is created. Their degree of success in an application will largely rest on the validity of the property assumptions. For defined conditions, they can give similar ranking to shear cell and permeability tests. The choice of use is less warranted than in the past due to the progress in automating shear cell and permeability tests, which has simplified their ease of use.

SOLIDS MIXING

GENERAL REFERENCES: Fan, Chen, and Lai, Recent Developments in Solids Mixing, *Powder Technology*, 61, 255–287 (1990); N. Harnby, M. F. Edwards, A. W. Nienow (eds.), *Mixing in the Process Industries*, 2d ed., Butterworth-Heinemann, 1992; B. Kaye, *Powder Mixing*, 1997; Ralf Weinekötter and Herman Gericke, *Mixing of Solids, Particle Technology Series*, Brian Scarlett (ed.), Kluwer Academic Publishers, Dordrecht 2000.

PRINCIPLES OF SOLIDS MIXING

Industrial Relevance of Solids Mixing The mixing of powders, particles, flakes, and granules has gained substantial economic importance in a broad range of industries, including, e.g., the mixing of human and animal foodstuff, pharmaceutical products, detergents, chemicals, and plastics. As in most cases the mixing process adds significant value to the product, the process can be regarded as a key unit operation to the overall process stream.

By far the most important use of mixing is the production of a homogeneous blend of several ingredients which neutralizes variations in concentration. But if the volume of material consists of one ingredient or compound exhibiting fluctuating properties caused by an upstream production process, or inherent to the raw material itself, the term *homogenization* is used for the neutralization of these fluctuations. By mixing, a new product or intermediate is created for which the quality and price are very often dependent upon the efficiency of the mixing process. This efficiency is determined both by the materials to be mixed, e.g., particle size and particle-size distribution, density, and surface roughness, and by the process and equipment used for performing the mixing. The design and operation of the mixing unit itself have a strong influence on the quality produced, but upstream material handling process steps such as feeding, sifting, weighing, and transport determine also both the quality and the capacity of the mixing process. Downstream processing may also destroy the product quality due to segregation (demixing). Continuous mixing is one solution which limits segregation by avoiding storage equipment.

The technical process of mixing is performed by a multitude of equipment available on the market. However, mixing processes are not always designed with the appropriate care. This causes a significant financial loss, which arises in two ways:

1. *The quality of the mix is poor:* In cases where the mixing produces the end product, this will be noticed immediately at the product's quality inspection. Frequently, however, mixing is only one in a series of further processing stages. In this case, the effects of unsatisfactory blending are less apparent, and might possibly be overlooked to the detriment of final product quality.

2. *The homogeneity is satisfactory but the effort employed is too great (overmixing):* Overmixing in batch blending is induced by an overlong mixing time or too long a residence time in the case of continuous blending. This leads to increased strain on the mixture, which can have an adverse effect on the quality of sensitive products. Furthermore, larger or more numerous pieces of equipment must be used than would be necessary in the case of an optimally configured mixing process.

Mixing Mechanisms: Dispersive and Convective Mixing The mixing process can be observed in diagrammatic form as an over-

lap of **dispersion** and **convection** (Fig. 21-42). Movement of the particulate materials is a prerequisite of both mechanisms. **Dispersion** is understood to mean the completely random change of place of the individual particles. The frequency with which the particles of *ingredient A* change place with those of another is related to the number of particles of the other ingredients in the *direct vicinity* of the particles of *ingredient A*. Dispersion is therefore a local effect (**micromixing**) taking place in the case of premix systems where a number of particles of different ingredients are in proximity, leading to a fine mix localized to very small areas. If the ingredients are spatially separated at the beginning of the process, long times will be required to mix them through dispersion alone, since there is a very low number of *assorted neighbors*. Dispersion corresponds to diffusion in liquid mixtures. However, in contrast to diffusion, mixing in the case of dispersion is not caused by any concentration gradient. The particles have to be in motion to get dispersed. **Convection** causes a movement of large groups of particles relative to each other (**macromixing**). The whole volume of material is continuously divided up and then mixed again after the portions have changed places (Fig. 21-42). This forced convection can be achieved by rotating elements. The dimension of the groups, which are composed of just one unmixed ingredient, is continuously reduced splitting action of the rotating paddles. Convection increases the number of *assorted neighbors* and thereby promotes the exchange processes of dispersive mixing. A material mass is divided up

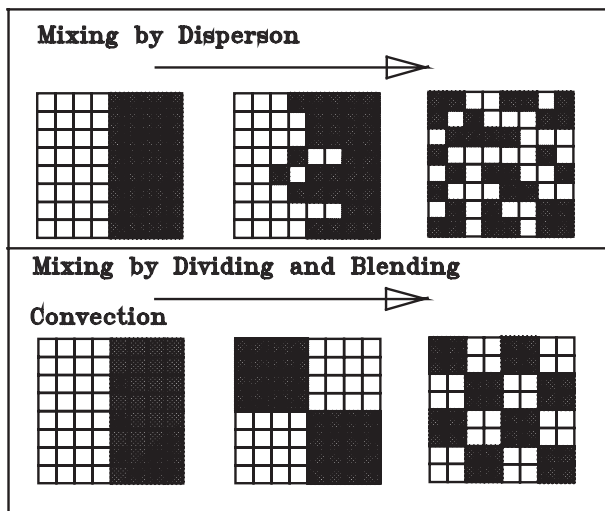


FIG. 21-42 The mixing process can be observed in diagrammatic form as an overlap of *dispersion* and *convection*. Mixture consists of two components A and B; A is symbolized by the white block and B by the hatched block. Dispersion results in a random arrangement of the particles; convection results in a regular pattern.

or convectively mixed through the rearrangement of a solid's layers by rotating devices in the mixer or by the fall of a stream of material in a static gravity mixer, as discussed below.

Segregation in Solids and Demixing If the ingredients in a solids mixture possess a selective, individual motional behavior, the mixture's quality can be reduced as a result of segregation. As yet only a partial understanding of such behavior exists, with particle movement behavior being influenced by particle properties such as size, shape, density, surface roughness, forces of attraction, and friction. In addition, industrial mixers each possess their own specific flow conditions. Particle size is, however, the dominant influence in segregation (J. C. Williams, *Mixing, Theory and Practice*, vol. 3, V. W. Uhl and J. B. Gray, (eds.), Academic Press, Orlando, Fla., 1986). Since there is a divergence of particle sizes in even a single ingredient, nearly all industrial powders can be considered as solid mixtures of particles of different size, and segregation is one of the characteristic problems of solids processing which must be overcome for successful processing. If mixtures are unsuitably stored or transported, they will separate according to particle size and thus segregate. Figure 21-43 illustrates typical mechanisms of segregation.

Agglomeration segregation arises through the preferential self-agglomeration of one component in a two-ingredient mixture (Fig. 21-43a). Agglomerates form when there are strong interparticle forces, and for these forces to have an effect, the particles must

be brought into close contact. In the case of agglomerates, the particles stick to one another as a result, e.g., of liquid bridges formed in solids, if a small quantity of moisture or other fluid is present. Electrostatic and van der Waals forces likewise induce cohesion of agglomerates. Van der Waals forces, reciprocal induced and dipolar, operate particularly upon finer grains smaller than $30\text{ }\mu\text{m}$ and bind them together. High-speed impellers or knives are utilized in the mixing chamber to create shear forces during mixing to break up these agglomerates. Agglomeration can, however, have a positive effect on mixing. If a solids mix contains a very fine ingredient with particles in the submicrometer range (e.g., pigments), these fine particles coat the coarser ones. An **ordered mixture** occurs, which is stabilized by the van der Waals forces and is thereby protected from segregation.

Flotation segregation can occur if a solids mix is *vibrated*, where the coarser particles *float up* against the gravity force and collect near the top surface, as illustrated in Fig. 21-43b for the case of a large particle in a mix of finer material. During vibration, smaller particles flow into the vacant space created underneath the large particle, preventing the large particle from reclaiming its original position. If the large particle has a higher density than the fines, it will compact the fines, further reducing their mobility and the ability of the large particle to sink. Solely because of the blocking effect of the larger particle's geometry there is little probability that this effect will run in reverse and that a bigger particle will take over the place left by a smaller one which has been lifted up. The large particle in this case would also have to displace several smaller ones. As a result the probability is higher that coarse particles will climb upward with vibration.

Percolation segregation is by far the most important segregational effect, which occurs when finer particles trickle down through the gaps between the larger ones (Fig. 21-43c). These gaps act as a sieve. If a solids mixture is moved, gaps briefly open up between the grains, allowing finer particles to selectively pass through the particle bed. Granted a single layer has a low degree of separation, but a bed of powder consists of many layers and interconnecting grades of particles which taken together can produce a significant division between fine and coarse grains (see Fig. 21-43), resulting in widespread segregation. Furthermore, percolation occurs even where there is but a small difference in the size of the particles (250- and $300\text{-}\mu\text{m}$ particles) [J. C. Williams, *Fuel Soc. J.*, University of Sheffield, **14**, 29 (1963)]. The most significant economical example is the poured heap appearing when filling and discharging bunkers or silos. A mobile layer with a high-speed gradient forms on the surface of such a cone, which, like a sieve, bars larger particles from passing into the cone's core. Large grains on the cone's mantle obviously slide or roll downward. But large, poorly mixed areas occur even inside the cone. Thus filling a silo or emptying it from a central discharge point is particularly critical. Remixing of such segregated heaps can be achieved through **mass flow** discharge; i.e., the silo's contents move downward in blocks, slipping at the walls, rather than emptying from the central core (**funnel flow**).

Transport Segregation This encompasses several effects which share the common factor of a gas contributing to the segregation processes. Trajectory and fluidized segregation can be defined, first, as occurring in cyclones or conveying into a silo where the particles are following the individual trajectories and, second, in fluidization. During fluidization particles are exposed to drag and gravity forces, which may lead to a segregation.

Williams (see above) gives an overview of the literature on the subject and suggests the following measures to counter segregation: The addition of a small quantity of water forms water bridges between the particles, reducing their mobility and thus stabilizing the condition of the mixture. Because of the cohesive behavior of particles smaller than $30\text{ }\mu\text{m}$ ($\rho_s = 2\text{ to }3\text{ kg/L}$) the tendency to segregate decreases below this grain size. Inclined planes down which the particles can roll should be avoided. In general, having ingredients of a uniform grain size is an advantage in blending.

Mixture Quality: The Statistical Definition of Homogeneity To judge the efficiency of a solids blender or of a mixing process in general, the status of mixing has to be quantified; thus a *degree of*

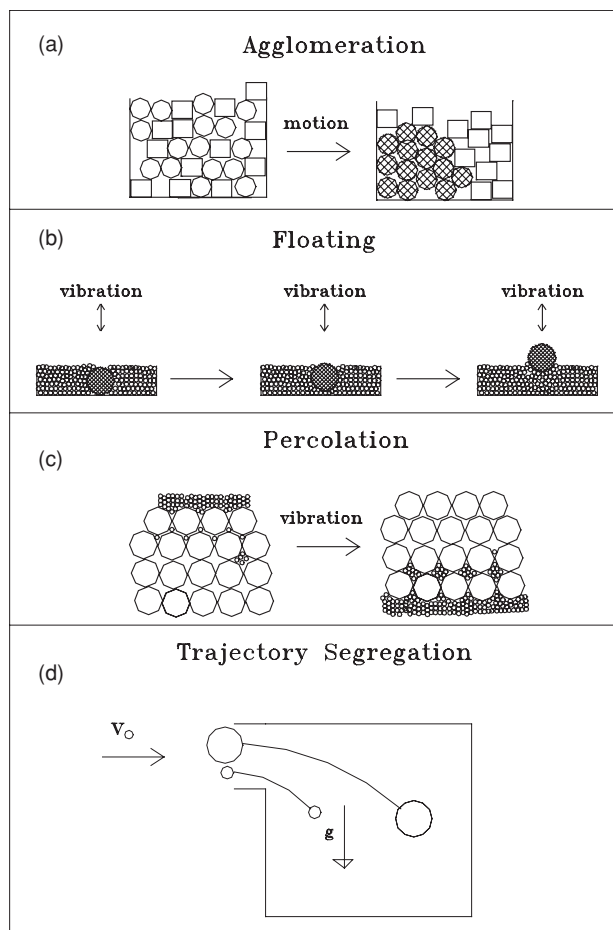


FIG. 21-43 Four mechanisms of segregation, following Williams.

mixing has to be defined. Here one has to specify what *property* characterizes a mixture, examples being composition, particle size, and temperature. The end goal of a mixing process is the uniformity of this property throughout the volume of material in the mixer. There are circumstances in which a good mix requires uniformity of several properties, e.g., particle size and composition. The mixture's condition is traditionally checked by taking a number of samples, after which these samples are examined for uniformity of the property of interest. The quantity of material sampled, or **sample size**, and the location of these samples are essential elements in evaluating a solids mixture.

Sample size thus represents the resolution by which a mixture can be judged. The smaller the size of the sample, the more closely the condition of the mixture will be scrutinized (Fig. 21-44). Dankwerts terms this the **scale of scrutiny** [P. V. Dankwerts, *The Definition and Measurement of Some Characteristics of Mixtures*, *Appl. Sci. Res.*, 279ff (1952)]. Specifying the size of the sample is therefore an essential step in analyzing a mixture's quality, since it quantifies the mixing task from the outset. The size of the sample can only be meaningfully specified in connection with the mixture's further application. In pharmaceutical production, active ingredients must be equally distributed; e.g., within the individual tablets in a production batch, the sample size for testing the condition of a mixture is one tablet. In less critical industries the sample size can be in tons. The traditional and general procedure is to take *identically sized* samples of the mixture from various points *at random* and to analyze them in an off-line analysis. Multielement mixtures can also be described as twin ingredient mixes when a particularly important ingredient, e.g., the active agent in pharmaceutical products, is viewed as a tracer element and all the other constituents are combined into one common ingredient. This is a simplification of the statistical description of solids mixtures. When two-element mixtures are being examined, it is sufficient to trace the concentration path of just one ingredient, the tracer. There will be a complementary concentration of the other ingredients. The description is completely analogous when the property or characteristic feature in which we are interested is not the concentration but is, e.g., moisture, temperature, or the particle's shape. If the tracer's concentration in the mixture is p and that of the other ingredients is q , we have the following relationship: $p + q = 1$. If you take samples of a specified size from the mixture and analyze them for their content of the tracer, the concentration of tracer x_i in the samples will fluctuate randomly around that tracer's concentration p in the whole mixture (the "base whole"). Therefore a mixture's quality can only be

described by using statistical means. The smaller the fluctuations in the samples' concentration x_i around the mixture's concentration p , the better its quality. This can be quantified by the statistical **variance** of sample concentration σ^2 , which consequently is frequently defined as the **degree of mixing**.

There are many more definitions of mix quality in literature on the subject, but in most instances these relate to an initial or final variance and are frequently too complicated for industrial application (K. Sommer, *Mixing of Solids*, in *Ullmann's Encyclopaedia of Industrial Chemistry*, vol. B4, Chap. 27, VCH Publishers Inc., 1992). The theoretical variance for finite sample numbers is calculated as follows:

$$\sigma^2 = \frac{1}{N_g} \sum_{i=1}^{N_g} (x_i - p)^2 \quad (21-58)$$

The relative standard deviation (RSD) is used as well for judging mixture quality. It is defined by

$$\text{RSD} = \sqrt{\sigma^2}/p \quad (21-59)$$

The variance is obtained by dividing up the whole mix, the base whole, into N_g samples of the same size and determining the concentration x_i in each sample. Figure 21-44 illustrates that smaller samples will cause a larger variance or degree of mixing.

If one analyzes not the whole mix but a number n of *randomly* distributed samples across the base whole, one determines instead the **sample variance** S^2 . If this procedure is repeated several times, a new value for the sample variance will be produced on each occasion, resulting in a statistical distribution of the sample variance. Thus each S^2 represents an estimated value for the unknown variance σ^2 . In many cases the concentration p is likewise unknown, and the random sample variance is then defined by using the **arithmetical average** μ of the sample's concentration x_i .

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2 \quad \mu = \frac{1}{n} \sum_{i=1}^n x_i \quad (21-60)$$

Random sample variance data are of little utility without knowing how accurately they describe the unknown, true variance σ^2 . The variance is therefore best stated as a desired **confidence interval** for σ^2 . The confidence interval used in mixing is mostly a unilateral one, derived by the χ^2 distribution. Interest is focused on the upper confidence limit, which, with a given degree of probability, will not be exceeded by the variance [Eq. (21-61)] [J. Raasch and K. Sommer, *The Application of Statistical Test Procedures in the Field of Mixing Technology*, in German, *Chemical Engineering*, **62**(1), 17-22 (1990)], which is given by

$$W\left(\sigma^2 < (n-1) \frac{S^2}{\chi^2_1}\right) = 1 - \Phi(\chi^2_1) \quad (21-61)$$

Figure 21-45 illustrates how the size of the confidence interval normalized with the sample variance decreases as the number of random samples n increases. The confidence interval depicts the accuracy of the analysis. The smaller the interval, the more exactly the mix quality can be estimated from the measured sample variance. If there are few samples, the mix quality's confidence interval is very large. An evaluation of the mix quality with a high degree of accuracy (a small confidence interval) requires that a large number of samples be taken and analyzed, which can be expensive and can require great effort. Accuracy and cost of analysis must therefore be balanced for the process at hand.

Example 3: Calculating Mixture Quality Three tons of a sand (80 percent by weight) and cement (20 percent by weight) mix has been produced. The quality of this mix has to be checked. Thirty samples at 2 kg of the material mixture have been taken at random, and the sand content in these samples established.

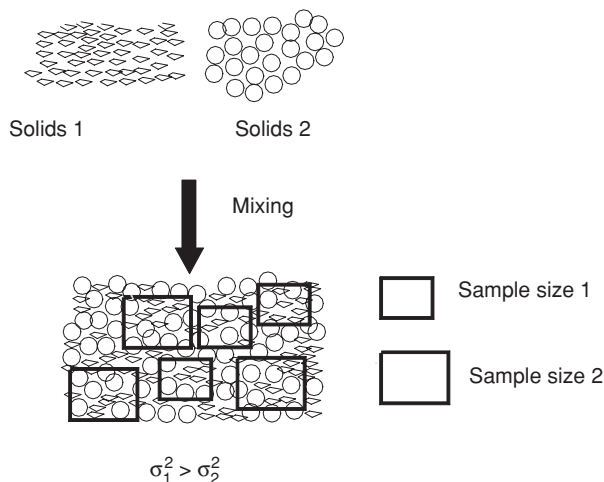


FIG. 21-44 The influence of the size of the sample on the numerical value of the degree of mixing.

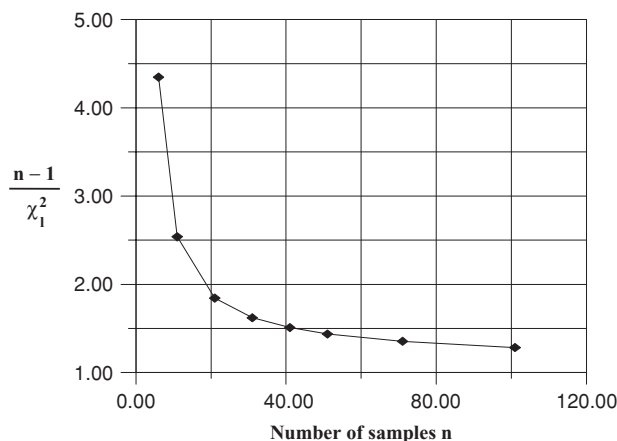


FIG. 21-45 The size of the unilateral confidence interval (95 percent) as a function of the number n of samples taken, measured in multiples of S^2 [cf. Eq. (21-62)]. Example: If 31 samples are taken, the upper limit of the variance's confidence interval assumes a value of 1.6 times that of the experimental sample variance S^2 .

The mass fraction of the sand x_i (kg_{sand}/kg_{mix}) in the samples comes to

3 samples @ 0.75; 7 @ 0.77; 5 @ 0.79; 6 @ 0.81; 7 @ 0.83; 2 @ 0.85

The degree of mixing defined as the variance of the mass fraction of sand in the mix needs to be determined. It has to be compared with the variance for a fully segregated system and the ideal variance of a random mix. First, the random sample variance S^2 [Eq. (21-60)] is calculated, and with it an upper limit for the true variance σ^2 can then be laid down. The sand's average concentration μ in the whole 3-ton mix is estimated by using the random sample average μ :

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{30} \sum_{i=1}^{30} x_i = 0.797$$

$$\begin{aligned} S^2 &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{29} \sum_{i=1}^{30} (x_i - 0.797)^2 \\ &= \frac{1}{29} (3 \cdot 0.047^2 + 7 \cdot 0.027^2 + 5 \cdot 0.007^2 + 6 \cdot 0.013^2 + 7 \cdot 0.033^2 + 2 \cdot 0.053^2) \\ &= 9.04 \times 10^{-4} \end{aligned}$$

Ninety-five percent is set as the probability W determining the size of the confidence interval for the variance σ^2 . An upper limit (unilateral confidence interval) is then calculated for variance σ^2 :

$$W \left[\sigma^2 < (n-1) \frac{S^2}{\chi^2_1} \right] = 0.95 = 1 - \Phi(\chi^2_1) \Rightarrow \Phi(\chi^2_1) = 0.05$$

From the table of the χ^2 distribution summation function (in statistical teaching books) $\Phi(\chi^2_1; n-1)$ the value 17.7 is derived for 29 degrees of freedom. Figure 21-45 allows a fast judgment of these values without consulting statistical tables. Values for $(n-1)/\chi^2_1$ are shown for different number of samples n .

$$\sigma^2 < (n-1) \frac{S^2}{\chi^2_1} = 29 \cdot \frac{9.04 \times 10^{-4}}{17.7} = 14.8 \times 10^{-4} \quad (21-62)$$

It can therefore be conclusively stated with a probability of 95 percent that the mix quality σ^2 is better (equals less) than 14.8×10^{-4} .

Ideal Mixtures A perfect mixture exists when the concentration at any randomly selected point in the mix in a sample of any size is the same as that of the overall concentration. The variance of a perfect mixture has a value of 0. This is only possible with gases and liq-

uids which can be mixed molecularly and where sample volumes of the mixture are many times larger than its ingredients, i.e., molecules. In the case of solids mixtures, particle size must be considered in comparison to both sample size and sensor area. Thus σ^2 depends on the size of the sample (Fig. 21-46). There are two limiting conditions of maximum homogeneity which are the equivalent of a minimum variance: an **ordered** and a **random mixture**.

Ordered Mixtures The components align themselves according to a defined pattern. Whether this ever happens in practice is debatable. There exists the notion that because of interparticle processes of attraction, this mix condition can be achieved. The interparticle forces find themselves in an interplay with those of gravity and other dispersive forces, which would prevent this type of ordered mix in the case of coarser particles. Interparticle forces predominate in the case of finer particles, i.e., cohesive powders. Ordered agglomerates or layered particles can arise. Sometimes not only the mix condition but also the mixing of powders in which these forces of attraction are significant is termed **ordered mixing** [H. Egermann and N. A. Orr, Comments on the paper "Recent Developments in Solids Mixing" by L. T. Fan et al., *Powder Technology*, **68**, 195–196 (1991)]. However, Egermann [L. T. Fan, Y. Chen, and F. S. Lai, Recent Developments in Solids Mixing, *Powder Technology* **61**, 255–287 (1990)] points to the fact that one should only use ordered mixing to describe the condition and not the mixing of fine particles using powerful interparticle forces.

Random Mixtures A random mixture also represents an ideal condition. It is defined as follows: A uniform random mix occurs when the probability of coming across an ingredient of the mix in any subsection of the area being examined is equal to that of any other point in time for all subsections of the same size, provided

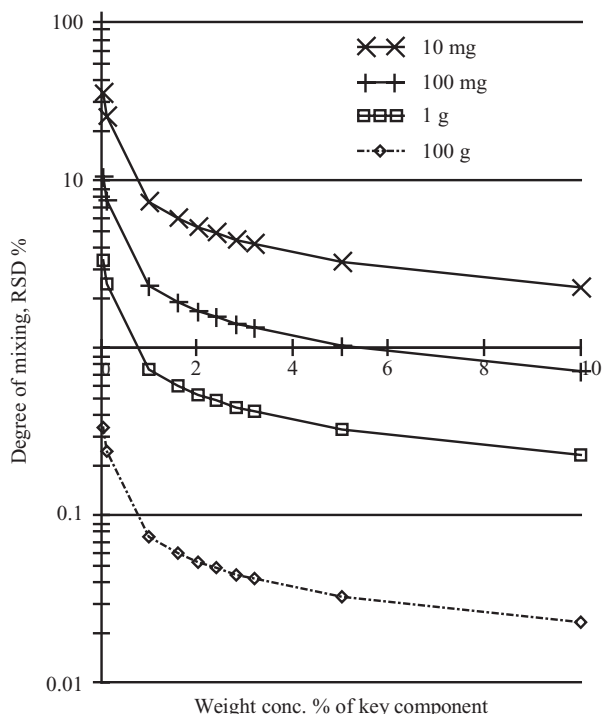


FIG. 21-46 Degree of mixing expressed as $RSD = \sqrt{\sigma^2}/P$ for a random mixture calculated following Sommer. The two components have the same particle-size distribution, $dp_{50} = 50 \mu\text{m}$, $d_{\text{max}} = 130 \mu\text{m}$, $m = 0.7$ (exponent of the power density distribution of the particle size) parameter: sample size ranging from 10 mg to 100 g (R. Weinekötter, Degree of Mixing and Precision for Continuous Mixing Processes, Proceedings Partec, Nuremberg, 2007).

that the condition exists that the particles can move freely. The variance of a random mixture is calculated as follows for a two-ingredient blend in which the particles are of the same size [P. M. C. Lacey, *The Mixing of Solid Particles*, *Trans. Instn. Chem. Engrs.*, **21**, 53-59 (1943)]:

$$\sigma^2 = \frac{p \cdot q}{n_p} \quad (21-63)$$

where p is the concentration of one of the ingredients in the mix, q is the other ($q = 1 - p$), and n_p is number of particles in the sample. Note that the variance of the random mix grows if the sample size decreases. The variance for a **completely segregated system** is given by

$$\sigma_{\text{segregated}}^2 = p \cdot q \quad (21-64)$$

Equation (21-63) is a highly simplified model, for no actual mixture consists of particles of the same size. It is likewise a practical disadvantage that the number of particles in the sample has to be known in order to calculate variance, rather than the usually specified sample volume. Stange calculated the variance of a random mix in which the ingredients possess a distribution of particle sizes. His approach is based on the fact that an ingredient possessing a distribution in particle size by necessity also has a distribution in particle mass. He made an allowance for the average mass m_p and m_q of the particles in each component and the particle mass's standard deviation σ_p and σ_q [K. Stange, *Die Mischgüte einer Zufallmischung als Grundlage zur Beurteilung von Mischversuchen* (The mix quality of a random mix as the basis for evaluating mixing trials), *Chem. Eng.*, **26**(6), 331-337 (1954)]. He designated the **variability** c as the quotient of the standard deviation and average particle mass, or

$$c_p = \frac{\sigma_p}{m_p} \quad c_q = \frac{\sigma_q}{m_q} \quad (21-65)$$

Variability is a measure for the width of the particle-size distribution. The higher the value of c , the broader the particle-size distribution.

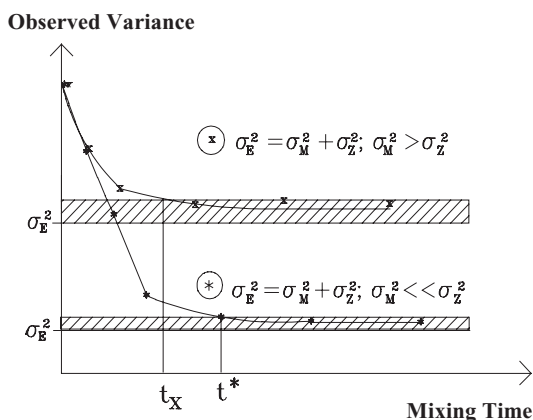


FIG. 21-47 Illustration of the influence of the measurement's accuracy on the variance as a function of the mixing time [following K. Sommer, *How to Compare the Mixing Properties of Solids Mixers* (in German), *Prep. Technol.* no. 5, 266-269 (1982)]. A set of samples have been taken at different mixing times for computing the sample variance. Special attention has to be paid whether the experimental sample variance monitors the errors of the analysis procedure (x) or detects really the mixing process (*). Confidence intervals for the final status σ_E^2 are shown as hatched sections.

The size of the sample is now specified in practice by its mass M and no longer by the number of particles n_p , as shown in Eq. (21-63). The variance in random mixture for the case of two-component mixes can be given by

$$\sigma^2 = \frac{pq}{M} [pm_q(1 + c_q^2) + qm_p(1 + c_p^2)] \quad (21-66)$$

Equation (21-66) estimates the variance of a random mixture, even if the components have different particle-size distributions. If the components have a small size (i.e., small mean particle mass) or a narrow particle-size distribution, that is, c_q and c_p are low, the random mix's variance falls. Sommer has presented mathematical models for calculating the variance of random mixtures for particulate systems with a particle-size distribution (Karl Sommer, *Sampling of Powders and Bulk Materials*, Springer-Verlag, Berlin, 1986, p. 164). This model has been used for deriving Fig. 21-46.

Measuring the Degree of Mixing The mixing process uniformly distributes one or more properties within a quantity of material. These can be physically recordable properties such as size, shape, moisture, temperature, or color. Frequently, however, it is the mixing of **chemically** differing components which forms the subject under examination. Off-line and on-line procedures are used for this examination (compare to subsection "Particle-Size Analysis"). Off-line procedure: A specified portion is (randomly or systematically) taken from the volume of material. These samples are often too large for a subsequent analysis and must then be split. Many analytical processes, e.g., the chemical analysis of solids using infrared spectroscopy, require the samples to be prepared beforehand. At all these stages there exists the danger that the mix status within the samples will be changed. As a consequence, when examining a mixing process whose efficiency can be characterized by the variance expression $\sigma_{\text{process}}^2$, all off- and on-line procedures give this variance only indirectly:

$$\sigma_{\text{observed}}^2 = \sigma_{\text{process}}^2 + \sigma_{\text{measurement}}^2 \quad (21-67)$$

The observed variance $\sigma_{\text{observed}}^2$ also contains the variance $\sigma_{\text{measurement}}^2$ resulting from the test procedure and which arises out of errors in the systematic or random taking, splitting, and preparation of the samples and from the actual analysis. A lot of attention is often paid to the accuracy of an analyzer when it is being bought. However, the preceding steps of sampling and preparation also have to fulfil exacting requirements so that the following can apply:

$$\sigma_{\text{process}}^2 \gg \sigma_{\text{measurement}}^2 \Rightarrow \sigma_{\text{process}}^2 = \sigma_{\text{observed}}^2 \quad (21-68)$$

Figure 21-47 illustrates the impact of precision of the determination of mixing time for batch mixers. It is not yet possible to theoretically forecast mixing times for solids, and therefore these have to be ascertained by experiments. The traditional method of determining mixing times is once again sampling followed by off-line analysis. The mixer is loaded and started. After the mixer has been loaded with the ingredients in accordance with a defined procedure, it is run and samples are taken from it at set time intervals. To do this the mixer usually has to be halted. The concentration of the tracer in the samples is established, and the random sample variance S^2 ascertained. This random sample variance serves as an estimated value for the variance σ_p^2 , which defines the mixture's condition. All analyses are burdened by errors, and this is expressed in a variance σ_m^2 derived from the sampling itself and from the analysis procedure. Initially there is a sharp fall in the random sample variance, and it runs asymptotically toward a final value of σ_E^2 as the mixing time increases. This stationary end value σ_E^2 is set by the variance of the mix in the stationary condition σ_Z^2 , for which the minimum would be the variance of an ideal random mix, and the variance σ_M^2 caused by errors in the analyzing process. The **mixing time** denotes that period in which the experimental random sample variance S^2 falls within the confidence interval of the stationary final condition σ_E^2 . Two cases can be considered. In the first case with large measurement errors, σ_E^2 is determined by the analyzing process itself since for sufficient mixing time the mixing process's fluctuations in

its stationary condition are much smaller than those arising out of the analysis or $\sigma_p^2 \ll \sigma_M^2$. In this case, the mixing process can only be tracked at its commencement, where $\sigma_p^2 > \sigma_M^2$. The "mixing time" t_x obtained under these conditions does not characterize the process. In the second case where the measurement errors are small, or $\sigma_p^2 \gg \sigma_M^2$, the analyzing process is sufficiently accurate for the mixing process to be followed through to its stationary condition. This allows an accurate determination of the true mixing time t^* . The "mixing time" t_x obtained on the basis of an unsatisfactory analysis is always deceptively shorter than the true time t^* .

On-line Procedures Advances in sensor technology and data processing are enabling an increased number of procedures to be completely monitored using **on-line procedures**. The great leap forward from off-line to on-line procedures lies in the fact that the whole process of preparing and analyzing samples has been automated. As a result of this automation, the amount of collectible test data has risen considerably, thereby enabling a more comprehensive statistical analysis and, in ideal cases, even regulation of the process. On-line procedures in most cases must be precisely matched to the process, and the expense in terms of equipment and investment is disparately higher. The accuracy of laboratory analyses in the case of off-line procedures cannot be produced by using on-line processes. There are as yet few on-line procedures for chemically analyzing solids. Near-infrared spectrometers fitted with fiber-optic sensors are used solely in the field of foodstuffs and for identifying raw materials in the pharmaceuticals industry and have also been applied to mixtures [Phil Williams and Karl Norris (eds.), *Near-Infrared Technology in the Agricultural and Food Industries*, American Association of Cereal Chemists, St. Paul, Minn., 1987; R. Weinekötter, R. Davies, and J. C. Steichen, *Determination of the Degree of Mixing and the Degree of Dispersion in Concentrated Suspensions*, Proceedings of the Second World Congress, Particle Technology, pp. 239–247, September 19–22, 1990, Kyoto, Japan]. For pharmaceutical mixes the NIR method has been proposed for the control of mixing efficiency (A. Niemöller, *Conformity Test for Evaluation of Near Infrared Data*, Proc. Int. Meeting on Pharmaceuticals, Biopharmaceutics and Pharmaceutical Technology, Nuremberg, March 15–18, 2004). This method records the specific adsorption of groups of chemicals on a particle's surface. If these spectrometers are based on modern diode array technology, a spectrum covering the whole wave range is obtained in a fraction of a second.

Sampling Procedures The purpose of taking samples is to record the properties of the whole volume of material from a small, analyzed portion of it. This is difficult to achieve with solids since industrial mixes in particular always present a distribution of grain sizes, shape, or density and can also separate out when samples are being taken, on account of the ingredients' specific motional behavior (see the subsection "Sampling").

EQUIPMENT FOR MIXING OF SOLIDS

A wide variety of equipment is commercially available to suit a multiplicity of mixing tasks. In this overview mixers and devices for mixing solids are divided into four groups: (1) mixed stockpiles, (2) bunker mixers, (3) rotating mixers or mixers with rotating tools, and (4) direct mixing of feeding streams.

Mixed Stockpiles Many bulk goods that are often stored in very large stockpiles do not possess uniform material properties within these stockpiles. In the case of raw materials, this may be caused by natural variations in deposits; or in the case of primary material, by variations between different production batches. In the iron and steel industry, e.g., there are fluctuations in the ore and carbon content of the finished material. If these stockpiles are emptied in the "first-in, first-out" principle, material with a variance in properties will find its way into the subsequent process and reduce its efficiency. To provide a uniform finished material, a mix is obtained by following a defined scheme for building up and emptying large stockpiles (Fig. 21-48). Such mixing processes are also called *homogenization*. As in any mixing process, the volume of material is homogenized by moving portions of it relative to each other. A long

stockpile is built up by a movable conveyor belt or other corresponding device traveling *lengthwise*. During loading the belt continuously travels up and down the whole length. In the strata thereby created is stored a temporal record of the material's delivery. If the material is now systematically removed *crosswise* to these layers, each portion removed from the stockpile (Fig. 21-48) will contain material from all the strata and therefore from the times it was supplied. Since such bins are built up over days or weeks, mixed stockpiles reduce the degree of long-term fluctuations in the material's properties.

Bunker and Silo Mixers Bunker and silo mixers (Fig. 21-49) are sealed vessels, the biggest of which may likewise serve to homogenize large quantities of solids. They are operated batchwise, continuously or with partial recirculation of the mixture. Their sealed construction also enables material to be conditioned, e.g., humidified, granulated, dried, or rendered inert, as well as mixed. In gravity mixers, granular material is simultaneously drawn off by a system of tubes at various heights and radial locations, brought together, and mixed. Other types of construction use a central takeoff tube into which the solids travel through openings arranged at various heights up this pipe. If the quality of the mix does not meet requirements, the withdrawn material is fed back into the bunker (Eichler and Dau, *Geometry and Mixing of Gravity Discharge Silo Mixers*, The First European Congress on Chemical Engineering, Florence, Italy, 1997, *Proceedings*, 2, 971–974). In this fashion the bunker's entire contents are recirculated several times and thus homogenized. The material drawn off in most cases is carried to the top of the bunker by air pressure (using an external circulation system).

Gravity mixers are designed for free-flowing powders and are offered in sizes ranging between 5 and 200 m³. The specific energy consumption, i.e., the energy input per product mass, is very low at under 1 to 3 kWh/t. Silo screw mixers are silos with a special funnel mixer at their outlet and are grouped with the gravity mixers. A concentric double cone gives a different residence time period for the material in the inner and outer cones, inducing remixing. Such mixers are available for quantities of material between 3 and 100 m³. In the case of granulate mixers, material from various areas of the vessel is brought together in its lower section and then carried upward by air pressure in a central pipe (using an internal circulating system) where the solids are separated from the gas and at the same time distributed on the surface. Design sizes reach up to 600 m³, and the specific energy input, like that of gravity mixers, is low. The rotating screw of a **conical screw mixer** transports the material upward from the bottom. This screw is at the same time driven along the wall of the vessel by a swiveling arm. This type of mixer also processes both pastes and cohesive powders. The solids at the container wall are continuously replaced by the action of the screw so that the mix can be indirectly heated or cooled through the container's outer wall. It is also used for granulation and drying. Mixers of this design are offered in capacities of between 25 L and 60 m³. In **blast air** or **air jet mixers**, air is blown in through jets arranged around the circumference of a mixing head placed in the bottom of the vessel. The specific air consumption is 10 to 30 N·m³/t, and the largest mixers have a capacity of 100 m³. If a fluid flowing through a bed of particles against the force of gravity reaches a critical speed (minimum fluidization velocity), the particles become suspended or fluidized by the fluid (see Sec. 17, "Gas-Solid Operations and Equipment").

Through increased particle mobility, **fluidized beds** possess excellent mix properties for solids in both a vertical and radial axis. In **circulating fluidized beds** often used in reaction processes, this is combined with elevated heat transfer and material circulation as a result of the high relative velocities of the gas and solids. Lower fluidizing speeds to limit air consumption are generally used if the fluidized bed serves only the purpose of mixing. Furthermore, differing volumes of air are fed to the air-permeable segments installed in the container's floor which serve to distribute air. The largest fluidized bed mixers as used in cement making reach a capacity of 10⁴ m³. The material must be fluidizable, i.e., free-flowing (with a particle size greater than 50 µm), and dry. The specific



FIG. 21-48 Recovery of the fine homogenized coal by system Chevron (Central Coking Plant, Saar GmbH, Germany); width of the bridge scraper is 57.5 m; capacity is 1200 t/h. (Courtesy of PWH-Krupp Engineering.)

power input lies between 1 and 2 kWh/t, but air consumption rises sharply in the case of particle sizes above 500 μm . **Fluidized-bed granulators** utilize the mixing properties of fluidization for granulation, atomized fluid distribution, and drying (see "Size Enlargement Equipment: Fluidized-Bed Granulators").

Rotating Mixers or Mixers with Rotating Component

Figure 21-50 shows four categories of mixers where the mix is agitated by rotating the whole unit or where movement in the mix is produced by rotating components built into the apparatus. These mixers are classified according to their Froude number (Fr):

$$Fr = \frac{r\omega^2}{g} = \frac{rn^2 4\pi^2}{g} \quad (21-69)$$

Here r denotes the mixer's radius or that of the mixer's agitators, g the gravitational acceleration, and ω the angular velocity. The Froude number therefore represents a dimensionless rotating frequency. The Froude number is the relationship between centrifugal and gravitational acceleration. No material properties are accounted for in the Froude number: Subject to this limitation, a distinction is drawn in Fig. 21-50 between $Fr < 1$, $Fr > 1$, and $Fr \gg 1$. **Free-fall mixers** are only suitable for free-flowing solids. Familiar examples of free-fall units are drum mixers and V-blenders. However, as the solids are generally free-flowing, demixing and segregation may also occur, leading to complete separation of the ingredients. Since drums are also used in related processes such as rotary tubular kilns or granulating drums for solids, these processes may also be prone to size segregation. In some cases, this may even be intentional, such as

with rotating disc granulators common in iron ore processing. Despite these risks of segregation, mixers without built-in agitators are particularly widely used in the pharmaceuticals and foodstuffs industries since they can be cleaned very thoroughly. **Asymmetrically moved mixers** in which, e.g., a cylinder is tilted obliquely to the main axis, turning over the mix, also belong in the free-fall category, e.g., being fertilizer drum granulation processes. Mixing is done gently. Because of the material's distance from the central axis, high torques have to be applied by the drive motor, and these moments have to be supported by the mixer's bearings and bed. Units with a capacity of 5000 L are offered. There are also mixers with operating range $Fr < 1$ where the work of moving the mix is undertaken by rotating agitators. The particles of solids are displaced relative to one another by agitators inside the mixer. This design is suitable for both cohesive, moist products and those which are free-flowing. Examples of **displacement mixers** are ribbon blenders or paddle mixers. Because of their low rpm the load on the machine is slight, but the mixing process is relatively slow. The specific energy input is low and lies under 5 kW/m³.

Ploughshear and centrifugal mixers operate in a range with $Fr > 1$. The consequence is that, at least in the vicinity of the outer edge of the agitator, the centrifugal forces exceed that of gravity and the particles are spun off. Thus instead of a pushing motion there is a flying one. This accelerates the mixing process both radially and axially. If the ingredients still need to be disagglomerated, high-speed cutters are brought into the mixing space to disagglomerate the mix by impact. At very high Froude number ranges ($Fr > 7$) there is a sharp increase in the shear forces acting on the mix. The impact load is large and sufficient to heat the product as a result of

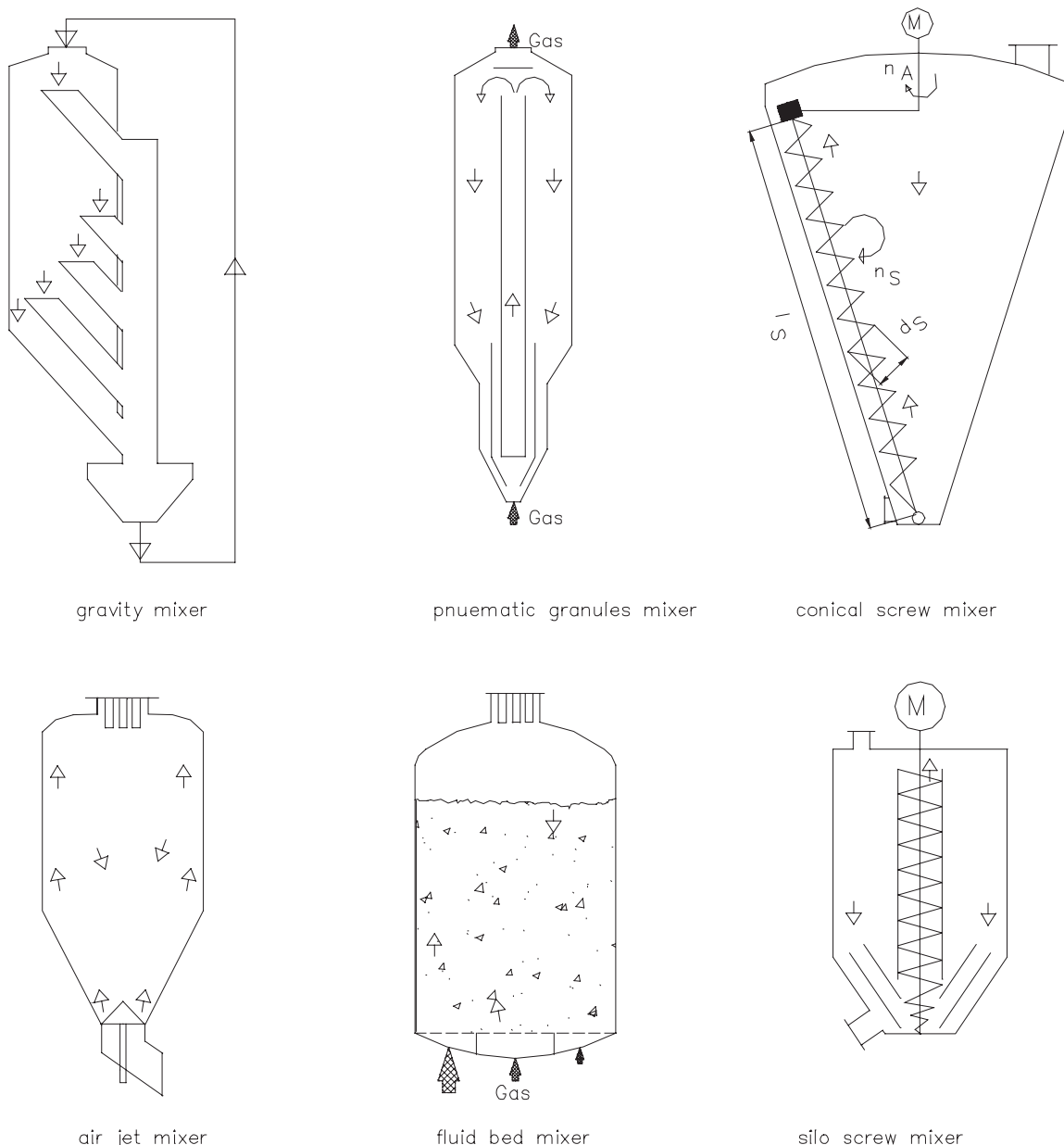


FIG. 21-49 Classification of bunker or silo mixers following Müller [W. Müller, Methoden und derzeitiger Kenntnisstand für Auslegungen beim Mischen von Feststoffen [Methods and the current state of the art in solids mixing configurations], *Chem. Eng.*, **53**, 831-844 (1981)].

dissipated energy. The heat is caused by friction between the mixer's tools and the solids as well as by friction among the solids' particles. As well as simple mixing, here the mixer's task is often disagglomeration, agglomeration, moistening, and sintering. Such mixers are especially used for producing plastics and in the pharmaceutical industry for granulation.

Mixing by Feeding Direct mixing of feed streams represents a continuous mixing process (Fig. 21-51). The solids are blended by metering in each ingredient and bringing these streams of solids together locally. There is no axial mixing (transverse or back mixing), or as such it is very low, with the result that the quality of the meter-

ing determines the mix's homogeneity. Metered feeder units should therefore ideally be used, preferably operated **gravimetrically** with appropriate feedback control of weight loss. According to the requirements of the case in question, mixing is also required obliquely to the direction of travel. If the ingredients are brought together in a perpendicular fall, this is achieved by their merging together. If this oblique mixing is not sufficient, **static mixers** can be used for free-flowing powders or granules where, e.g., the stream of solids is repeatedly divided up and brought back together by baffles as it drops down a tube. The energy input into the mixer is very low, but such systems need sufficient height to achieve mix quality.

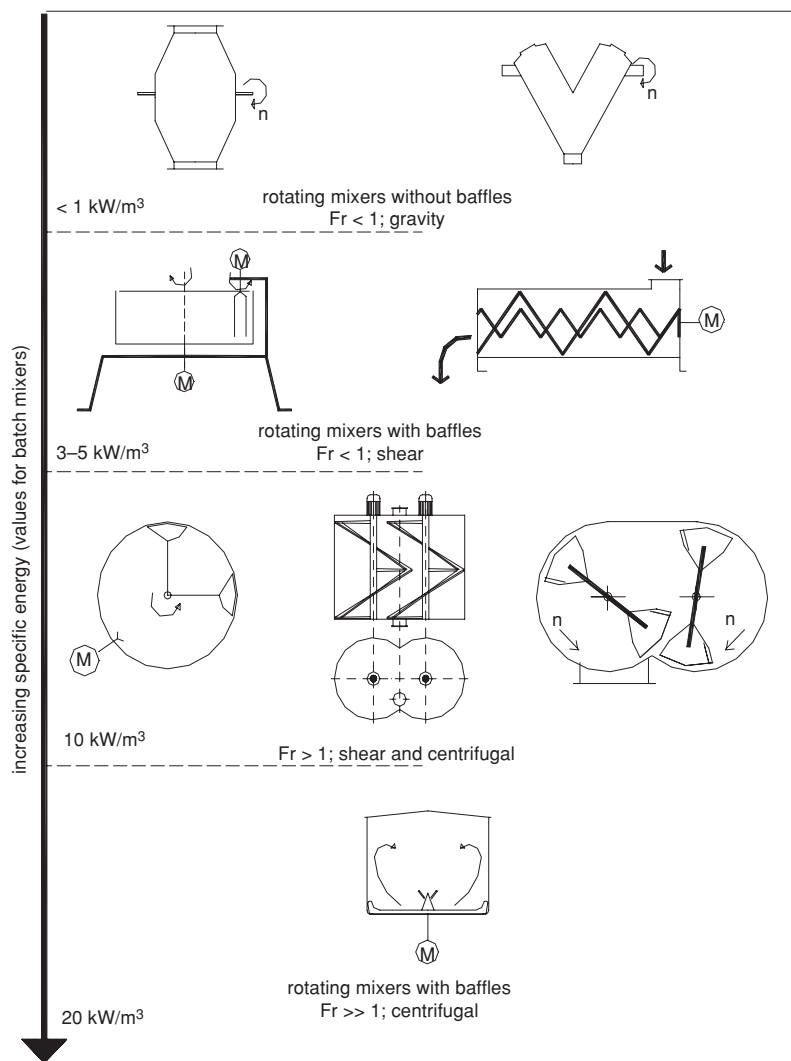


FIG. 21-50 Classification of mixers—movement of material by rotating agitators or revolving containers. [W. Müller, Methoden und derzeitiger Kenntnisstand für Auslegungen beim Mischen von Feststoffen [Methods and the current state of the art in solids mixing configurations], *Chem. Eng.*, **53**, 831–844 (1981)].

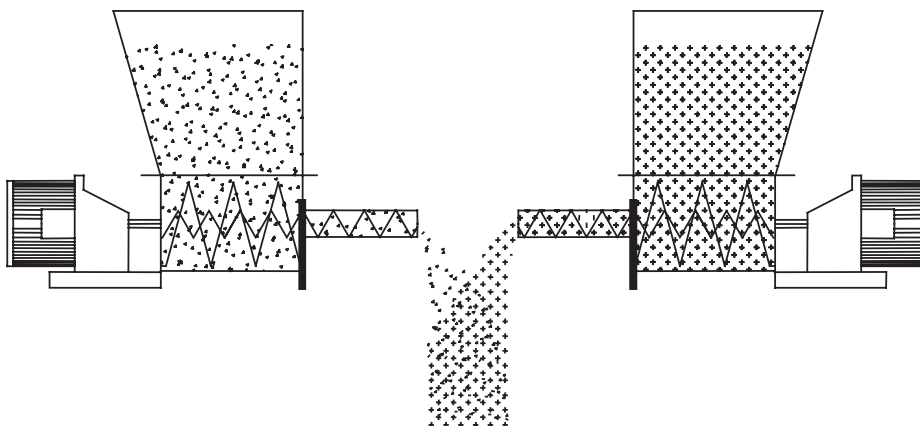


FIG. 21-51 Direct mixing of feeder streams.

It was shown that the efficiency for radial mixing depends on the gas phase as well (O. Eichstädt, Continuous Mixing of Fine Particles within Fluid Dynamic Vertical Tube Mixers, Dissertation, in German, ETH-Zurich, 1997). At best they operate with low volume concentration and for particles between 20 and 200 μm . Static mixers have been used for very abrasive free-flow materials such as silicon carbide. Since any rotating equipment is avoided inside static mixers, abrasion is limited. As will be shown below, mixture quality is dependent on feed consistency and residence time within the static mixer. Since the latter is very short in static mixers (seconds or fractions of a second), short-time feeding precision has to be very high to achieve high-quality mix.

DESIGNING SOLIDS MIXING PROCESSES

Goal and Task Formulation An essential prerequisite for the efficient design of a mixing process is a clear, exact, and comprehensive formulation of the task and objective. Applying Table 21-7a as a checklist guarantees a systematic formulation of the *mixing task* along with the major *formative conditions*. Priority *objectives* covering the economic requirements, quality targets, and operating conditions have to be met when one is engineering a mixing system. Besides a definition of the stipulated quality of the mix and an average production throughput (minimum or maximum), the *quality target* can include additional physical (moisture, grain size, temperature) and chemical properties required of the mixed product. Furthermore, the general principles of quality assurance frequently demand production documentation. This means that material batches must be coded, mixture recipes recorded, and the flow of materials in and out balanced out against their inventories and consumption. Clearly *formative economic conditions* such as investment, maintenance requirements, and utilization of existing space often determine the actual technical features of a design when it is put into practice. Specifications arising from the mixing system's operation are grouped under *formative operating conditions*. These set the requirements on

- Staff numbers and training
- Process monitoring, process management system design, and the degree of automation
- Operating, cleaning, and maintenance
- Safety, dust, explosion, and emission protection and the alarm system

Sometimes raw material costs exceed the processing cost by far; or manufacturing contributes a negligible part of the overall cost; e.g., the marketing and R&D determine the manufacturing cost of a newly patented pharmaceutical product.

The Choice: Mixing with Batch or Continuous Mixers Mixing processes can be designed as a batch or a continuous process.

TABLE 21-7a Checklist for Formulating a Mixing Task

<i>Mix recipes (mixture composition)</i>	
• Number and designation of the recipes • The preparation's composition (the ingredients' percentages and margins of accuracy to be observed, particularly in the case of low-dosage ingredients) • The percentage of each recipe as part of the total production output • The frequency with which the recipe is changed and any desired sequence • Cleaning operations when a recipe is changed [dry, wet, cleaning in place (CIP)]	
• Sampling and analyses	
<i>Ingredients</i>	
• Designation • Origin, supplier, packaging • Bulk density, solids density • Grain size (grain size distribution) and shape • Flow properties, gradient • Abrasiveness • Moistness (damp, hygroscopic, dry) • Temperature, sensitivity to thermal stress • Sensitivity to mechanical stress (crushing, abrasion, fracture)	
<i>Product (mixture)</i>	
• Mix quality • Bulk density • Fluidizability (air take-up during mixing) • Tendency to segregation • The mix's flow properties • Agglomeration, disagglomeration required	
<i>The mixer performance</i>	
• Mix performance: production volume per unit of production (average, minimum, maximum) • For batch mixers: Batch mix size (final volume after mixing); start-up filling level; the filled mixer's idle time • For continuous mixers: The production volume with an unchanged recipe; feed/mix output tolerance range	
<i>Integrating the mixers into the system</i>	
• Material flow diagram (average, maximum, and minimum figures) • The ingredients' inflow and outflow • Spatial requirements, height, layout • The mixture's usage • Storing, feeding, and weighing devices • The type of process inspection, process control, storage, and data exchange • Safety requirements	
<i>Mixer design</i>	
• Raw material, surfaces, and the inflow and outflow configuration • Heating, cooling, inertizing, pressurization, vacuum • The addition of liquid into the mixer • Disagglomeration • Current, steam, and water connections, adjuncts, types of protection, protection against explosion	
<i>Formative economic conditions</i>	
• Investment costs • Maintenance, running, and staff costs • Profitability	

TABLE 21-7b Comparison of Discontinuous and Continuous Mixing Processes

Implementation data	Discontinuous	Continuous
The number of ingredients	As many as wanted	2–10; any more ingredients are usually combined in a premix
Frequency with which the recipe is changed	Several times per hour	A recipe must remain unchanged for several hours
Cleaning frequency or idle time	Several times a day	Once a day or less
Production output, throughput	Any rate	More than 100 kg/h. Exception: feeding laboratory extrusions
Risk of separation	Present, therefore there must be short transportation paths, few intermediate silos	Low risk when the material is taken directly to the next processing stage or directly drawn off
Spatial requirement	Large amount of space and intermediate silos required for machines with a throughput greater than 5000 kg/h	Low spatial requirement even for machines with a high throughput
Requirements placed on the equipment	Simple feeding but high demands on the mixer	Accurate continuous feeding (feeding scales necessary) but low demands on the mixer
Safety	Steps have to be taken in the case of materials with a risk of explosion	The small quantities of material present during processing have a low potential risk, which simplifies safety design
Automation	Variable degree of automation	Contained in the processing



FIG. 21-52 Classical automated batch mixing installation. The components are stored in small silos shown at the top of picture. The materials are extracted from these hoppers in a downstream weighing hopper according to the recipe. Once all components are fed into this weighing hopper, a valve is opened and the exact batch falls into the downstream batch mixer.

Table 21-7b gives a detailed comparison of discontinuous and continuous mixing processes, to help guide the selection of a mixing method.

Batch Mixing Batch or discontinuous mixing is characterized by the fact that the mixer is filled with the ingredients, and after a certain **mixing time** the mixture is discharged. The feeding (or filling), mixing, and discharging operations are performed *one after the other*. Batch processing presents advantages for small quantities of material because of its lower investment costs and greater flexibility. Batch mixers are used even when very large volumes of material are being homogenized since continuous mixers are limited by their lower volume. However, in the batch mixer's very flexibility lies the danger that it is not being optimally utilized. For example, overmixing can occur, whereby the product could be damaged and the process's effectiveness suffers.

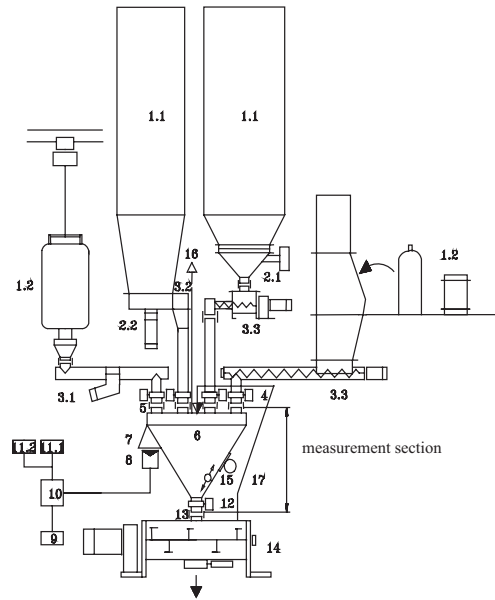


FIG. 21-53 Weighing hopper with additive weighing for feeding a batch mixer. 1.1 Storage silos; 1.2 big bag, bag, drum; 2.1-2.2 dischargers; 3.1-3.3 feeder units; 4 cutoff; 5 flexible connections; 6 weighing hopper; 7 support for gravity force; 8 gravity-operated sensor (load cell); 9 set point; 10 weighing analysis and regulation; 11.1 measured value indicator or output; 11.2 recorder (printer); 12 cutoff; 13 flexible connection; 14 mixer; 15 discharger; 16 dust extraction and weighing hopper ventilation; 17 mixer ventilation.

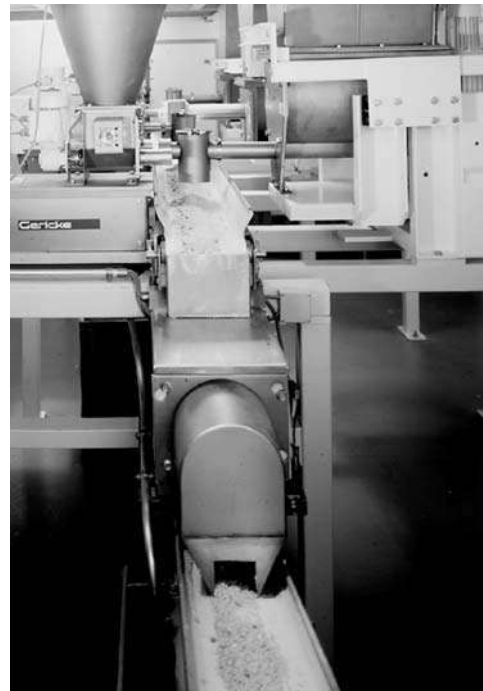


FIG. 21-54 Continuous mixing for the production of Muesli: Continuous gravimetric solids feeder (loss-in-weight feeding) supplies the components (raisins, flakes, etc.) at constant rate onto a belt, which delivers the components to the continuous mixer (bottom of the picture). The continuous mixer discharges onto a second belt.

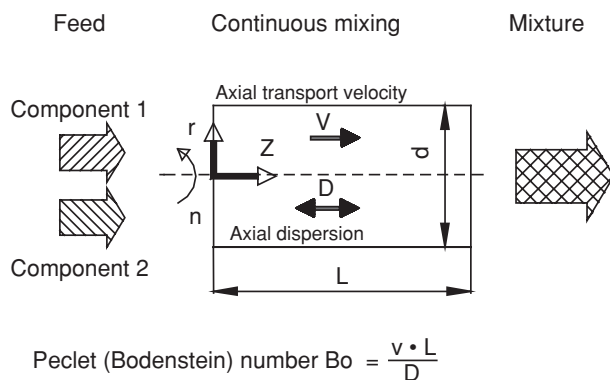


FIG. 21-55 The continuous mixing of two ingredients: Axial mixing or dispersion shows up as well as residence time distribution of the product inside the mixer.

Feeding and Weighing Equipment for a Batch Mixing Process The number of mix cycles multiplied by the usable mixer capacity gives the set mixture output per hour. The mix cycle consists of the filling, mixing, discharge, and idle times (Fig. 21-52). To this is added in special cases the time taken for sampling and analysis and that for associated processes such as disagglomeration and granulation. The capacity (throughput rate) of a batch mixing process having a mixture charge with a mass M is shown in Eq. (21-70):

$$\dot{m} = \frac{M}{t_f + t_m + t_d + t_i} \quad \frac{\text{kg}}{\text{s}} \quad (21-70)$$

The mixing time t_m depends on the selected mixer design and size, the filling time t_f on the system's configuration, while the discharge time t_d depends on both the mixer's design and the system's layout. The choice of feed and weighing devices is determined by the number of ingredients, their mass and proportions, the throughput volume, the stocking and mode of delivery, the spatial circumstances, degree of automation, etc. In the simplest case the ingredients are manually weighed into the mixer. In some cases, *sandwiching* of specific ingredients may be desirable, i.e., staged delivery of multiple layers of key ingredients between other excipients. Where there are higher

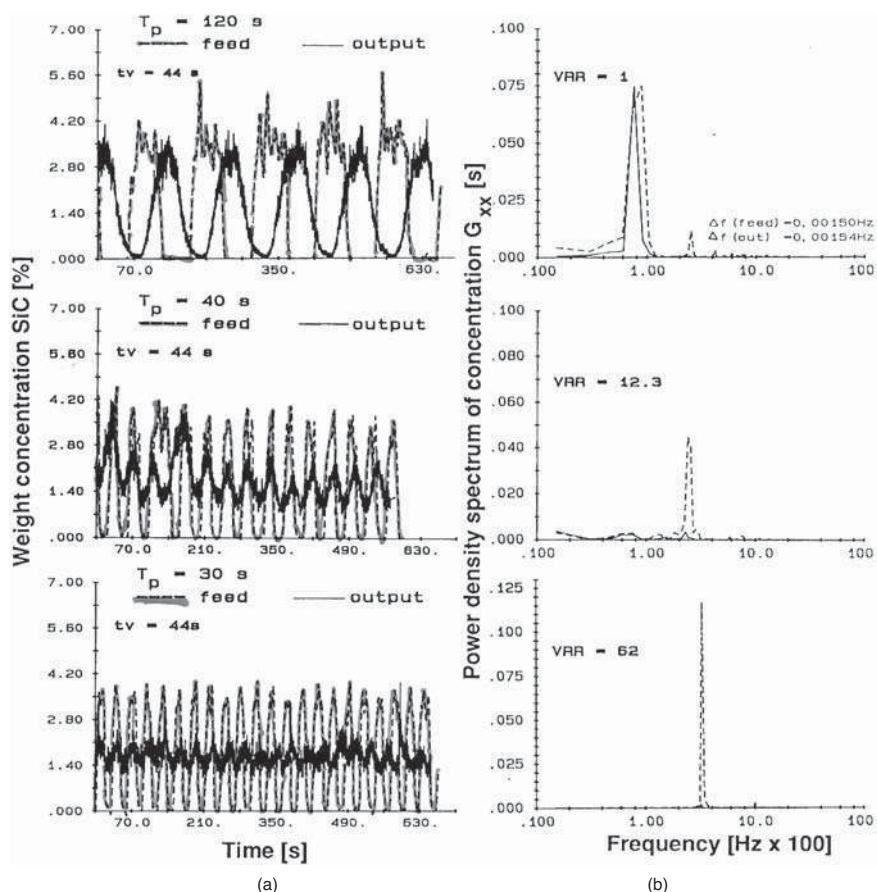


FIG. 21-56 Dampening of feed fluctuation in a continuous mixer—variance reduction ratio (VRR). The efficiency of continuous mixing processes is described by the variance reduction ratio. The variances in concentration of inlet and outlet are compared. Tracer-feed oscillating with different periods T_p , main component feed at constant rate (20 g/s), mean residence time in the continuous mixer $t_c = 44$ s. (a) Variation in time of SiC concentration: dotted line at the entrance of the continuous mixer, bold line at the outlet of the continuous mixer. (b) Power density spectrum of SiC concentration. High variance reduction ratios are achieved if the period of the tracer feed is small compared to the mean residence time in the mixer.

requirements in respect of accuracy, safety, and recording, a hopper scale represents a simple device for weighing and releasing the components into the mixing equipment (Fig. 21-53).

Continuous Mixing In a **continuous mixing** process (compare Figs. 21-52 and 21-54) the ingredients are continuously fed into the mixer, then mixed and prepared for the next processing stage. The operations of feeding, mixing, and discharging follow each other locally but occur simultaneously. In continuous mixing, the weighing and filling of a batch mixer are replaced by the ingredients' controlled continuous addition. The blending time in a continuous mixer is in fact the material's **residence time**, which is determined by the feed rate to the mixer. Losses of product during start-up or shutdown added to this lower degree of flexibility come as further disadvantages of the continuous process. Yet it possesses considerable advances over batch processing both in financial terms and in respect of process control: Even high-throughput continuous mixers are compact. A smaller-volume scale provides short mixing paths and ease of mixing. When integrated into a continuous production system, a continuous mixing process saves on reservoirs or silos and automating the course of the process is simplified. In the case of dangerous products or base materials, there is less potential risk with a continuous process since only a small quantity of material accumulates in the mixer. Segregation can be limited in a continuous mixer by its smaller required scale. A continuous mixer, which on account of its compact construction can be positioned before the next station in the processing chain, guarantees that a mix of a higher quality will in fact be made available to that next stage of the process, with smaller material handling distances.

The continuous mixer has principally two tasks (Fig. 21-55): The ingredients, which in an extreme case arrive in the mixer side by side, have to be **radially mixed** (*r*). In this case *radial* means lateral to the direction of the material's conveyance into the mixer. If in addition there are large feed rate fluctuations or the ingredients are themselves unho-

mogenized, the mixer must also minimize any differences in concentration in an axial direction (*z*), i.e., in the direction of the material's conveyance, or the mixture must be **axially mixed** as well. If a mixer only has to perform its task radially, it can have a very compact structure, since slim-line mixers with a high rpm very quickly equalize concentrations radially over short mixing paths. Feed fluctuations (Fig. 21-56) are damped by the residence time distribution of the material inside the mixer [R. Weinekötter and L. Reh, *Continuous Mixing of Fine Particles, Part. Part. Syst. Charact.*, **12**, 46–53 (1995)]. The residence time distribution describes the degree of axial dispersion occurring in the mixer. The Peclet (=Bodenstein) number *Bo* (Fig. 21-55) characterizes the ratio of axial transport velocity and axial dispersion coefficient *D*. The capability to reduce incoming fluctuations (thus variance) inside continuous mixers depends on the ratio of period of entrance fluctuation to the mean residence time as well as the residence time distribution. Besides the number of ingredients in the mix, a decisive feature in selecting the process is the individual component's flow volumes. Since the feed's constancy can only be maintained with a limited degree of accuracy at continuous feeding rates below 300 g/h, ingredients with low flow volumes necessitate a premixing operation. There is an increasing trend toward continuous mixing installations. Widely used are continuous processes in the plastics industry, detergents, and foodstuffs. Although less common, **pharmaceutical processes utilizing continuous mixing are growing in appeal** due to the small volume of the apparatus. The U.S. Food and Drug Administration, e.g., has promoted a **Process Analytical Technology (PAT) Initiative** with the objective of facilitating continuous processing to improve efficiency and manage variability (<http://www.fda.gov/cder/ops/Pat.htm>; Henry Berthiaux et al., *Continuous Mixing of Pharmaceutical Powder Mixtures*, 5th World Congress on Particle Technology, 2006; Marcos Llusa and Fernando Muzzio, *The Effect of Shear Mixing on the Blending of Cohesive Lubricants and Drugs*, *Pharmaceutical Technol.*, Dec. 2005).

PRINCIPLES OF SIZE REDUCTION

GENERAL REFERENCES: Annual reviews of size reduction, *Ind. Eng. Chem.*, October or November issues, by Work from 1934 to 1965, by Work and Snow in 1966 and 1967, and by Snow in 1968, 1969, and 1970; and in *Powder Technol.*, **5**, 351 (1972), and **7** (1973); Snow and Luckie, **10**, 129 (1973), **13**, 33 (1976), **23**(1), 31 (1979). *Chemical Engineering Catalog*, Reinhold, New York, annually. Cremer-Davies, *Chemical Engineering Practice*, vol. 3: *Solid Systems*, Butterworth, London, and Academic, New York, 1957. *Crushing and Grinding: A Bibliography*, Chemical Publishing, New York, 1960. European Symposia on Size Reduction: 1st, Frankfurt, 1962, publ. 1962, Rumpf (ed.), *Verlag Chemie*, Düsseldorf; 2d, Amsterdam, 1966, publ. 1967, Rumpf and Pietsch (eds.), *DECHEMA-Monogr.*, **57**; 3d, Cannes, 1971, publ. 1972, Rumpf and Schönert (eds.), *DECHEMA-Monogr.*, **69**. Gaudin, *Principles of Mineral Dressing*, McGraw-Hill, New York, 1939. International Mineral Processing Congresses: *Recent Developments in Mineral Dressing*, London, 1952, publ. 1953, Institution of Mining and Metallurgy; *Progress in Mineral Dressing*, Stockholm, 1957, publ. London, 1960, Institution of Mining and Metallurgy; 6th, Cannes, 1962, publ. 1965, Roberts (ed.), Pergamon, New York; 7th, New York, 1964, publ. 1965, Arbiter (ed.), vol. 1: *Technical Papers*, vol. 2: *Milling Methods in the Americas*, Gordon and Breach, New York; 8th, Leningrad, 1968; 9th, Prague, 1970; 10th, London, 1973; 11th, Cagliari, 1975; 12th, São Paulo, 1977. Lowrison, *Crushing and Grinding*, CRC Press, Cleveland, Ohio, 1974. *Pit and Quarry Handbook*, Pit & Quarry Publishing, Chicago, 1968. Richards and Locke, *Text Book of Ore Dressing*, 3d ed., McGraw-Hill, New York, 1940. Rose and Sullivan, *Ball, Tube and Rod Mills*, Chemical Publishing, New York, 1958. Snow, *Bibliography of Size Reduction*, vols. 1 to 9 (an update of the previous bibliography to 1973, including abstracts and index). U.S. Bur. Mines Rep. SO122069, available IIT Research Institute, Chicago, Ill. 60616. Stern, Guide to Crushing and Grinding Practice, *Chem. Eng.*, **69**(25), 129 (1962). Taggart, *Elements of Ore Dressing*, McGraw-Hill, New York, 1951. Since a large part of the literature is in German, availability of English translations is important. Translation numbers cited in this section refer to translations available through the National Translation Center, Library of Congress, Washington, D.C. Also, volumes of selected papers in English translation are available from the Institute for Mechanical Processing Technology, Karlsruhe Technical University, Karlsruhe, Germany.

INTRODUCTION

Industrial Uses of Grinding Grinding operations are critical to many industries, including mining cement manufacture, food processing, agricultural processes, and many chemical industries. Nearly every solid material undergoes size reduction at some point in its processing cycle. Grinding equipment is used both to reduce the size of a solid material by fracture and to intimately mix materials, *usually a solid and a liquid (dispersion)*.

Some of the common reasons for size reduction are to liberate a desired component for subsequent separation, as in separating ores from gangue; to prepare the material for subsequent chemical reaction, i.e., by enlarging the specific surface as in cement manufacture; to meet a size requirement for the quality of the end product, as in fillers or pigments for paints, plastics, agricultural chemicals, etc.; and to prepare wastes for recycling.

Types of Grinding: Particle Fracture vs. Deagglomeration There are two primary types of size reduction that occur in grinding equipment: deagglomeration and particle fracture. In deagglomeration, an aggregate of smaller particles (often with a fractal structure) is size-reduced by breaking clusters of particles off the main aggregate without breaking any of the "primary particles" that form the aggregates. In particle fracture, individual particles are broken rather than simply separating individual particles. Most operations involving particles larger than 10 μm (including materials thought of as rocks and stones) usually involve at least some particle fracture, whereas finer grinding is often mostly deagglomeration. At similar particle scales, deagglomeration requires much less energy than particle fracture. For example, fracture of materials down to a size of 0.1 μm is extremely difficult, whereas deagglomeration of materials in this size range is commonly practiced

in several industries, including the automotive paint industry and several electronics industries.

Wet vs. Dry Grinding Grinding can occur either wet or dry. Some devices, such as ball mills, can be fed either slurries or dry feeds. In practice, it is found that finer size can be achieved by wet grinding than by dry grinding. In wet grinding by media mills, product sizes of 0.5 μm are attainable with suitable surfactants, and deagglomeration can occur down to much smaller sizes. In dry grinding, the size in ball mills is generally limited by ball coating (Bond and Agthe, *Min. Technol.*, AIME Tech. Publ. 1160, 1940) to about 15 μm . In dry grinding with hammer mills or ring-roller mills, the limiting size is about 10 to 20 μm . Jet mills are generally limited to a mean product size of 10 μm . However, dense particles can be ground to 2 to 3 μm because of the greater ratio of inertia to aerodynamic drag. Dry processes can sometimes deagglomerate particles down to about 1 μm .

Typical Grinding Circuits There are as many different configurations for grinding processes as there are industries that use grinding equipment; however, many processes use the circuit shown in Fig. 21-57a. In this circuit a process stream enters a mill where the particle size is reduced; then, upon exiting the mill, the stream goes to some sort of classification device. There a stream containing the oversized particles is recycled back to the mill, and the product of desired size exits the circuit. Some grinding operations are simply one-pass without any recycler or classifier. For very fine grinding or dispersion (under 1 μm), classifiers are largely unavailable, so processes are either single-pass or recirculated through the mill and tested off-line until a desired particle size is obtained.

The fineness to which a material is ground has a marked effect on its production rate. Figure 21-57b shows an example of how the capacity decreases while the specific energy and cost increase as the product is ground finer. Concern about the rising cost of energy has led to publication of a report on this issue [National Materials Advisory Board, *Comminution and Energy Consumption*, Publ. NMAB-364, National Academy Press, Washington, 1981; available from National Technical Information Service, Springfield, Va. 22151]. This has shown that U.S. industries use approximately 32 billion kWh of electrical energy per annum in size-reduction operations. More than one-half of this energy is consumed in the crushing and grinding of minerals, one-quarter in the production of cement, one-eighth in coal, and one-eighth in agricultural products.

THEORETICAL BACKGROUND

Introduction The theoretical background for size reduction is often introduced with particle breakage (or, equivalently, droplet

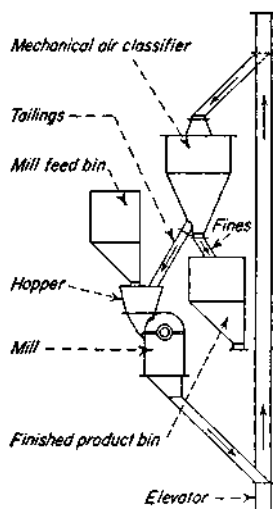


FIG. 21-57a Hammer mill in closed circuit with an air classifier.

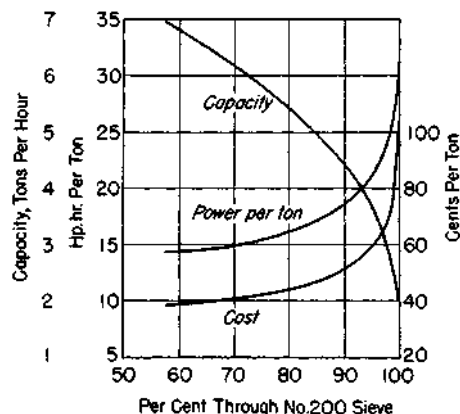


FIG. 21-57b Variation in capacity, power, and cost of grinding relative to fineness of product.

breakup for liquid-liquid system and bubble breakup for gas-liquid systems). It is relatively easy to write down force balances around a particle (or droplet) and make some predictions about how particles might break. Of particular interest in size reduction processes are predictions about the size distribution of particles after breakage and the force/energy required to break particles of a given size, shape, and material.

It has, however, proved difficult to relate theories of particle fracture to properties of interest to the grinding practitioner. This is so, in part, because single particle testing machines, although they do exist, are expensive and time-consuming to use. To get any useful information, many particles must be tested, and it is unclear that these tests reflect the kind of forces encountered in a given piece of grinding equipment. Even if representative fracture data can be obtained, this information needs to be combined with information on the force distribution and particle mechanics inside a particular grinding device to be useful for scale-up or predicting the effectiveness of a device. Most of this information (force distribution and particle motion inside devices) has not been studied in detail from either a theoretical or an empirical point of view, although this is beginning to change with the advent of more powerful computers combined with advances in numerical methods for fluid mechanics and discrete element models.

The practitioner is therefore limited to scale-up and scale-down from testing results of geometrically similar equipment (see "Energy Required and Scale-up," below) and using models which treat the devices as empirical "black boxes" while using a variety of population balance and grind rate theories to keep track of the particle distributions as they go into and out of the mills (see "Modeling of Milling Circuits," below).

Single-Particle Fracture The key issue in all breakage processes is the creation of a stress field inside the particle that is intense enough to cause breakage. The state of stress and the breakage reaction are affected by many parameters that can be grouped into both particle properties and loading conditions, as shown in Fig. 21-58.

The reaction of a particle to the state of stress is influenced by the material properties, the state of stress itself, and the presence of microcracks and flaws. Size reduction will start and continue as long as energy is available for the creation of new surface. The stresses provide the required energy and forces necessary for the crack growth on the inside and on the surface of the particle. However, a considerable part of the energy supplied during grinding will be wasted by processes other than particle breakage, such as the production of sound and heat, as well as plastic deformation.

The breakage theory of spheres is a reasonable approximation of what may occur in the size reduction of particles, as most size-reduction processes involve roughly spherical particles. An equation for the force required to crush a single particle that is spherical near the contact regions is given by the equation of Hertz (Timoshenko and Goodier, *Theory of*

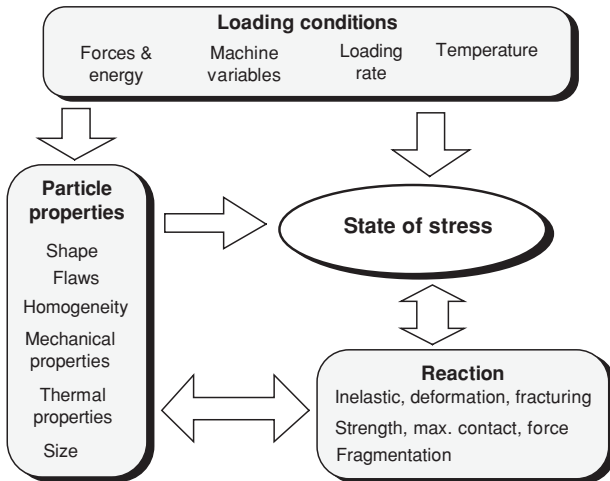


FIG. 21-58 Factors affecting the breakage of a particle. (After Heiskanen, 1995.)

Elasticity, 2d ed., McGraw-Hill, New York, 1951). In an experimental and theoretical study of glass spheres, Frank and Lawn [*Proc. R. Soc. (London)*, A299(1458), 291 (1967)] observed the repeated formation of ring cracks as increasing load was applied, causing the circle of contact to widen. Eventually a load is reached at which the ring crack deepens to form a cone crack, and at a sufficient load this propagates across the sphere to cause breakage into fragments. The authors' photographs show how the size of flaws that happen to be encountered at the edge of the circle of contact can result in a distribution of breakage strengths. Thus the mean value of breakage strength depends partly on intrinsic strength and partly on the extent of flaws present. Most industrial solids contain irregularities such as microscopic cracks and weaknesses caused by dislocations, nonstoichiometric composition, solid solutions, gas- and liquid-filled voids, or grain boundaries.

Inglis showed that these irregularities play a predominant role in particle breakage as the local stresses σ_i generated at the tips of the crack, as shown in Fig. 21-59, were much higher than the gross applied stress σ_N . The effect is expressed by stress concentration factor k

$$k = \frac{\sigma_i}{\sigma_N} = \frac{l}{r} \quad (21-71)$$

which is a function of the crack length l and the tip radius r .

Griffith found that tensile stresses always occur in the vicinity of crack tips, even when the applied gross stresses are compressive. He also showed that the largest tensile stresses are produced at cracks having a 30° angle to the compressive stress. Thus cracks play a key

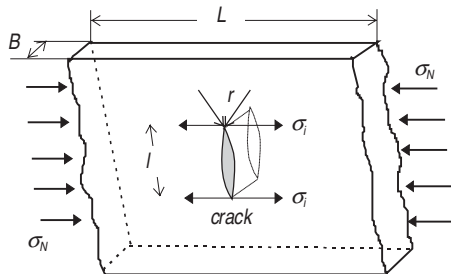


FIG. 21-59 A microcrack in an infinitely large plate.

role in propagation, and their effects greatly overshadow the theoretically calculated values for breakage of spheres or other ideal particles.

ENERGY REQUIRED AND SCALE-UP

Energy Laws Fracture mechanics expresses failure of materials in terms of both stress intensity and fracture toughness, in terms of energy to failure. Due to the difficulty of calculating the stresses on particles in grinding devices, many theoreticians have relied on energy-based theories to connect the performance of grinding devices to the material properties of the material being ground. In these cases, the energy required to break an ensemble of particles can be estimated without making detailed assumptions about the exact stress state of the particles, but rather by calculating the energy required to create fresh surface area with a variety of assumptions.

A variety of energy laws have been proposed. These laws are encompassed in a general differential equation (Walker et al., *Principles of Chemical Engineering*, 3d ed., McGraw-Hill, New York, 1937):

$$dE = -C dX/X^n \quad (21-72)$$

where E is the work done, X is the particle size, and C and n are constants.

For $n = 1$ the solution is Kick's law (Kick, *Das Gesetz der proportionalen Widerstande und seine Anwendung*, Leipzig, 1885). The law can be written

$$E = C \log (X_F/X_P) \quad (21-73)$$

where X_F is the feed-particle size, X_P is the product size, and X_F/X_P is the reduction ratio. For $n > 1$ the solution is

$$E = \left(\frac{C}{n-1} \right) \left(\frac{1}{X_P^{n-1}} - \frac{1}{X_F^{n-1}} \right) \quad (21-74)$$

For $n = 2$ this becomes *Rittinger's law*, which states that the energy is proportional to the new surface produced (Rittinger, *Lehrbuch der Aufbereitungskunde*, Ernst and Korn, Berlin, 1867).

The *Bond law* corresponds to the case in which $n = 1.5$ [Bond, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **193**, 484 (1952)]:

$$E = 100E_i \left(\frac{1}{\sqrt{X_P}} - \frac{1}{\sqrt{X_F}} \right) \quad (21-75)$$

where E_i is the **Bond work index**, or work required to reduce a unit weight from a theoretical infinite size to 80 percent passing $100 \mu\text{m}$. Extensive data on the work index have made this law useful for rough mill sizing especially for ball mills. Summary data are given in Table 21-8. The work index may be found experimentally from laboratory crushing and grinding tests or from commercial mill operations. Some rules of thumb for extrapolating the work index to conditions different from those measured are that for dry grinding the index must be increased by a factor of 1.34 over that measured in wet grinding; for open-circuit operations another factor of 1.34 is required over that measured in closed circuit; if the product size X_P is extrapolated below $70 \mu\text{m}$, an additional correction factor is $(10.3 + X_P)/1.145X_P$. Also for a jaw or gyratory crusher, the work index may be estimated from

$$E_i = 2.59C_s/\rho_s \quad (21-76)$$

where C_s = impact crushing resistance, (ft-lb)/in of thickness required to break; ρ_s = specific gravity, and E_i is expressed in kWh/ton.

The relation of energy expenditure to the size distribution produced has been thoroughly examined [Arbiter and Bhrany, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **217**, 245–252 (1960); Harris, *Inst. Min. Metall. Trans.*, **75**(3), C37 (1966); Holmes, *Trans. Inst. Chem. Eng. (London)*, **35**, 125–141 (1957); and Kelleher, *Br. Chem. Eng.*, **4**, 467–477 (1959); **5**, 773–783 (1960)].

The energy laws have not proved very successful in practice, most likely because only a very small amount of energy used in milling

TABLE 21-8 Average Work Indices for Various Materials*

Material	No. of tests	Specific gravity	Work index†	Material	No. of tests	Specific gravity	Work index†
All materials tested	2088	—	13.81	Taconite	66	3.52	14.87
Andesite	6	2.84	22.13	Kyanite	4	3.23	18.87
Barite	11	4.28	6.24	Lead ore	22	3.44	11.40
Basalt	10	2.89	20.41	Lead-zinc ore	27	3.37	11.35
Bauxite	11	2.38	9.45	Limestone	119	2.69	11.61
Cement clinker	60	3.09	13.49	Limestone for cement	62	2.68	10.18
Cement raw material	87	2.67	10.57	Manganese ore	15	3.74	12.46
Chrome ore	4	4.06	9.60	Magnesite, dead burned	1	5.22	16.80
Clay	9	2.23	7.10	Mica	2	2.89	134.50
Clay, calcined	7	2.32	1.43	Molybdenum	6	2.70	12.97
Coal	10	1.63	11.37	Nickel ore	11	3.32	11.88
Coke	12	1.51	20.70	Oil shale	9	1.76	18.10
Coke, fluid petroleum	2	1.63	38.60	Phosphate fertilizer	3	2.65	13.03
Coke, petroleum	2	1.78	73.80	Phosphate rock	27	2.66	10.13
Copper ore	308	3.02	13.13	Potash ore	8	2.37	8.88
Coral	5	2.70	10.16	Potash salt	3	2.18	8.23
Diorite	6	2.78	19.40	Pumice	4	1.96	11.93
Dolomite	18	2.82	11.31	Pyrite ore	4	3.48	8.90
Emery	4	3.48	58.18	Pyrrhotite ore	3	4.04	9.57
Feldspar	8	2.59	11.67	Quartzite	16	2.71	12.18
Ferrochrome	18	6.75	8.87	Quartz	17	2.64	12.77
Ferromanganese	10	5.91	7.77	Rutile ore	5	2.84	12.12
Ferrosilicon	15	4.91	12.83	Sandstone	8	2.68	11.53
Flint	5	2.65	26.16	Shale	13	2.58	16.40
Fluorspar	8	2.98	9.76	Silica	7	2.71	13.53
Gabbro	4	2.83	18.45	Silica sand	17	2.65	16.46
Galena	7	5.39	10.19	Silicon carbide	7	2.73	26.17
Garnet	3	3.30	12.37	Silver ore	6	2.72	17.30
Glass	5	2.58	3.08	Sinter	9	3.00	8.77
Gneiss	3	2.71	20.13	Slag	12	2.93	15.76
Gold ore	209	2.86	14.83	Slag, iron blast furnace	6	2.39	12.16
Granite	74	2.68	14.39	Slate	5	2.48	13.83
Graphite	6	1.75	45.03	Sodium silicate	3	2.10	13.00
Gravel	42	2.70	25.17	Spodumene ore	7	2.75	13.70
Gypsum rock	5	2.69	8.16	Syenite	3	2.73	14.90
Ilmenite	7	4.27	13.11	Tile	3	2.59	15.53
Iron ore	8	3.96	15.44	Tin ore	9	3.94	10.81
Hematite	79	3.76	12.68	Titanium ore	16	4.23	11.88
Hematite—specular	74	3.29	15.40	Trap rock	49	2.86	21.10
Oolitic	6	3.32	11.33	Uranium ore	20	2.70	17.93
Limanite	2	2.53	8.45	Zinc ore	10	3.68	12.42
Magnetite	83	3.88	10.21				

*Allis-Chalmers Corporation.

†Caution should be used in applying the average work index values listed here to specific installations since individual variations between materials in any classification may be quite large.

devices is actually used for breakage. A great deal of energy input into a mill is used to create noise and heat as well as simply move the material around the device. Although few systematic studies have been done, less (often, much less) than 5 percent of the energy input into a typical grinding device actually goes into breaking the material. The majority of the remaining energy is eventually converted to frictional heat, most of which heats up the product and the mill.

Mill efficiency can be judged in terms of energy input into the device as compared to the particle size achieved for a given material. It is rare that one grinding device will be more than twice as energy-efficient as another device in order to achieve the same particle size for the same material, and there are usually other tradeoffs for the more energy-efficient device. In particular, more energy-efficient devices have a tendency to have large, heavy mechanical components that cause great damage to equipment when moved, swung, etc. These, however, tend to be much more costly for the same capacity and harder to maintain than smaller, high-speed devices. For example, for many materials, roll mills are more energy-efficient than hammer mills, but they are also significantly more costly and have higher maintenance costs.

Fine Size Limit (See also “Single-Particle Fracture” above.) It has long been thought that a limiting size is attainable, and, in fact, it is almost a logical necessity that grinding cannot continue down to the molecular level. Nonetheless, recent results suggest that stirred

media mills are capable of grinding many materials down to particle sizes near 100 nm, finer than many predicted limits [see, e.g., S. Mende et al., *Powder Tech.*, **132**, 64–73 (2003) or F. Stenger et al., *Chem. Eng. Sci.*, **60**, 4557–4565 (2005)]. The requirements to achieve these sizes are high-energy input per unit volume, very fine media, a slurry formulated with dispersants designed to prevent deagglomeration of the very fine particles, and a great deal of energy and time. With improved technology and technique, finer grinds than ever before are being achieved, at least on the laboratory scale. The energy requirements of these processes are such that it is unlikely that many will be cost-effective. From a practical point of view, if particles much under 1 μm are desired, it is much better to synthesize them close to this size than to grind them down.

Breakage Modes and Grindability Different materials have a greater or lesser ease of grinding, or grindability. In general, soft, brittle materials are easier to grind than hard or ductile materials. Also, different types of grinding equipment apply forces in different ways, and this makes them more suited to particular classes of materials. Figure 21-60 lists the modes of particle loading as they occur in industrial mills. This loading can take place either by slow compression between two planes or by impact against a target. In these cases the force is normal to the plane. If the applied normal forces are too weak to affect the whole of the particle and are restricted to a partial volume at the surface of the particle, the mode is attrition. An alternative way



	COMPRESSION	IMPACT	ATTRITION	ABRASION
<u>COARSE</u>				
crushers	XX			
hammer crusher		XX		
<u>MEDIUM</u>				
roller mills	XX			X
high pressure	XX			X
rolls	XX	XX	XX	X
tumbling mills				
<u>FINE</u>				
vibrating mills	X	XX	XX	
planetary mills	X	XX	X	
hammer mills		XX		
cutter mills		XX	X	
<u>SUPER FINE</u>				
pin mills		XX	XX	
micro impact mills		XX	XX	X
opposed jet mills		XX	XX	X
spiral jet mills		X	XX	X
stirred ball mills			XX	XX

FIG. 21-60 Breakage modes in industrial mills. (Heiskanen, 1995.)

of particle loading is by applying a shear force by moving the loading planes horizontally. The table indicates that compression and impact are used more for coarse grinding, while attrition and abrasion are more common in fine and superfine grinding.

Hard materials (especially Mohs hardness 7 and above) are usually ground by devices designed for abrasion/attrition modes. For example, roll mills would rarely, if ever, be used for grinding of quartz, but media mills of various sorts have been successfully used to grind industrial diamonds. This is so primarily because both compression and high-energy impact modes have substantial contact between the mill and the very hard particles, which causes substantial wear of the device. Many attrition and abrasion devices, on the other hand, are designed so that a large component of grinding occurs by impact of particles on one another, rather than impact with the device. Wear still occurs, but its less dramatic than with other devices.

Ductile materials are an especially difficult problem for most grinding devices. Almost all grinding devices are designed for brittle materials and have some difficulties with ductile materials. However, devices with compression or abrasion modes tend to have the greatest difficulty with these kinds of materials. Mills with a compression mode will tend to flatten and flake these materials. Flaking can also occur in mills with a tangential abrasion mode, but smearing of the material across the surface of the mill is also common. In both cases, particle agglomeration can occur, as opposed to size reduction. Impact and attrition devices tend to do somewhat better with these materials, since their high-speed motion tends to cause more brittle fracture.

Conversely, mills with impact and attrition modes often do poorly with heat-sensitive materials where the materials become ductile as they heat up. Impact and attrition mills cause significant heating at the point of impact, and it is not uncommon for heat-sensitive materials (e.g., plastics) to stick to the device rather than being ground. In the worst cases, cryogenic grinding can be necessary for highly ductile or heat-sensitive materials.

Grindability Methods Laboratory experiments on single particles have been used to correlate grindability. In the past it has usually

been assumed that the total energy applied could be related to the grindability whether the energy is applied in a single blow or by repeated dropping of a weight on the sample [Gross and Zimmerly, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **87**, 27, 35 (1930)]. In fact, the results depend on the way in which the force is applied (Axelson, Ph.D. thesis, University of Minnesota, 1949). In spite of this, the results of large mill tests can often be correlated within 25 to 50 percent by a simple test, such as the number of drops of a particular weight needed to reduce a given amount of feed to below a certain mesh size. Two methods having particular application for coal are known as the ball-mill and Hardgrove methods. In the ball-mill method, the relative amounts of energy necessary to pulverize different coals are determined by placing a weighed sample of coal in a ball mill of a specified size and counting the number of revolutions required to grind the sample so that 80 percent of it will pass through a No. 200 sieve. The grindability index in percent is equal to 50,000 divided by the average of the number of revolutions required by two tests (ASTM designation D-408).

In the Hardgrove method, a prepared sample receives a definite amount of grinding energy in a miniature ball-ring pulverizer. The unknown sample is compared with a coal chosen as having 100 grindability. The Hardgrove grindability index = $13 + 6.93W$, where W is the weight of material passing the No. 200 sieve (see ASTM designation D-409).

Chandler [*Bull. Br. Coal Util. Res. Assoc.*, **29**(10), 333; (11), 371 (1965)] finds no good correlation of grindability measured on 11 coals with roll crushing and attrition, and so these methods should be used with caution. The Bond grindability method is described in the subsection "Capacity and Power Consumption." Manufacturers of various types of mills maintain laboratories in which grindability tests are made to determine the suitability of their machines. When grindability comparisons are made on small equipment of the manufacturers' own class, there is a basis for scale-up to commercial equipment. This is better than relying on a grindability index obtained in a ball mill to estimate the size and capacity of different types such as hammer or jet mills.

OPERATIONAL CONSIDERATIONS

Mill Wear Wear of mill components costs nearly as much as the energy required for comminution—hundreds of millions of dollars a year. The finer stages of comminution result in the greatest wear, because the grinding effort is greatest, as measured by the energy input per unit of feed. Parameters that affect wear fall under three categories: (1) the ore, including hardness, presence of corrosive minerals, and particle size; (2) the mill, including composition, microstructure, and mechanical properties of the material of construction, size of mill, and mill speed; and (3) the environment, including water chemistry and pH, oxygen potential, slurry solids content, and temperature [Moore et al., *Int. J. Mineral Processing*, **22**, 313–343 (1988)]. An abrasion index in terms of kilowatt-hour input per pound of metal lost furnishes a useful indication. In *wet grinding*, a synergy between mechanical wear and corrosion results in higher metal loss than with either mechanism alone [Iwasaki, *Int. J. Mineral Processing*, **22**, 345–360 (1988)]. This is due to removal of protective oxide films by abrasion, and by increased corrosion of stressed metal around gouge marks (Moore, loc. cit.). Wear rate is higher at lower solids content, since ball coating at high solids protects the balls from wear. This indicates that the mechanism is different from dry grinding. The rate of wear without corrosion can be measured with an inert atmosphere such as nitrogen in the mill. Insertion of marked balls into a ball mill best measures the wear rate at conditions in industrial mills, so long as there is not a galvanic effect due to a different composition of the balls. The mill must be cleared of dissimilar balls before a new composition is tested.

Sulfide ores promote corrosion due to galvanic coupling by a chemical reaction with oxygen present. Increasing the pH generally reduces corrosion. The use of harder materials enhances wear resistance, but this conflicts with achieving adequate ductility to avoid catastrophic brittle failure, so these two effects must be balanced. Wear-resistant materials can be divided into three groups: (1) abrasion-resistant steels, (2) alloyed cast irons, and (3) nonmetallics [see Durman, *Int. J. Mineral Processing*, **22**, 381–399 (1988) for a detailed discussion of these].

Cast irons of various sorts are often used for structural parts of large mills such as large ball mills and jaw crushers, while product contact parts such as ball-mill liners and cone crusher mantels are made from a variety of steels.

In many milling applications, mill manufacturers offer a choice of steels for product-contact surfaces (such as mill liner), usually at least one low-alloy “carbon” steel, and higher-alloy stainless steels. The exact alloys vary significantly with mill type. Stainless steels are used in applications where corrosion may occur (many wet grinding operations, but also high-alkali or high-acid minerals), but are more expensive and have lower wear resistance.

Nonmetallic materials include natural rubber, polyurethane, and ceramics. Rubber, due to its high resilience, is extremely wear-resistant in low-impact abrasion. It is inert to corrosive wear in mill liners, pipe linings, and screens. It is susceptible to cutting abrasion, so that wear increases in the presence of heavy particles, which penetrate, rather than rebound from, the wear surface. Rubber can also swell and soften in solvents. Advantages are its low density, leading to energy savings, ease of installation, and soundproofing qualities. Polyurethane has similar resilient characteristics. Its fluidity at the formation stage makes it suitable for the production of the wearing surface of screens, diaphragms, grates, classifiers, and pump and flotation impellers. The low heat tolerance of elastomers limits their use in dry processing where heat may build up.

Ceramics fill a niche in comminution where metal contamination cannot be tolerated such as pigments, cement, electronic materials, and pharmaceuticals (where any sort of contamination must be minimized). Use of ceramics has greatly increased in recent years, in part due to finer grinding requirements (and therefore higher energy and higher wear) for many industries and in part due to an increased production of electronic materials and pharmaceuticals. Also, the technology to produce mill parts from very hard ceramics such as tungsten carbide and yttria-stabilized zirconia have advanced, making larger parts available (although these are often expensive). Ceramic tiles

have been used for lining roller mills and chutes and cyclones, where there is a minimum of impact.

Safety The explosion hazard of nonmetallic materials such as sulfur, starch, wood flour, cereal dust, dextrin, coal, pitch, hard rubber, and plastics is often not appreciated [Hartmann and Nagy, U.S. Bur. Mines Rep. Invest., **3751** (1944)]. Explosions and fires may be initiated by discharges of static electricity, sparks from flames, hot surfaces, and spontaneous combustion. Metal powders also present a hazard because of their **flammability**. Their combustion is favored during grinding operations in which ball, hammer, or ring-roller mills are employed and during which a high grinding temperature may be reached. Many finely divided metal powders in suspension in air are potential **explosion hazards**, and causes for ignition of such dust clouds are numerous [Hartmann and Greenwald, *Min. Metall.*, **26**, 331 (1945)]. Concentration of the dust in air and its particle size are important factors that determine explosibility. Below a lower limit of concentration, no explosion can result because the heat of combustion is insufficient to propagate it. Above a maximum limiting concentration, an explosion cannot be produced because insufficient oxygen is available. The finer the particles, the more easily is ignition accomplished and the more rapid is the rate of combustion. This is illustrated in Fig. 21-61.

Isolation of the mills, use of nonsparking materials of construction, and magnetic separators to remove foreign magnetic material from the feed are useful **precautions** [Hartman, Nagy, and Brown, U.S. Bur. Mines Rep. Invest., **3722** (1943)]. Stainless steel has less sparking tendency than ordinary steel or forgings. Reduction of the oxygen content of air present in grinding systems is a means for preventing dust explosions in equipment [Brown, *U.S. Dep. Agri. Tech. Bull.* **74** (1928)]. Maintenance of oxygen content below 12 percent should be safe for most materials, but 8 percent is recommended for sulfur grinding. The use of **inert gas** has particular adaptation to pulverizers equipped with air classification; flue gas can be used for this purpose, and it is mixed with the air normally present in a system (see subsection “Chemicals and Soaps” for sulfur grinding). Despite the protection afforded by the use of inert gas, equipment should be provided with explosion vents, and structures should be designed with venting in mind [Brown and Hanson, *Chem. Metall. Eng.*, **40**, 116 (1933)].

Hard rubber presents a fire hazard when reduced on steam-heated rolls (see subsection “Organic Polymers”). Its dust is explosive [Twiss

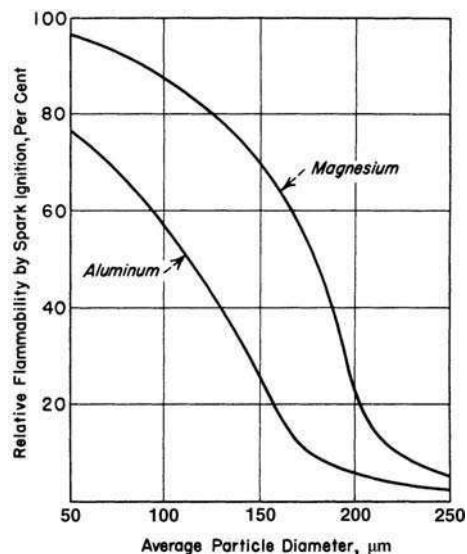


FIG. 21-61 Effect of fineness on the flammability of metal powders. (Hartmann, Nagy, and Brown, U.S. Bur. Mines Rep. Invest. 3722, 1943.)

and McGowan, *India Rubber J.*, **107**, 292 (1944)]. The annual publication *National Fire Codes for the Prevention of Dust Explosions* is available from the National Fire Protection Association, Quincy, Massachusetts, and should be of interest to those handling hazardous powders.

Temperature Stability Many materials are temperature-sensitive and can tolerate temperatures only slightly above room temperature, including many food products, polymers, and pharmaceuticals. This is a particular problem in grinding operations, as grinding inevitably adds heat to the ground material. The two major problems are that either the material will simply be damaged or denatured in some way, such as food products, or the material may melt or soften in the mill, usually causing significant operational problems.

Ways to deal with heat-sensitive materials include choosing a less energy-intensive mill, or running a mill at below optimum energy input. Some mills run naturally cooler than others. For example, jet mills can run cool because they need high gas flow for operation, and this has a significant cooling effect despite their high-energy intensity. Variable-speed drives are commonly used in stirred media mills to control the energy input to heat-sensitive slurries as energy input (and therefore temperature) is a strong function of stirrer speed.

Adding more cooling capability is often effective, but it can be expensive. Compositions containing fats and waxes are pulverized and blended readily if refrigerated air is introduced into their grinding systems (U.S. Patents 1,739,761 and 2,098,798; see also subsection "Organic Polymers" and Hixon, loc. cit., for flow sheets).

Hygroscopicity Some materials, such as salt, are very hygroscopic; they pick up water from air and deposit on mill surfaces, forming a hard cake. Mills with air classification units may be equipped so that the circulating air can be conditioned by mixing with hot or cold air, gases introduced into the mill, or dehumidification to prepare the air for the grinding of hygroscopic materials. Flow sheets including air dryers are also described by Hixon.

Dispersing Agents and Grinding Aids Grinding aids are helpful under some conditions. For example, surfactants make it possible to ball-mill magnesium in kerosene to 0.5- μm size [Fochtman, Bitten, and Katz, *Ind. Eng. Chem. Prod. Res. Dev.*, **2**, 212–216 (1963)]. Without surfactants the size attainable was 3 μm ; the rate of grinding was very slow at sizes below this. Also, the water in wet grinding may be considered to act as an additive.

Chemical agents that increase the rate of grinding are an attractive prospect since their cost is low. However, despite a voluminous literature on the subject, there is no accepted scientific method to choose such aids; there is not even agreement on the mechanisms by which they work. The subject has been reviewed [Fuerstenau, *KONA Powder and Particle*, **13**, 5–17 (1995)]. In wet grinding there are several theories, which have been reviewed [Somasundaran and Lin, *Ind. Eng. Chem. Process Des. Dev.*, **11**(3), 321 (1972); Snow, annual reviews, op. cit., 1970–1974; see also Rose, *Ball and Tube Milling*, Constable, London, 1958, pp. 245–249]. Additives can alter the rate of wet ball milling by changing the slurry viscosity or by altering the location of particles with respect to the balls. These effects are discussed under "Tumbling Mills." In conclusion, there is still no theoretical way to select the most effective additive. Empirical investigation, guided by the principles discussed earlier, is the only recourse. There are a number of commercially available grinding aids that may be tried. Also, a kit of 450 surfactants that can be used for systematic trials (Model SU-450, Chem Service Inc., West Chester, Pa. 19380) is available. Numerous experimental studies lead to the conclusion that dry grinding is limited by ball coating and that additives function by reducing the tendency to coat (Bond and Agthe, op. cit.). Most materials coat if they are ground finely enough, and softer materials coat at larger sizes than do hard materials. The presence of more than a few percent of soft gypsum promotes ball coating in cement-clinker grinding. The presence of a considerable amount of coarse particles above 35 mesh inhibits coating. Balls coat more readily as they become scratched. Small amounts of moisture may increase or decrease ball coating. Dry materials also coat. Materials used as grinding aids include solids such as graphite, oleoresinous liquid materials, volatile solids, and vapors. The complex effects of vapors have been extensively studied [Goette and Ziegler, *Z. Ver. Dtsch. Ing.*,

98, 373–376 (1956); and Locher and von Seebach, *Ind. Eng. Chem. Process Des. Dev.*, **11**(2), 190 (1972)], but water is the only vapor used in practice. The most effective additive for dry grinding is fumed silica that has been treated with methyl silazane [Tulis, *J. Hazard. Mater.*, **4**, 3 (1980)].

Cryogenic Grinding Cryogenic grinding is increasingly becoming a standard option for grinding of rubbers and plastics (especially powder coatings, but also some thermoplastics), as well as heat-sensitive materials such as some pharmaceuticals and chemicals. Many manufacturers of fine-grinding equipment have equipment options for cryogrinding, especially manufacturers of hammer mills and other rotary impact mills.

Cryogrinding adds to operating expenses due to the cost and recovery of liquid nitrogen, but capital cost is a more significant drawback to these systems. Modified mills, special feeders, as well as enhanced air handling and recovery systems are required and these tend to add significant cost to cryogenic systems. Partly for this reason, there is a healthy toll industry for cryogrinding where specialty equipment can be installed and used for a variety of applications to cover its cost. Many manufacturers of liquid nitrogen have information on cryogrinding applications on their Web sites.

SIZE REDUCTION COMBINED WITH OTHER OPERATIONS

Size Reduction Combined with Size Classification Grinding systems are batch or continuous in operation (Fig. 21-62). Most large-scale operations are continuous; batch ball or pebble mills are used only when small quantities are to be processed. Batch operation involves a high labor cost for charging and discharging the mill. Continuous operation is accomplished in open or closed circuit, as illustrated in Figs. 21-62 and 21-57a. **Operating economy** is the object of closed-circuit grinding with size classifiers. The idea is to remove the material from the mill before all of it is ground, separate the fine product in a classifier, and return the coarse for regrinding with the new feed to the mill. A mill with the fines removed in this way performs much more efficiently. Coarse material returned to a mill by a classifier is known as the **circulating load**; its rate may be from 1 to 10 times the production rate. The ability of the mill to transport material may limit the recycle rate; tube mills for use in such circuits may be designed with a smaller length-to-diameter ratio and hence a larger hydraulic gradient for greater flow or with compartments separated by diaphragms with lifters.

Internal size classification plays an essential role in the functioning of machines for dry grinding in the fine-size range; particles are retained in the grinding zone until they are as small as required in the finished product; then they are allowed to discharge. By closed-circuit operation the product size distribution is narrower and will have a larger proportion of particles of the desired size. On the other hand, making a *product size within narrow limits* (such as between 20 and 40 μm) is often requested but usually is not possible regardless of the grinding circuit used. The reason is that particle breakage is a random process, both as to the probability of breakage of particles and as to the sizes of fragments produced from each breakage event. The narrowest size distribution ideally attainable is one that has a slope of 1.0 when plotted on Gates-Gaudin-Schumann coordinates [$Y = (\bar{X}/k)^m$]. This can be demonstrated by examining the Gaudin-Meloy size distribution [$Y = 1 - (1 - X/X^*)^r$]. This is the distribution produced in a mill when particles are cut into pieces of random size, with r cuts per event. The case in which r is large corresponds to a breakage event producing many fines. The case

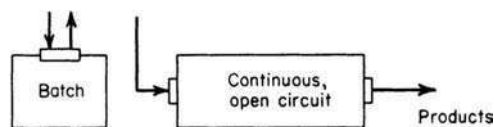


FIG. 21-62 Batch and continuous grinding systems.

in which r is 1 corresponds to an ideal case such as a knife cutter, in which each particle is cut once per event and the fragments are removed immediately by the classifier. The Meloy distribution with $r = 1$ reduces to the Schumann distribution with a slope of 1.0. Therefore, no practical grinding operation can have a slope greater than 1.0. Slopes typically range from 0.5 to 0.7. The specified product may still be made, but the finer fraction may have to be disposed of in some way. Within these limits, the size distribution of the classifier product depends both on the recycle ratio and on the sharpness of cut of the classifier used.

Size Classification The most common objective is **size classification**. Often only a particular range of product sizes is wanted for a given application. Since the particle breakage process always yields a spectrum of sizes, the product size cannot be directly controlled; however, mill operation can sometimes be varied to produce fewer fines at the expense of producing more coarse particles. By recycling the classified coarse fraction and regrinding it, production of the wanted size range is optimized. Such an arrangement of classifier and mill is called a **mill circuit** and is dealt with further below. More complex systems may include several unit operations such as mixing (Sec. 18), drying (Sec. 12), and agglomerating (see "Size Enlargement," in this section). Inlet and outlet silencers are helpful to reduce noise from high-speed mills. Chillers, air coolers, and explosion proofing may be added to meet requirements. Weighing and packaging facilities complete the system. Batch ball mills with low ball charges can be used in dry mixing or standardizing of dyes, pigments, colors, and insecticides to incorporate wetting agents and inert extenders. Disk mills, hammer mills, and other high-speed disintegration equipment are useful for final intensive blending of insecticide compositions, earth colors, cosmetic powders, and a variety of other finely divided materials that tend to agglomerate in ribbon and conical blenders. Liquid sprays or gases may be injected into the mill or airstream, for mixing with the material being pulverized to effect chemical reaction or surface treatment.

Other Systems Involving Size Reduction Industrial applications usually involve a number of processing steps combined with size reduction [Hixon, *Chem. Eng. Progress*, **87**, 36–44 (May 1991)].

Drying The **drying** of materials while they are being pulverized or disintegrated is known variously as *flash* or *dispersion* drying; a generic term is *pneumatic conveying* drying.

Beneficiation Ball and pebble mills, batch or continuous, offer considerable opportunity for combining a number of **processing steps** that include grinding [Underwood, *Ind. Eng. Chem.*, **30**, 905 (1938)]. Mills followed by air classifiers can serve to **separate components of mixtures** because of differences in specific gravity and the component that is pulverized readily. Grinding followed by *froth flotation* has become the beneficiation method most widely used for metallic ores and for nonmetallic minerals such as feldspar. Magnetic separation is the chief means used for upgrading taconite iron ore (see subsection "Ores and Minerals"). Magnetic separators frequently are employed to remove tramp magnetic solids from the feed to high-speed hammer and disk mills.

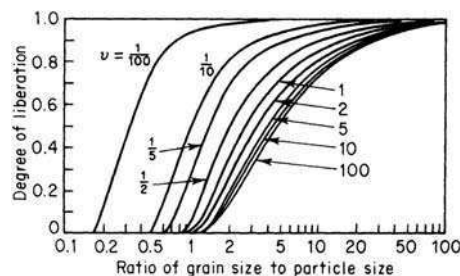


FIG. 21-63 Fraction of mineral B that is liberated as a function of volumetric abundance ratio v of gangue to mineral B (1/grade), and ratio of grain size to particle size of broken fragments (1/fineness). [Wiegel and Li, *Trans. Soc. Min. Eng.-Am. Inst. Min. Metall. Pet. Eng.*, **238**, 179 (1967).]

Liberation Most ores are heterogeneous, and the objective of grinding is to release the valuable mineral component so that it can be separated. Calculations based on a random-breakage model assuming no preferential breakage [Wiegel and Li, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **238**, 179–191 (1967)] agreed, at least in general trends, with plant data on the efficiency of release of mineral grains. Figure 21-63 shows that the desired mineral B can be liberated by coarse grinding when the grade is high so that mineral A becomes a small fraction and mineral B a large fraction of the total volume; mineral B can be liberated only by fine grinding below the grain size, when the grade is low so that there is a small proportion of grains of B . Similar curves, somewhat displaced in size, resulted from a more detailed integral geometry analysis by Barbery [*Minerals Engg.*, **5**(2), 123–141 (1992)]. There is at present no way to measure grain size on-line and thus to control liberation. A review of liberation modeling is given by Mehta et al. [*Powder Technol.*, **58**(3), 195–209 (1989)]. Many authors have assumed that breakage occurs preferentially along grain boundaries, but there is scant evidence for this. On the contrary, Gorski [*Bull. Acad. Pol. Sci. Ser. Sci. Tech.*, **20**(12), 929 (1972); CA 79, 20828k], from analysis of microscope sections, finds an intercrystalline character of comminution of dolomite regardless of the type of crusher used. The liberation of a valuable constituent does not necessarily translate directly into recovery in downstream processes. For example, flotation tends to be more efficient in intermediate sizes than at coarse or fine sizes [McIvor and Finch, *Minerals Engg.*, **4**(1), 9–23 (1991)]. For coarser sizes, failure to liberate may be the limitation; finer sizes that are liberated may still be carried through by the water flow. A conclusion is that overgrinding should be avoided by judicious use of size classifiers with recycle grinding.

MODELING AND SIMULATION OF GRINDING PROCESSES

MODELING OF MILLING CIRCUITS

Grinding processes have not benefited as much as some other types of processes from the great increase in computing power and modeling sophistication in the 1990s. Complete simulations of most grinding processes that would be useful to practicing engineers involve breakage mechanics and gas-phase or liquid-phase particle motion coupled in a complex way that is not yet practical to study. However, with the continuing increase of computing power, it is unlikely that this state will continue much longer. Fluid mechanics modeling is well advanced, and the main limitation to modeling many devices is having

enough computer power to keep track of a large number of particles as they move and are size-reduced. Traditionally, particle breakage is modeled by using variations of population balance methodology described below, but more recent models have tended to use discrete elements models which track the particles individually. The latter requires greater computing power, but may provide a more realistic way of accounting for particle dynamics in a device.

Computer simulation, based on population-balance models [Bass, *Z. Angew. Math. Phys.*, **5**(4), 283 (1954)], traces the breakage of each size of particle as a function of grinding time. Furthermore, the simulation models separate the breakage process into two aspects: a

breakage rate and a mean fragment-size distribution. These are both functions of the size of particle being broken. They usually are not derived from knowledge of the physics of fracture but are empirical functions fitted to milling data. The following formulation is given in terms of a discrete representation of size distribution; there are comparable equations in integro-differential form.

BATCH GRINDING

Grinding Rate Function Let w_k = the weight fraction of material retained on each screen of a nest of n screens; w_k is related to P_k , the fraction coarser than size X_k , by

$$w_k = (\partial P_k / \partial X_k) \Delta X_k \quad (21-77a)$$

where ΔX_k is the difference between the openings of screens k and $k + 1$. The **grinding-rate function** S_u is the rate at which the material of upper size u is selected for breakage in an increment of time, relative to the amount of that size present:

$$dw_u/dt = -S_u w_u \quad (21-77b)$$

Breakage Function The **breakage function** $\Delta B_{k,u}$ gives the size distribution of product breakage of size u into all smaller sizes k . Since some fragments from size u are large enough to remain in the range of size u , the term $\Delta B_{u,u}$ is not zero, and

$$\sum_{k=u}^u \Delta B_{k,u} = 1 \quad (21-78)$$

The differential equation of batch grinding is deduced from a balance on the material in the size range k . The rate of accumulation of material of size k equals the rate of production from all larger sizes minus the rate of breakage of material of size k :

$$\frac{dw_k}{dt} = \sum_{u=1}^k [w_u S_u(t) \Delta B_{k,u}] - S_k(t) w_k \quad (21-79)$$

In general, S_u is a function of all the milling variables. Also $\Delta B_{k,u}$ is a function of breakage conditions. If it is assumed that these functions are constant, then relatively simple solutions of the grinding equation are possible, including an analytical solution [Reid, *Chem. Eng. Sci.*, **20**(11), 953-963 (1965)] and matrix solutions [Broadbent and Callcott, *J. Inst. Fuel*, **29**, 524-539 (1956); **30**, 18-25 (1967); and Meloy and Bergstrom, *7th Int. Min. Proc. Congr. Tech. Pap.*, 1964, pp. 19-31].

Solution of Batch-Mill Equations In general, the grinding equation can be solved by numerical methods, e.g., the Euler technique (Austin and Gardner, *1st European Symposium on Size Reduction*, 1962) or the Runge-Kutta technique. The matrix method is a particularly convenient formulation of the Euler technique. Reid's **analytical solution** is useful for calculating the product as a function of time t for a constant feed composition. It is

$$w_{L,k} = \sum_{n=1}^k a_{k,n} \exp(-\bar{S}_n \Delta t) \quad (21-80)$$

where the subscript L refers to the discharge of the mill, 0 to the entrance, and $\bar{S}_n = 1$ "corrected" rate function defined by $\bar{S}_n = (1 - \Delta B_{n,n})$ and B is then normalized with $\Delta B_{n,n} = 0$. The coefficients are

$$a_{k,k} = w_{0k} - \sum_{n=1}^{k-1} a_{k,n} \quad (21-81a)$$

and

$$a_{k,n} = \sum_{u=n}^{k-1} \frac{S_u \Delta B_{k,u} a_{u,n}}{\bar{S}_k - \bar{S}_n} \quad (21-81b)$$

The coefficients are evaluated in order since they depend on the coefficients already obtained for larger sizes.

The basic idea behind the **Euler method** is to set the change in w per increment of time as

$$\Delta w_k = (dw_k/dt) \Delta t \quad (21-82)$$

where the derivative is evaluated from Eq. (21-79). Equation (21-82) is applied repeatedly for a succession of small time intervals until the desired duration of milling is reached. In the matrix method a modified rate function is defined $S'_k = S_k \Delta t$ as the amount of grinding that occurs in some small time Δt . The result is

$$\mathbf{w}_L = (\mathbf{I} + \mathbf{S}'\mathbf{B} - \mathbf{S}')\mathbf{w}_F = \mathbf{M}\mathbf{w}_F \quad (21-83)$$

where the quantities $\mathbf{w}$ are vectors, $\mathbf{S}'$ and $\mathbf{B}$ are the matrices of rate and breakage functions, and $\mathbf{I}$ is the unit matrix. This follows because the result obtained by multiplying these matrices is just the sum of products obtained from the Euler method. Equation (21-83) has a physical meaning. The unit matrix times $\mathbf{w}_F$ is simply the amount of feed that is not broken. $\mathbf{S}'\mathbf{B}\mathbf{w}_F$ is the amount of feed that is selected and broken into the vector of products; $\mathbf{S}'\mathbf{w}_F$ is the amount of material that is broken out of its size range and hence must be subtracted from this element of the product. The entire term in parentheses can be considered as a mill matrix $\mathbf{M}$. Thus the milling operation transforms the feed vector to the product vector. Meloy and Bergstrom (op. cit.) pointed out that when Eq. (21-83) is applied over a series of p short-time intervals, the result is

$$\mathbf{w}_L = \mathbf{M}^p \mathbf{w}_F \quad (21-83a)$$

Matrix multiplication happens to be cumulative in this special case. It is easy to raise a matrix to a power on a computer since three multiplications give the eighth power, etc. Therefore the matrix formulation is well adapted to computer use.

CONTINUOUS-MILL SIMULATION

Residence Time Distribution Batch-grinding experiments are the simplest type of experiments to produce data on grinding coefficients. But scale-up from batch to continuous mills must take into account the **residence-time distribution** in a continuous mill. This distribution is apparent if a tracer experiment is carried out. For this purpose, background ore is fed continuously, and a pulse of tagged feed is introduced at time t_0 . This tagged material appears in the effluent distributed over a period of time, as shown by a typical curve in Fig. 21-64. Because of this distribution some portions are exposed to grinding for longer times than others. Levenspiel (*Chemical Reaction Engineering*, Wiley, New York, 1962) shows several types of residence time distribution that can be observed. Data on large mills indicate that a curve like that of Fig. 21-64 is typical (Keienberg et al., *3d European Symposium on Size Reduction*, op. cit., 1972, p. 629). This curve

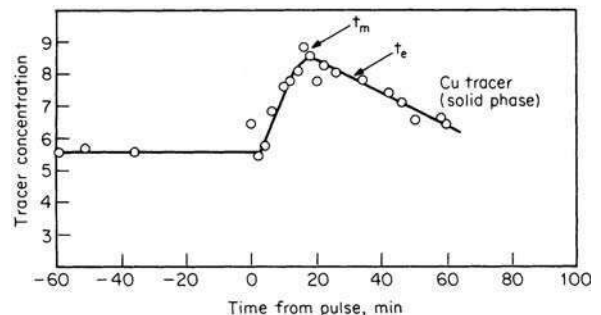


FIG. 21-64 Ore transit through a ball mill. Feed rate is 500 lb/h. (Courtesy Phelps Dodge Corporation.)

can be accurately expressed as a series of arbitrary functions (Merz and Molerus, *3d European Symposium on Size Reduction*, op. cit., 1972, p. 607). A good fit is more easily obtained if we choose a function that has the right shape since then only the first two moments are needed. The log-normal probability curve fits most available mill data, as was demonstrated by Mori [*Chem. Eng. (Japan)*, **2**(2), 173 (1964)]. Two examples are shown in Fig. 21-65. The log-normal plot fails only when the mill acts nearly as a perfect mixer. To measure a residence time distribution, a pulse of tagged feed is inserted into a continuous mill and the effluent is sampled on a schedule. If it is a dry mill, a soluble tracer such as salt or dye may be used and the samples analyzed conductimetrically or colorimetrically. If it is a wet mill, the tracer must be a solid of similar density to the ore. Materials such as copper concentrate, chrome brick, or barites have been used as tracers and analyzed by X-ray fluorescence. To plot results in log-normal coordinates, the concentration data must first be normalized from the form of Fig. 21-64 to the form of cumulative percent discharged, as in Fig. 21-65. For this, one must either know the total amount of pulse feed or determine it by a simple numerical integration using a computer. The data are then plotted as in Fig. 21-65, and the coefficients in the log-normal formula of Mori can be read directly from the graph. Here $t_e = t_{50}$ is the time when 50 percent of the pulse has emerged. The standard deviation σ is the time between t_{16} and t_{50} or between t_{50} and t_{84} . Knowing t_e and σ , one can reconstruct the straight line in log-normal coordinates. One can also calculate the vessel dispersion number Dt_e/L^2 , which is a measure of the sharpness of the pulse (Levenspiel, *Chemical Reactor Omnibook*, Oregon State University Bookstores Inc., 1979, p. 100.6). This number has erroneously been called by some the Peclet number. Here D is the particle diffusivity. A few available data are summarized (Snow, *International Conference on Particle Technology*, IIT Research Institute, Chicago, 1973, p. 28) for wet mills. Other experiments are presented for dry mills [Hogg et al., *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **258**, 194 (1975)]. The most important variables affecting the vessel dispersion number are L /diameter of the mill, ball size, mill speed, scale expressed either as diameter or as throughput, degree of ball filling, and degree of material filling.

Solution for Continuous Milling In the method of Mori (op. cit.), the residence time distribution is broken up into a number of

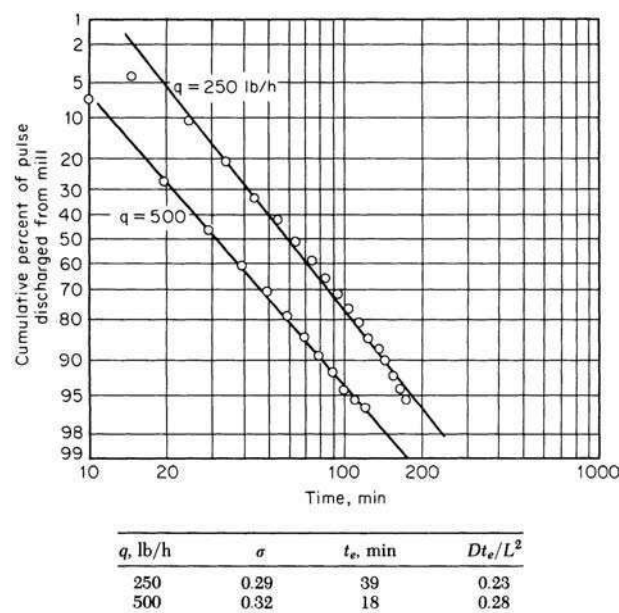


FIG. 21-65 Log-normal plot of residence-time distribution in Phelps Dodge mill.

segments, and the batch-grinding equation is applied to each of them. The resulting size distribution at the mill discharge is

$$\mathbf{w}(L) = \mathbf{w}(t) \Delta \phi \quad (21-84)$$

where $\mathbf{w}(t)$ is a matrix of solutions of the batch equation for the series of times t , with corresponding segments of the cumulative residence time curve. Using the Reid solution, Eq. (21-80), this becomes

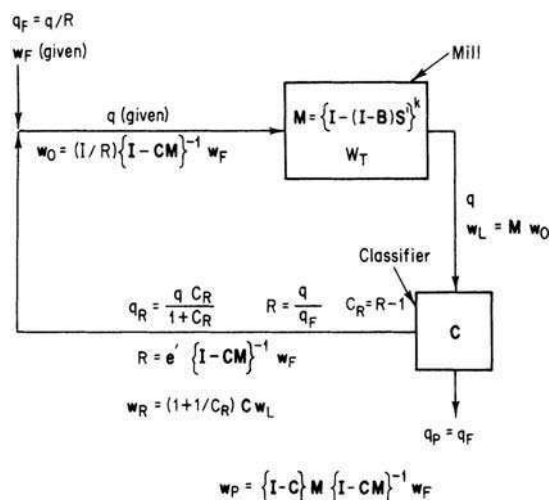
$$\mathbf{w}(L) = \mathbf{RZ} \Delta \phi \quad (21-85)$$

since the Reid solution [Eq. (21-80)] can be separated into a matrix $\mathbf{Z}$ of exponentials $\exp(-St)$ and another factor $\mathbf{R}$ involving only particle sizes. Austin, Klimpel, and Luckie (*Process Engineering of Size Reduction: Ball Milling*, Society of Mining Engineers of AIME, 1984) incorporated into this form a tanks-in-series model for the residence time distribution.

CLOSED-CIRCUIT MILLING

In closed-circuit milling, the tailings from a classifier are mixed with fresh feed and recycled to the mill. Calculations can be based on a material balance and an explicit solution such as Eq. (21-83a). Material balances for the normal circuit arrangement (Fig. 21-66) give

$$q = q_F + q_R \quad (21-86)$$



Nomenclature

- C_R = circulating load, $R - 1$
 C = classifier selectivity matrix, which has classifier selectivity-function values η on diagonal zeros elsewhere
 I = identity matrix, which has ones on diagonal, zeros elsewhere
 M = mill matrix, which transforms mill-feed-size distribution into mill-product-size distribution
 q = flow rate of a material stream
 R = recycle ratio q/q_F
 w = vector of differential size distribution of a material stream
 W_T = holdup, total mass of material in mill
 Subscripts:
 0 = inlet to mill
 F = feed stream
 L = mill-discharge stream
 P = product stream
 R = recycle stream, classifier tailings

FIG. 21-66 Normal closed-circuit continuous grinding system with stream flows and composition matrices, obtained by solving material-balance equations. [Callcott, *Trans. Inst. Min. Metall.*, **76**(1), C1-11 (1967).]

where q = total mill throughput, q_f = rate of feed of new material, and q_R = recycle rate. A material balance on each size gives

$$w_{0,k} = \frac{q_f w_{fk} + \frac{q_R}{R} \eta_k w_{L,k}}{q} \quad (21-87)$$

where $w_{0,k}$ = fraction of size k in the mixed feed streams, R = recycle ratio, and η_k = classifier selectivity for size k . With these conditions, a calculation of the transient behavior of the mill can be performed by using any method of solving the milling equation and iterating over intervals of time τ = residence time in the mill. This information is important for evaluating mill circuit control stability and strategies. If the throughput q is controlled to be a constant, as is often the case, then τ is constant, and a closed-form matrix solution can be found for the steady state [Calcott, *Trans. Inst. Min. Metall.*, **76**(1), C1-11 (1967)]. The resulting flow rates and composition vectors are given in Fig. 21-66. Calcott (loc. cit.) gives equations for the reverse-circuit case, in which the feed is classified before it enters the mill. These results can be used to investigate the effects of changes in feed composition on the product. Separate calculations can be made to find the effects of classifier selectivity, mill throughput or recycle, and grindability (rate function) to determine optimum mill-classifier combinations [Lynch, Whiten, and Draper, *Trans. Inst. Min. Metall.*, **76**, C169, 179 (1967)]. Equations such as these form the basis for computer codes that are available for modeling mill circuits (Austin, Klimpel, and Luckie, loc. cit.).

DATA ON BEHAVIOR OF GRINDING FUNCTIONS

Several breakage functions were early suggested [Gardner and Austin, *1st European Symposium on Size Reduction*, op. cit., 1962, p. 217; Broadbent and Calcott, *J. Inst. Fuel*, **29**, 524 (1956); 528 (1956); 18 (1957); **30**, 21 (1957)]. The simple Gates-Gaudin-Schumann equation has been most widely used to fit ball-mill data. For example, this form was assumed by Herbst and Fuerstenau [*Trans. Am. Inst. Min. Metall. Pet. Eng.*, **241**(4), 538 (1968)] and Kelsall et al. [*Powder Technol.*, **1**(5), 291 (1968); **2**(3), 162 (1968); **3**(3), 170 (1970)]. More recently it has been observed that when the Schumann equation is used, the amount of coarse fragments cannot be made to agree with the mill-product distribution regardless of the choice of rate function. This observation points to the need for a breakage function that has more coarse fragments, such as the function used by Reid and Stewart (Chemica meeting, 1970) and Stewart and Restarick [*Proc. Australas. Inst. Min. Metall.*, **239**, 81 (1971)] and shown in Fig. 21-67. This graph can be fitted by a double Schumann equation

$$B(X) = A \left(\frac{X}{X_0} \right)^s + (1 - A) \left(\frac{X}{X_0} \right)^r \quad (21-88)$$

where A is a coefficient less than 1.

In the investigations mentioned earlier, the breakage function was assumed to be normalizable; i.e., the shape was independent of X_0 . Austin and Luckie [*Powder Technol.*, **5**(5), 267 (1972)] allowed the coefficient A to vary with the size of particle breaking when grinding soft feeds.

Grinding Rate Functions These were determined by tracer experiments in laboratory mills by Kelsall et al. (op. cit.) and in similar work by Szanthy and Fuhrmann [*Aufbereit. Tech.*, **9**(5), 222 (1968)]. These curves can be fitted by the following equation:

$$\frac{S}{S_{\max}} = \left(\frac{X}{X_{\max}} \right)^{\alpha} \exp \left(- \frac{X}{X_{\max}} \right) \quad (21-89)$$

That a maximum must exist should be apparent from the observation of Coghill and Devaney (U.S. Bur. Mines Tech. Pap., 1937, p. 581) that there is an optimum ball size for each feed size. The position of this maximum depends on the ball size. In fact, the feed size for

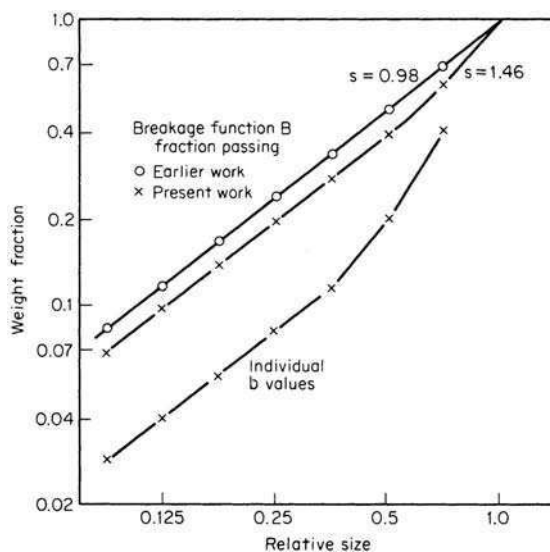


FIG. 21-67 Experimental breakage functions. (Reid and Stewart, *Chemical meeting*, 1970.)

which S is a maximum can be estimated by inverting the formula for optimum ball size given by Coghill and Devaney under "Tumbling Mills."

SCALE-UP AND CONTROL OF GRINDING CIRCUITS

Scale-up Based on Energy Since large mills are usually sized on the basis of power draft (see subsection "Energy Laws"), it is appropriate to scale up or convert from batch to continuous data by

$$S(X)_{\text{cont}} = S(X)_{\text{batch}} \frac{(W_T/KW)_{\text{batch}}}{(W_T/KW)_{\text{cont}}} \quad (21-90)$$

Usually W_T is not known for continuous mills, but it can be determined from $W_T = t_e Q$, where t_e is determined by a tracer measurement. Equation (21-90) will be valid if the holdup W_T is geometrically similar in the two mills or if operating conditions are in the range in which total production is independent of holdup. Studies of the kinetics of milling [Patat and Mempel, *Chem. Ing. Tech.*, **37**(9), 933; (11), 1146; (12), 1259 (1965)] indicate that there is a range of holdup in which this is true. More generally, Austin, Luckie, and Klimpel (loc. cit.) developed empirical relations to predict S as holdup varies. In particular, they observe a slowing of grinding rate when mill filling exceeds ball void volume due to cushioning.

Parameters for Scale-up Before simulation equations can be used, the parameter matrices **S** and **B** must be back-calculated from experimental data, which turns out to be difficult. One reason is that **S** and **B** occur as a product, so they are to some extent indeterminate; errors in one tend to be compensated by the other. Also, the number of parameters is larger than the number of data values from a single size-distribution measurement; but this is overcome by using data from grinding tests at a series of grinding times. This should be done anyway, since the empirical parameters should be determined to be valid over the experimental range of grinding times.

It may be easier to fit the parameters by forcing them to follow specified functional forms. In earliest attempts it was assumed that the forms should be normalizable (have the same shape whatever the size being broken). With complex ores containing minerals of different friability, the grinding functions **S** and **B** exhibit complex behavior

near the grain size (Choi et al., *Particulate and Multiphase Processes Conference Proceedings*, **1**, 903–916). Grinding function **B** is not normalizable with respect to feed size, and **S** does not follow a simple power law.

There are also experimental problems: When a feed-size distribution is ground for a short time, there is not enough change in the size distribution in the mill to distinguish between particles being broken into and out of intermediate sizes, unless individual feed-size ranges are tagged. Feeding narrow-size fractions alone solves the problem, but changes the milling environment; the presence of fines affects the grinding of coarser sizes. Gupta et al. [*Powder Technol.*, **28**(1), 97–106 (1981)] ground narrow fractions separately, but subtracted out the effect of the first 3 min of grinding, after which the behavior had become steady. Another experimental difficulty arises from the recycle of fines in a closed circuit, which soon “contaminates” the size distribution in the mill; it is better to conduct experiments in open circuit, or in batch mills on a laboratory scale.

There are few data demonstrating scale-up of the grinding-rate functions **S** and **B** from pilot- to industrial-scale mills. Weller et al. [*Int. J. Mineral Processing*, **22**, 119–147 (1988)] ground chalcopryite ore in pilot and plant mills and compared predicted parameters with laboratory data of Kelsall [*Electrical Engg. Trans.*, Institution of

Engineers Australia, **EE5**(1), 155–169 (1969)] and Austin, Klimpel, and Luckie (*Process Engineering of Size Reduction, Ball Milling*, Society of Mining Engineers, New York, 1984) for quartz. Grinding function **S** has a maximum for a particle size that depends on ball size, which can be expressed as $X_s/X_t = (d_s/d_t)^{2.4}$, where s = scaled-up mill, t = test mill, d = ball size, and X = particle size of maximum rate. Changing ball size also changes the rates according to $S_s/S_t = (d_s/d_t)^{0.55}$. These relations shift one rate curve onto another and allow scale-up to a different ball size. Mill diameter also affects rate by a factor $(D_s/D_t)^{0.5}$. Lynch (*Mineral Crushing and Grinding Circuits, Their Simulation Optimization Design and Control*, Elsevier Scientific Publishing Co., Amsterdam, 1977) and Austin, Klimpel, and Luckie (loc. cit.) developed scale-up factors for ball load, mill filling, and mill speed. In addition, slurry solids content is known to affect the rate, through its effect on slurry rheology. Austin, Klimpel, and Luckie (loc. cit.) present more complete simulation examples and compare them with experimental data to study scale-up and optimization of open and closed circuits, including classifiers such as hydrocyclones and screen bends. Differences in the classifier will affect the rates in a closed circuit. For these reasons scale-up is likely to be uncertain unless conditions in the large mill are as close as possible to those in the test mill.

CRUSHING AND GRINDING EQUIPMENT: DRY GRINDING—IMPACT AND ROLLER MILLS

JAW CRUSHERS

Design and Operation These crushers may be divided into two main groups, the *Blake* (Fig. 21-68), with a movable jaw pivoted at the top, giving greatest movement to the smallest lumps; and the *overhead eccentric*, which is also hinged at the top, but through an *eccentric-driven* shaft which imparts an elliptical motion to the jaw. Both types have a removable crushing plate, usually corrugated, fixed in a vertical position at the front end of a hollow rectangular frame. A similar plate is attached to the swinging movable jaw. The Blake jaw is moved through a knuckle action by the rising and falling of a second lever (pitman) car-

ried by an eccentric shaft. The vertical movement is communicated horizontally to the jaw by double-toggle plates. Because the jaw is pivoted at the top, the throw is greatest at the discharge, preventing choking.

The *overhead eccentric jaw crusher* falls into the second type. These are single-toggle machines. The lower end of the jaw is pulled back against the toggle by a tension rod and spring. The choice between the two types of jaw crushers is generally dictated by the feed characteristics, tonnage, and product requirements (Pryon, *Mineral Processing*, Mining Publications, London, 1960; Wills, *Mineral Processing Technology*, Pergamon, Oxford, 1979). Greater wear caused by the elliptical motion of the overhead eccentric and direct transmittal of

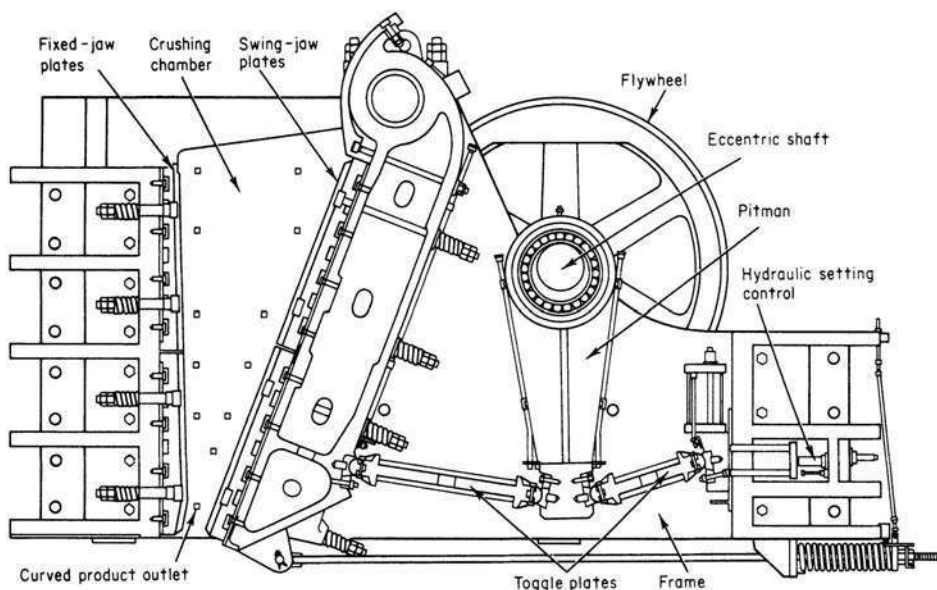


FIG. 21-68 Blake jaw crusher. (Allis Mineral Systems Grinding Div., Svedala Industries, Inc.)

shocks to the bearing limit use of this type to readily breakable material. Overhead eccentric crushers are generally preferred for crushing rocks with a hardness equal to or lower than that of limestone. Operating costs of the overhead eccentric are higher for the crushing of hard rocks, but its large reduction ratio is useful for simplified low-tonnage circuits with fewer grinding steps. The double-toggle type of crushers cost about 50 percent more than the similar overhead-eccentric type of crushers.

Comparison of Crushers The jaw crusher can accommodate the same size rocks as a gyratory, with lower capacity and also lower capital and maintenance costs, but similar installation costs. Therefore they are preferred when the crusher gape is more important than the throughput. Relining the gyratory requires greater effort than for the jaw, and also more space above and below the crusher.

Performance Jaw crushers are applied to the primary crushing of hard materials and are usually followed by other types of crushers. In smaller sizes they are used as single-stage machines. Typical capabilities and specifications are shown in Table 21-9a.

GYRATORY CRUSHERS

The development of improved supports and drive mechanisms has allowed gyratory crushers to take over most large hard-ore and

mineral-crushing applications. The largest expense of these units is in relining them. Operation is intermittent; so power demand is high, but the total power cost is not great.

Design and Operation The gyratory crusher consists of a cone-shaped pestle oscillating within a larger cone-shaped mortar or bowl. The angles of the cones are such that the width of the passage decreases toward the bottom of the working faces. The pestle consists of a mantle which is free to turn on its spindle. The spindle is oscillated from an eccentric bearing below. Differential motion causing attrition can occur only when pieces are caught simultaneously at the top and bottom of the passage owing to different radii at these points. The circular geometry of the crusher gives a favorably small **nip angle** in the horizontal direction. The nip angle in the vertical direction is less favorable and limits feed acceptance. The vertical nip angle is determined by the shape of the mantle and bowl liner; it is similar to that of a jaw crusher.

Primary crushers have a steep cone angle and a small reduction ratio. **Secondary** crushers have a wider cone angle; this allows the finer product to be spread over a larger passage area and also spreads the wear over a wider area. Wear occurs to the greatest extent in the lower, fine-crushing zone. These features are further extended in cone crushers; therefore secondary gyratories are much less popular than secondary cone crushers, but they can be used as primaries when

TABLE 21-9a Performance of Nordberg C Series Eccentric Jaw Crushers

		C95	C105	C 0	C100	C 05	C110	C125	C1 0	C1 5	C1 0	C200
Feed opening width mm (in)		930 (37)	1060 (42)	800 (32)	1000 (40)	1375 (54)	1100 (44)	1250 (49)	1400 (55)	1400 (55)	1600 (63)	2000 (79)
Feed opening depth mm (in)		580 (23)	700 (28)	510 (20)	760 (30)	760 (30)	850 (34)	950 (37)	1070 (42)	1100 (43)	1200 (47)	1500 (59)
Power kW (HP)		90 (125)	110 (150)	75 (100)	110 (150)	160 (200)	160 (200)	160 (200)	200 (250)	200 (300)	250 (350)	400 (500)
Speed (rpm)		330	300	350	260	260	230	220	220	220	220	200
Product size mm (in)	Closed side setting mm (in)	tph (Stph)	tph (Stph)	tph (Stph)	tph (Stph)	tph (Stph)	tph (Stph)	tph (Stph)	tph (Stph)	tph (Stph)	tph (Stph)	tph (Stph)
0-30	20			*								
0-1 1/8	3/4			*								
0-35	25	*	*	*								
0-1 3/8	1	*	*	*								
0-45	30	*	*	*								
0-1 3/4	1 1/8	*	*	*								
0-60	40	*	*	55 - 75	*	*	*					
0-2 3/8	1 5/8	*	*	60 - 80	*	*	*					
0-75	50	*	*	65 - 95	*	*	*					
0-3	2	*	*	75 - 100	*	*	*					
0-90	60	105 - 135	*	80 - 110	*	*	*					
0-3 1/2	2 3/8	115 - 150	*	90 - 120	*	*	*					
0-105	70	125 - 155	135 - 175	95 - 135	125 - 175	210 - 270	160 - 220					
0-4 1/8	2 3/4	135 - 170	150 - 190	110 - 145	140 - 190	230 - 295	175 - 240					
0-120	80	140 - 180	155 - 195	110 - 150	145 - 200	240 - 300	175 - 245	*				
0-4 3/4	3 1/8	155 - 200	170 - 215	120 - 165	160 - 215	260 - 330	195 - 270	*				
0-135	90	160 - 200	175 - 225	125 - 175	160 - 220	260 - 330	190 - 275	*	*	*		
0-5 3/8	3 1/2	175 - 220	195 - 245	140 - 190	175 - 240	285 - 360	215 - 300	*	*	*		
0-150	100	175 - 225	195 - 245	140 - 190	180 - 250	285 - 365	215 - 295	245 - 335	*	*	*	
0-6	4	195 - 250	210 - 270	150 - 210	200 - 275	315 - 400	235 - 325	270 - 370	*	*	*	
0-185	125	220 - 280	245 - 315	175 - 245	220 - 310	345 - 435	260 - 360	295 - 405	325 - 445	335 - 465	*	*
0-7	5	240 - 310	270 - 345	195 - 270	245 - 340	375 - 480	285 - 395	325 - 445	355 - 490	370 - 510	*	*
0-225	150	265 - 335	295 - 375	210 - 290	265 - 365	405 - 515	310 - 430	345 - 475	380 - 530	395 - 545	430 - 610	*
0-9	6	290 - 370	325 - 410	230 - 320	290 - 400	445 - 565	340 - 470	380 - 525	420 - 580	435 - 600	475 - 670	*
0-260	175	310 - 390	345 - 435	245 - 335	310 - 430	465 - 595	350 - 490	395 - 545	435 - 605	455 - 625	495 - 695	630 - 890
0-10	7	340 - 430	380 - 480	270 - 370	340 - 470	515 - 650	390 - 540	435 - 600	480 - 665	500 - 690	545 - 765	695 - 980
0-300	200		390 - 500		355 - 490	530 - 670	405 - 555	445 - 615	495 - 685	510 - 710	560 - 790	710 - 1000
0-12	8		430 - 550		390 - 535	580 - 740	445 - 610	490 - 675	545 - 750	565 - 780	615 - 870	780 - 1100
0-340	225							495 - 685	550 - 760	570 - 790	625 - 880	785 - 1100
0-13	9							545 - 750	605 - 835	630 - 870	685 - 965	860 - 1210
0-375	250							545 - 755	610 - 840	630 - 870	685 - 965	865 - 1210
0-15	10							600 - 830	670 - 925	695 - 960	755 - 1060	950 - 1340
0-410	275									690 - 950	745 - 1055	940 - 1320
0-16	11									760 - 1045	820 - 1160	1030 - 1450
0-450	300										815 - 1145	1015 - 1430
0-18	12										895 - 1260	1120 - 1570

*Smaller closed side settings can be often used depending on application and production requirements.

(From Metso Minerals brochure.)

quarrying produces suitable feed sizes. The three general types of gyratory crusher are the **suspended-spindle**, **supported-spindle**, and **fixed-spindle** types. Primary gyratories are designated by the size of feed opening, and secondary or reduction crushers by the diameter of the head in feet and inches. There is a close opening and a wide opening as the mantle gyrates with respect to the concave ring at the outlet end. The close opening is known as the close setting or the closed-side setting, while the wide opening is known as the wide-side or open-side setting. Specifications usually are based on closed settings. The setting is adjustable by raising or lowering the mantle.

The length of the crushing stroke greatly affects the capacity and the screen analysis of the crushed product. A very short stroke will give a very evenly crushed product but will not give the greatest capacity. A very long stroke will give the greatest capacity, but the product will contain a wider product-size distribution.

Performance Crushing occurs through the full cycle in a gyratory crusher, and this produces a higher crushing capacity than a similar-sized jaw crusher, which crushes only in the shutting half of the cycle. Gyratory crushers also tend to be easier to operate. They operate most efficiently when they are fully charged, with the main shaft fully buried in charge. Power consumption for gyratory crushers is also lower than that of jaw crushers. These are preferred over jaw crushers when capacities of 800 Mg/h (900 tons/h) or higher are required.

Gyratories make a product with open-side settings of 5 to 10 in at discharge rates from 600 to 6000 tons/h, depending on size. Most manufacturers offer a throw from 1/4 to 2 in. The throughput and power draw depend on the throw and the hardness of the ore, and on the amount of undersized material in the feed. Removal of undersized material (which can amount to one-third of the feed) by a stationary grizzly can reduce power draw. See Table 21-9b.

Gyratory crushers that feature wide-cone angles are called **cone crushers**. These are suitable for secondary crushing, because crushing of fines requires more work and causes more wear; the cone shape provides greater working area than primary or jaw crushers for grinding of the finer product. Crusher performance is harmed by sticky material in the feed, more than 10 percent fines in the feed smaller than the crusher setting, excessive feed moisture, feed-size segregation, uneven distribution of feed around the circumference, uneven feed control, insufficient capacity of conveyors and closed-circuit screens, extremely hard or tough feed material, and operation at less than recommended

speed. Rod mills are sometimes substituted for crushing of tough ore, since they provide more easily replaceable metal for wear.

Control of Crushers The objective of crusher control is usually to maximize crusher throughput at some specified product size, without overloading the crusher. Usually only three variables can be adjusted: feed rate, crusher opening, and feed size in the case of a secondary crusher. Four modes of control for a crusher are:

1. Setting overload control, where the gape setting is fixed except that it opens when overload occurs. A hardness change during high throughput can cause a power overload on the crusher, which control should protect against.
2. Constant power setting control, which maximizes throughput.
3. Pressure control, which provides settings that give maximum crusher force, and hence also throughput.
4. Feeding-rate control, for smooth operation. Setting control influences mainly product size and quality, while feed control determines capacity. Flow must also be synchronized with the feed requirements of downstream processes such as ball mills, and improved crusher efficiency can reduce the load on the more costly downstream grinding.

IMPACT BREAKERS

Impact breakers include heavy-duty hammer crushers, rotor impact breakers, and cage mills. They are generally coarse breakers which reduce the size of materials down to about 1 mm. Fine hammer mills are described in a following subsection. Not all rocks shatter well by impact. Impact breaking is best suited for the reduction of relatively nonabrasive and low-silica-content materials such as limestone, dolomite, anhydrite, shale, and cement rock, the most popular application being on limestone. Most of these devices, such as the hammer crusher shown in Fig. 21-69, have top-fed rotors (of various types) and open bottoms through which product discharge occurs. Some hammer crushers have screens or grates.

Hammer Crusher Pivoted hammers are mounted on a horizontal shaft, and crushing takes place by impact between the hammers and breaker plates. Heavy-duty hammer crushers are frequently used in the quarrying industry, for processing municipal solid waste, and to scrap automobiles.

The rotor of these machines is a cylinder to which is affixed a tough steel bar. Breakage can occur against this bar or on rebound from the

TABLE 21-9b Performance of Nordberg Superior MK-II Gyratory Crusher [in mtph (stph)]*

			Open Side Settings of Discharge Opening — Millimeters (Inches)											
Size	Feed Opening mm (in.)	Pinion RPM	Max. KW (HP)	125mm (5.0")	140mm (5 1/2")	150mm (6")	165mm (6 1/2")	175mm (7")	190mm (7 1/2")	200mm (8")	215mm (8 1/2")	230mm (9")	240mm (9 1/2")	250mm (10")
42-65	1065 (42)	600	375 (500)		1635 (1800)	1880 (2075)	2100 (2315)	2320 (2557)						
50-65	1270 (50)	600	375 (500)			2245 (2475)	2625 (2895)	2760 (3040)						
54-75	1370 (54)	600	450 (600)			2555 (2820)	2855 (3145)	3025 (3335)	3215 (3545)	3385 (3735)				
62-75	1575 (62)	600	450 (600)			2575 (2840)	3080 (3395)	3280 (3615)	3660 (4035)	3720 (4205)				
60-89	1525 (60)	600	600 (800)				4100 (4520)	4360 (4805)	4805 (5295)	5005 (5520)	5280 (5820)	5550 (6115)		
60-110	1525 (60)	514	1000 (1400)					5575 (6150)	5845 (6440)	6080 (6705)	6550 (7220)	6910 (7620)	7235 (7975)	7605 (8385)

*From Metso Minerals brochure.

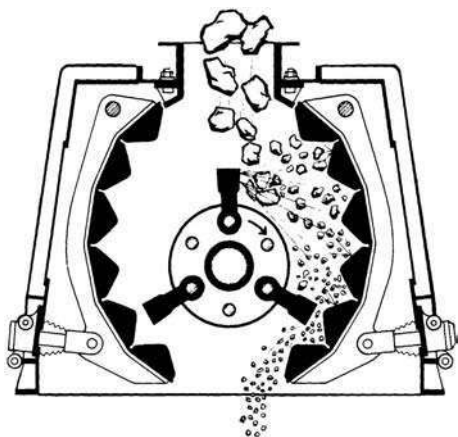


FIG. 21-69 Reversible impactor. (Pennsylvania Crusher Corp.)

walls of the device. Free impact breaking is the principle of the rotor breaker, and it does not rely on pinch crushing or attrition grinding between rotor hammers and breaker plates. The result is a high reduction ratio and elimination of secondary and tertiary crushing stages. By adding a screen on a portable mounting, a complete, compact mobile crushing plant of high capacity and efficiency is provided for use in any location.

The ring granulator features a rotor assembly with loose crushing rings, held outwardly by centrifugal force, which chop the feed. It is suitable for highly friable materials which may give excessive fines in an impact mill. For example, bituminous coal is ground to a product below 2 cm ($\frac{3}{4}$ in.). They have also been successfully used to grind abrasive quartz to sand size, due to the ease of replacement of the ring impact elements.

Cage Mills In a cage mill, cages of one, two, three, four, six, and eight rows, with bars of special alloy steel, revolving in opposite directions produce a powerful impact action that pulverizes many materials. Cage mills are used for many materials, including quarry rock, phosphate rock, and fertilizer and for disintegrating clays, colors, press cake, and bones. The advantage of multiple-row cages is the achievement of a greater reduction ratio in a single pass, and these devices can produce products significantly finer than other impactors in many cases, as fine as 325 mesh. These features and the low cost of the mills make them suitable for medium-scale operations where complicated circuits cannot be justified.

Prebreakers Aside from the normal problems of grinding, there are special procedures and equipment for breaking large masses of feed to smaller sizes for further grinding. There is the breaking or shredding of bales, as with rubber, cotton, or hay, in which the compacted mass does not readily come apart. There also is often caking in bags of plastic or hygroscopic materials which were originally fine. Although crushers are sometimes used, the desired size-reduction ratio often is not obtainable. Furthermore, a lower capital investment may result through choosing a less rugged device which progressively attacks the large mass to remove only small amounts at a time. Typically, these devices are toothed rotating shafts in casings.

HAMMER MILLS

Operation Hammer mills for fine pulverizing and disintegration are operated at high speeds. The rotor shaft may be vertical or horizontal, generally the latter. The shaft carries hammers, sometimes called beaters. The hammers may be T-shaped elements, stirrups, bars, or rings fixed or pivoted to the shaft or to disks fixed to the shaft. The grinding action results from **impact** and **attrition** between lumps or particles of the material being ground, the housing, and the grinding elements. A cylindrical screen or grating usually encloses all or part of the rotor. The fineness of product can be regulated by changing rotor speed, feed rate, or clearance between hammers and

grinding plates, as well as by changing the number and type of hammers used and the size of discharge openings.

The **screen** or **grating discharge** for a hammer mill serves as an internal classifier, but its limited area does not permit effective usage when small apertures are required. A larger external screen may then be required. The feed must be nonabrasive with a hardness of 1.5 or less. Hammer mills can reduce many materials so that substantially all the product passes a 200-mesh screen.

One of the subtleties of operating a hammer mill is that, in general, screen openings should be sized to be much larger than the desired product size. The screen serves to retain very large particles in the mill, but particles that pass through the screen are usually many times smaller than the screen opening. Thus, changing the screen opening size may strongly affect the coarse end of a product-size distribution, but will have limited effect on the median particle size and very little effect at all on the fines. These are more strongly affected by the speed, number, and type of hammers, and, most of all, the speed of the hammers. Screens with very fine openings (500 μ m and less) can be used in smaller laboratory mills to produce very fine product, but tend not to be rugged enough for large-scale use. Particle-size distribution in hammer mill products tends to be very broad, and in cases where relatively narrow product-size distribution is desired, some sort of grinding circuit with an external classifier is almost always needed.

There are a large number of hammer mill manufacturers. The basic designs are very similar, although there are subtle differences in performance and sturdiness that can lead to varying performance. For example, some machines have lower maximum rotation speeds than others. Less rugged and powerful machines might be fully adequate for vegetable materials (e.g., wood), but not suitable for fine mineral grinding. Occasionally, vendors are particularly experienced in a limited set of products and have designs which are especially suited for these. For relatively common materials, it is usually better to use vendors with practical experience in these materials.

Pin Mills In contrast to peripheral hammers of the rigid or swing types, there is a class of high-speed mills having pin breakers in the grinding circuit. These may be on a rotor with stator pins between circular rows of pins on the rotor disk, or they may be on rotors operating in opposite directions, thereby securing an increased differential of speed. There are machines with both vertical and horizontal shafts. In the devices with horizontal shafts, feed is through the top of the mill similar to hammer mills. In devices with vertical shafts, feed is along the shaft, and centrifugal force helps impact the outer ring of pins.

Unlike hammer mills, pin mills do not have screens. Pin mills have a higher energy input per pass than hammer mills and can generally grind softer materials to a finer particle size than hammer mills, while hammer mills perform better on hard or coarse materials. Because they do not have retaining screens, residence time in pin mills is shorter than in hammer mills, and pin mills are therefore more suitable for heat-sensitive materials or cryogrinding.

Universal Mills Several manufacturers are now making "universal mills," which are essentially hammer mill-style devices with fairly narrow chambers that can be fitted with either a variety of hammer mill type of hammers and screens (although usually only fixed hammers) or set up as a pin mill. These are useful where frequent product changes are made and it is necessary to be able to rapidly change the grind characteristics of the devices, such as small lot manufacturing or grinding research.

Hammer Mills with Internal Air Classifiers A few mills are designed with internal classifiers. These are generally capable of reducing products to particle sizes below 45 μ m, down to about 10 μ m, depending on the material. A good example of this type of mill is the Hosokawa Mikro-ACM mill, which is a pin mill fitted with an air classifier. There are also devices more like hammermills, such as the Raymond vertical mill, which do not grind quite as fine as the pin mill-based machine but can handle slightly more abrasive materials.

The **Mikro-ACM pulverizer** is a pin mill with the feed being carried through the rotating pins and recycled through an attached vane classifier. The classifier rotor is separately driven through a speed control which may be adjusted independently of the pin-rotor speed. Oversize particles are carried downward by the internal circulating airstream and are returned to the pin rotor for further reduction. The

constant flow of air through the ACM maintains a reasonable low temperature, which makes it ideal for handling heat-sensitive materials, and it is commonly used in the powder coating and pharmaceutical industries for fine grinding.

ROLL CRUSHERS

Once popular for coarse crushing in the minerals industry, these devices long ago lost favor to gyratory and jaw crushers because of their poorer wear characteristics with hard rocks. Roll crushers are still commonly used for grinding of agricultural products such as grains, and for both primary and secondary crushing of coal and other friable rocks such as oil shale and phosphate. The roll surface is smooth, corrugated, or toothed, depending on the application. *Smooth rolls* tend to wear ring-shaped corrugations that interfere with particle nipping, although some designs provide a mechanism to move one roll from side to side to spread the wear. *Corrugated rolls* give a better bite to the feed, but wear is still a problem. *Toothed rolls* are still practical for rocks of not too high silica content, since the teeth can be regularly resurfaced with hard steel by electric arc welding. Toothed rolls are frequently used for crushing coal and chemicals. For further details, see Edition 6 of this handbook.

The **capacity** of roll crushers is calculated from the ribbon theory, according to the formula

$$Q = dLs/2.96 \quad (21-91)$$

where Q = capacity, cm^3/min ; d = distance between rolls, cm ; L = length of rolls, cm ; and s = peripheral speed, cm/min . The denominator becomes 1728 in engineering units for Q in cubic feet per minute, d and L in inches, and s in inches per minute. This gives the theoretical capacity and is based on the rolls discharging a continuous, solid uniform ribbon of material. The actual capacity of the crusher depends on roll diameter, feed irregularities, and hardness and varies between 25 and 75 percent of theoretical capacity.

ROLL PRESS

One of the newer comminution devices, the **roll press**, has achieved significant commercial success, especially in the cement industry. It is used for fine crushing, replacing the function of a coarse ball mill or of tertiary crushers. Unlike ordinary roll crushers, which crush individual particles, the roll press is choke-fed and acts on a thick stream or ribbon of feed. Particles are crushed mostly against other particles, so wear is very low. A roll press can handle a hard rock such as quartz. Energy efficiency is also greater than in ball mills.

The product is in the form of agglomerated slabs. These are broken up in either a ball mill or an impact or hammer mill running at a speed too slow to break individual particles. Some materials may even deagglomerate from the handling that occurs in conveyors. A large proportion of fines is produced, but a fraction of coarse material survives. This makes recycle necessary.

From experiments to grind cement clinker to $-80\ \mu\text{m}$, as compression is increased from 100 to 300 MPa, the required recycle ratio decreases from 4 to 2.8. The energy required per ton of throughput increases from 2.5 to 3.5 kWh/ton. These data are for a 200-mm-diameter pilot-roll press. Status of 150 installations in the cement industry is reviewed [Strasser et al., *Rock Products*, **92**(5), 60-72 (1989)]. In cement clinker milling, wear is usually from 0.1 to 0.8 g/ton, and for cement raw materials it is between 0.2 and 1.2 g/ton, whereas it may be 20- to 40-in ball mills.

The size of the largest feed particles should not exceed $0.04 \times$ roll diameter D according to Schoenert (loc. cit.). However, it has been found [Wuestner et al., *Zement-Kalk-Gips*, **41**(7), 345-353 (1987); English edition, 207-212] that particles as large as 3 to 4 times the roll gap may be fed to an industrial press.

Machines with up to 2500-kW installed power and 1000-ton/h (900-ton/h) capacity have been installed. The largest presses can supply feed for four or five ball mills. Operating experience (Wuestner et al., loc. cit.) has shown that roll diameters of about 1 m are preferred, as a compromise between production rate and stress on the equipment.

The press must be operated choke-fed, with a substantial depth of feed in the hopper; otherwise it will act as an ordinary roll crusher.

ROLL RING-ROLLER MILLS

Roll ring-roller mills (Fig. 21-70) are equipped with rollers that operate against grinding rings. Pressure may be applied with heavy springs or by centrifugal force of the rollers against the ring. Either the ring or the rollers may be stationary. The grinding ring may be in a vertical or horizontal position. Ring-roller mills also are referred to as ring roll mills or roller mills or medium-speed mills. The ball-and-ring and bowl mills are types of ring-roller mill. Ring-roller mills are more energy-efficient than ball mills or hammer mills. The energy to grind coal to 80 percent passing 200 mesh was determined (Luckie and Austin, *Coal Grinding Technology—A Manual for Process Engineers*) as ball mill, 13 hp/ton; hammer mill, 22 hp/ton; roller mill, 9 hp/ton.

Raymond Ring-Roller Mill The Raymond ring-roller mill (Fig. 21-70) is a typical example of a ring-roller mill. The base of the mill carries the grinding ring, rigidly fixed in the base and lying in the horizontal plane. Underneath the grinding ring are tangential air ports through which the air enters the grinding chamber. A vertical shaft driven from below carries the roller journals. Centrifugal force urges the pivoted rollers against the ring. The raw material from the feeder drops between the rolls and ring and is crushed. Both centrifugal air motion and plows move the coarse feed to the nips. The air entrains fines and conveys them up from the grinding zone, providing some classification at this point. An air classifier is also mounted above the grinding zone to return oversize particles. The method of classification used with Raymond mills depends on the fineness desired. If a medium-fine product is required (up to 85 or 90 percent through a No. 100 sieve), a single-cone air classifier is used.

This consists of a housing surrounding the grinding elements with an outlet on top through which the finished product is discharged. This is known as the low-side mill. For a finer product and when frequent changes in fineness are required, the whizzer-type classifier is

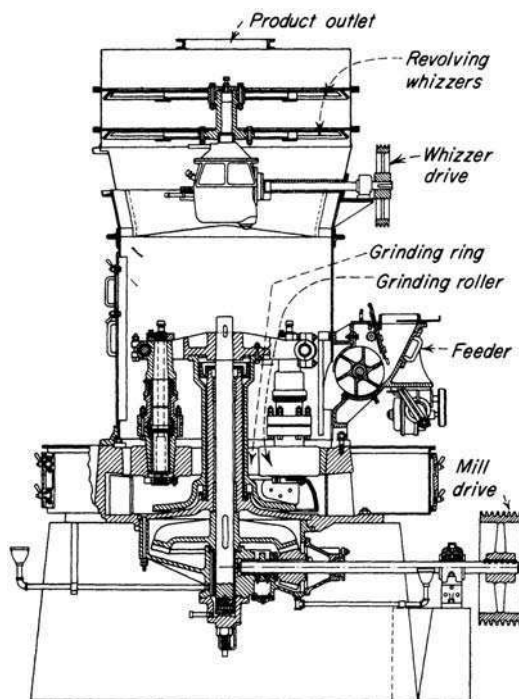


FIG. 21-70 Raymond high-side mill with an internal whizzer classifier. (ABB Raymond Div., Combustion Engineering Inc.)

used. This type of mill is known as the *high-side mill*. The Raymond ring-roll mill with internal air classification is used for the large-capacity fine grinding of most of the softer nonmetallic minerals. Materials with a Mohs-scale hardness up to and including 5 are handled economically on these units. Typical natural materials handled include barites, bauxite, clay, gypsum, magnesite, phosphate rock, iron oxide pigments, sulfur, talc, graphite, and a host of similar materials. Many of the manufactured pigments and a variety of chemicals are pulverized to high fineness on such units. Included are such materials as calcium phosphates, sodium phosphates, organic insecticides, powdered cornstarch, and many similar materials. When properly operated under suction, these mills are entirely dust-free and automatic.

PAN CRUSHERS

Design and Operation The pan crusher consists of one or more grinding wheels or mullers revolving in a pan; the pan may remain stationary and the mullers be driven, or the pan may be driven while the

mullers revolve by friction. The mullers are made of tough alloys such as Ni-Hard. Iron scrapers or plows at a proper angle feed the material under the mullers.

Performance The dry pan is useful for crushing medium-hard and soft materials such as clays, shales, cinders, and soft minerals such as barites. Materials fed should normally be 7.5 cm (3 in) or smaller, and a product able to pass No. 4 to No. 16 sieves can be delivered, depending on the hardness of the material. High reduction ratios with low power and maintenance are features of pan crushers. Production rates can range from 1 to 54 Mg/h (1 to 60 tons/h) according to pan size and hardness of material as well as fineness of feed and product.

The **wet pan** is used for developing plasticity or molding qualities in ceramic feed materials. The abrasive and kneading actions of the mullers blend finer particles with the coarser particles as they are crushed [Greaves-Walker, *Am. Refract. Inst. Tech. Bull.* 64 (1937)], and this is necessary so that a high packing density can be achieved to result in strength.

CRUSHING AND GRINDING EQUIPMENT: FLUID-ENERGY OR JET MILLS

DESIGN

Jet milling, also called fluid-energy grinding, is an increasingly used process in the chemical industry for processing brittle, heat-sensitive materials into very fine powders with a narrow size distribution. For more than 90 years jet mills have been built and applied successfully on a semilarge scale in the chemical industry. A number of famous designs are extensively described in a number of patents and publications.

Most such mills are variations on one of the fundamental configurations depicted in Fig. 21-71. The designs differ from each other by the arrangement of the nozzles and the classification section. In the following paragraphs the jet mill types are briefly discussed.

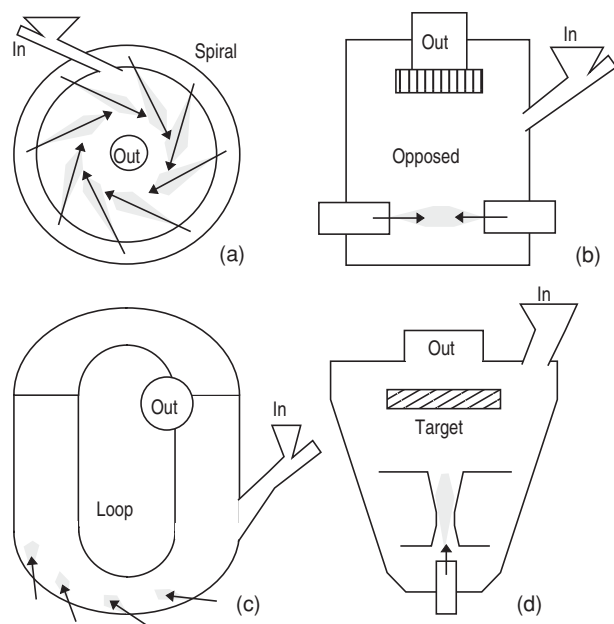


FIG. 21-71 Schematic representation of basic jet mill designs: (a) spiral; (b) opposed; (c) loop; (d) target.

The key feature of jet mills is the conversion of high pressure to kinetic energy. The operating fluid enters the grinding chamber through nozzles placed in the wall. The feed particles brought into the mill through a separate inlet are entrained by expanding jets and accelerated to velocities as high as the velocity of sound. In fact three collision geometries can be distinguished:

Interparticle collisions due to turbulence in a free jet

Collisions between particles accelerated by opposed jets

Impact of particles on a target

The turbulent nature of the jets causes particles to have differences in velocities and directions. Particle breakage in jet mills is mainly a result of interparticle collisions: wall collisions are generally thought to be of minor importance only, except in mill type D (Fig. 21-71). Fluid-energy-driven mills are a class of impact mills with a considerable degree of attrition due to eccentric and gliding interparticle impacts. The grinding mechanism via mutual collisions means that jet mills operate with virtually no product contact. In other words, the contamination grade is low.

The classification of product leaving the mill depends on a balance between centrifugal forces and drag forces in the flow field around the mill outlet. Mill types A and C create a free vortex at the outlet, while jet mill D makes use of gravity. Type B has an integrated rotor. The final product quality is largely determined by the success of classification.

TYPES

Spiral Jet Mill The original design of the spiral jet mill, also called a pancake mill, is shown in Fig. 21-71. This design was first described by Andrews in 1936 and patented under the name Micronizer. A number of nozzles are placed in the outer wall of the mill through which the grinding medium, a gas or steam, enters the mill.

A spiral jet mill combines both grinding and classification by the same jets. The vortex causes coarse particles of the mill contents to be transferred to the outer zone, as fines can leave through the central outlet. The solid feed is brought into the mill by an air pusher. The outlet is placed in the center of the mill chamber. The working principle of this mill was extensively investigated by Rumpf.

Spiral jet mills are notable for their robust design and compactness. Their direct air operation avoids the need for separate drive units. Another significant argument for the use of jet mills is the lower risk for dust explosions.

Opposed Jet Mill Opposed jet mills are fluid-energy-driven mills that contain two or more jets aligned toward each other (see Fig. 21-71b). Different versions are on the market, based on a design patented by Willoughby (1917). In this type of jet mill, opposed gas

streams entrain the mill holdup. At the intersection of the jets the coarse particles hit one another. The grinding air carries the particles upward in a kind of fluidized bed to the classification zone.

Adjustment of the rotor speed allows a direct control of the particle size of the end product. The feed is entered by a rotary valve. Drawbacks are the higher cost of investment and maintenance. These types of mills are described by Vogel and Nied.

Other Jet Mill Designs Figure 21-71d shows one of the earliest jet mill designs (around 1880), but it is still in use today. In this mill a jet loaded with particles is impacting on an anvil. Consequently the impact

efficiency is high for relatively large particles. Very fine grinding becomes difficult as small particles are decelerated in the stagnant zone in front of the target. Fines are dragged out in an airstream by a fan, as coarse material is recirculated to the jet entry. Points of improvement have included better classification and abrasive-resistant target material. This device is suitable to incorporate as a pregrinder.

The loop mill (Fig. 21-71c), also called Torus mill, was designed by Kidwell and Stephanoff (1940). The grinding fluid is brought into the grinding section. The fines leave the mill through the classification section.

CRUSHING AND GRINDING EQUIPMENT: WET/DRY GRINDING—MEDIA MILLS

OVERVIEW

Another class of grinding mills is media mills. These are mills which grind materials primarily through the action of mechanically agitated balls made out of metals (mostly steel) or various ceramics. Different mills use different methods of agitation. Some are more commonly used for dry grinding, others for wet grinding, and still others can be used in both modes. Types of media mills include tumbling mills, stirred media mills, and vibratory mills.

MEDIA SELECTION

A key to the performance of media mills is the selection of an appropriate grinding medium. Jorg Schwedes and his students have developed correlations which are effective in determining optimal media size for stirred media mills [Kwade et al., *Powder Technol.*, **86** (1996); and Becker et al., *Int. J. Miner. Process.*, **61** (2001)]. Although these correlations were developed for stirred media mills, the principles developed apply to all media mills.

In this methodology, energy input is broken up into stress intensity (SI) and stress frequency (SF), defined as:

$$SI = (\rho_m - \rho) D_m^3 V_t^2$$

$$SF = \omega (D_m/D)^2 t$$

where ρ is slurry density, ρ_m is media density, D_m is media diameter, ω is the rotational speed of a rotating mill, D is the rotor diameter of a rotating mill, and V_t is the tip speed of a rotating mill.

Stress intensity is related to the kinetic energy of media beads, and stress frequency is related to the frequency of collisions.

When stress intensity is plotted versus media particle size achieved at constant grinding energy (such as Fig. 21-72) for limestone, it can be seen that a large number of experimental data can be collapsed onto a single curve. There is a relatively narrow range of stress intensity which gives the smallest particle size, and larger or smaller stress intensities give increasingly larger particle sizes at the same energy input.

This can be explained in physical terms in the following way. For each material, there is a critical stress intensity. If the stress intensity applied during grinding is less than the critical stress intensity, then very little grinding occurs. If the applied stress intensity is much greater than the critical stress intensity, then unnecessary energy is being used in bead collisions, and a greater grinding rate could be obtained by using smaller beads that would collide more frequently. This has a very practical implication for choosing the size and, to some extent, the density of grinding beads. At a constant stirring rate (or tumbling rate or vibration rate), a small range of media sizes give an optimal grinding rate for a given material in a given mill. In practice, most mills are operated using media slightly larger than the optimal size, as changes in feed and media quality can shift the value of the

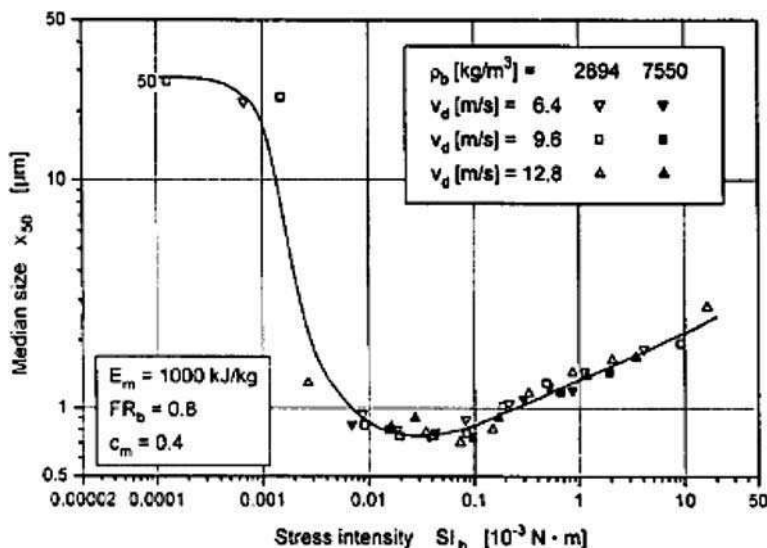


FIG. 21-72 Influence of stress intensity on the size of limestone for a specific energy input of 1000 kJ/kg. [From A. Kwade et al., *Powder Technol.* **86** (1996).]

critical stress intensity over the lifetime of an industrial process, and the falloff in grinding rate when one is below the critical stress intensity is quite dramatic.

Another important factor when choosing media mills is media and mill wear. Most media mills have fairly rapid rates of media wear, and it is not uncommon to have to replace media monthly or at least add partial loads of media weekly. Media wear will reduce the grind rate of a mill and can cause significant product contamination. Very hard media materials often have low wear rates, but can cause very rapid mill wear. Media with a good balance of properties tend to be specialty ceramics. Commonly used ceramics include glass, specialty sand, alumina, zirconia (although this is higher in mill wear), zirconia-silica composites, and yttria- or Ceria-stabilized zirconia. Yttria-stabilized zirconia is particularly wear resistant but is very expensive. Steel is often used as a medium and has a very good combination of low cost, good wear life, and gentle mill wear if a product can handle slight discoloration and iron content from the medium.

TUMBLING MILLS

Ball, pebble, rod, tube, and compartment mills have a cylindrical or conical shell, rotating on a horizontal axis, and are charged with a grinding medium such as balls of steel, flint, or porcelain or with steel rods. The **ball mill** differs from the tube mill by being short in length; its length, as a rule, is not far from its diameter (Fig. 21-73). Feed to ball mills can be as large as 2.5 to 4 cm (1 to 1½ in) for very fragile materials, although the top size is generally 1 cm (½ in). Most ball mills operate with a reduction ratio of 20:1 to 200:1. The largest balls are typically 13 cm (5 in) in diameter. The tube mill is generally long in comparison with its diameter, uses smaller balls, and produces a finer product. The compartment mill consists of a cylinder divided into two or more sections by perforated partitions; preliminary grinding takes place at one end and finish grinding at the charge end. These mills have a length-to-diameter ratio in excess of 2 and operate with a reduction ratio of up to 600:1.

Rod mills deliver a more uniform granular product than other revolving mills while minimizing the percentage of fines, which are sometimes detrimental. The **pebble mill** is a tube mill with flint or ceramic pebbles as the grinding medium and may be lined with

ceramic or other nonmetallic liners. The **rock-pebble mill** is an autogenous mill in which the medium consists of larger lumps scalped from a preceding step in the grinding flow sheet.

Design The conventional type of **batch mill** consists of a cylindrical steel shell with flat steel-flanged heads. Mill length is equal to or less than the diameter [Coghill, De Vaney, and O'Meara, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **112**, 79 (1934)]. The discharge opening is often opposite the loading manhole and for wet grinding usually is fitted with a valve. One or more vents are provided to release any pressure developed in the mill, to introduce inert gas, or to supply pressure to assist discharge of the mill. In dry grinding, the material is discharged into a hood through a grate over the manhole while the mill rotates. Jackets can be provided for heating and cooling.

Material is fed and discharged through hollow trunnions at opposite ends of **continuous mills**. A grate or diaphragm just inside the discharge end may be employed to regulate the slurry level in wet grinding and thus control retention time. In the case of **air-swept mills**, provision is made for blowing air in at one end and removing the ground material in air suspension at the same end or the other end. Ball mills usually have **liners** which are replaceable when they wear. Both all-rubber liners and rubber liners with metal lifter bars are currently used in large ball mills [McTavish, *Mining Engg.*, **42**, 1249–1251 (Nov. 1990)]. Lifters must be at least as high as the ball radius, to key the ball charge and ensure that the balls fall into the toe area of the mill [Powell, *Int. J. Mineral Process.*, **31**, 163–193 (1991)]. Special operating problems occur with smooth-lined mills owing to erratic slip of the charge against the wall. At low speeds the charge may **surge** from side to side without actually tumbling; at higher speeds tumbling with **oscillation** occurs. The use of lifters prevents this [Rose, *Proc. Inst. Mech. Eng. (London)*, **170**(23), 773–780 (1956)].

Pebble mills are frequently lined with nonmetallic materials when iron contamination would harm a product such as a white pigment or cement. Belgian sillex (silica) and porcelain block are popular linings. Silica linings and ball media have proved to wear better than other nonmetallic materials. Smaller mills, up to about 50-gal capacity, are made in one piece of ceramic with a cover.

Multicompartmented Mills Multicompartmented mills feature grinding of coarse feed to finished product in a single operation, wet or dry. The primary grinding compartment carries large grinding balls

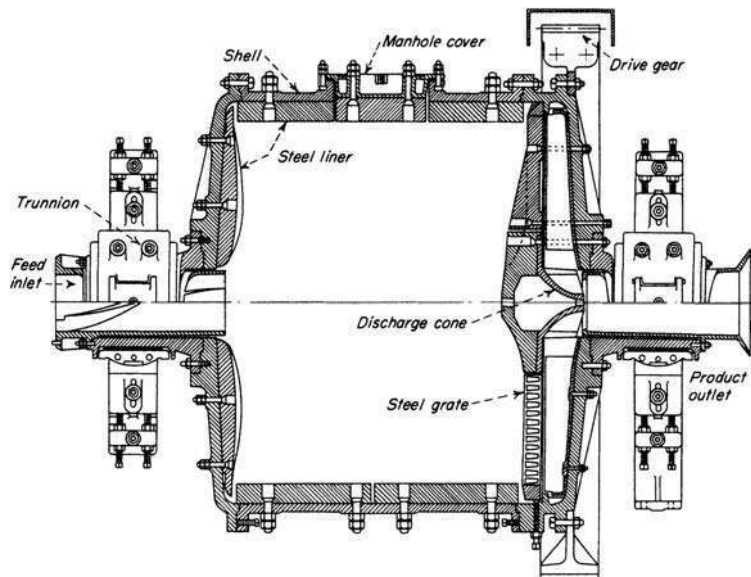


FIG. 21-73 Marcy grate-type continuous ball mill. (Allis Mineral Systems, Svedala Inc.)

or rods; one or more secondary compartments carry smaller media for finer grinding.

Operation Cascading and cataracting are the terms applied to the motion of grinding media. The former applies to the rolling of balls or pebbles from top to bottom of the heap, and the latter refers to the throwing of the balls through the air to the toe of the heap. The criterion by which the ball action in mills of various sizes may be compared is the concept of **critical speed**. It is the theoretical speed at which the centrifugal force on a ball in contact with the mill shell at the height of its path equals the force on it due to gravity:

$$N_c = 42.3/\sqrt{D} \quad (21-92)$$

where N_c is the critical speed, r/min, and D is diameter of the mill, m (ft), for a ball diameter that is small with respect to the mill diameter. The numerator becomes 76.6 when D is expressed in feet. **Actual mill speeds** range from 65 to 80 percent of critical. It might be generalized that 65 to 70 percent is required for fine wet grinding in viscous suspension and 70 to 75 percent for fine wet grinding in low-viscosity suspension and for dry grinding of large particles up to 1-cm (½-in) size. Unbaffled mills can run at 105 percent of critical to compensate for slip. The chief factors determining the size of **grinding balls** are fineness of the material being ground and maintenance cost for the ball charge. A coarse feed requires a larger ball than a fine feed. The need for a calculated ball-size feed distribution is open to question; however, methods have been proposed for calculating a rationed ball charge [Bond, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **153**, 373 (1943)]. The recommended optimum size of makeup rods and balls is [Bond, *Min. Eng.*, **10**, 592–595 (1958)]

$$D_b = \sqrt{\frac{X_f E_i}{K n_r}} \sqrt{\frac{\rho_s}{\sqrt{D}}} \quad (21-93)$$

where D_b = rod or ball diameter, cm (in); D = mill diameter, m (ft); E_i is the work index of the feed; n_r is speed, percent of critical; ρ_s is feed specific gravity; and K is a constant = 214 for rods and 143 for balls. The constant K becomes 300 for rods and 200 for balls when D_b and D are expressed in inches and feet, respectively. This formula gives reasonable results for production-sized mills but not for laboratory mills. The ratio between the recommended ball and rod sizes is 1.23.

Material and Ball Charges The load of a grinding medium can be expressed in terms of the percentage of the volume of the mill that it occupies; i.e., a bulk volume of balls half filling a mill is a 50 percent ball charge. The void space in a static bulk volume of balls is approximately 41 percent. The amount of material in a mill can be expressed conveniently as the ratio of its volume to that of the voids in the ball load. This is known as the **material-to-void ratio**. If the solid material and its suspending medium (water, air, etc.) just fill the ball voids, the M/V ratio is 1, for example. Grinding-media loads vary from 20 to 50 percent in practice, and M/V ratios are usually near 1.

The material charge of continuous mills, called the **holdup**, cannot be set directly. It is indirectly determined by operating conditions. There is a maximum throughput rate that depends on the shape of the mill, the flow characteristics of the feed, the speed of the mill, and the type of feed and discharge arrangement. Above this rate the holdup increases unstably. The holdup of material in a continuous mill determines the mean residence time, and thus the extent of grinding. Gupta et al. [*Int. J. Mineral Process.*, **8**, 345–358 (Oct. 1981)] analyzed published experimental data on a 40·40-cm grate discharge laboratory mill, and determined that holdup was represented by $H_w = (4.020 - 0.176WI) F_w + (0.040 + 0.01237WI) S_w - (4.970 + 0.395WI)$, where WI is Bond work index based on 100 percent passing a 200-mesh sieve, F_w is the solids feed rate, kg/min, and S_w is weight percent of solids in the feed. This represents experimental data for limestone, feldspar, sulfide ore, and quartz. The influence of WI is believed to be due to its effect on the amount of fines present in the mill. Parameters that did not affect H_w are specific gravity of feed material and feed size over the narrow range studied. Sufficient data were not available to develop a correlation for overflow mills, but the data indicated a linear variation of H_w with F as well. The mean resi-

dence time τ (defined as H_w/F) is the most important parameter since it determines the time over which particles are exposed to grinding. Measurements of the water (as opposed to the ore) of several industrial mills (Weller, *Automation in Mining Mineral and Metal Processing*, 3d IFAC Symposium, 303–309, 1980) showed that the maximum mill filling was about 40 percent, and the maximum flow velocity through the mill was 40 m/h. Swaroop et al. [*Powder Technol.*, **28**, 253–260 (Mar.–Apr. 1981)] found that the material holdup is higher and the vessel dispersion number $D\tau/L^2$ (see subsection “Continuous-Mill Simulation”) is lower in the rod mill than in the ball mill under identical dimensionless conditions. This indicates that the known narrow-product-size distribution from rod mills is partly due to less mixing in the rod mill, in addition to different breakage kinetics.

The holdup in grate-discharge mills depends on the grate openings. Kraft et al. [*Zement-Kalk-Gips Int.*, **42**(7), 353–359 (1989); English edition, 237–239] measured the effect of various hole designs in wet milling. They found that slots tangential to the circumference gave higher throughput and therefore lower holdup in the mill. Total hole area had little effect until the feed rate was raised to a critical value (30 m/h in a mill with 0.26-m diameter and 0.6 m long); above this rate the larger area led to lower holdup. The open area is normally specified between 3 and 15 percent, depending on the number of grinding chambers and other conditions. The slots should be 1.5 to 16 mm wide, tapered toward the discharge side by a factor of 1.5 to 2 to prevent blockage by particles.

Dry vs. Wet Grinding The choice between wet and dry grinding is generally dictated by the end use of the product. If the presence of liquid with the finished product is not objectionable or the feed is moist or wet, wet grinding generally is preferable to dry grinding, but power consumption, liner wear, and capital costs determine the choice. Other factors that influence the choice are the performance of subsequent dry or wet classification steps, the cost of drying, and the capability of subsequent processing steps for handling a wet product. The net production in wet grinding in the Bond grindability test varies from 145 to 200 percent of that in dry grinding depending on mesh [Maxson, Cadena, and Bond, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **112**, 130–145, 161 (1934)]. Ball mills have a large field of application for wet grinding in closed circuit with size classifiers, which also perform advantageously wet.

Dry Ball Milling In fine dry grinding, surface forces come into action, causing cushioning and ball coating, resulting in a less efficient use of energy. Grinding media and liner-wear consumption per ton of ground product are lower for a dry-grinding system. However, power consumption for dry grinding is about 30 percent larger than for wet grinding. Dry grinding requires the use of dust-collecting equipment.

Wet Ball Milling See Fig. 21-74. The **rheological properties** of the slurry affect the grinding behavior in ball mills. Rheology depends on solids content, particle size, and mineral chemical properties [Kawatra and Eisele, *Int. J. Mineral Process.*, **22**, 251–259 (1988)]. Above 50 vol. % solids, a mineral slurry may become pseudoplastic, i.e., it exhibits a yield value (Austin, Klimpel, and Luckie, *Process Engineering of Size Reduction: Ball Milling*, AIME, 1984). Above the yield value the grinding rate decreases, and this is believed to be due to adhesion of grinding media to the mill wall, causing centrifuging [Tangsatitkulchai and Austin, *Powder Technol.*, **59**(4), 285–293 (1989)]. Maximum power draw and fines production is achieved when the solids content is just below that which produces the critical yield. The solids concentration in a pebble-mill slurry should be high enough to give a slurry viscosity of at least 0.2 Pa·s (200 cP) for best grinding efficiency [Creyke and Webb, *Trans. Br. Ceram. Soc.*, **40**, 55 (1941)], but this may have been required to key the charge to the walls of the smooth mill used.

Since viscosity increases with amount of fines present, mill performance can often be improved by closed-circuit operation to remove fines. Chemicals such as surfactants allow the solids content to be increased without increasing the yield value of the pseudoplastic slurry, allowing a higher throughput. They may cause foaming problems downstream, however. Increasing temperature lowers the viscosity of water, which controls the viscosity of the slurry under high-shear conditions such as those encountered in the cyclone, but does not

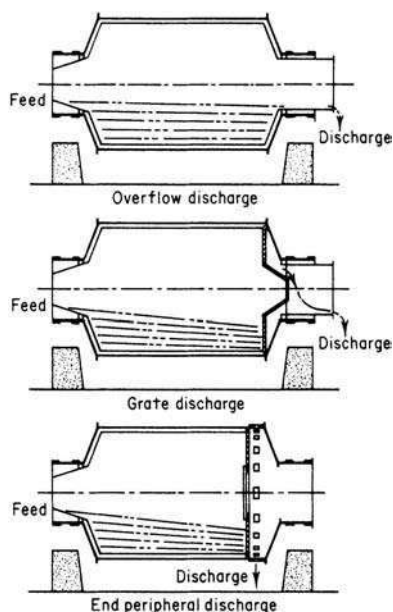


FIG. 21-74 Continuous ball-mill discharge arrangements for wet grinding.

greatly affect chemical forces. Slurry viscosity can be most directly controlled by controlling solids content.

MILL EFFICIENCIES

In summary, controlling factors for cylindrical mills are as follows:

1. Mill speed affects capacity, as well as liner and ball wear, in direct proportion up to 65 to 75 percent of critical speed.
2. Ball charge equal to 35 to 50 percent of the mill volume gives the maximum capacity.
3. Minimum-size balls capable of grinding the feed give maximum efficiency.
4. Bar-type lifters are essential for smooth operation.
5. Material filling equal to ball-void volume is optimum.
6. Higher-circulating loads tend to increase production and decrease the amount of unwanted fine material.
7. Low-level or grate discharge with recycle from a classifier increases grinding capacity over the center or overflow discharge; but liner, grate, and media wear is higher.
8. Ratio of solids to liquids in the mill must be considered on the basis of slurry rheology.

Capacity and Power Consumption One of the methods of mill sizing is based on the observation that the amount of grinding depends on the amount of energy expended, if one assumes comparable good practice of operation in each case. The energy applied to a ball mill is primarily determined by the size of mill and load of balls. Theoretical considerations show the net power to drive a ball mill to be proportional to $D^{2.5}$, but this exponent may be used without modification in comparing two mills only when operating conditions are identical [Gow, et al., *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **112**, 24 (1934)]. The net power (the gross power draw of the mill minus the power to turn an empty mill) to drive a ball mill was found to be

$$E = [(1.64L - 1)K + 1][(1.64D)^{2.5}E_2] \quad (21-94)$$

where L is the inside length of the mill, m (ft); D is the mean inside diameter of the mill, m (ft); E_2 is the net power used by a 0.6- by 0.6-m (2- by 2-ft) laboratory mill under similar operating condi-

tions; and K is 0.9 for mills less than 1.5 m (5 ft) long and 0.85 for mills over 1.5 m long. This formula may be used to scale up pilot milling experiments in which the diameter and length of the mill are changed, but the size of balls and the ball loading as a fraction of mill volume are unchanged. More accurate computer models are now available.

Morrell [*Trans. Instn. Min. Metall.*, Sect. C, **101**, 25–32 (1992)] established equations to predict power draft based on a model of the shape of the rotating ball mass. Photographic observations from laboratory and plant-sized mills, including autogenous, semiautogenous, and ball mills, showed that the shape of the material charge could roughly be represented by angles that gave the position of the toe and shoulder of the charge. The power is determined by the angular speed and the torque to lift the balls. The resulting equations show that power increases rapidly with mill filling up to 35 percent, then varies little between 35 and 50 percent. Also, net power is related to mill diameter to an exponent less than 2.5. This agrees with Bond [*Brit. Chem. Engr.*, 378–385 (1960)] who stated from plant experience that power increases with diameter to the 2.3 exponent or more for larger mills. Power input increases faster than volume, which varies with diameter squared. The equations can be used to estimate holdup for control of autogenous mills.

STIRRED MEDIA MILLS

Stirred media mills have a wide range of applications. They are often found in minerals processing grinding circuits for grinding in the size range of 5 to 50 μm , and they are the only mill capable of reliably grinding materials to submicrometer sizes. They are very commonly used for grinding and dispersion of dyes, clays, and pigments and are also used for biological cell disruption.

Stirred media mills are also the dominant process equipment used for dispersing fine powders into liquid, e.g., pigment dispersions, and have largely displaced ball mills in these applications. In these applications, they are capable of dispersing powders down to particle sizes below 100 nm effectively and reliably.

Stirred media mills are used almost exclusively for wet grinding. In general, the higher the tip speed of the rotor, the lower the viscosity that can be tolerated by the mill. At high viscosity, very little bead motion occurs. Similarly, mills with lower tip speeds can tolerate the use of larger, heavier media, since gravity will cause additional motion in this case.

Design In stirred mills, a central paddle wheel or disc ed armature stirs the media at speeds from 100 to 3000 r/min (for some lab units). Stirrer tip speeds vary from 2 m/s for some attritors to 18 m/s for some high-energy mills.

Attritors In the *Attritor* (Union Process Inc.) a single vertical armature rotates several long radial arms. The rotation speeds are much slower than with other stirred media mills, and the grinding behavior in these mills tends to be more like that in tumbling mills than in other stirred media mills. They can be used for higher-viscosity applications. These are available in batch, continuous, and circulation types.

Vertical Mills Vertical mills are, generally speaking, older designs whose chief advantage is that they are inexpensive. They are vertical chambers of various shapes with a central agitator shaft. The media are stirred by discs or pegs mounted on the shaft. Some mills are open at the top, while others are closed at the top. Most mills have a screen at the top to retain media in the mill.

The big drawback to vertical mills is that they have a limited flow rate range due to the need to have a flow rate high enough to help fluidize the media and low enough to avoid carrying media out of the top of the mill. The higher the viscosity of the slurry in the mill, the more difficult it is to find the optimal flow rate range. Slurries that change viscosity greatly during grinding, such as some high solid slurries, can be particularly challenging to grind in vertical mills.

Horizontal Media Mills Horizontal media mills are the most common style of mill and are manufactured by a large number of companies. Figure 21-75 illustrates the Drais continuous stirred media mill. The mill has a horizontal chamber with a central shaft. The media are stirred by discs or pegs mounted on the shaft. The

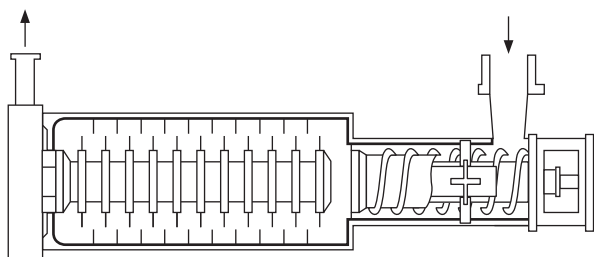


FIG. 21-75 Drais wet-grinding and dispersing system (U.S. patent 3,957,210) Draiswerke GmbH. [Stehr, International J. Mineral Processing, **22**(1-4), 431-444 (1988).]

advantage of horizontal machines is the elimination of gravity segregation of the feed. The feed slurry is pumped in at one end and discharged at the other where the media are retained by a screen or an array of closely spaced, flat discs. Most are useful for slurries up to about 50 Pa·s (50,000 cP). Also note that slurries with very low viscosities (under 1 Pa·s) can sometimes cause severe mill wear problems. Several manufacturers have mill designs where either the screen rotates or the mill outlet is designed in such a way as to use centrifugal force to keep media off the screen. These mills can use media as fine as 0.2 mm. They also have the highest flow rate capabilities. Hydrodynamically shaped screen cartridges can sometimes accommodate media as fine as 0.2 mm.

Agitator discs are available in several forms: smooth, perforated, eccentric, and pinned. The effect of disc design has received limited study, but pinned discs are usually reserved for highly viscous materials. Cooling water is circulated through a jacket and sometimes through the central shaft. The working speed of disc tips ranges from 5 to 18 m/s regardless of mill size. A series of mills may be used with decreasing media size and increasing rotary speed to achieve desired fine particle size.

Annular Gap Mills Some mills are designed with a large interior rotor that has a narrow gap between the rotor and the inner chamber wall. These annular gap mills generally have higher energy input per unit volume than do the other designs. Media wear tends to be correspondingly higher as well. Despite this, these mills can be recommended for heat-sensitive slurries, because the annular design of the mills allows for a very large heat-transfer surface.

Manufacturers There are many manufacturers of stirred media mills worldwide. Major manufacturers of stirred media mills include Netzsch, Buhler, Drais (now part of Buhler), Premier (now part of SPX), Union Process, and MorehouseCowles. Many of these manufacturers have devices specifically adapted for specific industries. For example, Buhler has some mills specifically designed to handle higher-viscosity inks, and Premier has a mill designed specifically for milling/flaking of metal powders.

PERFORMANCE OF BEAD MILLS

Variables affecting the milling process are listed below:

- Agitator speed
- Feed rate
- Size of beads
- Bead charge, percent of mill volume
- Feed concentration
- Density of beads
- Temperature
- Design of blades
- Shape of mill chamber
- Residence time

The availability of more powerful, continuous machines has extended the possible applications to both lower and higher size ranges, from 5- to 200- μ m product size, and to a feed size as large as 5 mm. The energy density may be 50 times larger than that in tumbling-ball mills, so that a smaller mill is required (Fig. 21-76). Mills range in size from

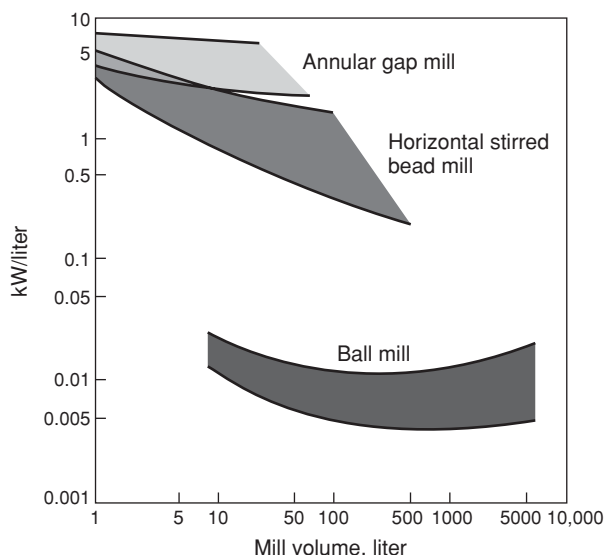


FIG. 21-76 Specific power of bead and ball mills [Kolb, Ceramic Forum International, **70**(5), 212-216 (1993)].

1 to 1000 L, with installed power up to 320 kW. Specific power ranges from 10 to 200 or even 2000 kWh/t, with feed rates usually less than 1 t. For stirred media mills, an optimum media size is about 20 times greater than the material to be ground. It is possible to relate Reynolds number to mill power draw in the same way that this is done for rotating mixers (see Fig. 21-77).

In vertical disc-stirred mills, the media should be in a fluidized condition (White, *Media Milling*, Premier Mill Co., 1991). Particles can pack in the bottom if there is not enough stirring action or feed flow; or in the top if flow is too high. These conditions are usually detected by experiment. A study of bead milling [Gao and Forssberg, *Int. J. Mineral Process.*, **32**(1-2), 45-59 (1993)] was done in a continuous Drais mill of 6-L capacity having seven 120·10-mm horizontal discs. Twenty-seven tests were done with variables at three levels. Dolomite was fed with 2 m²/g surface area in a slurry ranging from 65 to 75 percent solids by weight, or 39.5 to 51.3 percent by volume. Surface area produced was found to increase linearly with grinding time or specific-energy consumption. The variables studied strongly affected the milling rate; two extremes differed by a factor of 10. An optimum bead density for this feed material was 3.7. Evidently the discs of the chosen design could not effectively stir the denser beads. Higher slurry concentration above 70 wt % solids reduced the surface production per unit energy. The power input increased more than proportionally to speed.

Residence Time Distribution Commercially available bead mills have a diameter-to-length ratio ranging from 1:2.5 to 1:3.5. The ratio is expected to affect the *residence time distribution* (RTD). A wide distribution results in overgrinding some feed and undergrinding others. Data from Kula and Schuette [*Biotechnol. Progress*, **3**(1), 31-42 (1987)] show that in a Netzsch LME20 mill, RTD extends from 0.2 to 2.5 times the nominal time, indicating extensive stirring. (See "Biological Materials—Cell Disruption.") The RTD is even more important when the objective is to reduce the top size of the product as Stadler et al. [*Chemie-Ingenieur-Technik*, **62**(11), 907-915 (1990)] showed, because much of the feed received less than one-half the nominal residence time. A narrow RTD could be achieved by rapidly flowing material through the mill for as many as 10 passes.

VIBRATORY MILLS

The dominant form of industrial vibratory mill is the type with two horizontal tubes, called the **horizontal tube mill**. These tubes are mounted on springs and given a circular vibration by rotation of a

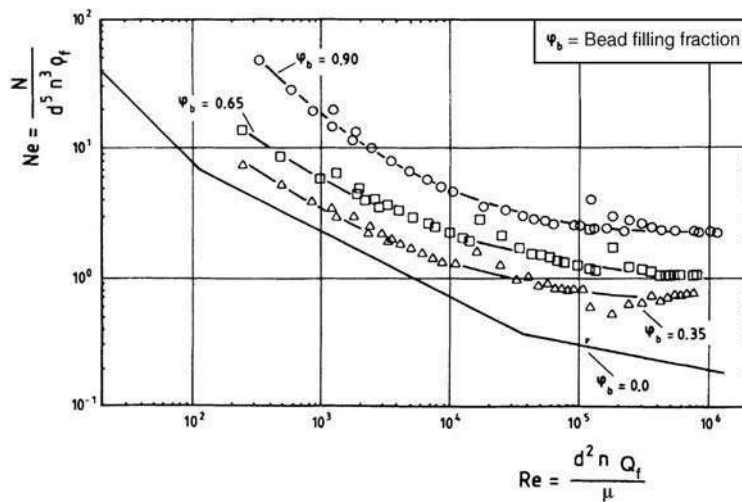


FIG. 20-77 Newton number as a function of Reynolds number for a horizontal stirred bead mill, with fluid alone and with various filling fractions of 1-mm glass beads [Weit and Schvedes, *Chemical Engineering and Technology*, **10**(6), 398–404 (1987)]. (N = power input, W; d = stirrer disk diameter, m; n = stirring speed, 1/s; μ = liquid viscosity, Pa·s; Q_f = feed rate, m³/s.)

counterweight. Many feed flow arrangements are possible, adapting to various applications. Variations include polymer lining to prevent iron contamination, blending of several components, and milling under inert gas and at high and low temperatures.

The vertical vibratory mill has good wear values and a low-noise output. It has an unfavorable residence time distribution, since in continuous operation it behaves as a well-stirred vessel. Tube mills are better for continuous operation. The mill volume of the vertical mill cannot be arbitrarily scaled up because the static load of the upper media, especially with steel beads, prevents thorough energy introduction into the lower layers. Larger throughputs can therefore only be obtained by using more mill troughs, as in tube mills. The primary applications of vibratory mills are in fine milling of medium to hard minerals primarily in dry form, producing particle sizes of 1 μ m and finer. Throughputs are typically 10 to 20 t/h. Grinding increases with residence time, active mill volume, energy density and vibration frequency, and media filling and feed charge.

The amount of energy that can be applied limits the tube size to 600 mm, although one design reaches 1000 mm. Larger vibratory amplitudes are more favorable for comminution than higher frequency. The development of larger vibratory mills is unlikely in the near future because of excitation problems. This has led to the use of mills with as many as six grinding tubes.

Performance The grinding-media diameter should preferably be 10 times that of the feed and should not exceed 100 times the feed diameter. To obtain improved efficiency when reducing size by several orders of magnitude, several stages should be used with different media diameters. As fine grinding proceeds, rheological factors alter the charge ratio, and power requirements may increase. Size availability varies, ranging from 1.3 cm (1/2 in) down to 325 mesh (44 μ m).

Advantages of vibratory mills are (1) simple construction and low capital cost, (2) very fine product size attainable with large reduction ratio in a single pass, (3) good adaptation to many uses, (4) small space and weight requirements, and (5) ease and low cost of maintenance. Disadvantages are (1) limited mill size and throughput, (2) vibration of the support and foundation, and (3) high-noise output, especially when run dry. The vibratory-tube mill is also suited to wet milling. In fine wet milling, this narrow residence time distribution lends itself to a simple open circuit with a small throughput. But for tasks of grinding to colloid-size range, the stirred media mill has the advantage.

Residence Time Distribution Hoeffl [Freiberger, *Forschungshefte A*, **750**, 119 pp. (1988)] carried out the first investigations of

residence time distribution and grinding on vibratory mills, and derived differential equations describing the motion. In vibratory horizontal tube mills, the mean axial transport velocity increases with increasing vibrational velocity, defined as the product $r_s\Omega$, where r_s = amplitude and Ω = frequency. Apparently the media act as a filter for the feed particles and are opened by vibrations. Nevertheless, good uniformity of transport is obtained, indicated by *vessel dispersion numbers* $D\tau/L^2$ (see "Simulation of Milling Circuits" above) in the range 0.06 to 0.08 measured in limestone grinding under conditions where both throughput and vibrational acceleration are optimum.

HICOM MILL

The Hicom mill is technically a vertical vibratory mill, but its design allows much higher energy input than do typical vibratory mills. The Hicom mill uses an irregular "nutating" motion to shake the mills, which allows much higher than normal g forces. Consequently, smaller media can be used and much higher grinding rates can be achieved. Hicom mill dry grinding performance tends to be competitive with jet mills, a substantial improvement over other vibratory mills. The Hicom mill is primarily used for dry grinding although it can also be used for wet grinding.

PLANETARY BALL MILLS

In planetary ball mills, several ball mill chambers are mounted on a frame in a circular pattern. The balls are all rotated in one direction (clockwise or counterclockwise), and the frame is rotated in the opposite direction, generating substantial centrifugal forces (10 to 50 g , depending on the device).

Planetary ball mills are difficult to make at large scale due to mechanical limitations. The largest mills commercially available have volumes in the range of 5 gal. Larger mills have been made, but they have tended to have very significant maintenance difficulties.

DISK ATTRITION MILLS

The disk or attrition mill is a modern counterpart of the early buhrstone mill. Stones are replaced by steel disks mounting interchangeable metal or abrasive grinding plates rotating at higher speeds, thus permitting a much broader range of application. They have a place in the grinding of tough organic materials, such as wood pulp and corn

grits. Grinding takes place between the plates, which may operate in a vertical or a horizontal plane. One or both disks may be rotated; if both, then in opposite directions. The assembly, comprising a shaft, disk, and grinding plate, is called a **runner**. Feed material enters a chute near the axis, passes between the grinding plates, and is discharged at the periphery of the disks. The grinding plates are bolted to the disks; the distance between them is adjustable.

DISPERSERS AND EMULSIFIERS

Media Mills and Roll Mills Both media mills and roll mills are commonly used for powder dispersion, especially in the paint and ink industries. Media mills used for these operations are essentially the same as described above, although finer media are used than are common in particle-grinding operations (down to 0.2 mm). Often, some sort of high-speed mixer is needed to disperse the powder into a liquid before trying to disperse powder in the media mill. Otherwise, large clumps of powder in the slurry can clog the mill.

Paint-grinding roller mills (Fig. 21-78) consist of two to five smooth rollers operating at differential speeds. A paste is fed between the first two rollers (low-speed) and is discharged from the final roller (high-speed) by a scraping blade. The paste passes from the surface of one roller to that of the next because of the differential speed, which also applies shear stress to the film of material passing between the rollers. Roll mills are sometimes heated so that higher-viscosity pastes can be ground and, in some cases, so that solvent can be removed.

Both of these mills can achieve very small particle-size dispersion (below 100 nm, if the primary particle size of the powder is small enough). However, formulation with surfactants is absolutely necessary to achieve fine particle dispersions. Otherwise, the particles will simply reagglomerate after leaving the shear field of the machine.

Dispersion and Colloid Mills Colloid mills have a variety of designs, but all have a rotating surface, usually a cone or a disc, with another surface near the rotor that forms a uniform gap (e.g., two discs parallel to each other). The liquid to be emulsified is pumped between the gaps. Sometimes, the design allows some pumping action between the rotor and the stator, and some machines of this type resemble centrifugal pumps in design. Colloid mills are relatively easy to clean and can handle materials with viscosity. For this reason, they are very common in the food and cosmetic industries for emulsifying pastes, creams, and lotions.

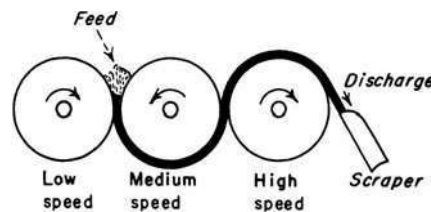


FIG. 21-78 Roller mill for paint grinding.

Pressure Homogenizers These are the wet grinding equivalents to jet mills, but they are used almost exclusively for emulsion and disagglomeration. There are several different styles of these, but all operate by generating pressures between 1000 and 50,000 psi using high-pressure pumps, with all the pressure drop occurring in a very small volume, such as flowing through an expansion valve. Some devices also have liquid jets which impinge on each other, similar to certain kinds of jet mills.

A **high-pressure valve homogenizer** such as the Gaulin and Rannie (APV Gaulin Group) forces the suspension through a narrow orifice. The equipment has two parts: a high-pressure piston pump and a homogenizer valve [Kula and Schuette, *Biotechnol. Progress*, **3**(1), 31–42 (1987)]. The pump in production machines may have up to six pistons. The valve opens at a preset or adjustable value, and the suspension is released at high velocity (300 m/s) and impinges on an impact ring. The flow changes direction twice by 90°, resulting in turbulence. There is also a two-stage valve, but it has been shown that it is better to expend all the pressure across a single stage. The temperature of the suspension increases about 2.5°C per 10-MPa pressure drop. Therefore intermediate cooling is required for multiple passes. Submicrometer-size emulsions can be achieved with jet homogenizers.

Microfluidizer The microfluidizer operates much the same as the valve homogenizers, but has a proprietary interaction chamber rather than an expansion valve. While valve homogenizers often have difficulties with particle slurries due to wear and clogging of the homogenizing valves, microfluidizers are much more robust and are often used in pharmaceutical processing. Interaction chambers for these applications must be made of specialized materials and can be expensive. Slurry particle sizes similar in size to those in media mill operations can be achieved with the microfluidizer.

CRUSHING AND GRINDING PRACTICE

CEREALS AND OTHER VEGETABLE PRODUCTS

Hammer mills or roll mills are used for a wide variety of vegetable products, from fine flour products to pulping for ethanol fermentation. Choice of mill usually depends on the exact nature of the feed and the desired product. For example, although usually cheaper to install and easier to operate, hammer mills cannot handle moist feeds as easily as roll mills, and roll mills tend to produce products with narrower size distributions.

Flour and Feed Meal The roller mill is the traditional machine for grinding wheat and rye into high-grade flour. A typical mill used for this purpose is fitted with two pairs of rolls, capable of making two separate reductions. After each reduction, the product is taken to a bolting machine or classifier to separate the fine flour; the coarse product is returned for further reduction. Feed is supplied at the top where a vibratory shaker spreads it out in a thin stream across the full width of the rolls. Rolls are made with various types of corrugation. Two standard types are generally used: the dull and the sharp. The former is mainly used on wheat and rye, and the latter on corn and feed. Under ordinary conditions, a sharp roll is used against a sharp roll for very tough wheat. A sharp, fast roll is used against a dull, slow roll for mod-

erately tough wheat; a dull, fast roll against a sharp, slow roll for slightly brittle wheat; and a dull roll against a dull roll for very brittle wheat. The speed ratio usually is 2½:1 for corrugated rolls and 1¼:1 for smooth rolls. By examining the marks made on the grain fragments, it has been concluded (Scott, *Flour Milling Processes*, Chapman & Hall, London, 1951) that the differential action of the rolls actually can open up the berry and strip the endosperm from the hulls.

High-speed hammer or pin mills result in some *selective grinding*. Such mills combined with air classification can produce fractions with controlled protein content. Flour with different protein content is needed for the baking of breads and cakes; these types of flour were formerly available only by selection of the type of wheat, which is limited by growing conditions prevailing in particular locations [Wichser, *Milling*, **3**(5), 123–125 (1958)].

Soybeans, Soybean Cake, and Other Pressed Cakes After granulation on rolls, the granules are generally treated in presses or solvent-extracted to remove the oil. The product from the presses goes to attrition mills or flour rolls and then to bolters, depending upon whether the finished product is to be a feed meal or a flour. The method used for grinding pressed cakes depends upon the nature of the cake, its purity, its residual oil, and its moisture content. If the whole cake is to be

In the **wet process**, the clay is masticated in a pug mill to break up lumps and is then dispersed with a dispersing aid and water to make a 40 percent solids slurry of low viscosity. A high-speed agitator such as a Cowles dissolver is used for this purpose. Sands are settled out, and then the clay is classified into two size fractions in either a hydrosettler or a continuous Sharples or Bird centrifuge. The fine fraction, with sizes of less than 1 μm , is used as a pigment and for paper coating, while the coarser fraction is used as a paper filler. A process for upgrading kaolin by grinding in a stirred bead mill has been reported [Stanczyk and Feld, *U.S. Bur. Mines Rep. Invest.*, **6327** and **6694** (1965)]. By this means the clay particles are delaminated, and the resulting platelets give a much improved surface on coated paper.

Talc and Soapstone Generally these are easily pulverized. Certain fibrous and foliated talcs may offer greater resistance to reduction to impalpable powder, but these are no longer produced because of their asbestos content. Talc milling is largely a grinding operation accompanied by air separation. Most of the industrial talcs are dry-ground. Dryers are commonly employed to predry ahead of the milling operation because the wet material reduces mill capacity by as much as 30 percent. Conventionally, in talc milling, rock taken from the mines is crushed in primary and then in secondary crushers to at least 1.25 cm ($\frac{1}{2}$ in) and frequently as fine as 0.16 cm ($\frac{1}{16}$ in). Ring-roll mills with internal air separation are widely used for the large-capacity fine grinding of the softer talcs. High-speed hammer mills with internal air separation have also had outstanding success on some of the softer high-purity talcs for very fine fineness. Talcs of extreme fineness and high surface area are used for various purposes in the paint, paper, plastics, and rubber industries.

Carbonates and Sulfates Carbonates include limestone, calcite, marble, marls, chalk, dolomite, and magnesite; the most important sulfates are barite, celestite, anhydrite, and gypsum. These are used as fillers in paint, paper, and rubber. (Gypsum and anhydrite are discussed below as part of the cement, lime, and gypsum industries.)

Silica and Feldspar These very hard minerals can be ground in ball/pebble mills with silex linings and flint balls. A feldspar mill is described in *U.S. Bur. Mines Cir.* **6488** (1931). It uses pebble mills with a Gayco air classifier. They can also be processed in ring-roller mills as the rings are easily replaced as they wear. Feldspar is also ground in continuous-tube mills with classification. Feldspar for the ceramic and chemical industries is ground finer than for the glass industry.

Asbestos and Mica Asbestos is no longer mined in the United States because of the severe health hazard. See previous editions of this handbook for process descriptions.

The micas, as a class, are difficult to grind to a fine powder; one exception is disintegrated schist, in which the mica occurs in minute flakes. For dry grinding, hammer mills equipped with an air transport system are generally used. Maintenance is often high. It has been established that the method of milling has a definite effect on the particle characteristics of the final product. Dry grinding of mica is customary for the coarser sizes down to 100 mesh. Micronized mica, produced by high-pressure steam jets, is considered to consist of highly delaminated particles.

Refractories Refractory bricks are made from fireclay, alumina, magnesite, chrome, forsterite, and silica ores. These materials are crushed and ground, wetted, pressed into shape, and fired. To obtain the maximum brick density, furnishes of several sizes are prepared and mixed. Thus a magnesia brick may be made from 40 percent coarse, 40 percent middling, and 20 percent fines. Preliminary crushing is done in jaw crushers or gyratories, intermediate crushing in pan mills or ring rolls, and fine grinding in open-circuit ball mills. Since refractory plants must make a variety of products in the same equipment, pan mills and ring rolls are preferred over ball mills because the former are more easily cleaned.

Sixty percent of refractory magnesite is made synthetically from Michigan brines. When calcined, this material is one of the hardest refractories to grind. Gyratory crushers, jaw crushers, pan mills, and ball mills are used. Alumina produced by the Bayer process is precipitated and then calcined [Krawczyk, *Ceram. Forum Int.*, **67**(7-8), 342-348 (1990)]. Aggregates are typically 20 to 70 μm and have to be reduced. The standard product is typically made in continuous dry

ball or vibratory mills to give a product d_{50} size of 3 to 7 μm , 98 percent finer than 45 μm . The mills are lined with wear-resistant alumina blocks, and balls or cylinders are used with an alumina content of 80 to 92 percent. The products containing up to 96 percent Al_2O_3 are used for bricks, kiln furniture, grinding balls and liners, high-voltage insulators, catalyst carriers, etc.

Ultrafine grinding is carried out batchwise in vibratory or ball mills, either dry or wet. The purpose of batch operation is to avoid the residence time distribution which would pass less-ground material through a continuous mill. The energy input is 20 to 30 times greater than that for standard grinding, with inputs of 1300 to 1600 kWh/ton compared to 40 to 60. Jet milling is also used, followed by air classification, which can reduce the top size below 8 μm . Among new mill developments, annular-gap bead mills and stirred bead mills are being used. These have a high cost, but result in a steep particle-size distribution when used in multipass mode [Kolb, *Ceram. Forum Int.*, **70**(5), 212-216 (1993)]. Costs for fine grinding typically exceed the cost of raw materials. Products are used for high-performance ceramics. Silicon carbide grains were reduced from 100 to 200 mesh to 80 percent below 1 μm in a version of stirred bead mill, using 20- to 30-mesh silicon carbide as media [Hoyer, *Rep. Investigations U.S. Bur. Mines*, **9097**, 9 pp. (1987)].

Crushed Stone and Aggregate In-pit crushing is increasingly being used to reduce the rock to a size that can be handled by a conveyor system. In quarries with a long, steep haul, conveyors may be more economic than trucks. The primary crusher is located near the quarry face, where it can be supplied by shovels, front-end loaders, or trucks. The crusher may be fully mobile or semimobile. It can be of any type listed below. The choices depend on individual quarry economics and are described by Faulkner [*Quarry Management and Products*, **7**(6), 159-168 (1980)]. Primary crushers used are jaw, gyratory, impact, and toothed roll crushers. Impact mills are limited to limestone and softer stone. With rocks containing more than 5 percent quartz, maintenance of hammers may become prohibitive. Gyratory and cone crushers dominate the field for secondary crushing of hard and tough stone. Rod mills have been employed to manufacture stone sand when natural sands are not available. Crushed stone for road building must be relatively strong and inert and must meet specifications regarding size distribution and shape. Both size and shape are determined by the crushing operation. The purpose of these specifications is to produce a mixture where the fines fill the voids in the coarser fractions, thus to increase load-bearing capacity. (See "Refractories" above.) Sometimes a product that does not meet these requirements must be adjusted by adding a specially crushed fraction. No crushing device available will give any arbitrary size distribution, and so crushing with a small reduction ratio and recycle of oversize is practiced when necessary.

FERTILIZERS AND PHOSPHATES

Fertilizers Many of the materials used in the fertilizer industry are pulverized, such as those serving as sources for calcium, phosphorus, potassium, and nitrogen. The most commonly used for their lime content are limestone, oyster shells, marls, lime, and, to a small extent, gypsum. Limestone is generally ground in hammer mills, ring-roller mills, and ball mills. Fineness required varies greatly from a No. 10 sieve to 75 percent through a No. 100 sieve.

Phosphates Phosphate rock is generally ground for one of two major purposes: for direct application to the soil or for acidulation with mineral acids in the manufacture of fertilizers. Because of larger capacities and fewer operating-personnel requirements, plant installations involving production rates over 900 Mg/h (100 tons/h) have used ball-mill grinding systems. Ring-roll mills are used in smaller applications. Rock for direct use as fertilizer is usually ground to various specifications, ranging from 40 percent minus 200 mesh to 70 percent minus 200 mesh. For manufacture of normal and concentrated superphosphates, the fineness of grind ranges from 65 percent minus 200 mesh to 85 percent minus 200 mesh.

Inorganic salts often do not require fine pulverizing, but they frequently become lumpy. In such cases, they are passed through a double-cage mill or some type of hammer mill.

Basic slag is often used as a source of phosphorus. Its grinding resistance depends largely upon the way in which it has been cooled; slowly cooled slag generally is more easily pulverized. The most common method for grinding basic slag is in a ball mill, followed by a tube mill or a compartment mill. Both systems may be in closed circuit with an air classifier. A 2.1- by 1.5-m (7- by 5-ft) mill, requiring 94 kW (125 hp), operating with a 4.2-m (14-ft) 22.5-kW (30-hp) classifier, gave a capacity of 4.5 Mg/h (5 tons/h) from the classifier, 95 percent through a No. 200 sieve. Mill product was 68 percent through a No. 200 sieve, and circulating load 100 percent.

CEMENT, LIME, AND GYPSUM

Portland Cement Portland cement manufacture requires grinding on a very large scale and entails a large use of electric power. Raw materials consist of sources of lime, alumina, and silica and range widely in properties, from crystalline limestone with silica inclusions to wet clay. Therefore a variety of crushers are needed to handle these materials. Typically a crushability test is conducted by measuring the product size from a laboratory impact mill on core samples [Schaefer and Gallus, *Zement- Kalk-Gips*, 41(10), 486-492 (1988); English ed., 277-280]. Abrasiveness is measured by the weight loss of the hammers. The presence of 5 to 10 percent silica can result in an abrasive rock, but only if the silica grain size exceeds 50 μm . Silica inclusions can also occur in soft rocks. The presence of sticky clay will usually result in handling problems, but other rocks can be handled even if moisture reaches 20 percent. If the rock is abrasive, the first stage of crushing may use gyratory or jaw crushers, otherwise a rotor-impact mill. Their reduction ratio is only 1:12 to 1:18, so they often must be followed by a hammer mill, or they can feed a roll press. Rotor crushers have become the dominant primary crusher for cement plants because of the characteristics. All these types of crushers may be installed in movable crusher plants. In the grinding of raw materials, two processes are used: the dry process in which the materials are dried to less than 1 percent moisture and then ground to a fine powder, and the wet process in which the grinding takes place with addition of water to the mills to produce a slurry.

Dry-Process Cement After crushing, the feed may be ground from a size of 5 to 6 cm (2 to 2½ in) to a powder of 75 to 90 percent passing a 200-mesh sieve in one or several stages. The first stage, reducing the material size to approximately 20 mesh, may be done in vertical, roller, ball-race, or ball mills. The last named rotate from 15 to 18 r/min and are charged with grinding balls 5 to 13 cm (2 to 5 in) in diameter. The second stage is done in tube mills charged with grinding balls of 2 to 5 cm (¾ to 2 in). Frequently ball and tube mills are combined into a single machine consisting of two or three compartments, separated by perforated steel diaphragms and charged with grinding media of different sizes. Rod mills are hardly ever used in cement plants. The compartments of a tube mill may be combined in various circuit arrangements with classifiers, as shown in Fig. 21-80. A dry-process plant has been described by Bergstrom [*Rock Prod.*, 59-62 (August 1968)].

Wet-Process Cement Ball, tube, and compartment mills of essentially the same construction as for the dry process are used for grinding. A water or clay slip is added at the feed end of the initial grinder, together with the roughly proportioned amounts of limestone and other components. In modern installations wet grinding is sometimes accomplished in ball mills alone, operating with excess water in

closed circuit with classifiers and hydroseparators. The circuits of Fig. 21-80 may also be used as a closed-circuit wet-grinding system incorporating a liquid solid cyclone as the classifier. A wet-process plant making cement from shale and limestone has been described by Bergstrom [*Rock Prod.*, 64-71 (June 1967)]. There are separate facilities for grinding each type of stone. The ball mill operates in closed circuit with a battery of Dutch State Mines screens. Material passing the screens is 85 percent minus 200 mesh.

Finish-Grinding of Cement Clinker Typically the hot clinker is first cooled and then ground in a compartment mill in a closed circuit with an air classifier. To crush the clinkers, balls as large as 5 in may be needed in the first compartment. A roll press added before the ball mill can reduce clinkers to a fine size and thus reduce the load on the ball mills. The main reason for adding a roll press has been to increase capacity of the plant and to lower cost. Installation of roll presses in several cement plants is described (31st IEEE Cement Industry Technical Conference, 1989). Considerable modification of the installation was required because of the characteristics of the press. A roll press is a constant-throughput machine, and the feed rate cannot easily be reduced to match the rate accepted by the ball mill that follows it. Several mills attempted to control the rate by increasing the recycle of coarse rejects from the air classifier, but the addition of such fine material was found to increase the pulling capacity of the rolls, e.g., from 180 to 250 t/h. With the resulting high recycle ratio of 5:1, the roll operation became unstable, and power peaks occurred. Deaeration of fines occurs in the nip, and this also interferes with feeding fines to the rolls. In some plants these problems were overcome by recirculating slabs of product directly from the roll discharge. In other cases the rolls were equipped with variable-speed drives to allow more versatile operation when producing several different grades/finenesses of cement. The roll press was found to be 2.5 times as efficient as the ball mill, in terms of new surface per unit energy. Tests showed that the slab from pressing of clinker at 120 bar and 20 percent recycle contained 97 percent finer than 2.8 mm, and 39 percent finer than 48 μm . Current operation is at 160 bar. The wear was small; after 4000 h of operation and 1.5 million tons of throughput, the wear rate was less than 0.1 g/ton, or 0.215 g/ton of finished cement. There is some wear of the working parts of the press, requiring occasional maintenance. The press is controlled by four control loops. The main control adjusts the gates that control slab recycle. Since this adjustment is sensitive, the level in the feed bin is controlled by adjusting the clinker-feed rate to ensure choke-feed conditions. Hydraulic pressure is also controlled. Separator reject rate is fixed. The investment cost was only \$42,000 per ton of increased capacity. Energy savings is 15 kWh/ton. This together with off-peak power rates results in energy cost savings of \$500,000/yr.

Lime Lime used for agricultural purposes generally is ground in hammer mills. It includes burned, hydrated, and raw limestone. When a fine product is desired, as in the building trade and for chemical manufacture, ring-roller mills, ball mills, and certain types of hammer mills are used.

Gypsum When gypsum is calcined in rotary kilns, it is first crushed and screened. After calcining it is pulverized. Tube mills are usually used. These impart plasticity and workability. Occasionally such calcined gypsum is passed through ring-roller mills ahead of the tube mills.

COAL, COKE, AND OTHER CARBON PRODUCTS

Bituminous Coal The grinding characteristics of bituminous coal are affected by impurities it contains, such as inherent ash, slate, gravel, sand, and sulfur balls. The grindability of coal is determined by grinding it in a standard laboratory mill and comparing the results with those obtained under identical conditions on a coal selected as a standard. This standard coal is a low-volatile coal from Jerome Mines, Upper Kintanning bed, Somerset County, Pennsylvania, and is assumed to have a grindability of 100. Thus a coal with a grindability of 125 could be pulverized more easily than the standard, while a coal with a grindability of 70 would be more difficult to grind. (Grindability and grindability methods are discussed under "Energy Required and Scale-up.")

Anthracite Anthracite is harder to reduce than bituminous coal. It is pulverized for foundry-facing mixtures in ball mills or hammer mills

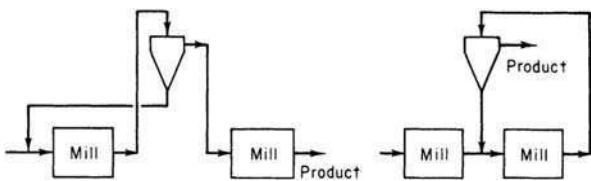


FIG. 21-80 Two cement-milling circuits. [For others, see Tonry, Pit Quarry (February-March 1959).]

followed by air classifiers. A 3- by 1.65-m (10-ft by 66-in) Hardinge mill in closed circuit with an air classifier, grinding 4 mesh anthracite with 3.5 percent moisture, produced 10.8 Mg/h (12 tons/h), 82 percent through No. 200 sieve. The power required for the mill was 278 kW (370 hp); for auxiliaries, 52.5 kW (70 hp); speed of mill, 19 r/min; ball load, 25.7 Mg (28.5 tons). Anthracite for use in the manufacture of electrodes is calcined, and the degree of calcination determines the grinding characteristics. Calcined anthracite is generally ground in ball and tube mills or ring-roller mills equipped with air classification.

Coke The grinding characteristics of coke vary widely. By-product coke is hard and abrasive, while certain foundry and retort coke is extremely hard to grind. For certain purposes it may be necessary to produce a uniform granule with minimum fines. This is best accomplished in rod or ball mills in closed circuit with screens. Petroleum coke is generally pulverized for the manufacture of electrodes; ring-roller mills with air classification and tube mills are generally used.

Other Carbon Products **Pitch** may be pulverized as a fuel or for other commercial purposes; in the former case the unit system of burning is generally employed, and the same equipment is used as described for coal. Grinding characteristics vary with the melting point, which may be anywhere from 50 to 175°C.

Natural graphite may be divided into three grades in respect to grinding characteristics: flake, crystalline, and amorphous. Flake is generally the most difficult to reduce to fine powder, and the crystalline variety is the most abrasive. Graphite is ground in ball mills, tube mills, ring-roller mills, and jet mills with or without air classification. Beneficiation by flotation is an essential part of most current procedures. Artificial graphite has been ground in ball mills in a closed circuit with air classifiers. For lubricants the graphite is ground wet in a paste in which water is eventually replaced by oil. The colloid mill is used for production of graphite paint.

Mineral black, a type of shale sometimes erroneously called *rotten stone*, contains a large amount of carbon and is used as a filler for paints and other chemical operations. It is pulverized and classified with the same equipment as shale, limestone, and barite.

Bone black is sometimes ground very fine for paint, ink, or chemical uses. A tube mill often is used, the mill discharging to a fan, which blows the material to a series of cyclone collectors in tandem.

Decolorizing carbons of vegetable origin should not be ground too fine. Standard fineness varies from 100 percent through No. 30 sieve to 100 percent through No. 50, with 50 to 70 percent on No. 200 sieve as the upper limit. Ball mills, hammer mills, and rolls, followed by screens, are used. When the material is used for filtering, a product of uniform size must be used.

Charcoal usually is ground in hammer mills with screen or air classification. For absorption of gases it is usually crushed and graded to about No. 16 sieve size. Care should be taken to prevent it from igniting during grinding.

Gilsonite sometimes is used in place of asphalt or pitch. It is easily pulverized and is generally reduced on hammer mills with air classification.

CHEMICALS, PIGMENTS, AND SOAPS

Colors and Pigments **Dry colors and dyestuffs** generally are pulverized in hammer mills. The jar mill or a large pebble mill is often used for small lots. There is a special problem with some dyes, which are coarsely crystalline. These are ground to the desired fineness with hammer or jet mills using air classification to limit the size. Synthetic pigments (mineral or organic) are usually fine agglomerates produced from aqueous crystallization processes. They are often lightly ground in media mills prior to drying. Dried pigments can be ground in hammer or jet mills to disintegrate agglomeration that occurs during grinding.

Dispersion of pigments into liquids is done predominantly by stirred media mills in the ink and paint industries. Roll mills are sometimes used for very fine dispersion or for very viscous materials such as some inks. Some grades of pigments disperse readily, or go into products with less stringent particle-size requirements, such as house-paints, and these require only high-speed dispersing mixers or colloid mills. Very difficult to disperse pigments, such as carbon black, are usually processed with a combination of these two processes, where a

high-speed disperser is used to premix the carbon black into the paint vehicle prior to processing in a media mill.

White pigments are basic commodities processed in large quantities. Titanium dioxide is the most important. The problem of cleaning the mill between batches does not exist as with different colors. These pigments are finish-ground to sell as dry pigments using mills with air classification. For the denser, low-oil-absorption grades, roller and pebble mills are employed. For looser, fluffier products, hammer and jet mills are used. Often a combination of the two mill actions is used to set the finished quality.

Chemicals Fine powder organic chemicals (herbicides are one example) can be processed similar to fine pigments: media mills for wet slurries of crystals, followed by drying and hammer mills or jet mill for dry material.

Sulfur The ring-roller mill can be used for the fine grinding of sulfur. Inert gases are supplied instead of hot air (see "Properties of Solids: Safety" for use of inert gas).

Soaps Soaps in a finely divided form may be classified as soap powder, powdered soap, and chips or flakes. The term *soap powder* is applied to a granular product, No. 12 to No. 16 sieve size with a certain amount of fines, which is produced in hammer mills with perforated or slotted screens. The oleates and erucates are best pulverized by multicage mills; laurates and palmitates, in cage mills and also in hammer mills if particularly fine division is not required. Stearates may generally be pulverized in multicage mills, screen mills, and air classification hammer mills.

POLYMERS

The grinding characteristics of various resins, gums, waxes, hard rubbers, and molding powders depend greatly upon their softening temperatures. When a finely divided product is required, it is often necessary to use a water-jacketed mill or a pulverizer with an air classifier in which cooled air is introduced into the system. Hammer and cage mills are used for this purpose. Some low-softening-temperature resins can be ground by mixing with 15 to 50 percent by weight of dry ice before grinding. Refrigerated air sometimes is introduced into the hammer mill to prevent softening and agglomeration [Dorris, *Chem. Metall. Eng.*, 51, 114 (July 1944)].

Gums and Resins Most **gums and resins**, natural or artificial, when used in the paint, varnish, or plastic industries, are not ground very fine, and hammer or cage mills will produce a suitable product. Roll crushers will often give a sufficiently fine product. Ring mills are sometimes used.

Rubber **Hard rubber** is one of the few combustible materials which is generally ground on heavy steam-heated rollers. The raw material passes to a series of rolls in closed circuit with screens and air classifiers. Farrel-Birmingham rolls are used extensively for this work. There is a differential in the roll diameters. The motor should be separated from the grinder by a firewall.

Molding Powders Specifications for **molding powders** vary widely, from a No. 8 to a No. 60 sieve product; generally the coarser products are No. 12, 14, or 20 sieve material. Specifications usually prescribe a minimum of fines (below No. 100 and No. 200 sieve). Molding powders are produced with hammer mills, either of the screen type or equipped with air classifiers. The following materials may be ground at ordinary temperatures if only the regular commercial fineness is required: amber, arabac, tragacanth, rosin, olibanum, gum benzoin, myrrh, guaiacum, and montan wax. If a finer product is required, hammer mills or attrition mills in closed circuit, with screens or air classifiers, are used.

Powder Coatings Powder coatings are quite fine, often 40 µm or less, and tend to be heat-sensitive. Also, to give a good finish, large particles, which have a detrimental effect on gloss, must be minimized. These are typically ground in air classifying mills or jet mills.

PROCESSING WASTE

In flow sheets for processing *municipal solid waste* (MSW), the objective is to separate the waste into useful materials, such as scrap metals, plastics, and *refuse-derived fuels* (RDFs). Usually size reduction is the

first step, followed by separations with screens or air classifiers, which attempt to recover concentrated fractions [Savage and Diaz, *Proc. ASME National Waste Processing Conference*, Denver, Colo., 361–373 (1986)]. Many installed circuits proved to be ineffective or not cost-effective, however. Begnaud and Noyon [*Biocycle*, **30**(3), 40–41 (1989)] concluded from a study of French operations that milling could not grind selectively enough to separate different materials. Size reduction uses either hammer mills or blade cutters (shredders). Hammer mills are likely to break glass into finer sizes, making it hard to separate. Better results may be obtained in a flow sheet where size reduction follows separation (Savage, Seminar on the Application of U.S. Water and Air Pollution Control Technology to Korea, Korea, May 1989). Wear is also a major cost, and wear rates are shown in Fig. 21-81. The maximum capacity of commercially available hammer mills is about 100 tons/h.

PHARMACEUTICAL MATERIALS

Specialized modification of fine grinding equipment for pharmaceutical grinding has become increasingly common. Most grinding is accomplished using a variety of air classifying mills and jet mills. Wet grinding with homogenizers and bead mills is becoming more common. Equipment for grinding pharmaceuticals must be readily cleaned to very high standards; many materials are very poisonous, and many materials are quite heat-sensitive. To meet cleanliness requirements, mills are often fitted with extra seals, stainless-steel parts of high-quality finish, and other expensive modifications. Modified mills can cost 5 times what a standard mill of the same type would cost.

BIOLOGICAL MATERIALS—CELL DISRUPTION

Mechanical disruption is the most practical first step in the release and isolation of proteins and enzymes from microorganisms on a commercial scale. The size-reduction method must be gently tuned to the strength of the organisms to minimize formation of fine fragments that interfere with subsequent clarification by centrifugation or filtration. Typically, fragments as fine as 0.3 μm are produced. High-speed stirred-bead mills and high-pressure homogenizers have been applied for cell disruption [Kula and Schuetz, *Biotechnol. Progress*, **3**(1), 31–42 (1987)]. There are two limiting cases in the operation of bead mills for disruption of bacterial cells. When the energy imparted by

collision of beads is insufficient to break all cells, the rate of breakage is proportional to the specific energy imparted [Bunge et al., *Chem. Engg. Sci.*, **47**(1), 225–232 (1992)]. On the other hand, when the energy is high due to higher speed above 8 m/s, larger beads above 1 mm, and low concentrations of 10 percent, each bead impact has more than enough energy to break any cells that are captured, which causes problems during subsequent separations. The strength of cell walls differs among bacteria, yeasts, and molds. The strength also varies with the species and the growth conditions, and must be determined experimentally. Beads of 0.5 mm are typically used for yeast and bacteria. Recommended bead charge is 85 percent for 0.5-mm beads and 80 percent for 1-mm beads [Schuette et al., *Enzyme Microbial Technol.*, **5**, 143 (1983)]. Residence time distribution is important in continuous mills.

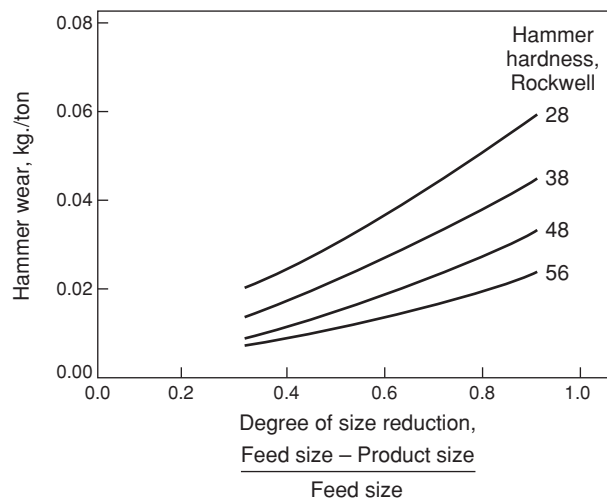


FIG. 21-81 Hammer wear as a consequence of shredding municipal solid waste. (Savage and Diaz, *Proceedings ASME National Waste Processing Conference*, Denver, CO, pp 361–373, 1986.)

PRINCIPLES OF SIZE ENLARGEMENT

GENERAL REFERENCES: Benbow and Bridgwater, *Paste Flow and Extrusion*, Oxford University Press, 1993. Ennis, *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, E&G Associates, Nashville, Tenn., 2006. Ennis, On the Mechanics of Granulation, Ph.D. thesis 1990, The City College of the City University of New York, University Microfilms International, 1991. Ennis, *Powder Technology*, June 1996. Kapur, *Adv. Chem. Eng.*, **10**, 55 (1978). Kristensen, *Acta Pharm. Suec.*, **25**, 187 (1988). Lütster and Ennis, *The Science and Engineering of Granulation Processes*, Kluwer Academic Publishers, 2005. Masters, *Spray Drying Handbook*, Wiley, 1979. Masters, *Spray Drying in Practice*, SprayDryConsult International, 2002. Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005. Pietsch, *Size Enlargement by Agglomeration*, Wiley, Chichester, 1992. Randolph and Larson, *Theory of Particulate Processes*, Academic Press, San Diego, 1988. Stanley-Wood (ed.), *Enlargement and Compaction of Particulate Solids*, Butterworth & Co. Ltd., 1983. Ball et al., *Agglomeration of Iron Ores*, Heinemann, London, 1973. Capes, *Particle Size Enlargement*, Elsevier, New York, 1980. King, "Tablets, Capsules and Pills," in *Remington's Pharmaceutical Sciences*, Mack Pub. Co., Easton, Pa., 1970. Knepper (ed.), *Agglomeration*, Interscience, New York, 1962. Mead (ed.), *Encyclopedia of Chemical Process Equipment*, Reinhold, New York, 1964. Pietsch, *Roll Pressing*, Heyden, London, 1976. Sastry (ed.), *Agglomeration 77*, AIME, New York, 1977. Sauchelli (ed.), *Chemistry and Technology of Fertilizers*, Reinhold, New York, 1960. Sherrington and Oliver, *Granulation*, Heyden, London, 1981.

SCOPE AND APPLICATIONS

Size enlargement is any process whereby small particles are agglomerated, compacted, or otherwise brought together into larger, relatively permanent masses in which the original particles can still be distinguished. Size enlargement processes are employed by a wide range of industries, including pharmaceutical and food processing, consumer products, fertilizer and detergent production, and the mineral processing industries. The term encompasses a variety of unit operations or processing techniques dedicated to particle agglomeration. **Agglomeration** is the formation of aggregates through the sticking together of feed and/or recycle material. These processes can be loosely broken down into agitation and compression methods. Although terminology is industry-specific, agglomeration by agitation will be referred to as **granulation**. As depicted in Fig. 21-82, a particulate feed is introduced to a process vessel and is agglomerated, either batchwise or continuously, to form a granulated product. **Agitative agglomeration** processes or granulation include fluid-bed, pan (or disc), drum, and mixer granulators as well as many hybrid designs. Such processes are also used as coating operations for controlled release, taste masking, and cases where solid cores may act as a

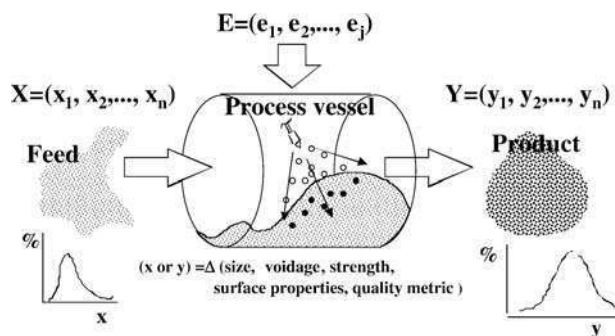


FIG. 21-82 The unit operation of agitative agglomeration, or granulation. (Reprinted from *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

carrier for a drug coating. The feed typically consists of a mixture of solid ingredients, referred to as a **formulation**, which include, an active or key ingredient, binders, diluents, disintegrants, flow aids, surfactants, wetting agents, lubricants, fillers, or end-use aids (e.g. sintering aids, colors or dyes, taste modifiers). The active ingredient is often referred to as the **technical** or **API** (active product ingredient), and it is the end-use ingredient of value, such as a drug substance, fertilizer, pesticide, or a key detergent agent. Agglomeration can be induced in several ways. A solvent or slurry can be atomized onto the bed of particles that coats either the particle or granule surfaces promoting agglomeration, or the spray drops can form small nuclei in the case of a powder feed that subsequently can agglomerate. The solvent or slurry may contain a binder, or a solid binder may be present as one component of the feed. Alternatively, the solvent may induce dissolution and recrystallization in the case of soluble particles. Slurries often contain the same particulate matter as the dry feed, and granules may be formed, either completely or partially, as the droplets solidify in flight prior to reaching the particle bed. **Spray-drying** is an extreme case where no further, intended agglomeration takes place after granule formation. Agglomeration may also be induced by heat, which either leads to controlled **sintering** of the particle bed or induces sintering or partial melting of a binder component of the feed, e.g., a polymer. Product forms generally include agglomerated or layered granules, coated carrier cores, or spray-dried product consisting of agglomerated solidified drops.

An alternative approach to size enlargement is by **compressive agglomeration** or **compaction** processes, where the mixture of particulate matter is fed to a compression device which promotes agglomeration due to pressure as depicted in Fig. 21-83. Either continuous sheets or strands of solid material are produced or some solid form such as a **briquette** or **tablet**. Either continuous sheets or strands may break down in subsequent handling to form a *granulated* material, or the material may be further processing through a variety of chopping or forced screening methods. Heat or cooling may be applied, in addition to induced frictional heating and particle deformation, and reaction may be induced such as with sintering processes. Carrier fluids may be present, either added or induced by melting, in which case the product is **wet-extruded**. Compaction processes range from confined compression devices such as tableting, briquetting machines, and ram extrusion to unconfined devices such as roll presses and extrusion and a variety of pellet mills. Capsule, vial, and blister pack filling operations could also be considered low-pressure compaction processes.

At the level of a manufacturing plant, the size-enlargement process involves several peripheral, unit operations such as milling, blending, drying or cooling, and classification, referred to generically as an **agglomeration circuit** (Fig. 21-84). In addition, more than one agglomeration step may be present, as in the case of pharmaceutical or detergent processes. In the case of pharmaceutical granulation,

granulated material formed by an agitative process is generally an intermediate product form, which is then followed by the compressive process of tableting. Upstream of this circuit might also involve spray-drying or crystallization of an active ingredient, or multiple granulation steps may be employed, as is the case with detergent and mineral processing, respectively.

In troubleshooting process upsets or product quality deviations, it is important to consider the high degree of interaction between the unit operations, which is much higher in the case of solids processing operations. Tableting failures might often be the result of granule properties originating in the upstream granulation step, or further still, due to production deviations of ingredients by spray-drying or crystallization, or blending and grinding steps.

Numerous benefits result from size-enlargement processes, as will be appreciated from Table 21-10. A wide variety of size-enlargement methods are available; a classification of these is given in Table 21-11 with key process attributes as well as typical subsequent processing. A primary purpose of wet granulation in the case of pharmaceutical processing is to create free-flowing, nonsegregating blends of ingredients of controlled strength, which may be reproducibly metered in subsequent tableting or for vial- or capsule-filling operations. The wet granulation process must generally achieve desired granule properties within some prescribed range. These attributes clearly depend on the application at hand. However, common to most processes is a specific granule size distribution and granule voidage. Size distribution affects flow and segregation properties as well as compaction behavior. Granule voidage controls strength, and impacts capsule and tablet dissolution behavior, as well as compaction behavior and tablet hardness. Control of granule size and voidage is discussed in detail. The approach taken here relies heavily on attempting to understand interactions at a particle level, and scaling this understanding to bulk effects. Developing an understanding of these microlevel processes of agglomeration allows a rational approach to the design, scale-up, and control of agglomeration processes. Although the approach is difficult, qualitative trends are uncovered along the way that aid in formulation development and process optimization, and that emphasize powder characterization as an integral part of product development and process design work.

MECHANICS OF SIZE-ENLARGEMENT PROCESSES

Granulation Rate Processes Granulation is controlled by four key **rate processes**, as outlined by Ennis [On the Mechanics of Granulation, Ph.D. thesis, The City College of the City University of New York, University Microfilms International No. 1416, 1990, printed 1991; *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, E&G Associates, 2006; *Theory of Granulation: An Engineering Perspective*, in Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005]. These include (1) wetting and nucleation,

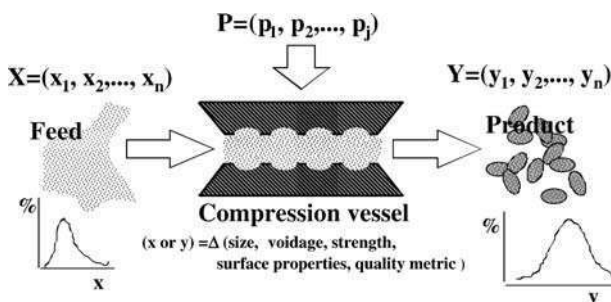


FIG. 21-83 The unit operation of compressive agglomeration, or compaction. (Reprinted from *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

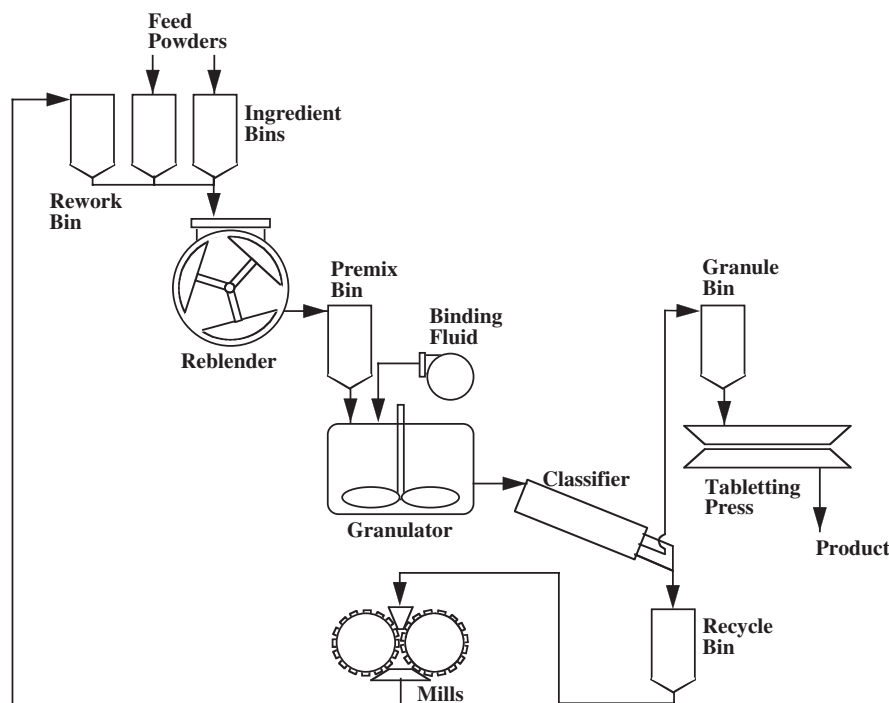


FIG. 21-84 A typical agglomeration circuit utilized in the processing of pharmaceutical or agricultural chemicals involving both granulation and compaction techniques. (Reprinted from *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

(2) coalescence or growth, (3) consolidation and densification, and (4) breakage or attrition (Fig. 21-85). Initial **wetting** of the feed powder and existing granules by the binding fluid is strongly influenced by spray rate or fluid distribution as well as feed formulation properties, in comparison with mechanical mixing. Wetting promotes **nucleation** of fine powders, or **coating** in the case of feed particle size in excess of drop size. Often wetting agents such as surfactants are carefully chosen to enhance poorly wetting feeds. In the **coales-**

cence or **growth** stage, partially wetted primary particles and larger nuclei coalesce to form granules composed of several particles. The term **nucleation** is typically applied to the initial coalescence of primary particles in the immediate vicinity of the larger wetting drop, whereas the more general term of **coalescence** refers to the successful collision of two granules to form a new, larger granule. In addition, the term **layering** is applied to the coalescence or layering of granules by primary feed powder. Nucleation is promoted from some initial distribution of moisture, such as a drop distribution or from the homogenization of a fluid feed to the bed, as with high-shear mixing, or by any maldistribution fluid such as dripping nozzles or flaking of caked wall material. The nucleation process is strongly linked with the wetting stage. As granules grow, they are consolidated by compaction forces due to bed agitation. This **consolidation** or densification stage strongly influences internal granule voidage or granule porosity, and therefore end-use properties such as granule strength, hardness, or dissolution. Formed granules may be particularly susceptible to **attrition** if they are inherently weak or if flaws develop during drying.

These mechanisms can occur simultaneously in all granulation operations, ranging from spray-drying to fluidized beds to high-shear mixers. However, certain mechanisms may dominate in a particular process. For example, fluidized-bed granulators are strongly influenced by the wetting process, whereas mechanical redispersion of binding fluid by impellers and particularly high-intensity choppers diminish the wetting contributions to granule size in high-shear mixing. On the other hand, granule consolidation is far more pronounced in high-shear mixing than fluidized-bed granulation. These simultaneous rate processes taken as a whole—and sometimes competing against one another—determine the final granule size distribution and granule structure and voidage resulting from a process, and therefore the final end-use or product quality attributes of the granulated product.

TABLE 21-10 Objectives of Size Enlargement

Production of useful structural forms, as in pressing of intricate shapes in powder metallurgy.
Provision of a defined quantity to facilitate dispensing and metering, as in agricultural chemical granules or pharmaceutical tablets.
Elimination of dust-handling hazards or losses, as in briquetting of waste fines.
Improved product appearance, or product renewal.
Reduced caking and lump formation, as in granulation of fertilizer.
Improved flow properties, generally defined as enhanced flow rates with improved flow rate uniformity, as in granulation of pharmaceuticals for tabletting or ceramics for pressing.
Increased bulk density for storage and tabletting feeds.
Creation of nonsegregating blends of powder ingredients with ideally uniform distribution of key ingredients, as in sintering of fines for steel or agricultural chemical or pharmaceutical granules.
Control of solubility, as in instant food products.
Control of porosity and surface-to-volume ratio, as with catalyst supports.
Improvement of heat-transfer characteristics, as in ores or glass for furnace feed.
Remove of particles from liquid, as with polymer additives, which induce clay flocculation.

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TABLE 21-11 Size Enlargement Methods and Application

Method	Product size (mm)	Granule density	Scale of operation	Additional comments and processing	Typical applications
Tumbling granulators Drums Discs	0.2–20	Moderate	0.5–800 tons/h	Very spherical granules Fluid-bed or rotary kiln drying	Fertilizers, iron and other ores, agricultural chemicals
Mixer-granulators Continuous high-shear (e.g., Shugi mixer) Batch high-shear (e.g., vertical mixer)	0.1–0.5 0.1–2	Low Moderate to high	Up to 50 tons/h Up to 500-kg batch	Handles cohesive materials, both batch and continuous, as well as viscous binders and nonwetable powders Fluid-bed, tray, or vacuum/microwave on-pot drying	Chemicals, detergents, clays, carbon black Pharmaceuticals, ceramics, clays
Fluidized granulators Fluidized beds Spouted beds Wurster coaters	0.1–1	Low (agglomerated) Moderate (layered)	100–900 kg batch 50 tons/h continuous	Flexible, relatively easy to scale, difficult for nonwetable powders and viscous binders, good for coating applications Same vessel drying, air handling requirements	Continuous: fertilizers, inorganic salts, food, detergents Batch: pharmaceuticals, agricultural chemicals, nuclear wastes
Centrifugal granulators	0.3–3	Moderate to high	Up to 200-kg batch	Powder layering and coating applications. Fluid-bed or same-pot drying	Pharmaceuticals, agricultural chemicals
Spray methods Spray drying Prilling	0.05–0.2 0.7–2	Low Moderate		Morphology of spray-dried powders can vary widely. Same vessel drying	Instant foods, dyes, detergents, ceramics, pharmaceuticals Urea, ammonium nitrate
Pressure compaction Extrusion Roll press Tablet press Molding press Pellet mill	>0.5 >1 10	High to very high	Up to 5 tons/h Up to 50 tons/h Up to 1 ton/h	Very narrow size distributions, very sensitive to powder flow and mechanical properties Often subsequent milling and blending operations	Pharmaceuticals, catalysts, inorganic chemicals, organic chemicals, plastic preforms, metal parts, ceramics, clays, minerals, animal feeds
Thermal processes Sintering	2–50	High to very high	Up to 100 tons/h	Strongest bonding	Ferrous and nonferrous ores, cement clinker, minerals, ceramics
Liquid systems Immiscible wetting in mixers Sol-gel processes Pellet flocculation	<0.3	Low	Up to 10 tons/h	Wet processing based on flocculation properties of particulate feed, subsequent drying	Coal fines, soot, and oil removal from water Metal dicarbide, silica hydrogels Waste sludges and slurries

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Compaction Microlevel Processes Compaction is a forming process controlled by mechanical properties of the feed in relationship to applied stresses and strains. Microlevel processes are controlled by particle properties such as friction, hardness, size, shape, surface energy, and elastic modulus (Fig. 21-86). The performance of compaction techniques is controlled by the ability of the particulate phase to uniformly **transmit stress**, and the relationship between applied stress and the compaction and strength characteristics of the final compacted particulate phase.

Key steps in any compaction process include (1) powder filling or feeding, (2) stress application and removal, and (3) compact ejection in the case of confined compression techniques. Powder filling and compact weight variability are strongly influenced by bulk density and powder flowability (cf. subsection “Solids Handling”), as well as any contributing segregation tendencies of the feed. The steps of **stress application** and **removal** consist of several competing mechanisms, as depicted in Fig. 21-86. Powders do not transmit stress uniformly. Wall friction impedes the applied load, causing a drop in stress as one moves away from the point of the applied load, e.g., a punch face in tableting or roll surface in roll pressing. Therefore, the applied load and resulting density are not uniform throughout the compact, and powder frictional properties control the **stress transmission** and distribution in the com-

pact [cf. subsection “Bulk Powder Characterization” in Brown and Richards, *Principles of Powder Mechanics*, Pergamon Press Ltd., Oxford, 1970; Stanley-Wood (ed.), *Enlargement and Compaction of Particulate Solids*, Butterworth & Co. Ltd., 1983]. For a local level of applied stress, particles deform at their point contacts, including plastic deformation for forces in excess of the particle surface **hardness**. This allows intimate contact at surface point contacts, allowing cohesion/adhesion to develop between particles, and therefore interfacial bonding, which is a function of their **interfacial surface energy**. During the short time scale of the applied load, any entrapped air must escape, which is a function of feed **permeability**, and a portion of the elastic strain energy is converted to permanent plastic deformation. Upon stress removal, the compact expands due to remaining elastic recovery of the matrix, which is a function of **elastic modulus**, as well as any expansion of remaining entrapped air. This can result in loss of particle bonding and flaw development, and this is exacerbated for cases of wide distributions in compact stress due to poor stress transmission. The final step of stress removal involves **compact ejection**, where any remaining radial elastic stresses are removed. If recovery is substantial, it can lead to capping or delamination of the compact.

These microlevel processes of compaction control the final flaw and density distribution throughout the compact, whether it is a roll

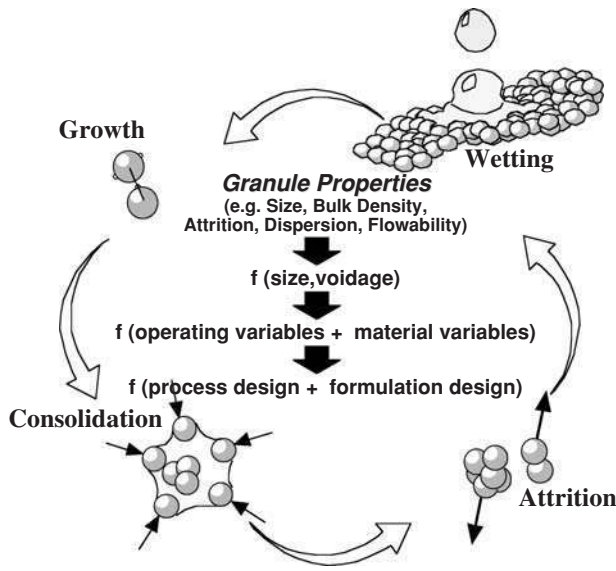


FIG. 21-85 The rate processes of agitative agglomeration, or granulation, which include powder wetting, granule growth, granule consolidation, and granule attrition. These processes combined control granule size and porosity, and they may be influenced by formulation or process design changes. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

pressed, extruded, or tableted product, and as such, control compact strength, hardness, and dissolution behavior.

Process vs. Formulation Design The end-use properties of granulated material are primarily controlled by granule size and inter-

nal granule **voidage** or **porosity**. Internal granule voidage ϵ_g and bed voidage ϵ_b , or voidage between granules, are related by

$$\rho_b = \rho_g(1 - \epsilon_b) = \rho_s(1 - \epsilon_g)(1 - \epsilon_g) \quad (21-95)$$

where ρ_b , ρ_g , and ρ_s are bulk, granule (or apparent), and skeletal primary particle density, respectively. Here, granule voidage and granule porosity are used interchangeably. Granule structure may also influence properties. Similar linkages exist in the case of compaction processes where hardness, voidage, and distribution of compact voidage are critical. To achieve a desired product quality as defined by metrics of end-use properties, granule size and voidage or compact properties may be manipulated by changes in either process operating variables or product material variables (Figs. 21-85 and 21-86), as initially outlined by Ennis (loc. cit., 2005, 2006). The first approach is the realm of traditional **process engineering**, whereas the second is **product engineering**. Both approaches are critical and must be integrated to achieve a desired endpoint in product quality. **Operating variables** are defined by the chosen granulation technique and peripheral processing equipment, as illustrated for a fluidized-bed and mixer-granulator in Fig. 21-87. In addition, the choice of agglomeration technique dictates the **mixing pattern** of the vessel. **Material variables** include parameters such as binder viscosity, surface tension, feed particle size distribution, powder friction, wall friction and lubrication, hardness, elastic modulus, and the adhesive properties of the solidified binder. Material variables are specified by the choice of ingredients, or **product formulation**. Both operating and material variables together define the granulation kinetic mechanisms and rate constants of wetting, growth, consolidation, and attrition, or the compaction processes for compressive techniques. Overcoming a given size-enlargement problem often requires changes in both processing conditions and product formulation.

The importance of granule voidage or density to final product quality is illustrated in Figs. 21-88 to 21-90 for a variety of formulations. Here, bulk density is observed to decrease, granule attrition to increase, and dissolution rate to increase with an increase in granule voidage. Bulk density is clearly a function of both granule size distribution, which controls bed voidage or porosity between granules, and

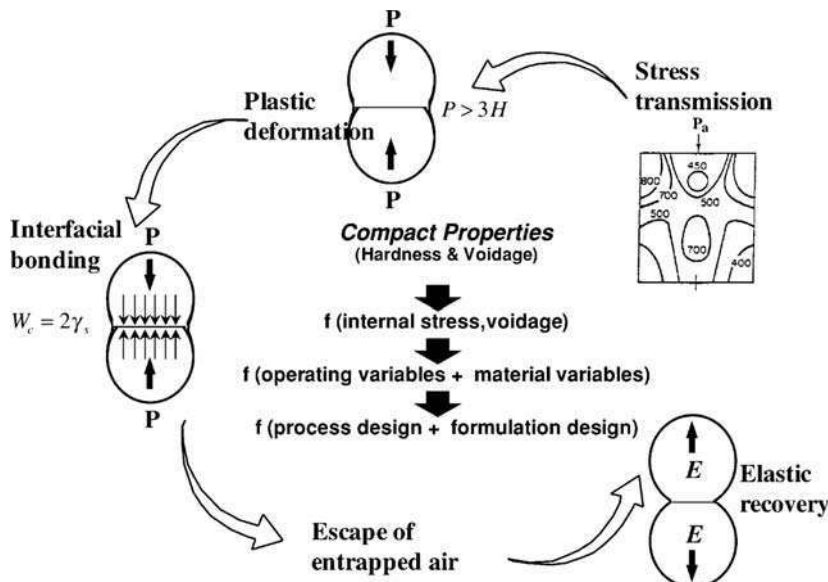


FIG. 21-86 The microlevel processes of compressive agglomeration, or compaction. These processes combined control compact strength, hardness, and porosity, and they may be influenced by formulation or process design changes. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

Batch Fluid-Bed Granulator

Nozzle height
Drop size
Spray rate
Bed height & temperature
Gas velocity & temperature
Distributor design
Granulation time

Batch High-Shear Granulator

Binding liquid
Impeller
Chopper
Liquid addition method
Liquid loading & rate
Impeller & chopper speed
Impeller design
Chopper location & size
Bed weight
Granulation time
Fluid-bed drying conditions

FIG. 21-87 Typical operating variable for granulation processes. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

the voidage within the granule itself. The data of Fig. 21-88 are normalized with respect to its zero-intercept, or its effective bulk density at zero granule voidage. The granule attrition results of Fig. 21-89 are based on a CIPAC test method, which is effectively the percentage of fines passing a fine mesh size following attrition in a tumbling apparatus. Granules weaken with increased voidage. The dissolution results of Fig. 21-90 measure the length required for granule dissolution in a long tube, or disintegration length also based on CIPAC test method. Increased granule voidage results in increased dissolution rate and shorter disintegration length. All industries have their own specific quality and in-process evaluation tests. However, what they have in common are the important contributing effects of granule size and granule voidage.

An example of the importance of distinguishing the effects of process and formulation changes can be illustrated with the help of Figs. 21-89 and 21-90. Let us assume the particular formulation and current process conditions produce a granulated material with a given attrition resistance and dissolution behavior (indicated as *current product*). If one desires instead to reach a given *target*, either formulation or process variables may be changed. Changes to the process, or operating variables, generally readily alter granule voidage. Examples to decrease voidage might include increased bed height, increased processing time, or increased peak bed moisture. However, only a range of such changes in voidage, and therefore attrition resistance and dissolution, are possible. The various curves in Figs. 21-89 and 21-90 are due to changes in formulation properties. Therefore, it may

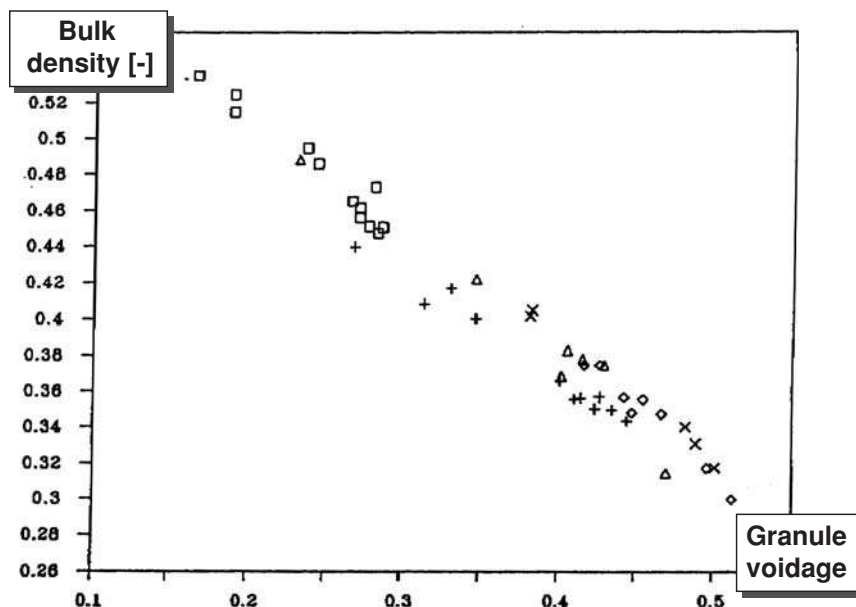


FIG. 21-88 Impact of granule voidage on bulk density. Normalized bulk density as a function of granule voidage. (After Maroglou, reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

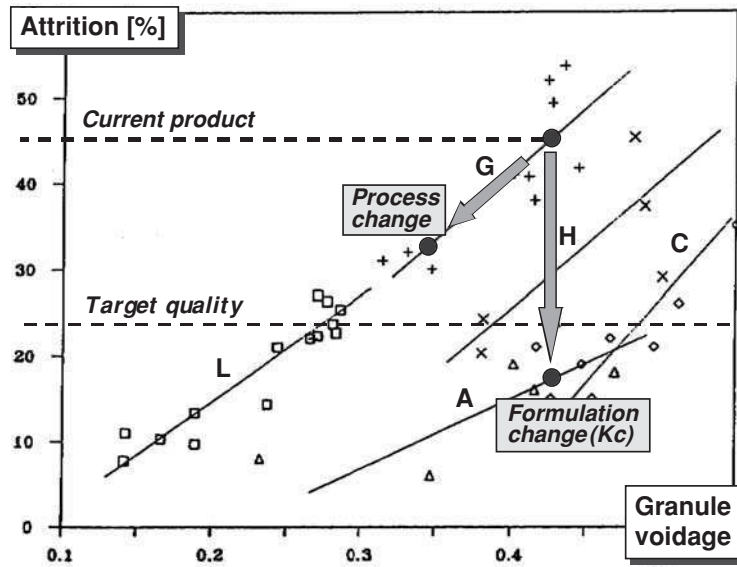


FIG. 21-89 Impact of granule voidage on strength and attrition. Illustration of process changes vs. formulation changes. (After Maroglou, reprinted from *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

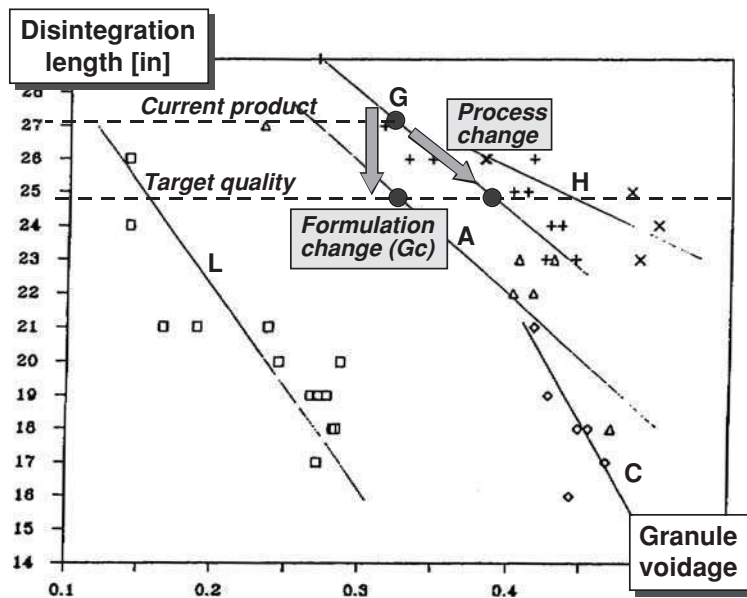


FIG. 21-90 Impact of granule voidage on dissolution. (After Maroglou, reprinted from *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

not be possible to reach a target change in dissolution without changes in formulation, or material variables. Examples of a key material variable affecting voidage would include feed primary particle size, inherent formulation bond strength, and binder solution viscosity, as discussed in detail in the following subsections. This critical interaction between and manipulation of operating and material variables is crucial for successful formulation development, and requires substantial collaboration between processing and formulation groups and a clear knowledge of the effect of scale-up on this interaction.

Key Historical Investigations A range of historical investigations have been undertaken involving the impact of operating variables on granulation behavior [cf. Ennis, loc. cit., 1991, 2006; Ennis, *Powder Technol.*, **88**, 203 (1996); Litster and Ennis, *The Science and Engineering of Granulation Processes*, Kluwer Academic Publishers, 2005; Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005; Turton et al., *Fluidized Bed Coating and Granulation*, Noyes Publications, 1999, p. 331; Pietsch, *Size Enlargement by Agglomeration*, Wiley, Chichester, 1992]. Typical variables have included the effects of bed hydrodynamics and agitation intensity, pan angle and speed, fluid-bed excess gas velocity, mixer impeller and chopper speeds, drum rotation speed, spray method, drop size, nozzle location, and binder and solvent feed rates. While such studies are important, their general application and utility to studies beyond the cited formulations and process conditions can be severely limited. Often the state of mixing, moisture distribution and rates, and material properties such as formulation size distribution, powder frictional properties, and solution viscosity are insufficiently defined. As such, these results should be used judiciously and with care. Often even the directions of the impact of operating variables on granule properties are altered by formulation changes.

Two key pieces of historical investigation require mention. The first involves growth and breakage mechanisms that control the evolution of the granule size distribution [Sastry and Fuerstenau, *Agglomeration '77*, Sastry (ed.), AIME, New York, 1977, p. 381], as illustrated in Fig. 21-91. These include the nucleation of fine powder to form initial primary granules, the coalescence of existing granules, and the layering of raw material onto previously formed nuclei or granules. Granules may be simultaneously compacted by consolidation and reduced

in size by breakage. There are strong interactions between these rate processes. In addition, these mechanisms in various forms have been incorporated into population balances modeling to predict granule size in the work of Sastry (loc. cit.) and Kapur [*Adv. Chem. Eng.*, **10**, 55 (1978); *Chem. Eng. Sci.*, **26**, 1093 (1971); *Ind. Eng. Chem. Eng. (Proc. Des. & Dev.)*, **5**, 5 (1966); *Chem. Eng. Sci.*, **27**, 1863 (1972)]. See subsection "Modeling and Simulation of Grinding Processes" for details. Given the progress made in connecting rate constants to formulation properties, the utility of population balance modeling has increased substantially.

The second important area of contribution involves the work of Rumpf [*The Strength of Granules and Agglomerates*, Knepper (ed.), *Agglomeration*, Interscience, New York, 1962, pp. 379–414; and *Particle Adhesion*, Sastry (ed.), *Agglomeration '77*, AIME, New York, 1977, pp. 97–129], which studied the impact of interparticle force H on granule static tensile strength, or

$$\sigma_T = \frac{9}{8} \left(\frac{1 - \epsilon_g}{\epsilon_g} \right) \frac{H}{a^2} = A \left(\frac{1 - \epsilon_g}{\epsilon_g} \right) \frac{\gamma \cos \theta}{a} \quad (21-96)$$

with $A = 9/4$ for pendular state
 $A = 6$ for capillary state

Forces of a variety of forms were studied, including viscous, semisolid, solid, electrostatic, and van der Waals forces. Of particular importance was the contribution of pendular bridge force between primary particles of size a arising from surface tension γ with a contact angle θ . This force summed over the granule area results in a granule static tensile strength σ_T , which is a function of pore saturation S as experimentally plotted (Fig. 21-92, with $U = 0$). The states of pore filling have been defined as pendular (single bridges), funicular (partial complete filling and single bridges), capillary (nearly complete filling $S \sim 80$ to 100 percent), followed by drop formation and loss of static strength. This approach is extended in subsequent subsections to include viscous forces and dynamic strength behavior ($U \neq 0$).

The approach taken here follows that of Rumpf and Kapur, namely, relating granule and particle level interactions to bulk behavior through the development of the rate processes of wetting and nucleation, granule growth and consolidation, and granule breakage and attrition.

PRODUCT CHARACTERIZATION

Powders are agglomerated to modify physical or physicochemical properties. Effective measurement of **agglomerate properties** is vital. However, many tests are industry-specific and take the form of empirical indices based on standardized protocols. Such tests as described below are useful for quality control, if used with care. However, since they often reflect an end use rather than a specific defined agglomerate property, they often are of little developmental utility for recommending process or formulation changes. Significant improvements have been made in the ability to measure real agglomerate properties. Key agglomerate properties are **size**, **porosity**, and **strength** and their associated distributions because these properties directly affect end-use attributes of the product, such as attrition resistance, flowability, bulk solid permeability, wettability and dispersibility, appearance, or the active agent release rate.

Size and Shape Agglomerate **mean size** and **size distribution** are both important properties. (See "Particle-Size Analysis"). For granular materials, sieve analysis is the most common sizing technique. Care is needed in sizing wet granules. Handling during sampling and sieving can cause changes in the size distribution through coalescence or breakage. Sieves are also easily blinded. Snap freezing the granules with liquid nitrogen prior to sizing overcomes these problems [Hall, *Chem. Eng. Sci.*, **41**, 187 (1986)]. **On-line** or **in-line measurement** of granules as large as 9 mm is now available by laser diffraction techniques, making improved granulation control schemes possible [Ogunnaike et al., *I.E.C. Fund.*, 1996]. Modern methods of rapid imaging also provide a variety of shape assessments (see "Particle-Size Analysis").

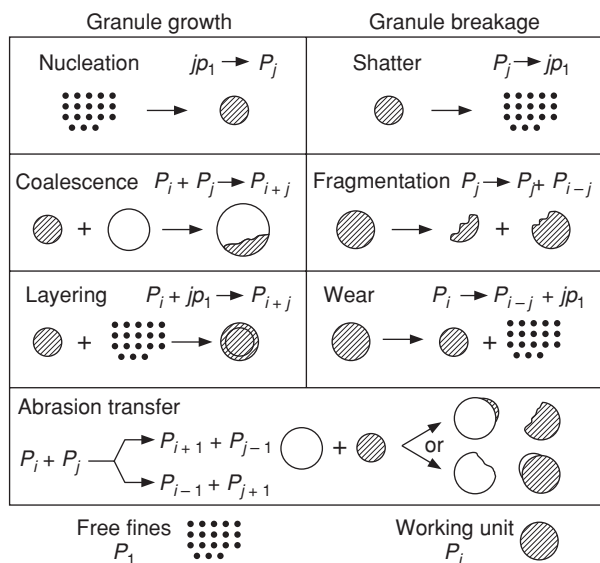


FIG. 21-91 Growth and breakage mechanisms in granulation processes. [After Sastry and Fuerstenau, *Agglomeration '77*, Sastry (ed.), AIME, New York, 1977, p. 381.]

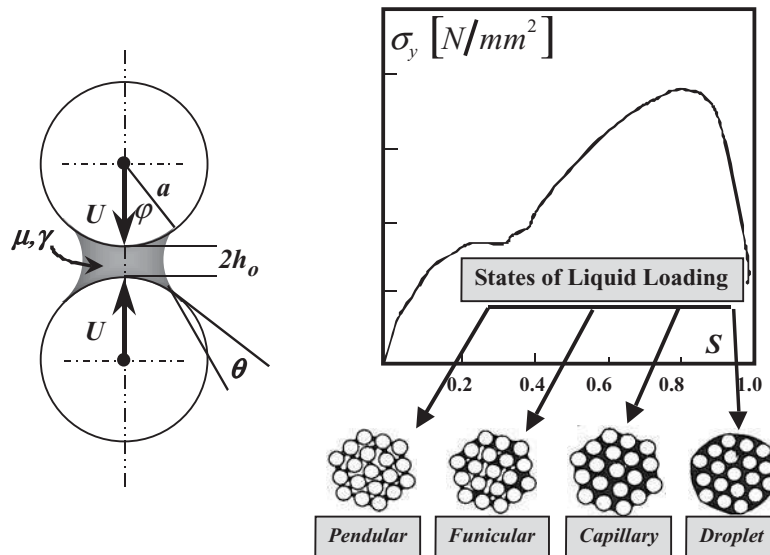


FIG. 21-92 Static yield strength of wet agglomerates versus pore saturation (collisional velocity $U = 0$). Here a is the size of a primary particle within the granule, and S is pore saturation resulting from the filling angle ϕ . [After Rumpf (*loc. cit.*), reprinted from *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, Ennis, 2006, with permission of E&G Associates. All rights reserved. Also, Newitt and Conway-Jones, *Trans. Inst. Chem. Engineers (London)*, **36**, 422 (1958)].

Porosity and Density There are three important densities of granular or agglomerated materials: **bulk density** ρ_b (related to the volume occupied by the bulk solid), the apparent or **agglomerate density** ρ_a (related to the volume occupied by the agglomerate including internal porosity), and the true or **skeletal solids density** ρ_s . These densities are related to one another and the interagglomerate **voidage** ϵ_b and the intraagglomerate **porosity** ϵ_a [Eq. (21-96)].

Bulk density is easily measured from the volume occupied by the bulk solid and is a strong function of sample preparation. True density is measured by standard techniques using liquid or gas pycnometry. Apparent (agglomerate) density is difficult to measure directly. Hinkley et al. [*Int. J. Min. Proc.*, **41**, 53–69 (1994)] describe a method for measuring the apparent density of wet granules by kerosene displacement. Agglomerate density may also be inferred from direct measurement of true density and porosity by using Eq. (21-96).

Agglomerate porosity can be measured by gas adsorption or mercury porosimetry. However, any breakage or compression of the granules under high pressure during porosimetry will invalidate the results. Often raw curves must be carefully analyzed to correct for penetration between granules and possible deformation. Some progress has also been made in the use of tomography to evaluate pore structure and distribution from X-ray images (Farber et al., *Powder Technol.*, **132**, 57 (2003)).

Strength of Agglomerates Agglomerate bonding mechanisms may be divided into five major groups [Rumpf, in Knepper (ed.), *Agglomeration*, op. cit., p. 379]. More than one mechanism may apply during a given size-enlargement operation. (In addition, see Krupp [*Adv. Colloid. Int. Sci.*, **1**, 111 (1967)] for a review of adhesion mechanisms.)

Solid bridges can form between particles by the sintering of ores, the crystallization of dissolved substances during drying as in the granulation of fertilizers, and the hardening of bonding agents such as glue and resins.

Mobile liquid binding produces cohesion through interfacial forces and capillary suction. Three states can be distinguished in an assembly of particles held together by a mobile liquid (Fig. 21-92).

Small amounts of liquid are held as discrete lens-shaped rings at the points of contact of the particles; this is the **pendular state**. As the liquid content increases, the rings coalesce and there is a continuous network of liquid interspersed with air; this is the **funicular state**. When all the pore spaces in the agglomerate are completely filled, the **capillary state** has been reached. When a mobile liquid bridge fails, it constricts and divides without fully exploiting the adhesion and cohesive forces in the bridge in the absence of viscous effects. Binder viscosity markedly increases the strength of the pendular bridge due to dynamic lubrication forces, and aids the transmission of adhesion. For many systems, viscous forces outweigh interfacial capillary effects, as demonstrated by Ennis et al. [*Chem. Eng. Sci.*, **45**, 3071 (1990)].

In the limit of high viscosity, **immobile liquid bridges** formed from materials such as asphalt or pitch fail by tearing apart the weakest bond. Then adhesion and/or cohesion forces are fully exploited, and binding ability is much larger.

Intermolecular and electrostatic forces bond very fine particles without the presence of material bridges. Such bonding is responsible for the tendency of particles less than about 1 mm in diameter to form agglomerates spontaneously under agitation. With larger particles, however, these short-range forces are insufficient to counterbalance the weight of the particle, and adhesion does not occur without applied pressure. High compaction pressures act to plastically flatten interparticle contacts and substantially enhance short-range forces.

Mechanical interlocking of particles may occur during the agitation or compression of, for example, fibrous particles, but it is probably only a minor contributor to agglomerate strength in most cases.

Equation (21-96) gives the tensile strength of an agglomerate of equal-sized spherical particles for an interparticle bonding force H (Rumpf, *loc. cit.*, p. 379). Figure 21-93 indicates values of tensile strength to be expected in various size-enlargement processes for a variety of binding mechanisms. In particular, note that viscous mechanisms of binding (e.g., adhesives) can exceed capillary effects in determining agglomerate strength.

Strength Testing Methods Compressed agglomerates often fail in **tension** along their diameter. This is the basis of the commonly

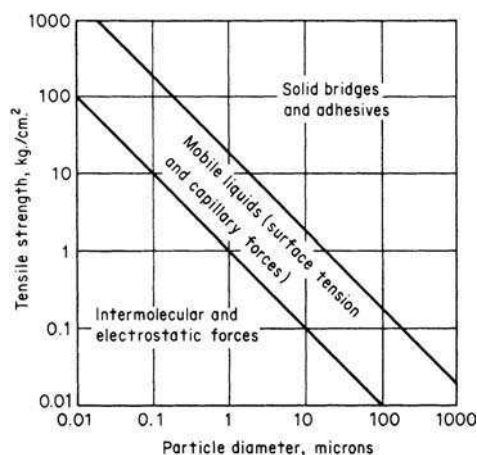


FIG. 21-93 Theoretical tensile strength of agglomerates. [Adapted from Rumpf, "Strength of Granules and Agglomerates," in Knepper (ed.), Agglomeration, Wiley, New York, 1962.]

used measurement of **crushing strength** of an agglomerate as a method to assess tensile strength. However, the brittle failure of a granule depends on the flaw distribution as well as the inherent tensile strength of bonds as given by the Griffith crack theory (Lawn, *Fracture of Brittle Solids*, 2d ed., Cambridge University Press, 1975). Therefore, it is more appropriate to characterize granule strength by **fracture toughness** K_{IC} [Kendall, *Nature*, **272**, 710 (1978); see also subsections "Theoretical Background" and "Breakage and Attrition"].

Several strength-related indices are measured in different industries which give some measure of resistance to attrition. These tests do not measure strength or toughness directly, but rather the size distribution of fragments after handling the agglomerates in a defined way. The handling could be repeated drops, tumbling in a drum, fluidizing, circulating in a pneumatic conveying loop, etc. These indices should only be used for quality control if the test procedure simulates the actual handling of the agglomerates during processing and transportation.

Flow Property Tests **Flowability** of the product granules can be characterized by **unconfined yield stress** and **angle of friction** by shear cell tests as used generally for bulk solids (see subsection "Powder Compaction"). **Caking** refers to deterioration in the flow properties of the granules due to chemical reaction or hydroscopic effects. Caking tests as used for fertilizer granules consist of two parts [Bookey and Raistrick, in Sauchelli (ed.), *Chemistry and*

Technology of Fertilizers, p. 454 year]. A cake of the granules is first formed in a compression chamber under controlled conditions of humidity, temperature, etc. The crushing strength of the cake is then measured to determine the degree of caking.

The propensity to cake may also be assessed by caking and thermal **dilatometry**, which assess compaction of powder and thermal softening under a variety of loading, temperature, and humidity conditions [Ennis et al., *Chem. Engg. Progress* (2007)].

Redispersion Tests Agglomerated products are often **redispersed** in a fluid by a customer. Examples include the dispersion of fertilizer granules in spray-tank solutions or of tablets within the gastrointestinal tract of the human body. The mechanisms comprising this redispersion process of product **wetting**, agglomerate **disintegration**, and final **dispersion** are related to interfacial properties (for details, see subsection "Wetting"). There are a wide range of industry-specific empirical indices dealing with redispersion assessment.

Disintegration height tests consist of measuring the length required for complete agglomerate disintegration in a long, narrow tube. Small fragments may still remain after initial agglomerate disintegration. The residual of material which remains undispersed is measured by a related test, or **long-tube sedimentation test**. The residual undispersed material is reported by the level in the bottom tip of the tube. A variation of this test is the **wet screen test**, which measures the residual remaining on a fine mesh screen (e.g., 350 mesh) following pouring the beaker solution through the screen.

Tablet-disintegration tests consist of cyclical immersion in a suitable dissolving fluid of pharmaceutical tablets contained in a basket. Acceptable tablets disintegrate completely by the end of the specified test period (*United States Pharmacopeia*, 17th rev., Mack Pub. Co., Easton, Pa., 1965, p. 919).

Permeability **Bulk solid permeability** is important in the iron and steel industry where gas-solid reactions occur in the sinter plant and blast furnace. It also strongly influences compaction processes, where entrapped gas can impede compaction, and solids-handling equipment, where restricted gas flow can impede product flowability. The permeability of a granular bed is inferred from measured pressure drop under controlled gas-flow conditions.

Physiochemical Assessments A variety of methods remain to assess both the chemical and physical nature of granulated and compacted product. Some of these include nitrogen adsorption measurements of surface area; adsorption isotherm measures of humidity and gas interactions; surface chemical assessment by inverse gas chromatography and near infrared and Rauman spectroscopy; X-ray powder diffraction measurements of polymorphism; and measurements of electrostatic charge. [See Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005; Stanley-Wood (ed.), *Enlargement and Compaction of Particulate Solids*, Butterworth & Co. Ltd., 1983.]

AGGLOMERATION RATE PROCESSES AND MECHANICS

GENERAL REFERENCES: Adetayo et al., *Powder Technol.*, **82**, 37 (1995). Benbow and Bridgwater, *Paste Flow and Extrusion*, Oxford University Press, 1993. Brown and Richards, *Principles of Powder Mechanics*, Pergamon Press, 1970. Ennis, *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, E&G Associates, Nashville, Tenn., 2006. Ennis, On the Mechanics of Granulation, Ph.D. thesis, 1990, The City College of the City University of New York, University Microfilms International, 1991. Ennis et al., *Powder Technol.*, **65**, 257 (1991). Ennis and Sunshine, *Tribology Int.*, **26**, 319 (1993). Ennis, *Powder Technol.*, June 1996. Holm et al. Parts V and VI, *Powder Technol.*, **43**, 213–233 (1985). Kristensen, *Acta Pharm. Suec.*, **25**, 187 (1988). Lawn, *Fracture of Brittle Solids*, 2d ed., Cambridge University Press, 1975. Litster and Ennis, *The Science and Engineering of Granulation Processes*, Kluwer Academic Publishers, 2005. Owens and Wendt, *J. Appl. Polym. Sci.*, **13**, 1741 (1969). Parfitt (ed.), *Dispersion of Powders in Liquids*, Elsevier Applied Science Publishers Ltd., 1986. Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005. Stanley-Wood (ed.), *Enlargement and Compaction of Particulate Solids*, Butterworth & Co. Ltd., 1983.

WETTING

The initial distribution of binding fluid can have a pronounced influence on the size distribution of seed granules or **nuclei** that are formed from fine powder. Both the final **extent** of and **rate** at which the fluid wets the particulate phase are important. Poor wetting results in drop coalescence and fewer, larger nuclei with ungranulated powder and overwettted masses, leading to broad nuclei distributions. Granulation can retain a memory, with nuclei size distribution impacting final granule size distribution. Therefore, initial wetting can be critical to uniform nuclei formation and often a narrow, uniform product. Wide nuclei distributions can lead to wide granule-size distributions. When the size of a particulate feed material is larger than drop size, wetting dynamics controls the distribution of **coating** material, which has a strong influence on the later stages of growth. Wetting phenomena also influence redistribution of individual ingredients

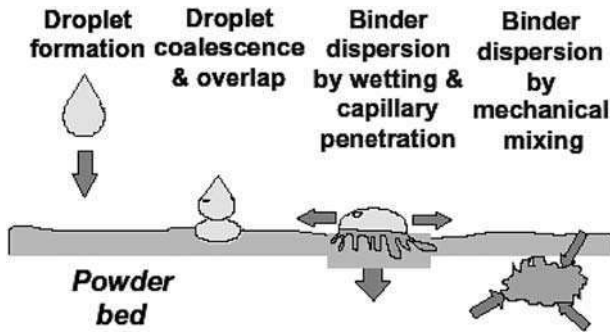


FIG. 21-94 Stages of wetting for fine powder compared to drop size.

within a granule, drying processes, and redispersion of granules in a fluid phase. Other granule properties such as voidage, strength, and attrition resistance may be influenced as well. **Preferential wetting** of certain formulation ingredients can cause component segregation with granule size. Extensive reviews of wetting research are available [Parfitt (ed.), *Dispersion of Powders in Liquids*, Elsevier Applied Science Publishers Ltd., 1986; Hapgood, Nucleation and Binder Dispersion in Wet Granulation, Ph.D. thesis, University of Queensland, 2000].

Mechanics of the Wetting Rate Process As outlined previously, wetting is the first stage in wet granulation involving liquid binder distribution to the feed powder. There are two extremes: (1) Liquid drop size is large compared to unit or primary particle size of the feed, and (2) particle size is large compared to the drop size. For the first case as depicted in Fig. 21-94 for fine feeds compared to drop size, the wetting process consists of several steps. First, droplets are formed related to spray distribution, or spray flux defined as the wetting area of the bed per unit time. Important operating variables include nozzle position, spray area, spray rate, and drop size. Second, droplets impact and coalesce on the powder bed surface if mixing or wet-in time is slow. Third, droplets spread and penetrate into the moving powder bed to form loose nuclei, again coalescing if wet-in is slow. In the case of high-shear processes, shear forces break down overwet clumps, also producing nuclei. For the second case of small drop size compared to the primary particle size, the liquid will coat the particles as depicted in Fig. 21-95. Coating is produced by collisions between the drop and the particle followed by spreading of the liquid over the particle surface. If the particle is porous, then liquid will also suck into the pores by capillary action. The wetting dynamics control the distribution of coating material, which has a strong influence on the later stages of growth as well as coating quality.

Methods of Measurement Methods of characterizing the rate process of wetting include four approaches, as illustrated in Table 21-12. The first considers the ability of a drop to spread across the powder. This approach involves the measurement of a **contact angle** of a drop on a powder compact. The contact angle is a measure of the affinity of the fluid for the solid as given by the Young-Dupré equation, or

$$\gamma^{sv} - \gamma^{sl} = \gamma^{lv} \cos \theta \quad (21-97)$$

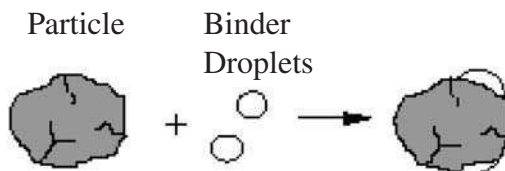


FIG. 21-95 Stages of wetting for coarse powder compared to drop size.

where γ^{sv} , γ^{sl} , and γ^{lv} are the solid-vapor, solid-liquid, and liquid-vapor interfacial energies, respectively, and θ is the contact angle measured through the liquid, as illustrated in Fig. 21-96. When the solid-vapor interfacial energy exceeds the solid-liquid energy, the fluid wets the solid with a contact angle less than 90° . In the limit of $\gamma^{sv} - \gamma^{sl} \geq \gamma^{lv}$, the contact angle equals 0° and the fluid **spreads** on the solid. The extent of wetting is controlled by the group $\gamma^{lv} \cos \theta$, which is referred to as the **adhesion tension**. Sessile drop studies of contact angle can be performed on powder compacts in the same way as on planar surfaces. As illustrated in Fig. 21-97, methods involve (1) direct measurement of the contact angle from the tangent to the air-binder interface, (2) solution of the Laplace-Young equation involving the contact angle as a boundary condition, or (3) indirect calculations of the contact angle from measurements of, e.g., drop height. Either the compact can be saturated with the fluid for static measurements, or dynamic measurements may be made through a computer imaging goniometer (Pan et al., *Dynamic Properties of Interfaces and Association Structure*, American Oil Chemists' Society Press, 1995).

For granulation processes, the dynamics of wetting are often crucial, requiring that powders be compared on the basis of a short time scale, **dynamic contact angle**. Important factors are the physical nature of the powder surface (particle size, pore size, porosity, environment, roughness, pretreatment). Powders which are **formulated** for granulation often are composed of a combination of ingredients. The dynamic wetting process is therefore influenced by the rates of ingredient dissolution and surfactant adsorption and desorption kinetics (Pan et al., loc. cit.).

The second approach to characterize wetting considers the ability of the fluid to penetrate a powder bed, as illustrated in Fig. 21-98. It involves the measurement of the extent and rate of fluid rise by capillary suction into a column of powder, better known as the **Washburn test**. Considering the powder to consist of capillaries of radius R , the equilibrium height of rise h_e is determined by equating capillary and gravimetric pressures, or

$$h_e = \frac{2\gamma^{lv} \cos \theta}{\Delta \rho g R} \quad (21-98a)$$

where $\Delta \rho$ is the fluid density with respect to air, g is gravity, and $\gamma^{lv} \cos \theta$ is the adhesion tension as before. In addition to the equilibrium height of rise, the dynamics of penetration can be equally important. By ignoring gravity and equating viscous losses with the capillary pressure, the rate dh/dt and dynamic height of rise h are given by

$$\frac{dh}{dt} = \frac{R\gamma^{lv} \cos \theta}{4\mu h} \quad \text{or} \quad h = \sqrt{\left(\frac{R\gamma^{lv} \cos \theta}{2\mu}\right)t} \quad (21-98b)$$

where t is time and μ is binder fluid viscosity [Parfitt (ed.), *Dispersion of Powders in Liquids*, Elsevier Applied Science Publishers Ltd., 1986, p. 10]. The grouping of terms in parentheses involves the material properties which control the dynamics of fluid penetration, namely, average pore radius, or **tortuosity** R (related to particle size and void distribution of the powder), adhesion tension, and binder viscosity.

The contact angle or adhesion tension of a binder solution with respect to a powder can be determined from the slope of the penetration profile. Washburn tests can also be used to investigate the

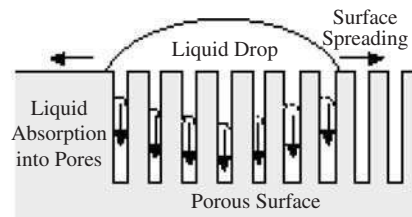
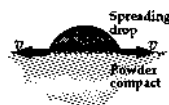
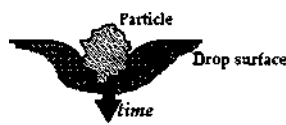
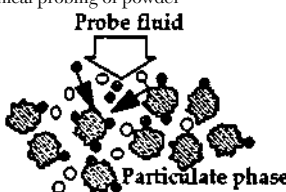


TABLE 21-12 Methods of Characterizing Wetting Dynamics of Particulate Systems

Mechanism of wetting	Characterization method
Spreading of drops on powder surface 	Contact angle goniometer Contact angle Drop height or volume Spreading velocity References: Kossen and Heertjes, <i>Chem. Eng. Sci.</i> , 20 , 593 (1965). Pan et al., <i>Dynamic Properties of Interfaces and Association Structure</i> , American Oil Chemists' Society Press, 1995.
Penetration of drops into powder bed 	Washburn test Rate of penetration by height or volume Bartell cell Capillary pressure difference References: Parfitt (ed.), <i>Dispersion of Powders in Liquids</i> , Elsevier Applied Science Publishers Ltd., 1986. Washburn, <i>Phys. Rev.</i> , 17 , 273 (1921). Bartell and Osterhof, <i>Ind. Eng. Chem.</i> , 19 , 1277 (1927).
Penetration of particles into fluid 	Flotation tests Penetration time Sediment height Critical solid surface energy distribution References: R. Ayala, Ph.D. thesis, Chemical Engineering, Carnegie Mellon University, 1985. Fuerstaneau et al., <i>Colloids and Surfaces</i> , 60 , 127 (1991). Vargha-Butler et al., in <i>Interfacial Phenomena in Coal Technology</i> , Botsaris & Glazman (eds.), Chap. 2, 1989.
Chemical probing of powder 	Inverse gas chromatography Preferential adsorption with probe gases Electrokinetics Zeta potential and charge Surfactant adsorption Preferential adsorption with probe surfactants References: Lloyd et al. (eds.), <i>ACS Symposium Series 391</i> , ACS, Washington, 1989. Aveyard and Haydon, <i>An Introduction to the Principles of Surface Chemistry</i> , Cambridge University Press, 1973. Shaw, <i>Introduction to Colloid and Surface Chemistry</i> , Butterworths & Co. Ltd., 1983.

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influence of powder preparation on penetration rates. The **Bartell cell** is related to the Washburn test, except that here adhesion tension is determined by a variable gas pressure which opposes penetration [Bartell and Osterhof, *Ind. Eng. Chem.*, **19**, 1277 (1927)].

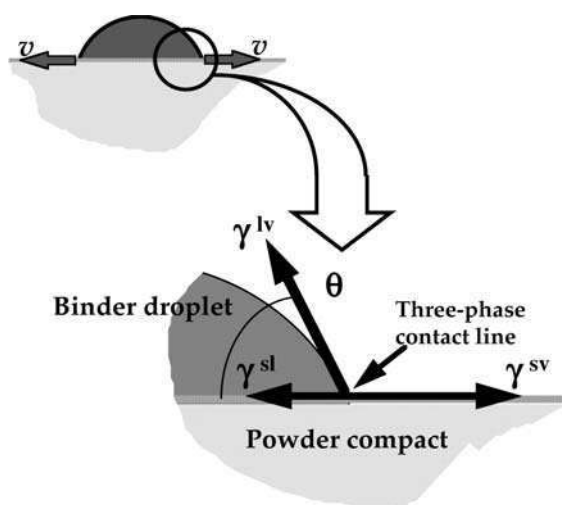


FIG. 21-96 Contact angle on a powder surface, where γ^{lv} , γ^{sl} , and γ^{sv} are the solid-vapor, solid-liquid, and liquid-vapor interfacial energies, and θ is the contact angle measured through the liquid.

The contact angle of a binder-particle system is not itself a primary thermodynamic quantity, but rather is a reflection of individual interfacial energies [Eq. 21-97]), which are a function of the molecular interactions of each phase with respect to one another. An interfacial energy may be broken down into its **dispersion** and **polar** components. These components reflect the chemical character of the interface, with the polar component due to hydrogen bonding and other polar interactions and the dispersion component due to van der Waals interactions. These components may be determined by the wetting tests described here, where a variety of solvents are chosen as the wetting fluids to probe specific molecular interactions as described by Zisman [Contact Angle, Wettability, and Adhesion, *Advances in Chemistry Series*, ACS, **43**, 1 (1964)]. These components of interfacial energy are strongly influenced by trace impurities, which often arise in crystallization of the active ingredient, or other forms of processing such as grinding, and they may be modified by judicious selection of **surfactants** (R. Ayala, Ph.D. thesis, Chemical Engineering, Carnegie Mellon University, 1985). Charges may also exist at interfaces. In the case of solid-fluid interfaces, these may be characterized by **electrokinetic** studies (Shaw, *Introduction to Colloid & Surface Chemistry*, Butterworths & Co. Ltd., 1983).

The total solid-fluid interfacial energy (i.e., both dispersion and polar components) is also referred to as the **critical solid surface energy** of the particulate phase. It is equal to the surface tension of a fluid which just wets the solid with zero contact angle. This property of the particle feed may be determined by a third approach to characterize wetting, involving the penetration of particles into a series of fluids of varying surface tension [R. Ayala, Ph.D. thesis, Chemical Engineering, Carnegie Mellon University, 1985; Fuerstaneau et al., *Colloids & Surfaces*, **60**, 127 (1991)]. The critical surface energy may also be determined from the variation of sediment height with the surface tension of the solvent [Vargha-Butler et al., *Colloids & Surfaces*,

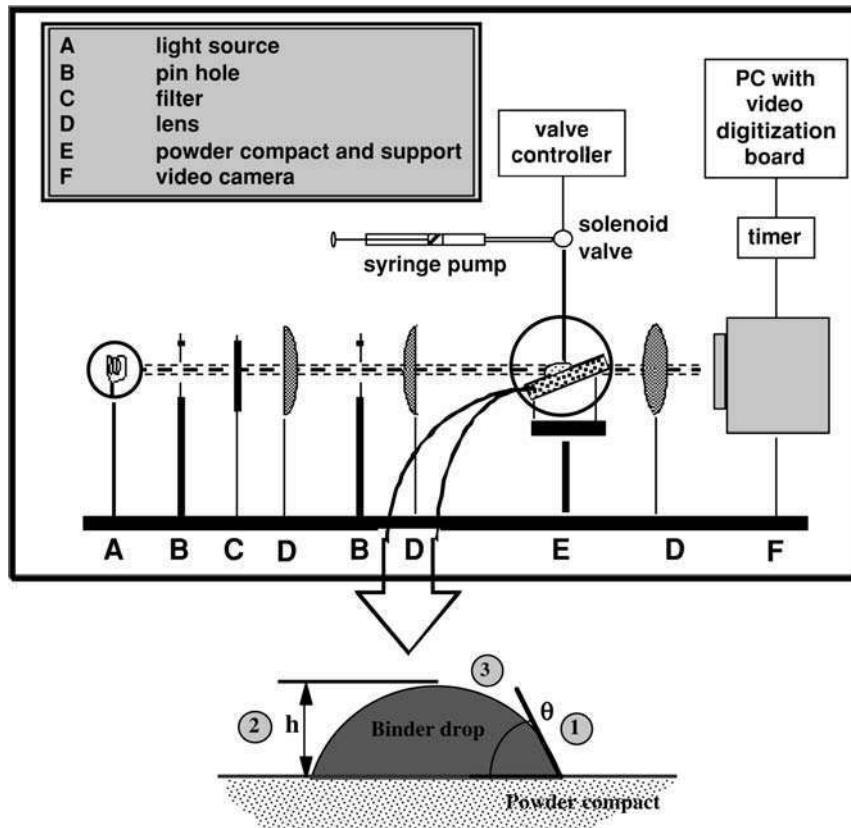


FIG. 21-97 Characterizing wetting by dynamic contact angle goniometry. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

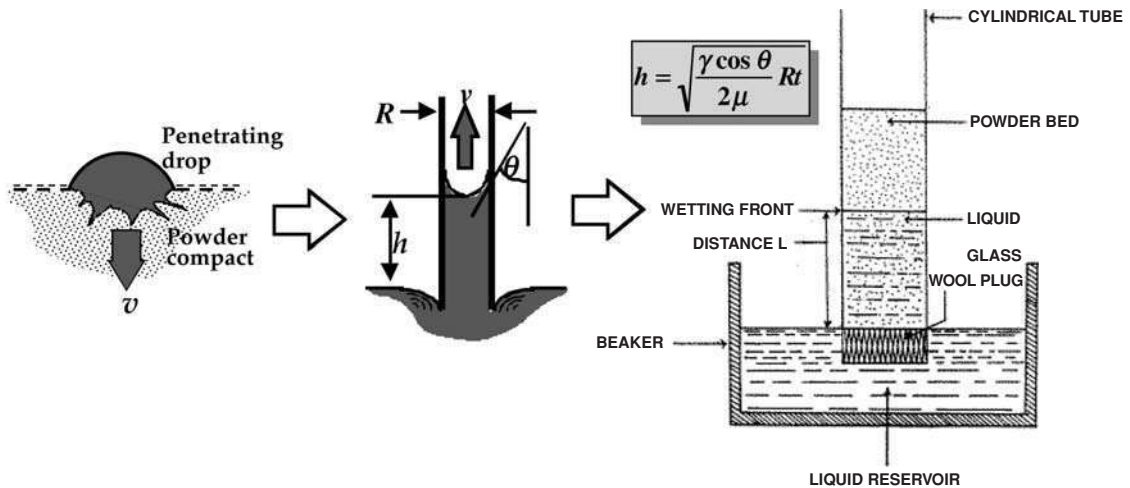


FIG. 21-98 Characterizing wetting by Washburn test and capillary rise. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

24, 315 (1987)]. Distributions in surface energy and its components often exist in practice, and these may be determined by the wetting measurements described here.

The last approach to characterizing wetting involves chemical probing of properties which control surface energy. As an example, **inverse gas chromatography** (IGC) uses the same principles and equipment as standard gas chromatography. In IGC, however, the mobile phase is comprised of probe gas molecules that move through a column packed with the powder of interest, which is the stationary phase. Surface energies of the powder are determined from the adsorption kinetics of both alkane and various polar probes. A distinct advantage of IGC over other methods is reproducible measurements of physical chemical surface properties, which control adhesion tension.

Examples of the Impact of Wetting Wetting dynamics have a pronounced influence on the initial nuclei distribution formed from fine powder. The influence of powder contact angle on the average size of nuclei formed in fluid-bed granulation is illustrated in Fig. 21-99, where the contact angle of the powder with respect to water was varied by changing the weight ratios of the ingredients of lactose and salicylic acid, which are hydrophilic and hydrophobic, respectively (Aulton and Banks, *Proceedings of Powder Technology in Pharmacy Conference*, Powder Advisory Centre, Basel, Switzerland, 1979). Note that granule size in this study is actually *nuclei* size, since little growth has taken place in the process. Nuclei size is seen to improve with contact angle. In addition, the x coordinate would more appropriately be replaced with adhesion tension. Aulton et al. [*J. Pharm. Pharmacol.*, **29**, 59P (1977)] also demonstrated the influence of surfactant concentration on shifting nuclei size due to changes in adhesion tension.

Figure 21-100a illustrates an example of dynamic wetting, where a drop is imaged as it wets in to a formulation tablet. The time scale of wetting is 2 s, with nearly complete wet-in occurring in 1 s. This particular formulation was granulated on a continuous pan system in excess of 2 tons/h. Figure 21-100b compares differences in lots of the formulation. Note that a second lot, referred to as problem technical, experiences significantly degraded granule strength and reduction in production rates. This is associated with nearly twice the initial con-

tact angle (120°) and slower-spreading velocity when compared with the good technical. Poor wetting in practice can translate to reduced production rates to compensate for increased time for drops to work into the powder bed surface. Weaker granules are also often observed, since poor wetting leads to repulsive bonding and high granule voidage. Note that differences in the lots are only observed over the first $\frac{1}{4}$ to $\frac{1}{2}$ s, illustrating the importance of comparing dynamic behavior of formulations, after which time surfactant adsorption/desorption reduces contact angle.

As an example of Washburn approaches, the effect of fluid penetration rate and the extent of penetration on granule-size distribution for drum granulation was shown by Gluba et al. [*Powder Hand. & Proc.*, **2**, 323 (1990)]. Increasing penetration rate, as reflected by Eq. (21-98b), increased granule size, and decreased asymmetry of the granule-size distribution as shown in Fig. 21-101.

Regimes of Nucleation and Wetting Two key features control this wetting and nucleation process. One is the time required for a drop to wet into the moving powder bed, in comparison to some circulation time of the process. As discussed previously, this wet-in time is strongly influenced by formulation properties [e.g., Eq. (21-98b)]. The second is the actual spray rate or spray flux, in comparison to solids flux moving through the spray zones. Spray flux is strongly influenced by manufacturing and process design.

One can envision that drop penetration time and spray flux define **regimes of nucleation** and wetting. If the wet-in is rapid and spray fluxes are low, individual drops will form discrete nuclei somewhat larger than the drop size, defining a **droplet-controlled regime**. At the other extreme, if drop penetration is slow and spray flux is large, drop coalescence and pooling of binder material will occur throughout the powder bed, which must be broken down by mechanical dispersion. In this **mechanical dispersion regime** of nucleation, shear forces control the breakdown of wetting clumps, independent of drop distribution.

Following between these two extreme regimes, drop overlap and coalescence occur to varying extent, defining an **intermediate regime** of nucleation, being a function of penetration time and spray flux. To better define wetting, particularly in the sense of process engineering and scale-up, we consider drop penetration or wet-in time and spray flux in greater detail.

Beginning with penetration time, Eq. (21-98b) defines key formulation properties controlling capillary rise in powder beds. From considering a distribution of macro- and micropores in the moving powder bed as shown in Fig. 21-102, Hapgood (loc. cit.) determined a total drop penetration time t_p of

$$t_p = 1.35 \frac{V_d^{2/3}}{\epsilon_{\text{eff}}^2} \left(\frac{\mu}{R_{\text{eff}} \gamma \cos \theta} \right) \quad (21-99)$$

As shown previously, drop wet-in time decreases with increasing pore radius R_{eff} , decreasing binder viscosity and increasing adhesion tension. In addition, drop penetration time decreases with decreasing drop size V_d and increasing bed porosity ϵ_{eff} . Effective pore radius R_{eff} is related to the surface-volume average particle size d_{32} , particle shape, and effective porosity of packing ϵ_{eff} by

$$R_{\text{eff}} = \frac{\phi d_{32}}{3} \left(\frac{\epsilon_{\text{eff}}}{1 - \epsilon_{\text{eff}}} \right) \quad (21-100)$$

To remain within a droplet-controlled regime of nucleation, the penetration time given by Eq. (21-99) should be much less than some characteristic circulation time t_c of the granulator in question. Circulation time is a function of mixing and bed weight, and it can change with scale-up.

In the case of spray flux, Fig. 21-103 illustrates an idealized powder bed of width B moving past a flat spray of spray rate dV/dt at a solids velocity of w . For a given spray rate, the number of drops is determined by drop volume, which in turn defines the drop area a per unit time that will be covered by the spray, giving a spray flux of

$$\frac{da}{dt} = \frac{dV/dt}{V_d} \left(\frac{\pi d_d^2}{4} \right) = \frac{3}{2} \frac{dV/dt}{d_d} \quad (21-101)$$

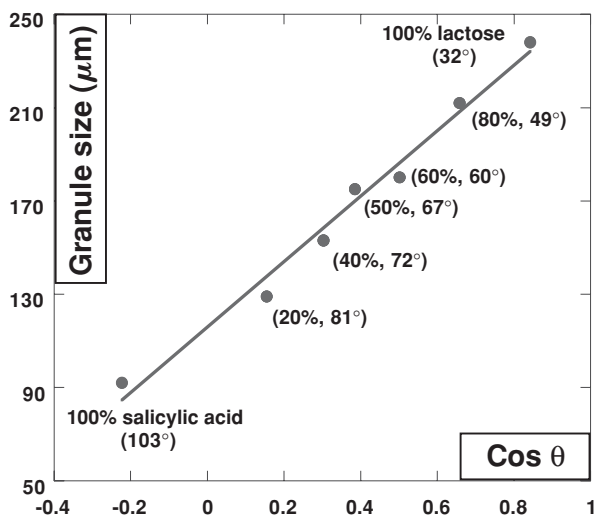


FIG. 21-99 The influence of contact angle on nuclei size formed in fluid-bed granulation of lactose/salicylic acid mixtures. Formulations ranged from hydrophobic (100% salicylic acid) to hydrophilic (100% lactose). Powder contact angle θ determined by goniometry and percent lactose of each formulation are given in parentheses. (Aulton and Banks, *Proceedings of Powder Technology in Pharmacy Conference*, Powder Advisory Centre, Basel, Switzerland, 1979.)

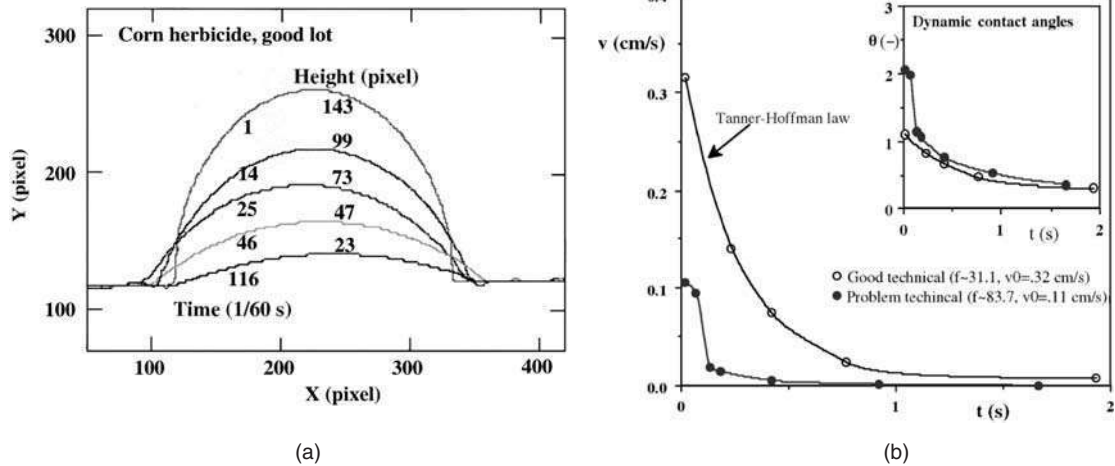


FIG. 21-100 Dynamic imaging of wetting, and its impact on continuous pan granulation. (a) Dynamic images of a drop wetting into a formulation with good active ingredient. (b) Comparison of surface spreading velocity and dynamic contact angle versus time for good and bad active ingredients or technical. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

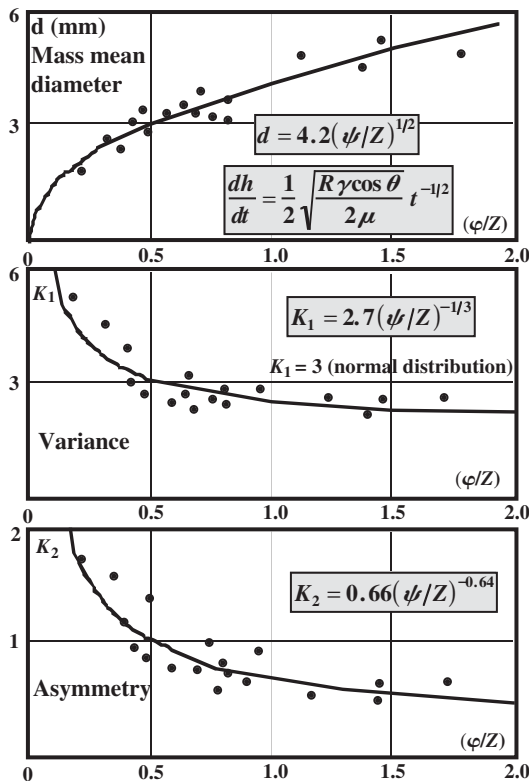


FIG. 21-101 Influence of capillary penetration on drum granule size. Increasing penetration rate increases granule size and decreases asymmetry of the granule-size distribution. [After Gluba *et al.*, Powder Hand. & Proc., 2, 323 (1990).]

As droplets contact the powder bed at a certain rate, the powder moves past the spray zone at its own velocity, or at solids flux given for this simple example by

$$\frac{dA}{dt} = Bw \quad (21-102)$$

The ratio of the droplet spray flux to the solids flux defines a dimensionless spray flux given by

$$\psi_a = \frac{da/dt}{dA/dt} = \frac{3}{2} \frac{dV/dt}{d_a(dA/dt)} \quad (21-103)$$

The dimensionless spray flux is the ratio of the rate at which wetted area is covered by droplets to the area of flux of powder through the spray zone, and it is a measure of the density of drops falling on the powder surface. As with drop penetration time, it plays a role in defining the regimes of nucleation as illustrated in Fig. 21-104. For small spray flux ($\psi_a \ll 1$), drops will not overlap on contact and will form separate discrete nuclei for fast penetration time. For large spray flux ($\psi_a \approx 1$), however, significant drop overlap occurs, forming nuclei much larger than drop size, and, in the limit, independent of drop size.

For the case of random drop deposition as described by a Poisson distribution, Hapgood (*loc. cit.*) showed the fraction of surface

$$R_{\text{eff}} = \frac{\phi d_{32}}{3} \frac{\epsilon_{\text{eff}}}{1 - \epsilon_{\text{eff}}} \quad \epsilon_{\text{eff}} = \epsilon_{\text{tap}} (1 - \epsilon + \epsilon_{\text{tap}})$$

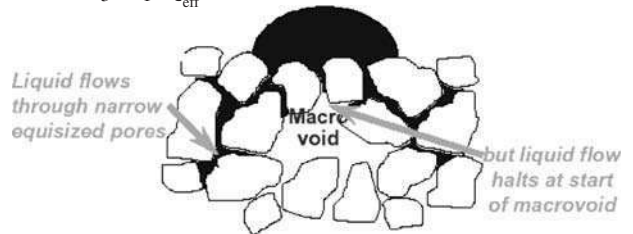


FIG. 21-102 Drop penetration in a moving powder bed. [After Hapgood (*loc. cit.*).]

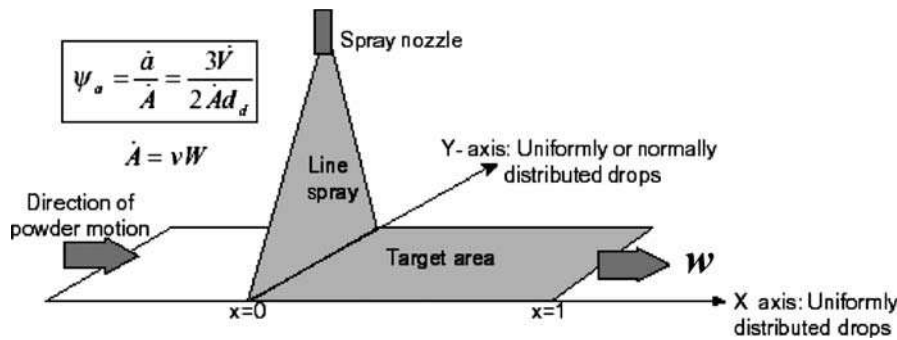


FIG. 21-103 Idealized flat spray zone in a spinning riffle granulator. [After Hapgood (loc. cit.).]

covered by spray was given by

$$f_{\text{single}} = 1 - \exp(-\psi_a) \quad (21-104)$$

In addition, the fraction of single drops forming individual nuclei (assuming rapid drop penetration) versus the number of agglomerates formed was given by

$$f_{\text{single}} = \exp(-4\psi_a) \quad (21-105)$$

$$f_{\text{agglom}} = 1 - \exp(-4\psi_a) \quad (21-106)$$

Examples of the above as applied to nucleation are depicted in Fig. 21-105. Here, nuclei distributions were studied as a function of drop size and spray flux. Lactose was sprayed with a flat spray in a spinning riffle granulator, mimicking the geometry of Fig. 21-103. For a small spray flux of $\psi_a = 0.22$, a clear relationship is seen between nuclei size and spray distribution, with nuclei formed somewhat larger than drop size. However, as the speed of the riffle is slowed (i.e., solids velocity and solids flux are decreased, and spray flux increased), the nuclei distribution widens with the formation of agglomerates.

The spray flux captures the impact of equipment operating variables on nucleation, and as such is very useful for scale-up if nucleation rates and nuclei sizes are to be maintained constant. The overall impact of dimensionless spray flux on nucleation and agglomerate formation is illustrated in Fig. 21-106, with agglomerates increasing with increased spray flux as clearly governed by Eq. (21-106) for the case of rapid drop penetration.

Regimes of nucleation may be defined (Fig. 21-107) with the help of dimensionless drop penetration time τ_p and spray flux ψ_a , or

$$\tau_p = \frac{t_p}{t_c} = \frac{\text{penetration time}}{\text{circulation time}} \quad \text{and} \quad \psi_a = \frac{da/dt}{dA/dt} \quad (21-107)$$

$$= \frac{\text{spray flux}}{\text{solids flux}}$$

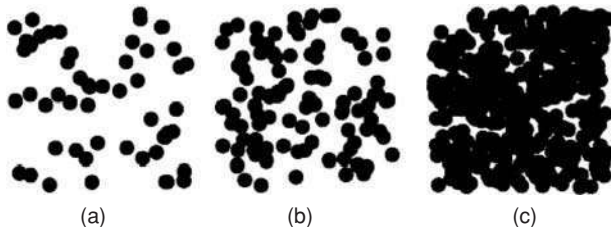


FIG. 21-104 Monte Carlo simulations of drop coverage: (a) 50 discs, $\psi_a = 0.29$, $f_{\text{covered}} = 0.26$; (b) 100 discs, $\psi_a = 0.59$, $f_{\text{covered}} = 0.45$; (c) 400 discs, $\psi_a = 2.4$, $f_{\text{covered}} = 0.91$. Image, 500×500 pixels; disc radius, 20 pixels. [After Hapgood (loc. cit.).]

A droplet-controlled nucleation regime occurs when there is both low spray flux (relatively few drops overlap) and fast droplet penetration—drops wet into the bed completely before bed mixing allows further drop contact. Nuclei will be formed of the order of drop size. A mechanical dispersion regime occurs at the other extreme of high spray flux, giving large drop overlap and coalescence, and large drop penetration times, promoted by poor wet-in rates and slow circulation

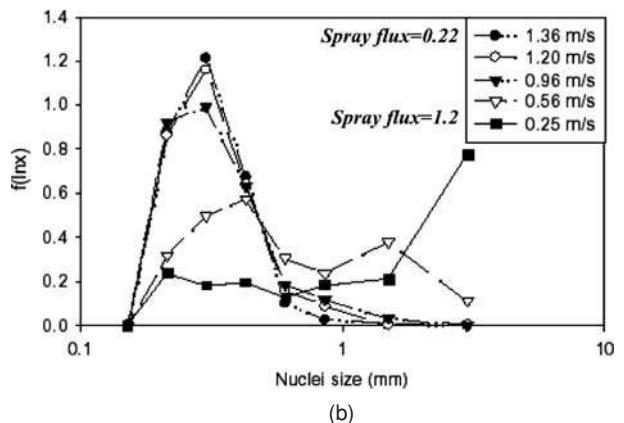
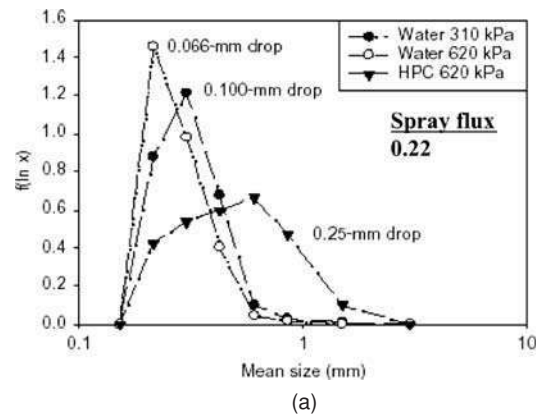


FIG. 21-105 Effect of (a) spray drop distribution (b) (low spray flux—water and HPC) and (b) powder velocity (variable spray flux—water) on nuclei size distribution. Lactose feed powder in spinning granulator. [Litster and Emnis, loc. cit.]

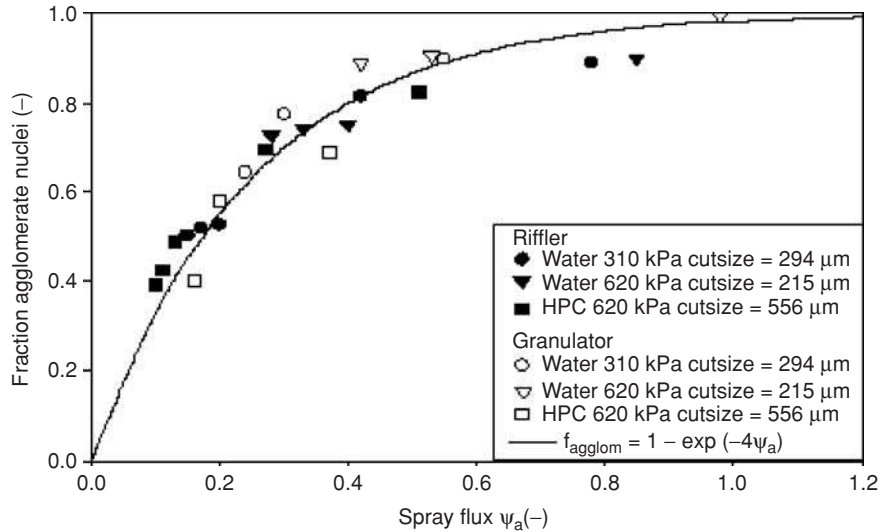


FIG. 21-106 Agglomerate formation vs. spray flux. Lactose powder with water and HPLC solutions. [After Hapgood (loc. cit.).]

times and poor mixing. In the regime, nucleation and binder dispersion occurs by mechanical agitation. Viscous, poorly wetting binders are slow to flow through pores in the powder bed in the case of poor penetration time. Drop coalescence on the powder surface occurs (also known as *pooling*), creating very broad nuclei size distributions. The binder solution delivery method (drop size, nozzle height) typically has minimal effect on the nuclei size distribution, although interfacial properties may affect nuclei and final granule strength. An intermediate regime exists for moderate drop penetration times and moderate spray flux, with the resulting nuclei regime narrowing with decreases in both.

There are several implications with regard to the nucleation regime map of Fig. 21-107 with regard to troubleshooting of wetting and nucleation problems. If drop penetration times are large, making adjustments to spray may not be sufficient to narrower granule size distributions if remaining in the mechanical regime. Significant changes to wetting and nucleation occur only if changes take the system across a regime boundary. This can occur in an undesirable way if processes are not scaled with due attention to remaining in the drop-controlled regime.

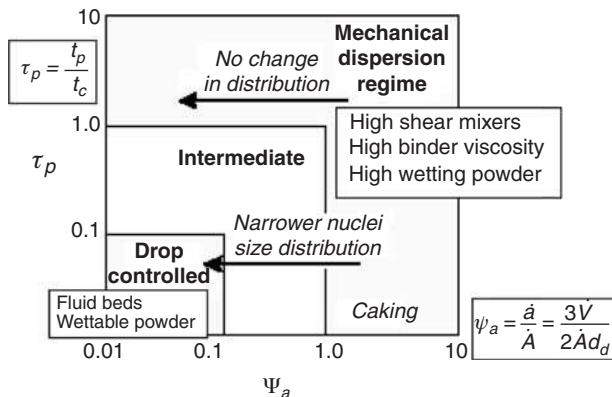


FIG. 21-107 A possible regime map of nucleation, relating spray flux, solids mixing (solids flux and circulation time), and formulation properties.

GROWTH AND CONSOLIDATION

The evolution of the granule-size distribution of a particulate feed in a granulation process is controlled by several mechanisms, as illustrated in Figs. 21-85 and 21-91. These include the **nucleation** of fine powder to form initial primary granules, the **coalescence** of existing granules, and the **layering** of raw material onto previously formed nuclei or granules. The breakdown of wet clumps into a stable nuclei distribution can also be included among coalescence mechanisms. As granules grow by coalescence, they are simultaneously compacted by **consolidation** mechanisms, which reduce internal granule voidage or porosity. Lastly, granules may be reduced in size by **breakage**. Dominant mechanisms of growth and consolidation are dictated by the relationship between critical particle properties and operating variables as well as by mixing, size distribution, and the choice of processing.

There are strong interactions between the growth and consolidation mechanisms, as illustrated for the case of drum granulation of fine feed (Fig. 21-108). Granule size progresses through three stages of growth, including rapid, exponential growth in the initial **nucleation** stage, followed by linear growth in the **transition** stage, and finishing with very slow growth in a final **balling** stage. Simultaneously with growth, granule porosity or voidage decreases with time as the granules are compacted. Granule growth and consolidation are intimately connected; increases in granule size are shown here to be associated with a decrease in granule porosity. This is a dominant theme in wet granulation.

As originally outlined in Ennis (loc. cit., 1991), these growth patterns are common throughout fluidized-bed, drum, pan, and high-shear mixer processes for a variety of formulations. Specific mechanisms of growth may dominate for a process—sometimes to the exclusion of others. However, what all processes have in common is that the prevailing mechanisms are dictated by a balance of critical particle level properties, which control formulation deformability, and operating variables, which control the local level of shear, or bed agitation intensity.

Granule Deformability In order for two colliding granules to coalesce rather than break up, the **collisional kinetic energy** must first be dissipated to prevent rebound, as illustrated in Fig. 21-109. In addition, the strength of the bond must resist any subsequent breakup forces in the process. The ability of the granules to deform during processing may be referred to as the formulation's **deformability**, and

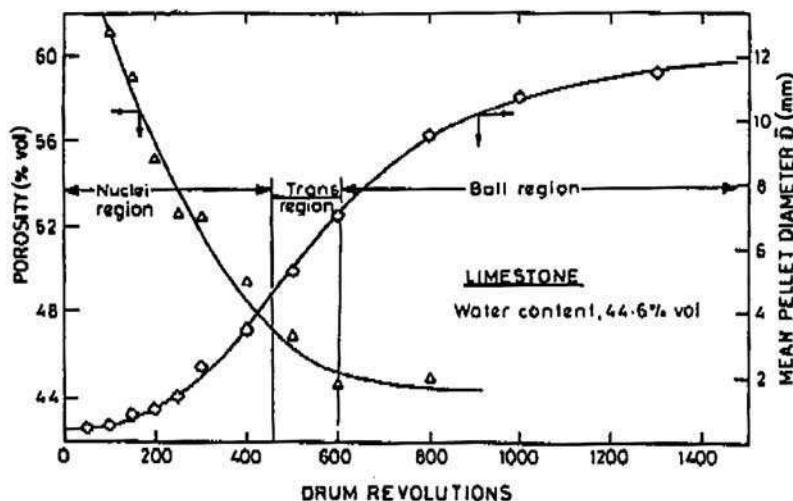


FIG. 21-108 Granule porosity and mean (pellet) size. Typical regimes of granule growth and consolidation. [After Kapur, *Adv. Chem. Eng.*, **10**, 55 (1978); *Chem. Eng. Sci.*, **26**, 1093 (1971).]

deformability has a large effect on growth rate. Increases in deformability increase the bonding or contact area, thereby dissipating and resisting breakup forces. From a balance of binding and separating forces and torque acting within the area of granule contact, Ouchi-yama and Tanaka [*IE&C Proc. Des. & Dev.*, **21**, 29 (1982)] derived a **critical limit of size** above which coalescence becomes impossible, or a maximum growth limit given by

$$D_c = (AQ^{3/2}K^{3/2}\sigma_T)^{1/4 - (3/2)\eta} \quad (21-108)$$

where K is deformability, a proportionality constant relating the maximum compressive force Q to the deformed contact area; A is a constant with units of L^3/F , which relates granule volume to impact compression force; and σ_T is the **tensile strength** of the granule bond [see Eq. (21-98)]. Granules are compacted as they collide. This expels pore fluid to the granule surface, thereby increasing local liquid satu-

ration in the contact area of colliding granules. This surface fluid (1) increases the tensile strength of the liquid bond σ_T and (2) increases surface plasticity and deformability K .

Here D_c represents the largest granule that may be grown in a granulation process, and it is a harmonic *average* granule size. Therefore, it is possible for the collision of two large granules to be unsuccessful, their average being beyond this critical size, whereas the collision of a large granule and a small granule leads to successful coalescence. The growth limit D_c is seen to increase with increased formulation deformability K (which will be shown to be a strong function of moisture and primary particle-size distribution), increased compressive forces Q (which are related to local shear levels in the process), and increased tensile forces σ_T (which are related to interparticle forces). The parameters ζ and η depend on the deformation mechanism within the contact area. For plastic deformation, $\zeta = 1$, $\eta = 0$, and $K \propto 1/H$, where H is hardness. For elastic, hertzian deformation, $\zeta = 2/3$, $\eta = 2/3$, and $K \propto (1/E^*)^{2/3}$, where E^* is the reduced elastic modulus. Granule deformation is generally dominated by **inelastic** behavior of the contacts during collision, with such deformation treated by the area of **inelastic contact mechanics** (Johnson, *Contact Mechanics*, Cambridge University Press, 1985).

Types of Granule Growth The importance of deformability to the growth process depends on **bed agitation intensity**. If little deformation takes place during granule collisions, the system is referred to as a **low-deformability** or **low-agitation-intensity** process. This generally includes fluid-bed, drum, and pan granulators. Growth is largely controlled by the extent of any surface fluid layer and surface deformability, with surface fluid playing a large role in dissipating collisional kinetic energy. Growth generally occurs at a faster time scale than overall granule deformation and consolidation. This is depicted in Fig. 21-110, where smaller granules can still be distinguished as part of a larger granule structure, or a popcorn-type appearance, as often occurs in fluid-bed granulation. Note that such a structure may not be observed if layering and nucleation alone dominate with little coalescence of large granules; in addition, the surface structure may be compacted and smoother over time due to the longer time-scale process of consolidation. In this case, granule coalescence and consolidation have less interaction than they do with high-deformability systems, making low deformability-low agitation systems easier to scale and model.

For high-shear rates or bed agitation intensity, large granule deformation occurs during granule collisions, and granule growth and consolidation occur on the same time scale. Such a system is referred to

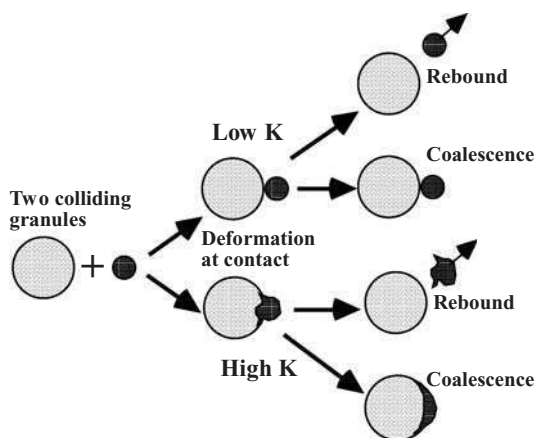


FIG. 21-109 Mechanisms of granule coalescence for low- and high-deformability systems. Rebound occurs for average granule sizes greater than the critical granule size D_c . K = deformability. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&C Associates. All rights reserved.)

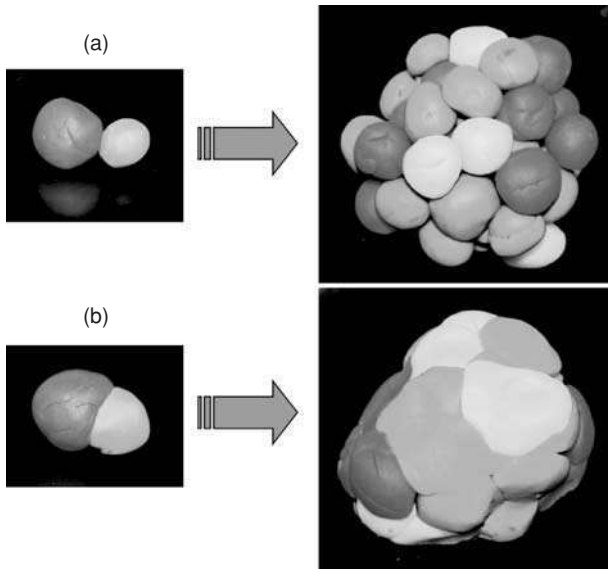


FIG. 21-110 Granule structures resulting from (a) low- and (b) high-deformability systems, typical for fluid-bed and high-shear mixer-granulators, respectively. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

as a **deformable** or **high-agitation-intensity** process, and this generally includes continuous pin and plow shear-type mixers, as well as batch high-shear pharmaceutical mixers. In these cases, substantial collisional kinetic energy is dissipated with deformation of the wet mass composing the granule. Rather than a *sticking-together* process

as often occurs in the low-deformability process of fluid beds, granules are *smashed* or *kneaded* together as with a high-shear mixer, and smaller granules are not distinguishable with the granule structure, as depicted in Fig. 21-110. High-agitation, highly deformable processes generally produce denser granules than low-deformability, low-agitation-intensity ones. In addition, the combined and competing effects of granule coalescence and consolidation make high-agitation processes difficult to model and scale. Both coalescence and consolidation initially increase with both increases in shear level and deformability, while at the same time as granules densify, they become less deformable, which works to lower coalescence in the later stages of growth.

Bed agitation intensity is controlled by mechanical variables of the process such as fluid-bed excess gas velocity or mixer impeller and chopper speed. Agitation intensity controls the relative collisional and shear velocities of granules within the process and therefore growth, breakage, consolidation, and final product density. Figure 21-111 summarizes typical characteristic velocities, agitation intensities and compaction pressures, and product relative densities achieved for a variety of size-enlargement processes.

Lastly, note that the process or formulation itself *cannot* uniquely define whether it falls into a low- or high-agitation-intensity process. As discussed more fully below, it is a function of *both* the level of shear and the formulation deformability. A very stiff formulation with low deformability may behave as a low-deformability system in a high-shear mixer; or a very pliable formulation may act as a high deformable system in a fluid-bed granulator.

Granule deformability and limiting size D_c are a strong function of moisture, as illustrated in Fig. 21-112. Deformability K is related to both the **yield strength** of the material σ_y , i.e., the ability of the material to resist stresses, and the ability of the surface to be strained without degradation or rupture of the granule, with this maximum allowable **critical deformation strain** denoted by $(\Delta L/L)_c$. Figure 21-113 illustrates the low-shear-rate stress-strain behavior of agglomerates during compression as a function of liquid saturation, with strain denoted by $\Delta L/L$. In general, high deformability K requires low yield strength σ_y and high critical strain $(\Delta L/L)_c$. Increasing moisture

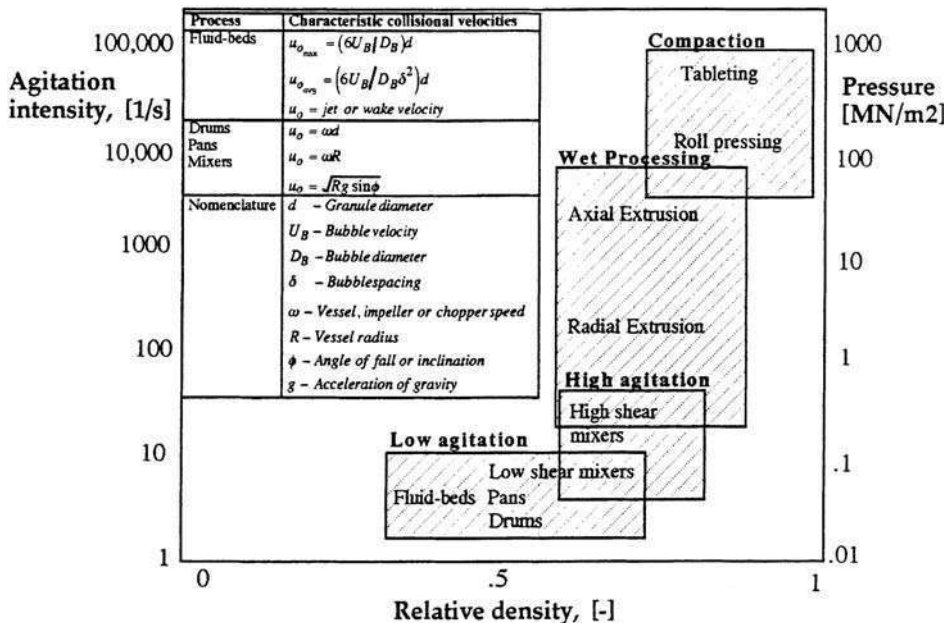


FIG. 21-111 Classification of agglomeration processes by agitation intensity and compaction pressure. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

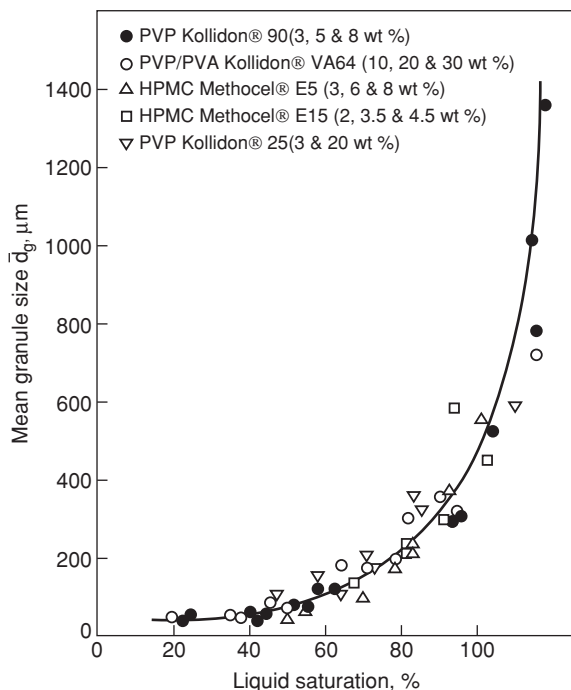


FIG. 21-112 Effect of granule saturation on mean granule diameter, indicating the marked increase in granule deformability with increased moisture. Mean granule diameter is a measure of the critical limit of size D_c . Granulation of calcium hydrogen phosphate with aqueous binder solutions in a Fielder PMAT 25 VG, high-shear mixer. [Ritala et al., Drug Dev. & Ind. Pharm., **14**(8), 1041 (1988).]

increases deformability by lowering interparticle frictional resistance, leading to an increase in mean granule size (Fig. 21-112). Saturation S is defined here as the volumetric percent of pore volume filled with moisture, with this pore volume controlled by granule porosity or voidage.

Deformability and Interparticle Forces In most cases, granule deformability increases with increasing moisture, decreasing binder viscosity, decreasing surface tension, decreasing interparticle friction, and increasing average primary particle size, as well as increasing bed agitation intensity. Interstitial fluid leads to pendular bridges between the primary particles composing a granule, giving rise to capillary and viscous interparticle forces. In addition, frictional forces develop as primary particles come into contact. Interparticle forces and their impact on deformability warrant further attention. Figure 21-114 illustrates two particles of radius a separated by a gap distance $2h_o$ (or in contact) approaching each other at a velocity U , bound by a pendular bridge of viscosity μ , density ρ , and surface tension γ . The two particles may represent two primary particles within the granule, in which case we are concerned about the contribution of interparticle forces to granule strength and deformability. Or they may represent two colliding granules, in which case we are concerned with the ability of the pendular bridge to dissipate granule kinetic energy and resist breakup forces in the granulation process. The pendular bridge consists of the binding fluid in the process, which includes the added solvent and any solubilized components. In some cases, it may also be desirable to include very fine solid components within the definition of the binding fluid and, therefore, consider instead a suspension viscosity and surface tension. These material parameters vary on a local level throughout the process and are time-dependent and a function of drying conditions.

For the case of a static liquid bridge of contact angle θ , surface tension induces an attractive capillary force F_{cap} between the two particles

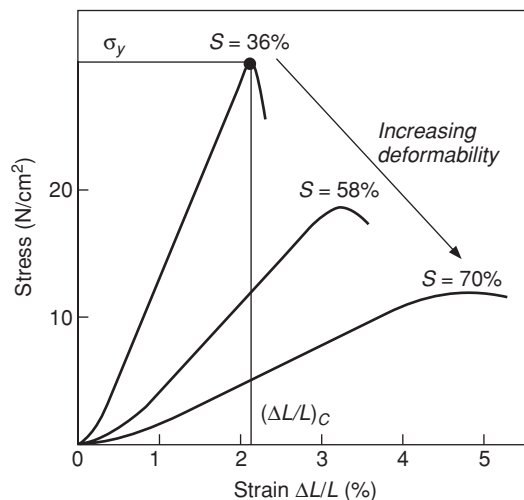


FIG. 21-113 The influence of sample saturation S on granule deformability. Deformation strain $(\Delta L/L)$ is measured as a function of applied stress, with the peak stress and strain denoted by tensile strength σ_y and critical strain $(\Delta L/L)_c$ of the material. Dicalcium phosphate with 15 wt % binding solution of PVP/PVA Kollidon® VA64, 50% compact porosity. [Holm et al., Powder Technol., **43**, 213 (1985), with kind permission from Elsevier Science SA, Lausanne, Switzerland.]

due to a three-phase contact line force and a pressure deficiency arising from interfacial curvature H_o and filling angle ϕ , given by

$$F_{cap} = \pi\gamma a (2 \cos \theta - 2H_o) \sin^2 \phi \quad (21-109)$$

The impact of this static pendular bridge force on static granule strength has been studied extensively, as illustrated in Fig. 21-92 (Ennis, loc. cit., 1991; Rumpf, loc. cit.; Kapur, loc. cit.). It is important to recognize that in most processes, however, the particles are moving relative to one another and, therefore, the bridge liquid is in motion. This gives rise to viscous lubrication forces F_{vis} that can contribute significantly to the total bridge strength, given by

$$F_{vis} = 3\pi\mu Ua/\epsilon \quad (21-110)$$

This viscous force increases with increasing binder viscosity μ and collision velocity U , and decreasing dimensionless gap distance $\epsilon = 2h_o/a$ [Ennis, loc. cit., 1991; Ennis et al., *Chem. Eng. Sci.*, **45** (10), 3071 (1990); Mazzone et al., *J. Colloid Interface Sci.*, **113**, 544 (1986)]. Written in dimensionless form, total dynamic bridge strength for newtonian fluids for particles in close contact is given by

$$F^* = \frac{1}{\pi\gamma a} (F_{cap} + F_{vis}) = F_o + 3Ca/\epsilon$$

$$\text{where } F_o = (2 \cos \theta - 2H_o) \sin^2 \phi \quad (21-111)$$

$$Ca = \mu U/\gamma$$

where Ca is a capillary number representing the ratio of viscous-to-capillary forces and is proportional to velocity. Dynamic bridge force consists of an initial constant, static bridge strength for small Ca (or near zero velocity) and then increases linearly with Ca (or velocity). This is confirmed experimentally as illustrated in Fig. 21-115 for the case of two spheres approaching axially. Extensions of the theory have also been conducted for nonnewtonian fluids, shearing motions, particle roughness, wettability, and time-dependent drying binders (Ennis, loc. cit., 1991).

For small velocities, small binder viscosity, and large gap distances, the strength of the bridge will approximate a static pendular bridge, or

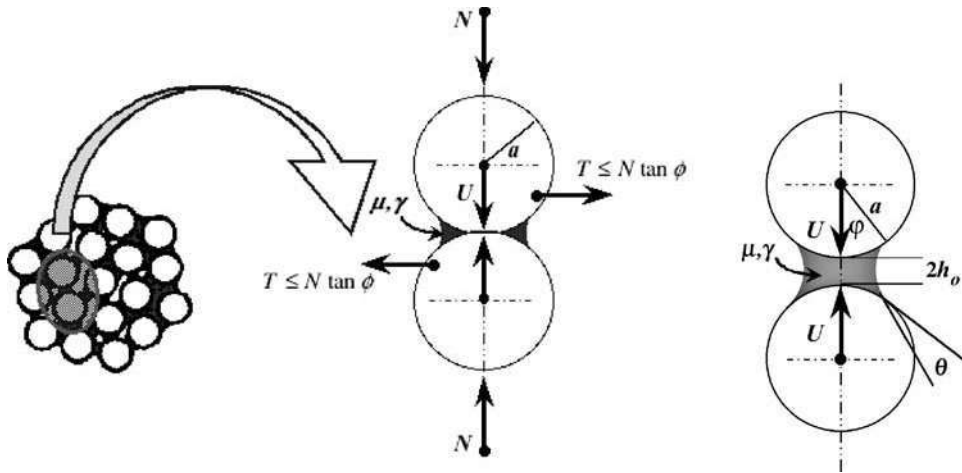


FIG. 21-114 Interparticle forces and granule deformability. Interparticle forces include capillary forces, viscous lubrication forces, and frictional forces. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

F_{cap} , which is proportional to and increases with increases in surface tension. This force is equivalent to the static pendular force H previously given in Eq. (21-96) as studied by Rumpf (loc. cit.). On the other hand, for large binder viscosities and velocities, or small gap distances, the bridge strength will approximately be equal to F_{vis} , which is proportional to and increases with increases in binder viscosity and velocity. This viscous force is singular in the gap distance and increases dramatically for small separation of the particles. It is important to note that as granules are consolidated, resulting in decreases in effective interparticle gap distance, and binders dry, resulting in large increases in binder viscosity, the dynamic bridge strength can exceed the static strength by orders of magnitude.

The important contributions of binder viscosity and friction to granule deformability are illustrated by fractions of energy dissipated during computer simulations of granule collisions, as depicted in Fig. 21-116.

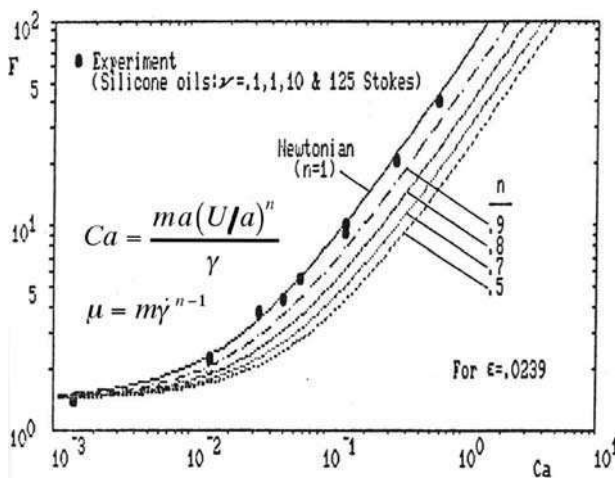


FIG. 21-115 Maximum strength of a liquid bridge between two axial moving particles as a function of Ca for newtonian and shear thinning fluids. (After Ennis, *On the Mechanics of Granulation*, Ph.D. thesis, 1990, The City College of the City University of New York, University Microfilms International, 1991, with permission.)

Of the energy, 60 percent is dissipated through viscous losses, with the majority of the remainder through interparticle friction. Very little loss is due to capillary forces. Therefore, modern approaches to granule coalescence rest in understanding the impact of granule deformability on growth, rather than the original framework put for regarding pendular and funicular forces due to interparticle liquid bridges alone.

Deformability and Wet Mass Rheology The static yield stress of wet compacts has previously been reported in Fig. 21-113. However, the dependence of interparticle forces on shear rate clearly impacts **wet mass rheology** and therefore deformability. Figure 21-117 illustrates the dynamic stress-strain response of compacts, demonstrating that the peak flow or yield stress increases proportionally with compression velocity [Iveson et al., *Powder Technol.*, **127**, 149 (2002)]. Peak flow stress of wet unsaturated compacts (initially pendular state) can be seen to also increase with Ca as follows (Fig. 21-118):

$$\frac{\sigma_y^{\text{Peak}}}{\gamma a} = \sigma_0 + A \overline{Ca}^B \quad \text{where} \quad \sigma_0 = 5.0 - 5.3$$

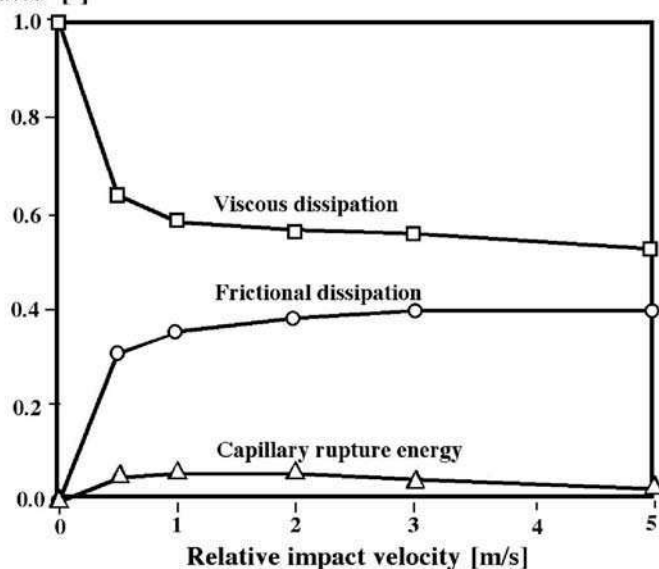
$$A = 280 - 320 \quad B = 0.58 - 0.64$$

$$\overline{Ca} = \mu \dot{\epsilon} a / \gamma \quad (21-112)$$

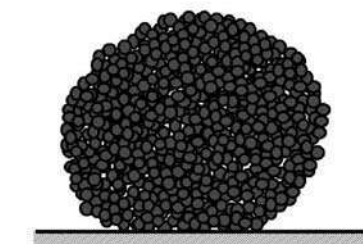
There are several important issues worth noting with regard to these results. First is the similarity between the strength of the assembly or compact [Eq. (21-112)] and the strength of the individual dynamic pendular bridge given by Eq. (21-111); both curves are similar in shape with a capillary number dependency. As with the pendular bridge, two regions may be defined. In region 1 for a bulk capillary number of $Ca < 10^{-4}$, the strength or yield stress of the compact depends on the static pendular bridge, and therefore on surface tension, particle size, and liquid loading. In region 2 for $Ca > 10^{-4}$, the strength depends on the viscous contribution to bridge strength, and therefore on binder viscosity and strain rate, in addition to particle size.

Second is that the results of Figs. 21-117 and 21-118 do not clearly depict the role of saturation and compact porosity. Decreases in compact porosity generally increase compact strength through increases in interparticle friction, whereas increases in saturation lower strength (e.g., Figs. 21-112 and 21-113 and Holm et al. [Parts V and VI, *Powder Technol.*, **43**, 213-233 (1985)]). Hence, the curve of Fig. 21-118 should be expected to shift with these variables, particularly since the viscous force for axial approach is singular in the interparticle gap distance [Eq. (21-111)].

Fractional energy dissipated [-]



Impacted granule (inviscid binder)



Impacted granule (viscous binder)

FIG. 21-116 Distribution of energy dissipation during agglomerate collisions, with granular simulations of wall impact for 128- μ s duration for inviscid and viscous binder agglomerates. (After Adams, Thornton, and Lian, *Agglomeration and Size Enlargement*, Proc. 1st Int. Particle Technology Forum, vol. I, Denver, Colo., AIChE, New York, 1994, pp. 155–286, with permission.)

Last is that the mechanism of compact failure also depends on strain rate. Figure 21-118 illustrates schematically the crack behavior observed in compacts as a function of capillary number. At low Ca , compacts fail by brittle fracture with macroscopic crack propagation, whereas at high Ca , compacts fail by plastic flow, which is more desirable to promote growth.

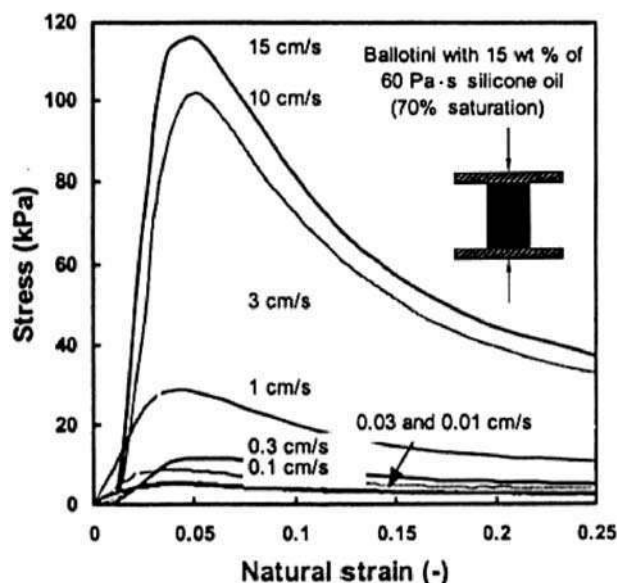


FIG. 21-117 Typical compact stress response for fast compression vs. crosshead compression velocity for glass ballotini ($d_{32} = 35 \mu\text{m}$) and compact diameter 20 mm, length 25 mm. [After Iveson et al., *Powder Technol.*, **127**, 149 (2002), with permission.]

Within the context of granulation, small yield stresses at low Ca may result in unsuccessful growth when these stresses are compared with large breakup forces. With increased yield stress come stronger granules but also decreased deformability. Therefore, high strength might imply a low-deformability growth mechanism for low-shear processes such as a fluid-bed. On the other hand, it might imply smaller growth rates for high-shear processes, which are able to overcome this yield stress and bring about kneading action and plastic flow in the process. Therefore, it is important to bear in mind that increased liquid saturation may initially lower yield stress, allowing greater plastic deformation during granule collisions. However, as granules grow and consolidate and decrease in voidage, they also strengthen and rise in yield stress, becoming less deformable with

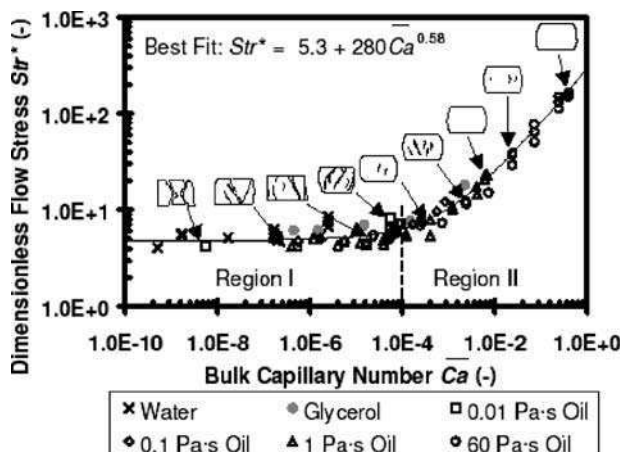


FIG. 21-118 Dimensionless peak flow stress of Fig. 21-154 vs. bulk capillary number, for various binder solutions. [After Iveson et al., *Powder Technol.*, **127**, 149 (2002), with permission.]

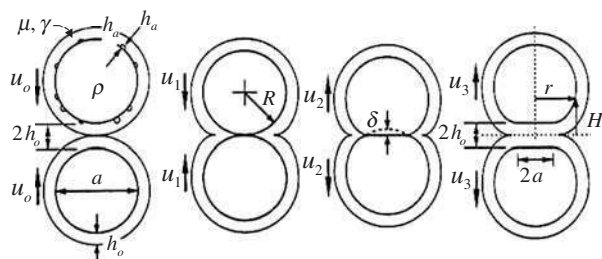


FIG. 21-119 Collisions between surface wet granules, beginning with approach and ending with separation [Liu et al., *AIChE J.*, **46**(3), 529 (2000)]. Note that no deformation takes place in the original Stokes model [Ennis et al., *Powder Technol.*, **65**, 257 (1991)].

time and withstanding shear forces in the granulator. Hence, the desired granule strength and deformability is linked in a complex way to granulator shear forces and consolidation behavior, and is the subject of current investigations.

Low Agitation Intensity—Low Deformability Growth For low-agitation processes or for formulations which allow little granule deformation during granule collisions, consolidation of the granules occurs at a much slower rate than growth, and granule deformation can be ignored to a first approximation. The growth process can be modeled by the collision of two nearly stiff granules, each coated by a liquid layer of thickness h (Fig. 21-119). For the case of zero plastic deformation and neglecting capillary contributions to bridge strength, the probability of successful coalescence is governed by a dimensionless energy of collision, or **viscous Stokes number** St_v , given by

$$St_v = \frac{4\rho u_o d}{9\mu} \quad (21-113)$$

where u_o is the relative collisional velocity of the granules, ρ is granule density, d is the harmonic average of granule diameter, and μ is the solution-phase binder viscosity. The Stokes number represents the ratio of **initial collisional kinetic energy** to the **energy dissipated by viscous lubrication** forces, and it is one measure of normalized bed agitation energy. Successful growth by coalescence or layering requires that

$$St_v < St^* \quad \text{where} \quad St^* = \left(1 + \frac{1}{e_r}\right) \ln\left(\frac{h}{h_a}\right) \quad (21-114)$$

where St^* is a **critical Stokes number** representing the energy required for rebound. The binder layer thickness h is related to liquid loading, d is the harmonic average of granule diameter, and μ is the solution-phase binder viscosity. The Stokes number represents the ratio of **initial collisional kinetic energy** to the **energy dissipated by viscous lubrication** forces, and it is one measure of normalized bed agitation energy. Successful growth by coalescence or layering requires that

Both the **binder solution viscosity** μ and the granule density are largely properties of the feed. Binder viscosity is a function of local temperature, collisional strain rate (for nonnewtonian binders), and binder concentration, which is dictated by drying rate and local mass transfer and bed moisture. Viscosity can be manipulated in formulation through judicious selection of binding and surfactant agents and measured by standard rheological techniques (Bird et al., *Dynamics of Polymeric Liquids*, vol.1, Wiley, 1977). The **collisional velocity** is a function of process design and operating variables, and is related to bed agitation intensity and mixing. Possible choices of u_o are summarized in Fig. 21-111, and discussed further below. Note that u_o is an **interparticle collisional velocity**, which is not necessarily the local average granular flow velocity.

Three regimes of granule growth may be identified for low-agitation-intensity, low-deformability processes [Ennis et al., *Powder Technol.*, **65**, 257 (1991)], as depicted for fluid-bed granulation in Fig. 21-120. For small granules or high binder viscosity lying within a **noninertial regime** of granulation, all values of St_v will lie below the critical value St^* and therefore all granule collisions result in successful coalescence and growth *provided* binder is present. Growth rate is independent of granule kinetic energy, particle size, and binder viscosity (provided other rate processes are constant). *Distribution of binding fluid and degree of mixing* then control growth, and this is strongly coupled with the rate process of wetting. (See subsection "Wetting".) As shown in Fig. 21-120, both binders have the same initial growth rate for similar spray rates, independent of binder viscosity. Increases in bed moisture (e.g., spray rate, drop rate) and increases in granule collisions in the presence of binder will increase the overall rate of growth. Bear in mind, however, that there is a 100 percent success of these collisions, since dissipation of energy far exceeds collisional kinetic energy.

As granules grow in size, their collisional momentum increases, leading to localized regions in the process where St_v exceeds the critical value St^* . In this **inertial regime** of granulation, the granule size, binder viscosity, and collision velocity determine the proportion of the bed in which granule rebound or unsuccessful coalescence is possible. Increases in binder viscosity and decreases in agitation intensity increase the **extent of granule growth**, i.e., the largest granule that may be grown [for example, D_c of Eq. (21-108)]. This is confirmed in Fig. 21-120 with the CMC binder continuing to grow, whereas the PVP system with lower viscosity slows in growth. However, note that binder distribution and mixing, and not binder viscosity, control the rate of growth. For example, increasing binder viscosity will not affect growth rate, or initial granule size, but it will result in an increased growth limit. For deformable systems, the opposite will hold true.

When the spatial average of St_v exceeds St^* , growth is balanced by granule disruption or breakup, leading to the **coating regime** of granulation. Growth continues by coating of granules by binding fluid alone. The PVP system with lower viscosity is seen to reach its growth limit and therefore coating regime in Fig. 21-120.

Transitions between granulation regimes depend on bed hydrodynamics. As demonstrated by Fig. 21-120, granulation of an initially fine powder may exhibit characteristics of all three granulation regimes as time progresses, since St_v increases with increasing granule size. Implications and additional examples regarding the regime analysis are highlighted by Ennis [loc. cit., 2006; *Powder Technol.*, **88**, 203 (1996)]. In particular, increases in fluid-bed excess gas velocity exhibit a similar but opposite effect on growth rate to binder viscosity; namely, it is observed to not affect growth rate in the initial inertial regime of growth, but instead lowers the growth limit in the inertial regime.

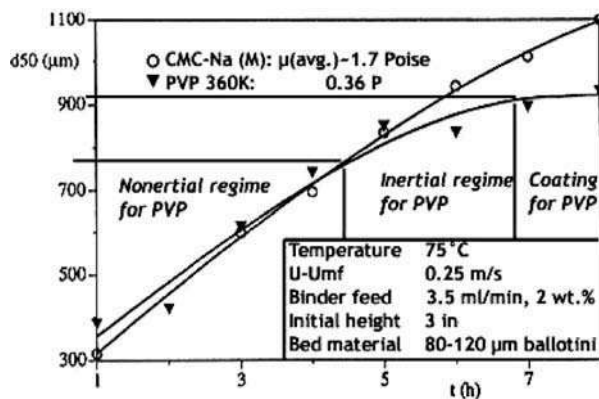


FIG. 21-120 Median granule diameter for fluid-bed granulation of ballottini with binders of different viscosity indicating regimes of growth [Ennis et al., *Powder Technol.*, **65**, 257 (1991)].

Example 4: Extent of Noninertial Growth Growth by coalescence in granulation processes may be modeled by the population balance. (See "Modeling and Simulation of Granulation Processes" subsection.) It is necessary to determine both the mechanism and kernels—or rate constants—which describe growth. For fine powders within the noninertial regime of growth, all collisions result in successful coalescence *provided* binder is present. Coalescence occurs via a random, size-independent kernel, which is only a function of liquid loading y , since all collisions are successful in the presence of binder, or

$$\beta(u, v) = k = k^* f(y) \quad (21-115)$$

The dependence of growth on liquid loading $f(y)$ strongly depends on wetting properties, spray distribution, and mixing. For random growth and in the presence of sufficient binding fluid, it may be rigorously proved that the average granule size increases exponentially with time, or

$$d = d_0 e^{kt} \quad (21-116)$$

This exponential increase in size with time is confirmed experimentally in Fig. 21-121a, where increases in liquid loading $f(y)$ increase growth rate. (Note granule saturation S is connected to liquid loading y and porosity.) Based on the regime analysis above, growth will continue in a process while the conditions of Eq. (21-114) are met; i.e., dissipation exceeds collisional kinetic energy, or put another way, granules do not have sufficient momentum based on their current size to exceed the energy dissipated during the collision. Examples of these growth limits are seen in the drum granulation work of Kapur (loc. cit.) in Fig. 21-121a, as well as fluid beds (Fig. 21-120) and mixers (Fig. 21-122). It may be shown that the **maximum extent** of granulation $(kt)_{\max}$ occurring within the noninertial regime is given by

$$(kt)_{\max} = \ln\left(\frac{d_{\max}}{d_0}\right) = 6 \ln(St^*/St_0) f(y) \propto \ln \frac{\mu}{\rho u_c d_0} \quad (21-117)$$

where St_0 is the Stokes number based on initial nuclei diameter d_0 [Adetayo et al., *Powder Technol.*, **82**, 37 (1995)]. Extent $(kt)_{\max}$ is taken as the logarithm of the growth limit in the first random stage of growth, or $d_{\max}$. The growth limits $d_{\max}$ of Fig. 21-121a are replotted as extents in Fig. 21-121b. Here, $(kt)_{\max}$ is observed to depend linearly on liquid loading y . Therefore, the maximum granule size depends exponentially on liquid loading, as observed experimentally (Fig. 21-112).

From Eq. (21-117), it is possible to scale or normalize a variety of drum granulation data to a common drum speed and binder viscosity. Maximum granule size $d_{\max}$ and extent $(kt)_{\max}$ depend linearly and logarithmically, respectively, on binder viscosity and the inverse of agitation velocity. This is illustrated based on the data of Fig. 21-121b, where the slope of each formulation line depends linearly on binder viscosity. Figure 21-121c provides the normalization of extent $(kt)_{\max}$ for the drum granulation of limestone and fertilizers, correcting for differences in binder viscosity, granule density, and drum rotation speed, with the data collapsing onto a common line.

High Agitation Intensity Growth For high-agitation processes involving **high-shear mixing** or for readily deformable formulations, granule deformability, plastic deformation, and granule consolidation can no longer be neglected as they occur at the same rate as granule growth. Typical growth profiles for high-shear mixers are illustrated in Fig. 21-122. Two stages of growth are evident, which reveal the possible effects of binder viscosity and impeller speed, as shown for data replotted vs. impeller speed in Fig. 21-123. The **initial, nonequilibrium stage** of growth is controlled by granule deformability and is of greatest practical significance in manufacturing for high-shear mixers for deformable formulations. Increases in St due to lower viscosity or higher impeller speed increase the rate of growth, as shown in Fig. 21-122, since the system becomes more deformable and easier to **knead** into larger granule structures. These effects are contrary to what is predicted

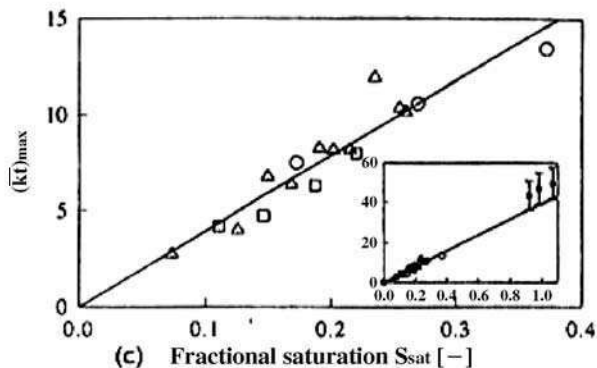
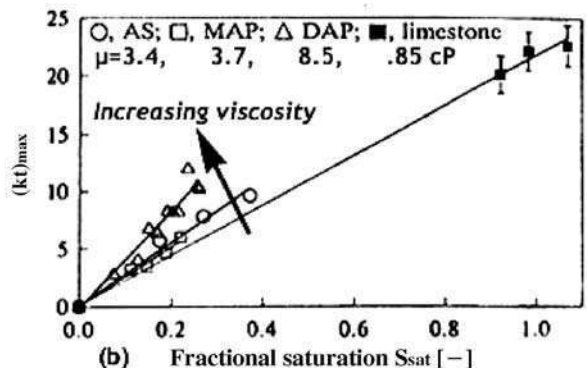
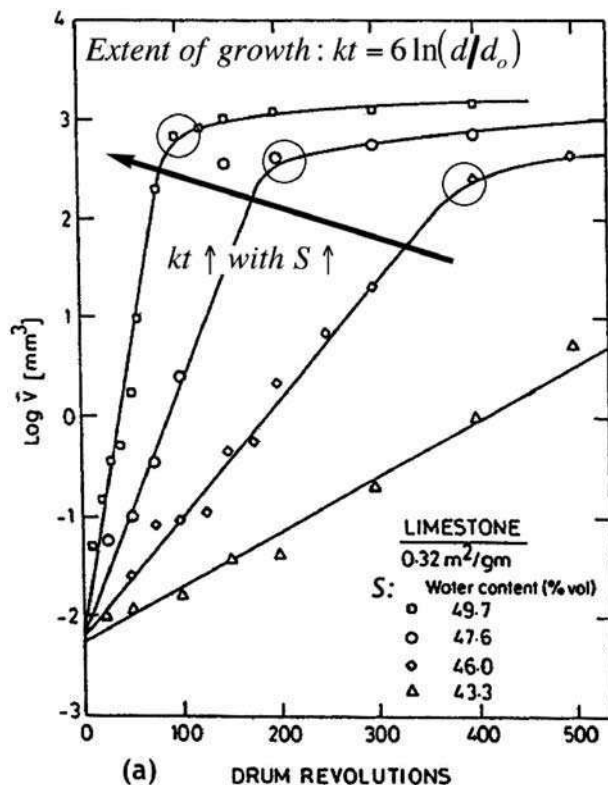


FIG. 21-121 (a) Exponential growth in drum granulation reaching a growth limit $d_{\max}$ or maximum extent of growth $(kt)_{\max}$, which are functions of moisture saturation (Kapur, loc. cit.). (b) Maximum extent of noninertial growth $(kt)_{\max}$ as a linear function of saturation of the powder feed and binder viscosity. (c) Maximum extent normalized for differences in binder viscosity, drum speed, and granule density by Stokes number. [Adetayo et al., *Powder Technol.*, **82**, 37 (1995).]

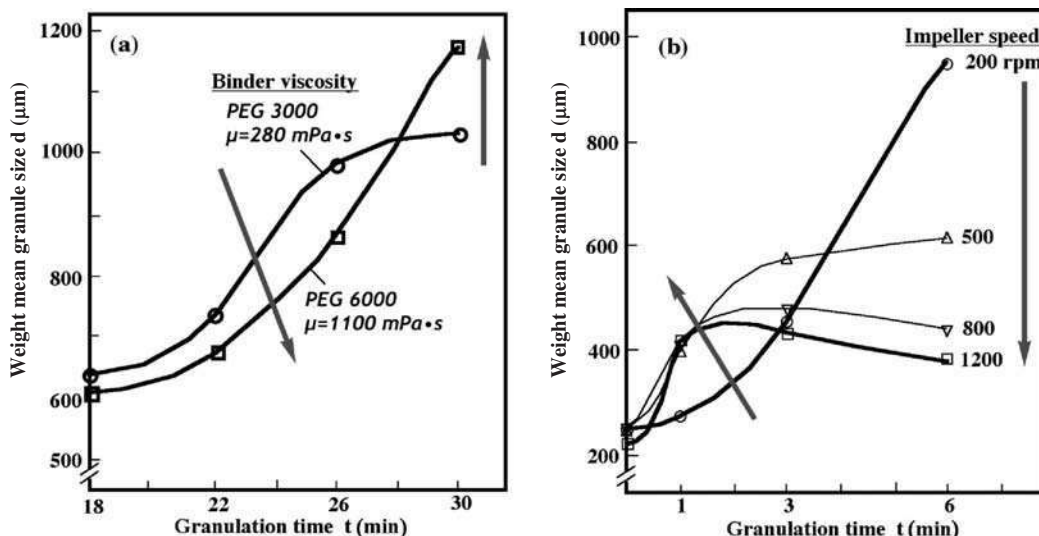


FIG. 21-122 Granule diameter as a function of time for high-shear mixer granulation, illustrating the influence of deformability on growth behavior. Directions of increasing viscosity and impeller speed are indicated by arrows. (a) A 10-L vertical high-shear melt granulation of lactose with liquid loading of 15 wt % binder and impeller speed of 1400 rpm for two different viscosity grades of polyethylene glycol binders. [Schaefer et al., *Drug Dev. & Ind. Pharm.*, **16**(8), 1249 (1990), with permission.] (b) A 10-L vertical high-shear mixer granulation of dicalcium phosphate with 15 wt % binder solution of PVP/PVA Kollidon® VA64, liquid loading of 16.8 wt %, and chopper speed of 1000 rpm for varying impeller speed. [Schaefer et al., *Pharm. Ind.*, **52**(9), 1147 (1990), with permission.]

from the Stokes analysis based on rigid, low deformable granules [Eq. (21-114)], where high viscosity and low velocity increase the growth limit. In this nonequilibrium deformable stage, high viscosity and low velocity give less growth due to less kneading action.

Growth continues until disruptive and growth forces are balanced in the process, similar to a coating stage of growth. This last **equilibrium stage** of growth represents a balance between dissipation and

collisional kinetic energy, and so increases in St_t decrease the final granule size, as expected from the Stokes analysis [Eq. (21-114)]. Note that the equilibrium granule diameter decreases with the inverse square root of the impeller speed, as it should based on $St = St^*$, with $u_o = d \cdot (du/dx) = \omega d$.

The Stokes analysis is used to determine the effect of operating variables and binder viscosity on *equilibrium* growth, where disruptive and growth forces are balanced. In the early stages of growth for high-shear mixers, the Stokes analysis in its present form is inapplicable. Freshly formed, uncompacted granules are easily deformed, and as growth proceeds and consolidation of granules occur, they will surface-harden and become more resistant to deformation. This increases the importance of the elasticity of the granule assembly. Therefore, in later stages of growth, older granules approach the ideal Stokes model of rigid collisions. For these reasons, the Stokes approach has had reasonable success in providing an overall framework with which to compare a wide variety of granulating materials (Ennis, *Powder Technol.*, 1996). In addition, the Stokes number controls in part the degree of deformation occurring during a collision since it represents the importance of collision kinetic energy in relation to viscous dissipation, although the exact dependence of deformation on St is presently unknown.

The Stokes coalescence criteria of Eq. (21-114) must be generalized to account for substantial plastic deformation to treat the initial nonequilibrium stages of growth in high-agitation systems such as high-shear mixers. In this case, granule growth and deformation are controlled by a generalization of St_t , or a deformation Stokes number St_{def} , as originally defined by Tardos et al. [Tardos and Khan, AIChE Annual Meeting, Miami, 1995; Tardos et al., *Powder Technol.*, **94**, 245 (1997)]:

$$St_{def} = \frac{\rho u_o^2}{\sigma_y} \text{ (impact)} \quad \text{or} \quad \frac{\rho (du/dx)^2 d^2}{\sigma_y} \text{ (shear)} \quad (21-118)$$

Viscosity has been replaced by a generalized form of plastic deformation controlled by the yield stress σ_y , which may be determined by compression experiments (e.g., Fig. 21-117). As shown previously, yield stress is related to deformability of the wet mass and is a function of shear rate, binder viscosity, and surface tension (captured by a bulk

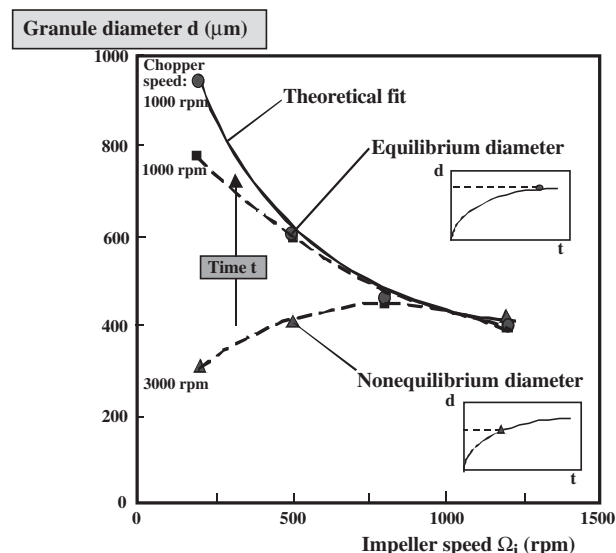


FIG. 21-123 Granule diameter as a function of impeller speed for both initial nonequilibrium and final equilibrium growth limits for high-shear mixer granulation, data from Fig. 21-124. [Ennis, *Powder Technol.*, **88**, 203 (1996), with permission.]

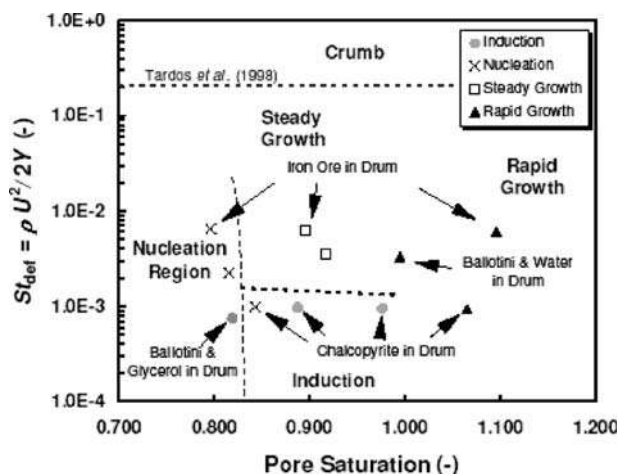


FIG. 21-124 Regime map of growth mechanisms, based on moisture level and deformability of formulations [Iveson *et al.*, *Powder Technol.*, **117**, 83 (2001)].

capillary number), as well as primary particle size, friction, saturation, and voidage as previously presented [cf. Eq. (21-112)].

Critical conditions required for granule coalescence may be defined in terms of the viscous and deformation Stokes numbers, or St_v and St_{def} , respectively. These represent a complex generalization of the critical Stokes number given by Eq. (21-114) and are discussed in detail elsewhere [Litster and Ennis, *The Science and Engineering of Granulation Processes*, Kluwer Academic, 2004; Iveson *et al.*, *Powder Technol.*, **88**, 15 (1996)].

An overall view of the impact of deformability of growth behavior may be gained from Fig. 21-124, where types of granule growth are plotted vs. deformability in a regime map, and yield stress has been measured by compression experiments [Iveson *et al.*, *Powder Technol.*, **117**, 83 (2001)]. Growth mechanism depends on the competing effects of high shear promoting growth by deformation, on the one hand, and the breakup of granules giving a growth limit, on the otherhand. For high velocities that exceed the dissipation energy [Eq. (21-114)] or significantly exceed the dynamic strength of the granule, growth is not possible by deformation due to high shear or high St_{def} , and the material remains in a crumb state. For low pore saturation and lower St_{def} , growth is possible by initial wetting and nucleation, with surrounding powder remaining ungranulated and the formed nuclei surviving breakup forces. At intermediate levels of moisture, growth occurs at a steady rate for moderate deformability, where larger granules grow preferentially or by crushing and layering [Newitt and Conway Jones, *loc. cit.*; Capes and Danckwerts, *Trans. Inst. Chem. Eng.*, **43**, T116 (1965); Linkson *et al.*, *Trans. Inst. Chem. Eng.*, **51**, 251 (1973)]. Linear or power law behavior as observed is shown by Kapur (*loc. cit.*), where for preferential growth

$$d^m - d_0^m = m(kt) \quad (21-119)$$

For nondeformable systems, random exponential growth is expected for sufficient saturation (Fig. 21-121). However, for lower levels of saturation, a delay with little or no growth may be observed. This delay, or induction time, is related to the time required to work moisture to the surface to promote growth, and in some cases, the growth can be rapid and unstable, which also occurs in all cases of high moisture. Pore saturation may be calculated by

$$S = \frac{w\rho_s(1 - \epsilon_g)}{\rho\epsilon_g} \quad (21-120)$$

where w is the liquid-solid mass ratio. The current regime map, while providing a starting point, requires considerable development.

Overall growth depends on the mechanics of local growth, as well as the overall mixing pattern and local/overall moisture distribution. Levels of shear are poorly understood in high-shear processes. In addition, growth by both deformation and the rigid growth model is possible. Lastly, deformability is intimately linked to both voidage and moisture. They are not a constant for a formation, but depend on time and the growth process itself through the interplay of growth and consolidation.

Determination of St^* The extent of growth is controlled by some **limit of granule size**, reflected either by the critical Stokes number St^* or by the critical limit of granule size D_c . There are three possible methods to determine this critical limit. The first involves measuring the critical rotation speed for the survival of a series of liquid binder drops during drum granulation (Ennis, *On the Mechanics of Granulation*, Ph.D. thesis, 1990, The City College of the City University of New York, University Microfilms International, 1991). A second refined version involves measuring the survival of granules in a couette-fluidized shear device (Tardos and Khan, *loc. cit.*; Tardos *et al.*, *loc. cit.*). Both the onset of granule deformation and complete granule rupture are determined from the dependence of granule shape and the number of surviving granules, respectively, on shear rate (Fig. 21-125). The critical shear rate describing complete granule rupture defines St^* , whereas the onset of deformation and the beginning of granule breakdown define an additional critical value $St_{def} = St^*$. The third approach is to measure the deviation in the growth rate curve from random exponential growth (Adetayo and Ennis, *AIChE J.*, 1996). The deviation from random growth indicates a value of w^* , or the **critical granule diameter** at which noninertial growth ends (Fig. 21-126). This value is related to D_c . (See the "Modeling and Simulation" subsection for further discussion.) The last approach is through the direct measurement of the yield stress through compression experiments.

Example 5: High-Shear Mixer Growth An important case study for high-deformability growth was conducted by Holm *et al.* [Parts V and VI, *Powder Technol.*, **43**, 213 (1985)] for high-shear mixer granulation. Lactose, dicalcium phosphate, and dicalcium phosphate/starch mixtures (15 and 45 percent starch) were granulated in a Fielder PMAT 25 VG laboratory-scale mixer. Granule size, porosity, power level, temperature rise, and fines disappearance were monitored during liquid addition and wet massing phases. Impeller and chopper speeds were kept constant at 250 and 3500 rpm, respectively, with 7.0 to 7.5 kg of starting material. Liquid flow rates and amount of binder added were varied according to the formulation. Figure 21-127 illustrates typical power profiles during granulation, whereas Fig. 21-128 illustrates the resulting granule size and voidage (or porosity). Note that wet massing time (as opposed to total process time) is defined as the amount of time following the end of liquid addition, and the beginning of massing time is indicated in Fig. 21-127.

Clear connections may be drawn between granule growth, consolidation, power consumption, and granule deformability (Figs. 21-127 and 21-128). For the case of lactose, there is no further rise in power following the end of water addition (beginning of wet massing), and this corresponds to no further changes in granule size and porosity. In contrast, dicalcium phosphate continues to grow through the wet massing stage, with corresponding continual increases in granule size and porosity. Lastly, the starch formulations are noted to have power increase for approximately 2 min into the wet massing stage, corresponding to 2 min of growth; however, growth ceased when power consumption leveled off. Therefore, power clearly tracks growth and consolidation behavior.

Further results connecting power and growth to compact deformability are provided in Holm (*loc. cit.*). The deformability of lactose compacts, as a function of saturation and porosity, is shown to increase with moisture in a stable fashion. In other words, the lactose formulation is readily deformable, and growth begins immediately with water addition. This steady growth is consistent with values observed in drum granulation. Growth rates and power rise do not lag behind spray addition, and growth ceases with the end of spraying. Dicalcium phosphate compacts, on the other hand, remain undeformable until a critical moisture is reached, after which they become extremely deformable and plastic. This unstable behavior is reflected by an inductive lag in growth and power after the end of spray addition (consistent with data for drum granulation), ending by unstable growth and bowl sticking as moisture is finally worked to the surface.

In closing, a comment should be made with regard to using power for control and scale-up. While it is true the power is reflective of the growth process, it is a dependent variable in many respects. *Different* lots of a *set* formulation, e.g., may have different yield properties and deformability, and a different dependence on moisture. This may be due to minute particle property changes

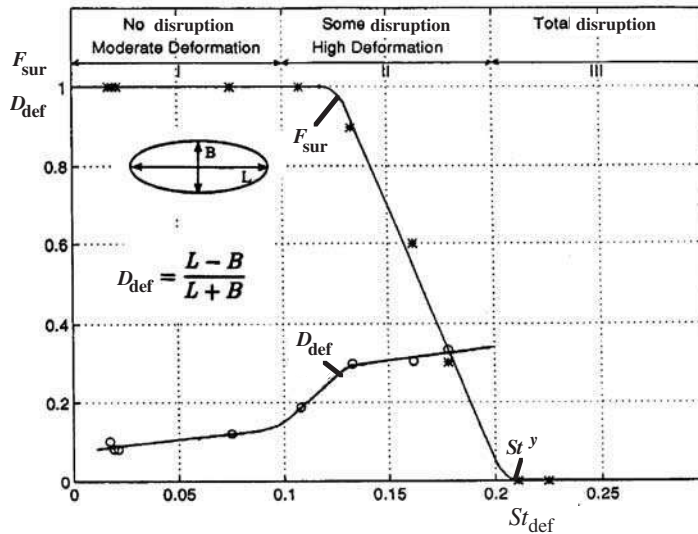


FIG. 21-125 Determination of the onset of granule deformation and complete granule breakdown with the fluidized-couette constant-shear device. St_{def} is a deformation yield, F_{sur} is the fraction of surviving granules, and D_{def} is the average degree of granule deformation; $St_{def} = St^y$ and $F_{sur} = 0$ complete granule breakdown. [Tardos and Khan, *AIChE Annual Meeting*, 1995; Tardos et al., *Powder Technol.*, **94**, 245 (1997).]

controlling the rate processes. Therefore, there is not a unique relationship between power and growth. However, power measurements might be useful to indicate a shift in formation properties. Lastly, **specific power** should be used for scale-up, where power is normalized by the *active* portion of the powder bed, which could change over wet massing time. The impact of scale-up on mixing and distribution of power in a wet mass, however, is only partly understood at this point.

Granule Consolidation and Densification Consolidation or densification of granules determines **granule porosity** and hence **granule density**. Granules may consolidate over extended times and achieve high densities if there is no simultaneous drying to stop the consolidation process. The extent and rate of consolidation are determined by the balance between the collision energy and the granule resistance to deformation, as described by the Stokes numbers previously defined.

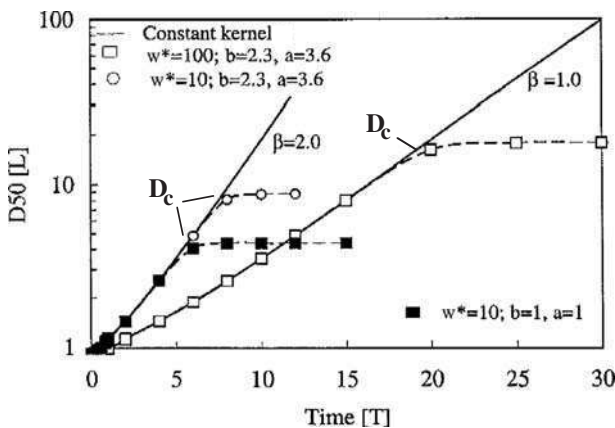


FIG. 21-126 Determination of critical granule diameter, or growth limit, from the evolution of the granule-size distribution (Adetayo and Ennis, *AIChE J.*, 1996).

The voidage ϵ_g may be shown to depend on time as follows:

$$\frac{\epsilon_g - \epsilon_{\min}}{\epsilon_o - \epsilon_{\min}} = \exp(-\beta t) \quad \text{where} \quad \beta = f_n(S, St, St_{def}) \quad (21-121)$$

Here S is granule saturation related to liquid loading; ϵ_o and $\epsilon_{\min}$ are the beginning and final (minimum) granule porosity, respectively [Iveson et al., *Powder Technol.*, **88**, 15 (1996)]. The consolidation process and final granule voidage control the granule strength, dissolution behavior, and attrition resistance (cf. Figs. 21-88 to 21-90), in addition to controlling the growth process through its impact on deformability. Granule voidage also impacts bulk density, mass flow rates for feeding, and possible subsequent compact properties such as hardness or compact uniformity.

The effects of binder viscosity and liquid content are complex and interrelated. For low-viscosity binders, consolidation *increases* with

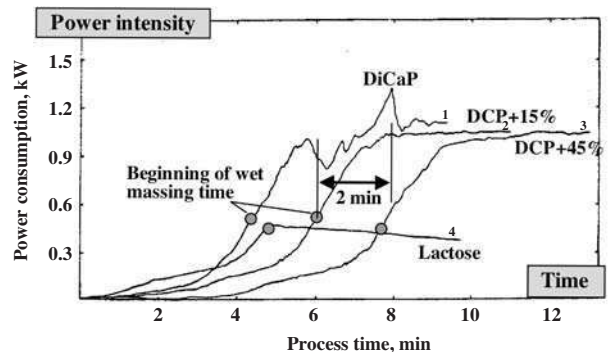


FIG. 21-127 Power consumption for lactose, dicalcium phosphate, and dicalcium phosphate/starch mixtures (15 and 45 percent starch) granulated in a Fielder PMAT 25 VG. Impeller speed is 250 rpm, chopper speed 3000 rpm. [Holm et al., Parts V and VI, *Powder Technol.*, **43**, 213 (1985); Kristensen et al., *Acta Pharm. Sci.*, **25**, 187 (1988).]

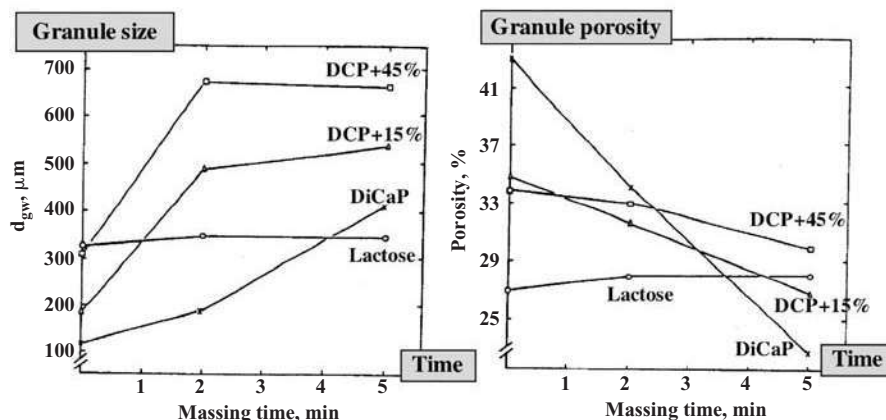


FIG. 21-128 Granule size and porosity vs. wet massing time for lactose, dicalcium phosphate, and dicalcium phosphate/starch mixtures (15 and 45 percent starch) granulated in a Fielder PMAT 25 VG. Impeller speed is 250 rpm, chopper speed 3000 rpm. [Holm *et al.*, *Parts V and VI, Powder Technol.*, **43**, 213 (1985); Kristensen *et al.*, *Acta Pharm. Sci.*, **25**, 187 (1988).]

liquid content, as shown in Fig. 21-129. This is the predominant effect for the majority of granulation systems, with liquid content related to peak bed moisture on average. Increased drop size and spray flux are also known to increase consolidation. Drying affects peak bed moisture and consolidation as well by varying both moisture level and binder viscosity; generally increased drying slows the consolidation process. For very viscous binders, consolidation *decreases* with increasing liquid content (Fig. 21-130). As a second important effect, decreasing feed particle size decreases the rate of consolidation due to the high specific surface area and low permeability of fine powders, thereby decreasing granule voidage. Lastly, increasing agitation intensity and process residence time increases the degree of consolidation by increasing the energy of collision and compaction time. The exact combined effect of formulation properties is determined by the balance between viscous dissipation and particle frictional losses, and therefore the rate is expected to depend on the viscous and deformation Stokes numbers.

BREAKAGE AND ATTRITION

Dry granule strength impacts three key areas of processing. These include the physical attrition or breakage of granules during the granulation and drying processes, the breakage of granules in subsequent material handling steps such as conveying or feeding, and lastly the deformation and breakdown of granules in compaction processes such as tableting. (Note that breakage also includes breakdown of wet granules or overmassed wet cake in granulation, which is outside the scope of this subsection.) Modern approaches to granule strength rely on **fracture mechanics** (Lawn, *Fracture of Brittle Solids*, 2d ed., Cambridge University Press, 1975). In this context, a granule is viewed as a nonuniform **physical composite** possessing certain macroscopic mechanical properties, such as a generally anisotropic **yield stress**, as well as an inherent **flaw distribution**. Hard materials may fail in tension, with the breaking strength being much less than the inherent tensile strength of bonds because of the existence of flaws. Flaws act to

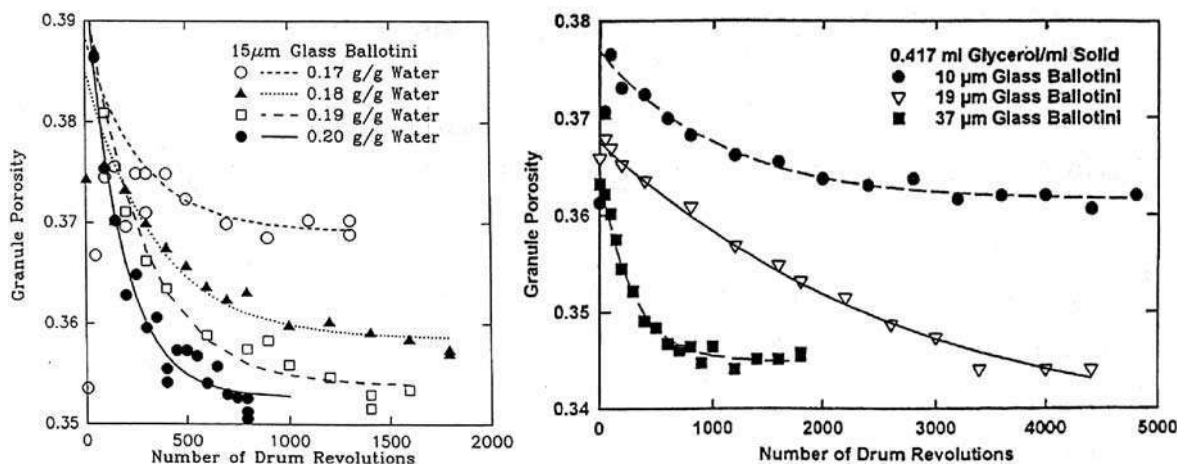


FIG. 21-129 Effect of binder liquid content and primary feed particle size on granule porosity for the drum granulation of glass ballotini. Decreasing granule porosity corresponds to increasing extent of granule consolidation. [Iveson *et al.*, *Powder Technol.*, **88**, 15 (1996).]

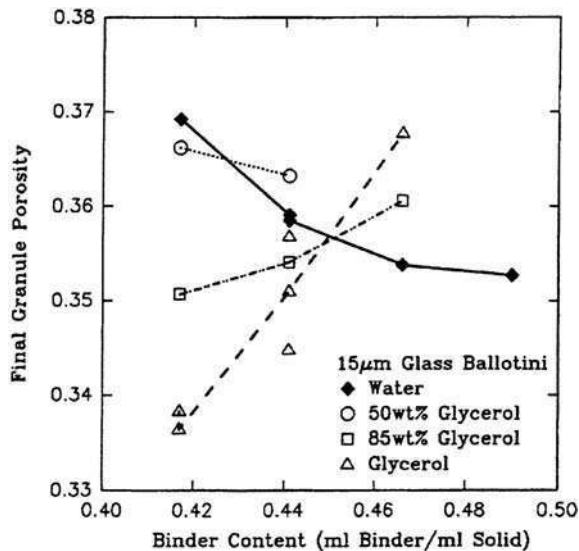


FIG. 21-130 Effect of binder viscosity and liquid content on final granule porosity for the drum granulation of 15- μ m glass ballotini. Decreasing granule porosity corresponds to increasing extent of granule consolidation. (Iveson et al., Powder Technol., 1996.)

concentrate stress, as depicted in Fig. 21-131 for commercial Metamucil tablets. Here, razor scores or notches have been added to the tablets, which were subsequently broken under three-point bend loading described below. In all cases, the tablets break at the razor score—which acts as a sharp flaw to concentrate stress—rather than at the tableted original indentation notch.

Bulk breakage tests of granule strength measure both the inherent bond strength of the granule and its flaw distribution [Ennis, loc. cit., 1991; Ennis and Sunshine, *Tribology Int.*, **26**, 319 (1993)]. Figure 21-89 previously illustrated granule attrition results for a variety of formulations. Attrition clearly increases with increasing voidage; note that this voidage is a function of granule consolidation discussed previously. Different formulations fall on different curves, due to inherently differing interparticle bond strengths. It is often important to separate the impact of bond strength vs. voidage on attrition and granule strength. Processing influences flaw distribution and granule voidage, whereas inherent bond strength is controlled by formulation properties.

The **mechanism of granule breakage** (Fig. 21-91) is a strong function of the **materials properties** of the granule itself as well as the type of loading imposed by the test conditions [Bemros and Bridgwater, *Powder Technol.*, **49**, 97 (1987)]. Ranking of product breakage

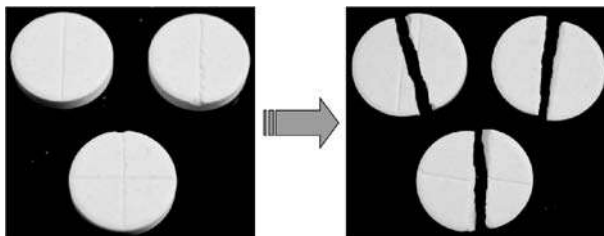


FIG. 21-131 Breakage of Metamucil tablets under three-point loading with razor scoring. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E-Z-G Associates. All rights reserved.)

resistance by ad hoc tests may be test-specific, and in the worst case differs from actual process conditions. Instead, material properties should be measured by standardized mechanical property tests which minimize the effect of flaws and loading conditions under well-defined geometries of internal stress, as described below.

Fracture Properties **Fracture toughness** K_{Ic} defines the stress distribution in the body (Fig. 21-132) just before fracture and is given by

$$K_{Ic} = Y_{cf} \sigma_f \sqrt{\pi c} \quad (21-122)$$

where σ_f is the applied fracture stress, c is the length of the crack in the body, and Y_{cf} is a calibration factor introduced to account for different body geometries (Lawn, loc. cit.). The elastic stress is increased dramatically as the crack tip is approached. In practice, however, the elastic stress cannot exceed the **yield stress** of the material, implying a region of local yielding at the crack tip.

To nevertheless apply the simple framework of linear elastic fracture mechanics, Irwin [*J. Applied Mech.*, **24**, 361 (1957)] proposed that this **process zone size** r_p be treated as an **effective** increase in crack length δ_c . Fracture toughness is then given by

$$K_{Ic} = Y_{cf} \sigma_f \sqrt{\pi(c + \delta_c)} \quad \text{with} \quad \delta_c \sim r_p \quad (21-123)$$

The process zone is a measure of the yield stress or **plasticity** of the material in comparison to its brittleness. Yielding within the process zone may take place either plastically or by diffuse microcracking, depending on the brittleness of the material. For plastic yielding, r_p is also referred to as the **plastic zone size**.

The **critical strain energy release rate** G_c is the energy equivalent to fracture toughness, first proposed by Griffith [*Phil. Trans. Royal Soc.*, **A221**, 163 (1920)]. With an elastic modulus of E , toughness and release rate are related by

$$G_c = K_{Ic}^2 / E \quad (21-124)$$

Fracture Measurements To ascertain fracture properties in any reproducible fashion, very specific test geometries must be used since it is necessary to know the stress distribution at predefined, induced cracks of known length. Three traditional methods are (1) the **three-point bend test**, (2) **indentation fracture testing**, and (3) **hertzian contact compression** between two spheres of the material (see "Fracture" under "Size Reduction"). Figure 21-133 illustrates a typical geometry and force response for the case of a three-point bend test. By breaking a series of dried formulation bars under three-point bend loading of varying crack length, the fracture toughness is determined from the variance of fracture stress on crack length, as given by Eq. (21-123). Here, δ_c is initially taken as zero and determined in addition to toughness (Ennis and Sunshine, loc. cit.).

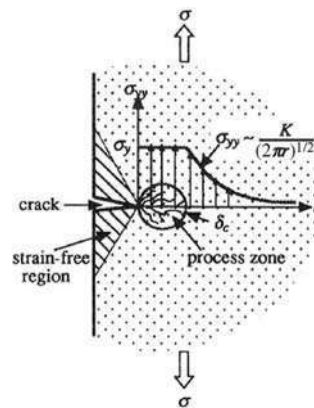


FIG. 21-132 Fracture of a brittle material by crack propagation. [Ennis and Sunshine, *Tribology Int.*, **26**, 319 (1993), with permission.]

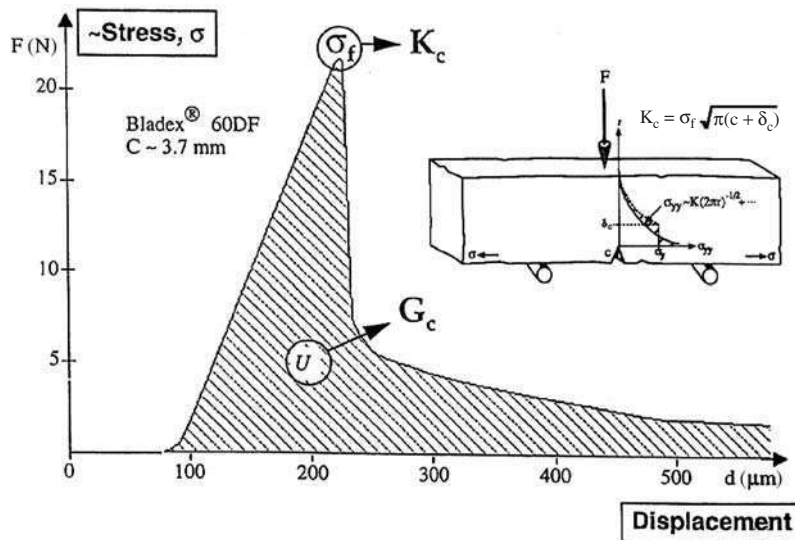


FIG. 21-133 Typical force-displacement curve for three-point bend semistable failure. [Ennis and Sunshine, Tribology Int., 26, 319 (1993), with permission.]

In the case of indentation fracture (Fig. 21-134), one determines the **hardness** H from the area of the residual plastic impression and the fracture toughness from the lengths of cracks propagating from the indent as a function of indentation load F (Johnsson and Ennis, *Proc. First International Particle Technology Forum*, vol. 2, AICHE, Denver, 1994, p. 178). Hardness is a measure of the yield strength of the material. Toughness and hardness in the case of indentation are given by

$$K_c = \beta \sqrt{\frac{E}{H}} \frac{F}{c^{3/2}} \quad \text{and} \quad H \sim \frac{F}{A} \quad (21-125)$$

Table 21-13 compares typical fracture properties of agglomerated materials. Fracture toughness K_c is seen to range from 0.01 to 0.06 $\text{Mpa} \cdot \text{m}^{1/2}$, less than that typical for polymers and ceramics, presumably due to the high agglomerate voidage. Critical strain energy release rates G_c from 1 to 200 J/m^2 , are typical for ceramics but less than that for polymers. Process zone sizes δ_c are seen to be large and of the order of 0.1 to 1 mm, values typical for polymers. Ceramics, however typically have process zone sizes less than 1 μm . Critical displacements

required for fracture may be estimated by the ratio G_c/E , which is an indication of the **brittleness** of the material. This value was of the order of 10^{-7} to 10^{-8} mm for polymer-glass agglomerates, similar to polymers, and of the order of 10^{-9} mm for herbicide bars, similar to ceramics. In summary, granulated materials behave similar to brittle ceramics which have small critical displacements and yield strains but also similar to ductile polymers which have large process or plastic zone sizes.

Mechanisms of Attrition and Breakage The process zone plays a large role in determining the mechanism of granule breakage (Fig. 21-91). (Ennis and Sunshine, loc. cit.). Agglomerates with process zones small in comparison to granule size break by a brittle fracture mechanism into smaller fragments, or **fragmentation** or **fracture**. However, for agglomerates with process zones of the order of their size, there is insufficient volume of agglomerate to concentrate enough elastic energy to propagate gross fracture during a collision. The mechanism of breakage for these materials is one of **wear**, **erosion**, or **attrition** brought about by diffuse microcracking. In the limit of very weak bonds, agglomerates may also **shatter** into small fragments or primary particles.

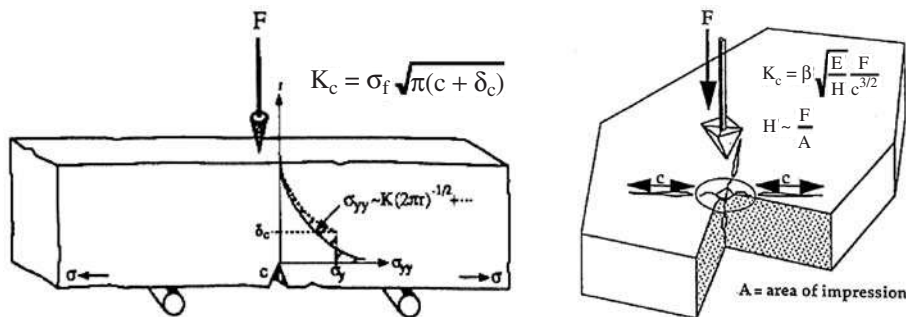


FIG. 21-134 Three-point bend and indentation testing for fracture properties. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

TABLE 21-13 Fracture Properties of Agglomerated Materials

Material	Id	K_c (MPa·m ^{1/2})	G_c (J/m ²)	δ_c (μm)	E (MPa)	G_c/E (m)
Bladex 60 [®]	B60	0.070	3.0	340	567	5.29e-09
Bladex 90 [®]	B90	0.014	0.96	82.7	191	5.00e-09
Glean [®]	G	0.035	2.9	787	261	1.10e-08
Glean [®] Aged [†]	GA	0.045	3.2	3510	465	6.98e-09
CMC-Na (M) [†]	CMC	0.157	117.0	641	266	4.39e-07
Klucel GF [†]	KGF	0.106	59.6	703	441	1.35e-07
PVP 360K [†]	PVP	0.585	199.0	1450	1201	1.66e-07
CMC 2% 1kN [†]	C2/1	0.097	16.8	1360	410	4.10e-08
CMC 2% 5kN [†]	C2/5	0.087	21.1	1260	399	5.28e-08
CMC 5% 1kN [†]	C5/1	0.068	15.9	231	317	5.02e-08

[®]Dupont corn herbicides.

[†]50-μm glass beads with polymer binder.

Ennis and Sunshine, *Tribology Int.*, **26**, 319 (1993).

Each mechanism of breakage implies a different functional dependence of breakage rate on material properties. Granules generally have been found to have a large process zone (Table 21-13), which suggests granule wear as a dominant mechanism or breakage or attrition. For the case of **abrasive wear** of ceramics due to surface scratching by loaded indentors, Evans & Wilshaw [*Acta Metallurgica*, **24**, 939 (1976)] determined a volumetric wear rate V of

$$V = \frac{d_i^{1/2}}{A^{1/4} K_c^{3/4} H^{1/2}} P^{5/4} l \quad (21-126)$$

where d_i is indenter diameter, P is applied load, l is wear displacement of the indenter, and A is apparent area of contact of the indenter with the surface. Therefore, wear rate depends inversely on fracture toughness. For the case of fragmentation, Yuregir et al. [*Chem. Eng. Sci.*, **42**, 843 (1987)] have shown that the fragmentation rate of organic and inorganic crystals is given by

$$V \sim \frac{H}{K_c^2} \rho u^2 a \quad (21-127)$$

where a is crystal length, ρ is crystal density, and u is impact velocity. Note that hardness plays an opposite role for fragmentation than for wear, since it acts to concentrate stress for fracture. Fragmentation rate is a stronger function of toughness as well.

Drawing on analogies with this work, the breakage rates by wear B_w and fragmentation B_f for the case of fluid-bed granulation and drying

processes should be of the forms

$$B_w = \frac{d_0^{1/2}}{K_c^{3/4} H^{1/2}} h_b^{5/4} (U - U_{mf}) \quad (21-128)$$

$$B_f \sim \frac{H}{K_c^2} \rho (U - U_{mf})^2 a \quad (21-129)$$

where d is granule diameter, d_0 is primary particle diameter, $U - U_{mf}$ is fluid-bed excess gas velocity, and h_b is bed height. Figure 21-135 illustrates the dependence of erosion rate on material properties for bars and fluid-bed granules undergoing a wear mechanism of breakage, as governed by Eqs. (21-126 and 21-128).

POWDER COMPACTION

Compressive or **compaction** techniques of agglomeration encompass a variety of unit operations with varying degrees of confinement (Fig. 21-136), ranging from completely confined as in the case of tableting to unconfined as in the case of roll pressing. Regardless of the unit of operation, the ability of powders to freely flow, easily compact, forming permanent interparticle bonding, and maintain strength during stress unloading determines the success of compaction. As opposed to the kinetic rate processes of granulation, compaction is a forming process consisting of a variety of microlevel powder processes (Fig. 21-86) strongly influenced by mechanical properties of the feed. These key areas are now discussed.

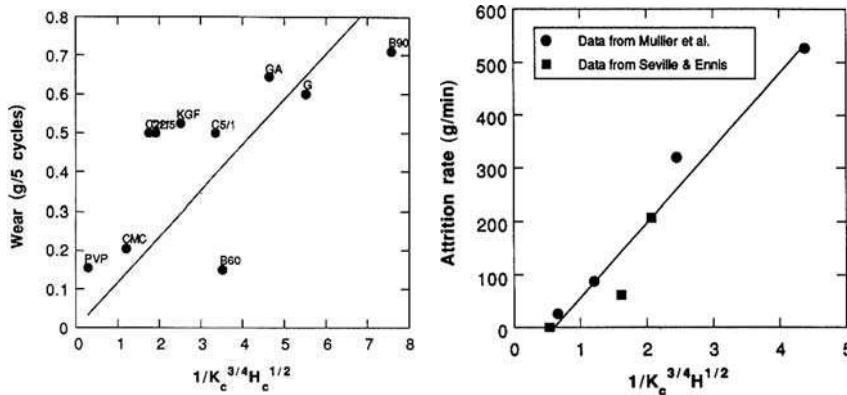


FIG. 21-135 Bar wear rate and fluid-bed erosion rate as a function of granule material properties. K_c is fracture toughness and H is hardness as measured by three-point bend tests. [Ennis and Sunshine, *Tribology Int.*, **26**, 319 (1993), with permission.]

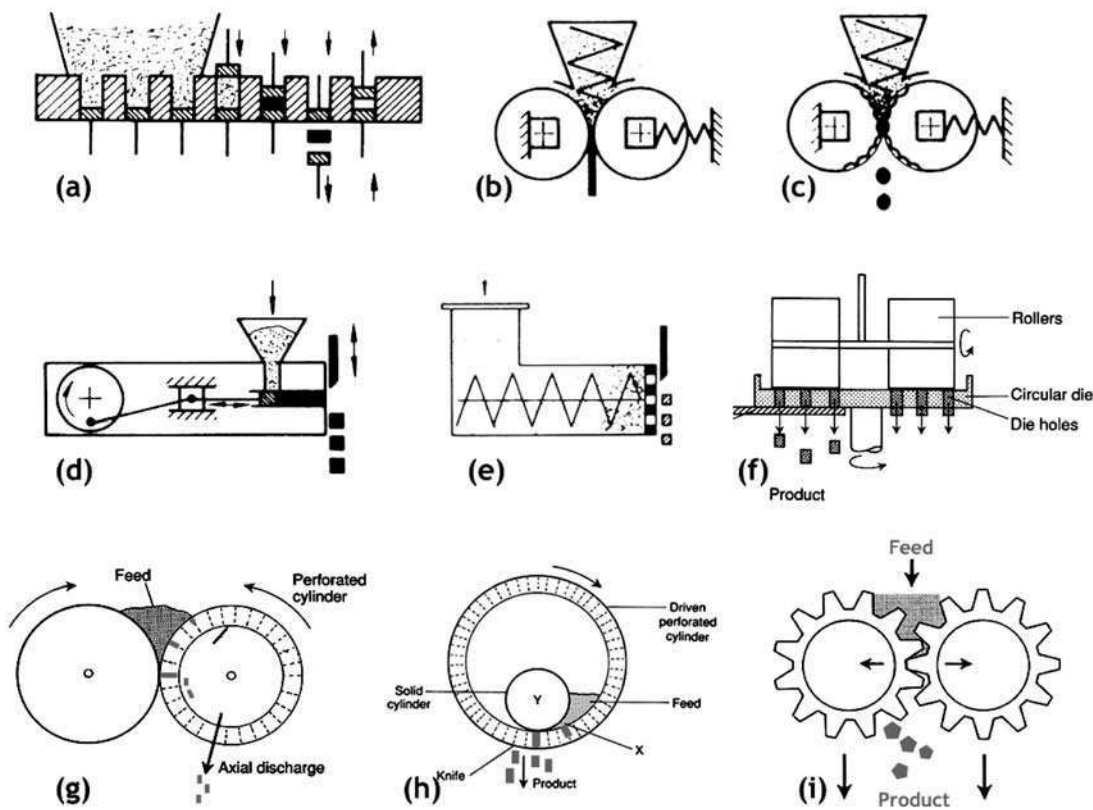


FIG. 21-136 Examples of compressive agglomeration, or compaction, processes. Dry compaction: (a) tableting, (b) roll pressing, (c) briquetting, (d) ram extrusion. Paste extrusion: (e) screw extrusion, (f) table pelletizing, (g) double-roll pelletizing, (h) concentric-roll pelletizing, and (i) tooth extrusion. (After Pietsch, *Size Enlargement by Agglomeration*, Wiley, Chichester, 1992; and Benbow and Bridgwater, *Paste Flow and Extrusion*, Oxford University Press, New York, 1993.)

Powder Feeding Bulk density control of feed materials and reproducible powder feeding are crucial to the smooth operation of compaction techniques. **Flowability data** developed from bulk shear cell and permeability measurements are invaluable in designing machine hoppers for device filling.

As an example, mass flow rates W_s out of openings of diameter B for coarse materials may be estimated by

$$W_s = 0.785 \rho_p (B - 1.4d_p)^{2.5} \sqrt{\frac{g}{2m \tan \alpha} \left(1 - \frac{ff}{\text{Rel}}\right)} \quad (21-130)$$

Here, ff is the **flow factor** determining the stress at the opening which is a function of wall and powder friction, Rel is a **relative flow index** giving the ratio of opening stress to the powder's **cohesive strength**, α is the hopper angle, d_p is particle diameter, and $m = 1$ or 2 for slot or conical hoppers, respectively (cf. "Solids Handling"). The relative flow index is indirectly proportional to powder strength. Increasing powder strength lowers feed rate both here and in a general sense. Examples would include hopper discharge, flow into dies, or screw feeding of roll presses. Powder strength generally increases with decreasing particle size, increasing size distribution, decreasing particle hardness, increasing surface energy and increasing shape factor.

Lubricants or glidants are also added in small amounts to improve flow properties. **Glidants** such as fumed silica are often added to lower powder strength. Also note that in contrast, **lubricants** such as magnesium stearate may actually increase powder strength and adversely lower the flow rate. The primary purpose of

lubricants is to modify die friction as discussed below, rather than alter powder flow rates.

Equally important to powder strength and bulk density of the feed is bulk **gas permeability**. Permeability controls the gas pressure developed within the bulk powder during feeding. Lower powder permeability means greater time is required for gas depressurization after movement of the powder, e.g., filling of a roll press gap or tablet die. Permeability is given by the dependence of gas pressure drop in a powder bed on gas velocity. In addition, as powders discharge from feed hoppers, they undergo expansion during movement, requiring in turn that gas flow into the powder. This concurrent flow of gas impedes the powder discharge, with mass discharge rate as given by, e.g., Eq. (21-130) decreasing with decreasing gas permeability. Permeability decreases with decreasing particle size.

Production rate of compaction processes, and the associated quality issues of pushing production limits, is intimately linking to flow properties and permeability. Poor flow properties associated with large cohesive strength will lower filling of dies and presses. In addition to impeding feed rate, low-permeability powders entrap gas, which later becomes pressurized in compaction, leading to compact flaws during stress removal and compact ejection. Low-permeability powders therefore require larger dwell times to allow escape of entrapped gas, if such gas is not removed prior to filling. Feeding problems are most acute for direct **powder filling** of compression devices, as opposed to **granular feeds**. Although industry- and process-specific, gravimetric feeding is preferred, in which variations in flow rate are used in feedback control to modify screw rate, which helps compensate for variations in feed bulk density and cohesive strength. In addition,

complex-force feeding and vacuum-assisted systems have been developed to aid filling and ensure uniform bulk density; such designs aid immensely in compensating for low feed permeability.

Compact Density Compact strength depends on the number and strength of interparticle bonds [Eq. (21-96)] created during consolidation, and both generally increase with increasing **compact density**. Compact density is in turn a function of the **maximum pressure** achieved during compaction. The **mechanisms of compaction** have been discussed by Cooper and Eaton [J. Am. Ceramic Soc., 45, 97 (1962)] in terms of two largely independent probabilistic processes. The first is the filling of large holes with particles from the original size distribution. The second is the filling of holes smaller than the original particles by **plastic flow** or **fragmentation**. Additional possible mechanisms include the low-pressure elimination of arches and cavities created during die filling due to wall effects, and the final high-pressure consolidation of the particle phase itself. As these mechanisms manifest themselves over different pressure ranges, four stages of compression are generally observed in the **compressibility diagram** when density is measured over a wide pressure range (Fig. 21-137). The slope of the intermediate- and high-pressure regions is defined as $1/\kappa$, where κ is the **compressibility** of the powder. The density at an arbitrary pressure σ is given by a **compaction equation** of the form

$$\rho = \rho_0 \left[\frac{\sigma}{\sigma_0} \right]^{1/\kappa} \quad (21-131)$$

where ρ_0 is the density at an arbitrary pressure or stress σ_0 . Table 21-14 gives a summary of common compaction relations. For a complete review of compaction equations, see Kawakita and Lüdde [Powder Technol., 4, 61 (1970/71)] and Hersey et al. [Proc. First International Conf. of Compaction & Consolidation of Particulate Matter, Brighton, 165 (1972)].

Compact Strength Both **particle size** and **bond strength** control final compact strength for a given compact density or voidage [Eq. (21-96), Fig. 21-93]. Krupp [Adv. Colloidal Interface Sci., 1, 111 (1967)] has shown the adhesive force between two compressed particles varies inversely with **hardness**, and is proportional to the initial compressive force and **surface energy** of the particles. Although surface energy and elastic deformation play a role, increasing **plastic deformation** at particle contacts with decreasing hardness is likely the major mechanism contributing to large **permanent** bond formation and successful compaction in practice. Figure 21-138 illustrates the strength of mineral compacts of varying hardness and size cut. To obtain significant strength, Benbow (Enlargement and Compaction of Particulate Solids, Stanley-Wood (ed.), Butterworths, 1983, p. 169) found that a **critical yield pressure** must be exceeded which was independent of size but found to increase linearly with particle **hardness**. Strength also increases linearly with compaction pressure, with the slope inversely related to particle size. Similar results were obtained by others for ferrous powder, sucrose, sodium chloride, and coal [Hardman and Lilly, Proc. Royal Soc. A., 333, 183 (1973)]. Particle hardness and elasticity may be characterized directly by nanoindentation [Johnson and Ennis,

TABLE 21-14 Common Compaction Relations*

Equation	Authors
$\ln \frac{\rho_i - \rho_c}{\rho_i - \rho_c} = KP_A$	Athy, Shapiro, Heckel, Konopicky, Seelig
$\ln \frac{\rho_c}{\rho_i} \left(\frac{\rho_c - \rho_i}{\rho_i - \rho_c} \right) = KP_A$	Ballhausen
$\ln \frac{\rho_i}{\rho_i} \left(\frac{\rho_i - \rho_i}{\rho_i - \rho_c} \right) = KP_A$	Spencer
$\ln \frac{\rho_c}{\rho_i} = KP_A^a$	Nishihara, Nutting
$\ln \frac{\rho_i - \rho_c}{\rho_i} + K \left(\frac{\rho_c - \rho_i}{\rho_i - \rho_c} \right)^{1/3} = aP_A$	Murray
$\ln \frac{\rho_i}{\rho_c} \left(\frac{\rho_c - \rho_i}{\rho_i - \rho_i} \right) = \ln Ka - (b + c)P_A$	Cooper and Eaton
$\frac{\rho_i}{\rho_c} = 1 - KP_A^a$	Umeya
$\rho_c = KP_A^a$	Jaky
$\rho_c = K(1 - P_A)^a$	Jenike
$\rho_c - \rho_i = KP_A^{1/3}$	Smith
$\rho_c - \rho_i = KP_A^2$	Shaler
$\frac{\rho_c - \rho_i}{\rho_c} = \frac{K \times aP_A}{1 + aP_A}$	Kawakita
$\frac{\rho_i}{\rho_c} \left(\frac{\rho_c - \rho_i}{\rho_i - \rho_i} \right) = \frac{KP_A}{1 + KP_A}$	Aketa
$\frac{1}{\rho_c} = K - a \ln P_A$	Walker, Bal'shin, Williams, Higuchi, Terzaghi
$\rho_c = K + a \ln P_A$	Gurnham
$\frac{1}{\rho_c} = K - a \ln P_A$	Jones
$\frac{1}{\rho_c} = K - a \ln (P_A - b)$	Mogami
$\frac{\rho_c - \rho_i}{\rho_c} = KP_A \rho_i + a \left(\frac{P_A}{P_A + b} \right)$	Tanimoto
$\frac{\rho_c - \rho_i}{\rho_c} = \ln (KP_A + b)$	Rieschel

* ρ_0 , density of powder; ρ_i , initial apparent density of powder; ρ_c , density of powder applied pressure P_A ; K , a , b and c are constants.

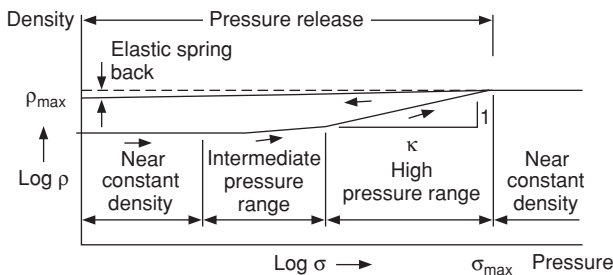


FIG. 21-137 Compressibility diagram of a typical powder illustrating four stages of compaction.

Proc. First International Particle Technology Forum, vol. 2, AICHE, Denver, 1994, p. 178), whereas surface energy can be characterized by inverse gas chromatography and other adsorption techniques. Particle yield pressures and elastic moduli of the powder feeds can also be determined by uniaxial compaction experiments which monitor deformation and pressure throughout the compaction cycles. In addition, rate effects are investigated, as plastic and elastic properties can be rate-dependent for some materials.

Compaction Pressure The minimum compaction pressure is the pressure that induces significant plastic deformation or yielding of the feed particles or granules; i.e., particle/granule strength must be exceeded, such that this pressure exceeds any unloading forces inducing compact failure. Plastic deformation is necessary to produce some measure of final compact strength. While brittle fragmentation may also help increase compact density and points of interparticle bonding as well, in the end some degree of plastic deformation and

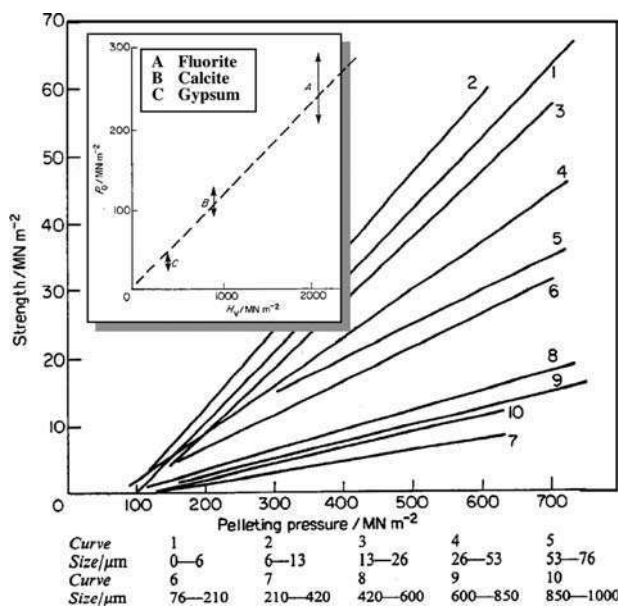


FIG. 21-138 Effect of pelleting pressure on axial crushing strength of compacted calcite particles of different sizes, demonstrating existence of a critical yield pressure. Inset shows the effect of hardness of critical yield pressure. [Benbow, Enlargement and Compaction of Particulate Solids, Stanley-Wood (ed.), Butterworths, 1983, p. 169.]

interlocking is required to achieve some minimum compact strength. Lastly, keep in mind that low powder permeability and entrapped gas may act to later destroy permanent bonding. At the other extreme, compaction pressure is limited since as pressure is raised (e.g., roll load or tableting pressure), elastic effects also increase. During pressure unloading, elastic recovery and gas expansion can induce flaw formation by destroying bonding that was originally created by plastic deformation and adhesion. Therefore, most materials have an allowable compaction pressure range. This range may be narrowed further by other product quality attributes, e.g., desired conveying strength, storage, or redispersion properties. Compacts can be

produced through an automated die compaction simulator, and then the compacts are tested for quality attributes. Such simulators measure all relevant die forces, allowing a connection between powder properties and compaction behavior and product quality. This approach helps identify specific shortcomings in feed properties that require reformulation or improvement. Various quality tests may be employed including compact hardness testing, uniaxial compaction of compacts, shear testing, conveying tests, dust tests, and wetting and dissolution tests.

The development of **flaws** and the loss of interparticle bonding during decompression substantially weaken compacts (see "Breakage and Attrition" subsection). **Delamination** during load removal involves the fracture of the compact into layers, and it is induced by strain recovery in excess of the elastic limit of the material, which cannot be accommodated by plastic flow. Delamination also occurs during compact ejection, where the part of the compact which is clear of the die elastically recovers in the radial direction while the lower part remains confined. This differential strain sets up shear stresses, causing fracture along the top of the compact referred to as **capping**.

Stress Transmission After determination of the necessary compaction pressure range, the compactor must be designed to achieve this desired pressure within the compact geometry for a given loading and dwell time. In this regard, it is key to realize that powders do *not* uniformly transmit stress with fluids (see "Bulk Powder Characterization Methods: Powder Mechanics"). As pressure is applied to a powder in a die or roll press, various zones in the compact are subjected to **differing intensities of pressure** and shear. Typical pressure and density distributions for uniaxial die compaction are shown in Fig. 21-139. High- and low-density annuli are apparent along the die corners, with a dense axial core in the lower part of the compact and a low-density core just below the moving upper punch. These density variations are due to the formation of a dense conical wedge acting along the top punch (A) with a resultant force directed toward the center of the compact (B). The wedge is densified to the greatest extent by the shearing forces developed by the axial motion of the upper punch along the stationary wall whereas the corners along the bottom stationary die are densified the least (C). The lower axial core (B) is densified by the wedge, whereas the upper low-density region (D) is shielded by the wedge from the full axial compressive force. These variations in pressure lead to local variations in compact density and strength as well as differential zones of expansion upon compact unloading, which in turn can lead to flaws in the compact.

From another point of view, the relationships between compaction pressure and compact strength and density discussed [Table 21-14, Eq. (21-96), Fig. 21-93) and the controlling compaction mechanisms

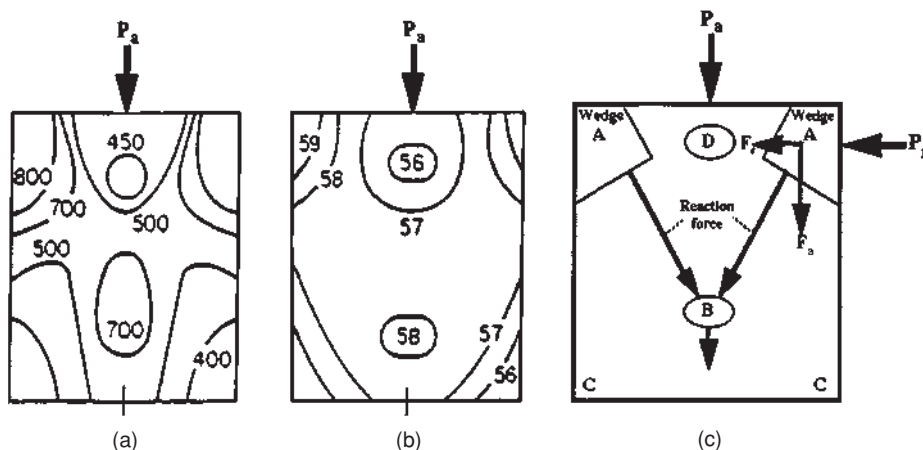


FIG. 21-139 Reaction in compacts of magnesium carbonate when pressed ($P_a = 671 \text{ kg/cm}^2$). (a) Stress contour levels in kilograms per square centimeter. (b) Density contours in percent solids. (c) Reaction force developed at wedge responsible for stress and density patterns. [Train, Trans. Inst. Chem. Eng. (London), **35**, 258 (1957).]

are in reality *local relationships* restricted to a small region of the tablet for the given localized pressure. These local volume regions taken together form a compact. The *uniformity of pressure* across these regions is absolutely critical to successful compaction. Applied compaction pressure in fact must be sufficient to induce deformation and bonding in the regions of lowest pressure and weakest resultant strength. If there are wide variations in local pressure, this will by necessity result in high compaction in other regions with associated large elastic recovery during unloading, possibly inducing compact failure. If compact pressure is uniform, less applied average compaction pressure will be required overall, minimizing flaw development and compact ejection forces.

Compaction stress decreases exponentially with axial distance from the applied pressure [Stribos et al., *Powder Technol.*, **18**, 187, 209 (1977)] due to **frictional properties** of the powder and die wall. As originally demonstrated by Janseen [Zeits. D. Vereins Deutsch Ing., **39**(35), 1045 (1895)], the axial stress experienced within a cylindrical die due to an applied axial load σ_o may be estimated by

$$\sigma_z = \sigma_o e^{-(4\mu_w K_\phi/D)z} \quad (21-132)$$

where D is die diameter, z is axial distance from the applied load, K_ϕ is a lateral stress transmission coefficient (**Janseen coefficient**), and μ_w is the **wall friction coefficient** (see "Bulk Characterization Methods: Powder Mechanics"). The explanation for this drop in compaction pressure may be demonstrated in Fig. 21-140. The given applied load σ_o results in a radial pressure σ_r acting at the wall given by

$$\sigma_r = K_\phi \sigma_o \quad \text{with} \quad K_\phi = (1 - \sin \phi_w)/(1 + \sin \phi_w) \quad (21-133)$$

Radial pressure is therefore controlled by the **effective angle of powder friction** ϕ_w . Typical values range from 40 to 60°, with *increases* in powder friction leading to a decrease in radial pressure for a given loading. Further, note the contrast with typical fluids that develop an isotropic pressure under load. The radial pressure σ_r in turn produces a wall shear stress τ_w which acts to oppose the applied load σ_o , given by:

$$\tau_w = \mu_w \sigma_r \quad \text{with} \quad \mu_w = \tan \phi_w \quad (21-134)$$

and ϕ_w is the **effective angle of wall friction**. Decreasing wall friction lowers the wall shear stress acting to decrease the compaction pressure, for a given radial wall pressure.

The ratio σ_r/σ_o may be taken as a measure of stress uniformity. In practice, it increases toward unity with decreasing aspect ratio of the compact, decreasing diameter, increasing powder friction, and, most important, decreasing wall friction, as controlled by the addition of lubricants. Low stress transmission results in not only poor compact uniformity, but also large residual radial stresses after stress unloading, giving rise to flaws and delamination as well as large die ejection forces.

Equation (21-132) provides only an approximate relation for determining stress distribution during compaction. With modern finite element codes based on soils and plasticity models of powder behavior

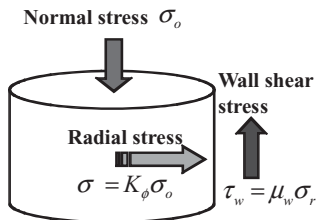


FIG. 21-140 Stresses developed in a column of powder with applied load as a function of powder frictional properties, neglecting gravity. [After Janseen, Zeits. D. Vereins Deutsch Ing., **39**(35), 1045 (1895)].

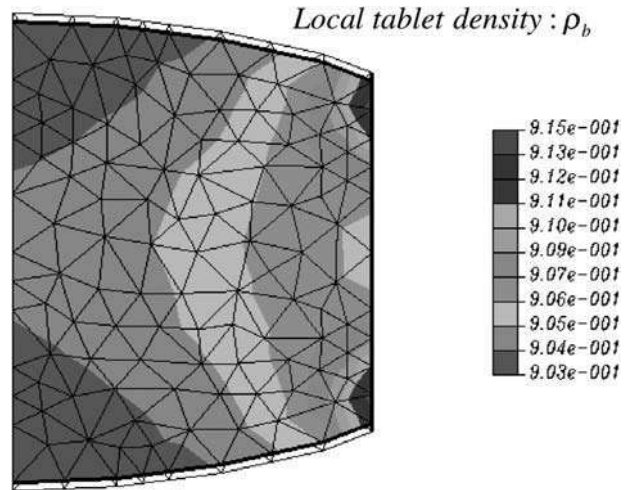


FIG. 21-141 Density developed in one-half of a tablet during compression, based on plasticity and compaction models. (Lewis et al., *Casting and Powder Compaction Group*, Department of Mechanical Engineering, University of Wales Swansea, <http://www.swan.ac.uk/nfa/>, with permission.)

using the above frictional properties, compact density and stress may be determined for any geometry, as illustrated in Fig. 21-141.

Hiestand Tableting Indices Likelihood of failure during decompression depends on the ability of the material to relieve elastic stress by plastic deformation without undergoing brittle fracture, and this is time-dependent. Those which relieve stress rapidly are less likely to cap or delaminate. Hiestand and Smith [*Powder Technol.*, **38**, 145 (1984)] developed three pharmaceutical **tableting indices**, which are applicable for general characterization of powder compactability. The **strain index** (SI) is a measure of the elastic recovery following plastic deformation, the **bonding index** (BI) is a measure of plastic deformation at contacts and bond survival, and the **brittle fracture index** (BFI) is a measure of compact brittleness.

Compaction Cycles Insight into compaction performance is gained from direct analysis of pressure/density data over the cycle of axial compact compression and decompression. Figure 21-142 illustrates typical **Heckel profiles** for plastic and brittle deforming materials which are determined from density measurements of *unloading* compacts. The slope of the curves gives an indication of the yield pressure of the particles. The contribution of fragmentation and rearrangement to densification is indicated by the low-pressure deviation from linearity. In addition, elastic recovery contributes to the degree of hysteresis which occurs in the *at-pressure* density curve during compression followed by decompression [Doelker, *Powder*

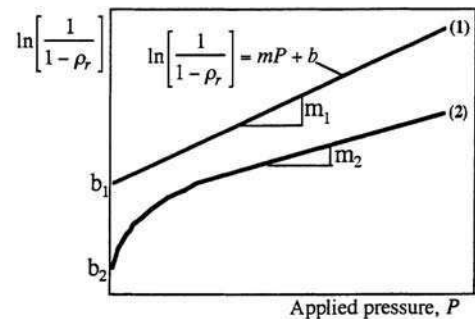


FIG. 21-142 Heckel profiles of the unloaded relative compact density for (1) a material densifying by pure plastic deformation and (2) a material densifying with contributions from brittle fragmentation and particle rearrangement.

Technology and Pharmaceutical Processes, Chulia et al. (eds.), Elsevier, 1994, p. 403].

Controlling Powder Compaction Compaction properties of powders are generally improved by improving flow properties. In particular, stress transmission improves with either lowering the wall friction angle or increasing the angle of friction of the powder. **Internal lubricants** may be mixed with the feed material to be compacted. They aid stress transmission by reducing the wall friction, but may also weaken bonding properties and the unconfined yield stress of the powder as well as lower powder friction, which acts to lower stress transmission. **External lubricants** are applied to the die surface, to impact wall friction alone.

Binders improve the strength of compacts through increased plastic deformation or chemical bonding. They may be classified as **matrix type**, **film type**, and **chemical**. Komarek [*Chem. Eng.*, 74(25), 154 (1967)] provides a classification of binders and lubricants used in the tableting of various materials. See also Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005, and Stanley-Wood (ed.), *Enlargement and Compaction of Particulate Solids*, Butterworth & Co. Ltd., 1983.

Particle properties such as size, shape, elastic/plastic properties, and surface properties are equally important. Generally decreased particle or granule hardness, increased surface energy, and raising particle size improve flow properties. Increasing particle size, which raises powder permeability, and applied vacuum and forces loading (e.g., screw or ram designs) help aid powder deaeration. Improved deaeration, powder flowability, and improved stress transmission generally improve all compaction processes, eliminate delamination and flaw formation, and improve production rates.

PASTE EXTRUSION

As in dry compaction processes, size-enlargement processes involving **paste extrusion** are also dominated by powder friction, including, e.g., both radial and axial extrusion in addition to some pressing operations. To illustrate the impact of frictional properties, we consider here an example of axial extrusion, as illustrated in Fig. 21-143 for a single screw extruder. Three key regions may be identified in this case: (1) a metering, mixing, and kneading zone; (2) a solids conveying zone where material is compacted and transported, largely in a plug flow fashion; and (3) the die plate extrusion region. In the conveying region, pressure increases as one moves down the barrel to reach some **maximum backpressure**, which is a function of screw speed, barrel and flight friction, and rheological properties of the paste. In other words, the extruder acts as a pump that can develop a certain total pumping or backpressure. In addition, the plug velocity, and hence throughput, is also a function of friction and rheological properties of the paste. The relationship between this maximum pressure and throughput is referred to as an **extruder characteristic**. The second region is the extrusion process through the die. Given the backpressure developed in the first conveying region, and again the paste rheology and friction, the paste will extrude at a certain rate through the die holes. The relationship between die plate pressure drop and throughput is referred to as a **die plate characteristic**. Therefore, the two regions are coupled through the operating backpressure of the extruder.

Compaction in a Channel Consider a powder being compacted in a channel of wetted perimeter C and cross-sectional area A , as shown in Fig. 21-144. The pressure which develops at the end of the

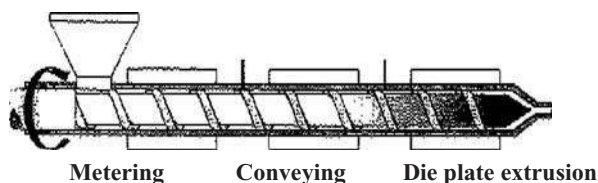


FIG. 21-143 Typical single-screw extruder, identifying key regions.

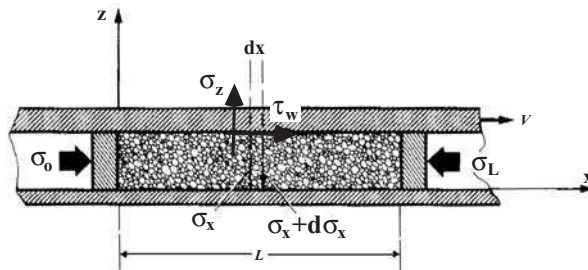


FIG. 21-144 Powder compaction in a channel, and associated force balance.

channel σ_L of width W and length $x = L$ will be given by

$$\sigma_L = \sigma_0 e^{\pm(\mu K C/A)L} = \sigma_0 e^{\pm\mu K_0 L(2W+2H)/(WH)} \quad \text{for } v = \pm \text{value} \quad (21-135)$$

where σ_0 is the applied feed pressure and K_0 and μ_w are the stress transmission and wall friction coefficients defined above. Note in comparison Eq. (21-132), where in contrast the sign of the exponential coefficient here depends on the direction of velocity v in Fig. 21-144. For positive velocity, the movement of the walls forward acts to increase the applied feed pressure, with the degree of pressurization increases with increasing wall friction coefficient and aspect ratio (CL/A), as well as increasing K_0 or decreasing powder friction [Eq. (21-133)]. This degree of pressurization is a key source in the driving pressure of extrusion.

Drag-Induced Flow in Straight Channels Consider now a rectangular channel sliding over an infinite plate (Fig. 21-145). The channel represents the unwound flight of a screw of width W and depth H ; and the plate, a barrel moving at a linear velocity V at an angle θ to the down channel direction x . The solids plug formed within the channel moves forward in the down channel direction x at a velocity u due to the friction of the moving upper plate, which conveys it forward as the screw moves backward. The vectorial difference between the plate velocity V and the plug velocity u gives the relative velocity at which the plate slides over the moving plug, or V^* . This produces a frictional wall stress τ_w acting at this top plate (or barrel) on the plug in the same direction as V^* . The angle difference between the plate velocity V and shear stress τ_w is referred to as the solids conveying angle, which is easily shown to be given by

$$\tan \Theta = \frac{u \sin \theta}{V - u \cos \theta} \quad (21-136)$$

The solids conveying angle Θ is zero for stationary solids ($u = 0$) and increases with increasing flow rate or throughput (increasing u). Neglecting the impact of cross-channel modification in friction, a force balance on the plug allows us to determine a relation akin to Eq. (21-135) for compaction in a channel, or

$$\sigma_L = \sigma_0 e^{[C_b \mu_b \cos(\Theta+\theta) - C_s \mu_s] K_d L/A} = \sigma_0 e^{[W \mu_b \cos(\Theta+\theta) - (W+2H) \mu_s] K_d L/WH} \quad (21-137)$$

where μ_b and μ_s are the barrel and screw flight friction, respectively. As conveying angle Θ increases, $\cos(\Theta + \theta)$ decreases, and therefore the overall pressure rise in the extruder *decreases*. Since conveying angle increases with increasing throughput [Eq. (21-136)], an inverse relationship exists between throughput and pressure rise. This suggests the following potential implications with regard to extruder operation: (1) Increasing pressure rise *decreases* conveying throughput for constant frictional coefficients, (2) increasing barrel friction or lowering flight friction increases pressure for constant throughput, and (3) increasing barrel friction or lowering flight friction increases throughput for constant pressure rise. Barrel friction acts to increase extruder pressurization, whereas flight friction works against this pressurization. Note also that the exact operating pressure must be determined in conjunction with die face pressure drop.

Paste Rheology Paste frictional and rheological properties control the flow rate through the final extrusion die face or basket. One

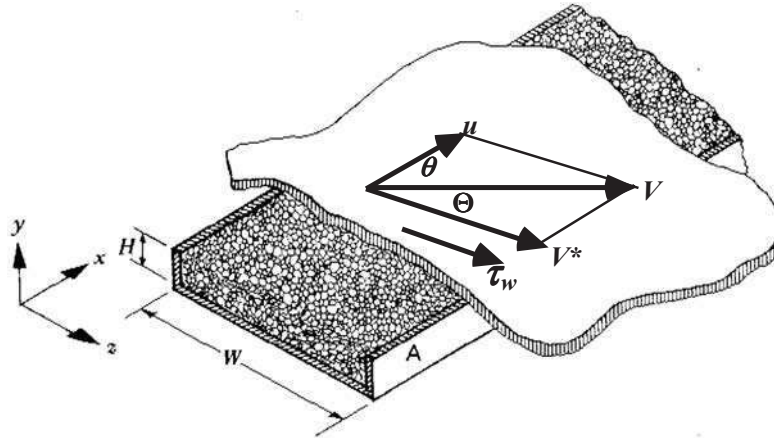


FIG. 21-145 The unwound screw channel, illustrating the barrel moving at a velocity V and at an angle θ with respect to the down channel direction x . The barrel slides over the solids at a relative velocity V^* , resulting in a frictional shear stress along the wall of τ_w and a solids conveying angle of Θ .

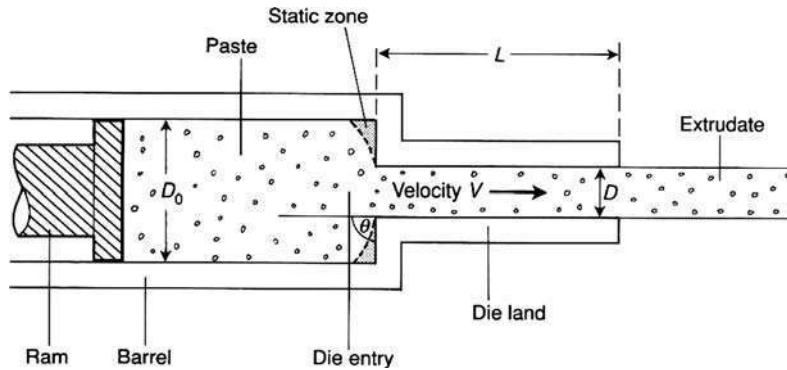


FIG. 21-146 Paste extrusion through a cylindrical die land with square entry. Note that the pressure drop consists of both an orifice pressure drop due to changing area (yielding within the paste) and a die land pressure drop (yielding along the wall). (Benbow and Bridgwater, *Paste Flow and Extrusion*, Oxford University Press, New York, 1993.)

approach to paste rheology is that of **capillary rheometry**, where parameters are determined for the paste as a function of die geometry and velocity, which can be used to determine die pressure drop for the production geometry, or the die characteristic. Figure 21-146 illustrates a typical die geometry, for which the pressure drop is typically modeled by a relationship of the form

$$P_e = P_o + P_f = (\sigma_o + \alpha V^B) \ln A_o/A_1 + (\tau_o + \alpha' V^B) \frac{4L}{D} \quad (21-138)$$

As proposed in the work of Benbow and Bridgwater (*Paste Flow and Extrusion*, Oxford University Press, New York, 1993), the first term P_o represents an orifice pressure in a converging section, required to overcome internal yield stresses within the material as the cross-sectional area is reduced from A_o to A_1 . The second term P_f represents frictional stresses that must be overcome to extrude through the die

land of aspect ratio L/D and constant diameter D . Through capillary rheometry experiments of varying die geometry and throughput, the various constants of Eq. (21-138) are estimated from measured pressure drops. These parameters are then used to calculate the die characteristic for the extruder at hand, namely, the die pressure drop vs. rate.

Although relatively unexplored, an alternative approach is to determine frictional yield properties by high-pressure shear and triaxial cells, and to incorporate these properties into soils or plasticity models for finite element simulations of flow within the extruder body, as has been done for compaction (cf. Fig. 21-141).

With increasing pressure, the conveying throughput provided by the screw will be less, whereas the possible throughput through the die face will be more. The intersection of the extruder and die characteristics determines the critical output of the extruder. (See "Screw and Other Paste Extruders" subsection for details.)

CONTROL AND DESIGN OF GRANULATION PROCESSES

ENGINEERING APPROACHES TO DESIGN

Advances in the understanding of granulation phenomena rest heavily in engineering process design. A change in granule size or voidage is akin to a change in chemical species, and so analogies exist between growth kinetics and chemical kinetics and the unit operations of size enlargement and chemical reaction (Fig. 21-147), where several scales of analysis must be considered for successful process design.

Scales of Analysis Consider the molecular or primary particle/single granule interactions occurring within a small volume element of material A within a mixing process, as shown in Fig. 21-148. On this molecular or **granule scale of scrutiny**, the designs of chemical reactors and of granulation processes differ conceptually in that the former deals with **chemical transformations** whereas the latter deals primarily with **physical transformations** controlled by **mechanical processing**. Here, the rate processes of granulation are controlled by a set of key physicochemical interactions defined in the following sections. These mechanisms or rate processes compete to control granule growth on a **granule volume scale of scrutiny**, as shown for the small volume element of material A of Fig. 21-148. Within this small volume of material, one is concerned with the rate at which one or more chemical species is converted to a product in the case of chemical kinetic. This is generally dictated by a reaction rate or **kinetic constant**, which is in turn a local function of temperature, pressure, and the concentration of

feed species, as was established from previous physicochemical considerations. These local variables are in turn a function of overall heat, mass, and momentum transfer of the vessel controlled by mixing and heating/cooling. The chemical conversion occurring within this local volume element may be integrated over the entire vessel to determine the chemical yield or extent of conversion for the reactor vessel; the impact of mixing and heat transfer is generally considered in this step at the **process volume scale of scrutiny**. In the case of a granulation process, an identical mechanistic approach exists for design, where chemical kinetics is replaced by **granulation kinetics**. The performance of a *granulator* may be described by the **extent of granulation** of a species. Let $(x_1, x_2, \dots, x_n)$ represent a list of attributes such as average granule size, porosity, strength, and any other generic quality metric and associated variances. Alternatively, $(x_1, x_2, \dots, x_n)$ might represent the concentrations or numbers of certain granule size or density classes, just as in the case of chemical reactors. The proper design of a chemical reactor or a *granulator* then relies on understanding and controlling the evolution (both *time* and *spatial*) of the feed vector X to the desired product vector Y . Inevitably, the reactor or granulator is contained within a larger plant-scale process chain, or **manufacturing circuit**, with overall plant performance being dictated by the interaction between individual unit operations. At the **plant scale of scrutiny**, understanding interactions between unit operations can be critical to plant performance and product quality. These interactions are far more

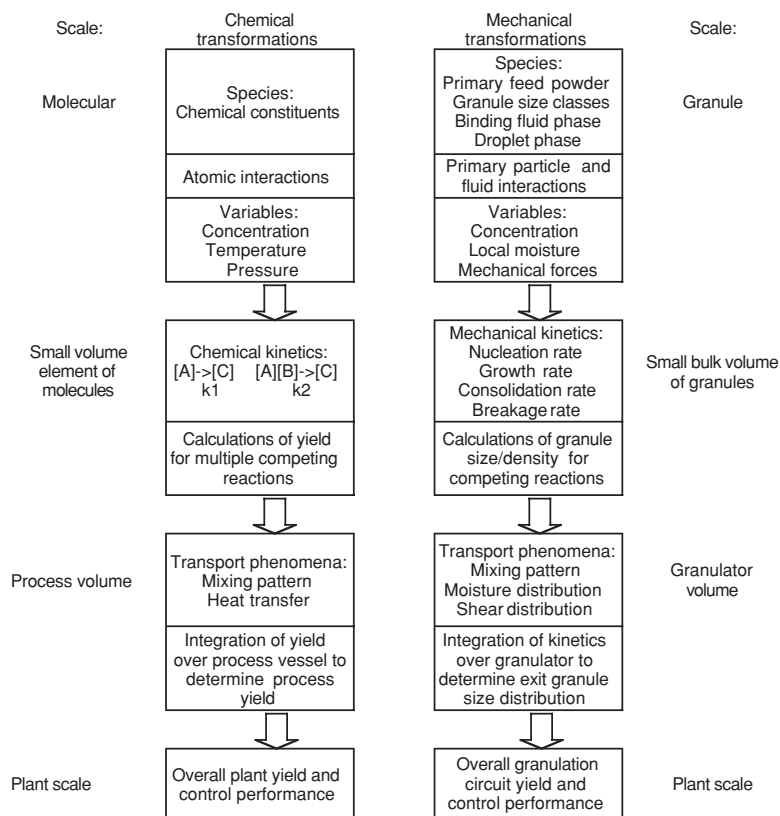


FIG. 21-147 Changes in state as applied to granulator and reactor kinetics and design. [Emmis, *Theory of Granulation: An Engineering Approach*, in Handbook of Pharmaceutical Granulation Technology, 2d ed., Parikh (ed.), Taylor & Francis, 2005. With permission.]

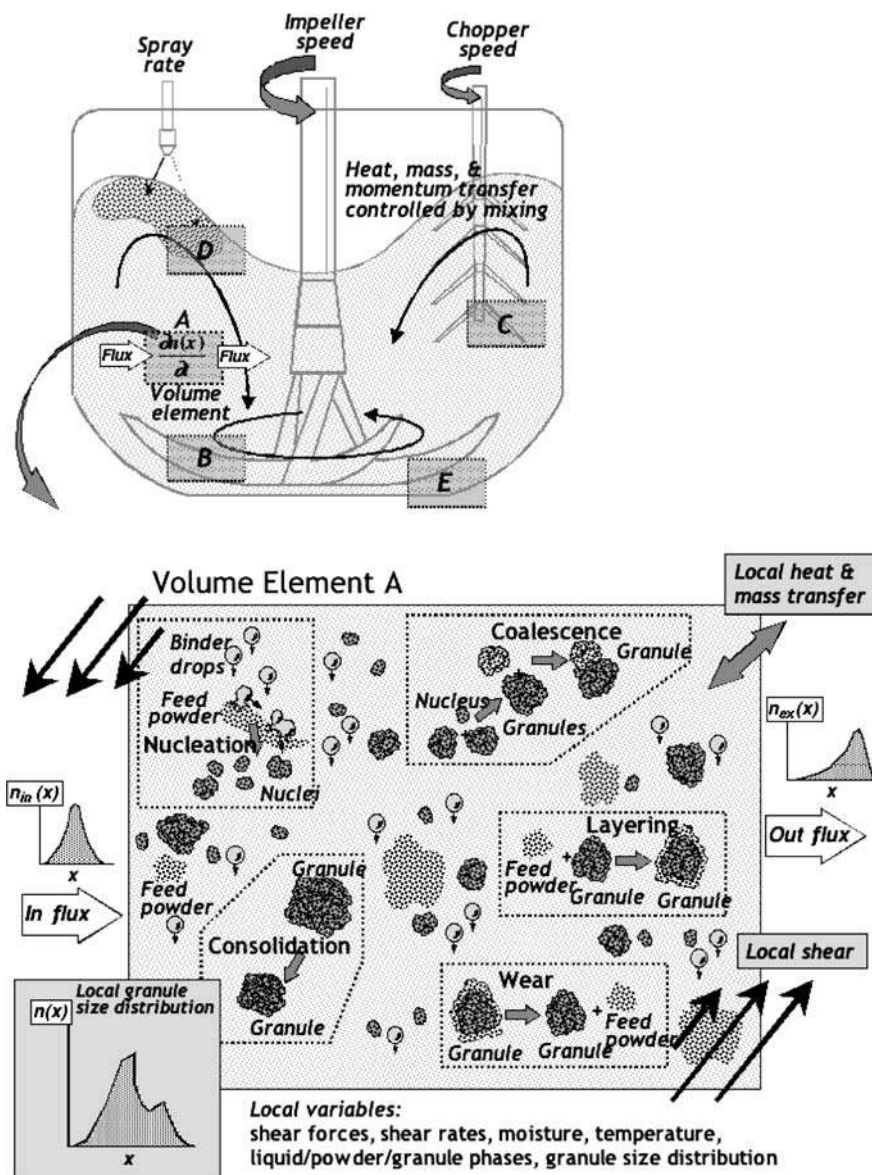


FIG. 21-148 Granulation within a local volume element, as a subvolume of a process granulator volume, which controls local size distribution. [Ennis, *Theory of Granulation: An Engineering Approach*, in *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Parikh (ed.), Taylor & Francis, 2005. With permission.]

substantial with solids processing than with liquid-gas processing. Ignoring these interactions often leads processing personnel to misdiagnose sources of poor plant performance. We consider each of these scales in greater detail below.

There are several important points worth noting with regard to this approach. First, the design of chemical reactors is well developed and an integral part of traditional chemical engineering education. (See, e.g., Levenspiel, *Chemical Reaction Engineering*, 2d ed., Wiley, New York, 1972.) In contrast, only the most rudimentary elements of reaction kinetics have been applied to granulator design. Second, an appreciation of this engineering approach is absolutely vital to properly scale up granulation processes for difficult formulations. Lastly, this perspective provides a logical framework with which to approach and unravel

complex processing problems, often involving several competing phenomena. Significant progress had been made with this approach in crystallization (Randolph and Larson, *Theory of Particulate Processes*, Academic Press, 1988) and grinding (Prasher, *Crushing and Grinding Process Handbook*, Wiley, 1987). Many complexities arise when applying the results of the previous subsections detailing granulation mechanisms to granulation processing. The purpose of this subsection is to summarize approaches to controlling these rate processes by placing them within the context of actual granulation systems and granulator design. See also "Modeling and Simulation of Granulation Processes."

Scale: Granule Size and Primary Feed Particles When considering a scale of scrutiny of the order of granules, we ask what controls the rate processes. This step links formulation or material

variables to the process operating variables, and successful granulator design hinges on this understanding. Two key local variables of the volume element A include the *local bed moisture* and the local level of shear (both *shear rate* and *shear forces*). These variables play an analogous role of species concentration and temperature in controlling kinetics in chemical reaction. In the case of chemical reaction, increased temperature or concentration of a feed species generally increases the reaction rate. For the case of granulation considered here, increases in shear rate and moisture result in increased granule/powder collisions in the presence of binding fluid, resulting in an increased frequency of successful growth events and increases in granule growth rate. Increases in shear forces also increase the granule consolidation rate and aid growth for deformable formulations. In the limit of very high shear (e.g., due to choppers), they promote wet and dry granule breakage, or limit the growth rate. Lastly, in the case of simultaneous drying, bed and gas-phase moisture and temperature control heat and mass transfer and the resulting drying kinetics, which can be important in fluid-bed granulation and temperature-induced drying in high-shear mixing.

Scale: Granule Volume Element A small bulk volume element A of granules (Fig. 21-148) has a particular granule size distribution, which is controlled by the local granulation rate processes shown pictorially in Fig. 21-149. In the wetting and nucleation rate process, droplets interact with fine powder to form initial nuclei, either directly or through mechanical breakdown of pooled overwetted regions. It is generally useful to consider the initial **powder phase** and **drop phases** as independent feed phases to the **granule phase**. In addition, the granule phase can be broken down into separate **species**, each species corresponding to a particular granule mesh size cut. Nucleation therefore results in a loss of powder and drop phases and the **birth** of granules. Granules and initial nuclei collide within this volume element with one another and with the surrounding powder phase, resulting in both granule growth and consolidation due to compaction forces. Granule growth by coalescence also results in the discrete birth of granules to a new granule size class or species, as well as loss or **death** of granules from the originating size classes. On the other hand, granule growth by layering and granule consolidation result in a slow differential increase and decrease in granule size, respectively. Granule breakage by fracture and attrition (or wear) act in a similar but opposite fashion to granule coalescence and layering,

and increase the powder phase and species of smaller granules. Lastly, this volume element of granules interacts with surrounding bed material, as granulated, powder, and drop phases flow to and from surrounding volume elements. The rate processes of granulation and the mass exchange with surrounding elements combine to control the local granule size distribution and growth rate within this small volume element.

As illustrated in Fig. 21-149, conducting an inventory of all granules entering and leaving a given size class $n(x)$ leads to a microlevel population balance over the volume element A , which governs the **local** average growth rate, given by

$$\frac{\partial n_a}{\partial t} + \frac{\partial}{\partial x_i} (n_a u_i) = G_a = B_a - D_a \quad (21-139)$$

where $n(x,t)$ is the instantaneous granule size distribution, which varies with time and position; G , B , and D are growth, birth, and death rates due to granule coalescence and granule fracture. The second term on the left side reflects contributions to the distribution from layering and wear as well as exchanges of granules with surrounding volume elements. Nucleation rate would be considered a boundary condition of Eq. (21-139), providing a source of initial granules or nuclei.

Solutions to this population balance are described in greater detail in the subsection "Modeling and Simulation of Grinding Processes." Analytical solutions are only possible in the simplest of cases. Although actual processes would require specific examination, some general comments are warranted. Beginning with nucleation, in the case of fast drop penetration into fine powders and for small spray flux, new granules will be formed of the order of the drop size distribution, and will contribute to those particular size cuts or granule species. If spray is stopped at low moisture levels, one might obtain a bimodal distribution of nuclei size superimposed on the original feed distribution. Very little growth may occur for these low moisture levels. This should not be confused with induction-type growth, which is a result of low overall formulation deformability. In fact, the moisture level of the nuclei themselves will be found to be high and nearly saturated. Moisture, however, is locked up within these nuclei, surrounded by large amounts of fine powder. Therefore, it is important not to confuse granule moisture, local moisture, and the overall

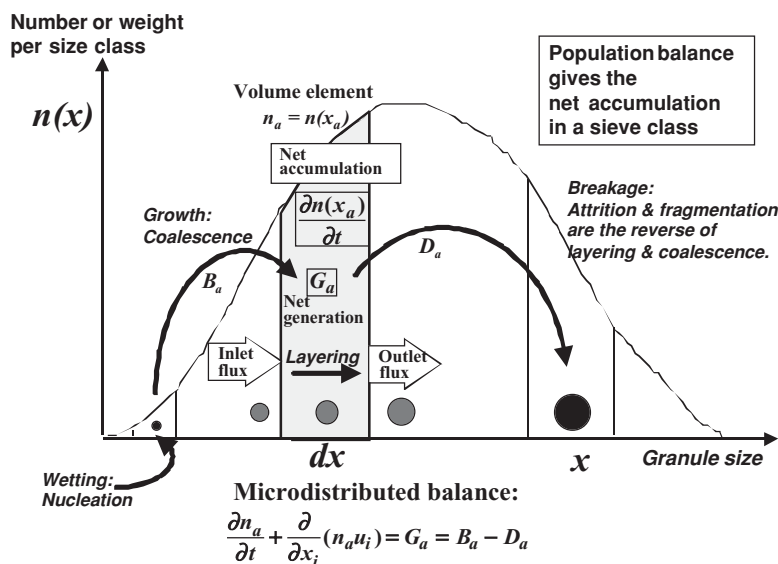


FIG. 21-149 The population balance over a sieve class, over specific granule size class. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

average peak bed moisture of the process; they are very much not the same and are influenced by proper vessel design and operation. As moisture levels increase and the concentration of the *ungranulated* powder phase decreases, the portion of the granule phase increases. As granules begin to interact more fully due to decreased surrounding powder and greater chances to achieve wet granule interaction, granule coalescence begins to occur. This in turn results in a decrease in granule number, and a rapid, often exponential, increase in granule size as previously demonstrated. Coalescence generally leads to an initial widening of the granule-size distribution until the **granule growth limit** is reached. As larger granules begin to exceed this growth limit, they can no longer coalesce with granules of similar size. Their growth rate drops substantially as they can continue to grow only by coalescence with fine granules or by layering with any remaining fine powder. At this point, the granule-size distribution generally narrows with time. Note that this is a *local* description of growth, whereas the overall growth rate of the process depends greatly on mixing, described next, as controlled by process design.

Scale: Granulator Vessel The local variables of moisture and shear level vary with volume element, or position in the granulator, which leads to the kinetics of nucleation, growth, consolidation, and breakage being dependent on position in the vessel, leading to a scale of scrutiny of the vessel size. As shown in Fig. 21-147, moisture levels and drop phase concentration and nucleation will be high at position *D*. Significant growth will occur at position *B* due to increased shear forces and granule deformation, as well as increased contacting. Significant breakage can occur at position *C* in the vicinity of choppers. Each of these positions or volume elements will have its own specific granule-size distribution.

Solids mixing impacts overall granulation in several ways, with mixing details given in subsection "Solids Mixing." (See also Weinekötter and Gericke, *Mixing of Solids*, Kluwer Academic, 2000.) First, it controls the local shear. Local shear rates and forces are a function of shear stress transfer through the powder bed, which is in turn a function of mixer design and bed bulk density, granule size distribution, and frictional properties. Local shear rates determine granule collisional velocities. This first area is possibly one of the least understood areas of powder processing, and it requires additional research to establish the connection between operating variables and local shear rates and forces. It is also a very important scale-up consideration. Second, solids mixing controls the interchange of moisture, powder phase, and droplet phases among the local volume elements. Third, it controls the interchange of the granulated phase.

Within the context of reaction kinetics (Levenspiel, loc.cit.), one considers extremes of mixing between well-mixed continuous and plug flow continuous or well-mixed batch processes. The impact of mixing on reaction kinetics is well understood, and similar implications exist for granulation growth kinetics. In particular, well-mixed continuous processes would be expected to provide the widest granule size distribution (deep continuous fluidized beds are an example), whereas plug flow or well-mixed batch processes should result in narrower distributions. *All else being equal*, plug flow continuous and batch well-mixed processes should produce identical size distributions. An example might include comparing a continuous shallow to a batch fluid-bed granulator. In addition, it is possible to narrow the distribution further by purposely segregating the bed by granule size, or staging the addition of ingredients, although this is a less explored area of granulator design. Pan granulation is a specific process promoting segregation by granule size. Since large granules interact less with smaller granule size classes, layering can be promoted at the expense of coalescence, thereby narrowing the granule-size distribution. Lastly, it should be possible to predict effects of dispersion, backmixing, and dead/stagnant zones on granule-size distribution, based on previous chemical kinetic studies.

Equation (21-139) reflects the evolution of granule size distribution for a particular volume element. When integrating this equation over the entire vessel, one is able to predict the granule-size distribution vs. time and position within the granulator. Lastly, it is important to understand the complexities of scaling rate processes on a local level to overall growth rate of the granulator. If such considerations are not

taken into account, misleading conclusions with regard to granulation behavior may be drawn. Wide distributions in moisture and shear level, as well as granule size, and how this interacts with scale-up must be kept in mind when applying the detailed description of rate processes discussed in the previous subsections.

CONTROLLING PROCESSING IN PRACTICE

Tables 21-15 and 21-16 summarize formulation and operating variables and their impact on granulation. From a processing perspective, we begin with the uniformity of the process in terms of solids mixing. Approaching a uniform state of mixing as previously described will ensure equal moisture and shear levels and therefore uniform granulation kinetics throughout the bed; however, poor mixing will lead to differences in local kinetics. If not accounted for in design, these local differences will lead to a wider distribution in granule-size distribution and properties than is necessary, and often in unpredictable fashions—particularly with scale-up.

Increasing fluid-bed excess gas velocity ($U - U_{mf}$) will increase solids flux and decrease circulation time. This can potentially narrow nuclei distribution for intermediate drop penetration times. Growth rates will be minimally affected due to increased contacting; however, the growth limit will decrease. There will be some increase in granule consolidation, and potentially a large increase in attrition. Lastly, initial drying kinetics will increase. Impeller speed in mixers will play a similar role in increasing solids flux. However, initial growth rates and granule consolidation are likely to increase substantially with an increase in impeller speed. The growth limit will decrease, partly controlled by chopper speed.

Fluidized beds can be one of the most uniform processes in terms of mixing and temperature. Powder frictional forces are overcome as drag forces of the fluidizing gas support bed weight, and gas bubbles promote rapid and intensive mixing. In the case of mixers, impeller speed in comparison to bed mass promotes mixing, with choppers eliminating any gross maldistribution of moisture and overgrowth.

With regard to bed weight, forces in fluid beds and therefore consolidation and granule density generally scale with bed height. As a gross rule of thumb, ideally the power input per unit mass should be maintained with mixer scale-up. However, cohesive powders can be ineffective in transmitting stress, meaning that only a portion of the bed may be activated with shear at large scale, whereas the entire bowl may be in motion at a laboratory scale. Therefore, mixing may not be as uniform in mixers as it is in fluidized beds. Equipment design also plays a large role, including air distributor and impeller/chopper design for fluid beds and mixers, respectively.

Increasing bed moisture and residence time increases overall growth and consolidation. However, it also increases the chances of bed defluidization or overmassing/bowl buildup in fluid beds and mixers, respectively. Increasing bed temperature normally acts to lower bed moisture due to drying. This acts to raise effective binder viscosity and lower granule consolidation and density, as well as initial growth rates for the case of high-shear mixers. This effect of temperature and drying generally offsets the inverse relationship between viscosity and temperature.

Spray distribution generally has a large effect in fluid beds, but in many cases, a small effect in mixers. In fact, fluid-bed granulation is only practical for wettable powders with short drop penetration time, since otherwise defluidization of the bed would be promoted to local pooling of fluid. Mechanical dispersion counteracts this in mixers. There may be a benefit, however, to slowing the spray rate in mixers for formulation with inductive growth behavior, as this will minimize the lag between spray and growth, as discussed previously.

In summary, for the case of fluid-bed granulation, growth rate is largely controlled by spray rate and distribution and consolidation rate by bed height and peak bed moisture. For the case of mixers, growth and consolidation are controlled by impeller and chopper speed. From a formulation perspective, we now turn to each rate process.

Controlling Wetting in Practice Table 21-17 summarizes typical changes in material and operating variables which improve wetting uniformity. Also listed are appropriate routes to achieve these changes

TABLE 21-15 Impact of Key Operating Variables in Fluid-Bed and Mixer Granulation

Effect of changing key process variables	Fluidized beds (including coating and drying)	High-shear mixers
Increasing solids mixing, solids flux, and bed agitation	Increasing excess gas velocity: Improves bed uniformity Increases solids flux Decreases solids circulation time Potentially improves nucleation Has no effect on noninertial growth rate Lowers growth limit Shows some increase in granule consolidation Increases granule attrition Increases initial drying kinetics Distributor design: Impacts attrition and defluidization	Increasing impeller/chopper speed: Improves bed uniformity Increases solids flux Decreases solids circulation time Potentially improves nucleation Increases growth rate Lowers growth limit Increases granule consolidation Increases granule attrition Impeller/chopper design: Improvements needed to improve shear transmission for cohesive powders
Increasing bed weight	Increasing bed height: Increases granule consolidation, density, and strength	Increasing bed weight: Generally lowers power per unit mass in most mixers, lowering growth rate Also increases nonuniformity of cohesive powders, and lowers solids flux and increases circulation time
Increasing bed moisture (Note: Increasing bed temperature normally acts to lower bed moisture due to drying.) Increasing residence time	Increases rates of nucleation, growth, and consolidation, giving larger, denser granules with generally a wider distribution. Distribution can narrow if growth limit is reached. Increases chances of defluidization	Increases rates of nucleation, growth, and consolidation giving larger, denser granules with generally a wider distribution Distribution can narrow if growth limit is reached. Increases chances of over massing and bowl buildup
Increasing spray distribution: Lowers liquid feed or spray rate Lowers drop size Increases number of nozzles Increases air pressure (two-fluid nozzles) Increases solids mixing (above)	Largely affected Wettable powders and short penetration times generally required For fast penetration: Decreases growth rate Decreases spread of size distribution Decreases granule density and strength For slow penetration: Poor process choice Defluidization likely	Less affected Poorly wetting powders and longer penetration time possible For fast penetration: Decreases growth rate Decreases spread of size distribution Decreases granule density and strength For slow penetration: Mechanical dispersion of fluid Little effect on distribution; however, slowing rate of addition minimizes lag in growth rate
Increasing feed particle size (can be controlled by milling)	Requires increase in excess gas velocity Has minimal effect on growth rate Increases in granule consolidation and density	Increases growth rate Increases granule consolidation and density

Ennis, *Theory of Granulation: An Engineering Approach*, in *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Parikh (ed.), Taylor & Francis, 2005. With permission.

in a given variable through changes in either the formulation or processing. Improved wetting uniformity generally implies a tighter granule size distribution and improved product quality. Equations (21-99), (21-103), and (21-107) provide basic trends of the impact of material variables on wetting dynamics and extent, as described by the dimensionless spray flux and drop penetration time.

Since drying occurs simultaneously with wetting, the effect of drying can substantially modify the expected impact of a given process variable, and this should not be overlooked. In addition, simultaneously drying often implies that the *dynamics* of wetting are far more important than the *extent*.

Adhesion tension should be maximized to increase the rate and extent of both binder spreading and binder penetration. Maximizing adhesion tension is achieved by minimizing contact angle and maximizing surface tension of the binding solution. These two aspects work against each other as surfactant is added to a binding fluid, and in general, there is an optimum surfactant concentration for the formulation. Surfactant type influences adsorption and desorption kinetics at the three-phase contact line. Inappropriate surfactants can lead to

Marangoni interfacial stresses which slow the dynamics of wetting. Additional variables which influence adhesion tension include (1) impurity profile and particle habit/morphology typically controlled in the particle formation stage such as crystallization, (2) temperature of granulation, and (3) technique of grinding, which is an additional source of impurity as well.

Decreases in binder viscosity enhance the rate of both binder spreading and binder penetration. The prime control over the viscosity of the binding solution is through binder concentration. Therefore, liquid loading and drying conditions strongly influence binder viscosity. For processes *without simultaneous drying*, binder viscosity generally decreases with increasing temperature. For processes *with simultaneous drying*, however, the dominant observed effect is that lowering temperature lowers binder viscosity and enhances wetting due to decreased rates of drying and increased liquid loading.

Changes in particle-size distribution affect the pore distribution of the powder. Large pores between particles enhance the *rate* of binder penetration, whereas they decrease the final *extent*. In addition, the particle size distribution affects the ability of the particles to pack

TABLE 21-16 Summary of Governing Groups for Granulation and Compaction

Rate process and governing groups	Key formulation properties governing group increases with:	Key process parameters governing group increases with:
Wetting and nucleation Spray flux ψ_s (small flux desirable)	Decreasing binder viscosity, per its effect on atomization	Increasing spray volume or rate Decreasing number on nozzles Decreasing solids flux Decreasing solids velocity, e.g., impeller or drum speed, fluidization velocity Decreasing spray area Decreasing spray time Decrease bed circulation time Increasing drop size
Drop penetration time τ_p (small time desirable)	Decreasing adhesion tension Increasing binder viscosity Decreasing effective powder pore size	
Growth and consolidation Viscous and deformation Stokes numbers St_v and St_{def} Granule saturation S Bulk capillary number Ca	Decreasing formulation yield stress Decreasing binder viscosity Increasing granule density Increasing granule size Increasing primary particle size or granule voidage Increasing binder viscosity Increasing particle friction Decreasing surface tension	Increase bed moisture or saturation Increasing granule collision velocity (Table 21-41) Increasing impeller and chopper speeds, drum speed, fluidization velocity Increasing bed height or scale Increasing granule collision velocity
Breakage Viscous and deformation Stokes numbers St_v and St_{def} for wet breakage Some relationship between toughness, hardness, and energy (yet undefined)	As above for growth and consolidation Decreasing fracture toughness Mechanism-dependent, hardness Increasing granule density Increasing granule voidage	As above for growth and consolidation Increasing granule collision velocity Increasing bed turnover and erosion displacement
Solids mixing Froude number Fr Some measure of frictional shear to inertia (yet undefined)	Increasing granule density Increasing interparticle friction Decreasing granule density	Increasing impeller diameter or vessel scale Increasing impeller speed Increasing collision velocity
Compaction Stress transmission ratio (high desirable) (low desirable) Relative deaeration time (low desirable) Relative permanent adhesion	Decreasing wall friction Increasing powder friction Decreasing bulk permeability Decreasing particle hardness Increasing surface energy (See also Hiestand indices)	Decreasing gap and die aspect ratio Increasing production rate Increasing compaction pressure

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TABLE 21-17 Controlling Wetting in Granulation Processes

Typical changes in material or operating variables that improve wetting uniformity	Appropriate routes to alter variable through formulation changes	Appropriate routes to alter variable through process changes
Increase adhesion tension. Maximize surface tension. Minimize contact angle.	Alter surfactant concentration or type to maximize adhesion tension and minimize Marangoni effects. Precoat powder with wettable monolayers, e.g., coatings or steam.	Control impurity levels in particle formation. Alter crystal habit in particle formation. Minimize surface roughness in milling.
Decrease binder viscosity.	Lower binder concentration. Change binder. Decrease any diluents and polymers that act as thickeners.	Raise temperature for processes without simultaneous drying. Lower temperature for processes with simultaneous drying since binder concentration will decrease due to increased liquid loading.
Increase pore size to increase rate of fluid penetration. Decrease pore size to increase extent of fluid penetration.	Modify particle-size distribution of feed ingredients.	Alter milling, classification or formation conditions of feed if appropriate to modify particle size distribution.
Improve spray distribution (related to dimensionless spray flux, given by ratio of spray to solid fluxes).	Improve atomization by lowering binder fluid viscosity.	Increase wetted area of the bed per unit mass per unit time by increasing the number of spray nozzles, lowering spray rate; increase air pressure or flow rate of two fluid nozzles.
Increase solids mixing (related to dimensionless spray flux).	Improve powder flowability of feed.	Increase agitation intensity (e.g., impeller speed, fluidization gas velocity, or rotation speed).
Minimize moisture buildup and losses.	Avoid formulations that exhibit adhesive characteristics with respect to process walls.	Maintain spray nozzles to avoid caking and nozzle drip. Avoid spray entrainment in process airstreams and spraying process walls.

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21-116 SOLID-SOLID OPERATIONS AND PROCESSING

TABLE 21-18 Controlling Growth and Consolidation in Granulation Processes

Typical changes in material or operating variables that maximize growth and consolidation	Appropriate routes to alter variable through formulation changes	Appropriate routes to alter variable through process changes
Rate of growth (low deformability): Increase rate of nuclei formation. Increase collision frequency. Increase residence time.	Improve wetting properties. (See "Wetting" subsection.) Increase binder distribution.	Increase spray rate and number of drops. Increase mixer impeller or drum rotation speed or fluid-bed gas velocity. Increase batch time or lower feed rate.
Rate of growth (high deformability): Decrease binder viscosity. Increase agitation intensity. Increase particle density. Increase rate of nuclei formation, collision frequency, and residence time, as above for low-deformability systems.	Decrease binder concentration or change binder. Decrease any diluents and polymers that act as thickeners.	Decrease operating temperature for systems with simultaneous drying. Otherwise increase temperature. Increase mixer impeller or drum rotation speed or fluid-bed gas velocity.
Extent of growth: Increase binder viscosity. Decrease agitation intensity. Decrease particle density. Increase liquid loading.	Increase binder concentration, change binder, or add diluents and polymers as thickeners.	Increase operating temperature for systems with simultaneous drying. Otherwise decrease temperature. Decrease mixer impeller or drum rotation speed or fluid-bed gas velocity. Extent observed to increase linearly with moisture.
Rate of consolidation: Decrease binder viscosity. Increase agitation intensity. Increase particle density. Increase particle size.	As above for high-deformability systems. Particle size and friction strongly interact with binder viscosity to control consolidation. Feed particle size may be increased and fine tail of distribution removed.	As above for high-deformability systems. In addition, increase compaction forces by increasing bed weight, or altering mixer impeller or fluid-bed distributor plate design. Size is controlled in milling and particle formation.

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TABLE 21-19 Controlling Breakage in Granulation Processes

Typical changes in material or operating variables which minimize breakage	Appropriate routes to alter variable through formulation changes	Appropriate routes to alter variable through process changes
Increase fracture toughness. Maximize overall bond strength. Minimize agglomerate voidage.	Increase binder concentration or change binder. Bond strength is strongly influenced by formulation and compatibility of binder with primary particles.	Decrease binder viscosity to increase agglomerate consolidation by altering process temperatures (usually decrease for systems with simultaneous drying). Increase bed agitation intensity (e.g., increase impeller speed, increase bed height) to increase agglomerate consolidation. Increase granulation residence time to increase agglomerate consolidation, but minimize drying time.
Increase hardness to reduce wear: Minimize binder plasticity. Minimize agglomerate voidage.	Increase binder concentration or change voidage. binder. Binder plasticity is strongly influenced by binder type.	See above effects which decrease agglomerate voidage.
Decrease hardness to reduce fragmentation: Maximize binder plasticity. Maximize agglomerate voidage.	Change binder. Binder plasticity is strongly influenced by binder type. Apply coating to alter surface hardness.	Reverse the above effects to increase agglomerate voidage.
Decrease load to reduce wear.	Lower formulation density.	Decrease bed agitation and compaction forces (e.g., mixer impeller speed, fluid-bed height, bed weight, fluid-bed excess gas velocity).
Decrease contact displacement to reduce wear.		Decrease contacting by lowering mixing and collision frequency (e.g., mixer impeller speed, excess fluid-bed gas velocity, drum rotation speed).
Decrease impact velocity to reduce fragmentation.	Lower formulation density.	Decrease bed agitation intensity (e.g., mixer impeller speed, fluid-bed excess gas velocity, drum rotation speed). Also it is strongly influenced by distributor plate design in fluid beds, or impeller and chopper design in mixers.

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within the drop as well as the final degree of saturation [Waldie, *Chem. Eng. Sci.*, **46**, 2781 (1991)].

The drop distribution and spray rate of binder fluid have a major influence on wetting. Generally, finer drops will enhance wetting as well as the distribution of binding fluid. The more important question, however, involves how large may the drops be or how high a spray rate is possible. The answer depends on the wetting propensity of the feed. If the liquid loading for a given spray rate exceeds the ability of the fluid to penetrate and spread on the powder, maldistribution in binding fluid will develop in the bed. This maldistribution increases with increasing spray rate, increasing drop size, and decreasing spray area (due to, e.g., bringing the nozzle closer to the bed or switching to fewer nozzles). The maldistribution will lead to large granules on one hand and fine ungranulated powder on the other. In general, the width of the granule-size distribution will increase, and generally the average size will decrease. Improved spray distribution can be aided by increases in agitation intensity (e.g., mixer impeller or chopper speed, drum rotation rate, or fluidization gas velocity) and by minimizing moisture losses due to spray entrainment, dripping nozzles, or powder caking on process walls.

Controlling Growth and Consolidation in Practice Table 21-18 summarizes typical changes in material and operating variables which maximize granule growth and consolidation. Also listed are appropriate routes to achieve these changes in a given variable through changes in either the formulation or processing. Growth and consolidation of granules are strongly influenced by rigid (especially fluid-beds) and deformability (especially mixers) Stokes numbers. Increasing St increases energy with respect to dissipation during deformation of granules. Therefore, the rate of growth for deformable

systems (e.g., deformable formulation or high-shear mixing) and the rate of consolidation of granules generally increase with increasing St . St may be increased by decreasing binder viscosity or increasing agitation intensity. Changes in binder viscosity may be accomplished by formulation changes (e.g., the type or concentration of binder) or by operating temperature changes. In addition, simultaneous drying strongly influences the effective binder concentration and viscosity. The *maximum* extent of growth increases with decreasing St and increased liquid loading. Increasing particle size also increases the rate of consolidation, and this can be modified by upstream milling or crystallization conditions.

Controlling Breakage in Practice Table 21-19 summarizes typical changes in material and operating variables necessary to minimize breakage. Also listed are appropriate routes to achieve these changes in a given variable through changes in either the formulation or processing. Both fracture toughness and hardness are strongly influenced by the compatibility of the binder with the primary particles, as well as the elastic and plastic properties of the binder. In addition, hardness and toughness increase with decreasing voidage and are influenced by previous consolidation of the granules. While the direct effect of increasing gas velocity and bed height is to increase breakage of dried granules, increases in these variables may also act to increase consolidation of wet granules, lower voidage, and therefore lower the final breakage rate. Granule structure also influences breakage rate; e.g., a layered structure is less prone to breakage than a raspberry-shaped agglomerate. However, it may be impossible to compensate for extremely low toughness by changes in structure. Measurements of fracture properties help define expected breakage rates for a product and aid product development of formulations.

SIZE ENLARGEMENT EQUIPMENT AND PRACTICE

GENERAL REFERENCES: Ball et al., *Agglomeration of Iron Ores*, Heinemann, London, 1973. Benbow and Bridgwater, *Paste Flow and Extrusion*, Oxford University Press, New York, 1993. Ennis, *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, E&G Associates, Nashville, Tenn., 2006. Kristensen, *Acta Pharm. Suec.*, **25**, 187 (1988). Lister and Ennis, *The Science and Engineering of Granulation Processes*, Kluwer Academic Publishers, 2005. Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005. Masters, *Spray Drying in Practice*, SprayDryConsult International ApS, 2002. Master, *Spray Drying Handbook*, 5th ed., Longman Scientific Technical, 1991. Pietsch, *Size Enlargement by Agglomeration*, Wiley, Chichester, 1992. Pietsch, *Roll Pressing*, Heyden, London, 1976. Stanley-Wood, *Enlargement and Compaction of Particulate Solids*, Butterworth & Co. Ltd., 1983.

Particle-size enlargement equipment can be classified into several groups, with typical objectives summarized in Table 21-10 and advantages and applications summarized in Table 21-11. See "Scope and Applications." Comparisons of bed agitation intensity, compaction pressures, and product bulk density for select agglomeration processes are highlighted in Fig. 21-111. Terminology is industry-specific. In the following discussion, particle-size enlargement in tumbling, mixer, and fluidized-bed granulators is referred to as **granulation**. Granulation processes vary from low to medium levels of applied shear and stress, producing granules of low to medium density. The presence of liquid binder is essential for granule growth and green strength. Granulation includes pelletization or balling as used in the iron ore industry, but does not include the breakdown of compacts by screening as used in some tableting industries. The term *pelleting* or *pelletization* is used for extrusion processes only. **Spray processes** include slurry atomization operations such as prilling and spray drying. **Pressure compaction processes** include dry compaction techniques such as roll pressing and tableting and wet techniques such as radial and axial extrusion. Compaction processes rely on pressure to

increase agglomerate density and give sufficient compact strength, either with or without liquid binder, with a resulting high compact density.

In fluidized-bed granulators, the bed of solids is supported and mixed by fluidization gas, generally with simultaneously drying. With small bed agitation intensities and high binder viscosities due to drying, **fluidized-bed granulators** can produce one of the lowest-density granules of all processes, with the exception of spray drying. Fluidized and spouted fluidized beds are also used for coating or layering applications from solution or melt feeds, which can produce spherical, densely layered granules. At the other extreme of granulation processes are **high-shear mixer-granulators**, where mechanical blades and choppers induce binder distribution and growth, producing dense, sometimes irregular granules. Fluidized beds are generally a low-agitation, low-deformability process where spray distribution is critical, whereas mixer-granulators lie at the other end of bed agitation as a high-agitation, high-deformability process dominated by shear forces and formulation deformability. (See "Growth and Consolidation," Figs. 21-110 and 21-111.) **Tumbling granulators** such as rotating discs and drums produce spherical granules of low to medium density, and lie between fluidized bed and mixer in terms of bed agitation intensity and granule density. They have the highest throughput of all granulation processes, often with high recycle ratios. Preferential segregation in, e.g., disc granulators can produce very tight size distributions of uniform spherical granules. Mixer and fluid-bed granules operate as both continuous and batch-processed, dependent on the specific industry. See "Growth and Consolidation" and "Controlling Wetting in Practice" subsections for a discussion of granulation mechanisms and controlling properties.

Extrusion processes can operate wet or dry to produce narrowly sized, dense agglomerates or pellets. Wet extrusion is also often followed by spheronization techniques to round the product. Extrusion

operates on the principle of forcing powder in a plastic state through a die, perforated plate, or screen. Material undergoes substantial shear in the equipment, and operation and product attributes are strongly influenced by the frictional interaction between the powder and wall. For wet extrusion, the rheology of the wet mass or paste is also important. In compaction processes, material is directly consolidated between two opposing surfaces, with varying degrees of powder confinement. These processes exert the highest applied force of any size enlargement device to give the highest density product. Successful operation depends on good transmission of the applied force through the powder, escape of any entrapped gases, and development of strong interparticle bonds. Both extrusion and compaction processes are very sensitive to powder flow and mechanical properties of the feed. See "Powder Compaction" and "Powder Extrusion" for a discussion of feed property impact on equipment performance.

The choice of size enlargement equipment for a product at hand is subject to a variety of constraints, some of which are listed in Table 21-20. Ideally, the choice of equipment should be made on the basis of the desired final product attributes, making allowances for any special processing requirements (e.g., heat, moisture sensitivity, polymorphism). In practice, however, the dominant driver behind technology selection for a company relies heavily on historical process experience. Unfortunately, this can lead to long-term challenges if a new product is envisioned which differs significantly in formulation feed and final product attributes. Mixers do not generally produce porous, low-density granules, can be difficult to scale over large volume changes, and can produce significant frictional heating. On the other hand, fluid beds cannot process hydrophobic formulations or produce dense granules, with layering for slurry sprays being an exception, but are easier to scale up in practice and are robust to feed property changes. Compaction of fine powder (direct compression) and extrusion processes are sensitive to frictional properties and cannot tolerate large upstream variations in size distribution of a formulation. In many cases tradeoffs must be made. For example, a desire to eliminate solvent and dust handling in fluid-bed processing must be balanced by the fact that this process produces porous granules that might be highly desirable for their fast dissolution behavior. Lastly, in choosing and designing a granulation process, one must consider both product and process engineering, as discussed above ("Process vs. Formulation Design"). The range of possible product attributes is set during feed powder formulation, or **product engineering**, and control within this range is specific to the operating variables chosen during **process engineering**. The degree of interaction of these endeavors governs the success of the size-enlargement process as well as any scale-up efforts. We now discuss the myriad equipment choices avail-

able. Related granulation and compaction mechanisms have been discussed previously within the context of formulation and product engineering (see "Agglomeration Rate Processes and Mechanics.")

TUMBLING GRANULATORS

In **tumbling granulators**, particles are set in motion by the tumbling action caused by the balance between gravity and centrifugal forces. The most common types of tumbling granulators are **drum** and inclined **disc granulators**. Their use is widespread including the iron ore industry (where the process is sometimes called **balling** or **wet pelletization**), fertilizer manufacturer, and agricultural chemicals.

Tumbling granulators generally produce granules in the size range of 1 to 20 mm and are not suitable for making granules smaller than 250 μm . Granule density generally falls between that of fluidized-bed and mixer granulators (Fig. 21-111), and it is difficult to produce highly porous agglomerates in tumbling granulators. Tumbling equipment is also suitable for coating large particles, but it is difficult to coat small particles, as growth by **coalescence** of the seed particles is hard to control.

Drum and disc granulators generally operate in **continuous feed** mode. A key advantage to these systems is the ability to run at large scale. Drums with diameters up to 4 m and throughputs up to 100 tons/h are widely used in the mineral industry.

Disc Granulators Figure 21-150 shows the elements of a **disc granulator**. It is also referred to as a **pelletizer** in the iron ore industry or a **pan granulator** in the agricultural chemical industry. The equipment consists of a rotating tilted disc or pan with a rim. Solids and fluid agents are continuously added to the disc. A coating of the feed material builds up on the disc, and the thickness of this layer is controlled by scrapers or a plow, which oscillate mechanically. The surface of the pan may also be lined with expanded metal or an abrasive coating to promote proper lifting and cascading of the particulate bed, although this is generally unnecessary for fine materials. Solids are typically introduced to the disc by either volumetric or gravimetric feeders, preferable at the bottom edge of the rotating granular bed. Gravimetric feeding generally improves granulation performance due to smaller fluctuations in feed rates. Such fluctuations act to disrupt rolling action in the disc and can lead to maldistributions in moisture and local buildup on the disc surface. Wetting fluids that promote growth are generally applied by a series of single-fluid spray nozzles distributed across the face of the bed. Solids feed and spray nozzle locations have a pronounced effect on granulation performance and granule structure.

Variations of the simple disc shape include (1) an outer reroll ring which allows granules to be simultaneously coated or densified without further growth, (2) multisteped sidewalls, and (3) a pan in the form of a truncated cone (Capes, *Particle Size Enlargement*, Elsevier, 1980). Discs in the form of deep pans running close to horizontal with internal blades and choppers are also available, as a hybrid disc-mixer system.

The required disc rotation speed is given in terms of the **critical speed**, i.e., the speed at which a single particle is held stationary on the rim of the disc due to centripetal forces. The critical speed N_c is given by

$$N_c = \sqrt{\frac{g \sin \delta}{2\pi^2 D}} \quad (21-140)$$

where g is the gravitational acceleration, δ is the angle of the disc to the horizontal, and D is the disc diameter. The typical operating range for discs is 50 to 75 percent of critical speed, with angles δ of 45 to 55°. This range ensures a good tumbling action. If the speed is too low, sliding will occur. If the speed is too high, particles are thrown off the disc or openings develop in the bed, allowing spray *blow-through* and uneven buildup on the disc bottom. Proper speed is influenced by flow properties of the feed materials, bed moisture, and pan angle, in addition to granulation performance.

Discs range in size from laboratory units of 30 cm in diameter up to production units of 10 m in diameter with throughputs ranging from 1000 lb/h up to 100 tons/h in the iron ore industry. Figure 21-151

TABLE 21-20 Considerations for Choice of Size-Enlargement Process

Final product attributes, in particular agglomerate size, size distribution, voidage, strength, and dissolution behavior
Form of the active ingredient (dry powder, melt, slurry, or solution), and its amount and nature (hydrophobic, hydrophilic, moisture or heat sensitivity, polymorphic changes)
Need for moisture-sensitive (dry processing) formulations or heat-sensitive formulations
Robustness of a process to handle a wide range of formulations, as opposed to a dedicated product line
Air and solvent handling requirements as well as degree of unit containment due to dust or solvent hazards
Desired scale of operation, and type (batch vs. continuous). Ease of process scale-up and scale-down, as well as range of granule property control at one scale
Multiple unit operations in one vessel (e.g., granulation, drying, coating in a fluidized bed)
Process monitoring capabilities and ease of integration into process control schemes
Maintenance and utility requirements; ease of cleaning to prevent product cross-contamination
Integration of size enlargement equipment into existing process plant
Existing company and supporting vendor experience with specific granulation equipment

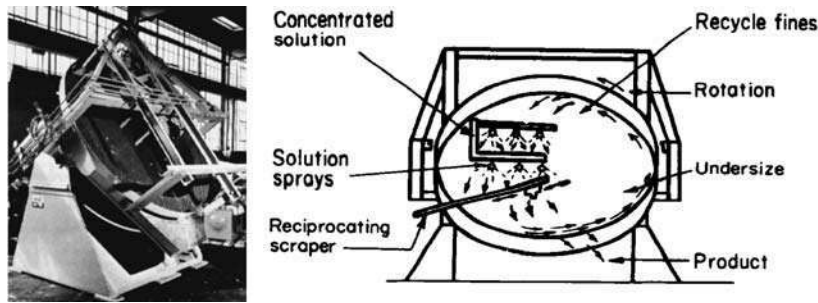


FIG. 21-150 A typical disc granulator [Capes, Particle Size Enlargement, Elsevier, 1980].

shows throughput capacities for discs of varying diameters for different applications and formulation feed densities. When scaling up from laboratory or pilot tests, it is usual to keep the same disc angle and fraction of critical speed. Power consumption and throughput are approximately proportional to the square of disc diameters, and disc height is typically 10 to 20 percent of diameter. It should be emphasized that these relationships are best used as a guide and in combination with actual experimental data on the system in question to indicate the approximate effect of scale-up.

A key feature of disc operation is the inherent **size classification** (Fig. 21-152). Centripetal forces throw small granules and ungranulated feed high on the disc, whereas large granules remain in the **eye** and exit as product. In addition, the granular bed generally sits on a bed of ungranulated powder and freshly formed nuclei. **Size segregation** leads to exit of only product granules from the eye at the rim of the disc. This classification effect substantially narrows exit granule-size distribution, as compared to drum granulators, and discs typically operate with little or no pellet recycle. Due to this segregation, positioning of the feed and spray nozzles is key in controlling the balance of granulation rate processes and resultant granule structure. Disc granulators produce the narrowest first-pass granule size distribution

of all granulation systems, second only to compaction processes of wet extrusion or fluid-bed coating systems.

Total holdup and granule residence time distribution vary with changes in operating parameters, which affect granule motion on the disc. Total holdup (mean residence time) increases with decreasing pan angle, increasing speed, and increasing moisture content. The residence time distribution for a disc lies between the mixing extremes of plug flow and completely mixed, and can have a marked effect on granule-size distribution and structure (e.g., layered vs. agglomerated). Increasing the disc angle narrows the residence time distribution and promotes layered growth. Several mixing models for disc granulators have been proposed. One- to two-minute residence times are common.

Drum Granulators Granulation drums are common in the metallurgical and fertilizer industries and are primarily used for very large throughput applications (see Table 21-21). In contrast to discs, there is no output size classification and high recycle rates of off-size product are common. As a first approximation, granules can be considered to flow through the drum in plug flow, although back mixing to some extent is common.

As illustrated in Fig. 21-153, a granulation drum consists of an inclined cylinder, which may be either open-ended or fitted with annular retaining rings. Either feeds may be premoistened by mixers to form granule nuclei, or liquid may be sprayed onto the tumbling bed via nozzles or distributor pipe systems. Drums are usually tilted longitudinally a few degrees from the horizontal (0 to 10°) to assist flow of granules through the drum. The critical speed for the drum is calculated from Eq. (21-140) with $\delta = 80$ to 90° . To achieve a cascading, tumbling motion of the load, drums operate at lower fractions of critical speed than discs, typically 30 to 50 percent of N_c . If drum speed is too low, intermittent sliding of the bed will occur with poor tumbling motion; if too high, material will be pinned to the drum wall, increasing the likelihood of bed cataracting and spray blow-through. Scrapers of various designs are often employed to control buildup of the drum wall. Holdup in the drum is between 10 and 20 percent of the drum volume. Drum length

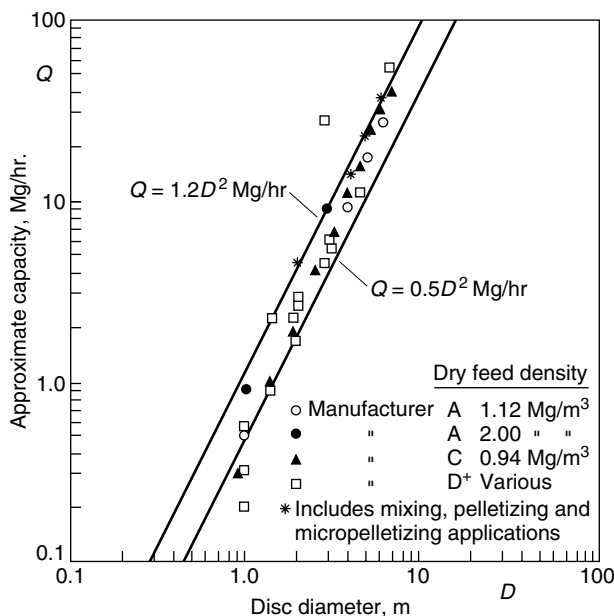


FIG. 21-151 Capacity of inclined disc granulators of varying diameter and formulation feed densities. [Capes, Particle Size Enlargement, Elsevier, 1980.]

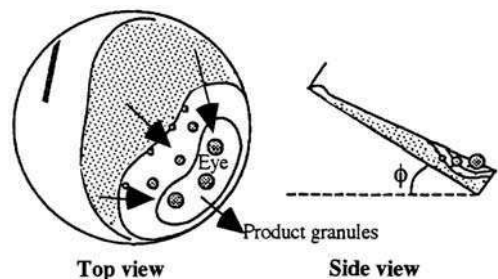


FIG. 21-152 Granule segregation on a disc granulator, illustrating a size classified granular bed sitting on ungranulated feed powder.

TABLE 21-21 Characteristics of Large-Scale Granulation Drums

Diameter (ft)	Length (ft)	Power (hp)	Speed (rpm)	Approximate capacity (tons/h)*
Fertilizer granulation				
5	10	15	10–17	7.5
6	12	25	9–16	10
7	14	30	9–15	20
8	14	60	20–14	25
8	16	75	20–14	40
10	20	150	7–12	50
Iron ore balling				
9	31	60	12–14	54
10	31	60	12–14	65
12	33	75	10	98

*Capacity excludes recycle. Actual drum throughput may be much higher.

NOTE: To convert feet to centimeters, multiply by 30.48; to convert tons per hour to megagrams per hour, multiply by 0.907; and to convert horsepower to kilowatts, multiply by 0.746.

From Capes, *Particle Size Enlargement*, Elsevier, 1980.

ranges from 2 to 5 times diameter, and power and capacity scale with drum volume. Holdup and mean residence time are controlled by drum length, with difficult systems requiring longer residence times than those that agglomerate readily. One- to two-minute residence times are common.

Variations of the basic cylindrical shape are the multicone drum, which contains a series of compartments formed by annular baffles [Stirling, in Knepper (ed.), *Agglomeration*, Interscience, New York, 1962], falling curtain and fluidized drum granulators (having an internal distributor running the length of the drum), the Sacket star granulator, and deep disc granulators with internal screens and recycle.

Drum granulation plants often have significant **recycle** of under-size, and sometimes crushed oversize, granules. Recycle ratios between 2:1 and 5:1 are common in iron ore balling and fertilizer **granulation circuits**. This large recycle stream has a major effect on circuit operation, stability, and control. A surge of material in the recycle stream affects both the moisture content and the size distribution in the drum. Surging and limit cycle behavior are common. There are several possible reasons for this, including:

1. A shift in controlling mechanism from coalescence to layering when the ratio of recycled pellets to new feed changes [Sastri and Fuerstenau, *Trans. Soc. Mining Eng., AIME*, **258**, 335–340 (1975)]

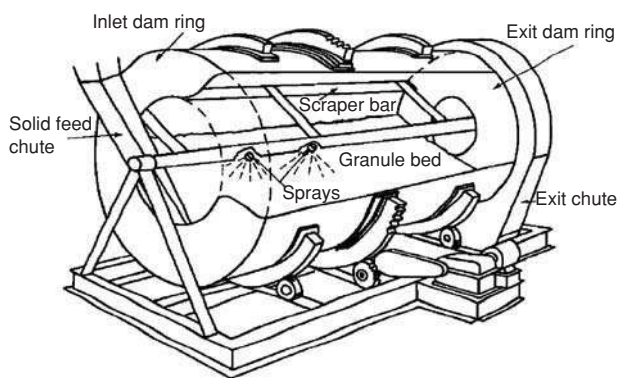


FIG. 21-153 A rolling drum granulator [Capes, *Particle Size Enlargement*, Elsevier, 1980].

2. Significant changes in the moisture content in the drum due to recycle fluctuations (recycle of dry granules in fertilizer granulation) [Zhang et al., *Control of Particulate Processes IV* (1995)]. In many cases, plants simply live with these problems. However, use of modern model-based control schemes in conjunction with improved methods for on-line moisture and particle-size analysis can help overcome these effects [Ennis (ed.), *Powder Technol.*, **82** (1995); Zhang et al., *Control of Particulate Processes IV* (1995)].

Controlling Granulation Rate Processes Granulation rate processes have been discussed in detail above (see “Agglomeration Rate Processes and Mechanics” subsection). Nucleation, coalescence, consolidation, and layering are all important processes in tumbling granulation, which could be considered a low- to medium-agitation-intensity process. See also Tables 21-15 to 21-19.

Nucleation, or the formation of seed granules, is critically controlled by spray distribution and interfacial properties of the particulate feed. **Nuclei** are generated from liquid spray drops, scraper bars, or initial coalescence of feed particles. Bed agitation intensity is low to moderate and has only a secondary effect in breaking up/down large nuclei or overwet regions. Therefore, tumbling systems should be maintained in a droplet-controlled regime of nucleation. Spray flux ψ_s should be maintained at less than 0.2, where the solids flux may be estimated from the width of the spray zone and the drum peripheral speed DN [see Eqs. (21-102) and (21-103)]. Fast drop penetration times are most suitable, with low binder viscosities, wetting powder, and larger feed. Fine powders are also possible with layered growth or high recycle. Poor wetting inhibits capacity, particularly in disc granulation. Poor wetting limits production rate to prevent overwet masses. In addition, nucleation determines the initial granule-size distribution and is therefore critical in low-agitation-intensity processes. Wetting and nucleation can be enhanced by increasing temperature and feed particle size, by decreasing binder viscosity, or by improved spray distribution, e.g., by multiple nozzles.

Granule coalescence or growth in tumbling granulators can be complex for a number of reasons:

1. Granules remain wet and can deform and consolidate. The behavior of a granule is therefore a function of its history.
2. Different granulation behavior is observed for broad and narrow feed-size distributions.

3. There is often complex competition between growth mechanisms. As a general rule, growth is linked to consolidation. For a batch process beginning with fine feed, random exponential growth initially occurs followed by a transition to a slower preferential balling stage of growth [Eqs. (21-116) and (21-119)]. This is tied to a similar decrease in granule voidage through the consolidation process.

For less deformable systems, an induction time may be observed, with time required to work moisture to the surface. Such systems are often unstable. For highly deformable and weak formulations, initial linear, preferential growth may be observed with large granules crushing weaker small granules, which are then layered onto the surviving large granules, referred to as crushing and layering.

The granule-size distribution generally narrows with residence time for broad feed-size distribution, whereas fine feeds widen until reaching the critical limiting size of the formulation, after which they will narrow. This limiting size of growth depends linearly on binder viscosity and inversely on agitation velocity and granule density, or

$$d_{\max} \propto \frac{\mu}{\rho u_o} \quad (21-141)$$

Table 21-22 gives possible choices for the collision velocity. Figure 21-121 demonstrates that successful scaling of these effects has been achieved in practice.

Note that many of the above observations are based on batch experiments, whereas in most drum granulation systems, very high recycle ratios are present. This recycle material is often composed of well-formed granules, and so the above observations may be masked.

Growth rate is very sensitive to liquid content for narrow initial-size distributions, with increases in liquid content for fine powders leading to an approximate exponential increase in granule size. For low-viscosity liquids, granulation occurs when very close to the saturation of the

TABLE 21-22 Possible Choices of Impact Velocity u_0 or u_c for Stokes Numbers

Granulation process	Collision velocity (maxima)	Shear velocity (averages)
Tumbling (pans and drums)	ND (N is drum/disk speed) (D is drum/disk diameter)	Nd (N is drum/disk speed)
Mixers	$N_i D_i$ (D_i is impeller diameter) $N_c D_c$ (D_c is chopper diameter)	$N_i d$ (i is impeller) $N_c d$ (c is chopper)
Fluidized beds	$(6U_B/D_B)d$ (U_B, D_B is bubble velocity & diameter) U_j (U_j is distributor jet velocity)	$N\delta$ (δ is impeller wall gap) $(6U_B/\delta D_B)d$ (δ is bubble gap)

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granule. This leads to the following equation to estimate moisture requirements (Capes, *Particle Size Enlargement*, Elsevier, 1980):

$$w = \frac{\epsilon \rho_l}{\epsilon \rho_l + (1 - \epsilon) \rho_s}$$

$$w = \frac{1}{1 + 1.85(\rho_s/\rho_l)} \quad d_p < 30 \mu\text{m} \quad (21-142)$$

$$w = \frac{1}{1 + 2.17(\rho_s/\rho_l)} \quad d_p > 30 \mu\text{m}$$

where w is the weight fraction of the liquid, ϵ is the porosity of the close-packed material, ρ_s is true particle density, ρ_l is liquid density, and d_p is the average size of the feed material. Equation (21-142) is suitable for preliminary mass balance requirements for liquid binders with similar properties to water. If possible, however, the liquid requirements should be measured in a balling test on the material in question, since unusual packing and wetting effects, particle internal porosity and solubility, air inclusions, etc., may cause error. Approximate moisture requirements for balling several systems are given in Table 21-23. In addition, for materials containing soluble constituents, such as fertilizer formulations, the total solution phase ratio controls growth, and not simply the amount of binding fluid used.

When fines are recycled as in iron ore sinter feed or fertilizer drum granulation, fines are rapidly granulated and removed from the distribution up to some critical size, which is a function of both moisture content and binder viscosity. Changing the initial-size distribution changes the granule porosity and hence moisture requirements [Adetayo et al., *Chem. Eng. Sci.*, **48**, 3951 (1993)]. Since recycle rates in drum systems are high, differences in size distribution between feed and recycle streams are one source of the limit cycle behavior observed in practice.

Growth by **layering** is important for the addition of fine powder feed to recycled, well-formed granules in drum granulation circuits and for disc granulators. In each case, layering will compete with **nuclei formation** and coalescence as growth mechanisms. Layered growth leads to a smaller number of larger, denser granules with a narrower size distribution than growth by coalescence. Layering is favored by a high ratio of pellets to new feed, low moisture, and positioning powder feed to fall onto tumbling granules.

Mechanisms of growth in disc granulation may be altered by spray location, as illustrated in Fig. 21-154. Spraying toward the eye and granule region promotes agglomerated growth with wide size distribution and low bulk density, whereas spraying feed powder promotes denser, layered growth with narrow size distribution and high bulk density, largely due to the fact that the formed granules have a larger effective residence time. Similar implications would apply in drum granulation as well, and staged moisture addition or dry feed addition is yet relatively unexplored.

Consolidation of the granules in tumbling granulators directly determines granule density and porosity. Since there is typically no in

situ drying to stop the consolidation process, granules consolidate over extended times. Consolidation rates are controlled by Eq. (21-121) (cf. "Granule Consolidation and Densification" subsection.) The maximum collision velocity u_c increases with both drum or disc speed as well as size, with $u_c = ND/2$. Increasing bed moisture and size and speed and angle of drums and discs will increase the rate of consolidation. Increasing residence time through lower feed rates will increase the extent of consolidation. With disc granulators, residence time can be increased by increasing bed depth (controlled by bottom inserts), raising disc speed, or lowering disc angle. With drum granulators, residence time can be influenced by internal baffling.

Moisture Control in Tumbling Granulation Maldistributions in moisture often occur in granulation systems. There are two key sources. One is caused by local variations in spray rate, poor wetting, and fluctuations in solids feed rate. The other is due to induction-like growth (cf. Fig. 21-128), where time is required to work moisture to the surface of granules, and when such moisture is finally available, it is often too much for stable growth to occur. In addition, for such instable formulations, operators may inherently overspray the process, or during scale-up, greater consolidation of granules occurs, again providing excess moisture.

Additives have been explored particularly in the minerals industry to damp out moisture maldistribution. Figure 21-155 illustrates the

TABLE 21-23 Moisture Requirements for Granulating Various Materials

Raw material	Approximate size of raw material, less than indicated mesh	Moisture content of balled product, wt % H ₂ O
Precipitated calcium carbonate	200	29.5–32.1
Hydrated lime	325	25.7–26.6
Pulverized coal	48	20.8–22.1
Calcined ammonium metavanadate	200	20.9–21.8
Lead-zinc concentrate	20	6.9–7.2
Iron pyrite calcine	100	12.2–12.8
Specular hematite concentrate	150	8.0–10.0
Taconite concentrate	150	8.7–10.1
Magnetic concentrate	325	9.8–10.2
Direct-shipping open-pit iron ores	10	10.3–10.9
Underground iron ore	¼ in.	10.4–10.7
Basic oxygen converter fume	1 µ	9.2–9.6
Raw cement meal	150	13.0–13.9
Fly ash	150	24.9–25.8
Fly ash-sewage sludge composite	150	25.7–27.1
Fly ash-clay slurry composite	150	22.4–24.9
Coal-limestone composite	100	21.3–22.8
Coal-iron ore composite	48	12.8–13.9
Iron ore-limestone composite	100	9.7–10.9
Coal-iron ore-limestone composite	14	13.3–14.8

Dravo Corp.

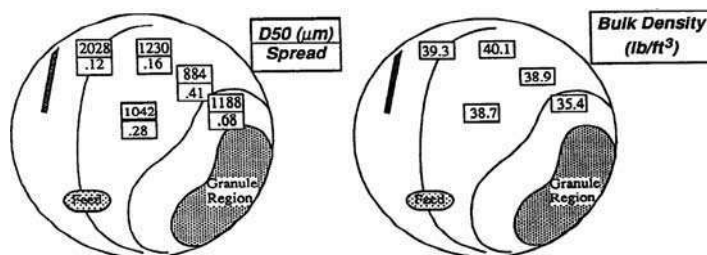


FIG. 21-154 Impact of single nozzle location on granule-size distribution and bulk density for disc granulation, 3-ft diameter, 200 lb/h. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

decrease in drum growth rate for taconite ores that occurs with increasing amounts of bentonite clay [Kapur et al., *Chem. Eng. Sci.*, **28**, 1535 (1973)]. However, what merits particular mention is that the decrease in balling rate is disproportionate with the level of moisture, as illustrated in Fig. 21-156. In other words, high moisture levels are more affected by bentonite level. As illustrated for a bentonite level of 0.23 wt %, a moisture level of 46 vol % behaves or is converted to an equivalent balling rate of 43 vol % at zero bentonite, whereas a moisture level of 44 percent is converted to an equivalent rate of 42.5 percent. In other words, a 2 percent deviation of moisture is converted to a 0.5 percent deviation in moisture in the presence of the bentonite, which as a result narrows the wide variations in balling rate that might otherwise be possible due to moisture maldistribution. In addition, the overall balling rate is slowed, making the granulation process more controllable. Although unexplored, similar effects would be expected in fluid-bed and mixer granulation.

Granulator-Dryers for Layering and Coating Some designs of tumbling granulators also act as driers specifically to encourage layered growth or coating and discourage coalescence or agglomeration, e.g., the fluidized drum granulator [Anon, *Nitrogen*, **196**, 3-6 (1992)]. These systems have drum internals designed to produce a falling curtain of granules past an atomized feed solution or slurry. Layered granules are dried by a stream of warm air before circulating through the coating zone again. Applications are in fertilizer and industrial chemicals manufacture. Analysis of these systems is similar to that of fluidized-bed granulator-dryers.

In the pharmaceutical industry, pan granulators are still widely used for coating application. Pans are only suitable for coating relatively

large granules or tablets. For smaller particles, the probability of coalescence is too high.

Relative Merits of Disc vs. Drum Granulators The principal difference between disc and drum granulators is the classifying action of the disc, resulting in disc granulators having narrower exit granule-size distributions than do drums. This can alleviate the need for product screening and recycle for disc granulators in some industries. For industries with tight granule-size specifications, however, recycle rates are rarely more than 1:2 compared to drum recycle rates often as high as 5:1. The classified mixing action of the disc affects product bulk density, growth mechanisms, and granule structure as well. Generally, drum granulators produce denser granules than disks. Control of growth mechanisms on discs is complex, since regions of growth overlap and mechanisms compete. Both layered and partially agglomerated structures are therefore possible in disc granulators (Fig. 21-154). Other advantages claimed for the disc granulator include low equipment cost, sensitivity to operating controls, and easy observation of the granulation/classification action, all of which lend versatility in agglomerating many different materials. Dusty materials and chemical reactions such as the ammonization of fertilizer are handled less readily in the disc granulator than in the drum.

Advantages claimed for the drum granulator over the disc are greater capacity, longer retention time for materials difficult to agglomerate or of poor flow properties, and less sensitivity to upsets in the system due to the damping effect of a large recirculated load.

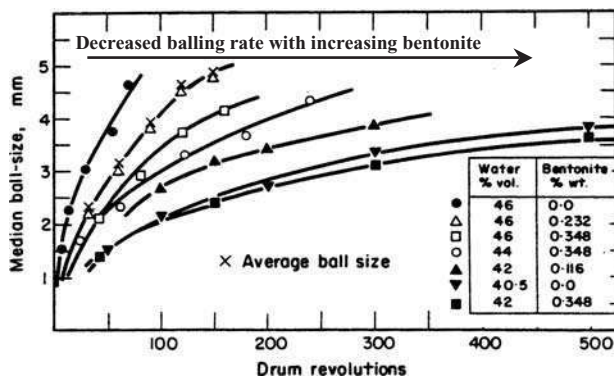


FIG. 21-155 Median ball size vs. drum revolutions for the granulation of taconite ore for varying moisture and bentonite clay levels. [Kapur et al., *Chem. Eng. Sci.*, **28**, 1535 (1973).]

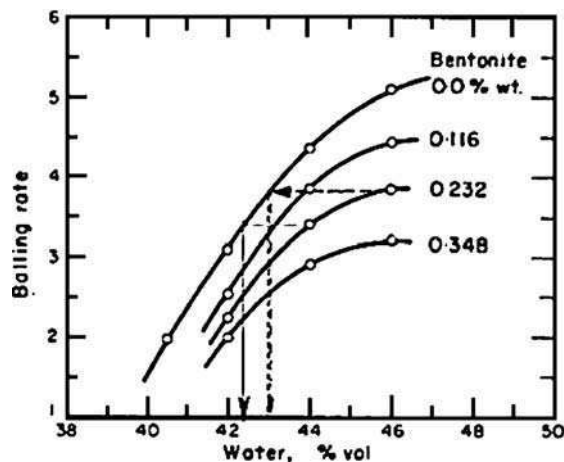


FIG. 21-156 Balling rate for drum granulation of taconite ore as a function of moisture and bentonite clay levels. [Kapur et al., *Chem. Eng. Sci.*, **28**, 1535 (1973).]

TABLE 21-24 Dimensions and Scale-up Rules for Tumbling Granulators

Inclined disc	Drum
Throughput or capacity: $Q \propto D^2$ (area)	$Q \propto LD^2$ (10–20% of volume)
Power consumption: $P \propto D^2$ (area)	$P \propto LD^2$ (volume)
Speed: 50–75% of N_c	30–50% of N_c
Angle: 40–70° from vertical (45–55° most common)	0–10° from horizontal
Height or Length: 20% of D (10–30% of D range)	2–5 times D

Disadvantages are high recycle rates that can promote limit cycle behavior or degradation of properties of the product.

Scale-up and Operation Typical dimensions and scale-up rules for discs and drums are summarized in Table 21-24. Table 21-16 summarizes key groups governing scale-up. For drum granulators, powder consumption and capacity scale with volume, allowing greater throughput than is possible with discs, where capacity scales with area. Although powder properties are not strictly taken into account, similar mixing patterns may be maintained to a first approximation with scale-up by maintaining constant Froude number. In this case, the peripheral speed and velocity of the drum scale with diameter as

$$Fr = \frac{DN^2}{g} = \text{const} \Rightarrow N \propto D^{-0.5} \quad \text{and} \quad u_c = \frac{ND}{2} \propto D^{0.5} \quad (21-143)$$

As a first approximation to scale-up, it is desirable to maintain similar spray flux, which controls nucleation, and similar Stokes numbers, which control growth (see “Mechanics of Size Enlargement Processes”). Both depend for tumbling granulators on the peripheral velocity. In the case of wetting and nucleation, powder or solids flux past the nozzle will scale with the peripheral speed, or with $D^{0.5}$. On the other hand, spray rate will scale with volume to maintain similar moisture (for geometric similarity D^3), which requires a larger spray area to maintain the same dimensionless spray flux [Eq. (21-103)]. This is most easily achieved by increasing the number of nozzles, which can be estimated for constant flux to be given by

$$\psi_a = \frac{\text{spray flux}}{\text{solid flux}} \propto \frac{D^3}{nBu_c} \propto \frac{D^{2.5}}{nB} = \text{const} \leq 0.2$$

$$\text{or} \quad n \propto \frac{D^{2.5}}{B} \quad (21-144)$$

where n is the number of nozzles and B is the spray width. In practice, fewer nozzles than predicted by Eq. (21-144) will likely be required due to greater consolidation forces or residence time with scale-up.

The impact of scale on growth and consolidation depends upon the choice of collisional velocity (Table 21-22). Granule consolidation and density likely increase with the maximum peripheral speed, or scale-up by $D^{0.5}$. For growth, on the basis of a critical viscous Stokes number with a shear assumption for collisional velocity, or $u_c = Nd$, the maximum growth limit scales with

$$d_{\max} \propto \frac{\mu}{\rho u_c} \propto \sqrt{\frac{\mu}{\rho N}} \propto \sqrt{\frac{\mu D^{0.5}}{\rho}} \quad (21-145)$$

when constant Froude number is maintained.

Other key operational issues are to employ gravimetric feeding preferred to volumetric to control moisture and upsets in flow patterns.

Fluctuations in mass feed rate may also be matched by changes in spray rates through load cell measurements of flow and feedforward control. Dripping nozzles should be prevented through pressure monitoring as well as wetting of process walls. On-line measurements of moisture by near-infrared and of granule size (up to 9 mm) are possible by laser diffraction and imaging.

MIXER GRANULATORS

Mixer granulators contain an agitator to mix particles and liquid and cause granulation. In fact, mixing any wet solid will cause some granulation, even if unintentionally. Mixer granulators have a wide range of applications including ceramics, pharmaceuticals, agrichemicals, and detergents (Table 21-11), and they have the following advantages:

- They can process plastic, sticky materials and can spread viscous binders. That is, they can operate in the mechanical dispersion regime of wetting and the deformable regime of growth (see “Granulation Rate Processes”).
- They are less sensitive to operating conditions than tumbling and fluid-bed granulators, although associated with this is less understanding of control and scale-up of granulation mechanisms.
- High-intensity mixers are the only type of granulator that can produce small (<2 mm) high-density granules.

Power and maintenance costs are higher than for tumbling granulators. Outside of high-intensity continuous systems (e.g., the Schugi inline mixer), mixers are not feasible for very large throughput applications if substantial growth is required. Granules produced in mixer granulators may not be as spherical as those produced in tumbling granulators, and are generally denser due to higher agitation intensity (see Fig. 21-111). Control of the amount of liquid phase and the intensity and duration of mixing determine agglomerate size and density. Due to greater compaction and kneading action, generally less liquid is required in mixers than in tumbling and fluid-bed granulation. As opposed to tumbling and fluidized-bed granulators, an extremely wide range of mixer granulator equipment is available. The equipment can be divided somewhat arbitrarily into low- and high-shear mixers, although there is considerable overlap in shear rates, and actual growth mechanisms also depend on wet mass rheology in addition to shear rates.

Low-Speed Mixers Low-speed mixers include (1) ribbon or paddle blenders, (2) planetary mixers, (3) orbiting screw mixers, (4) sigma blade mixers, and (5) double-cone or V blenders, operating with rotation rates or impeller speeds less than 100 rpm. (See Fig. 21-157 and subsection “Solids Mixing”.) **Pug mills** and **paddle mills** as well as ribbon blenders are used for both batch and continuous applications. These devices have horizontal troughs in which rotate central shafts with attached mixing blades of bar, rod, paddle, and other designs. The vessel may be of single- or double-trough design. The rotating blades throw material forward and to the center to achieve a kneading, mixing action. Characteristics of a range of pug mills available for fertilizer granulation are given in Table 21-25. These mills have largely been replaced by tumbling granulators in metallurgical and fertilizer applications, but they are still used as a premixing step for blending very different raw materials, e.g., filter cake with dry powder.

Batch planetary mixers are used extensively in the pharmaceutical industry for powder granulation. A typical batch size of 100 to 200 kg has a power input of 10 to 20 kW. Mixing times in these granulators are quite long (20 to 40 min), and many have been replaced with batch high-shear mixers.

High-Speed Mixers High-speed mixers include continuous shaft mixers and batch high-speed mixers. **Continuous shaft mixers** have blades or pins rotating at high speed on a central shaft. Both horizontal and vertical shaft designs are available (Figs. 21-158 and 21-159). Examples include the vertical **Schugi™ mixer** and the horizontal **pin or peg mixers**. These mixers operate at high speed (200 to 3500 rpm) to produce granules of 0.5 to 1.5 mm with a residence time of a few seconds, during which intimate mixing of a sprayed liquid binder and fine cohesive feed powder is achieved. However, little time is available for substantial product growth or densification, and the granulated product is generally fine, irregular, and fluffy with low bulk density. Schugi™ and pin mixer capacities may range up to 200 tons/h with

TABLE 21-25 Characteristics of Pug Mixers for Fertilizer Granulation*

Model	Material bulk density, lb/ft ³	Approximate capacity, tons/h	Size (width × length), ft	Plate thickness, in	Shaft diameter, in	Speed, r/min	Drive, hp
A	25	8	2 × 8	¼	3	56	15
	50	15	2 × 8	¼	3	56	20
	75	22	2 × 8	¼	3	56	25
	100	30	2 × 8	¼	3	56	30
B	25	30	4 × 8	¾	4	56	30
	50	60	4 × 8	¾	4	56	50
	75	90	4 × 8	¾	4	56	75
	100	120	4 × 8	¾	5	56	100
C	25	30	4 × 12	¾	5	56	50
	50	60	4 × 12	¾	5	56	100
	75	90	4 × 12	¾	6	56	150
	100	120	4 × 12	¾	6	56	200
	125	180	4 × 12	¾	7	56	300

*Feeco International, Inc. To convert pounds per cubic foot to kilograms per cubic meter, multiply by 16; to convert tons per hour to megagrams per hour, multiply by 0.907; to convert feet to centimeters, multiply by 30.5; to convert inches to centimeters, multiply by 2.54; and to convert horsepower to kilowatts, multiply by 0.746.

power requirements up to 200 kW. Typical plant capacities of pug mixers are 10 to 20 tons/h (Capes, *Particle Size Enlargement*, 1980). Examples of applications include detergents, agricultural chemicals, foodstuffs, clays, ceramics, and carbon black.

Batch high-shear mixer granulators are used extensively in the pharmaceutical industry, where they are valued for their robustness to processing a range of powders as well as their ease of enclosure. Plow-shaped mixers rotate on a horizontal shaft at 60 to 800 rpm, with impeller tip speeds of the order of 10 ms^{-1} . Most designs incorporate an off-center high-speed cutter or chopper rotating at much higher speed (500 to 3500 rpm), which breaks down overwetting powder mass and limits the maximum granule size. Scale ranges from 10 to 1200 L, with granulation

times on the order of 5 to 10 min, which includes both wet massing and granulation stages operating at low and high impeller speed, respectively. Several designs with both vertical and horizontal shafts are available (Figs. 21-160). [Schaefer, *Acta. Pharm. Sci.*, **25**, 205 (1988)].

Variations in equipment, impeller, and chopper geometry result in very wide variations in shear rate and powder flow patterns among manufacturers (Fig 21-161). Therefore, great caution should be exercised in transferring formulations and empirical knowledge between mixer designs. For example, the effect of chopper on granule attributes has been observed to be small for Fielder™ and Diosna™ vertical bottom-driven mixer designs, provided the chopper is on (Litster et al., loc. cit., 2002), whereas chopper effects are large in the

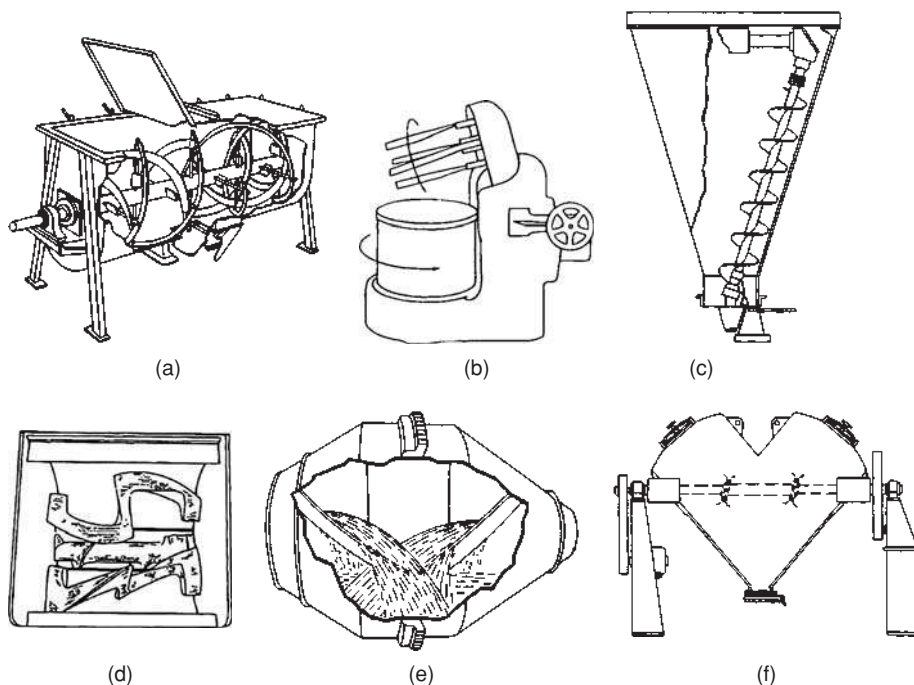


FIG. 21-157 Examples of low-shear mixers used in granulation. (a) Ribbon blender; (b) planetary mixer; (c) orbiting screw mixer; (d) sigma blender; (e) double-cone blender with baffles; (f) v blender with breaker bar. (See also "Solids Mixing.") [(b) and (d), Chirkot and Propst, in Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005.]



FIG. 21-158 The Schugi Flexomix® vertical high-shear continuous granulator. (Courtesy Bepex Corp.)

Lödige™ horizontal plough shear design where tools push material into relatively large chopper zones (Iveson et al., loc. cit., 2000), or in Gral™ top-driven designs, which possess much larger choppers.

The Eirich mixer granulator is a unique design commonly employed in ceramics and clay industries (Fig. 21-161*d*). Here both impeller blades and the chopper rotate on eccentrically mounted vertical shafts, while in addition the cylindrical bowl rotates. This provides a hybrid of mixer and tumbling granulation, and generally more spherical granules are produced than those achieved in other batch mixer processes.

Powder Flow Patterns and Scaling of Mixing Powder flow properties have a significant impact on high-shear mixing, particularly

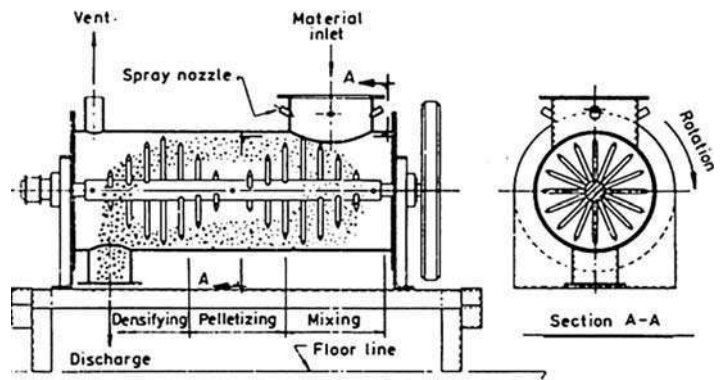
in the presence of cohesive powders and moisture. Frictional properties play a key role as in bulk solids flow (cf. “Bulk Flow Properties” subsection), as well as particle inertia and two-phase interactions with interstitial gas. As yet, the rheological behavior of rapid powder flow under shear is only partly understood in an engineering design sense. However, as with bulk solids flow, powder beds under shear do not readily transmit shear stress, and often develop shear zones in which large shear velocities exist between more relatively stagnant masses of powder. Put another way, traditional scaling approaches based on liquid mixing should be used with great caution. This greatly impacts scale-up since the entire mass of the powder bed is not activated, with only a portion of the bed being sheared and mixed, and this activated portion generally varies over the life of the granulation. In some cases, small shear zones between blades and vessel walls could dominate growth. Furthermore, such powder mixing is very sensitive to equipment geometry, varying widely among manufacturers.

Key visual studies and mixing studies have been undertaken to attempt to understand equipment interactions [Litster et al., *Powder Technol.*, **124**, 272 (2002); Ramaker et al., *Int. J. Pharm.*, **166**, 89 (1998); Knight et al., *Chem. Eng. Sci.*, **56**, 4457 (2001); Forest et al., *Proc. World Congress of Chem. Eng.*, Melbourne, 2001]. In the case of horizontal plough shears, large masses were found to cycle around the shaft undergoing little internal shear, with high shear occurring between the masses and at the wall. In the case of vertical batch mixers, two regimes were observed as illustrated in Figs. 21-162 and 21-163. At low impeller speed, the blade slips along the bottom surface of the powder, with powder being displaced upward momentarily in a **bumping flow** regime. Little turnover of the bed occurs, with rotational motion or **powder surface velocity** increasing with increasing impeller speed. At high impeller speed, material is forced upward along the walls tumbling downward along the bed surface toward the central impeller axis in a **roping flow** regime. There is significant rotational motion of the bed with good turnover, with the entire bed being activated. In the case of Fielder™ mixer design, the powder surface velocity was observed to remain approximately constant (Litster et al., loc. cit.) indicating that slip and substantial shear may still be occurring between the impeller blade and solid mass. In addition, note that the surface velocity of the powder is typically no more than 10 percent of the impeller tip speed (Fig. 21-163).

The transition between regimes occurs at a critical velocity N_c or critical Froude number $Fr_c = DN_c^2/g$, representing a shift in the balance between gravity and powder rotational inertia. Fr_c is expected to be an increasing function of dimensionless bed height and a decreasing function of powder cohesion, although this has yet to be confirmed. Generally it should decrease with increasing bed moisture due to sprayed binder fluid, potentially giving a shift between regimes occurring during the granulation. Furthermore, Knight et al. (loc. cit.) found the mixer torque measurements scaled with



(a)



(b)

FIG. 21-159 Examples of horizontal high-shear mixers. (a) CB 75 horizontal pin mixer (Courtesy Lödige GmbH). (b) Peg granulator [Capes, Size Enlargement, 1985].

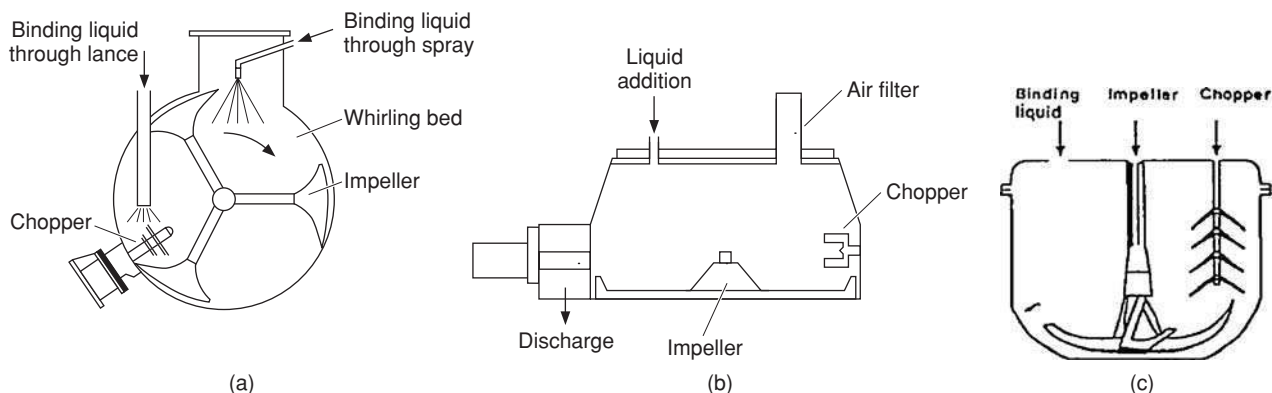


FIG. 21-160 High-shear mixer granulators for pharmaceutical granule preparation for subsequent tableting. (a) Horizontal plough shear, (b) vertical bottom-driven shear, and (c) vertical top-driven shear.

Froude number, potentially providing a scale-up rule for mixer granulation.

Controlling Granulation Rate Processes Despite their robustness with respect to processing a wide variation in formulation types, mixers present the greatest challenge for predicting granule properties of all granulation techniques. All rate processes often play a role, and they are often intertwined (see “Agglomeration Rate Processes and Mechanics”). Granule deformation is important due to the high-shear forces existing in mixers as compared with tumbling and fluid-bed granulators. As deformability is linked to granule saturation and voidage, consolidation and growth are highly coupled. High-speed choppers also bring about significant granule breakage, both wet and dry, often providing a limit on granule growth. Furthermore, a very wide range of shear rates and impact velocities exist throughout the bed, which vary over the life of the granulation, are strongly impacted by powder flow properties, and are equipment-specific (see “Powder Flow Patterns and Scaling of Mixing”). To further complicate matters, geometric similarity is often not preserved for commercial equipment and dominates rate processes by shift with scale-up. It is not surprising there are wide opinions about control of such processes and their method of scale-up.

Some key questions to be addressed as part of formulation, processing, and scale-up efforts include these:

1. Is the formulation wettable with fast drop penetration time? Is there possible preferential wetting of active vs. excipients? Is little growth required, applying only a nucleation stage of granulation?
2. If substantial growth is required, what is the deformability of the formulation? Does it readily grow, or is the formulation stiffer, requiring an induction time to work moisture to the surface? How much moisture is required, and how does it impact deformability?
3. What is the relative volume occupied by the chopper? If large growth is desired, does the formulation reach its maximum limit of granule size during latter stages of growth?
4. What granule density is required, and what is its impact on downstream processing such as tableting and final product quality? Is additional processing required to densify wet granules, e.g., by second-stage mixing or fluid-bed drying?
5. Is dry granule attrition occurring in the process, or is the product prone to dust formation?

These questions are now addressed within the context of rate processes. See also Tables 21-15 to 21-19.

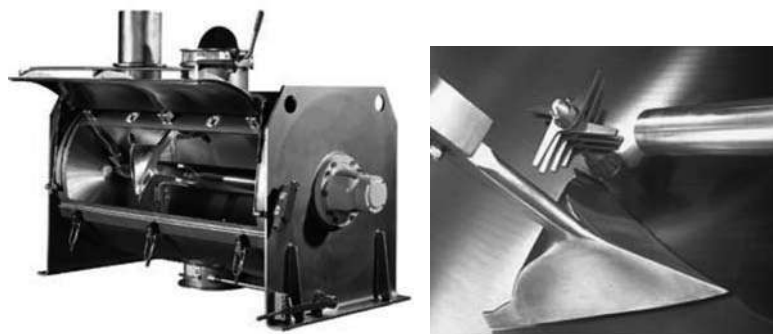
If little growth is desired, granulation may be halted early in growth, or limited to **nucleation**. If the formulation is wettable by the binding fluid, this is best achieved in the **drop-controlled** regime of nucleation, which requires fast drop penetration and low spray flux [Eq. (21-107)], Fig. 21-107. Small drop penetration time t_p is possible for low binder viscosity, high adhesion tension, coarse feed powders, and

fine drops for a given bed circulation time [Eq. 21-99]. Small dimensionless spray flux $\psi_a < 0.2$ occurs for low spray rate for a given area and high solids flux past the spray zone, which is influenced by the number of nozzles and surface solids velocity. Drop-controlled growth gives the tightest nuclei distribution, as illustrated in Fig. 21-164 for commercial mixers. For large spray rates, pumping of binding solution, viscous binders, unwettable powder, or poor powder mixing, nucleation will occur by **mechanical dispersion** of the fluid and breakdown of overwet masses by impellers and choppers, resulting in wider nuclei size distributions.

Large solids fluxes may be necessary for drop-controlled nucleation, implying operating the mixer in the roping regime of mixing for vertical mixers in excess of a critical velocity or Froude number Fr_c (see “Powder Flow Patterns and Scaling of Mixing”). However, it is quite possible that this might also set off simultaneous growth. In practice, binding fluid is often added at low impeller speed (bumping flow), followed by wet massing at high speed (roping fluid). This interaction between initial nucleation, subsequent growth, and powder mixing is product-specific and illustrates the source of current disagreement behind the method of binder addition. For a readily growing formulation, operating in roping flow during binder addition will readily initiate growth, whereas operating in bumping flow may lead to wider nuclei distributions.

In most mixers, granule grows by a **high-agitation, high-deformability** mechanism, where deformability cannot be ignored (see “Growth and Consolidation” and Figs. 21-109 and 21-110). An example of such growth in mixers is detailed above (see Example 5 High-Shear Mixer Growth.) **Granule growth** and **granule consolidation** are initially controlled by the deformability of the wet mass. This **deformability** is a strong function of saturation [Eq. (21-120), Figs. 21-112 and 21-113] as well as shear level as represented by the deformation Stokes number St_{def} [Eq. (21-118)]. Saturations of 80 to 100 percent are generally required to initiate growth, although this is formulation-dependent. Ideally plastic deformation of the wet mass is desired without crumbling. Overall deformability increases with increasing primary particle size, decreasing binder viscosity, and increasing granule voidage, implying fresh granules are more deformable and capable of growth than older, compacted granules.

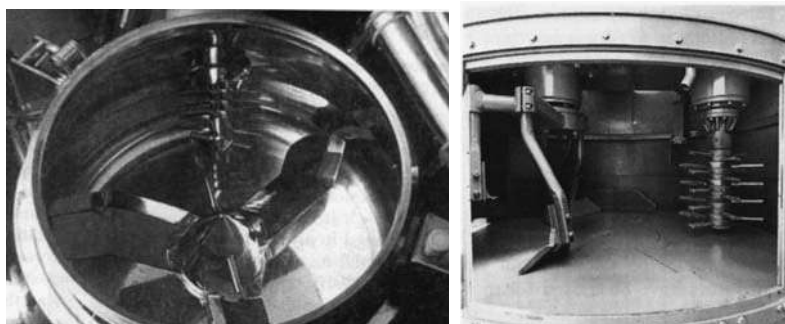
Initial growth in mixing is associated with compaction and decreases in granule porosity (Fig. 21-128). Granule size should increase with increasing St_{def} , implying increasing impeller speed, granule density, and bed moisture and decreasing solution viscosity and wet mass deformability. In this stage, granule-size distribution is expected to increase in proportion to granule size in a **self-preserving** fashion (Adetayo and Ennis, loc. cit., 1996). However, the material may reach a **limiting granule size**, in which case the distribution will then narrow. This limit is strongly dependent on chopper and impeller speed and mixer design. In the later stages of growth, granules



(a)



(b)



(c)

(d)

FIG. 21-161 Commercial examples of available high-shear granulators. (a) Horizontal plough shear; FKM 1200 with plough shovel tools, chopper, and liquid addition lance. (Courtesy Loedige Corp.) (b) Vertical top-driven shear; GMX top-driven mixer with mixing impeller blade. (Courtesy Vector Corp.) (c) Vertical bottom-driven shear; Glatt VG bottom-driven mixer with mixing impeller and chopper blade. (Courtesy Glatt GmbH.) (d) Eccentric vertical shear mixer; Eirich vertical mixer with rotation bowl and eccentric rotating blade and chopper. (Courtesy Eirich GmbH.)

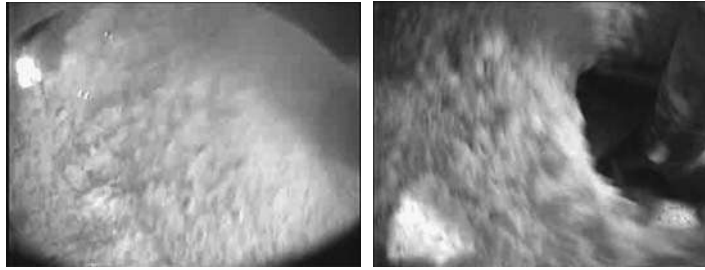


FIG. 21-162 Examples of bumping and roping flow in commercial batch high-shear mixers. *Left:* Bumping flow at low impeller speed during spray cycles. *Right:* Roping flow at high impeller speed in wet massing, GMX top-driven mixer. (Courtesy Vector Corporation.)

may become less deformable, switching to a nondeformable method of growth as with tumbling granulation. Here, the limit of growth will increase with increasing binder viscosity and decrease with increasing impeller speed [cf. Eq. (21-145) and Figs. 21-121 and 21-122].

Scale-up and Operation As discussed above, scale-up of high-shear mixers is difficult due to complex powder flow patterns, wide variations in shear rates among mixers, and competition among granulation rate processes. Ideally, the following should be preserved with scale-up.

1. Geometric similarity, including dimensionless powder bed height
2. Dimensionless spray flux and drop penetration time
3. Collisional shear velocities in terms of viscous and deformation Stokes numbers
4. Constant maximum collisional velocities controlling breakage

In practice, it may not be possible to meet all these objectives, and only limited understanding exists with regard to the best choice of shear rates in scaling. Table 21-16 summarizes key governing groups governing scale-up.

For many commercial mixers, since geometric similarity is often not preserved—even within a given manufacturer—Kristensen recommends constant relative swept volume ratio $\dot{V}$ as a scale-up parameter as a starting point, defined as

$$\dot{V}_R = \frac{\dot{V}_{\text{imp}}}{V_{\text{tot}}} \quad (21-146)$$

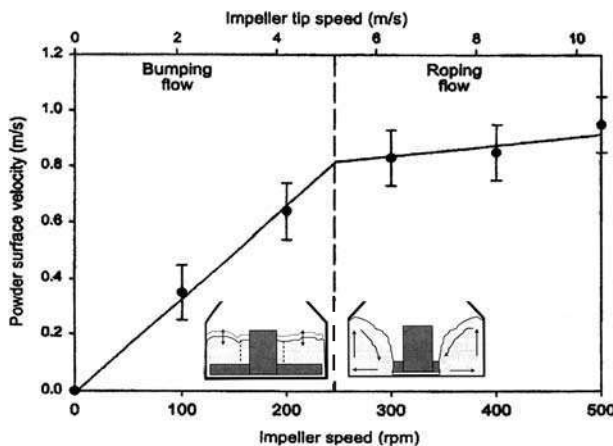


FIG. 21-163 Powder surface velocities as a function of impeller tip speed. Dry lactose in a 25-L Fielder™. (Litster and Ennis, *The Science and Engineering of Granulation Processes*, Kluwer, 2004.)

where $\dot{V}_{\text{imp}}$ is the volume rate swept by the impeller and V_{tot} is the total volume of the granulator. For constant geometric similarity (where impeller diameter would scale with bed diameter), maintaining constant relative swept volume is equivalent to maintaining constant tip speed, or

$$u_o = ND = \text{const} \quad \text{or} \quad N \propto 1/D \quad (21-147)$$

where N and D are impeller speed and diameter, respectively. Depending on mixer design, relative swept volume may decrease significantly with scale when similarity is not preserved (Schaefer, loc. cit.), requiring increases in impeller speed over Eq. (21-147) with scale-up to compensate. Alternatively, dimensionless bed height might be reduced to maintain a constant swept mass rate. In reality, maintaining a constant percentage of swept mass rate is required, but this is complicated by complex powder flow patterns and cannot be easily achieved in practice with many impeller designs. In general, scale-up leads to poorer liquid distribution, higher-porosity granules, and wider granule-size distributions. Required granulation time may increase with scale although this depends on the importance of consolidation kinetics, as discussed above.

To improve upon maintaining swept volume as a scale-up criterion, one must also consider the impact of scale-up on wetting, nucleation, growth, and granule consolidation. To maintain bed moisture with scale-up requires that the sprayed binding fluid increase with the bed volumetric scaling ratio β , or

$$V_2 = \dot{V}_2 \Delta t_2 = \beta V_1 = \beta \dot{V}_1 \Delta t_1 \quad \text{where} \quad \beta = V_2^B / V_1^B \quad (21-148)$$

where V^B is the bed volume. This requires that either the spray time Δt_s or the spray rate $\dot{V}$ be increased. To maintain constant wetting in the case of drop-controlled nucleation, spray flux and penetration time must be maintained constant [Eqs. (21-103) and (21-107)]. If drop size is maintained, a constant spray flux constrains the number of nozzles n required upon scale-up, or

$$\Psi_a = \frac{3}{2} \frac{\dot{V}}{d_a w B} = \text{const} \Rightarrow \Psi_{a2} = \frac{\beta \dot{V}_{s1} (n_1 \Delta t_1 / n_2 \Delta t_2)}{w_2 B_2} = \Psi_{a1} \quad (21-149)$$

Therefore to maintain constant spray flux with increasing scale, one may increase the number of nozzles, spray time, spray width, or solids velocity through the spray zone. Spray width would be altered by nozzle height on nozzle design, whereas solids velocity is related to impeller speed. In bumping flow, solids velocity most likely increases with tip speed for horizontal mixers, whereas in roping flow it is constant. A decision about which regime of mixing is desired must be reached. If roping flow is chosen, $Fr > Fr_c$ (see "Powder Flow Patterns and Scaling of Mixing") maintains this regime of mixing with scale-up.

Equation (21-149) assumes a very fast drop penetration time, and that spray zones do not interact, or multiple passes do not cause drop overlap on previous nuclei. To compensate for this, smaller fluxes may

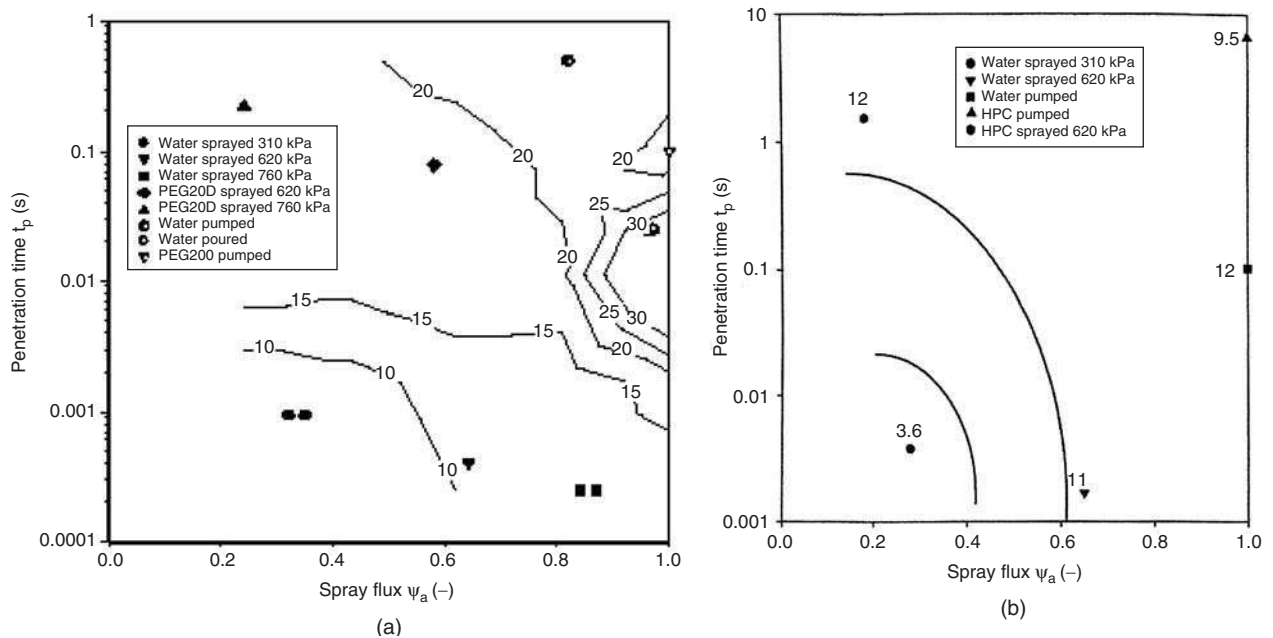


FIG. 21-164 Nucleation regime maps for high-shear batch mixing of lactose. Standard deviations in granule-size distribution are indicated by contour and directly. (a) 6L Hobart for 10 s [Hapgood, *loc. cit.*, 2000]. (b) 25L Fielder, 15% liquid content [Hapgood *et al.*, *AIChE J.*, **49**, 350, (2003)].

be desirable with scale-up. Furthermore, greater compaction forces at large scale may require less moisture, or an effectively smaller β , requiring further adjustment in spray parameters.

For nonwetttable powders or viscous binders, nucleation will occur through mechanical distribution of the fluid, and spray nozzles are likely unwarranted. Instead fluid should be added upstream and in the vicinity of chopper locations, where choppers will break down over-wetted nonuniform masses. Little is known at present about such scaling, although maintaining constant chopper swept volume with scale-up is desirable.

For deformable growth and consolidation where a limit of growth is not achieved, a constant deformation Stokes number St_{def} [Eq. (21-118)] should be maintained with scale-up. Depending upon the choice of impact velocity u_o (Table 21-22) and assuming a constant profile of developing yield stress Y , this leads to different scaling criteria as follows:

$$\text{Constant tip speed: } u_o = ND \Rightarrow N \propto 1/D \quad (21-150a)$$

$$\text{Constant granular shear: } u_o = Nd \Rightarrow N = \text{const} \quad (21-150b)$$

$$\text{Constant Froude number: } Fr = DN^2/g \Rightarrow N \propto 1/\sqrt{D} \quad (21-150c)$$

The last criterion of maintaining constant Froude number will lead to an increase in deformation Stokes number if the tip speed controls growth, or a decrease if a granular shear velocity is a more appropriate choice. A conservative recommendation is to scale with constant Fr for deformable growth systems far from a limit of growth. Note that if scaling is performed on the basis of constant tip speed, Froude number will decrease with scale-up, and could alter mixing regimes from roping to bumping flow (see “Bulk Flow Properties”). Lastly, any changes in speed could alter bulk capillary number and dynamic yield stress Y [Eq. (21-112), Fig. 21-118].

For systems approaching a limiting granule growth with less deformation, constant viscous Stokes number should be maintained. This does not alter the scaling criteria based on the deformation Stokes

number [Eq. (21-118)]. Depending on the most appropriate measure of impact velocity, the limit of growth is given by

$$\text{Constant tip speed: } u_o = ND \Rightarrow d_{max} \propto \mu/\rho ND \quad (21-151a)$$

$$\text{Constant granular shear: } u_o = Nd \Rightarrow d_{max} \propto \sqrt{\mu/\rho N} \quad (21-151b)$$

If constant Froude number scaling is chosen, these limits become:

$$d_{max} \propto \begin{cases} \mu/\rho\sqrt{D} & \text{tip speed} \\ \sqrt{\mu D^{0.5}/\rho} & \text{granular shear} \end{cases} \quad (21-152)$$

It is also possible to stage the growth process, allowing for changes in both impeller and chopper speed. Three stages to consider would be nucleation, initial deformable growth, and final equilibrium growth. For example, one might choose constant-Froude-number scaling for nucleation and initial deformable growth, followed by a decrease in impeller speed to constant tip speed in the later stages of growth.

Staged growth is also possible by staging granulation processes as performed in detergent manufacture (Fig. 21-165), consisting of a nucleation stage at high shear and short residence time in a pin mixer followed by a growth and consolidation stage at moderate shear and long residence time in a plough shear mixer. Feed and exit granule distributions are given in Fig. 21-166.

Power dissipation can lead to temperature increases of up to 40°C in the mass. Note that evaporation of liquid as a result of this increase needs to be accounted for in determining liquid requirements for granulation.

Impeller shaft **power intensity** (kW/kg) has been used both as a rheological tool to characterize formulation deformability and as a control technique to judge granulation endpoint, primarily due to its relationship to granule deformation [see Kristensen *et al.*, *Acta. Pharm. Sci.*, **25**, 187 (1988), and Holm *et al.*, *Powder Technol.*, **43**, 225 (1985)]. Swept volume ratio is a preliminary estimate of expected

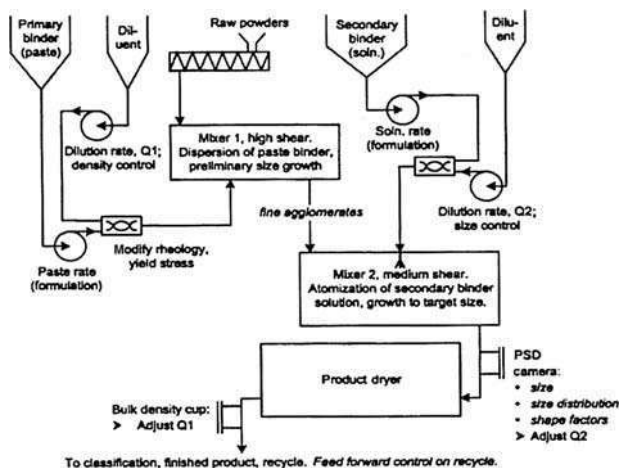


FIG. 21-165 Two-stage continuous granulation process, consisting of high-shear nucleation (pin mixer) and moderate shear growth and consolidation (plough shear). [Mort et al., Powder Technol., **117**, 173 (2003).]

power intensity. There are two key issues associated with this approach: (1) The portion of the bed mass activated may only be local to the impeller, and this percentage of the bed changes with impeller speed, life of the granulation, and scale-up. (2) The relationship between deformability, growth, and power will vary from lot to lot if there are variations in physical properties of the formulation such as surface chemistry or size distribution. There have been attempts to account for these variations through the use of specific work [Sirois and Craig, *Pharm Dev. Technol.*, **5**, 365 (2000)].

FLUIDIZED-BED AND RELATED GRANULATORS

In fluidized granulators (**fluidized beds** and **spouted beds**), particles are set in motion by air rather than by mechanical agitation. Applications include fertilizers, industrial chemicals, agricultural chemicals, pharmaceutical granulation, and a range of coating processes. Fluidized granulators produce either high-porosity granules due to the agglomeration of powder feeds or high-strength, layered granules due to coating of seed particles or granules by liquid feeds.

Figure 21-167 shows a typical production-size **batch** fluid-bed granulator. The air handling unit dehumidifies and heats the inlet air. Heated fluidization air enters the processing zone through a distributor, which also supports the particle bed. Liquid binder is sprayed through an air atomizing nozzle located above, in, or below the bed. Bag filters or cyclones are needed to remove dust from the exit air. Other fluidization gases such as nitrogen are also used in place of or in combination with air to avoid potential explosion hazards due to fine powders. **Continuous** fluid-bed granulators are used in the fertilizer, food, and detergent industries. For fertilizer applications, near-size granules are recycled to control the granule-size distribution. Dust is not recycled directly, but is first remelted or slurried in the liquid feed.

Advantages of fluidized beds over other granulation systems include high volumetric intensity, simultaneous drying and granulation, high heat- and mass-transfer rates, and robustness with respect to operating variables on product quality. Disadvantages include high operating costs with respect to air handling and dust containment and the potential of defluidization due to uncontrolled growth, making them unsuitable generally for very viscous fluid binders or unwettable powders. [See Parikh (ed.) *Handbook of Pharmaceutical Granulation Technology*, 2d ed., (2005) for additional details.]

Hydrodynamics The hydrodynamics of fluidized beds is covered in detail in Sec. 17. Only aspects specifically related to particle-size enlargement are discussed here. Granular products from

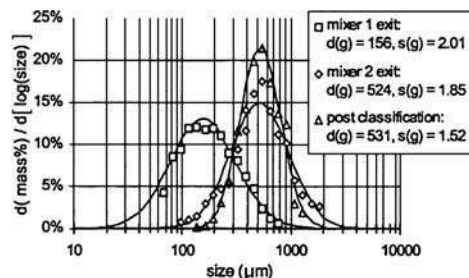


FIG. 21-166 Two-stage continuous granulation process, consisting of high-shear nucleation (pin mixer) and moderate shear growth and consolidation (plough shear). Exit granule size distributions are determined by in-line imaging. [Mort et al., Powder Technol., **117**, 173 (2003).]

fluidized beds are generally **group B** or **group D** particles under Geldart's powder classification. However, for batch granulation, the bed may initially consist of a **group A** powder. For granulation, fluidized beds typically operate in the range $1.5U_{mf} < U < 5U_{mf}$, where U_{mf} is the minimum fluidization velocity and U is the operating superficial gas velocity. For batch granulation, the gas velocity may need to be increased significantly during operation to maintain the velocity in this range as the bed particle size increases.

For groups B and D particles, nearly all the excess gas velocity $U - U_{mf}$ flows as bubbles through the bed. The flow of bubbles controls particle mixing, attrition, and elutriation. Therefore, elutriation and attrition rates are proportional to excess gas velocity. Readers should refer to Sec. 17 for important information and correlations on Geldart's powder classification, minimum fluidization velocity, bubble growth and bed expansion, and elutriation.

In summary however, it is important that mixing, bed turnover, solids flux, bed expansion, shear within the dense phase of the bed, and heat and mass transfer control drying scale with fluid-bed excess gas velocity $U - U_{mf}$.

Mass and Energy Balances Due to the good mixing and heat-transfer properties of fluidized beds, the exit gas temperature is assumed to be the same as the bed temperature, when operating with proper fluidization. Fluidized-bed granulators also act simultaneously as dryers and therefore are subject to the same **mass and energy balance limits** as dryers, namely:

1. Solvent concentration of the atomized binding fluid in the **exit air** cannot exceed the saturation value for the solvent in the fluidizing gas at the bed temperature.

2. The supplied energy in the **inlet air** must be sufficient to evaporate the solvent and maintain the bed at the desired temperature. Both these limits restrict the maximum rate of liquid feed or binder addition for given inlet gas velocity and temperature. The liquid feed rate, however, is generally further restricted to avoid excess coalescence or **quenching**, as defined by low spray flux ψ_a .

Controlling Granulation Rate Processes Table 21-26 summarizes the typical effect of feed properties (material variables) and operating variables on fluidized-bed granulation. (See also Fig. 21-168.) Due to the range of mechanisms operating simultaneously, the combined effect of these variables can be complex. Understanding individual rate processes allows at least semiquantitative analysis to be used in design and operation. See also Tables 21-15 to 21-19 on controlling the individual granulation rate processes of wetting, coalescence and consolidation, and breakage, respectively.

Nucleation in fluidized-bed granulation by necessity occurs within a **drop-controlled** regime, which requires fast drop penetration and low spray flux [Eq. (21-107), Figure. 21-107]. Spray flux ψ_a should be no more than 0.2, and quite possibly much lower. Increasing wettability has been shown to increase nuclei size, presumably due to more stable operation (Fig. 21-99). Figure 21-168 illustrates the impact of increasing spray flux and fluid-bed gas velocity on size distribution. Decreasing dimensional spray flux (which is inverse to

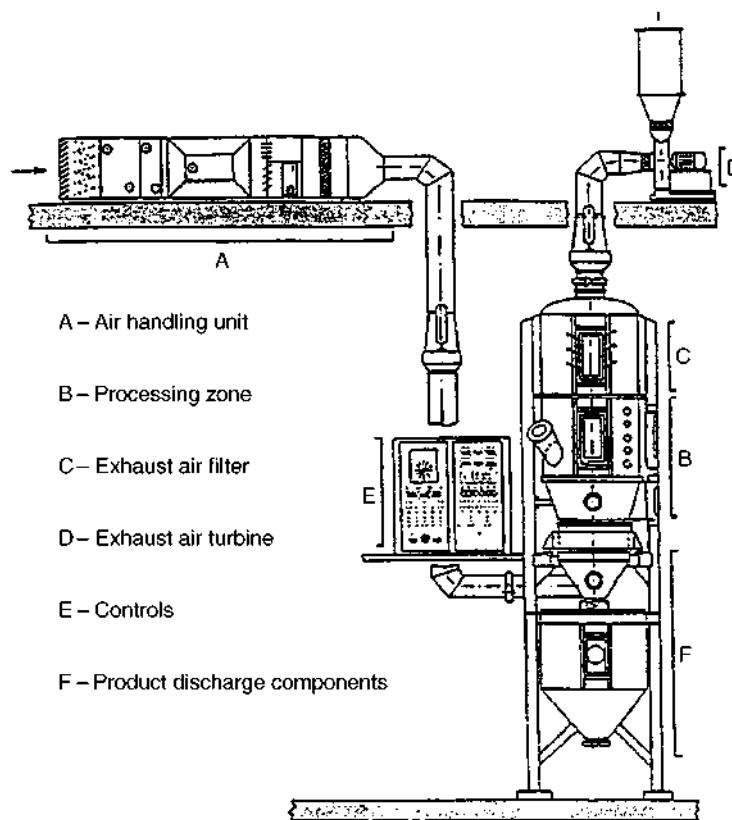


FIG. 21-167 Fluid-bed granulator for batch processing of powder feeds [Ghebre-Sellasie (ed.), *Pharmaceutical Pelletization Technology*, Marcel Dekker, 1989].

TABLE 21-26 Effect of Variables on Fluidized-Bed Granulation

Operating or material variable	Effect of increasing variable
Liquid feed or spray rate	Increases size and spread of granule size distribution Increases granule density and strength Increases chance of defluidization due to quenching
Liquid droplet size (decreases with increasing atomization air or nozzle atomization ratio NAR)	Increases size and spread of granule size distribution
Gas velocity	Increases attrition and elutriation rates (major effect) Decreases coalescence for inertial growth Has no effect on coalescence for noninertial growth, unless altering bed moisture through drying Increases granule consolidation and density
Bed height	Increases granule density and strength
Bed temperature	Decreases granule density and strength
Binder viscosity	Increases coalescence for inertial growth Has no effect on coalescence for noninertial growth Decreases granule density
Particle or granule size	Decreases chance of coalescence Increases required gas velocity to maintain fluidization

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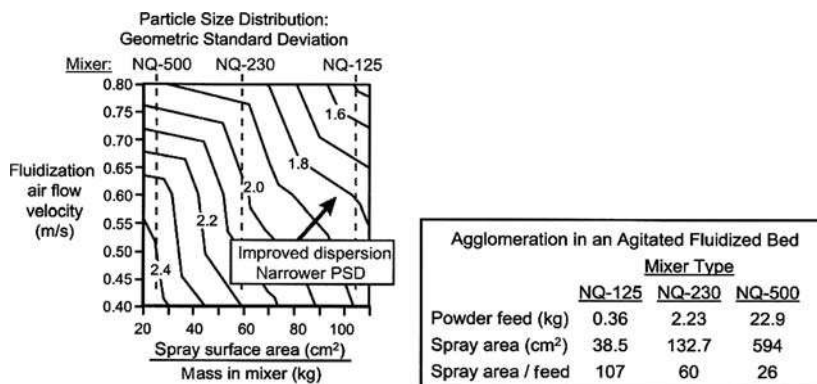


FIG. 21-168 Geometric standard deviation of granule size distribution in an agitated fluid-bed granulator as a function of spray flux, as controlled by fluidization velocity and spray surface area. (Mort et al., 1998; Tardos et al., 1997.)

spray area) leads to a narrower granule-size distribution, as well as increasing fluidization velocity, which increases solids flux through the nozzle zone. In general, bed turnover rate and solids flux increase with increasing $U - U_{mf}$. Dimensionless spray flux ψ_a decreases with decreasing nozzle spray flux and increasing solids flux. In addition, drop penetration time increases with finer drops or increasing atomization ratio.

Air atomizing nozzles are commonly used to control the droplet size distribution independently of the liquid feed rate (Fig. 21-169), as controlled by the nozzle atomization ratio (NAR), given by the volumetric ratio of air-to-liquid flow. **Defluidization** may occur due to large drops and growth. At the other extreme, fine drops with large air volumes may be entrained into the freeboard of the fluid bed, coating bags, vessel wall, leading to caked material, or promoting nozzle dripping.

The formation of large, wet agglomerates that dry slowly is called **wet quenching**, and it is brought about by too high a spray flux and poor drop penetration time. Large, wet agglomerates defluidize, causing channeling, and poor mixing and ultimately leading to shutdown. Sources of wet quenching include high liquid spray rates,

large spray droplets, dripping nozzles, or insufficient mixing due to low excess gas velocity $U - U_{mf}$. It is also linked to the drying capacity of the gas, which should be considered during scale-up. **Dry quenching** (uncontrolled coalescence) is the formation and defluidization of large, stable dry agglomerates, which also may ultimately lead to shutdown and can be related again to high spray flux and/or rapid granule coalescence, particularly in the last stages of growth where the bed is dominated by granules with little available fine powder.

Competing mechanisms of growth include **layering**, which results in dense, strong granules with a very tight size distribution, and **coalescence**, which results in raspberry-like agglomerates of higher voidage. Growth rates range from 10 to 100 $\mu\text{m}/\text{h}$ to 100 to 1000 $\mu\text{m}/\text{h}$ for growth by layering and coalescence, respectively. Layering is generally more prevalent in coating processes or continuous processes with seeded recycle. In terms of coalescence, fluid-bed granulation follows a nondeformable growth model and generally remains within a noninertial regime of growth. Here, growth rate is *not* a function of gas velocity or binder viscosity (Fig. 21-120), and the distribution of spray and the design of the spray zone dominate successful operation. The *final* maximum growth limit, however, does vary in proportion to binder solution viscosity and inversely with gas velocity, as controlled by relative collision velocities within the bed. Possible choices of collision velocity include the relative shear occurring within the dense phase and entrance velocities existing at the distributor. In terms of consolidation, bed height is also critical in terms of compaction forces discussed below. From two-phase fluidization theory (see Dalton et al., 1973, and Sec. 17), the relative shear collisional velocity u_c occurring in the dense phase between bubbles is controlled by bubble velocity U_B and diameter D_B , or

$$u_c = \left(\frac{6U_B}{D_B} \right) d \quad \text{max} \quad \text{and} \quad u_c = \left(\frac{6U_B}{8D_B} \right) d \quad (\text{average}) \quad (21-153)$$

where

$$U_B = 0.71 \sqrt{gD_B} + (U - U_{mf})$$

$$\text{and} \quad D_B = 0.54(U - U_{mf})^{0.4} H^{0.8} / g^{0.2} \quad (21-154)$$

In most cases it may be shown that this collisional velocity is a weakly increasing function of excess gas velocity and bed height. As a general rule, increasing excess velocity $U - U_{mf}$ decreases overall growth for a number of reasons. It limits the maximum diameter as predicted by Stokes criteria [Eqs. (21-117) and (21-118)]; it lowers spray flux, giving less drop overlap and finer nuclei, but with a tighter distribution; and it increases drying rate during the spray cycle.

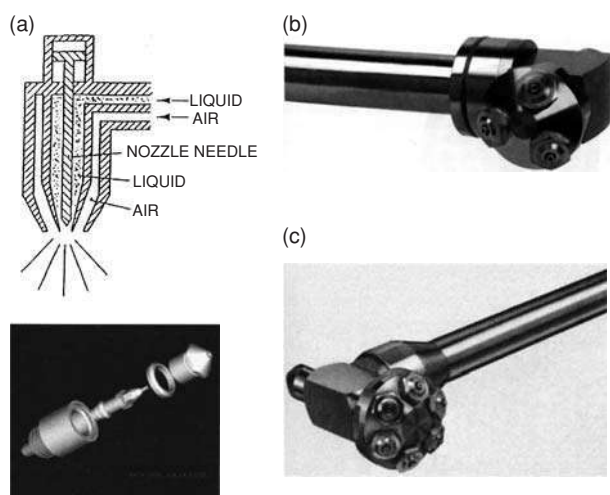


FIG. 21-169 Examples of atomization nozzles. (a) Schematic of single-port nozzle (courtesy Niro Pharma Systems). (b) Three-port nozzle (courtesy Vector Corporation). (c) Six-port nozzle (courtesy Glatt Group). [After Parikh (ed.), Handbook of Pharmaceutical Granulation Technology, 2d ed., 2005.]



FIG. 21-170 Niro™ Precision Granulation process. Agglomeration in bottom-spray draft tube fluid-bed granulation. Scale-up is accomplished by replicating draft tube geometry. (Courtesy Niro Pharma Systems.)

Granule **consolidation**, which controls granule voidage, and hence indirectly granule **breakage**, generally increases with increasing peak bed moisture, increasing bed height, increasing gas velocity, decreasing primary particle size, increasing spray flux and drop size, and decreasing bed temperature. Increasing bed temperature and gas velocity increases the drying rate, thereby lowering bed moisture, leading to more porous, weaker granules. Weaker granules, increased bed turnover rate, and greater distributor velocities with increasing $U - U_{mf}$ can increase attrition and lower average growth. Note that high collisional velocities and bed height can also promote nonuniform granule density, or shell formation, which can impact redispersion and attrition behavior.

Scale-up and Operation Spray nozzles suffer from caking on the outside and clogging on the inside. When the nozzle is below the bed surface, fast capture of the liquid drops by bed particles, as well as scouring of the nozzle by particles, prevents caking. Blockages inside the nozzle are also common, particularly for slurries. The nozzle design should be as simple as possible, and provision for in situ cleaning or easy removal is essential.

Early detection of quenching is important. The initial stages of defluidization are detected by monitoring the bed temperature just above the distributor. A sudden increase (dry quenching) or decrease (wet quenching) indicates the onset of bed defluidization. Wet quenching is avoided by reducing the liquid feed rate and improving the nozzle operation. In situ jet grinding is sometimes used to limit the maximum stable size of dry agglomerates.

Control is accomplished by monitoring bed temperature as well as granule size and density of samples. Temperature may be controlled by adjusting the liquid feed rate or inlet air temperature. For batch granulation, the fluidizing air velocity should be increased during the batch to maintain constant $U - U_{mf}$, accomplished most easily by staged changes. **Bed pressure fluctuations** can be used to monitor the quality of fluidization and to indicate when gas velocity increases are required. In addition, intermittent sampling systems may be employed with **on-line size analyzers** to monitor granule size.

Scale-up of fluid-bed granulators relies heavily on pilot-scale tests. The pilot plant fluid bed should be at least 0.3 m in diameter so that bubbling rather than slugging fluidization behavior occurs. Key in scale-up is the increase in agitation intensity with increasing bed height. In particular, granule density and attrition resistance increase linearly with operating bed height, whereas the rate of granule dispersion decreases.

There are a variety of approaches to scale-up. One is to maintain constant bed height, resulting in similar compaction forces. In this case, peak bed moisture should be maintained to produce similar rates of granule consolidation and growth. In addition, nucleation rate may be kept constant by maintaining spray flux ψ_n . This is accomplished by increasing the number of nozzles in proportion to scale to maintain spray per unit area, and by maintaining constant solids flux through the

spray zone by maintaining constant excess gas velocity $U - U_{mf}$. In essence, this approach simply replicates the bed to achieve scale-up and the basis of cross-sectional area. A recent batch commercial example is the Niro™ Precision Granulation process illustrated in Fig. 21-170. This approach is also directly applicable to continuous processes.

In many batch processes, however, fluid-bed height increases with scale, complicating scale-up. Competing changes in spray flux, drying rate which controls peak bed moisture, and compaction forces scaling with bed height must be balanced. If peak bed moisture is maintained, denser granules will likely result. This may be compensated by increasing gas velocity or raising bed temperature to lower bed moisture with scale-up. The dependence of granule voidage on bed height and moisture must be explored at small scale to determine how much compensation is required. To maintain constant hydrodynamics, Horio et al. (*Proc. Fluidization V*, New York, Engineering Foundation, 1986, p. 151) suggested that excess velocity be increased as

$$(U - U_{mf})_2 = \sqrt{m}(U - U_{mf})_1 \quad (21-155)$$

$$m = (H_2/H_1) \quad (dU_{p,gs}/U_{p,gs}) > 30$$

where geometric similarity of the bed is maintained. The spray rate should be adjusted on the basis of the modified drying rate and desired peak bed moisture desired at scale-up. The number of nozzles can then be determined by maintaining similar spray flux.

Draft Tube Designs and Spouted Beds A **draft tube** is often employed to regulate particle circulation patterns. The most common design is the **Wurster** draft tube fluid bed employed extensively in the pharmaceutical industry, usually for coating and layered growth applications. The **Wurster coater** uses a bottom positioned spray, but other variations are available (Table 21-27).

The **spouted-bed granulator** consists of a central high-velocity **spout** surrounded by a **moving bed annular region**. All air enters

TABLE 21-27 Sizes and Capacities of Wurster Coaters*

Bed diameter, in	Batch size, kg
7	3–5
9	7–10
12	12–20
18	35–55
24	95–125
32	200–275
46	400–575

*Ghebre-Sellase (ed.), *Pharmaceutical Pelletization Technology*. Marcel Dekker, 1989.

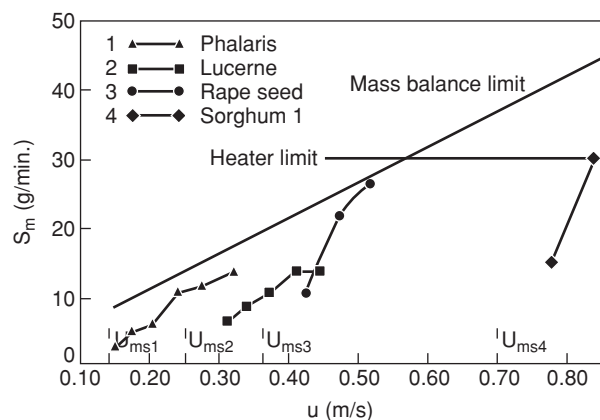


FIG. 21-171 Effect of gas velocity on maximum liquid rate for a spouted-bed seed coater. [Liu and Litster, *Powder Technol.*, **74**, 259 (1993). With permission from Elsevier Science SA, Lausanne, Switzerland.]

through the orifice at the base of the spout. Particles entrained in the spout are carried to the bed surface and rain down on the annulus as a fountain. Bottom-sprayed designs are the most common. Due to the very high gas velocity in the spout, granules grow by layering only. Therefore, spouted beds are good for coating applications. However, attrition rates are also high, so the technique is not suited to weak granules. Spouted beds are well suited to group D particles and are more tolerant of nonspherical particles than a fluidized bed. Particle circulation is better controlled than in a fluidized bed, unless a draft tube design is employed. Spouted beds are difficult to scale past two meters in diameter.

The liquid spray rate to a spouted bed may be limited by agglomerate formation in the spray zone causing spout collapse [Liu and Litster, *Powder Tech.*, **74**, 259 (1993)]. The maximum liquid spray rate increases with increasing gas velocity, increasing bed temperature, and decreasing binder viscosity (see Fig. 21-171). The maximum

liquid flow rate is typically between 20 and 90 percent of that required to saturate the exit air, depending on operating conditions. Elutriation of fines from spouted-bed granulators is due mostly to the attrition of newly layered material, rather than spray drying. The elutriation rate is proportional to the kinetic energy in the inlet air [see Eq. (21-158)].

CENTRIFUGAL GRANULATORS

In the pharmaceutical industry, a range of centrifugal granulator designs are used. In each of these, a horizontal disc rotates at high speed causing the feed to form a rotating **rope** at the walls of the vessel (see Fig. 21-172). There is usually an allowance for drying air to enter around the edge of the spinning disc. Applications of such granulators include spheronization of extruded pellets, dry-powder layering of granules or sugar spheres, and coating of pellets or granules by liquid feeds.

Centrifugal Designs Centrifugal granulators tend to give denser granules or powder layers than fluidized beds and more spherical granules than mixer granulators. Operating costs are reasonable but capital cost is generally high compared to other options. Several types are available including the **CF granulator** (Fig. 21-172) and **rotary fluidized-bed** designs, which allow high gas volumes and therefore significant drying rates (Table 21-28). CF granulator capacities range from 3 to 80 kg with rotor diameters of 0.36 to 1.3 m and rotor speeds of 45 to 360 rpm [Ghebre-Selassie (ed.), *Pharmaceutical Pelletization Technology*, Marcel Dekker, 1989].

Particle Motion and Scale-up Very little fundamental information is published on centrifugal granulators. Qualitatively, good operation relies on maintaining a smoothly rotating **stable rope** of tumbling particles. Operating variables which affect the particle motion are disc speed, peripheral air velocity, and the presence of baffles.

For a given design, good rope formation is only possible for a small range of disc speeds. If the speed is too low, a rope does not form. If the speed is too high, very high attrition rates can occur. Scale-up on the basis of either **constant peripheral speed** ($DN = \text{const.}$), or **constant Froude number** ($DN^2 = \text{const.}$) is possible. Increasing peripheral air velocity and baffles helps to increase the rate of rope turnover. In designs with tangential powder or liquid feed tubes, additional baffles are usually not necessary. The motion of particles in the equipment is also a function of the frictional properties of the feed, so the optimum operating conditions are feed specific.

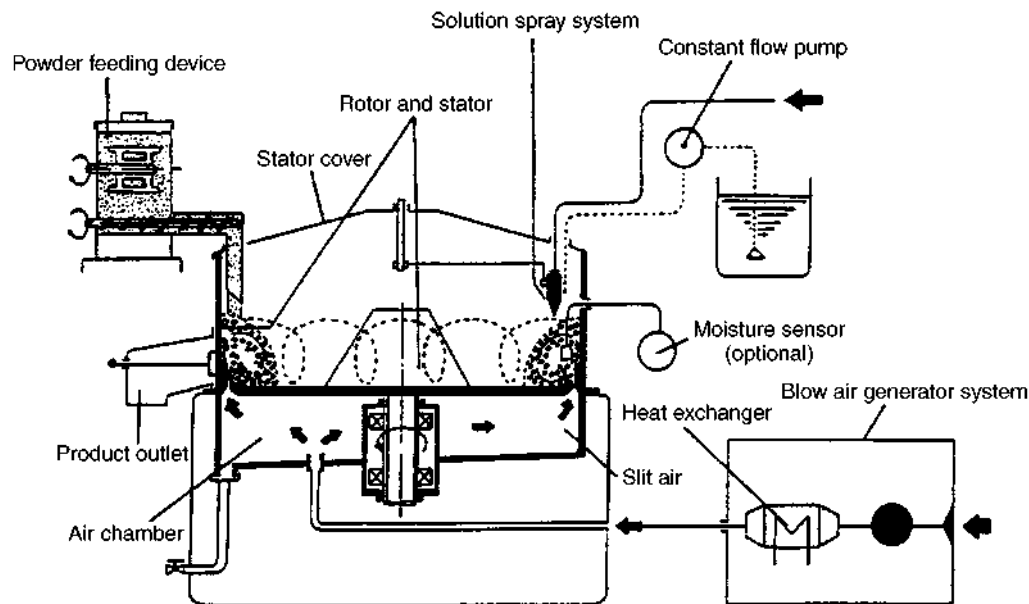


FIG. 21-172 Schematic of a CF granulator. (Ghebre-Selassie, 1989.)

TABLE 21-28 Specifications of Glatt Rotary Fluid-Bed Granulators*

Parameter	15	60	200	500
Volume, L	45	220	670	1560
Fan				
Power, kW	11	22	37	55
Capacity, m ³ /h	1500	4500	8000	12000
Heating capacity, kW	37	107	212	370
Diameter, m	1.7	2.5	3.45	4.0

*Glatt Company, in Ghebree-Selassie (ed.), *Pharmaceutical Pelletization Technology*, Marcel Dekker, 1989.

Granulation Rate Processes Possible granulation processes occurring in centrifugal granulators are extrudate breakage, consolidation, rounding (spheronization), coalescence, powder layering and coating, and attrition. Very little information is available about these processes as they occur in centrifugal granulators; however, similar principles from tumbling and fluid-bed granulators will apply.

SPRAY PROCESSES

Spray processes include **spray dryers**, **prilling towers**, **spouted and fluid beds**, and **flash dryers**. Feed solids in a fluid state (solution, gel, paste, emulsion, slurry, or melt) are dispersed in a gas and converted to granular solid products by heat and/or mass transfer. In spray processes, the size distribution of the particulate product is largely set by the drop size distribution; i.e., **nucleation** is the dominant granulation process, or more precisely **particle formation**. Exceptions are where fines are recycled to coalesce with new spray droplets and where spray-dried powders are rewet in a second tower to encourage agglomeration. For spray drying, a large amount of solvent must be evaporated whereas prilling is a spray-cooling process. Fluidized or spouted bed may be used to capture nucleated fines as hybrid granulator designs, e.g., **fluid-bed spray dryers**.

Product diameter is small and bulk density is low in most cases, except prilling. Feed liquids must be pumpable and capable of atomization or dispersion. Attrition is usually high, requiring fines recycle or recovery. Given the importance of the droplet size distribution, nozzle design and an understanding of the fluid mechanics of drop formation are critical. In addition, heat- and mass-transfer rates during drying can strongly affect the particle morphology, of which a wide range of characteristics are possible.

Spray Drying Detailed descriptions of spray dispersion dryers, together with application, design, and cost information, are given in Sec. 12. Product quality is determined by a number of properties such as particle form, size, flavor, color, and heat stability. Particle size and size distribution, of course, are of greatest interest from the point of view of size enlargement.

Figures 21-173 and 21-174 illustrate typical process and the stages of spray atomization, spray-air contacting and evaporation, and final product collection. A range of particle structures may be obtained, depending on the tower temperature in comparison to the boiling point and rheological properties of the feed (Fig. 21-175). Particles sizes ranging from 3 to 200 μm are possible with two-fluid atomizers producing the finest material, followed by rotary wheel and pressure nozzles.

In general, particle size is a function of atomizer operating conditions and of the solids content, liquid viscosity, liquid density, and feed rate. Coarser, more granular products can be made by increasing viscosity (through greater solids content, lower temperature, etc.), by increasing feed rate, and by the presence of binders to produce greater agglomeration of semidry droplets. Less-intense atomization and spray-air contact also increase particle size, as does a lower exit temperature, which yields a moister (and hence a more coherent) product. This latter type of spray-drying agglomeration system has been described by Masters and Stoltze [*Food Eng.*, 64 (February 1973)] for the production of instant skim-milk powders in which the completion of drying and cooling takes place in vibrating conveyors (see Sec. 17) downstream of the spray dryer.

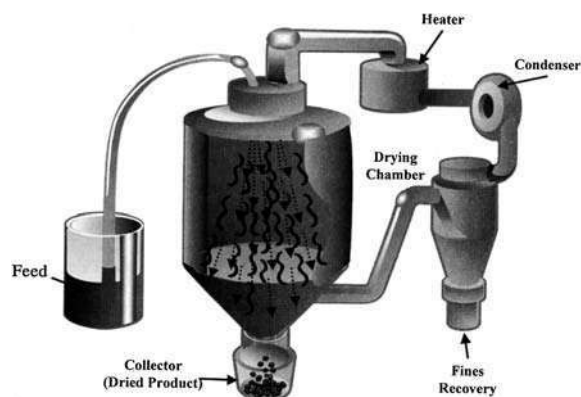


FIG. 21-173 Schematic of a typical spray-drying process. [Çelik and Wendel, in Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005. With permission.]

Prilling The prilling process is similar to spray drying and consists of spraying droplets of liquid into the top of a tower and allowing these to fall against a countercurrent stream of air. During their fall the droplets are solidified into approximately spherical particles or prills which are up to about 3 mm in diameter, or larger than those formed in spray drying. The process also differs from spray drying since the

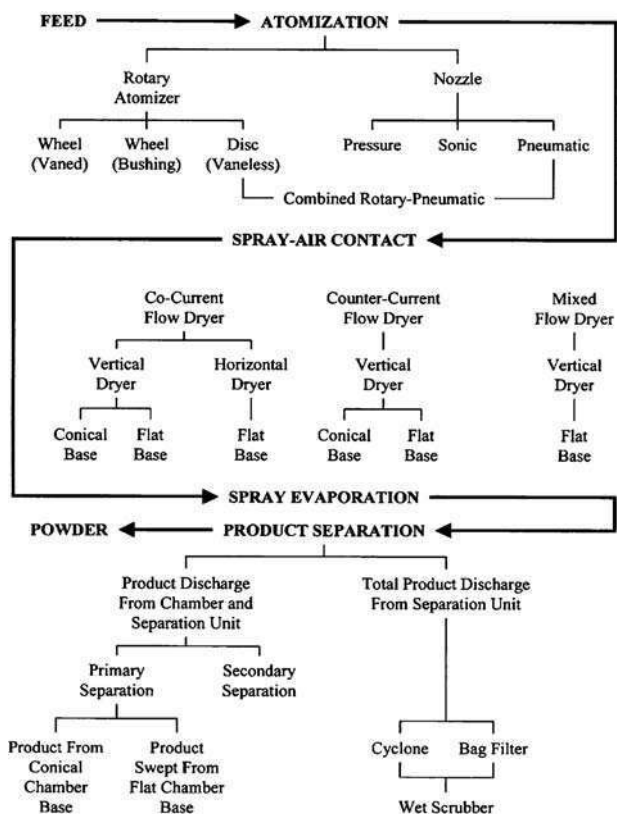


FIG. 21-174 Typical stages of a spray-drying process: atomization, spray-air contact/evaporation, and product collection. (Masters, *Spray Drying Handbook*, 5th ed., Longman Scientific Technical, 1991. With permission.)

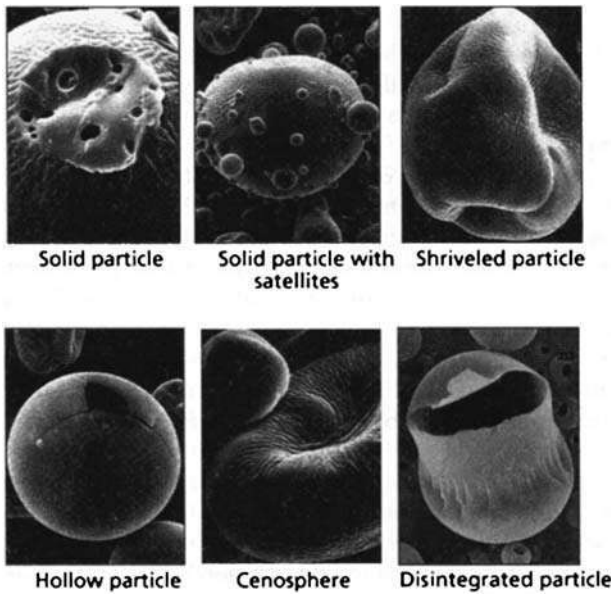


FIG. 21-175 Types of spray-dried particles, depending on drying conditions and feed boiling point. (Courtesy Niro Pharma Systems.)

droplets are formed from a melt which solidifies primarily by cooling with little, if any, contribution from drying. Traditionally, ammonium nitrate, urea, and other materials of low viscosity and melting point and high surface tension have been treated in this way. Improvements in the process now allow viscous and high-melting-point materials and slurries containing undissolved solids to be treated as well.

The design of a prilling unit first must take into account the properties of the material and its sprayability before the tower design can proceed. By using data on the melting point, viscosity, surface tension, etc., of the material, together with laboratory-scale spraying tests, it is possible to specify optimum temperature, pressure, and orifice size for the required prill size and quality. Tower sizing basically consists of specifying the cross-sectional area and the height of fall. The former is determined primarily by the number of spray nozzles necessary for the desired production rate. Tower height must be sufficient to accomplish solidification and is dependent on the heat-transfer characteristics of the prills and the operating conditions (e.g., air temperature). Because of relatively large prill size, narrow but very tall towers are used to ensure that the prills are sufficiently solid when they reach

the bottom. Table 21-29 describes the principal characteristics of a typical prilling tower.

Theoretical calculations are possible to determine tower height with reasonable accuracy. Simple parallel streamline flow of both droplets and air is a reasonable assumption in the case of prilling towers compared with the more complex rotational flows produced in spray dryers. For velocity of fall, see, for example, Becker [*Can. J. Chem.*, **37**, 85 (1959)]. For heat transfer, see, e.g., Kramers [*Physica*, **12**, 61 (1946)]. Specific design procedures for prilling towers are available in the *Proceedings of the Fertilizer Society (England)*; see Berg and Hallie, no. 59, 1960; and Carter and Roberts, no. 110, 1969.

Recent developments in nozzle design have led to drastic reductions in the required height of prilling towers. However, such nozzle designs are largely proprietary, and little information is openly available.

Flash Drying Special designs of pneumatic conveyor dryers, described in Sec. 12, can handle filter and centrifuge cakes and other sticky or pasty feeds to yield granular size-enlarged products. The dry product is recycled and mixed with fresh, cohesive feed, followed by disintegration and dispersion of the mixed feed in the drying air stream.

PRESSURE COMPACTION PROCESSES

The success of **compressive agglomeration** or **pressure compaction** processes depends on the effective utilization and transmission of the applied external force and on the ability of the material to form and maintain permanent interparticle bonds during **pressure compaction** (or **consolidation**) and **decompression**. Both of these aspects are controlled in turn by the geometry of the confined space, the nature of the applied loads, and the physical properties of the particulate material and of the confining walls.

Pressure compaction is carried out in two classes of equipment (Fig. 21-136). These are **dry confined-pressure devices** (molding, piston, tableting, briquetting, and roll presses), in which material is directly consolidated in closed molds or between two opposing surfaces, where the degree of confinement varies with design; and **paste extrusion devices** (pellet mills, screw extruders, table and cylinder pelletizers), in which material undergoes considerable shear and mixing as it is consolidated while being pressed through a die. See Table 21-11 for examples of use. Product densities and pressures are substantially higher than with agitative agglomeration techniques, as shown in Fig. 21-111. For detailed equipment discussion, see also Pietsch (*Size Enlargement by Agglomeration*, Wiley, Chichester, 1992) and Benbow and Bridgwater (*Paste Flow and Extrusion*, Oxford University Press, New York, 1993).

Powder hardness, friction, particle size, and permeability have a considerable impact on process performance and developed compaction pressures. As a general rule, the success of **dry compaction** improves with the following (Table 21-16):

TABLE 21-29 Some Characteristics of a Typical Prilling Operations*

Tower size		
Prill tube height, ft	130	
Rectangular cross section, ft	11 by 21.4	
Cooling air		
Rate, lb/h	360,000	
Inlet temperature	Ambient	
Temperature rise, °F	15	
Melt		
Type	Urea	Ammonium nitrate
Rate, lb/h	35,200 (190 lb H ₂ O)	43,720 (90 lb H ₂ O)
Inlet temperature, °F	275	365
Prills		
Outlet temperature, °F	120	225
Size, mm	Approximately 1 to 3	

*HPD Incorporated. To convert feet to centimeters, multiply by 30.5; to convert pounds per hour to kilograms per hour, multiply 0.4535; °C = (°F - 32) × 5%.

1. **Increased stress transmission**, improving uniformity of pressure throughout the compact. For the case of die-type compaction, transmission increases with decreasing wall or die friction, increasing powder friction, decreasing aspect ratio, and decreasing compact size. Increased stress transmission improves the uniformity of compact density, and decreases residual radial stresses after compact unloading, which in turn lowers the likelihood of capping and delamination and lowers ejection forces.

2. **Decreased deaeration time** of the powder feed. If large deaeration is required, it becomes more likely that air will be entrapped within the die or feed zone, which not only can lower the powder feed rate, but also can result in gas pressurization during compact formation, which can create flaws and delamination during unloading. Relative deaeration time improves with decreasing production rate, increased bulk powder permeability, increased vacuum and forced feeding, and any upstream efforts to densify the product, one example being granulation.

3. **Increased permanent bonding**. Generally this increases with applied force (given good stress transmission), decreasing particle hardness, increased elastic modulus, and decreasing particle size. See also Hiestand tableting indices under "Powder Compaction" subsection.

4. **Increase powder flowability**. Powder feed rates improve with decreasing powder cohesive strength, increasing flow gaps, and increasing bulk permeability. See "Solids Handling: Bulk Solids Flow Characterization."

A range of compaction processes are discussed below, and these rules of thumb generally extend to all such processes in one form or another. See "Powder Compaction" for detailed discussion.

Piston and Molding Presses Piston or molding presses are used to create uniform and sometimes intricate compacts, especially in powder metallurgy and plastics forming. Equipment comprises a mechanically or hydraulically operated press and, attached to the platens of the press, a two-part mold consisting of top (male) and bottom (female) portions. The action of pressure and heat on the particulate charge causes it to flow and take the shape of the cavity of the mold. Compacts of metal powders are then sintered to develop metallic properties, whereas compacts of plastics are essentially finished products on discharge from the molding machine.

Tableting Presses Tableting presses are employed in applications having strict specifications for weight, thickness, hardness, density, and appearance in the agglomerated product. They produce simpler shapes at higher production rates than do molding presses. A single-punch press is one that will take one station of tools consisting of an upper punch, a lower punch, and a die. A rotary press employs a rotating round die table with multiple stations of punches and dies. Older rotary machines are single-sided; that is, there is one fill station and one compression station to produce one tablet per station at every revolution of the rotary head. Modern high-speed rotary presses are double-sided; that is, there are two feed and compression stations to produce two tablets per station at every revolution of the rotary head. Some characteristics of tableting presses are shown in Table 21-30.

For successful tableting, a material must have suitable flow properties to allow it to be fed to the tableting machine. Wet or dry granulation is used to improve the flow properties of materials. In the case of **wet granulation**, agitative granulation techniques such as fluidized beds or mixer granulators as discussed above are often employed.

In **dry granulation**, the blended dry ingredients are first densified in a heavy-duty rotary tableting press which produces "slugs" 1.9 to 2.5 cm ($\frac{3}{4}$ to 1 in) in diameter. These are subsequently crushed into particles of the size required for tableting. Predensification can also be accomplished by using a rotary compactor-granulator system. A third technique, direct compaction, uses sophisticated devices to feed the blended dry ingredients to a high-speed rotary press.

Figure 21-176 illustrates the stations of a typical rotary tablet press of die filling, weight adjustment, compaction, punch unloading, tablet ejection, and tablet knockoff. See "Powder Compaction" for detailed discussion of the impact of powder properties on die filling, compaction, and ejection forces. As discussed above, these stages of compaction improve with increased stress transmission (controlled by lubrication and die geometry), decreased deaeration time (increasing powder permeability and decreasing production rate), increased plastic, permanent deformation, and increased powder flowability (decreasing powder cohesion, increasing flow index, and increased die diameter and clearances).

Excellent accounts of tableting in the pharmaceutical industry have been given by Kibbe [*Chem. Eng. Prog.*, **62**(8), 112 (1966)], Carstensen (*Handbook of Powder Science & Technology*, Fayed & Otten (eds.), Van Nostrand Reinhold Inc., 1983, p. 252), Stanley-Wood (ed.) (loc. cit.), and Doelker (loc. cit.).

Figure 21-177 illustrates typical **defects** that occur in tableting as well as other compaction processes. Lamination or more specifically **delamination** occurs during compact ejection where the compact or tablet breaks into several layers perpendicular to its compression axis. **Capping** is a specific case where a conical endpiece dislodges from the surface of the compact. **Weak equators** are similar to delamination, where failure occurs at the midline of the compact. A key cause of these flaws is poor stress transmission resulting in large radial stresses and wall shear stresses, and it can be improved through lowering wall friction with lubrication or changing the compact aspect ratio. Such flaws occur during compact ejection but also within the compact itself, and they may be hidden, thereby weakening overall compact strength. Note that delamination can often be prevented in split dies, where the residual radial stress is relieved radially rather than by axial ejection. Sticking to punch surfaces or **die fouling** may also contribute to capping and delamination, and it can be assessed through wall friction and adhesion measurements. **Localized cracks** form in complex geometries during both compression and unloading, again due to nonuniform compression related to stress transmission. Small amounts of very hard or elastic material differing from the overall powder bed matrix can cause irregular **spontaneous fracture** of the compact, and it is often caused by recycle material, nonuniform feed, or entrapped air due to high production rate and low feed permeability. **Flashing** and **skirting** leading to a ring of weak material around edges are due to worn punches.

Roll Presses Roll presses compact raw material as it is carried into the gap between two rolls rotating at equal speeds (Fig. 21-178). The size and shape of the agglomerates are determined by the geometry of the roll surfaces. Pockets or indentations in the roll surfaces form briquettes the shape of eggs, pillows, teardrops, or similar forms from a few grams up to 2 kg (5 lb) or more in weight. Smooth or corrugated rolls produce a solid sheet, which can be granulated or broken down into the desired particle size on conventional grinding equipment.

Roll presses can produce large quantities of materials at low cost, but the product is less uniform than that from molding or tableting presses. The introduction of the proper quantity of material into each of the rapidly rotating pockets in the rolls is the most difficult problem in the briquetting operation. Various types of feeders have helped to overcome much of this difficulty.

The impacting rolls can be either solid or divided into segments. Segmented rolls are preferred for hot briquetting, as the thermal expansion of the equipment can be controlled more easily.

Roll presses provide a **mechanical advantage** in amplifying the feed pressure P_0 to some maximum value P_m . This maximum pressure P_m and the roll compaction time control compact density. Generally speaking, as compaction time decreases (e.g., by increasing roll speed), the minimum necessary pressure for quality compacts increases. There may be an upper limit of pressure as well for friable materials or elastic materials prone to delamination.

TABLE 21-30 Characteristics of Tableting Presses*

	Single-punch	Rotary
Tablets per minute	8-140	72-6000
Tablet diameter, in.	$\frac{1}{8}$ -4	$\frac{3}{4}$ -2 $\frac{1}{2}$
Pressure, tons	1 $\frac{1}{2}$ -100	4-100
Horsepower	$\frac{1}{4}$ -15	1 $\frac{1}{2}$ -50

*Browning, *Chem. Eng.*, **74**(25), 147 (1967).

NOTE: To convert inches to centimeters, multiply by 2.54; to convert tons to megagrams, multiply by 0.907; and to convert horsepower to kilowatts, multiply by 0.746.

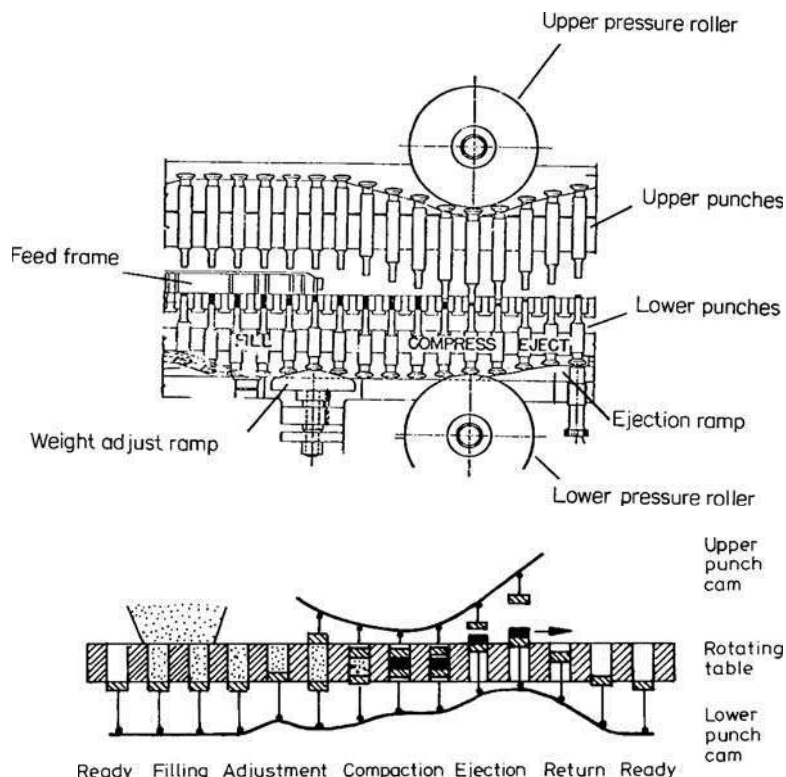


FIG. 21-176 Typical multistation rotary tableting press, indicating stages of tableting for one station. (Pietsch, Size Enlargement by Agglomeration, Wiley, Chichester, 1992.)

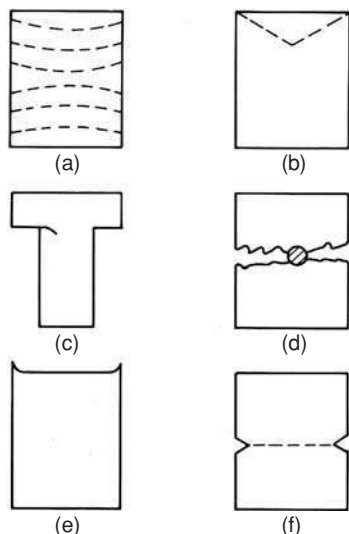


FIG. 21-177 Common defects occurring during tableting and compaction: (a) lamination, (b) capping, (c) localized cracks, (d) spontaneous cracking, (e) flashing or skirting, and (f) weak equator. (Benbow and Bridgwater, *Paste Flow and Extrusion*, Oxford University Press, New York, 1993.)

Pressure amplification occurs in two regions of the press (Fig. 21-178). Above the **angle of nip**, sliding occurs between the material and roll surface as material is forced into the rolls, with intermediate pressure ranging from 1 to 10 psi. Energy is dissipated primarily through overcoming particle friction and cohesion. Below the angle of nip, no slip occurs as the powder is compressed into a compact and pressure may increase up to several thousand psi. Both of these intermediate and high-pressure regions of densification are indicated in the compressibility diagram of Fig. 21-137.

The overall performance of the press and its mechanical advantage (P_m/P_0) depend on the mechanical and frictional properties of the powder. (See "Powder Compaction" subsections.) For design procedures, see Johanson [*Proc. Inst. Briquet. Agglom. Bien. Conf.*, **9**, 17 (1965).] Nip angle α generally increases with decreasing compressibility κ , or with increasing roll friction angle ϕ_w and effective angle of friction ϕ . Powders compress easily and have high-friction grip high in the rolls. The mechanical advantage pressure ratio (P_m/P_0) increases and the time of compaction decreases with decreasing nip angle since the pressure is focused over a smaller roll area. In addition, the mechanical advantage generally increases with increasing compressibility and roll friction.

The most important factor that must be determined in a given application is the pressing force required for the production of acceptable compacts. Roll loadings (i.e., roll separating force divided by roll width) in commercial installations vary from 4.4 MN/m to more than 440 MN/m (1000 lb/in to more than 100,000 lb/in). Roll sizes up to 91 cm (36 in) in diameter by 61 cm (24 in) wide are in use.

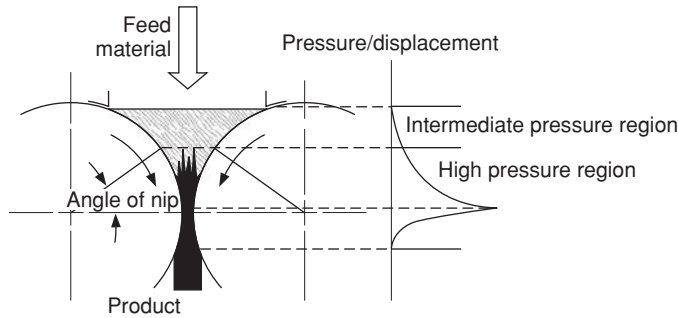


FIG. 21-178 Regions of compression in roll presses. Slippage and particle rearrangement occur above the angle of nip, and powder compaction at high pressure occurs in the nonslip region below the angle of nip.

The roll loading L is related to the maximum developed pressure and roll diameter by

$$L = \frac{F}{W} = \frac{1}{2} f P_m D \propto P_m D^{1/2} (h + d)^{1/2} \quad (21-156)$$

where F is the **roll-separating force**, D and W are the roll diameter and width, f is a **roll-force factor** dependent on compressibility κ and gap thickness as given in Fig. 21-179, h is the gap thickness, and $d/2$ is the pocket depth for briquette rolls. (Pietsch, *Size Enlargement by Agglomeration*, Wiley, Chichester, 1992.) The maximum pressure P_m is established on the basis of required compact density and quality, and it is a strong function of roll gap distance and powder properties as discussed above, particularly compressibility. Small variations in feed properties can have a pronounced effect on maximum pressure P_m and press performance. Roll presses are scaled on the basis of constant maximum pressure. The required roll loading increases approximately with the square root of increasing roll diameter or gap width.

The appropriate roll force then scales as follows:

$$F_2 = F_1 \sqrt{\frac{D_2}{D_1}} \sqrt{\frac{(h + d)_2}{(h + d)_1}} \left(\frac{W_2}{W_1} \right) \quad (21-157)$$

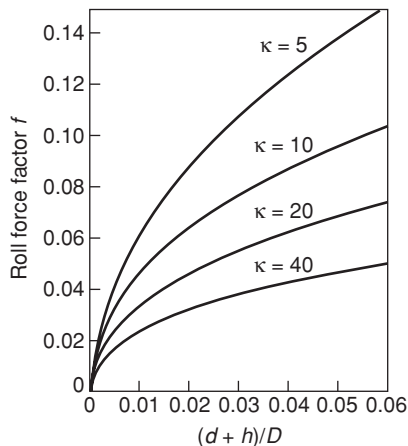


FIG. 21-179 Roll force factor as a function of compressibility κ and dimensionless gap distance $(d + h)/D$. [Pietsch (ed.), *Roll Pressing*, Powder Advisory Centre, London, 1987.]

It may be difficult to achieve geometric scaling of gap distance in practice. In addition, the impact of entrapped air and deaeration must be considered as part of scale-up, and this is not accounted for in the scaling work of Johanson (loc. cit.).

The allowable roll width is inversely related to the required pressing force because of mechanical design considerations. The throughput of a roll press at constant roll speed decreases as pressing force increases since the allowable roll width is less. Machines with capacities up to 45 Mg/h (50 tons/h) are available. Some average figures for the pressing force and energy necessary to compress a number of materials on roll-type briquette machines are given in Table 21-31. Typical capacities are given in Table 21-32.

During compression in the slip region, escaping air may induce fluidization or erratic pulsating of the feed. This effect, which is controlled by the permeability of the powder, limits the allowable roll speed of the press, and may also induce compact delamination. Increases in roll speed or decreases in permeability require larger feed pressures.

Recent advances in roll press design focus heavily on achieving rapid deaeration of the feed, screw design (double or single), screw loading, and vacuum considerations to remove entrapped air. Fluctuations in screw feed pressure have been shown to correlate with frequency of turns, which brings about density variations in the sheets exiting the rolls. See Miller [in Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., 2005] for a review.

Pellet Mills Pellet mills operate on the principle shown in Fig. 21-180. Moist, plastic feed is pushed through holes in dies of various shapes. The friction of the material in the die holes supplies the resistance necessary for compaction. Adjustable knives shear the rodlike extrudates into pellets of the desired length. Although several designs are in use, the most commonly used pellet mills operate by applying power to the die and rotating it around a freely turning roller with fixed horizontal or vertical axis. Concentric cylinder, double-roll cylinder, and table roll are commonly available designs (Fig. 21-136).

Pellet quality and capacity vary with properties of the feed such as moisture, lubricating characteristics, particle size, and abrasiveness, as well as die characteristics and speed. A readily pelleted material will yield about 122 kg/kWh [200 lb/(hp·h)] by using a die with 0.6-cm (1/4-in) holes. Some characteristics of pellet mills are given in Table 21-33.

Wet mass rheology heavily impacts performance through controlling both developed pressures and extrusion through the die, as in the case of paste extrusion (see "Paste Extrusion" and "Screw and Other Paste Extruders" subsections). In addition, the developed pressure in the roller nips behaves in a similar fashion to roll presses (see "Roll Presses" subsection).

Screw and Other Paste Extruders Screw extruders employ a screw to force material in a plastic state continuously through a die. If the die hole is round, a compact in the form of a rod is formed, whereas if the hole is a thin slit, a film or sheet is formed.

21-140 SOLID-SOLID OPERATIONS AND PROCESSING

TABLE 21-31 Pressure and Energy Requirements to Briquette Various Materials*

Pressure range, lb/in ²	Approximate energy required, kWh/ton	Type of material being briquetted or compacted		
		Without binder	With binder	Hot
Low 500–20,000	2–4	Mixed fertilizers, phosphate ores, shales, urea	Coal, charcoal, coke, lignite, animal feed, candy	Phosphate ores, urea
Medium 20,000–50,000	4–8	Acrylic resins, plastics, PVC, ammonium chloride, DMT, copper compounds, lead	Ferroalloys, fluorspar, nickel	Iron, potash, glass-making mixtures
High 50,000–80,000	8–16	Aluminum, copper, zinc, vanadium, calcined dolomite, lime, magnesia, magnesium carbonates, sodium chloride, sodium and potassium compounds	Flue dust, natural and reduced iron ores	Flue dust, iron oxide, natural and reduced iron ores, scrap metals
Very high >80,000	>16	Metal powders, titanium	—	Metal chips

*Courtesy Bepex Corporation. To convert pounds per square inch to newtons per square meter, multiply by 6895; to convert kilowatthours per ton to kilowatthours per megagram, multiply by 1.1.

TABLE 21-32 Some Typical Capacities (tons/h) for a Range of Roll Presses*

Roll diameter, in	10	16	12	10.3	13	20.5	28	36
Maximum roll-face width, in	3.25	6	4	6	8	13.5	27	10
Roll-separating force, tons	25	50	40	50	75	150	300	360
Carbon								
Coal, coke		2	1		3	6	25	
Charcoal			8			13		
Activated					3	7		
Metal and ores								
Alumina					5	10	28	
Aluminum				2	4	8	20	
Brass, copper	0.5			1.5	3	6	16	
Steel-mill waste					5	10		
Iron				3	6	15	40	
Nickel powder					2.5	5.0		
Nickel ore						20	40	
Stainless steel				2	5	10		
Steel								25
Bauxite		1.5				10	20	
Ferrometals						10		
Chemicals								
Copper sulfate	0.5	1.5		1	3	6	15	
Potassium hydroxide				1	4	8		
Soda ash	0.5				3	6	15	
Urea	0.25					10		
DMT	0.25				2	6		
Minerals								
Potash						20	80	
Salt				2	5	9		
Lime					4	8	15	
Calcium sulfate							13	40
Fluorspar						5	10	28
Magnesium oxide						1.5	5	
Asbestos						1.5	3	
Cement						5		
Glass batch						5	12	

*Courtesy Bepex Corporation. To convert inches to centimeters, multiply by 2.54; to convert tons to megagrams, multiply by 0.907; and to convert tons per hour to megagrams per hour, multiply by 0.907.



FIG. 21-180 Operating principle of a pellet mill.

TABLE 21-33 Characteristics of Pellet Mills

Horsepower range	10–250
Capacity, lb/(hp·h)	75–300
Die characteristics	
Size	Up to 26 in inside diameter × approximately 8 in wide
Speed range	75–500 r/min
Hole-size range	1/16–1 1/4 in inside diameter
Rollers	As many as three rolls; up to 10-in diameter

NOTE: To convert horsepower to kilowatts, multiply by 0.746; to convert pounds per horsepower-hour to kilograms per kilowatthour, multiply by 0.6; and to convert inches to centimeters, multiply by 2.54.

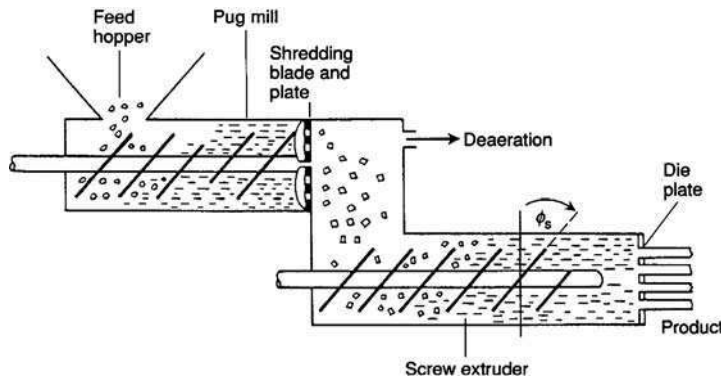


FIG. 21-181 Screw extruder with upstream pug mill, shredding plate, and deaeration stage. (Benbow and Bridgwater, *Paste Flow and Extrusion*, Oxford University Press, 1993, with permission.)

Basic types of extruders include both screw extruders such as axial endplate, radial screen, and basket designs, as well as pelletization equipment described above, such as rotary cylinder or gear and ram or piston extruders (Fig. 21-136). See Newton [*Powder Technology and Pharm. Processes*, Chulia et al. (eds.), Elsevier, 1994, p. 391], Pietsch (*Size Enlargement by Agglomeration*, Wiley, 1992), and Benbow and Bridgwater (*Paste Flow and Extrusion*, Oxford University Press, 1993).

Figure 21-181 illustrates a typical extruder layout, with upstream pug mill, shredding plate, and deaeration stage. Premixing and extrusion through a pug mill help achieve initial densification prior to final screw extrusion. As with all compaction processes, deaeration must be accounted for, which often occurs under vacuum. A wider variety of single- and twin-screw designs are available, which vary in screw and barrel geometry, the degrees of intermeshing, and rotation direction (Fig. 21-182).

Both wet and dry extrusion techniques are available, and both are strongly influenced by the frictional properties of the particulate phase and wall. In the case of wet extrusion, rheological properties

of the liquid phase are equally important. See Pietsch (*Size Enlargement by Agglomeration*, Wiley, Chichester, 1992, p. 346), and Benbow et al. [*Chem. Eng. Sci.*, **422**, 2151 (1987)] for a review of design procedures for dry and wet extrusion, respectively. Die face throughput increases with increasing pressure developed at the die, whereas the developed pressure from the screw decreases with increasing throughput. These relationships are referred to as the **die and screw characteristics** of the extruder, respectively, as illustrated in Fig. 21-183 (see "Screw and Other Paste Extruders" subsection), and in addition to rheology and wall friction, they are influenced by wear of dies, screws, and barrels over equipment life, which modify wall friction properties and die entrance effects. The intersection of these characteristics determines the **operating point**, or throughput, of the extruder.

The formation of **defects** and **phase separation** is an important consideration in paste extrusion. Typical defects include lamination or **delamination** (occurring with joining of adjacent past streams) and **surface fracture**, often referred to as shark-skin formation. Surface fracture generally increases with decreasing paste liquid

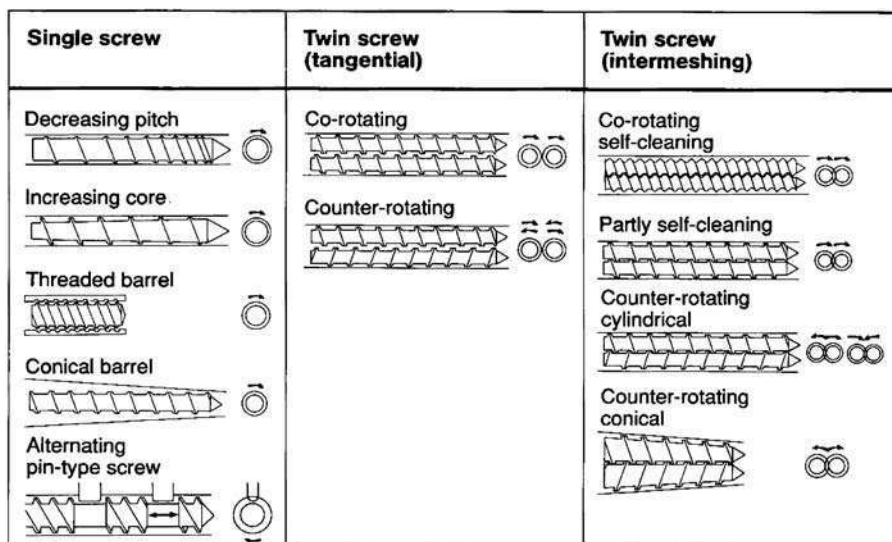


FIG. 21-182 Available screw extruder systems, illustrated barrel and screw type, as well as rotation. (After Benbow and Bridgwater, *Paste Flow and Extrusion*, Oxford University Press, 1993. Courtesy Werner and Pfleiderer.)

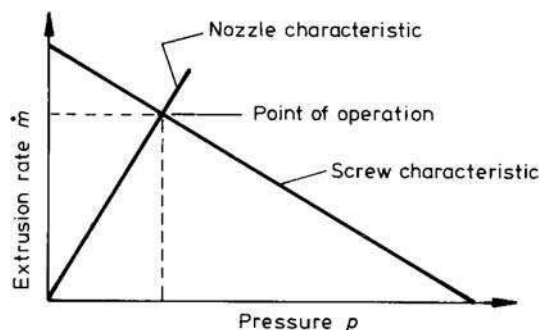


FIG. 21-183 Determination of extruder capacity or throughput based on the intersection of screw and nozzle (die face) characteristics. (From Pietsch, *Size Enlargement by Agglomeration*, Wiley, 1992.)

content (Fig. 21-184), increasing extrusion die velocity, and decreasing die length. Phase separation can lead to extruder or die failure, with rapid rise in pressures associated with the fluid phase separating from the powder matrix. The chance of phase separation increases with increased operating pressure, increased bulk powder permeability (or increasing particle size), and decreasing liquid viscosity. See Benbow and Bridgwater (loc. cit.) for detailed discussions of these effects.

A common use of screw extruders is in the forming and compounding of plastics. Table 21-34 shows typical outputs that can be expected per horsepower for various plastics and the characteristics of several popular extruder sizes.

Deairing pugmill extruders, which combine mixing, densification, and extrusion in one operation, are available for agglomerating clays, catalysts, fertilizers, etc. Table 21-35 gives data on screw extruders for the production of catalyst pellets.

THERMAL PROCESSES

Bonding and agglomeration by temperature elevation or reduction are applied either in conjunction with other size-enlargement processes or as a separate process. Agglomeration occurs through one or more of the following mechanisms:

1. Drying of a concentrated slurry or wet mass of fines
2. Fusion
3. High-temperature chemical reaction
4. Solidification and/or crystallization of a melt or concentrated slurry during cooling

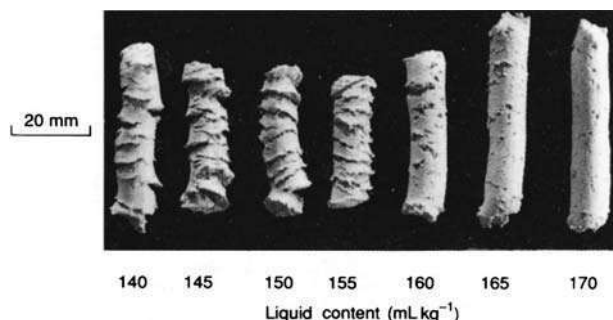


FIG. 21-184 Effect of liquid content of surface fracture. α -alumina and 5 wt % Celacol in water, with die ($D = 9.5$ mm, $L = 3.14$ mm) at a velocity $V = 1.2$ mm/s. (Benbow and Bridgwater, *Paste Flow and Extrusion*, Oxford University Press, 1993, with permission.)

TABLE 21-34 Characteristics of Plastics Extruders*

Efficiencies			lb/(hp-h)
Rigid PVC			7–10
Plasticized PVC			10–13
Impact polystyrene			8–12
ABS polymers			5–9
Low-density polyethylene			7–10
High-density polyethylene			4–8
Polypropylene			5–10
Nylon			8–12
Relation of size, power, and output			
hp	Diameter		Output, lb/h, low-density polyethylene
	in	mm	
15	2	45	Up to 125
25	2½	60	Up to 250
50	3½	90	Up to 450
100	4½	120	Up to 800

*The *Encyclopedia of Plastics Equipment*, Simonds (ed.), Reinhold, New York, 1964.

NOTE: To convert inches to centimeters, multiply by 2.54; to convert horsepower to kilowatts, multiply by 0.746; to convert pounds per hour to kilograms per hour, multiply by 0.4535; to convert pounds per horsepower-hour to kilograms per kilowatt-hour, multiply by 0.6.

Sintering and Heat Hardening In powder metallurgy compacts are sintered with or without the addition of binders. In ore processing the agglomerated mixture is either sintered or indurated. Sintering refers to a process in which fuel is mixed with the ore and burned on a grate. The product is a porous cake. Induration, or heat hardening, is accomplished by combustion of gases passed through the bed. The aim is to harden the pellets without fusing them together, as is done in the sintering process.

Ceramic bond formation and grain growth by diffusion are the two prominent reactions for bonding at the high temperature (1100 to 1370°C, or 2000 to 2500°F, for iron ore) employed. The minimum temperature required for sintering may be measured by modern **dilatometry** techniques, as well as by differential scanning calorimetry. See Compo et al. [*Powder Tech.*, **51**(1), **87** (1987); *Particle Characterization*, **1**, 171 (1984)] for reviews.

In addition to agglomeration, other useful processes may occur during sintering and heat hardening. For example, carbonates and sulfates, may be decomposed, or sulfur may be eliminated. Although the major application is in ore beneficiation, other applications, such as the preparation of lightweight aggregate from fly ash and the formation of clinker from cement raw meal, are also possible. Nonferrous sinter is produced from oxides and sulfides of manganese, zinc, lead, and nickel. An excellent account of the many possible applications is given by Ban et al. [Knepper (ed.), *Agglomeration*, op. cit., p. 511] and Ball et al. (*Agglomeration of Iron Ores*, 1973). The highest tonnage use at present is in the beneficiation of iron ore.

TABLE 21-35 Characteristics of Pelletizing Screw Extruders for Catalysts*

Screw diameter, in	Drive hp	Typical capacity, lb/h
2.25		60
4	7.5–15	200–600
6	Up to 60	600–1500
8	75–100	Up to 2000

*Courtesy The Bonnot Co. To convert inches to centimeters, multiply by 2.54; to convert horsepower to kilowatts, multiply by 0.746; and to convert pounds per hour to kilograms per hour, multiply by 0.4535.

NOTE:

1. Typical feeds are high alumina, kaolin carriers, molecular sieves, and gels.
2. Water-cooled worm and barrel, variable-speed drive.
3. Die orifices as small as ¼ in.
4. Vacuum-deairing option available.

The machine most commonly used for sintering iron ores is a traveling grate, which is a modification of the Dwight-Lloyd continuous sintering machine formerly used only in the lead and zinc industries. Modern sintering machines may be 4 m (13 ft) wide by 60 m (200 ft) long and have capacities of 7200 Mg/day (8000 tons/day).

The productive capacity of a sintering strand is related directly to the rate at which the burning zone moves downward through the bed. This rate, which is of the order of 2.5 cm/min (1 in/min), is controlled by the air rate through the bed, with the air functioning as the heat-transfer medium.

Heat hardening of green iron-ore pellets is accomplished in a vertical shaft furnace, a traveling-grate machine, or a grate-plus-kiln combination (see Ball et al., op. cit.).

Drying and Solidification Granular free-flowing solid products are often an important result of the drying of concentrated slurries and pastes and the cooling of melts. Size enlargement of originally finely divided solids results. Pressure agglomeration including extrusion, pelleting, and briquetting is used to preform wet material into forms suitable for drying in through-circulation and other types of dryers. Details are given in Sec. 12 in the account of solids-drying equipment.

Rotating-drum-type and belt-type heat-transfer equipment forms granular products directly from fluid pastes and melts without intermediate preforms. These processes are described in Sec. 5 as examples of indirect heat transfer to and from the solid phase. When solidification results from melt freezing, the operation is known as *flaking*. If evaporation occurs, solidification is by drying.

MODELING AND SIMULATION OF GRANULATION PROCESSES

For granulation processes, granule size distribution is an important if not the most important property. The evolution of the granule size distribution within the process can be followed using population balance modeling techniques. This approach is also used for other size-change processes including crushing and grinding. (See section "Principles of Size Reduction.") The use of the **population balance (PB)** is outlined briefly below. For more in-depth analysis see Randolph and Larson (*Theory of Particulate Processes*, 2d ed., Academic Press, 1991), Ennis and Litster (*The Science and Engineering of Granulation Processes*, Chapman-Hall, 1997), and Sastry and Loftus [*Proc. 5th Int. Symp. Agglom.*, IChemE, 623 (1989)]. See also Cameron and Wang for a recent review of modeling and control [in Parikh (ed.), *Handbook of Pharmaceutical Granulation Technology*, 2d ed., Taylor & Francis, 2005].

- The key uses of PB modeling of granulation processes are
- Critical evaluation of data to determine controlling granulation mechanisms
 - In design, to predict the mean size and size distribution of product granules

- Sensitivity analysis: to analyze quantitatively the effect of changes to operating conditions and feed variables on product quality
- Circuit simulation, optimization, and process control

The use of PB modeling by practitioners has been limited for two reasons. First, in many cases the kinetic parameters for the models have been difficult to predict and are very sensitive to operating conditions. Second, the PB equations are complex and difficult to solve. However, recent advances in understanding of granulation micromechanics, as well as better numerical solution techniques and faster computers, means that the use of PB models by practitioners should expand.

THE POPULATION BALANCE

The PB is a statement of continuity for particulate systems. It includes a **kinetic expression** for each mechanism which changes a particle property. Consider a section of a granulator as illustrated in Fig. 21-185. The PB follows the change in the granule size distribution as granules are born, die, grow, and enter or leave the control

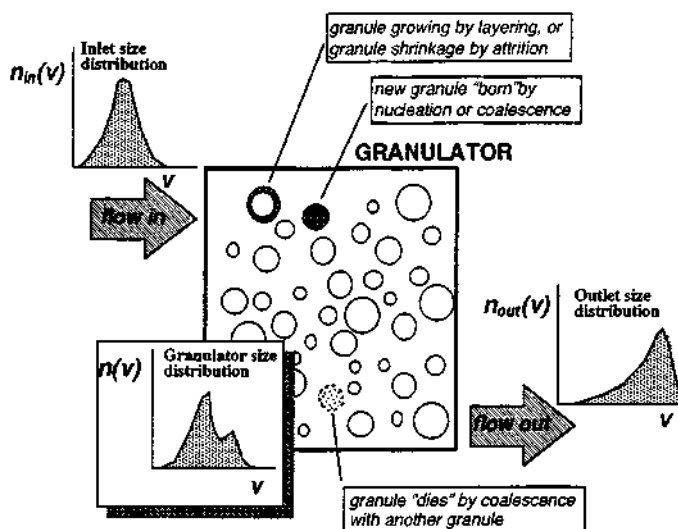


FIG. 21-185 Changes to the granule size distribution due to granulation-rate processes as particles move through the granulator. (Reprinted from *Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance*, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

volume. As discussed in detail previously ("Agglomeration Rate Processes and Mechanics"), the granulation mechanisms which cause these changes are nucleation, layering, coalescence, and attrition (Fig. 21-91 and Table 21-36). The number of particles-per-unit volume of granulator between size volume v and $v + dv$ is $n(v) dv$, where $n(v)$ is the **number frequency size distribution** by size volume, having dimensions of number per unit granulator and volume per unit size volume. For constant granulator volume, the macroscopic PB for the granulator in terms of $n(v)$ is:

$$\begin{aligned} \frac{\partial n(v,t)}{\partial t} = & \frac{Q_{in}}{V} n_{in}(v) - \frac{Q_{ex}}{V} n_{ex}(v) - \frac{\partial(G^\circ - A^\circ)n(v,t)}{\partial v} \\ & + B_{nuc}(v) + \frac{1}{2N_t} \int_0^v \beta(u, v-u) n(u,t) n(v-u,t) du \\ & - \frac{1}{N_t} \int_0^\infty \beta(u, v,t) n(u,t) n(v,t) du \quad (21-158) \end{aligned}$$

where V is the volume of the granulator; Q_{in} and Q_{ex} are the inlet and exit flow rates from the granulator; $G(v)$, $A(v)$, and $B_{nuc}(v)$ are the layering, attrition, and nucleation rates, respectively; $B(u, v, t)$ is the coalescence kernel and N_t is the total number of granules-per-unit volume of granulator. The left-hand side of Eq. (21-158) is the accumulation of particles of a given size volume. The terms on the right-hand side are in turn: the bulk flow into and out of the control volume, the convective flux along the size axis due to layering and attrition, the birth of new particles due to nucleation, and birth and death of granules due to coalescence. Equation (21-158) is written in terms of granule volume v , but could also be written in terms of granule size x or could also be expanded to follow changes in other granule properties, e.g., changes in granule density or porosity due to consolidation.

MODELING INDIVIDUAL GROWTH MECHANISMS

The **granule size distribution (GSD)** is a strong function of the balance between different mechanisms for size change shown in Table 21-33—layering, attrition, nucleation, and coalescence. For example, Fig. 21-186 shows the difference in the GSD for a doubling in mean granule size due to (1) layering only, or (2) coalescence only for batch, plug-flow, and well-mixed granulators. Table 21-36 describes how four key rate mechanisms effect the GSD.

Nucleation Nucleation increases both the mass and number of the granules. For the case where new granules are produced by liquid

feed, which dries or solidifies, the nucleation rate is given by the new feed, droplet size n_s and the volumetric spray rate S :

$$B(v)_{nuc} = S n_s(v) \quad (21-159)$$

In processes where new powder feed has a much smaller particle size than the smallest granular product, the feed powder can be considered as a **continuous phase** that can nucleate to form new granules [Sastri & Fuerstenau, *Powder Technol.*, **7**, 97 (1975)]. The size of the nuclei is then related to nucleation mechanism. In the case of nucleation by spray, the size of the nuclei is of the order of the droplet size and proportional to $\cos\theta$, where θ is binder fluid-particle contact angle (see Fig. 21-99).

Layering Layering increases granule size and mass by the progressive coating of new material onto existing granules, but it does not alter the number of granules in the system. As with nucleation, the new feed may be in liquid form (where there is simultaneous drying or cooling) or may be present as a fine powder. Where the feed is a powder, the process is sometimes called **pseudolayering** or **snowballing**. It is often reasonable to assume a linear-growth rate $G(x)$ which is independent of granule size. For batch and plug-flow granulators, this causes the initial feed distribution to shift forward in time with the shape of the GSD remaining unaltered and governed by a traveling-wave equation (Table 21-37). As an example, Fig. 21-187 illustrates size-independent growth of limestone pellets by snowballing in a batch drum. Size-independent linear growth rate implies that the volumetric growth rate $G^\circ(v)$ is proportional to projected granule surface area, or $G^\circ(v) \propto v^{2/3} \propto x^2$. This assumption is true only if all granules receive the same exposure to new feed. Any form of segregation will invalidate this assumption [Liu and Litster, *Powder Technol.*, **74**, 259 (1993)]. The growth rate $G^\circ(v)$ by layering only can be calculated directly from the mass balance:

$$\dot{V}_{feed} = (1 - \epsilon) \int_0^\infty G^\circ(v) n(v) dv \quad (21-160)$$

where $\dot{V}_{feed}$ is the volumetric flow rate of new feed and ϵ is the granule porosity.

Coalescence Coalescence is the most difficult mechanism to model. It is easiest to write the population balance [Eq. (21-158)] in terms of number distribution by volume $n(v)$ because granule volume is conserved in a coalescence event. The key parameter is the **coalescence kernel** or rate constant $\beta(u, v)$. The kernel dictates the overall rate of coalescence, as well as the effect of granule size on coalescence rate. The **order** of the kernel has a major effect on the shape and evolution of the granule size distribution. [See Adetayo & Ennis, *AIChE J.* 1997.] Several empirical kernels have been proposed and used (Table 21-38).

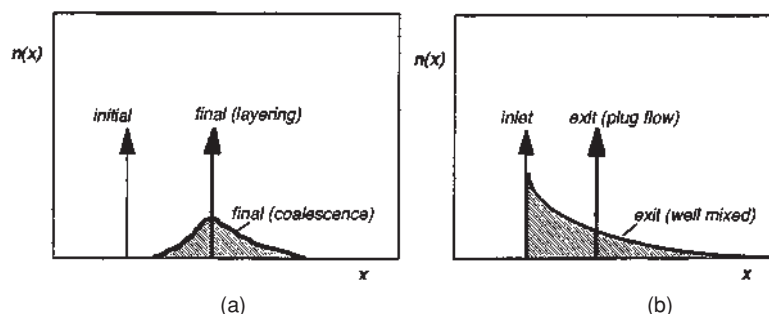
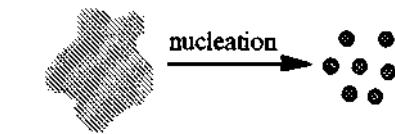
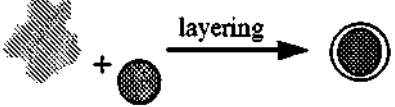
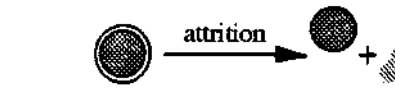


FIG. 21-186 The effect of growth mechanism and mixing on product granule size distribution for (a) batch growth by layering or coalescence, and (b) layered growth in well-mixed or plug-flow granulators. (Reprinted from Design and Optimization of Granulation and Compaction Processes for Enhanced Product Performance, Ennis, 2006, with permission of E&G Associates. All rights reserved.)

TABLE 21-36 Impact of Granulation Mechanisms on Size Distribution

Mechanism	Changes number of granules?	Changes mass of granules?	Discrete or differential?
	yes	yes	discrete
	no	yes	differential
	yes	no	discrete
	no	yes	differential

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All the kernels are empirical, or semiempirical and must be fitted to plant or laboratory data. The kernel proposed by Adetayo and Ennis is consistent with the granulation regime analysis described above (see section on growth) and is therefore recommended:

$$\beta(u, v) = \begin{cases} k, w < w^* \\ 0, w > w^* \end{cases} \quad w = \frac{(uv)^a}{(u+v)^b} \quad (21-161)$$

where w^* is the **critical average granule volume** in a collision corresponding to $St = St^*$, and it is related to the critical cutoff diameter defined above. For fine powders in the noninertial regime (see section "Growth and Consolidation") where $St \ll St^*$, this kernel collapses to the simple **random** or size-independent kernel $\beta = k$ for

which the mean granule size increases exponentially with time. Where deformation is unimportant, coalescence occurs only in the noninertial regime and stops abruptly when $St_v = St^*$. Based on the granulation regime analysis, the effects of feed characteristics and operating variables on granulation extent has been predicted [Adetayo et al., *Powder Tech.*, **82**, 47–59 (1995)].

Modeling growth where deformation is significant is more difficult. It can be assumed that a critical cutoff size exists w^* , which determines which combination of granule sizes are capable of coalescence, based on their inertia. When the harmonic average of sizes of two colliding granules w is less than this critical cutoff size w^* , coalescence is successful, or

$$w = \frac{(uv)^b}{(u+v)^a} = w^* = \frac{\pi}{6} \left(\frac{16\mu}{\rho u_0} St^* \right)^3 \quad (21-162)$$

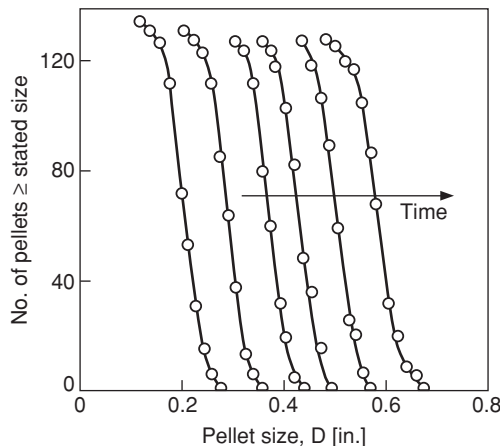


FIG. 21-187 Batch drum growth of limestone pellets by layering with a size-independent linear growth rate [Capes, Chem. Eng., **45**, CE78 (1967).]

where a and b are model parameters expected to vary with granule deformability, and u and v are granule volumes. To be dimensionally consistent, $2b - a = 1$. w^* and w involving the parameters a and b represent a generalization of the Stokes analysis for nondeforming systems, for which case $a = b = 1$. For deformable systems, the kernel is then represented by Eq. (21-161). Figure 21-188 illustrates the evolution of the granule size distribution as predicted by this cutoff-based kernel that accounts for deformability. The cutoff kernel is seen to clearly track the experimental average granule size over the life of the granulation, illustrating that multiple kernels are not necessary to describe the various stages of granule growth, including the initial stage of random noninertial coalescence and the final stage of nonrandom preferential inertial growth by balling or crushing and layering (see Fig. 21-91).

Attrition The wearing away of granule surface material by attrition is the direct opposite of layering. It is an important mechanism when drying occurs simultaneously with granulation and granule velocities are high, e.g., fluidized beds and spouted beds. In a fluid bed [Ennis and Sunshine, *Tribology Int.*, **26**, 319 (1993)], attrition rate is proportional to excess gas velocity $U - U_{mf}$ and approximately inversely proportional to granule-fracture toughness K_c , or $A \propto (U - U_{mf})/K_c$. For

TABLE 21-37 Some Analytical Solutions to the Population Balance*

Mixing state	Mechanisms operating	Initial or inlet size distribution	Final or exit size distribution
Batch	Layering only: $G(x) = \text{constant}$	Any initial size distribution, $n_0(x)$	$n(x) = n_0(x - \Delta x)$ where $\Delta x = Gt$
Continuous & well-mixed	Layering only: $G(x) = \text{constant}$	$n_{in}(x) = N_{in} \delta(x - x_{in})$	$n(x) = \frac{N_0 G}{\tau} \exp\left(-\frac{\tau(x - x_m)}{G}\right)$
Batch	Coalescence only, size independent: $\beta(u, v) = \beta_0$	$n_0(v) = N_0 \delta(v - v_0)$	$n(v) = \frac{N_0}{\bar{v}} \exp\left(-\frac{v}{\bar{v}}\right)$ where $\bar{v} = v_0 \exp\left(\frac{\beta_0 t}{6}\right)$
Batch	Coalescence only, size independent: $\beta(u, v) = \beta_0$	$n_0(v) = \frac{N_0}{v_0} \exp\left(-\frac{v}{v_0}\right)$	$n(v) = \frac{4 N_0}{v_0(N_0 \beta_0 t + 2)^2} \exp\left[\frac{-2v/v_0}{N_0 \beta_0 t + 2}\right]$

*Randolph and Larson, *Theory of Particulate Processes*, 2d ed., Academic Press, New York (1988); Gelbart and Seinfeld, *J. Computational Physics*, **28**, 357 (1978).

spouted beds, most attrition occurs in the spout and the attrition rate may be expressed as

$$A \propto \frac{A_i U_i^3}{K} \tag{21-163}$$

where A_i and U_i are the inlet orifice area and gas velocity, respectively. Attrition rate also increases with increasing slurry feed rate [Liu and Lister, *Powder Tech.*, **74**, 259 (1993)]. Granule breakage by fragmentation

is also possible, with its rate being described by an **on function**, which plays a similar role as the coalescence kernel does for growth. (See "Principles of Size Reduction" and "Breakage Modes and Grindability" sections for additional details.)

SOLUTION OF THE POPULATION BALANCE

Effects of Mixing As with chemical reactors, the **degree of mixing** within the granulator has an important effect on the final granule size distribution because of its influence on the **residence time distribution**. Figure 21-186 shows the difference in exit size distribution for a plug-flow and well-mixed granulator for growth by layering only. In general, the exit size distribution is broadened and the extent of growth (for constant rate constants) is diminished for an increased degree of mixing in the granulator. With layering and attrition rates playing the role of generalized velocities, coalescence, and fragmentation rates, the role of reaction rate constants, methodologies of traditional reaction engineering may be employed to design granulation systems or optimize the granule size distribution. [For the related example of crystallization, see Randolph and Larson (*Theory of Particulate Processes*, 2d ed., Academic Press, 1991).] Table 21-39 lists some mixing models that have been used for several types of granulators.

Analytical Solutions Solution of the population balance is not trivial. Analytical solutions are available for only a limited number of special cases, of which some examples of practical importance are summarized in Table 21-37. For other analytical solutions, see general references on population balances given above.

In general, analytical solutions are only available for specific initial or inlet size distributions. However, for batch granulation where the only growth mechanism is coalescence, at long times the size distribution may become **self-preserving**. The size distribution is self-preserving if the normalized size distributions $\phi = \phi(\eta)$ at long times are independent of mean size $\bar{v}$, or

$$\phi = \phi(\eta) \quad \text{only where} \quad \eta = v/\bar{v}$$
$$\bar{v} = \int_0^\infty v \cdot n(v,t) \, dv \tag{21-164}$$

Analytical solutions for self-preserving growth do exist for some coalescence kernels and such behavior is sometimes seen in practice (Fig. 21-189). Roughly speaking, self-preserving growth implies that the width of the size distribution increases in proportion to mean granule size, i.e., the width is uniquely related to the mean of the distribution.

Numerical Solutions For many practical applications, numerical solutions to the population balance are necessary. Several numerical solution techniques have been proposed. It is usual to break the

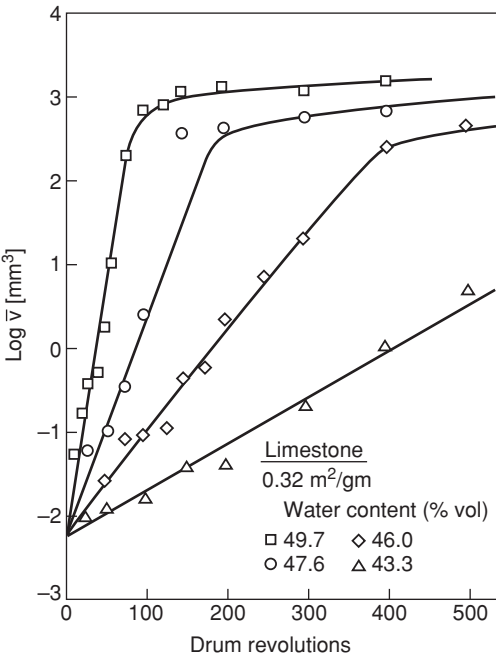


FIG. 21-188 Batch drum growth of limestone by coalescence. Note granule size increases exponentially with time in the first stage of noninertial growth. Experimental data of Kapur [Adv. Chem. Eng., **10**, 56 (1978)] compared with single deformable granulation kernel [Eqs. (21-161), and (21-162)]. [Adetayo & Ennis, AIChEJ. (In press).] Reproduced with permission of the American Institute of Chemical Engineers. Copyright AIChE. All rights reserved.

TABLE 21-38 Coalescence Kernels for Granulation

Kernel	Reference and comments
$\beta = \beta_0$	Kapur & Fuerstenau [<i>I & EC Proc. Des. & Dev.</i> , 8 (1), 56 (1969)], size-independent kernel.
$\beta = \beta_0 \frac{(u+v)^a}{(uv)^b}$	Kapur [<i>Chem. Eng. Sci.</i> , 27 , 1863 (1972)], preferential coalescence of limestone.
$\beta = \beta_0 \frac{(u^{2/3} + v^{2/3})}{1/u + 1/v}$	Sastry [<i>Int. J. Min. Proc.</i> , 2 , 187 (1975)], preferential balling of iron ore and limestone.
$\beta(u,v) = \begin{cases} k, & w < w^* \\ 0, & w > w^* \end{cases} \quad w = \frac{(uv)^a}{(u+v)^b}$	Adetayo & Ennis [<i>AIChE J.</i> , (1997)], based on granulation regime analysis.

TABLE 21-39 Mixing Models for Continuous Granulators

Granulator	Mixing model	Reference
Fluid bed	Well-mixed	See Sec. 17
Spouted bed	Well-mixed	Liu and Litster, <i>Powder Tech.</i> , 74 , 259 (1993)
	Two-zone model	Litster et al. [<i>Proc. 6th Int. Symp. Agglom., Soc. Powder Tech., Japan</i> , 123 (1993)].
Drum	Plug-flow	Adetayo et al., <i>Powder Tech.</i> , 82 , 47-59 (1995)
Disc	Two well-mixed tanks in series with classified exit	Sastry & Loftus [<i>Proc. 5th Int. Symp. Agglom.</i> , IChemE, 623 (1989)]
	Well-mixed tank and plug-flow in series with fines bypass	Ennis, Personal communication (1986)

size range into discrete intervals and then solve the resulting series of ordinary differential equations. A geometric discretization reduces the number of size intervals (and equations) that are required. Litster et al. [*AIChE J.*, (1995)] give a general discretized PB for nucleation,

growth, and coalescence with a geometric discretization of $v_j = 2^{1/q}v_{j-1}$ where q is an integer. Accuracy is increased (at the expense of computational time) by increasing the value of q . Their discretized PB is recommended for general use.

SIMULATION OF GRANULATION CIRCUITS WITH RECYCLE

When granulation circuits include recycle streams, both steady-state and dynamic responses can be important. Computer simulation packages are now widely used to design and optimize many process flow sheets, e.g., comminution circuits, but simulation of granulation circuits is much less common. Commercial packages do not contain library models for granulators. Some researchers have developed simulations and used these for optimization and control studies [Sastry, *Proc. 3d Int. Symp. Agglom.* (1981); Adetayo et al., *Computers Chem. Eng.*, **19**, 383 (1995); Zhang et al., *Control of Part. Processes IV* (1995)]. For these simulations, dynamic population-balance models have been used for the granulator. Standard literature models are used for auxiliary equipment such as screens, dryers, and crushers. These simulations are valuable tools for optimization studies and development of control strategies in granulation circuits, and may be employed to investigate the effects of transient upsets in operating variables, particularly moisture level and recycle ratio, on circuit performance.

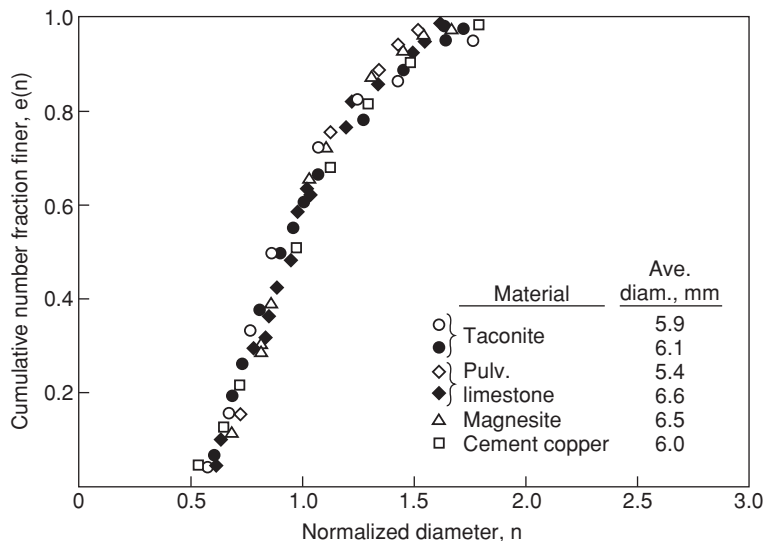


FIG. 21-189 Self-preserving size distributions for batch coalescence in drum granulation. [Sastry, *Int. J. Min. Proc.*, **2**, 187 (1975).] With kind permission of Elsevier Science -NL, 1055 KV Amsterdam, the Netherlands.

SECTION 22

Waste
Management

PERRY'S CHEMICAL ENGINEERS' HANDBOOK

8TH EDITION



LOUIS THEODORE, KENNETH N. WEISS
JOHN D. MCKENNA, F. LEE SMITH
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List of Abbreviations

Abbreviation	Definition	Abbreviation	Definition
3P	Pollution prevention pays	LCC	Life cycle costing
ABS	Alkyl benzene sulfonate	LOX	Liquid oxygen
ACC	Annualized capital costs	MACT	Maximum achievable control technology
BACT	Best available control technology	MSDA	Material safety data sheets
BAT	Best available technology	MSW	Municipal solid waste
BCOD	Biodegradable chemical oxygen demand	MWC	Municipal waste combustors
BCT	Best conventional technology	MWI	Medical waste incinerators
BOD	Biochemical oxygen demand	NBOD	Nitrogenous biochemical oxygen demand
BSRT	Biomass solids retention time	NIMBY	Not in my back yard
BTEX	Benzene, toluene, xylene	NPDES	National pollutant discharge elimination system
CAA	Clean Air Act	NSPS	New source performance standards
CAAA	Clean Air Act Amendments	PCB	Polychlorinated biphenyl
CCP	Comprehensive costing procedures	PIES	Pollution prevention information exchange systems
CFR	Code of federal regulations	PM	Particulate matter
CKD	Cement kiln dust	POTW	Publicly owned treatment work
COD	Chemical oxygen demand	PPIC	Pollution prevention information clearinghouse
CPI	Chemical process industries	PSD	Prevention of significant deterioration
CRF	Capital recovery factor	RCRA	Resource Conservation and Recovery Act
CTDMPLUS	Complex terrain dispersion model plus algorithms for unstable situations	RDF	Refuse-derived fuel
CWRT	Center for Waste Reduction Technologies	SARA	Superfund Amendments and Reauthorization Act
DCF	Direct installation cost factor	SCR	Selective catalytic reduction
DO	Dissolved oxygen	SCS	Stationary-container systems
DRE	Destruction and removal efficiency	SE	Strength of the treated waste
EBCT	Empty bed contact time	SMART	Save money and reduce toxics
EMAS	Eco-management and audit scheme	SO	Strength of the untreated waste
EMS	Environmental management system	SS	Suspended solids
EPA	Environmental Protection Agency	TCC	Total capital cost
FML	Flexible membrane liner	TCP	Traditional costing procedures
GAX	Granular activated carbon	TGNMO	Total gas nonmethane organics
HAPS	Hazardous air pollutants	TOC	Total organic carbon
HAZWOPER	Hazardous waste operators	TSA	Total systems approach
HCS	Hauled-container systems	TSCA	Toxic Substances Control Act
HRT	Reactor hydraulic retention time	TSD	Treatment, storage, and disposal
HSWA	Hazardous and Solid Waste Act	UASB	Upflow anaerobic sludge blanket
I-TEF	International toxic equivalency factor	VOC	Volatile organic compound
ICF	Indirect installation cost factor	VOST	Volatile organic sampling train
ISO	International Organization for Standardization	VSS	Volatile suspended solids
LCA	Life cycle assessment	WRAP	Waste reduction always pays
		WTE	Waste-to-energy (systems)

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INTRODUCTION TO WASTE MANAGEMENT AND REGULATORY OVERVIEW

In this section, a number of references are made to laws and procedures that have been formulated in the United States with respect to waste management. An engineer handling waste-management problems in another country would well be advised to know the specific laws and regulations of that country. Nevertheless, the treatment given here is believed to be useful as a general guide.

MULTIMEDIA APPROACH TO ENVIRONMENTAL REGULATIONS IN THE UNITED STATES

Among the most complex problems to be faced by industry during the 1990s is the proper control and use of the natural environment. In the 1970s the engineering profession became acutely aware of its responsibility to society, particularly for the protection of public health and welfare. The decade saw the formation and rapid growth of the U.S. Environmental Protection Agency (EPA) and the passage of federal and state laws governing virtually every aspect of the environment. The end of the decade, however, brought a realization that only the more simplistic problems had been addressed. A limited number of

large sources had removed substantial percentages of a few readily definable air pollutants from their emissions. The incremental costs to improve the removal percentages would be significant and would involve increasing numbers of smaller sources, and the health hazards of a host of additional toxic pollutants remained to be quantified and control techniques developed.

Moreover, in the 1970s, air, water, and waste were treated as separate problem areas to be governed by their own statutes and regulations. Toward the latter part of the decade, however, it became obvious that environmental problems were closely interwoven and should be treated in concert. The traditional type of regulation—command and control—had severely restricted compliance options.

The 1980s began with EPA efforts redirected to take advantage of the case-specific knowledge, technical expertise, and imagination of those being regulated. Providing plant engineers with an incentive to find more efficient ways of abating pollution would greatly stimulate innovation in control technology. This is a principal objective, for example, of EPA's "controlled trading" air pollution program, established in the Offsets Policy Interpretative Ruling issued by the EPA in 1976, with statutory foundation given by the Clean Air Act Amendments of 1977. The Clean Air Act Amendments of

1990 expanded the program even more to the control of sulfur oxides under Title IV. In effect, a commodities market on “clean air” was developed.

The rapidly expanding body of federal regulation presents an awesome challenge to traditional practices of corporate decision-making, management, and long-range planning. Those responsible for new plants must take stock of the emerging requirements and construct a fresh approach.

The full impact of the Clean Air Act Amendments of 1990, the Clean Water Act, the Safe Drinking Water Act, the Resource Conservation and Recovery Act, the Comprehensive Environmental Responsibility, Compensation and Liability (Superfund) Act, and the Toxic Substances Control Act is still not generally appreciated. The combination of all these requirements, sometimes imposing conflicting demands or establishing differing time schedules, makes the task of obtaining all regulatory approvals extremely complex.

One of the dominant impacts of environmental regulations is that the lead time required for the planning and construction of new plants is substantially increased. When new plants generate major environmental complexities, the implications can be profound. Of course, the exact extent of additions to lead time will vary widely from one case to another, depending on which permit requirements apply and on what difficulties are encountered. For major expansions in any field of heavy industry, however, the delay resulting from federal requirements could conceivably add 2 to 3 years to total lead time. Moreover, there is always the possibility that regulatory approval will be denied. So, contingency plans for fulfilling production needs must be developed.

The 1990s saw the emergence of environmental management systems (EMSs) across the globe including the ISO 14001 environmental management system and the European Union's Eco-Management and Audit Scheme (EMAS). Any EMS is a continual cycle of planning, implementing, reviewing, and improving the processes and actions that an organization undertakes to meet its business and environmental goals. EMSs are built on the “Plan, Do, Check, Act” model (Fig. 22-1) that leads to continual improvement. Planning includes identifying environmental aspects and establishing goals (plan); implementing includes training and operational controls (do); checking includes monitoring and corrective action (check); and reviewing includes progress review and acting to make needed changes to the EMS (act). Organizations that have implemented an EMS often require all their suppliers to become EMS-certified as environmental programs become part of everyday business.

Any company planning a major expansion must concentrate on environmental factors from the outset. Since many environmental approvals require a public hearing, the views of local elected officials and the community at large are extremely important. To an unprecedented degree, the political acceptability of a project can now be crucial.

PLANT STRATEGIES

At the plant level, a number of things can be done to minimize the impact of environmental quality requirements. These include:

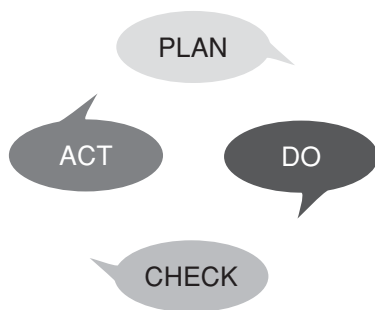


FIG. 22-1 “Plan, Do, Check, Act” model.

1. Maintaining an accurate source-emission inventory
2. Continually evaluating process operations to identify potential modifications that might reduce or eliminate environmental impacts
3. Ensuring that good housekeeping and strong preventive-maintenance programs exist and are followed
4. Investigating available and emerging pollution-control technologies
5. Keeping well informed of the regulations and the directions in which they are moving
6. Working closely with the appropriate regulatory agencies and maintaining open communications to discuss the effects that new regulations may have
7. Keeping the public informed through a good public-relations program
8. Implementing an EMS

It is unrealistic to expect that at any point in the foreseeable future Congress will reverse direction, reduce the effect of regulatory controls, or reestablish the preexisting legal situation in which private companies are free to construct major industrial facilities with little or no restraint by federal regulation.

CORPORATE STRATEGIC PLANNING

Contingency planning represents an essential component of sound environmental planning for a new plant. The environmental uncertainties surrounding a large capital project should be specified and related to other contingencies (such as marketing, competitive reactions, politics, foreign trade, etc.) and mapped out in the overall corporate strategy.

Environmental factors should also be incorporated into a company's technical or research and development program. Since the planning horizons for new projects may now extend to 5 to 10 years, R&D programs can be designed for specific projects. These may include new process modifications or end-of-pipe control technologies.

Another clear need is to integrate environmental factors into financial planning for major projects. It must be recognized that strategic environmental planning is as important to the long-range goal of the corporation as is financial planning. Trade-off decisions regarding financing may have to change as the project goes through successive stages of environmental planning and permit negotiations. For example, requirements for the use of more expensive pollution control technology may significantly increase total project costs; or a change from end-of-pipe to process modification technology may preclude the use of industrial revenue bond financing under Internal Revenue Service (IRS) rules. Regulatory delays can affect assumptions as to both the rate of expenditure and inflation factors. Investment, production, environmental, and legal factors are all interrelated and can have a major impact on corporate cash flow.

Most companies must learn to deal more creatively with local officials and public opinion. The social responsibility of companies can become an extremely important issue. Companies should apply thoughtfulness and skill to the timing and conduct of public hearings. Management must recognize that local officials have views and constituencies that go beyond attracting new jobs.

From all these factors, it is clear that the approval and construction of major new industrial plants or expansions is a far more complicated operation than it has been in the past, even the recent past. Stringent environmental restrictions are likely to preclude construction of certain facilities at locations where they otherwise might have been built. In other cases, acquisition of required approvals may generate a heated technical and political debate that can drag out the regulatory process for several years.

In many instances, new requirements may be imposed while a company is seeking approval for a proposed new plant. Thus, companies intending to expand their basic production facilities should anticipate their needs far in advance, begin preparation to meet the regulatory challenge they will eventually confront, and select sites with careful consideration of environmental attributes. It is the objective of this section to assist the engineer in meeting this environmental regulatory challenge.

UNITED STATES AIR QUALITY LEGISLATION AND REGULATIONS

Although considerable federal legislation dealing with air pollution has been enacted since the 1950s, the basic statutory framework now in effect was established by the Clean Air Act of 1970; amended in 1974 to deal with energy-related issues; amended in 1977, when a number of amendments containing particularly important provisions associated with the approval of new industrial plants were adopted; and amended in 1990 to address toxic air pollutants and ozone nonattainment areas.

Clean Air Act of 1970 The Clean Air Act of 1970 was founded on the concept of attaining National Ambient Air Quality Standards (NAAQS). Data were accumulated and analyzed to establish the quality of the air, identify sources of pollution, determine how pollutants disperse and interact in the ambient air, and define reductions and controls necessary to achieve air-quality objectives.

EPA promulgated the basic set of current ambient air-quality standards in April 1971. The specific regulated pollutants were particulates, sulfur dioxide, photochemical oxidants, hydrocarbons, carbon monoxide, and nitrogen oxides. In 1978, lead was added. Table 22-1 enumerates the present standards.

To provide basic geographic units for the air-pollution control program, the United States was divided into 247 air quality control regions (AQCRs). By a standard rollback approach, the total quantity of pollution in a region was estimated, the quantity of pollution that could be tolerated without exceeding standards was then calculated, and the degree of reduction called for was determined. States were required by EPA to develop state implementation plans (SIPs) to achieve compliance.

The act also directed EPA to set new source performance standards (NSPS) for specific industrial categories. New plants were required to use the best system of emission reduction available. EPA gradually issued these standards, which now cover a number of basic industrial categories (as listed in Table 22-2). The 1977 amendments to the Clean Air Act directed EPA to accelerate the NSPS program and included a regulatory program to prevent significant deterioration in those areas of the country where the NAAQS were being attained.

Finally, Sec. 112 of the Clean Air Act required that EPA promulgate National Emission Standards for Hazardous Air Pollutants

(NESHAPs). Between 1970 and 1989, standards were promulgated for asbestos, beryllium, mercury, vinyl chloride, benzene, arsenic, radionuclides, and coke-oven emissions.

Prevention of Significant Deterioration (PSD) Of all the federal laws placing environmental controls on industry (and, in particular, on new plants), perhaps the most confusing and restrictive are the limits imposed for the prevention of significant deterioration (PSD) of air quality. These limits apply to areas of the country that are already cleaner than required by ambient air-quality standards. This regulatory framework evolved from judicial and administrative action under the 1970 Clean Air Act and subsequently was given full statutory foundation by the 1977 Clean Air Act Amendments.

EPA established an area classification scheme to be applied in all such regions. The basic idea was to allow a moderate amount of industrial development but not enough to degrade air quality to a point at which it barely complied with standards. In addition, states were to designate certain areas where pristine air quality was especially desirable. All air-quality areas were categorized as Class I, Class II, or Class III. Class I areas were pristine areas subject to the tightest control. Permanently designated Class I areas included international parks, national wilderness areas, memorial parks exceeding 5000 acres, and national parks exceeding 6000 acres. Although the nature of these areas is such that industrial projects would not be located within them, their Class I status could affect projects in neighboring areas where meteorological conditions might result in the transport of emissions into them. Class II areas were areas of moderate industrial growth. Class III areas were areas of major industrialization. Under EPA regulations promulgated in December 1974, all areas were initially categorized as Class II. States were authorized to reclassify specified areas as Class I or Class III.

The EPA regulations also established another critical concept known as the *increment*. This was the numerical definition of the amount of additional pollution that may be allowed through the combined effects of all new growth in a particular locality (see Table 22-3). To assure that the increments would not be used up hastily, EPA specified that each major new plant must install best available control technology (BACT) to limit emissions. BACT is determined based on a case-by-case engineering analysis and is more stringent than NSPS.

To implement these controls, EPA requires that every new source undergo preconstruction review. The regulations prohibited a company

TABLE 22-1 National Ambient Air Quality Standards

Pollutant	Primary stds.	Averaging times	Secondary stds.
Carbon monoxide	9 ppm (10 mg/m ³) 35 ppm (40 mg/m ³)	8-h ^a 1-h ^a	None None
Lead	1.5 µg/m ³	Quarterly average	Same as primary
Nitrogen dioxide	0.053 ppm (100 µg/m ³)	Annual (arithmetic mean)	Same as primary
Particulate matter (PM ₁₀)	50 µg/m ³ 150 µg/m ³	Annual ^b (arithmetic mean) 24-h ^c	Same as primary Same as primary
Particulate matter (PM _{2.5})	15.0 µg/m ³ 65 µg/m ³	Annual ^c (arithmetic mean) 24-h ^d	Same as primary Same as primary
Ozone	0.08 ppm 0.12 ppm	8-h ^e 1-h ^f	Same as primary Same as primary
Sulfur oxides	0.03 ppm 0.14 ppm	Annual (arithmetic mean) 24-h ^e 3-h ^e	— — 0.5 ppm (1300 µg/m ³)

^aNot to be exceeded more than once per year.

^bTo attain this standard, the expected annual arithmetic mean PM₁₀ concentration at each monitor within an area must not exceed 50 µg/m³.

^cTo attain this standard, the 3-year average of the annual arithmetic mean PM_{2.5} concentrations from single or multiple community-oriented monitors must not exceed 15.0 µg/m³.

^dTo attain this standard, the 3-year average of the 98th percentile of 24-h concentrations at each population-oriented monitor within an area must not exceed 65 µg/m³.

^eTo attain this standard, the 3-year average of the fourth-highest daily maximum 8-h average ozone concentrations measured at each monitor within an area over each year must not exceed 0.08 ppm.

^f(1) The standard is attained when the expected number of days per calendar year with maximum hourly average concentrations above 0.12 ppm is ≤ 1.

(2) The 1-h NAAQS will no longer apply to an area 1 year after the effective date of the designation of that area for the 8-h ozone NAAQS. The effective designation date for most areas is June 15, 2004. [40 CFR 50.9; see Federal Register of April 30, 2004 (69 FR 23996)].

TABLE 22-2 New Source Performance Standards (NSPS) from 40 CFR Part 60 as of June 2004

40 CFR 60	NSPS
Subpart C	Emission Guidelines and Compliance Times
Subpart Ca	(Reserved)
Subpart Cb	Emissions Guidelines and Compliance Times for Large Municipal Waste Combustors That Are Constructed On or Before September 20, 1994
Subpart Cc	Emission Guidelines and Compliance Times for Municipal Solid Waste Landfills
Subpart Cd	Emission Guidelines and Compliance Times for Sulfuric Acid Production Units
Subpart Ce	Emission Guidelines and Compliance Times for Hospital/Medical/Infectious Waste Incinerators
Subpart D	Standards of Performance for Fossil-Fuel-Fired Steam Generators for Which Construction is Commenced After August 17, 1971
Subpart Da	Standards of Performance for Electric Utility Steam Generating Units for Which Construction Is Commenced After September 18, 1978
Subpart Db	Standards of Performance for Industrial-Commercial-Institutional Steam Generating Units
Subpart Dc	Standards of Performance for Small Industrial-Commercial-Institutional Steam Generating Units
Subpart E	Standards of Performance for Incinerators
Subpart Ea	Standards of Performance for Municipal Waste Combustors for which Construction is Commenced after December 20, 1989 and on or before September 20, 1994
Subpart Eb	Standards of Performance for Large Municipal Waste Combustors for Which Construction is Commenced after September 20, 1994 or for which Modification of Reconstruction is Commenced after June 19, 1996
Subpart Ec	Standards of Performance for Hospital/Medical/Infectious Waste Incinerators for which Construction Is Commenced after June 20, 1996
Subpart F	Standards of Performance for Portland Cement Plants
Subpart G	Standards of Performance for Nitric Acid Plants
Subpart H	Standards of Performance for Sulfuric Acid Plants
Subpart I	Standards of Performance for Hot Mix Asphalt Facilities
Subpart J	Standards of Performance for Petroleum Refineries
Subpart K	Standards of Performance for Storage Vessels for Petroleum Liquids Constructed After June 11, 1973 and Prior to May 19, 1978
Subpart Ka	Standards of Performance for Storage Vessels for Petroleum Liquids for Which Construction, Reconstruction, or Modification Commenced After May 18, 1978, and Prior to July 23, 1984
Subpart Kb	Standards of Performance for Volatile Organic Liquid Storage Vessels (Including Petroleum Liquid Storage Vessels) for Which Construction, Reconstruction, or Modification Commenced after July 23, 1984
Subpart L	Standards of Performance for Secondary Lead Smelters
Subpart M	Standards of Performance for Secondary Brass and Bronze Production Plants
Subpart N	Standards of Performance for Primary Emissions from Basic Oxygen Process Furnaces for Which Construction Is Commenced After June 11, 1973
Subpart Na	Standards of Performance for Secondary Emissions From Basic Oxygen Process Steelmaking Facilities for Which Construction Is Commenced After January 20, 1983
Subpart O	Standards of Performance for Sewage Treatment Plants
Subpart P	Standards of Performance for Primary Copper Smelters
Subpart Q	Standards of Performance for Primary Zinc Smelters
Subpart R	Standards of Performance for Primary Lead Smelters
Subpart S	Standards of Performance for Primary Aluminum Reduction Plants
Subpart T	Standards of Performance for the Phosphate Fertilizer Industry: Wet-Process Phosphoric Acid Plants
Subpart U	Standards of Performance for the Phosphate Fertilizer Industry: Superphosphoric Acid Plants
Subpart V	Standards of Performance for the Phosphate Fertilizer Industry: Diammonium Phosphate Plants
Subpart W	Standards of Performance for the Phosphate Fertilizer Industry: Triple Superphosphate Plants
Subpart X	Standards of Performance for the Phosphate Fertilizer Industry: Granular Triple Superphosphate Storage Facilities
Subpart Y	Standards of Performance for Coal Preparation Plants
Subpart ZA	Standards of Performance for Ferroalloy Production Facilities
Subpart AA	Standards of Performance for Steel Plants: Electric Arc Furnaces Constructed After October 21, 1974, and On or Before August 17, 1983
Subpart AAa	Standards of Performance for Steel Plants: Electric Arc Furnaces and Argon-Oxygen Decarburization Vessels Constructed After August 17, 1983
Subpart BB	Standards of Performance for Kraft Pulp Mills
Subpart CC	Standards of Performance for Glass Manufacturing Plants
Subpart DD	Standards of Performance for Grain Elevators
Subpart EE	Standards of Performance for Surface Coating of Metal Furniture
Subpart FF	(Reserved)
Subpart GG	Standards of Performance for Stationary Gas Turbines
Subpart HH	Standards of Performance for Lime Manufacturing Plants
Subpart KK	Standards of Performance for Lead-Acid Battery Manufacturing Plants
Subpart LL	Standards of Performance for Metallic Mineral Processing Plants
Subpart MM	Standards of Performance for Automobile and Light Duty Truck Surface Coating Operations
Subpart NN	Standards of Performance for Phosphate Rock Plants
Subpart PP	Standards of Performance for Ammonium Sulfate Manufacture
Subpart QQ	Standards of Performance for the Graphic Arts Industry: Publication Rotogravure Printing
Subpart RR	Standards of Performance for Pressure Sensitive Tape and Label Surface Coating Operations
Subpart SS	Standards of Performance for Industrial Surface Coating: Large Appliances
Subpart TT	Standards of Performance for Metal Coil Surface Coating
Subpart UU	Standards of Performance for Asphalt Processing and Asphalt Roofing Manufacture
Subpart VV	Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry
Subpart WW	Standards of Performance for the Beverage Can Surface Coating Industry
Subpart XX	Standards of Performance for Bulk Gasoline Terminals
Subpart AAA	Standards of Performance for New Residential Wood Heaters

TABLE 22-2 New Source Performance Standards (NSPS) from 40 CFR Part 60 as of June 2004 (Concluded)

40 CFR 60	NSPS
Subpart BBB	Standards of Performance for the Rubber Tire Manufacturing Industry
Subpart CCC	(Reserved)
Subpart DDD	Standards of Performance for Volatile Organic Compound (VOC) Emissions from the Polymer Manufacturing Industry
Subpart EEE	(Reserved)
Subpart FFF	Standards of Performance for Flexible Vinyl and Urethane Coating and Printing
Subpart GGG	Standards of Performance for Equipment Leaks of VOC in Petroleum Refineries
Subpart HHH	Standards of Performance for Synthetic Fiber Production Facilities
Subpart III	Standards of Performance for Volatile Organic Compound Emissions From the Synthetic Organic Chemical Manufacturing Industry (SOCMI) Air Oxidation Unit Processes
Subpart JJJ	Standards of Performance for Petroleum Dry Cleaners
Subpart KKK	Standards of Performance for Equipment Leaks of VOC From Onshore Natural Gas Processing Plants
Subpart LLL	Standards of Performance for Onshore Natural Gas Processing: SO(2) Emissions
Subpart MMM	(Reserved)
Subpart NNN	Standards of Performance for Volatile Organic Compound Emissions from Synthetic Organic Chemical Manufacturing Industry Distillation Operations
Subpart OOO	Standards of Performance for Nonmetallic Mineral Processing Plants
Subpart PPP	Standards of Performance for Wool Fiberglass Insulation Manufacturing Plants
Subpart QQQ	Standards of Performance for VOC Emissions From Petroleum Refinery Wastewater Systems
Subpart RRR	Standards of Performance for Volatile Organic Compound Emissions From Synthetic Organic Chemical Manufacturing Industry (SOCMI) Reactor Processes
Subpart SSS	Standards of Performance for Magnetic Tape Coating Facilities
Subpart TTT	Standards of Performance for Industrial Surface Coating; Coating of Plastic Parts for Business Machines
Subpart UUU	Standards of Performance for Calciners and Dryers in Mineral Industries
Subpart VVV	Standards of Performance for Polymeric Coating of Supporting Substrates Facilities
Subpart WWW	Standards of Performance for Municipal Solid Waste Landfills
Subpart AAAA	Standards of Performance for Small Municipal Waste Combustion Units for Which Construction Is Commenced After August 30, 1999 or for Which Modification or Reconstruction Is Commenced After June 6, 2001
Subpart BBBB	Emission Guidelines and Compliance Times for Small Municipal Waste Combustion Units Constructed On or Before August 30, 1999
Subpart CCCC	Standards of Performance for Commercial and Industrial Solid Waste Incineration Units for Which Construction Is Commenced After November 30, 1999 or for Which Modification or Reconstruction Is Commenced on or After June 1, 2001
Subpart DDDD	Emissions Guidelines and Compliance Times for Commercial and Industrial Solid Waste Incineration Units that Commenced Construction On or Before November 30, 1999

from commencing construction on a new source until the review had been completed and provided that, as part of the review procedure, public notice should be given and an opportunity provided for a public hearing on any disputed questions.

Sources Subject to Prevention of Significant Deterioration (PSD) Sources subject to PSD regulations (40 CFR, Sec. 52.21) are major stationary sources and major modifications located in attainment areas and unclassified areas. A major stationary source was defined as any source listed in Table 22-4 with the potential to emit 100 tons per year or more of any pollutant regulated under the Clean Air Act (CAA) or any other source with the potential to emit 250 tons per year or more of any CAA pollutant. The "potential to emit" is defined as the maximum capacity to emit the pollutant under applicable emission standards and permit conditions (after application of any air pollution control equipment) excluding secondary emissions. A "major modification" is defined as any physical or operational change of a major stationary source producing a "significant net emissions increase" of any CAA pollutant (see Table 22-5).

Ambient monitoring is required of all CAA pollutants with emissions greater than or equal to Table 22-5 values for which there are

NAAQS. Continuous monitoring is also required for other CAA pollutants for which the EPA or the state determines that monitoring is necessary. The EPA or the state may exempt any CAA pollutant from these monitoring requirements if the maximum air-quality impact of the emissions increase is less than the values in Table 22-6 or if present concentrations of the pollutant in the area that the new source would affect are less than the Table 22-6 values. The EPA or the state may accept representative existing monitoring data collected within 3 years of the permit application to satisfy monitoring requirements.

EPA regulations provide exemption from BACT and ambient air-impact analysis if the modification that would increase emissions is accompanied by other changes within the plant that would net a zero increase in total emissions. This exemption is referred to as the "no net increase" exemption.

A full PSD review would include a case-by-case determination of the controls required by BACT, an ambient air-impact analysis to determine whether the source might violate applicable increments or air-quality standards; an assessment of the effect on visibility, soils, and vegetation; submission of monitoring data; and full public review.

Nonattainment (NA) Those areas of the United States failing to attain compliance with ambient air-quality standards are considered nonattainment areas. New plants could be constructed in nonattainment areas only if stringent conditions are met. Emissions have to be controlled to the greatest degree possible, and more than equivalent offsetting emission reductions have to be obtained from other sources to assure progress toward achievement of the ambient air-quality standards. Specifically, (1) the new source must be equipped with pollution controls to assure lowest achievable emission rate (LAER), which in no case can be less stringent than any applicable NSPS; (2) all existing sources owned by an applicant in the same region must be in compliance with applicable state implementation plan requirements or be under an approved schedule or

TABLE 22-3 PSD Air Quality Increments ($\mu\text{g}/\text{m}^3$)

	Class I area	Class II area	Class III area
Sulfur dioxide			
Annual	2	20	40
24-h	5	91	182
3-h	25	512	700
PM ₁₀			
Annual	4	17	34
24-h	8	30	60
Nitrogen dioxide			
Annual	2.5	25	50

TABLE 22-4 Sources Subject to PSD Regulation If Their Potential to Emit Equals or Exceeds 100 Tons per Year

Fossil-fuel-fired steam electric plants of more than 250 million Btu/h heat input
Coal-cleaning plants (with thermal dryers)
Kraft-pulp mills
Portland-cement plants
Primary zinc smelters
Iron and steel mill plants
Primary aluminum-ore-reduction plants
Primary copper smelters
Municipal incinerators capable of charging more than 250 tons of refuse per day
Hydrofluoric, sulfuric, and nitric acid plants
Petroleum refineries
Lime plants
Phosphate-rock-processing plants
Coke-oven batteries
Sulfur-recovery plants
Carbon-black plants (furnace process)
Primary lead smelters
Fuel-conversion plants
Sintering plants
Secondary metal-production plants
Chemical-process plants
Fossil-fuel boilers (or combinations thereof) totaling more than 250 million Btu/h heat input
Petroleum-storage and -transfer units with total storage capacity exceeding 300,000 bbl
Taconite-ore-processing plants
Glass-fiber-processing plants
Charcoal-production plants

an enforcement order to achieve such compliance; (3) the applicant must have sufficient offsets to more than make up for the emissions to be generated by the new source (after application of LAER); and (4) the emission offsets must provide “a positive net air quality benefit in the affected area.”

LAER was deliberately a technology-forcing standard of control. The statute stated that LAER must reflect (1) the most stringent emission limitation contained in the implementation plan of any state for such category of sources unless the applicant can demonstrate that such a limitation is not achievable, or (2) the most stringent limitation achievable in practice within the industrial category, whichever is more stringent. In no event could LAER be less stringent than any applicable NSPS. While the statutory language defining BACT directed that “energy, environmental, and economic impacts and other costs” be taken into account, the comparable provision on LAER provided no instruction that economics be considered.

TABLE 22-5 Significant Net Emissions Increase

Pollutant	Tons/yr
PM ₁₀	15
SO ₂	40
NO _x	40
VOC	40
CO	100
Lead	0.6
PM	25
Fluorides	3
Sulfuric acid mist	7
Hydrogen sulfide	10
Total reduced sulfur	10
Municipal waste combustor	
Acid gases	40
Metals	15
Organics	3.5 × 10 ⁻⁶
CFCs (11, 12, 113, 114 & 115)	Any increase
Halons (1211, 1301, 2402)	Any increase
Any emissions increase resulting in a >1 μ/m ³ 24-h impact in a Class I area	

TABLE 22-6 Concentration Impacts below Which Ambient Monitoring May Not Be Required

	μg/m ³	Average time
CO	575	8-h maximum
NO ₂	14	Annual
PM ₁₀	10	24-h maximum
SO ₂	13	24-h maximum
Lead	0.1	3-month
Fluorides	0.25	24-h maximum
Total reduced sulfur	10	1-h maximum
H ₂ S	0.2	1-h maximum
Reduced-sulfur compounds	10	1-h maximum

For existing sources emitting pollutants for which the area is nonattainment, reasonable available control technology (RACT) would be required. EPA defines RACT by industrial category.

Minor Source Preconstruction Permits Commonly after the application of air pollution controls the emissions associated with new facilities or modifications of existing facilities will be less than the emissions thresholds necessary to trigger PSD or nonattainment new source review. In such instances, a minor source preconstruction permit is required from the state permitting authority and new source performance standards (NSPS) apply.

Controlled-Trading Program The legislation enacted under the Clean Air Act Amendments of 1977 provided the foundation for EPA's controlled-trading program, the essential elements of which include:

- No-net-increase provisions of PSD
- Offsets policy (under nonattainment)
- Banking and brokerage (under nonattainment)

While these different policies vary broadly in form, their objective is essentially the same: to substitute flexible economic-incentive systems for the current rigid, technology-based regulations that specify exactly how companies must comply. These market mechanisms have made regulating easier for EPA and less burdensome and costly for industry.

PSD No-Net-Increase Provisions Under the PSD program, the affected source is the entire plant. This source definition allows a company to determine the most cost-effective means to control pollution any time a plant is modified. Emission increases associated with a new process or production line may be compensated for by emission increases at other parts of the plant. As long as the entire site net emissions increase is maintained below the levels identified in Table 22-5, a PSD review is not required.

Offsets Policy Offsets were EPA's first application of the concept that one source could meet its environmental protection obligations by getting another source to assume additional control actions. In nonattainment areas, pollution from a proposed new source, even one that controls its emissions to the lowest possible level, would aggravate existing violations of ambient air-quality standards and trigger the statutory prohibition. The offsets policy provided these new sources with an alternative. The source could proceed with construction plans, provided that:

1. The source would control emissions to the lowest achievable level.
2. Other sources owned by the applicant were in compliance or on an approved compliance schedule.

3. Existing sources were persuaded to reduce emissions by an amount at least equal to the pollution that the new source would add.

Banking and Brokerage Policy EPA's banking policy is aimed at providing companies with incentives to find more offsets. Under the original offset policy, a firm shutting down or modifying a facility could apply the reduction in emissions to new construction elsewhere in the region only if the changes were made simultaneously. However, with banking a company can “deposit” the reduction for later use or sale. Such a policy will clearly establish that clean air (or the right to use it) has direct economic value.

Clean Air Act of 1990 In November 1990, Congress adopted the Clean Air Act Amendments of 1990, providing substantial

TABLE 22-7 Ozone Nonattainment Area Classifications and Associated Requirements

Nonattainment area classification	One-hour ozone concentration design value, ppm	Attainment date	Major source threshold level, tons VOCs/yr	Offset ratio for new/modified sources
Marginal	0.121–0.138	Nov. 15, 1993	100	1.1 to 1
Moderate	0.138–0.160	Nov. 15, 1996	100	1.15 to 1
Serious	0.160–0.180	Nov. 15, 1999	50	1.2 to 1
Severe	0.180–0.190	Nov. 15, 2005	25	1.3 to 1
	0.190–0.280	Nov. 15, 2007	25	1.3 to 1
Extreme	0.280 and up	Nov. 15, 2010	10	1.5 to 1

changes to many aspects of the existing CAAA. The concepts of NAAQS, NSPS, and PSD remain virtually unchanged. However, significant changes have occurred in several areas that directly affect industrial facilities and electric utilities and air-pollution control at these facilities. These include changes and additions in the following major areas:

- Title I Nonattainment areas
- Title III Hazardous air pollutants
- Title IV Acid deposition control
- Title V Operating permits
- Title VI Stratospheric ozone protection

Title I: Nonattainment Areas The existing regulations for nonattainment areas were made more stringent in several areas. The CAAA of 1990 requires the development of comprehensive emission inventory-tracking for all nonattainment areas and establishes a classification scheme that defined nonattainment areas into levels of severity. For example, ozone nonattainment areas were designated as marginal, moderate, serious, severe (two levels), and extreme, with compliance deadlines of 3, 6, 9, 15–17, and 20 years, respectively, with each classification having more stringent requirements regarding strategies for compliance (see Table 22-7). Volatile organic compound (VOC) emissions reductions of 15 percent were required in moderate areas by 1996 and 3 percent a year thereafter for severe or extreme areas until compliance was achieved. In addition, the definition of a major source of ozone precursors (previously 100 tons per year of NO_x, CO, or VOC emissions) was redefined to as little as 10 tons per year in the extreme classification, with increased offset requirements of 1.5 to 1 for new and modified sources. In 2004, EPA adopted a new 8-h ozone ambient air quality standard. Ultimately the 8-h standard will replace the 1-h standard although both standards apply during a several year transition period. The new standard adopted the same regulatory approach as previously existed for the 1-h standard. New extended deadlines for compliance are being adopted based on the same marginal to extreme area approach previously described with the new deadlines identified in Table 22-8. These requirements place major constraints on affected industries in these nonattainment areas. A similar approach is being taken in PM₁₀ and CO nonattainment areas.

Title III: Hazardous Air Pollutants The Title III provisions on hazardous air pollutants (HAPs) represent a major departure from the previous approach of developing NESHAPs. While only eight HAPs were designated in the 20 years since enactment of the CAAA of 1970, the new CAAA of 1990 designated 189 pollutants as HAPs requiring regulation. The state allows EPA to list additional HAPs or to delist any HAP. As of 2004 there were 188 HAPs identified in Table 22-9. Major sources are defined as any source (new or existing) that emits (after control) 10 tons a year or more of any regulated HAP or 25 tons a year or more of any combination of HAPs. A major source of HAPs must comply with emissions standards known as MACT standards that EPA has been adopting since 1992. Table 22-10 provides a list of MACT standards as of 2004 that affect all named existing and new major HAP sources. If a proposed new source is not of a type identified in Table 22-10, then the new source must undergo a case-by-case MACT evaluation to identify the appropriate level of control which is generally the most stringent achieved in practice.

For existing sources, MACT is defined as a stringency equivalent to the average of the best 12 percent of the sources in the category. For

new sources, MACT is defined as the best controlled system. New sources are required to meet MACT immediately, while existing sources have three years from the date of promulgation of the appropriate MACT standard.

Title IV: Acid Deposition Control The Acid Deposition Control Program is designed to reduce emissions of SO₂ in the United States by 10 million tons per year, resulting in a net yearly emission of 8.9 million tons by the year 2000. Phase I of the program requires 111 existing uncontrolled coal-fired power plants (≥ 100 MW) to reduce emissions to 2.5 pounds of SO₂ per 10⁶ Btu by 1995 (1997 if scrubbers are used to reduce emissions by at least 90 percent). The reduction is to be accomplished by issuing all affected units emission “allowances” equivalent to what their annual average SO₂ emissions would have been in the years 1985–1987 based on 2.5 pounds SO₂ per 10⁶ Btu coal. The regulations represent a significant departure from previous regulations where specified SO₂ removal efficiencies were mandated; rather, the utilities will be allowed the flexibility of choosing which strategies will be used (e.g., coal washing, low-sulfur coal, flue gas desulfurization, etc.) and which units will be controlled, as long as the overall “allowances” are not exceeded. Any excess reduction in SO₂ by a utility will create “banked” emissions that can be sold or used at another unit.

Phase II of Title IV limits the majority of plants ≥ 20 MW and all plants ≥ 75 MW to maximum emissions of 1.2 pounds of SO₂ per 10⁶ Btu after the year 2000. In general, new plants would have to acquire banked emission allowances in order to be built. Emission allowances will be traded through a combination of sell/purchase with other utilities, EPA auctions and direct sales.

Control of NO_x under the CAAA of 1990 will be accomplished through the issuance of a revised NSPS in 1994, with the objective of reducing emissions by 2 million tons a year from 1980 emission levels. The technology being considered is the use of low-NO_x burners (LNBs). The new emission standards will not apply to cyclone and wet bottom boilers, unless alternative technologies are found, as these cannot be retrofitted with existing LNB technologies.

Title V: Operating Permits Title V of the 1990 Clean Air Act Amendments established a new operating permit program to be administered by state agencies in accordance with federal guidelines. The program basically requires a source to obtain a permit that covers

TABLE 22-8 The 8-h Ozone NAAQS Classifications and Attainment Dates*

Area classification ¹	8-h ozone concentration, ppm	Attainment date ¹
Marginal	0.081–0.092	2007
Moderate	0.092–0.107	2010
Serious	0.107–0.120	2013
Severe-15	0.120–0.127	2019
Severe-17	0.127–0.187	2021
Extreme	0.187 and up	2024

*Areas newly designated as nonattainment under the 8-h standard which were never previously nonattainment must comply with the standard within 5 years of being designated nonattainment or generally in 2009.

¹These classifications and attainment dates apply to areas previously designated nonattainment under the 1-h ozone NAAQS.

TABLE 22-9 Regulated Hazardous Air Pollutants (HAPs) from 40 CFR Part 63 as of June 2004

CAS no.	HAP	CAS no.	HAP
75070	Acetaldehyde	108101	Methyl isobutyl ketone (Hexone)
60355	Acetamide	624839	Methyl isocyanate
75058	Acetonitrile	80626	Methyl methacrylate
98862	Acetophenone	1634044	Methyl tert butyl ether
53963	Acetylaminofluorene	101144	4,4-Methyene bis(2-chloroaniline)
107028	Acrolein	75092	Methylene chloride (Dichloromethane)
79061	Acrylamide	101688	Methylene diphenyl diisocyanate (MDI)
79107	Acrylic acid	101779	4,4-Methylenedianiline
107131	Acrylonitrile	91203	Naphthalene
107051	Allyl chloride	98953	Nitrobenzene
92671	4-Aminobiphenyl	92933	4-Nitrobiphenyl
62533	Aniline	100027	4-Nitrophenol
90040	o-Anisidine	79469	2-Nitropropane
1332214	Asbestos	684935	N-Nitroso-N-methylurea
71432	Benzene (including benzene from gasoline)	62759	N-Nitrosodimethylamine
92875	Benidine	59892	N-Nitrosomorpholine
98077	Benzotrichloride	56382	Parathion
100447	Benzyl chloride	82688	Pentachloronitrobenzene (Quintobenzene)
92524	Biphenyl	87865	Pentachlorophenol
117817	Bis(2-ethylhexyl)phthalate (DEHP)	108952	Phenol
54288	Bis(chloromethyl)ether	106503	p-Phenylenediamine
75252	Bromoform	75445	Phosgene
106990	1,3-Butadiene	7803512	Phosphine
156627	Calcium cyanamide	7723140	Phosphorus
133062	Captan	85449	Phthalic anhydride
63252	Carbaryl	1336363	Polychlorinated biphenyls (Aroclors)
75150	Carbon disulfide	1120714	1,3-Propane sultone
56235	Carbon tetrachloride	57578	beta-Propiolactone
463581	Carbonyl sulfide	123386	Propionaldehyde
120809	Catechol	114261	Propoxur (Baygon)
133904	Chloramben	78875	Propylene dichloride (1,2-Dichloropropane)
57749	Chlordane	542756	1,3-Dichloropropene
7782505	Chlorine	62737	Dichlorvos
79118	Chloroacetic acid	111422	Diethanolamine
532274	2-Chloroacetophenone	121697	N,N-Diethyl aniline (N,N-Dimethylaniline)
108907	Chlorobenzene	64675	Diethyl sulfate
510156	Chlorobenzilate	119904	3,3-Dimethoxybenzidine
67663	Chloroform	60117	Dimethyl aminoazobenzene
107302	Chloromethyl methyl ether	119937	3,3'-Dimethyl benzidine
126998	Chloroprene	79447	Dimethyl carbamoyl chloride
1319773	Cresols/Cresylic acid (isomers and mixture)	68122	Dimethyl formamide
95487	o-Cresol	57147	1,1-Dimethyl hydrazine
108394	m-Cresol	131113	Dimethyl phthalate
106445	p-Cresol	77781	Dimethyl sulfate
98828	Cumene	534521	4,6-Dinitro-o-cresol, and salts
94757	2,4-D, salts and esters	51285	2,4-Dinitrophenol
3547044	DDE	121142	2,4-Dinitrotoluene
334883	Diazomethane	123911	1,4-Dioxane (1,4-Diethyleneoxide)
132649	Dibenzofurans	122667	1,2-Diphenylhydrazine
96128	1,2-Dibromo-3-chloropropane	106898	Epichlorohydrin (1-Chloro-2,3-epoxypropane)
84742	Dibutylphthalate	106887	1,2-Epoxybutane
106467	1,4-Dichlorobenzene(p)	140885	Ethyl acrylate
91941	3,3-Dichlorobenzidine	100414	Ethyl benzene
111444	Dichloroethyl ether (Bis (2-chloroethyl)ether)	51796	Ethyl carbamate (Urethane)
76448	Heptachlor	75003	Ethyl chloride (Chloroethane)
118741	Hexachlorobenzene	106934	Ethylene dibromide (Dibromoethane)
87683	Hexachlorobutadiene	107062	Ethylene dichloride (1,2-Dichloroethane)
77474	Hexachlorocyclopentadiene	107211	Ethylene glycol
67721	Hexachloroethane	151564	Ethylene imine (Aziridine)
822060	Hexamethylene-1,6-diisocyanate	75218	Ethylene oxide
680319	Hexamethylphosphoramide	96457	Ethylene thiourea
110543	Hexane	75343	Ethylidene dichloride (1,1-Dichloroethane)
302012	Hydrazine	50000	Formaldehyde
7647010	Hydrochloric acid	75569	Propylene oxide
7664393	Hydrogen fluoride (Hydrofluoric acid)	75558	1,2-Propylenimine (2-Methyl aziridine)
123319	Hydroquinone	91225	Quinoline
78591	Isophorone	106514	Quinone
58899	Lindane (all isomers)	100425	Styrene
108316	Maleic anhydride	96093	Styrene oxide
67561	Methanol	1746016	2,3,7,8-Tetrachlorodibenzo-p-dioxin
72435	Methoxychlor	79345	1,1,2,2-Tetrachloroethane
74839	Methyl bromide (Bromomethane)	127184	Tetrachloroethylene (Perchloroethylene)
74873	Methyl chloride (Chloromethane)	7550450	Titanium tetrachloride
71556	Methyl chloroform (1,1,1-Trichloroethane)	108883	Toluene
78933	Methyl ethyl ketone (2-Butanone)	95807	2,4-Toluene diamine
60344	Methyl hydrazine	584849	2,4-Toluene diisocyanate
74884	Methyl iodide (Iodomethane)	95534	o-Toluidine

22-12 WASTE MANAGEMENT

TABLE 22-9 Regulated Hazardous Air Pollutants (HAPs) from 40 CFR Part 63 as of June 2004 (Concluded)

CAS no.	HAP	CAS no.	HAP
8001352	Toxaphene (chlorinated camphene)	—	Antimony compounds
120821	1,2,4-Trichlorobenzene	—	Arsenic compounds (inorganic including arsine)
79005	1,1,2-Trichloroethane	—	Beryllium compounds
79016	Trichloroethylene	—	Cadmium compounds
95954	2,4,5-Trichlorophenol	—	Chromium compounds
88062	2,4,6-Trichlorophenol	—	Cobalt compounds
121448	Triethylamine	—	Coke oven emissions
1582098	Trifluralin	—	Cyanide compounds ¹
540841	2,2,4-Trimethylpentane	—	Glycol ethers ²
108054	Vinyl acetate	—	Lead compounds
593602	Vinyl bromide	—	Manganese compounds
75014	Vinyl chloride	—	Mercury compounds
75354	Vinylidene chloride (1,1-Dichloroethylene)	—	Fine mineral fibers ³
1330207	Xylenes (isomers and mixture)	—	Nickel compounds
95476	o-Xylenes	—	Polycyclic organic matter ⁴
108383	m-Xylenes	—	Radionuclides (including radon) ⁵
106423	p-Xylenes	—	Selenium compounds

NOTE: For all listings above which contain the word "compounds" and for glycol ethers, the following applies: Unless otherwise specified, these listings are defined as including any unique chemical substance that contains the named chemical (i.e., antimony, arsenic, etc.) as part of that chemical's infrastructure.

¹XCN where X = H' or any other group where a formal dissociation may occur, for example, KCN or Ca (CN)₂.

²Includes mono- and di-ethers of ethylene glycol, diethylene glycol, and triethylene glycol R-(OCH₂CH₂)_n-OR' where n = 1, 2, or 3; R = alkyl or aryl groups; R' = R, H, or groups which, when removed, yield glycol ethers with the structure R-(OCH₂CH)_n-OH. Polymers are excluded from the glycol category.

³Includes mineral fiber emissions from facilities manufacturing or processing glass, rock, or slag fibers (or other mineral derived fibers) of average diameter 1 micrometer or less.

⁴Includes organic compounds with more than one benzene ring, and which have a boiling point greater than or equal to 100°C.

⁵A type of atom which spontaneously undergoes radioactive decay.

each and every requirement applicable to the source under the Clean Air Act. A major impact of the Title V program is that sources are required to implement measures to demonstrate routinely that they are operating in compliance with permit terms. This represents a dramatic shift from previous practices where, to bring an enforcement action, regulatory authorities were required to demonstrate that a source was in noncompliance.

Initially, all "major" sources of air pollution are required to obtain an operating permit. However, any state permitting authority may extend the applicability of the operating permit to minor sources as well. Once a source is subject to the permit program as a major source for any one pollutant, emissions of every regulated air pollutant must be addressed in the permit application.

The operating permit must outline specifically how and when a source will be allowed to operate over the five-year term of the permit. The permit the state develops from an application becomes the principal mechanism for enforcement of all air-quality regulations. As such, it is critically important to submit an application that allows maximum operating flexibility.

Sources must also include in their permit applications monitoring protocols sufficient to document compliance with each permit term and condition.

The Title V operating permit requires the submission of at least five types of reports:

1. The initial compliance report
2. The annual compliance certification
3. Monitoring reports submitted at least every six months
4. Progress reports for sources not in compliance when the application is submitted
5. Prompt reports on any deviations from the permit terms

Permit fees are a mandatory element of the Title V program. In most cases, fees are assessed for emissions on a dollars-per-ton-emitted basis.

Title VI: Stratospheric Ozone Protection Title VI of the 1990 Clean Air Act Amendments established a program to implement the provisions of the Montreal Protocol, a worldwide agreement to reduce the use and emission of ozone-depleting substances. EPA's regulations adopted in response to Title VI outline a series of requirements for facilities that use equipment containing ODS compounds. Facilities must be certain that they handle and manage ODS compounds as prescribed in the rules. Only certified technicians and staff may maintain

affected equipment, and detailed records must be maintained. Table 22-11 identifies common ODS compounds.

Regulatory Direction The current direction of regulations and air-pollution control efforts is clearly toward significantly reducing the emissions to the environment of a broad range of compounds, including:

1. Volatile organic compounds and other ozone precursors (CO and NO_x)
2. Hazardous air pollutants, including carcinogenic organic emissions and heavy metal emissions
3. Acid rain precursors, including SO_x and NO_x

In addition, the PM_{2.5} NAAQS will continue to place emphasis on quantifying and reducing particulate emissions in the less than 2.5 µm particle-size range. Particle size-specific emission factors have been developed for many sources, and size-specific emission standards have been developed in a number of states. These standards are addressing concerns related to nitrates and suitable condensates in the atmosphere.

Although it is not possible to predict the future, it is possible to prepare for it and influence it. It is highly recommended that maximum flexibility be designed into new air-pollution control systems to allow for increasingly more stringent emission standards for both particulates and gases. Further, it is everyone's responsibility to provide a thorough review of existing and proposed new processes and to make every attempt to identify economical process modifications and/or material substitutions that reduce or, in some cases, eliminate both the emissions to the environment and the overdependency on retrofitted or new end-of-pipe control systems.

UNITED STATES WATER QUALITY LEGISLATION AND REGULATIONS

Federal Water Pollution Control Act In 1948, the original Federal Water Pollution Control Act (FWPCA) was passed. This act and its various amendments are often referred to as the Clean Water Act (CWA). It provided loans for treatment plant construction and temporary authority for federal control of interstate water pollution. The enforcement powers were so heavily dependent on the states as to make the act almost unworkable. In 1956, several amendments to the FWPCA were passed that made federal enforcement procedures less cumbersome. The provision for state consent was removed by amendments passed in 1961, which also extended federal authority to include navigable waters in the United States.

TABLE 22-10 Maximum Achievable Control Technology (MACT) Standards from 40 CFR Part 63 as of June 2004

40 CFR 63	MACT Standard	40 CFR 63	MACT Standard
Subpart F	National Emission Standards for Organic Hazardous Air Pollutants from the Synthetic Organic Chemical Manufacturing Industry	Subpart RR	National Emission Standards for Individual Drain Systems
Subpart G	National Emission Standards for Organic Hazardous Air Pollutants From Synthetic Organic Chemical Manufacturing Industry Process Vents, Storage Vessels, Transfer Operations, and Wastewater	Subpart SS	National Emission Standards for Closed Vent Systems, Control Devices, Recovery Devices and Routing to a Fuel Gas System or a Process
Subpart H	National Emission Standards for Organic Hazardous Air Pollutants for Equipment Leaks	Subpart TT	National Emission Standards for Equipment Leaks—Control Level 1
Subpart I	National Emission Standards for Organic Hazardous Air Pollutants for Certain Processes Subject to the Negotiated Regulation for Equipment Leaks	Subpart UU	National Emission Standards for Equipment Leaks—Control Level 2 Standards
Subpart J	National Emission Standards for Organic Hazardous Air Pollutants for Polyvinyl Chloride and Copolymers Production	Subpart VV	National Emission Standards for Oil-Water Separators and Organic-Water Separators
Subpart K	(Reserved)	Subpart WW	National Emission Standards for Storage Vessels (Tanks)—Control Level 2
Subpart L	National Emission Standards for Coke Oven Batteries	Subpart XX	National Emission Standards for Ethylene Manufacturing Process Units: Heat Exchange Systems and Waste Operations
Subpart M	National Perchloroethylene Air Emission Standards for Dry Cleaning Facilities	Subpart YY	National Emission Standards for Hazardous Air Pollutants for Source Categories: Generic Maximum Achievable Control Technology Standards
Subpart N	National Emission Standards for Chromium Emissions from Hard and Decorative Chromium Electroplating and Chromium Anodizing Tanks	Subpart CCC	National Emission Standards for Hazardous Air Pollutants for Steel Pickling—HCI Process Facilities and Hydrochloric Acid Regeneration Plants
Subpart O	Ethylene Oxide Emissions Standards for Sterilization Facilities	Subpart DDD	National Emission Standards for Hazardous Air Pollutants for Mineral Wool Production
Subpart Q	National Emission Standards for Hazardous Air Pollutants for Industrial Process Cooling Towers	Subpart EEE	National Emission Standards for Hazardous Air Pollutants From Hazardous Waste Combustors
Subpart R	National Emission Standards for Gasoline Distribution Facilities (Bulk Gasoline Terminals and Pipeline Breakout Stations)	Subpart GGG	National Emission Standards for Pharmaceuticals Production
Subpart S	National Emission Standards for Hazardous Air Pollutants from the Pulp and Paper Industry	Subpart HHH	National Emission Standards for Hazardous Air Pollutants From Natural Gas Transmission and Storage Facilities
Subpart T	National Emission Standards for Halogenated Solvent Cleaning	Subpart III	National Emission Standards for Hazardous Air Pollutants for Flexible Polyurethane Foam Production
Subpart U	National Emission Standards for Hazardous Air Pollutant Emissions: Group I Polymers and Resins	Subpart JJJ	National Emission Standards for Hazardous Air Pollutant Emissions: Group IV Polymers and Resins
Subpart W	National Emission Standards for Hazardous Air Pollutants for Epoxy Resins Production and Non-Nylon Polyamides Production	Subpart LLL	National Emission Standards for Hazardous Air Pollutants From the Portland Cement Manufacturing Industry
Subpart X	National Emission Standards for Hazardous Air Pollutants From Secondary Lead Smelting	Subpart MMM	National Emission Standards for Hazardous Air Pollutants for Pesticide Active Ingredient Production
Subpart Y	National Emission Standards for Marine Tank Vessel Tank Loading Operations	Subpart NNN	National Emission Standards for Hazardous Air Pollutants for Wool Fiberglass Manufacturing
Subpart AA	National Emission Standards for Hazardous Air Pollutants From Phosphoric Acid Manufacturing Plants	Subpart OOO	National Emission Standards for Hazardous Air Pollutant Emissions: Manufacture of Amino/Phenolic Resins
Subpart BB	National Emission Standards for Hazardous Air Pollutants From Phosphate Fertilizers Production Plants	Subpart PPP	National Emission Standards for Hazardous Air Pollutant Emissions for Polyether Polyols Production
Subpart CC	National Emission Standards for Hazardous Air Pollutants From Petroleum Refineries	Subpart QQQ	National Emission Standards for Hazardous Air Pollutants for Primary Copper Smelting
Subpart DD	National Emission Standards for Hazardous Air Pollutants from Off-Site Waste and Recovery Operations	Subpart RRR	National Emission Standards for Hazardous Air Pollutants for Secondary Aluminum Production
Subpart EE	National Emission Standards for Magnetic Tape Manufacturing Operations	Subpart TTT	National Emission Standards for Hazardous Air Pollutants for Primary Lead Smelting
Subpart GG	National Emission Standards for Aerospace Manufacturing and Rework Facilities	Subpart UUU	National Emission Standards for Hazardous Air Pollutants for Petroleum Refineries: Catalytic Cracking Units, Catalytic Reforming Units, and Sulfur Recovery Units
Subpart HH	National Emission Standards for Hazardous Air Pollutants From Oil and Natural Gas Production Facilities	Subpart VVV	National Emission Standards for Hazardous Air Pollutants: Publicly Owned Treatment Works
Subpart II	National Emission Standards for Shipbuilding and Ship Repair (Surface Coating)	Subpart XXX	National Emission Standards for Hazardous Air Pollutants for Ferroalloys Production: Ferromanganese and Silicomanganese
Subpart JJ	National Emission Standards for Wood Furniture Manufacturing Operations	Subpart AAAA	National Emission Standards for Hazardous Air Pollutants: Municipal Solid Waste Landfills
Subpart KK	National Emission Standards for the Printing and Publishing Industry	Subpart CCCC	National Emission Standards for Hazardous Air Pollutants: Manufacturing of Nutritional Yeast
Subpart LL	National Emission Standards for Hazardous Air Pollutants for Primary Aluminum Reduction Plants	Subpart EEEE	National Emission Standards for Hazardous Air Pollutants: Organic Liquids Distribution (Non-Gasoline)
Subpart MM	National Emission Standards for Hazardous Air Pollutants for Chemical Recovery Combustion Source at Kraft, Soda, Sulfite, and Stand-Alone Semichemical Pulp Mills	Subpart FFFF	National Emission Standards for Hazardous Air Pollutants: Miscellaneous Organic Chemical Manufacturing
Subpart OO	National Emission Standards for Tanks—Level 1	Subpart GGGG	National Emission Standards for Hazardous Air Pollutants: Solvent Extraction for Vegetable Oil Production
Subpart PP	National Emission Standards for Containers		
Subpart QQ	National Emission Standards for Surface Impoundments		

TABLE 22-10 Maximum Achievable Control Technology (MACT) Standards from 40 CFR Part 63 as of June 2004 (Concluded)

40 CFR 63	MACT Standard	40 CFR 63	MACT Standard
Subpart HHHH	National Emission Standards for Hazardous Air Pollutants for Wet-Formed Fiberglass Mat Production	Subpart CCCCC	National Emission Standards for Hazardous Air Pollutants for Coke Ovens: Pushing, Quenching, and Battery Stacks
Subpart JJJJ	National Emission Standards for Hazardous Air Pollutants: Paper and Other Web Coating	Subpart FFFFF	National Emission Standards for Hazardous Air Pollutants for Integrated Iron and Steel Manufacturing Facilities
Subpart KKKK	National Emission Standards for Hazardous Air Pollutants: Surface Coating of Metal Cans	Subpart GGGGG	National Emission Standards for Hazardous Air Pollutants: Site Remediation
Subpart MMMM	National Emission Standards for Hazardous Air Pollutants for Surface Coating of Miscellaneous	Subpart HHHHH	National Emission Standards for Hazardous Air Pollutants: Miscellaneous Coating Manufacturing
Subpart NNNN	National Emission Standards for Hazardous Air Pollutants: Surface Coating of Large Appliances	Subpart IIIII	National Emission Standards for Hazardous Air Pollutants: Mercury Emissions from Mercury Cell Chlor-Alkali Plants
Subpart OOOO	National Emission Standards for Hazardous Air Pollutants: Printing, Coating, and Dyeing of Fabrics and Other Textiles	Subpart JJJJJ	National Emission Standards for Hazardous Air Pollutants for Brick and Structural Clay Products Manufacturing
Subpart QQQQ	National Emission Standards for Hazardous Air Pollutants: Surface Coating of Wood Building Products	Subpart KKKKK	National Emission Standards for Hazardous Air Pollutants for Clay Ceramics Manufacturing
Subpart RRRR	National Emission Standards for Hazardous Air Pollutants: Surface Coating of Metal Furniture	Subpart LLLLL	National Emission Standards for Hazardous Air Pollutants: Asphalt Processing and Asphalt Roofing Manufacturing
Subpart SSSS	National Emission Standards for Hazardous Air Pollutants: Surface Coating of Metal Coil	Subpart MMMM	National Emission Standards for Hazardous Air Pollutants: Flexible Polyurethane Foam Fabrication Operations
Subpart TTTT	National Emission Standards for Hazardous Air Pollutants for Leather Finishing Operations	Subpart NNNNN	National Emission Standards for Hazardous Air Pollutants: Hydrochloric Acid Production
Subpart UUUU	National Emission Standards for Hazardous Air Pollutants for Cellulose Products Manufacturing	Subpart PPPPP	National Emission Standards for Hazardous Air Pollutants for Engine Test Cells/Stands
Subpart VVVV	National Emission Standards for Hazardous Air Pollutants for Boat Manufacturing	Subpart QQQQQ	National Emission Standards for Hazardous Air Pollutants for Friction Materials Manufacturing Facilities
Subpart WWWW	National Emission Standards for Hazardous Air Pollutants: Reinforced Plastic Composites Production	Subpart RRRRR	National Emission Standards for Hazardous Air Pollutants: Taconite Iron Ore Processing
Subpart XXXX	National Emission Standards for Hazardous Air Pollutants: Rubber Tire Manufacturing	Subpart SSSSS	National Emission Standards for Hazardous Air Pollutants for Refractory Products Manufacturing
Subpart YYYY	National Emission Standards for Hazardous Air Pollutants for Stationary Combustion Turbines	Subpart TTTTT	National Emission Standards for Hazardous Air Pollutants for Primary Magnesium Refining
Subpart AAAAA	National Emission Standards for Hazardous Air Pollutants for Lime Manufacturing Plants		
Subpart BBBB	National Emission Standards for Hazardous Air Pollutants for Semiconductor Manufacturing		

In 1965 the Water Quality Act established a new trend in water pollution control. It provided that the states set water quality standards in accordance with federal guidelines. If the states failed to do so, the federal government, subject to a review hearing, would set the standards. In 1966, the Clean Water Restoration Act transferred the Federal Water Pollution Control Administration from the Department of Health, Education, and Welfare to the Department of the Interior. It also gave the Interior Department the responsibility for the Oil Pollution Act.

After the creation of EPA in 1970, the EPA was given the responsibility previously held by the Department of the Interior with respect to water pollution control. In subsequent amendments to the FWPCA in 1973, 1974, 1975, 1976, and 1977, additional federal programs were established. The goals of these programs were to make waterways of the United States fishable and swimmable by 1983 and to achieve zero discharge of pollutants by 1985. The National Pollutant Discharge Elimination System (NPDES) was established as the basic regulatory mechanism for water pollution control. Under this program, the states

were given the authority to issue permits to "point-source" dischargers provided the dischargers gave assurance that the following standards would be met:

1. Source-specific effluent limitations [including new source performance standards (NSPS)]
 2. Toxic pollutant regulations (for specific substances regardless of source)
 3. Regulations applicable to oil and hazardous substance liability
- To achieve that stated water quality goal of fishable and swimmable waters by 1983, each state was required by EPA to adopt water quality standards that met or exceeded the federal water quality criteria. After each state submitted its own water quality standards, which were subsequently approved by EPA, the federal criteria were removed from the Code of Federal Regulations. The state water quality standards are used as the basis for establishing both point-source-based effluent limitations and toxic pollutant limitations used in issuing NPDES permits to point-source discharges.

Source-Based Effluent Limitations Under the FWPCA, EPA was responsible for establishing point-source effluent limitations for municipal dischargers, industrial dischargers, industrial users of municipal treatment works, and effluent limitations for toxic substances (applicable to all dischargers).

Standards promulgated or proposed by EPA under 40 CFR, Parts 402 through 699, prescribe effluent limitation guidelines for existing sources, standards of performance for new sources, and pretreatment standards for new and existing sources. Effluent limitations and new source performance standards apply to discharges made directly into receiving bodies of water. The new standards require best-available technology (BAT) and are to be used by the states when issuing NPDES permits for all sources 18 months after EPA makes them final. Pretreatment standards apply to waste streams from industrial sources that are sent to publicly owned treatment works (POTW) for final treatment. These regulations are meant to

TABLE 22-11 Common ODS Compounds

Class I-CFCs	Class II-HCFCs
CFC-11	HCFC-22
CFC-12	HCFC-123
CFC-13	HCFC-124
CFC-111	HCFC-141
CFC-112	HCFC-142
CFC-113	
CFC-114	
CFC-115	
Halons	Others
Halon-1211	Carbon tetrachloride
Halon-1301	Methyl chloroform
Halon-1202	Methyl bromide

protect the POTW from any materials that would either harm the treatment facility or pass through untreated. They are to be enforced primarily by the local POTW. These standards are applicable to particular classes of point sources and pertain to discharges into navigable waters without regard to the quality of the receiving water. Standards are specific for numerous subcategories under each point-source category.

Limitations based upon application of the best practicable control technology currently available (BPT) apply to existing point sources and should have been achieved by July 1, 1977. Limitations based upon application of the BATEA (best-available technology economically achievable) that will result in reasonable further progress toward elimination of discharges had to be achieved by July 1, 1984.

Clean Water Act of 1977 The 1977 Clean Water Act directed EPA to review all BAT guidelines for conventional pollutants in those industries not already covered.

On August 23, 1978 (43 FR 37570), the EPA proposed a new approach to the control of conventional pollutants by effluent guideline limitations. The new guidelines were known as best conventional (pollutants control) technology (BCT). These guidelines replaced the existing BAT limitations, which were determined to be unreasonable for certain categories of pollutants.

To determine if BCT limitations would be necessary, the cost-effectiveness of conventional pollutant reduction to BAT levels beyond BPT levels had to be determined and compared to the cost of removal of this same amount of pollutant by a publicly owned treatment works of similar capacity. If it was equally cost-effective for the industry to achieve the reduction required for meeting the BAT limitations as the POTW, then the BCT limit was made equal to the BAT level. When this test was applied, the BAT limitations set for certain categories were found to be unreasonable. In these subcategories EPA proposed to remove the BAT limitations and revert to the BPT limitations until BCT control levels could be formulated.

Control of Toxic Pollutants Since the early 1980s, EPA's water quality standards guidance placed increasing importance on toxic pollutant control. The agency urged states to adopt criteria into their standards for priority toxic pollutants, particularly those for which EPA had published criteria guidance. EPA also provided guidance to help and support state adoption of toxic pollutant standards with the *Water Quality Standards Handbook* (1983) and the *Technical Support Document for Water Quality Toxics Control* (1985 and 1991).

Despite EPA's urging and guidance, state response was disappointing. A few states adopted large numbers of numeric toxic pollutant criteria, primarily for the protection of aquatic life. Most other states adopted few or no water quality criteria for priority toxic pollutants. Some relied on "free from toxicity" criteria and "action levels" for toxic pollutants or occasionally calculated site-specific criteria. Few states addressed the protection of human health by adopting numeric human health criteria.

State development of case-by-case effluent limits using procedures that did not rely on the statewide adoption of numeric criteria for the priority toxic pollutants frustrated Congress. Congress perceived that states were failing to aggressively address toxics and that EPA was not using its oversight role to push the states to move more quickly and comprehensively. Many in Congress believed that these delays undermined the effectiveness of the act's framework.

1987 CWA Amendments In 1987, Congress, unwilling to tolerate further delays, added Section 303(c)(2)(B) to the CWA. The section provided that whenever a state reviews water quality standards or revises or adopts new standards, the state has to adopt criteria for all toxic pollutants listed pursuant to Section 307(a)(1) of the act for which criteria have been published under Section 304(a), discharge or presence of which in the affected waters could reasonably be expected to interfere with those designated uses adopted by the state, as necessary to support such designated uses. Such criteria had to be specific numerical criteria for such toxic pollutants. When numerical criteria are not available, wherever a state reviews water quality standards, or revises or adopts new standards, the state has to adopt criteria based on biological monitoring or assessment methods consistent with information published pursuant to Section 304(a)(8).

Nothing in this section was to be construed to limit or delay the use of effluent limitations or other permit conditions based on or involving biological monitoring or assessment methods or previously adopted numerical criteria.

In response to this new congressional mandate, EPA redoubled its efforts to promote and assist state adoption of numerical water quality standards for priority toxic pollutants. EPA's efforts included the development and issuance of guidance to the states on acceptable implementation procedures. EPA attempted to provide the maximum flexibility in its options that complied not only with the express statutory language but also with the ultimate congressional objective: prompt adoption of numeric toxic pollutant criteria. The agency believed that flexibility was important so that each state could comply with Section 303(c)(2)(B) within its resource constraints. EPA distributed final guidance on December 12, 1988. This guidance was similar to earlier drafts available for review by the states. The availability of the guidance was published in the Federal Register on January 5, 1989 (54 FR 346).

The structure of Section 303(c) is to require states to review their water quality standards at least once each 3-year period. Section 303(c)(2)(B) instructs states to include reviews for toxics criteria whenever they initiate a triennial review. EPA initially looked at February 4, 1990, the 3-year anniversary of the 1987 CWA amendments, as a convenient point to index state compliance. The April 17, 1990, Federal Register Notice (55 FR 14350) used this index point for the preliminary assessment of state compliance. However, some states were very nearly completing their state administrative processes for ongoing reviews when the 1987 amendments were enacted and could not legally amend those proceedings to address additional toxics criteria. Therefore, in the interest of fairness, and to provide such states a full 13-year review period, EPA's FY 1990 Agency Operating Guidance provided that states should complete adoption of the numeric criteria to meet Section 303(c)(2)(B) by September 30, 1990.

Section 303(c) does not provide penalties for states that do not complete timely water quality standard reviews. In no previous case had an EPA Administrator found that state failure to complete a review within 3 years jeopardized the public health or welfare to such an extent that promulgation of federal standards pursuant to Section 303(c)(4)(8) was justified. However, the pre-1987 CWA never mandated state adoption of priority toxic pollutants or other specific criteria. EPA relied on its water quality standards regulation (40 CFR 131.11) and its criteria and program guidance to the states on appropriate parametric coverage in state water quality standards, including toxic pollutants. With congressional concern exhibited in the legislative history for the 1987 Amendments regarding undue delays by states and EPA, and because states have been explicitly required to adopt numeric criteria for appropriate priority toxic pollutants since 1983, the agency is proceeding to promulgate federal standards pursuant to Section 303(c)(4)(B) of the CWA and 40 CFR 131.22(b).

States have made substantial recent progress in the adoption, and EPA approval, of toxic pollutant water quality standards. Furthermore, virtually all states have at least proposed new toxics criteria for priority toxic pollutants since Section 303(c)(2)(B) was added to the CWA in February 1987. Unfortunately, not all such state proposals address, in a comprehensive manner, the requirements of Section 303(c)(2)(B). For example, some states have proposed to adopt criteria to protect aquatic life, but not human health; other states have proposed human health criteria that do not address major exposure pathways (such as the combination of both fish consumption and drinking water). In addition, in some cases final adoption of proposed state toxics criteria that would be approved by EPA has been substantially delayed due to controversial and difficult issues associated with the toxic pollutant criteria adoption process.

Biological Criteria While the overall mandate of the Clean Water Act may now be more clearly stated and understood, the tools needed are still under development, and their full application is being worked out. The direction is toward a more comprehensive approach to water quality protection, which might be more appropriately termed *water resource protection* to encompass the living resources and their habitat along with the water itself.

In 1991, EPA directed states to adopt biological criteria into their water quality standards by September 30, 1993. To assist the states, EPA issued its "Policy on the Use of Biological Assessments and Criteria in the Water-Quality Program," which sets forth the key policy directions governing the shift to the use of biological criteria:

- "Biological surveys shall be fully integrated with toxicity and chemical-specific assessment methods in state water-quality programs."
- "Biological surveys should be used together with whole-effluent and ambient toxicity testing, and chemical-specific analyses to assess attainment/nonattainment of designated aquatic life uses in state water-quality standards."
- "If any one of the three assessment methods demonstrate that water-quality standards are not attained, it is EPA's policy that appropriate action should be taken to achieve attainment, including use of regulatory authority" (the independent applicability policy).
- "States should designate aquatic life uses that appropriately address biological integrity and adopt biological criteria necessary to protect those uses."

These policy statements are founded on the existing language and authorities in Clean Water Act Sections 303(c)(2)(A) and (B). EPA defined biological criteria as "numerical values or narrative expressions used to describe the expected structure and function of the aquatic community."

Most states currently conduct biological surveys. These consist of the collection and analysis of resident aquatic community data and a subsequent determination of the aquatic community's structure and function. A limiting factor in the number of surveys conducted is simply the funding to support field sampling. Most states also take their survey data through the next level of analysis, which is a biological assessment, in which the biological condition of the water body is evaluated. A state needs to conduct a biological assessment to justify designated waters as having uses less than the Clean Water Act goal of "fishable/swimmable" [Sec. 101(a)(2)].

Implementing biological criteria requires that the state establish what the expected condition of a biological community should be. This expected condition (or reference condition) is based on measurements at either unimpacted sites or sites that are minimally impacted by human activities. A key element of developing and adopting biological criteria is to develop a set of metrics (biological measurements) and indices (summations of those metrics) that are sensitive to reflecting the community changes that occur as a result of human perturbation. Metrics may include measures of species richness, percentages of species of a particular feeding type, and measures of species abundance and condition. The development of metrics and indices has been a major technical advance that effectively brought the field to the point of implementing biological integrity measures.

Aside from the need for additional monitoring and data as a basis for biological criteria, the largest impediment is currently related to the question of how or whether biological criteria should be translated directly to NPDES permit requirements. A strict reading of the NPDES regulations leads some individuals to conclude that every water quality standard potentially affected by a discharge must have a specific effluent limit. In the case of biological criteria, the direct translation isn't always possible. The preferred approach is to use biological criteria as a primary means to identify impaired waters that will then need additional study. Appropriate discharge controls can then be applied.

The other area of contention in adopting biocriteria is the EPA policy of *independent applicability*. This policy recognizes that chemical-specific criteria, toxicity testing (WET tests), and biological criteria each have unique as well as overlapping attributes, limitations, and program applications. No single approach is superior to another; rather, the approaches are complementary. When any one of these approaches indicates that water quality is impaired, appropriate regulatory action is needed. Critics of this approach most frequently cite biological criteria as the superior, or determining, measure and would use it to override an exceedance of chemical-specific criteria.

EPA and many states are now moving toward a watershed or basin management approach to water management. This is a holistic approach that looks at the multiple stresses and activities that affect a basin and evaluates these effects in the context of the survival of ecological systems. Biological criteria are recognized by states as a necessary

tool for detecting impacts to important aquatic resources that would otherwise be missed, particularly those caused by non-point-source activities, and for defining the desired endpoint of environmental restoration activities.

To assist states, EPA has issued programmatic guidance (*Biological Criteria: National Program Guidance for Surface Waters*, April 1990; *Procedures for Initiating Narrative Biological Criteria*, October 1992). In addition, EPA will be issuing over the next couple of years technical guidance specific to developing biological criteria for streams and small rivers, lakes and reservoirs, estuaries, and wetlands.

Metal Bioavailability and Toxicity Another area of policy and regulatory change that bears directly on questions of biological integrity is the application of toxic metal water quality criteria. EPA is in the midst of reconsidering its approach to implementing toxic metals criteria. This was prompted in part by the difficulty that some dischargers are having in meeting the state ambient water quality criteria for metals, which generally are expressed as total recoverable metals (a measurement that includes the metal dissolved in water plus metal that becomes dissolved when the sample is treated with acid). At issue is whether the criteria can be expressed in a form that more accurately addresses the toxicity of the metal to aquatic life, thereby providing the desired level of aquatic life protection, while allowing greater leeway in the total amount of metal discharged. This is a technically complex issue that will require long-term research even if short-term solutions are implemented.

The bioavailability, and hence the toxicity, of metal depends on the physical and chemical form of the metal, which in turn depends on the chemical characteristics of the surrounding water. The dissolved form of the metal is generally viewed as more bioavailable and therefore more toxic than the particulate form. Particulate matter and dissolved organic matter can bind the metal, making it less bioavailable. What is not well known or documented is the various chemical transformations that occur both within the effluent stream and when the effluent reaches and mixes with the receiving water. Metal that is not bioavailable in the effluent may become bioavailable under ambient chemical conditions.

The NPDES regulations (40 CFR 122.45) require effluent limits to be expressed as total recoverable metal. This requirement makes sense as a means to monitor and regulate both the total metal loading and the effectiveness of wastewater treatment that involves chemical precipitation of the metal.

From the perspective of ecological integrity called for in the Clean Water Act, any adjustment to the implementation of toxic metals criteria needs to be integrated with both sediment criteria and biological criteria to provide ecosystem protection envisioned by the act.

Total Maximum Daily Load (TMDL) Under Section 303(d) of the 1972 Clean Water Act, states, territories, and authorized tribes were required to develop a list of impaired waters. The impaired waters do not meet water quality standards that states, territories, and authorized tribes have set for them, even after point sources of pollution have installed the minimum required levels of pollution control technology. The law requires that these jurisdictions establish priority rankings for waters on the lists and develop TMDLs for these waters.

This part of the Clean Water Act was relatively neglected until 1996. A Federal Advisory Committee was convened and produced in 1998 a report and subsequent proposed changes for implementation of the TMDL program and associated changes in the National Pollutant Discharge Elimination System for point sources. A number of court orders were also motivating factors in the implementation and proposed changes to the rule.

The TMDL specifies the amount of a particular pollutant that may be present in a water body, allocates allowable pollutant loads among sources, and provides the basis for attaining or maintaining water quality standards.

The TMDL regulations were issued as draft in 1999 and finally published on July 13, 2000.

Water Quality Trading Within the approach to TMDL rules and subsequent management policy, the emphasis on *Pollutant trading or water quality trading* has evolved. Trading allows sources with responsibility for discharge reductions the flexibility to determine where reduction will occur. Within the trading approach, the economic advantages are emphasized. This is driven by the TMDL or more

stringent water quality-based requirement in an NPDES permit and the potential fact that discharge sources have significantly different costs to control the pollutant of concern.

In January 2003, EPA published the *Water Quality Trading Policy*. The policy outlines the trading objectives, requirements, and elements of a trading program. While barriers exist to the implementation of an effective water quality trading program, the cost of discharge quality compliance would warrant consideration of the approach.

Bioterrorism Act of 2003 While not directly related to water quality regulations, the security and vulnerability of community drinking water systems were addressed in the Public Health Security and Bioterrorism Preparedness and Response Act of 2002 (Bioterrorism Act). The vulnerability assessments were intended to examine a facility's ability to defend against adversarial actions that might substantially disrupt the ability of a water system to provide a safe and reliable supply of drinking water.

For community drinking water systems serving more than 3300 persons, it was required to conduct a vulnerability assessment (VA), certify and submit a copy of the VA to the EPA Administrator, prepare or revise an emergency response plan based on the results of the VA, and within 6 months certify to the EPA Administrator that an emergency response plan has been completed or updated. The VA requirement was to be completed by all facilities in June 2004.

Cooling Water Intake Regulation The EPA established requirements for facilities with cooling water intake structure to implement best-available technologies to minimize adverse environmental impacts. Under Section 316(b) of the CWA, the protection of aquatic organisms from being killed or injured by impingement or entrainment was established. The rule was divided into three phases. Phase I was published in December 2001 and addressed new facilities. Phase II was published in July 2004. The Phase II rule addressed existing electric generating plants that withdraw greater than 50 million gallons per day and use at least 25 percent of their withdrawn water for cooling purposes only. Phase III was prepublished in November 2004. The Phase III rule addresses other electric generating facilities and industrial sectors that withdraw water. It is anticipated that Phase III will be published within 1 to 2 years.

The rule provides that the facility may choose one of five compliance alternatives for establishing best available technology for minimizing adverse environmental impact at the site. Under current NPDES program regulations, the 316(b) requirements would occur when an existing NPDES permit was reissued or when an existing permit was modified or revoked and reissued.

Water Reuse With increasing water sustainability and scarcity issues in the United States, the U.S. Environmental Protection Agency (EPA) and the U.S. Agency for International Development (USAID) are revising portions of the 1992 *EPA Guidelines for Water Reuse* to reflect significant technical advancements and institutional developments since 1992. The guidelines will also address new areas, including national reclaimed water use trends, endocrine disrupters, and approaches to integrated water resources management.

Across the world, reclaimed water is becoming a critical water source, and reuse strategies are recognized in many U.S. states as an integral part of water resources management. The original guidelines, published in 1980, gave many state agencies direction in establishing reuse permits and helped to foster state water reuse regulations. The revised guidelines will include an updated inventory of state reuse regulations.

Regulatory Direction EPA and states are directed by the Clean Water Act to develop programs to meet the act's stated objective: "to restore and maintain the chemical, physical, and biological integrity of the Nation's waters" [Section 101 (a)]. Efforts to date have emphasized "clean water" quite literally, by focusing on the chemical makeup of discharges and their compliance with chemical water quality standards established for surface water bodies. These programs have successfully addressed many water pollution problems, but they are not sufficient to identify and address all of them. A continuing large gap in the current regulatory scheme is the absence of a direct measure of the condition of the biological resources that we are intending to protect (the biological integrity of the water body). Without such a measure it will be difficult to determine whether our water management

approaches are successful in meeting the intent of the act. Taken together, chemical, physical, and biological integrity is equivalent to the *ecological integrity* of a water body. It is highly likely the future will see the interpretation of biological integrity and ecological integrity, and the management of ecological systems, assume a much more prominent role in water quality management.

To ensure technically sound decision making, EPA should review its water quality criteria at a minimum of every 10 years. EPA's water quality criteria consist of information regarding the concentrations of chemicals or levels of parameters in water that protect aquatic life and human health, and act as regulatory thresholds for determination of impaired waters. Many of EPA's current water quality criteria are outdated, and have not been developed using consistent methodologies.

To the extent that EPA does pursue changes or additions to the current water quality standards program, it is critical that such modifications not be simply added onto existing requirements as independent new program elements. In pursuing any modifications EPA should carefully examine opportunities for coordinating and streamlining requirements within the context of the overall water quality standards program to maximize efficiency in the face of substantial resource constraints.

UNITED STATES SOLID WASTE LEGISLATION AND REGULATIONS

Much of the current activity in the field of solid waste management, especially with respect to hazardous wastes and resources recovery, is a direct consequence of recent legislation. Therefore, it is important to review the principal legislation that has affected the entire field of solid-waste management.

What follows is a brief review of existing legislation that affects the management of solid wastes. The actual legislation must be consulted for specific detail. Implementation of the legislation is accomplished through regulations adopted by federal, state, and local agencies. Because these regulations are revised continuously, they must be monitored continuously, especially when design and construction work is to be undertaken.

Rivers and Harbors Act, 1899 Passed in 1899, the Rivers and Harbors Act directed the U.S. Army Corps of Engineers to regulate the dumping of debris in navigable waters and adjacent lands.

Solid Waste Disposal Act, 1965 Modern solid-waste legislation dates from 1965, when the Solid Waste Disposal Act, Title II of Public Law 88-272, was enacted by Congress. The principal intent of this act was to promote the demonstration, construction, and application of solid waste management and resource-recovery systems that preserve and enhance the quality of air, water, and land resources.

National Environmental Policy Act, 1969 The National Environmental Policy Act (NEPA) of 1969 was the first federal act that required coordination of federal projects and their impacts with the nation's resources. The act specified the creation of the Council on Environmental Quality in the Executive Office of the President. This body has the authority to force every federal agency to submit to the council an environmental impact statement on every activity or project which it may sponsor or over which it has jurisdiction.

Resource Recovery Act, 1970 The Solid Waste Disposal Act of 1965 was amended by Public Law 95-512, the Resources Recovery Act of 1970. This act directed that the emphasis of the national solid-waste-management program should be shifted from disposal as its primary objective to that of recycling and reuse of recoverable materials in solid wastes or the conversion of wastes to energy.

Resource Conservation and Recovery Act, 1976 RCRA is the primary statute governing the regulation of solid and hazardous waste. It completely replaced the Solid Waste Disposal Act of 1965 and supplemented the Resource Recovery Act of 1970; RCRA itself was substantially amended by the Hazardous and Solid Waste Amendments of 1984 (HSWA). The principal objectives of RCRA as amended are to:

- Promote the protection of human health and the environment from potential adverse effects of improper solid and hazardous waste management
- Conserve material and energy resources through waste recycling and recovery

- Reduce or eliminate the generation of hazardous waste as expeditiously as possible

To achieve these objectives, RCRA authorized EPA to regulate the generation, treatment, storage, transportation, and disposal of hazardous wastes. The structure of the national hazardous waste regulatory program envisioned by Congress is laid out in Subtitle C of RCRA (Sections 3001 through 3019), which authorized EPA to:

- Promulgate standards governing hazardous waste generation and management
 - Promulgate standards for permitting hazardous waste treatment, storage, and disposal facilities
 - Inspect hazardous waste management facilities
 - Enforce RCRA standards
 - Authorize states to manage the RCRA Subtitle C program, in whole or in part, within their respective borders, subject to EPA oversight
- Federal RCRA hazardous waste regulations are set forth in 40 CFR Parts 260 through 279. The core of the RCRA regulations establishes the "cradle to grave" hazardous waste regulatory program through seven major sets of regulations:
- Identification and listing of regulated hazardous wastes (Part 261)
 - Standards for generators of hazardous waste (Part 262)
 - Standards for transporters of hazardous waste (Part 263)
 - Standards for owners/operators of hazardous waste treatment, storage, and disposal facilities (Parts 264, 265, and 267)
 - Standards for the management of specific hazardous wastes and specific types of hazardous waste management facilities (Part 266)
 - Land disposal restriction standards (Part 268)
 - Requirements for the issuance of permits to hazardous waste facilities (Part 270)
 - Approved state hazardous waste management programs (Part 272)
 - Standards for unusual waste management (Part 273)
 - Standards for management of used oil (Part 279)
 - Standards and procedures for authorizing state hazardous waste programs to be operated in lieu of the federal program (Part 271)

EPA, under Section 3006 of RCRA, may authorize a state to administer and enforce a state hazardous waste program in lieu of the federal Subtitle C program. To receive authorization, a state program must:

- Be equivalent to the federal Subtitle C program
- Be consistent with, and no less stringent than, the federal program and other authorized state programs
- Provide adequate enforcement of compliance with Subtitle C requirements

Toxic Substances Control Act, 1976 The two major goals of the Toxic Substances Control Act (TSCA), passed by Congress in 1976, are (1) the acquisition of sufficient information to identify and evaluate potential hazards from chemical substances and (2) the regulation of the manufacture, processing, distribution, use, and disposal of any substance that presents an unreasonable risk of injury to health of the environment.

Under TSCA, the EPA has issued a ban on the manufacture, processing, and distribution of products containing PCBs. Exporting of PCBs has also been banned. TSCA also required that PCB mixtures containing more than 50 ppm PCBs must be disposed of in an acceptable incinerator or chemical waste landfill. All PCB containers or products containing PCBs had to be clearly marked and records maintained by the operator of each facility handling at least 45 kilograms of PCBs. These records include PCBs in use in transformers and capacitors, PCBs in transformers and capacitors removed from service, PCBs stored for disposal, and a report on the ultimate disposal of the PCBs. Additionally, PCB transformers, must be registered, and records must be kept of hydraulic systems, natural gas pipeline systems, voltage regulators, electromagnetic switches, and ballasts.

TSCA also placed restrictions on the use of chlorofluorocarbons, asbestos, and fully halogenated chlorofluoroalkanes such as aerosol propellants.

Regulatory Direction There is no doubt that pollution prevention continues to be the regulatory direction. The development of new and more efficient processes and waste minimization technologies will be essential to support this effort.

An area certain to receive regulatory attention in the future is nanotechnology. The applications and implications of nanotechnology not only are changing lives every day, but also are moving so fast that many

have not yet grasped their tremendous impact (Theodore, L., and R. Kunz, *Nanotechnology: Environmental Implications and Solutions*, Wiley, Hoboken, N.J., 2002). It is clear that nanotechnology will revolutionize the way industry operates, creating chemical processes and products that are more efficient and less expensive. A primary obstacle to achieving this goal will be the need to control, reduce, and ultimately eliminate the environmental and related problems associated with nanotechnology, or dilemmas that may develop through its misuse.

Completely new legislation and regulatory rulemaking may be necessary for environmental control of nanotechnology. However, in the meantime, one may speculate on how the existing regulatory framework might be applied to the nanotechnology area as this emerging field develops over the next several years.

Commercial applications of nanotechnology are likely to be regulated under the Toxic Substances Control Act (TSCA), which authorized EPA to review and establish limits on the manufacture, processing, distribution, use, and/or disposal of new materials the EPA determines to pose "an unreasonable risk of injury to human health or the environment." The term *chemicals* is defined broadly by TSCA. Unless qualifying for an exemption under the law [R&D (a statutory exemption requiring no further approval by EPA), low-volume production, low environmental releases along with low volume, or plans for limited test marketing], a prospective manufacturer is subject to the full-blown premanufacture notice (PMN) procedure. This requires submittal of said notice, along with toxicity and other data to EPA at least 90 days before commencing production of the chemical substance. (Theodore, L., and R. Kunz, *Nanotechnology: Environmental Implications and Solutions*, Wiley, Hoboken, N.J., 2002).

Approval then involves recordkeeping, reporting, and other requirements under the statute. Requirements will differ, depending on whether EPA determines that a particular application constitutes a "significant new use" or a "new chemical substance." EPA can impose limits on production, including an outright ban when it is deemed necessary for adequate protection against "an unreasonable risk of injury to health or the environment." EPA may revisit a chemical's status under TSCA and change the degree or type of regulation when new health and environmental data warrant. EPA is expected to be issuing several new TSCA test rules in 2004 [Bergeson, L. L., "Expect a Busy Year at EPA," *Chem. Processing* 17 (April 14, 2004)]. If the experience with genetically engineered organisms is any indication, there will be a push for EPA to update regulations in the future to reflect changes, advances, and trends in nanotechnology [Bergeson, L. L., "Genetically Engineered Organisms Face Changing Regulations," *Chem. Processing* (March 2004)].

Workplace exposure to chemical substances and the potential for pulmonary toxicity are subject to regulation by the Occupational Safety and Health Administration under the Occupational Safety and Health Act (OSHA), including the requirement that potential hazards be disclosed on material safety data sheets (MSDS). (An interesting question arises as to whether carbon nanotubes, chemically carbon but with different properties because of their small size and structure, are indeed to be considered the same as or different from carbon black for MSDS purposes.) Both government and private agencies can be expected to develop the requisite threshold limit values (TLVs) for workplace exposure. Also, EPA may once again utilize TSCA to assert its own jurisdiction, appropriate or not, to minimize exposure in the workplace.

Wastes from a commercial-scale nanotechnology facility would be treated under the Resource Conservation and Recovery Act (RCRA), provided that they meet the criteria for RCRA waste. RCRA requirements could be triggered by a listed manufacturing process or the act's specified hazardous waste characteristics. The type and extent of regulation would depend on how much hazardous waste is generated and whether the wastes generated are treated, stored, or disposed of on-site.

Finally, opponents of nanotechnology, especially, may be able to use the National Environmental Policy Act (NEPA) to impede nanotechnology research funded by the U.S. government. A "major Federal action significantly affecting the quality of the human environment" is subject to the environmental impact provision under NEPA. (Various states also have environmental impact assessment requirements that could delay or put a stop to construction of nanotechnology facilities.) Time will tell.

POLLUTION PREVENTION

This subsection is drawn in part from "Pollution Prevention Overview," prepared by B. Wainwright and L. Theodore, copyright 1993.

FURTHER READING: American Society of Testing and Materials, *Standard Guide for Industrial Source Reduction*, draft copy dated June 16, 1992. American Society of Testing and Materials, *Pollution Prevention, Reuse, Recycling and Environmental Efficiency*, June, 1992. California Department of Health Services, *Economic Implications of Waste Reduction, Recycling, Treatment, and Disposal of Hazardous Wastes: The Fourth Biennial Report*, California, 1988. Citizen's Clearinghouse for Hazardous Waste, *Reduction of Hazardous Waste: The Only Serious Management Option*, Falls Church, CCHW, 1986. Congress of the United States. Office of Technology Assessment, *Serious Reduction of Hazardous Waste: For Pollution Prevention and Industrial Efficiency*, Washington, D.C., GPO, 1986. Dupont, R., L. Theodore, and K. Ganesan, *Pollution Prevention: The Waste Management Option for the 21st Century*, Lewis Publishers, 2000. Friedlander, S., "Pollution Prevention—Implications for Engineering Design, Research, and Education," *Environment*, May 1989, p. 10. Theodore, L., and R. Kunz, *Nanotechnology: Environmental Implications and Solutions*, Wiley, 2005. Theodore, L. and R. Allen, *Pollution Prevention: An ETS Theodore Tutorial*, Roanoke, VA, ETS International, Inc., 1994. Theodore, L., *A Citizen's Guide to Pollution Prevention*, East Williston, NY, 1993. United States EPA, *Facility Pollution Prevention Guide*, (EPA/600/R-92/088), Washington, D.C., May 1992. United States EPA, "Pollution Prevention Fact Sheets," Washington, D.C., GPO, 1991. United States EPA, *1987 National Biennial RCRA Hazardous Waste Report—Executive Summary*, Washington, D.C., GPO, 1991. United States EPA, Office of Pollution Prevention, *Report on the U.S. Environmental Protection Agency's Pollution Prevention Program*, Washington, D.C., GPO, 1991. Wilcox, J., and L. Theodore, *Engineering and Environmental Ethics*, Wiley, 1998. World Wildlife Fund, *Getting at the Source—Executive Summary*, 1991.

INTRODUCTION

The amount of waste generated in the United States has reached staggering proportions; according to the United States Environmental Protection Agency (EPA), 250 million tons of solid waste alone are generated annually. Although both the Resource Conservation and Recovery Act (RCRA) and the Hazardous and Solid Waste Act (HSWA) encourage businesses to minimize the wastes they generate, the majority of the environmental protection efforts are still centered around treatment and pollution clean-up.

The passage of the Pollution Prevention Act of 1990 has redirected industry's approach to environmental management; pollution prevention has now become the environmental option of the 21st century. Whereas typical waste-management strategies concentrate on "end-of-pipe" pollution control, pollution prevention attempts to handle waste at the source (i.e., source reduction). As waste handling and disposal costs increase, the application of pollution prevention measures is becoming more attractive than ever before. Industry is currently exploring the advantages of multimedia waste reduction and developing agendas to *strengthen* environmental design while *lessening* production costs.

There are profound opportunities for both industry and the individual to prevent the generation of waste; indeed, pollution prevention is today primarily stimulated by economics, legislation, liability concerns, and the enhanced environmental benefit of managing waste at the source. The Pollution Prevention Act of 1990 established pollution prevention as a national policy, declaring that "waste should be prevented or reduced at the source wherever feasible, while pollution that cannot be prevented should be recycled in an environmentally safe manner" (Ref. 1). The EPA's policy establishes the following hierarchy of waste management:

1. Source reduction
2. Recycling/reuse
3. Treatment
4. Ultimate disposal

The hierarchy's categories are prioritized so as to promote the examination of each individual alternative prior to the investigation of subsequent options (i.e., the most preferable alternative should be thoroughly evaluated before consideration is given to a less accepted option). Practices that decrease, avoid, or eliminate the generation of

waste are considered source reduction and can include the implementation of procedures as simple and economical as good house-keeping. Recycling is the use, reuse, or reclamation of wastes and/or materials that may involve the incorporation of waste recovery techniques (e.g., distillation, filtration). Recycling can be performed at the facility (i.e., on-site) or at an off-site reclamation facility. Treatment involves the destruction or detoxification of wastes into nontoxic or less toxic materials by chemical, biological, or physical methods, or any combination of these control methods. Disposal has been included in the hierarchy because it is recognized that residual wastes will exist; the EPA's so-called "ultimate disposal" options include land-filling, land farming, ocean dumping, and deep-well injection. However, the term *ultimate disposal* is a misnomer, but is included here because of its earlier adaptation by the EPA.

Table 22-12 provides a rough timetable demonstrating the United States' approach to waste management. Note how waste management has begun to shift from pollution *control*-driven activities to pollution *prevention* activities.

The application of waste-management practices in the United States has recently moved toward securing a new pollution prevention ethic. The performance of pollution prevention assessments and their subsequent implementation will encourage increased activity into methods that will further aid in the reduction of hazardous wastes. One of the most important and propitious consequences of the pollution-prevention movement will be the development of life-cycle design and standardized life-cycle cost-accounting procedures. These two consequences are briefly discussed in the two paragraphs that follow. Additional information is provided in a later subsection.

The key element of life-cycle design is Life-Cycle Assessment (LCA). LCA is generally envisioned as a process to evaluate the environmental burdens associated with the cradle-to-grave life cycle of a product, process, or activity. A product's life cycle can be roughly described in terms of the following stages:

1. Raw material
2. Bulk material processing
3. Production
4. Manufacturing and assembly
5. Use and service
6. Retirement
7. Disposal

Maintaining an objective process while spanning this life cycle can be difficult given the varying perspective of groups affected by different parts of that cycle. LCA typically does not include any direct or indirect monetary costs or impacts to individual companies or consumers.

Another fundamental goal of life-cycle design is to promote sustainable development at the global, regional, and local levels. There is significant evidence that suggests that current patterns of human and industrial activity on a global scale are not following a sustainable path. Changes to achieve a more sustainable system will require that environmental issues be more effectively addressed in the future. Principles for achieving sustainable development should include (Ref. 2):

TABLE 22-12 Waste Management Timetable

Prior to 1945	No control
1945–1960	Little control
1960–1970	Some control
1970–1975	Greater control (EPA is founded)
1975–1980	More sophisticated control
1980–1985	Beginning of waste-reduction management
1985–1990	Waste-reduction management
1990–1995	Formal pollution prevention programs (Pollution Prevention Act)
1995–2000	Widespread acceptance of pollution prevention
After 2000	The sky's the limit

1. *Sustainable resource use (conserving resources, minimizing depletion of nonrenewable resources, using sustainable practices for managing renewable resources).* There can be no product development or economic activity of any kind without available resources. Except for solar energy, the supply of resources is finite. Efficient designs conserve resources while also reducing impacts caused by material extraction and related activities. Depletion of nonrenewable resources and overuse of otherwise renewable resources limits their availability to future generations.

2. *Maintenance of ecosystem structure and function.* This is a principal element of sustainability. Because it is difficult to imagine how human health can be maintained in a degraded, unhealthy natural world, the issue of ecosystem health should be a more fundamental concern. Sustainability requires that the health of all diverse species as well as their interrelated ecological functions be maintained. As only one species in a complex web of ecological interactions, humans cannot separate their success from that of the total system.

3. *Environmental justice.* The issue of environmental justice has come to mean different things to different people. Theodore (Ref. 3) has indicated that the subject of environmental justice contains four key elements that are interrelated: environmental racism, environmental health, environmental equity, and environmental politics. (Unlike many environmentalists, Theodore has contended that only the last issue, politics, is a factor in environmental justice.) A major challenge in sustainable development is achieving both intergenerational and intersocietal environmental justice. Overconsuming resources and polluting the planet in such a way that it enjoins future generations from access to reasonable comforts irresponsibly transfer problems to the future in exchange for short-term gains. Beyond this intergenerational conflict, enormous inequities in the distribution of resources continue to exist between developed and less developed countries. Inequities also occur within national boundaries.

Life cycle is a perspective that can consider the true costs of product production and/or services provided and utilized by analyzing the price associated with potential environmental degradation and energy consumption, as well as more customary costs like capital expenditure and operating expenses. A host of economic and economic-related terms have appeared in the literature. Some of these include total cost assessment, life-cycle costing, and full-cost accounting. Unfortunately, these terms have come to mean different things to different people at different times. In an attempt to remove this ambiguity, the following three economic terms are defined below.

1. *Traditional costing procedure (TCP).* This accounting procedure *only* takes into account capital and operating (including environmental) costs.

2. *Comprehensive costing procedure (CCP).* The economic procedure includes not only the traditional capital and operating costs but also peripheral costs such as liability, regulatory related expenses, borrowing power, and social considerations.

3. *Life-cycle costing (LCC).* This type of analysis requires that all the traditional costs of project or product system, from raw-material acquisition to end-result product disposal, be considered.

The TCP approach is relatively simple and can be easily applied to studies involving comparisons of different equipment, different processes, or even parts of processes. CCP has now emerged as the most realistic approach that can be employed in economic project analyses. It is the recommended procedure for pollution-prevention studies. The LCC approach is usually applied to the life-cycle analysis (LCA) of a product or service. It has found occasional application in project analysis.

The remainder of this subsection on pollution prevention will be concerned with providing the reader with the necessary background to understand the meaning of pollution prevention and its useful implementation. Assessment procedures and the economic benefits derived from managing pollution at the source are discussed along with methods of cost accounting for pollution prevention. Additionally, regulatory and nonregulatory methods to promote pollution prevention and overcome barriers are examined, and ethical considerations are presented. By eliminating waste at the source, all can participate in the protection of the environment by reducing the amount of waste mate-

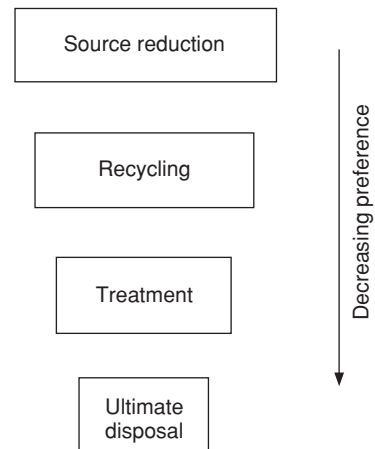


FIG. 22-2 Pollution prevention hierarchy.

rial that would otherwise need to be treated or ultimately disposed; accordingly, attention is also given to pollution prevention in both the domestic and business office environments.

POLLUTION-PREVENTION HIERARCHY

As discussed in the Introduction, the hierarchy set forth by the EPA in the Pollution Prevention Act establishes an order to which waste-management activities should be employed to reduce the quantity of waste generated. The preferred method is source reduction, as indicated in Fig. 22-2. This approach actually precedes traditional waste management by addressing the source of the problem prior to its occurrence.

Although the EPA's policy does not consider recycling or treatment as actual pollution prevention methods per se, these methods present an opportunity to reduce the amount of waste that might otherwise be discharged into the environment. Clearly, the definition of pollution prevention and its synonyms (e.g., waste minimization) must be understood to fully appreciate and apply these techniques.

Waste minimization generally considers all of the methods in the EPA hierarchy (except for disposal) appropriate to reduce the volume or quantity of waste requiring disposal (e.g., source reduction). The definition of *source reduction* as applied in the Pollution Prevention Act, however, is "any practice that reduces the amount of any hazardous substance, pollutant, or contaminant entering any waste stream or otherwise released into the environment . . . prior to recycling, treatment or disposal" (Ref. 1). Source reduction reduces the amount of waste generated; it is therefore considered true pollution prevention and has the highest priority in the EPA hierarchy.

Recycling (or reuse) refers to the use (or reuse) of materials that would otherwise be disposed of or treated as a waste product. A good example is a rechargeable battery. Wastes that cannot be directly reused may often be recovered on-site through methods such as distillation. When on-site recovery or reuse is not feasible due to quality specifications or the inability to perform recovery on-site, off-site recovery at a permitted commercial recovery facility is often a possibility. Such management techniques are considered secondary to source reduction and should only be used when pollution cannot be prevented.

The treatment of waste is the third element of the hierarchy and should be utilized only in the absence of feasible source reduction or recycling opportunities. Waste treatment involves the use of chemical, biological, or physical processes to reduce or eliminate waste material. The incineration of wastes is included in this category and is considered "preferable to other treatment methods (i.e., chemical, biological, and physical) because incineration can permanently destroy the

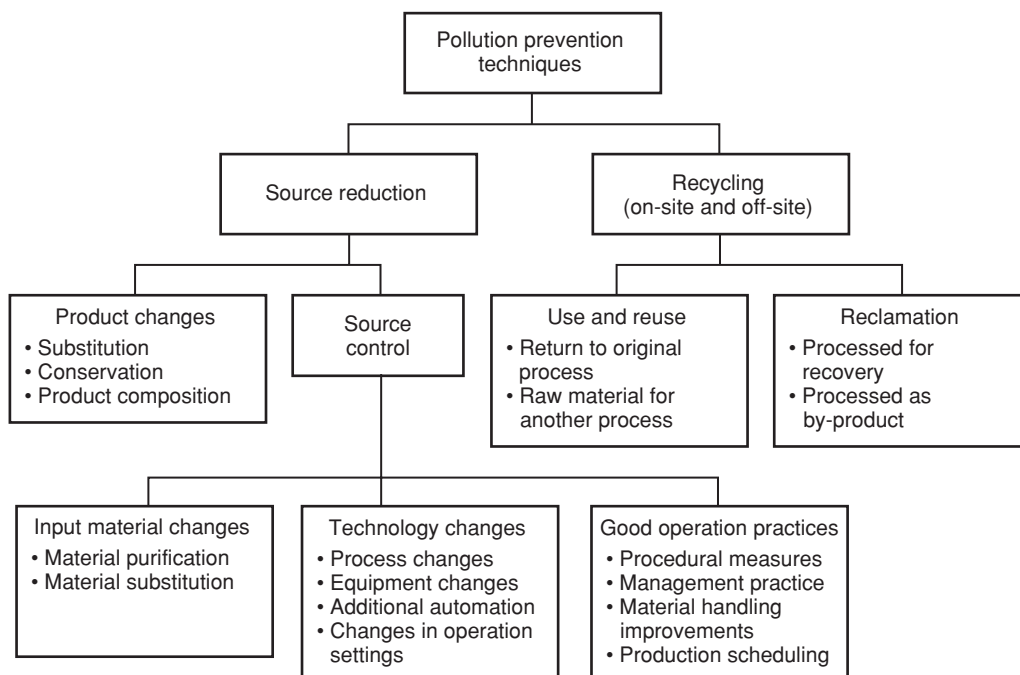


FIG. 22-3 Pollution prevention techniques.

hazardous components in waste materials” (Ref. 4). It can also be employed to reduce the volume of waste to be treated.

Several of these pollution-prevention elements are used by industry in combination to achieve the greatest waste reduction. Residual wastes that cannot be prevented or otherwise managed are then disposed of only as a last resort.

Figure 22-3 provides a more detailed schematic representation of the two preferred pollution prevention techniques (i.e., source reduction and recycling).

MULTIMEDIA ANALYSIS AND LIFE-CYCLE ANALYSIS

Multimedia Analysis In order to properly design and then implement a pollution prevention program, sources of all wastes must be fully understood and evaluated. A multimedia analysis involves a multifaceted approach. It must not only consider one waste stream but all potentially contaminated media (e.g., air, water, land). Past waste-management practices have been concerned primarily with treatment. All too often, such methods solve one waste problem by transferring a contaminant from one medium to another (e.g., air stripping); such waste shifting is *not* pollution prevention or waste reduction.

Pollution prevention techniques must be evaluated through a thorough consideration of all media, hence the term multimedia. This approach is a clear departure from previous pollution treatment or control techniques where it was acceptable to transfer a pollutant from one source to another in order to solve a waste problem. Such strategies merely provide short-term solutions to an ever increasing problem. As an example, air pollution control equipment prevents or reduces the discharge of waste into the air but at the same time can produce a solid (hazardous) waste problem.

Life-Cycle Analysis The aforementioned multimedia approach to evaluating a product’s waste stream(s) aims to ensure that the treatment of one waste stream does not result in the generation or increase in an additional waste output. Clearly, impacts resulting during the production of a product or service must be evaluated over its entire

history or life cycle. This life-cycle analysis or total systems approach (Ref. 3) is crucial to identifying opportunities for improvement. As described earlier, this type of evaluation identifies “energy use, material inputs, and wastes generated during a product’s life: from extraction and processing of raw materials to manufacture and transport of a product to the marketplace and finally to use and dispose of the product” (Ref. 5).

During a forum convened by the World Wildlife Fund and the Conservation Foundation in May 1990, various steering committees recommended that a three-part life-cycle model be adopted. This model consists of the following:

1. An inventory of materials and energy used, and environmental releases from all stages in the life of a product or process
2. An analysis of potential environmental effects related to energy use and material resources and environmental releases
3. An analysis of the changes needed to bring about environmental improvements for the product or process under evaluation

Traditional cost analysis often fails to include factors relevant to future damage claims resulting from litigation, the depletion of natural resources, the effects of energy use, and the like. As such, waste-management options such as treatment and disposal may appear preferential if an overall life-cycle cost analysis is not performed. It is evident that environmental costs from cradle to grave have to be evaluated together with more conventional production costs to accurately ascertain genuine production costs. In the future, a total systems approach will most likely involve a more careful evaluation of pollution, energy, and safety issues. For example, if one was to compare the benefits of coal versus oil as a fuel source for an electric power plant, the use of coal might be considered economically favorable. In addition to the cost issues, however, one must be concerned with the environmental effects of coal mining (e.g., transportation and storage prior to use as a fuel). Society often has a tendency to overlook the fact that there are serious health and safety matters (e.g., miner exposure) that must be considered along with the effects of fugitive emissions. When these effects are weighed alongside standard economic factors, the full cost benefits of coal

usage may be eclipsed by environmental costs. Thus, many of the economic benefits associated with pollution prevention are often unrecognized due to inappropriate cost-accounting methods. For this reason, economic considerations are briefly detailed later.

POLLUTION-PREVENTION ASSESSMENT PROCEDURES

The first step in establishing a pollution prevention program is the obtainment of management commitment. This is necessary given the inherent need for project structure and control. Management will determine the amount of funding allotted for the program as well as specific program goals. The data collected during the actual evaluation is then used to develop options for reducing the types and amounts of waste generated. Figure 22-4 depicts a systematic approach that can be used during the procedure. After a particular waste stream or area of concern is identified, feasibility studies are performed involving both economic and technical considerations. Finally, preferred alternatives are implemented. The four phases of the assessment (i.e., planning and organization, assessment, feasibility, and implementation) are introduced in the following subsections. Sources of additional information as well as information on industrial programs are also provided.

Planning and Organization The purpose of this phase is to obtain management commitment, define and develop program goals, and assemble a project team. Proper planning and organization are crucial to the successful performance of the pollution-prevention assessment. Both managers and facility staff play important roles in the assessment procedure by providing the necessary commitment and familiarity with the facility, its processes, and current waste-management operations. It is the benefits of the program, including economic advantages, liability

reduction, regulatory compliance, and improved public image that often leads to management support.

Once management has made a commitment to the program and goals have been set, a program task force is established. The selection of a team leader will be dependent upon many factors including the ability to effectively interface with both the assessment team and management staff.

The task force must be capable of identifying pollution reduction alternatives as well as be cognizant of inherent obstacles to the process. Barriers frequently arise from the anxiety associated with the belief that the program will negatively affect product quality or result in production losses. According to an EPA survey, 30 percent of industry comments responded that they were concerned that product quality would decline if waste minimization techniques were implemented (Ref. 6). As such, the assessment team, and the team leader in particular, must be ready to react to these and other concerns (Ref. 2).

Assessment Phase The assessment phase aims to collect data needed to identify and analyze pollution-prevention opportunities. Assessment of the facility's waste-reduction needs includes the examination of (hazardous) waste streams, process operations, and the identification of techniques that often promise the reduction of waste generation. Information is often derived from observations made during a facility walk-through, interviews with employees (e.g., operators, line workers), and review of site or regulatory records. One professional organization suggests the following sources of information be reviewed, as available (Ref. 7):

1. Product design criteria
2. Process flow diagrams for all solid waste, wastewater, and air emissions sources
3. Site maps showing the location of all pertinent units (e.g., pollution-control devices, points of discharge)
4. Environmental documentation, including: Material Safety Data Sheets (MSDS), military specification data, permits (e.g., NPDES, POTW, RCRA), SARA Title III reports, waste manifests, and any pending permits or application information
5. Economic data, including: cost of raw materials; cost of air, wastewater, and (hazardous) waste treatment; waste management operating and maintenance costs; and waste disposal costs
6. Managerial information: environmental policies and procedures; prioritization of waste-management concerns; automated or computerized waste-management systems; inventory and distribution procedures; maintenance scheduling practices; planned modifications or revisions to existing operations that would impact waste-generation activities; and the basis of source reduction decisions and policies

The use of process flow diagrams and material balances are worthwhile methods to quantify losses or emissions and provide essential data to estimate the size and cost of additional equipment, other data to evaluate economic performance, and a baseline for tracking the progress of minimization efforts (Ref. 3). Material balances should be applied to individual waste streams or processes and then utilized to construct an overall balance for the facility. Details on these calculations are available in the literature (Ref. 8). In addition, an introduction to this subject is provided in the next section.

The data collected is then used to prioritize waste streams and operations for assessment. Each waste stream is assigned a priority based on corporate pollution-prevention goals and objectives. Once waste origins are identified and ranked, potential methods to reduce the waste stream are evaluated. The identification of alternatives is generally based on discussions with the facility staff; review of technical literature; and contacts with suppliers, trade organizations, and regulatory agencies.

Alternatives identified during this phase of the assessment are evaluated using screening procedures so as to reduce the number of alternatives requiring further exploration during the feasibility analysis phase. Some criteria used during this screening procedure include: cost-effectiveness; implementation time; economic, compliance, safety and liability concerns; waste-reduction potential; and whether the technology is proven (Refs. 4, 8). Options that meet established criteria are then examined further during the feasibility analysis.

Feasibility Analysis The selection procedure is performed by an evaluation of technical and economic considerations. The technical

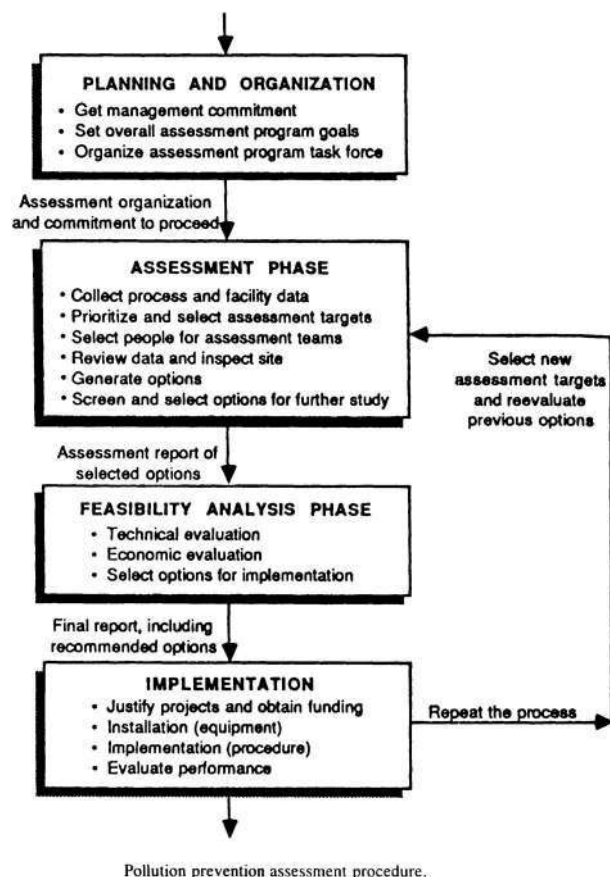


FIG. 22-4 Pollution prevention assessment procedure.

evaluation determines whether a given option will work as planned. Some typical considerations follow:

1. Safety concerns
2. Product quality impacts or production delays during implementation
3. Labor and/or training requirements
4. Creation of new environmental concerns
5. Waste reduction potential
6. Utility and budget requirements
7. Space and compatibility concerns

If an option proves to be technically ineffective or inappropriate, it is deleted from the list of potential alternatives. Either following or concurrent with the technical evaluation, an economic study is performed, weighing standard measures of profitability such as payback period, investment returns, and net present value. Many of these costs (or, more appropriately, cost savings) may be substantial yet are difficult to quantify. (Refer to *Economic Considerations Associated with Pollution Prevention*.)

Implementation The findings of the overall assessment are used to demonstrate the technical and economic worthiness of program implementation. Once appropriate funding is obtained, the program is implemented not unlike any other project requiring new procedures or equipment. When preferred waste-pollution-prevention techniques are identified, they are implemented and should become part of the facility's day-to-day management and operation. Subsequent to the program's execution, its performance should be evaluated in order to demonstrate effectiveness, generate data to further refine and augment waste-reduction procedures, and maintain management support.

It should be noted that waste reduction, energy conservation, and safety issues are interrelated and often complementary to each other. For example, the reduction in the amount of energy a facility consumes usually results in reduced emissions associated with the generation of power. Energy expenditures associated with the treatment and transport of waste are similarly reduced when the amount of waste generated is lessened; at the same time, worker safety is elevated due to reduced exposure to hazardous materials. However, this is not always the case. Addition of air-pollution control systems at power plants decreases net power output due to the power consumed by the equipment. This in turn requires more fuel to be combusted for the same power exported to the grid. This additional fuel increases pollution from coal mining, transport, ash disposal, and the like. In extreme cases, very high recovery efficiencies for some pollutants can raise, not lower, overall total emissions. Seventy percent removal might produce a 2 percent loss in power output; 90 percent recovery could lead to a 3 percent loss; 95 percent, a 5 percent loss; 99 percent, a 10 percent loss, and so on. The point to be made is that pollution control is generally not a "free lunch."

Sources of Information The successful development and implementation of any pollution prevention program is not only dependent on a thorough understanding of the facility's operations but also requires an intimate knowledge of current opportunities and advances in the field. In fact, 32 percent of industry respondents to an earlier EPA survey identified the lack of technical information as a major factor delaying or preventing the implementation of a waste-minimization program (Ref. 6). One of EPA's positive contributions has been the development of a national Pollution Prevention Information Clearinghouse (PPIC) and the Pollution Prevention Information Exchange System (PIES) to facilitate the exchange of information needed to promote pollution prevention through efficient information transfer (Ref. 2).

PPIC is operated by the EPA's Office of Research and Development and the Office of Pollution Prevention. The clearinghouse is comprised of four elements:

1. Repository, including a hard copy reference library and collection center and an on-line information retrieval and ordering system.
2. PIES, a computerized conduit to databases and document ordering, accessible via modem and personal computer: (703) 506-1025.
3. PPIC uses the RCRA/Superfund and Small Business Ombudsman Hotlines as well as a PPIC technical assistance line to answer pollution-prevention questions, access information in the PPIC, and

assist in document ordering and searches. To access PPIC by telephone, call:

RCRA/Superfund Hotline, (800) 242-9346
Small Business Ombudsman Hotline, (800) 368-5888
PPIC Technical Assistance, (703) 821-4800

4. PPIC compiles and disseminates information packets and bulletins, and initiates networking efforts with other national and international organizations.

Additionally, the EPA publishes a newsletter entitled *Pollution Prevention News* that contains information including EPA news, technologies, program updates, and case studies. The EPA's Risk Reduction Engineering Laboratory and the Center for Environmental Research Information has published several guidance documents, developed in cooperation with the California Department of Health Services. The manuals supplement generic waste reduction information presented in the EPA's *Waste Minimization Opportunity Assessment Manual* (Ref. 9).

Pollution prevention or waste minimization programs have been established at the State level and as such are good sources of information. Both Federal and State agencies are working with universities and research centers and may also provide assistance. For example, the American Institute of Chemical Engineers has established the Center for Waste Reduction Technologies (CWRT), a program based on targeted research, technology transfer, and enhanced education.

Industry Programs A significant pollution-prevention resource may very well be found with the "competition." Several large companies have established well-known programs that have successfully incorporated pollution-prevention practices into their manufacturing processes. These include, but are not limited to: 3M—Pollution Prevention Pays (3P); Dow Chemical—Waste Reduction Always Pays (WRAP); Chevron—Save Money and Reduce Toxics (SMART); and, the General Dynamics Zero Discharge Program.

Smaller companies can benefit by the assistance offered by these larger corporations. It is clear that access to information is of major importance when implementing efficient pollution-prevention programs. By adopting such programs, industry is affirming pollution prevention's application as a good business practice and not simply a "noble" effort.

ASSESSMENT PHASE: MATERIAL BALANCE CALCULATIONS

(The reader is directed to Refs. 4 and 10 for further information.)

One of the key elements of the assessment phase of a pollution prevention program involves mass balance equations. These calculations are often referred to as material balances; the calculations are performed via the conservation law for mass. The details of this often-used law are described below.

The conservation law for mass can be applied to any process or system. The general form of the law follows:

$$\text{mass in} - \text{mass out} + \text{mass generated} = \text{mass accumulated}$$

This equation can be applied to the total mass involved in a process or to a particular species, on either a mole or mass basis. The conservation law for mass can be applied to steady-state or unsteady-state processes and to batch or continuous systems. A steady-state system is one in which there is no change in conditions (e.g., temperature, pressure) or rates of flow with time at any given point in the system; the accumulation term then becomes zero. If there is no chemical reaction, the generation term is zero. All other processes are classified as unsteady state.

To isolate a system for study, the system is separated from the surroundings by a boundary or envelope that may either be real (e.g., a reactor vessel) or imaginary. Mass crossing the boundary and entering the system is part of the mass-in term. The equation may be used for any compound whose quantity does not change by chemical reaction or for any chemical element, regardless of whether it has participated in a chemical reaction. Furthermore, it may be written for one piece of equipment, several pieces of equipment, or around an entire process (i.e., a total material balance).

The conservation of mass law finds a major application during the performance of pollution-prevention assessments. As described earlier,

a pollution-prevention assessment is a systematic, planned procedure with the objective of identifying methods to reduce or eliminate waste. The assessment process should characterize the selected waste streams and processes (Ref. 11)—a necessary ingredient if a material balance is to be performed. Some of the data required for the material balance calculation may be collected during the first review of site-specific data; however, in some instances, the information may not be collected until an actual site walk-through is performed.

Simplified mass balances should be developed for each of the important waste-generating operations to identify sources and gain a better understanding of the origins of each waste stream. Since a mass balance is essentially a check to make sure that what goes into a process (i.e., the total mass of all raw materials) is what leaves the process (i.e., the total mass of the products and by-products), the material balance should be made individually for all components that enter and leave the process. When chemical reactions take place in a system, there may be an advantage to performing “elemental balances” for specific chemical elements in a system. Material balances can assist in determining concentrations of waste constituents where analytical test data are limited. They are particularly useful when there are points in the production process where it is difficult or uneconomical to collect analytical data.

Mass-balance calculations are particularly useful for quantifying fugitive emissions such as evaporative losses. Waste stream data and mass balances will enable one to track flow and characteristics of the waste streams over time. Since in most cases the accumulation equals zero (steady-state operation), it can then be assumed that any buildup is actually leaving the process through fugitive emissions or other means. This will be useful in identifying trends in waste/pollutant generation and will also be critical in the task of measuring the performance of implemented pollution prevention options. The result of these activities is a catalog of waste streams that provides a description of each waste, including quantities, frequency of discharge, composition, and other important information useful for material balance. Of course, some assumptions or educated estimates will be needed when it is impossible to obtain specific information.

By performing a material balance in conjunction with a pollution prevention assessment, the amount of waste generated becomes known. The success of the pollution prevention program can therefore be measured by using this information on baseline generation rates (i.e., that rate at which waste is generated without pollution prevention considerations) (Ref. 12).

BARRIERS AND INCENTIVES TO POLLUTION PREVENTION

As discussed previously, industry is beginning to realize that there are profound benefits associated with pollution prevention, including cost effectiveness, reduced liability, enhanced public image, and regulatory compliance. Nevertheless, there are barriers or disincentives identified with pollution prevention. This section will briefly outline both barriers and incentives that may need to be confronted or considered during the evaluation of a pollution prevention program.

Barriers to Pollution Prevention (“The Dirty Dozen”) There are numerous reasons why more businesses are not reducing the wastes they generate. The following “dirty dozen” are common disincentives:

1. *Technical limitations.* Given the complexity of present manufacturing processes, waste streams exist that cannot be reduced with current technology. The need for continued research and development is evident.

2. *Lack of information.* In some instances, the information needed to make a pollution-prevention decision may be confidential or is difficult to obtain. In addition, many decision makers are simply unaware of the potential opportunities available regarding information to aid in the implementation of a pollution-prevention program.

3. *Consumer preference obstacles.* Consumer preference strongly affects the manner in which a product is produced, packaged, and marketed. If the implementation of a pollution-prevention program results in an increase in the cost of a product or decreased convenience or availability, consumers might be reluctant to use it.

4. *Concern over product quality decline.* The use of a less hazardous material in a product’s manufacturing process may result in decreased life, durability, or competitiveness.

5. *Economic concerns.* Many companies are unaware of the economic advantages associated with pollution prevention. Legitimate concerns may include decreased profit margins or the lack of funds required for the initial capital investment.

6. *Resistance to change.* The unwillingness of many businesses to change is rooted in their reluctance to try technologies that may be unproven or based on a combination of the barriers discussed in this section.

7. *Regulatory barriers.* Existing regulations that have created incentives for the control and containment of wastes are at the same time discouraging the exploration of pollution-prevention alternatives. Moreover, since regulatory enforcement is often intermittent, current legislation can weaken waste-reduction incentives.

8. *Lack of markets.* The implementation of pollution-prevention processes and the production of environmentally friendly products will be of no avail if markets do not exist for such goods. As an example, the recycling of newspaper in the United States has resulted in an overabundance of waste paper without markets prepared to take advantage of this raw material.

9. *Management apathy.* Many managers capable of making decisions to begin pollution-prevention activities do not realize the potential benefits of pollution prevention.

10. *Institutional barriers.* In an organization without a strong infrastructure to support pollution-prevention plans, waste-reduction programs will be difficult to implement. Similarly, if there is no mechanism in place to hold individuals accountable for their actions, the successful implementation of a pollution-prevention program will be limited.

11. *Lack of awareness of pollution prevention advantages.* As mentioned in reason no. 5, decision makers may merely be uninformed of the benefits associated with pollution reduction.

12. *Concern over the dissemination of confidential product information.* If a pollution-prevention assessment reveals confidential data pertinent to a company’s product, fear may exist that the organization will lose a competitive edge with other businesses in the industry.

Pollution-Prevention Incentives (“A Baker’s Dozen”) Various means exist to encourage pollution prevention through regulatory measures, economic incentives, and technical assistance programs. Since the benefits of pollution prevention can surpass prevention barriers, a “baker’s dozen” incentives is presented below:

1. *Economic benefits.* The most obvious economic benefits associated with pollution prevention are the savings that result from the elimination of waste storage, treatment, handling, transport, and disposal. Additionally, less tangible economic benefits are realized in terms of decreased liability, regulatory compliance costs (e.g., permits), legal and insurance costs, and improved process efficiency. Pollution prevention almost always pays for itself, particularly when the time required to comply with regulatory standards is considered. Several of these economic benefits are discussed separately below.

2. *Regulatory compliance.* Quite simply, when wastes are not generated, compliance issues are not a concern. Waste-management costs associated with record keeping, reporting, and laboratory analysis are reduced or eliminated. Pollution prevention’s proactive approach to waste management will better prepare industry for future regulation of many hazardous substances and wastes that are currently unregulated. Regulations have, and will continue to be, a moving target.

3. *Liability reduction.* Facilities are responsible for their wastes from “cradle-to-grave.” By eliminating or reducing waste generation, future liabilities can also be decreased. Additionally, the need for expensive pollution liability insurance requirements may be abated.

4. *Enhanced public image.* Consumers are interested in purchasing goods that are safer for the environment, and this demand, depending on how they respond, can mean success or failure for many companies. Businesses should therefore be sensitive to consumer demands and use pollution-prevention efforts to their utmost advantage by producing goods that are environmentally friendly.

5. *Federal and state grants.* Federal and state grant programs have been developed to strengthen pollution-prevention programs initiated by states and private entities. The EPA's "Pollution Prevention by and for Small Business" grant program awards grants to small businesses to assist their development and demonstration of new pollution-prevention technologies.

6. *Market incentives.* Public demand for environmentally preferred products has generated a market for recycled goods and related products; products can be designed with these environmental characteristics in mind, offering a competitive advantage. In addition, many private and public agencies are beginning to stimulate the market for recycled goods by writing contracts and specifications that call for the use of recycled materials.

7. *Reduced waste-treatment costs.* As discussed in reason no. 5 of the dirty dozen, the increasing costs of traditional end-of-pipe waste-management practices are avoided or reduced through the implementation of pollution-prevention programs.

8. *Potential tax incentives.* In an effort to promote pollution prevention, taxes may eventually need to be levied to encourage waste generators to consider reduction programs. Conversely, tax breaks could be developed for corporations that utilize pollution-prevention methods to foster pollution prevention.

9. *Decreased worker exposure.* By reducing or eliminating chemical exposures, businesses benefit by lessening the potential for chronic workplace exposure and serious accidents and emergencies. The burden of medical monitoring programs, personal exposure monitoring, and potential damage claims are also reduced.

10. *Decreased energy consumption.* As mentioned previously, methods of energy conservation are often interrelated and complementary to each other. Energy expenditures associated with the treatment and transport of waste are usually but not always reduced when the amount of waste generated is lessened, while at the same time the pollution associated with energy consumed by these activities is abated.

11. *Increased operating efficiencies.* A potential beneficial side effect of pollution-prevention activities is a concurrent increase in operating efficiency. Through a pollution-prevention assessment, the assessment team can identify sources of waste that result in (hazardous) waste generation and loss in process performance. The implementation of a reduction program will often rectify such problems through modernization, innovation, and the implementation of good operating practices.

12. *Competitive advantages.* By taking advantage of the many benefits associated with pollution prevention, businesses can gain a competitive edge.

13. *Reduced negative environmental impacts.* Through an evaluation of pollution-prevention alternatives, which consider a total systems approach, consideration is given to the negative impact of environmental damage to natural resources and species that occur during raw-material procurement and waste disposal. The performance of pollution-prevention endeavors will therefore result in enhanced environmental protection.

ECONOMIC CONSIDERATIONS ASSOCIATED WITH POLLUTION-PREVENTION PROGRAMS

The purpose of this subsection is to outline the basic elements of a pollution-prevention cost-accounting system that incorporates both traditional and less tangible economic variables. The intent is not to present a detailed discussion of economic analysis but to help identify the more important elements that must be considered to properly quantify pollution-prevention options.

The greatest driving force behind any pollution-prevention plan is the promise of economic opportunities and cost savings over the long term. Pollution prevention is now recognized as one of the lowest-cost options for waste/pollutant management. Hence, an understanding of the economics involved in pollution prevention programs/options is quite important in making decisions at both the engineering and management levels. Every engineer should be able to execute an economic evaluation of a proposed project. If the project cannot be justified economically after *all* factors—and these will be discussed in

TABLE 22-13 Economic Analysis Timetable

Prior to 1945	Capital costs only
1945–1960	Capital and some operating costs
1960–1970	Capital and operating costs
1970–1975	Capital, operating, and some environmental control costs
1975–1980	Capital, operating, and environmental control costs
1980–1985	Capital, operating, and more sophisticated environmental control costs
1985–1990	Capital, operating, and environmental controls, and some life-cycle analysis (Total Systems Approach)
1990–1995	Capital, operating, and environmental control costs and life-cycle analysis (Total Systems Approach)
1995–2000	Widespread acceptance of Total Systems Approach
After 2000	To be detailed at a later date

more detail below—have been taken into account, it should obviously not be pursued. The earlier such a project is identified, the fewer resources will be wasted.

Before the true cost or profit of a pollution-prevention program can be evaluated, the factors contributing to the economics must be recognized. There are two traditional contributing factors (capital costs and operating costs), but there are also other important costs and benefits associated with pollution prevention that need to be quantified if a meaningful economic analysis is going to be performed. Table 22-13 demonstrates the evolution of various cost-accounting methods. Although Tables 22-12 (see Introduction) and 22-13 are not directly related, the reader is left with the option of comparing some of the similarities between the two.

The Total Systems Approach (TSA) referenced in Table 22-13 aims to quantify not only the economic aspects of pollution prevention but also the social costs associated with the production of a product or service from cradle to grave (i.e., life cycle). The TSA attempts to quantify less tangible benefits such as the reduced risk derived from not using a hazardous substance. The future is certain to see more emphasis placed on the TSA approach in any pollution-prevention program. As described earlier, a utility considering the option of converting from a gas-fired boiler to coal-firing is usually not concerned with the environmental effects and implications associated with such activities as mining, transporting, and storing the coal prior to its usage as an energy feedstock. Pollution-prevention approaches since the mid-to-late 1990s have become more aware of this need.

The economic evaluation referred to above is usually carried out using standard measures of profitability. Each company and organization has its own economic criteria for selecting projects for implementation. (For example, a project can be judged on its payback period. For some companies, if the payback period is more than 3 years, it is a dead issue.) In performing an economic evaluation, various costs and savings must be considered. As noted earlier, the economic analysis presented in this subsection represents a preliminary, rather than a detailed, analysis. For smaller facilities with only a few (and perhaps simple) processes, the entire pollution-prevention assessment procedure will tend to be much less formal. In this situation, several obvious pollution-prevention options such as the installation of flow controls and good operating practices may be implemented with little or no economic evaluation. In these instances, no complicated analyses are necessary to demonstrate the advantages of adopting the selected pollution-prevention option. A proper perspective must also be maintained between the magnitude of savings that a potential option may offer and the amount of manpower required to do the technical and economic feasibility analyses. A short description of the various aforementioned economic factors—including capital and operating costs and other considerations—follows.

Once identified, the costs and/or savings are placed into their appropriate categories and quantified for subsequent analysis. Equipment cost is a function of many variables, one of the most significant of which is capacity. Other important variables include operating temperature and/or pressure conditions, and degree of equipment sophistication. Preliminary estimates are often made using simple cost-capacity relationships that are valid when the other variables are confined to a narrow range of values.

The usual technique for determining the capital costs (i.e., total capital costs, which include equipment design, purchase, and installation) for the facility is based on the factored method of establishing direct and indirect installation costs as a function of the known equipment costs. This is basically a modified Lang method, whereby cost factors are applied to known equipment costs (Refs. 13 and 14). The first step is to obtain from vendors the purchase prices of the primary and auxiliary equipment. The total base price, designated by X , which should include instrumentation, control, taxes, freight costs, and so on, serves as the basis for estimating the direct and indirect installation costs. These costs are obtained by multiplying X by the cost factors, which are available in the literature (see Refs. 13–19).

The second step is to estimate the direct installation costs by summing all the cost factors involved in the direct installation costs, which include piping, insulation, foundation and supports, and so on. The sum of these factors is designated as the DCF (direct installation cost factor). The direct installation costs are then the product of the DCF and X . The third step consists of estimating the indirect installation cost. Here all the cost factors for the indirect installation costs (engineering and supervision, startup, construction fees, and so on) are added; the sum is designated by ICF (indirect installation cost factor). The indirect installation costs are then the product of ICF and X . Once the direct and indirect installation costs have been calculated, the total capital cost (TCC) may be evaluated as follows:

$$TCC = X + (DCF)(X) + (ICF)(X)$$

This can then be converted to annualized capital costs (ACC) with the use of the capital recovery factor (CRF). The CRF can be calculated from the following equation:

$$CRF = \frac{(i)(1+i)^n}{(1+i)^n - 1}$$

where n = projected lifetime of the project, years
 i = annual interest rate, expressed as a fraction

The annualized capital cost (ACC) is the product of the CRF and TCC and represents the total annualized installed equipment cost distributed over the lifetime of the project. The ACC reflects the cost associated with the initial capital outlay over the depreciable life of the system. Although investment and operating costs can be accounted for in other ways such as present-worth analysis, the capital recovery method is often preferred because of its simplicity and versatility. This is especially true when comparing somewhat similar systems having different depreciable lives. In such decisions, there are usually other considerations besides economic, but if all other factors are equal, the alternative with the lowest total annualized cost should be the most viable.

Operating costs can vary from site to site since these costs reflect local conditions (e.g., staffing practices, labor, utility costs). Operating costs, like capital costs, may be separated into two categories: direct and indirect costs. Direct costs are those that cover material and labor and are directly involved in operating the facility. These include labor, materials, maintenance and maintenance supplies, replacement parts, wastes, disposal fees, utilities, and laboratory costs. However, the major direct operating costs are usually those associated with labor and materials. Indirect costs are those operating costs associated with, but not directly involved in, operating the facility; costs such as overhead (e.g., building/land leasing and office supplies), administrative fees, property taxes, and insurance fees fall into this category.

The main problem with the traditional type of economic analysis is that it is difficult—nay, in some cases impossible—to quantify some of the not-so-obvious economic merits of a pollution-prevention program. Several considerations have just recently surfaced as factors that need to be taken into account in any meaningful economic analysis of a pollution-prevention effort. What follows is a summary listing of these considerations, most of which have been detailed earlier.

1. Decreased long-term liabilities
2. Regulatory compliance
3. Regulatory recordkeeping
4. Dealings with the EPA

5. Dealings with state and local regulatory bodies
6. Elimination or reduction of fines and penalties
7. Potential tax benefits
8. Customer relations
9. Stockholder support (corporate image)
10. Improved public image
11. Reduced technical support
12. Potential insurance costs and claims
13. Effect on borrowing power
14. Improved mental and physical well-being of employees
15. Reduced health-maintenance costs
16. Employee morale
17. Other process benefits
18. Improved worker safety
19. Avoidance of rising costs of waste treatment and/or disposal
20. Reduced training costs
21. Reduced emergency response planning

Many proposed pollution-prevention programs have been squelched in their early stages because a comprehensive economic analysis was not performed. Until the effects described above are included, the true merits of a pollution-prevention program may be clouded by incorrect and/or incomplete economic accounting. Can something be done by industry to remedy this problem? One approach is to use a modified version of the standard Delphi panel. In order to estimate these other economic benefits of pollution prevention, several knowledgeable individuals within and perhaps outside the organization are asked to independently provide estimates, with explanatory details, on these economic benefits. Each individual in the panel is then allowed to independently review all responses. The cycle is then repeated until the group's responses approach convergence.

Finally, pollution-prevention measures can provide a company with the opportunity of looking their neighbors in the eye and truthfully saying that all that can reasonably be done to prevent pollution is being done. In effect, the company is doing right by the environment. Is there an economic advantage to this? It is not only a difficult question to answer quantitatively but also a difficult one to answer qualitatively. Industry is left to ponder the answer to this question.

POLLUTION PREVENTION AT THE DOMESTIC AND OFFICE LEVELS

Concurrent with the United States' growth as an international economic superpower during the years following World War II, a new paradigm was established whereby society became accustomed to the convenience and ease with which goods could be discarded after a relatively short useful life. Individuals have come to expect these everyday comforts with what may be considered an unconscious ignorance towards the ultimate effect of the throwaway lifestyle. In fact, many, while fearful of environmental degradation, are not aware of the ill effect these actions have on the world as a whole. Many individuals who abide by the "not in my back yard" (NIMBY) mind-set also feel pollution prevention does not have to occur "in my house."

The past three decades have seen an increased social awareness of the impact of modern-day lifestyles on the environment. Public environmental concerns include issues such as waste disposal; hazardous-material regulations; depletion of natural resources; and air, water, and land pollution. Nevertheless, roughly one-half of the total quantity of waste generated each year can be attributed to domestic sources!

More recently, concern about the environment has begun to stimulate environmentally correct behavior. After all, the choices made today affect the environment of tomorrow. Simple decisions can be made at work and at home that conserve natural resources and lessen the burden placed on a waste-management system. By eliminating waste at the source, society is participating in the protection of the environment by reducing the amount of waste that would otherwise need to be treated or ultimately disposed.

There are numerous areas of environmental concern that can be directly influenced by the consumer's actions. The first issue, which is described above, is that of waste generation. Second, energy conservation has significantly affected Americans and has resulted in cost-saving

measures that have directly reduced pollution. As mentioned previously, energy conservation is directly related to pollution prevention since a reduction in energy use usually corresponds to less energy production and, consequently, less pollution output. A third area of concern is that of accident and emergency planning. Relatively recent accidents like Chernobyl and Bhopal have increased public awareness and helped stimulate regulatory policies concerned with emergency planning. Specifically, Title III of the Superfund Amendments and Reauthorization Act (SARA) of 1986 established the Emergency Planning and Community Right-to-Know Act, which has forever changed the concept of environmental management. This law attempts to avert potential emergencies through careful planning and the development of contingency plans. Planning for emergency situations can help protect human health and the environment (Ref. 3).

Based on the above three areas of potential concern, a plethora of waste reduction activities can be performed at home or in an office environment. One can easily note the similarities between these activities and those pollution-prevention activities performed in industry. The following few examples are identified by category according to this notation (Ref. 3):

- Waste reduction
- ★ Accidents, health, and safety
- Energy conservation

At home

- Purchase products with the least amount of packaging
- Borrow items used infrequently
- ★ Handle materials to avoid spills (and slips, trips)
- ★ Keep hazardous materials out-of-reach of children
- Use energy-efficient lighting (e.g., fluorescent)
- Install water-flow restriction devices on sink faucets and showerheads

At the office

- Pass on verbal memos when written correspondence isn't required
- Reuse paper before recycling it
- ★ Know building evacuation procedures
- ★ Adhere to company medical policies (e.g., annual physicals)
- Don't waste utilities simply because you are not paying for them
- Take public transportation to the office

The use of pollution-prevention principles on the home front clearly does not involve the use of high-technology equipment or major lifestyle changes; success is only dependent upon active and willing public participation. All can help to make a difference.

Before pollution prevention becomes a fully accepted way of life, considerable effort still needs to be expended to change the way one looks at waste management. A desire to use "green" products and services will be of no avail in a market where these goods are not available. Participation in pollution-prevention programs will increase through continued education, community efforts, and lobbying for change. Market incentives can be created and strengthened by tax policies, price preferences, and packaging regulations created at the federal, state, and local levels. Individuals should take part by communicating with industry and expressing their concerns. For example, citizens should feel free to write letters to decision makers at both business organizations and government institutions regarding specific products or legislation. Letters can be positive, demonstrating personal endorsement of a green product, or disapproving, expressing discontent and reluctance to use a particular product because of its negative environmental effect (Ref. 3).

By starting now, society will learn through experience to better manage its waste while providing a safer and cleaner environment for future generations.

ETHICAL CONSIDERATIONS

Given the evolutionary nature of pollution prevention, it is evident that as technology changes and continued progress is achieved, society's

opinion of both what is possible and desirable will also change. Government officials, scientists, and engineers will face new challenges to fulfill society's needs while concurrently meeting the requirements of changing environmental regulations. It is now apparent that attention should also be given to ethical considerations and their application to pollution prevention policy. How one makes decisions on the basis of ethical beliefs is clearly a personal issue but one that should be addressed. In order to examine this issue, the meaning of ethics must be known, although the intent here is not to provide a detailed discussion regarding the philosophy of ethics or morality.

Ethics can simply be defined as the analysis of the rightness and wrongness of an act or actions. According to Dr. Andrew Varga, director of the Philosophical Resources Program at Fordham University, in order to discern the morality of an act, it is customary to look at the act on the basis of four separate elements: its object, motive, circumstances, and consequences (Ref. 21). Rooted in this analysis is the belief that if one part of the act is bad, then the overall act itself cannot be considered good. Of course, there are instances where the good effects outweigh the bad, and therefore there may be a reason to permit the evil. The application of this principle is not cut and dry and requires decisions to be made on a case-by-case basis. Each must make well-judged decisions rooted in an understanding of the interaction between technology and the environment. After all, the decisions made today will have an impact not only on this generation but on many generations to come. If one chooses today not to implement a waste-reduction program in order to meet a short-term goal of increased productivity, this might be considered a good decision since it benefits the company and its employees. However, should a major release occur that results in the contamination of a local sole-source of drinking water, what is the good?

As an additional example, toxicological studies have indicated that test animals exposed to small quantities of toxic chemicals had better health than control groups that were not exposed. A theory has been developed that says that a low-level exposure to the toxic chemical results in a challenge to the animal to maintain homeostasis; this challenge increases the animal's vigor and, correspondingly, its health. However, larger doses seem to cause an inability to adjust, resulting in negative health effects. Based on this theory, some individuals would believe that absolute pollution reduction might not be necessary.

FUTURE TRENDS

The reader should keep an open mind when dealing with the types of issues discussed above and facing challenges perhaps not yet imagined. Clearly, there is no simple solution or answer to many of the questions. The EPA is currently *attempting* to develop a partnership with government, industry, and educators to produce and distribute pollution-prevention educational materials. Given EPA's past history and performance, there are understandable doubts as to whether this program will succeed (Ref. 22).

Finally, no discussion on pollution prevention would be complete without reference to the activities of the 50-year-old International Organization for Standardization (ISO) and the general subject of nanotechnology (Ref. 23). Both are briefly described below.

ISO created Technical Committee 207 (TC 207) to develop new standards for environmental management systems (EMS). The ramifications, especially to the chemical industry, which has become heavily involved in the continued development and application of these standards, will be great. TC 207's activities are being scrutinized closely. The new environmental management set of standards is entitled ISO 14000. With this new standard in place, it is expected that customers will require their product suppliers to be certified by ISO 14000. In addition, some service suppliers who have an impact on the environment are expected to obtain an ISO 14000 certification. Certification will imply to the customer that, when the product or service was prepared, the environment was not significantly damaged in the process. This will effectively require that the life-cycle design mentality discussed earlier be applied to all processes. Implementing the life-cycle design framework will require significant organizational and operational changes in business. To effectively promote the goals of

sustainable development, life-cycle designs will have to successfully address cost, performance, and cultural and legal factors.

Nanotechnology is concerned with the world of invisible miniscule particles that are dominated by forces of physics and chemistry that cannot be applied at the macro or human-scale level. These particles have come to be defined by some as nanomaterials, and these materials possess unusual properties not present in traditional and/or ordinary materials. The new technology allows the engineering of matter by systems and/or processes that effectively deal with atoms. Interestingly, the classic laws of science are different at the nanoscale. Nanoparticles possess large surface areas and essentially no inner mass; i.e., their surface-to-mass ratio is extremely high. This new "science" is based on the knowledge that particles in the nanometer range, and nanostructures or nanomachines that are developed from these nanoparticles, possess special properties and exhibit unique behavior; and can significantly impact physical, chemical, electrical, biological, mechanical, etc., functional qualities. These new characteristics can be harnessed and exploited by applied scientists to engineer "Industrial Revolution II" processes. Present-day and future applications include chemical products, e.g., parts, plastics, specialty metals, powders; computer chips and computer systems; pollution prevention areas that can include energy conservation, environmental control, and health/safety issues, plus addressing crime and terrorism concerns. In effect, the sky's the limit regarding efforts in this area. As far as the environment is concerned, this new technology can terminate pollution as it is known today. By ingeniously controlling the building

of small and large products via a process of precise controlled building, there is no waste since environmental engineering is built in by design. However, activists have begun to organize against the science, calling for a moratorium on nanotechnology products until the social (health) and environmental risks are better understood. At the time of the preparation of this manuscript, justifiably (in the opinion of the author) no nanoregulations had been put into place. One of the main obstacles to achieving this goal will be to control, reduce, and ultimately eliminate environmental and related problems, or problems that may develop through the misuse or use of nanotechnology.

The impact of all of the above on both the industry and the consumer will be significant. All have heard the expression "I've met the enemy . . . and we're it." After all, it is the consumer who can and will ultimately have the final say. Once the consumer refuses to buy and/or accept products and/or services that damage the environment, that industry is either out of business or must change its operations to environmentally acceptable alternatives. With the new standards, customers (in addition to other organizations) of certified organizations will be assured that the products or services they purchase have been produced in accordance with universally accepted standards of environmental management. Organizational claims, which today can be misleading or erroneous, will, under the standards, be backed up by comprehensive and detailed environmental management systems that must withstand the scrutiny of intense audits.

Can all of the above be achieved in the near future? If it can, then society need only key its environmental efforts on educating the consumer.

AIR-POLLUTION MANAGEMENT OF STATIONARY SOURCES

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INTRODUCTION

Air pollutants may be classified into two broad categories: (1) natural and (2) human-made. Natural sources of air pollutants include:

- Windblown dust
- Volcanic ash and gases

- Ozone from lightning and the ozone layer
- Esters and terpenes from vegetation
- Smoke, gases, and fly ash from forest fires
- Pollens and other aeroallergens
- Gases and odors from biologic activities
- Natural radioactivity

Such sources constitute background pollution and that portion of the pollution problem over which control activities can have little, if any, effect.

Human-made sources cover a wide spectrum of chemical and physical activities and are the major contributors to urban air pollution. Air pollutants in the United States pour out from over 10 million vehicles, the refuse of approximately 300 million people, the generation of billions of kilowatts of electricity, and the production of innumerable products demanded by everyday living. Hundreds of millions of tons of air pollutants are generated annually in the United States alone. The six pollutants identified in the Clean Air Act are shown in Table 22-14. Annual emission statistics for these six pollutants are considered major indicators of the U.S. air quality. During the 1970 to 2003 period, the total emissions of the six pollutants declined by 51 percent, while at the same time the gross domestic product increased by 176 percent, the population by 39 percent, and energy consumption by 45 percent. Total emissions in the United States are summarized by source category for the year 1998 in Table 22-15.

Air pollutants may also be classified as to the origin and state of matter:

1. Origin

- a. Primary. Emitted to the atmosphere from a process
- b. Secondary. Formed in the atmosphere as a result of a chemical reaction

2. State of matter

- a. Gaseous. True gases such as sulfur dioxide, nitrogen oxide, ozone, carbon monoxide, etc.; vapors such as gasoline, paint solvent, dry cleaning agents, etc.
- b. Particulate. Finely divided solids or liquids; solids such as dust, fumes, and smokes; and liquids such as droplets, mists, fogs, and aerosols

Gaseous Pollutants Gaseous pollutants may be classified as inorganic or organic. Inorganic pollutants consist of:

1. *Sulfur gases.* Sulfur dioxide, sulfur trioxide, hydrogen sulfide
2. *Oxides of carbon.* Carbon monoxide, carbon dioxide

TABLE 22-14 Six Principal Pollutants' Annual Emission Statistics, Millions of Tons per Year

	1970	1975	1980	1985 ^a	1990	1995	2000 ^a	2002	2003 ^b
Carbon monoxide (CO)	197.3	184.0	177.8	169.6	143.6	120.0	102.4	96.4	93.7
Nitrogen oxides (NO _x) ^c	26.9	26.4	27.1	25.8	25.1	24.7	22.3	20.8	20.5
Particulate matter (PM) ^d									
PM ₁₀	12.2 ^e	7.0	6.2	3.6	3.2	3.1	2.3	2.4	2.3
PM _{2.5} ^e	NA	NA	NA	NA	2.3	2.2	1.8	1.8	1.8
Sulfur dioxide (SO ₂)	31.2	28.0	25.9	23.3	23.1	18.6	16.3	15.3	15.8
Volatile organic compounds (VOC)	33.7	30.2	30.1	26.9	23.1	21.6	16.9	15.8	15.4
Lead ^f	0.221	0.16	0.074	0.022	0.005	0.004	0.003	0.003	0.003
Totals ^g	301.5	275.8	267.2	249.2	218.1	188.0	160.2	150.2	147.7

NOTES:

^a In 1985 and 1996 EPA refined its methods for estimating emissions. Between 1970 and 1975, EPA revised its methods for estimating particulate matter emissions.

^b The estimates for 2003 are preliminary.

^c NO_x estimates prior to 1990 include emissions from fires. Fires would represent a small percentage of the NO_x emissions.

^d PM estimates do not include condensable PM, or the majority of PM_{2.5} that is formed in the atmosphere from "precursor" gases such as SO₂ and NO_x.

^e EPA has not estimated PM_{2.5} emissions prior to 1990.

^f The 1999 estimate for lead is used to represent 2000 and 2003 because lead estimates do not exist for these years.

^g PM_{2.5} emissions are not added when calculating the total because they are included in the PM₁₀ estimate.

SOURCE: EPA 2003 Emissions Report (www.epa.gov/air/urbanair/6poll.html).

3. *Nitrogen gases.* Nitrous oxide, nitric oxide, nitrogen dioxide, other nitrous oxides

4. *Halogens, halides.* Hydrogen fluoride, hydrogen chloride, chlorine, fluorine, silicon tetrafluoride

5. *Photochemical products.* Ozone, oxidants

6. *Cyanides.* Hydrogen cyanide

7. *Ammonium compounds.* Ammonia

8. *Chlorofluorocarbons.* 1,1,1-trichloro-2,2,2-trifluoroethane; trichlorofluoromethane, dichlorodifluoromethane; chlorodifluoromethane; 1,2-dichloro-1,1,2,2-tetrafluoroethane; chloropentafluoroethane

Organic pollutants consist of:

1. Hydrocarbons

a. Paraffins. Methane, ethane, octane

b. Acetylene

c. Olefins. Ethylene, butadiene

d. Aromatics. Benzene, toluene, benzpyrene, xylene, styrene

2. Aliphatic oxygenated compounds

a. Aldehydes. Formaldehyde

b. Ketones. Acetone, methyl ethyl ketone

c. Organic acids

d. Alcohols. Methanol, ethanol, isopropanol

e. Organic halides. Cyanogen chloride bromobenzyl cyanide

f. Organic sulfides. Dimethyl sulfide

g. Organic hydroperoxides. Peroxyacetyl nitrite or nitrate (PAN)

The most common gaseous pollutants and their major sources and significance are presented in Table 22-16.

Particulate Pollutants Particulates may be defined as solid or liquid matter whose effective diameter is larger than a molecule but smaller than approximately 100 μm . Particulates dispersed in a gaseous medium are collectively termed an *aerosol*. The terms *smoke*, *fog*, *haze*, and *dust* are commonly used to describe particular types of aerosols, depending on the size, shape, and characteristic behavior of the dispersed particles. Aerosols are rather difficult to classify on a scientific basis in terms of their fundamental properties such as settling rate under the influence of external forces, optical activity, ability to absorb an electrical charge, particle size and structure, surface-to-volume ratio, reaction activity, physiological action, and so on. In general, particle size and settling rate have been the most characteristic properties for many purposes. On the other hand, particles on the order of 1 μm or less settle so slowly that, for all practical purposes, they are regarded as permanent suspensions. Despite possible advantages of scientific classification schemes, the use of popular descriptive terms such as smoke, dust, and mist, which are essentially based on the mode of formation, appears to be a satisfactory and convenient method of classification. In addition, this approach is so well established and understood that it undoubtedly would be difficult to change.

Dust is typically formed by the pulverization or mechanical disintegration of solid matter into particles of smaller size by processes such as grinding, crushing, and drilling. Particle sizes of dust range from a lower limit of about 1 mm up to about 100 or 200 μm and larger. Dust particles are usually irregular in shape, and particle size refers to some average dimension for any given particle. Common examples include

TABLE 22-15 1998 National Point and Area Emissions by Source Category and Pollutant, 1000 Short Tons

Source category	CO			NO _x			VOC			SO ₂		
	Point	Area	Total	Point	Area	Total	Point	Area	Total	Point	Area	Total
Fuel comb. elec. util.	413	4	417	6,095	8	6,103	54	0	54	13,217	0	13,217
Fuel comb. industrial	908	206	1,114	2,142	827	2,969	144	17	161	2,075	820	2,895
Fuel comb. other	83	3,760	3,843	142	975	1,117	12	666	678	191	418	609
Chemical & allied product mfg.	1,129	0	1,129	152	0	152	312	84	396	299	0	299
Metals processing	1,494	1	1,495	88	0	88	75	0	76	444	0	444
Petroleum & related industries	365	3	368	122	16	138	223	273	496	344	0	345
Other industrial processes	629	3	632	402	6	408	388	62	450	366	3	370
Solvent utilization	2	0	2	2	0	2	639	4,640	5,278	1	0	1
Storage & transport	80	0	80	7	0	7	298	1,025	1,324	3	0	3
Waste disposal & recycling	31	1,123	1,154	38	59	97	27	406	433	19	24	42
Highway vehicles	0	50,386	50,386	0	7,765	7,765	0	5,325	5,325	0	326	326
Off-highway	0	19,914	19,914	0	5,280	5,280	0	2,461	2,461	0	1,084	1,084
Natural sources	0	0	0	0	0	0	0	14	14	0	0	0
Miscellaneous	0	8,920	8,920	1	327	328	2	770	772	0	12	12
Total	5,134	84,319	89,454	9,190	15,264	24,454	2,174	15,743	17,917	16,960	2,688	19,647

SOURCE: National Air Pollution Trends: 1900–1998, EPA 454/R-00-002, March 2000 (www.epa.gov/ttn/chieftrends/index.html).

TABLE 22-16 Typical Gaseous Pollutants and Their Principal Sources and Significance

Air pollutants	From manufacturing sources such as these	In typical industries	Cause these damaging effects
Alcohols	Used as a solvent in coatings	Surface coatings, printing	Sensory and respiratory irritation
Aldehydes	Results from thermal decomposition of fats, oil, or glycerol; used in some glues and binders	Food processing, light process, wood furniture, chip board	An irritating odor, suffocating, pungent, choking; not immediately dangerous to life; can become intolerable in a very short time
Ammonia	Used in refrigeration, chemical processes such as dye making, explosives, lacquer, fertilizer	Textiles, chemicals	Corrosive to copper, brass, aluminum, and zinc; high concentration producing chemical burns on wet skin
Aromatics	Used as a solvent in coatings	Surface coatings, printing	Irritation of mucous membranes, narcotic effects; some are carcinogens
Arsine	Any soldering, pickling, etching, or plating process involving metals or acids containing arsenic	Chemical processing, smelting	Breakdown of red cells in blood
Carbon dioxide	Fuel combustion; calcining	Industrial boilers, cement and lime production	Greenhouse gas
Carbon monoxide	Fuming of metallic oxides, gas-operated fork trucks	Primary metals; steel and aluminum	Reduction in oxygen-carrying capacity of blood
Chlorine	Manufactured by electrolysis, bleaching cotton and flour; by-product of organic chemicals	Textiles, chemicals	Attacks entire respiratory tract and mucous membrane of eye
Chlorofluorocarbons	Used in refrigeration and production of porous foams; degreasing agent	Refrigeration, plastic foam production, metal fabricating	Attacks stratospheric ozone layer; greenhouse gas
Hydrochloric acid	Combustion of coal or wastes containing chlorinated plastics	Coal-fired boilers, incinerators	Irritant to eyes and respiratory system
Hydrogen cyanide	From metal plating, blast furnaces, dyestuff works	Metal fabricating, primary metals, textiles	Capable of affecting nerve cells
Hydrogen fluoride	Catalyst in some petroleum refining, etching glass, silicate extraction; by-product in electrolytic production of aluminum	Petroleum, primary metals, aluminum	Strong irritant and corrosive action on all body tissue; damage to citrus plants, effect on teeth and bones of cattle from eating plants
Hydrogen sulfide	Refinery gases, crude oil, sulfur recovery, various chemical industries using sulfur compounds	Petroleum and chemicals; Kraft pulping process	Foul odor of rotten eggs; irritating to eyes and respiratory tract; darkening exterior paint
Ketones	Used as a solvent in coatings	Surface coatings, printing	Sensory and respiratory irritation
Lead	Incineration, smelting and casting, transportation	Copper and lead smelting, MSWs	Neurological impairments; kidney, liver, and heart damage.
Nitrogen oxides	High-temperature combustion: metal cleaning, fertilizer, explosives, nitric acid; carbon-arc combustion; manufacture of H_2SO_4	Metal fabrication, heavy chemicals	Irritating gas affecting lungs; vegetation damage
Odors	Slaughtering and rendering animals, tanning animal hides, canning, smoking meats, roasting coffee, brewing beer, processing toiletries	Food processing, allied industries	Objectionable odors
Ozone	Reaction product of VOC and nitrogen oxides	Not produced directly	Irritant to eyes and respiratory system
Phosgenes	Thermal decomposition of chlorinated hydrocarbons, degreasing, manufacture of dyestuffs, pharmaceuticals, organic chemicals	Metal fabrication, heavy chemicals	Damage capable of leading to pulmonary edema, often delayed
Sulfur dioxide	Fuel combustion (coal, oil), smelting and casting, manufacture of paper by sulfite process	Primary metals (ferrous and nonferrous); pulp and paper	Sensory and respiratory irritation, vegetation damage, corrosion, possible adverse effect on health

fly ash, rock dusts, and ordinary flour. *Smoke* implies a certain degree of optical density and is typically derived from the burning of organic materials such as wood, coal, and tobacco. Smoke particles are very fine, ranging in size from less than $0.01\text{ }\mu\text{m}$ up to $1\text{ }\mu\text{m}$. They are usually spherical in shape if of liquid or tarry composition and irregular in shape if of solid composition. Owing to their very fine particle size, smokes can remain in suspension for long periods of time and exhibit lively brownian motion.

Fumes are typically formed by processes such as sublimation, condensation, or combustion, generally at relatively high temperatures. They range in particle size from less than $0.1\text{ }\mu\text{m}$ to $1\text{ }\mu\text{m}$. Similar to smokes, they settle very slowly and exhibit strong brownian motion.

Mists or fogs are typically formed either by the condensation of water or other vapors on suitable nuclei, giving a suspension of small liquid droplets, or by the atomization of liquids. Particle sizes of natural fogs and mists lie between 2 and $200\text{ }\mu\text{m}$. Droplets larger than $200\text{ }\mu\text{m}$ are more properly classified as drizzle or rain. Many of the important properties of aerosols that depend on particle size are presented in Sec. 17, Fig. 17-34.

When a liquid or solid substance is emitted to the air as particulate matter, its properties and effects may be changed. As a substance is broken up into smaller and smaller particles, more of its surface area is exposed to the air. Under these circumstances, the substance, whatever its chemical composition, tends to combine physically or chemically

with other particles or gases in the atmosphere. The resulting combinations are frequently unpredictable. Very small aerosol particles (from 0.001 to 0.1 μm) can act as condensation nuclei to facilitate the condensation of water vapor, thus promoting the formation of fog and ground mist. Particles less than 2 or 3 μm in size (about half by weight of the particles suspended in urban air) can penetrate the mucous membrane and attract and convey harmful chemicals such as sulfur dioxide. In order to address the special concerns related to the effects of very fine, inhalable particulates, EPA now has ambient standards in place for both PM_{10} and $\text{PM}_{2.5}$.

By virtue of the increased surface area of the small aerosol particles and as a result of the adsorption of gas molecules or other such properties that are able to facilitate chemical reactions, aerosols tend to exhibit greatly enhanced surface activity. Many substances that oxidize slowly in their massive state will oxidize extremely fast or possibly even explode when dispersed as fine particles in the air. Dust explosions, for example, are often caused by the unstable burning or oxidation of combustible particles, brought about by their relatively large specific surfaces. Adsorption and catalytic phenomena can also be extremely important in analyzing and understanding the problems of particulate pollution. The conversion of sulfur dioxide to corrosive sulfuric acid assisted by the catalytic action of iron oxide particles, for example, demonstrates the catalytic nature of certain types of particles in the atmosphere. Finally, aerosols can absorb radiant energy and rapidly conduct heat to the surrounding gases of the atmosphere. These are gases that ordinarily would be incapable of absorbing radiant energy by themselves. As a result, the air in contact with the aerosols can become much warmer.

Estimating Emissions from Sources Knowledge of the types and rates of emissions is fundamental to evaluation of any air pollution problem. A comprehensive material balance on the process can often assist in this assessment. Estimates of the rates at which pollutants are discharged from various processes can also be obtained by utilizing published emission factors. See *Compilation of Air Pollution Emission Factors (AP-42)*, 4th ed., U.S. EPA, Research Triangle Park, North Carolina, September, 1985, with all succeeding supplements and the EPA Technology Transfer Network's CHIEF. The emission factor is a statistical average of the rate at which pollutants are emitted from the burning or processing of a given quantity of material or on the basis of some other meaningful parameter. Emission factors are affected by the techniques employed in the processing, handling, or burning operations, by the quality of the material used, and by the efficiency of the air-pollution control. Since the combination of these factors tends to be unique to a source, emission factors appropriate for one source may not be satisfactory for another source. Hence, care and good judgment must be exercised in identifying appropriate emission factors. If appropriate emission factors cannot be found or if air-pollution control equipment is to be designed, specific source sampling should be conducted. The major industrial sources of pollutants, the air contaminants emitted, and typical control techniques are summarized in Table 22-17.

Effects of Air Pollutants

Materials The damage that air pollutants can do to some materials is well known: ozone in photochemical smog cracks rubber, weakens fabrics, and fades dyes; hydrogen sulfide tarnishes silver; smoke dirties laundry; acid aerosols ruin nylon hose. Among the most important effects are discoloration, corrosion, the soiling of goods, and impairment of visibility.

1. **Discoloration.** Many air pollutants accumulate on and discolor buildings. Not only does sooty material blacken buildings, but it can accumulate and become encrusted. This can hide lines and decorations and thereby disfigure structures and reduce their aesthetic appeal. Another common effect is the discoloration of paint by certain acid gases. A good example is the blackening of white paint with a lead base by hydrogen sulfide.

2. **Corrosion.** A more serious effect and one of great economic importance is the corrosive action of acid gases on building materials. Such acids can cause stone surfaces to blister and peel; mortar can be reduced to powder. Metals are also damaged by the corrosive action of some pollutants. Another common effect is the deterioration of tires

and other rubber goods. Cracking and apparent "drying" occur when these goods are exposed to ozone and other oxidants.

3. **Soiling of goods.** Clothes, real estate, automobiles, and household goods can easily be soiled by air contaminants, and the more frequent cleaning thus required can become expensive. Also, more frequent cleaning often leads to a shorter life span for materials and to the need to purchase goods more often.

4. **Impairment of visibility.** The impairment of atmospheric visibility (i.e., decreased visual range through a polluted atmosphere) is caused by the scattering of sunlight by particles suspended in the air. It is not a result of sunlight being obscured by materials in the air. Since light scattering, and not obscuration, is the main cause of the reduction in visibility, reduced visibility due to the presence of air pollutants occurs primarily on bright days. On cloudy days or at night there may be no noticeable effect, although the same particulate concentration may exist at these times as on sunny days. Reduction in visibility creates several problems. The most significant are the adverse effects on aircraft, highway, and harbor operations. Reduced visibility can reduce quality of life and also cause adverse aesthetic impressions that can seriously affect tourism and restrict the growth and development of any area. Extreme conditions such as dust storms or sandstorms can actually cause physical damage by themselves.

Vegetation Vegetation is more sensitive than animals to many air contaminants, and methods have been developed that use plant response to measure and identify contaminants. The effects of air pollution on vegetation can appear as death, stunted growth, reduced crop yield, and degradation of color. It is interesting to note that in some cases of color damage such as the silverying of leafy vegetables by oxidants, the plant may still be used as food without any danger to the consumer; however, the consumer usually will not buy such vegetables on aesthetic grounds, so the grower still sustains a loss. Among the pollutants that can harm plants are sulfur dioxide, hydrogen fluoride, and ethylene. Plant damage caused by constituents of photochemical smog has been studied extensively. Damage has been attributed to ozone and peroxyacetyl nitrates, higher aldehydes, and products of the reaction of ozone with olefins. However, none of the cases precisely duplicates all features of the damage observed in the field, and the question remains open to some debate and further study.

Animals Considerable work continues to be performed on the effects of pollutants on animals, including, for a few species, experiments involving mixed pollutants and mixed gas-aerosol systems. In general, such work has shown that mixed pollutants may act in several different ways. They may produce an effect that is additive, amounting to the sum of the effects of each contaminant acting alone; they may produce an effect that is greater than the simply additive (synergistic) or less than the simply additive (antagonistic); or they may produce an effect that differs in some other way from the simply additive.

The mechanism by which an animal can become poisoned in many instances is completely different from that by which humans are affected. As in humans, inhalation is an important route of entry in acute air-pollution exposures such as the Meuse Valley and Donora incidents (see the paragraph on **humans** below). However, probably the most common exposure for herbivorous animals grazing within a zone of pollution will be the ingestion of feed contaminated by air pollutants. In this case, inhalation is of secondary importance.

Air pollutants that present a hazard to livestock, therefore, are those that are taken up by vegetation or deposited on the plants. Only a few pollutants have been observed to cause harm to animals. These include arsenic, fluorides, lead, mercury, and molybdenum.

Humans There seems to be little question that, during many of the more serious episodes, air pollution can have a significant effect on health, especially upon the young, elderly, or people already in ill health. Hundreds of excess deaths have been attributed to incidents in London in 1952, 1956, 1957, and 1962; in Donora, Pennsylvania, in 1948; in New York City in 1953, 1963, and 1966; and Bhopal, India in 1989. Many of the people affected were in failing health, and they were generally suffering from lung conditions. In addition, hundreds of thousands of persons have suffered from serious discomfort and inconvenience, including eye irritation and chest pains, during these and other such incidents. Such acute problems are actually the lesser of the health problems. There is considerable evidence of a chronic threat to

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TABLE 22-17 Control Techniques Applicable to Unit Processes at Important Emission Sources

Industry	Process of operation	Air contaminants emitted	Control techniques
Aluminum reduction plants	Materials handling: Buckets and belt Conveyor or pneumatic conveyor	Particulates (dust)	Exhaust systems and baghouse
	Anode and cathode electrode preparation: Cathode (baking) Anode (grinding and blending) Baking	Hydrocarbon emissions from binder Particulates (dust) Particulates (dust), CO, SO ₂ , hydrocarbons, and fluorides	Exhaust systems and mechanical collectors High-efficiency cyclone, electrostatic precipitators, scrubbers, catalytic combustion or incinerators, flares, baghouse
	Pot charging	Particulates (dust), CO, HF, SO ₂ , CF ₄ , and hydrocarbons	High-efficiency cyclone, baghouse, spray towers, floating-bed scrubber, electrostatic precipitators, chemisorption, wet electrostatic precipitators
	Metal casting	Cl ₂ , HCl, CO, and particulates (dust)	Exhaust systems and scrubbers
Asphalt plants	Materials handling, storage and classifiers: elevators, chutes, vibrating screens Drying: rotary oil- or gas-fired	Particulates (dust) Particulates, SO ₂ , NO _x , VOC, CO, and smoke	Wetting; exhaust systems with a scrubber or baghouse Proper combustion controls, fuel-oil preheating where required; local exhaust system, cyclone and a scrubber or baghouse
	Truck traffic	Dust	Paving, wetting down truck routes
Cement plants	Quarrying: primary crusher, secondary crusher, conveying, storage Dry processes: materials handling, air separator (hot-air furnace) Grinding Pneumatic, conveying and storage Wet process: materials handling, grinding, storage Kiln operations: rotary kiln	Particulates (dust) Particulates (dust) Particulates (dust) Wet materials, no dust Particulates (dust), CO, SO _x , NO _x , hydrocarbons, aldehydes, ketones Particulates (dust)	Wetting; exhaust systems with fabric filters Local exhaust system with mechanical collectors and baghouse Local exhaust system with cyclones and baghouse Electrostatic precipitators, acoustic horns and baghouses, scrubber
	Clinker cooling: materials handling	Particulates (dust)	Local exhaust system and electrostatic precipitators or fabric filters
	Grinding and packing, air separator, grinding, pneumatic conveying, materials handling, packaging	Particulates (dust)	Local exhaust system and fabric filters
Coal-preparation plants	Materials handling: conveyors, elevators, chutes Sizing: crushing, screening, classifying Dedusting	Particulates (dust) Particulates (dust) Particulates (dust)	Local exhaust system and cyclones Local exhaust system and cyclones Local exhaust system, cyclone precleaners, and baghouse
	Storing coal in piles Refuse piles	Blowing particulates (dust) H ₂ S, particulates, and smoke from burning storage piles	Wetting, plastic-spray covering Digging out fire, pumping water onto fire area, blanketing with incombustible material
	Coal drying: rotary, screen, suspension, fluid-bed, cascade	Dust, smoke, particulates, sulfur oxides, H ₂ S	Exhaust systems with cyclones and fabric filters
Coke plants	By-product-ovens charging	Smoke, particulates (dust)	Pipe-line charging, careful charging techniques, portable hooding and scrubber or baghouses
	Pushing	Smoke, particulates (dust), SO ₂	Minimizing green-coke pushing, scrubbers and baghouses
	Quenching	Smoke, particulates (dust and mists), phenols, and ammonia	Baffles and spray tower
	By-product processing	CO, H ₂ S, methane, ammonia, H ₂ , phenols, hydrogen cyanide, N ₂ , benzene, xylene, etc.	Electrostatic precipitator, scrubber, flaring
	Material storage (coal and coke)	Particulates (dust)	Wetting, plastic spray, fire-prevention techniques
Electric utilities (industrial and commercial facilities)	Coal-fired boilers	Particulates SO _x , NO _x	Baghouses Precipitators, SCR, and SNCR Acid gas scrubbers
Fertilizer industry (chemical)	Phosphate fertilizers: crushing, grinding, and calcining Hydrolysis of P ₂ O ₅ Acidulation and curing Granulation Ammoniation	Particulates (dust) PH ₃ , P ₂ O ₅ PO ₄ mist HF, SiF ₄ Particulates (dust)/(product recovery) NH ₃ , NH ₄ Cl, SiF ₄ , HF	Exhaust system, scrubber, cyclone, baghouse Scrubbers, flare Scrubbers Exhaust system, scrubber, or baghouse Cyclone, electrostatic precipitator, baghouse, high-energy scrubber

TABLE 22-17 Control Techniques Applicable to Unit Processes at Important Emission Sources (Continued)

Industry	Process of operation	Air contaminants emitted	Control techniques
Fertilizer industry (chemical)	Nitric acid acidulation Superphosphate storage and shipping Ammonium nitrate reactor Prilling tower	NO _x , gaseous fluoride compounds Particulates (dust) NH ₃ , NO _x NH ₄ , NO ₃	Scrubber, addition of urea Exhaust system, cyclone or baghouse Scrubber Proper operation control, scrubbers
Foundries: Iron	Melting (cupola): Charging Melting Pouring Bottom drop	Smoke and particulates Smoke and particulates, fume Oil, mist, CO Smoke and particulates	Closed top with exhaust system, CO afterburner, gas-cooling device and scrubbers, baghouse or electrostatic precipitator, wetting to extinguish fire
Brass and bronze	Melting: Charging Melting Pouring	Smoke particulates, oil mist Zinc oxide fume, particulates, smoke Zinc oxide fume, lead oxide fume	Low-zinc-content red brass: use of good combustion controls and slag cover; high-zinc-content brass: use of good combustion controls, local exhaust system, and baghouse or scrubber
Aluminum	Melting: charging, melting, pouring	Smoke and particulates	Charging clean material (no paint or grease); proper operation required; no air-pollution-control equipment if no fluxes are used and degassing is not required; dirty charge requiring exhaust system with scrubbers and baghouses
Zinc	Melting: Charging Melting Pouring Sand-handling shakeout Magnetic pulley, conveyors, and elevators, rotary cooler, screening, crusher-mixer Coke-making ovens	Smoke and particulates Zinc oxide fume Oil mist and hydrocarbons from diecasting machines Particulates (dust), smoke, organic vapors Particulates (dust) Organic acids, aldehydes, smoke, hydrocarbons	Exhaust system with cyclone and baghouse, charging clean material (no paint or grease) Careful skimming of dross Use of low-smoking die-casting lubricants Exhaust system, cyclone, and baghouse Use of binders that will allow ovens to operate at less than 204°C (400°F) or exhaust systems and afterburners
Galvanizing operations	Hot-dip-galvanizing-tank kettle: dipping material into the molten zinc; dusting flux onto the surface of the molten zinc	Fumes, particulates (liquid), vapors: NH ₄ Cl, ZnO, ZnCl ₂ , Zn, NH ₃ , oil, and carbon	Close-fitting hoods with high in-draft velocities (in some cases, the hood may not be able to be close to the kettle, so the in-draft velocity must be very high), baghouses, electrostatic precipitators
Kraft pulp mills	Digesters: batch and continuous Multiple-effect evaporators Recovery furnace Weak and strong black-liquor oxidation Smelt tanks Lime kiln	Mercaptans, methanol (odors) H ₂ S, other odors H ₂ S, mercaptans, organic sulfides, and disulfides H ₂ S Particulates (mist or dust) Particulates (dust), H ₂ S	Condensers and use of lime kiln, boiler, or furnaces as afterburners Caustic scrubbing and thermal oxidation of noncondensables Proper combustion controls for fluctuating load and unrestricted primary and secondary air flow to furnace and dry-bottom electrostatic precipitator; noncontact evaporator Packed tower and cyclone Demisters, venturi, packed tower, or impingement-type scrubbers Venturi scrubbers
Municipal and industrial incinerators	Single-chamber incinerators Multiple-chamber incinerators (retort, inline): Flue-fed Wood waste Municipal incinerators (50 tons and up per day):	Particulates, smoke, volatiles, CO, SO _x , ammonia, organic acids, aldehydes, NO _x , dioxins, hydrocarbons, odors, HCl, furans Particulates, smoke, and combustion contaminants Particulates, smoke, and combustion contaminants Particulates, smoke, and combustion contaminants Particulates, smoke, volatiles, CO, ammonia, organic acids, aldehydes, NO _x , furans, hydrocarbons, SO _x ,	Afterburner, combustion controls Operating at rated capacity, using auxiliary fuel as specified, and good maintenance, including timely cleanout of ash Use of charging gates and automatic controls for draft; afterburner Continuous-feed systems; operation at design load and excess air; cyclones Preparation of materials, including weighing, grinding, shredding; control of tipping area, furnace design with

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TABLE 22-17 Control Techniques Applicable to Unit Processes at Important Emission Sources (Continued)

Industry	Process of operation	Air contaminants emitted	Control techniques
Municipal and industrial incinerators	Municipal incinerators (50 tons and up per day): Pathological incinerators Industrial waste	hydrogen chloride, dioxins and odors Odors, hydrocarbons, HCl, dioxins, furans Particulates, smoke, and combustion contaminants	proper automatic controls; proper start-up techniques; maintenance of design operating temperatures; use of electrostatic precipitators, scrubbers, and baghouses; proper ash cleanout Proper charging, acid gas scrubber, baghouse Modified fuel feed, auxiliary fuel and dryer systems, cyclones, scrubbers
Nonferrous smelters, primary: Copper	Roasting Reverberatory furnace Converters: charging, slag skim, pouring, air or oxygen blow	SO ₂ , particulates, fume Smoke, particulates, metal oxide fumes, SO ₂ Smoke, fume, SO ₂	Exhaust system, settling chambers, cyclones or scrubbers and electrostatic precipitators for dust and fumes and sulfuric acid plant for SO ₂ Exhaust system, settling chambers, cyclones or scrubbers and electrostatic precipitators for dust and fumes and sulfuric acid plant for SO ₂ Exhaust system, settling chambers, cyclones or scrubbers and electrostatic precipitators for dust and fumes and sulfuric acid plant for SO ₂
Lead	Sintering Blast furnace Dross reverberatory furnace Refining kettles	SO ₂ , particulates, smoke SO ₂ , CO, particulates, lead oxide, zinc oxide SO ₂ , particulates, fume SO ₂ , particulates	Exhaust system, cyclones and baghouse or precipitators for dust and fumes, sulfuric acid plant for SO ₂ Exhaust system, settling chambers, afterburner and cooling device, cyclone, and baghouse Exhaust system, settling chambers, cyclone and cooling device, baghouse Local exhaust system, cooling device, baghouse or precipitator
Cadmium	Roasters, slag, fuming furnaces, deleading kilns	Particulates	Local exhaust system, baghouse or precipitator
Zinc	Roasting Sintering Calcining Retorts: electric arc	Particulates (dust) and SO ₂ Particulates (dust) and SO ₂ Zinc oxide fume, particulates, SO ₂ , CO	Exhaust system, humidifier, cyclone, scrubber, electrostatic precipitator, and acid plant Exhaust system, humidifier, electrostatic precipitator, and acid plant Exhaust system, baghouse, scrubber or acid plant
Nonferrous smelters, secondary	Blast furnaces and cupolas-recovery of metal from scrap and slag Reverberatory furnaces Sweat furnaces Wire reclamation and autobody burning	Dust, fumes, particulates, oil vapor, smoke, CO Dust, fumes, particulates, smoke, gaseous fluxing materials Smoke, particulates, fumes Smoke, particulates	Exhaust systems, cooling devices, CO burners and baghouses or precipitators Exhaust systems, and baghouses or precipitators, or venturi scrubbers Precleaning metal and exhaust systems with afterburner and baghouse Scrubbers and afterburners
Paint and varnish manufacturing	Resin manufacturing: closed reaction vessel Varnish: cooking-open or closed vessels Solvent thinning	Acrolein, other aldehydes and fatty acids (odors), phthalic anhydride (sublimed) Ketones, fatty acids, formic acids, acetic acid, glycerine, acrolein, other aldehydes, phenols and terpenes; from tall oils, hydrogen sulfide, alkyl sulfide, butyl mercaptan, and thiofen (odors) Olefins, branched-chain aromatics and ketones (odors), solvents	Exhaust systems with scrubbers and fume burners Exhaust system with scrubbers and fume burners; close-fitting hoods required for open kettles Exhaust system with fume burners
Rendering plants	Feedstock storage and housekeeping Cookers and percolators Grinding	Odors SO ₂ , mercaptans, ammonia, odors Particulates (dust)	Quick processing, washdown of all concrete surfaces, paving of dirt roads, proper sewer maintenance, enclosure, packed towers Exhaust system, condenser, scrubber, or incinerator Exhaust system and scrubber
Roofing plants (asphalt saturators)	Felt or paper saturators: spray section, asphalt tank, wet looper	Asphalt vapors and particulates (liquid)	Exhaust system with high inlet velocity at hoods (3658 m/s [>200 ft/min]) with either scrubbers, baghouses, or two-stage low-voltage electrostatic precipitators

TABLE 22-17 Control Techniques Applicable to Unit Processes at Important Emission Sources (Concluded)

Industry	Process of operation	Air contaminants emitted	Control techniques
Roofing plants (asphalt saturators)	Crushed rock or other minerals handling	Particulates (dust)	Local exhaust system, cyclone or multiple cyclones
Steel mills	Blast furnaces: charging, pouring	CO, fumes, smoke, particulates (dust)	Good maintenance, seal leaks; use of higher ratio of pelletized or sintered ore; CO burned in waste-heat boilers, stoves, or coke ovens; cyclone, scrubber, and baghouse
	Electric steel furnaces: charging, pouring, oxygen blow	Fumes, smoke, particulates (dust), CO	Segregating dirty scrap; proper hooding, baghouses or electrostatic precipitator
	Open-hearth furnaces: oxygen blow, pouring	Fumes, smoke, SO _x , particulates (dust), CO, NO _x	Proper hooding, settling chambers, waste-heat boiler, baghouse, electrostatic precipitator, and wet scrubber
	Basic oxygen furnaces: oxygen blowing	Fumes, smoke, CO, particulates (dust)	Proper hooding (capturing of emissions and dilute CO), scrubbers, or electrostatic precipitator
	Raw material storage Pelletizing Sintering	Particulates (dust) Particulates (dust) Smoke, particulates (dust), SO ₂ , NO _x	Wetting or application of plastic spray Proper hooding, cyclone, baghouse Proper hooding, cyclones, wet scrubbers, baghouse, or precipitator

human health from air pollution. This evidence ranges from the rapid rise of emphysema as a major health problem, through identification of carcinogenic compounds in smog, to statistical evidence that people exposed to polluted atmospheres over extended periods of time suffer from a number of ailments and a reduction in their life span. There may even be a significant indirect exposure to air pollution. As noted above, air pollutants may be deposited onto vegetation or into bodies of water, where they enter the food chain. The impact of such indirect exposures is still under review.

A large body of evidence is available to indicate that atmospheric pollution in varying degrees does affect health adversely. [Amdur, Melvin, and Drinker, "Effect of Inhalation of Sulfur Dioxide by Man," *Lancet*, **2**, 758 (1953); Barton, Corn, Gee, Vassallo, and Thomas, "Response of Healthy Men to Inhaled Low Concentrations of Gas-Aerosol Mixtures," *Arch. Environ. Health*, **18**, 681 (1969); Bates, Bell, Burnham, Hazucha, and Mantha, "Problems in Studies of Human Exposure to Air Pollutants," *Can. Med. Assoc. J.*, **103**, 833 (1970); Ciocco and Thompson, "A Follow-Up of Donora Ten Years After: Methodology and Findings," *Am. J. Public Health*, **51**, 155 (1961); Daly, "Air Pollution and Causes of Death," *Br. J. Soc. Med.*, **13**, 14 (1959); Jaffe, "The Biological Effect of Photochemical Air Pollutants on Man and Animals," *Am. J. Public Health*, New York, **57**, 1269 (1967); New York Academy of Medicine, Committee on Public Health, "Air Pollution and Health," *Bull. N.Y. Acad. Med.*, **42**, 588 (1966); Pemberton and Goldberg, "Air Pollution and Bronchitis," *Br. Med. J.*, London, **2**, 567 (1954); Snell and Luchsinger, "Effect of Sulfur Dioxide on Expiratory Flowrates and Total Respiratory Resistance in Normal Human Subjects," *Arch. Environ. Health*, Chicago, **18**, 693 (1969); Speizer and Frank, "A Comparison of Changes in Pulmonary Flow Resistance in Healthy Volunteers Acutely Exposed to SO₂ by Mouth and by Nose," *J. Ind. Med.*, **23**, 75 (1966); Stocks, "Cancer and Bronchitis Mortality in Relation to Atmospheric Deposit and Smoke," *Br. Med. J. London*, **1**, 74 (1959); Toyama, "Air Pollution and Its Health Effects in Japan," *Arch. Environ. Health*, Chicago, **8**, 153 (1963); U.K. Ministry of Health, "Mortality and Morbidity During London Fog of December 1952," Report on Public Health and Medical Subjects No. 95, London, 1954; U.S. Public Health Service, "Air Pollution in Donora, Pa.: Preliminary Report," Public Health Bull. 306.] It contributes to excesses of death, increased morbidity, and earlier onset of chronic respiratory diseases. There is evidence of a relationship between the intensity of the pollution and the severity of attributable health effects and a consistency of the relationship between these environmental stresses and diseases of the target organs. Air pollutants can both initiate and aggravate a variety of respiratory diseases including asthma. In fact, the clinical presentation of asthma may be considered an air pollution host-defense disorder brought on by specific airborne irritants: pollens, infectious agents, and gaseous and particulate chemicals. The bronchopulmonary response to these foreign irritants is bronchospasm and hypersecretion; the airways are intermittently and reversibly obstructed.

Air-pollutant effects on neural and sensory functions in humans vary widely. Odorous pollutants cause only minor annoyance; yet, if persistent, they can lead to irritation, emotional upset, anorexia, and mental depression. Carbon monoxide can cause death secondary to the depression of the respiratory centers of the central nervous system. Short of death, repeated and prolonged exposure to carbon monoxide can alter sensory protection, temporal perception, and higher mental functions. Lipid-soluble aerosols can enter the body and be absorbed in the lipids of the central nervous system. Once there, their effects may persist long after the initial contact has been removed. Examples of agents of long-term chronic effects are organic phosphate pesticides and aerosols carrying the metals lead, mercury, and cadmium.

The acute toxicological effects of most air contaminants are reasonably well understood, but the effects of exposure to heterogeneous mixtures of gases and particulates at very low concentrations are only beginning to be comprehended. Two general approaches can be used to study the effects of air contaminants on humans: epidemiology, which attempts to associate the effect in large populations with the cause, and laboratory research, which begins with the cause and attempts to determine the effects. Ideally, the two methods should complement each other.

EPA recognizes that scientific studies show a link between inhalable PM (alone and in combination with other pollutants in the air) and a series of health effects. While both coarse and fine particles accumulated in the respiratory system and are associated with numerous adverse health effects, fine particles have been more clearly tied to the most serious health effects. Fine particles have been associated with decreased lung functions, disease, and even premature death. The elderly and children are notable in the sensitive groups that appear to be at greatest risk (JAWMA Special Issues, July and August 2000; PM2000, *Particulate Matter and Health—The Scientific Basis for Regulatory Decision Making*).

Epidemiology, the more costly of the two, requires great care in planning and often suffers from incomplete data and lack of controls. One great advantage, however, is that moral barriers do not limit its application to humans as they do with some kinds of laboratory research. The method is therefore highly useful and has produced considerable information. Laboratory research is less costly than epidemiology, and its results can be checked against controls and verified by experimental repetition.

A SOURCE-CONTROL-PROBLEM STRATEGY

Strategy Control technology is self-defeating if it creates undesirable side effects in meeting objectives. Air pollution control must be considered in terms of regulatory requirements, total technological systems (equipment and processes), and ecological consequences,

such as the problems of treatment and disposal of collected pollutants. It should be noted that the 1990 Clean Air Act Amendments (CAAA) have impacted on the control approach in a significant manner. In particular, the CAAA have placed an increased emphasis on control technology by requiring Best Available Control Technology (BACT) on new major sources and modifications, and by requiring Maximum Achievable Control Technology (MACT) on new and *existing* major sources of Hazardous Air Pollutants (HAPs).

The control strategy for environmental-impact assessment often focuses on five alternatives whose purpose would be the reduction and/or elimination of pollutant emissions:

1. Elimination of the operation entirely or in part
2. Modification of the operation
3. Relocation of the operation
4. Application of appropriate control technology
5. Combinations thereof

In light of the relatively high costs often associated with pollution-control systems, engineers are directing considerable effort toward process modification to eliminate as much of the pollution problem as possible at the source. This includes evaluating alternative manufacturing and production techniques, substituting raw materials, and improving process-control methods. Unfortunately, if there is no alternative, the application of the correct pollution-control equipment is essential. The equipment must be designed to comply with regulatory emission limitations on a continual basis, interruptions being subject to severe penalty depending upon the circumstances. The requirement for design performance on a continual basis places very heavy emphasis on operation and maintenance practices. The escalating costs of energy, labor, and materials can make operation and maintenance considerations even more important than the original capital cost.

Factors in Control-Equipment Selection In order to solve an air-pollution problem, the problem must be defined in detail. A number of factors must be considered prior to selecting a particular piece of air-pollution-control equipment. In general, these factors can be grouped into three categories: environmental, engineering, and economic.

Environmental Factors These include (1) equipment location, (2) available space, (3) ambient conditions, (4) availability of adequate utilities (i.e., power, water, etc.) and ancillary-system facilities (i.e., waste treatment and disposal, etc.), (5) maximum allowable emission (air pollution codes), (6) aesthetic considerations (i.e., visible steam or water-vapor plume, etc.), (7) contributions of the air-pollution-control system to wastewater and land pollution, and (8) contribution of the air-pollution-control system to plant noise levels.

Engineering Factors These include:

1. Contaminant characteristics (e.g., physical and chemical properties, concentration, particulate shape and size distribution [in the case of particulates], chemical reactivity, corrosivity, abrasiveness, and toxicity)
2. Gas-stream characteristics (e.g., volume flow rate, temperature, pressure, humidity, composition, viscosity, density, reactivity, combustibility, corrosivity, and toxicity)
3. Design and performance characteristics of the particular control system (i.e., size and weight, fractional efficiency curves [in the case of particulates]), mass-transfer and/or contaminant-destruction capability (in the case of gases or vapors), pressure drop, reliability, turndown capability, power requirements, utility requirements, temperature limitations, maintenance requirements, operating cycles (including startup and shutdown) and flexibility toward complying with more stringent air-pollution codes.

Economic Factors These include capital cost (equipment, installation, engineering, etc.), operating cost (utilities, maintenance, etc.), emissions fees, and life-cycle cost over the expected equipment lifetime.

Comparing Control-Equipment Alternatives The final choice in equipment selection is usually dictated by the equipment capable of achieving compliance with regulatory codes at the lowest uniform annual cost (amortized capital investment plus operation and maintenance costs). To compare specific control-equipment alternatives, knowledge of the particular application and site is essential. A preliminary screening, however, may be performed by reviewing the advantages and disadvantages of each type of air-pollution-control equipment. General advantages and disadvantages of the most popular types of air-pollution

TABLE 22-18 Advantages and Disadvantages of Cyclone Collectors

Advantages

1. Low cost of construction
2. Relatively simple equipment with few maintenance problems
3. Relatively low operating pressure drops (for degree of particulate removal obtained) in the range of approximately 2- to 6-in water column
4. Temperature and pressure limitations imposed only by the materials of construction used
5. Collected material recovered dry for subsequent processing or disposal
6. Relatively small space requirements

Disadvantages

1. Relatively low overall particulate collection efficiencies, especially on particulates below 10 μm in size
2. Inability to handle tacky materials

equipment for gases and particulates are presented in Tables 22-18 through 22-30. Other activities that must be accomplished before final compliance is achieved are presented in Table 22-31.

In addition to using annualized cost comparisons in evaluating an air-pollution-control (APC) equipment installation, the impact of the 1990 Clean Air Act Amendments (CAAA) and resulting regulations also must be included in the evaluation. The CAAA prescribes specific pollution-control requirements for particular industries and locations. As an example, the CAAA requires that any major stationary source or major modification plan that is subject to Prevention of Significant Deterioration (PSD) requirements must undergo a Best Available Control Technology (BACT) analysis. The BACT analysis is done on a case-by-case basis. The process involves evaluating the possible types of air-pollution-control equipment that could be used for technology and energy and in terms of the environment and economy. The analysis uses a top-down approach that lists all available control technologies in descending order of effectiveness. The most stringent technology is chosen unless it can be demonstrated that, due to technical, energy, environmental, or economic considerations, this type of APC technology is not feasible.

The 1990 CAAA introduced a new level of control for hazardous (toxic) air pollutants (HAPs). As a result, EPA has identified 188 HAPs for regulation. Rather than rely upon ambient air quality standards to set acceptable exposures to HAPs, the CAAA requires that EPA promulgate through the end of the decade Maximum Achievable Control Technology (MACT) standards for controlling HAPs emitted from specified industries. These standards are based on the level of control established by the best performing 12 percent of industries in each of the categories identified by EPA.

TABLE 22-19 Advantages and Disadvantages of Wet Scrubbers

Advantages

1. No secondary dust sources
2. Relatively small space requirements
3. Ability to collect gases as well as particulates (especially "sticky" ones)
4. Ability to handle high-temperature, high-humidity gas streams
5. Capital cost low (if wastewater treatment system not required)
6. For some processes, gas stream already at high pressures (so pressure-drop considerations may not be significant)
7. Ability to achieve high collection efficiencies on fine particulates (however, at the expense of pressure drop)
8. Ability to handle gas streams containing flammable or explosive materials

Disadvantages

1. Possible creation of water-disposal problem
2. Product collected wet
3. Corrosion problems more severe than with dry systems
4. Steam plume opacity and/or droplet entrainment possibly objectionable
5. Pressure-drop and horsepower requirements possibly high
6. Solids buildup at the wet-dry interface possibly a problem
7. Relatively high maintenance costs
8. Must be protected from freezing
9. Low exit gas temperature reduces exhaust plume dispersion
10. Moist exhaust gas precludes use of most additional controls

TABLE 22-20 Advantages and Disadvantages of Dry Scrubbers*Advantages*

1. No wet sludge to dispose of
2. Relatively small space requirements
3. Ability to collect acid gases at high efficiencies
4. Ability to handle high-temperature gas streams
5. Dry exhaust allows addition of fabric filter to control particulate

Disadvantages

1. Acid gas control efficiency not as high as with wet scrubber
2. No particulate collection—dry scrubber generates particulate
3. Corrosion problems more severe than with dry systems
4. Solids buildup at the wet-dry interface possibly a problem
5. Relatively high maintenance costs
6. Must be protected from freezing
7. Low exit gas temperature reduces exhaust plume dispersion

DISPERSION FROM STACKS

Stacks discharging to the atmosphere have long been the most common industrial method of disposing of waste gases. The concentrations to which humans, plants, animals, and structures are exposed at ground level can be reduced significantly by emitting the waste gases from a process at great heights. Although tall stacks may be effective in lowering the ground-level concentration of pollutants, they do not in themselves reduce the amount of pollutants released into the atmosphere. However, in certain situations, their use can be the most practical and economical way of dealing with an air-pollution problem.

Preliminary Design Considerations To determine the acceptability of a stack as a means of disposing of waste gases, the acceptable ground-level concentration (GLC) of the pollutant or pollutants must be determined. The topography of the area must also be considered so that the stack can be properly located with respect to buildings and hills that might introduce a factor of air turbulence into the operation of the stack. Awareness of the meteorological conditions prevalent in the area, such as the prevailing winds, humidity, and rainfall, is also essential. Finally, an accurate knowledge of the constituents of the waste gas and their physical and chemical properties is paramount.

TABLE 22-21 Advantages and Disadvantages of Electrostatic Precipitators*Advantages*

1. Extremely high particulate (coarse and fine) collection efficiencies attainable (at a relatively low expenditure of energy)
2. Collected material recovered dry for subsequent processing or disposal
3. Low pressure drop
4. Designed for continuous operation with minimum maintenance requirements
5. Relatively low operating costs
6. Capable of operation under high pressure (to 150 lb/in²) or vacuum conditions
7. Capable of operation at high temperatures [to 704°C(1300°F)]
8. Relatively large gas flow rates capable of effective handling

Disadvantages

1. High capital cost
2. Very sensitive to fluctuations in gas-stream conditions (in particular, flows, temperature, particulate and gas composition, and particulate loadings)
3. Certain particulates difficult to collect owing to extremely high- or low-resistivity characteristics
4. Relatively large space requirements required for installation
5. Explosion hazard when treating combustible gases and/or collecting combustible particulates
6. Special precautions required to safeguard personnel from the high voltage
7. Ozone produced by the negatively charged discharge electrode during gas ionization
8. Relatively sophisticated maintenance personnel required
9. Gas ionization may cause dissociation of gas stream constituents and result in creation of toxic byproducts
10. Sticky particulates may be difficult to remove from plates
11. Not effective in capturing some contaminants that exist as vapors at high temperatures (e.g., heavy metals, dioxins)

TABLE 22-22 Advantages and Disadvantages of Fabric-Filter Systems*Advantages*

1. Extremely high collection efficiency on both coarse and fine (sub-micrometer) particles
2. Relatively insensitive to gas-stream fluctuation; efficiency and pressure drop relatively unaffected by large changes in inlet dust loadings for continuously cleaned filters
3. Filter outlet air capable of being recirculated within the plant in many cases (for energy conservation)
4. Collected material recovered dry for subsequent processing or disposal
5. No problems with liquid-waste disposal, water pollution, or liquid freezing
6. Corrosion and rusting of components usually not problems
7. No hazard of high voltage, simplifying maintenance and repair and permitting collection of flammable dusts
8. Use of selected fibrous or granular filter aids (precoating), permitting the high-efficiency collection of submicrometer smokes and gaseous contaminants
9. Filter collectors available in a large number of configurations, resulting in a range of dimensions and inlet and outlet flange locations to suit installment requirements
10. Relatively simple operation

Disadvantages

1. Temperatures much in excess of 288°C (550°F) requiring special refractory mineral or metallic fabrics that are still in the developmental stage and can be very expensive
2. Certain dusts possibly requiring fabric treatments to reduce dust seeping or, in other cases, assist in the removal of the collected dust
3. Concentrations of some dusts in the collector (~50 g/m³) forming a possible fire or explosion hazard if a spark or flame is admitted by accident; possibility of fabrics burning if readily oxidizable dust is being collected
4. Relatively high maintenance requirements (bag replacement, etc.)
5. Fabric life possibly shortened at elevated temperatures and in the presence of acid or alkaline particulate or gas constituents
6. Hygroscopic materials, condensation of moisture, or tarry adhesive components possibly causing crusty caking or plugging of the fabric or requiring special additives
7. Replacement of fabric, possibly requiring respiratory protection for maintenance personnel
8. Medium pressure-drop requirements, typically in the range 4- to 10-in water column

Wind Direction and Speed Wind direction is measured at the height at which the pollutant is released, and the mean direction will indicate the direction of travel of the pollutants. In meteorology, it is conventional to consider the wind direction as the direction from which the wind blows; therefore, a northwest wind will move pollutants to the southeast of the source.

The effect of wind speed is twofold: (1) Wind speed will determine the travel time from a source to a given receptor; and (2) wind speed will affect dilution in the downwind direction. Generally, the concentration of air pollutants downwind from a source is inversely proportional to wind speed.

Wind speed has velocity components in all directions so that there are vertical motions as well as horizontal ones. These random motions of widely different scales and periods are essentially responsible for the movement and diffusion of pollutants about the mean downwind path. These motions can be considered atmospheric turbulence. If the scale of a turbulent motion (i.e., the size of an eddy) is larger than the size of the pollutant plume in its vicinity, the eddy will move that portion of the plume. If an eddy is smaller than the plume, its effect will be to diffuse or spread out the plume. This diffusion caused by eddy motion is widely variable in the atmosphere, but even when the effect of this diffusion is least, it is in the vicinity of three orders of magnitude greater than diffusion by molecular action alone.

Mechanical turbulence is the induced-eddy structure of the atmosphere due to the roughness of the surface over which the air is passing. Therefore, the existence of trees, shrubs, buildings, and terrain features will cause mechanical turbulence. The height and spacing of the elements causing the roughness will affect the turbulence. In general, the higher the roughness elements, the greater the mechanical turbulence. In addition, mechanical turbulence increases as wind speed increases.

TABLE 22-23 Advantages and Disadvantages of Absorption Systems (Packed and Plate Columns)*Advantages*

1. Relatively low pressure drop
2. Standardization in fiberglass-reinforced plastic (FRP) construction permitting operation in highly corrosive atmospheres
3. Capable of achieving relatively high mass-transfer efficiencies
4. Increasing the height and/or type of packing or number of plates capable of improving mass transfer without purchasing a new piece of equipment
5. Relatively low capital cost
6. Relatively small space requirements
7. Ability to collect particulates as well as gases
8. Collected substances may be recovered by distillation

Disadvantages

1. Possibility of creating water (or liquid) disposal problem
2. Product collected wet
3. Particulates deposition possibly causing plugging of the bed or plates
4. When FRP construction is used, sensitive to temperature
5. Relatively high maintenance costs
6. Must be protected from freezing

Thermal turbulence is turbulence induced by the stability of the atmosphere. When the Earth's surface is heated by the sun's radiation, the lower layer of the atmosphere tends to rise and thermal turbulence becomes greater, especially under conditions of light wind. On clear nights with wind, heat is radiated from the Earth's surface, resulting in the cooling of the ground and the air adjacent to it. This results in extreme stability of the atmosphere near the Earth's surface. Under these conditions, turbulence is at a minimum. Attempts to relate different measures of turbulence of the wind (or stability of the atmosphere) to atmospheric diffusion have been made for some time. The measurement of atmospheric stability by temperature-difference measurements on a tower is frequently utilized as an indirect measure of turbulence, particularly when climatological estimates of turbulence are desired.

Lapse Rate and Atmospheric Stability Apart from mechanical interference with the steady flow of air caused by buildings and other obstacles, the most important factor that influences the degree of turbulence and hence the speed of diffusion in the lower air is the variation of temperature with height above the ground, referred to as the "lapse rate." The dry-adiabatic lapse rate (DALR) is the temperature change for a rising parcel of dry air. The dry-adiabatic lapse rate can be approximated as -1°C per 100 m, or $dT/dz = -10^{-2}^{\circ}\text{C}/\text{m}$ or $-5.4^{\circ}\text{F}/1000$ ft. If the rising air contains water vapor, the cooling due to adiabatic expansion will result in the relative humidity being increased, and saturation may be reached. Further ascent would then lead to condensation of water vapor, and the latent heat thus released would reduce the rate of cooling of the rising air. The buoyancy force on a warm-air parcel is caused by the difference between its density and that of the surrounding air. The perfect-gas law shows that, at a fixed pressure (altitude), the temperature and density of an air parcel are inversely related; temperature is normally used to determine buoyancy because it is easier to measure than density. If the temperature gradient (lapse rate) of the atmosphere is the same as the adiabatic lapse rate, a parcel of air displaced from its original position will expand or contract in such a manner that its density and temperature remain the same as its surroundings. In this case there will be no buoyancy forces on the displaced parcel, and the atmosphere is termed "neutrally stable."

TABLE 22-24 Comparison of Plate and Packed Columns*Packed column*

1. Lower pressure drop
2. Simpler and cheaper to construct
3. Preferable for liquids with high-foaming tendencies

Plate column

1. Less susceptible to plugging
2. Less weight
3. Less of a problem with channeling
4. Temperature surge resulting in less damage

TABLE 22-25 Advantages and Disadvantages of Adsorption Systems*Advantages*

1. Possibility of product recovery
2. Excellent control and response to process changes
3. No chemical-disposal problem when pollutant (product) recovered and returned to process
4. Capability of systems for fully automatic, unattended operation
5. Capability to remove gaseous or vapor contaminants from process streams to extremely low levels

Disadvantages

1. Product recovery possibly requiring an exotic, expensive distillation (or extraction) scheme
2. Adsorbent progressively deteriorating in capacity as the number of cycles increases
3. Adsorbent regeneration requiring a steam or vacuum source
4. Relatively high capital cost
5. Prefiltering of gas stream possibly required to remove any particulate capable of plugging the adsorbent bed
6. Cooling of gas stream possibly required to get to the usual range of operation (less than 49°C [120°F])
7. Relatively high steam requirements to desorb high-molecular-weight hydrocarbons
8. Spent adsorbent may be considered a hazardous waste
9. Some contaminants may undergo a violent exothermic reaction with the adsorbent

If the atmospheric temperature decreases faster with increasing altitude than the adiabatic lapse rate (superadiabatic), a parcel of air displaced upward will have a higher temperature than the surrounding air. Its density will be lower, giving it a net upward buoyancy force. The opposite situation exists if the parcel of air is displaced downward, and the parcel experiences a downward buoyancy force. Once a parcel of air has started moving up or down, it will continue to do so, causing unstable atmospheric conditions. If the temperature decreases more slowly with increasing altitude than the adiabatic lapse rate, a displaced parcel of air experiences a net restoring force. The buoyancy forces then cause stable atmospheric conditions (see Fig. 22-5).

Strongly stable lapse rates are commonly referred to as *inversions*. The strong stability inhibits mixing across the inversion layer. Normally these conditions of strong stability extend for only several hundred meters vertically. The vertical extent of the inversion is referred to as the *inversion depth*. Two distinct types are observed: the ground-level inversion, caused by radiative cooling of the ground at night, and inversions aloft, occurring between 500 and several thousand meters above the ground (see Fig. 22-6). Some of the more common lapse-rate profiles with the corresponding effect on stackplumes are presented in Fig. 22-7.

From the viewpoint of air pollution, both stable surface layers and low-level inversions are undesirable because they minimize the rate of dilution of contaminants in the atmosphere. Even though the surface layer may be unstable, a low-level inversion will act as a barrier to vertical mixing, and contaminants will accumulate in the surface layer

TABLE 22-26 Advantages and Disadvantages of Combustion Systems*Advantages*

1. Simplicity of operation
2. Capability of steam generation or heat recovery in other forms
3. Capability for virtually complete destruction of organic contaminants

Disadvantages

1. Relatively high operating costs (particularly associated with fuel requirements)
2. Potential for flashback and subsequent explosion hazard
3. Catalyst poisoning (in the case of catalytic incineration)
4. Incomplete combustion, possibly creating potentially worse pollution problems
5. Even complete combustion may produce SO_2 , NO_x , and CO_2
6. High temperature components and exhaust may be hazardous to maintenance personnel and birds
7. High maintenance requirements—especially if operation is cyclic

TABLE 22-27 Advantages and Disadvantages of Condensers

Advantages

1. Pure product recovery (in the case of indirect-contact condensers)
2. Water used as the coolant in an indirect-contact condenser (i.e., shell-and-tube heat exchanger), not in contact with contaminated gas stream, and is reusable after cooling
3. May be used to produce vacuum to remove contaminants from process

Disadvantages

1. Relatively low removal efficiency for gaseous contaminants (at concentrations typical of pollution-control applications)
2. Coolant requirements possibly extremely expensive
3. Direct-contact condenser may produce water discharge problems

below the inversion. Stable atmospheric conditions tend to be more frequent and longest in persistence in the autumn, but inversions and stable lapse rates are prevalent at all seasons of the year.

Design Calculations For a given stack height, the calculational sequence begins by first estimating the effective height of the emission, employing an applicable plume-rise equation. The maximum GLC may then be determined by using an appropriate atmospheric-diffusion equation. A simple comparison of the calculated GLC for the particular pollutant with the maximum GLC permitted by the local air-pollution codes dictates whether the stack is operating satisfactorily. Conversely, with knowledge of the maximum acceptable GLC standards, a stack that will satisfy these standards can be properly designed.

Effective Height of an Emission The effective height of an emission rarely corresponds to the physical height of the stack. If the plume is caught in the turbulent wake of the stack or of buildings in the vicinity of the stack, the effluent will be mixed rapidly downward toward the ground. If the plume is emitted free of these turbulent zones, a number of source emission characteristics and meteorological factors influence the rise of the plume. The source emission characteristics include the gas flow rate and the temperature of the effluent at the top of the stack and the diameter of the stack opening. The meteorological factors influencing plume rise include wind speed, air temperature, shear of the wind speed with height, and atmospheric stability. No theory on plume rise presently takes into account all of these variables. Most of the equations that have been formulated for computing the effective height of an emission are semi-empirical. When considering any of these plume-rise equations, it is important to evaluate each in terms of the assumptions made and the circumstances existing at the time that the particular correlation was formulated. The formulas generally are not applicable to tall stacks (above 305 m [1000 ft] effective height).

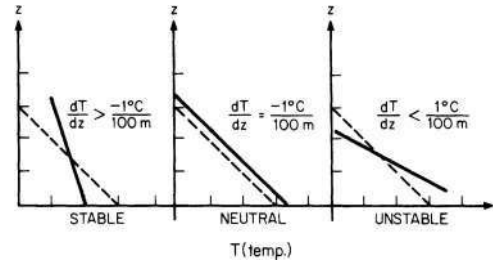


FIG. 22-5 Stability criteria with measured lapse rate.

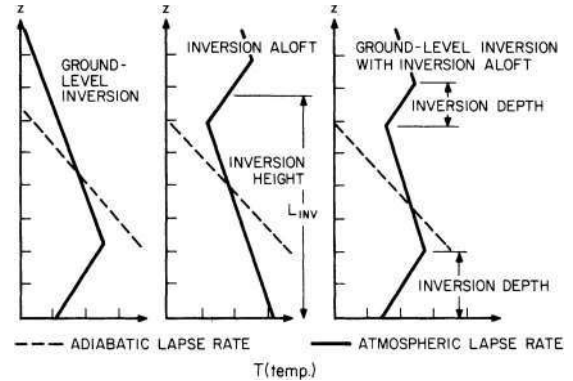


FIG. 22-6 Characteristic lapse rates under inversion conditions.

The effective stack height (equivalent to the effective height of the emission) is the sum of the actual stack height, the plume rise due to the exhaust velocity (momentum) of the issuing gases, and the buoyancy rise, which is a function of the temperature of the gases being emitted and the atmospheric conditions.

Some of the more common plume-rise equations have been summarized by Buonicore and Theodore (*Industrial Control Equipment for Gaseous Pollutants*, vol. 2, CRC Press, Boca Raton, Florida, 1975) and include:

Temperature gradient	Observation	Description
		Strong lapse—looping (unstable) This is usually a fair-weather daytime condition, since strong solar heating of the ground is required. Looping is not favored by cloudiness, snow cover, or strong winds.
		Weak lapse—coning (slightly unstable or neutral) This is usually favored by cloudy and windy conditions and may occur day or night. In dry climates it may occur infrequently, and in cloudy climates it may be the most frequent type observed.
		Inversion—fanning (stable) This is principally a nighttime condition. It is favored by light winds, clear skies, and snow cover. The condition may persist in some climates for several days at a time during winter, especially in the higher latitudes.
		Inversion below, lapse aloft—lofting (transition from unstable to stable) This condition occurs during transition from lapse to inversion and should be observed most frequently near sunset; it may be very transitory or persist for several hours. The shaded zone of strong effluent concentration is caused by trapping by the inversion of effluent carried into the stable layer by turbulent eddies that penetrate the layer for a short distance.
		Lapse below, inversion aloft—fumigation (transition from stable to unstable) This occurs when the nocturnal inversion is dissipated by heat from the morning sun. The lapse layer usually starts at the ground and works its way upward (less rapidly in winter than in summer). Fumigation may also occur in sea-breeze circulations during late morning or early afternoon. The shaded zone of strong concentration is that portion of the plume which has not yet been mixed downward.

FIG. 22-7 Lapse-rate characteristics of atmospheric-diffusion transport of stack emissions.

TABLE 22-28 Advantages and Disadvantages of Biofiltration**Advantages**

1. Uses natural biological processes and materials
2. Relatively simple and economical
3. High destruction efficiencies for oxygen-rich, low-contaminant concentration gas streams
4. Waste products are CO₂ and water

Disadvantages

1. Raw gas must not be lethal to microorganisms
2. Gas stream must be maintained at proper temperature and humidity
3. Heavy particulate loadings can damage pore structure of filter bed

- ASME, *Recommended Guide for the Prediction of the Dispersion of Airborne Effluents*, ASME, New York, 1968.
 - Bosanquet-Carey-Halton, *Proc. Inst. Mech. Eng. (London)*, **162**, 355 (1950).
 - Briggs, *Plume Rise*, AEC Critical Review ser., U.S. Atomic Energy Commission, Div. Tech. Inf.
 - Brummage et al., *The Calculation of Atmospheric Dispersion from a Stack*, CONCAWE, The Hague, 1966.
 - Carson and Moses, *J. Air Pollut. Control Assoc.*, **18**, 454 (1968) and **19**, 862 (1969).
 - Csanady, *Int. J. Air Water Pollut.*, **4**, 47 (1961).
 - Davison-Bryant, *Trans. Conf. Ind. Wastes*, 14th Ann. Meet. Ind. Hyg. Found. Am., **39**, 1949.
 - Holland, *Workbook of Atmospheric Dispersion Estimates*, U.S. EPA Publ. AP-26, 1970.
 - Lucas, Moore, and Spurr, *Int. J. Air Water Pollut.*, **7**, 473 (1963).
 - Montgomery et al., *J. Air Pollut. Control Assoc.*, **22**(10), 779, TVA (1972).
 - Stone and Clarke, *British Experience with Tall Stacks for Air Pollution Control on Large Fossil-Fueled Power Plants*, American Power Conference, Chicago, 1967.
 - Stumke, *Staub*, **23**, 549 (1963).
- See also:
- Briggs, G. A., *Plume Rise Predications: Lectures on Air Pollution and Environmental Impact Analyses*, Workshop Proceedings, American Meteorological Society, Boston, Massachusetts, 1975, pp. 59-111.
 - Randerson, Darryl (ed.), *Plume Rise and Buoyancy Effects, Atmospheric Science and Power Production*, DOE Report DOE/TIC-27601, 1981.

Maximum Ground-Level Concentrations The effective height of an emission having been determined, the next step is to study its path downward by using the appropriate atmospheric-dispersion formula. Some of the more popular atmospheric-dispersion calculational procedures have been summarized by Buonicore and Theodore (op. cit.) and include:

- Bosanquet-Pearson model [Pasquill, *Meteorol. Mag.*, **90**, 33, 1063 (1961), and Gifford, *Nucl. Saf.*, **2**, 4, 47 (1961).]
 - Sutton model [Q.J.R. *Meteorol. Soc.*, **73**, 257 (1947).]
 - TVA Model [Carpenter et al., *J. Air Pollut. Control Assoc.*, **21**(8), (1971), and Montgomery et al., *J. Air Pollut. Control Assoc.*, **23**(5), 388 (1973).]
- See also:
- Bornstein et al., *Simulation of Urban Barrier Effects on Polluted Urban Boundary Layers Using the Three-Dimensional URBMET/IVM Model with Urban Topography*, Air Pollution Proceedings, 1993.

TABLE 22-29 Advantages and Disadvantages of Membrane Filtration**Advantages**

1. Low energy utilization
2. Modular design
3. Low capital costs
4. Low maintenance
5. Superior separation ability

Disadvantages

1. Potential particulate fouling problems if not properly designed

TABLE 22-30 Advantages and Disadvantages of Selective Catalytic Reduction of Nitrogen Oxides**Advantages**

1. Capable of 90% NO_x removal
2. Uses readily available urea reagent
3. Exhaust products are N₂ and water

Disadvantages

1. Spent catalyst may be considered hazardous waste
2. Gas stream must be maintained at proper temperature and humidity
3. Heavy particulate loadings can damage pore structure of filter bed

- EPA, *Guidance on the Application of Refined Dispersion Models for Air Toxins Releases*, EPA-450/4-91-007.
- Zannetti, Paolo, *Air Pollution Modeling: Theories, Computational Methods, and Available Software*, Van Nostrand, Reinhold, New York, 1990.
- Zannetti, Paolo, *Numerical Simulation Modeling of Air Pollution: An Overview*, Ecological Physical Chemistry, 2d International Workshop, May 1992.

Miscellaneous Effects

Evaporative Cooling When effluent gases are washed to absorb certain constituents prior to emission, the gases are cooled and become saturated with water vapor. Upon release of the gases, further cooling due to contact with cold surfaces of ductwork or stack is likely. This cooling causes water droplets to condense in the gas stream. Upon release of the gases from the stack, the water droplets evaporate, withdrawing the latent heat of vaporization from the air and cooling the plume. The resulting negative buoyancy reduces the effective stack height. The result may be a plume (with a greater density than that of the ambient atmosphere) that will fall to the ground. If any pollutant remains after scrubbing, its full effect will be felt on the ground in the vicinity of the stack.

Aerodynamic Downwash Should the stack exit velocity be too low as compared with the speed of the crosswind, some of the effluent can be pulled downward by the low pressure on the lee side of the stack. This phenomenon, known as "stack-tip downwash," can be minimized by keeping the exit velocity greater than the mean wind speed (i.e., typically twice the mean wind speed). Another way to minimize stack-tip downwash is to fit the top of the stack with a flat disc that extends for at least one stack diameter outward from the stack.

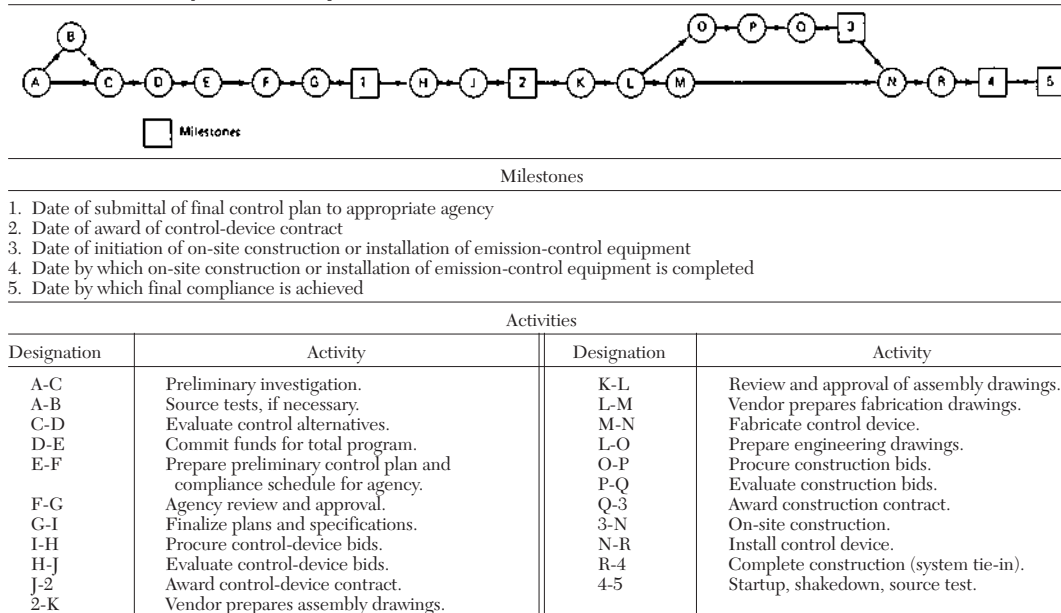
If it becomes necessary to increase the stack-gas exit velocity to avoid downwash, it may be necessary to remodel the stack exit. A venturi-nozzle design has been found to be the most effective. This design also keeps pressure losses to a minimum.

Building Downwash A review must be conducted for each stack to determine if building downwash effects need to be considered. Atmospheric flow is disrupted by aerodynamic forces in the immediate vicinity of structures or terrain obstacles. The disrupted flow near either building structures or terrain obstacles can both enhance the vertical dispersion of emissions from the source and reduce the effective height of the emissions from the source, resulting in an increase in the maximum GLC.

EPA Air Dispersion Models EPA addresses modeling techniques in the "Guideline on Air Quality Models" [Appendix W (July 2003) of 40 CFR Part 51]. The guideline provides a common basis for estimating the air quality concentrations of criteria pollutants used in assessing control strategies and developing emission limits. The continuing development of new air quality models in response to regulatory requirements and the expanded requirements for models to cover ever more complex problems have emphasized the need for periodic review and update of guidance on these techniques. Information on the current status of modeling guidance can always be obtained from EPA's regional offices.

The guideline recommends air quality modeling techniques that should be applied to State Implementation Plan (SIP) revisions for existing sources and to new source reviews (NSRs), including prevention of significant deterioration (PSD). In addition the guideline serves to identify, for all interested parties, those techniques and databases that EPA considers acceptable. Dispersion models, while uniquely filling

TABLE 22-31 Compliance Activity and Schedule Chart



one program need, have become a primary analytical tool in most air quality assessments. Air quality measurements can be used in a complementary manner to dispersion models, with due regard for the strengths and weaknesses of both analysis techniques. The diversity of the nation's topography and climate, and variations in source configurations and operating characteristics dictate against a strict modeling "cookbook." There is no one model capable of properly addressing all conceivable situations even within a broad category such as point sources.

The guideline provides a consistent basis for selection of the most accurate models and databases for use in air quality assessments. There are two levels of sophistication of models. The first level consists of relatively simple estimation techniques that generally use pre-set, worst-case meteorological conditions to provide conservative estimates of the air quality impact of a specific source, or source category. These are called screening techniques or screening models. The purpose of such techniques is to eliminate the need for more detailed modeling for those sources that clearly will not cause or contribute to ambient concentrations in excess of either the National Ambient Air Quality Standards (NAAQS) or the allowable PSD concentration increments. The second level consists of those analytical techniques that provide more detailed treatment of physical and chemical atmospheric processes, require more detailed and precise input data, and provide more specialized concentration estimates. Most of the screening and refined models discussed in the guideline, codes, associated documentation, and other useful information are available for download from EPA's Support Center for Regulatory Air Modeling (SCRAM) Internet web site at <http://www.epa.gov/scram001>.

CONTROL OF GASEOUS EMISSIONS

There are four chemical-engineering unit operations commonly used for the control of gaseous emissions:

1. **Absorption.** See Sec. 4, "Thermodynamics"; and Sec. 18, "Liquid Gas Systems." For plate columns, see Sec. 18, "Gas-Liquid Contacting: Plate Columns." For packed columns, see Sec. 18, "Gas-Liquid Contacting: Packed Columns."
2. **Adsorption.** See Sec. 16, "Adsorption and Ion Exchange."
3. **Combustion.** See Sec. 9, "Heat Generation" and "Fired Process Equipment." For incinerators, see material under these subsections.
4. **Condensation.** See Sec. 10, "Heat Transmission"; Sec. 11, "Heat-Transfer Equipment"; and Sec. 12, "Evaporative Cooling." For direct-

contact condensers, See Sec. 11, "Evaporator Accessories"; and Sec. 12, "Evaporative Cooling." For indirect-contact condensers, see Sec. 10, "Heat Transfer with Change of Phase" and "Thermal Design of Heat-Transfer Equipment"; and Sec. 11, "Shell and Tube Heat Exchangers" and "Other Heat Exchangers for Liquids and Gases."

These operations, which are routine chemical engineering operations, have been treated extensively in other sections of this handbook.

There are three additional chemical engineering unit operations that are increasing in use in recent years. They are:

1. *Biofiltration*
2. *Membrane filtration*
3. *Selective catalytic reduction*

and are discussed further in this section.

Absorption The engineering design of gas absorption equipment must be based on a sound application of the principles of diffusion, equilibrium, and mass transfer as developed in Secs. 5 and 14 of the handbook. The main requirement in equipment design is to bring the gas into intimate contact with the liquid; that is, to provide a large interfacial area and a high intensity of interface renewal and to minimize resistance and maximize driving force. This contacting of the phases can be achieved in many different types of equipment, the most important being packed and plate columns. The final choice between them rests with the various criteria that must be met. For example, if the pressure drop through the column is large enough that compression costs become significant, a packed column may be preferable to a plate-type column because of the lower pressure drop.

In most processes involving the absorption of a gaseous pollutant from an effluent gas stream, the gas stream is the processed fluid; hence, its inlet condition (flow rate, composition, and temperature) are usually known. The temperature and composition of the inlet liquid and the composition of the outlet gas are usually specified. The main objectives in the design of an absorption column, then, are the determination of the solvent flow rate and the calculation of the principal dimensions of the equipment (column diameter and height to accomplish the operation). These objectives can be obtained by evaluating, for a selected solvent at a given flow rate, the number of theoretical separation units (stage or plates) and converting them into practical units, column heights or number of actual plates by means of existing correlations.

The general design procedure consists of a number of steps to be taken into consideration. These include:

1. Solvent selection
2. Equilibrium-data evaluation
3. Estimation of operating data (usually consisting of a mass and energy in which the energy balance decides whether the absorption balance can be considered isothermal or adiabatic)
4. Column selection (should the column selection not be obvious or specified, calculations must be carried out for the different types of columns and the final based on economic considerations)
5. Calculation of column diameter (for packed columns, this is usually based on flooding conditions, and, for plate columns, on the optimum gas velocity or the liquid-handling capacity of the plate)
6. Estimation of column height or number of plates (for packed columns, column height is obtained by multiplying the number of transfer units, obtained from a knowledge of equilibrium and operating data, by the height of a transfer unit; for plate columns, the number of theoretical plates determined from the plot of equilibrium and operating lines is divided by the estimated overall plate efficiency to give the number of actual plates, which in turn allows the column height to be estimated from the plate spacing)
7. Determination of pressure drop through the column (for packed columns, correlations dependent of packing type, column-operating data, and physical properties of the constituents involved are available to estimate the pressure drop through the packing; for plate columns, the pressure drop per plate is obtained and multiplied by the number of plates)

Solvent Selection The choice of a particular solvent is most important. Frequently, water is used, as it is very inexpensive and plentiful, but the following properties must also be considered:

1. *Gas solubility.* A high gas solubility is desired, since this increases the absorption rate and minimizes the quantity of solvent necessary. Generally, solvents of a chemical nature similar to that of the solute to be absorbed will provide good solubility.
2. *Volatility.* A low solvent vapor pressure is desired, since the gas leaving the absorption unit is ordinarily saturated with the solvent, and much may thereby be lost.
3. *Corrosiveness.*
4. *Cost.*
5. *Viscosity.* Low viscosity is preferred for reasons of rapid absorption rates, improved flooding characteristics, lower pressure drops, and good heat-transfer characteristics.
6. *Chemical stability.* The solvent should be chemically stable and, if possible, nonflammable.
7. *Toxicity.*
8. *Low freezing point.* If possible, a low freezing point is favored, since any solidification of the solvent in the column makes the column inoperable.

Equipment The principal types of gas-absorption equipment may be classified as follows:

1. Packed columns (continuous operation)
2. Plate columns (staged operations)
3. Miscellaneous

Of the three categories, the packed column is by far the most commonly used for the absorption of gaseous pollutants. Miscellaneous gas-absorption equipment could include acid gas scrubbers that are commonly classified as either "wet" or "dry." In wet scrubber systems, the absorption tower uses a lime-based sorbent liquor that reacts with the acid gases to form a wet/solid by-product. Dry scrubbers can be grouped into three categories: (1) spray dryers; (2) circulating spray dryers; and (3) dry injection. Each of these systems yields a dry product that can be captured with a fabric filter baghouse downstream and thus avoids costly wastewater treatment systems. The baghouse is highly efficient in capturing the particulate emissions, and a portion of the overall acid gas removal has been found to occur within the baghouse. Additional information may be found by referring to the appropriate sections in this *handbook* and the many excellent texts available, such as McCabe and Smith, *Unit Operations of Chemical Engineering*, 3d ed., McGraw-Hill, New York, 1976; Sherwood and Pigford, *Absorption and Extraction*, 2d ed., McGraw-Hill, New York, 1952; Smith, *Design of Equilibrium Stage Processes*, McGraw-Hill, New York, 1963; Treybal, *Mass Transfer Operations*, 3d ed., McGraw-Hill, New York, 1980; McKenna, Mycock, and Theodore, *Handbook of Air Pollution Control Engineering and Technology*, CRC Press, Boca

Raton, Florida, 1995; Theodore and Allen, *Air Pollution Control Equipment*, ETSI Prof. Training Inst., Roanoke, Virginia, 1993.

Adsorption The design of gas-adsorption equipment is in many ways analogous to the design of gas-absorption equipment, with a solid adsorbent replacing the liquid solvent (see Secs. 16 and 19). Similarity is evident in the material- and energy-balance equations as well as in the methods employed to determine the column height. The final choice, as one would expect, rests with the overall process economics.

Selection of Adsorbent Industrial adsorbents are usually capable of adsorbing both organic and inorganic gases and vapors. However, their preferential adsorption characteristics and other physical properties make each of them more or less specific for a particular application. General experience has shown that, for the adsorption of vapors of an organic nature, activated carbon has superior properties, having hydrocarbon-selective properties and high adsorption capacity for such materials. Inorganic adsorbents, such as activated alumina or silica gel, can also be used to adsorb organic materials, but difficulties can arise during regeneration. Activated alumina, silica gel, and molecular sieves will also preferentially adsorb any water vapor with the organic contaminant. At times this may be a considerable drawback in the application of these adsorbents for organic-contaminant removal.

The normal method of regeneration of adsorbents is by use of steam, inert gas (i.e., nitrogen), or other gas streams, and in the majority of cases this can cause at least slight decomposition of the organic compound on the adsorbent. Two difficulties arise: (1) incomplete recovery of the adsorbate, although this may be unimportant; and (2) progressive deterioration in capacity of the adsorbent as the number of cycles increases owing to blocking of the pores from carbon formed by hydrocarbon decomposition. With activated carbon, a steaming process is used, and the difficulties of regeneration are thereby overcome. This is not feasible with silica gel or activated alumina because of the risk of breakdown of these materials when in contact with liquid water.

In some cases, none of the adsorbents has sufficient retaining capacity for a particular contaminant. In these applications, a large-surface-area adsorbent can be impregnated with an inorganic compound or, in rare cases, with a high-molecular-weight organic compound that can react chemically with the particular contaminant. For example, iodine-impregnated carbons are used for removal of mercury vapor and bromine-impregnated carbons for ethylene or propylene removal. The action of these impregnants is either catalytic conversion or reaction to a nonobjectionable compound or to a more easily adsorbed compound. For this case, general adsorption theory no longer applies to the overall effects of the process. For example, mercury removal by an iodine-impregnated carbon proceeds faster at a higher temperature, and a better overall efficiency can be obtained than in a low-temperature system.

Since adsorption takes place at the interphase boundary, the adsorption surface area becomes an important consideration. Generally, the higher the adsorption surface area, the greater its adsorption capacity. However, the surface area has to be "available" in a particular pore size within the adsorbent. At low partial pressure (or concentration) a surface area in the smallest pores in which the adsorbate can enter is the most efficient. At higher pressures the larger pores become more important; at very high concentrations, capillary condensation will take place within the pores, and the total micropore volume becomes the limiting factor.

The action of molecular sieves is slightly different from that of other adsorbents in that selectivity is determined more by the pore-size limitations of the particular sieve. In selecting molecular sieves, it is important that the contaminant to be removed be smaller than the available pore size. Hence, it is important that the particular adsorbent not only have an affinity for the contaminant in question but also have sufficient surface area available for adsorption.

Design Data The adsorbent having been selected, the next step is to calculate the quantity of adsorbent required and eventually consider other factors such as the temperature rise of the gas stream due to adsorption and the useful life of the adsorbent under operating conditions. The sizing and overall design of the adsorption system depend on the properties and characteristics of both the feed gas to be treated and the adsorbent. The following information should be known or available for design purposes:

1. Gas stream
 - a. Adsorbate concentration.
 - b. Temperature.
 - c. Temperature rise during adsorption.
 - d. Pressure.
 - e. Flow rate.
 - f. Presence of adsorbent contaminant material.
2. Adsorbent
 - a. Adsorption capacity as used on stream.
 - b. Temperature rise during adsorption.
 - c. Isothermal or adiabatic operation.
 - d. Life, if presence of contaminant material is unavoidable.
 - e. Possibility of catalytic effects causing an adverse chemical reaction in the gas stream or the formation of solid polymerizates on the adsorbent bed, with consequent deterioration.
 - f. Bulk density.
 - g. Particle size, usually reported as a mean equivalent particle diameter. The dimensions and shape of particles affect both the pressure drops through the adsorbent bed and the diffusion rate into the particles. All things being equal, adsorbent beds consisting of smaller particles, although causing a higher pressure drop, will be more efficient.
 - h. Pore data, which are important because they may permit elimination from consideration of adsorbents whose pore diameter will not admit the desired adsorbate molecule.
 - i. Hardness, which indicates the care that must be taken in handling adsorbents to prevent the formation of undesirable fines.
 - j. Regeneration information.

The design techniques used include both stagewise and continuous-contacting methods and can be applied to batch, continuous, and semicontinuous operations.

Adsorption Phenomena The adsorption process involves three necessary steps. The fluid must first come in contact with the adsorbent, at which time the adsorbate is preferentially or selectively adsorbed on the adsorbent. Next the fluid must be separated from the adsorbent-adsorbate, and, finally, the adsorbent must be regenerated by removing the adsorbate or by discarding used adsorbent and replacing it with fresh material. Regeneration is performed in a variety of ways, depending on the nature of the adsorbate. Gases or vapors are usually desorbed by either raising the temperature (thermal cycle) or reducing the pressure (pressure cycle). The more popular thermal cycle is accomplished by passing hot gas through the adsorption bed in the direction opposite to the flow during the adsorption cycle. This ensures that the gas passing through the unit during the adsorption cycle always meets the most active adsorbent last and that the adsorbate concentration in the adsorbent at the outlet end of the unit is always maintained at a minimum.

In the first step, in which the molecules of the fluid come in contact with the adsorbent, an equilibrium is established between the adsorbed fluid and the fluid remaining in the fluid phase. Figures 22-8 through 22-10 show several experimental equilibrium adsorption isotherms for a number of components adsorbed on various adsorbents. Consider Fig. 22-8, in which the concentration of adsorbed gas on the solid is plotted against the equilibrium partial pressure p^0 of the vapor or gas at constant temperature. At 40°C, for example, pure propane vapor at a pressure of 550 mm Hg is in equilibrium with an adsorbate concentration at point *P* of 0.04 lb adsorbed propane per pound of silica gel. Increasing the pressure of the propane will cause more propane to be adsorbed, while decreasing the pressure of the system at *P* will cause propane to be desorbed from the carbon.

The adsorptive capacity of activated carbon for some common solvent vapors is shown in Table 22-32.

Adsorption-Control Equipment If a gas stream must be treated for a short period, usually only one adsorption unit is necessary, provided, of course, that a sufficient time interval is available between adsorption cycles to permit regeneration. However, this is usually not the case. Since an uninterrupted flow of treated gas is often required, it is necessary to employ one or more units capable of operating in this fashion. The units are designed to handle gas flows without interruption and are characterized by their mode of contact, either staged or continuous. By far the most common type of adsorption system used to remove an objectionable pollutant from a gas stream consists of a

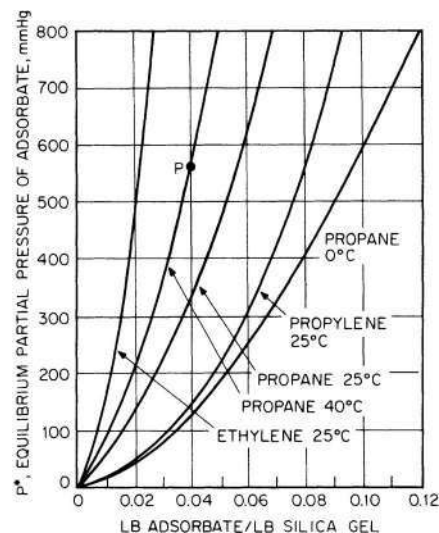


FIG. 22-8 Equilibrium partial pressures for certain organics on silica gel.

number of fixed-bed units operating in such a sequence that the gas flow remains uninterrupted. A two- or three-bed system is usually employed, with one or two beds bypassed for regeneration while one is adsorbing. A typical two-bed system is shown in Fig. 22-11a, while a typical three-bed system is shown in Fig. 22-11b. The type of system best suited for a particular job is determined from several factors, including the amount and rate of material being adsorbed, the time between cycles, the time required for regeneration, and the cooling time, if required.

Typical of continuous-contact operation for gaseous-pollutant adsorption is the use of a fluidized bed. During steady-state staged-contact operation, the gas flows up through a series of successive fluidized-bed stages, permitting maximum gas-solid contact on each stage. A typical arrangement of this type is shown in Fig. 22-12 for multistage countercurrent adsorption with regeneration. In the upper part of the tower, the particles are contacted countercurrently

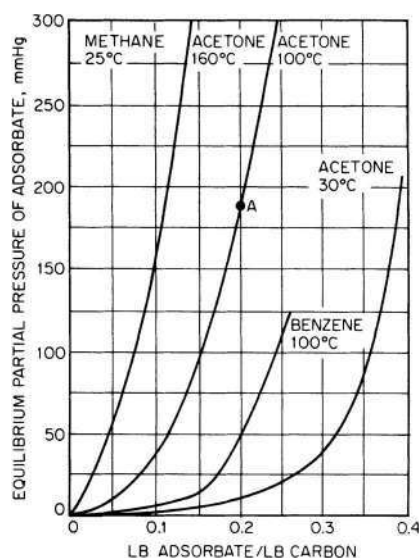


FIG. 22-9 Equilibrium partial pressures for certain organics on carbon.

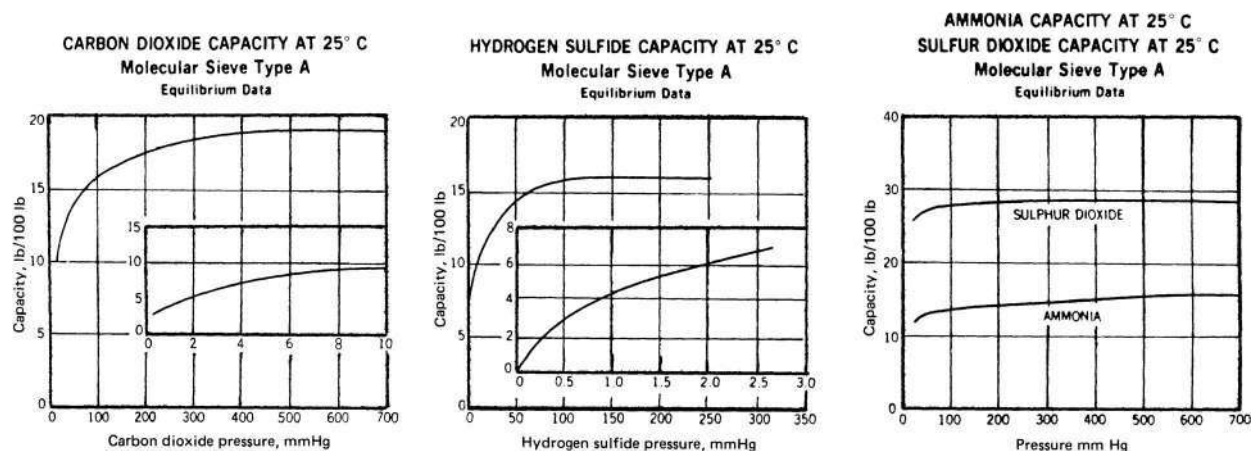


FIG. 22-10 Equilibrium partial pressures for certain gases on molecular sieves. (A. J. Buonicore and L. Theodore, *Industrial Control Equipment for Gaseous Pollutants*, vol. I, CRC Press, Boca Raton, Fla., 1975.)

on perforated trays in relatively shallow beds with the gas stream containing the pollutant, the adsorbent solids moving from tray to tray through downspouts. In the lower part of the tower, the adsorbent is regenerated by similar contact with hot gas, which desorbs and carries off the pollutant. The regenerated adsorbent is then recirculated by an airlift to the top of the tower.

Although the continuous-countercurrent type of operation has found limited application in the removal of gaseous pollutants from process streams (for example, the removal of carbon dioxide and sulfur compounds such as hydrogen sulfide and carbonyl sulfide), by far the most common type of operation presently in use is the fixed-bed adsorber. The relatively high cost of continuously transporting solid particles as required in steady-state operations makes fixed-bed adsorption an attractive, economical alternative. If intermittent or batch operation is practical, a simple one-bed system, cycling alternately between the adsorption and regeneration phases, will suffice.

Additional information may be found by referring to the appropriate sections of this *handbook*. A comprehensive treatment of adsorber design principles is given in Buonicore and Theodore, *Industrial Control Equipment for Gaseous Pollutants*, vol. 1, 2d ed., CRC press, Boca Raton, Florida, 1992.

Combustion Many organic compounds released from manufacturing operations can be converted to innocuous carbon dioxide and water by rapid oxidation (chemical reaction): combustion. However, combustion of gases containing halides may require the addition of acid gas treatment to the combustor exhaust.

Three rapid oxidation methods are typically used to destroy combustible contaminants: (1) flares (direct-flame-combustion), (2) thermal combustors, and (3) catalytic combustors. The thermal and flare methods are characterized by the presence of a flame during combus-

tion. The combustion process is also commonly referred to as "after-burning" or "incineration."

To achieve complete combustion (i.e., the combination of the combustible elements and compounds of a fuel with all the oxygen that they can utilize), sufficient space, time, and turbulence and a temperature high enough to ignite the constituents must be provided.

The three T's of combustion—time, temperature, and turbulence—govern the speed and completeness of the combustion reaction. For complete combustion, the oxygen must come into intimate contact with the combustible molecule at sufficient temperature and for a sufficient length of time for the reaction to be completed. Incomplete reactions may result in the generation of aldehydes, organic acids, carbon, and carbon monoxide.

Combustion-Control Equipment Combustion-control equipment can be divided into three types: (1) flares, (2) thermal incinerators, (3) catalytic incinerators.

Flares In many industrial operations and particularly in chemical plants and petroleum refineries, large volumes of combustible waste gases are produced. These gases result from undetected leaks in the operating equipment, from upset conditions in the normal operation of a plant in which gases must be vented to avoid dangerously high pressures in operating equipment, from plant startups, and from emergency shutdowns. Large quantities of gases may also result from off-specification product or from excess product that cannot be sold. Flows are typically intermittent, with flow rates during major upsets of up to several million cubic feet per hour.

The preferred control method for excess gases and vapors is to recover them in a blowdown recovery system. However, large quantities of gas, especially those during upset and emergency conditions, are difficult to contain and reprocess. In the past all waste gases were vented directly to the atmosphere. However, widespread venting caused safety and environmental problems, and, in practice, it is now customary to collect such gases in a closed flare system and to burn them as they are discharged.

Although flares can be used to dispose of excess waste gases, such systems can present additional safety problems. These include the explosion potential, thermal-radiation hazards from the flame, and the problem of toxic asphyxiation during flameout. Aside from these safety aspects, there are several other problems associated with flaring that must be dealt with during the design and operation of a flare system. These problems include the formation of smoke, the luminosity of the flame, noise during flame, and the possible emission of by-product air pollutants during flaring.

The heat content of the waste stream to be disposed is another important consideration. The heat content of the waste gas falls into two classes. The gases can either support their own combustion or not. In general, a waste gas with a heating value greater than 7443 kJ/m³

TABLE 22-32 Adsorptive Capacity of Common Solvents on Activated Carbons*

Solvent	Carbon bed weight, %†
Acetone	8
Heptane	6
Isopropyl alcohol	8
Methylene chloride	10
Perchloroethylene	20
Stoddard solvent	2-7
1,1,1-Trichloroethane	12
Trichloroethylene	15
Trichlorotrifluoroethane	8
VM&P naphtha	7

* Assuming steam desorption at 5 to 10 psig.

† For example, 8 lb of acetone adsorbed on 100 lb of activated carbon.

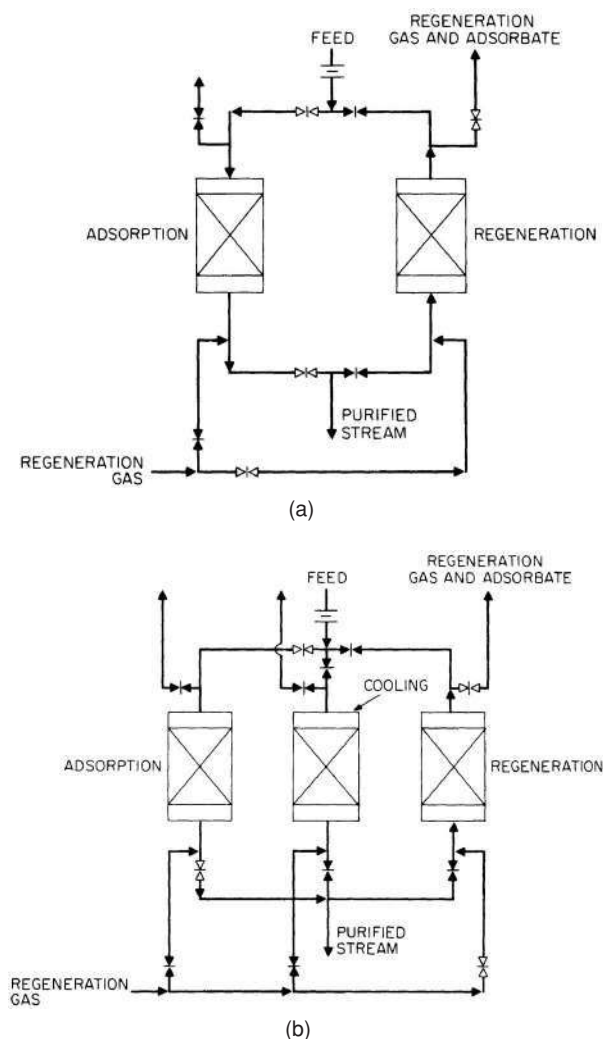


FIG. 22-11 (a) Typical two-bed adsorption system. (b) Typical three-bed adsorption system.

(200 Btu/ft³) can be flared successfully. The heating value is based on the lower heating value of the waste gas at the flare. Below 7443 kJ/m³, enriching the waste gas by injecting another gas with a higher heating value may be necessary. The addition of such a rich gas is called "endothermic flaring." Gases with a heating value as low as 2233 kJ/m³ (60 Btu/ft³) have been flared but at a significant fuel demand. It is usually not feasible to flare a gas with a heating value below 3721 kJ/m³ (100 Btu/ft³). If the flow of low-Btu gas is continuous, thermal or catalytic incineration can be used to dispose of the gas. For intermittent flows, however, endothermic flaring may be the only possibility.

Although most flares are used to dispose of intermittent waste gases, some continuous flares are in use, but generally only for relatively small volumes of gases. The heating value of large-volume continuous-flow waste gases is usually too valuable to lose in a flare. Vapor recovery or the use of the vapor as a fuel in a process heater is preferred over flaring. Since auxiliary fuel must be added to the gas in order to flare, large continuous flows of a low-heating-value gas are usually more efficient to burn in a thermal incinerator than in the flame of a flare.

Flares are mostly used for the disposal of hydrocarbons. Waste gases composed of natural gas, propane, ethylene, propylene, butadi-

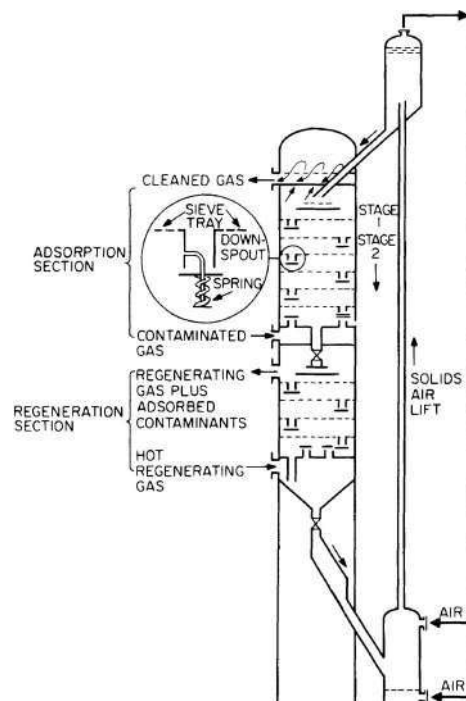


FIG. 22-12 Multistage countercurrent adsorption with regeneration.

ene, and butane probably constitute over 95 percent of the material flared. Flares have been used successfully to control malodorous gases such as mercaptans and amines, but care must be taken when flaring these gases. Unless the flare is very efficient and gives good combustion, obnoxious fumes can escape unburned and cause a nuisance.

Flaring of hydrogen sulfide should be avoided because of its toxicity and low odor threshold. In addition, burning relatively small amounts of hydrogen sulfide can create enough sulfur dioxide to cause crop damage or a local nuisance. For gases whose combustion products may cause problems, such as those containing hydrogen sulfide or chlorinated hydrocarbons, flaring is not recommended.

Thermal Incinerators Thermal incinerators or afterburners can be used over a fairly wide but low range of organic vapor concentration. The concentration of the organics in air must be substantially below the lower flammable level (lower explosive limit). As a rule, a factor of four is employed for safety precautions. Reactions are conducted at elevated temperatures to ensure high chemical-reaction rates for the organics. To achieve this temperature, it is necessary to preheat the feed stream using auxiliary energy. Along with the contaminant-laden gas stream, air and fuel are continuously delivered to the incinerator (see Fig. 22-13). The fuel and contaminants are combusted with air in a firing unit (burner). The burner may utilize the air in the process-waste stream as the combustion air for the auxiliary fuel, or it may use a separate source of outside air. The products of combustion and the unreacted feed stream are intensely mixed and enter the reaction zone of the unit. The pollutants in the process-gas stream are then reacted at the elevated temperature. Thermal incinerators generally require operating temperatures in the range of 650 to 980°C (1200 to 1800°F) for combustion of most organic pollutants (see Table 22-33). A residence time of 0.2 to 1.0 is often recommended, but this factor is dictated primarily by complex kinetic considerations. The kinetics of hydrocarbon (HC) combustion in the presence of excess oxygen can be simplified into the following first-order rate equation:

$$\frac{d(\text{HCl})}{dt} = -k[\text{HCl}] \quad (22-1)$$

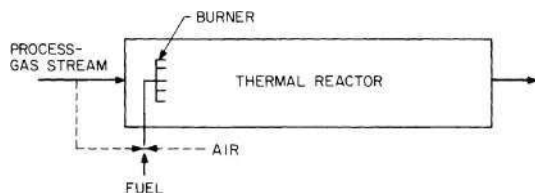


FIG. 22-13 Thermal-combustion device.

where k = pseudo-first-order rate constant (s^{-1}). If the initial concentration is $C_{A,0}$, the solution of Eq. (22-1) is:

$$\ln\left(\frac{C_A}{C_{A,0}}\right) = -kt \quad (22-2)$$

Equation (22-2) is frequently used for a kinetic modeling of a burner using mole fractions in the range of 0.15 and 0.001 for oxygen and HC, respectively. The rate constant is generally of the following Arrhenius form:

$$k = Ae^{-E/RT} \quad (22-3)$$

where: A = pre-exponential factor, s^{-1} (see Table 2, *Air Pollution Engineering Manual*, Van Nostrand Reinhold, New York, 1992, p. 62)

E = activation energy, cal/gmol (see Table 2, *Air Pollution Engineering Manual*, Van Nostrand Reinhold, New York, 1992, p. 62)

R = universal gas constant, 1.987 cal/gmol $^{\circ}K$

T = absolute temperature, $^{\circ}K$

The referenced table for A and E is a summary of first-order HC combustion reactions.

The end combustion products are continuously emitted at the outlet of the reactor. The average gas velocity can range from as low as 3 m/s (10 ft/s) to as high as 15 m/s (50 ft/s). These high velocities are required to prevent settling of particulates (if present) and to minimize the dangers of flashback and fire hazards. Space velocity calculations are given in the "Incinerator Design and Performance Equation" section.

The fuel is usually natural gas. The energy liberated by reaction may be directly recovered in the process or indirectly recovered by suitable external heat exchange (see Fig. 22-14).

Because of the high operating temperatures, the unit must be constructed of metals capable of withstanding this condition. Combustion

TABLE 22-33 Thermal Afterburners: Conditions Required for Satisfactory Performance in Various Abatement Applications

Abatement category	Afterburner residence time, s	Temperature, $^{\circ}F$
Hydrocarbon emissions: 90 + % destruction of HC	0.3–0.5	1100–1250*
Hydrocarbons + CO: 90 + % destruction of HC + CO	0.3–0.5	1250–1500
Odor		
50–90% destruction	0.3–0.5	1000–1200
90–99% destruction	0.3–0.5	1100–1300
99 + % destruction	0.3–0.5	1200–1500
Smokes and plumes		
White smoke (liquid mist)		
Plume abatement	0.3–0.5	800–1000†
90 + % destruction of HC + CO	0.3–0.5	1250–1500
Black smoke (soot and combustible particulates)	0.7–1.0	1400–2000

*Temperatures of 1400 to 1500 $^{\circ}F$ (760 to 816 $^{\circ}C$) may be required if the hydrocarbon has a significant content of any of the following: methane, cello-solve, and substituted aromatics (e.g., toluene and xylenes).

†Operation for plume abatement only is not recommended, since this merely converts a visible hydrocarbon emission into an invisible one and frequently creates a new odor problem because of partial oxidation in the afterburner.

devices are usually constructed with an outer steel shell that is lined with refractory material. Refractory-wall thickness is usually in the 0.05- to 0.23-m (2- to 9-in) range, depending upon temperature considerations.

Some of the advantages of the thermal incinerators are:

1. Removal of organic gases
2. Removal of submicrometer organic particles
3. Simplicity of construction
4. Small space requirements

Some of the disadvantages are:

1. High operating costs
2. Fire hazards
3. Flashback possibilities

Catalytic Incinerators Catalytic incinerators are an alternative to thermal incinerators. For simple reactions, the effect of the presence of a catalyst is to (1) increase the rate of the reaction, (2) permit the reaction to occur at a lower temperature, and (3) reduce the reactor volume.

In a typical catalytic incinerator for the combustion of organic vapors, the gas stream is delivered to the reactor continuously by a fan at a velocity in the range of 3 to 15 m/s (10 to 30 ft/s), but at a lower temperature, usually in the range of 350 to 425 $^{\circ}C$ (650 to 800 $^{\circ}F$), than the thermal unit. (Design and performance equations used for calculating space velocities are given in the next section). The gases, which may or may not be preheated, pass through the catalyst bed, where the combustion reaction occurs. The combustion products, which again are made up of water vapor, carbon dioxide, inerts and unreacted vapors, are continuously discharged from the outlet at a higher temperature. Energy savings can again be effected by heat recovery from the exit stream.

Metals in the platinum family are recognized for their ability to promote combustion at low temperatures. Other catalysts include various oxides of copper, chromium, vanadium, nickel, and cobalt. These catalysts are subject to poisoning, particularly from halogens, halogen and sulfur compounds, zinc, arsenic, lead, mercury, and particulates. It is therefore important that catalyst surfaces be clean and active to ensure optimum performance.

Catalysts may be porous pellets, usually cylindrical or spherical in shape, ranging from 0.16 to 1.27 cm ($\frac{1}{16}$ to $\frac{1}{2}$ in) in diameter. Small sizes are recommended, but the pressure drop through the reactor increases. Among other shapes are honeycombs, ribbons, and wire mesh. Since catalysis is a surface phenomenon, a physical property of these particles is that the internal pore surface is nearly infinitely greater than the outside surface.

The following sequence of steps is involved in the catalytic conversion of reactants to products:

1. Transfer of reactants to and products from the outer catalyst surface
2. Diffusion of reactants and products within the pores of the catalyst
3. Activated adsorption of reactants and the desorption of the products on the active centers of the catalyst

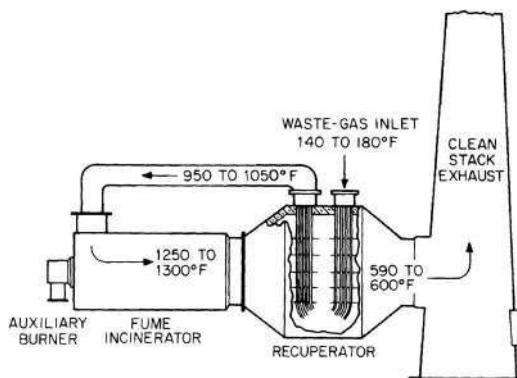


FIG. 22-14 Thermal combustion with energy (heat) recovery.

4. Reaction or reactions on active centers on the catalyst surface
At the same time, energy effects arising from chemical reaction can result in the following:

1. Heat transfer to or from active centers to the catalyst-particle surface
2. Heat transfer to and from reactants and products within the catalyst particle
3. Heat transfer to and from moving streams in the reactor
4. Heat transfer from one catalyst particle to another within the reactor

5. Heat transfer to or from the walls of the reactor

Some of the advantages of catalytic incinerators are:

1. Lower fuel requirements as compared with thermal incinerators
2. Lower operating temperatures
3. Minimum insulation requirements
4. Reduced fire hazards
5. Reduced flashback problems

The disadvantages include:

1. Higher initial cost than thermal incinerators
2. Catalyst poisoning
3. Necessity of first removing large particulates
4. Catalyst-regeneration problems
5. Catalyst disposal

Incinerator Design and Performance Equations The key incinerator design and performance calculations are the required fuel usage and physical dimensions of the unit. The following is a general calculation procedure to use in solving for these two parameters, assuming that the process gas stream flow, inlet temperature, combustion temperature, and required residence time are known. The combustion temperature and residence time for thermal incinerators can be estimated using Table 22-33 and Eqs. (22-1) through (22-3).

1. The heat load needed to heat the inlet process gas stream to the incinerator operating temperature is:

$$Q = \Delta H \quad (22-4)$$

2. Correct the heat load for radiant heat losses, RL :

$$Q = (1 + RL)(\Delta H); RL = \text{fractional basis} \quad (22-5)$$

3. Assuming that natural gas is used to fire the burner with a known heating value of HV_G , calculate the available heat at the operating temperature. A shortcut method usually used for most engineering purposes is:

$$HA_T = (HV_G) \left(\frac{HA_T}{HV_G} \right)_{\text{ref}} \quad (22-6)$$

where the subscript "ref" refers to a reference fuel. For natural gas with a reference HV_G of 1059 Btu/scf, the heat from the combustion using no excess air would be given by:

$$(HA_T)_{\text{ref}} = -0.237(t) + 981; T = F \quad (22-7)$$

4. The amount of natural gas needed as fuel (NG) is given by:

$$NG = \frac{Q}{HA}; \text{consistent units} \quad (22-8)$$

5. The resulting volumetric flow rate is the sum of the combustion of the natural gas q and the process gas stream p at the operating temperature:

$$q_T = q_p + q_c \quad (22-9)$$

A good estimate for q_c is:

$$q_c = (11.5)(NG) \quad (22-10)$$

6. The diameter of the combustion device is given by:

$$S = \frac{q_T}{v_t} \quad (22-11)$$

where v_t is defined as the velocity of the gas stream at the incinerator operating temperature.

Condensation Frequently in air-pollution-control practice, it becomes necessary to treat an effluent stream consisting of a con-

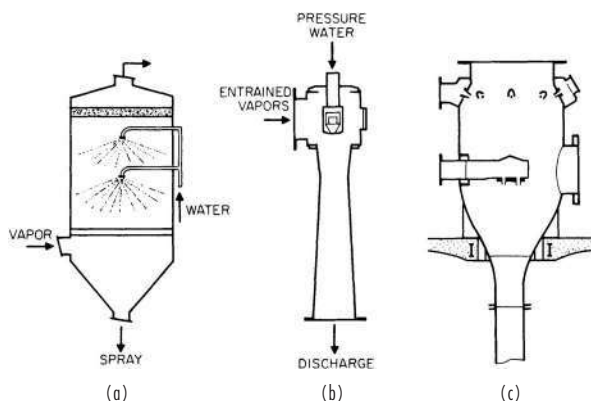


FIG. 22-15 Typical direct-contact condensers. (a) Spray chamber. (b) Jet. (c) Barometric.

densable pollutant vapor and a noncondensable gas. One control method to remove such pollutants from process-gas streams that is often overlooked is condensation. Condensers can be used to collect condensable emissions discharged to the atmosphere, particularly when the vapor concentration is high. This is usually accomplished by lowering the temperature of the gaseous stream, although an increase in pressure will produce the same result. The former approach is usually employed by industry, since pressure changes (even small ones) on large volumetric gas-flow rates are often economically prohibitive.

Condensation Equipment There are two basic types of condensers used for control: contact and surface. In contact condensers, the gaseous stream is brought into direct contact with a cooling medium so that the vapors condense and mix with the coolant (see Fig. 22-15). The more widely used system, however, is the surface condenser (or heat exchanger), in which the vapor and the cooling medium are separated by a wall (see Fig. 22-16). Since high removal efficiencies cannot be obtained with low-condensable vapor concentrations, condensers are typically used for pretreatment prior to some other more efficient control device such as an incinerator, absorber, or adsorber.

Contact Condensers Spray condensers, jet condensers, and barometric condensers all utilize water or some other liquid in direct contact with the vapor to be condensed. The temperature approach between the liquid and the vapor is very small, so the efficiency of the condenser is high, but large volumes of the liquid are necessary. If the vapor is soluble in the liquid, the system is essentially an absorptive one. If the vapor is not soluble, the system is a true condenser, in which case the temperature of the vapor must be below the dew point. Direct-contact condensers are seldom used for the removal of organic solvent vapors because the condensate will contain an organic-water mixture that must be separated or treated before disposal. They are, however, the most effective method of removing heat from hot gas streams when the recovery of organics is not a consideration.

In a direct-contact condenser, a stream of water or other cooling liquid is brought into direct contact with the vapor to be condensed. The liquid stream leaving the chamber contains the original cooling liquid plus the condensed substances. The gaseous stream leaving the

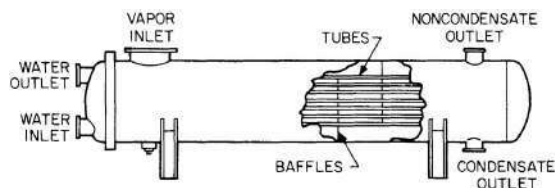


FIG. 22-16 Typical surface condenser (shell-and-tube).

chamber contains the noncondensable gases and such condensable vapor as did not condense; it is reasonable to assume that the vapors in the exit gas stream are saturated. It is then the temperature of the exit gas stream that determines the collection efficiency of the condenser.

The advantages of contact condensers are that (1) they can be used to produce a vacuum, thereby creating a draft to remove odorous vapors and also reduce boiling points in cookers and vats; (2) they usually are simpler and less expensive than the surface type; and (3) they usually have considerable odor-removing capacity because of the greater condensate dilution (13 lb of 60°F water is required to condense 1 lb of steam at 212°F and cool the condensate to 140°F). The principal disadvantage is the large water requirement. Depending on the nature of the condensate, odor in the wastewater can be offset by using treatment chemicals.

Direct-contact condensers involve the simultaneous transfer of heat and mass. Design procedures available for absorption, humidification, cooling towers, and the like may be applied with some modifications.

Surface Condensers Surface condensers (indirect-contact condensers) are used extensively in the chemical-process industry. They are employed in the air-pollution-equipment industry for recovery, control, and/or removal of trace impurities or contaminants. In the surface type, coolant does not contact the vapor condensate. There are various types of surface condensers including the shell-and-tube, fin-fan, finned-hairpin, finned-tube-section, and tubular. The use of surface condensers has several advantages. Salable condensate can be recovered. If water is used for coolant, it can be reused, or the condenser may be air-cooled when water is not available. Also, surface condensers require less water and produce 10 to 20 times less condensate. Their disadvantage is that they are usually more expensive and require more maintenance than the contact type.

Biological APC Technologies

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Bioscrubbers, biotrickling filters, and biofilters are biological APC technologies which use enzymatic catalytic oxidation to completely break down (metabolize) biodegradable air pollutants to H₂O, CO₂, and salts. These pollutants, present in the airstream as vapors, can be either organic or inorganic (H₂S, NH₃, CO, etc.). The oxidation catalysts (enzymes) are produced and maintained within moist, living, active biomass, which operates near ambient temperature and pressure. In practice, these biological APC processes represent a subclass of gas adsorption processes, in which pollutant mass transfer from the gas phase to the liquid phase is enhanced by a fast chemical reaction in the liquid phase, which contains the biomass. Unlike incineration, biodegradation of compounds containing sulfur, nitrogen, and halogens produces benign by-products, biomass, and inorganic ions (SO₄²⁻, NO₃⁻, Cl⁻, etc.), which require little or no subsequent treatment prior

to their release into the environment. Biological processes can be particularly desirable because they neither transfer the identified pollutants into another phase (as with carbon adsorption, condensation, etc.) nor produce additional, collateral air pollution from fuel combustion [as does incineration (SO₂, NO₂, CO, CO₂, particulates, etc.)].

Biomass: The Community of Microorganisms* The microorganisms which make up the biomass in these devices are similar to those found in soil and in biological wastewater treatment plants: bacteria, often with smaller populations of fungi, ciliates, and other protists. Bacteria are extremely diverse in their biochemical degradation abilities. They all lack cellular nuclei. Individually, a bacterium has limited metabolic capabilities, because a single bacterium has limited quantities of DNA. However, individual bacteria can modify their DNA by exchange and transfer with other bacteria present. Nevertheless, this limited DNA restricts the ability of one kind of bacterium to produce all the different enzymes needed to completely break down all available substrates. Consequently, at steady state, differing bacteria live together in mixed consortia (communities) such that, taken together, these consortia can produce all the enzymes needed to completely break down the available substrates to CO₂, H₂O, salts, etc.

In bioscrubbers, biotrickling filters, and biofilters, the principal microbial populations metabolize the primary substrates, the biodegradable air pollutants. The secondary and generally smaller microbial populations assist the process of total breakdown by consuming bacterial metabolic intermediates from the primary substrates as well as other waste products produced by the other bacteria. In biotrickling filters and biofilters, the biomass exists within the airstream as a fixed biofilm, a thin layer of microbes growing on an inert attachment surface. An important advantage of an attached biofilm is that the microbes tend to become distributed along chemical gradients by depth within the biofilm. Often, the most rapidly growing microbes are concentrated near the outer surface, where both oxygen and the primary substrate are most abundant, and the slower-growing, secondary microbes are concentrated near the attachment surface. This configuration protects the secondary microbes with an outer layer of expendable microbes. It also provides a nearly infinite mean cell residence time for the often slow-growing but essential secondary microbes, needed for complete breakdown of the pollutants. These bacterial populations are consumed by still other predatory microbial populations, which decompose them, recycling their substances to the consortia. Acting together, these microbial populations finish the complete breakdown of the primary substrates.

Biofiltration: Theoretical Considerations

Maximum biofilter inlet pollutant concentration For an aerobic fixed-film device, the local total flux for all pollutants into the biofilm must not exceed the local flux of oxygen into the biofilm, on a stoichiometric basis. Therefore, the maximum acceptable total pollutant concentration is fixed, at the biofilter inlet, by the low maximum oxygen solubility in the biofilm, at the biofilm interface. This oxygen concentration is about 8 mg/L (8 ppm) for air at ambient conditions. If the total pollutant concentration exceeds this maximum limit, aerobic biological conditions cannot be maintained throughout the biofilter, a condition which can result in biofilter failure.

For a single pollutant biofilter application, using the stoichiometric coefficients v for complete pollutant oxidation, diffusivities D , and molecular weights (MW), the maximum acceptable inlet biofilm interface pollutant POL concentration S_{POL} can be estimated from the maximum available biofilm interface oxygen concentration S_{O_2} :

$$S_{POL} = \left[\frac{v_{POL}}{v_{O_2}} \right] \left[\frac{D_{O_2}}{D_{POL}} \right] \left[\frac{MW_{POL}}{MW_{O_2}} \right] S_{O_2} \quad (22-12)$$

Pollutant solubility At these very low biofilm interface concentrations, biodegradable pollutant solubility in water can be described by using Henry's law:

$$p_i = H_i C_i \quad (22-13)$$

where p is the partial pressure of the solute i in the gas phase; C is the concentration of the solute in the aqueous liquid phase; and H is the

*The contributions to this subsection by Prof. Lynn Margulis, University of Massachusetts, Amherst, are gratefully appreciated.

proportionality constant, generally referred to as the Henry's law constant. (Henry's law constants are not true constants but rather functions of temperature.) Experimental data are needed to estimate H , however, because pollutant-water mixtures are highly nonideal. The best reference for these solubilities is corroborated data from a computerized data bank that specializes in environmental aqueous systems. Such data banks are especially helpful for calculating solubilities for uncommon pollutants of environmental interest, as well as for compounds that dissociate in water, i.e., are acidic or basic.

Estimating Henry's law constants Henry's law constants can be estimated from extensive data published for solute-water LLE (liquid-liquid equilibria):

$$H_i = \frac{(1 - x_{\text{water}})P_i^{\text{sat}}}{x_i} \quad (22-14)$$

where x_{water} is the equilibrium mole fraction of water in the organic phase (often negligible), P_i^{sat} is the vapor pressure of the organic solute, and x_i is the equilibrium mole fraction of the solute in the water phase.

Extensive, reviewed tabulated pollutant data also exist (see subsection "Solubilities" in Sec. 2 of this handbook). Any single tabulated Henry's law constant H is published with its measurement temperature, typically 20 or 25°C. The constant H for a subcritical pollutant (i.e., toluene) is proportional to its pure component vapor pressure P^{sat} , and therefore H is an exponential function of temperature T . Using Henry's and Raoult's laws, a pollutant's constant H for a given ambient design temperature can be estimated by using its tabulated value and its vapor pressure, by extrapolating $\leq 25^\circ\text{C}$ over the ambient temperature range ($4^\circ\text{C} < T < 50^\circ\text{C}$), from T_1 to T_2 :

$$H_2 = \frac{\gamma_2 P_2^{\text{sat}}}{\gamma_1 P_1^{\text{sat}}} H_1 \equiv \frac{P_2^{\text{sat}}}{P_1^{\text{sat}}} H_1 \quad (22-15a)$$

For a more soluble (or completely miscible) pollutant (i.e., ethanol), a Henry's law constant can be similarly estimated by using its tabulated infinite dilution activity coefficient γ_∞ , measured at T_1 , by extrapolation using P_2^{sat} , from T_1 to T_2 :

$$H_2 = \gamma_{\infty 2} P_2^{\text{sat}} \equiv \gamma_{\infty 1} P_1^{\text{sat}} \quad (22-15b)$$

Such estimations for H_2 are single-point extrapolations of nonlinear functions, and therefore should be used with caution. In both cases estimation error is introduced due to the nonlinear temperature dependence of the activity coefficient.

Biofilter kinetics and design model These biological systems are complex, so that normal simplifying assumptions are approximate at best: biomass homogeneity, independent and uniform multisubstrate kinetics, etc. Nevertheless, for understanding the basics of these bioabsorbers, it is helpful to consider them in terms of a single substrate system, with uniform classic kinetics and mass transfer, and no liquid layer external to the fixed biofilm. Chemical and environmental engineers use different expressions to describe the kinetics of substrate (pollutant) utilization—the shifting order form and the Monod (or Michaelis-Menten) form:

$$\frac{d(S_f)}{dt} = \frac{k_1 S_f}{1 + k_2 S_f} = \frac{k X_f S_f}{K_s + S_f} \quad (22-16)$$

where the variables are rate-limiting substrate concentration S_f , first-order reaction rate coefficient $k^1 = k_1$, zero-order rate coefficient $k^0 = k_1/k_2$, maximum utilization rate coefficient for the rate-limiting substrate k , density of the active substrate degraders X_f , and Monod half-velocity coefficient K_s .

Biofilters are chemically enhanced absorbers, and therefore mass transfer limited (see "Absorption with Chemical Reaction" in Sec. 14). The magnitudes for the Hatta [= Damkohler II = (Thiele modulus)²] numbers are quite low, perhaps below 5. Nevertheless, for design simplicity, mass-transfer limitation is generally assumed to be in the liquid phase (the biofilm). For a single-component biofilter, the simplified biofilter model and design equation is

$$\tau = \frac{H}{k_L a_f} \ln \left| \frac{S_{g(\text{IN})}}{S_{g(\text{OUT})}} \right| \quad (22-17)$$

where the variables are the empty-bed residence time τ , a dimensionless variable H , liquid-phase local mass-transfer coefficient k_L , and air-liquid phase effective specific surface a_f . (The units used for the dimensionless H must be the same as those used for S_g [Eq. (22-17)].) If the mass-transfer coefficient were for an ideal, single-component, fast first-order reaction system, for a biofilm having no external liquid layer of any significant thickness (i.e., not more than a moist surface), and all other variables are assumed to have constant magnitudes, then

$$k_L = \sqrt{k^1 D_f} = \sqrt{\frac{k X_f D_f}{K_s}} \quad (22-18a)$$

where D_f is the pollutant diffusion coefficient within the biofilm and k_L has a single magnitude, a function of temperature. For a single component following Monod reaction kinetics, k_L is defined as

$$k_L = \sqrt{\frac{2k X_f D_f}{S_i^2} \left(S_i - S_{\text{min}} - K_s \ln \frac{K_s + S_i}{K_s + S_{\text{min}}} \right)} \quad (22-18b)$$

where S_i and S_{min} are the pollutant biofilm concentrations, at the biofilm interface (maximum) and at the biofilm attachment surface (minimum), respectively. All other variables are assumed to have constant magnitudes throughout the biofilter. (For a "deep biofilm," S_{min} is assumed to approach the limit of zero.) Therefore, k_L is a function of the local pollutant concentration as well as a function of temperature. Note that Eq. (22-18b) predicts a lower value for k_L than does Eq. (22-18a), until the pollutant concentration approaches the limit of zero, where they are equal.

General Process Description For any APC design it is difficult to accurately characterize variations of temperature, flow rate, and composition for the polluted source stream, especially during the engineering design of the upstream process source. For biological APC design is added the state of the art, the imperfect scientific understanding of the behavior of biomass under long-term steady-state operation, but more importantly as the biomass and media respond to stream variations, especially significant step change variations. This APC field remains a technical art. The magnitude of $k_L a_f$ cannot be calculated with confidence for any media formulation from first principles, especially for multiple pollutants. Consequently, and especially for multiple components, responsible vendors can be expected to recommend field pilot testing for unfamiliar applications, as an appropriate expense for achieving a minimum annualized system cost design. What follows is written to assist chemical engineers to effectively discuss their project needs with responsible vendors of biological APC technologies, and to comprehend what they receive from such vendors.

Most natural, and most artificial, organic compounds can be biodegraded. However, only some biodegradable compounds are candidates for economical biological APC, which typically means a residence time measured in minutes [Eq. (22-17)]. Although $k_L a_f$ is important, its practical range of magnitudes is modest, and candidate suitability for treatment is principally determined by its solubility, coupled with the ratio of the design inlet and outlet concentrations. Note, for instance, that economical designs are achievable for pollutants with low concentrations and low solubility (flavors and fragrances) or for high concentrations and high solubility (ethanol, butyric acid). In practice, compounds such as ethanol, acetone, formaldehyde, acetic acid, and diethylether are good candidates; BTEX, pinenes, and similarly soluble compounds are acceptable candidates; and largely nonpolar hydrocarbons such as alkanes (and highly halogenated compounds in general) are poor candidates. The most expensive media typically can offer the highest values for $k_L a_f$ and are generally justified through reduction of the required EBRT (reduced media volume) needed for a specific application.

An ID fan, which minimizes leakage of polluted air, is normally used to maintain airflow through the process and to overcome the total system pressure drop, typically measured in centimeters of water. The optimum design temperature is determined by the trade-off between solubility, which decreases with temperature, and biological kinetics, which increases. Another optimization factor is how much heating or cooling is necessary for the inlet stream. Operation is

generally controlled to within a narrow range of a couple of degrees. The selected design temperature may range from perhaps 20 to 45°C, commonly 30 to 35°C. Particulates or aerosols, including organic liquid droplets, should be removed (perhaps $> 15 \mu\text{m}$). The inlet stream is usually prehumidified to 95 to nearly 100 percent relative humidity. Finally, temperature, particulate, and humidity control is often managed in a single step.

Biofilters Biofilters fall into two categories: soil bed biofilter (open bed) and artificial media biofilters (confined media).

Soil bed biofilters consist of an air distribution system of perforated piping buried under a layer of soil media, typically exposed outdoors. The treated air is emitted directly to atmosphere (Fig. 22-17). The soil media used can consist of a blend of natural soil components and other materials. This is a fixed-film bioreactor, with the biofilm attached to other inert soil materials. Soil media are biologically self-sustaining, hydrophilic, and self-buffering and often last for decades of use. Over a hundred are in operation. Although most soil beds are employed for odor control, any otherwise suitable candidate pollutant should be treatable in a soil bed. Soil bed operating temperature is maintained by inlet air temperature control, and soil biofilters are practical for any climate, including very cold conditions (i.e., Calgary, Alberta, Canada).

Artificial media biofilters employ a biologically active medium housed in a containment vessel (Fig. 22-18). The support media can consist of natural and synthetic materials, including compost, peat, bark, wood chips, etc., and polymer foam, activated carbon, rubber tire waste, ceramic pellets, etc., respectively. The media are typically blended, wetted, and biologically seeded (inoculated) before installation. Inert materials, such as polystyrene beads, may be blended with the media to prevent compaction and pressure drop increase and to maintain biofilm specific surface. Media moisture content can be a critical operating parameter. Secondary humidification is often employed, via water spraying, which can be accurately controlled by weigh cell measurement of the total media bed mass. Media replacement may be required 1 to 2 times per decade. Depending on the media and the pollutants, nutrients and/or pH buffers may be applied with the water spray. Some biofilters are maintained wet enough to drip, to purge soluble waste salts. Generally, artificial media biofilters can be designed for higher loadings than can soil bed biofilters. Upflow and downflow designs are practical.

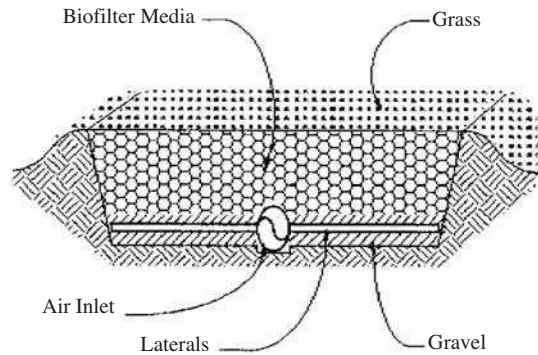


FIG. 22-17 Pretreated contaminated air flows into an inlet manifold, lateral piping, and into a continuous layer of gravel or crushed stone, providing even air distribution across the bottom of the media. Air is biofiltered, flowing upward through the biofilter media, exiting to atmosphere through the media surface or a grass cover. (Courtesy of Bohn Biofilter, www.bohnbiofilter.com.)

Bioscrubbers Bioscrubbers combine physical absorption with biological degradation, using two separate unit operations. The pollutants are absorbed into water within the scrubber. This water is then directed to an external biological reactor, where the pollutants are degraded. The treated liquid is recycled to the bioscrubber. One configuration employs water and a fixed-film bioreactor, while a second configuration employs MLSS and a slurry reactor. In both cases, separating absorption from reaction permits scrubbing pollutant concentrations far above those which a fixed-film system would survive (due to O_2 limitation). Scrubbing can be designed to attenuate substantial and rapid composition change in the inlet, assisting the performance of a downstream biotreatment step. The scrubber can be multistaged (packed column, Fig. 22-18) or single-staged (a venturi scrubber or spray chamber, Fig. 22-19). Scrubbers with horizontal airflow are also available. Nutrients, pH buffers, and makeup water are added, and biomass and waste salts are purged, as needed. Bioscrubbing is commonly combined with

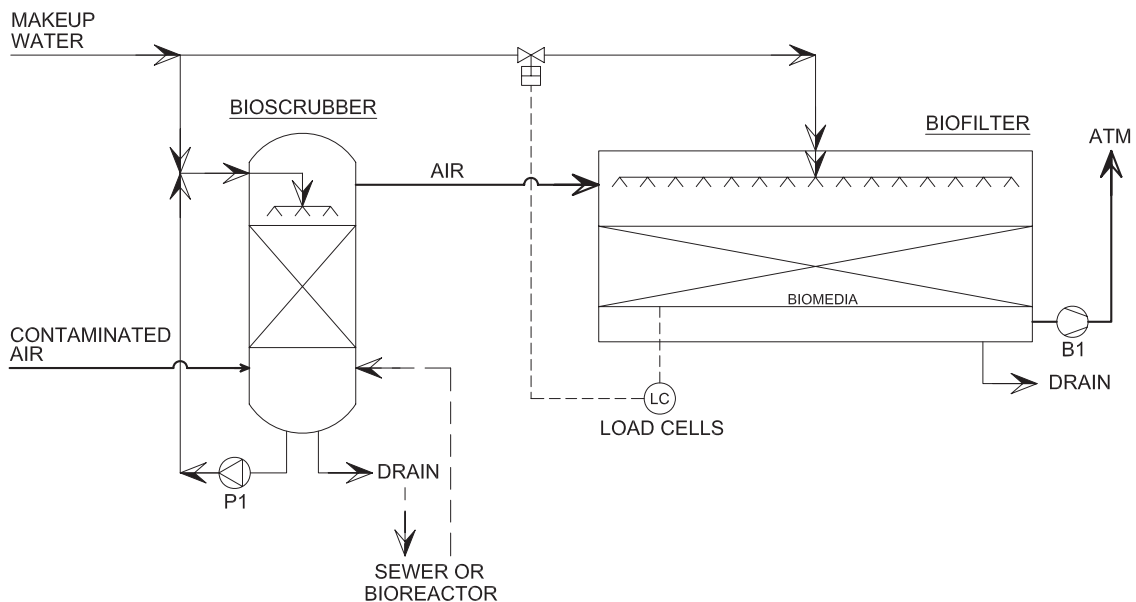


FIG. 22-18 Contaminated air is pretreated (shown: a countercurrent bioscrubber), before passing into the biofilter. Air is biofiltered, flowing downward through the media. Biomedia moisture content is accurately controlled via load-cell control of water sprays on media surface. (Courtesy of Waterleau Biofilter-Bioton, Bel-Air, www.waterleau.com.)

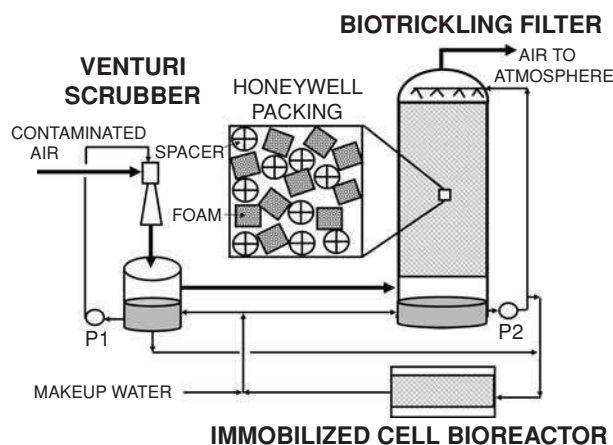


FIG. 22-19 Contaminated air is pretreated (shown: a Venturi scrubber), before passing into the biotrickling filter. Air is biofiltered, flowing downward through the media. (Courtesy of Honeywell PAI—Biological Air Treatment System, www.honeywellpai.com.)

temperature, particulate, and humidity control as feed pretreatment to biotrickling filters and biofilters. A combined multipollutant plot of $\log H = f(1/T)$ is a helpful design aid in the selection of the optimum stream composition for the scrubber effluent stream.

Biotrickling Filters Biotrickling filters are a hybrid of bioscrubbers and biofilters. A fixed biofilm (often a deep biofilm) is employed, typically attached to a synthetic medium containing blocks of polymer foam, lava rock, sintered ceramics, etc. (Fig. 22-19). Water permeability is preferred for the biofilm support media. Inert materials, such as Pall rings or other distillation-type packing materials, may be blended with the media to prevent compaction and pressure drop increase and to maintain biofilm specific surface. Water is recirculated through the filter. Although the water layer forms a resistance to pollutant mass transfer into the fixed biofilm, the water stream also absorbs pollutants which can be removed from the filter for separate biological treatment, as described above for bioscrubbers. As for a bioscrubber, this hybrid can respond more effectively to rapid changes in the feed, and can survive

higher inlet concentrations, than can a biofilter. This feature, in turn, can assist the performance of a downstream biotreatment step. Nutrients, pH buffers and makeup water are added, and biomass and waste salts are purged, as needed. Upflow, downflow, and horizontal airflow designs are practical.

Use of Activated Carbon Activated carbon (AC) can be used to enhance these processes in at least three ways. Positioned in front of the process, AC can provide pollutant concentration capacitance, reducing concentration peak magnitudes and rapid rate of variation and providing minimum concentrations during periods of no or low source pollutant flow. Positioned behind the process, AC can increase the overall process elimination efficiency, as well as scavenging nonbiologically degradable compounds (i.e., poly-halogenated organics). Mixed with the media, AC is reported to combine these two functions, enhancing biological removal and protecting the system during periods of high pollutant inlet variation. Note that AC positioned in front can present a fire hazard for certain pollutants, such as acetone. AC positioned in back is exposed to very high humidity, which restricts adsorption capacity.

Relative Biofiltration Treatment Costs The true final annualized cost for treating a polluted airstream with any specific APC technology is complex and will be affected by many factors, including project-specific factors. Nevertheless, Fig. 22-20 can provide helpful guidance to the relationships between the probable final costs for treating a polluted airstream using different APC technologies.

Membrane Filtration Membrane systems have been used for several decades to separate colloidal and molecular slurries by the chemical process industries (CPI). Membrane filtration was not viewed as a commercially viable pollution control technology until recently. This conventional wisdom was due to fouling problems exhibited when handling process streams with high solid content. This changed with the advent of membranes composed of high-flux cellulose acetate. In Europe and Asia, membrane filtration systems have been used for several years, primarily for ethanol dewatering for synthetic fuel plants. Membrane systems were selected in these cases for their (1) low energy utilization; (2) modular design; (3) low capital costs; (4) low maintenance; and (5) superior separations. In the United States, as EPA regulations become increasingly stringent, a renewed interest in advanced membrane filtration systems has occurred. In this case, the driving factors are the membrane system's ability to operate in a pollution-free, closed-loop manner with minimum wastewater output. Low capital and maintenance costs also result from the small amount of moving parts within the membrane systems.

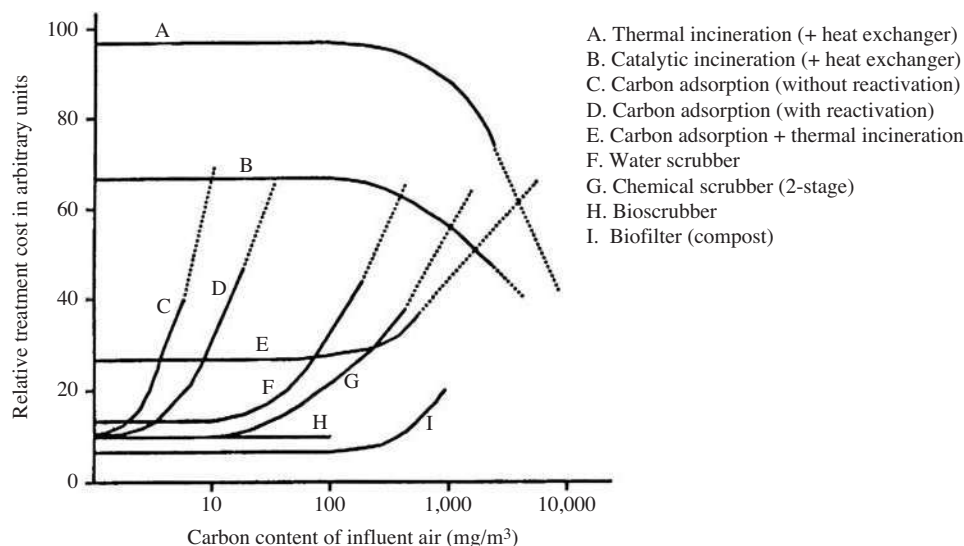


FIG. 22-20 Relative (and approximate) treatment costs for removing VOCs from air (Kosky and Neff, 1988). Assumption: 40,000 m³/h at 1982 price level. (Note: Current costs for chemical scrubbing can be 2 or more times greater than for biofiltration.)

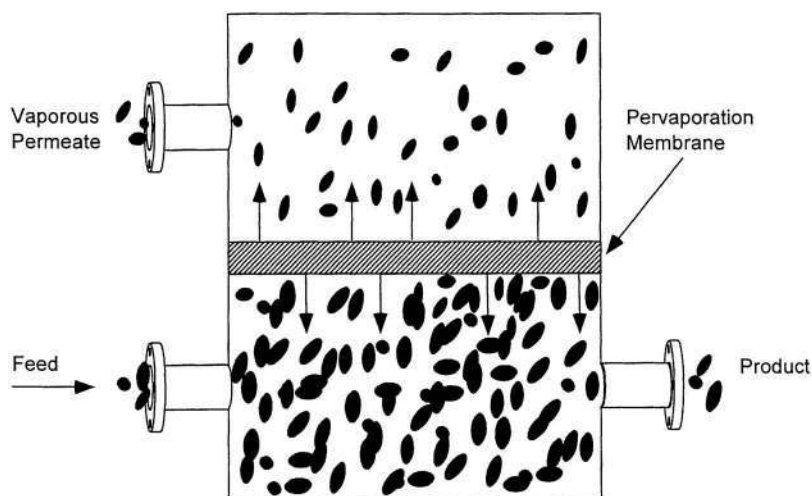


FIG. 22-21 Pervaporation of gas from liquid feed across membrane to vaporous permeate. (SOURCE: Redrawn from Ref. 24.)

Process Descriptions Selectively permeable membranes have an increasingly wide range of uses and configurations as the need for more advanced pollution control systems are required. There are four major types of membrane systems: (1) pervaporation; (2) reverse osmosis (RO); (3) gas absorption; and (4) gas adsorption. Only membrane pervaporation is currently commercialized.

Membrane Pervaporation Since 1987, membrane pervaporation has become widely accepted in the CPI as an effective means of separation and recovery of liquid-phase process streams. It is most commonly used to dehydrate liquid hydrocarbons to yield a high-purity ethanol, isopropanol, and ethylene glycol product. The method basically consists of a selectively-permeable membrane layer separating a liquid feed stream and a gas phase permeate stream as shown in Fig. 22-21. The permeation rate and selectivity is governed by the physicochemical composition of the membrane. Pervaporation differs from reverse osmosis systems in that the permeate rate is not a function of osmotic pressure, since the permeate is maintained at saturation pressure (Ref. 24).

Three general process groups are commonly used when describing pervaporation: (1) water removal from organics; (2) organic removal from water (solvent recovery); and (3) organic/organic separation. Organic/organic separations are very uncommon and therefore will not be discussed further.

Ethanol Dehydration The membrane-pervaporation process for dehydrating ethanol was first developed by GFT in West Germany in the mid 1970s, with the first commercial units being installed in Brazil and the Philippines. At both sites, the pervaporation unit was coupled to a continuous sugarcane fermentation process that produced ethanol at concentrations up to 96 percent after vacuum pervaporation. The key advantages of the GFT process are (1) no additive chemicals are required; (2) the process is skid-mounted (low capital costs and small footprint); (3) and there is a low energy demand. The low energy requirement is achieved because only a small fraction of the water is actually vaporized and that the required permeation driving force is provided by only a small vacuum pump. The basic ethanol dehydration process schematic is shown in Fig. 22-22. A key advantage of all pervaporation processes is that vapor-liquid equilibria and possible resulting azeotropic effects are irrelevant (see Fig. 22-23 and Ref. 24).

Solvent Recovery The largest current industrial use of pervaporation is the treatment of mixed organic process streams that have become contaminated with small (10 percent) quantities of water. Pervaporation becomes very attractive when dehydrating streams down to less than 1 percent water. The advantages result from the small operating costs relative to distillation and adsorption. Also, dis-

tillation is often impossible, since azeotropes commonly form in multicomponent organic/water mixtures.

Pervaporation occurs in three basic steps:

1. Preferential sorption of chemical species
2. Diffusion of chemical species through the membrane
3. Desorption of chemical species from the membrane

Steps 1 and 2 are controlled by the specific polymer chemistry and its designed interaction with the liquid phase. The last step, consisting of evaporation of the chemical species, is considered to be a fast, nonselective process. Step 2 is the rate-limiting step. The development by the membrane manufacturers of highly selective, highly permeable composite membranes that resist fouling from solids has subsequently been the key to commercialization of pervaporation systems. The membrane composition and structure are designed in layers, with each layer fulfilling a specific requirement. Using membrane dehydration as an example, a membrane filter would be composed of a support layer of nonwoven porous polyester below a layer of polyacrylonitrile (PAN) or polysulfone ultrafiltration membrane and a layer of 0.1- μm -thick crosslinked polyacrylate, polyvinyl alcohol (PVA). Other separations membranes generally use the same two sublayers. The top layer is interchanged according to the selectivity desired.

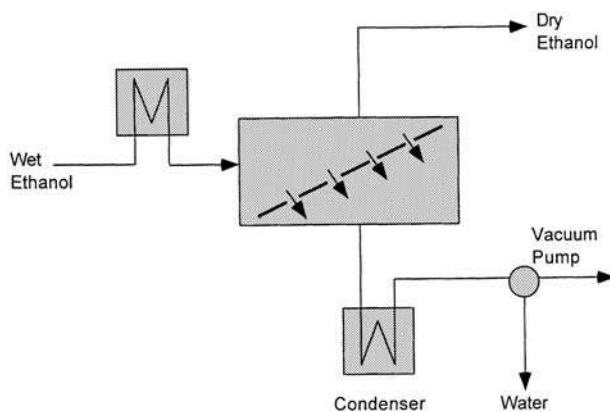


FIG. 22-22 Ethanol dehydration using pervaporation membrane. (SOURCE: Redrawn from Ref. 24.)

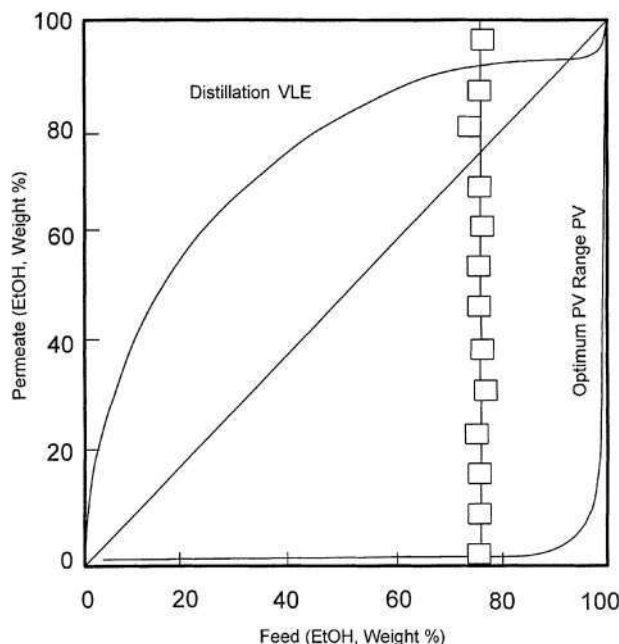


FIG. 22-23 Comparison of two types of pervaporation membranes to distillation of ethanol-water mixtures. (SOURCE: Redrawn from Ref. 24.)

Emerging Membrane Control Technologies The recent improvements in membrane technology have spawned several potentially commercial membrane filtration uses.

Reverse Osmosis (RO) Membranes A type of membrane system for treating oily wastewater is currently undergoing commercialization by Bend Research, Inc. The system uses a tube-side feed module that yields high fluxes while being able to handle high-solids-content waste streams (Ref. 25). Another type of reverse osmosis technique is being designed to yield ultrapurified HF recovered from spent etching solutions. It is estimated that 20,000 tons of spent solution is annually generated in the United States and that using membrane RO could save 1 million bbl/yr of oil.

In Situ Filter Membranes In situ membranes are being fitted into incinerator flue-gas stacks in an attempt to reduce hydrocarbon emissions. Two types of commercially available gas separation membranes are being studied: (1) flat cellulose acetate sheets; and (2) hollow-tube fiber modules made of polyamides.

Vibratory Shear-Enhanced Membranes The vibratory shear-enhancing process (VSEP) is just starting commercialization by Logic International, Emeryville, CA. It employs the use of intense sinusoidal shear waves to ensure that the membrane surfaces remain active and clean of solid matter. The application of this technology would be in the purification of wastewater (Ref. 2).

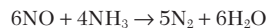
Vapor Permeation Vapor permeation is similar to vapor pervaporation except that the feed stream for permeation is a gas. The future commercial viability of this process is based upon energy and capital costs savings derived from the feed already being in the vapor-phase, as in fractional distillation, so no additional heat input would be required. Its foreseen application areas would be the organics recovery from solvent-laden vapors and pollution treatment. One commercial unit was installed in Germany in 1989 (Ref. 26).

Selective Catalytic Reduction of Nitrogen Oxides The traditional approach to reducing ambient ozone concentrations has been to reduce VOC emissions, an ozone precursor. In many areas, it has now been recognized that elimination of persistent exceedances of the National Ambient Air Quality Standard for ozone may require more

attention to reductions in the other ingredients in ozone formation, nitrogen oxides (NO_x). In such areas, ozone concentrations are controlled by NO_x rather than VOC emissions.

Selective catalytic reduction (SCR) has been used to control NO_x emissions from utility boilers in Europe and Japan for over a decade. Applications of SCR to control process NO_x emissions in the chemical industry are becoming increasingly common. A typical SCR system is shown in Fig. 22-24.

NO_x -laden fumes are preheated by effluent from the catalyst vessel in the feed/effluent heat exchanger and then heated by a gas- or oil-fired heater to over 600°F. A controlled quantity of ammonia is injected into the gas stream before it is passed through a metal oxide, zeolite, or promoted zeolite catalyst bed. The NO_x is reduced to nitrogen and water in the presence of ammonia in accordance with the following exothermic reactions:



NO_x analyzers at the preheater inlet and catalyst vessel outlet monitor NO_x concentrations and control the ammonia feed rate. The effluent gives up much of its heat to the incoming gas in the feed/effluent exchanger. The vent gas is discharged at about 350°F.

SOURCE CONTROL OF PARTICULATE EMISSIONS

There are four conventional types of equipment used for the control of particulate emissions:

1. Mechanical collectors
2. Wet scrubbers
3. Electrostatic precipitators
4. Fabric filters

Each is discussed in Sec. 17 of this handbook under "Gas-Solids Separations." The effectiveness of conventional air-pollution-control equipment for particulate removal is compared in Fig. 22-25. These fractional efficiency curves indicate that the equipment is least efficient in removing particulates in the 0.1- to 1.0- μm range. For wet scrubbers and fabric filters, the very small particulates (0.1 μm) can be efficiently removed by brownian diffusion. The smaller the particulates, the more intense their brownian motion and the easier their collection by diffusion forces. Larger particulates (>1 μm) are collected principally by impaction, and removal efficiency increases with particulate size. The minimum in the fractional efficiency curve for scrubbers and filters occurs in the transition range between removal by brownian diffusion and removal by impaction.

A somewhat similar situation exists for electrostatic precipitators. Particulates larger than about 1 μm have high mobilities because they are highly charged. Those smaller than a few tenths of a micrometer can achieve moderate mobilities with even a small charge because of aerodynamic slip. A minimum in collection efficiency usually occurs in the transition range between 0.1 and 1.0 μm . The situation is further complicated because not all particulates smaller than about 0.1 μm acquire charges in an ion field. Hence, the efficiency of removal of very small particulates decreases after reaching a maximum in the submicrometer range.

The selection of the optimum type of particulate collection device (i.e., ESP or fabric filter baghouse) is often not obvious without conducting a site-specific economic evaluation. This situation has been brought about by both the recent reductions in the allowable emissions levels and advancements with fabric filter and ESP technologies. Such technoeconomic evaluations can result in application and even site-specific differences in the final optimum choice (see *Precip Newsletter*, 220, June, 1994 and *Fabric Filter Newsletter*, 223, June, 1994).

Improvements in existing control technology for fine particulates and the development of advanced techniques are top-priority research goals. Conventional control devices have certain limitations. Precipitators, for example, are limited by the magnitude of charge on

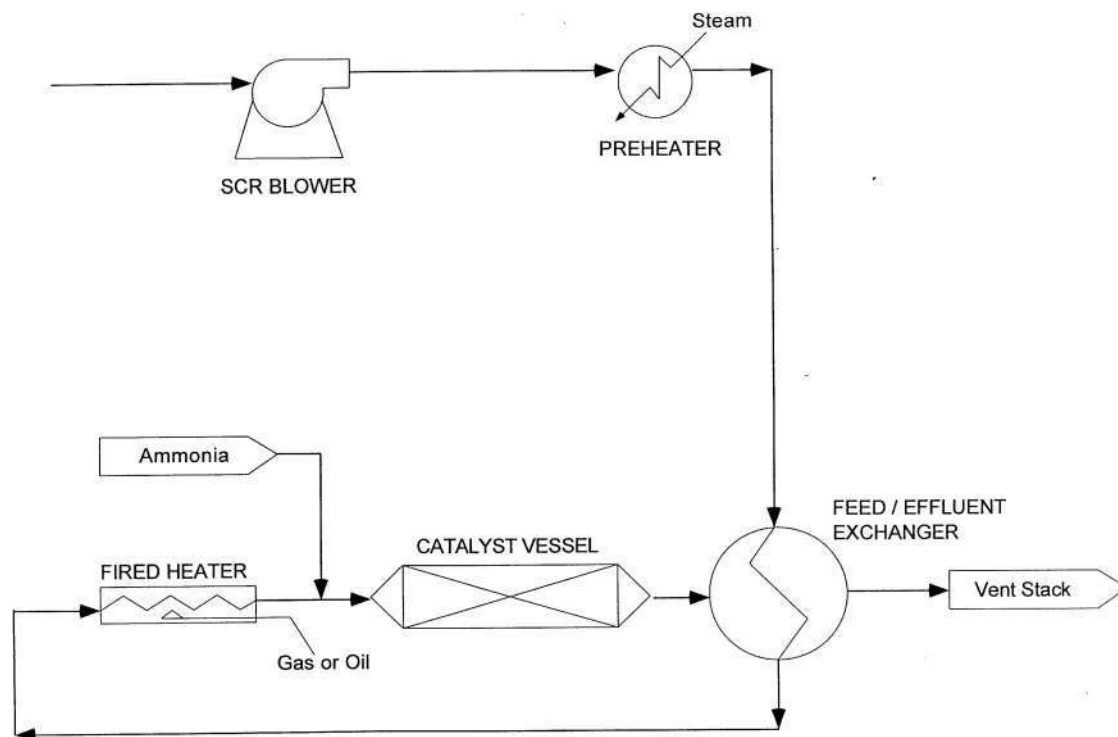


FIG. 22-24 Selective catalytic reduction of nitrogen oxides.

the particulate, the electric field, and dust reentrainment. Also, the resistivity of the particulate material may adversely affect both charge and electric field. Advances are needed to overcome resistivity and extend the performance of precipitators not limited by resistivity (see Buonicore and Theodore, "Control Technology for Fine Particulate Emissions," DOE Rep. ANL/ECT-5, Argonne National Laboratory, Argonne, Illinois, October 1978). Recent design developments with the potential to improve precipitator performance include pulse energization, electron beam ionization, wide plate spacing, and precharged units [Balakrishnan et al., "Emerging Technologies for Air Pollution Control," *Pollut. Eng.*, **11**, 28-32 (Nov. 1979); "Pulse Energization," *Environ. Sci. Technol.* **13**(9), 1044 (1974); and Midkaff, "Change in Precipitator Design Expected to Help Plants Meet Clean Air Laws," *Power*, **126**(10), 79 (1979)]; Robert A. Mastropietro, "Impact of ESP Performance on PM_{2.5}," European Particulate Control Users Group Meeting, Pisa, Italy, Nov. 7, 2000.

Fabric filters are limited by physical size and bag-life considerations. Some sacrifices in efficiency might be tolerated if higher air-cloth ratios could be achieved without reducing bag life (improved pulse-jet systems). Improvements in fabric filtration may also be possible by enhancing electrostatic effects that may contribute to rapid formation of a filter cake after cleaning. J.C. Mycock, J. Turner, and J. Farmer, "Baghouse Filtration Products Verification Testing, How It Benefits the Boiler Baghouse Operator," Council of Industrial Boiler Owners Conference, May 6, 2002; C. Jean Bustard et al. "Long Term Evaluation of Activated Carbon Injection for Mercury Control Upstream of a Cohnpac Fabric Filter," Air Quality IV, Arlington, Va., Sept. 22, 2003.

Scrubber technology is limited by scaling and fouling, overall reliability, and energy consumption. The use of supplementary forces acting on particulates to cause them to grow or otherwise be more easily collected at lower pressure drops is being closely investigated. The development of electrostatic and flux-force-condensation scrubbers is a step in this direction.

The electrostatic effect can be incorporated into wet scrubbing by charging the particulates and/or the scrubbing-liquor droplets. Electrostatic scrubbers may be capable of achieving the same efficiency for fine-particulate removal as is achieved by high-energy scrubbers, but at substantially lower power input. The major drawbacks are increased maintenance of electrical equipment and higher capital cost.

Flux-force-condensation scrubbers combine the effects of flux force (diffusiophoresis and thermophoresis) and water-vapor condensation. These scrubbers contact hot, humid gas with subcooled liquid, and/or they inject steam into saturated gas, and they have demonstrated that a number of these novel devices can remove fine particulates (see Fig. 22-26). Although limited in terms of commercialization, these systems may find application in many industries.

EMISSIONS MEASUREMENT

Introduction An accurate quantitative analysis of the discharge of pollutants from a process must be determined prior to the design and/or selection of control equipment. If the unit is properly engineered by utilizing the emission data as input to the control device and the code requirements as maximum-effluent limitations, most pollutants can be successfully controlled.

Sampling is the keystone of source analysis. Sampling methods and tools vary in their complexity according to the specific task; therefore, a degree of both technical knowledge and common sense is needed to design a sampling function. Sampling is done to measure quantities or concentrations of pollutants in effluent gas streams, to measure the efficiency of a pollution-abatement device, to guide the designer of pollution-control equipment and facilities, and/or to appraise contamination from a process or a source. A complete measurement requires a determination of the concentration and contaminant characteristics as well as the associated gas flow. Most statutory limitations require mass rates of emissions; both concentration and volumetric-flow-rate data are therefore required.

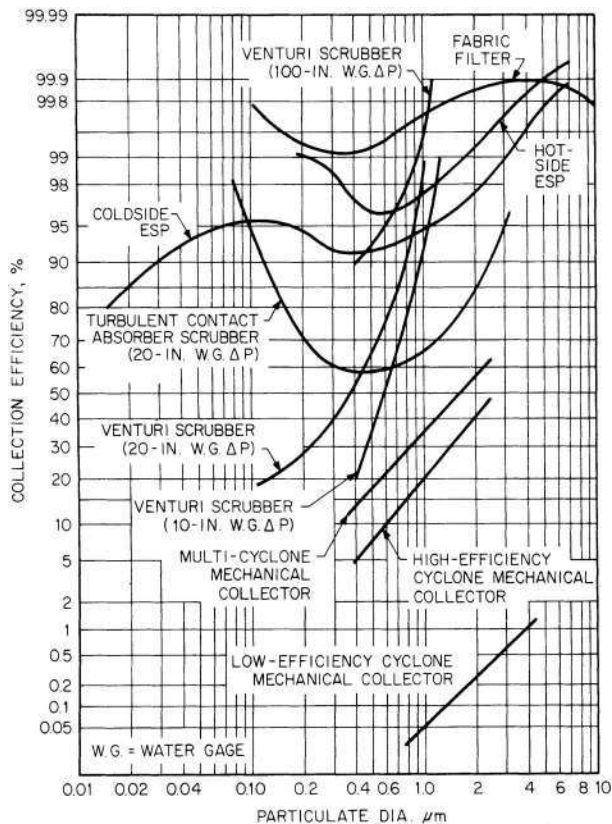


FIG. 22-25 Fractional efficiency curves for conventional air-pollution-control devices. [Chem. Eng., 87(13), 83 (June 30, 1980).]

The selection of a sampling site and the number of sampling points required are based on attempts to get representative samples. To accomplish this, the sampling site should be at least eight stack or duct diameters downstream and two diameters upstream from any flow disturbance, such as a bend, expansion, contraction, valve, fitting, or visible flame.

Once the sampling location has been decided on, the flue cross section is laid out in a number of equal areas, the center of each being the point where the measurement is to be taken. For rectangular stacks, the cross section is divided into equal areas of the same shape, and the traverse points are located at the center of each equal area, as shown in Fig. 22-27. The ratio of length to width of each elemental area should be selected. For circular stacks, the cross section is divided into equal annular areas, and the traverse points are located at the centroid of each area. The location of the traverse points as a percentage of diameter from the inside wall to the traverse point for circular-stack sampling is given in Table 22-34. The number of traverse points necessary on each of two perpendiculars for a particular stack may be estimated from Fig. 22-28.

Once these traverse points have been determined, velocity measurements are made to determine gas flow. The stack-gas velocity is usually determined by means of a pitot tube and differential-pressure gauge. When velocities are very low (less than 3 m/s [10 ft/s]) and when great accuracy is not required, an anemometer may be used. For gases moving in small pipes at relatively high velocities or pressures, orifice-disk meters or venturi meters may be used. These are valuable as continuous or permanent measuring devices.

Once a flow profile has been established, sampling strategy can be considered. Since sampling collection can be simplified and greatly

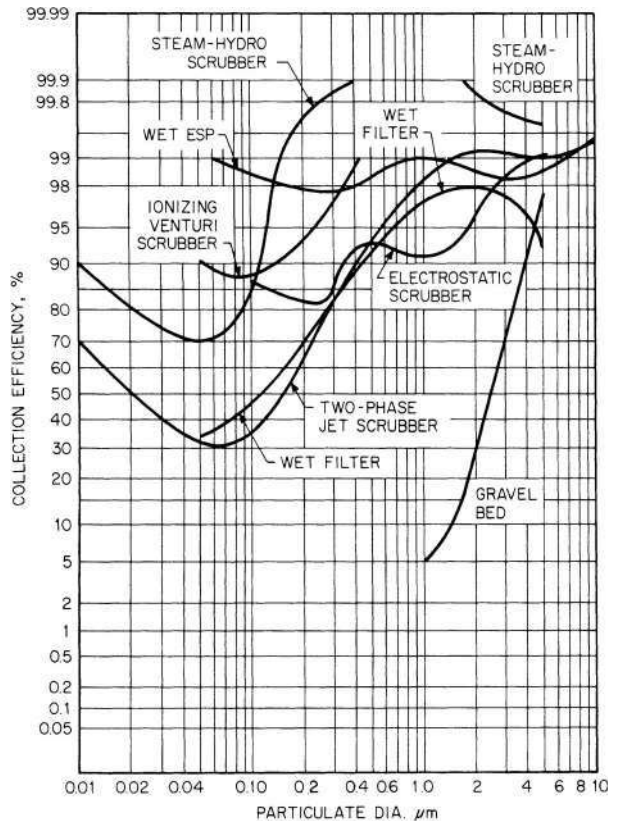


FIG. 22-26 Fractional efficiency curves for novel air-pollution-control devices. [Chem. Eng., 87(13), 85 (June 30, 1980).]

reduced depending on flow characteristics, it is best to complete the flow-profile measurement before sampling or measuring pollutant concentrations.

Sampling Methodology The following subsections review the methods specified for sampling commonly regulated pollutants as well as sampling for more exotic volatile and semivolatile organic compounds. In all sampling procedures, the main concern is to obtain a representative sample; the U.S. EPA has published

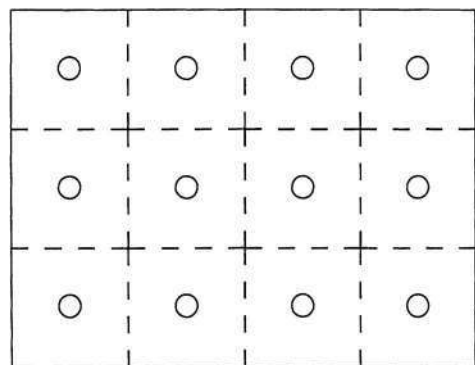


FIG. 22-27 Example showing rectangular stack cross section divided into 12 equal areas, with a traverse point at centroid of each area.

TABLE 22-34 Location of Traverse Points in Circular Stacks

(Percent of stack diameter from inside wall to traverse point)

Traverse point number on a diameter	Number of traverse points on a diameter										
	2	4	6	8	10	12	14	18	20	22	24
1	14.6	6.7	4.4	3.2	2.6	2.1	1.8	1.4	1.3	1.1	1.1
2	85.4	25.0	14.6	10.5	8.2	6.7	5.7	4.4	3.9	3.5	3.2
3		75.0	29.6	19.4	14.6	11.8	9.9	7.5	6.7	6.0	5.5
4		93.3	70.4	32.3	22.6	17.7	14.6	10.9	9.7	8.7	7.9
5			85.4	67.7	34.2	25.0	20.1	14.6	12.9	11.6	10.5
6			95.6	80.6	65.8	35.6	26.9	18.8	16.5	14.6	13.2
7				89.5	77.4	64.4	36.6	23.6	20.4	18.0	16.1
8				96.8	85.4	75.0	63.4	29.6	25.0	21.8	19.4
9					91.8	82.3	73.1	38.2	30.6	26.2	23.0
10					97.4	88.2	79.9	61.8	38.8	31.5	27.2
11						93.3	85.4	70.4	61.2	39.3	32.3
12						97.9	90.1	76.4	69.4	60.7	39.8
13							94.3	81.2	75.0	68.5	60.2
14							98.2	85.4	79.6	73.8	67.7
15								89.1	83.5	78.5	72.8
16								92.5	87.1	82.0	77.0
17								95.6	90.3	85.4	80.8
18								98.6	93.3	88.4	83.9
19									96.1	91.3	86.8
20									96.7	94.0	89.5
21										96.5	92.1
22										98.9	94.5
23											96.8
24											96.9

reference sampling methods for measuring emissions of specific pollutants so that uniform procedures can be applied in testing to obtain a representative sample. Table 22-35 provides a list of some of the most commonly employed test methods. A complete listing of all EPA test methods can be found by going online to the USEPA Technology Transfer Network Emission Measurement Center at www.epa.gov/ttn/emc/promgate.html. These are the CFR Promulgated test methods published in the Federal Register and are the federal government's official legal versions. The test methods reviewed in the following subsections address measuring the emissions of the following pollutants: particulate matter, sulfur dioxide, nitrogen oxides, carbon monoxide, fluorides, hydrogen chloride, total gaseous organics, multiple metals, volatile organic compounds, and semivolatile organic compounds.

Each sampling method requires the use of complex sampling equipment that must be calibrated and operated in accordance with specified reference methods. Additionally, the process or source that is being tested must be operated in a specific manner, usually at rated capacity, under normal procedures.

Velocity and Volumetric Flow Rate The U.S. EPA has published Method 2 as a reference method for determining stack-gas velocity and volumetric flow rate. At several designated sampling points, which represent equal portions of the stack volume (areas in the stack), the velocity and temperature are measured with instrumentation shown in Fig. 22-29.

Measurements to determine volumetric flow rate usually require approximately 30 min. Since sampling rates depend on stack-gas velocity, a preliminary velocity check is usually made prior to testing for pollutants to aid in selecting the proper equipment and in determining the approximate sampling rate for the test.

The volumetric flow rate determined by this method is usually within ± 10 percent of the true volumetric flow rate.

Sources of Methods and Information For the person trying to obtain current information or to enter into the field of source measurements, several particularly helpful information sources are available. The U.S. EPA methods fall in two groups—those used by EPA's Office of Air Quality Planning and Standards (OAQPS) and those used by EPA's Office of Solid Waste (OSW).

The Emission Measurement Technical Information Center (EMTIC) at Research Triangle Park, N.C., is supported by EPA's Office of Air Quality Planning and Standards. Perhaps the most efficient of several available forms of assistance is the EMTIC Bulletin Board System (BBS). Test methods are included along with announcements, utility programs, miscellaneous documents, and other information. The EMTIC/BBS may be reached through TTN 2000 on the Internet at <http://www.epa.gov/ttn>. An EMTIC representative can be reached by telephone at 919-541-0200. EMTIC sponsors workshops and training courses jointly with EPA's Air Pollution Training Institute. Training videotapes, a newsletter, and other mailings are also available from EMTIC.

An excellent source for information concerning OSW's SW-846 Methods is the Methods Information Communication Exchange (MICE). MICE can be reached on the Internet at mice@lan828.ehsg.saic.com. A telephone call to the MICE line, at 703-821-4690, will put the information seeker in touch with an automated information service or with a live representative. Although the function of MICE is to provide information, they will usually send copies of up to three methods. They will not provide copies of the entire SW-846 Methods Manual. The SW-846 Methods Manual may be obtained on CD-ROM or hard copy from National Technical Information Service (NTIS). The NTIS order number for the CD-ROM which includes the third edition and updates 1-3 is PB97-501928INQ. NTIS has a web site at <http://www.ntis.gov> and may also be reached by telephone at 703-487-4650. SW-846 may also be obtained from the Government Printing Office (GPO). Ordering information for GPO is *Test Methods for Evaluating Solid Waste, Physical/Chemical Methods, SW-846 Manual*, 3d ed., Document No. 955-001-000001. Available from Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., November 1986.

The full document is available from the U.S. Government Printing Office, telephone 202-783-3238. GPO also has a web site at <http://www.access.gpo.gov>.

For more information or copies of the California Environmental Protection Agency, Air Resources Board Methods (a.k.a. CARB Methods), contact <http://www.arb.ca.gov/testmeth/testmeth.htm> or telephone the Engineering and Laboratory Branch at 916-263-1630.

EPA reports may be ordered from NTIS at the web site or telephone number given above.

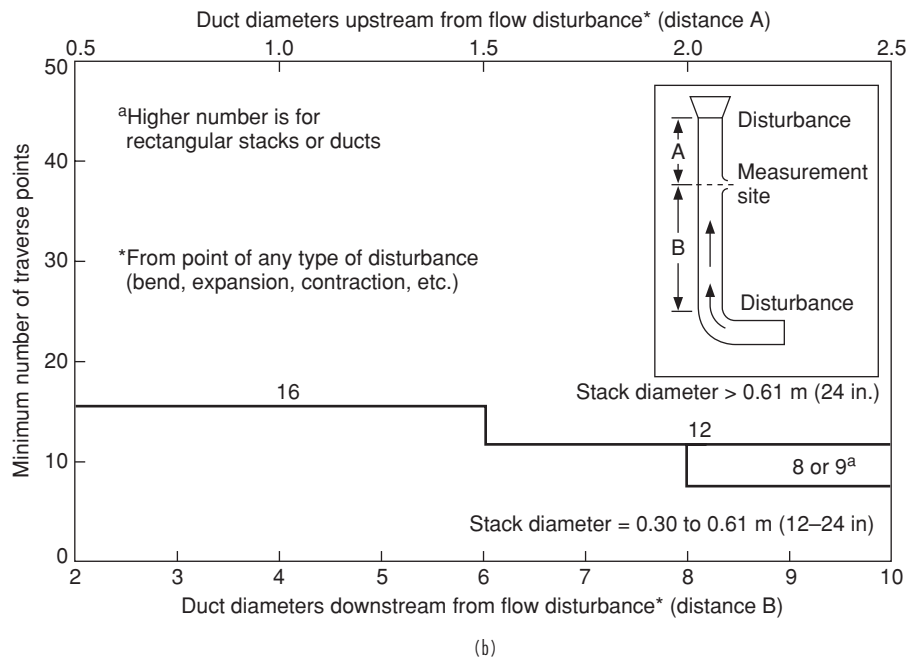
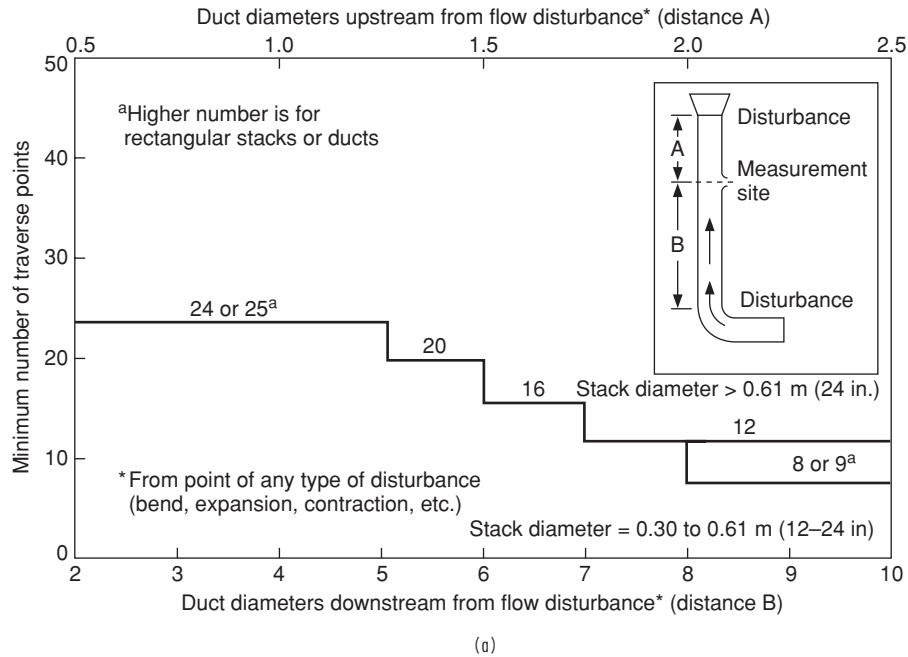
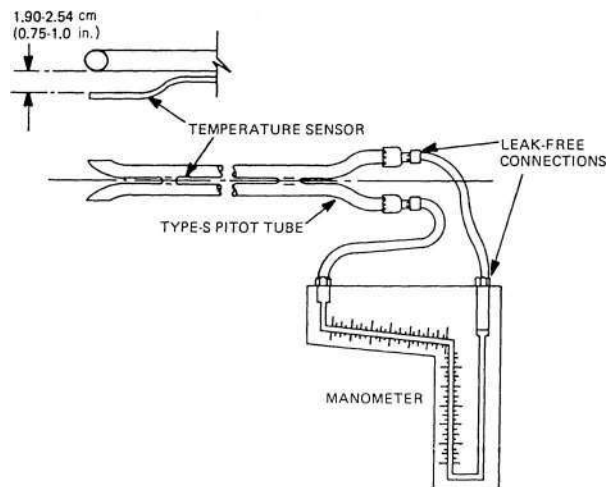


FIG. 22-28 Minimum number of traverse points for (a) particulate traverse and (b) velocity (nonparticulate) traverse.

TABLE 22-35 Selected EPA Test Methods

Commonly Used in Stationary Source Compliance Tests

Method	Test parameter(s)
Office of Air Quality Planning and Standards (OAQPS)	
Methods 1–4	Test location, volumetric flow, gas composition, moisture content
Method 5	Particulate matter
Method 6	Sulfur dioxide
Method 7	Nitrogen oxides
Method 9	Opacity
Method 10	Carbon monoxide
Method 13	Total fluoride
Method 23	Dioxins and furans
Method 25	VOCs
Method 26	Halogens, halides
Method 29	Metals
Method 201	Particulate PM ₁₀
Method 202	Condensible particulate matter
Office of Solid Waste (OSW)	
Method 0010	Modified Method 5, semivolatile organics
Method 0030	Volatile organics

**FIG. 22-29** Velocity-measurement system.

INDUSTRIAL WASTEWATER MANAGEMENT

FURTHER READING: Eckenfelder, W. W., *Industrial Water Pollution Control*, 2d ed., McGraw-Hill, 1989. Metcalf & Eddy, Inc., *Wastewater Engineering, Treatment, Disposal and Reuse*, 3d ed., McGraw-Hill, 1990. Nemerow, N. L. and A. Dasgupta, *Industrial and Hazardous Waste Treatment*, Van Nostrand Reinhold, New York, 1991.

INTRODUCTION

All industrial operations produce some wastewaters which must be returned to the environment. Wastewaters can be classified as (1) domestic wastewaters, (2) process wastewaters, and (3) cooling wastewaters. Domestic wastewaters are produced by plant workers, shower facilities, and cafeterias. Process wastewaters result from spills, leaks, product washing, and noncooling processes. Cooling wastewaters are the result of various cooling processes and can be once-pass systems or multiple-recycle cooling systems. Once-pass cooling systems employ large volumes of cooling waters that are used once and returned to the environment. Multiple-recycle cooling systems have various types of cooling towers to return excess heat to the environment and require periodic blowdown to prevent excess buildup of salts.

Domestic wastewaters are generally handled by the normal sanitary-sewerage system to prevent the spread of pathogenic microorganisms which might cause disease. Normally, process wastewaters do not pose the potential for pathogenic microorganisms, but they do pose potential damage to the environment through either direct or indirect chemical reactions. Some process wastes are readily biodegraded and create an immediate oxygen demand. Other process wastes are toxic and represent a direct health hazard to biological life in the environment. Cooling wastewaters are the least dangerous, but they can contain process wastewaters as a result of leaks in the cooling systems. Recycle cooling systems tend to concentrate both inorganic and organic contaminants to a point at which damage can be created.

Recently, concern for subtle aspects of environmental damage has started to take precedence over damage referred to above. It has been realized that the presence of substances in water at concentrations far below those that will produce overt toxicity or excessive reduction of dissolved oxygen levels can have a major impact by altering the predomination of organisms in the aquatic ecosystem. This is beginning to have an effect on water-quality standards and, consequently, allow-

able discharges. Unfortunately, the extent of knowledge is not sufficient upon which to base definitive standards for these subtle effects. Emerging chemicals of concern include pharmaceutically active agents, endocrine disrupters, and low concentrations of heavy metals and metalloids. Over the next decade, it is anticipated that accumulation of knowledge will make it possible to delineate defensible standards. Another recent concern is that of contaminated storm water from industrial sites. Federal guidelines to control this wastewater are presently being developed. Most concern is for product spills on plant property but outside the production facility and from rain contact with spoil piles.

UNITED STATES LEGISLATION, REGULATIONS, AND GOVERNMENT AGENCIES

Federal Legislation Public Law 92-500 promulgated in 1972 created the primary framework for management of water pollution in the United States. This act has been amended many times since 1972. Major amendments occurred in 1977 (Public Laws 95-217 and 95-576), 1981 (Public Laws 97-117 and 97-164), and 1987 (Public Law 100-4). This law and its various amendments are referred to as the Clean Water Act. The Clean Water Act addresses a large number of issues of water pollution management. A general review of these issues has been presented in an earlier chapter in this handbook. Primary with respect to control of industrial wastewater is the National Pollutant Discharge Elimination System (NPDES), originally established by PL 92-500. Any municipality or industry that discharges wastewater to the navigable waters of the United States must obtain a discharge permit under the regulations set forth by the NPDES. Under this system, there are three classes of pollutants (conventional pollutants, priority pollutants, and nonconventional/nonpriority pollutants). Conventional pollutants are substances such as biochemical oxygen demand (BOD), suspended solids (SS), pH, oil and grease, and coliforms. Priority pollutants are a list of 129 substances originally set forth in a consent decree between the Environmental Protection Agency and several environmental organizations. This list was incorporated into the 1977 amendments. Most of the substances on this list are organics, but it does include most of the heavy metals. These substances are generally considered to be toxic. However, the toxicity is not absolute; it depends on the concentration. In addition, many of

the organics on the list are under appropriate conditions biodegradable. In reality, substances for inclusion on the priority pollutant list were chosen on the basis of a risk assessment rather than only a hazard assessment. The third class of pollutants could include any pollutant not in the first two categories. Examples of substances that are presently regulated in the third category are nitrogen, phosphorous, sodium, and chlorine residual. The overall goals of the Clean Water Act are to restore and maintain water quality in the navigable waters of the United States. The initial standard was to ensure that these waters would be clean enough for recreation (swimmable) and ecologically secure (fishable). Initially this was to be achieved by curtailment of the discharge of pollutants. Eventually the regulatory system was essentially to phase out discharge of any pollutant into water. Obviously, if no discharge of pollutants occurs, the waters will be maintained in close to a pristine condition. However, the early reduction of pollutant discharge to zero was realized as impractical. Thus, at present, the NPDES has prescribed the limits on discharges as a function of the type of pollutant, the type of industry discharging, and the desired water quality. The specific requirements for each individual discharger are established in the NPDES permit issued to the discharger. These permits are reviewed every five years and are subject to change at the time of review.

A major tactic that was adopted in the Clean Water Act was to establish uniform technology standards, by class of pollutant and specific industry type, which applied nationwide to all dischargers. Thus, a kraft mill in Oregon would have to meet essentially the same discharge standards as a kraft mill in New York. In establishing these standards, the EPA took into account the state of the art of waste treatment in each particular industry as well as cost and ecological effectiveness. These discharge standards have been published in the *Federal Register* for more than thirty industrial categories and several hundred subcategories (the Commerce Department Industrial Classification System was used to establish these categories) in the *Federal Register*. These have been promulgated over an extensive period of time. The reader is advised to consult the index to this document to ascertain regulations that apply to a particular industry. Table 22-36 presents the major industrial categories.

TABLE 22-36 Industry Categories

1. Adhesives and sealants
2. Aluminum forming
3. Asbestos manufacturing
4. Auto and other laundries
5. Battery manufacturing
6. Coal mining
7. Coil coating
8. Copper forming
9. Electric and electronic components
10. Electroplating
11. Explosives manufacturing
12. Ferroalloys
13. Foundries
14. Gum and wood chemicals
15. Inorganic chemicals manufacturing
16. Iron and steel manufacturing
17. Leather tanning and finishing
18. Mechanical products manufacturing
19. Nonferrous metals manufacturing
20. Ore mining
21. Organic chemicals manufacturing
22. Pesticides
23. Petroleum refining
24. Pharmaceutical preparations
25. Photographic equipment and supplies
26. Plastic and synthetic materials manufacturing
27. Plastic processing
28. Porcelain enamelling
29. Printing and publishing
30. Pulp and paperboard mills
31. Soap and detergent manufacturing
32. Steam electric power plants
33. Textile mills
34. Timber products processing

As indicated above, not only are the discharge standards organized on an industry-by-industry basis, but they are different depending upon which of the three classes of pollutants is being regulated. The standards for conventional pollutants are referred to as best conventional technology (BCT). The standards for priority pollutants are referred to as best available technology (BAT), as are those for non-conventional, nonpriority pollutants. These standards envision that the technology will not be limited to treatment but may include revision in the industrial processing and/or reuse of effluents (pollution prevention). They are usually presented as mass of pollutant discharged per unit of product produced.

In some situations it is anticipated that application of BCT and BAT may not be sufficient to ensure that water-quality standards are achieved in a stream segment. Studies have indicated that approximately 10 percent of the stream segments in the United States will have their water-quality standards violated, even if all the dischargers to that stream segment meet BCT and BAT regulations. These segments are referred to as water quality limited segments. The NPDES permit for those who discharge into water quality limited segments must provide for pollutant removal in excess of that required by BCT and BAT so that water-quality standards (which are jointly established by each state and EPA) are achieved. The stream segments in which water-quality standards will be met by application of BCT and BAT are referred to as "effluent quality limited segments."

For industrial discharges that enter municipal sewers and thus eventually municipal treatment plants, NPDES regulations are nominally the same as for industries that discharge directly to navigable waters; i.e., they must meet BCT and BAT standards. These are referred to as "categorical industrial pretreatment standards." However if it can be demonstrated that the municipal treatment plant can remove a pollutant in the industrial waste, a "removal credit" can be assigned to the permit, thus lowering the requirement on the industrial discharger. The monetary charge to the industry by the municipality for this service is negotiable but must be within parameters established by EPA for industrial user charges if the municipality received a federal grant for construction of the treatment plant. A removal credit cannot be assigned if removal of the industrial pollutant results in difficulty in ultimate disposal of sludge from the municipal plant. In addition, an industry cannot discharge any substance that can result in physical damage to the municipal sewer and/or interfere with any aspect of treatment plant performance even if the substance is not covered by BCT or BAT regulations.

The EPA has established a National Pretreatment Program to address the growing impacts industrial effluents have on publicly owned treatment works (POTWs). These impacts include biological toxicity and inhibition causing biological upsets that can reduce plant efficiency; accumulation of heavy metals in primary and secondary sludge that may impact beneficial reuse applications; increase in whole effluent toxicity (WET) which can impact plant discharge permit requirements; and increasing organic and solid loadings on plants that are reaching their design capacity. The EPA has established two sets of rules that are the basis of the National Pretreatment Program, the categorical pretreatment standards and prohibited discharge standards. The categorical pretreatment standards are industry- or technology-based limitations on pollutant discharges to POTWs promulgated by USEPA in accordance with Section 307 of the Clean Water Act. These standards apply to specific process wastewaters of particular industrial categories (see 40 CFR 403.6 and 40 CFR Parts 405 to 471). The prohibited discharge standards forbid certain chemicals and discharges by any sewer system. These specific discharges are described in full in the *Federal Register* and include 126 specific chemicals. The program requires all POTWs with a flow greater than 5 mgd or those receiving significant industrial flow to establish an industrial pretreatment program. States and individual municipalities can develop their own pretreatment programs to address system-specific industrial wastewater impacts. Currently, over 1500 municipalities have established their own pretreatment programs that either meet or exceed the EPA pretreatment standards. Along with setting standards for pretreatment, the EPA identifies best-available technologies to economically control industrial effluents. In addition to industrial discharges, the EPA has initiated a program to better control

stormwater discharges associated with industrial activity. This program is part of the national NPDES Phase I Stormwater Program.

Environmental Protection Agency President Nixon created the EPA in 1970 to coordinate all environmental pollution-control activities at the federal level. The EPA was placed directly under the Office of the President so that it could be more responsive to the political process. In the succeeding decade, the EPA produced a series of federal regulations increasing federal control over all wastewater-pollution-control activities. In January 1981, President Reagan reversed the trend of greater federal regulation and began to decrease the role of the federal EPA. However, during the 1980s, an equilibrium was achieved between those who wish less regulation and those who wish more. In general, industry and the EPA reached agreement on the optimum level of regulation.

State Water-Pollution-Control Offices Every state has its own water-pollution-control office. Some states have reorganized along the lines of the federal EPA with state EPA offices, while others have kept their water-pollution-control offices within state health departments. Prior to 1965, each state controlled its own water-pollution-control programs. Conflicts between states and uneven enforcement of state regulations resulted in the federal government's assuming the leadership role. Unfortunately, conflicts between states shifted to being conflicts between the states and the federal EPA. By 1980, the state water-pollution-control offices were primarily concerned with handling most of the detailed permit and paperwork for the EPA and in furnishing technical assistance to industries at the local level. In general, most of the details of regulation are carried out by state water-pollution-control agencies with oversight by EPA.

WASTEWATER CHARACTERISTICS

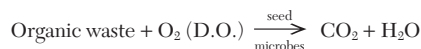
Wastewater characteristics vary widely from industry to industry. Obviously, the specific characteristics will affect the treatment techniques chosen for use in meeting discharge requirements. Some general characteristics that should be considered in planning are given in Table 22-37. Because of the large number of pollutant substances, wastewater characteristics are not usually considered on a substance-by-substance basis. Rather, substances of similar pollution effects are grouped together into classes of pollutants or characteristics as indicated below.

Priority Pollutants Recently, greatest concern has been for this class of substances for the reasons given previously. These materials are treated on an individual-substance basis for regulatory control. Thus, each industry could receive a discharge permit that lists an acceptable level for each priority pollutant. Table 22-38 presents a list of these substances; most are organic, but some inorganics are included. All are considered toxic, but, as indicated previously, there is wide variation in their toxicity. Most of the organics are biologically degradable despite their toxicity (Refs. 30 and 31). USEPA has collected data on the occurrence of these substances in various industrial

wastes and their treatability. A recent trend has been to avoid their use in industrial processing.

Organics The organic composition of industrial wastes varies widely, primarily due to the different raw materials used by each specific industry. These organics include proteins, carbohydrates, fats and oils, petrochemicals, solvents, pharmaceutical, small and large molecules, solids and liquids. Another complication is that a typical industry produces many diverse wastestreams. Good practice is to conduct a material balance throughout an entire production facility. This survey should include a flow diagram, location and sizes of piping, tanks and flow volumes, as well as an analysis of each stream. Results of an industrial waste survey for an industry are given in Table 22-39. Noteworthy is the range in waste sources, including organic soap, toilet articles, ABS (alkyl benzene sulfonate), and the relatively clean but hot condenser water that makes up half the plant flow, while the strongest wastes—spent caustic and fly ash—have the lowest flows. See Tables 22-39 and 22-40 for information on the average characteristics of wastes from specific industries.

An important measure of the waste organic strength is the 5-day biochemical oxygen demand (BOD₅). As this test measures the demand for oxygen in the water environment caused by organics released by industry and municipalities, it has been the primary parameter in determining the strength and effects of a pollutant. This test determines the oxygen demand of a waste exposed to biological organisms (controlled seed) for an incubation period of five days. Usually this demand is caused by degradation of organics according to the following simplified equation, but reduced inorganics in some industries may also cause demand (i.e., Fe²⁺, S²⁻, and SO₃²⁻).



This wet lab test measures the decrease in dissolved oxygen (D.O.) concentration in 5 days, which is then related to the sample strength. If the test is extended over 20 days, the BOD₂₀ (ultimate BOD) is obtained and corresponds more closely to the Chemical Oxygen Demand (COD) test. The COD test uses strong chemical oxidizing agents with catalysts and heat to oxidize the wastewater and obtain a value that is almost always larger than the 5- and 20-day BOD values. Some organic compounds (like pyridene, a ring structure containing nitrogen) resists chemical oxidation giving a low COD. A major advantage of the COD test is the completion time of less than 3 hours, versus 5 days for the BOD₅ test. Unfortunately, state and federal regulations generally require BOD₅ values, but approximate correlations can be made to allow computation of BOD from COD. A more rapid measure of the organic content of a waste is the instrumental test for total organic carbon (TOC), which takes a few minutes and may be correlated to both COD and BOD for specific wastes. Unfortunately, BOD₅ results are subject to wide statistical variations and require close scrutiny and experience. For municipal wastewaters, BOD₅ is about 67 percent of the ultimate BOD and 40–45 percent of the COD, indicating a large amount of nonbiodegradable COD and the continuing need to run BOD as well as COD and TOC. An example of BOD, COD, and TOC relationships for chemical industry wastewater is given in Table 22-40. The concentrations of the wastewaters vary by two orders of magnitude, and the BOD/COD, COD/TOC, and BOD/TOC ratios vary less than twofold. The table indicates that correlation/codification is possible, but care and continual scrutiny must be exercised.

Another technique for organics measurement that overcomes the long period required for the BOD test is the use of continuous respirometry. Here the waste (full-strength rather than diluted as in the standard BOD test) is contacted with biomass in an apparatus that continuously measures the dissolved oxygen consumption. This test determines the ultimate BOD in a few hours if a high level of biomass is used. The test can also yield information on toxicity, the need to develop an acclimated biomass, and required rates of oxygen supply.

In general, low-molecular-weight water-soluble organics are biodegraded readily. As organic complexity increases, solubility and biodegradability decrease. Soluble organics are metabolized more easily than insoluble organics. Complex carbohydrates, proteins, and

TABLE 22-37 Wastewater Characteristics

Property	Characteristic	Example	Size or concentration
Solubility	Soluble	Sugar	>100 gm/L
	Insoluble	PCB	<1 mg/L
Stability, biological	Degradable	Sugar	
	Refractory	DDT, metals	
Solids	Dissolved	NaCl	<10 ⁻⁹ m
	Colloidal	Carbon	>10 ⁻⁶ –<10 ⁻⁹ m
	Suspended	Bacterium	>10 ⁻⁶ m
	Carbon	Alcohol	
Organic	Inorganic	Cu ²⁺	
	Acidic	HNO ₃	
	Neutral	Salt (NaCl)	1–12
	Basic	NaOH	
Temperature	High–low	Cooling	>5°
		Heat exchange	>30°
Toxicity	Biological effect	Heavy metals	Varies
		Priority compounds	
Nutrients	N	NH ₃	Varies
	P	PO ₄ ³⁻	

TABLE 22-38 List of Priority Chemicals*

Compound name	Compound name	Compound name
1. Acenaphthene†	37. 1,2-Diphenylhydrazine†	83. Indeno (1,2,3-cd) pyrene (2,3- <i>o</i> -phenylene)pyrene)
2. Acrolein†	38. Ethylbenzene†	84. Pyrene
3. Acrylonitrile†	39. Fluoranthene†	85. Tetrachloroethylene†
4. Benzene†		86. Toluene†
5. Benzidine†		87. Trichloroethylene†
6. Carbon tetrachloride† (tetrachloromethane)	Haloethers† (other than those listed elsewhere)	88. Vinyl chloride† (chloroethylene)
Chlorinated benzenes (other than dichlorobenzenes)	40. 4-Chlorophenyl phenyl ether	
7. Chlorobenzene	41. 4-Bromophenyl phenyl ether	Pesticides and metabolites
8. 1,2,4-Trichlorobenzene	42. Bis(2-chloroisopropyl) ether	89. Aldrin†
9. Hexachlorobenzene	43. Bis(2-chloroethoxy) methane	90. Dieldrin†
Chlorinated ethanes† (including 1,2-dichloroethane, 1,1,1-trichloroethane, and hexachloroethane)	Halomethanes† (other than those listed elsewhere)	91. Chlordane† (technical mixture and metabolites)
10. 1,2-Dichloroethane	44. Methylene chloride (dichloromethane)	
11. 1,1,1-Trichloroethane	45. Methyl chloride (chloromethane)	DDT and metabolites†
12. Hexachloroethane	46. Methyl bromide (bromomethane)	92. 4,4'-DDT
13. 1,1-Dichloroethane	47. Bromoform (tribromomethane)	93. 4,4'-DDE (<i>p,p'</i> -DDX)
14. 1,1,2-Trichloroethane	48. Dichlorobromomethane	94. 4,4'-DDD (<i>p,p'</i> -TDE)
15. 1,1,2,2-Tetrachloroethane	49. Trichlorofluoromethane	
16. Chloroethane (ethyl chloride)	50. Dichlorodifluoromethane	Endosulfan and metabolites†
Chloroalkyl ethers† (chloromethyl, chloroethyl, and mixed ethers)	51. Chlorodibromomethane	95. α -Endosulfan- α
17. Bis(chloromethyl) ether	52. Hexachlorobutadiene†	96. β -Endosulfan- β
18. Bis(2-chloroethyl) ether	53. Hexachlorocyclopentadiene†	97. Endosulfan sulfate
19. 2-Chloroethyl vinyl ether (mixed)	54. Isophorone†	
Chlorinated naphthalene†	55. Naphthalene†	Endrin and metabolites†
20. 2-Chloronaphthalene	56. Nitrobenzene†	98. Endrin
Chlorinated phenols† (other than those listed elsewhere; includes trichlorophenols and chlorinated cresols)	Nitrophenols† (including 2,4-dinitrophenol and dinitrocresol)	99. Endrin aldehyde
21. 2,4,6-Trichlorophenol	57. 2-Nitrophenol	
22. <i>para</i> -Chloro- <i>meta</i> -cresol	58. 4-Nitrophenol	Heptachlor and metabolites†
23. Chloroform (trichloromethane)†	59. 2,4-Dinitrophenol†	100. Heptachlor
24. 2-Chlorophenol†	60. 4,6-Dinitro- <i>o</i> -cresol	101. Heptachlor epoxide
Dichlorobenzenes†	Nitrosamines†	
25. 1,2-Dichlorobenzene	61. <i>N</i> -Nitrosodimethylamine	Hexachlorocyclohexane (all isomers)†
26. 1,3-Dichlorobenzene	62. <i>N</i> -Nitrosodiphenylamine	102. α -BHC- α
27. 1,4-Dichlorobenzene	63. <i>N</i> -Nitrosodi- <i>n</i> -propylamine	103. β -BHC- β
Dichlorobenzidine†	64. Pentachlorophenol†	104. γ -BHC (lindane)- γ
28. 3,3'-Dichlorobenzidine	65. Phenol†	105. δ -BHC- δ
Dichloroethylenes† (1,1-dichloroethylene and 1,2-dichloroethylene)	Phthalate esters†	
29. 1,1-Dichloroethylene	66. Bis(2-ethylhexyl) phthalate	Polychlorinated biphenyls (PCB)†
30. 1,2- <i>trans</i> -Dichloroethylene	67. Butyl benzyl phthalate	106. PCB-1242 (Arochlor 1242)
31. 2,4-Dichlorophenol†	68. Di- <i>n</i> -butyl phthalate	107. PCB-1254 (Arochlor 1254)
Dichloropropane and dichloropropene†	69. Di- <i>n</i> -octyl phthalate	108. PCB-1221 (Arochlor 1221)
32. 1,2-Dichloropropane	70. Diethyl phthalate	109. PCB-1232 (Arochlor 1232)
33. 1,2-Dichloropropylene (1,2-dichloropropene)	71. Dimethyl phthalate	110. PCB-1248 (Arochlor 1248)
34. 2,4-Dimethylphenol†		111. PCB-1260 (Arochlor 1260)
Dinitrotoluene†	Polynuclear aromatic hydrocarbons (PAH)†	112. PCB-1016 (Arochlor 1016)
35. 2,4-Dinitrotoluene	72. Benzo(a)anthracene (1,2-benzanthracene)	113. Toxaphene†
36. 2,6-Dinitrotoluene	73. Benzo(a)pyrene (3,4-benzopyrene)	114. antimony (total)
	74. 3,4-Benzofluoranthene	115. arsenic (total)
	75. Benzo(k)fluoranthene (11,12-benzofluoranthene)	116. asbestos (fibrous)
	76. Chrysene	117. beryllium (total)
	77. Acenaphthylene	118. cadmium (total)
	78. Anthracene	119. chromium (total)
	79. Benzo(ghi)perylene (1,12-benzoperylene)	120. copper (total)
	80. Fluorene	121. cyanide (total)
	81. Phenanthrene	122. lead (total)
	82. Dibenzo(a,h)anthracene (1,2,5,6-dibenzanthracene)	123. mercury (total)
		124. nickel (total)
		125. selenium (total)
		126. silver (total)
		127. thallium (total)
		128. zinc (total)
		129. 2,3,7,8-Tetrachlorodibenzo- <i>p</i> -dioxin (TCDD)

*Adapted from Eckenfelder, W. W. Jr., *Industrial Water Pollution Control*, 2d ed., McGraw-Hill, New York, 1989.

†Specific compounds and chemical classes as listed in the consent degree.

TABLE 22-39 Industrial Waste Components of a Soap, Detergents, and Toilet Articles Plant

Waste source	Sampling station	COD, mg/L	BOD, mg/L	SS, mg/L	ABS, mg/L	Flow, gal/min
Liquid soap	D	1,100	565	195	28	300
Toilet articles	E	2,650	1,540	810	69	50
Soap production	R	29	16	39	2	30
ABS production	S	1,440	380	309	600	110
Powerhouse	P	66	10	50	0	550
Condenser	C	59	21	24	0	1100
Spent caustic	B	30,000	10,000	563	5	2
Tank bottoms	A	120,000	150,000	426	20	1.5
Fly ash	F			6750		10
Main sewer		450	260	120	37	2150

NOTE: gal/min = 3.78×10^{-3} m³/min.SOURCE: Eckenfelder, W. W., *Industrial Water Pollution Control*, 2d ed., McGraw-Hill, New York, 1989.

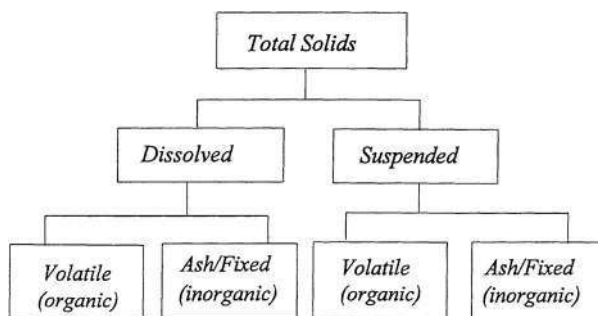
fats and oils must be hydrolyzed to simple sugars, aminos, and other organic acids prior to metabolism. Petrochemicals, pulp and paper, slaughterhouse, brewery, and numerous other industrial wastes containing complex organics have been satisfactorily treated biologically, but proper testing and evaluation is necessary.

Inorganics The inorganics in most industrial wastes are the direct result of inorganic compounds in the carriage water. Soft-water sources will have lower inorganics than hard-water or saltwater sources. However, some industrial wastewaters can contain significant quantities of inorganics which result from chemical additions during plant operation. Many food processing wastewaters are high in sodium. While domestic wastewaters have a balance in organics and inorganics, many process wastewaters from industry are deficient in specific inorganic compounds. Biodegradation of organic compounds requires adequate nitrogen, phosphorus, iron, and trace salts. Ammonium salts or nitrate salts can provide the nitrogen, while phosphates supply the phosphorus. Either ferrous or ferric salts or even normal steel corrosion can supply the needed iron. Other trace elements needed for biodegradation are potassium, calcium, magnesium, cobalt, molybdenum, chloride, and sulfur. Carriage water or demineralizer wastewaters or corrosion products can supply the needed trace elements for good metabolism. Occasionally, it is necessary to add specific trace elements or nutrient elements.

pH and Alkalinity Wastewaters should have pH values between 6 and 9 for minimum impact on the environment. Wastewaters with pH values less than 6 will tend to be corrosive as a result of the excess hydrogen ions. On the other hand, raising the pH above 9 will cause some of the metal ions to precipitate as carbonates or as hydroxides at higher pH levels. Alkalinity is important in keeping pH values at the right levels. Bicarbonate alkalinity is the primary buffer in wastewaters. It is important to have adequate alkalinity to neutralize the acid waste components as well as those formed by partial metabolism of organics. Many neutral organics such as carbohydrates, aldehydes, ketones, and alcohols are biodegraded through organic acids which must be neutralized by the available alkalinity. If alkalinity is inadequate, sodium carbonate is a better form to add than lime. Lime tends to be hard to control accurately and results in high pH levels and pre-

TABLE 22-40 BOD, COD, and TOC Relationships

Type of waste	BOD ₅ , mg/L	COD, mg/L	TOC, mg/L	BOD ₅ /COD	COD/TOC	BOD ₅ /TOC
Chemical	700	1,400	450	0.50	3.12	1.55
Chemical	850	1,900	580	0.45	3.28	1.47
Chemical	8,000	17,500	5,800	0.46	3.02	1.38
Chemical	9,700	15,000	5,500	0.65	2.72	1.76
Chemical	24,000	41,300	9,500	0.58	4.35	2.53
Chemical	60,700	78,000	26,000	0.78	3.00	2.34
Chemical	62,000	143,000	48,140	0.43	2.96	1.28

Adapted from Eckenfelder, W.W. and D.L. Ford, *Water Pollution Control*, Pemberton Press, Austin and New York, 1970.**FIG. 22-30 Solids identification.** Abbreviations: TS, total solids; SS, suspended solid; D, dissolved; V, volatile.

cipitation of the calcium which forms part of the alkalinity. In a few instances, sodium bicarbonate may be the best source of alkalinity.

Temperature Most industrial wastes tend to be on the warm side. For the most part, temperature is not a critical issue below 37°C if wastewaters are to receive biological treatment. It is possible to operate thermophilic biological wastewater-treatment systems up to 65°C with acclimated microbes. Low-temperature operations in northern climates can result in very low winter temperatures and slow reaction rates for both biological treatment systems and chemical treatment systems. Increased viscosity of wastewaters at low temperatures makes solid separation more difficult. Efforts are generally made to keep operating temperatures between 10 and 30°C if possible.

Dissolved Oxygen Oxygen is a critical environmental resource in receiving streams and lakes. Aquatic life requires reasonable dissolved-oxygen (DO) levels. EPA has set minimum stream DO levels at 5 mg/L during summer operations, when the rate of biological metabolism is a maximum. It is important that wastewaters have maximum DO levels when they are discharged and have a minimum of oxygen-demanding components so that DO remains above 5 mg/L. DO is a poorly soluble gas in water, having a solubility around 9.1 mg/L at 20°C and 101.3-kPa (1-atm) air pressure. As the temperature increases and the pressure decreases with higher elevations above sea level, the solubility of oxygen decreases. Thus, DO is a minimum when BOD rates are a maximum. Lowering the temperature yields higher levels of DO saturation, but the biological metabolism rate decreases. Warm-wastewater discharges tend to aggravate the DO situation in receiving waters.

Solids Total solids is the residue remaining from a wastewater dried at 103–105°C. It includes the fractions shown in Fig. 22-30. The first separation is the portion that passes through a 2-μm filter (dissolved) and those solids captured on the filter (suspended). Combustion at 500°C further separates the solids into volatile and ash (fixed) solids. Although ash and volatile solids do not distinguish inorganic from organic solids exactly, due to loss of inorganics on combustion, the volatile fraction is often used as an approximate representation of the organics present. Another type of solids, settleable solids, refers to solids that settle in an Imhoff cone in one hour. Industrial wastes vary substantially in these types of solids and require individual wastewater treatment process analysis. An example of possible variation is given in Table 22-41.

Nutrients and Eutrophication Nitrogen and phosphorus cause significant problems in the environment and require special attention in industrial wastes. Nitrogen, phosphorus, or both may cause aquatic

TABLE 22-41 Solids Variation in Industrial Wastewater

Type of solids	Plating	Pulp and paper
Total	High	High
Dissolved	High	High
Suspended	Low	High
Organic	Low	High
Inorganic	High	High

TABLE 22-42 Nutrient Forms

Parameter	Example
Organic N	protein
Ammonia	NH ₃
Ammonium	NH ₄ ⁺
Nitrite	NO ₂ ⁻
Nitrate	NO ₃ ⁻
Organic P	malathion
Ortho-P	PO ₄ ³⁻
Poly-P	(PO ₄ ³⁻) _x

biological productivity to increase, resulting in low dissolved oxygen and eutrophication of lakes, rivers, estuaries, and marine waters. Table 22-42 gives the primary nutrient forms causing problems, while the following equation shows the biological oxidation or oxygen-consuming potential of the most common nitrogen forms.

When organics containing reduced nitrogen are degraded, they usually produce ammonium, which is in equilibrium with ammonia. As the pK for $\text{NH}_3 \leftrightarrow \text{NH}_4^+$ is 9.3, the ammonium ion is the primary form present in virtually all biological treatment systems, as they operate at pH < 8.5 and usually in the pH range of 6.5–7.5. In aerobic reactions, ammonium is oxidized by nitrifying bacteria (nitrosomonas) to nitrite and each mg of NH_4^+-N oxidized will require 3.43 mg D.O. Further oxidation of nitrite by nitrobacter yields nitrate and uses an additional 1.14 mg of D.O. for a total D.O. consumption of 4.57 mg. Thus, organic and ammonium nitrogen can exert significant biochemical oxygen demand in the water environment. This nitrogen demand is referred to as nitrogenous or NBOD, whereas organic BOD is CBOD (carbonaceous). In treatment of wastewaters with organics and ammonium, the total oxygen demand (TOD) may have to be satisfied in accordance with the approximate formula:

$$\text{TOD} \cong 1.5 \text{ BOD}_5 + 4.5 (\text{TKN})$$

where TKN (total Kjeldahl nitrogen) = Organic N + NH_4^+-N .

Phosphorus is not oxidized or reduced biologically, but ortho-P may be formed from organic and poly-P. Ortho-P may be removed by chemical precipitation or biologically with sludges and will be covered in a later section.

While many industrial wastes are so low in nitrogen and phosphorus that these must be added if biologically based treatment is to be used, others contain very high levels of these nutrients. For example, paint-production wastes are high in nitrogen, and detergent production wastes are high in phosphorus. Treatment for removal of these nutrients is required in areas where eutrophication is a problem.

Whole Effluent Toxicity (WET) Whole effluent toxicity testing is an important part of the EPA's approach to protecting receiving waters from toxic effluents. The EPA has established 17 analytical methods for testing acute and chronic toxicity of point sources (municipal and industrial wastewater effluents) and their impact on surface water environments. WET is a term used to describe the toxic effects imposed on a species or population of aquatic organisms caused by exposure to an effluent. WET is determined analytically by exposing sensitive indigenous organisms to effluents using WET test protocols established by the EPA. WET tests report on the general acute and chronic toxicity of all constituents in a complex effluent and do not represent the toxicity of specific chemicals. To this end, the WET test results could represent the combined toxic effect of heavy metals, un-ionized ammonia, and other toxic constituents in an effluent. The EPA and state regulators have been using WET as an analytical tool for controlling industrial discharges and developing and enforcing industrial pretreatment programs.

Oil and Grease These substances are found in many industrial wastes (i.e., meat packing, petrochemical, and soap production). They tend to float on the water surface, blocking oxygen transfer, interfering with recreation, and producing an aesthetically poor appearance in the water. Measurement is by a solvent extraction procedure. Many somewhat different substances will register as oil and grease in this test. Often oil and grease interfere with other treatment operations, so they must be removed as part of the initial stages of treatment.

WASTEWATER TREATMENT

As indicated above, industrial wastewater contains a vast array of pollutants in soluble, colloidal, and particulate forms, both inorganic and organic. In addition, the required effluent standards are also diverse, varying with the industrial and pollutant class. Consequently, there can be no standard design for industrial water-pollution control. Rather, each site requires a customized design to achieve optimum performance. However, each of the many proven processes for industrial waste treatment is able to remove more than one type of pollutant and is in general applicable to more than one industry. In the sections that follow, waste-treatment processes are discussed more from the broad-based generalized perspective than with narrow specificity. Generally, a combination of several processes is utilized to achieve the degree of treatment required at the least cost.

Much of the experience and data from wastewater treatment has been gained from municipal treatment plants. Industrial liquid wastes are similar to wastewater but differ in significant ways. Thus, typical design parameters and standards developed for municipal wastewater operations must not be blindly utilized for industrial wastewater. It is best to run laboratory and small pilot tests with the specific industrial wastewater as part of the design process. It is most important to understand the temporal variations in industrial wastewater strength, flow, and waste components and their effect on the performance of various treatment processes. Industry personnel in an effort to reduce cost often neglect laboratory and pilot studies and depend on waste characteristics from similar plants. This strategy often results in failure, delay, and increased costs. Careful studies on the actual waste at a plant site cannot be overemphasized.

PRETREATMENT

Many industrial-wastewater streams should be pretreated prior to discharge to municipal sewerage systems or even to a central industrial sewerage system. Many POTWs have pretreatment programs that follow the EPA program by regulating specific industries and specific chemical constituents. Product substitution, process alteration, or pretreatment should be considered to address wastewater streams that may fall under the local or federal pretreatment program. Pretreatment may include single- or multiple-unit processes aimed at reducing the overall toxicity and adverse impact of the waste stream on the total treatment system.

Equalization Equalization is one of the most important pretreatment devices. The batch discharge of concentrated wastes is best suited for equalization. It may be important to equalize wastewater flows, wastewater concentrations, or both. Periodic wastewater discharges tend to overload treatment units. Flow equalization tends to level out the hydraulic loads on treatment units. It may or may not level out concentration variations, depending upon the extent of mixing within the equalization basin. Mechanical mixing may be adequate if the wastes are purely chemical in their reactivity. Biodegradable wastes normally require aeration mixing so that the microbes are kept aerobic and nuisance odors are prevented. Diffused aeration systems offer better mixing under variable load conditions than mechanical surface aeration equipment. Mixing and oxygen transfer are both important with biodegradable wastewaters. Operation on regular cycles determines the size of the equalization basin. There is no advantage in making the equalization basin any larger than necessary to level out wastewater variations. Industrial operation on a 5-day, 40-h week will normally make a 2-day equalization basin as large as needed for continuous operation of the wastewater-treatment system under uniform conditions.

Neutralization Acidic or basic wastewaters must be neutralized prior to discharge. If an industry produces both acidic and basic wastes, these wastes may be mixed together at the proper rates to obtain neutral pH levels. Equalization basins can be used as neutralization basins. When separate chemical neutralization is required, sodium hydroxide is the easiest base material to handle in a liquid form and can be used at various concentrations for in-line neutralization with a minimum of equipment. Yet, lime remains the most widely used base for acid neutralization. Limestone is used when reaction

rates are slow and considerable time is available for reaction. Sulfuric acid is the primary acid used to neutralize high-pH wastewaters unless calcium sulfate might be precipitated as a result of the neutralization reaction. Hydrochloric acid can be used for neutralization of basic wastes if sulfuric acid is not acceptable. For very weak basic wastewaters carbon dioxide can be adequate for neutralization.

Grease and Oil Removal Grease and oils tend to form insoluble layers with water as a result of their hydrophobic characteristics. These hydrophobic materials can be easily separated from the water phase by gravity and simple skimming, provided they are not too well mixed with the water prior to separation. If the oils and greases form emulsions with water as a result of turbulent mixing, the emulsions are difficult to break. Separation of oil and grease should be carried out near the point of their mixing with water. In a few instances, air bubbles can be added to the oil and grease mixtures to separate the hydrophobic materials from the water phase by flotation. Chemicals have also been added to help break the emulsions. American Petroleum Institute (API) separators have been used extensively by the petroleum industry to remove oils from wastewaters. The food industries use grease traps to collect the grease prior to its discharge. Unfortunately, grease traps are designed for regular cleaning of the trapped grease. Too often they are allowed to fill up and discharge the excess grease into the sewer or are flushed with hot water and steam to fluidize the grease for easy discharge to the sewer. A grease trap should be designed for a specific volume of grease to be collected over specific time periods. Care should be taken to design the trap so that the grease can easily be removed and properly handled. Neglected or poorly designed grease traps are worse than no grease traps at all.

Toxic Substances Recent federal legislation has made it illegal for industries to discharge toxic materials in wastewaters. Each industry is responsible for determining if any of its wastewater components are toxic to the environment and to remove them prior to the wastewater discharge. The EPA has identified a number of priority pollutants which must be removed and kept under proper control from their origin to their point of ultimate disposal. Major emphasis has recently been placed on heavy metals and on complex organics that have been implicated in possible cancer production. Pretreatment is essential to reduce heavy metals below toxic levels and to prevent discharge of any toxic organics. Many industries will be required to pretreat waste streams under the local or federal pretreatment program. In the past, these pretreatment programs have focused on heavy metals and toxic organics. More recently they have begun to concentrate on whole effluent toxicity, with future emphasis likely to be on low concentrations of heavy metals and metalloids, pharmaceutical agents, endocrine disrupters, and other emerging pollutants of concern.

PRIMARY TREATMENT

Wastewater treatment is directed toward removal of pollutants with the least effort. Suspended solids are removed by either physical or chemical separation techniques and handled as concentrated solids.

Screens Fine screens such as hydroscreens are used to remove moderate-size particles that are not easily compressed under fluid flow. Fine screens are normally used when the quantities of screened particles are large enough to justify the additional units. Mechanically cleaned fine screens have been used for separating large particles. A few industries have used large bar screens to catch large solids that could clog or damage pumps or equipment following the screens.

Grit Chambers Industries with sand or hard, inert particles in their wastewaters have found aerated grit chambers useful for the rapid separation of these inert particles. Aerated grit chambers are relatively small, with total volume based on 3-min retention at maximum flow. Diffused air is normally used to create the mixing pattern shown in Fig. 22-31, with the heavy, inert particles removed by centrifugal action and friction against the tank walls. The air flow rate is adjusted for the specific particles to be removed. Floatable solids are removed in the aerated grit chamber. It is important to provide for regular removal of floatable solids from the surface of the grit chamber; otherwise, nuisance conditions will be created. The settled grit is normally removed with a continuous screw and buried in a landfill.

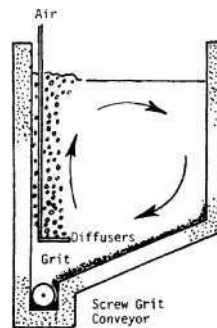


FIG. 22-31 Schematic diagram of an aerated grit chamber.

Gravity Sedimentation Slowly settling particles are removed with gravity sedimentation tanks. For the most part, these tanks are designed on the basis of retention time, surface overflow rate, and minimum depth. A sedimentation tank can be rectangular or circular. The important factor affecting its removal efficiency is the hydraulic flow pattern through the tank. The energy contained in the incoming wastewater flow must be dissipated before the solids can settle. The wastewater flow must be distributed properly through the sedimentation volume for maximum settling efficiency. After the solids have settled, the settled effluent should be collected without creating serious hydraulic currents that could adversely affect the sedimentation process. Effluent weirs are placed at the end of rectangular sedimentation tanks and around the periphery of circular sedimentation tanks to ensure uniform flow out of the tanks. Once the solids have settled, they must be removed from the sedimentation-tank floor by scraping and hydraulic flow. Conventional sedimentation tanks have sludge hoppers to collect the concentrated sludge and to prevent removal of excess volumes of water with the settled solids. Cross-sectional diagrams of conventional sedimentation tanks are shown in Figs. 22-32 and 22-33.

Design criteria for gravity sedimentation tanks normally provide for 2-h retention based on average flow, with longer retention periods used for light solids or inert solids that do not change during their retention in the tank. Care should be taken that sedimentation time is not too long; otherwise, the solids will compact too densely and affect

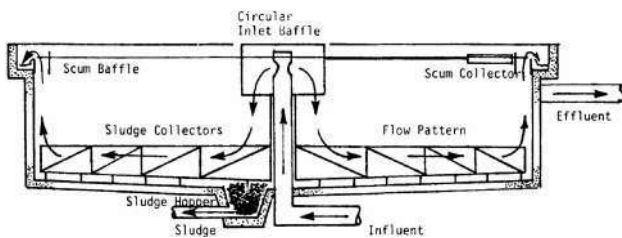


FIG. 22-32 Schematic diagram of a circular sedimentation tank.

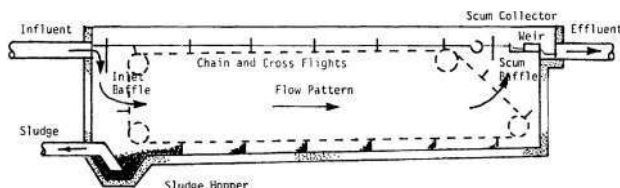


FIG. 22-33 Schematic diagram of a rectangular sedimentation tank.

solids collection and removal. Organic solids generally will not compact to more than 5 to 10 percent. Inorganic solids will compact up to 20 or 30 percent. Centrifugal sludge pumps can handle solids up to 5 or 6 percent, while positive-displacement sludge pumps can handle solids up to 10 percent. With solids above 10 percent the sludge tends to lose fluid properties and must be handled as a semi-solid rather than a fluid. Circular sedimentation tanks have steel truss boxes with angled sludge scrapers on the lower side. As the sludge scrapers rotate, the solids are pushed toward the sludge hopper for removal on a continuous or semicontinuous basis. The rectangular sedimentation tanks employ chain-and-flight sludge collectors or rail-mounted sludge collectors. When floating solids can occur in primary sedimentation tanks, surface skimmers are mounted on the sludge scrapers so that the surface solids are removed at regular intervals.

The surface overflow rate (SOR) for primary sedimentation is normally held close to $40.74 \text{ m}^3/(\text{m}^2\cdot\text{day})$ [$1000 \text{ gal}/(\text{ft}^2\cdot\text{day})$] for average flow rates, depending upon the solids characteristics. Lowering the SOR below $40.74 \text{ m}^3/(\text{m}^2\cdot\text{day})$ does not produce improved effluent quality in proportion to the reduction in SOR. Generally, the minimum depth of sedimentation tanks is 3.0 m (10 ft), with circular sedimentation tanks having a minimum diameter of 6.0 m (20 ft) and rectangular sedimentation tanks having length-to-width ratios of 5:1. Chain-and-flight limitations generally keep the width of rectangular sedimentation tanks to increments of 6.0 m (20 ft) or less. While hydraulic overflow rates have been limited on the effluent weirs, operating experience has indicated that the recommended limit of $186 \text{ m}^3/(\text{m}\cdot\text{day})$ [$15,000 \text{ gal}/(\text{ft}\cdot\text{day})$] is lower than necessary for good operation. A circular sedimentation tank with a single-edge weir provides adequate weir length and is easier to adjust than one with a double-sided weir. More problems appear to be created from improper adjustment of the effluent weirs than from improper length.

Chemical Precipitation Lightweight suspended solids and colloidal solids can be removed by chemical precipitation and gravity sedimentation. In effect, the chemical precipitate is used to agglomerate the tiny particles into large particles that settle rapidly in normal sedimentation tanks. Aluminum sulfate, ferric chloride, ferrous sulfate, lime, and polyelectrolytes have been used as coagulants. The choice of coagulant depends upon the chemical characteristics of the particles being removed, the pH of the wastewaters, and the cost and availability of the precipitants. While the precipitation reaction results in removal of the suspended solids, it increases the amount of sludge to be handled. The chemical sludge must be considered along with the characteristics of the original suspended solids in evaluating sludge-processing systems.

Normally, chemical precipitation requires a rapid mixing system and a flocculation system ahead of the sedimentation tank. With a rectangular sedimentation tank, the rapid-mixer and flocculation units are added ahead of the tank. With a circular sedimentation tank the rapid-mixer and flocculation units are built into the tank. Schematic diagrams of chemical treatment systems are shown in Figs. 22-34 and 22-35. Rapid mixers are designed to provide 30-s retention at average flow with sufficient turbulence to mix the chemicals with the incoming wastewaters. The flocculation units are designed for slow mixing at 20-min retention. These units are designed to cause the particles to collide and increase in size without excessive shearing. Care must be taken to move the flocculated mixture from the flocculation unit to the sedimentation unit without disrupting the large floc particles.

The parameter used to design rapid mix and flocculation systems is the root mean square velocity gradient G , which is defined by equation

$$G = \left(\frac{P}{VU} \right)^{1/2} \left(\frac{1}{s} \right)$$

where P = power input to the water (ft·lb/s)
 V = mixer or flocculator volume (ft³)
 U = absolute viscosity of water (lb·s/ft²)

Optimum mixing usually requires a G value of greater than 1000 inverse seconds. Optimum flocculation occurs when G is in the range 10–100 inverse seconds.

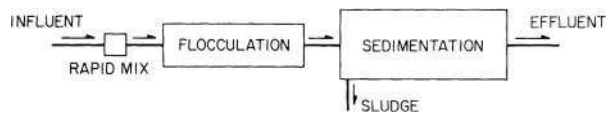


FIG. 22-34 Schematic diagram of a chemical precipitation system for rectangular sedimentation tanks.

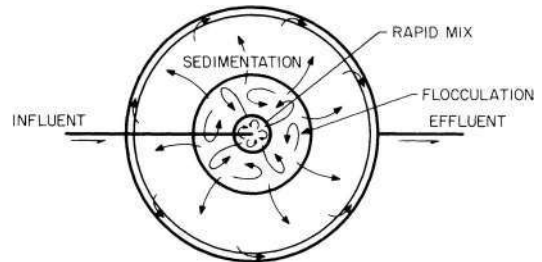
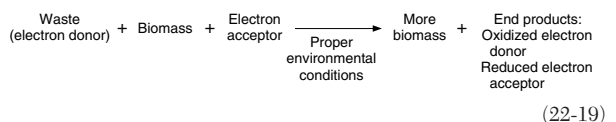


FIG. 22-35 Schematic diagram of a chemical precipitation system for circular sedimentation tanks.

Chemical precipitation can remove 95 percent of the suspended solids, up to 50 percent of the soluble organics and the bulk of the heavy metals in a wastewater. Removal of soluble organics is a function of the coagulant chemical, with iron salts yielding best results and lime the poorest. Metal removal is primarily a function of pH and the ionic state of the metal. Guidance is available from solubility product data.

SECONDARY TREATMENT

Secondary treatment utilizes processes in which microorganisms, primarily bacteria, stabilize waste components. The mixture of microorganisms is usually referred to as biomass. A portion of the waste is oxidized, releasing energy, the remainder is utilized as building blocks of protoplasm. The energy released by biomass metabolism is utilized to produce the new units of protoplasm. Thus, the incentive for the biomass to stabilize waste is that it provides the energy and basic chemical components required for reproduction. The process of biological waste conversion is illustrated by Eq. (22-19).



As this equation indicates, the waste generally serves as an electron donor, necessitating that an electron acceptor be supplied. A variety of substances can be utilized as electron acceptors, including molecular oxygen, carbon dioxide, oxidized forms of nitrogen, sulfur, and organic substances. The characteristics of the end products of the reaction are determined by the electron acceptor. Table 22-43 is a list of typical end products as a function of the electron acceptor. In general, the end products of this reaction are at a much lower energy level than the waste components, thus resulting in the release of energy referred to above. Although this process is usually utilized for the stabilization of organic substances, it can also be utilized for oxidation of inorganics. For example, biomass-mediated oxidation of iron, nitrogen, and sulfur is known to occur in nature and in anthropogenic processes.

Equation (22-19) describes the biomass-mediated reaction and indicates that proper environmental conditions are required for the reaction to take place. These conditions are required by the biomass, not the electron donor or acceptor. The environmental conditions include pH, temperature, nutrients, ionic balance, and so on. In general, biomass can function over a wide pH range generally from 5 to 9. However,

TABLE 22-43 Electron Acceptors and End Products for Biological Reactions

Electron acceptors	End product
Molecular oxygen	Water, CO ₂ , oxidized nitrogen
Oxidized nitrogen	N ₂ , N ₂ O, NO, CO ₂ , H ₂ O
Oxidized sulfur	H ₂ S, S, CO ₂ , H ₂ O
CO ₂ , acetic acid, formic acid	CH ₄ , CO ₂ , H ₂
Complex organics	H ₂ , simple organics, CO ₂ , H ₂ O

some microbes require a much narrower pH range; i.e., effective methane fermentation requires a pH in the range of 6.5–7.5. It is just as important to maintain a relatively constant pH in the process as it is to stay within the range given above. Microorganisms can function effectively at the extremes of their pH range provided they are given the opportunity to acclimate to these conditions. Continual changes in pH are detrimental, even if the organisms are on the average near the middle of their effective pH range. A similar situation prevails for temperature. Most organisms can function well over a broad range of temperature but do not adjust well to frequent fluctuations of even a few degrees. There are three major temperature ranges in which microorganisms function. The psychrophilic range (5 to 20°C), the mesophilic range (20 to 45°C), and the thermophilic range (45 to 70°C). In general the microbes that function in one of these temperature ranges cannot function efficiently in the other ranges. As it is generally uneconomical to adjust the temperature of a waste, most processes are operated in the mesophilic range. If the normal temperature of the waste is above or below the mesophilic range, the process will be operated in the psychrophilic or thermophilic range as appropriate. However, occasionally the temperature of the waste is altered to improve performance. For example, some anaerobic treatment processes are operated under thermophilic conditions, even though the waste must be heated to achieve this temperature range. This is carried out in order to speed up the degradation of complex organics and/or to achieve kill of mesophilic pathogens. It should be noted that any time the biological operation of a process moves away from its optimum or most effective range, be it pH, temperature, nutrients, or what have you, the rate of biological processing is reduced.

All microorganisms require varying amounts of a large number of nutrients. These are required because they are necessary components of bacterial protoplasm. The nutrients can be divided into three groups: macro, minor, and micro. The macronutrients are those that comprise most of the biomass. These are given by the commonly accepted formula for biomass (C₆₀H₈₇O₂₃N₁₂P). The carbon, hydrogen, and oxygen are normally supplied by the waste and water, but the nitrogen and phosphorous must often be added to industrial wastes to ensure that a sufficient amount is present. A good rule is that the mass of nitrogen should be at least 5 percent of the BOD, and the mass of phosphorous should be at least 20 percent of the mass of nitrogen. One of the major operational expenses is the purchase of nitrogen and phosphorous for addition to biologically based treatment processes. The quantities of nitrogen and phosphorous referred to above as required are actually in excess of the minimum amounts needed. The actual amount required depends upon the quantity of excess biomass wasted from the system and the amount of N and P available in the waste. This will be expanded upon later in this section. The minor nutrients include the typical inorganic components of water. These are given in Table 22-44. The range of concentrations required in the wastewater for the minor nutrients is 1–100 mg/L. The micronutrients include the substances that we normally refer to as trace metals and vitamins. It is interesting to note that the trace metals include virtually all of the toxic heavy metals. This reinforces the statement made above that toxicity is a function of concentration and not an absolute parameter. Whether or not the substances referred to as vitamins will be required depends upon the type of microorganisms required to stabilize the waste materials. Many microorganisms have the ability to make their own vitamins from the waste components; thus, a supplement is not needed. However, occasionally the addition of an external source of vitamins is essential to the success of a biologically based waste-treatment sys-

TABLE 22-44 Minor and Micro Nutrients Required for Biologically Mediated Reactors

Minor 1–100 mg/L
Sodium, potassium, calcium, magnesium, iron, chloride, sulfate
Micro 1–100 µg/L
Copper, cobalt, nickel, manganese, boron, vanadium, zinc, lead, molybdenum, various organic vitamins, various amino acids

tem. In general, the trace nutrients must be present in a waste at a level of a few micrograms per liter.

One aspect of the basic equation describing biological treatment of waste that has not been referred to previously is that biomass appears on both sides of the equation. As was indicated above, the only reason that microorganisms function in waste-treatment systems is because it enables them to reproduce. Thus, the quantity of biomass in a waste-treatment system is higher after the treatment process than before it. This is favorable in that there is a continual production of the organisms required to stabilize the waste. Thus, one of the major reactants is, in effect, available free of charge. However, there is an unfavorable side in that unless some organisms are wasted from the system, an excess level will build up, and the process could choke on organisms. The wasted organisms are referred to as sludge. A major cost component of all biologically based processes is the need to provide for the ultimate disposal of this sludge.

Biologically based treatment processes probably account for the majority of the treatment systems used for industrial waste management because of their low cost and because most substances are amenable to biological breakdown. However, some substances are difficult to degrade biologically. Unfortunately, it is not possible at present to predict *a priori* the biodegradability of a specific organic compound; rather, we must depend upon experience and testing. The collective experience of the field has been put into compendia by EPA in a variety of documents. However, these data are primarily qualitative. There have been some attempts to develop a system of prediction of biodegradability based on a number of compound parameters such as solubility, presence or absence of certain functional groups, compound polarity, and so on. Unfortunately, none of these systems has advanced to the point where reliable quantitative predictions are possible. Another complication is that some organics that are easily biodegradable at low concentration exert a toxic effect at high concentration. Thus, literature data can be confusing. Phenol is a typical compound that shows ease of biodegradation when the concentration is below 500 mg/L but poor biodegradation at higher concentrations. Another factor affecting both biodegradation and toxicity is whether or not a substance is in solution. In general, if a substance is not in solution, it is not available to affect the biomass. Thus, the presence of a waste in substances that can precipitate, complex, or absorb other waste components can have a significant effect on reports of biodegradability and/or toxicity. A quantitative estimate of toxicity can be obtained in terms of the change in kinetic parameters of a system. These kinetic parameters are discussed below.

Design of Biological Treatment Systems In the past, the design of biologically based waste-treatment systems has been derived from rules of thumb. During the past two decades, however, a more fundamental system has been developed and is presently widely used to design such systems. This system is based upon a fundamental understanding of the kinetics and stoichiometry of biological reactions. The system is codified in terms of equations in which four pseudo constants appear. These pseudo constants are k_m , the maximum substrate utilization rate (1/time); K_s , the half maximal velocity concentration (mg/L); Y , the yield coefficient; and b , the endogenous respiration rate (1/time). These are referred to as pseudo constants because, in the mathematical manipulation of the equations in which they appear, they are treated as constants. However, the value of each is a function of the nature of the microbes, the pH, the temperature, and the components of the waste. It is important to remember that if any of these change, the value of the pseudo

constants may change as well. The kinetics of biological reactions are described by Eq. (22-20).

$$\frac{dS}{dt} = \frac{k_m SX}{K_s + S} \quad (22-20)$$

where t = time (days)
 S = waste concentration (mg/L)
 X = biomass concentration (mg/L)
 k_m = (mg/L substrate ÷ mg/L biomass) – time

The accumulation or growth of biosolids is given by Eq. (22-21).

$$\frac{dX}{dt} = Y \frac{dS}{dt} - bX \quad (22-21)$$

where X = biomass level (mg/L).

The equations that have been developed for design using these pseudo constants are based on steady-state mass balances of the biomass and the waste components around both the reactor of the system and the device used to separate and recycle microorganisms. Thus, the equations that can be derived will be dependent upon the characteristics of the reactor and the separator. It is impossible here to present equations for all the different types of systems. As an illustration, the equations for a common system (a complete mix – stirred tank reactor with recycle) are presented below.

$$S_e = \frac{K_s[1 + \text{BSRT}(b)]}{\text{BSRT}(YK_m - b) - 1} \quad (22-22)$$

$$X = \frac{(\text{BSRT})(Y)(S_o - S_e)}{\text{HRT}[1 + (b)\text{BSRT}]} \quad (22-23)$$

where S_e = influent waste concentration (mg/L)
 S_o = treated waste concentration (mg/L)
 HRT = reactor hydraulic retention time

Note that these equations predict some unexpected results. The strength of the untreated waste (S_o) has no effect on the strength of the treated waste (S_e). Neither does the size of the reactor (HRT). Rather, a parameter referred to as the biomass solids retention time (BSRT) is the key parameter determining the system performance. This is illustrated in Fig. 22-36. As the BSRT increases, the concentration of untreated waste in the effluent decreases irrespective of the reactor size or the waste strength. However, the reactor size and waste strength have a significant effect on the level of biomass (X) that is maintained in the system at steady state. Since the development of an excess level of biomass in the system can lead to system upset, it is important to take cognizance of the waste strength and the reactor size. But treatment performance with respect to removal of the waste components is again a function only of BSRT and the value of the

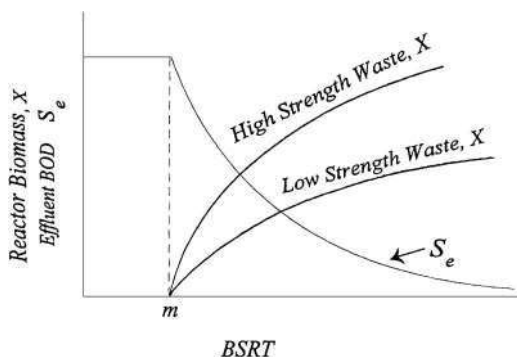


FIG. 22-36 Effect of BSRT on biological treatment process performance. m = minimum BSRT.

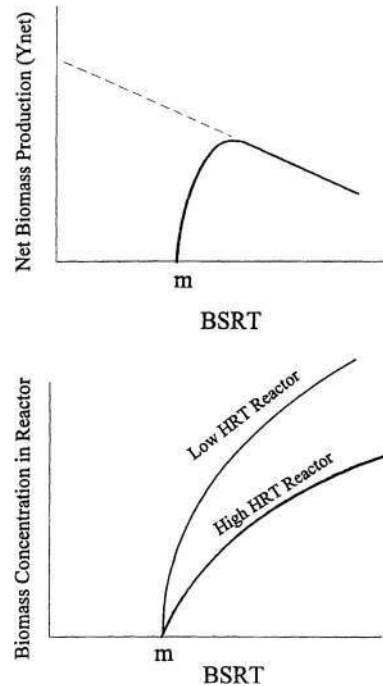


FIG. 22-37 Biomass production.

pseudo constants referred to above. The BSRT also has an effect on the level of biomass in the system and the quantity of excess biomass produced (Fig. 22-37). The latter will determine the quantity of waste sludge which must be dealt with as well as the N and P requirement. The N and P in the biomass removed from the system each day must be replaced. From the formula given previously for biomass, N is 12 percent and P is 2.3 percent by weight in biomass grown under ideal conditions (see Fig. 22-37). Also, note that m , the minimum BSRT, is the minimum time required for the microorganisms to double in mass. Below this minimum, washout occurs and substrate removal approaches zero.

Although identical results are not obtained for other reactor configurations, the design equations yield similar patterns. The dominant parameters in determining system performance are again the BSRT and the pseudo constants. The latter are not under the control of the design engineer as they are functions of the waste and the microorganisms that developed in the system. The BSRT is the major design parameter under the control of the design engineer. This parameter has been defined as the ratio of the biomass in the reactor to the biomass produced from the waste each day. At steady state, the level of biomass in the system is constant; thus, the biomass produced must equal the biomass wasted. The minimum BSRT that can be utilized is that which will produce the degree of treatment required. Generally, this minimum value is less than the range of BSRT used in design and operation. This provides not only a safety factor, but, in addition, it is necessary in order to foster the growth of certain favorable groups of microorganisms in the system. Generally, BSRT values in the range of 3–15 days are utilized in most systems, although BSRT values of less than 1 day for high rate systems are usually adequate to ensure greater than 95 percent destruction of waste components in all but anaerobic systems. For the latter, a minimum BSRT of 8 days is needed for 95 percent destruction. BSRT is controlled by the concentration of biomass (X) in the system and the quantity wasted each day; thus, the treatment plant operator can alter BSRT by altering the rate of biomass wasting.

Reactor Concepts A large number of reactor concepts seem to be used in biologically based treatment systems. However, this diversity

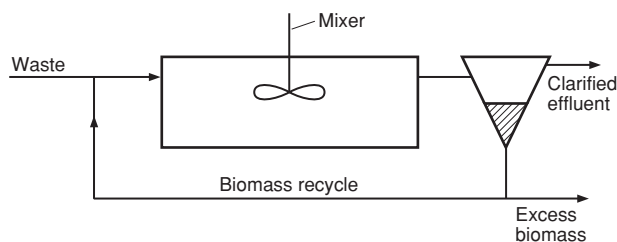


FIG. 22-38 Diagram of a suspended growth system.

is more apparent than real. In reality, there are only two major reactor types that are used. One is referred to as a suspended growth reactor, the other is referred to as a fixed film reactor. In the former, the waste and the microorganisms move through the reactor, with the microorganisms constantly suspended in the flow. After exiting the reactor, the suspension flows through a separator, which separates the organisms from the liquid. Some of the organisms are wasted as sludge, while the remainder are returned to the reactor. The supernatant is discharged either to the environment or to other treatment units (Fig. 22-38). The functioning of the separator is very important, as poor performance of the separator will result in high solids in the effluent and reduction of organisms in the recycle and eventually also in the reactor. As indicated above, the level of the BSRT used in design of suspended growth systems is set to produce a biomass of the proper level in the system; i.e., a level at which waste degradation is rapid but not so high that excess loading on the separator will occur. BSRT also influences the ability of biomass to self flocculate and thus be removed in the separator (usually a clarifier).

In the fixed-film reactor, the organisms grow on an inert surface that is maintained in the reactor. The inert surface can be granular material, proprietary plastic packing, rotating discs, wood slats, mass-transfer packing, or even a sponge-type material. The reactor can be flooded or have a mixed gas-liquid space (Fig. 22-39). The biomass level on the packing is controlled by hydraulic scour produced as the waste liquid flows through the reactor. There is no need of a separator to ensure that the biomass level in the reactor is maintained. The specific surface area of the packing is the design parameter used to ensure an adequate level of biomass. However, a separator is usually supplied to capture biomass washed from the packing surface by the flow of the waste, thus providing a clarified effluent. As with the suspended growth system, if this biomass escapes the system, it will result

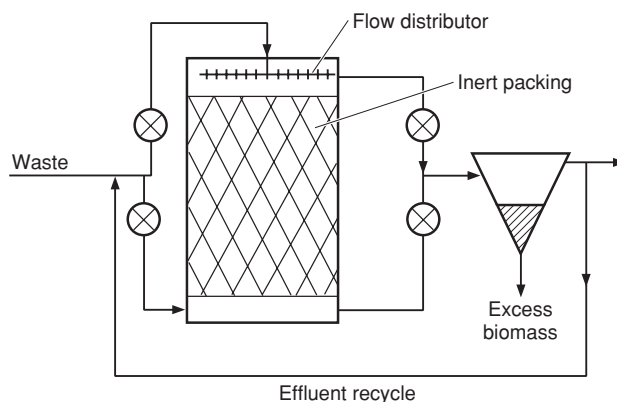


FIG. 22-39 Diagram of a fixed-film system.

in a return to the environment of organic laden effluent, thus negating the effectiveness of the waste treatment. As indicated above, biomass level in a fixed film reactor is maintained by a balance between the rate of growth on the packing (which is a function of the strength of the waste, the yield coefficient and the BSRT), and the rate of hydraulic flushing by the waste flow. Recycle of treated wastewater is used to control the degree of hydraulic flushing. Thus, typical design parameters for fixed film reactors include both an organic loading and a hydraulic loading. Details on the typical levels for these will be given in a later section.

The reactor concepts described above can be utilized with any of the electron acceptor systems previously discussed, although some reactor types perform better with specific electron acceptors. For example, suspended growth systems are generally superior to fixed film systems when molecular oxygen is the electron acceptor because it is easier to supply oxygen to suspended growth systems.

On the other hand, fixed-film systems that use nitrate as the electron acceptor can be superior to activated sludge systems since nitrogen bubbles produced won't affect sludge settling and oxygen diffusion limitations are not an issue. Different reactor types are better suited for different strengths and types of wastewater. Weak wastewaters are generally easier to treat with fixed-film systems since the appropriate amount of biomass is inherently maintained in the system. Very strong wastes are typically better treated using anaerobic systems to eliminate oxygen limitation and significantly reduce waste sludge production. In addition, the yield coefficient Y is much lower for anaerobic systems, reducing sludge production and excessive biomass accumulation. Different types and combinations of activated and fixed-film reactor types can be utilized to perform tertiary treatment on wastewaters and to reclaim hazardous waste streams that are contaminated with recalcitrant chemicals such as PCBs, halogenated aliphatics, and PAHs. Table 22-45 summarizes some of the advantages and disadvantages of various combinations of reactor type, electron acceptor, and waste strength.

Determination of Kinetic and Stoichiometric Pseudo Constants As indicated above, these parameters are most important for predicting the performance of biologically based treatment systems. It would be ideal if tabulations of these were available for various industrial wastes as a function of pH temperature and nutrient levels. Unfortunately, little reliable data has been codified. Only certain trends have been established, and these are primarily the result of studies on municipal wastewater. For example, the yield coefficient Y has been shown to be much higher for systems that are aerobic (molecular oxygen as the electron acceptor) than for anaerobic systems (sulfate or carbon dioxide as the electron acceptors). Systems where oxidized nitrogen is the electron acceptor (termed *anoxic*) exhibit yield values intermediate between aerobic and

TABLE 22-45 Favorable (F) and Unfavorable (U) Combinations of Electron Acceptor, Waste Strength, and Reactor Type

Suspended growth reactor		
Electron acceptor	Waste strength	Condition
Aerobic	Low–modest	F
Aerobic	High	U
Anoxic	Low–modest	F
Anoxic	High	U
Anaerobic	Low–modest	U
Anaerobic	High	F
Fixed film reactor		
Electron acceptor	Waste strength	Condition
Aerobic	Low	F
Aerobic	Modest–high	U
Anoxic	Low–modest	F
Anoxic	High	U
Anaerobic	Low–modest	F
Anaerobic	High	F

anaerobic systems. The endogenous respiration rate is higher for aerobic and anoxic systems than anaerobic systems. However, no trends have been established for values of the maximum specific growth rate or the half maximal velocity concentration. Thus, the values applicable to a specific waste must be determined from laboratory studies.

The laboratory studies utilized small-scale (1–5-L) reactors. These are satisfactory because the reaction rates observed are independent of reactor size. Several reactors are operated in parallel on the waste, each at a different BSRT. When steady state is reached after several weeks, data on the biomass level (X) in the system and the untreated waste level in the effluent (usually in terms of BOD or COD) are collected. These data can be plotted for equation forms that will yield linear plots on rectangular coordinates. From the intercepts and the slope of the lines, it is possible to determine values of the four pseudo constants. Table 22-46 presents some available data from the literature on these pseudo constants. Figure 22-40 illustrates the procedure for their determination from the laboratory studies discussed previously.

Activated Sludge This treatment process is the most widely used aerobic suspended growth reactor system. It will consistently produce a high-quality effluent (BOD₅ and SS of 20–30 mg/L). Operational costs are higher than for other secondary treatment processes primarily because of the need to supply molecular oxygen using energy-intensive mechanical aerator- or sparger-type equipment. Removal of soluble organics, colloidal, particulates, and inorganics are achieved in this system through a combination of biological metabolism, adsorption, and entrapment in the biological floc. Indeed, many pollutants that are not biologically degradable are removed during activated sludge treatment by adsorption or entrapment by the floc. For example, most heavy metals form hydroxide or carbonate precipitates under the pH conditions maintained in activated sludge, and most organics are easily adsorbed to the surface of the biological floc. A qualitative guide to the latter is provided by the octanol-water partition coefficient of a compound.

All activated sludge systems include a suspended growth reactor in which the wastewater, recycled sludge, and molecular oxygen are mixed. The latter must be dissolved in the water; thus the need for an energy-intensive pure oxygen or air supply system. Usually, air is the source of the molecular oxygen rather than pure oxygen. Energy for mixing of the reactor contents is supplied by the aeration equipment. All systems include a separator and pump station for sludge recycle and sludge wasting. The separator is usually a sedimentation tank that is designed to function as both a clarifier and a thickener. Many modifications of the activated sludge process have been developed over the years and are described below. Most of these involve differences in the way the reactor is compartmentalized with respect to introduction of waste, recycle, and/or oxygen supply.

Modifications The modifications of activated sludge systems offer considerable choice in processes. Some of the most popular modifications of the activated sludge process are illustrated in Fig. 22-41. Conventional activated sludge uses a long narrow reactor with air supplied along the length of the reactor. The recycle sludge and waste are introduced at the head end of the reactor producing a zone of high waste to biomass concentration and high oxygen demand.

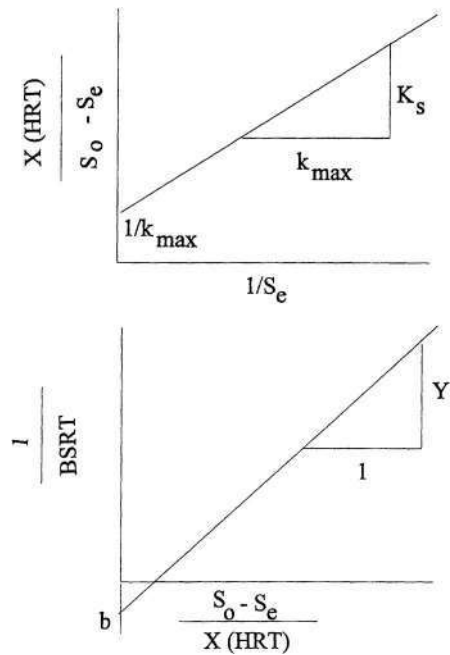


FIG. 22-40 Determination of pseudo-kinetic constants.

This modification is used for relatively dilute wastes such as municipal wastewater. Step aeration systems distribute the waste along the length of the aerator, thus reducing the oxygen demand at the head end of the reactor and spreading the oxygen demand more uniformly over the whole reactor. In the complete mix system, the waste and sludge recycle are uniformly distributed over the whole reactor, resulting in the waste load and oxygen demand being uniform in the entire reactor. Complete-mixing activated sludge is the most popular system for industrial wastes because of its ability to absorb shock loads better than other modifications. Contact stabilization is a modification of activated sludge that is best suited to wastewaters having high suspended solids and low soluble organics. Contact stabilization employs a short-term mixing tank to adsorb the suspended solids and metabolize the soluble organics, a sedimentation tank for solids separation, and a reaeration tank for stabilization of the suspended organics. Extended-aeration systems are actually long-term-aeration, completely mixed activated-sludge systems. They employ 24- to 48-h aeration periods and high mixed-liquor suspended solids to provide complete stabilization of the organics and aerobic digestion of the activated sludge in the same aeration tank. The oxidation ditch is a popular form of the extended-aeration system employing mechanical aeration. Pure-oxygen systems are designed to treat strong industrial

TABLE 22-46 Typical of Values for Pseudo Constants

Biomass type	Substrate	$k_{\max}$ mg substrate mg biomass-day	Y , mg biomass mg substrate	K_s , mg substrate liter	b , 1 day	Remarks
Mixed culture	Sewage (COD)	5–10	0.5	50	0.05	Aerobic 20°C
Mixed culture	Glucose (COD)	7.5	0.6	10	0.07–.1	Aerobic 20°C
Mixed culture	Skim milk (COD)	5	—	100	0.05	Aerobic 20°C
Mixed culture	Soybean waste (COD)	12	—	355	0.144	Aerobic 20°C
Methane bacteria	Acetic acid (COD)	8.7	.04	165	0.035	35°C
Anaerobic mixed	Propanoic acid (COD)	7.7	.04	60	0.035	35°C
Anaerobic mixed	Sewage sludge (COD)	6.7	.04	1,800	0.03	35°C
Anoxic	NO ₃ as N	0.375	0.8	0.1	0.04	Methanol feed
Aerobic	NH ₄ as N	5	0.2	1.4	.05	20°C

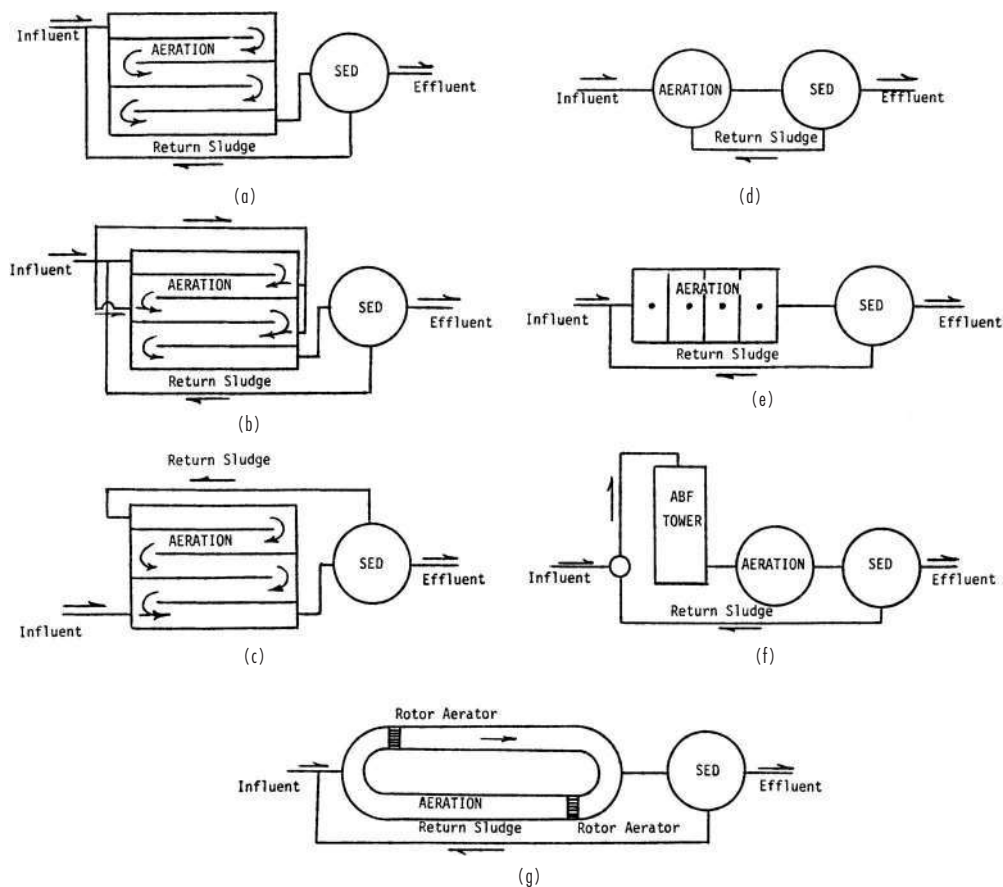


FIG. 22-41 Schematic diagrams of various modifications of the activated-sludge process. (a) Conventional activated sludge. (b) Step aeration. (c) Contact stabilization. (d) Complete mixing. (e) Pure oxygen. (f) Activated biofiltration (ABF). (g) Oxidation ditch.

wastes in a series of completely mixed units having relatively short contact periods. One of the latest modifications of activated sludge employs powdered activated carbon to adsorb complex organics and assist in solids separation. Another modification employs a redwood-medium trickling filter ahead of a short-term aeration tank with mixed liquor recycled over the redwood-medium tower to provide heavy microbial growth on the redwood as well as in the aeration tank.

As indicated previously, success with the activated sludge process requires that the biomass have good self-flocculating properties. Significant research effort has been expended to determine the conditions that favor the development of good settling biomass cultures. These have indicated that nutritional deficiency, levels of dissolved oxygen between 0 to 0.5 mg/L, and pH values below 6.0 will favor the predomination of filamentous biomass. Filamentous organisms settle and compact poorly and thus are difficult to separate from liquid. By avoiding the above conditions, and with the application of selector technology, the predomination of filaments in the biomass can be eliminated. A selector is often a short contact (15–30 min) reactor set ahead of the main activated sludge reactor. All of the recycle sludge and all of the waste are routed to the selector. In the selector, either a high rate of aeration (aerobic selector) is used to keep the dissolved oxygen above 2 mg/L, or no aeration occurs (anaerobic selector) so that the dissolved oxygen is zero. Because filamentous organisms are microaerophilic, they cannot predominate when the dissolved oxygen level is zero or high. The anaerobic selector

not only selects in favor of nonfilamentous biomass but also fosters luxury uptake of phosphorous and is used in systems where phosphorous removal is desired.

Aeration Systems These systems control the design of aeration tanks. Aeration equipment has two major functions: mixing and oxygen transfer. Diffused-aeration equipment employs either a fixed-speed positive-displacement blower or a high-speed turbine blower for readily adjustable air volumes. Air diffusers can be located along one side of the aeration tank or spread over the entire bottom of the tank. They can be either fine-bubble or coarse-bubble diffusers. Fine-bubble diffusers are more efficient in oxygen transfer but require more extensive air-cleaning equipment to prevent them from clogging as a result of dirty air. Mechanical-surface-aeration equipment is more efficient than diffused-aeration equipment but is not as flexible. Economics has dictated the use of large-power aerators, but tank configuration has tended to favor the use of greater numbers of lower-power aerators. Oxidation ditches use horizontal rotor-type aerators. Mixing is a critical problem with mechanical-surface aerators since they are a point-source pump of limited capacity. Experience has indicated that bearings are a serious problem with mechanical-aeration equipment. Wave action generated within the aeration tank tends to produce lateral stresses on the bearings and has resulted in failures and increased maintenance costs. Slow-speed mechanical-surface-aeration units present fewer problems than the high-speed mechanical-surface-aeration units. Deep tanks, greater than 3.0 m (10 ft), require draft tubes to ensure proper hydraulic flow through the aeration tank.

Short-circuiting is one of the major problems associated with mechanical aeration equipment. Combined mechanical- and diffused-aeration systems have enjoyed some popularity for industrial-waste systems that treat variable organic loads. The mechanical mixers provide the fluid mixing with the diffused aeration varied for different oxygen-transfer rates.

Diffused-aeration systems transfer from 20 to 40 mg/(L O₂·h). Combined mechanical- and diffused-aeration systems can transfer up to 65 mg/(L O₂·h), while mechanical-surface aerators can provide up to 90 mg/(L O₂·h). Pure-oxygen systems can provide the highest oxygen-transfer rate, up to 150 mg/(L O₂·h). Aeration equipment must provide sufficient oxygen to meet the peak oxygen demand; otherwise, the system will fail to provide proper treatment. For this reason, the peak oxygen demand and the rate of transfer for the desired equipment determine the size of the aeration tank in terms of retention time. Economics dictates a balance between the size of the aeration tank and the size of the aeration equipment. As the cost of power increases, economics will favor constructing a larger aeration tank and smaller aerators. It is equally important to examine the hydraulic flow pattern around each aerator to ensure maximum efficiency of oxygen transfer. Improper spacing of aeration equipment can waste energy.

There is no standard aeration-tank shape or size. Aeration tanks can be round, square, or rectangular. Shallow aeration tanks are more difficult to mix than deeper tanks. Yet aeration-tank depths have ranged from 0.6 m (2 ft) to 18 m (60 ft). The oxidation-ditch systems tend to be shallow, while some high-rate diffused-aeration systems have used very deep tanks to provide more efficient oxygen transfer.

Regardless of the aeration equipment employed, oxygen-transfer rates must provide from 0.6 to 1.4 kg of oxygen/kg BOD₅ (0.6 to 1.4 lb oxygen/lb BOD₅) stabilized in the aeration tank for carbonaceous-oxygen demand. Nitrogen oxidation can increase oxygen demand at the rate of 4.3 kg (4.3 lb) of oxygen/kg (lb) of ammonia nitrogen oxidized. At low oxygen-transfer rates more excess activated sludge must be removed from the system than at high oxygen-transfer rates. Here again the economics of sludge handling must be balanced against the cost of oxygen transfer. The quantity of waste activated sludge will depend upon wastewater characteristics. The inert suspended solids entering the treatment system must be removed with the excess activated sludge. The soluble organics are stabilized by converting a portion of the organics into suspended solids, producing from 0.3 to 0.8 kg (0.3 to 0.8 lb) of volatile suspended solids/kg (lb) of BOD₅ stabilized. Biodegradable suspended solids in the wastewaters will result in destruction of the original suspended solids and their conversion to a new form. Depending upon the chemical characteristics of the biodegradable suspended solids, the conversion factor will range from 0.7 to 1.2 kg (0.7 to 1.2 lb) of microbial solids produced/kg (lb) of suspended solids destroyed. If the suspended solids produced by metabolism are not wasted from the system, they will eventually be discharged in the effluent. While considerable efforts have been directed toward developing activated-sludge systems which totally consume the excess solids, no such system has proved to be practical. The concept of total oxidation of excess sludge is fundamentally unsound and should be recognized as such.

A definitive determination of the waste sludge production and the oxygen requirement can be obtained using the pseudo constants referred to previously. The ultimate BOD in the waste will be accounted for by the sum of the oxidation and sludge synthesis [Eq. (22-24)].

Thus:

$$\text{Waste BOD}_u = \text{oxygen required} + \text{BOD}_u \text{ of sludge produced} \quad (22-24)$$

where: BOD_u of sludge = $(Y_{\text{net}})(1.42)/(\text{Waste BOD}_u)$

$$Y_{\text{net}} = \frac{Y}{1 + b(\text{BSRT})}$$

1.42 = Factor converting biomass to oxygen units; i.e.,
BOD_u ≈ COD

Sedimentation Tanks These tanks are an integral part of any activated-sludge system. It is essential to separate the suspended solids from the treated liquid if a high-quality effluent is to be produced. Circular sedimentation tanks with various types of hydraulic sludge collectors have become the standard secondary sedimentation system. Square tanks have been used with common-wall construction for compact design with multiple tanks. Most secondary sedimentation tanks use center-feed inlets and peripheral-weir outlets. Recently, efforts have been made to employ peripheral inlets with submerged-orifice flow controllers and either center-weir outlets or peripheral-weir outlets adjacent to the peripheral-inlet channel.

Aside from flow control, basic design considerations have centered on surface overflow rates, retention time, and weir overflow rate. Surface overflow rates have been slowly reduced from 33 m³/(m²·day) [800 gal/(ft²·day)] to 24 m³/(m²·day) to 16 m³/(m²·day) [600 gal/(ft²·day) to 400 gal/(ft²·day)] and even to 12 m³/(m²·day) [300 gal/(ft²·day)] in some instances, based on average raw-waste flows. Operational results have not demonstrated that lower surface overflow rates improve effluent quality, making 33 m³/(m²·day) [800 gal/(ft²·day)] the design choice in most systems. Retention time has been found to be an important design factor, averaging 2 h on the basis of raw-waste flows. Longer retention periods tend to produce rising sludge problems, while shorter retention periods do not provide for good solids separation with high-return sludge flow rates. Effluent-weir overflow rates have been limited to 186 m³/(m·day) [15,000 gal/(ft·day)] with a tendency to reduce the rate to 124 m³/(m·day) [10,000 gal/(ft·day)]. Lower effluent-weir overflow rates are obtained by using dual-sided effluent weirs cantilevered from the periphery of the tank. Unfortunately, proper adjustment of dual-side effluent weirs has created more hydraulic problems than the weir overflow rate. Field data have shown that effluent quality is not really affected by weir overflow rates up to 990 m³/(m·day) [80,000 gal/(ft·day)] or even 1240 m³/(m·day) [100,000 gal/(ft·day)] in a properly designed sedimentation tank. A single peripheral weir, being easy to adjust and keep clean, appears to be optimal for secondary sedimentation tanks from an operational point of view.

Depth tends to be determined from the retention time and the surface overflow rate. As surface overflow rates were reduced, the depth of sedimentation tanks was reduced to keep retention time from being excessive. It was recognized that depth was a valid design parameter and was more critical in some systems than retention time. As mixed-liquor suspended-solids (MLSS) concentrations increase, the depth should also be increased. Minimum sedimentation-tank depths for variable operations should be 3.0 m (10 ft) with depths to 4.5 m (15 ft) if 3000 mg/L MLSS concentrations are to be maintained under variable hydraulic conditions. With MLSS concentrations above 4000 mg/L, the depth of the sedimentation tank should be increased to 6.0 m (20 ft). The key is to keep a definite freeboard over the settled-sludge blanket so that variable hydraulic flows do not lift the solids over the effluent weir.

Scum baffles around the periphery of the sedimentation tank and radial scum collectors are standard equipment to ensure that rising solids or other scum materials are removed as quickly as they form. Hydraulic sludge-collection tubes have replaced the center sludge well, but they have caused a new set of operational problems. These tubes were designed to remove the settled sludge at a faster rate than conventional sludge scrapers. To obtain good hydraulic distribution in the sludge-collection tubes, it was necessary to increase the rate of return sludge flow and decrease the concentration of return sludge. The higher total inflow to the sedimentation tank created increased forces that lifted the settled-solids blanket at the wall, causing loss of excessive suspended solids and lower effluent quality. Operating data tend to favor conventional secondary sedimentation tanks over hydraulic sludge-collection systems. Return-sludge rates normally range from 25 to 50 percent for MLSS concentrations up to 3300 mg/L. Most return-sludge pumps are centrifugal pumps with capacities up to 100 percent raw-waste flow.

Gravity settling can concentrate activated sludge to 10,000 mg/L, but hydraulic sludge-collecting tubes tend to operate best below 8,000 mg/L. The excess activated sludge can be wasted either from the

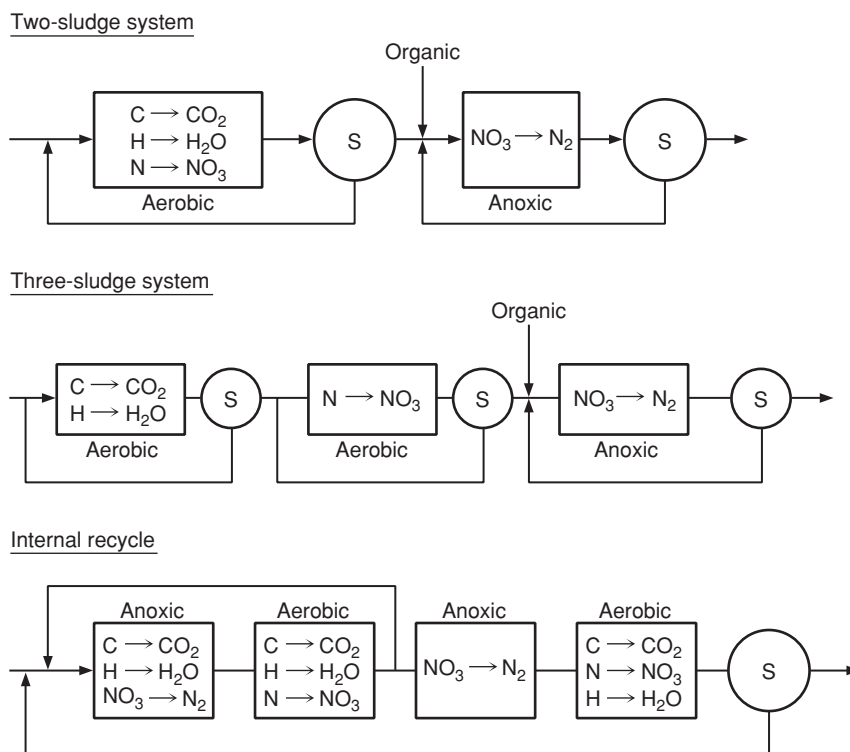


FIG. 22-42 Nitrogen removal systems.

return sludge or from a separate waste-sludge hopper near the center of the tank. The low solids concentrations result in large volumes of waste activated sludge in comparison with primary sludge. Unfortunately, the physical characteristics of waste activated sludge prevent significant concentration without the expenditure of considerable energy. Gravity thickening can produce 2 percent solids, while air flotation can produce 4 percent solids concentration. Centrifuges are able to concentrate activated sludge from 10 to 15 percent solids, but the capture is limited. Vacuum filters can equal the performance of centrifuges if the sludge is chemically conditioned. Filter presses and belt-press filters can produce cakes with 15 to 25 percent solids. It is very important that the excess activated sludge formed in the aeration tanks be wasted on a regular basis; otherwise, effluent quality will deteriorate. Care should be taken to ensure that sludge-thickening systems do not control activated-sludge operations. Alternative sludge-handling provisions should be available during maintenance on sludge-thickening equipment. At no time should final sedimentation tanks be used for the storage of sludge beyond that required by daily operational variations.

Anaerobic/Anoxic Activated Sludge The activated sludge concept (i.e., suspended growth reactor) can be used for anaerobic or anoxic systems in which no oxygen or air is added to the reactor. An anoxic activated sludge is used for systems in which removal of nitrate is a goal or where nitrate is used as the electron acceptor. These systems (denitrification) will be successful if the nitrate is reduced to low levels in the reactor so that nitrate reduction to nitrogen gas does not take place in the clarifier-thickener. Nitrogen gas production in the clarifier will result in escape of biological solids with the effluent as nitrogen bubbles floating sludge to the clarifier surface. For nitrogen reduction, a source of organics (electron donor) is required. Any inexpensive carbohydrate can be effectively used for nitrate removal. Many systems utilize methanol as the donor because it is rapidly

metabolized; others use the organics in sewage in order to reduce chemical costs. Nitrate reduction is invariably used as part of a system in which organics and nitrogen removal are goals. In such systems, the nitrogen in the waste is first oxidized to nitrate and then reduced to nitrogen gas. Figure 22-42 presents some flow sheets for such systems. Table 22-47 presents some design parameters for nitrate-reduction-activated sludge systems. Anaerobic activated sludge has been used for strong industrial wastes high in degradable organic solids. In these systems, a high rate of gasification takes place in the sludge separator so that a highly clarified effluent is usually not obtained. A vacuum degasifier is incorporated in such systems to reduce solids loss. Such systems have been used primarily with meat packing wastes that are warm, high in BOD and yield a high level of bicarbonate buffer as a result of ammonia release from protein breakdown. All of these conditions favor anaerobic processing. The use of this process scheme provides a high BSRT (15–30 days), usually required for anaerobic treatment, at a low HRT (1–2 days).

Lagoons Lagoons are low-cost, easy-to-operate wastewater-treatment systems capable of producing satisfactory effluents. Nominally, a lagoon is a suspended-growth no-recycle reactor with a variable degree of mixing. In lagoons in which mechanical or diffused aeration is used, mixing may be sufficient to approach complete mixing

TABLE 22-47 Design Parameter for Nitrogen Removal

System	BSRT, d	HRT, h	X, mg/L	pH
Carbon removal†	2–5	1–3	1000–2000	6.5–8.0
Nitrification†	10–20	0.5–3	1000–2000	7.4–8.6
Denitrification†	1–5	0.2–2	1000–2000	6.5–7.0
Internal recycle*	10–40	8–20	2000–4000	6.5–8.0

*Total for all reactors

†Separate reactors

(i.e., solids maintained in suspension). In other types of lagoons, most solids settle and remain on the lagoon bottom, but some mixing is achieved as a result of gas production from bacterial metabolism and wind action. Lagoons are categorized as aerobic, facultative, or anaerobic on the basis of degree of aeration. Aerobic lagoons primarily depend on mechanical or diffused air supply. Significant oxygen supply is also realized through natural surface aeration. A facultative lagoon is dependent primarily on natural surface aeration and oxygen generated by algal cells. These two lagoon types are relatively shallow to encourage surface aeration and provide for maximum algal activity. The third type of lagoon is maintained under anaerobic conditions to foster methane fermentation. This system is often covered with floating polystyrene panels to block surface aeration and help prevent a drop in temperature. Anaerobic lagoons are several meters deeper than the other two types. Lagoon flow schemes can be complex, employing lagoons in series and recycle from downstream to upstream lagoons. The major effect of recycle is to maintain control of the solids. If solids escape the lagoon system, a poor effluent is produced. Periodically controlled solids removal must take place or solids will escape.

Lagoons are, in effect, inexpensive reactors. They are shallow basins either cut below grade or formed by dikes built above grade or a combination of a cut and dike. The bottom must be lined with an impermeable barrier and the sides protected from wind erosion. These systems are best used where large areas of inexpensive land are available.

Facultative Lagoons These lagoons have been designed to use both aerobic and anaerobic reactions. Normally, facultative lagoons consist of two or more cells in series. The settleable solids tend to settle out in the first cell and undergo anaerobic metabolism with the production of organic acids and methane gas, which bubbles out to the atmosphere. Algae at the surface of the lagoon utilize sunlight for their energy in converting carbon dioxide, water, and ammonium ions into algal protoplasm with the release of oxygen as a waste product. Aerobic bacteria utilize the oxygen released by the algae to stabilize the soluble and colloidal organics. Thus, the bacteria and algae form a symbiotic relationship as shown in Fig. 22-43. The interesting aspect of facultative lagoons is that the organic matter in the incoming wastewaters is not stabilized but rather is converted to microbial protoplasm, which has a slower rate of oxygen demand. In fact, in some facultative lagoons inorganic compounds in the wastewaters are converted to organic compounds with a total increase in organics within the lagoon system.

Facultative lagoons are designed on the basis of organic load in relationship to the potential sunlight availability. In the northern part of the United States facultative lagoons are designed on the basis of $2.2 \text{ g}/(\text{m}^2\cdot\text{day})$ [$20 \text{ lb BOD}_5/(\text{acre}\cdot\text{day})$]. In the middle part of the United States the organic load can be increased to 3.4 to $4.5 \text{ g}/(\text{m}^2\cdot\text{day})$ [30 to $40 \text{ lb BOD}_5/(\text{acre}\cdot\text{day})$], while in the southern part the organic load can be increased to $6.7 \text{ g}/(\text{m}^2\cdot\text{day})$ [$60 \text{ lb BOD}_5/(\text{acre}\cdot\text{day})$]. The depth of lagoons is normally maintained between 1.0 and 1.7 m (3 and 5 ft). A depth less than 1.0 m (3 ft) encourages the growth of aquatic

weeds and permits mosquito breeding. In dry areas the maximum depth may be increased above 1.7 m (5 ft) depending upon evaporation. Most facultative lagoons depend upon natural wind action for mixing and should not be placed in screened areas where wind action is blocked.

Effluent quality from facultative lagoons is related primarily to the suspended solids created by living and dead microbes. The long retention period in the lagoons allows the microbes to die off, leaving a small particle that settles slowly. The release of nutrients from the dead microbes permits the algae to survive by recycling the nutrients. Thus, the algae determine the ultimate effluent quality. Use of series ponds with well-designed transfer structures between ponds permits maximum retention of algae within the ponds and the best-quality effluent. Normally the soluble BOD_5 is under 5 or 10 mg/L with a total effluent BOD_5 under 30 mg/L . The effluent suspended solids will vary widely during the different seasons of the year, being a maximum of 70 to 100 mg/L in the summer months and a minimum of 10 to 20 mg/L in the winter months. If suspended-solids removal is essential, chemical precipitation is the best method available at the present time. Slow sand filters and rock filters have been studied for suspended-solids removal; they work well as long as the effluent suspended solids are relatively low, 40 to 70 mg/L .

Aerated Lagoons These lagoons originated from efforts to control overloaded facultative lagoons. Since the lagoons were deficient in oxygen, additional oxygen was supplied by either mechanical surface aerators or diffused aerators. Mechanical surface aerators were quickly accepted as the primary aerators because they could be quickly added to existing ponds and moved to strategic locations. Unfortunately, the high-speed, floating surface aeration units were not efficient, and large numbers were required for existing lagoons. The problem was simply one of poor mixing in a very shallow lagoon.

Eventually, diffused aeration equipment was added to relatively deep lagoons [3.0 to 6.0 m (10 to 20 ft)]. Mixing became the most significant parameter for good oxygen transfer in aerated lagoons. From an economical point of view, it was found that a completely mixed aerated lagoon with 24-h retention provided the best balance between mixing and oxygen transfer. As the organic load increased, the fluid-retention time also increased. Short-term aeration permitted metabolism of the soluble organics by the bacteria, but time did not permit metabolism of the suspended solids. The suspended solids were combined with the microbial solids produced from metabolism and discharged from the aerated lagoon to a solids-separation pond. Data from the short-term aerated lagoon indicated that 50 percent BOD_5 stabilization occurred, with conversion of the soluble organics to microbial cells. The problem was separation and stabilization of the microbial cells. Short-term sedimentation ponds permitted separation of the solids without significant algae growths but required cleaning at frequent intervals to keep them from filling with solids and flowing into the effluent. Long-term lagoons permitted solids separation and stabilization but also permitted algae to grow and affect effluent quality.

Aerated lagoons were simply dispersed microbial reactors which permitted conversion of the organic components in the wastewaters to microbial solids without stabilization. The residual organics in solution were very low, less than 5 mg/L BOD_5 . By adding oxygen and improving mixing, the microbial metabolism reaction was speeded up, but the stabilization of the microbial solids has remained a problem to be solved.

Anaerobic Lagoons These lagoons were developed when a major fraction of the organic contaminants consisted of suspended solids that could be removed easily by gravity sedimentation. The anaerobic lagoons are relatively deep [8.0 to 6.0 m (10 to 20 ft)], with a short fluid-retention time (3 to 5 days) and a high BOD_5 loading rate, up to $3.2 \text{ kg}/(\text{m}^3\cdot\text{day})$ [$200 \text{ lb}/(1000 \text{ ft}^3\cdot\text{day})$]. Microbial metabolism in the settled-solids layer produces methane and carbon dioxide, which quickly rise to the surface, carrying some of the suspended solids. A scum layer that retards oxygen transfer and release of obnoxious gases is quickly produced in anaerobic lagoons. Mixing with a grinder pump can provide a better environment for metabolism of the suspended solids. The key for anaerobic lagoons is adequate buffer to keep the

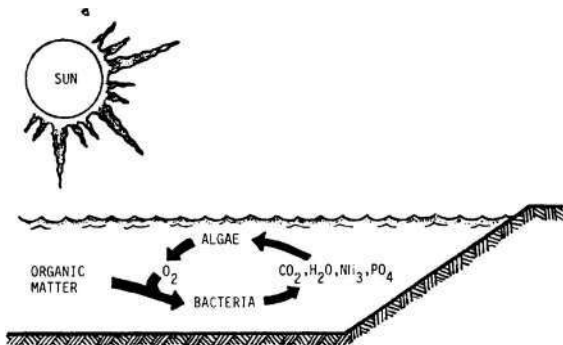


FIG. 22-43 Schematic diagram of oxidation-pond operations.

pH between 6.5 and 8.0. Protein wastes have proved to be the best pollutants to be treated by anaerobic lagoons, with the ammonium ions reacting with carbon dioxide and water to form ammonium bicarbonate as the primary buffer. High-carbohydrate wastes are poor in anaerobic lagoons since they produce organic acids without adequate buffer, making it difficult to maintain a suitable pH for good microbial growth.

Anaerobic lagoons do not produce a high-quality effluent but are able to reduce the BOD load by 80 to 90 percent with a minimum of effort. Since anaerobic lagoons work best on strong organic wastes, their effluent must be treated by either aerated lagoons or facultative lagoons. An anaerobic lagoon is simply the first stage in the treatment of strong organic wastewaters.

Fixed-Film Reactor Systems A major advantage of fixed film systems is that a flocculent-type biomass is not necessary as the biomass remains in the reactor attached to inert packing. Biomass does periodically slough off or break away from the packing, usually in large chunks that can be easily removed in a clarifier. On the other hand, the time of contact between the biomass and the waste is much shorter than in suspended growth systems, making it difficult to achieve the same degree of treatment especially in aerobic systems. Aerobic, anoxic, and anaerobic fixed film systems are utilized for waste treatment.

Aerobic systems including trickling filters and rotating biological contactors (RBC) are operated in a nonflooded mode to ensure adequate oxygen supply. Other aerobic, anoxic, and anaerobic systems employ flooded reactors. The most common systems are packed beds (anaerobic trickling filter) and fluidized or expanded bed systems.

Trickling Filters For years trickling filters were the mainstay of biological wastewater treatment systems because of their simplicity of design and operation. Trickling filters were displaced as the primary biological treatment system by activated sludge because of better effluent quality. Trickling filters are simply fixed-medium biological reactors with the wastewaters being spread over the surface of a solid medium where the microbes are growing. The microbes remove the organics from the wastewaters flowing over the fixed medium. Oxygen from the air permits aerobic reactions to occur at the surface of the microbial layer, but anaerobic metabolism occurs at the bottom of the microbial layer where oxygen does not penetrate.

Originally, the medium in trickling filters was rock, but rock has largely been replaced by plastic, which provides greater void space per unit of surface area and occupies less volume within the filter. A plastic medium permitted trickling filters to be increased from a medium depth of 1.8 m (6 ft) to one of 4.2 m (14 ft) and even 6.0 m (20 ft). The wastewaters are normally applied by a rotary distributor or a fixed-spray nozzle. The spraying or discharging of wastewaters above the trickling-filter medium permits better distribution over the medium and oxygen transfer before reaching the medium. The effluent from the trickling-filter medium is captured in a clay-tile underdrain system or in a tank below the plastic medium. It is important that the bottom of the trickling filter be open for air to move quickly through the filter and bring adequate oxygen for the microbial reactions.

If a high-quality effluent is required, trickling filters must be operated at a low hydraulic-loading rate and a low organic-loading rate. Low-rate trickling filters are operated at hydraulic loadings of 2.2×10^{-5} to 4.3×10^{-5} m³/(m²·s) [2 million to 4 million gal/(acre·day)]. High-rate trickling filters are designed for 10.8×10^{-5} to 40.3×10^{-5} m³/(m²·s) [10 million to 40 million gal/(acre·day)] hydraulic loadings and organic loadings up to 1.4 kg/(m³·day) [90 lb BOD₅/(1000 ft³·day)]. Plastic-medium trickling filters have been designed to operate up to 108×10^{-5} m³/(m²·s) [100 million gal/(acre·day)] or even higher, with organic loadings up to 4.8 kg/(m³·day) [300 lb BOD₅/(1000 ft³·day)]. Low-rate trickling filters will produce better than 90 percent BOD₅ and suspended-solids reductions, while high-rate trickling filters will produce from 65 to 75 percent BOD₅ reduction. Plastic-medium trickling filters will produce from 59 to 85 percent BOD₅ reduction depending upon the organic-loading rate. It is important to recognize that concentrated industrial wastes will require considerable hydraulic recirculation around the trickling filter to obtain the proper hydraulic-loading rate without excessive organic loads. With high recirculation rates the organic load is distributed over the entire volume of the trickling filter for maximum organic removal.

The short fluid-retention time within the trickling filter is the primary reason for the low treatment efficiency.

Rotating Biological Contactors (RBCs) The newest form of trickling filter is the rotating biological contactor with a series of circular plastic disks, 3.0 to 3.6 m (10 to 12 ft) in diameter, immersed to approximately 40 percent diameter in a shaped contact tank. The RBC disks rotate at 2 to 5 r/min. As the disks travel through the wastewaters, a small layer adheres to them. As the disks travel into the air, the microbes on the disk surface oxidize the organics. Thus, only a small amount of energy is required to supply the required oxygen for wastewater treatment. As the microbes build up on the plastic disks, the shearing velocity that is created by the movement of the disks through the water causes the excess microbes to be removed from the disks and discharged to the final sedimentation tank.

Rotating biological contactors have been very popular in treating industrial wastes because of their relatively small size and their low energy requirements. Unfortunately, there have occurred a number of problems which should be recognized prior to using RBCs. Strong industrial wastes tend to create excessive microbial growths which are not easily sheared off and which create high oxygen-demand rates with the production of hydrogen sulfide and other obnoxious odors. The heavy microbial growths have damaged some of the disks and caused some shaft failures. The disks are currently being covered with plastic shells to prevent nuisance odors from occurring. Air must be forced through the covered RBC systems and be chemically treated before being discharged back into the environment. Recirculation of wastewater flow around the RBC units can distribute the load over all the units and reduce the heavy initial microbial growths. RBC units also work best under uniform organic loads, requiring surge tanks for many industrial wastes. The net result has been for the cost of RBC units to approach that of other treatment units in terms of organic matter stabilized.

RBCs should be designed on both a hydraulic-loading rate and an organic-loading rate. Normally, hydraulic-loading rates of up to 0.16 m³/(m²·day) [4 gal/(ft²·day)] of surface area are used with organic loading rates up to 44 kg/(m²·day) [9 lb BOD₅/(ft²·day)]. Treatment efficiency is primarily a function of the fluid-retention time and the organic-loading rate. At low organic-loading rates the RBC units will produce nitrification in the same way as low-rate trickling filters.

Packed-Bed Fixed-Film Systems These systems were originally termed anaerobic trickling filters because the first systems were submerged columns filled with stones run under anaerobic conditions. A wide variety of packed media is now used ranging in size from granules 40 mesh to 7.5-cm (3-in) stones. Many systems use open structure plastic packing similar to that used in aerobic trickling filters.

The systems using granular media packing are used for anoxic denitrification. They are usually downflow, thus serving the dual function of filtration and denitrification. Contact times are short (EBCT < 15 min), but excellent removal is achieved due to the high level of biomass retained in the reactor. Pacing the methanol dose to the varying feed nitrate concentration is crucial. Frequent, short-duration backwash (usually several times per day) is required or the nitrogen bubbles formed will bind the system, causing poor results. Extended backwash every two to three days is required or the system will clog on the biomass growth. Thus, several units in parallel or a large holding tank are needed to compensate for the down time during backwash. Backwash does not remove all the biomass; a thin film remains coating the packing. Thus, denitrification begins immediately when the flow is restored.

The systems using the larger packing are used in the treatment of relatively strong, low-suspended-solids industrial waste. These systems are closed columns usually run in an upflow mode with a gas space at the top. These are operated under anaerobic conditions with waste conversion to methane and carbon dioxide as the goal. Effluent recycle is often used to help maintain the pH in the inlet zone in the correct range 6.5–7.5 for the methane bacteria. Some wastes require the addition of alkaline material to prevent a pH drop. Sodium bicarbonate is often recommended for pH control because it is easier to handle than lime or sodium hydroxide, and because an overdose of bicarbonate will only raise the pH modestly. An overdose of lime or sodium hydroxide can easily raise the pH above 8.0. Table 22-48 gives

TABLE 22-48 Anaerobic Process Performance on Industrial Wastewater UASB, Submerged Filter (SF), FBR

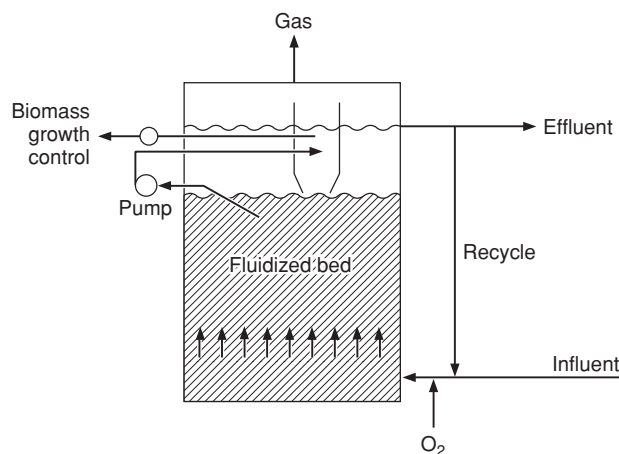
Process	Wastewater	Reactor size, MG	COD, g/l	OVL, kg/m ³ d	%COD _r	HRT-d	°C
SF	Rum slops	3.5	80–105	15	71	7.8	35
SF	Modified guar	0.27	9.1	7.5	60	1.0	37
SF	Chemical	1.5	14	11	90	0.7	H
SF	Milk	0.2	3	7.5	60	0.5	32
SF	PMFC	10 ⁻⁵	13.7	23	72	0.6	36
UASB	Potato	0.58	2.5	3	85	0.7	35
UASB	Sugar beet	0.21	3	16	88	—	—
UASB	Brewery	1.16	1.6–2.2S	4.4	83	0.4	30
UASB	Brewery	1.16	2.0–2.4S	8.7	78	0.2	30
UASB	PMFC	10 ⁻⁵	13.7	4–5	87	2.9	36
FBR	PMFC	10 ⁻⁵	13.7	35–48	88	0.4	36
FBR	Soft drink	0.04	3.0	6–7	75	0.5	35
FBR	Chemical	0.04	35	14	95	2.5	35

OVL = organic volumetric load (COD); S = settled effluent; H = heated; PMFC = paper mill foul condensate (5,6).

NOTE: MG = 3785 m³.

some performance data with systems treating industrial wastes. HRTs of 1 to 2 days are used, as the buildup of growth on the packing ensures a BSRT of 20–50 days. It should be possible to lower the HRT further, but in practice this has not been successful because biomass starts to escape from the system or plugging occurs. Some escape is due to high gasification rates, and some is due to the fact that anaerobic sludge attaches less tenaciously to packing than aerobic or anoxic sludge. These systems can handle wastes with moderate solids levels. Periodically, solids must be removed from the reactor to prevent plugging of the packing or loss of solids in the effluent.

Biological Fluidized Beds This high-rate process has been used successfully for aerobic, anoxic, and anaerobic treatment of municipal and industrial wastewaters. Numerous small- and large-scale applications for hazardous waste, contaminated groundwater, nontoxic industrial waste and municipal wastewater have been reported (Refs. 32 and 33). The basic element of the process is a bed of solid carrier particles, such as sand or granular activated carbon, placed in a reactor through which wastewater is passed upflow with sufficient velocity to impart motion or fluidize the carrier. An active growth of biological organisms grow as firmly attached mass surrounding each of the carrier particles. As the wastewater contaminants pass by the biologically covered carrier, they are removed from the wastewater through biological and adsorptive mechanisms. Figure 22-44 is a schematic of the process.

**FIG. 22-44** Schematic of fluidized bed process.

The influent wastewater enters the reactor through a pipe manifold and is introduced downflow through nozzles that distribute the flow uniformly at the base of the reactor. Reversing direction at the bottom, the flow fluidizes the carrier when the fluid drag overcomes the buoyant weight of the carrier and its attached biomass layer. During startup (before much biomass has accumulated), the flow velocity required to achieve fluidization is higher than after the biomass attaches. Recycle of treated effluent is adjusted to achieve the desired degree of fluidization. As biomass accumulates, the particles of coated biomass will separate to a greater extent at constant flow velocity. Thus, as the system ages and more biomass accumulates, the extent of bed expansion increases (the volume of voids increases). This phenomenon is advantageous because it prevents clogging of the bed with biomass. Consequently, higher levels of biomass attachment are possible than in other types of fixed film systems. However, eventually, the degree of bed expansion may become excessive. Reduction of recycle will reduce expansion but may not be feasible because recycle has several purposes (i.e., supply of nutrients, alkalinity, and dilution of waste strength). Control of the expanded bed surface level is automatically accomplished using a sensor that activates a biomass growth control system at a prescribed level and maintains the bed at the proper depth. A pump removes a portion of the attached biomass, separates the biomass and inert carrier by abrasion, and pumps the mixture into a separator. Here the heavy carrier settles back into the fluidized bed and the abraded biomass, which is less dense, is removed from the system by gravity or a second pump. Other growth control designs are also used. Effluent is withdrawn from the supernatant layer above the fluidized bed. The reactor is usually not covered unless it is operating under anaerobic conditions and methane, odorous gases, or other safety precautions are mandated.

When aerobic treatment is to be provided to high concentrations of organics, pure oxygen or hydrogen peroxide may be injected into the wastewater prior to entering the reactor. Liquid oxygen (LOX) or pressure swing absorption (PSA) systems have been used to supply oxygen. Air may be used at low D.O. demands.

In full-scale applications, this process has been found to operate at significantly higher volumetric loading rates for wastewater treatment than other processes. The primary reasons for the very high rates of contaminant removal is the high biologically active surface area available (approximately 1000 ft²/ft³ of reactor) and the high concentration of reactor biological solids (8,000–40,000 mg/L) that can be maintained (Table 22-49). Because of these atypical high values, designs usually indicate a 200–500 percent reduction in reactor volume when compared to other fixed film and suspended growth treatment processes. In Table 22-50, is a list of full-scale commercial applications of the process operated at high wastewater concentrations including aerobic oxidation of organics, anoxic denitrification and anaerobic treatment systems.

TABLE 22-49 Process Comparisons

Biological process	MLVSS, mg/L	Surface area, ft ² /ft ³
Activated sludge	1500–3000	—
Pure oxygen activated sludge	2000–5000	—
Suspended growth nitrification	1000–2000	—
Suspended growth denitrification	1500–3000	—
Fluidized bed-CBOD removal	12,000–20,000	800–1200
Fluidized bed-nitrification	8,000–12,000	800–1200
Fluidized bed-denitrification	25,000–40,000	800–1200
Trickling filter-CBOD removal	—	12–50
RBC-CBOD removal	—	30–40

Of special note is the enhancement to the process when granular activated carbon (GAC) is used as the carrier. Because GAC has adsorptive properties, organic compounds present in potable waters and wastewater at low concentrations, often less than ten mg/L, are removed by adsorption and subsequently consumed by the biological organisms that grow in the fluidized bed. The BTEX compounds, methylene chloride, chlorobenzene, plastics industry toxic effluent, and many others are removed in this manner. BTEX contamination of groundwater from leaking gasoline storage tanks is a major problem, and sixteen full-scale fluidized bed process applications have been made. Contaminated groundwater is pumped to the ground surface for treatment using the fluidized bed in the aerobic mode. Often the level of BTEX is 1–10 mg/L and about 99 percent is removed in less than ten minutes detention time. Installations in operation range in size from 30 to 3000 gpm. The smaller installations are often skid-mounted and may be moved from location to location at a given site. A major advantage of this process over stripping towers and vacuum systems for treating volatile organics (VOCs) is the elimination of effluent gas treatment.

Pilot-plant studies and full-scale plants have shown successful treatment of wastewaters with 5000 to 50,000 mg/L COD from dairy, brewery, other food preparation wastes; paper pulp wastes; deicing fluids; and other hazardous and nonhazardous materials. Design criteria for biological fluidized-bed systems are included in Table 22-51.

A third type of reactor system is the upflow anaerobic sludge blanket (UASB) reactor system. The UASB is a continuous-flow system that maintains a dense flocculated biomass that is granular. The wastewater passes through this dense sludge blanket to achieve desired treatment efficiency. The UASB, anaerobic filter, and fluidized-bed systems have all been applied successfully to treat a variety of industrial waste streams. Table 22-44 summarizes some of the applications of these treatment technologies.

Design criteria for satisfactory biological fluidized bed treatment systems include the major parameters given in Table 22-51.

TABLE 22-50 Full-Scale Commercial Applications of Fluidized Bed Process

Application	Type	Reactor volume, m ³
CBOD—paint—General Motors	Aerobic	108
C.NBOD—sanitary and automotive—GM	Aerobic	165
Chemical—Grindsted Products Denmark	Aerobic	730
Nitrification—fish hatchery—Idaho	Aerobic	820
BTEX, groundwater—Ohio	Aerobic	250
Chemical, Toxicity reduction—Texas	Aerobic	225
Denitrification, Nuclear Fuel, DOE, Fernald, OH	Anoxic	53
CBOD, Denitrification—municipal, Nevada	Anoxic	1450
Petrochemical—Reliance Ind. India	Anaerobic	850
Soft drink, Delaware	Anaerobic	166
Brewery—El Aguila, Spain	Anaerobic	1570

TABLE 22-51 Typical Design Parameters for Fluidized Beds

Waste	Mode	Influent concentration, mg/L	Volumetric load, lb/ft ³ ·d	Hydraulic detention time, hr	Volatile solids (biosolids), g/L
Organic COD	Aerobic	<2000	0.1–0.6	0.5–2.5	12–20
Organic COD	Aerobic	<10	0.02–0.1	0.1–0.5	4–8
NH ₃ -N	Aerobic	<25	0.05–0.3	0.1–0.5	7–14
NO ₃ -N	Anoxic	<5,000	0.4–2	0.1–2.4	20–40
Organic COD	Anaerobic	>4000	0.5–4	2–24	20–40

NOTE: lb/ft³·d = 16 kg/m³·d.

PHYSICAL-CHEMICAL TREATMENT

Processes and/or unit operations that fall under this classification include adsorption, ion exchange, stripping, chemical oxidation, and membrane separations. All of these are more expensive than biological treatment but are used for removal of pollutants that are not easily removed by biomass. Often these are utilized in series with biological treatment; but sometimes they are used as stand-alone processes.

Adsorption This is the most widely used of the physical-chemical treatment processes. It is used primarily for the removal of soluble organics with activated carbon serving as the adsorbent. Most liquid-phase-activated carbon adsorption reactions follow a Freundlich Isotherm [Eq. (22-25)].

$$y = kc^{1/n} \quad (22-25)$$

where: y = adsorbent capacity, mass pollutant/mass carbon
 c = concentration of pollutant in waste, mass/volume
 k and n = empirical constants

EPA has compiled significant data on values of k and n for environmentally significant pollutants with typical activated carbons. Assuming equilibrium is reached, the isotherm provides the dose of carbon required for treatment. In a concurrent contacting process, the capacity is set by the required effluent concentration. In a countercurrent process, the capacity of the carbon is set by the untreated waste pollutant concentration. Thus countercurrent contacting is preferred.

Activated carbon is available in powdered form (200–400 mesh) and granular form (10–40 mesh). The latter is more expensive but is easier to regenerate and easier to utilize in a countercurrent contactor. Powdered carbon is applied in well-mixed slurry-type contactors for detention times of several hours after which separation from the flow occurs by sedimentation. Often coagulation, flocculation, and filtration are required in addition to sedimentation. As it is difficult to regenerate, powdered carbon is usually discarded after use. Granular carbon is used in column contactors with EBCT of 30 minutes to 1 hour. Often several contactors are used in series, providing for full countercurrent contact. A single contactor will provide only partial countercurrent contact. When a contactor is exhausted, the carbon is regenerated either by a thermal method or by passing a solvent through the contactor. For waste-treatment applications where a large number of pollutants must be removed but the quantity of each pollutant is small, thermal regeneration is favored. In situations where a single pollutant in large quantity is removed by the carbon, solvent extraction regeneration can be used, especially where the pollutant can be recovered from the solvent and reused. Thermal regeneration is a complex operation. It requires removal of the carbon from the contactor, drainage of free water, transport to a furnace, heating under controlled conditions of temperature, oxygen, time, water vapor partial pressure, quenching, transport back to the reactor, and reloading of the column. Five to ten percent of the carbon is lost in this regeneration process due to burning and attrition during each regeneration cycle. Multiple hearth, rotary kiln, and fluidized bed furnaces have all been successfully used for carbon regeneration.

Pretreatment prior to carbon adsorption is usually for removal of suspended solids. Often this process is used as tertiary treatment after

primary and biological treatment. In either situation, the carbon columns must be designed to provide for backwash. Some solids will escape pretreatment, and biological growth will occur on the carbon, even with extensive pretreatment. Originally carbon treatment was viewed only as applicable for removal of toxic organics or those that are difficult to degrade biologically. Present practice applies carbon adsorption as a procedure for removal of all types of organics. It is realized that some biological activity will occur in virtually any activated carbon unit, so the design must be adjusted accordingly.

Ion Exchange This process has been employed for many years for treatment of industrial water supplies but not for wastes. However, some new ion-exchange materials have recently been developed that can be used for removal of specific pollutants. These new resins are primarily useful for selective removal of heavy metals, even though the target metals are present at low concentration in a wastewater containing many other inorganics. An ion-exchange process is usually operated with the waste being run downflow through a series of columns containing the appropriate ion-exchange resins. EBCT contact times of 30 min to 1 hour are used. Pretreatment for suspended solids and organics removal is practiced as well as pH adjustment.

The capacity of an ion-exchange resin is a function of the type and concentration of regenerant. Because ion-exchange resins are so selective for the target compound, a significant excess of the regenerant must be used. Up to a point, the more regenerant, the greater the capacity of the resin. Unfortunately, this results in waste of most of the regenerant. In fact, the bulk of the operational cost for this process is for regenerant purchase and its disposal; thus, regeneration must be optimized. Regenerants used (selection depends on the ion exchange resin) include sodium chloride, sodium carbonate, sulfuric acid, sodium hydroxide, and ammonia. Table 22-52 gives information on some of the new highly selective resins. These resins may provide the only practical method for reduction of heavy metals to the very low levels required by recent EPA regulations.

Stripping Air stripping is applied for the removal of volatile substances from water. Henry's law is the key relationship for use in design of stripping systems. The minimum gas-to-liquid ratio required for stripping is given by:

$$\frac{G}{L} = \frac{S_{in} - X_{out}}{(H)X_{out}}$$

for a concurrent process and

$$\frac{G}{L} = \frac{X_{in} - X_{out}}{(H)X_{in}}$$

for a countercurrent process where:

G = airflow rate, mass/time

L = waste flow rate, volume/time

X = concentration of pollutant in waste, mass/volume

H = Henry's constant for the pollutant in water, volume/mass

Higher ratios of gas to liquid flow are required than those computed above because mass-transfer limitations must be overcome. Stripping

can occur by sparging air into a tank containing the waste. Indeed, stripping of organics from activated sludge tanks is of concern because of the possibility that the public will be exposed to airborne pollutants including odors. A much more efficient stripping procedure is to use counterflow or crossflow contact towers. Procedures for design of these and mass-transfer characteristics of various packings are available elsewhere in this handbook.

Stripping has been successfully and economically employed for removal of halogenated organics from water and wastes with dispersion of the effluent gas to the atmosphere. However, recent EPA regulations have curtailed this practice. Now removal of these toxic organics from the gas stream is also required. Systems employing activated carbon (prepared for use with gas streams) are employed as well as systems to oxidize the organics in the gas stream. However, the cost of cleaning up the gas stream often exceeds the cost of stripping these organics from the water.

Chemical Oxidation This process has not been widely utilized because of its high cost. Only where the concentration of the target compound is very low will the quantity of oxidant required be low enough to justify treatment by chemical oxidation. The efficiency of this process is also low, as many side reactions can occur that will consume the oxidant. In addition, complete oxidation of organics to carbon dioxide and water often will not occur unless a significant overdose is used. However, renewed interest has recently occurred for two reasons.

1. It has been found that partial oxidation is satisfactory as a pretreatment to biological or carbon adsorption treatment. Partial oxidation often seems to make recalcitrant organics easier to degrade biologically and easier to adsorb.

2. A combination of ozone oxidation with simultaneous exposure to ultraviolet light seems to produce a self-renewing chain reaction that can significantly reduce the dose of ozone needed to accomplish oxidation.

Oxidants commonly used include ozone, permanganate, chlorine, chlorine dioxide, and ferrate, often in combination with catalysts. Standard-type mixed reactors are used with contact times of several minutes to an hour.

Advanced Oxidation Processes Advanced oxidation processes (AOPs) utilize the extremely strong oxidizing power of hydroxyl radicals to oxidize recalcitrant organic compounds to the mineral end products of CO_2 , H_2O , HCl , etc. The hydroxyl radicals are produced on-site using a combination of traditional oxidation processes such as ozone/hydrogen peroxide, ultraviolet radiation/ozone, ultraviolet radiation/hydrogen peroxide, Fenton's reagent (ferrous iron and hydrogen peroxide) with ultraviolet radiation, and titanium dioxide/ultraviolet radiation (catalytic oxidation). The hydroxyl radical is a powerful nonselective oxidant, which can effectively and rapidly oxidize most organic compounds. As shown in Table 22-53, the hydroxyl radical has greater oxidizing power than do traditional oxidizing reagents. AOPs can be used to remove overall organic content (COD) or to destroy specific compounds. Table 22-53 shows the relative oxidation power of a select number of oxidizing species compared to chlorine. AOPs are most economical for treating industrial and hazardous waste streams that have a low level of recalcitrant contamination (<50 ppm). AOPs are emerging as an efficient and cost-effective treatment technology for specific industrial wastewaters and are likely to have broader applications as the various technologies are proved in the field.

TABLE 22-52 Selective Ion-Exchange Resins

Application	Exchanger type	Composition	Regenerant
Softening	Cation	Polystyrene matrix Sulfonic acid functional groups	NaCl
Heavy metals	Cation	Polystyrene matrix Chelating functional groups	Mineral acids
Chromate	Anion	Polystyrene matrix Tertiary or quaternary ammonium functional groups	Sodium carbonate or alkaline NaCl
Nitrate	Anion	Polystyrene matrix Tributyl ammonium functional group	NaCl

TABLE 22-53 Relative Oxidation Power of Select Oxidizing Reagents

Oxidizing species	Relative oxidation power
Chlorine	1.0
Hypochlorous acid	1.1
Permanganate	1.24
Hydrogen peroxide	1.31
Ozone	1.52
Atomic oxygen	1.78
Hydroxyl radical	2.05
Titanium dioxide	2.35

Membrane Processes These processes use a selectively permeable membrane to separate pollutants from water. Most of the membranes are formulated from complex organics that polymerize during membrane preparation. This allows the membrane to be tailored to discriminate by molecular size or by degree of hydrogen bonding potential. Ultrafiltration membranes discriminate by molecular size or weight, while reverse osmosis membranes discriminate by hydrogen-bonding characteristics. The permeability of these membranes is low: from 0.38–3.8 m/day (10–100 gal/d/ft²). The apparatus in which they are used must provide a high surface area per unit volume.

Membrane Bioreactors (MBRs) Membrane bioreactors are a technology that combines biological degradation of waste products with membrane filtration (Fig. 22-45). Typically, MBRs are operated as activated sludge systems with high biomass concentrations (10,000 to 15,000 mg/L MLSS) and long solids retention times (30 to 60 days). The membranes act as a solid-liquid separation to replace secondary clarifiers and granular media polishing filters. With the development of more economical, durable, and efficient membrane components, MBR systems have become feasible options for aerobic treatment of municipal wastewaters, and potentially for anaerobic treatment of industrial (i.e. low- to medium-strength) wastewaters. Two different process configurations for MBRs (sidestream or submerged modules) are used within wastewater treatment. The sidestream process is operated with high velocities and high transmembrane pressures (up to 5 bar). This results in high energy consumption and excessive shear forces within the reactor. The process with submerged modules was especially designed for biological wastewater treatment. The modules are submerged in the aerobic activated sludge zone where flow across the membrane surface is attained by aeration and mixing, resulting in a low flux rate with low transmembrane and vacuum pressures. Submerged membranes typically last longer (8+ years) and require less frequent cleaning and maintenance. With recent advances in membrane materials and process technology, MBRs have become more reliable, durable, and affordable. MBRs can be utilized for any level of treatment, but are most cost-effective when high-quality effluent is required.

The submerged MBR system is a small-footprint, single-process unit that can achieve a high-quality standard with low solids production. The membrane systems typically utilize either micro- or ultramembranes configured as hollow fibers, flat sheets, or tubes depending upon the manufacturer. Experience over the past decade has proved MBRs to be reliable and cost-effective for high-quality effluent applications, and relatively easy to maintain and operate. The key parameters for MBR

design and operation are flux rate, transmembrane pressure, fouling rate, and membrane cleaning schedule. MBRs have many possible applications in industrial and advanced municipal wastewater treatment and are becoming the preferred technology for wastewater reuse applications.

Industrial Reuse Many industries that use large volumes of freshwater are turning toward industrial reuse to minimize costs associated with attaining freshwater and disposing of wastewater. The need to minimize water consumption is a function of economics, water shortages caused by drought and overdevelopment, increasingly stringent regulations, and environmental awareness. Industrial reuse water is wastewater that is produced on-site that does not contain sewage and has been adequately treated for use in some part of the industrial operation. Reuse has become more economically feasible as a result of new treatment technologies (MBRs, AOPs, etc.) and rising costs of freshwater and wastewater disposal.

There are many industrial sources and uses of industrial reuse water. Common industrial processes that require large quantities of water, such as evaporative cooling towers and power stations, can be supplied using reclaimed wastewater. Other industrial applications for reuse water include boilers; stack scrubbing; washing of vehicles, buildings, and mechanical parts; and process water. The key to effective industrial reuse is to match the proper treatment processes (chemical, physical, biological) to achieve the desired quality of reuse water. Zero discharge industrial systems have recently been implemented in the United States and appear to be a growing trend, sustaining water resources and promoting smart industrial development. Some common industries that have implemented water reuse include the paper pulp industry, the pharmaceutical industry, chemical manufacturers, power plants, and refineries.

SLUDGE PROCESSING

Objectives Sludges consist primarily of the solids removed from liquid wastes during their processing. Thus, sludges could contain a wide variety of pollutants and residuals from the application of treatment chemicals; i.e., large organic solids, colloidal organic solids, metal sulfides, heavy-metal hydroxides and carbonates, heavy-metal organic complexes, calcium and magnesium hydroxides, calcium carbonate, precipitated soaps and detergents, and biomass and precipitated phosphates. As sludge even after extensive concentration and dewatering is still greater than 50 percent by weight water, it can also contain soluble pollutants such as ammonia, priority pollutants, and nonbiologically degradable COD.

The general treatment or management of sludge involves stabilization of biodegradable organics, concentration and dewatering, and ultimate disposal of the stabilized and dewatered residue. A large number of individual unit processes and unit operations are used in a sludge-management scheme. Those most frequently used are discussed below. Occasionally, only one of these is needed, but usually several are used in a series arrangement.

Because of the wide variability in sludge characteristics and the variation in acceptability of treated sludges for ultimate disposal (this is a function of the location and characteristics of the ultimate disposal site), it is impossible to prescribe any particular sludge-management plan. In the sections below, general performance of individual sludge-treatment processes and operations is presented.

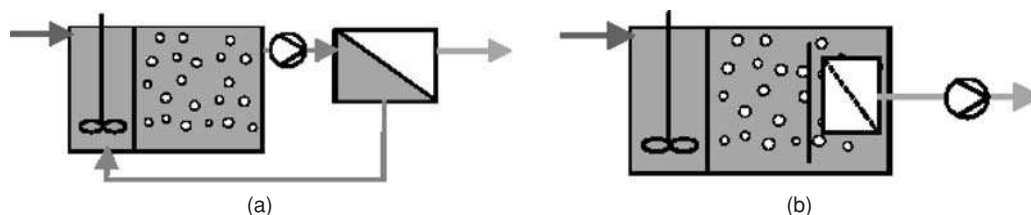


FIG. 22-45 Membrane bioreactor configurations for wastewater treatment. (a) With sidestream module; (b) with submerged module.

Concentration: Thickening and Flotation Generated sludges are often dilute (1–2 percent solids by weight). In order to reduce the volumetric loading on other processes, the first step in sludge processing is often concentration. The most popular process is gravity thickening which is carried out in treatment units similar to circular clarifiers. Organic sludges from primary treatment can usually be concentrated to 5–8 percent solids. Sludges from secondary treatment can be thickened to 2 percent solids. The potential concentration with completely inorganic sludges is higher (greater than 10 percent solids) except for sludges high in metal hydroxides. Polymers are often used to speed up thickening and increase concentration. Thickening is enhanced by long retention of the solids in the thickening apparatus. However, when biodegradable organics are present, solids retention time must be at a level that will not foster biological activity, lest odors, gas generation, and solids hydrolysis occur. Loading rates on thickeners range from 50–122 kg/m²/d (10–25 lb/ft²/d) for primary sludge to 12–45 kg/m²/d (2.5–9 lb/ft²/d). Solids detention time is 0.5 days in summer to several days in winter.

Flotation Air flotation has proved to be successful in concentrating secondary sludges to about 4 percent solids. The incoming solids are normally saturated with air at 275 to 350 kPa (40 to 50 psig) prior to being released in the flotation tank. As the air comes out of solution, the fine bubbles are trapped under the suspended solids and carry them to the surface of the tank. The air bubbles compact the solids as a floating mass. Normally, the air-to-solids ratio is about 0.01–0.05 L/g (0.16–0.8 ft³/lb). The thickened solids are scraped off the surface, while the effluent is drawn off the middle of the tank and returned to the treatment system. In large flotation tanks with high flows the effluent rather than the incoming solids is pressurized and recycled to the influent. The size of the flotation tanks is determined primarily by the solids-loading rate, directly or indirectly. A solids loading of 25 to 97 kg/(m²·day) [5 to 20 lb/(ft²·day)] has been found to be adequate. On a flow basis this translates into 0.14 to 2.7 L/(m²·day) [0.2 to 4 gal/(ft²·min)] surface area.

As with thickening, air flotation is enhanced by the addition of polymers. Flotation has been successfully used with wholly inorganic metal hydroxide sludges. Polymers and surfactants are used as additives. Engineering details on air flotation equipment has been developed by and is available from various equipment manufacturing companies. Liquid removed during thickening and flotation is usually returned to the head end of the plant.

Stabilization (Anaerobic Digestion, Aerobic Digestion, High Lime Treatment) Sludges high in organics can be stabilized by subjecting them to biological treatment. The most popular system is anaerobic digestion.

Anaerobic Digestion Anaerobic digesters are large covered tanks with detention times of 30 days, based on the volume of sludge added daily. Digesters are usually heated with an external heat exchanger to 35–37°C to speed the rate of reaction. Mixing is essential to provide good contact between the microbes and the incoming organic solids. Gas mixing and mechanical mixers have been used to provide mixing in the anaerobic digester. Following digestion, the sludge enters a holding tank, which is basically a solids-separation unit and is not normally equipped for either heating or mixing. The supernatant is recycled back to the treatment plant, while the settled sludge is allowed to concentrate to 3–6 percent solids before being further processed.

Anaerobic digestion results in the conversion of the biodegradable organics to methane, carbon dioxide, and microbial cells. Because of the energy in the methane, the production of microbial mass is quite low, less than 0.1 kg/kg (0.1 lb volatile suspended solids (VSS)/lb) BOD metabolized except for carbohydrate wastes. The production of methane is 0.35 m³/kg (5.6 ft³/lb) BOD destroyed. Digester gases range from 50 to 80 percent methane and 20 to 50 percent carbon dioxide, depending on the chemical characteristics of the waste organics being digested. The methane is often used on site for heat and power generation.

There are three major groups of bacteria that function in anaerobic digestion. The first group hydrolyzes large soluble and nonsoluble organic compounds such as proteins, fats and oils (grease), and carbohydrates, producing smaller water-soluble compounds. These are

then degraded by acid-forming bacteria, producing simple volatile organic acids (primarily acetic acid) and hydrogen. The last group (the methane bacteria) split acetic acid to methane and carbon dioxide and produce methane from carbon dioxide and hydrogen. Good operation requires the destruction of the volatile acids as quickly as they are produced. If this does not occur, the volatile acids will build up and depress the pH, which will eventually inhibit the methane bacteria. To prevent this from occurring, feed of organics to the digester should be as uniform as possible.

If continuous addition of solids is not possible, additions should be made at as short intervals as possible. Alkalinity levels are normally maintained at about 3000 to 5000 mg/L to keep the pH in the range 6.5–7.5 as a buffer against variable organic-acid production with varying organic loads. Proteins will produce an adequate buffer, but carbohydrates will require the addition of alkalinity to provide a sufficient buffer. Sodium bicarbonate should be used to supply the buffer.

An anaerobic digester is a no-recycle complete mix reactor. Thus, its performance is independent of organic loading but is controlled by hydraulic retention time (HRT). Based on kinetic theory and values of the pseudo constants for methane bacteria, a minimum HRT of 3 to 4 days is required. To provide a safety factor and compensate for load variation as indicated earlier, HRT is kept in the range 10 to 30 days. Thickening of feed sludge is used to reduce the tank volume required to achieve the long HRT values. When the sludge is high in protein, the alkalinity can increase to greater than 5000 mg/L, and the pH will rise past 7.5. This can result in free ammonia toxicity. To avoid this situation, the pH should be reduced below 7.5 with hydrochloric acid. Use of nitric or sulfuric acid will result in significant operational problems.

Aerobic Digestion Waste activated sludge can be treated more easily in aerobic treatment systems than in anaerobic systems. The sludge has already been partially aerobically digested in the aeration tank. For the most part, only about 25 to 35 percent of the waste activated sludge can be digested. An additional aeration period of 15 to 20 days should be adequate to reduce the residual biodegradable mass to a satisfactory level for dewatering and return to the environment. One of the problems in aerobic digestion is the inability to concentrate the solids to levels greater than 2 percent. A second problem is nitrification. The high protein concentration in the biodegradable solids results in the release of ammonia, which can be oxidized during the long retention period in the aerobic digester. Limiting oxygen supply to the aerobic digester appears to be the best method to handle nitrification and the resulting low pH.

A new concept is to use an on/off air supply cycle. During aeration, nitrates are produced. When the air is shut off, nitrates are reduced to nitrogen gas. This prevents acid buildup and removes nitrogen from the sludge. High power cost for aerobic digestion restricts the applicability of this process.

High Lime Treatment Uses doses of lime sufficient to raise the pH of sludge to 12 or above. As long as the pH is maintained at this level, biological breakdown will not occur. In this sense, the sludge is stable. However, any reduction of pH as a result of contact with CO₂ in the air will allow biological breakdown to begin. Thus, this technique should only be used as temporary treatment until further processing can occur. It is not permanent stabilization as is provided by anaerobic or aerobic digestion.

Sludge Dewatering Dewatering is different from concentration in that the latter still leaves a substance with the properties of a liquid. The former produces a product which is essentially a friable solid. When the water content of sludge is reduced to <70–80 percent, it forms a porous solid called sludge cake. There is no free water in the cake, as the water is chemically combined with the solids or tightly adsorbed on the internal pores. The operations below which are used to dewater sludge can be applied at any stage of the sludge management process, but often they follow concentration and/or biological stabilization. Chemical conditioning is almost always used to aid dewatering.

Lime, alum, and various ferric salts have been used to condition sludge prior to dewatering. Lime reacts to form calcium carbonate crystals, which act as a solid matrix to hold the sludge particles apart and allow the water to escape during dewatering. Alum and iron salts

help displace some of the bound water from hydrophilic organics and form part of the inorganic matrix. Chemical conditioning increases the mass of sludge to be ultimately handled from 10 to 25 percent, depending upon the characteristics of the individual sludge. Chemical conditioning can also help remove some of the fine particles by incorporating them into insoluble chemical precipitates. The water (supernatant) removed from the sludge during dewatering is often high in suspended solids and organics. Addition of polymers prior to, during, or after dewatering will often reduce the level of pollutants in the supernatant.

Centrifugation Both basket and solid-bowl centrifuges have been used to concentrate waste sludges. Field data have shown that it is possible to obtain 10 to 20 percent solids with waste activated sludge, 15 to 30 percent solids with a mixture of primary and waste activated sludge, and up to 30 to 35 percent solids with primary sludge alone. Centrifuges result in 85 to 90 percent solids capture with good operation. The problem is that the centrate contains the fine solids not easily removed. The centrate is normally returned to the treatment process, where it may or may not be removed. Economics do not favor centrifuges unless the sludge cake produced is at least 25 to 30 percent solids. For the most part, centrifuges are designed by equipment manufacturers from field experience. With varying sludge characteristics centrifuge characteristics will also vary widely.

Vacuum Filtration Vacuum filtration has been the most common method employed in dewatering sludges. Vacuum filters consist of a rotary drum covered with a cloth-filter medium. Various plastic fibers as well as wool have been used for the filter cloth. The filter operates by drawing a vacuum as the drum rotates into chemically conditioned sludge. The vacuum holds a thin layer of sludge, which is dewatered as the drum rotates through the air after leaving the vat. When the drum rotates the cloth to the opposite side of the apparatus, air-pressure jets replace the vacuum, causing the sludge cake to separate from the cloth medium as the cloth moves away from the drum. The cloth travels over a series of rollers, with the sludge being separated by a knife edge and dropping onto a conveyor belt by gravity. The dewatered sludge is moved on the conveyor belt to the next concentration point, while the filter cloth is spray-washed and returned to the drum prior to entering the sludge vat. Vacuum filters yield the poorest results on waste activated sludge and the best results on primary sludge. Waste activated sludge will concentrate to between 12 and 18 percent solids at a rate of 4.9 to 9.8 kg dry cake/(m²·h) [1 to 2 lb/(ft²·h)]. Primary sludge can be dewatered to 25 to 30 percent solids at a rate of 49 kg dry cake/(m²·h) [10 lb/(ft²·h)].

Pressure Filtration Pressure filtration has been used increasingly since the early 1970s because of its ability to produce a drier sludge cake. The pressure filters consist of a series of plates and frames separated by a cloth medium. Sludge is forced into the filter under pressure, while the filtrate is drawn off. When maximum pressure is reached, the influent-sludge flow is stopped and the pressure filter is allowed to discharge the residual filtrate prior to opening the filter and allowing the filter cake to drop by gravity to a conveyor belt below the filter press. The pressure filter operates at a pressure between 689 and 1380 kPa (100 and 200 psig) and takes 1.5 to 4 h for the pressure cycle. Normally, 20 to 30 min is required to remove the filter cake. The sludge cakes will vary from 20 to 25 percent for waste activated sludge to 50 percent for primary sludge. Chemical conditioning is necessary to obtain good dewatering of the sludges.

Belt-Press Filters The newest filter for handling waste activated sludge is the belt-press filter. The belt press utilizes a continuous cloth-filter belt. Waste activated sludge is spread over the filter medium, and water is removed initially by gravity. The open belt with the sludge moves into contact with a second moving belt, which squeezes the sludge layer between rollers with ever-increasing pressure. The sludge cake is removed at the end of the filter press by a knife blade, with the sludge dropping by gravity to a conveyor belt. Belt-press filters can produce sludge with 20–30 percent solids.

Sand Beds Sand filter beds can be used to dewater either anaerobically or aerobically digested sludges. They work best on relatively small treatment systems located in relatively dry areas. The sand bed consists of coarse gravel graded to fine sand in a series of layers to a depth of 0.45 to 0.6 m (1.5 to 2 ft). The digested sludge is placed over the entire filter

surface to a depth of 0.3 m (12 in) and allowed to sit until dry. Free water will drain through the sand bed to an open pipe underdrain system and be removed from the filter. Air drying will slowly remove the remaining water. The sludge must be cleaned from the bed by hand prior to adding a second layer of sludge. The sludge layer will drop from an initial thickness of 3 m (12 in) to about 0.006 m (¼ in). An open sand bed can generally handle 49 to 122 kg dry solids/(m²·year) [10 to 25 lb/(ft²·year)]. Covered sand beds have been used in wet climates as well as in cold climates, but economics does not favor their use.

SLUDGE DISPOSAL

Incineration Incineration has been used to reduce the volume of sludge after dewatering. The organic fractions in sludges lend themselves to incineration if the sludge does not have an excessive water content. Multiple-hearth and fluid-bed incinerators have been extensively used for sludge combustion.

A multiple-hearth incinerator consists of several hearths in a vertical cylindrical furnace. The dewatered sludge is added to the top hearth and is slowly pushed through the incinerator, dropping by gravity to the next lower layer until it finally reaches the bottom layer. The top layer is used for drying the sludge with the hot gases from the lower layers. As the temperature of the furnace increases, the organics begin to degrade and undergo combustion. Air is used to add the necessary oxygen and to control the temperature during combustion. It is very important to keep temperatures above 600°C to ensure complete oxidation of the volatile organics. One of the problems with the multiple-hearth incinerator is volatilization of odorous organics during the drying phase before the temperature reaches combustion levels. Even afterburners on the exhaust-gas line may not be adequate for complete oxidation. Air-pollution-control devices are required on all incinerators to remove fly ash and corrosive gases. The ash from the incinerator must be cooled, collected, and conveyed back to the environment, normally to a sanitary landfill for burial. The residual ash will weigh from 10 to 30 percent of the original dry weight of the sludge. Supplemental fuels are needed to start the incinerator and to ensure adequate temperatures with sludges containing excessive moisture, such as activated sludge. Heat recovery from wastes is being given more consideration. It is possible to combine the sludges with other wastes to provide a better fuel for the incinerator.

A fluid-bed incinerator uses hot sand as a heat reservoir for dewatering the sludge and combusting the organics. The turbulence created by the incoming air and the sand suspension requires the effluent gases to be treated in a wet scrubber prior to final discharge. The ash is removed from the scrubber water by a cyclone separator. The scrubber water is normally returned to the treatment process and diluted with the total plant effluent. The ash is normally buried.

Sanitary Landfills Dewatered sludge, either raw or digested, is often buried in a sanitary landfill to minimize the environmental impact. Increased concern over sanitary landfills has made it more difficult simply to bury dewatered sludge. Sanitary landfills must be made secure from leachate and be monitored regularly to ensure that no environmental damage occurs. The moisture content of most sludges makes them a problem at sanitary landfills designed for solid wastes, requiring separate burial even at the same landfill.

Beneficial Reuse of Biosolids Biosolids are biological treatment sludges that are stabilized by using a variety of methods including alkaline stabilization, anaerobic digestion, aerobic digestion, composting, or heat drying and pelletization. The disposal of biosolids has typically been done through incineration and landfilling. With the severe decline in the number of landfills and the difficulty with incineration meeting stringent Clean Air Act standards, biosolids are increasingly being land-applied as a beneficial reuse. The specific type of land application is a function of the quality of biosolids being applied. Beneficial reuse of biosolids is regulated by the EPA (Part 503 Rules), which has established guidelines and specific criteria for reuse application. The EPA biosolids rules regulate chemical contaminants in the biosolids (metals), pathogen and pathogen indicator content, sludge stabilization methodology, land application site access and setback distances, vector attraction reduction methods, and the crop type and harvesting schedule.

EPA Class A or Class B biosolids can be land-applied. Land application serves two purposes: (1) to provide an inexpensive fertilizer and soil conditioner and (2) to continue treatment of the sludge through exposure to sunlight, plant uptake of metals, and desiccation of pathogens. The benefits of biosolids land application include addition

of nutrients (micro and macro) to soil, replacing agricultural dependency on chemical fertilizer; improving soil texture and increasing biological activity to promote root growth; slowly releasing nutrients to reduce off-site transport of nutrients; and lower cost than landfilling or incineration.

MANAGEMENT OF SOLID WASTES

INTRODUCTION

Solid wastes are all the wastes arising from human and animal activities that are normally solid and that are discarded as useless or unwanted. The term as used in this subsection is all-inclusive, and it encompasses the heterogeneous mass of throwaways. The three R's should be applied to solid wastes: reuse, recycle, and reduce. When these techniques have been implemented, management of the remaining residual solid waste can be addressed.

Functional Elements The activities associated with the management of solid wastes from the point of generation to final disposal have been grouped into the functional elements identified in Fig. 22-46. By considering each fundamental element separately, it is possible to (1) identify the fundamental element and (2) develop, when possible,

quantifiable relationships for the purpose of making engineering comparisons, analyses, and evaluations.

Waste Reduction Processes can be redesigned to reduce the amount of waste generated. For example, transfer lines between processes can be blown clear pneumatically to drive liquid into the batch mix tank.

Waste Generation Waste generation encompasses those activities in which materials are identified as no longer being of value and are either thrown away or gathered together for disposal. From the standpoint of economics, the best place to sort waste materials for recovery is at the source of generation.

Reuse Waste may be diverted to reuse. For example, containers may be cleaned and reused, or the waste from one process may become the feedstock for another.

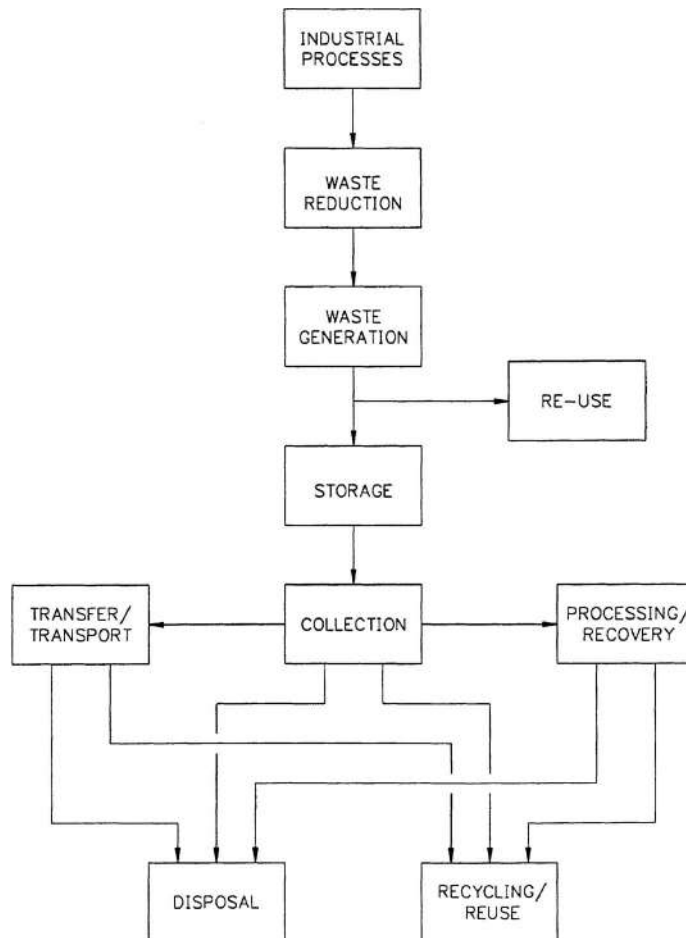


FIG. 22-46 Functional elements in a solid-waste management system. (Updated from G. Tchobanoglous, H. Theisen, and R. Eliassen, *Solid Wastes: Engineering and Management Issues*, McGraw-Hill, New York, 1977.)

On-Site Handling, Storage, and Processing The functional element encompasses those activities associated with the handling, storage, and processing of solid wastes at or near the point of generation. On-site storage is of primary importance because of the aesthetic considerations, public health, public safety, and economics involved.

Collection The functional element of collection includes the gathering of solid wastes and the hauling of wastes after collection to the location where the collection vehicle is emptied. As shown in Fig. 22-46, this location may be a transfer station, a processing station, or a landfill disposal site.

Transfer and Transport The functional element of transfer and transport involves two steps: (1) the transfer of wastes from the smaller collection containers to the larger transport equipment and (2) the subsequent transport of the wastes, usually over long distances, to the disposal site.

Processing and Recovery The functional element of processing and recovery includes all the techniques, equipment, and facilities used both to improve the efficiency of the other functional elements and to recover usable materials, conversion products, or energy from solid wastes. Materials that can be recycled are exported to facilities equipped to do so. Residues go to disposal.

Disposal The final functional element in the solid waste management system is disposal. Disposal is the ultimate fate of all solid wastes, whether they are wastes collected and transported directly to a landfill site, semisolid wastes (sludge) from industrial treatment plants and air-pollution-control devices, incinerator residue, compost, or other substances from various solid waste processing plants that are of no further use.

Solid-Waste-Management Systems Practical aspects associated with solid-waste-management systems not covered in the presentation include financing, operations, equipment management, personnel, reporting, cost accounting and budgeting, contract administration ordinances and guidelines, and public communications.

UNITED STATES LEGISLATION, REGULATIONS, AND GOVERNMENT AGENCIES

Much of the current activity in the field of solid-waste management, especially with respect to hazardous wastes and resources recovery, is a direct consequence of legislation. It is imperative to have a working knowledge of waste regulations, including RCRA and MACT (for EPA hazardous waste); TSCA (Toxic Substances Control Act) for PCBs and toxic waste; Solid Waste Disposal Act; the Clean Air Act; and PSD (prevention of significant deterioration) regulations. The Clean Air Act has far-reaching implications for any waste-related operation (e.g., incineration or landfill) that emits particulates or gaseous emissions. Primary elements of the act are Title I, dealing with fugitive emissions monitoring; Title III, with reduction of organic hazardous air pollutants; and Title V, with operating permits. The 1990 amendments have added a significant permitting burden for industrial plants and require additional controls for HAPs (hazardous air pollutants). In addition, state and local regulations apply to waste operations. The process of initiating major technology improvements in poor air quality areas, proving performance, and then spreading that technology elsewhere steadily ratcheted the emission limitations for new plants down to ever-lower levels.

Particulate emissions are an example of the ratcheting process. In the 1960s, burning 1 ton of MSW emitted about 30 lb of smoke. In 1971, the NSPS for incinerators reduced PM emissions to approximately 1.9 lb of PM per ton of MSW burned, a 94 percent reduction. In the 1980s, the focus changed to minimizing sulfur dioxide and PM. New plant PM levels have been below 0.2 lb/ton of MSW burned since the late 1980s—a 99 percent reduction from 30 lb in about two decades. Incremental improvements came from technology meeting the more stringent emission criteria. Each incremental improvement also used much more resources—energy, money and human talent—than the previous step.

Dioxin and furan emissions are another example of the ratcheting process. When first measured in the 1970s, concentrations were 1000

times greater than those routinely found 10 years later in state-of-the-art plants. The technology to control sulfur dioxide and hydrogen chloride emissions also reduced dioxin and furan emissions. Multipathway health risk assessments routinely show negligible risk from this level of dioxin and furan emissions. Even so, the 1995 Emissions Guidelines call for these low-emitting systems to demonstrate dioxin emissions another 50 times lower. This reduction is the anticipated by-product of using activated carbon to reduce mercury emissions, as it will also adsorb dioxins and furans.

The Maximum Achievable Control Technology (MACT) emission limitations required by the 1990 Clean Air Act Amendments (CAAA) show the ultimate effect of the ratcheting process. After a little more than two decades of ratcheting, MACTs have become a comparatively minor source of combustion-related air pollution. Other artificial and natural sources such as automobiles, trucks, power plants, fireplaces, wood stoves, metal production furnaces, industrial manufacturing processes, volcanoes, forest fires, and backyard trash burning are now the major known sources of combustion-related pollutants.

Regulations have been adopted by federal, state, and local agencies to implement the legislation. Before any design and construction work is undertaken, these regulations must be reviewed and care taken to factor in the nearly continuous regulatory revisions, as noted above.

GENERATION OF SOLID WASTES

Solid wastes, as noted previously, include all solid or semisolid materials that are no longer considered of sufficient value to be retained in a given setting. The types and sources of solid wastes, the physical and chemical composition of solid wastes, and typical solid-waste generation rates are considered in this subsection.

Types of Solid Waste The term *solid wastes* is all-inclusive and encompasses all sources, types of classifications, compositions, and properties. As a basis for subsequent discussions, it will be helpful to define the various types of solid wastes that are generated. It is important to note that the definitions of solid-waste terms and the classifications vary greatly in practice and in the literature. Consequently, the use of published data requires considerable care, judgment, and common sense. The following definitions are intended to serve as a guide.

1. **Food wastes.** Food wastes are the animal, fruit, or vegetable residues (also called *garbage*) resulting from the handling, preparation, cooking, and eating of foods. The most important characteristic of these wastes is that they are putrescible and will decompose rapidly, especially in warm weather.

2. **Rubbish.** Rubbish consists of combustible and noncombustible solid wastes, excluding food wastes or other putrescible materials. Typically, combustible rubbish consists of materials such as paper, cardboard, plastics, textiles, rubber, leather, wood, furniture, and garden trimmings. Noncombustible rubbish consists of items such as glass, crockery, tin cans, aluminum cans, ferrous and other nonferrous metals, dirt, and construction wastes.

3. **Ashes and residues.** These are the materials remaining from the burning of wood, coal, coke, and other combustible wastes. Residues from power plants normally are composed of fine powdery materials, cinders, clinkers, and small amounts of burned and partially burned materials. Fly ash from coal boilers and CKD (cement kiln dust) are frequently sold for stabilization of waste, waste bulking operations, and incorporation into building products such as gypsum from sulfur dioxide scrubbing.

4. **Construction and demolition wastes.** C&D wastes that come from razed buildings and other structures are classified as demolition wastes. Wastes from the construction, remodeling, and repair of commercial and industrial buildings and other similar structures are classified as construction wastes. These wastes may include dirt, stones, concrete, bricks, plaster, lumber, shingles, and plumbing, heating, and electrical parts.

5. **Special wastes.** Wastes such as street sweepings, roadside litter, catch-basin debris, dead animals, and abandoned vehicles are classified as special wastes.

6. **Treatment plant wastes.** The solid and semisolid wastes from water, wastewater, and industrial waste-treatment facilities are included in this classification.

TABLE 22-54 Sources and Types of Industrial Wastes*

Code	SIC group classification	Waste-generating processes	Expected specific wastes
19	Ordnance and accessories	Manufacturing, assembling	Metals, plastic, rubber, paper, wood, cloth, chemical residues
20	Food and kindred products	Processing, packaging, shipping	Meats, fats, oils, bones, offal, vegetables, fruits, nuts and shells, cereals
22	Textile mill products	Weaving, processing, dyeing, shipping	Cloth and filter residues
23	Apparel and other finished products	Cutting, sewing, sizing, pressing	Cloth, fibers, metals, plastics, rubber
24	Lumber and wood products	Sawmills, millwork plants, wooden containers, miscellaneous wood products, manufacturing	Scrap wood, shavings, sawdust; in some instances, metals, plastics, fibers, glues, sealers, paints, solvents
25a	Furniture, wood	Manufacture of household and office furniture, partitions, office and store fixtures, mattresses	Those listed under Code 24, in addition, cloth and padding residues
25b	Furniture, metal	Manufacture of household and office furniture, lockers, springs, frames	Metals, plastics, resins, glass, wood, rubber, adhesives, cloth paper
26	Paper and allied products	Paper manufacture, conversion of paper and paperboard, manufacture of paperboard boxes and containers	Paper and filter residues, chemicals, paper coatings and filters, inks, glues, fasteners
27	Printing and publishing	Newspaper publishing, printing, lithography, engraving, bookbinding	Paper, newsprint, cardboard, metals, chemicals, cloth, inks, glues
28	Chemicals and related products	Manufacture and preparation of inorganic chemicals (ranging from drugs and soaps to paints and varnishes and explosives)	Organic and inorganic chemicals, metals, plastics, rubber, glass, oils, paints, solvents, pigments
29	Petroleum refining and related industries	Manufacture of paving and roofing materials	Asphalt and tars, felts, paper, cloth, fiber
30	Rubber and miscellaneous plastic products	Manufacture of fabricated rubber and plastic products	Scrap rubber and plastics, lampblack, curing compounds, dyes
31	Leather and leather products	Leather tanning and finishing, manufacture of leather belting and packaging	Scrap leather, thread, dyes, oils, processing and curing compounds
32	Stone, clay, and glass products	Manufacture of flat glass, fabrication or forming of glass; manufacture of concrete, gypsum, and plaster products; forming and processing of stone products, abrasives, asbestos, and miscellaneous non-mineral products	Glass, cement, clay, ceramics, gypsum, asbestos, stone, paper, abrasives
33	Primary metal industries	Melting, casting, forging, drawing, rolling, forming, extruding operations	Ferrous and non-ferrous metals scrap, slag, sand, cores, patterns, bonding agents
34	Fabricated metal products	Manufacture of metal cans, hand tools, general hardware, non-electrical heating apparatus, plumbing fixtures, fabricated structural products, wire, farm machinery and equipment, coating and engraving of metal	Metals, ceramics, sand, slag, scale, coatings, solvents, lubricants, pickling liquors
35	Machinery (except electrical)	Manufacture of equipment for construction, elevators, moving stairways, conveyors, industrial trucks, trailers, stackers, machine tools, etc.	Slag, sand, cores, metal scrap, wood, plastics, resins, rubber, cloth, paints, solvents, petroleum products
36	Electrical	Manufacture of electrical equipment, appliances and communication apparatus, machining, drawing, forming, welding, stamping, winding, painting, plating, baking, firing operations	Metal scrap, carbon, glass, exotic metals, rubber, plastics, resins, fibers, cloth residues, PCBs
37	Transportation equipment	Manufacture of motor vehicles, truck and bus bodies, motor-vehicle parts and accessories, aircraft and parts, ship and boat building, repairing motorcycles and bicycles and parts, etc.	Metal scrap, glass, fiber, wood, rubber, plastics, cloth, paints, solvents, petroleum products
38	Professional scientific controlling instruments	Manufacture of engineering, laboratory and research instruments and associated equipment	Metals, plastics, resins, glass, wood, rubber, fibers, abrasives
39	Miscellaneous manufacturing	Manufacture of jewelry, silverware, plate ware, toys, amusements, sporting and athletic goods, costume novelties, buttons, brooms, brushes, signs, advertising displays	Metals, glass, plastics, resin, leather, rubber, composition, bone, cloth, straw, adhesives, paints, solvents

*From C. L. Mantell (ed.), *Solid Wastes: Origin, Collection, Processing, and Disposal*, Wiley-Interscience, New York, 1975. PCBs added to item 36.

7. Agricultural wastes. Wastes and residues resulting from diverse agricultural activities, such as the planting and harvesting of row, field, and tree and vine crops, the production of milk, the production of animals for slaughter, and the operation of feedlots are collectively called agricultural wastes.

Hazardous Wastes The US EPA has defined hazardous waste in RCRA regulations, CFR Parts 260 and 261. A waste may be hazardous if it exhibits one or more of the following characteristics: (1) ignitability, (2) corrosivity, (3) reactivity, and (4) toxicity. A detailed definition of these terms was first published in the Federal Register on May 19, 1980, pp. 22, 121–122. A waste may be hazardous if listed in Appendix VIII.

In the past, hazardous wastes were often grouped into the following categories: (1) radioactive substances, (2) chemicals, (3) biological wastes, (4) flammable wastes, and (5) explosives. The chemical category included wastes that were corrosive, reactive, and toxic. The

principal sources of hazardous biological wastes are hospitals and biological research facilities.

All hazardous waste combustors (HWCs), including hazardous waste incinerators (HWIs), cement kilns (CKs), and lightweight aggregate kilns (LWAKs), are subject to the MACT standards regardless of size or major source status. HWCs are required to be in compliance with the standards by September 30, 2003, and are required to demonstrate compliance by March 30, 2004, through a Comprehensive Performance Test (CPT). The rule allows for a 1-year extension of the compliance date for facilities that need to install pollution control equipment to meet standards.

The HWC MACT standards regulate emissions of the following pollutants:

- Dioxins and furans (D/F)
- Mercury (Hg)

- Total chlorine (HCl/Cl₂)
- Semivolatile metals (SVMs)—lead and cadmium
- Low-volatility metals (LVMs)—arsenic, beryllium, and chromium
- Particulate matter (PM)
- Carbon monoxide (CO)
- Hydrocarbons (HC)

The HWC MACT standards require continuous monitoring of both emissions and operating parameters to demonstrate compliance at all times. Continuous emissions monitoring is required for CO or HC. In the future, facilities will also be required to use continuous emission monitors for PM. In addition, cement kilns may be required to use an opacity monitor.

The Federal Register (vol. 69, no. 76, April 20, 2004) issued Proposed Rules for Existing and New Sources of HWCs. This included not only the HWIs, CKs, and LWAKs, but also for the first time the solid-fuel-fired boilers, liquid-fuel-fired boilers, and hydrochloric acid production furnaces. The boilers and HCl furnaces had been covered by the Boilers and Industrial Furnaces standards issued as Boiler and Industrial Furnace Act (BIF) in 1991. When MACT was enacted for HWCs on September 30, 1999, the BIFs were not included. The proposed standards were listed in Tables 1 and 2 of the FR of April 20, 2004.

Sources of Industrial Wastes Knowledge of the sources and types of solid wastes, along with data on the composition and rates of generation, is basic to the design and operation of the functional elements associated with the management of solid wastes.

Conventional Wastes Sources and types of industrial solid wastes generated by Standard Industrial Classification (SIC) group classification are reported in Table 22-54. The expected specific wastes in the table are most readily identifiable.

Hazardous Waste Hazardous wastes are generated in limited amounts throughout most industrial activities. In terms of generation, concern is with the identification of amounts and types of hazardous wastes developed at each source, with emphasis on those sources where significant waste quantities are generated.

The USEPA released "National Biennial RCRA Hazardous Waste Report" based on 2001 data. The three-volume report, which represents comprehensive data on the generation and management of hazardous waste in the United States during 2001, includes the National Analysis, the State Detail Analysis, and the List of Reported RCRA Sites. The report noted a decrease to 40.8 million

tons compared to 206 million tons of hazardous waste generated in 1991. The number of large-quantity generators decreased to 19,024 from 23,426 in 1991. The number of waste treatment, storage, and disposal (TSD) facilities decreased to 2479 from 3862 in 1991.

The majority—56.1 percent—of the hazardous wastes generated during 2001 were managed in aqueous waste treatment units. Land disposal accounted for another 4.6 percent of the total, divided as follows: 2.1 million tons sent to landfills and 65,000 tons managed by land farming. Resource recovery operations managed 3.7 percent of the 46 million tons, with solvent-recovery units managing 3.6 million tons; fuel-blending units, 1.4 million tons; metals recovery, 1.46 million tons; and other methods such as acid regeneration and waste oil recovery accounting for 9.8 million tons. Thermal treatment systems burned 6.3 percent of the hazardous wastes generated in 2001: 1.65 million tons was incinerated, and 1.7 million tons was used as fuel in boilers and industrial furnaces.

The generation of hazardous wastes by spillage must also be considered. The quantities of hazardous wastes that are involved in spillage usually are not known. After a spill, the wastes requiring collection and disposal are often significantly greater than the amount of spilled wastes, especially when an absorbing material, such as straw, sand, and "oil-dry," is used to soak up liquid hazardous wastes. Similarly, the volume and mass of contaminated soil requiring excavation are far greater than the waste volume involved in a spill. Both the adsorbing material and the liquid are classified as hazardous waste.

Properties of Solid Wastes Information on the properties of solid wastes is important in evaluating alternative equipment needs, systems, and management programs and plans.

Physical Composition Information and data on the physical composition of solid wastes including (1) identification of the individual components that make up industrial and municipal solid wastes, (2) density of solid wastes, and (3) moisture content are presented below.

1. *Individual components.* Components that typically make up most industrial and municipal solid wastes and their relative distribution are reported in Table 22-55. Although any number of components could be selected, those listed in the table have been chosen because they are readily identifiable, are consistent with component categories reported in the literature, and are adequate for the characterization of solid wastes for most applications.

2. *Density.* Typical densities for various wastes as found in containers are reported by source in Table 22-56. Because the densities of

TABLE 22-55 Typical Data on Distribution of Industrial Wastes Generated by Major Industries and Municipalities*

SIC code	SIC group classification	Percent by mass									
		Food wastes [†]	Paper	Wood	Leather	Rubber	Plastics	Metals	Glass	Textiles	Miscellaneous
20	Food and kindred products	15–20	50–60	5–10	0–2	0–2	0–5	5–10	4–10	0–2	5–15
22	Textile mill products	0–2	40–50	0–2	0–2	0–2	3–10	0–2	0–2	20–40	0–5
23	Apparel and other finished products	0–2	40–60	0–2	0–2	0–2	0–2	0–2	0–2	30–50	0–5
24	Lumber and wood products	0–2	10–20	60–80	0–2	0–2	0–2	0–2	0–2	0–2	5–10
25a	Furniture, wood	0–2	20–30	30–50	0–2	0–2	0–2	0–2	0–2	0–5	0–5
25b	Furniture, metal	0–2	20–40	10–20	0–2	0–2	0–2	20–40	0–2	0–5	0–10
26	Paper and allied products	0–2	40–60	10–15	0–2	0–2	0–2	5–15	0–2	0–2	10–20
27	Printing and publishing	0–2	60–90	5–10	0–2	0–2	0–2	0–2	0–2	0–2	0–5
28	Chemicals and related products	0–2	40–60	2–10	0–2	0–2	5–15	5–10	0–5	0–2	15–25
29	Petroleum refining and related industries	0–2	60–80	5–15	0–2	0–2	10–20	2–10	0–12	0–2	2–10
30	Rubber and miscellaneous plastic products	0–2	40–60	2–10	0–2	5–20	10–20	0–2	0–2	0–2	0–5
31	Leather and leather products	0–2	5–10	5–10	40–60	0–2	0–2	10–20	0–2	0–2	0–5
32	Stone, clay, and glass products	0–2	20–40	2–10	0–2	0–2	0–2	5–10	10–20	0–2	30–50
33	Primary metal industries	0–2	30–50	5–15	0–2	0–2	2–10	2–10	0–5	0–2	20–40
34	Fabricated metal products	0–2	30–50	5–15	0–2	0–2	0–2	15–30	0–2	0–2	5–15
35	Machinery (except electrical)	0–2	30–50	5–15	0–2	0–2	1–5	15–30	0–2	0–2	0–5
36	Electrical	0–2	60–80	5–15	0–2	0–2	2–5	2–5	0–2	0–2	0–5
37	Transportation equipment	0–2	40–60	5–15	0–2	0–2	2–5	0–2	0–2	0–2	15–30
38	Professional scientific controlling instruments	0–2	30–50	2–10	0–2	0–2	5–10	5–15	0–2	0–2	0–5
39	Miscellaneous manufacturing	0–2	40–60	10–20	0–2	0–2	5–15	2–10	0–2	0–2	5–15
Municipal	Municipal	10–20	40–60	1–4	0–2	0–2	2–10	3–15	4–16	0–4	5–30

*Adapted in part from D. G. Wilson (ed.), *Handbook of Solid Waste Management*, Van Nostrand Reinhold, New York, 1977.

†With the exception of food and kindred products, food wastes are from company cafeterias, canteens, etc.

TABLE 22-56 Typical Density and Moisture-Content Data for Domestic, Commercial, and Industrial Solid Waste

Item	Density, kg/m ³		Moisture content, % by mass	
	Range	Typical	Range	Typical
Residential (uncompacted)				
Food wastes (mixed)	130–480	290	50–80	70
Paper	40–130	85	4–10	6
Cardboard	40–80	50	4–8	6
Plastics	40–130	65	1–4	2
Textiles	40–100	65	6–15	10
Rubber	100–200	130	1–4	2
Leather	100–260	160	8–12	10
Garden trimmings	60–225	100	30–80	60
Wood	130–320	240	15–40	20
Glass	160–480	195	1–4	2
Tin cans	50–160	90	2–4	3
Nonferrous metals	65–240	160	2–4	2
Ferrous metals	130–1150	320	2–4	2
Dirt, ashes, etc.	320–1000	480	6–12	8
Ashes	650–830	745	6–12	6
Rubbish (mixed)	90–180	130	5–20	15
Residential (compacted)				
In compactor truck	180–450	300	15–40	20
In landfill (normally compacted)	360–500	450	15–40	30°
In landfill (well-compacted)	590–740	600	15–40	30°
Commercial				
Food wastes (wet)	475–950	535	50–85	75
Appliances	150–200	180	0–5	
Wooden crates	110–160	110	10–30	20
Tree trimmings	100–180	150	20–80	50
Rubbish (combustible)	50–180	120	5–25	15
Rubbish (noncombustible)	180–360	300	5–15	10
Rubbish (mixed)	140–180	160	5–20	12
Construction; demolition; remediation				
Mixed demolition (noncombustible)	1000–1600	1420	2–10	4
Mixed demolition (combustible)	300–400	360	4–15	8
Mixed construction (combustible)	180–360	260	4–15	8
Broken concrete	1200–1800	1540	0–5	—
Contaminated soil	1200–1900	1600	5–25	10
Industrial wastes				
Chemical sludges (wet)	800–1100	1000	75–99	80
Fly ash	700–900	800	2–10	4
Leather scraps	100–250	160	6–15	10
Metal scraps (heavy)	1500–2000	1780	0–5	
Metal scraps (light)	500–900	740	0–5	
Metal scraps (mixed)	700–1500	900	0–5	
Oils, tars, asphalt	800–1000	950	0–5	2
Sawdust	100–350	290	10–40	15
Textile wastes	100–220	180	6–15	10
Wood (mixed)	400–675	500	10–40	20
Agricultural wastes				
Agricultural (mixed)	400–750	560	48–80	50
Fruit wastes (mixed)	250–750	360	60–90	75
Manure (wet)	900–1050	1000	75–96	94
Vegetable wastes (mixed)	200–700	360	50–80	65

°Depends on degree of surface water infiltration.

solid waste vary markedly with geographic location, season of the year, and length of time in storage, great care should be used in selecting typical values.

3. *Moisture content.* The moisture content of solid wastes usually is expressed as the mass of moisture per unit mass of wet or dry material. In the wet-mass method of measurement, the moisture in a sample is expressed as a percentage of the wet mass of the material; in the dry-mass method, it is expressed as a percentage of the dry mass of the material. In equation form, the wet-mass moisture content is expressed as

$$\text{Moisture content (\%)} = \left\{ a - \frac{b}{a} \right\} \times 100 \quad (22-26)$$

where a = initial mass of sample as delivered and b = mass of sample after drying. Typical data on the moisture content for the solid-waste

components are given in Table 22-56. For most industrial solid wastes, the moisture content will vary from 10 to 25 percent.

4. *Particle size.* The material handling properties of solid wastes are dependent upon particle size. This applies as well to feed preparation and air pollution control which are affected by solid-waste particle size and cohesiveness. For wastes such as bulk soils, the amount of fines (from clay and silt) is critical for system design. Cohesiveness, which varies with moisture content, is important for bin and conveyor design.

Chemical Composition Information on the chemical composition of solid wastes is important in evaluating alternative processing and recovery options. If solid wastes are to be used as fuel, the six most important properties to be known are

1. Proximate analysis

a. Moisture (loss at 105°C for 1 h)

TABLE 22-57 Typical Proximate-Analysis and Energy-Content Data for Components in Domestic, Commercial, and Industrial Solid Waste*

Component	Proximate analysis, % by mass				Energy content, kJ/kg		
	Moisture	Volatile matter	Fixed carbon	Non-combustible	As collected	Dry	Moisture- and ash-free
Food and food products							
Fats	2.0	95.3	2.5	0.2	37,530	38,296	38,374
Food wastes (mixed)	70.0	21.4	3.6	5.0	4,175	13,917	16,700
Fruit wastes	78.7	16.6	4.0	0.7	3,970	18,638	19,271
Meat wastes	38.8	56.4	1.8	3.1	17,730	28,970	30,516
Paper products							
Cardboard	5.2	77.5	12.3	5.0	16,380	17,278	18,240
Magazines	4.1	66.4	7.0	22.5	12,220	12,742	16,648
Newsprint	6.0	81.1	11.5	1.4	18,550	19,734	20,032
Paper (mixed)	10.2	75.9	8.4	5.4	15,815	17,611	18,738
Waxed cartons	3.4	90.9	4.5	1.2	26,345	27,272	27,615
Plastics							
Plastics (mixed)	0.2	95.8	2.0	2.0	32,000	32,064	32,720
Polyethylene	0.2	98.5	<0.1	1.2	43,465	43,552	44,082
Polystyrene	0.2	98.7	0.7	0.5	38,190	38,266	38,216
Polyurethane	0.2	87.1	8.3	4.4	26,060	26,112	27,316
Polyvinyl chloride	0.2	86.9	10.8	2.1	22,690	22,735	23,224
Wood, trees, etc.							
Garden trimmings	60.0	30	9.5	0.5	6,050	15,125	15,316
Green wood	50.0	42.3	7.3	0.4	4,885	9,770	9,848
Hardwood	12.0	75.1	12.4	0.5	17,100	19,432	19,542
Wood (mixed)	20.0	67.9	11.3	0.8	15,444	19,344	19,500
Leather, rubber, textiles, etc.							
Leather (mixed)	10	68.5	12.5	9.0	18,515	20,572	22,858
Rubber (mixed)	1.2	83.9	4.9	9.9	25,330	25,638	28,493
Textiles (mixed)	10	66.0	17.5	6.5	17,445	19,383	20,892
Glass, metals, etc.							
Glass and mineral	2	—	—	96–99+	196†	200	200
Metal, tin cans	5	—	—	94–99+	1,425†	1,500	1,500
Metals, ferrous	2	—	—	96–99+	—	—	—
Metals, nonferrous	2	—	—	94–99+	—	—	—
Miscellaneous							
Office sweepings	3.2	20.5	6.3	70	8,535	8,817	31,847
Multiple wastes	20 (15–40)	53 (30–60)	7 (5–15)	20 (9–30)	10,470	13,090	17,450
Industrial wastes	15 (10–30)	58 (30–60)	7 (5–15)	20 (10–30)	11,630	13,682	17,892

*Adapted in part from D. G. Wilson (ed.), *Handbook of Solid Waste Management*, Van Nostrand Reinhold, New York, 1977, and G. Tchobanoglous, H. Theisen, and R. Eliassen, *Solid Wastes: Engineering Principles and Management Issues*, McGraw-Hill, New York, 1977.

†Energy content is from coatings, labels, and attached materials.

- b. Volatile matter (additional loss on heating to 950°C)
- c. Ash (residue after burning)
- d. Fixed carbon (remainder)
2. Fusion point of ash
3. Ultimate analysis, percent of C (carbon), H (hydrogen), O (oxygen), N (nitrogen), S (sulfur), and ash
4. Heating value
5. Organic chlorine
6. Organic sulfur

Typical proximate-analysis data for the combustible components of industrial and municipal solid wastes are presented in Table 22-57.

Typical data on the inert residue and energy values for solid wastes may be converted to a dry basis by using Eq. (22-27):

$$\frac{\text{kJ}}{\text{kg (dry basis)}} = \frac{\text{kJ}}{\text{kg (as discarded)}} \left(\frac{100}{100 - \% \text{ moisture}} \right) \quad (22-27)$$

The corresponding equation on an ash-free basis is

$$\frac{\text{kJ}}{\text{kg (ash-free dry basis)}} = \frac{\text{kJ}}{\text{kg (as discarded)}} \times \left(\frac{100}{100 - \% \text{ ash} - \% \text{ moisture}} \right) \quad (22-28)$$

Representative data on the ultimate analysis of typical industrial and municipal-waste components are presented in Table 22-58. If energy values are not available, approximate values can be determined by using Eq. (22-29), known as the modified Dulong formula, and the data in Table 22-58.

$$\text{kJ/kg} = 337C + 1428 \left(H - \frac{1}{8}O \right) + 95S \quad (22-29)$$

where C = carbon, percent
 H = hydrogen, percent
 O = oxygen, percent
 S = sulfur, percent

Quantities of Solid Wastes Representative data on the quantities of solid wastes and factors affecting the generation rates are considered briefly in the following paragraphs.

Typical Generation Rates Typical unit waste generation rates from selected industrial sources are reported in Table 22-59. Because waste generation practices are changing so rapidly, the presentation of “typical” waste generation data may not be reliable.

Factors That Affect Generation Rates Factors that influence the quantity of industrial wastes generated include (1) the extent of salvage and recycle operations, (2) company attitudes, and (3) legislation

TABLE 22-58 Typical Ultimate-Analysis Data for Components in Domestic, Commercial, and Industrial Solid Waste*

Components	Percent by mass (dry basis)					
	Carbon	Hydrogen	Oxygen	Nitrogen	Sulfur	Ash
Foods and food products						
Fats	73.0	11.5	14.8	0.4	0.1	0.2
Food wastes (mixed)	48.0	6.4	37.6	2.6	0.4	5.0
Fruit wastes	48.5	6.2	39.5	1.4	0.2	4.2
Meat wastes	59.6	9.4	24.7	1.2	0.2	4.9
Paper products	45.4	6.1	42.1	0.3	0.1	6.0
Cardboard	43.0	5.9	44.8	0.3	0.2	5.0
Magazines	32.9	5.0	38.6	0.1	0.1	23.3
Newsprint	49.1	6.1	43.0	< 0.1	0.2	23.3
Paper (mixed)	43.4	5.8	44.3	0.3	0.2	6.0
Waxed cartons	59.2	9.3	30.1	0.1	.1	1.2
Plastics						
Plastics (mixed)	60.0	7.2	22.8	—	—	10.0
Polyethylene	85.2	14.2	—	< 0.1	< 0.1	0.4
Polystyrene	87.1	8.4	4.0	0.2	—	0.3
Polyurethane†	63.3	6.3	17.6	6.0	< 0.1	4.3
Polyvinyl chloride‡	45.2	5.6	1.6	0.1	0.1	2.0
Wood, trees, etc.						
Garden trimmings	46.0	6.0	38.0	3.4	0.3	6.3
Green timber	50.1	6.4	42.3	0.1	0.1	1.0
Hardwood	49.6	6.1	43.2	0.1	< 0.1	0.9
Wood (mixed)	49.5	6.0	42.7	0.2	< 0.1	1.5
Wood chips (mixed)	48.1	5.8	45.5	0.1	< 0.1	0.4
Glass, metals, etc.						
Glass and mineral	0.5	0.1	0.4	< 0.1	—	98.9
Metals (mixed)	4.5	0.6	4.3	< 0.1	—	90.5
Leather, rubber, textiles						
Leather (mixed)	60.0	8.0	11.6	10.0	0.4	10.0
Rubber (mixed)	69.7	8.7	—	—	1.6	20.0
Textiles (mixed)	48.0	6.4	40.0	2.2	0.2	3.2
Miscellaneous						
Office sweepings	24.3	3.0	4.0	0.5	0.2	68.0
Oils, paints	66.9	9.6	5.2	2.0	—	16.3
Refuse-derived fuel (RAF)	44.7	6.2	38.4	0.7	< 0.1	9.9

*Adapted in part from D. G. Wilson (ed.), *Handbook of Solid Waste Management*, Van Nostrand Reinhold, New York, 1977, and G. Tchobanoglous, H. Theissen, and R. Eliassen, *Solid Wastes Engineering Principles and Management Issues*, McGraw-Hill, New York, 1977.

†Organic content is from coatings, labels and other attached materials.

‡Remainder is chlorine.

and regulations. The existence of salvage and recycling operations within an industry definitely affects the quantities of wastes collected. Whether such operations affect the quantities generated is another matter. The subsection "Pollution Prevention" by Louis Theodore points out actions taken by industrial plants as a result of the 1990 Pollution Prevention Act. Significant reductions in the quantities of solid wastes generated will occur when and if companies are willing to change—of their own volition—to conserve natural resources, to handle waste at its source rather than use "end-of-pipe" pollution control, and to reduce the economic burdens associated with the management of solid wastes. Perhaps the most important factor affecting the generation of certain types of waste is the existence of local, state, and federal regulations concerning the use and disposal of specific material. In general, the more regulated the waste, the higher the cost for treatment and disposal and the greater the incentive to reduce generation of the waste.

In 2001, U.S. residents, businesses, and institutions produced more than 229 million tons of MSW, which is approximately 4.4 lb of

waste per person per day, up from 2.7 lb per person per day in 1960. Several waste management practices, such as source reduction, recycling, and composting, prevent or divert materials from the waste stream. Source reduction involves altering the design, manufacture, or use of products and materials to reduce the amount and toxicity of what gets thrown away. A portion of each material category was recycled or composted in 2001. The highest rates of recovery were achieved with yard trimmings, paper products, and metal products. About 57 percent (5.8 million tons) of yard trimmings were recovered for composting in 2001. This represents a 4-fold increase from 1990. About 45 percent (36.7 million tons) of paper and paperboard was recovered for recycling in 2001. Recycling these organic materials alone diverted nearly 23 percent of MSW from landfills and combustion facilities. In addition, about 6.3 million tons, or about 35 percent, of metals were recovered for recycling. See Figs. 22-47 and 22-48.

EPA has ranked the most environmentally sound strategies for MSW. Source reduction (including reuse) is the most preferred

TABLE 22-59 Unit Solid-Waste Generation Rates for Selected Industrial Sources

Source	Unit	Range
Canned and frozen foods	Metric tons/metric tons of raw product	0.04–0.06
Printing and publishing	Metric tons/metric tons of raw paper	0.08–0.10
Automotive	Metric tons/vehicle produced	0.6–0.8
Petroleum refining	Metric tons/(employee day)	0.04–0.05
Rubber	Metric tons/metric tons of raw rubber	0.01–0.3

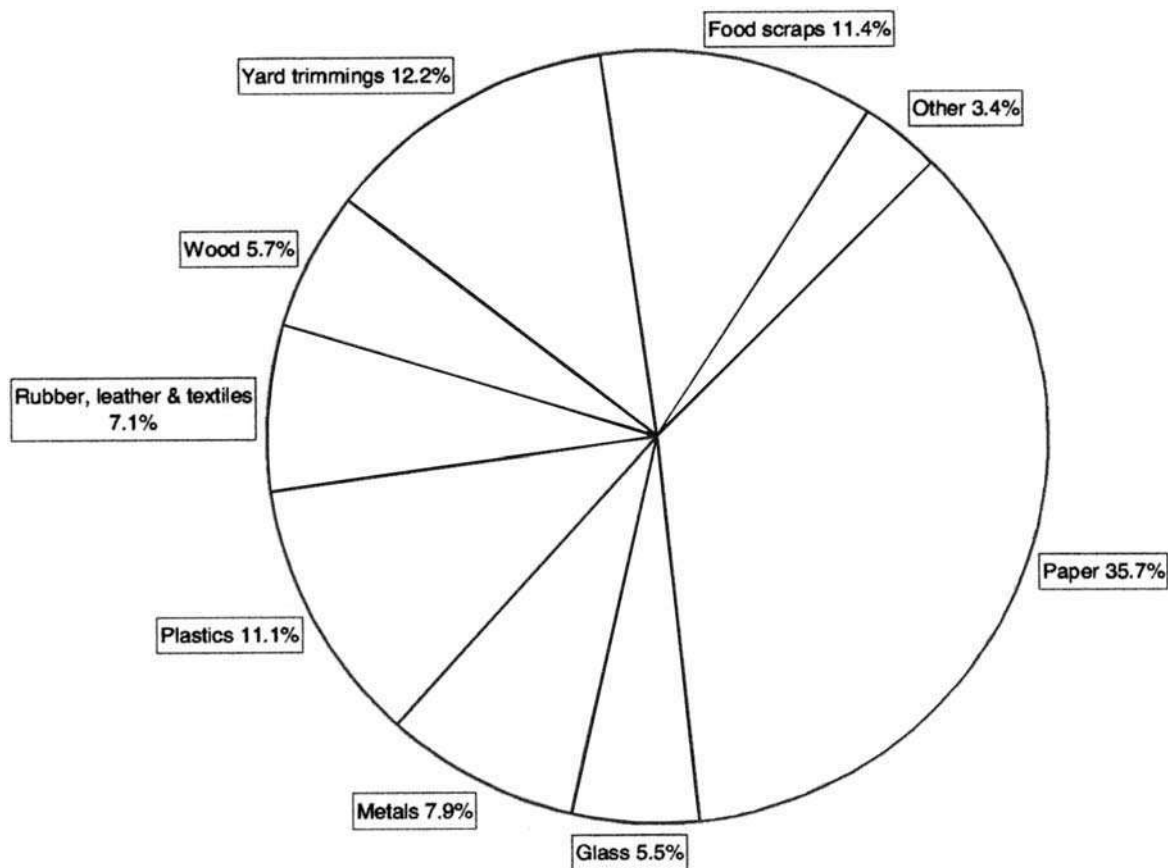


FIG. 22-47 2001 total MSW generation—229 million tons.

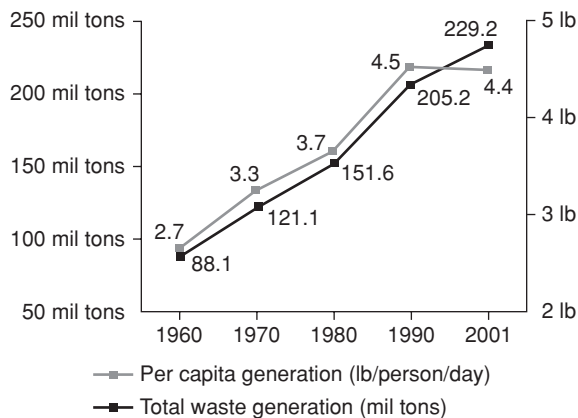


FIG. 22-48 Trends in MSW generation, 1960–2001.

method, followed by recycling and composting, and, lastly, disposal in MWCs and landfills. Currently, in the United States, 30 percent is recovered and recycled or composted, 15 percent is burned at combustion facilities (MWCs and WTEs), and the remaining 56 percent is disposed of in landfills. See Fig. 22-49.

Source reduction can be a successful method of reducing waste generation. Practices such as grass recycling, backyard composting, two-sided copying of paper, and transport packaging reduction by industry have yielded substantial benefits through source reduction. Source reduction has many environmental benefits. It prevents emissions of many greenhouse gases, reduces pollutants, saves energy, conserves resources, and reduces the need for new landfills and MWCs. Table 22-60 shows the total source reduction between 1992 and 2000 in millions of tons. Note the rapid rise in the first 4 years from 0.6 million to 31.0 million tons. From 1996 to 2000, a period of 4 years, a reduction of 77 percent has continued. The materials that made up these reductions are noted in Table 22-61.

Recycling, including composting, diverted 68 million tons of material away from landfills and MWCs in 2001, up from 34 million tons in 1990. See Fig. 22-50.

ON-SITE HANDLING, STORAGE, AND PROCESSING

The handling, storage, and processing of solid wastes at the source before they are collected are the second element of the six functional elements in the solid-waste-management system.

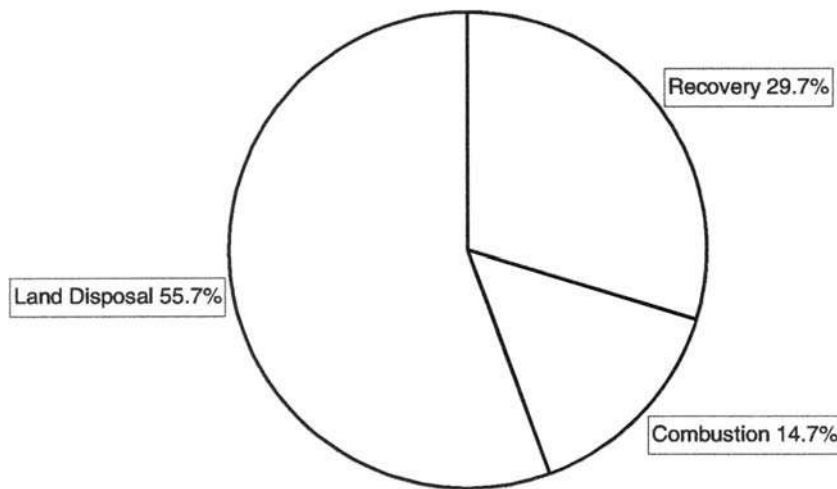


FIG. 22-49 Management of MSW in the United States, 2001.

On-Site Handling On-site handling refers to the activities associated with the handling of solid wastes until they are placed in the containers used for storage before collection. Depending on the type of collection service, handling may also be required to move loaded containers to the collection point and to return the empty containers to the point where they are stored between collections.

Conventional Solid Wastes In most office, commercial, and industrial buildings, solid wastes that accumulate in individual offices or work locations usually are collected in relatively large containers mounted on casters. Once filled, these containers are removed by means of the service elevator, if there is one, and emptied into (1) large storage containers, (2) compactors used in conjunction with the storage containers, (3) stationary compactors that can compress the material into bales or into specially designed containers, or (4) other processing equipment.

Hazardous Wastes When hazardous wastes are generated, special containers are usually provided, and trained personnel (OSHA

1910.120 required such workers to have HAZWOPER training) are responsible for the handling of these wastes. Hazardous wastes include solids, sludges, and liquids; hence, container requirements vary with the form of waste.

On-Site Storage Factors that must be considered in the on-site storage of solid wastes include (1) the type of container to be used, (2) the container location, (3) public health and aesthetics, (4) the collection methods to be used, and (5) future transport method.

Containers To a large extent, the types and capacities of the containers used depend on the characteristics of the solid wastes to be collected, the collection frequency, and the space available for the placement of containers.

1. *Containers for conventional wastes.* The types and capacities of containers now commonly used for on-site storage of solid wastes are summarized in Table 22-62. The small containers are used in individual offices and workstations. The medium-size and large containers are used at locations where large volumes are generated.

2. *Containers for hazardous wastes.* On-site storage practices are a function of the types and amounts of hazardous wastes generated and the time period over which waste generation occurs. Usually, when large quantities are generated, special facilities are used that have sufficient capacity to hold wastes accumulated over a period of

TABLE 22-60 Source Reduction of Municipal Solid Waste since 1990

Year	Million tons of source reduced
1992	0.6
1994	8.0
1995	21.4
1996	31.0
1997	31.8
1998	37.3
1999	42.8
2000	55.1

TABLE 22-61 Source Reduction by Major Material Categories, 2000

Waste stream	Million tons of source reduced
Durable goods (e.g., appliances, furniture)	5.4
Nondurable goods (e.g., newspapers, clothing)	9.3
Containers and packaging (e.g., bottles, boxes)	15.5
Other MSW (e.g., yard trimmings, food scraps)	25.0
Total source reduction (1990 baseline)	55.1

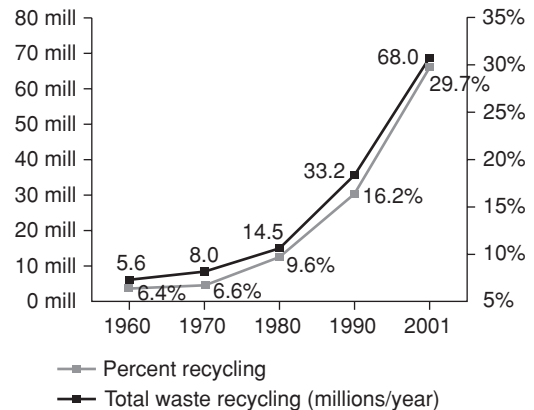


FIG. 22-50 Waste recycling rates, 1960–2001.

TABLE 22-62 Data on the Types and Sizes of Containers Used for On-Site Storage of Solid Wastes

Container type	Capacity			Dimensions*	
	Unit	Range	Typical	Unit	Typical
Small-capacity					
Plastic or metal (office type)	L	16–40	28	mm	$(180 \times 300)B \times (260 \times 380)T \times 380H$
Plastic or galvanized metal	L	75–150	120	mm	$510D \times 660H$
Barrel, plastic, aluminum, or fiber barrel	L	20–250	120	mm	$510D \times 660H$
Disposable paper bags (standard, leak-resistant, and leakproof)	L	75–210	120	mm	$380W \times 380d \times 1100H$
Disposal plastic bag	L	20–200	170	mm	$460W \times 380d \times 1000H$
Medium-capacity					
Side or top loading	m ³	0.75–9	3	mm	$1830W \times 1070d \times 1650H$
Bulk bags	m ³	0.3–2	1	mm	$1000W \times 1000d \times 1000H$
Large-capacity					
Open top, rolloff (also called debris bags)	m ³	9–38	27	mm	$2440W \times 1830H \times 6100L$
Used with stationary compactor	m ³	15–30	23	mm	$2440W \times 1830H \times 5490L$
Equipped with self-contained compaction mechanism	m ³	15–30	23	mm	$2440W \times 2440H \times 6710L$
Trailer-mounted					
Open top	m ³	15–38	27	mm	$2440W \times 3660H \times 6100L$
Enclosed, equipped with self-contained compaction mechanism	m ³	15–30	27	mm	$2440W \times 3660H \times 7320L$

*B = bottom, T = top, D = diameter, H = height, W = width, L = length, and d = depth.

several days. When only small amounts of hazardous wastes are generated on an intermittent basis, they may be containerized, and limited quantities may be stored for periods covering months or years. General information on the storage containers used for hazardous wastes and the conditions of their use is presented in Table 22-63.

Container Location The location of containers at existing commercial and industrial facilities depends on both the location of available space and service-access conditions. In newer facilities, specific service areas have been included for this purpose. Often, because the containers are not owned by the commercial or industrial activity, the locations and types of containers to be used for on-site storage must be worked out jointly between the industry and the public or private collection agency.

On-Site Processing of Solid Wastes On-site processing methods are used to (1) recover usable materials from solid wastes, (2) reduce the volume, or (3) alter the physical form. The most common on-site processing operations as applied to large commercial and industrial sources include manual sorting, compaction, and incineration. These and other processing operations are considered in the portion of this subsection dealing with processing and resource recovery. Factors that should be considered in the selection of on-site processing equipment are summarized in Table 22-64.

PROCESSING AND RESOURCE RECOVERY

The purpose of this subsection is to introduce the reader to the techniques and methods used to recover materials, conversion products,

and energy from solid wastes. Topics to be considered include (1) processing techniques for solid waste, (2) processing techniques for hazardous wastes, (3) materials-recovery systems, (4) recovery of biological conversion products, (5) thermal processes, and (6) waste-to-energy systems.

Because so many of the techniques, especially those associated with the recovery of materials and energy and the processing of solid hazardous wastes, are in a state of flux with respect to application and design criteria, the objective here is only to introduce them to the reader. If these techniques are to be considered in the development of waste-management systems, current engineering design and performance data must be obtained from consultants, operating records, field tests, equipment manufacturers, and available literature.

Processing Techniques for Solid Waste Processing techniques used in solid-waste-management systems (1) improve the efficiency of the systems, (2) recover resources (usable materials), and (3) prepare materials for recovery of conversion products and energy. The more important techniques used for processing solid wastes are summarized below:

- **Manual component separation.** The manual separation of solid-waste components can be accomplished at the source where solid wastes are generated, at a transfer station, at a centralized processing station, or at the disposal site. Manual sorting at the source of generation is the most positive way to achieve the recovery and reuse of materials. The number and types of components salvaged or sorted (e.g., cardboard and high-quality paper, metals, and wood) depend on the location, the opportunities for recycling, and the resale market.

TABLE 22-63 Typical Data on Containers Used for Storage and Transport of Hazardous Waste

Waste category	Container		Auxiliary equipment and conditions of use
	Type	Capacity	
Radioactive substances	Lead encased in concrete	Varies with waste	Isolated storage buildings; high-capacity hoists and lighting equipment; special container markings
Corrosive, reactive, and toxic chemicals	Lined metal drums	210 L	
	Metal drums	210 L	
	Plastic drums	210 L, up to 500 L	
Liquids and sludges	Lined metal drums	210 L	Washing facilities for empty containers, special blending precautions to prevent hazardous reactions; incompatible wastes stored separately
	Lined and unlined storage tanks	Up to 20 m ³	
	Drums	142–3400 L	
	Trucks	11–30 m ³	
Biological wastes	Vacuum tankers	11–15 m ³	Drums, hand-trucks, pallets, forklifts
	Sealed plastic bags	120 L	
	Lined boxes, lined metal drums	57 L	
Flammable wastes	Metal drums	210 L	Heat sterilization prior to bagging; special heavy-duty bags with hazard warning printed on sides
	Storage tanks	Up to 20 m ³	
Explosives	Shock-absorbing containers	Varies	Fume ventilation, temperature control
			Temperature control; special container markings

TABLE 22-64 Factors That Should Be Considered in Evaluating On-Site Processing Equipment

Factor	Evaluation
Capabilities	What will the device or mechanism do? Will its use be an improvement over conventional practices?
Reliability	Will the equipment perform its designated functions with little attention beyond preventive maintenance? Has the effectiveness of the equipment been demonstrated in use over a reasonable period of time or merely predicted?
Service	Will servicing capabilities beyond those of the local building maintenance staff be required occasionally? Are properly trained service personnel available through the equipment manufacturer or the local distributor?
Safety of operation	Is the proposed equipment reasonably foolproof so that it may be operated by tenants or building personnel with limited mechanical knowledge or abilities? Does it have adequate safeguards to discourage careless use?
Ease of operation	Is the equipment easy to operate by a tenant or building personnel? Unless functions and actual operations of equipment can be carried out easily, they may be ignored or bypassed by paid personnel and most often by "paying" tenants.
Efficiency	Does the equipment perform efficiently and with a minimum of attention? Under most conditions, equipment that completes an operational cycle each time that it is used should be selected.
Environmental effects	Does the equipment pollute or contaminate the environment? When possible, equipment should reduce environmental pollution presently associated with conventional functions.
Health hazards	Does the device, mechanism, or equipment create or amplify health hazards?
Aesthetics	Do the equipment and its arrangement offend the senses? Every effort should be made to reduce or eliminate offending sights, odors, and noises.
Economics	What are the economics involved? Both first and annual costs must be considered. Future operation and maintenance costs must be assessed carefully. All factors being equal, equipment produced by well-established companies, having a proven history of satisfactory operations should be given appropriate consideration.
Flexibility	Will the equipment or its placement allow future changes to the process to handle wastes with differing characteristics?

*From G. Tchobanoglous, H. Theissen, and R. Eliassen, *Solid Wastes: Engineering Principles and Management Issues*, McGraw-Hill, New York, 1977.

There has been an evolution in the solid-waste industry to combine manual and automatic separation techniques to reduce overall costs and produce a cleaner product, especially for recyclable materials. Component separation is the heart of recycling, and it is required to raise the concentration of the material to the point that it is worth reclaiming, and reduce the amount of tramp materials to meet specifications. Recycling occurs on a major scale, and for MSW, it accounted for 30 percent of the waste stream in 2001.

- **Storage and transfer.** When solid wastes are to be processed for material recovery, storage and transfer facilities should be considered an essential part of the processing operation. Important factors in the design of such facilities include (1) the size of the material before and after processing, (2) the density of the material, (3) the angle of repose before and after processing, (4) the abrasive characteristics of the materials, and (5) the moisture content.
- **Mechanical volume reduction.** Mechanical volume reduction is perhaps the most important factor in the development and opera-

tion of solid-waste-management systems. Vehicles equipped with compaction mechanisms are used for the collection of most industrial solid wastes. To increase the useful life of landfills, wastes are compacted there also via mobile compaction equipment. Paper for recycling is baled for shipping to processing centers. When compacting industrial solid wastes, it has been found that the final density (typically about 1100 kg/m³) is essentially the same regardless of the starting density and applied pressure. This fact is important in evaluating manufacturer's claims.

- **Chemical volume reduction.** Incineration has been the method commonly used to reduce the volume of wastes chemically. One of the most attractive features of the incineration process is that it can be used to reduce the original volume of combustible solid wastes by 80 to 90 percent. The technology of incineration has advanced with many mass burn facilities now having two or more combustors with capacities of 1000 tons/day of refuse per unit. However, regulations limiting low volatile and semivolatile metals, acid gases, free chlorine and hydrogen chloride, unburned hydrocarbons, polychlorinated dioxin and furan (PCDD/PCDF) emissions have resulted in higher costs and operating complexity.
- **Mechanical size alteration.** The objective of size reduction is to obtain a final product that is reasonably uniform and considerably reduced in size in comparison to its original form. It is important to note that size reduction does not necessarily imply volume reduction. In some situations, the total volume of the material after size reduction may be greater than the original volume. Shredding is most often used for size reduction. Two-stage (coarse, fine) low-speed shredders and a single-stage shredder with screen and recycle of oversize are the most common systems. The gain in ease of materials handling must be weighed against the substantial operating costs for shredding equipment. A substantial market for the technology lies in shredding tires to produce fuel for cement kilns and utility boilers. With finer shredding, the product can be used to make thick resilient mats for play areas or for inclusion in asphalt pavement.
- **Mechanical component separation.** Component separation is a necessary operation in the recovery of resources from solid wastes and in instances when energy and conversion products are to be recovered from process wastes.
- **Magnetic and electromechanical separation.** Magnetic separation of ferrous materials is a well-established technique. More recently, a variety of electromechanical techniques have been developed for the removal of several nonferrous materials.
- **Drying and dewatering.** In many solid-waste energy-recovery and incineration systems, the shredded light fraction is predried to decrease weight. Although energy requirements for drying wastes vary with the application, the required energy input can be established by using a value of about 4300 kJ/kg of water evaporated. Drying can frequently increase waste throughput in many treatment systems. For incinerators, it produces more stable combustion and better ash quality.
- **Bulking of liquid wastes.** Liquid wastes are prohibited from landfills. Wastes with free liquids are mixed with a bulking agent such as dry sawdust, cement kiln dust, or fly ash.

Processing of Hazardous Waste As with conventional solid wastes, the processing of hazardous wastes is undertaken for three purposes: (1) to recover useful materials, (2) to reduce the amount of wastes that must be disposed in landfills, and (3) to prepare the wastes for ultimate disposal.

Processing Techniques The processing of hazardous wastes on a batch basis can be accomplished by physical, chemical, thermal, and biological means. The various individual processes in each category are reported in Table 22-65. Clearly, the number of possible treatment-process combinations is staggering. In practice, the physical, chemical, and thermal treatment operations and processes are the ones most commonly used.

Identification of Waste Constituents In any processing (and disposal) scheme, the key item is knowledge of the characteristics of the waste being handled. Without this information, effective processing or treatment is impossible. For this reason, the characteristics of the wastes must be known before they are accepted and hauled to a

TABLE 22-65 Biological and Thermal Processes Used for Recovery of Conversion Products from Solid Waste

Process	Conversion product	Preprocessing required	Comments
Biological			
Composting	Humuslike material	Shredding, air separation	Lack of markets a primary shortcoming; technically proved in full-scale application
Anaerobic digestion	Methane gas	Shredding, air separation	Technology on laboratory scale only
Biological conversion to protein	Protein, alcohol	Shredding, air separation	Technology on pilot scale only
Biological fermentation	Glucose, furfural	Shredding, air separation	Used in conjunction with the hydrolytic process
Thermal			
Incineration with heat recovery	Energy in the form of steam	None	Markets for steam required; proved in numerous full-scale applications; air-quality regulations possibly prohibiting use
Thermal desorption	Clean soil used for fill or reclaimed raw materials such as deoiled metal turnings	Sizing, blending, dilution	In routine use for Superfund organic contaminated soil
Supplementary fuel firing in boilers	Energy in the form of steam	Shredding, air separation, magnetic separation	If at least capital investment desired, existing air-quality regulations possibly prohibiting use
Gasification	Energy in the form of low-energy gas	Shredding, air separation, magnetic separation	Gasification also capable of being used for codisposal for industrial sludges
Pyrolysis	Energy in the form of gas or oil and char	Shredding, magnetic separation	Technology proved only in pilot applications but full-scale use has rarely succeeded due to high operating costs and lack of materials for gas, oil, and char only.
Hydrolysis	Glucose, furfural	Shredding, air separation	Technology on pilot scale only
Chemical conversion	Oil, gas, cellulose acetate	Shredding, air separation	Technology on pilot scale only

treatment or disposal site. In most states, proper identification of the constituents of the waste is the responsibility of the waste generator.

Materials-Recovery Systems Paper, rubber, plastics, textiles, glass, metals, and organic and inorganic materials are the principal recoverable materials contained in industrial solid wastes.

Once a decision has been made to recover materials and/or energy, process flow sheets must be developed for the removal of the desired components, subject to predetermined materials specifications. A typical flow sheet for the recovery of specific components and the preparation of combustible materials for use as a fuel source is presented in Fig. 22-51. The light combustible materials are often identified as refuse-derived fuel (RDF).

The design and layout of the physical facilities that make up the processing-plant flow sheet are an important aspect in the implementation and successful operation of such systems. Important factors that must be considered in the design and layout of such systems include (1) process performance efficiency, (2) reliability and flexibility, (3) ease and economy of operation, (4) aesthetics, and (5) environmental controls.

Recovery of Biological Conversion Products Biological conversion products that can be derived from wastes include alcohols and a variety of other intermediate organic compounds. The principal processes that have been used are reported in Table 22-68. Composting and anaerobic digestion, the two most highly developed processes, are considered further. The recovery of gas from landfills is discussed in the portion of this subsection dealing with ultimate disposal.

Composting If the organic materials, excluding plastics, rubber, and leather, are separated from municipal solid wastes and subjected to bacterial decomposition, the end product remaining after dissimilatory and assimilatory bacterial activity is called *compost* or *humus*. The entire process involving both separation and bacterial conversion of the organic solid wastes is known as *composting*. Decomposition of the organic solid wastes may be accomplished either aerobically or anaerobically, depending on the availability of oxygen.

Most composting operations involve three basic steps—(1) preparation of solid wastes, (2) decomposition of the solid wastes, and (3) size reduction—and moisture and nutrient addition are part of the preparation step. Several techniques have been developed to accomplish the decomposition step. Once the solid wastes have been con-

verted to a humus, they are ready for the third step of product preparation and marketing. This step may include fine grinding, blending with various additives, granulation, bagging, storage, shipping, and, in some cases, direct marketing. The principal design considerations associated with the biological decomposition of prepared solid wastes are presented in Table 22-66.

Anaerobic Digestion Anaerobic digestion or anaerobic fermentation, as it is often called, is the process used for the production of methane from solid wastes. In most processes in which methane is to be produced from solid wastes by anaerobic digestion, three basic steps are involved. The first step involves preparation of the organic fraction of the solid wastes for anaerobic digestion and usually includes receiving, sorting, separation, and size reduction. The second step involves the addition of moisture and nutrients, blending, pH adjustment to about 6.7, heating of the slurry to between 327 and 333 K (130 and 140°F), and anaerobic digestion in a reactor with continuous flow, in which the contents are well mixed for a time varying from 8 to 15 days. The third step involves capture, storage, and, if necessary, separation of the gas components evolved during the digestion process. The fourth step is the disposal of the digested sludge, an additional task that must be accomplished. Some important design considerations are reported in Table 22-67. Because of the variability of the results reported in the literature, it is recommended that pilot-plant studies be conducted if the digestion process is to be used for the conversion of solid wastes.

Thermal Processes Conversion products that can be derived from solid wastes include heat, gases, a variety of oils, and various related organic compounds. The principal thermal processes that have been used for the recovery of usable conversion products from solid wastes are reported in Table 22-65.

Incineration with Heat Recovery Heat contained in the gases produced from the incineration of solid wastes can be recovered as steam. The low-level heat remaining in the gases after heat recovery can also be used to preheat the combustion air, boiler makeup water, or solid-waste fuel.

1. *In existing incinerators.* With existing incinerators, waste-heat boilers can be installed to extract heat from the combustion gases without introducing excess amounts of air or moisture. Typically,

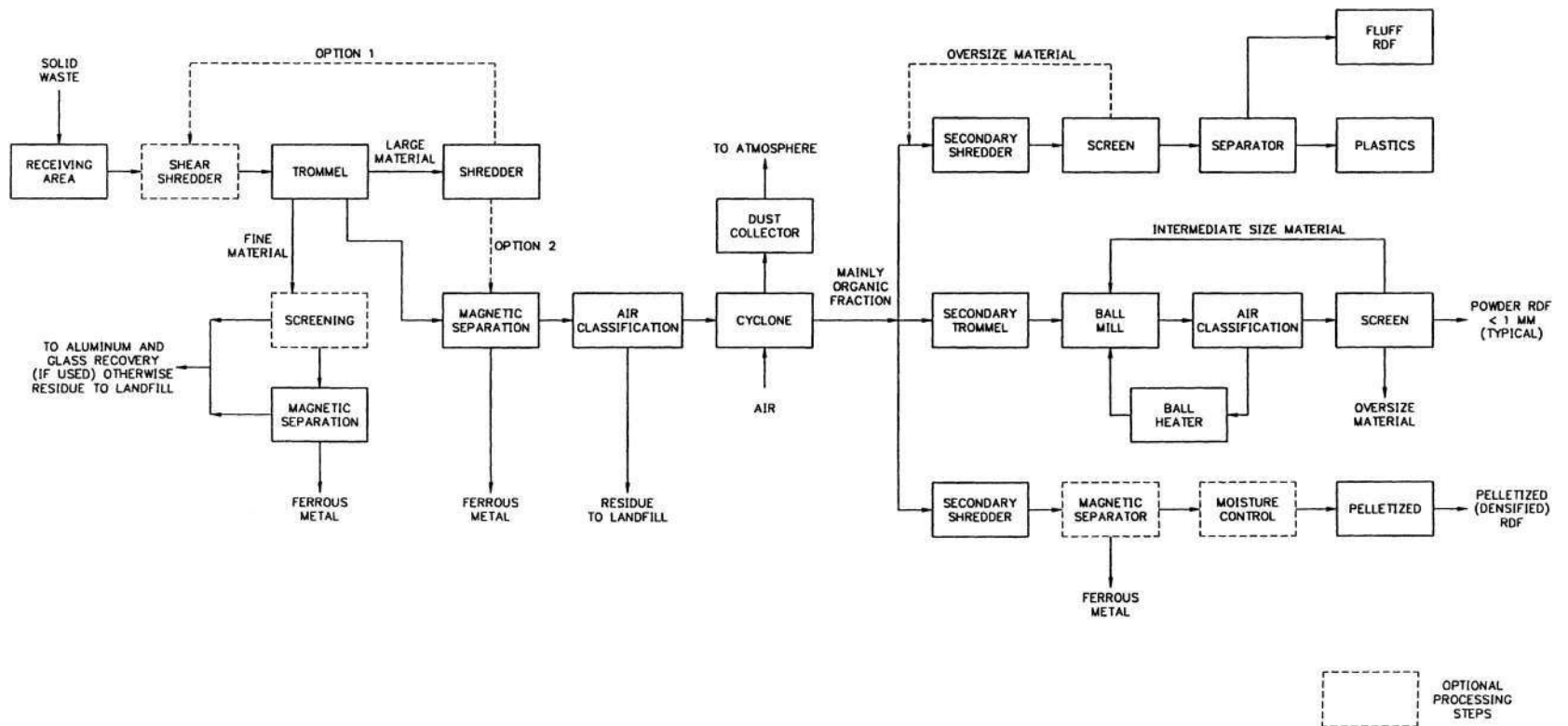


FIG. 22-51 Typical flow sheet for the recovery of materials and production of refuse-derived fuels (RDF). [Adapted in part from D. C. Wilson (ed.), *Waste Management: Planning, Evaluation, Technologies*, Oxford University Press, Oxford, 1981.]

TABLE 22-66 Important Design Considerations for Anaerobic Composting Processes*

Item	Comment
Particle size	For optimum results the size of solid waste should be between 25 and 75 mm (1 and 3 in).
Seeding and mixing	Composting time can be reduced by seeding with partially decomposed solid wastes to the extent of about 1 to 5 percent by weight. Sewage sludge can also be added to prepared solid wastes. When sludge is added, the final moisture content is the controlling variable.
Mixing or turning	To prevent drying, caking, and air channeling, material in the process of being composted should be mixed or turned on a regular schedule or as required. Frequency of mixing or turning will depend on the type of composting operation.
Air requirements	Air with at least 50 percent of the initial oxygen concentration remaining should reach all parts of the composting material for optimum results, especially in mechanical systems.
Total oxygen requirements	The theoretical quantity of oxygen required can be estimated.
Moisture content	Moisture content should be in the range between 50 and 60 percent during the composting process. The optimum value appears to be about 55 percent.
Temperature	For best results, temperature should be maintained between 322 and 327 K (130 and 140°F) for the first few days and between 327 and 333 K (130 and 140°F) for the remainder of the active composting period. If temperature goes beyond 339 K (150°F), biological activity is reduced significantly.
Carbon-nitrogen ratio	Initial carbon-nitrogen ratios (by mass) between 35 and 50 are optimum for aerobic composting. At lower ratios ammonia is given off. Biological activity is also impeded at lower ratios. At higher ratios nitrogen may be a limiting nutrient.
pH	To minimize the loss of nitrogen in the form of ammonia gas, pH should not rise above about 8.5.
Control of pathogens	If the process is properly conducted, it is possible to kill all the pathogens, weeds, and seeds during the composting process. To do this, the temperature must be maintained between 333 and 334 K (140 and 160°F) for 24 h.

*Adapted from G. Tchobanoglous, H. Theisen, and R. Eliassen, *Solid Wastes: Engineering Principles and Management Issues*, McGraw-Hill, New York, 1977.

incinerator gases will be cooled from a range of 1250 to 1375 K (1800 to 2000°F) to a range of 500 to 800 K (600 to 1000°F) before being discharged to the air-pollution-control system. Apart from the production of steam, the use of a boiler system is beneficial in reducing the volume of gas to be processed in the air-pollution-control equipment. The compounds in the waste stream will generate products of combustion and ash that may create serious corrosion and fouling problems in waste-heat boilers (WHBs). WHBs also tend to generate dioxins and furans when particulates, combined with free chlorine and other dioxin/furan precursors, attach to the tubes at proper reformation temperatures and time to cause the reformation of D/Fs.

2. *In water-wall incinerators.* The internal walls of the combustion chamber are lined with boiler tubes that are arranged vertically and welded together in continuous sections. When water walls are employed in place of refractory materials, they are not only useful for the recovery of steam but also extremely effective in controlling furnace temperature without introducing excess air; however, they are subject to corrosion by the hydrochloric acid produced from the burning of some plastic compounds and the molten ash containing salts (chlorides and sulfates) that attach to the tubes.

Combustion Combustion of industrial and municipal waste is an attractive waste management option because it reduces the volume of waste by 70 to 90 percent. In the face of shrinking landfill availability, municipal waste combustion capacity in the United States has grown at a rate significantly faster than the growth rate for municipal refuse generation.

Types of Combustors The three main classes of facilities used to combust municipal refuse are mass-burn, modular, and RDF-fired facilities. Mass-burn combustors are field-erected and generally range in size

from 50 to over 1000 tons/day of refuse feed per unit (Fig. 22-52). Modular combustors burn waste with little more preprocessing than do the mass-burn units. These range in size from 5 to 100 tons/day. Total U.S. MSW waste-to-energy plants had a design capacity of 98,719 tons/day per a 2004 survey. There were 72 plants in 2004. In addition to traditional WTE contained in the first two classes, the third major class of municipal waste combustor burns RDF. The types of waste-to-energy boilers used to combust RDF can include suspension, stoker, and fluidized-bed designs. RDF fuels can also be fired directly in large industrial and utility boilers that are now used for the production of power with pulverized or stoker coal, oil, and gas. Although the process is not well established with coal, it appears that about 15 to 20 percent of the heat input can be from RDF. With oil as the fuel, about 10 percent of the heat input can be from RDF. Depending on the degree of processing, suspension, spreader-stoker, and double-vortex firing systems have been used. RDF includes source-separated material, such as whole tires and chipped tires fired in cement kilns and boilers, and wood waste. Combustion of this material was estimated to be 2.4 million tons in 2001.

The system shown in Fig. 22-52 is a schematic of the Montenegro Montgomery LP located in Plymouth Township, Montgomery County, Pa. The facility specifications are as follows:

Waste throughput:	1200 tons/day (2 × 600)
Technology:	L&C Steinmuller, GmbH
Steam output:	324,000 lb/h (2 × 162,000)
Steam quality:	650 psig/750°F
Boiler mfr.:	L&C Steinmuller, GmbH
Extraction turbine:	36 MW gross
Turbine mfr.:	General Electric
Generator:	General Electric

TABLE 22-67 Important Design Considerations for Anaerobic Digestion*

Item	Comment
Size of material shredded	Wastes to be digested should be shredded to a size that will not interfere with the efficient functioning of pumping and mixing operations.
Mixing equipment	To achieve optimum results and to avoid scum buildup, mechanical mixing is recommended.
Percentage of solid wastes mixed with sludge	Although amounts of waste varying from 50 to 90+ percent have been used, 60 percent appears to be a reasonable compromise.
Hydraulic and mean cell residence time, $\Theta_h = \Theta_c$	Washout time is in the range of 3 to 4 days. Use 8 to 15 days for design of base design on results of pilot-plant studies.
Loading rate	0.6 to 1.6 kg/(m ³ ·day) [0.04 to 0.10 lb/(ft ³ ·day)]. It is not well defined at present. Significantly higher rates have been reported.
Temperature	Between 327 and 333 K (130 and 140°F)
Destruction of volatile solid wastes	Varies from about 60 to 80 percent; 70 percent can be used for estimating purposes.
Total solids destroyed	Varies from 40 to 60 percent, depending on amount of inert material present originally.
Gas production	0.5 to 0.75 m ³ /kg (8 to 12 ft ³ /lb) of volatile solids destroyed (CH ₄ = 60 percent; CO ₂ = 40 percent)

*From G. Tchobanoglous, H. Theisen, and R. Eliassen, *Solid Wastes: Engineering Principles and Management Issues*, McGraw-Hill, New York, 1977.

NOTE: Actual removal rates for volatile solids may be less, depending on the amount of material diverted to the scum layer.

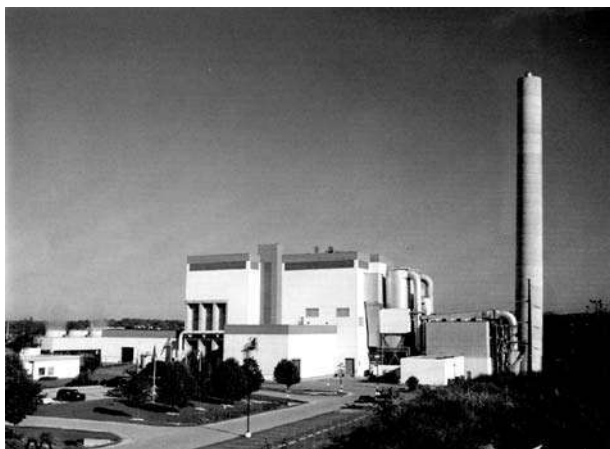


FIG. 22-52 1200-tons/day MSW plant of Montenay Montgomery.

Dry scrubber:	Research-Cottrell (slaked lime)
Baghouse:	Research-Cottrell (reverse air)
CEMs:	Environmental elements
DeNO _x :	Fuel tech
Mercury removal:	Sorbaline
Material recovery:	Ferrous metals

The construction of the facility began on May 23, 1989, and went into full operation on February 17, 1992, on an interim basis upon completion of the performance tests. The facility achieved full acceptance on May 23, 1994, and began a 20-year service agreement with the Eastern District of Montgomery County. The plant serves 24 municipalities in the Eastern District with a population of 425,000. The processing capacity is 1200 tons/day; 377,000 tons/yr with an electrical capacity of 36 MW. The electricity produced is sold to Philadelphia Electric Company (PECO).

See Fig. 22-53, which is typical of most waste-to-energy (WTE) plants burning MSW. Trucks deliver MSW to the scale house of the facility, where the truck is weighed and identified. At (2), the trucks unload waste into a storage pit in an enclosed tipping hall. The truck fumes and waste odors are drawn into the furnaces by large fans to provide combustion air and to prevent escape of odors. The waste is inspected in the storage pit (3), prior to feeding the waste into the fur-

nace by large overhead cranes. Waste is burned at high temperatures (over 2000°F) in the furnaces (4). The waste moves through the furnaces through drying, combustion, and final burnout stages over a 60-min period.

The heat of combustion generates steam in the boiler (5). The steam is used in making electricity in a steam turbine/generator set. After providing the in-plant electrical needs, excess energy is sold to the local electric utility and/or utilized by nearby process steam industries. After the combustion of the waste, ferrous metals are removed from the remaining residue by screening and magnetic separation (6). These ferrous metals are then recycled. The remaining residue can be beneficially reused or recycled, further reducing reliance on landfilling.

State-of-the-art pollution control equipment (7) removes particulates, acid gases, and other air emissions following the waste combustion process. Clean Air Act and other environmental standards are achieved through a combination of equipment such as spray dryers, baghouses, and carbon injection systems. Exhaust gases are drawn into an induced-draft fan which maintains a negative pressure throughout the entire system from waste feeding through the pollution controls. This prevents fugitive emissions from the process. These gases are then pushed into the exhaust stack and into the atmosphere. Stack gas sampling is conducted on platforms in the vertical stack section.

Gasification The gasification process involves the partial combustion of a carbonaceous or hydrocarbon fuel to generate a combustible fuel gas rich in carbon monoxide and hydrogen. A gasifier is similar to an incinerator operating under reducing conditions. Heat to sustain the process is derived from exothermic reactions, while the combustible components of the low-energy gas are primarily generated by endothermic reactions. The reaction kinetics of the gasification process are quite complex and still the subject of considerable debate.

When a gasifier is at atmospheric pressure with air as the partial oxidant, the end products of the gasification process are a low-energy gas typically containing (by volume) 10 percent CO₂, 20 percent CO, 15 percent H₂, and 2 percent CH₄, with the balance being N₂ and a carbon-rich ash. Because of the diluting effect of the nitrogen in the input air, the low-energy gas has an energy content in the range of the 5.2 to 6.0 MJ/m³ (140 to 160 Btu/ft³). Dual-bed gasifiers produce a gas with double this heating value. When pure oxygen is used as the oxidant, a medium-energy gas with an energy content in the range of 12.9 to 13.8 MJ/m³ (345 to 370 Btu/ft³) is produced. Gasifiers were in widespread use on coal and wood until natural gas displaced them in the 1930s through the 1950s. Some large coal gasifiers are in use today in the United States and worldwide. While the process can work on solid waste, incinerators (which gasify and burn in one chamber) are favored over gasifiers.

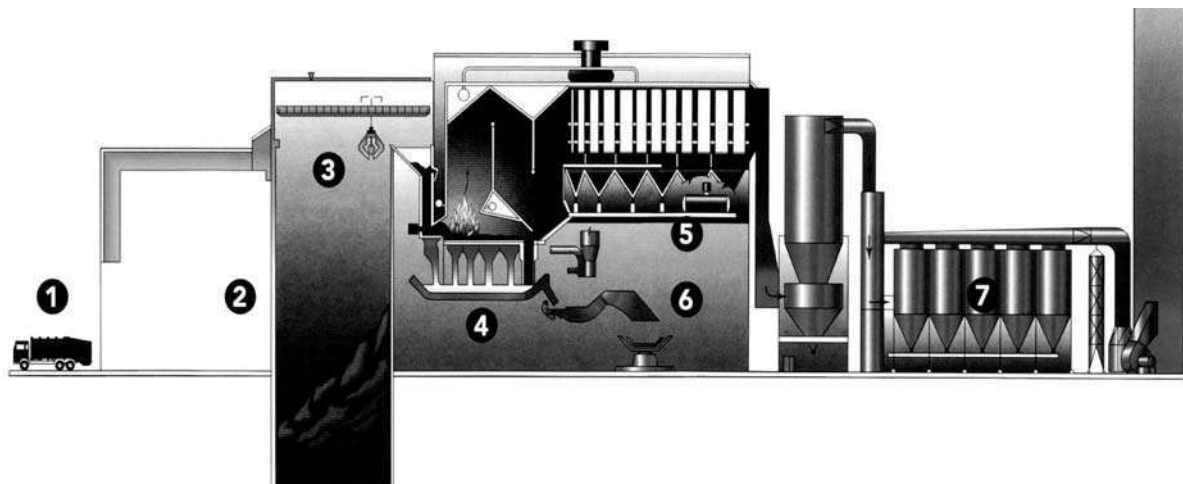


FIG. 22-53 1200-tons/day MSW plant schematic.

Pyrolysis Of the main alternative chemical conversion processes that have been investigated, pyrolysis has received the most attention. Pyrolysis has been tested in countless pilot plants, and many full-scale demonstration systems have operated. Few attained any long-term commercial use. Major issues were lack of market for the unstable and acidic pyrolytic oils and the char and high operating costs.

Depending on the type of reactor used, the physical form of solid wastes to be pyrolyzed can vary from unshredded raw wastes to the finely ground portion of the wastes remaining after two stages of shredding and air classification. Upon heating in an oxygen-free atmosphere, most organic substances can be split via thermal cracking and condensation reactions into gaseous, liquid, and solid fractions. *Pyrolysis* is the term used to describe the process while the term *destructive distillation* is also often used. In contrast to the combustion process, which is highly exothermic, the pyrolytic process involves both highly endothermic and exothermic reactions. For this reason, the term *destructive distillation* is often used as an alternative for *pyrolysis*.

The characteristics of the three major component fractions resulting from the pyrolysis are (1) a gas stream containing primarily hydrogen, methane, carbon monoxide, reduced sulfur compounds, carbon disulfide, and various other gases, depending on the organic characteristics of the material being pyrolyzed; (2) a fraction that consists of a tar and/or oil stream that is liquid at room temperature and has been found to contain hundreds of chemicals such as acetic acid, acetone, methanol, and phenols; and (3) a char consisting of almost pure carbon plus any inert material that may have entered the process. It has been found that distribution of the product fractions varies with the temperature at which the pyrolysis is carried out. Under conditions of maximum gasification, the energy content of pyrolytic oils has been estimated to be about 23.2 MJ/kg (10,000 Btu/lb).

Waste-to-Energy Systems The preceding subsection ended at the production of steam. WTE systems take over at this point, using high-pressure/high-temperature steam to drive turbines and produce shaft horsepower for prime movers at industrial plants or to generate electricity.

This fuel may be solid or gas or oil or fuels from a gasifier or pyrolysis system.

Typical flow sheets for alternative energy-recovery systems are shown in Fig. 22-54. Perhaps the most common flow sheet for the production of electric energy involves the use of a steam turbine-generator combination. As shown, when solid wastes are used as the basic fuel source, four operating modes are possible. A flow sheet using a gas turbine-generator combination is shown in Fig. 22-54. The low-energy gas is compressed under high pressure so that it can be used more effectively in the gas turbine. Use of low- or medium-Btu gas for gas turbines has been attempted, and success requires good design and operation of gas cleaning equipment prior to introduction into the combustor of the gas turbine.

Efficiency Factors Representative efficiency data for boilers, pyrolytic reactors, gas turbines, steam turbine-generator combinations, electric generators, and related plant use and loss factors are given in Table 22-68. In any installation in which energy is being produced, allowance must be made for the power needs of that station or process and for unaccounted-for process-heat losses. Typically, the auxiliary power allowance varies from 4 to 8 percent of the power produced. Process-heat losses usually will vary from 2 to 8 percent. In general, steam pressures of 600 psig and temperatures of 650°F are considered minimum for economical power generation. Industrial plants may choose a cogeneration topping cycle, with steam exhaust from the turbine at the plant's process steam pressure, typically in the 125 to 250 psig range. For commercial WTE plants, condensing turbines are the norm.

Determination of Energy Output and Efficiency for Energy-Recovery Systems An analysis of the amount of energy produced from a solid-waste energy conversion system using an incinerator boiler system turbine electric generator combination with a capacity of 1000 t/day of waste is presented in Table 22-69. If it is assumed that 10 percent of the power generated is used for the front-end processing system (typical values vary from 8 to 14 percent), then the net power for export is 24,604 kW and the overall efficiency is 17.4 percent.

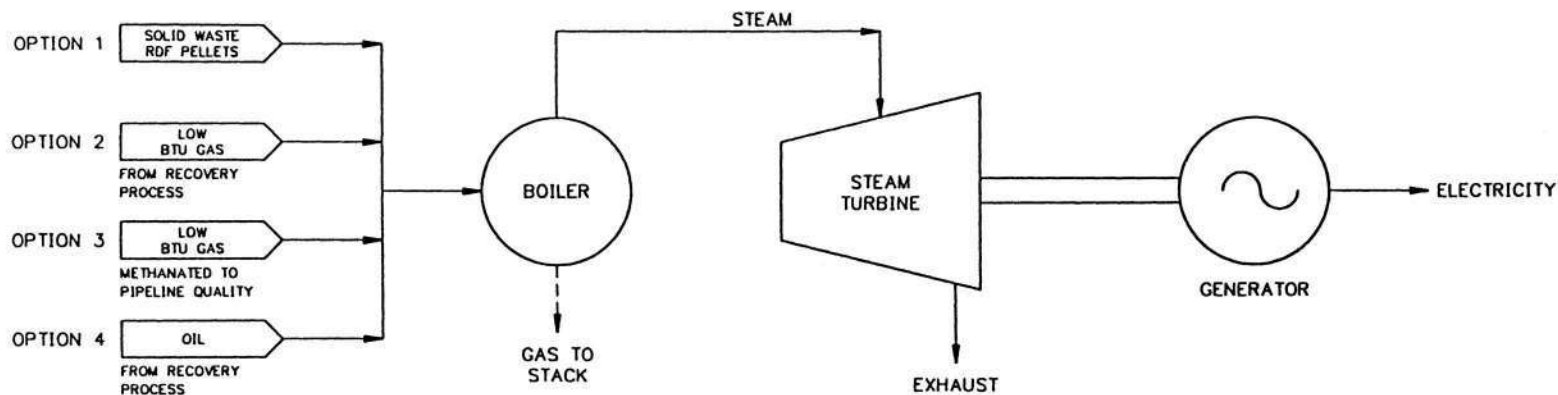
Concentration of WTE Incinerators The total number of municipal waste incinerator facilities as listed in the *Solid Waste Digest*, vol. 14, no. 6, June 2004 (a publication of Chartwell Information, EBI, Inc., of San Diego, Calif.), is 72. This is an increase of 10 facilities since September 1994 (a 10-year period). See Table 22-70. The wastes burned in these facilities total 8.96 percent of total municipal wastes managed in landfills, incinerators, and transfer stations. This amounts to 98,719 tons/day of combusted municipal waste. This also is an increase of 10,248 tons over the 88,471 tons in September 1994, an average of 1000 tons/yr.

One notes that the heavily populated areas of the country also have the highest number of WTE facilities as well as the highest intake of municipal waste into incinerators. This is also due to the lack of low-cost land and open space for landfills compared to the midwest and western states. The amount of waste combusted in the northeastern states is 25.6 percent of the total generated compared to 8.96 percent of all municipal wastes combusted nationwide. The WTE cost per ton averages \$62 in the northeastern states compared to \$59 for the nation. Incinerator costs are similar to landfill costs in the northeastern states. However, landfill costs are far lower than WTE tipping fees across the nation. For example, in states such as Idaho (landfill costs, \$18.80/ton) and Texas (landfill costs \$21.03/ton), WTE incinerator plants cannot compete at this time. However, it is interesting to note that the landfill costs have doubled in 10 years while WTE tipping fees have dropped about 10 percent. The stricter regulations applied to landfills have caused this increase in landfill tipping fees. The NIMBY (Not In My Back Yard) syndrome concerning incineration systems has also hurt the siting and permitting of many of these facilities. Figure 22-55 shows the range of costs for all solid-waste management in June 2004. The national index was \$39.49/ton in 2004, up from \$37.93/ton in 1994. This is an average increase of only 0.1 percent per year, which is well below the inflation rate for this 10-year period.

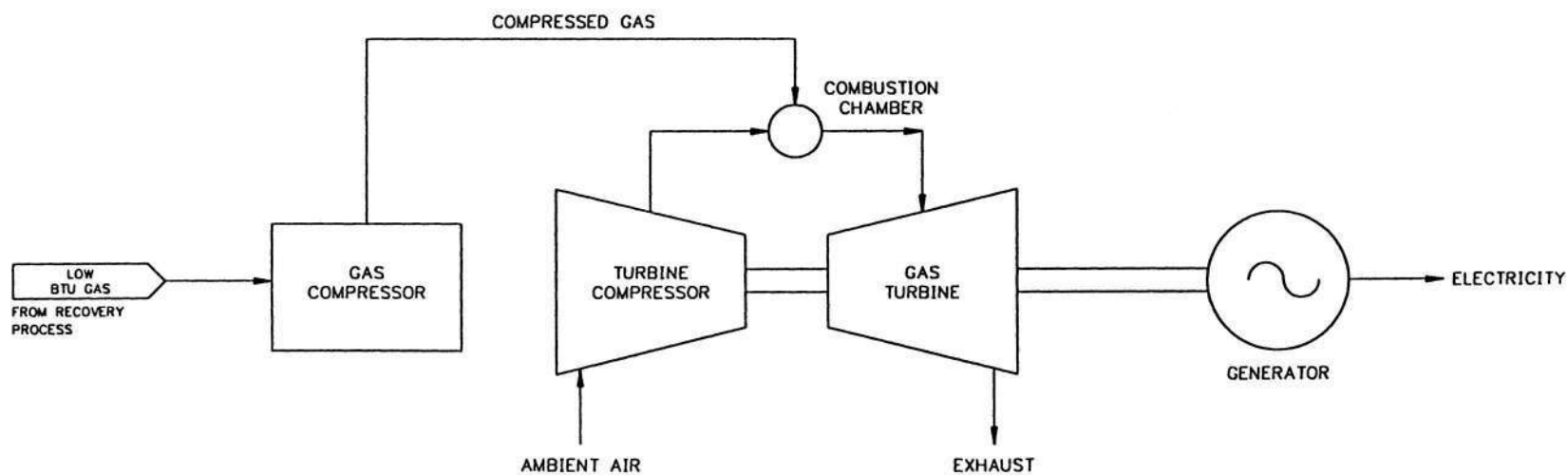
REGULATIONS APPLICABLE TO MUNICIPAL WASTE COMBUSTORS

New Source Performance Standards (NSPS) were promulgated under Sections 111(b) and 129 of the CAA Amendments of 1990. The NSPS applies to new municipal solid-waste combustors (MWCs) with capacity to combust more than 250 tons/day of municipal solid waste that commenced construction after September 20, 1994. The proposed standards and guidelines were published in the Federal Register on September 20, 1994. They are listed in Title 40, Protection of the Environment, Chapter 1, Part 60 Standards of Performance for New Stationary Sources (<http://a257.g.akamaitech.net/7/25/23422/14mar20010800/edocket.access.gpo.gov/cfr2001>). Section 129 of the CAA of 1990 applies to a range of solid-waste incinerators including MWCs, medical waste incinerators (MWIs), and industrial waste incinerators. On June 28, 2004, the Federal Register (vol. 69, no. 123) established that Section 129(a)(5) of the CAA requires EPA to review, and if necessary revise, those standards every 5 years. This rulemaking addresses those requirements and is the first 5-year review of the MACT standards. Implementation of these MACT standards has been highly effective and has reduced dioxin/furan emissions by more than 99 percent since 1990. Similar reductions have occurred for other CAA Section 129 pollutants. Incinerators for hazardous solid and liquid wastes are covered under RCRA and MACT regulations (40 CFR Parts 260 through 272) and TSCA, Toxic Substances Control Act, 1976 (40 CFR Parts 700 through 766).

Regulated Pollutants The NSPS regulates MWC emissions and nitrogen oxides (NO_x) emissions from individual MWC units larger than 250 tons/day capacity. MWC emissions are subcategorized as MWC metal emissions, MWC organic emissions, and MWC acid gas emissions. The NSPS establishes emission limits for organic emissions (measured as dioxins and furans), MWC metal emissions (measured as particulate matter PM, opacity, cadmium, lead, and mercury), and MWC acid gas emissions as sulfur dioxide SO₂ and hydrogen chloride HCl, as well as NO_x emission limits.



OPTIONS WITH STEAM-TURBINE-GENERATOR COMBINATION



OPTIONS WITH GAS-COMPRESSOR-GAS-TURBINE-GENERATOR COMBINATION

FIG. 22-54 Flow sheet—alternative energy recovery systems.

TABLE 22-68 Typical Thermal Efficiency and Plant Use and Loss Factors for Individual Components and Processes Used for Recovery of Energy from Solid Wastes

Component	Efficiency*		Comments
	Range	Typical	
Incinerator-boiler	40–68	63	Mass-fired
Boiler			
Solid fuel	60–75	72	Processed solid wastes (RDF)
Low-btu gas	60–80	75	Necessity to modify burners
Oil-fired	65–85	80	Oils produced from solid wastes possibly required to be blended to reduce corrosiveness
Gasifier	60–70	70	
Pyrolysis reactor	65–75	70	
Turbines			
Combustion gas			
Simple cycle	8–12	10	Including necessary appurtenances
Regenerative	20–26	24	
Expansion gas	30–50	40	
Steam turbine-generator system			
Less than 10 MW	24–40	29 ^{†‡}	Including condenser, heaters, and all other necessary appurtenances but not boiler
10 MW	28–32	31.6 ^{†‡}	
Electric generator			
Less than 10 MW	88–92	90	
Over 10 MW	94–98	96	

* Theoretical value for mechanical equivalent of heat = 3600 kJ/kWh.

† Efficiency varies with exhaust pressure. Typical value given is based on an exhaust pressure in the range of 50 to 100 mmHg.

‡ Heat rate = 11,395 kJ/kWh = (3600 kJ/kWh)/0.316.

MWCs Organic Emissions The NSPS limits organic emissions to a total dioxin plus furan emission mass limit of 13 ng/scfm (at 7 percent O₂ dry volume). This level is approximately equivalent to a toxic equivalent (TEQ) of 2.0 ng/dscm, using the 1990 international toxic equivalency factor (I-TEF) approach.

MWCs Metal Emissions The NSPS includes a PM emission limit of 0.015 grain per dry standard cubic feet (gr/dscf) at 7 percent oxygen dry/volume and an opacity limit of 10 percent (6-min average). Cadmium is limited to 0.020 ng/dnm³ and lead to 0.3 ng/dnm³, all at 7 percent O₂ dry volume.

MWCs Acid Gas Emissions The NSPS requires a 95 percent reduction of HCl emissions and an 80 percent reduction of SO₂ emissions for new MWCs or an emission limit of 25 ppmv for HCl and 30 ppmv for SO₂ (at 7 percent O₂ dry volume).

Nitrogen Oxides Emissions The NSPS limits NO_x emission to 150 ppmv (at 7 percent O₂ dry volume).

MWC Air-Pollution-Control Systems MWCs generate flue gas that contains particulates, acid gases, and trace amounts of organic and volatile metals. Particulates have traditionally been removed by use of cyclone separators, baghouses, and electrostatic precipitators. Acid gases require neutralization and removal from the gas stream. This can be accomplished by adding reagents or chemicals to the gas stream and removing products of the chemical reaction when these materials are mixed together. Two major types of APCs are employed for PM, metals and acid gas removal:

- Dry systems, where the gas stream is humidified and chemicals (lime, sodium hydroxide, or carbonate and activated carbon) are added to the system
- Wet systems, where large quantities of water-containing chemicals

(lime, sodium hydroxide or carbonates, activated carbon) wash the gas stream

MWC facilities are required to meet some of the toughest environmental air emissions standards in the country. Complying with these standards makes modern waste combustors among the cleanest producers of electricity—and may even provide a means of improving a community's overall air quality.

ULTIMATE DISPOSAL

Disposal on or in the earth's mantle is, at present, the only viable method for the long-term handling of (1) solid wastes that are collected and are of no further use, (2) the residual matter remaining after solid wastes have been processed, and (3) the residual matter remaining after the recovery of conversion products and/or energy has been accomplished. The three land disposal methods used most commonly are (1) landfilling, (2) landfarming, and (3) deep-well injection. Although incineration is being used more often as a disposal method, it is, in reality, a processing method. Recently, the concept of using muds in the ocean floor as a waste storage location also has received some attention.

Landfilling of Solid Waste Landfilling involves the controlled disposal of solid wastes on or in the upper layer of the earth's mantle. Important aspects in the implementation of sanitary landfills include (1) site selection, (2) landfilling methods and operations, (3) occurrence of gases and leachate in landfills, (4) movement and control of landfill gases and leachate, and (5) landfill design. Landfilling is a large-scale operation. The number of landfills decreased substantially over the period from 1988 to 2001. There were nearly 8000 in 1988,

TABLE 22-69 Energy Output and Efficiency for 1000 t/day of Waste Steam Boiler Turbine-Generator Energy-Recovery Plant Using Unprocessed Industrial Solid Wastes with Energy Content of 12,000 kJ/kg

Item	Value
Energy available in solid wastes, million kJ/h [(1000 t/day × 1000 kg/t × 12,000 kJ/kg) / (24 h/day × 10 ⁶ kJ/million kJ)]	500
Steam energy available, million kJ/h (500 million kJ/h × 0.7)	350
Electric power generation, kW (350 million kJ/h)/(11,395 kJ/kWh)	30,715
Station-service allowance, kW [30,715 (0.06)]	-1,843
Unaccounted heat losses, kW [30,715 (0.05)]	1,536
Net electric power for export, kW	27,336
Overall efficiency, percent [(27,336 kW)/[(500,000,000 kJ/h)/(3600 kJ/kWh)]](100)	19.7

TABLE 22-70 Summary of Waste Disposal Pricing and Volumes by Type of Facility and State and Region, June 2004

Region/state	Average tipping fees by facility type								Daily volume of waste by facility type							
	Landfills		Transfer		WTE		All types		Landfills		Transfer		WTE		All types	
	\$/ton	12-mo chg.	\$/ton	12-mo chg.	\$/ton	12-mo chg.	Tons/day	12-mo chg.	Tons/day	12-mo chg.	Tons/day	12-mo chg.	Tons/day	12-mo chg.	Tons/day	12-mo chg.
Northeast																
Connecticut	60.62	-11.6%	65.22	-4.0%	64.79	2.1%	64.60	-1.4%	1,060	-38.4%	4,310	-7.3%	7,830	0.5%	8,890	-6.5%
Delaware	58.48	0.2%	54.93	-6.0%	—	N/A	57.87	-0.8%	4,240	-7.6%	880	-10.2%	—	N/A	4,240	-7.6%
Maine	55.43	9.2%	38.53	11.5%	71.38	<1%	57.33	-10.2%	630	1.6%	1,880	382.1%	2,600	<1%	3,230	0.3%
Maryland	52.90	5.5%	36.16	-3.1%	65.21	0.1%	49.64	1.5%	7,340	-14.8%	4,620	16.1%	2,490	<1%	9,900	-11.6%
Massachusetts	67.66	8.7%	84.80	31.5%	67.30	-1.4%	73.47	13.8%	6,810	0.9%	11,220	-8.9%	10,150	-0.5%	17,750	2.8%
New Hampshire	72.38	-5.1%	90.28	8.6%	79.71	-6.0%	77.10	-2.3%	4,640	-1.7%	1,510	-20.9%	760	-1.3%	5,390	-1.8%
New Jersey	66.61	28.7%	77.61	5.4%	60.78	-7.0%	69.51	9.8%	13,250	-1.3%	12,500	-7.6%	6,360	-18.3%	20,300	-4.3%
New York	48.46	0.5%	61.46	7.9%	58.11	-0.1%	54.92	2.9%	26,120	-3.9%	20,740	-16.6%	11,220	-2.2%	37,420	-3.4%
Pennsylvania	56.47	2.8%	64.46	0.2%	56.76	<1%	57.86	2.1%	80,180	-1.5%	18,560	14.9%	10,380	<1%	90,600	-3.4%
Rhode Island	64.37	11.4%	68.18	-7.3%	—	N/A	64.98	7.2%	3,660	1.4%	690	-13.8%	—	N/A	3,660	1.4%
Vermont	70.46	9.7%	56.09	4.1%	42.83	<1%	60.49	6.8%	740	-23.7%	1,640	-37.6%	10	<1%	750	-23.5%
Northeast total	57.15	4.8%	66.67	6.6%	62.30	-1.1%	60.39	3.8%	148,663	-3.2%	78,556	-4.4%	51,792	-3.2%	202,120	-3.6%
Southern																
Alabama	26.72	4.1%	32.23	8.6%	39.90	<1%	27.54	4.4%	20,150	-16.4%	1,870	-21.8%	630	-8.7%	20,780	-16.1%
Arkansas	27.75	8.5%	23.98	-2.2%	9.97	-45.5%	27.00	6.6%	7,510	-15.1%	1,130	-49.3%	130	85.7%	7,640	-14.3%
District of Columbia	—	N/A	59.77	5.3%	—	N/A	59.77	5.3%	—	N/A	3,150	16.7%	—	N/A	—	N/A
Florida	37.00	-1.3%	40.95	5.3%	57.43	-1.0%	42.03	1.0%	49,950	33.3%	17,170	3.0%	17,480	-1.2%	69,870	21.6%
Georgia	33.13	6.7%	33.94	1.1%	60.00	<1%	33.26	4.6%	39,150	69.0%	6,290	-25.0%	390	29.0%	39,830	71.1%
Kentucky	33.01	6.6%	33.89	105.8%	—	N/A	33.16	17.8%	17,930	-4.1%	3,730	-16.9%	—	N/A	17,930	-4.1%
Louisiana	26.90	-1.7%	32.35	-0.5%	—	N/A	27.80	-1.4%	15,340	-4.2%	3,000	-2.9%	—	N/A	15,340	-4.2%
Mississippi	26.88	6.3%	32.34	-1.4%	—	N/A	27.51	5.0%	10,300	-16.1%	1,330	-22.7%	—	N/A	10,300	-16.1%
North Carolina	33.27	7.7%	38.57	1.1%	46.00	35.3%	34.63	5.6%	20,300	-8.2%	5,820	-27.9%	400	<1%	20,750	-8.6%
South Carolina	33.41	-4.2%	33.94	-0.5%	59.50	<1%	34.19	-3.3%	19,340	-6.6%	2,670	-21.0%	620	<1%	19,960	-6.4%
Tennessee	29.16	-0.5%	36.05	26.1%	30.56	85.8%	29.75	3.1%	22,280	-9.4%	1,750	-59.4%	420	-8.7%	22,830	-9.4%
Virginia	40.05	3.0%	47.24	55.7%	45.36	0.5%	42.04	12.6%	41,510	-7.8%	10,680	-14.3%	6,640	-2.8%	49,580	-6.9%
West Virginia	35.38	-1.0%	42.03	-17.6%	—	N/A	36.00	-3.7%	5,630	-15.0%	580	-26.6%	—	N/A	5,630	-15.0%
Southern total	33.43	3.2%	39.98	17.5%	53.28	0.3%	35.65	6.0%	269,370	3.8%	59,163	-16.3%	26,710	-0.6%	300,438	3.4%
Midwest																
Illinois	35.88	5.7%	48.66	65.7%	59.00	<1%	40.19	23.0%	63,880	7.3%	29,780	2.6%	1,200	<1%	65,450	7.1%
Indiana	31.19	9.0%	39.17	43.1%	28.11	8.3%	33.28	18.1%	22,660	-7.9%	9,750	14.2%	2,070	7.8%	24,920	-6.6%
Iowa	32.12	-3.0%	34.35	-9.1%	—	N/A	32.41	-4.0%	6,990	-21.2%	980	-31.9%	—	N/A	7,000	-21.1%
Kansas	31.87	5.4%	35.62	9.9%	—	N/A	32.63	6.0%	9,560	5.3%	2,420	-19.9%	—	N/A	9,560	5.3%
Michigan	35.32	7.3%	41.05	39.4%	54.43	-0.6%	37.04	9.3%	57,510	-1.1%	6,640	-10.3%	4,160	-2.8%	61,670	-1.2%
Minnesota	58.80	11.8%	52.77	31.7%	62.25	23.7%	57.23	21.8%	6,040	-9.9%	6,670	-12.0%	4,210	-18.9%	10,320	-13.7%
Missouri	36.17	4.3%	33.16	39.3%	—	N/A	35.48	11.2%	15,750	-2.2%	4,740	-14.6%	—	N/A	15,750	-2.2%
Nebraska	25.61	-0.2%	35.27	1.1%	—	N/A	26.73	-0.3%	7,470	-8.6%	980	-16.2%	—	N/A	7,470	-8.6%
North Dakota	27.89	11.6%	31.63	-10.5%	—	N/A	28.88	7.0%	1,690	-20.3%	500	-18.0%	—	N/A	1,710	-20.5%
Ohio	33.60	10.7%	38.07	41.1%	—	N/A	34.97	19.1%	45,100	-1.3%	19,660	6.7%	—	N/A	45,180	-1.2%
South Dakota	27.54	7.4%	60.67	18.3%	—	N/A	30.33	6.4%	1,380	-23.3%	130	-43.5%	—	N/A	1,380	-23.3%
Wisconsin	38.00	4.0%	37.77	1.8%	50.36	<1%	38.17	3.5%	18,690	-5.2%	5,140	-14.6%	420	<1%	19,110	-5.1%
Midwest total	35.00	6.9%	42.72	41.6%	52.95	8.1%	37.22	14.2%	256,724	-1.5%	87,394	-1.8%	12,058	-7.3%	269,524	-1.7%
Western																
Arizona	26.93	3.7%	31.90	-0.7%	—	N/A	28.42	2.0%	19,520	-9.1%	8,290	-13.2%	—	N/A	19,530	-9.1%
Colorado	22.27	8.5%	33.68	12.9%	25.00	<1%	23.83	8.5%	20,590	-0.7%	3,230	-14.3%	10	<1%	20,610	-0.6%
Idaho	18.80	-7.3%	30.86	-32.9%	—	N/A	21.45	-8.4%	4,700	-41.3%	1,320	18.9%	—	N/A	4,700	-41.3%
Montana	25.40	20.5%	45.28	9.4%	60.00	<1%	28.91	17.6%	2,470	<1%	410	-14.6%	50	<1%	2,570	<1%
Nevada	18.74	28.4%	31.51	37.5%	—	N/A	21.19	29.2%	8,230	-1.2%	1,960	-15.5%	—	N/A	8,230	-1.2%
New Mexico	20.65	0.4%	15.37	8.2%	—	N/A	19.77	2.4%	9,100	-10.5%	1,840	-26.7%	—	N/A	9,100	-10.5%
Oklahoma	22.22	3.2%	21.85	2.7%	55.00	<1%	24.71	4.0%	12,870	-12.3%	710	-30.4%	1,130	<1%	14,000	-11.4%
Texas	21.03	4.2%	25.64	46.2%	55.49	33.0%	21.52	7.6%	74,920	-2.1%	7,570	-5.5%	60	-25.0%	75,210	-2.2%
Utah	24.59	-2.2%	24.77	0.1%	25.00	<1%	24.68	-1.7%	13,210	20.2%	1,010	-17.2%	10	<1%	13,280	20.7%
Wyoming	25.03	13.9%	46.56	-3.9%	—	N/A	28.94	6.8%	1,180	-36.2%	260	-40.9%	—	N/A	1,180	-36.2%
Western total	22.14	4.7%	28.92	11.0%	54.79	1.7%	23.17	5.7%	166,788	-4.8%	26,590	-12.6%	1,251	-1.6%	168,406	-4.8%
Pacific																
Alaska	37.31	-24.4%	14.77	12.3%	140.81	-0.1%	40.63	-27.8%	2,100	18.6%	200	400.0%	120	-25.0%	2,220	15.0%
California	36.58	5.1%	43.86	9.0%	37.58	-0.9%	39.12	6.0%	122,110	-1.7%	45,940	-14.0%	3,260	-5.0%	126,800	-2.1%
Hawaii	62.00	2.3%	95.32	17.6%	81.27	<1%	71.22	2.3%	2,280	-6.9%	270	-41.3%	1,440	<1%	3,730	-4.1%
Oregon	30.79	-2.7%	32.67	36.9%	65.52	0.9%	32.35	8.1%	12,650	3.5%	6,040	-13.1%	530	-14.5%	13,190	2.6%
Washington	44.68	7.2%	80.38	3.2%	88.23	21.1%	60.73	3.5%	12,650	-2.8%	8,950	-22.3%	1,540	-8.9%	14,630	-3.3%
Pacific total	37.16	4.1%	48.21	7.4%	62.02	5.0%	41.30	4.9%	151,790	-1.3%	61,404	-15.2%	6,907	-5.9%	160,561	-1.7%
National total	36.06	4.8%	48.12	16.2%	58.60	0.7%	39.49	6.8%	993,335	-0.9%	313,107	-9.2%	98,719	-3.2%	1,101,049	-1.2%

SOURCE: Chartwell Information, a division of Environmental Business International, Inc.

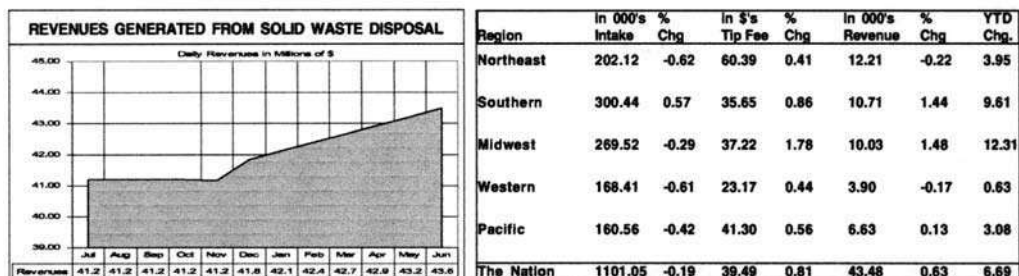
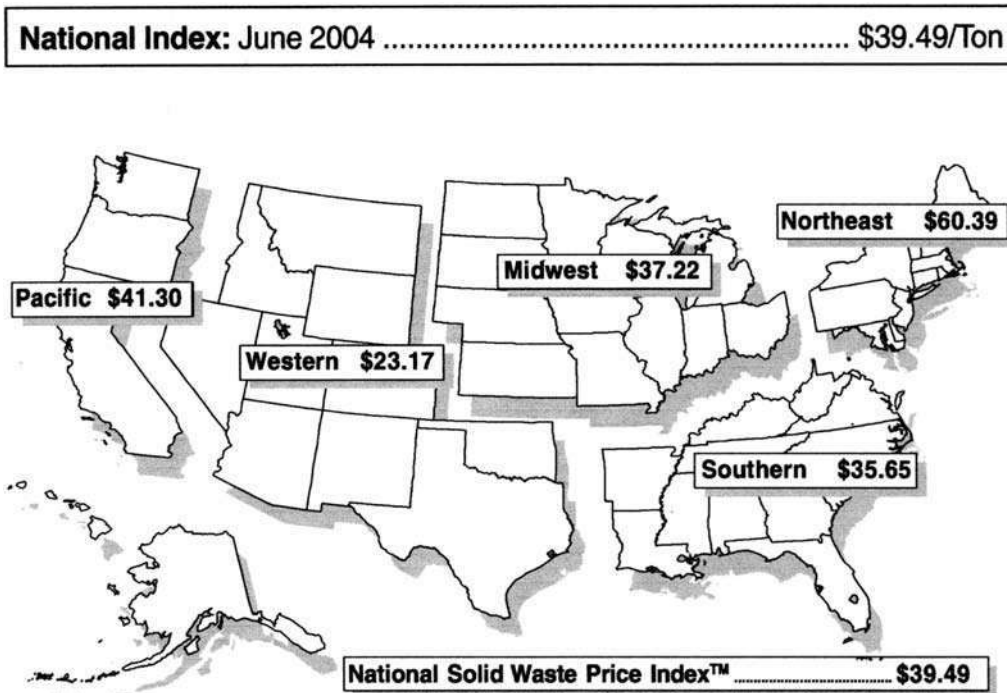


FIG. 22-55 Solid-waste management costs, June 2004.

down to 1858 landfills in service nationwide in 2001 accepting approximately 160 million tons of waste. The number of landfills is decreasing, and the capacity of remaining landfills is rising, as new landfills are larger. New landfills are more sophisticated than older ones and are therefore more costly to design and operate. Over the long term, the tonnage of MSW landfilled in 1990 was 140.1 million tons, but decreased to 122.4 million tons in 1995. The tonnage increased to 131.8 million tons in 1999, then declined to 127.6 in 2001. The tonnage landfilled results from an interaction among generation, recycling, and combustion, which do not necessarily rise and fall at the same time. Figure 22-56 shows the decrease of landfills that has occurred over the 13-year period from 1988 to 2001.

The landfilling of hazardous waste is considered separately.

Site Selection Factors that must be considered in evaluating potential solid-waste-disposal sites are summarized in Table 22-71. Final selection of a disposal site usually is based on the results of a preliminary site survey, results of engineering design and cost studies, and an environmental impact assessment.

Landfilling Methods and Operations To use the available area at a landfill site effectively, a plan of operation for the placement of solid wastes must be prepared. Various operational methods have been developed primarily on the basis of field experience. The principal methods used for landfilling dry areas may be classified as (1) area and (2) depression.

1. **Area method.** The area method is used when the terrain is unsuitable for the excavation of trenches in which to place the solid wastes. The filling operation usually is started by building an earthen levee against which wastes are placed in thin layers and then compacted (see Fig. 22-57). Each layer is compacted as the filling progresses until the thickness of the compacted wastes reaches a height varying from 2 to 3 m (6 to 10 ft). At that time, and at the end of each day's operation, a 150- to 300-mm (6- to 12-in) layer of cover material is placed over the completed fill. The cover material must be hauled in by truck or earthmoving equipment from adjacent land or from borrow-pit areas. In some newer landfill operations, the daily cover material is omitted, and reusable geotextile covers, similar to very

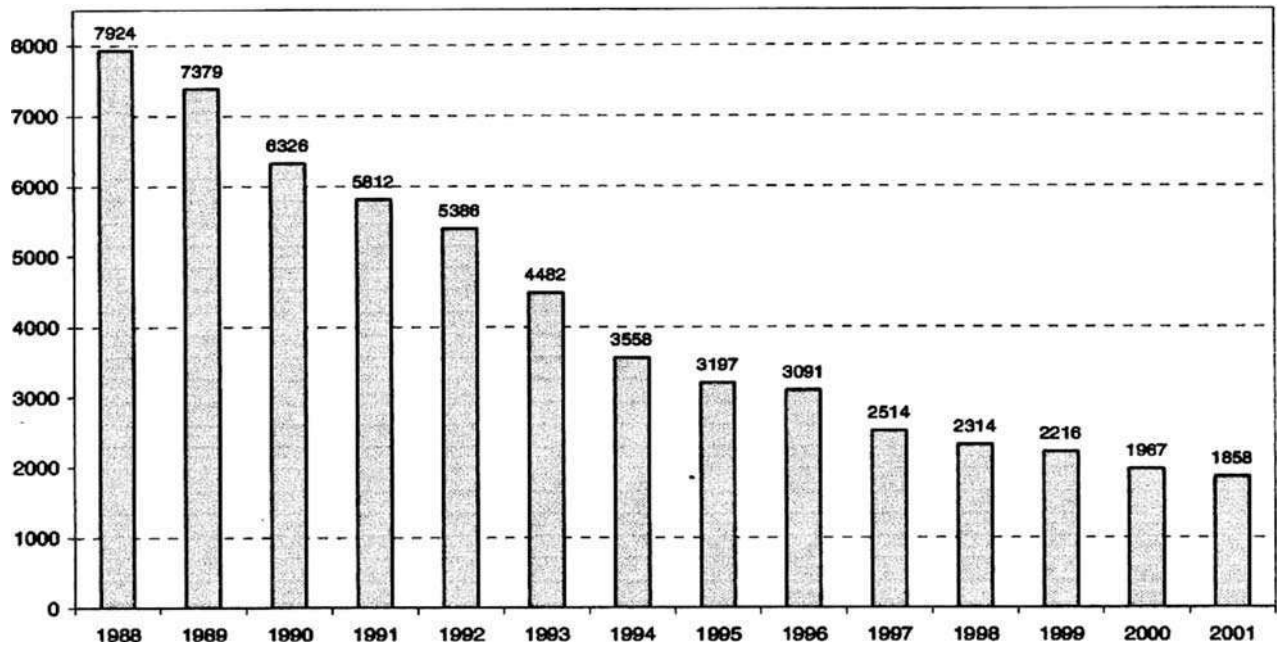


FIG. 22-56 Number of landfills, 1988–2001.

large tarps, are used. A completed lift including the cover material is called a *cell* (see Fig. 22-58). Successive lifts are placed on top of one another until the final grade called for in the ultimate development plan is reached. A final layer of cover material is used when the fill reaches the final design height.

2. *Depression method.* At locations where natural or artificial depressions exist, it is often possible to use them effectively for land-filling operations. Canyons, ravines, dry borrow pits, and quarries have been used for this purpose. The techniques to place and compact solid wastes in depression landfills vary with the geometry of the site,

the characteristics of the cover material, the hydrology and geology of the site, and access of the site. In a canyon site, filling starts at the head end of the canyon (see Fig. 22-59) and ends at the mouth. The practice prevents the accumulation of water behind the landfill. Wastes usually are deposited on the canyon floor and from there are pushed up against the canyon face at a slope of about 2 : 1. In this way, a high degree of compaction can be achieved.

3. *Landfills in wet areas.* Because of the problems associated with contamination of local groundwater, the development of odors, and structural stability, landfills must be avoided in wetlands. If wet

TABLE 22-71 Important Factors in Preliminary Selection of Landfill Sites

Factor	Remarks
Available land area	In selecting potential land disposal sites, it is important to ensure that sufficient land area is available. Sufficient area to operate for at least 1 year at a given site is needed to minimize costs.
Impact of processing and resource recovery	It is important to project the extent of resource-recovery processing activities that are likely to occur in the future and determine their impact on the quantity and condition of the residual materials to be disposed of.
Haul distance	Although minimum haul distances are desirable, other factors must also be considered. These include collection-route location, types of wastes to be hauled, local traffic patterns, and characteristics of the routes to and from the disposal site (condition of the routes, traffic patterns, and access conditions).
Soil conditions and topography	Because it is necessary to provide material for each day's landfill and a final layer of cover after the filling has been completed, data on the amounts and characteristics of the soils in the area must be obtained. Local topography will affect the type of landfill operation to be used, equipment requirements, and the extent of work necessary to make the site usable.
Climatological conditions	Local weather conditions must also be considered in the evaluation of potential sites. Under winter conditions where freezing is severe, landfill cover material must be available in stockpiles when excavation is impractical. Wind and wind patterns must also be considered carefully. To avoid blowing or flying papers, windbreaks must be established.
Surface-water hydrology	The local surface-water hydrology of the area is important in establishing the existing natural drainage and runoff characteristics that must be considered.
Geologic and hydrogeologic conditions	Geologic and hydrogeologic conditions are perhaps the most important factors in establishing the environmental suitability of the area for a landfill site. Data on these factors are required to assess the pollution potential of the proposed site and to establish what must be done to the site to control the movement of leachate or gases from the landfill.
Local environmental conditions	The proximity of both residential and industrial developments is extremely important. Great care must be taken in their operation if they are to be environmentally sound with respect to noise, odor, dust, flying paper, and vector control.
Ultimate uses	Because the ultimate use affects the design and operation of the landfill, this issue must be resolved before the layout and design of the landfill are started.

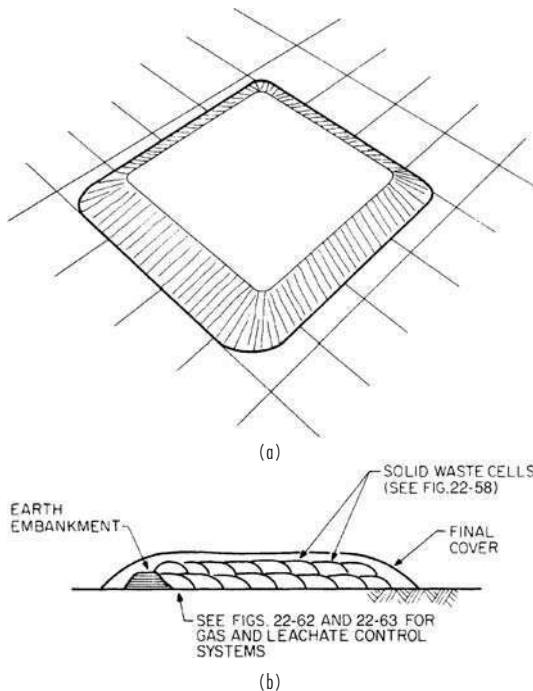


FIG. 22-57 Area method for landfilling solid wastes. (a) Pictorial view of completed landfill. (b) Section through landfill.

areas such as ponds, pits, or quarries must be used as landfill sites, special provisions must be made to contain or eliminate the movement of leachate and gases from completed cells. Usually this is accomplished by first draining the site and then lining the bottom with a clay liner or other appropriate sealants. If a clay liner is used, it is important to continue operation of the drainage facility until the site is filled to avoid the creation of uplift pressures that could cause the liner to rupture from heaving.

Occurrence of Gases and Leachate in Landfills The following biological, physical, and chemical events occur when solid wastes are placed in a sanitary landfill: (1) biological decay of organic materials, either aerobically or anaerobically, with the evolution of gases and liquids; (2) chemical oxidation of waste materials; (3) escape of gases from the fill; (4) movement of liquids caused by differential heads; (5) dissolving and leaching of organic and inorganic materials by water and leachate moving through the fill; (6) movement of dissolved material by concentration gradients and osmosis; and (7) uneven settlement caused by consolidation of material into voids.

With respect to item 1, bacterial decomposition initially occurs under aerobic conditions because a certain amount of air is trapped within the landfill. However, the oxygen in the trapped air is exhausted within days, and long-term decomposition occurs under anaerobic conditions.

1. *Gases in landfills.* Gases found in landfills include air, ammonia, carbon dioxide, carbon monoxide, hydrogen, hydrogen sulfide, methane, nitrogen, and oxygen. Data on the molecular weight and density of these gases are presented in Sec. 2. Carbon dioxide and methane are the principal gases produced from the anaerobic decomposition of the organic solid-waste components.

The anaerobic conversion of organic compounds is thought to occur in three steps. The first involves the enzyme-mediated transformation (hydrolysis) of higher-weight molecular compounds into compounds suitable for use as a source of energy and cell carbon; the second is associated with the bacterial conversion of the compounds resulting

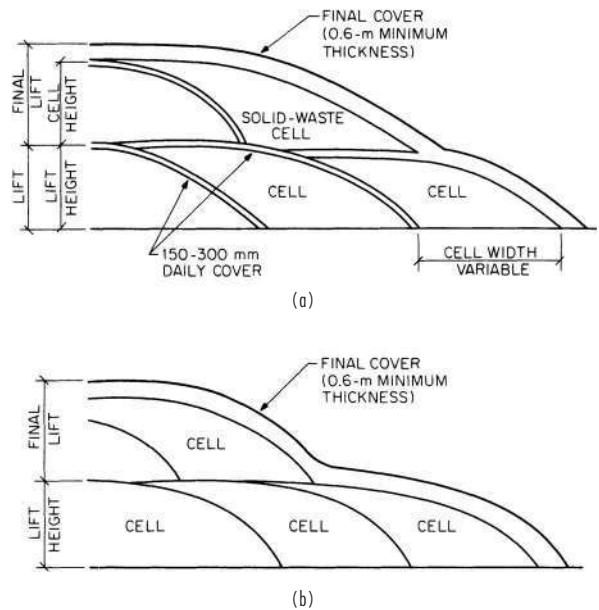


FIG. 22-58 Typical section through a landfill. (a) With daily or intermediate cover. (b) Without daily or intermediate cover.

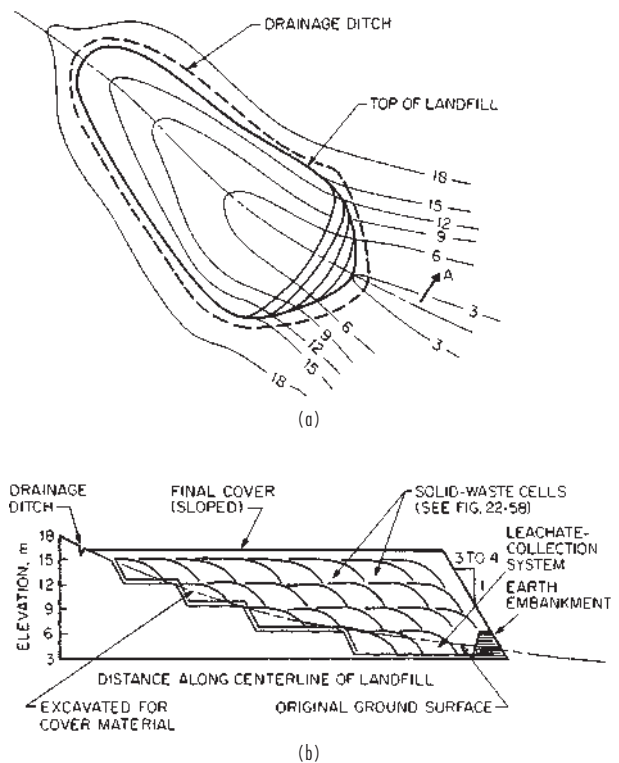
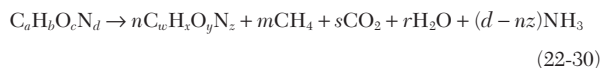


FIG. 22-59 Depression method for landfilling solid wastes. (a) Plan view: canyon-site landfill. (b) Section through landfill.

from the first step into identifiable lower-molecular-weight intermediate compounds; and the third step involves the bacterial conversion of the intermediate compounds into simpler end products, such as carbon dioxide (CO₂) and methane (CH₄). The overall anaerobic conversion of organic industrial wastes can be represented with the following equation:



where $s = a - nw = m$ and $r = c - ny - 2s$. The terms $C_aH_bO_cN_d$ and $C_uH_vO_xN_z$ are used to represent on a molar basis the composition of the material present at the start of the process.

The rate of decomposition in unmanaged landfills, as measured by gas production, reaches a peak within the first 2 years and then slowly tapers off, continuing in many cases for periods of up to 25 years or more. The total volume of the gases released during anaerobic decomposition can be estimated in a number of ways. If all the organic constituents in the wastes (with the exception of plastics, rubber, and leather) are represented with a generalized formula of the form $C_aH_bO_cN_d$, the total volume of gas can be estimated by using Eq. (22-30) with the assumption of completed conversion to carbon dioxide and methane.

2. Leachate in landfills. Leachate may be defined as liquid that has percolated through solid waste and has extracted dissolved or suspended materials from it. In most landfills, the liquid portion of the leachate is composed of the liquid produced from the decomposition of the wastes and liquid that has entered the landfill from external sources, such as surface drainage, rainfall, groundwater, and water from underground springs. Representative data on chemical characteristics of leachate are reported in Table 22-72.

Gas and Leachate Movement and Control Under ideal conditions, the gases generated from a landfill should be either vented to the atmosphere or, in larger landfills, collected for the production of energy. Landfills with >2.5 million cubic meters of waste or >50 Mg/yr NMOC (nonmethane organic compounds) emissions may require landfill-gas collection and flare systems, per EPA support WWW, CFR 60 Regulations. The leachate should be either contained within the landfill or removed for treatment.

1. Gas movement. In most cases, over 90 percent of the gas volume produced from the decomposition of solid wastes consists of methane and carbon dioxide. Although most of the methane escapes to the atmosphere, both methane and carbon dioxide have been found in concentrations of up to 40 percent at lateral distances of up to 120 m (400 ft) from the edges of landfills. Methane can accumulate below buildings or in other enclosed spaces or close to a sanitary landfill. With proper venting at the landfill source, methane should not pose a problem.

Because carbon dioxide is about 1.5 times as dense as air and 2.8 times as dense as methane, it tends to move toward the bottom of the landfill. As a result, the concentration of carbon dioxide in the lower portions of landfill may be high for years. Ultimately, because of its density, carbon dioxide will also move downward through the underlying formation until it reaches the groundwater. Because carbon dioxide is readily soluble in water, it usually lowers the pH, which in turn can increase the hardness and mineral content of the groundwater through the solubilization of calcium and magnesium carbonates.

Currently, MSW landfills are estimated to release approximately 14,300 tons/yr of NMOC. Methane and carbon dioxide contribute to the greenhouse effect, and the release of VOCs cause air quality issues such as smog and ozone formation. In the gas of a MSW landfill, VOCs comprise about 39 percent of NMOC.

2. Control of gas movement. The lateral movement of gases produced in a landfill can be controlled by installing vents made of materials that are more permeable than the surrounding soil. Typically, as shown in Fig. 22-60a, gas vents are constructed of gravel. The spacing of cell vents depends on the width of the waste cell but usually varies from 18 to 60 m (60 to 200 ft). The thickness of the gravel layer should be such that it will remain continuous even though there may be dif-

ferential settling; 300 to 450 mm (12 to 18 in) thick is recommended. Barrier vents (see Fig. 22-60b) also can be used to control the lateral movements of gases. Well vents are often used in conjunction with lateral surface vents buried below grade in a gravel trench (see Fig. 22-60c). Details of a gas vent are shown in Fig. 22-61. Control of the downward movement of gases can be accomplished by installing perforated pipes in the gravel layer at the bottom of the landfill. If the gases cannot be vented laterally, it may be necessary to install gas wells and vent the gas to atmosphere. This is considered a passive venting system. See Fig. 22-62. The movement of landfill gases through adjacent soil formation can be controlled by constructing barriers of materials that are more impermeable than soil. See Fig. 22-63a. Some of the landfill sealants that are available for this use are identified in Table 22-73. Of these, the use of compacted clays is the most common. The thickness will vary depending on the type of clay and the degree of control required; thicknesses ranging from 0.015 to 1.25 m (6 to 48 in) have been used. Covers of landfills are also typically multilayered and may include foundation layer, clay layer, membrane, drainage layer (synthetic or natural), root zone layer, or soil.

3. Control of gas movement by recovery. The movement of gases in landfills can also be controlled by installing gas-recovery wells in completed landfills (see Fig. 22-63b). This is considered an active venting system. Clay and other lines are used when landfill gas is to be

TABLE 22-72 Typical Leachate Quality of Municipal Waste

S1 number	Parameter	Overall range (mg/L except as indicated)
1	TDS	584–55,000
2	Specific conductance	480–72,500 μ mho/cm
3	Total suspended solids	2–140,900
4	BOD	ND–195,000
5	COD	6.6–99,000
6	TOC	ND–40,000
7	pH	3.7–8.9 units
8	Total alkalinity	ND–15,050
9	Hardness	0.1–225,000
10	Chloride	2–11,375
11	Calcium	3.0–2,500
12	Sodium	12–6,010
13	Total Kjeldahl nitrogen	2–3,320
14	Iron	ND–4,000
15	Potassium	ND–3,200
16	Magnesium	4.0–780
17	Ammonia–nitrogen	ND–1,200
18	Sulfate	ND–1,850
19	Aluminum	ND–85
20	Zinc	ND–731
21	Manganese	ND–400
22	Total phosphorus	ND–234
23	Boron	0.87–13
24	Barium	ND–12.5
25	Nickel	ND–7.5
26	Nitrate–nitrogen	ND–250
27	Lead	ND–14.2
28	Chromium	ND–5.6
29	Antimony	ND–3.19
30	Copper	ND–9.0
31	Thallium	ND–0.78
32	Cyanide	ND–6
33	Arsenic	ND–70.2
34	Molybdenum	0.07–1.43
35	Tin	ND–0.16
36	Nitrite–nitrogen	ND–1.46
37	Selenium	ND–1.85
38	Cadmium	ND–0.4
39	Silver	ND–1.96
40	Beryllium	ND–0.36
41	Mercury	ND–3.0
42	Turbidity	40–500 Jackson units

From McGinley, P. M., and P. Kmet, *Formation Characteristics, Treatment, and Disposal of Leachate from Municipal Solid Waste Landfills*, Bureau of Solid Waste Management, Wisconsin Department of Natural Resources, Madison, 1984.

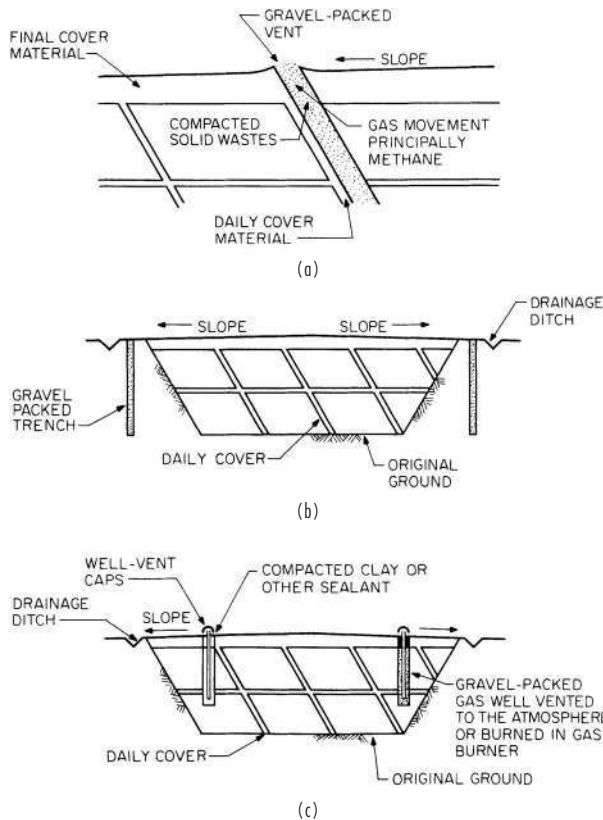


FIG. 22-60 Vents used to control the lateral movement of gases in landfills. (a) Cell. (b) Barrier. (c) Well. (From G. Tchobanoglous, H. Theisen, and R. Eliassen, *Solid Wastes: Engineering Principles and Management Issues*, McGraw-Hill, New York, 1977.)

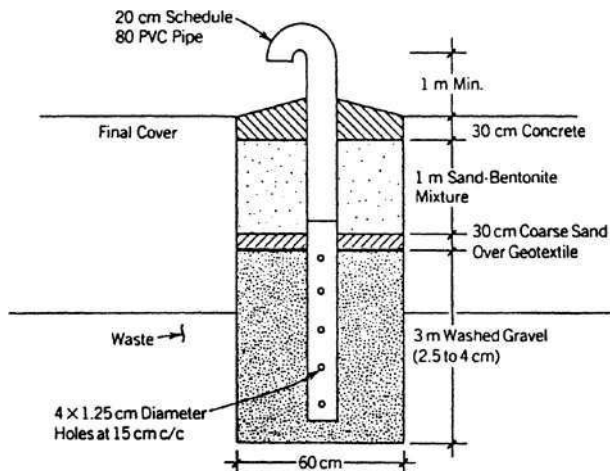


FIG. 22-61 Typical detail of an isolated gas vent. (From Bagchi, A., *Design, Construction, and Monitoring of Sanitary Landfill*, Wiley, 1990.)

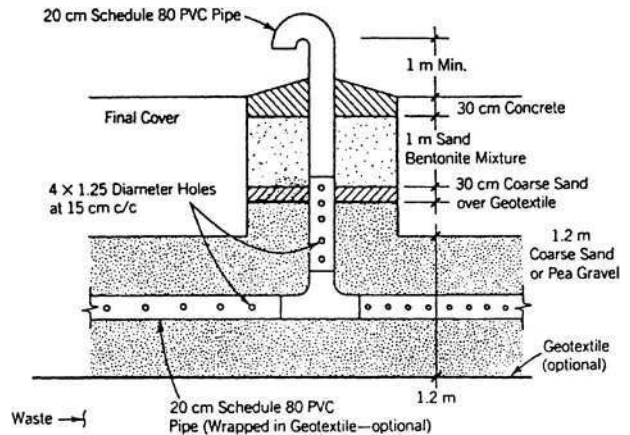


FIG. 22-62 Typical detail of a passive gas venting system with a header pipe. (From Bagchi, A., *Design, Construction, and Monitoring of Sanitary Landfill*, Wiley, 1990.)

recovered. In some gas-recovery systems, leachate is collected and recycled to the top of the landfill and reinjected through perforated lines located in drainage trenches. Typically, the rate of gas production is greater in leachate-recirculation systems.

Gas recovery systems have been installed in some large municipal landfills, with the gas flared used to run spark-ignition diesel engines to generate power for sale to the grid or the gas is cleaned up and piped to nearby users. The economics of such operations must be reviewed for each site. The end use of gas will affect the overall economics. The

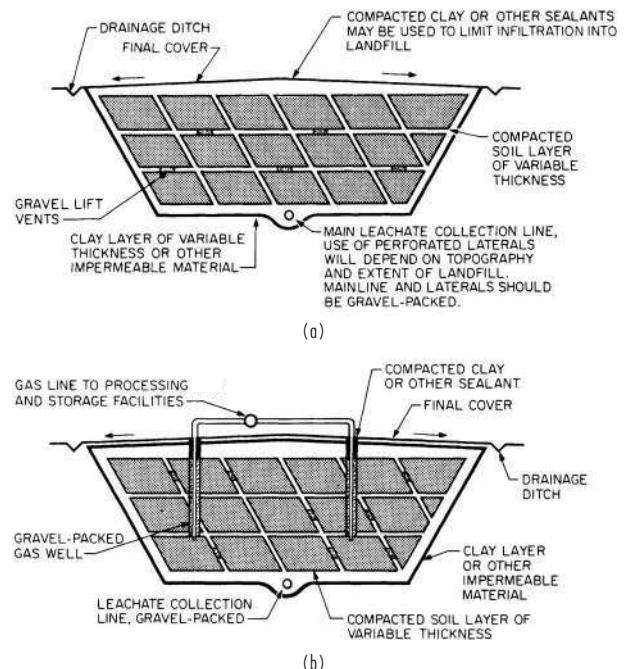


FIG. 22-63 Use of an impermeable liner to control the movement of gases and leachate in landfills. (a) Without gas recovery. (b) With gas recovery. (From G. Tchobanoglous, H. Theisen, and R. Eliassen, *Solid Wastes: Engineering Principles and Management Issues*, McGraw-Hill, New York, 1977.)

TABLE 22-73 Landfill Sealants for the Control of Gas and Leachate Movement

Classification	Sealant	
	Representative types	
Compacted soil Compacted clay	Bentonites, illites, kaolinites	
Inorganic chemicals Synthetic chemicals Synthetic membrane liners	Sodium carbonate, silicate, or pyrophosphate Polymers, rubber, latex Polyvinyl chloride, butyl rubber, Hypalon, polyethylene, nylon-reinforced liners	
Asphalt	Modified asphalt, asphalt-covered polypropylene fabric, asphalt concrete	
Others	Gunit concrete, soil, cement, plastic soil cement	

*From G. Tchobanoglous, H. Theisen, and R. Eliassen, *Solid Wastes: Engineering Principles and Management Issues*, McGraw-Hill, New York, 1977.

cost of gas cleanup and processing equipment will limit the recovery of landfill gases, especially from small landfills.

4. *Leachate movement.* Under normal conditions, leachate is found in the bottom of landfills. From there it moves through the underlying strata, although some lateral movement may also occur, depending on the characteristics of the surrounding material. The rate of seepage of leachate from the bottom of a landfill can be estimated by Darcy's law by assuming that the material below the landfill to the top of the water table is saturated and that a small layer of leachate exists at the bottom of the fill. Under these conditions, the leachate discharge rate per unit area is equal to the value of the coefficient of permeability K , expressed in meters per day. The computed value represents the maximum amount of seepage that would be expected, and this value should be used for design purposes. Under normal conditions, the actual rate would be less than this value because the soil column below the landfill would not be saturated. Models have been developed to aid in the estimation of leachate quantity. Bagchi has covered details of these models in the *Design, Construction, and Monitoring of Sanitary Landfill*.

5. *Control of leachate movement.* As leachate percolates through the underlying strata, many of the chemical and biological constituents originally contained in it will be removed by the filtering and adsorptive action of the material composing the strata. In general, the extent of this action depends on the characteristics of the soil, especially the clay content. Because of the potential risk involved in allowing leachate to percolate to the groundwater, best practice calls for its elimination or containment. Ultimately, it will be necessary to collect and treat the leachate.

The use of clay has been the favored method of reducing or eliminating the percolation of leachate (see Fig. 22-63 and Table 22-73). Membrane liners are used most often today but require care so that they will not be damaged during the filling operations. Equally important in controlling the movement of leachate is the elimination of surface-water infiltration, which is the major contributor to the total volume of leachate. With the use of an impermeable clay layer, membrane liners, an appropriate surface slope (1 to 2 percent), and adequate drainage, surface infiltration can be controlled effectively.

TABLE 22-74 Short- and Long-Term Actions for Effective Industrial Solid-Waste Management*

Actions	Remarks
Short-term	
1. Inventory wastes.	Document all types, quantities, and sources of wastes (both nonhazardous and hazardous wastes).
2. Inventory inactive sites.	All inactive sites where wastes have been disposed of in the past should be inventoried. Data should be gathered on buried wastes, including types, quantity, and sources. A groundwater-monitoring program should be developed.
3. Characterize wastes.	In addition to general information on the characteristics of the wastes, all hazardous wastes should be individually characterized.
4. Assign responsibilities.	Assign responsibilities and authority at plant and headquarters for the storage, collection, treatment, and disposal of all types of hazardous wastes.
5. Track the movement of wastes.	Develop a logging system for hazardous wastes containing the date, waste description, source, volume shipped or hauled, name of hauler, and destination. Follow through to be sure that wastes reach their destination.
6. Develop emergency procedures.	Develop procedures for dealing with emergency situations involving the storage, collection, treatment, and disposal of hazardous wastes.
7. Obtain permits.	Start obtaining the necessary waste-disposal permits as soon as possible.
Long-term	
1. Remove all accumulated wastes.	Develop a systematic program for removing all accumulated wastes stored on the plant site.
2. Separate wastes.	Institute a long-term program to separate wastes at the source of production.
3. Reduce wastes.	A systematic program should be undertaken to examine all sources of waste production and to develop alternative operations and processes to reduce waste generation.
4. Improve facilities.	Upgrade facilities to meet RCRA requirements. A data-collection program should be instituted to obtain any needed data.
5. Review all waste-management agreements.	Develop detailed contracts with outside waste-management firms. Define clearly the duties and responsibilities of plant personnel and waste-collection personnel.
6. Review and develop disposals-tie options.	Develop long-term projections for landfill requirements and initiate a program to secure the needed sites.
7. Secure appropriate engineering and consulting services.	Make sure that your engineering departments are involved early in the process.
8. Monitor legislative programs.	Retain outside consultants for specific tasks.
	Develop a program for monitoring new regulations and for inputting to appropriate federal, state, and local agencies on the modification and development of new regulations.

*Adapted in part from R. Sobel, "How Industry Can Prepare for RCRA," *Chem. Eng.*, **86**(1), 82 (Jan. 29, 1979).

6. *Settlement and structural characteristics of landfills.* The settlement of landfills depends on the initial compaction, characteristics of wastes, degree of decomposition, and effects of consolidation when the leachate and gases are formed in the landfill. The height of the completed fill will also influence the initial compaction and degree of consolidation.

PLANNING

Because of the ever-growing number of federal regulations governing the disposal of nonhazardous and hazardous solid wastes, it is prudent to develop both short-term and long-term action programs to deal with all aspects of solid-waste management. Important short- and long-term actions are identified in Table 22-74.

SECTION 23

Process Safety

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PROCESS SAFETY INTRODUCTION

GENERAL REFERENCES: AIChE/CCPS, *Guidelines for Chemical Process Quantitative Risk Analysis*, 2d ed., American Institute of Chemical Engineers, New York, 2000. AIChE/CCPS, *Guidelines for Hazards Evaluation Procedures*, 2d ed., American Institute of Chemical Engineers, New York, 1992. Crowl and Louver, *Chemical Process Safety: Fundamentals with Applications*, 2d ed., Prentice-Hall, Englewood Cliffs, N.J., 2002. Mannan, *Lees' Loss Prevention in the Process Industries*, 3d ed., Elsevier, Amsterdam.

Process safety differs from the traditional approach to accident prevention in several ways (Mannan, *Lees' Loss Prevention in the Process Industries*, 3d ed., Elsevier, 2005, p. 1/9):

- There is greater concern with accidents that arise out of the technology.
- There is greater emphasis on foreseeing hazards and taking action before accidents occur.
- There is greater emphasis on a systematic rather than a trial-and-error approach, particularly on systematic methods of identifying hazards and of estimating the probability that they will occur and their consequences.
- There is concern with accidents that cause damage to plant and loss of profit but do not injure anyone, as well as those that do cause injury.
- Traditional practices and standards are looked at more critically.

The term *loss prevention* can be applied in any industry but is widely used in the process industries where it usually means the same as *process safety*.

Chemical plants, and other industrial facilities, may contain large quantities of hazardous materials. The materials may be hazardous due to toxicity, reactivity, flammability, or explosivity. A chemical plant may also contain large amounts of energy—the energy either is required to process the materials or is contained in the materials themselves. An *accident* occurs when control of this material or energy is lost. An *accident* is defined as an unplanned event leading to undesired consequences. The consequences might include injury to people, damage to the environment, or loss of inventory and production, or damage to equipment.

A *hazard* is defined as a chemical or physical condition that has the potential for causing damage to people, property, or the environment (AIChE/CCPS, *Guidelines for Chemical Process Quantitative Risk Analysis*, 2d ed., American Institute of Chemical Engineers, New York, 2000, p. 6). Hazards exist in a chemical plant due to the nature of the materials processed or due to the physical conditions under which the materials are processed, i.e., high pressure or temperature. These hazards are present most of the time. An initiating event is required to begin the accident process. Once initiated, the accident follows a sequence of steps, called the event sequence, that results in an incident outcome. The consequences of the accident are the resulting effects of the incident. For instance, a rupture in a pipeline due to corrosion (initiating event) results in leakage of a flammable liquid from the process. The liquid evaporates and mixes with air to form a flammable cloud, which finds an ignition source (event sequence), resulting in a fire (incident outcome). The consequences of the accident are considerable fire damage and loss of production.

Risk is defined as a measure of human injury, environmental damage, or economic loss in terms of both the incident likelihood (probability) and the magnitude of the loss or injury (consequence) (AIChE/CCPS, *Guidelines for Chemical Process Quantitative Risk Analysis*, 2d ed., American Institute of Chemical Engineers, New York, 2000, pp. 5–6). It is important that both likelihood and consequence be included in risk. For instance, seat belt use is based on a reduction in the consequences of an accident. However, many people argue against seat belts based on probabilities, which is an incorrect application of the risk concept.

A good safety program identifies and removes existing hazards. An outstanding safety program prevents the existence of safety hazards in the

first place. An outstanding safety program is achieved by company commitment, visibility, and management support. This is usually achieved by a corporatewide safety policy. This safety policy usually includes the following items: (1) the company is very serious about safety, (2) safety cannot be prioritized and is a part of everyone's job function, (3) everyone is responsible for safety, including management.

To ensure that the safety program is working, most companies have a safety policy follow-through. This includes monthly safety meetings, performance reviews, and safety audits. The monthly safety meetings include a discussion of any accidents (and resolution of prevention means), training on specific issues, inspection of facilities, and delegation of work. Performance reviews within the company for all employees must have a visible safety performance component.

Safety audits are a very important means of ensuring that the safety program is operating as intended. Audits are usually done yearly by an audit team. The audit team is comprised of corporate and site safety people and other experts, as needed, including industrial hygiene, toxicology, and/or process safety experts. The audit team activities include (1) reviewing records (including accident reports, training, monthly meetings), (2) inspecting random facilities to see if they are in compliance, (3) interviewing the employees to determine how they participate in the safety program, (4) making recommendations on

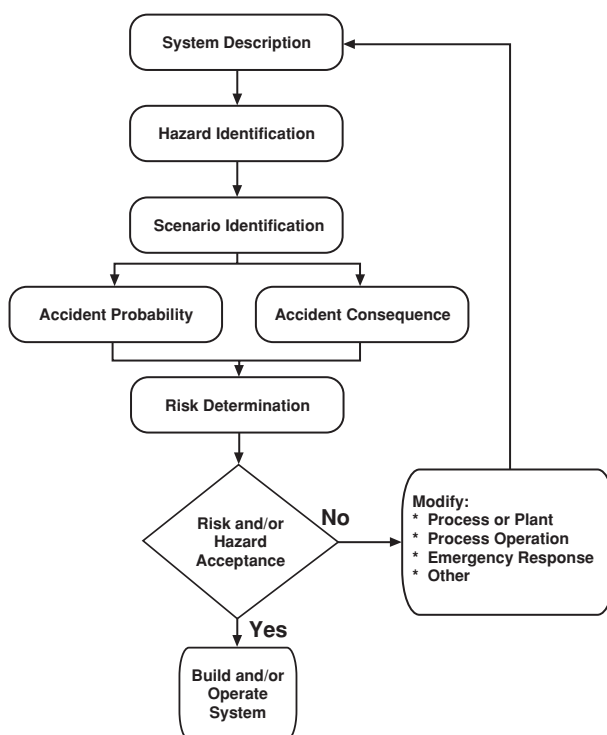


FIG. 23-1 The hazard identification and risk assessment procedure. [Guidelines for Hazards Evaluation Procedures, Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE); copyright 1985 AIChE and reproduced with permission.]

how the program can be improved, and (5) rating the performance of the unit. The audit results are reported to upper management with the expectation that the designated unit will implement improvements in short order. Many companies perform a combined audit, which may include environmental and quality issues.

Figure 23-1 shows the hazards identification and risk assessment procedure. The procedure begins with a complete description of the process. This includes detailed PFD and P&I diagrams, complete specifications on all equipment, maintenance records, operating procedures, and so forth. A hazard identification procedure is then selected (see Hazard Analysis subsection) to identify the hazards and their nature. This is followed by identification of all potential event sequences and potential incidents (scenarios) that can result in loss of control of energy or material. Next is an evaluation of both the consequences and the probability. The consequences are estimated by using source models (to describe the

release of material and energy) coupled with a consequence model to describe the incident outcome. The consequence models include dispersion, fire, and explosion modeling. The results of the consequence models are used to estimate the impacts on people, environment, and property. The accident probability is estimated by using fault trees or generic databases for the initial event sequences. Event trees may be used to account for mitigation and postrelease incidents. Finally, the risk is estimated by combining the potential consequence for each event with the event frequency and summing over all events.

Once the risk is determined, a decision must be made on risk acceptance. This can be done by comparison to a relative or absolute standard. If the risk is acceptable, then the decision is made to build and/or operate the process. If the risk is not acceptable, then something must be changed. This could include the process design, the operation, or maintenance, or additional layers of protection might be added.

CASE HISTORIES

GENERAL REFERENCES: *One Hundred Largest Losses: A Thirty Year Review of Property Damage Losses in the Hydrocarbon Chemical Industry*, 20th ed. (M&M Protection, Consultants, Chicago); Mannan, S., ed., *Lees' Loss Prevention in the Process Industries*, Elsevier, 2005; Kletz, T. A., *Learning from Accidents*, Gulf Professional Publishing, 2001; Kletz, T. A., *What Went Wrong? Case Histories of Process Plant Disasters*, Editions Technip, 1998; and Sanders, R. E., *Chemical Process Safety: Learning from Case Histories*, Editions Technip, 1999.

INTRODUCTION

Engineers must give significant thought to the consequences of their decisions and indecisions. A wise step during conceptual and design phases is to review previous negative experiences of others and within your own organization. Periodically review the status of recent chemical accidents. The U.S. Chemical Safety and Hazards Investigation Board web site, www.csb.gov, offers details on many investigations related to chemical industry accidents within the United States. Look for similarities and dissimilarities to your current practice, and carefully make appropriate changes and improvements to avoid repeating similar accidents.

HYDROCARBON FIRES AND EXPLOSIONS

The explosion and fires at the Texaco Refinery, Milford Haven, Wales, 24 July 1994. Reference: Health and Safety Executive (HSE); HSE Books, Her Majesty's Stationary Office, Norwich, England, 1997.

On July 24, 1994, an explosion followed by a number of fires occurred at 13:23 at the Texaco refinery in Milford Haven, Wales, England. Prior to this explosion, around 9 a.m., a severe coastal electrical storm caused plant disturbances that affected the vacuum distillation, alkylation, butamer, and FCC units. The explosion occurred due to a combination of failures in management, equipment, and control systems. Given its calculated TNT equivalent of at least 4 tons, significant portions of the refinery were damaged. That no fatalities occurred is attributed partially to the accident occurring on a Sunday, as well as the fortuitous location of those who were near the explosion.

As the plant attempted adjustments to the upsets caused by the electrical storm, liquid was continuously pumped into a process vessel with a closed outlet valve. The control system indicated that this valve was open. As the unit overfilled, the only means of exit was a relief system designed for vapor. When the liquid reached the relief system, its momentum was high enough to rip apart the ductwork and cause a massive release of hydrocarbons into the environment. Minutes prior to the explosion, operating personnel were responding to 275 alarms of which 80 percent had high priority. An ignition source was found 110 m away. Recommendations from the accident investigation included

the necessity of operating personnel having knowledge about simple volumetric and mass balances; that control systems be configured to provide an overview of the condition of the process; that safety critical alarms be distinguishable from other alarms; and that liquid knockout drums exist for relief systems designed for vapor.

DUST EXPLOSIONS

West Pharmaceutical Services Plant in Kinston, North Carolina, 29 January 2003, and CTA Acoustics Manufacturing Plant in Corbin, Kentucky, 20 February 2003. Reference: U.S. Chemical Safety Board (CSB); www.csb.gov/index.cfm?folder=completed_investigations&page=info&INV_ID=34 and [ID=35](http://www.csb.gov/index.cfm?folder=completed_investigations&page=info&INV_ID=35)

On January 29, 2003, the West Pharmaceutical explosion killed six workers and injured dozens more. The CSB determined that fine polyethylene dust particles, released during the production of rubber products, had accumulated above the tiles of a false ceiling, creating an explosion hazard at the plant. A similar incident occurred a few weeks later, at the CTA Acoustics manufacturing plant in Corbin, Kentucky, fatally injuring seven workers and injuring more than 30 others. This facility produced fiberglass insulation for the automotive industry. CSB investigators found that the explosion was fueled by resin dust accumulated in a production area, likely ignited by flames from a malfunctioning oven. The resin involved was a phenolic binder used in producing fiberglass mats.

CSB investigators determined that both disasters resulted from accumulations of combustible dust. Workers and workplaces need to be protected from this insidious hazard. The lesson learned here is the importance of housekeeping. Some companies will allow only $\frac{1}{32}$ in of dust to accumulate before cleaning. Suspended ceilings must be suspected as areas that can accumulate dust. Often the first explosion may be minor, but the dust dislodged can be explosive enough to level the building on the second ignition.

REACTIVE CHEMICALS

Explosion, Morton International, Inc., Paterson, New Jersey, 8 April 1998. Reference: CSB; www.csb.gov/completed_investigations/docs/MortonInvestigationReport.pdf

On April 8, 1998, at 20:18, an explosion and fire occurred during the production of Automate Yellow 96 Dye at Morton International, Inc. Yellow 96 dye was produced by mixing and reacting two chemicals, *ortho*-nitrochlorobenzene (*o*-NCB) and 2-ethylhexylamine (2-EHA). The explosion and fire were the consequence of a runaway reaction, which overpressurized a 2000-gal capacity chemical reactor vessel and released flammable material that ignited. Nine employees were injured, including two seriously, and potentially hazardous materials were released into

the community. The CSB investigation team determined that the reaction accelerated beyond the heat removal capability of the kettle. The resulting high temperature led to a secondary runaway reaction (decomposition of *o*-NCB). The initial runaway reaction was most likely caused by a combination of the following factors: (1) The reaction was started at a temperature higher than normal, (2) the steam used to initiate the reaction was left on for too long, and (3) the use of cooling water to control the reaction rate was not initiated soon enough. The Paterson facility was not aware of the decomposition reaction. A similar incident occurred with a process using *o*-NCB in Sauget, Illinois, in 1974 (Vincent, G. C., *Loss Prev.* 1971, 5: 46–52).

MATERIALS OF CONSTRUCTION

Ruptured chlorine hose. Reference: CSB; www.csb.gov/safety_publications/docs/ChlorineHoseSafetyAdvisory.pdf

On August 14, 2002, a 1-in chlorine transfer hose (CTH) used in a rail-car offloading operation at DPC Enterprises in Festus, Missouri, catastrophically ruptured and initiated a sequence of events that led to the release of 48,000 lb of chlorine into neighboring areas. The material of construction of the ruptured hose was incorrect. The distributor fabricated bulk CTH with Schedule 80 Monel 400 end fittings and a high-density polyethylene spiral guard. Three hoses were shipped directly to the Festus facility from the distributor; two were put into service on June 15, 2002. The hose involved in the incident failed after 59 days in service.

Most plastics react chemically with chlorine because of their hydrocarbon structural makeup. This reactivity is avoided with some plastics in which fluorine atoms have been substituted into the hydrocarbon molecule. The Chlorine Institute recommends that hoses constructed with such an inner lining “have a structural layer braid of polyvinylidene fluoride (PVDF) monofilament material or a structural braid of Hastelloy C-276.” An underlying lesson here is material compatibility. Material compatibility tables exist that engineers can consult, including in other sections within this volume.

TOXICOLOGY

Vessel explosion, D. D. Williamson & Co., Inc., Louisville, Kentucky, 11 April 2003. Reference: CSB; www.csb.gov/completed_investigations/docs/CSB_DDWWilliamsonReport.pdf

On April 11, 2003, at approximately 2:10 a.m., a 2200-gal stainless steel spray dryer feed tank at the D. D. Williamson & Co., Inc. (DDW), plant in Louisville, Kentucky, exploded. One operator was killed. The other four men working at the plant at the time of the incident were not injured. The incident was most likely initiated by overheating by a 130-psi steam supply. The feed tank was manually

controlled for temperature and pressure. The tank had a maximum working pressure of 40 psi. A concrete block wall to the east separated the feed tank from a 12,000-gal aqua ammonia storage tank (29.4% ammonia). After the explosion, the feed tank's shell split open in a vertical line. It was propelled through the wall and struck the ammonia storage tank, located 15 ft to the west. The ammonia storage tank was knocked off its foundation approximately 10 ft, and piping was ripped loose. This resulted in a 26,000-lb aqua ammonia leak. Metro Louisville Health Department obtained maximum ammonia readings of 50 parts per million (ppm) at the fence line and 35 ppm on a nearby street. No injuries were reported in the area of the ammonia release.

A number of management decisions factor into this case. There was no program to evaluate necessary layers of protection on the spray dryer feed tanks. Likewise, there was no recognition of the need to provide process control and alarm instrumentation on the two feed tanks. Reliance on a single local temperature indicator that must be read by operators is insufficient. On the morning of the incident, the operators were unaware that the system had exceeded normal operating conditions. The feed tanks were installed for use in the spray dryer process without a review of their design versus system requirements. Safety valves on the spray dryer feed tanks had been removed to transport the tanks to Louisville and were never reinstalled. Inadequate hazard analysis systems didn't identify feed tank hazards. The ASME Code, Section VIII (2001 ASME Boiler and Pressure Vessel Code: Design and Fabrication of Pressure Vessels, American Society of Mechanical Engineers, 2001), requires that all vessels having an internal operating pressure exceeding 15 psi be provided with pressure relief devices. Finally, equipment layout should always be considered in the design stage. Methods such as the Dow Fire and Explosion Index (AIChE, 1994) can assist in determining the optimum spacing between critical units.

NITROGEN ASPHYXIAION

Union Carbide Corporation, Hahnville, Louisiana, 27 March 1998. Reference: CSB; [www.csb.gov/completed_investigations/docs/Final Union Carbide Report.pdf](http://www.csb.gov/completed_investigations/docs/Final%20Union%20Carbide%20Report.pdf) and [/SB-Nitrogen-6-11-03.pdf](http://www.csb.gov/completed_investigations/docs/Final%20Union%20Carbide%20Report.pdf).

On March 27, 1998, at approximately 12:15 p.m., two workers at Union Carbide Corporation's Taft/Star Manufacturing Plant in Hahnville, Louisiana, were overcome by nitrogen gas while performing a black light inspection at an open end of a 48-in-wide horizontal pipe. One Union Carbide employee was killed, and an independent contractor was seriously injured due to nitrogen asphyxiation. Nitrogen was being injected into a nearby reactor to prevent contamination of a catalyst by oxygen and related materials. The nitrogen also flowed through some of the piping systems connected to the reactors. No warning sign was posted on the pipe opening identifying it as a confined space. Nor was there a warning that the pipe contained potentially hazardous nitrogen.

HAZARDOUS MATERIALS AND CONDITIONS

FLAMMABILITY

Nomenclature

K_G	deflagration index for gases (bar·m/s)
K_{St}	deflagration index for dusts (bar·m/s)
LFL	lower flammability limit (vol % fuel in air)
LOC	limiting oxygen concentration
n	number of combustible species
P	pressure
T	temperature (°C)
t	time (s)
UFL	upper flammability limit (vol. % fuel in air)
V	vessel volume (m ³)
y_i	mole fraction of component i on a combustible basis
z	stoichiometric coefficient for oxygen
ΔH_c	net heat of combustion (kcal/mol)

GENERAL REFERENCES: Crowl and Louvar, *Chemical Process Safety: Fundamentals with Applications*, 2d ed., Prentice-Hall, Upper Saddle River, N.J., 2002, Chaps. 6 and 7. Crowl, *Understanding Explosions*, American Institute of Chemical Engineers, New York, 2003. Eckoff, *Dust Explosions in the Process Industries*, 2d ed., Butterworth-Heinemann, now Elsevier, Amsterdam, 1997. Kinney and Graham, *Explosive Shocks in Air*, 2d ed., Springer-Verlag, New York, 1985. Lewis and von Elbe, *Combustion, Flames and Explosions of Gases*, 3d ed., Academic Press, New York, 1987. Mannan, *Lees' Loss Prevention in the Process Industries*, 3d ed., Elsevier, Amsterdam, 2005, Chap. 16: Fire, Chap. 17: Explosion.

Introduction Fire and explosions in chemical plants and refineries are rare, but when they do occur, they are very dramatic.

Accident statistics have shown that fires and explosions represent 97 percent of the largest accidents in the chemical industry (J. Coco, ed., *Large Property Damage Losses in the Hydrocarbon-Chemical Industry: A Thirty Year Review*, J. H. Marsh and McLennan, New York, 1997).

Prevention of fires and explosions requires

1. An understanding of the fundamentals of fires and explosions
2. Proper experimental characterization of flammable and explosive materials

3. Proper application of these concepts in the plant environment
The technology does exist to handle and process flammable and explosive materials safely, and to mitigate the effects of an explosion. The challenges to this problem are as follows:

1. Combustion behavior varies widely and is dependent on a wide range of parameters.
2. There is an incomplete fundamental understanding of fires and explosions. Predictive methods are still under development.
3. Fire and explosion properties are not fundamentally based and are an artifact of a particular experimental apparatus and procedure.
4. High-quality data from a standardized apparatus that produces consistent results are lacking.
5. The application of these concepts in a plant environment is difficult.

The Fire Triangle The fire triangle is shown in Fig. 23-2. It shows that a fire will result if fuel, oxidant, and an ignition source are present. In reality, the fuel and oxidant must be within certain concentration ranges, and the ignition source must be robust enough to initiate the fire. The fire triangle applies to gases, liquids, and solids. Liquids are volatilized and solids decompose prior to combustion in the vapor phase. For dusts arising from solid materials, the particle size, distribution, and suspension in the gas are also important parameters in the combustion—these are sometimes included in the fire triangle.

The usual oxidizer in the fire triangle is oxygen in the air. However, gases such as fluorine and chlorine; liquids such as peroxides and chlorates; and solids such as ammonium nitrate and some metals can serve the role of an oxidizer. Exothermic decomposition, without oxygen, is also possible, e.g., with ethylene oxide or acetylene.

Ignition arises from a wide variety of sources, including static electricity, hot surfaces, sparks, open flames, and electric circuits. Ignition sources are elusive and difficult to eliminate entirely, although efforts should always be made to reduce them.

If any one side of the fire triangle is removed, a fire will not result. In the past, the most common method for fire control was elimination of ignition sources. However, experience has shown that this is not robust enough. Current fire control prevention methods continue with elimination of ignition sources, while focusing efforts more strongly on preventing flammable mixtures.

Definition of Terms The following are terms necessary to characterize fires and explosions (Crowl and Louvar, *Chemical Process Safety: Fundamentals with Applications*, 2d ed. Prentice-Hall, Upper Saddle River, N.J., 2002, pp. 227–229).

Autoignition temperature (AIT) This is a fixed temperature above which adequate energy is available in the environment to provide an ignition source.

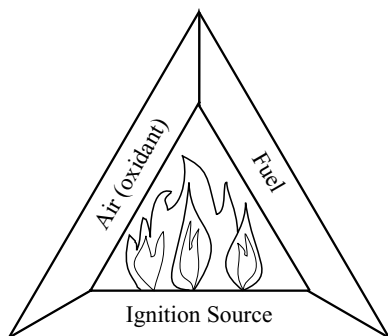


FIG. 23-2 The fire triangle showing the requirement for combustion of gases and vapors. [D. A. Crowl, *Understanding Explosions*, Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE); copyright 2003 AIChE and reproduced with permission.]

Boiling-liquid expanding-vapor explosion (BLEVE) A BLEVE occurs if a vessel that contains a liquid at a temperature above its atmospheric pressure boiling point ruptures. The subsequent BLEVE is the explosive vaporization of a large fraction of the vessel contents, possibly followed by combustion or explosion of the vaporized cloud if it is combustible. This type of explosion occurs when an external fire heats the contents of a tank of volatile material. As the tank contents heat, the vapor pressure of the liquid within the tank increases, and the tank's structural integrity is reduced because of the heating. If the tank ruptures, the hot liquid volatilizes explosively.

Combustion or fire Combustion or fire is a chemical reaction in which a substance combines with an oxidant and releases energy. Part of the energy released is used to sustain the reaction.

Confined explosion This explosion occurs within a vessel or a building.

Deflagration In this explosion the reaction front moves at a speed less than the speed of sound in the unreacted medium.

Detonation In this explosion the reaction front moves at a speed greater than the speed of sound in the unreacted medium.

Dust explosion This explosion results from the rapid combustion of fine solid particles. Many solid materials (including common metals such as iron and aluminum) become flammable when reduced to a fine powder and suspended in air.

Explosion An explosion is a rapid expansion of gases resulting in a rapidly moving pressure or shock wave. The expansion can be mechanical (by means of a sudden rupture of a pressurized vessel), or it can be the result of a rapid chemical reaction. Explosion damage is caused by the pressure or shock wave.

Fire point The fire point is the lowest temperature at which a vapor above a liquid will continue to burn once ignited; the fire point temperature is higher than the flash point.

Flammability limits Vapor-air mixtures will ignite and burn only over a well-specified range of compositions. The mixture will not burn when the composition is lower than the lower flammable limit (LFL); the mixture is too lean for combustion. The mixture is also not combustible when the composition is too rich, i.e., that is, when it is above the upper flammable limit (UFL). A mixture is flammable only when the composition is between the LFL and the UFL. Commonly used units are volume percent of fuel (percentage of fuel plus air).

Lower explosion limit (LEL) and upper explosion limit (UEL) are used interchangeably with LFL and UFL.

Flash point (FP) The flash point of a liquid is the lowest temperature at which it gives off enough vapor to form an ignitable mixture with air. At the flash point, the vapor will burn but only briefly; inadequate vapor is produced to maintain combustion. The flash point generally increases with increasing pressure.

There are several different experimental methods used to determine flash points. Each method produces a somewhat different value. The two most commonly used methods are open cup and closed cup, depending on the physical configuration of the experimental equipment. The open-cup flash point is a few degrees higher than the closed-cup flash point.

Ignition Ignition of a flammable mixture may be caused by a flammable mixture coming in contact with a source of ignition with sufficient energy or by the gas reaching a temperature high enough to cause the gas to autoignite.

Mechanical explosion A mechanical explosion results from the sudden failure of a vessel containing high-pressure, nonreactive gas.

Minimum ignition energy This is the minimum energy input required to initiate combustion.

Overpressure The pressure over ambient that results from an explosion.

Shock wave This is an abrupt pressure wave moving through a gas. A shock wave in open air is followed by a strong wind; the combined shock wave and wind is called a blast wave. The pressure increase in the shock wave is so rapid that the process is mostly adiabatic.

Unconfined explosion Unconfined explosions occur in the open. This type of explosion is usually the result of a flammable gas spill. The gas is dispersed and mixed with air until it comes in contact with an ignition source. Unconfined explosions are rarer than confined explosions because the explosive material is frequently diluted below the LFL by

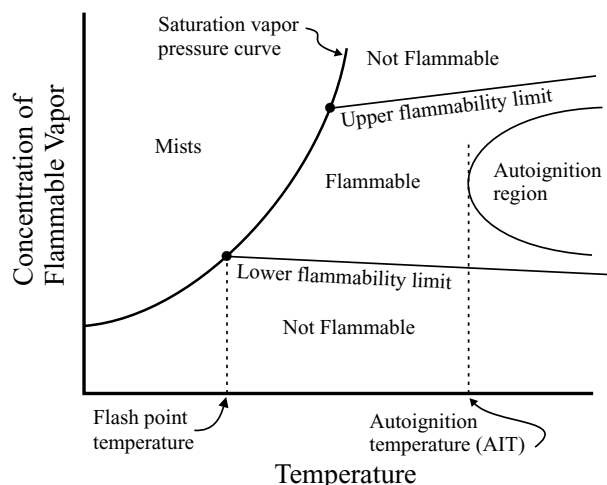


FIG. 23-3 The relationship between the various flammability properties. (D. A. Crowl and J. F. Louvar, *Chemical Process Safety: Fundamentals with Applications*, 2d ed., © 2002. Adapted by permission of Pearson Education, Inc., Upper Saddle River, N.J.)

wind dispersion. These explosions are destructive because large quantities of gas and large areas are frequently involved.

Figure 23-3 is a plot of concentration versus temperature and shows how several of these definitions are related. The exponential curve in Fig. 23-3 represents the saturation vapor pressure curve for the liquid material. Typically, the UFL increases and the LFL decreases with temperature. The LFL theoretically intersects the saturation vapor pressure curve at the flash point, although experimental data are not always consistent. The autoignition temperature is actually the lowest temperature of an autoignition region. The behavior of the autoignition region and the flammability limits at higher temperatures are not well understood.

The flash point and flammability limits are not fundamental properties but are defined only by the specific experimental apparatus and procedure used.

Section 2 provides flammability data for a number of compounds.

Combustion and Flammability Hazards

Vapor Mixtures Frequently, flammability data are required for vapor mixtures. The flammability limits for the mixture are estimated by using LeChatelier's rule [LeChatelier, "Estimation of Firedamp by Flammability Limits," *Ann. Mines* (1891), ser. 8, 19: 388–395, with translation in *Process Safety Progress*, 23(3): 172].

$$LFL_{\text{mix}} = \frac{1}{\sum_{i=1}^n y_i / LFL_i} \quad (23-1)$$

where LFL_i = lower flammability limit for component i (in volume %)
 y_i = mole fraction of component i on a combustible basis
 n = number of combustible species

An identical equation can be written for the UFL.

Note that Eq. (23-1) is only applied to the combustible species, and the mole fraction is computed using only the combustible species.

LeChatelier's rule is empirically derived and is not universally applicable. Mashuga and Crowl [Mashuga and Crowl, "Derivation of LeChatelier's Mixing Rule for Flammable Limits," *Process Safety Progress*, 19(2): 112–118 (2000)] determined that the following assumptions are present in LeChatelier's rule:

1. The product heat capacities are constant.
2. The number of moles of gas is constant.

3. The combustion kinetics of the pure species is independent of and unchanged by the presence of other combustible species.

4. The adiabatic temperature rise at the flammability limit is the same for all species.

These assumptions were found to be reasonably valid at the LFL and less so at the UFL.

Liquid Mixtures Flash point temperatures for mixtures of liquids can be estimated if only one component is flammable and the flash point temperature of the flammable component is known. In this case the flash point temperature is estimated by determining the temperature at which the vapor pressure of the flammable component in the mixture is equal to the pure component vapor pressure at its flash point. Estimation of flash point temperatures for mixtures of several flammable components can be done by a similar procedure, but it is recommended that the flash point temperature be measured experimentally.

Flammability Limit Dependence on Temperature In general, as the temperature increases, the flammability range widens, i.e., the LFL decreases and the UFL increases. Zabetakis et al. (Zabetakis, Lambiris, and Scott, "Flame Temperatures of Limit Mixtures," *7th Symposium on Combustion*, Butterworths, London, 1959) derived the following empirical equations, which are approximate for many hydrocarbons:

$$LFT_T = LFL_{25} - \frac{0.75}{\Delta H_c} (T - 25) \quad (23-2)$$

$$UFL_T = UFL_{25} + \frac{0.75}{\Delta H_c} (T - 25)$$

where ΔH_c is the net heat of combustion (kcal/mol) and T is the temperature ($^{\circ}\text{C}$).

Flammability Limit Dependence on Pressure Pressure has little effect on the LFL except at very low pressures (<50 mmHg absolute) where flames do not propagate.

The UFL increases as the pressure is increased. A very approximate equation for the change in UFL with pressure is available for some hydrocarbon gases (Zabetakis, "Fire and Explosion Hazards at Temperature and Pressure Extremes," *AICHE Inst. Chem. Engr. Symp.*, ser. 2, pp. 99–104, 1965):

$$UFL_P = UFL + 20.6 (\log P + 1) \quad (23-3)$$

where P is the pressure (megapascals absolute) and UFL is the upper flammability limit (vol % fuel in air at 1 atm).

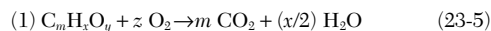
Estimating Flammability Limits There are a number of very approximate methods available to estimate flammability limits. However, for critical safety values, experimental determination as close as possible to actual process conditions is always recommended.

Jones [Jones, "Inflammation Limits and Their Practical Application in Hazardous Industrial Operations," *Chem. Rev.*, 22(1): 1–26 (1938)] found that for many hydrocarbon vapors the LFL and UFL can be estimated from the stoichiometric concentration of fuel:

$$\begin{aligned} LFL &= 0.55C_{\text{st}} \\ UFL &= 3.50C_{\text{st}} \end{aligned} \quad (23-4)$$

where C_{st} is the stoichiometric fuel concentration (vol % fuel in air).

For a stoichiometric combustion equation of the form



it follows that

$$z = m + \frac{x}{4} - \frac{y}{2} \quad (23-6)$$

and furthermore that

$$C_{\text{st}} = \frac{100}{1 + z/0.21} \quad (23-7)$$

TABLE 23-1 Limiting Oxygen Concentrations (Volume Percent Oxygen Concentrations above Which Combustion Can Occur)

Gas or vapor	N ₂ / Air	CO ₂ / Air
Methane	12	14.5
Ethane	11	13.5
Propane	11.5	14.5
<i>n</i> -Butane	12	14.5
Isobutane	12	15
<i>n</i> -Pentane	12	14.5
Isopentane	12	14.5
<i>n</i> -Hexane	12	14.5
<i>n</i> -Heptane	11.5	14.5
Ethylene	10	11.5
Propylene	11.5	14
1-Butene	11.5	14
Isobutylene	12	15
Butadiene	10.5	13
3-Methyl-1-butene	11.5	14
Benzene	11.4	14
Toluene	9.5	—
Styrene	9.0	—
Cyclopropane	11.5	14
Gasoline		
(73/100)	12	15
(100/130)	12	15
(115/145)	12	14.5
Kerosene	10 (150°C)	13 (150°C)
JP-1 fuel	10.5 (150°C)	14 (150°C)
Natural gas	12	14.5
Acetone	11.5	14
<i>t</i> -Butanol	NA	16.5 (150°C)
Carbon disulfide	5	7.5
Carbon monoxide	5.5	5.5
Ethanol	10.5	13
Ethyl ether	10.5	13
Hydrogen	5	5.2
Hydrogen sulfide	7.5	11.5
Isobutyl formate	12.5	15
Methanol	10	12
Methyl acetate	11	13.5

Data from NFPA 68, *Venting of Deflagrations* (Quincy, Mass.: National Fire Protection Association, 1994).

Equation (23-7) can be used with (23-4) to estimate the LFL and UFL.

Suzuki [Suzuki, "Empirical Relationship between Lower Flammability Limits and Standard Enthalpies of Combustion of Organic Compounds," *Fire and Materials*, 18: 333–336 (1994); Suzuki and Koide, "Correlation between Upper Flammability Limits and Thermochemical Properties of Organic Compounds," *Fire and Materials*, 18: pp. 393–397 (1994)] provides more detailed correlations for the UFL and LFL in terms of the heat of combustion.

Flammability limits can also be estimated by using calculated adiabatic flame temperatures and a chemical equilibrium program [Mashuga and Crowl, "Flammability Zone Prediction Using Calculated Adiabatic Flame Temperatures," *Process Safety Progress*, 18 (3) (1999)].

Limiting Oxygen Concentration (LOC) Below the limiting oxygen concentration it is not possible to support combustion, independent of the fuel concentration. The LOC is expressed in units of volume percent of oxygen. The LOC is dependent on the pressure and temperature, and on the inert gas. Table 23-1 lists a number of LOCs, and it shows that the LOC changes if carbon dioxide is the inert gas instead of nitrogen.

The LOC can be estimated for many hydrocarbons from

$$\text{LOC} = z \text{ LFL} \quad (23-8)$$

where z is the stoichiometric coefficient for oxygen [see Eq. (23-5)] and LFL is the lower flammability limit.

Flammability Diagram Figure 23-4 shows a typical flammability diagram. Point A shows how the scales are oriented—at any point on

the diagram the concentrations must add up to 100 percent. At point A we have 60% fuel, 20% oxygen, and 20% nitrogen. The air line represents all possible combinations of fuel and air—it intersects the nitrogen axis at 79% nitrogen which is the composition of air. The stoichiometric line represents all stoichiometric combinations of fuel and oxygen. If the combustion reaction is written according to Eq. (23-5), then the intersection of the stoichiometric line with the oxygen axis is given by

$$100 \left(\frac{z}{1+z} \right) \quad (23-9)$$

The LFL and UFL are drawn on the air line from the fuel axis values.

The flammability zone for most hydrocarbon vapors is shown as drawn in Fig. 23-4. Any concentration within the flammability zone is defined as flammable.

The LOC is the oxygen concentration at the very nose of the flammability zone. It is found from a line drawn from the nose of the flammability zone to the oxygen axis.

Crowl (*Understanding Explosions*, American Institute of Chemical Engineers, New York, 2003, App. A) derived a number of rules for using flammability diagrams:

1. If two gas mixtures R and S are combined, the resulting mixture composition lies on a line connecting points R and S on the flammability diagram. The location of the final mixture on the straight line depends on the relative moles in the mixtures combined: If mixture S has more moles, the final mixture point will lie closer to point S . This is identical to the lever rule used for phase diagrams.

2. If a mixture R is continuously diluted with mixture S , the mixture composition follows along the straight line between points R and S on the flammability diagram. As the dilution continues, the mixture composition moves closer and closer to point S . Eventually, at infinite dilution the mixture composition is at point S .

3. For systems having composition points that fall on a straight line passing through an apex corresponding to one pure component, the other two components are present in a fixed ratio along the entire line length.

Figure 23-5 shows how nitrogen can be used to avoid the flammable zone during the vessel preparation for maintenance. In this case nitrogen is pumped into the vessel until a concentration is reached at point S . Then air can be pumped in, arriving at point R . Figure 23-6 shows the reverse procedure. Now nitrogen is added until point S is reached, then fuel is pumped in until point R is reached. In both cases the flammable zone is avoided.

A complete flammability diagram requires hundreds of tests in a combustion sphere [Mashuga and Crowl, "Application of the Flammability Diagram for the Evaluation of Fire and Explosion Hazards of Flammable Vapors," *Proc. Safety Prog.*, 17 (3): 176–183 (1998)]. However, an approximate diagram can be drawn by using the LFL, UFL, LOC, and flammability limits in pure oxygen. The following procedure is used:

1. Draw the flammability limits in air as points on the air line, using the fuel axis values.

2. Draw the flammability limits in pure oxygen as points on the oxygen scale, using the fuel axis values. Table 23-2 provides a number of values for the flammability limits in pure oxygen. These are drawn on the oxygen axis using the fuel axis concentrations.

3. Use Eq. (23-9) to draw a point on the oxygen axis, and then draw the stoichiometric line from this point to the 100 percent nitrogen apex.

4. Locate the LOC on the oxygen axis. Draw a line parallel to the fuel axis until it intersects the stoichiometric line. Draw a point at the intersection.

5. Connect the points to estimate the flammability zone. In reality, not all the data are available, so a reduced form of the above procedure is used to draw a partial diagram (Crowl, *Understanding Explosions*, American Institute of Chemical Engineers, New York, 2003, p. 27).

Ignition Sources and Energy Table 23-3 provides a list of the ignition sources for major fires. As seen in Table 23-3, ignition sources are very common and cannot be used as the only method of fire prevention.

TABLE 23-2 Flammability Limits in Pure Oxygen

Compound	Formula	Limits of flammability in pure oxygen	
		Lower	Upper
Hydrogen	H ₂	4.0	94
Carbon monoxide*	CO	15.5	94
Ammonia	NH ₃	15.0	79
Methane	CH ₄	5.1	61
Ethane	C ₂ H ₆	3.0	66
Ethylene	C ₂ H ₄	3.0	80
Propylene	C ₃ H ₆	2.1	53
Cyclopropane	C ₃ H ₆	2.5	60
Diethyl ether	C ₄ H ₁₀ O	2.0	82
Divinyl ether	C ₄ H ₆ O	1.8	85

*The limits are insensitive to p_{H_2O} above a few millimeters of mercury.

Data from B. Lewis and G. von Elbe, *Combustion, Flames and Explosions of Gases* (New York: Harcourt Brace Jovanovich, 1987).

enough to remain suspended in air for prolonged periods. Mists are suspended liquid droplets produced by condensation of vapor into liquid or by the breaking up of liquid into a dispersed state by splashing, spraying, or atomizing.

For liquid droplets with diameters less than 0.01 mm, the LFL is virtually the same as the substance in vapor form. For mechanically formed mists with drop diameters between 0.001 and 0.2 mm, the LFL decreases as the drop diameter increases. In experiments with larger drop diameters the LFL was less than one-tenth of the vapor LFL. Thus, suspended droplets have a profound effect on flammability.

Explosions

Introduction Gas explosions depend on a large number of parameters, including temperature, pressure, gas composition, ignition source, geometry of surroundings, turbulence in the gas, mixing, time before ignition, and so forth. Thus, gas explosions are difficult to characterize and predict.

An explosion occurs when energy is released into the gas phase in a very short time, typically milliseconds or less. If the energy is released into the gas phase, the energy causes the gases to expand very rapidly, forcing back the surrounding gas and initiating a pressure wave that moves rapidly outward from the blast origin. The pressure wave contains energy which causes damage to the surroundings. A prediction of the damage effects from an explosion requires a clear understanding of how this pressure wave behaves.

TABLE 23-3 Ignition Sources of Major Fires*

Electrical (wiring of motors)	23%
Smoking	18
Friction (bearings or broken parts)	10
Overheated materials (abnormally high temperatures)	8
Hot surfaces (heat from boilers, lamps, etc.)	7
Burner flames (improper use of torches, etc.)	7
Combustion sparks (sparks and embers)	5
Spontaneous ignition (rubbish, etc.)	4
Cutting and welding (sparks, arcs, heat, etc.)	4
Exposure (fires jumping into new areas)	3
Incendiarism (fires maliciously set)	3
Mechanical sparks (grinders, crushers, etc.)	2
Molten substances (hot spills)	2
Chemical action (processes not in control)	1
Static sparks (release of accumulated energy)	1
Lightning (where lightning rods are not used)	1
Miscellaneous	1

**Accident Prevention Manual for Industrial Operations* (Chicago: National Safety Council, 1974).

TABLE 23-4 Minimum Ignition Energy (MIE) for Selected Gases

Chemical	Minimum ignition energy, mJ
Acetylene	0.020
Benzene	0.225
1,3-Butadiene	0.125
<i>n</i> -Butane	0.260
Cyclohexane	0.223
Cyclopropane	0.180
Ethane	0.240
Ethene	0.124
Ethylacetate	0.480
Ethylene oxide	0.062
<i>n</i> -Heptane	0.240
Hexane	0.248
Hydrogen	0.018
Methane	0.280
Methanol	0.140
Methyl acetylene	0.120
Methyl ethyl ketone	0.280
<i>n</i> -Pentane	0.220
2-Pentane	0.180
Propane	0.250

Data from I. Glassman, *Combustion*, 3d ed. (New York: Academic Press, 1996).

Detonation and Deflagration The difference between a detonation and deflagration depends on how fast the pressure wave moves out from the blast origin. If the pressure wave moves at a speed less than the speed of sound in the ambient gas, then a deflagration results. If the pressure wave moves at a speed greater than the speed of sound in the ambient gas, then a detonation results.

For ideal gases, the speed of sound is a function of temperature and molecular weight only. For air at 20°C the speed of sound is 344 m/s (1129 ft/s).

For a detonation, the reaction front moves faster than the speed of sound, pushing the pressure wave or shock front immediately ahead of it. For a deflagration, the reaction front moves at a speed less than the speed of sound, resulting in a pressure wave that moves at the speed of sound, moving away from the reaction front. A noticeable difference is found in the resulting pressure-time or pressure-distance plots.

The difference in behavior between a detonation and deflagration results in a significant difference in the damage. For a detonation, the damage is usually localized. However, for a deflagration, the damage is more widespread.

For high explosives, such as TNT, detonations are the normal result. However, for flammable vapors, deflagrations are more common.

Confined Explosions A confined explosion occurs in a building or process. Empirical studies on deflagrations (Tang and Baker, "A New Set of Blast Curves from Vapor Cloud Explosions," 33d Loss Prevention Symposium, AIChE, 1999; Mercx, van Wees, and Opschoor, "Current Research at TNO on Vapour Cloud Explosion Modeling," *Plant/Operations Progress*, October 1993) have shown that the behavior of the explosion is highly dependent on the degree of confinement. Confinement may be due to process equipment, buildings, storage vessels, and anything else that impedes the expansion of the reaction front.

These studies have found that increased confinement leads to flame acceleration and increased damage. The flame acceleration is caused by increased turbulence which stretches and tears the flame front, resulting in a larger flame front surface and an increased combustion rate. The turbulence is caused by two phenomena. First, the unburned gases are pushed and accelerated by the combustion products behind the reaction front. Second, turbulence is caused by the interaction of the gases with obstacles. The increased combustion rate results in additional turbulence and additional acceleration, providing a feedback mechanism for even more turbulence.

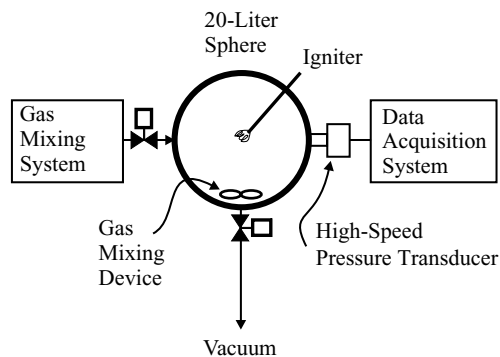


FIG. 23-7 An apparatus for collecting explosion data for gases and vapors. [D. A. Crowl, *Understanding Explosions*, Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE); copyright 2003 AIChE and reproduced with permission.]

Characterizing Explosive Behavior for Vapors and Dusts

Figure 23-7 is a schematic of a device used to characterize explosive vapors. This vessel is typically 3 to 20 L. It includes a gas handling and mixing system (not shown), an igniter to initiate the combustion, and a high-speed pressure transducer capable of measuring the pressure changes at the millisecond level.

The igniter can be of several types, including a fuse wire, spark, or chemical ignition system. A typical energy for ignition is 10 J, although gases can be ignited at much lower levels.

The gas is metered into the chamber to provide a mixture of a known composition. At a specified time the igniter is activated, and data are collected from the pressure transducer.

A typical pressure time plot is shown in Fig. 23-8. After ignition, the pressure increases rapidly, reaches a peak, and then diminishes as the reaction products are consumed and the gases are quenched and cooled by the vessel wall.

The experiment is repeated over a range of concentrations. A plot of the maximum pressure versus fuel concentration is used to determine the flammability limits, as shown in Fig. 23-9. A pressure increase of 7 percent over initial ambient pressure is used to define

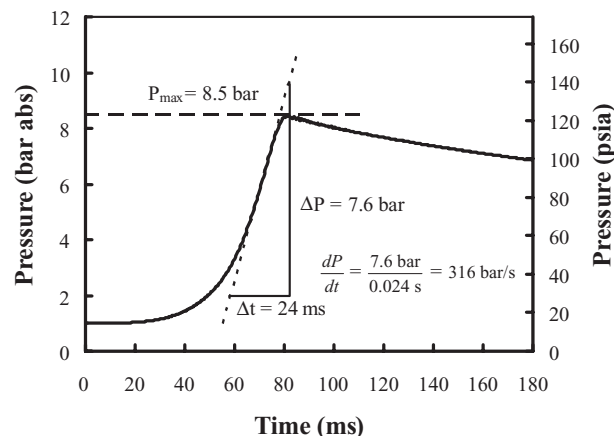
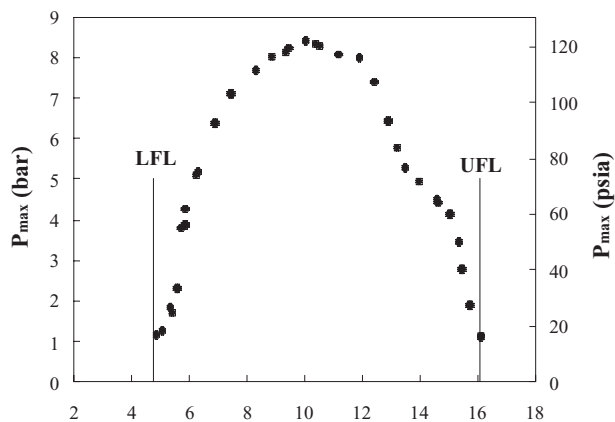


FIG. 23-8 Typical pressure versus time data obtained from gas explosion apparatus shown in Fig. 23-9. [Daniel A. Crowl and Joseph F. Louvar, *Chemical Process Safety: Fundamentals with Applications*, 2d ed., © 2002. Adapted by permission of Pearson Education, Inc., Upper Saddle River, N.J.]



Methane Concentration in Air

FIG. 23-9 Maximum pressure as a function of volume percent concentration for methane in air in a 20-L test sphere. The initial temperature and pressure are 25°C and 1 atm. The stoichiometric concentration is 9.51% methane. [C. V. Mashuga and D. A. Crowl, "Application of the Flammability Diagram for Evaluation of Fire and Explosion Hazards of Flammable Vapors," *Process Safety Progress*, vol. 17, no. 3; copyright 1998 American Institute of Chemical Engineers (AIChE) and reproduced with permission.]

the flammability limits (ASTM E918, *Standard Procedure for Determining Flammability Limits at Elevated Temperature and Pressure*, American Society for Testing and Materials, Philadelphia, 1992).

Figure 23-10 shows a device used to characterize the combustion of dusts. In this case, the dusts are initially contained in a small carrier external to the vessel, and the dust is blown in just prior to ignition. A typical pressure-time curve for the dust apparatus is shown in Fig. 23-11. The vessel is initially at a pressure less than atmospheric, but the pressure increases to atmospheric after the dust sample is blown in. After the dust is blown in, a delay time occurs in order for the dust to become quiescent but still suspended in the gas. The results are highly dependent on the delay time.

Two parameters are used to characterize the combustion for both the vapor and dust cases. The first is the maximum pressure during the combustion process, and the second is the maximum rate of

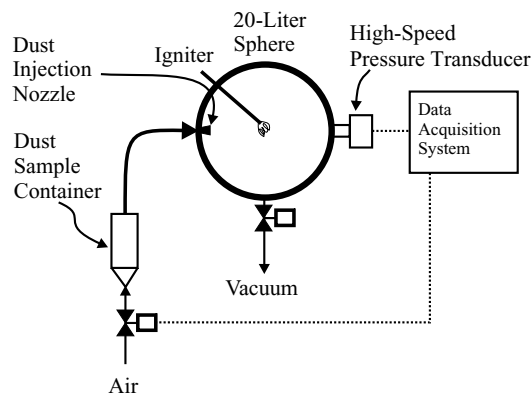


FIG. 23-10 An apparatus for collecting explosion data for dusts. [D. A. Crowl, *Understanding Explosions*, Center for Chemical Process Safety (CCPS) of the American Institute of Chemical Engineers (AIChE); copyright 2003 AIChE and reproduced with permission.]

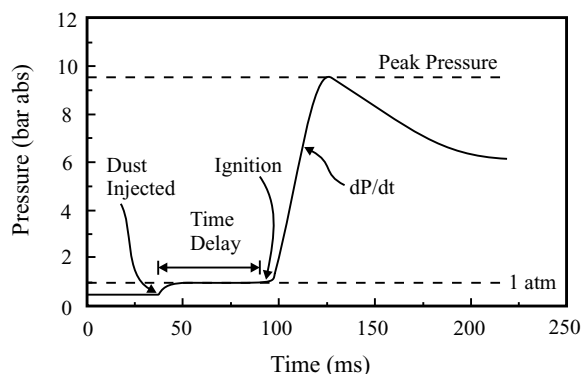


FIG. 23-11 Pressure data from dust explosion device. (Daniel A. Crowl and Joseph F. Louvar, *Chemical Process Safety: Fundamentals with Applications*, 2d ed., © 2002. Adapted by permission of Pearson Education, Inc., Upper Saddle River, N.J.)

pressure increase. Empirical studies have shown that a deflagration index can be computed from the maximum rate of pressure increase:

$$K_G \text{ or } K_{St} = \left(\frac{dP}{dt} \right)_{\max} V^{1/3} \quad (23-10)$$

where K_G = deflagration index for gases (bar·m/s)

K_{St} = deflagration index for dusts (bar·m/s)

P = pressure (bar)

t = time (s)

V = the vessel volume (m^3)

The higher the value of the deflagration index, the more robust the combustion. Table 23-5 contains combustion data for gases while Table 23-6 contains combustion data for dusts.

Vapor Cloud Explosions A vapor cloud explosion (VCE) occurs when a large quantity of flammable material is released, is mixed with enough air to form a flammable mixture, and is ignited. Damage from a VCE is due mostly to the overpressure, but significant damage to equipment and personnel may occur due to thermal radiation from the resulting fireball.

A VCE requires several conditions to occur (*Estimating the Flammable Mass of a Vapor Cloud*, American Institute of Chemical Engineers, New York, 1999):

1. The released material must be flammable.
2. A cloud of sufficient size must form prior to ignition.
3. The released material must mix with an adequate quantity of air to produce a sufficient mass in the flammable range.
4. The speed of the flame propagation must accelerate as the vapor cloud burns. This acceleration can be due to turbulence, as discussed in the section on confined explosions. Without this acceleration, only a flash fire will result.

Most VCEs involving flammable liquids or gases result only in a deflagration—detonations are unlikely. As the confinement of the vapor cloud increases, due to congestion from process equipment, the flame accelerates and higher overpressures are achieved. The higher overpressures may approach the severity of a detonation.

Four methods are available to estimate the damage from a VCE: TNT equivalency, TNO Multi-Energy method, Baker-Strehlow-Tang method, and computational fluid dynamics. The TNT equivalency method is discussed in the Estimation of Damage Effects section. The other methods are discussed elsewhere (*Guidelines for Evaluating the Characteristics of Vapor Cloud Explosions, Flash Fires and BLEVES*, American Institute of Chemical Engineers, New York, 1994; *Guidelines for Chemical Process Quantitative Risk Analysis*, 2d ed., American Institute of Chemical Engineers, 2000).

Boiling-Liquid Expanding-Vapor Explosions A boiling-liquid expanding-vapor explosion, commonly called a BLEVE (pronounced ble-vee), occurs when a vessel containing a liquid stored at a temperature above its normal boiling point fails catastrophically. After failure, a fraction of the liquid flashes almost instantaneously into vapor. Damage may be caused, in part, by the rapid expansion of the vapor and fragments from the failing vessel. The liquid may be water.

TABLE 23-5 Maximum Pressures and Deflagration Indices for a Number of Gases and Vapors

Chemical	Maximum pressure $P_{\max}$, barg			Deflagration index K_G , bar·m/s		
	NFPA 68 1997	Bartknecht 1993	Senecal 1998	NFPA 68 1997	Bartknecht 1993	Senecal 1998
Acetylene	10.6			109		
Ammonia	5.4			10		
Butane	8.0	8.0		92	92	
Carbon disulfide	6.4			105		
Diethyl ether	8.1			115		
Ethane	7.8	7.8	7.4	106	106	78
Ethyl alcohol	7.0			78		
Ethyl benzene	6.6	7.4		94	96	
Ethylene			8.0			171
Hydrogen	6.9	6.8	6.5	659	550	638
Hydrogen sulfide	7.4			45		
Isobutane			7.4			67
Methane	7.05	7.1	6.7	64	55	46
Methyl alcohol		7.5	7.2		75	94
Methylene chloride	5.0			5		
Pentane	7.65	7.8		104	104	
Propane	7.9	7.9	7.2	96	100	76
Toluene		7.8			94	

Data selected from:

NFPA 68: *Venting of Deflagrations* (Quincy, Mass.: National Fire Protection Association, 1997).

W. Bartknecht, *Explosionsschutz: Grundlagen und Anwendung* (New York: Springer-Verlag, 1993).

J. A. Senecal and P. A. Beaulieu, "K_G: Data and Analysis," *31st Loss Prevention Symposium* (New York: American Institute of Chemical Engineers, 1997).

TABLE 23-6 Combustion Data for Dust Clouds*

Dust	Median particle size, μm	Minimum explosive dust conc., g/m^3	P_{max} , barg	K_{St} , $\text{bar}\cdot\text{m/s}$	Minimum ignition energy, mJ
<i>Cotton, Wood, Peat</i>					
Cotton	44	100	7.2	24	—
Cellulose	51	60	9.3	66	250
Wood dust	33	—	—	—	100
Wood dust	80	—	—	—	7
Paper dust	<10	—	5.7	18	—
<i>Feed, Food</i>					
Dextrose	80	60	4.3	18	—
Fructose	200	125	6.4	27	180
Fructose	400	—	—	—	>4000
Wheat grain dust	80	60	9.3	112	—
Milk powder	165	60	8.1	90	75
Rice flour	—	60	7.4	57	>100
Wheat flour	50	—	—	—	540
Milk sugar	10	60	8.3	75	14
<i>Coal, Coal Products</i>					
Activated carbon	18	60	8.8	44	—
Bituminous coal	<10	—	9.0	55	—
<i>Plastics, Resins, Rubber</i>					
Polyacrylamide	10	250	5.9	12	—
Polyester	<10	—	10.1	194	—
Polyethylene	72	—	7.5	67	—
Polyethylene	280	—	6.2	20	—
Polypropylene	25	30	8.4	101	—
Polypropylene	162	200	7.7	38	—
Polystyrene (copolymer)	155	30	8.4	110	—
Polystyrene (hard foam)	760	—	8.4	23	—
Polyurethane	3	<30	7.8	156	—
<i>Intermediate Products, Auxiliary Materials</i>					
Adipinic acid	<10	60	8.0	97	—
Naphthalene	95	15	8.5	178	<1
Salicylic acid	—	30	—	—	—
<i>Other Technical, Chemical Products</i>					
Organic dyestuff (blue)	<10	—	9.0	73	—
Organic dyestuff (red)	<10	50	11.2	249	—
Organic dyestuff (red)	52	60	9.8	237	—
<i>Metals, Alloys</i>					
Aluminum powder	<10	60	11.2	515	—
Aluminum powder	22	30	11.5	110	—
Bronze powder	18	750	4.1	31	—
Iron (from dry filter)	12	500	5.2	50	—
Magnesium	28	30	17.5	508	—
Magnesium	240	500	7.0	12	—
Silicon	<10	125	10.2	126	54
Zinc (dust from collector)	<10	250	6.7	125	—
<i>Other Inorganic Products</i>					
Graphite (99.5% C)	7	<30	5.9	71	—
Sulfur	20	30	6.8	151	—
Toner	<10	60	8.9	196	4
St Classes for Dusts					
Deflagration index K_{St} , $\text{bar}\cdot\text{m/s}$		St class			
0		St-0			
1–200		St-1			
200–300		St-2			
>300		St-3			

*Data selected from R. K. Eckoff, *Dust Explosions in the Process Industries* (Oxford, England: Butterworth-Heinemann, 1997).

The most damaging BLEVE occurs when a vessel contains a flammable liquid stored at a temperature above its normal boiling point. The vessel walls below the liquid level are maintained at a low temperature due to the rapid heat transfer to the liquid. However, the vessel walls exposed to the fire above the liquid level will heat rapidly due to the much lower heat transfer to the vapor. The vessel wall temperature

will increase to a point where the strength of the vessel wall is significantly reduced. The vessel wall will fail catastrophically, resulting in the flashing of a large quantity of flammable liquid into vapor. Since a fire is already present, the resulting vapor cloud will ignite almost immediately. Overpressures from the vessel failure may result, but most of the damage is caused by radiation from the resulting large fireball.

Dust Explosions

Nomenclature

A	Area of a vent opening, m^2
A_W	Effective vent area, m^2
A_K	Geometric vent area, m^2
$(dP/dt)_{\max}$	Maximum rate of pressure rise, $\text{bar}\cdot\text{s}^{-1}$
$K_{\max}$	Maximum explosion constant, $\text{m}\cdot\text{bar}\cdot\text{s}^{-1}$
$K_{\min}$	Minimum explosion constant, $\text{m}\cdot\text{bar}\cdot\text{s}^{-1}$
ℓ	Length of a pipe, m
L_D	Length of vent duct, m
L_{DS}	Maximum length of vent duct, m
L_F	Flame length, m
P_a	Activation overpressure, bar
P_{ext}	Maximum external peak of overpressure, bar
$P_{\max}$	Maximum explosion overpressure, bar
P_{red}	Reduced explosion overpressure, bar
$P_{\text{red,max}}$	Maximum reduced explosion overpressure, bar
P_{stat}	Static activation overpressure, bar
R	Distance to vent area, m
$T_{\max}$	Maximum permissible surface temperature, $^{\circ}\text{C}$
V	Vessel volume, m^3
v_c	Relative circumferential speed, $\text{m}\cdot\text{s}^{-1}$
W_F	Width of flame, m
α	Angle between the horizontal and vertical axes of vent area, deg

GENERAL REFERENCES: Bartknecht, *Dust Explosions*, Springer, New York, 1989. Bartknecht, *Explosionsschutz* (Explosion Protection), Springer, Berlin, 1993. Crowl and Louvar, *Chemical Process Safety*, Prentice-Hall, New Jersey, 1990. "Dust Explosions," 28th Annual Loss Prevention Symposium, Atlanta, 1994. Eckhoff, *Dust Explosions in the Process Industries*, 2d ed., Butterworth-Heinemann, London, 1997. Health, Safety and Loss Prevention in the Oil, Chemical and Process Industries, Butterworth/Heinemann, Singapore, 1993. NFPA 69, *Standard on Explosion Prevention Systems*, Quincy, Mass., 1997. Hattwig and Steen, eds., *Handbook of Explosion Prevention and Protection*, Wiley-Vch Verlag GmbH & Co. KGaA, Weinheim, 2004. VDI-Report 1601, *Safe Handling of Combustible Dust*, VDI-Verlag GmbH, Düsseldorf, 2001. VDI-Guideline 2263, *Dust Fires and Dust Explosions*, Beuth Verlag, Berlin, 1992. European Standard EN 1127-1, *Explosives atmospheres: Explosion prevention and protection*, Pt. 1: *Basic Concepts and Methodology*, 1997.

Definition of Dust Explosion A dust explosion is the rapid combustion of a dust cloud. In a confined or nearly confined space, the explosion is characterized by relatively rapid development of pressure with flame propagation and the evolution of large quantities of heat and reaction products. The required oxygen for this combustion is mostly supplied by the combustion air. The condition necessary for a dust explosion is a simultaneous presence of a dust cloud of proper concentration in air that will support combustion and a suitable ignition source.

Explosions are either deflagrations or detonations. The difference depends on the speed of the shock wave emanating from the explosion. If the pressure wave moves at a speed less than or equal to the speed of sound in the unreacted medium, it is a deflagration; if it moves faster than the speed of sound, the explosion is a detonation.

The term *dust* is used if the maximum particle size of the solids mixture is below 500 μm .

In the following, dusts are called combustible in the airborne state only if they require oxygen from the air for exothermic reaction.

Glossary

Activation overpressure P_a That pressure threshold, above the pressure at ignition of the reactants, at which a firing signal is applied to the suppressor(s).

Cubic low The correlation of the vessel volume with the maximum rate of pressure rise. $V^{1/3} \cdot (dP/dt)_{\max} = \text{constant} = K_{\max}$.

Dust explosions class, St Dusts are classified in accordance with the $K_{\max}$ values.

Explosion Propagation of a flame in a premixture of combustible gases, suspended dust(s), combustible vapor(s), mist(s), or mixtures thereof, in a gaseous oxidant such as air, in a closed or substantially closed vessel.

Explosion pressure resistant (EPR) Design of a construction following the calculation and construction directions for pressure vessels.

Explosion pressure-shock resistant (EPSR) Design of a construction allowing greater utilization of the material strength than the EPR design.

Maximum reduced explosion overpressure $P_{\text{red,max}}$ The maximum pressure generated by an explosion of a dust-air mixture in a vented or suppressed vessel under systematically varied dust concentrations.

Minimum ignition energy (MIE) Lowest electrical energy stored in a capacitor which, upon discharge, is just sufficient to effect ignition of the most ignitable atmosphere under specified test conditions.

Minimum ignition temperature of a dust cloud (MIT_c) The lowest temperature of a hot surface on which the most ignitable mixture of the dust with air is ignited under specified test conditions.

Minimum ignition temperature of a dust layer (MIT_L) The lowest temperature of a hot surface on which a dust layer is ignited under specified test conditions.

Static activation overpressure P_{stat} Pressure at which the retaining element breaks or releases such that the venting element is able to open when the rate of pressure rise is $\leq 0.1 \text{ bar/min}$.

Vent area A Area of an opening for explosion venting.

Venting efficiency EF Ratio of the effective vent area A_e to the geometric vent area A_k .

Venting element That part of vent area device that covers the vent area and opens under explosion conditions.

Vessel length-to-diameter ratio (L/D) The ratio of the longest linear dimension L (length, height) of a round vessel to its geometric or equivalent diameter D .

Prevention and Protection Concept against Dust Explosions

Explosion protection encompasses the measures implemented against explosion hazards in the handling of combustible substances and the assessment of the effectiveness of protective measures for the avoidance or dependable reduction of these hazards. The explosions protection concept is valid for all mixtures of combustible substances and distinguishes between

1. Measures that prevent or restrict formation of a hazardous explosive atmosphere
2. Measures that prevent the ignition of a hazardous, explosive atmosphere
3. Constructional measures that limit the effects of an explosion to a harmless level

From a safety standpoint, priority must be given to the measures in item 1. Item 2 cannot be used as a sole protective measure for flammable gas or solvent vapors in industrial practice with sufficient reliability, but can be applied as the sole protective measure when only combustible dusts are present if the minimum ignition energy of the dusts is high ($\geq 10 \text{ mJ}$) and the operating area concerned can easily be monitored.

If the measures under items 1 and 2, which are also known as preventive measures, cannot be used with sufficient reliability, the constructional measures must be applied.

In practice, in most cases it is sufficient to determine and judge systematically the explosion risk with a sequence of specific questions, shown in Fig. 23-12.

During the evaluation it is assumed that ignition of an existing combustible atmosphere is always possible. The assessment is thus independent of the question of whether ignition sources are present.

Preventive Explosion Protection The principle of preventive explosion protection comprises the reliable exclusion of one of the requirements necessary for the development of an explosion. An explosion can thus be excluded with certainty by

- Avoiding the development of explosive mixtures
- Replacing the atmospheric oxygen by inert gas, working in a vacuum, or using inert dust
- Preventing the occurrence of effective ignition sources

Avoidance or Reduction of Explosive Combustible Fuel-Air Mixtures

Flammable gas or vapor-air mixtures This can be achieved if the flammable substance can be replaced by a nonflammable substance or the concentration of flammable substance can be kept so low that the gas or vapor-air mixture is too lean for an explosion.

The development of a hazardous atmosphere can also be prevented by ventilation measures. A distinction is made here between *natural* ventilation, which is usually sufficient only in the open air, and *artificial*

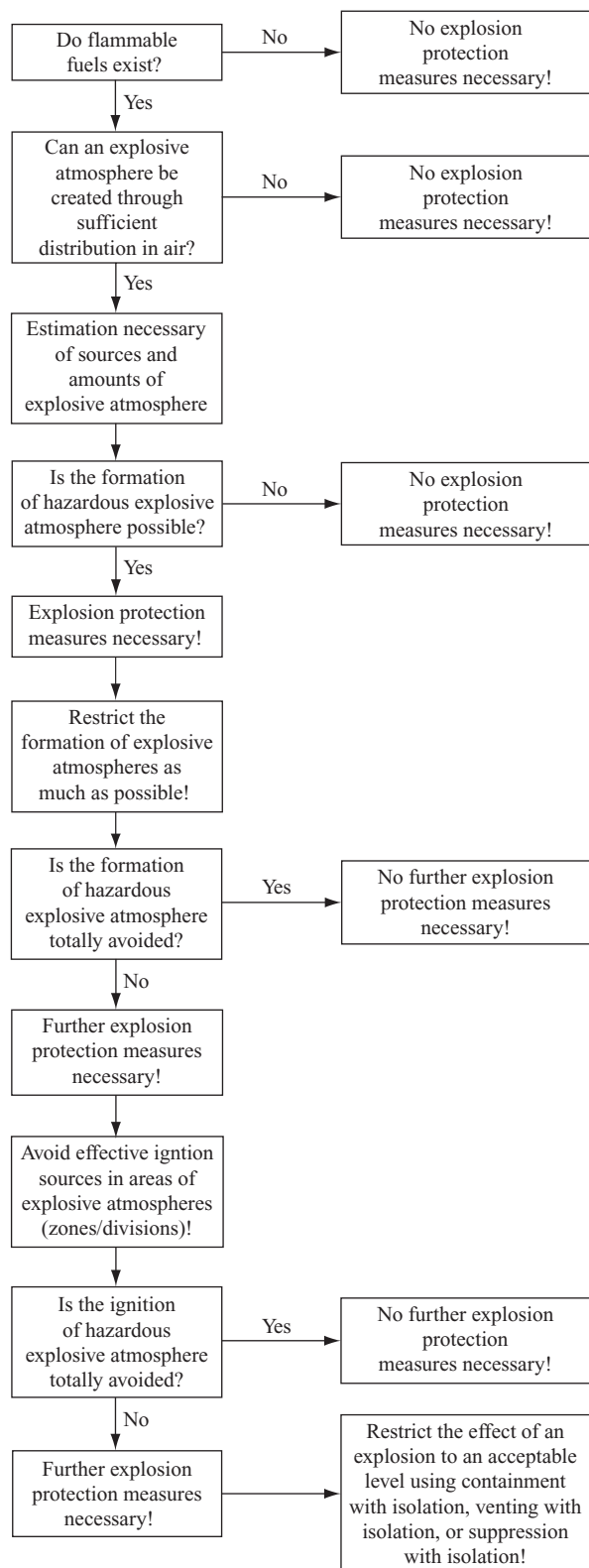


FIG. 23-12 Evaluation flowchart for preventing and/or limiting explosions.

ventilation (IEC 60079-10, "Electrical Apparatus for Explosive Atmospheres. Part 10: Classification of Hazardous Areas"). *Artificial* ventilation permits the use of greater amounts of air and the selective circulation of air in areas surrounding the equipment. Its use and the calculation of the minimum volume flow rate for the supply and exhaust air are subject to certain requirements.

The use of *gas alarm devices* in connection with, e.g., ventilation measures is also possible. The factors influencing the decision regarding setting of the required gas detectors include the relative density of the flammable gases and vapors. With gases and vapors that are heavier than air, the sensors and the waste air openings should be installed near the floor, with those that are lighter than air near the ceiling.

Combustible dust-air mixtures These mixtures can be avoided or restricted if the combustible dust can be replaced by a noncombustible dust or the dust concentration can be kept so low that an explosive dust-air mixture is never actually formed.

The explosibility limits do not have the same meaning as with flammable gases and flammable vapors, owing to the interaction between dust layers and suspended dust. This protective measure can, e.g., be used when dust deposits are avoided in operating areas or in the airstream of clean air lines after filter installation, where in normal operation the lower explosion limit (LEL) is not reached. However, dust deposits must be anticipated with time. When these dust deposits are whirled up in the air, an explosion hazard can arise. Such a hazard can be avoided by regular cleaning. The dust can be extracted directly at its point of origin by suitable ventilation measures.

Avoidance of Explosions through Inerting The introduction of inert gas in the area to be protected against explosions lowers the oxygen volume content below the limiting oxygen concentration (LOC) so that ignition of the mixture can no longer take place. This process is called *inerting* (CEN/TR 15281, "Guidance on Inerting for the Prevention of Explosions").

One has to be aware of the danger of asphyxiation from gases in inerted equipment. This is also important for surrounding areas in case of major leaks.

Inerting is not a protective measure to avoid exothermic decomposition. For the avoidance of (smoldering) fires, oxygen concentrations lower than the LOC must usually be adhered to and must be determined from case to case. In addition to the nitrogen normally used, all nonflammable gases which do not support combustion or react with the combustible dust can be considered for use as the inert gas. The inerting effect generally decreases in the following order: carbon dioxide → water vapor → flue gas → nitrogen → noble gases. In special cases, liquid nitrogen or dry ice is used.

The LOC depends upon the combustible material and the type of inert gas used. It decreases with increased temperature and pressure. A distinction has to be made between the determined LOC value and the concentration which results by subtracting a safety margin.

The maximum allowable oxygen concentration (MAOC), which is, in general, 2 vol % below the LOC, has to include the following considerations: Fluctuation in oxygen concentrations due to process and breakdown conditions per time and location, as well as the requirement for protective measures or emergency measures to become effective. In addition, a concentration level for an alarm has to be set below the MAOC.

Explosive dusts can also be changed into mixtures which are no longer explosive by the addition of inert dusts (e.g., rock salt, sodium sulfate). In general, inert dust additions of more than 50 wt % are necessary here. It is also possible to replace flammable solvents and cleaning agents by nonflammable halogenated hydrocarbons or water, or flammable pressure transmission fluids by halocarbon oils.

Avoidance of Effective Ignition Sources Explosions can be prevented if ignition sources capable of igniting combustible material-air mixtures can successfully be avoided. A distinction is made between trivial ignition sources (e.g., welding, smoking, cutting) and mechanically generated spark, mechanically generated hot surfaces, lumps of smoldering material, and static electricity. Trivial ignition sources can also reliably be excluded by organizational measures such as the systematic employment of permits. This measure should always be employed, even when constructional measures are applied, unless the explosive atmosphere is avoided with certainty.

TABLE 23-7 Zone Classification versus Minimum Extent of Preventive Measures against Danger of Ignition

Explosion hazard zone	Avoid ignition sources which can occur:
20/0	During normal operation, during frequent malfunction, and during rare malfunction
21/1	During normal operation and during frequent malfunction
22/2	During normal operation

Hazardous places are classified in terms of zones (divisions) on the basis of the frequency and duration of the occurrence of an explosive atmosphere (IEC 60079-10:200X, "Classification of Hazardous Areas"; European Standard EN 50281-3:200X, "Classification of Areas Where Combustible Dusts Are or May Be Present").

Fundamentally all 13 kinds of ignition sources mentioned and also described in detail in EN 1127-1 must be considered. In EN 1127-1, the descriptions refer to both the ignition mechanisms of the different kinds of ignition source and the necessary scope of protection (material- and zone-dependent). The ignition sources should be classified according to the likelihood of their occurrence in the following manner:

1. Sources of ignition which can occur *continuously or frequently*
2. Sources of ignition which can occur in *rare* situations
3. Sources of ignition which can occur in *very rare* situations

In terms of the equipment, protective systems, and components used, this classification must be considered equivalent to

1. Sources of ignition which can occur during *normal operation*
2. Sources of ignition which can occur solely as a result of *malfunctions*
3. Sources of ignition which can occur solely as a result of *rare malfunctions*

Ignition sources at devices, protective systems, and components must be avoided, depending on zones (Table 23-7); i.e., the zone determines the minimum extent of preventive measures against danger of ignition.

Flammable gas or vapor-air mixtures Due to their low minimum ignition energies (<1 mJ), avoidance of effective ignition sources in flammable gas or vapor-air mixtures is in principle possible only in exceptional cases. For *hot surfaces* a maximum permissible surface temperature $T_{\max}$ must be specified, with the help of the minimum ignition temperature (MIT) of flammable gases, vapors, or liquids, so that the temperature of all surfaces is not exceeded (Table 23-8).

Combustible dust-air mixtures For every installation a check has to be made to determine which ignition source may become effective and whether it can be prevented with a sufficient degree of safety. With more sensitive products and complex installations, it becomes more and more difficult to exclude ignition sources with ample safety.

With dusts the avoidance of effective ignition sources depends on the ignition sensitivity, i.e., on the temperature-related minimum ignition energy (MIE). Because a low MIE means that both the number of effective ignition sources rises and the probability of the release of an ignition of a dust-air mixture rises. Safe handling of particularly or extremely ignition-sensitive dusts requires in general increased use of preventive measures in comparison to normally ignitable sensitive dusts.

It is therefore usual to combine the evaluation of the ignition sensitivity (MIE) of the dusts with the type and extent of protective measures in accordance with Table 23-9.

TABLE 23-8 Measures against Ignition of Flammable Gas, Vapors, and Liquids by Hot Surfaces

Zone	Protection by limiting the hot surface temperature (EN 1127-1)
0	$T_{\max} \leq 0.8 \text{ MIT}$
1	$T_{\max} \leq 0.8 \text{ MIT}$
2	$T_{\max} \leq \text{MIT}$

TABLE 23-9 Ignition Sensitivity of Dusts as a Function of Protective Measures

Ignition sensitivity of dusts	Protective measures
Normally sensitive (MIE $\geq 10 \text{ mJ}$)	Avoidance of effective ignition sources.
Easily sensitive (3 mJ $\leq$ MIE $< 10 \text{ mJ}$)	Expert decision or Avoidance of effective ignition sources and additional protective measures.
Extremely sensitive (MIE $< 3 \text{ mJ}$)	Avoidance of effective ignition sources and additional protective measures.

Protective measures: Inerting or constructional explosion protection (e.g., containment).

For hot surfaces a maximum permissible surface temperature $T_{\max}$ must be specified, with the help of the minimum ignition temperature of a dust cloud MIT_C and the minimum ignition temperature of a dust layer MIT_L (usually 5-mm dust layer = glow temperature), so that the temperature of all surfaces is not exceeded. Independent of the zone the $T_{\max}$ for dust clouds is $\frac{2}{3}$ MIT_C and for dust layers MIT_L – 75 K.

Mechanically generated sparks and resultant *hot surfaces* together are regarded as one of the more important causes of ignition in industrial practice. The hot surfaces show considerably better incendiarity in comparison with the short-lived, mechanically generated sparks. Neither ignition source appears in industrial practice from the normal metallic materials of construction rubbing against each other or against stone if the relative circumferential speeds v_c are less than or equal to 1 m/s and the power requirement is no more than 4 kW (Table 23-10). This is not valid for cerium-iron, titanium, and zirconium.

Regarding *electrostatic ignition sources*, see the subsection "Static Electricity," later. In addition, see CENELEC CLC/TR 50404, "Electrostatics—Code of Practice for the Avoidance of Hazards due to Static Electricity," June 2003.

Explosion Protection through Design Measures Design measures which restrict the effects of an explosion to a safe level are always necessary when the goal of avoiding explosions cannot be achieved—or at least not with sufficient reliability—through the use of preventive explosion protection. This ensures that people are not injured and further that the protected equipment is usually ready for operation a short time after an explosion. In applying design measures, the possibility of an explosion is not prevented. Therefore, all exposed equipment has to be built to be explosion-pressure-resistant in order to withstand the anticipated explosion pressure. The anticipated explosion pressure may be the maximum explosion overpressure or the maximum reduced explosion overpressure. In addition, *any propagation of an explosion to other parts or process areas has to be prevented*. Depending on the anticipated explosion pressure, a distinction is made between the following explosion-pressure-resistant designs:

- Those capable of withstanding the maximum explosion overpressure
- Those capable of withstanding an explosion overpressure reduced by explosion suppression or explosion venting

The strength of the protected vessels or apparatus may be either explosion-pressure-resistant or explosion pressure shock-resistant.

Containment Containment is understood to mean the possibility of designing vessels and equipment for the full *maximum explosion overpressure*, which is generally from $P_{\max} = 7$ to 10 bar. The explosion-resistant vessel can then be designed as explosion-pressure-resistant or

TABLE 23-10 Influence of Relative Circumferential Speed v_c on Danger of Ignition for Combustible Dusts

$*v_c \leq 1 \text{ m/s}$	There is no danger of ignition.
$v_c > 1-10 \text{ m/s}$	Every case has to be judged separately, considering the product and material-specific characteristics.
$v_c > 10 \text{ m/s}$	In every case there is danger of ignition.

* In addition, low power requirements $W \leq 4 \text{ kW}$.